

11TH EDITION



Present Knowledge in Nutrition



Volume 1: Basic Nutrition and Metabolism

Edited by **Bernadette P. Marriott, Diane F. Birt,
Virginia A. Stallings and Allison A. Yates**



PRESENT KNOWLEDGE IN NUTRITION

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BASIC NUTRITION AND METABOLISM

VOLUME 1

ELEVENTH EDITION

Edited by

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We dedicate this 11th edition of Present Knowledge in Nutrition to the members of the global nutrition community who continue to seek the best science, interpret that science for the better good of people worldwide, and persist in countering nonscientifically based nutrition information with evidence-based approaches to better human health. We also dedicate this edition to our own scientific mentors and colleagues who have persuaded and occasionally pushed us in the direction of the highest quality nutrition science—no matter the cost.

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Editor Biographies

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Bernadette P. Marriott holds the position of Professor Emerita and Nutrition Section Director Emerita, Departments of Medicine and Psychiatry, Medical University of South Carolina. Bernadette has over 35 years of experience in the fields of nutrition, psychology, and comparative medicine with expertise in diet, nutrition, and chronic disease. Dr. Marriott has worked in scientific and administration positions in the federal government, the National Academies, universities, and foundations. She was founding director of the Office of Dietary Supplements, NIH and Deputy Director, Food and Nutrition Board, NAS. Her research has focused on both human and animal nutrition and related behavioral studies (in humans: diet and health research and food labeling; in animals: nonhuman primate nutrition and behavioral ecology). She is currently leading or has recently led research projects funded by the Army, DoD, NSF, NIH, USDA, industry, and foundations. Bernadette Marriott has a BSc in biology/immunology from Bucknell University (1970), a PhD in psychology from the University of Aberdeen, Scotland (1976), and postgraduate training in trace mineral nutrition, comparative medicine, and advanced statistics. She has published extensively, is on a number of national committees and university scientific advisory boards, and is a frequent speaker on diet, dietary supplements, and health. She is currently a member of the Food and Nutrition Board, US National Academy of Sciences, and the American Society for Nutrition Committee on Advocacy and Science Policy. In 2016, Dr. Marriott was inducted as a Fellow of the American Society for Nutrition.



DIANE F. BIRT, PHD

Diane F. Birt is a Distinguished Professor in Food Science and Human Nutrition at Iowa State University. She has BS degrees in Home Economics and Chemistry from Whittier College (1971) and a PhD in Nutrition from Purdue University (1975). Her expertise is in diet and cancer prevention and plant components and health promotion. She was at the University of Nebraska Medical Center (1976–97) before becoming Chair of the Department of Food Science and Human Nutrition (1997–2004) at Iowa State University. Dietary prevention of cancer has been a long-standing interest in the Birt laboratory. More recent research has focused on the prevention of colon cancer by slowly digested maize starches using cell culture and animal models that reflect particular genetic changes that are important in human colon cancer development. She was on the Board of Scientific Counselors for the National Toxicology Program (US Department of Health) and the Food and Nutrition Board of the Institute of Medicine, US National Academy of Sciences. In 2015, Dr. Birt was inducted as a Fellow of the American Society for Nutrition, and in 2016, she was inducted as a member of the National Academy of Medicine.



VIRGINIA A. STALLINGS, MD, MS

Virginia Stallings is a Professor of Pediatrics at the Children's Hospital of Philadelphia and the Perelman School of Medicine at the University of Pennsylvania and is the recipient of the Jean A. Cortner Endowed Chair in Pediatric Gastroenterology and Nutrition. She holds a BS in Nutrition and Food from Auburn University, MS in Nutrition and Biochemistry from Cornell University, and MD from the University of Alabama Birmingham. Her general pediatric residency was completed at the University of Virginia followed by a subspecialty fellowship in nutrition at the Hospital for Sick Children. Over her career at the Children's Hospital of Philadelphia, she contributed to clinical care, fellow and faculty training, and clinical and translational research in the abnormalities of growth, nutritional status, and health of children with chronic diseases and in those in good health. She has served on many National Academy of Sciences committees to advise on child health, nutrition, and federal nutrition programs and is a member of the National Academy of Medicine. The American Academy of Pediatrics and American Society for Nutrition have recognized her efforts with awards for science, mentoring, and service.



ALLISON A. YATES, PHD, MSPH, RD

Allison A. Yates holds Bachelor's and Master's degrees from the University of California at Los Angeles in public health and dietetics and a PhD from the University of California at Berkeley in nutrition, and is a registered dietitian having completed a dietetic internship at the VA Center in Los Angeles. She served on the faculties of the University of Texas Health Science Center in Houston, Emory University School of Medicine, and was the founding Dean of the College of Health and Human Sciences at the University of Southern Mississippi, where she led the development of the first accredited public health program in the state. Her research focused on human protein and energy requirements. In 1994, she was named Director of the Food and Nutrition Board of the Institute of Medicine of the US National Academy of Sciences, where, over a 10-year period, she led the expanded approach to establishing human requirements and recommendations for nutrients, termed Dietary Reference Intakes, for the United States and Canada. She then served as Director of the Beltsville Human Nutrition Research Center of the US Department of Agriculture, Agricultural Research Service (ARS), and served as Associate Director for the ARS Beltsville Area region, retiring from USDA in 2014, when she was also inducted as a fellow of the American Society for Nutrition. Since that time, she has led a volunteer effort to establish a framework for establishing reference values for bioactive components in foods.



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Foreword

Timely information regarding nutrition is critical to improving human health and well-being and safeguarding the environment. The mission of the International Life Sciences Institute (ILSI) is in part to provide such information, and we are pleased to present the 11th edition of *Present Knowledge in Nutrition*.

First published in 1953, *Present Knowledge in Nutrition* was launched with the goal of providing readers with the most comprehensive and current information covering the broad fields within the nutrition discipline. Reflecting the global relevance of nutrition, this edition's authors are from a variety of countries and reflect a "who's who" of nutritional science.

ILSI is a worldwide nonprofit organization that seeks to foster science for the public good and collaboration among scientists, all governed by ILSI's core principles of scientific integrity. With 16 entities worldwide, ILSI published 70 scientific articles globally in 2019 and hosted 152 workshops addressing nutrition, food safety, and sustainability. ILSI is a world leader in creating public–private partnerships that advance science for the betterment of public health and achieve positive, real-world impact.

We trust the two volumes of *Present Knowledge in Nutrition* will be valuable resources for researchers, health professionals, clinicians, educators, and advanced nutrition students. ILSI is proud of the contributions made by the authors and editors of this key reference, and we are excited to advance the discipline of nutrition with this publication.

Kerr Dow, ILSI Board of Trustees Co-Chair

A handwritten signature in black ink, appearing to read "Kerr Dow".

Michael Doyle, ILSI Board of Trustees Co-Chair

A handwritten signature in black ink, appearing to read "Michael P. Doyle".

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Preface

As editors, we feel privileged to have been asked to edit the 11th edition of *Present Knowledge in Nutrition*. The 11th edition moves this major nutrition reference source beyond its 65-year history and into an explosion of exciting new methodologies and understandings of the role of diet and nutrition in human health and well-being. In a global survey conducted in 2017, *Present Knowledge in Nutrition* was identified as a key resource for the latest information in the nutrition field for nutrition and dietetic professionals and clinicians. Specifically, survey participants stated that *Present Knowledge in Nutrition* was the source to which they turned when seeking the latest information in an area of nutrition that was outside their main expertise. Further, *Present Knowledge in Nutrition* is valued as an academic text in advanced nutrition courses. Recognizing the important role of the periodic updates of *Present Knowledge in Nutrition* among scientists and practitioners, as editors we have sought to maintain the long-standing tradition of identifying content-thought leaders to provide the most comprehensive and latest information in their fields in the chapters represented in this edition.

The 11th edition of *Present Knowledge in Nutrition* is presented in two companion volumes: *Volume 1: Basic Nutrition and Metabolism* and *Volume 2: Clinical and Applied Topics in Nutrition*. Provision of two volumes enables the reader to more quickly identify the location of relevant materials and makes the printed copies of this 72-chapter edition more physically portable. This 11th edition includes full color illustrations and other color-enhanced features. At the end of each chapter, authors clearly have identified important research gaps and needs for future research. Volume 1 includes chapters

that provide the latest scientific knowledge on requirements for specific nutrients and genomics and chapters that discuss important cross-disciplinary topics including systems biology, the microbiome, and the role of nutrition in regulation of immune function. Volume 2 provides the most recent information on life-stage nutrition, obesity, physical activity, and eating behavior; dietary guidance; and nutrition surveillance, as well as major topics in nutrition and disease processes and medical nutrition therapy.

In addition to print volumes, the 11th edition of *Present Knowledge in Nutrition* is available in electronic format through the website: <https://pkn11.org/where-to-buy/>. The electronic format provides broader access not only globally but also for educational use.

We believe the authors have done an outstanding job in presenting the latest information in their respective fields and hope this edition will continue the long tradition of being an essential resource broadly in the nutrition field.

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Acknowledgments

Development of a book of this in-depth and highly scientifically current content represents a large commitment of time and effort by many people. First, we would like to thank the authors of the 72 chapters for their commitment to this two-volume reference work and most importantly their dedication to presenting the best information of the current status of the science in their respective fields. Second, this edition would not have come to fruition without the untiring work and guidance of Allison Worden and James Cameron of the International Life Sciences Institute. We appreciated very much the early guidance of two of the editors of the 10th edition, John Erdman and Steven Zeisel, as we were forming the 11th edition concept. Key to any endeavor of this size is the support of family, colleagues, and friends to whom we owe our gratitude for their forbearance during the many hours devoted to this work's development and production.

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S E C T I O N A

Macronutrients

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ENERGY METABOLISM

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SUMMARY

Man covers energy expenditure by the oxidation of the nutrients carbohydrate, protein, fat, and sometimes alcohol. Energy expenditure is assessed with indirect calorimetry, measuring oxygen consumption and/or carbon dioxide production, and with direct calorimetry, measuring heat loss from nutrient oxidation. Energy expenditure is a function of (1) body composition determining resting energy expenditure (REE), (2) food intake determining the energy cost of food processing, and (3) body movement determining activity-induced energy expenditure (AEE). Energy expenditure, and thus energy requirement, is generally higher in larger individuals, e.g., overweight and obese subjects, through a higher REE. Subjects in negative energy balance show reduced energy expenditure, mainly through a reduction of AEE. Physical activity is rather a function than a determinant of energy balance. The main determinant of energy balance is energy intake.

Keywords: Body composition; Diet-induced energy expenditure; Doubly labeled water; Energy requirement; Indirect calorimetry; Physical activity level; Resting energy expenditure.

I. INTRODUCTION

Man, as a heterotrophic organism, derives his energy by the oxidation of nutrients. The oxidation yields energy to synthesize ATP for energy metabolism and is finally mainly transformed into heat. Total energy expenditure (TEE) consists of three components: energy expenditure for the processing of food—diet-induced energy expenditure (DEE); energy expenditure for maintenance—resting energy expenditure (REE); and energy expenditure for physical activity—activity-induced energy expenditure (AEE).

The following sections describe the measurement of nutrient and energy metabolism, components of energy expenditure, energy requirement, and disease-related alterations in energy requirement.

II. NUTRIENT AND ENERGY METABOLISM

A. Nutrient Metabolism

Energy expenditure is covered by the oxidation of carbohydrate, protein, and fat—taken together, the

macronutrients—and sometimes alcohol. Methods for the measurement of energy expenditure are based on oxygen consumption, carbon dioxide production, and heat production of the metabolized nutrients. As an example, oxidation of 1 mol of glucose requires 6 mol of oxygen and produces 6 mol of water, 6 mol of carbon dioxide, and 2.8 MJ heat. Carbohydrate, fat, and alcohol are usually completely oxidized to water and carbon dioxide, whereas in the oxidation of protein, the nitrogen component is excreted in the urine, mainly as urea.

Thus, the contribution of the three macronutrients, carbohydrate, protein, and fat, to energy metabolism is derived from the three parameters: oxygen consumption; carbon dioxide production; and urine nitrogen loss.

As an illustration, the formula of Brouwer calculates macronutrient oxidation based on the relationships of the three end products of metabolism ([Box 1.1](#)).

Energy of ingested macronutrients is not fully available to energy metabolism, since part of it is lost in urine and feces. The energy loss in urine is mainly a function of protein catabolism to excrete the nitrogen, as mentioned above. Fecal energy loss is mainly a function of the consumption of nondigestible fiber. Metabolizable energy,

BOX 1.1

Brouwer formula for calculation of the contribution of carbohydrate, protein, and fat to energy metabolism as derived from the three parameters: oxygen consumption, carbon dioxide production, and urine nitrogen loss.¹²

Carbohydrate oxidation (g/d) = $4.170 V_{\text{CO}_2} - 2.965 V_{\text{O}_2} - 0.390 P$;

Protein oxidation (g/d) = $6.25 N$;

Fat oxidation (g/d) = $1.718 V_{\text{O}_2} - 1.718 V_{\text{CO}_2} - 0.315 P$

where V_{CO_2} = carbon dioxide production (L/d), corrected for alcohol oxidation; V_{O_2} = oxygen consumption (L/d), corrected for alcohol oxidation; P = protein oxidation (g/d); N = total nitrogen excreted in urine (g/d).

ingested energy minus energy loss in urine and feces, of the typical Western diet varies around 90% of gross dietary energy intake, the energy as measured with a bomb calorimeter. Surprisingly, energy loss in urine and feces is not affected by the amount of food consumed.^{1,2}

Food composition tables for the calculation of energy intake have traditionally used the *Atwater factors* to adjust for energy losses in urine and feces and thus convert gross nutrient intake to metabolizable energy intake.³ The *Atwater factors* for carbohydrate, protein, fat, and alcohol are, respectively, 17, 17, 37, and 29 kJ (4, 4, 9, and 7 kcal)/g.

There is a hierarchy in nutrient metabolism after consumption, reflecting the storage capacity in the body.⁴ Alcohol, as a toxin, is readily absorbed and is principally eliminated by metabolism in the liver. Since ethanol is not stored in the body, it must be oxidized. Thus, alcohol ingestion results in short-term reduction of oxidation of the other nutrients. Carbohydrate and protein intake cause an increase in their oxidation by autoregulation. The increase in their oxidation is governed by the relative small storage capacity of carbohydrate as glycogen and protein in the labile protein and amino acid pool.⁴ Fat is at the bottom of the oxidation hierarchy. Alcohol, carbohydrate, and protein oxidation suppress fat oxidation.⁵ Fat oxidation is a function of the presence or absence of the other energy-providing nutrients. Fat oxidation is mainly a function of energy balance in that when intake exceeds expenditure, excess energy is stored by storing ingested fat. When intake is lower than expenditure, mobilized body fat, which in general is abundantly available, makes up the difference.

Humans are discontinuous eaters and continuous metabolizers. Thus, part of the energy intake has to be stored before it is used. Within a day, intervals with a positive energy balance after a meal alternate with intervals with a negative energy balance, such as overnight. There is a marked diurnal pattern of nutrient metabolism, with a higher rate of carbohydrate oxidation after meals and a higher rate of fat oxidation during the night and in the early morning, before food consumption.⁶ Energy intake

usually balances energy expenditure over several days up to a week. Energy intake strongly correlates with energy expenditure on a weekly basis.⁷ For example, military cadets did not show an increase in energy intake on days with higher energy expenditure; the matching energy intake came some days afterward.⁷ Similarly, a change in fat intake does not immediately affect fat oxidation. It took 7 days before fat oxidation was raised sufficiently in young lean subjects to match fat intake, when diet composition was isoenergetically switched from low-fat to high-fat diet.⁸

In conclusion, measuring oxygen consumption, carbon dioxide production, and urinary nitrogen loss allows assessment of energy and nutrient metabolism. Results are affected by energy balance, and thus, especially for the assessment of nutrient metabolism, measurements should cover at least a full 24 h cycle and possibly up to a week.

B. Energy Metabolism

Energy production for energy expenditure can be assessed by measuring heat loss from the oxidation of nutrients (direct calorimetry) or by measuring gas exchange in the oxidation of nutrients (indirect calorimetry). One of the first devices to measure energy expenditure was a direct calorimeter for small animals.

Direct calorimetry

As early as 1780, Lavoisier measured heat loss of a guinea pig by placing the animal in a wire cage surrounded by ice chunks. Energy expenditure was calculated from the amount of ice melted by the animal's body heat. Later, direct calorimetry systems, including human calorimeters, used airflow or water flow to measure heat exchange.⁹

An airflow calorimeter consists of a temperature-insulated ventilated space—for instance, a room large enough to house a subject. The temperature change of air flowing through the room multiplied by its mass flow rate and specific heat gives the rate of heat loss from the subject.⁹

BOX 1.2

Equations to calculate total energy expenditure (TEE) from oxygen consumption, carbon dioxide production, and urinary nitrogen loss.

deV Weir⁶⁷: $TEE \text{ (kJ)} = 16.32 V_{O_2} \text{ consumption (L)} + 4.60 V_{CO_2} \text{ production (L)} - 2.17 \text{ urinary nitrogen loss (g)}$

Brouwer¹²: $TEE \text{ (kJ)} = 16.20 V_{O_2} \text{ consumption (L)} + 5.00 V_{CO_2} \text{ production (L)} - 0.15 \text{ urinary nitrogen loss (g)}$

An example of a water flow calorimeter is a suit calorimeter. The subject is dressed in a close-fitting elastic undergarment, which carries a network of small plastic tubing over the entire body surface, except for the face, fingers, and soles of the feet. Water circulated through the tubing carries heat from the skin, which is measured as the product of the mass flow of water and the change in temperature across the suit. An insulating outside garment limits the exchange of heat with the environment.¹⁰

Indirect calorimetry

Presently, state of the art is the assessment of energy expenditure with indirect calorimetry because body heat loss, when measured with direct calorimetry, does not include power output for external work during exercise. Body heat loss can be up to 25% lower than TEE during endurance exercise.¹¹

In indirect calorimetry, energy expenditure is calculated from nutrient metabolism by measuring oxygen consumption, carbon dioxide production, and urinary nitrogen loss. Examples of equations that are used to calculate the relationships are the Weir equation and the Brouwer equation (Box 1.2).

Usually, for the calculation of energy expenditure, one does not measure urinary nitrogen loss, so the protein correction is neglected. It is difficult to measure urinary nitrogen loss, especially over intervals shorter than a day, and the resulting error is small. For a subject in energy balance consuming a regular diet containing 10%–15% of the energy from protein, the protein correction is smaller than 1% of energy expenditure.¹² Thus, energy expenditure is usually assessed by measuring oxygen consumption and carbon dioxide production.

In a recent variant of indirect calorimetry, the doubly labeled water method, energy expenditure is primarily calculated from measured carbon dioxide production.¹³ With this method, the energy equivalent of carbon dioxide used in the calculation of energy expenditure is based on an estimate of nutrient metabolism. The energy equivalent of carbon dioxide ranges between a lower value of 21.1 kJ/L for carbohydrate to a higher value of 27.8 kJ/L for fat.¹² For an average diet of energy from carbohydrate (55%), protein (10%–15%), and fat (30%–35%), the energy equivalent of carbon dioxide is 23.6 kJ/L.

The doubly labeled water method is based on the observation that oxygen in respiratory carbon dioxide is in isotopic equilibrium with oxygen in body water.¹⁴ After enriching body water with oxygen-18 (¹⁸O), a stable isotope of oxygen, ¹⁸O is lost as a function of normal water loss and carbon dioxide production. The ¹⁸O loss in water is measured by simultaneous enrichment of body water with hydrogen-2 (²H), a stable isotope of hydrogen, lost from the body in water only. Thus, after consumption of doubly labeled water (²H¹⁸O), carbon dioxide production can be calculated from the difference in washout between the two isotopes.

C. Application

The optimal technique for the indirect measurement of nutrient metabolism utilizes a respiration chamber,



FIGURE 1.1 Respiration chamber used for indirect calorimetry of subjects to measure energy expenditure for one or more days.

allowing measurements of oxygen consumption, carbon dioxide production, and urinary nitrogen loss over a 24-h cycle. Most respiration chambers have a volume of 10–30 m³ and are equipped with a bed, toilet, washbasin, and communication facilities like television and Internet (Fig. 1.1). A measurement in a respiration chamber typically covers one or more days. The optimal observation interval for application of the doubly labeled water method is 1–3 weeks: 1 week for subjects with a high energy turnover and 3 weeks when energy turnover is low as in sedentary older individuals.¹⁵

Energy metabolism techniques utilizing indirect calorimetry for the measurement of human energy expenditure also include measurement with a mouthpiece or facemask, or with a ventilated hood. Mouthpiece or facemask and ventilated hood are used for the measurement of gas exchange over limited time intervals, lasting one or more hours.

Mouthpiece and facemask have been applied for the measurement of energy expenditure for defined activities, including determining $\dot{V}_{O_2}$ max. A ventilated hood allows the measurement of REE and DEE. TEE can be measured in a respiration chamber or with doubly labeled water. TEE, as measured in a respiration chamber over 24 h, can be split in the three components: REE, DEE, and AEE, with an independent simultaneous measurement of body movement in the chamber.¹⁶ In contrast, the doubly labeled water method allows assessment of TEE without constraining the behavior of a subject.¹³

Indirect calorimetry systems for the measurement of energy expenditure require regular validation with a reference method. The standard option is a gas mixture of known composition to check the calibration of the gas analyzers and passing a known air volume through

the system to check the flow detector. A more sophisticated method is burning methanol, which checks the gas analyzers and the flow detector simultaneously.^{17,18} Alternatively, one or more healthy subjects can be measured, where the resulting energy expenditures should be comparable with the values derived from a prediction equation for the same subjects.

III. COMPONENTS OF ENERGY EXPENDITURE

TEE is mainly a function of food intake, body composition, and physical activity. Food intake determines energy expenditure for the processing of food. Body composition determines energy expenditure for maintenance or REE, generally the largest component of TEE. Physical activity determines AEE, the most variable component of TEE, showing large differences within a day and between days. Here, determinants of energy expenditure are described with respect to mechanisms and effect size.

A. Diet-Induced Energy Expenditure

Energy expenditure for the processing of food or DEE is the energy expenditure for intestinal absorption of nutrients, the initial steps of their metabolism and the storage of the absorbed but not immediately oxidized nutrients. Other names for the same component of TEE are specific dynamic action, thermic effect of food, and diet-induced thermogenesis. It is measured as the increase in energy expenditure above REE after consumption of a meal and expressed as the increase in energy expenditure divided by the energy content of the meal consumed. Measuring energy expenditure after a typical

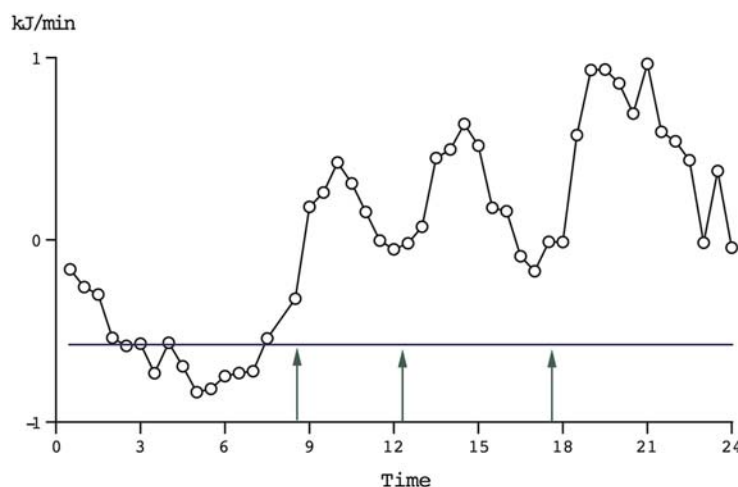


FIGURE 1.2 The mean pattern of diet-induced energy expenditure (DEE) throughout the day, calculated by plotting the residual of the individual relationship between energy expenditure and physical activity in time, as measured over 30-min intervals from a 24-h observation in a respiration chamber (blue line = level of resting energy expenditure [REE]; arrows = meal times). *Reproduced from Westerterp KR. Diet induced thermogenesis. Nutr Metab. 2004;1:5.*

single meal providing about 25% of estimated daily energy expenditure with 15% of the meal energy as protein showed that DEE lasts beyond 6 h.¹⁹ Measuring energy expenditure over 24 h in a respiration chamber showed the level of energy expenditure, after adjustment for AEE and thus including only REE and DEE, did not return to REE before lunch (at 4 h after breakfast), or before dinner at 5 h after lunch (Fig. 1.2¹⁶). Overnight, REE was reached at 8 h after dinner.¹⁶

The main determinants of DEE are the amount of food consumed and food composition. Theoretical values for DEE, based on the amount of ATP required for the initial steps of their metabolism and storage, are 0%–3% for fat, 5%–10% for carbohydrate, and 20%–30% for protein.²⁰ For a mixed diet with 30%–35% of energy from fat, 50%–55% from carbohydrate, and 10%–15% from protein, DEE is about 10% of the total amount of energy ingested. Thus, when a subject's energy intake meets the subject's energy requirement, DEE is 10% of TEE. In practice, reported DEE values are around 10% of energy intake with slightly higher values for diets higher in protein, as expected.¹⁶

B. Resting Energy Expenditure

Maintenance metabolism or REE is defined as the daily rate of energy expenditure to maintain and

preserve the integrity of vital functions. The measurement of REE must meet four conditions: the subject is awake; is measured in a thermoneutral environment to avoid heat production for the maintenance of body temperature; is fasted long enough to eliminate DEE; and is at rest to eliminate AEE. To meet measurement conditions, a subject usually stays overnight in the research facility where food intake and physical activity are controlled and REE is measured directly after waking up in the morning, 10–12 h after the last food intake and before getting physically active.

The major determinant of REE is body composition, i.e., fat-free body mass. Differences in REE between subjects are largely explained by differences in fat-free mass. REE in women is generally lower than REE in men because, for the same body weight, women generally have a lower fat-free mass. In a multivariate analysis with fat-free mass and gender, gender did not emerge as a significant contributor to the explained variation.²¹

The regression of REE on fat-free mass has a significant and positive intercept (Fig. 1.3²²). Thus, adjusting REE for differences in fat-free mass between subjects by dividing REE by the absolute fat-free mass value is not meaningful. Then, the smaller the fat-free mass, the higher the REE per kg fat-free mass.

When measurements of REE are not available, there are prediction equations for REE, based on an estimation of fat-free mass from subject sex, weight, height, and age. Examples are Harris–Benedict²³; Food and Agricultural Organization (FAO)²⁴; Owen^{25,26}; and Mifflin equations²⁷ (Table 1.1). They all were derived in Caucasian subjects and thus are applicable for Caucasian subjects. The more an individual shares characteristics with the group of people from whom the equation was developed, the better the estimate. For example, the Mifflin equation was derived from a population with a larger number of obese subjects and thus is usually applied for overweight and obese subjects.

Ethnicities differing in body build affecting body composition require ethnicity-specific equations to accurately predict REE from subject sex, weight, height, and age.

Alternatively, there are equations predicting REE directly from fat-free mass such as the Cunningham equation.²⁸ The Cunningham equation was based on data from eight studies, including more than 100 subjects each, women and men, lean and obese (Table 1.1). An exception, where even the Cunningham equation based on fat-free mass did not produce accurate estimates for REE, are elite athletes. Sjödin and colleagues measured REE values 16% higher in Olympic cross-country skiers than predicted from the weight-based FAO equation and 12% higher than in sedentary fat-free mass-matched control subjects.²⁹

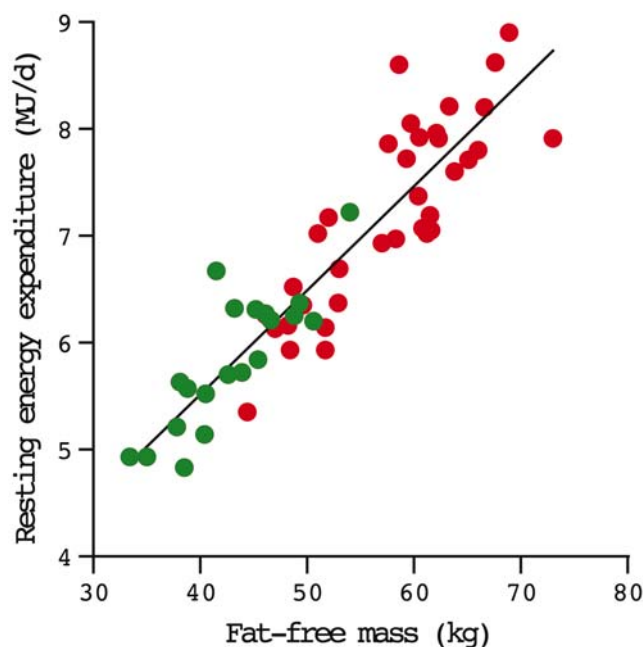


FIGURE 1.3 Resting energy expenditure (REE) plotted as a function of fat-free mass (FFM) with the linear regression line: $REE (MJ/d) = 0.097 FFM (kg) + 1.61$, $R^2 = 0.83$ (green dots, women; red dots men). Data from Pannemans DL, Westertep KR. Energy expenditure, physical activity and basal metabolic rate of elderly subjects. *Br J Nutr* 1995;73:571–581.

TABLE 1.1 Prediction equations for resting energy expenditure (REE, kJ/d) from body mass (BM, kg), height (H, m), age (A, y), and gender (0 for women and 1 for men), or from FFM (kg).

Reference	Subjects			Equation
	Women (n)	Men (n)	Total obese (n)	
Harris–Benedict ²³	99	132	6	$REE = 49.0BM + 2350H - 23.4A + 448G - 218^a$
FAO/WHO/UNU ²⁴	1239	3575	0	$REE = 55.6BM + 1397H + 146; A = 18-30; G = 0$ $REE = 36.4BM - 105H + 3619; A = 30-60; G = 0$ $REE = 38.5BM + 2665H - 1264; A > 60; G = 0$ $REE = 64.4BM - 113H + 3000; A = 18-30; G = 1$ $REE = 47.2BM + 67H + 3769; A = 30-60; G = 1$ $REE = 36.8BM + 4720H - 4481; A > 60; G = 1$
Owen et al. ^{25,26}	44	60	32	$REE = 33.7BM + 1852H - 14.9A + 1200G + 590^a$
Mifflin et al. ²⁷	247	251	234	$REE = 41.8BM + 2613H - 20.6A + 695G - 674$
Cunningham ²⁸	1622 lean and obese men and women			$REE \text{ (kcal/d)} = 370 + 21.6 \times FFM$

^aEquation derived by combining the original data for women and men.

C. Activity-Induced Energy Expenditure

Energy expenditure for physical activity or AEE is the energy produced by skeletal muscle, mainly for body movement. It is calculated from TEE by adjustment for REE and DEE. Measurement of TEE for calculation of AEE in daily life, without interference with the behavior of a subject, is performed with the doubly labeled water method, which was first applied in humans in 1982.³⁰

The physical activity level (PAL) can then be calculated from a measured or estimated REE value where $PAL = TEE/REE$.^{24,31} Alternatively, AEE can be calculated as $0.9 (TEE - REE)$, assuming DEE is a fixed

fraction of 10% of the consumed energy, and energy intake meets TEE (thus no contribution from body fat as an energy source; see [Section IV.A](#)).

Based on a compilation of doubly labeled water-measured TEE values over 20 years worldwide, the World Health Organization classified PAL values as 1.40–1.69 for a sedentary or light active lifestyle; 1.70–1.99 for active or moderately active lifestyles; and 2.00–2.40 for a vigorously active lifestyle.³¹ Similarly, the US Institute of Medicine classified PAL values as 1.0–1.39 for a sedentary lifestyle; 1.4–1.59 for a low active lifestyle; 1.6–1.89 for an active lifestyle; and 1.9–2.49 for a very active lifestyle.³² A PAL value of 1.4 is typical for one-year-olds, starting to move around by walking ([Fig. 1.4](#)^{31,33}). Subsequently, PAL increases to reach a mean adult level around 1.7, decreasing after reaching the age of 50 y in most subjects. At a PAL value of 1.7, AEE is about one-third of TEE, assuming DEE is 10% of TEE. Higher PAL values than 2.4 for a vigorously active lifestyle are observed in extremely active young adults, including athletes. However, in most studies with PAL values higher than 2.00–2.40, subjects were in a negative energy balance showing there is a limit to AEE for a sustainable lifestyle.³⁴

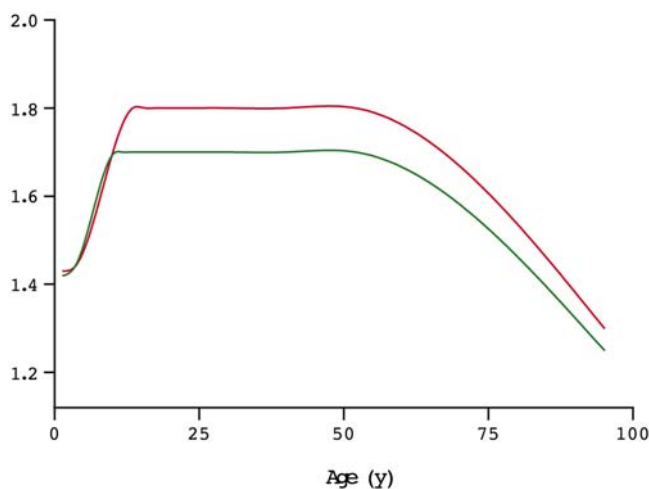


FIGURE 1.4 General model for the physical activity level (PAL), total energy expenditure (TEE) as a multiple of resting energy expenditure (REE), in relation to age (green = women; red line = men). Data from FAO/WHO/UNU. Human energy requirements. Joint FAO/WHO/UNU Expert Consultation. FAO Food and Nutrition Technical Report Series no. 1. Rome, 2004; Westerterp KR. Exercise, energy expenditure and energy balance, as measured with doubly labelled water. Proc Nutr Soc 2018;77: 4–10.

IV. ENERGY REQUIREMENT

Energy requirement is a function of body composition, food intake, and physical activity as determinants of, respectively, REE, DEE, and AEE, the components of TEE. For healthy subjects in energy balance, energy requirement is based on measured or estimated REE and PAL. Thus, energy requirement for an average adult woman is 11 MJ (2630 kcal)/d and for an average adult man 13 MJ (3107 kcal)/d ([Box 1.3](#)). The difference in

BOX 1.3

Example: Energy requirement for an average adult woman and man with a typical PAL of 1.7 (active lifestyle).

Subject characteristics, average adult woman: age 25 y; height 1.70 m; body mass 65 kg.

Subject characteristics, average adult man: age 25 y; height 1.80 m; body mass 75 kg.

Resting energy expenditure (REE) is estimated with the Harris–Benedict equation (Table 1.1) and TEE estimated based on PAL = 1.7:

Woman, REE = 6.4 MJ (1530 kcal)/d; TEE (1.7×6.4) = 11 MJ (2630 kcal)/d

Man, REE = 7.6 MJ (1816 kcal)/d; TEE (1.7×7.6) = 13 MJ (3107 kcal)/d

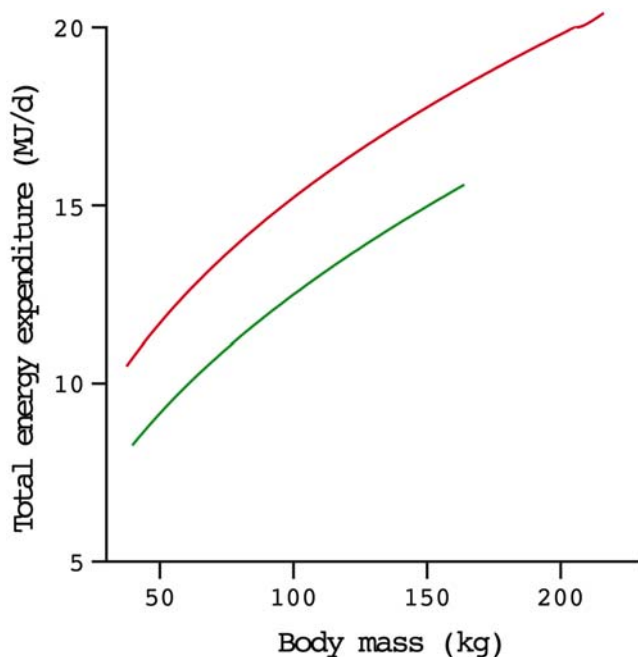


FIGURE 1.5 General model for total energy expenditure (TEE) as a function of body mass (green = women; red = men). Data from Westerterp KR. Exercise, energy expenditure and energy balance, as measured with doubly labelled water. *Proc Nutr Soc* 2018;77:4–10.

energy requirement between similar age women and men is mainly due to a difference in body composition, resulting in a lower REE for women. Women generally have lower fat-free mass than men, resulting in a lower energy requirement despite a similar PAL.

In addition to body composition, food intake, and physical activity, energy requirement is affected by adaptive thermogenesis and exercise economy.

A. Body Composition and Energy Requirement

Energy requirement increases with body mass. Energy expenditure and thus energy requirement is higher in overweight and obese than in lean subjects, an observation confirmed by doubly labeled water assessment of

TEE in free-living subjects.³⁵ The relationship between body mass and TEE is curvilinear (Fig. 1.5³³) and mainly reflects the higher REE in subjects with a larger body mass.³⁶ Overweight and obese subjects are characterized by a larger, metabolically inert, fat mass as well as a larger metabolically active body mass.

A larger maintenance requirement for a larger body mass requires more energy from food, and thus DEE is higher in larger subjects as well. There has been the suggestion of a reduced DEE in obese subjects, explained by insulin resistance.³⁷ However, a later review did not confirm DEE is reduced in obese subjects.³⁸ Thus, a larger body mass implies a higher REE and DEE in overweight and obese, as well as lean individuals.

A larger body mass requires more energy to perform the same physical activities, especially weight-bearing activities. Doubly labeled water studies have shown that PAL is similar at different levels of body mass index, except the very highest.^{39,40} Body mass index is defined as the weight (kg) of an individual divided by height (m)². Subjects with a body mass index higher than 40 kg/m², generally show a lower PAL. TEE increases with body mass as a function of fat-free mass and not as a function of total body mass^{41,42} because AEE per kg body mass is negatively related to body fat percentage.⁴² Thus, fatter subjects generally move less than lean subjects despite similar or higher measured AEE.⁴³

B. Food Intake and Energy Requirement

Underfeeding

The Minnesota Experiment is a classical study conducted to determine the effect of food intake on energy expenditure.⁴⁴ Famine in Europe during World War II was the impetus for a controlled experiment in America in 1944 to determine changes in energy balance induced by semistarvation. Thirty-two men, conscientious objectors to the war recruited from work camps in the United States, stayed in the laboratory for a total of 48 weeks. Following a 12-week baseline period, 24 weeks of

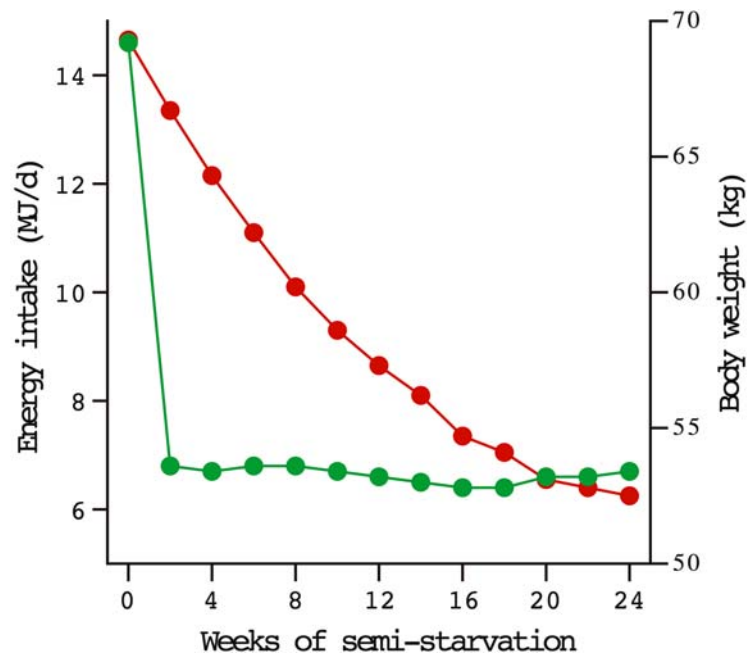


FIGURE 1.6 Mean daily energy intake (green dots) and mean body weight (red dots) of 32 men during 24 weeks of semistarvation. Data from Keys A, Brozek J, Henschel A, et al., *The Biology of Human Starvation*, 1950, University of Minnesota Press, Minneapolis.

semistarvation concluded with an additional 12 weeks of rehabilitation. During the study, subjects were assigned to specific maintenance tasks and outdoor activities. The diet in the baseline period was adjusted to body weight maintenance, providing an average of 14.6 MJ (3490 kcal)/d. In the semistarvation period, the diet was served in two meals per day at 8:30 and 17:00, providing an average of 6.6 MJ (1575 kcal)/d. The body weight of the subjects with an initial mean body mass index of 21.4 kg/m² decreased from a mean value of 69.4 – 52.6 kg, resulting in a final mean body mass index of 16.3 kg/m². The weight loss decreased progressively and nearly reached a plateau toward the end of the semistarvation period⁴⁴ (Fig. 1.6). Thus, the subjects adjusted energy expenditure to reach energy balance after 24 weeks at an average of 45% of what had been their ad libitum energy intake during the first 12 weeks. The main part of the 8.0 MJ (1912 kcal)/d adaptation of TEE stemmed from a reduction of AEE by 4.7 MJ (1123 kcal)/d, one-third of which resulted from the loss of weight making movement less energy demanding and the remaining two-thirds through reduction of body movement.

Reducing energy intake by 8 MJ (1912 kcal)/d would result in a reduction of DEE by 0.8 MJ (191 kcal)/d, 10% of the reduction of energy intake. The remaining 2.5 MJ (600 kcal)/d of the 8 MJ (1912 kcal)/d adaptation of TEE resulted from a reduction of REE, two-thirds of which came from a decrease in fat-free mass to be maintained and the remaining one-third from a lowering of tissue metabolism.

Together, the adaptation can be split into a “passive” component of about 4 MJ (950 kcal)/d (through the reduction of DEE due to lower food intake, a reduction of AEE due to moving less weight when physically active, and a reduction of REE due to having less fat-free mass to maintain), and an “active” component of 4 MJ (950 kcal)/d (through a reduction in physical activity and a reduction in tissue metabolism). Similar adaptations have been observed in overweight and obese subjects when subjected to an energy-restricted diet to lose weight.^{45,46}

Overfeeding

So far, overfeeding studies have not shown “active” adaptations of energy expenditure. Overfeeding only induces an increase in TEE through the increase of DEE due to a higher food intake and an increase of AEE due to moving more weight due to the resulting higher body weight when physically active and an increase of REE resulting from having more fat-free mass to maintain.⁴⁷ There are no indications for an overfeeding-induced increase in physical activity or an increase in tissue metabolism. One study which doubled energy intake for 9 weeks induced a mean body weight increase of 17 kg, while showing the opposite for AEE, with a reduction of PAL from 1.87 (an active lifestyle) to 1.45 (a very sedentary activity level).⁴⁸

C. Physical Activity and Energy Requirement

A typical study showing the effect of a change in PAL on energy expenditure is a report of an experiment

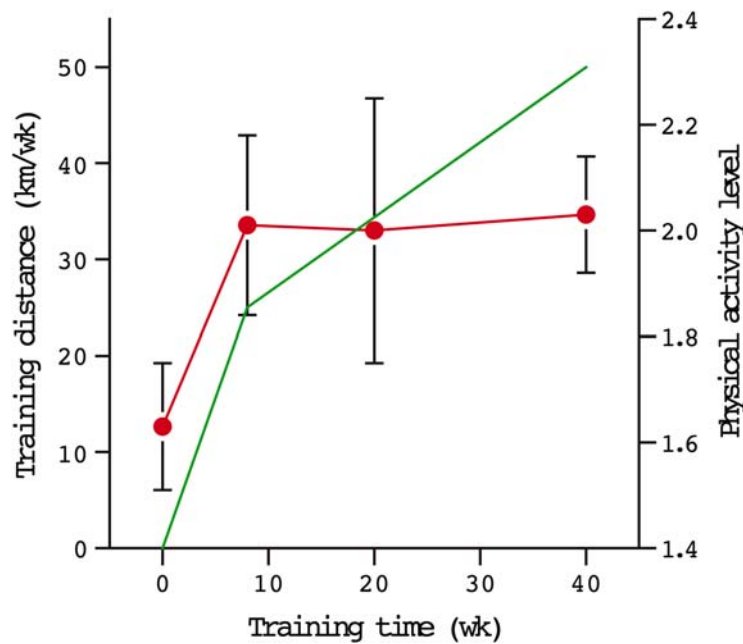


FIGURE 1.7 Training distance (green line) and physical activity level (PAL, red dots), total energy expenditure (TEE) as a multiple of resting energy expenditure (REE), in subjects completing a 40-week preparation to run a half marathon. Data from Westerterp KR, Meijer GA, Janssen EM, et al., Long-term effect of physical activity on energy balance and body composition. *Br J Nutr* 1992;68:21–30.

training young adults for 44 weeks to run a half marathon competition.⁴⁹ Before the training, subjects were not participating in any sports such as running or jogging and were not active in any other sport for more than 1 hour per week. Out of nearly 400 respondents, 16 women and 16 men were selected of comparable age (28–41 y) and normal body weight (reported body mass index 20–25 kg/m²). The training consisted of four training sessions per week, increasing the running time from 0 to 10–30 min, 20–60 min, and 30–90 min per session after 8, 20, and 40 weeks, respectively. Energy expenditure measurements included overnight REE in a respiration chamber and TEE over 2-week intervals with doubly labeled water, after 0, 8, 20, and 40 weeks of training. During the study, five women and four men withdrew from the study. All dropouts had an initial body mass index higher than the group mean, suggesting a higher body weight limited an ability to participate in weight-bearing activities like running. In those who completed the study, the training induced a change in PAL from an initial mean value below 1.7 for a sedentary or light active lifestyle to values around 2.0 for a vigorously active lifestyle⁴⁹ (Fig. 1.7). PAL increased initially and subsequently leveled off despite a further doubling of the training volume. The initial increase in PAL was higher than the predicted cost of the training and could not be explained by a decrease in nontraining activity, indicating a low exercise economy. The subsequent increase in training volume without a further increase in PAL shows a training-induced increase in exercise

economy in novice runners. At the group level, the training did not induce a change in body weight despite a 20%–30% increase in TEE, indicating a compensatory increase in energy intake. However, most subjects gained fat-free mass and lost fat mass. The loss in fat mass was positively related to initial fat mass, fatter individuals losing more than lean ones. Other studies have confirmed that exercise training results in a healthier body composition as reflected by a reduction of body fat, especially in overweight and obese subjects, with little or no long-term effect on body weight.⁵⁰

D. Adaptive Thermogenesis

Adaptive thermogenesis is a change in energy expenditure beyond that which can be predicted by the change in body weight and/or body composition. Thus, as described above, there was an adaptive reduction in REE and AEE through a lowering of tissue metabolism and a reduction in body movement, respectively, during long-term semistarvation.⁴⁴ So far, there is no evidence for a similar adaptive response to a positive energy balance, through an increase in energy expenditure.⁴⁷ There is a biological defense against weight loss and little or no defense against weight gain.⁵¹

Adaptive thermogenesis in response to energy restriction is one of the explanations for the limited success of attempts to lose weight in overweight and obese subjects. Studies suggest that a reduction in REE and in AEE occurs at as small a weight loss as >5% and >10%, respectively.⁵² The rate of weight loss does not seem to

affect the extent of adaptive thermogenesis. Adaptive thermogenesis was not different for obese subjects losing a similar amount of weight with a total fast, a very low energy diet providing 2.5 MJ (600 kcal)/d, or a low energy diet providing 5.2 MJ (1250 kcal)/d.⁵³ Most studies applying intermittent energy restriction—dieting on some days of the week—show no difference in weight loss with continuous dieting either.⁵⁴ So far, there is no dieting strategy to overcome energy restriction—induced adaptive thermogenesis.

The adaptive reduction in AEE in response to energy restriction was shown to recover when energy balance was reached during weight maintenance after weight loss.⁴⁶ However, the adaptive reduction in REE in response to energy restriction did not recover during weight maintenance after weight loss.⁴⁵ The adaptive reduction of REE was sustained up to 44 weeks after moderate weight loss.

Additionally, it appears that adaptive thermogenesis persists as long as weight loss is maintained.⁵⁵ Morbidly obese subjects showed a reduced REE when measured more than 8 years after successful weight loss and weight maintenance.

E. Exercise Economy

There are indications that TEE does not increase linearly with increasing physical activity. In a large cross-sectional sample of subjects where physical activity was measured with accelerometers, TEE plateaued at higher activity levels.⁵⁶ One explanation is a higher exercise economy in subjects with a higher activity level, allowing more physical activity for the same energy expenditure.

Training

Increasing physical activity by training increases exercise economy as shown above in novice runners training to run a half marathon.⁴⁹ After an initial training-induced increase in TEE, subjects could double the training volume without a further increase⁴⁹ (Fig. 1.7). The initial training-induced increase in TEE was twice as high as predicted from the training load, indicating a low exercise economy in this selected group of sedentary subjects.

Aging

Another example of a change in exercise economy is the effect of aging. Aging is associated with a reduction of AEE and PAL.⁵⁷ Lower physical activity in older adults is due to multiple factors, including an increased cost of daily life activities.⁵⁸ Exercise training was shown to increase exercise economy and thus preserve physical activity in older adults.⁵⁹ Multicomponent fitness training was shown to decrease the energetic cost of walking, possibly by improving motor control while

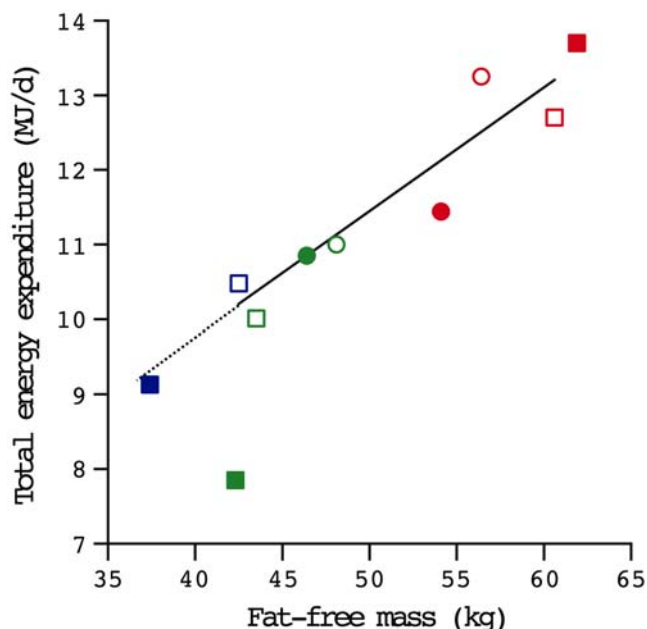


FIGURE 1.8 Total energy expenditure plotted as a function of fat-free mass (FFM) for patients (closed red square, reference 62; closed green dot, reference 63; closed blue square, reference 64; closed green square, reference 65; and closed red dot, reference 66) and control subjects from the same studies (open symbols), with the linear regression line for the data from control subjects.

walking.⁵⁹ Additional mechanisms regulating exercise economy in response to physical activity are subject for future research.

V. ENERGY REQUIREMENT AND DISEASE

In patients with a disease process, TEE is similar or often reduced even in the presence of increased REE because of decreased AEE.⁶⁰ There was no difference in TEE between clinically stable HIV-infected patients and matched controls despite REE, when adjusted for fat-free mass, being approximately 10% higher in the patients.⁶¹ Similarly, there was no difference in TEE between clinically stable patients with chronic obstructive pulmonary disease with a normal REE and those with an increased REE. The variation noted in TEE in patients with chronic obstructive pulmonary disease appeared to reflect differences in AEE, but not differences in REE.⁶²

A lower PAL was also the main determinant of lower-than-normal TEE in patients with rheumatoid arthritis.⁶³ Doubly labeled water-assessed PAL was significantly lower in short bowel syndrome patients (1.4 ± 0.1) than in matched nonshort bowel syndrome patients (1.75 ± 0.3).⁶⁴ Lack of adequate energy to move around and perform activities of daily life due to chronic malabsorption could be a main reason.

TEE, when adjusted for differences in fat-free mass, was similar for patients and control subjects in most studies performed so far^{61–66} (Fig. 1.8). Female patients with rheumatoid arthritis showed the lowest TEE and the lowest fat-free mass. TEE adjusted for fat-free mass was in line with the extrapolated regression of control subjects. The lower-than-normal PAL in these patients might be explained by a deficiency in fat-free mass. Only TEE in short bowel syndrome patients clearly fell below the regression of TEE on fat-free mass in control subjects, possibly through a lack of energy as explained above.⁶⁴

Disease is often suggested to induce alterations in energy expenditure, although not confirmed with objective methods like the doubly labeled water technique.

Thus, PAL, measured by the doubly labeled water technique, did not differ significantly between patients with chronic low back pain and healthy controls, matched for age and gender.⁶⁵ Mean PAL values were 1.66 ± 0.30 and 1.77 ± 0.32 in patients and controls, respectively. Thus the postulated presence of disuse in patients with chronic low back pain was not confirmed. Similarly, patients with Parkinson's disease commonly exhibit weight loss, attributed to various factors, including elevated energy expenditure. However, there was no difference in TEE between weight loss- and weight stable-Parkinson's disease patients measured with doubly labeled water.⁶⁶

RESEARCH GAPS

- What limits energy expenditure for a sustainable lifestyle? Total energy expenditure reaches a maximum value of 2.0–2.4 REE in subjects with strenuous work or high leisure activity.³³ Higher TEE values can be reached over shorter time intervals only. Extreme events reveal an alimentary limit.⁶⁸ One indication for an alimentary limit of TEE is the low TEE in short bowel syndrome patients (Fig. 1.8).
- How can the adaptive reduction in energy expenditure when a normal body weight has been reached after losing excess weight be overcome? Most adults gradually gain weight with increasing age. Subsequent weight loss results in adaptive thermogenesis with no indication of a reduction in adaptive thermogenesis for up to 1 y while weight loss is maintained.⁴⁵
- What regulates changes in exercise economy in response to physical activity? Sedentary subjects show a higher than predicted increase in TEE at the start of an exercise training intervention, indicating a low exercise economy.⁴⁹ Exercise training increases physical activity without affecting TEE, through an increase in exercise economy. Mechanisms regulating exercise economy require further study.

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PROTEIN AND AMINO ACIDS

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SUMMARY

This chapter briefly introduces basic biochemistry of protein and amino acids (AAs) and their chemical structures, followed by the discussion on the dynamic process of protein and amino digestion, absorption, and their turnover in whole body and specific tissues in vivo as the biochemical and physiological basis in assessing protein requirements. The latter part of the chapter provided an in-depth discussion of the latest knowledge of the indicators used to assess protein requirements and AA requirements. The chapter finishes with a discussion of the current recommended intakes for protein and AAs, along with aspects of their use in nutritional assessment and issues in developing recommendations of protein/AAs for health maintenance throughout the life cycle.

Keywords: Amino acids; Nitrogen; Protein; Protein requirement; Protein structure; Protein synthesis; Protein turnover; Recommended Dietary Allowance; Signaling.

I. INTRODUCTION

A. Background

The goal of the science of nutrition is to determine the adequate quality and quantity of the nutrients required by the hosts based on understanding how the nutrients are utilized through the complex biochemical processes in the body that generate energy, develop and maintain body mass, and regulate metabolic homeostasis. A unique feature of protein and amino acids (AAs) as nutrients is that ingested proteins must be degraded into their constituent peptides and AAs during digestion which is then followed by absorption; the body then utilizes the AAs as building blocks to form its own protein mass in muscle, visceral organs, and circulating proteins, to maintain physical and physiological functions. The average content of nitrogen in dietary protein is ~16% by weight, so nitrogen and protein metabolism are frequently considered together. In diseased conditions, altered protein and AA metabolism may serve as biomarkers of or may contribute to the pathophysiology of the disease process.

B. Nutrient Function and Structure

Amino acids

AAs are a group of organic molecules that consist of a basic amino (or imino) nitrogen group ($-\text{NH}_2$), an acidic carboxyl group ($-\text{COOH}$), and an organic R group (or side chain) linked through a central carbon. There are nearly 100 AAs in nature, but only 20 are used to make protein (see Fig. 2.1). The R group uniquely defines an AA and affects its chemical and physical nature. For example, some R groups are negatively or positively charged or contain branches or rings leading to unique groupings, such as the branched chain AA (BCAA; leucine, isoleucine, and valine) or aromatic AA (tyrosine, tryptophan, and phenylalanine). Like other classes of nutrients, AAs are categorized as essential (sometimes termed indispensable), nonessential (dispensable), or conditionally essential. The nine essential AAs (phenylalanine, isoleucine, leucine, lysine, methionine, threonine, tryptophan, valine, and histidine) must be consumed in the diet because the body

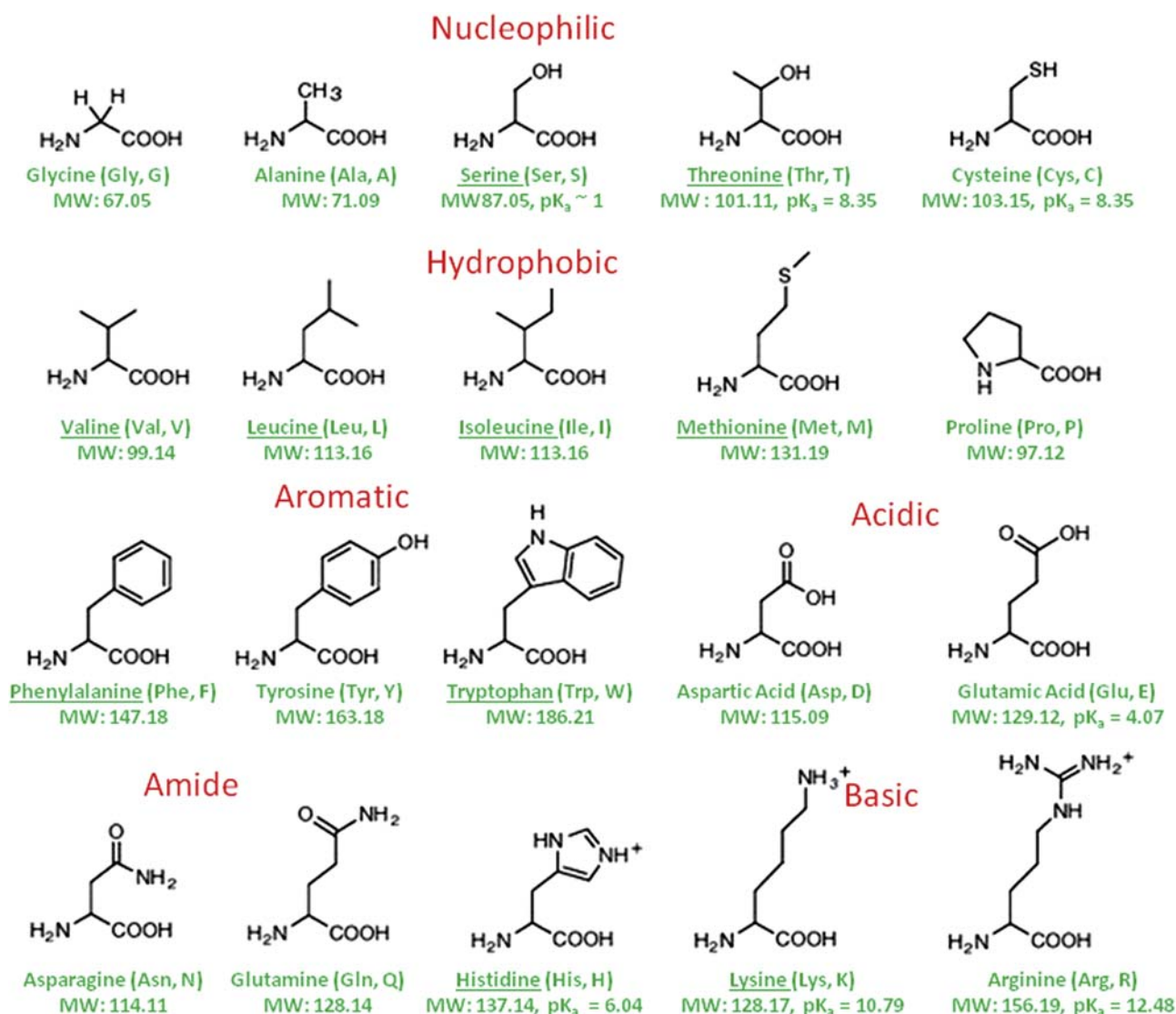


FIGURE 2.1 Structure of the 19 amino acids and 1 imino acid (proline) used in human protein synthesis. Names of amino acids essential (indispensable) for humans are underlined.

cannot synthesize them; histidine is essential primarily for children as there is some capacity for it to be synthesized when not required in high amounts such as during growth. The other 11, called nonessential, may be made by the body from essential AA or glucose. In this chapter, the terms essential and nonessential are used.

Many factors affect essentiality; for example, some infants, especially those born prematurely, or adults under certain health circumstances, cannot make adequate amounts of several of them, making them conditionally essential.^{1,2,3} It has been suggested that in the neonate, only five are truly dietarily nonessential or dispensable (alanine, aspartate, glutamate, serine, and probably asparagine).⁴

In addition to protein synthesis, AAs are oxidized as an energy source, converted to carbohydrates through

the process of gluconeogenesis, or converted to other AAs. In healthy subjects, AA oxidation provides about 15% of daily energy. In certain clinical conditions, such as severe burn injury or trauma, the oxidation of AAs is increased leading to increased breakdown of the body's own proteins and a state of protein wasting. Therefore, in many clinical and disease conditions, patients require increased protein intake.⁵

One can consider two pools representing AAs and body protein. The first, a free or labile AA pool, consists of free AAs dissolved in body fluids. The proteins in the tissues and circulation are grouped into a second pool. These two pools are in constant interaction, due to the continual degradation and resynthesis of proteins as proteins turn over at various rates, some of which may be short and others very long, or not degraded in the

body, such as keratin. The free or labile AA pool expands as AAs are supplied from absorbed AAs derived from dietary proteins and from de novo synthesis in cells (including those of the gut, which are also a source of nonessential AAs); this labile pool decreases with the loss of AAs by oxidation, excretion, or conversion to other metabolites.

Dietary protein is the primary source of AAs. Foods of animal origin, such as meat, poultry, fish, eggs, and dairy products, are good sources of essential AAs. Plant sources of protein often contain lesser amounts of one or more of the essential AAs so a single source of plant protein often cannot provide the full spectrum of essential AAs to meet requirements, particularly during growth. Soy is an exception as it contains all the essential AAs in amounts adequate for adults. Hence, vegetarians must consume a variety of plant proteins to balance the intake of essential AAs. For example, a grain such as wheat is low in lysine while corn has low amounts of tryptophan and lysine, but when either is combined with legumes such as garbanzo beans (low in methionine), the diet contains adequate amounts of the limiting AAs, assuring complementarity.⁶

Peptides

Peptides are composed of AAs chemically linked by peptide bonds. The peptide bond always involves a single covalent link between the α -carboxy (oxygen-bearing carbon) of one AA and the α -amino (or imino in the case of proline) nitrogen of a second AA. In the formation of a peptide bond from two AAs, a molecule of water is eliminated. Dipeptides are two AAs linked by a peptide bond, whereas three AAs bonded together by peptide bonds are called tripeptides. Polypeptides consist of 10

or more AAs bonded together by peptide bonds, and an intermediate string between 4 and 10 AAs is an oligopeptide. Peptides differ from proteins by virtue of their size. Traditionally, peptide chains that are short enough to be made synthetically from the constituent AAs are called peptides, rather than proteins.⁷

C. Functions of Proteins

Proteins are the most abundant organic molecules in cells and are fundamental to cell structure and function. All proteins in humans are constructed from the same basic set of 20 AAs linked by peptide bonds. In addition, some contain sulfur AAs and others have ligands with phosphorus, iron, zinc, and copper.

The roles of proteins are versatile; examples of proteins and their functions are described in [Box 2.1](#). These highlighted functions are only a fraction of the many roles that proteins play, but the list conveys some sense of the immense variety of proteins and their importance in the body.⁸ More recent research reveals increasingly complex roles for protein and AAs in regulation of body composition, maintenance of organ/tissues such as bone, gastrointestinal (GI) function, bacterial flora, glucose homeostasis, cell signaling, and satiety.⁹

Transport proteins

Transport proteins in plasma bind and carry specific molecules or ions from one organ to another. Some move around in the body fluids, carrying nutrients and other molecules. For example, hemoglobin carries oxygen from the lungs to the cells throughout the body; lipoproteins transport lipids in the blood; and specific proteins, such

BOX 2.1

Functions of Selected Proteins and Amino Acids

Growth, maintenance, and movement: Proteins form integral parts of most body structures such as skin, tendons, membranes, muscles, organs, and bones. As such, they support the growth and repair of body tissues and protect the body (skin, immunoglobulins, etc).

Enzymes: Proteins function as enzymes and modulate various biochemical reactions for energy production and body mass formation.

Hormones: Some, but not all hormones, are proteins or derived from an amino acid; hormones function as messengers and signals and regulate body processes (e.g., insulin, growth hormone, oxytocin).

Immune function/communication: The structure of antibodies, cytokines, chemokines, and regulators of gene transcription and translation through mTOR signaling pathway are proteins.

Fluid and electrolyte balance: Proteins in extracellular fluid help to maintain the fluid volume and the composition of body fluids (e.g., albumin).

Acid-base balance: Proteins help maintain the acid-base balance of body fluids by acting as buffers because of their charged amino acids.

Transportation and storage: Proteins transport substances, such as lipids, vitamins, minerals, and oxygen, throughout the body and storage of micronutrients (e.g., ferritin).

as retinol-binding protein, carry vitamins (vitamin A). Other kinds of transport proteins are present in cell membranes and are adapted to bind and transport glucose, AAs, and other nutrients across the membrane into cells.

Signaling proteins

Signaling proteins serve as signaling molecules that modulate multiple cellular processes, including protein synthesis and a cell's response to chemicals in the environment. Signal transduction plays an important role in controlling cell function and homeostasis through a set of chemical reactions that occur when a molecule, such as a hormone, attaches to a receptor on the cell membrane. This is followed by a cascade of biochemical reactions inside the cell that eventually reach the target molecule or reaction. The process thus triggers a series of intracellular events and invokes a cellular response. It is a method by which molecules inside the cell can be altered by molecules outside. Each step of these events involves biochemical reactions of proteins. An example of this sequence is the multiple set of proteins that mediate the insulin transduction pathway. Insulin binds to a cell receptor called "insulin receptor substrate" on cell surfaces; the receptor triggers a series of biochemical reactions through a chain of signal proteins, which eventually modulates the functions of insulin in the target cells, such as increasing or decreasing glucose transport into adipose and muscle cells.¹⁰

II. NUTRIENT METABOLISM

A. Protein Synthesis

A protein's structure and shape determine its functionality. Fundamentally, the process of protein synthesis consists of the following stages; a detailed review of synthesis, breakdown, and turnover is available¹¹:

Stage 1: Transcription: To inform a cell of the sequence of AAs for a needed protein, a stretch of deoxyribonucleic acid (DNA) in the cell nucleus serves as a template for making a strand of ribonucleic acid (RNA), called messenger RNA (mRNA), that carries a code which lists the order of the AAs that will be needed to make a given protein. Each mRNA strand copies exactly the instructions for making a specific protein that the cell needs.

Stage 2: Initiation of the polypeptide chain: The mRNA leaves the nucleus through the nuclear membrane bearing the code for the polypeptide to be made and attaches itself to the protein-making machinery of the cell, the ribosomes.

Stage 3: Elongation: Another form of RNA, transfer RNA (tRNA), collects specific free AAs from the cell

fluid. Each tRNA carries its AA to the mRNA, which dictates the sequence in which the AAs will be attached to form the protein strands. Thus, the mRNA ensures the AAs are lined up in the correct sequence. The sequence of AAs in each protein ultimately determines its configuration to support a specific function.

Stage 4: Termination and release: As the AAs are lined up in the right sequence and the ribosome moves along the mRNA, an enzyme connects one AA after another in a peptide bond to rapidly form the growing protein strand of 40 to over a 100 AAs. The tRNA is freed from its corresponding AA to pick another of its specific AA. When all the AAs have been attached based on the mRNA coding, the completed protein is released.

Stage 5: Folding and processing: To achieve its biologically active form, the polypeptide that was formed undergoes folding into its proper three-dimensional conformation. Before or after folding, the new polypeptide may undergo processing by enzymatic action to remove initiating AAs; to introduce phosphate, methyl, carboxyl, or other groups into certain AA residues on the polypeptide; or to attach oligosaccharides or other prosthetic groups.

Stages 2–5, occurring in the cytosol, together are called *translation*. Transcription and translation are summed up by the central dogma of molecular biology: DNA → RNA → Protein, with the formed protein peptide chains undergoing posttranslational modifications. The chemical properties of AAs in a protein molecule and their spatial configurations in a protein determine its biological activity.

The information necessary for its production resides in the cellular nucleus as DNA, which makes up a large component of genes. If a genetic error alters the mRNA code and thus the AA sequence of a protein, or if a mistake is made in copying the sequence, an altered protein will result, sometimes with dramatic consequences.

Resulting protein structure

Proteins are large polypeptide molecules with molecular weights of 6500 to 205,000 Da. The three-dimensional configuration of a protein molecule determines the function of the protein. Two classes of strong bonds (peptide and disulfide) and three classes of weak bonds (hydrogen, hydrophobic, and electrostatic or salt) stabilize most protein molecular configurations. A single protein molecule may contain one or more protein structure types: primary, secondary, tertiary, and quaternary structure.

Primary structure refers to the "linear" sequence of AAs in a polypeptide chain and the location of disulfide bonds, if present. Secondary structure refers to "local" ordered structure via hydrogen bonding, mainly within the

peptide backbone. Tertiary structure refers to the “global” folding of a single polypeptide chain. Hydrogen bonding involving groups from both the peptide backbone and the nonpolar side chains is important in stabilizing tertiary structure. Quaternary structure refers to the stable association of multiple polypeptide chains resulting in an active unit united by forces other than covalent bonds (not peptide or disulfide bonds). The forces that stabilize these aggregates are hydrogen bonds and electrostatic (or salt) bonds formed between AA residues on the surfaces of the polypeptide chains.

The protein structure is formed through the processes of translation and posttranslational modification. The configuration of a protein determines the physical property and the function of the protein. For example, the configuration of an enzyme protein determines the catalytic sites of action and the ability of the individual enzyme to function. Another example of such posttranslational modification is of the residues of proline and lysine in the protein procollagen, which are hydroxylated to hydroxyproline and hydroxylysine residues, permitting cross-linking of proteins to increase strength and stability of the resulting collagen.

B. Protein Breakdown

Protein breakdown or proteolysis is the biochemical process by which proteins are “disintegrated” into smaller polypeptides or AAs. Protein degradation occurs during the digestion and absorption of ingested and endogenous GI proteins as well as within the cells

of organs and tissues. Proteolysis is typically catalyzed by cellular enzymes called proteases and may also occur by intramolecular digestion. Low pH or high temperatures can also cause nonenzymatic proteolysis.

Intracellular proteolysis removes damaged and abnormal proteins and prevents their accumulation, thus serving to regulate cellular processes by removing enzymes and regulatory proteins that are no longer needed. Intracellular proteins are degraded into free AAs through two mechanisms: lysosomal proteolysis, which degrades cytoplasmic components via autophagy into the intracellular membrane-enclosed lysosome, and the ubiquitin-proteasome system. The autophagy-lysosomal pathway is normally a nonselective process, but it may become selective upon starvation whereby proteins containing peptide sequences biochemically related to Lys-Phe-Glu-Arg-Gln (KFERQ) are selectively broken down.¹² Ubiquitin is a basic 76-AA peptide that joins to lysine residues selectively on the proteins marked for degradation through the action of a multiple group of enzymes of the proteasome to degrade proteins into free AAs. The free AAs can then be oxidized, converted to other metabolic intermediates, or reused for protein synthesis in the cycle of protein turnover.^{13,14}

C. Digestion

Dietary protein is digested in the GI tract to free AAs and peptides by the action of gastric acid and enzymatic hydrolysis. Multiple AA transport systems exist for subsequent absorption of the freed AAs and peptides.¹⁵

TABLE 2.1 GI peptides (hormones) and their function.

Names	Secretory sites	Main target sites	Functions
Gastrin	G cell (pyloric antrum, duodenum and pancreas)	Parietal cell	Acid secretion, gastric contraction
Secretin	S cell (duodenum)	Pancreas	Water and bicarbonate secretion
Cholecystokinin	I cell (duodenum and jejunum)	Pancreas Gall bladder	Pancreatic enzyme secretion, gall bladder contraction
Gastric inhibitory peptide (GIP)	K cell (duodenum and jejunum)	Pancreas	Insulin secretion
Glucagon like peptide-1	L cell (small intestine)	Pancreas	Insulin secretion
Glucagon like peptide-2	L cell (small intestine)	Small intestine	Intestinal growth (crypt cell)
Peptide YY	L cell (small intestine)	Brain stem	Slows gastric emptying, reduces appetite
Somatostatin	Delta cell (pyloric antrum) Hypothalamus	GI tract Pituitary gland	Suppresses the release of gastrointestinal hormones

Gastrointestinal peptides

The GI tract produces a variety of chemical transmitters (GI peptides) that are involved in controlling GI motility, secretion, absorption, growth, and development.¹¹ While the cells that produce GI peptides are dispersed throughout the GI tract, regulatory peptides are found in the esophagus, stomach, small and large intestines, and the pancreas. Although GI peptides are typically considered hormones, most of them act on the same cells from which they are released, on neighboring cells via classic endocrine mechanisms, and on cells following their release from nerves.^{11,16} Many of the peptides in the GI tract are also found in the enteric nervous system and the central nervous system. The secretory site, target, and function of the nutritionally important GI peptides are listed in Table 2.1.

Dietary proteins and AAs enter the GI tract via both endogenous and exogenous routes. Endogenous sources include desquamated mucosal cells, digestive enzymes, and other glycoproteins such as mucus. Such endogenous protein entering the digestive tract and serving to also dilute dietary protein has been estimated as equal to or greater than the amount of dietary protein consumed.¹⁷ Exogenous or dietary protein sources in the form of meat, poultry, fish, milk products, grains, legumes, and vegetables provide both energy and AAs essential to the body.⁹

In the GI tract, the exogenous proteins are digested into peptides and free AAs through a series of hydrolytic reactions and feedback mechanisms (Fig. 2.2). Minimal protein digestion occurs within the mouth or esophagus. Once proteins reach the stomach, however, hydrochloric acid (HCl) is released by the parietal cells of the stomach. HCl release may be stimulated by the hormone gastrin, the neurotransmitter acetylcholine from vagal nerve stimulation, the neuropeptide gastrin-releasing peptide, or the amine, histamine.¹¹ HCl denatures protein structures to make them more susceptible to enzymatic action and converts pepsinogen, an inactive proenzyme or zymogen released by the stomach's chief cells, to its active form, pepsin.¹⁸ Pepsin in turn may activate other pepsinogen molecules or hydrolyze specific peptide bonds into large polypeptides, oligopeptides, and free AAs as reaction end products.¹¹

This mixture, known as acid chyme, next passes into the duodenum and small intestine where most of protein digestion takes place. The end products within the acid chyme stimulate the secretion of the hormones secretin and cholecystokinin (CCK), which in turn travel through the bloodstream to pancreatic acinar cells to stimulate the release of alkaline pancreatic juice. Secretin and CCK also stimulate the release of digestive proenzymes trypsinogen, procarboxypeptidases, chymotrypsinogen, and proelastase (Table 2.2). The higher pH of

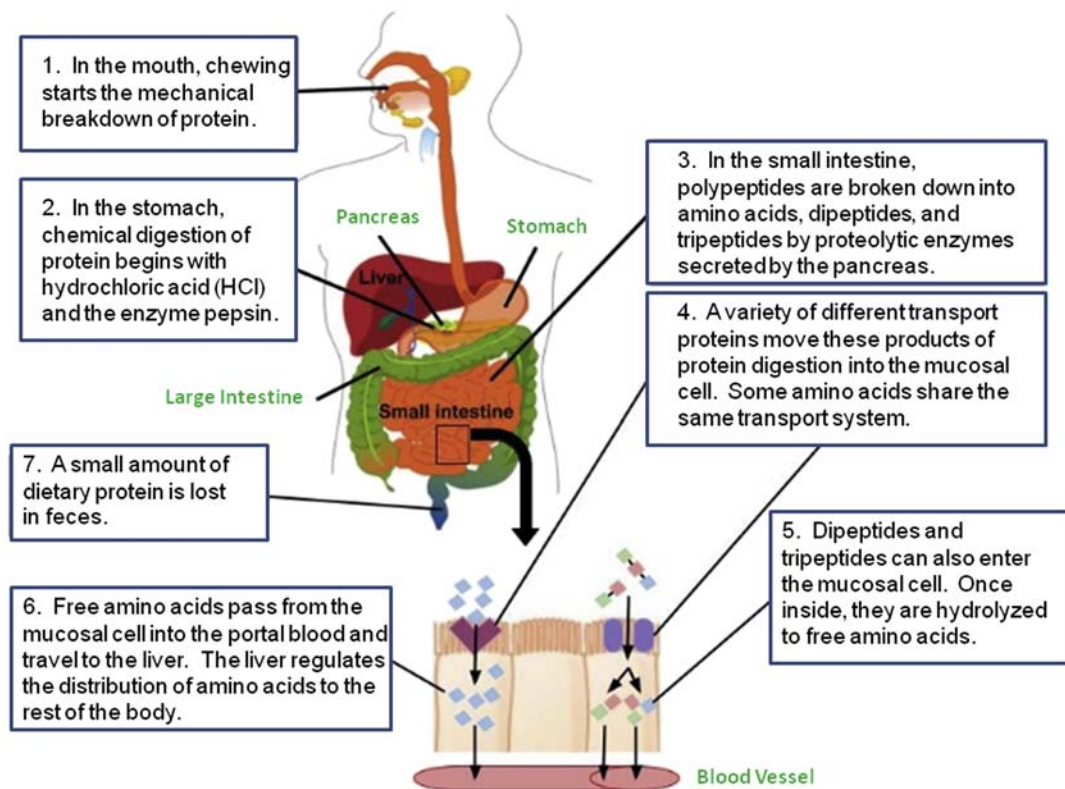


FIGURE 2.2 Protein digestion and absorption.

TABLE 2.2 Some proteolytic enzymes responsible for digesting protein.

Zymogen	Enzyme or activator	Enzyme	Site of activity
Pepsinogen	HCl or pepsin	Pepsin	Stomach
Trypsinogen	Enteropeptidase; trypsin	Trypsin	Intestine
Chymotrypsinogen	Trypsin	Chymotrypsin	Intestine
Procarboxypeptidases	Trypsin	Carboxypeptidases A, B Aminopeptidases	Intestine

the duodenum deactivates the pepsin and provides an optimal pH for the activity of the pancreatic enzymes. By releasing proteolytic enzymes initially in their inactive form, the enzyme-forming cells are protected from self-digestion.^{19,11}

Enzymatic hydrolysis

Enterokinase (also known as enteropeptidase) is an enzyme secreted from the brush border of the small intestine, also in response to secretin and CCK. Enterokinase serves to activate trypsinogen to trypsin, which in turn converts many of the other proenzymes into their respective enzymatic forms for ongoing protein hydrolysis.¹¹ At this point, endogenous protein sources from sloughing mucosal cells or other secretory proteins are also recycled through the digestive process to form the general pool of AAs.

The presence of protein within the gut signals further enzyme secretion. When the majority of the protein has been digested, the presence of unbound trypsin acts as a feedback mechanism to turn off further secretion of trypsinogen by the pancreas.¹⁹ At this point, the by-products of pancreatic protease digestion, free AAs and peptides of two to six AA components, now enter the final phases of digestion. The brush border of the small intestine produces a variety of peptidases, which allow peptide digestion to continue through the distal small intestine and ileum.^{20,21,11}

Role of gut microbes

In recent years, the importance of gut microbes in the digestion, absorption, metabolism, and transformation process of dietary protein in the GI tract has been recognized. Like all dietary components, AAs can be metabolized by gut microbes into numerous microbial metabolites that may affect gut microflora. It has been proposed that the ratio between protein and carbohydrate or even a low-protein diet may be advantageous compared to a diet high in dietary protein, which may lead to an increase of pathogenic microorganisms with associated higher risk of metabolic diseases due to cross talk between dietary protein and gut microbiota composition and function. It has been proposed that this is a

result of the cross talk between protein metabolism and host immune response.²²

D. Absorption

The final end products of protein digestion (e.g., free AAs, dipeptides, and tripeptides) are now ready for absorption within the small intestine. To a lesser extent, the absorption of intact proteins may occur through leaks at cell junctions, which accounts for the immune response to foreign proteins associated with food allergies.^{11,23} However, the vast majority of AAs are absorbed in the proximal small intestine, leaving less than 1% of dietary protein to be excreted in the feces.^{20,11} Through both sodium-dependent and sodium-independent active transport systems, AAs pass from the lumen of the small intestine into the enterocyte. The various AA carrier systems have an affinity for specific AAs.²⁴ The rate of AA absorption may vary depending upon the chemical properties of the AA. Essential AAs are absorbed faster than nonessential AAs: methionine and the BCAAs leucine, isoleucine, and valine are the most rapidly absorbed.²⁵ Competition can occur between AAs for transport by a carrier system.

The active transport of di- and tripeptides across the brush border membrane of the enterocyte also involves a competition for transporters but utilizes different carrier systems than those of the free AAs. In fact, peptide absorption occurs more rapidly than an equivalent mixture of free AAs.^{26,27} It is estimated that 67% of dietary protein is absorbed as di- and tripeptides with the remaining 33% absorbed as free AAs.^{11,28,29} Once inside the enterocyte, these peptides are finally converted to free AAs by peptide hydrolases. The final phase of AA absorption occurs across the basolateral membrane of the enterocyte by diffusion or active transport into the portal circulation (Fig. 2.2).

E. Protein and Amino Acid Metabolism

Body protein is in a constant dynamic state; the body breaks down protein into free AAs which enter the free AA pool. There are constant fluxes among different AA

pools, especially the flux between muscle and visceral organs. The free AAs are reutilized to synthesize proteins or serve as intermediates of energy metabolism, via gluconeogenesis and oxidation. Therefore, AAs are constantly trafficking between protein-bound and protein-free pools.

The turnover rates of individual proteins vary depending on the function of the protein and its location. For example, peptides and hormones have relatively high rates of synthesis and degradation, in contrast to structural or plasma proteins. In a normal subject during a 24-h period, the nitrogen in the dietary protein consumed matches nitrogen output, indicating that the rates of protein synthesis and breakdown reach a balance without net protein gain or loss. Almost all nitrogen is lost in urine, with small amounts excreted in feces and much smaller amounts as sweat and miscellaneous losses. Urinary nitrogen losses in healthy adults are about 11–15 g when consuming 70–100 g of protein. These losses are primarily urea, with small amounts of ammonia, creatinine, uric acid, and other nitrogen-containing end products of protein metabolism.

However, the balance between protein synthesis and breakdown is altered by disease and certain physiological states, as well as changes in nutrient intake.³⁰

The body of a 70-kg man is about 16% or 11 kg of protein. About 43% is in skeletal muscle, 15% in structural tissues and 15% in blood.³¹ About 50% of total body protein is made up of four proteins, actin and myosin in skeletal muscle, collagen, and hemoglobin, with collagen comprising 25% of the total protein. In malnutrition, this proportion can rise to 50% because of the substantial loss of noncollagen proteins, whereas collagen itself is retained.³² Visceral tissues (e.g., kidney and liver), while containing together only about 7 kg or 10% of the total protein, are very active in metabolism. Other organs such as the brain, lung, heart, and bone contribute the remainder. Newborn infants have less in muscle and much more in brain and visceral tissue than adults proportionally.³³ Protein turnover in infants on a body weight basis is greater than in young adults, which is greater than in elderly.³³ Visceral tissues such as liver and intestine together are thought to be responsible for up to 50% of whole body protein turnover^{34,35} while even though skeletal muscle contains the largest component of body protein, it is estimated to contribute only ~25% to daily turnover.^{34,36}

Methods to measure protein turnover

Protein turnover rates in both healthy individuals and critically ill patients may be measured by several different methods. Each of them has some limitations, but these in vivo measurements have provided most of the quantitative information of protein metabolism in living humans. Measurements of gene expression in

tissue may provide qualitative information on the possible changes occurring in either protein synthesis or breakdown in specific tissues (such as muscle), instead of the dynamic fluxes of intermediate metabolites. The methods developed to quantify the dynamic aspects of protein metabolism are listed in Table 2.3.

Nitrogen balance; Nitrogen balance is the difference between the amount of nitrogen taken in and the amount excreted or lost and has been the traditional approach to evaluate protein nutritional balance.

Mathematically,

$$\text{Nitrogen balance} = \text{Nitrogen intake} - \text{Nitrogen output}$$

where nitrogen intake is the total nitrogen administered either by an enteral or parenteral route, and nitrogen output is the sum of urinary, fecal, dermal, and other body fluid losses of nitrogen.

In a clinical setting (see the classical work of Wilmore³⁷), since the majority of nitrogen output is urine, and about 80% of urinary nitrogen content is from urea, approximate nitrogen output is estimated as

Nitrogen output (g/day) =

$$\text{Urinary urea nitrogen (mg/100 mL)} \times \text{urinary volume (L/day)} / 100 / 20\% \text{ of urinary urea losses} + 2 \text{ g}^{37}.$$

However, the concentration of urinary urea nitrogen is affected by stress and can be increased with increased urinary excretion of nonurea nitrogen coming from nonprotein compounds such as creatinine or nucleic acids. Stool, dermal, and other losses are usually estimated to be about 2 g nitrogen/day, unless there is diarrhea, ostomy/fistula losses, or skin exudate such as with large area burns.³⁸ There are also other losses not accounted for clinically, such as menstrual losses, seminal fluid, and nasal secretions. In clinical settings, a 24-h urine collection is required for urea nitrogen measurements. However, others have used shorter collection times and extrapolated those results to a 24-h collection. Adjustments need to be made in certain patients.^{39,40}

Many other factors affect the accuracy of nitrogen balance measurement. In addition to renal dysfunction, errors in estimating intake or incomplete collections of urine, stool, fistula, or ostomy losses may affect balance results. It is worth mentioning that the measurement of whole body nitrogen balance is relatively simple and noninvasive, is a rough estimate of the difference between body protein synthesis and breakdown, and is a poor indicator for characterizing inter-organ AA flux and changes in protein turnover.⁴¹ More sensitive tools exist for determining substrate balance across an organ and AA circulation such as isotopically labeled tracers.

TABLE 2.3 Methods in place to measure protein metabolism.

Methods	Characteristics
Nitrogen balance	• Measure of nitrogen input (diet) minus nitrogen output (urine + estimates of fecal and other endogenous losses) • Can be used in clinical setting • Decreased accuracy in certain clinical conditions (renal dysfunction, high output fistula, and ostomy) • Provides information on the net difference between protein synthesis and breakdown; however, the specific changes in either protein synthesis or protein breakdown are unknown
Tracer methodology using stable isotope-labeled amino acids	• The constant IV infusion of an amino acid(s) or bolus injection method is used in human studies • The subject is required to be at a metabolically steady state • Ideal for measuring acute changes in subjects • Important research tool that allows direct in vivo measurements of protein synthesis and breakdown rates in the whole body
Tracer plus imaging technology (27a)	• Using radioactive positron emission tomography (PET) with tracer amino acid, such as ^{11}C-methionine, which does not have degradation pathway in muscle, can image the kinetics of this tracer under PET camera to quantify the rate of protein synthesis in muscle tissue • Drawback to its use is the short-lived radioactivity of the gamma tracer • Does not measure protein breakdown
Arteriovenous amino acid differences across organs	• Invasive methods for subjects as a tracer amino acid must be infused. Samples of blood are obtained from both the arterial and venous catheters • These measurements are made across muscle beds. Samples of blood are obtained from both the arterial and venous catheter
Urinary creatinine excreted over a 24 h period	• Indirect measurement of muscle mass • Has high variation
3-Methyl histidine excretion	• Can be used as a measure of protein degradation as 3-methyl histidine is not oxidized • Compromised accuracy with organ dysfunction

Tracer methodology to measure protein turnover; To further trace how ingested protein is utilized and influences protein catabolism, nonradioactive, stable isotope tracer AAs to measure whole body protein turnover were first used in the 1950s but were fraught with methodological difficulties. It was not until 1969 that a simpler tracer method was developed by Picou and Taylor-Roberts using ^{15}N -glycine.⁴² It was based on the assumption that administered ^{15}N -glycine tracer is evenly distributed throughout the whole body nitrogen pool. The estimate was based on the following series of equations:

$$\begin{aligned} &^{15}\text{N-glycine administration rate}/ \\ &^{15}\text{N-nitrogen excretion in urine rate} \\ &= \text{whole body nitrogen turnover rate}/ \\ &\quad \text{total urinary nitrogen excretion} \end{aligned} \quad (2.1)$$

Therefore, from the rate of ^{15}N -glycine entry via administration into the body nitrogen pool, and the rate ^{15}N -nitrogen excretion in urine, and estimation of total urinary nitrogen excretion, the whole body nitrogen turnover rate, Q , can be calculated at a steady condition:

$$\text{N influx} = \text{N outflux} = Q \quad (2.2)$$

$$\begin{aligned} Q &= \text{N influx} \\ &= \text{N released from protein breakdown} \\ &\quad + \text{dietary N intake} \end{aligned} \quad (2.3)$$

$$\begin{aligned} Q &= \text{N outflux} \\ &= \text{N incorporated into protein (protein synthesis)} \\ &\quad + \text{urinary N excretion} \end{aligned} \quad (2.4)$$

Therefore, from the measurements of total isotope-labeled nitrogen intake, ^{15}N -output in the urine, the total dietary nitrogen intake rate, and the urinary nitrogen excretion rate, the rates of whole body protein breakdown (Eq. 2.3), and whole body protein synthesis (Eq. 2.4) can be determined.

The tracer methods thus can provide quantitative information on the rates of both protein synthesis and breakdown compared to the nitrogen balance method. The isotope tracer method provides the fate of the dietary protein within the body, in terms of its influence on protein anabolism and catabolism of the host.

Measurements of protein turnover in specific tissue organs in vivo; Understanding how nutrition supports individual organ protein metabolism is important in physiology and pathophysiology. There are a number of methods used to measure protein synthesis in specific organs.

1. One approach is to directly measure the rate of stable isotope-labeled AAs incorporated into specific proteins, into muscle or gut mucosa, or into multiple circulating proteins. The measurement is conducted by a constant infusion of stable isotope-labeled amino tracer until a steady state of the tracer in the protein is reached; the increment of tracer enrichment in the protein-bound AA pool is used to calculate the protein synthesis rate.⁴³
2. Another approach is the measurement of arteriovenous (A-V) difference in concentration of AAs in arterial blood inflow to a tissue or organ and the venous blood outflow of the same organ tissues; from the difference between the nitrogen inflow and outflow, the protein metabolism in the organ can be assessed. A positive nitrogen balance indicates net nitrogen accretion in the organ, while negative balance indicates net loss of protein in that organ.⁴⁴
3. More complicated models combine the A-V difference of concentration of stable isotope labeled-AA tracers.^{45,46} These models involve complicated mathematical calculations to quantify net protein balance and rates of protein synthesis and breakdown in specific tissues such as muscle, under different nutritional conditions, and disease conditions.⁴⁷
4. Positron emission tomography (PET) imaging is a possible new approach to measure protein metabolism in muscle. Because methionine is not degraded in muscle tissue, injection of the short-lived ($T_{1/2} = 20$ min) ^{11}C -labeled methionine followed by imaging the metabolic behavior of the methionine tracer in muscle provides protein synthesis rates in skeletal muscles of limbs. It has been validated in animal studies⁴⁸ and applied to human subjects.⁴⁹

Other approaches to measure protein breakdown; Protein turnover in skeletal muscle accounts for a substantial portion of whole-body protein turnover.³⁴ Muscle wasting has been a metabolic feature of multiple disease conditions, especially in critical illness.³⁹ Since N- τ -methylhistidine (3-methylhistidine) is a component of actin and myosin, it is not reutilized for protein synthesis following the breakdown of muscle protein but is quantitatively excreted in the urine. Its presence can thus indicate the degree of muscle breakdown occurring at the time of measurement.

F. Protein Metabolism in Specific Organs

Gastrointestinal tract

The GI tract is a metabolically active organ and is the primary vehicle by which dietary nutrients enter the body. It has been estimated that a minimum of 25% of protein consumed is used in its first pass within the

splanchnic bed—circulation in the GI tract, liver, spleen, and pancreas.⁵⁰ In the presence of a low-protein diet, GI tract growth is preserved over that of other peripheral tissues or muscle.^{51,52} A large proportion of the AAs utilized by the GI tract, especially the small intestine, are used for the synthesis of secretory proteins. Within the intestinal cells, AAs may also be used for synthesis of nucleic acids, the antioxidant glutathione, Apo proteins as components of lipoproteins, or other nitrogen-containing compounds.⁵¹

As an important energy source to the GI tract, dietary glutamate and aspartate account for a significant proportion of the energy used by the small intestine.⁵¹ Glutamine released by the lung and skeletal muscle into the circulation is a primary source of energy for the enterocyte and may have trophic effects on the mucosal cells of the GI tract. Glutamine, as a precursor to nucleotide synthesis, is also used by the immune system, which is abundant within the GI tract as gut-associated lymphoid tissue.⁵³

Liver

Liver is a key organ for protein and AA metabolism due to its high capacity for uptake and metabolism of AAs. Because of its anatomical location in the GI tract, the liver helps to regulate the flow of AAs and other nitrogenous compounds into the systemic circulation from ingested food or from endogenous protein degradation in tissues.⁴⁷ Approximately 57% of the AAs extracted by the liver are either oxidized or used to synthesize plasma proteins in the liver.¹¹ Important roles that the liver plays in metabolism are discussed below.

Degradation of amino acids; Most of the AAs are degraded in the liver. The biochemical processes include deamination and transamination of AAs, followed by conversion of the nonnitrogenous part of those molecules to glucose or lipids. Several of the enzymes used in these pathways (for example, alanine and aspartate aminotransferases) are commonly assayed in serum to assess liver damage.⁵⁴

Nonessential amino acid synthesis; Liver is the primary organ for synthesis of most of the nonessential AAs from essential AAs, such as tyrosine from phenylalanine and cysteine from methionine, or de novo synthesis of AAs from carbohydrates such as alanine from pyruvate and aspartate from oxaloacetate. Carrier systems also exist for AA transport into the hepatocyte as the only site for oxidation of five of the essential AAs, but not the BCAAs.^{20,11}

Protein synthesis; AAs for use in the liver are transported across the basolateral membrane of the enterocyte into the portal vein circulation (Fig. 2.2). Most of the plasma proteins, including apo proteins and acute-phase

proteins, are synthesized and released from liver to the general circulation. Albumin, the most abundant plasma protein, is synthesized almost exclusively by the liver. Some plasma proteins synthesized in the liver play important roles, including maintenance of oncotic pressure in the blood (albumin), transport functions (albumin, prealbumin, transferrin, lipoproteins, some globulins), clotting functions (fibrinogen, prothrombin), and enzymatic reactions (alanine aminotransferase, aspartate amino-transferase).

Urea synthesis; The liver removes ammonia from the body through the urea cycle (Krebs–Henseleit) pathway. Ammonia, a metabolite of nitrogen compounds, has high toxicity. It can accumulate from exogenous protein intake or from protein breakdown during catabolic states such as trauma or sepsis.⁵⁴ If not rapidly and efficiently removed from the circulation, it impairs the central nervous system function. The normal blood concentration of the ammonium ion (NH_4^+) is $<50 \mu\text{mol/L}$; a doubling to $100 \mu\text{mol/L}$ can lead to a disturbance in consciousness, and at $200 \mu\text{mol/L}$, coma and convulsions may occur. The major biochemical pathway to eliminate ammonia is via the urea cycle exclusively found in the liver. Amino groups donated by ammonia and L-aspartate are converted to urea in this cycle, while L-ornithine, citrulline, L-argininosuccinate, and L-arginine act as intermediates. Urea ($\text{NH}_2\text{—CO—NH}_2$) is subsequently released into the circulation and excreted in the urine by the kidneys.

Carbon metabolism; For many AAs, ketoacid analogues are formed when the amino group(s) is removed, and some are thus ready to enter oxidative metabolic pathways. For example, both α -ketoglutarate (from glutamate) and pyruvate (from alanine) are intermediates of the tricarboxylic acid pathway in glucose oxidation. The carbon skeletons of others participate in degradation pathways that result in intermediates that are then metabolized in these oxidative pathways. Thus, protein contributes to the energy supply of the body, with about 15%–16% of dietary energy consumed as protein in the United States.⁵⁵ This contribution to energy needs increases during periods of caloric restriction or when carbohydrate stores (such as from glycogen) are depleted.

Skeletal muscle

After passing through the splanchnic organs (mainly the liver), comparatively little of the BCAAs are oxidized versus other AAs that are metabolized within liver. Using stable isotope-labeled tracers, it was estimated that $\leq 2\%$ of ingested leucine was oxidized when it passed the splanchnic region, mainly the liver,

as compared to 80% for phenylalanine.⁵⁶ Therefore, a large portion of ingested BCAAs enter the systemic circulation for metabolism, largely by the muscle.⁵⁷

Skeletal muscle uptake of AAs following a protein-containing meal leads to net protein synthesis within the muscle.²⁰ The fed state supplies AAs from dietary protein as a source of energy and helps replenish AA stores. Muscle mass is maintained in a relative steady state with a fraction of the muscle broken down and resynthesized daily (protein turnover).¹¹ Most AAs are released from the muscle in proportion to their concentrations within the muscle.^{58,59} The release of BCAA is more conservative than that of other AAs. Alanine and glutamine are released from muscle at higher concentrations than their concentrations within muscle, playing a key role in muscle BCAA metabolism by participating in removal of the excess nitrogen produced during BCAA degradation, which mostly occurs in muscle.^{58,60}

The glucose–alanine cycle is another biochemical process for skeletal muscle to eliminate nitrogen waste while replenishing its energy supply. Glycolysis, the process of glucose oxidation within the muscle, produces alanine via the action of alanine transaminase or alanine aminotransferase. Alanine is then transported through the bloodstream to the liver where its carbon atoms are used in gluconeogenesis to produce glucose, which is then delivered back to the muscle. The amino group from alanine is converted in the liver to urea and eventually excreted by the kidneys.⁶¹

Within muscle, leucine is the only AA completely oxidized for energy. During a fast, leucine levels rise both in the bloodstream and within the muscle. Leucine oxidation by skeletal muscle in turn increases, producing acetyl coenzyme A that acts as an energy source for the muscle while simultaneously sparing pyruvate oxidation in the tricarboxylic acid cycle. Pyruvate is then reduced to lactate in the absence of oxygen (anaerobic respiration) by the action of lactate dehydrogenase and released by the muscle. The carbon skeleton of pyruvate is thus transferred to the liver in the form of lactate as the precursor for gluconeogenesis. Leucine oxidation by skeletal muscle may have some effect on sparing these essential gluconeogenic precursors.⁶²

Kidney

The kidney plays an important role in the interorgan exchange and synthesis of a variety of AAs.⁵¹ The kidney preferentially extracts specific AAs, such as glycine, alanine, glutamine, glutamate, phenylalanine, and aspartate. The kidney is also a major site for arginine, histidine, and serine production.²⁰ Additionally, the kidney plays an important role in the conversions of phenylalanine to tyrosine and glycine to serine.⁶² Aside from the liver, the kidney is the only organ with the

necessary enzymes for conversion of the gluconeogenic AAs to glucose during the process of gluconeogenesis.

Another major role of the kidney is to eliminate nitrogenous waste by urinary excretion. As previously mentioned, ~80% of the body's nitrogenous wastes are excreted as urinary urea nitrogen.²⁰ In addition to excreting urea, the hepatic metabolite of amino groups, ammonium ions are another important nitrogen excretory product, generated in the kidney by the action of the enzyme glutaminase on glutamine, yielding glutamate and ammonium ion, which can then be excreted. The generation of ammonium ions is important in acid–base homeostasis as ammonium excretion is the body's main way to excrete excess hydrogen ions, thus preventing acidosis.

Brain and central nervous system

AAs are actively transported into the neurons in the brain and are present in higher concentrations than within plasma. Essential AAs perform a variety of important functions or serve as precursors of neurotransmitters, initiating and transducing evoked signals between neurons and other cells. For example, tryptophan acts as a precursor to the sleep-regulating hormone, melatonin, and the excitatory neurotransmitter, serotonin.⁶³ Tyrosine is used to synthesize the catecholamines (dopamine, norepinephrine, and epinephrine), the blood levels of which rise during stress, causing an increase in heart rate, blood pressure, and blood glucose.

Neuropeptides are synthesized, stored, and secreted by neurons of the central and peripheral nervous system. After intracellular synthesis, they are processed in vesicles that are transported to axon terminals along microtubules via fast axonal transport. These peptides modulate neuronal communication by acting on cell surface receptors and are more potent than other neurotransmitters.⁶⁴

Neuropeptides are diverse and important in specific functions in the body. The enkephalins and endorphins are similar to opiates, affecting pain sensation, body temperature, and blood pressure regulation. CCK is one of the most abundant neuropeptides in the central nervous system, although it is also synthesized and secreted by enteroendocrine cells in the duodenum, stimulating the digestion of fat and protein. Though the function of CCK within the brain is not fully understood, it has been shown to play a role in the control of food intake or in the pathogenesis of certain types of anxiety and/or schizophrenia. Antidiuretic hormone (vasopressin) secreted by the pituitary gland also performs a variety of functions within the body, including specific hormonal actions in the brain with respect to memory formation, blood pressure and temperature regulation, and pair-bonding behavior.^{20,64}

III. DIETARY REQUIREMENTS

Dietary protein intake on a daily basis ranges widely in most individuals, while requirements are based on having both an adequate amount of each of the essential AAs and additional AAs to participate in interconversions to provide the specific AAs needed for synthesis of protein and other AA-based compounds. Due to continuous protein breakdown and resynthesis, the body can adapt over a period, usually 5–7 days, to lower protein intakes by metabolizing body protein to provide required essential AAs and nitrogen. After this adaptation to lower or higher intakes, a new steady state in the body pools develops.

In situations where there is little protein intake over a prolonged period of time, deficiency states develop. If there are adequate sources of energy from carbohydrates and fats, protein malnutrition develops. Usually seen in young children who have a comparatively high requirement for protein for growth, the condition is called kwashiorkor, characterized by fatty liver, abnormal hair color, diminished skeletal muscle, and edema. If both protein and energy intakes are severely inadequate such as in famine situations, marasmus develops in infants and small children, who show weight loss, stunted growth, and diarrhea. Starvation in adults results in depletion of the body's protein mass. These conditions are life-threatening, particularly in infants and young children. Worldwide, estimates of over 10% of the population are wasted or malnourished.⁶⁵

Protein–energy malnutrition is also found in non-famine situations: it has been identified in hospitals, associated usually with disease,^{66,67,68} and is frequently seen in older individuals.⁶⁹ Protein deficiency in early life has been shown to have adverse effects on the later cognitive function⁷⁰ and in children and adults, affects immune function, increasing risk for septicemia.⁷¹ Kidney function is also adversely affected by protein deficiency.⁷² Modern focus on meeting dietary protein needs in these circumstances through public programs and aggressive diet therapy has decreased the incidence of undernutrition in these situations.

A. Indicators and Criteria of Protein Adequacy

There is no one indicator to assess adequacy for dietary protein, and no single parameter is completely reliable to assess protein nutritional status. Borderline inadequate protein intakes in infants and children are reflected in failure to grow as estimated by length or height.^{33,73} However, weight–height relationships can be distorted by edema and ascites.⁷⁴ Midupper arm parameters such as arm muscle circumference have been used to measure protein status.⁷⁵ The triceps skinfold

is reflective of energy nutritional status, while the arm muscle circumference (or diameter) is reflective of protein nutritional status (unless a myopathy or neuropathy is present).⁷⁶ In addition, urinary creatinine excretion has been used as a reflection of muscle mass,^{74,77,75} but it is not very sensitive. Serum proteins such as albumin, prealbumin, transferrin, and retinol binding protein are often used clinically in evaluating protein status in hospitalized patients but often of limited use as values may be affected by fluid retention, drugs, etc. They may be of greater use as a screen for the general population.

In determining adequacy of requirements, defined as the lowest daily intake for a nutrient that meets the need of apparently healthy individuals, approaches have been taken to estimate requirements from short-term metabolic studies. Unlike most required nutrients, the body has many mechanisms to adapt to the varying levels of intake of dietary protein. This adaptation occurs due to the relatively large body pools that can be accessed when specific AAs or nitrogen is needed. When no protein is consumed, protein breakdown continues at a comparatively higher than protein synthesis, with a decrease in nitrogen excretion. Nitrogen excretion decreases to a relatively constant level provided energy is adequate; and this lowest level of nitrogen loss when not consuming protein has been termed basal or endogenous losses. These endogenous losses have been correlated with basal metabolic rate and body weight.^{78,79} In older individuals, these losses are the same as for younger individuals based on body weight but higher based on lean body mass.⁸⁰ Energy must be adequate to prevent the degradation of body protein to meet the needs of glucose and carbon skeletons of nonessential AAs.⁸¹ Unless the correct balance of essential AAs is present in the diet in the correct amounts (refer below to protein quality), utilization of dietary protein will be diminished.

B. Approaches to Assess Protein Adequacy/Status

Factorial method

This method has been employed to determine protein requirements in the past⁸² and is based on estimating endogenous nitrogen losses (also called obligatory nitrogen losses) as described above when a person is fed a diet over a period of time, usually >6 days, that is essentially free of protein, but with adequate energy intake to maintain body weight. Estimates (usually by measurement) of the amounts lost in feces, hair, seminal fluid, menstrual losses, sweat, and skin losses are added to the amount of nitrogen excreted in urine. Additional amounts for growth or losses due to lactation are

included when appropriate. Since the major losses of nitrogen under most conditions are via urine and feces, the protein requirement can be estimated by extrapolating the obligatory losses to a point where it is assumed that protein needs (as nitrogen) equal the obligatory protein lost as nitrogen (termed nitrogen equilibrium) (see [Table 2.3](#) for the various methods).

While this method was used in the 1960's through 1980's, the weakness in the method is that feeding the amount of protein estimated as being lost did not result in nitrogen equilibrium, in part due to the need for additional nitrogen required to metabolize dietary protein (considered the efficiency of utilizing dietary protein) and because of the curvilinear relationship between protein intake and nitrogen retention.^{83,84} An additional issue is how to estimate the additional nitrogen (as protein) needed to meet the needs for growth for infants and children and during pregnancy and lactation—amounts where efficiency of utilization is known to be less than 100%.

Nitrogen balance method

The balance method is typically used for nutrients that are not metabolized or whose excretion products can be obtained. A given intake is consumed, and after equilibrium is reached at this level of intake, all routes of excretion are measured or estimated. When intake is equal to excretion, or “0” balance, it is assumed that the individual's need for that nutrient is met at that level of intake. It does not provide an understanding of the mechanisms in place that result in changes in excretion. Since the major component of dietary protein is nitrogen, and for most proteins approximately 16% is nitrogen, nitrogen intake can be measured, as can excretion in carefully controlled studies in apparently healthy people. Nitrogen balance occurs when nitrogen intake equals the amount of nitrogen excreted in urine, feces, skin, and miscellaneous losses.

This method of estimating human protein requirements generated sufficient data for use in estimating total protein (nitrogen) requirements.⁸⁵ One assumption in using this method is that when adult needs are met or exceeded, after a period of 6–10 days consuming a specific intake level, nitrogen intake equals nitrogen excretion (and no growth occurs such as might with muscle building). When intakes are inadequate, negative nitrogen balance results, meaning that more nitrogen is excreted than is consumed, due to net losses of body protein. In order to ensure that the amount of protein is being evaluated and not the need for a specific essential AA, high-quality proteins are utilized as test proteins to prevent negative nitrogen balance due to a limiting essential AA^{86,87} ([Table 2.3](#)).

As mentioned, multiple days of consuming a constant level of dietary protein are needed for adaptation to reach a new steady state of nitrogen excretion.^{88,89} Such studies also require accurate nitrogen balance measurements: if not all, the dietary protein is consumed thus overestimating actual intake and not all the losses are obtained or accurately estimated thus underestimating nitrogen losses, false positive nitrogen balances may be obtained.⁹⁰

Statistical analysis of nitrogen balance data

While studies in healthy adults have shown that protein requirements can be met through the use of nitrogen balance methodology, the relationship between intake and requirements near the point of nitrogen equilibrium does not appear to be linear. In assessing the data available at the time, the Institute of Medicine (IOM) report on protein requirements released in 2002⁸⁵ evaluated three methods to compare data from multiple research studies that evaluated protein requirements in men, women, elderly, and from nonanimal protein-based diets.⁸⁵ Using only data from nitrogen balance studies in which levels of nitrogen (protein) intake that were found to be inadequate as well as adequate, three methods of interpolation were evaluated: a smooth nonlinear model,^{91,83} a two-phase linear model^{92,93} and a linear model.^{89,94} The linear interpolation model was thus used to estimate both the individual requirements (the intakes predicted to result in zero balance) and thus the distribution of protein requirements.⁸⁵

Some have argued that this may underestimate actual protein needs, as it depends on equilibration to a steady state at the lowest level of protein intake at which nitrogen balance may be achieved, and given that little is known about corresponding changes in body pools, lean body mass may be adversely affected at such minimal levels.^{95,96} In particular, reference has been made to the possible need for additional levels of dietary protein in the elderly,⁹⁷ where decreased muscle mass is frequently observed.

C. Assessment of Amino Acid Adequacy

Indicators for assessing essential amino acid needs

The primary method to determine the requirement for an individual essential AA focuses on measuring the relationship between the intake of the AA in an otherwise adequate diet and a marker of nutritional adequacy. Markers can be the effect of the AA on nitrogen balance or the concentration of the AA resulting from its metabolism and utilization. Research studies over the last three decades using nitrogen balance-derived values for essential AAs in adults are lower than values derived by the other methods.⁸⁵

Plasma amino acid response method

This method depends on the use of a test essential AA that is followed as the intake of the AA is increased from an inadequate amount to levels above adequacy; at levels below requirement, the circulating concentration of the essential AA is not only low but changes little in response to increases in intake of the AA; as the intake amount is near the requirement level, the level in plasma becomes more sensitive and starts to increase and continues to increase at higher levels. The point at which the relationship between intake and plasma concentration starts to increase (becomes a new linear relationship) is considered to be an estimate of the requirement. This method has been extended to measure the changes in plasma concentration of the essential AA from the postabsorptive to the fed state postconsumption,⁹⁸ expecting to rise only when the dietary supply of the AA is greater than the individual's requirement.

Amino acid oxidation methods

A number of methods have been developed to improve on nitrogen balance to determine essential AA requirements. With the direct amino acid oxidation method, the essential test AA is labeled with ¹³C, and production of the label as ¹³CO₂ is then taken as the measure of irreversible oxidative loss of the AA. For almost all of the essential AAs, the inability of the human to synthesize the carbon skeleton is responsible for its essentiality. When the requirement is reached, then the low constant oxidation rate starts to increase proportionately. The most salient problem arises from the reliance on the determination of a breakpoint in the oxidation of the test AA. One limitation of this method is that it can only be used for BCAA, phenylalanine, and lysine given their oxidation steps. A subsequent modification of this method was to conduct studies over 24 h (24 h AA balance method) to include a period of fasting and feeding.

Subsequently, the indicator amino acid oxidation method was developed and is based on measuring AA oxidation; it uses the catabolism of the carbon skeleton of an indicator (nonlimiting) AA as a carbon analogue of nitrogen balance. When a single essential AA is provided below its requirement, it acts to limit the ability to retain other nonlimiting AAs in body protein, which then are oxidized at higher rates until the essential AA is at a level at which protein synthesis and breakdown is constant.⁸⁵ For further discussion of these methods, see the DRI report.⁸⁵

D. Reference Intake Levels

Reference intakes for dietary protein established in 2002 as part of the Dietary Reference Intake (DRI)

TABLE 2.4 Dietary Reference Intakes for dietary protein for Canada and the United States.⁴⁴

Age group	EAR g protein/kg		EAR g protein/d		RDA g protein/d		AI g protein	
	Males	Females	Males	Females	Males	Females	/kg	/d
0–6 months							1.52	9.1
7–12 months			1.0/kg	1.0/kg	11	11		
1–4 years			0.8/kg	0.8/kg	13	13		
4–8 years			0.76/kg	0.76/kg	19	19		
9–13 years			0.76/kg	0.76/kg	34	34		
14–18 years			0.73/kg	0.71/kg	52	46		
18–30 years	0.66	0.66	47	38	52	46		
30–50 years	0.66	0.66	47	38	52	46		
50–70 years	0.66	0.66	47	38	52	46		
>70 years	0.66	0.66	47	38	52	46		
Pregnancy				+18		+21		
Lactation				+18		+25		

AI, Adequate Intake; EAR, Estimated Average Requirement; RDA, Recommended Dietary Allowance.

process of the IOM for the United States and Canada are listed in Table 2.4⁸⁵ The Recommended Dietary Allowance (RDA) is defined as the average daily level of intake sufficient to meet the nutrient requirements of nearly all (>97%) healthy individuals in a specific population group, while the Estimated Average Requirement (EAR) is the amount that is sufficient to meet the nutrient requirements of half of a population group and is used to evaluate adequacy of nutrient intakes.⁸⁵

Estimated average requirements

As part of the DRI process for the United States and Canada, a metaanalysis of available nitrogen balance studies⁹⁴ was used to estimate the EAR for nitrogen (and thus for dietary protein) in healthy adults⁸⁵ (see Table 2.4). The criterion of adequacy used for the protein EAR is based on the lowest continuing intake of dietary protein that is sufficient to achieve body nitrogen equilibrium (zero balance).

Although the data indicated that women had a lower nitrogen requirement than men per kg body weight, perhaps because women on average have a larger fat mass (15 vs. 28%, respectively) when controlled for lean body mass, no gender differences in protein requirements were found. The reference body weights of 70 and 57 kg for men and women, respectively, were used to establish the daily EAR (see Table 2.4). The metaanalysis included 19 published nitrogen balance studies from many countries for 235 individuals given at least three levels of nitrogen intake for periods of 10–14 days of nitrogen intake or that only provided group data for multiple levels of intake (n = 174 individuals).⁹⁴ While differences in

requirements were evaluated for elderly, for women, and for studies using only plant proteins, no differences were identified. Thus the EAR remains the same for these subgroups.

Given that protein requirements for children need to reflect needs for maintenance and for growth, a factorial approach was used to estimate the average requirement (EAR) for children.⁸⁵

Recommended dietary allowance

The RDA for the United States and Canada is calculated at 2 standard deviations above the EAR. Since the nitrogen balance data for adults were distributed in a log normal fashion, the CV was 12% based on the statistics employed, and the RDAs were rounded after multiplying the EAR by 1.24.⁸⁵ This CV of 12% was also used to increase the EARs to obtain the RDAs for other age groups (see Table 2.4).

Adequate intake

As with other nutrients, the DRI category established for young infants is an Adequate Intake (AI), based on the average amount in human milk consumed in the age group (0.78 L/d for 0–6 mo).⁸⁵ The average protein content of human milk for this age group was estimated to be 11.7 g/L. The resulting AI is 9.1 g/d or 1.52 g/kg for a 6 kg infant (Table 2.4).

While an AI was estimated to be 14.4 g/d or 1.6 g/kg/d based on the reference weight of 9 kg for older infants 7–12 mo, concern that other protein sources might not be equivalent to that in human milk resulted in an EAR being established for older infants 7–12 mo based

on the factorial method.⁸⁵ The EAR for older infants 7–12 mo is thus 1.0 g/kg with the RDA set at 11 g/d (Table 2.4).

E. Tolerable Upper Intake Level

As part of the DRI process, the scientific literature was evaluated regarding effects of consuming very high protein diets to determine if there was adequate information upon which to establish a Tolerable Upper Intake Level (UL). Populations worldwide have consumed significantly higher protein intakes above the typical 15%–16% of energy as seen in the United States; e.g., protein accounts for just under 50% of energy in the diets of Eskimos when eating only meat. There are a few nitrogen balance studies at high protein levels in the literature in which subjects were fed 200–300 g/d; they have all shown positive nitrogen balances^{99,100,101} but no detrimental effects on protein homeostasis were noted. Of concern is whether there is a maximum rate of urea synthesis to counter the effects of high ammonium ion concentrations. Although there are some reports of effects of high protein supplements, there was insufficient evidence upon which to base a UL.⁸⁵

F. Dietary Sources

Almost all protein in the diet is in the form of AAs linked by peptide bonds; free AAs are minimal except from tissues with large AA pools (such as liver or plasma). Other sources of nitrogen include nucleic acids and small nitrogen-containing molecules. Dietary proteins are usually assumed to be about 16% nitrogen by weight (thus a factor of 6.25 is used when converting nitrogen to equivalent protein), but this percentage varies with the protein's AA composition. For example, approximately 20%–27% of the nitrogen in human milk is from nonprotein nitrogenous substances such as urea, resulting in a lower protein content than would be expected when using a factor based on 16%. When the weights of analyzed AAs in a food (taking into account water of hydrolysis) are summed, the protein/nitrogen ratio is 5.38 for egg, 5.62 for whole milk, 4.86 for cooked ham, 5.70 for whole-meal wheat bread, and 6.07 for soymilk.⁸⁵ Thus, the factor used is important when converting the amount of nitrogen present in a specific foodstuff to total protein present.

All nine essential AAs are present in foods from animal sources such as milk, eggs, fish, and cheese; thus animal proteins are considered complete proteins. Foods that are derived from plants, such as grains, nuts, seeds, and vegetables, are usually low in one or more of the AAs essential for humans, and thus when consumed individually, they are considered

incomplete proteins, even if they have similar amounts of total protein per gram of the food item (meaning that they have higher amounts of nonessential AAs). For example, a serving of 3 oz of meat contains about 25 g of protein, whereas 8 oz of cooked oatmeal contains 6 g of protein and is limited in lysine based on comparison with protein scoring patterns (see Table 2.5 and discussion below).

Estimates of protein intake in the United States

The 2013–16 cohort of the What We Eat in America component of the NHANES survey found that 5% of men and 8% of adult women in the United States had inadequate diets when compared to the EAR of 0.66 g/kg body weight.⁵⁵ Of interest in this analysis, 12% of those over 70 years of age had inadequate intakes of dietary protein,⁵⁵ meaning that the estimates of their dietary intake were below 0.66 g protein/kg body weight.

It has been argued that the protein requirements of elderly are greater than that established as the RDA in 2002 by the IOM, in part due to improvements in lean body mass in elderly provided supplemental essential AAs.^{102,103} Subsequent studies have pointed to an increased need for total protein in elderly due to changes in protein turnover irrespective of the essential AA content.¹⁰⁴ Given that national surveys show that a comparatively large proportion of older adults in the United States (~12%) have intakes below the EAR, the minimal amount of protein need in the group, protein intake in this age group is of concern.

TABLE 2.5 Indispensable amino acid scoring patterns, mg/g protein.

Amino acid	FAO/WHO/ UNU 1985 pattern ^a	Recommended FNB/IOM pattern ^b	FAO/WHO/ UNU 2007 pattern ^c
Histidine	19	15	18
Isoleucine	28	25	31
Leucine	66	55	63
Lysine	58	51	52
Methionine + cysteine	25	25	25
Phenylalanine + tyrosine	63	47	46
Threonine	34	27	27
Tryptophan	11	7	7
Valine	35	32	41

^aBased on requirements for 2–5 years olds for protein (0.99 g/kg/d).⁹⁷

^bBased on estimated average requirements for 1–3 years olds for protein (0.66 g/kg/d) and indispensable amino acids.⁴⁴

^cBased on average requirements for 1–2 years olds for protein (0.86 g/kg/d) and indispensable amino acids.⁹⁶

Determining protein quality

Protein quality focuses on the relative composition of AAs present in a protein source compared to a protein pattern than is considered to represent the highest quality protein. Methods used in some countries for labeling protein content of a food item, or methods proposed for use for labeling, include the protein efficiency ratio (PER), required in Canada; the protein digestibility corrected amino acid score (PDCAAS), originally recommended by FAO/WHO in 1993 and in the United States by the Food and Drug Administration; and the digestible essential amino acid score (DIAAS), most recently recommended by FAO/WHO¹⁰⁷ which includes measurement of ileal digestibility.¹⁰⁷ While PER requires an animal growth study comparing utilization of a test protein with casein protein, other methods evaluate a food's protein quality by comparing its essential AA composition to the ratio of AAs estimated to be needed by humans. The practice has been to compare the amount of each essential AA in a food to a reference (scoring) pattern based on the estimated essential AA requirements of a 2- to 5-year-old child^{106,108} (see Table 2.5).

For example, when a protein source has essential AAs in which some are less than the scoring pattern per gram of protein, the AA with the lowest percent of the scoring pattern's amount is said to be the limiting AA. For example, when compared to the PDCAAS score, if an essential AA such as lysine has the lowest percentage, e.g., 40, then only 40% of that protein can be used for protein synthesis unless another protein is consumed that is high in lysine. The highest PDCAAS score any protein can achieve is 100. This would mean that 100% of that protein can be used for protein synthesis—that it contains all essential AAs in appropriate ratios per gram of protein. Given that typical diets contain many sources of dietary protein, foods can function as “complementary proteins” if consumed simultaneously and are higher in the AA that is limiting in the other protein.¹⁰⁹ Comparison with a scoring pattern indicates the overall quality of a protein because it represents the relative adequacy of its most limiting AA as needed by the most vulnerable individuals in the population, young children, whose needs for essential AAs for growth are great.

An important benefit of using the PDCAAS score is that this approach uses human AA requirements, whereas animal bioassays are used by the PER method, which was used in the past in the United States.⁵⁴ A step used in both methods is to multiply the lowest AA ratio by its true protein digestibility. While this ensures that the appropriate levels of essential AAs are available from the protein source, it does require determining digestibility in an animal model. This aspect has spurred research to obtain an improved method to evaluate

protein quality. The comparative ease of measuring ileal digestibility in the DIAAS method has advanced it as a possible successor to PDCAAS.¹¹⁰

As part of the DRI process, a scoring pattern to evaluate proteins was also proposed, based on the EARs for essential AAs established for 1–3 year old children⁸⁵ (see Table 2.5).

Methods to analyze for protein content in food

Historically, the protein content of foods was obtained by determining total nitrogen in a food sample as measured by the Kjeldahl method.¹¹¹ Nitrogen content is then multiplied by a factor $\times 6.25$ to arrive at the protein content, based on two assumptions: (1) carbohydrates and fats do not contain nitrogen and (2) almost all nitrogen in the diet is present as AAs in proteins. As mentioned earlier, the average nitrogen content of proteins was approximately 16%, hence the factor of 6.25.¹¹²

IV. ISSUES RELATED TO PROTEIN AND AMINO ACIDS IN HUMAN HEALTH

A. Adaptation to Fasting and Starvation

Fasting

Individuals are in a fasting state as blood glucose levels return to a baseline level before the next meal, with a decrease in insulin levels and rise in glucagon levels.¹¹³ Tissues such as the brain and red blood cells require a constant supply of glucose.

Starvation

In prolonged starvation, gluconeogenesis is the major source of circulating glucose. The major substrates for gluconeogenesis are AAs, which are derived primarily from the breakdown of skeletal muscle protein. Gluconeogenesis from AA substrates occurs in both the liver and the kidney.

B. Transition to Ketone Bodies as the Primary Fuel Source

Gluconeogenesis from AA substrates occurs in both the liver and the kidney. The biochemical process begins within 4–6 h of the last meal.^{114,61} After approximately 2 days of starvation, the brain gradually switches its fuel source from glucose to ketone bodies; however, blood cells and the adrenal medulla continue to rely on glucose as the primary energy substrate. This adaptation to starvation involves the conversion in the liver of free fatty acids to ketone bodies. As the metabolically active muscle mass is diminished with starvation, metabolic

processes occurring within these tissues are also reduced, accounting for a reduction in resting energy expenditure. Within approximately 1 week, adaptation to starvation

with a ketone-based fuel system minimizes gluconeogenesis and slows further protein breakdown.^{115,116}

RESEARCH GAPS

- How alternative protein sources can meet the needs of the growing population.
- Considerations about sustainability of present and future production of protein rich foods.
- Development of new biomarkers for adequacy that may rely on new approaches to data using nonlinear modeling.
- Additional focus on whether there are problems with consuming too much protein.
- Effective methods to teach consumers about complementarity of proteins as many move toward plant-based diets.
- Increased focus on how gut microbiota affect protein/AA metabolism in nutrition and neurological function.

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CARBOHYDRATES

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SUMMARY

Carbohydrates are ubiquitous in the human dietary pattern and provide a widely accepted, often affordable, food for many consumers. Because of this, carbohydrates often serve as the primary energy source for many individuals, and their role in developing and maintaining a healthy eating pattern cannot be overlooked. In this chapter, the structure and function of carbohydrates is discussed, as well as their digestion, absorption, and metabolism. Dietary recommendations for carbohydrates, as well as the food sources that provide them in the diet, are discussed, as are the various health outcomes for which carbohydrates are thought to influence. Lastly, current trends in carbohydrate consumption are detailed, and thoughts for future research in carbohydrate science are presented.

Keywords: Added sugar; Carbohydrates; Dietary fiber; Glycemic response; Lactose intolerance; Prebiotics; Starch; Sugar; Whole grains.

I. INTRODUCTION

A. Background

Carbohydrates account for more than half of the calories we consume. In general, most staple crops and many foods recommended as part of a healthy diet are high in carbohydrate. Plant foods are naturally high in carbohydrates, and current recommendations for increased consumption of whole grains, fruits, vegetables, pulses, and dairy products that are low in fat suggest that carbohydrates will continue to represent a significant source of energy in the diet. While some fruits and vegetables are ready to consume in their natural form, many foods with high carbohydrate concentrations require some form of processing before they are generally acceptable for consumption. This is because humans, as monogastric animals, are not able to digest and absorb the unprocessed carbohydrates found in many vegetables, grains, and pulses. This has led to a multitude of processing

methods that have not only made foods more digestible for consumers but also have ushered in an era of highly processed foods that have caused concerns about the healthfulness of consumers' diets.

The role of processed carbohydrates in the diet is complex. Sugars can be added to some foods to serve as an energy source for yeast, to retain water and prevent some types of bacterial growth, and to preserve foods such as jams and jellies. Starches can be used as a thickening agent to form critical gluten structures in breads and rolls and as coatings for certain packaged foods. While these properties and functions in the food supply are generally beneficial, processing can also contribute some negative aspects to foods in the supply chain. Grains that have been overly refined may be preferred by consumers, but they are often stripped of their component parts that include dietary fibers, vitamins, and minerals that have been linked to a host of positive health outcomes.¹ The use of excessive added

sugars in processed products might meet consumer desires for sweetness, but there is growing concern that these additional empty calories are contributing to negative health outcomes. Certainly, processing has been shown to have both its advantages and its drawbacks.

The consumption of carbohydrates as a major source of energy has largely been supported by dietary guidelines provided to consumers by governmental agencies and health organizations, often with a focus on meeting nutrient needs through dietary pattern focused on moderation and variety. However, not all high carbohydrate foods are nutrient dense, especially those that have undergone extensive processing. This has resulted in recent recommendations that have supported limiting the amount of “added sugars” in the diet.² Carbohydrates represent an incredibly broad and diverse group of foods, and, as such, our present knowledge of their role in the diet and in human health is constantly expanding and adapting based on current research.

B. Nutrient Function and Structure

Carbohydrates are polyhydroxy aldehydes, ketones, or substances that, when hydrolyzed, produce these compounds. Carbohydrates are subdivided into several categories that are based upon the number of sugar units present, and in the way in which the sugar units are chemically bonded to one another. They contain carbon, hydrogen, and oxygen, typically in a 1:2:1 proportion that approximately resembles that of a hydrate of carbon, which is where the term “carbohydrate” is derived. Carbohydrates are subdivided into several categories that are based upon the number of sugar units present, and in the way in which the sugar units are chemically

bonded to one another. These classifications generally fall under two classes: simple carbohydrates or complex carbohydrates. These classes are further subdivided based on the number of saccharide units a particular carbohydrate contains (Fig. 3.1).

Monosaccharides

Monosaccharides represent the simplest structural form of carbohydrates, as they consist of a single saccharide unit and are not able to be further hydrolyzed into smaller units of carbohydrates. Each of the three common monosaccharides is hexoses, containing 6 carbon, 12 hydrogen, and 6 oxygen molecules in slightly varied configurations. The three most common monosaccharides are glucose, fructose, and galactose. In addition to these three sugars, the monosaccharide category also includes sugar alcohols and pentoses.

Glucose, sometimes referred to as dextrose or blood sugar, is the most abundant monosaccharide but, on its own, represents only a very small amount of the carbohydrate consumed in the typical diet. Instead, glucose is usually consumed when it is linked to other sugars as part of a di- or polysaccharide. Fructose, found commonly in fruits, some vegetables, and in honey, is also a common part of the average diet. A major source of fructose in the diet comes from the increased use of high-fructose corn syrup, which is used frequently in industry as a relatively inexpensive sweetener in foods and sweetened beverages.³ Galactose, the last of the three major monosaccharides, is typically consumed as part of the disaccharide lactose and is relatively scarce on its own in nature.

The pentoses, ribose, deoxyribose, and ribitol are also monosaccharides but are not thought of as major sources

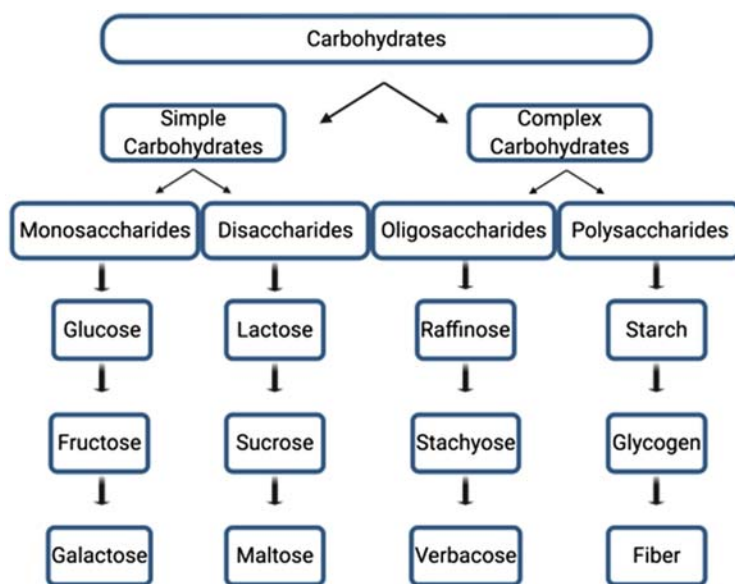


FIGURE 3.1 The hierarchy of carbohydrate nomenclature.

of dietary energy. They are called “pentoses” because, unlike glucose, fructose, and galactose, they are comprised of a five-carbon backbone rather than a six-carbon backbone. While the pentoses are not major sources of energy, they do serve a critical function in many critical metabolic processes like the maintenance of both deoxyribonucleic acid and ribonucleic acid.

Disaccharides

Disaccharides are two monosaccharides that have been joined to one another through a glycosidic bond, which is also sometimes called an acetal bond, and occurs between hydroxyl groups on each of the individual monosaccharides. Disaccharides are a major source of energy in the diet and are typically thought of as the following three major compounds: sucrose, lactose, and maltose.

Sucrose, commonly thought of as table sugar, is the most widely available disaccharide and is the most commonly used naturally occurring sweetener. Sucrose is composed of glucose and fructose, and while it occurs naturally in some fruits and vegetables, it is most commonly thought of in its refined form, which is primarily derived through the processing of sugarcane or beets. Due to its unique structure, sucrose cannot act as a reducing sugar. Reducing sugars are those sugars that have a free aldehyde group or a free ketone group that allows them to act as a reducing agent. The ability of a sugar to act as a reducing agent is important in food processing and production because reducing sugars are necessary for the Maillard reaction to take place. In the Maillard reaction, reducing sugars and amino acids react and create desirable flavors, colors, and textures in some food products. Lactose, which comprises of glucose bonded to galactose, occurs naturally in milk and milk products. As discussed later, lactose is a carbohydrate of concern for some individuals for whom it is not well tolerated. Lastly, the compound maltose contains two glucose molecules that are joined by an alpha bond. Maltose is primarily formed through the partial hydrolysis of starches and other polysaccharides. Because this process is needed in the production of many alcoholic beverages (malted beverages), maltose is typically found in beer and malt liquor.

Trehalose is another disaccharide, though it is a rather insignificant carbohydrate in the diet. Trehalose is a nonreducing sugar that is found primarily in fungi, yeasts, insects, and shellfish.

Oligosaccharides

Oligosaccharides represent carbohydrates that contain between 3 and 10 single sugar residues and are not relatively abundant in the diet when compared to other more common carbohydrates like those in the disaccharide category. Common oligosaccharides include raffinose, stachyose, and verbascose. These oligosaccharides can be found in relatively abundant levels

in legumes, whole grains, some cruciferous vegetables, and some fruits. Humans lack the ability to properly digest these carbohydrates because we lack the digestive enzyme α -galactosidase, thus oligosaccharides are not hydrolyzed and are instead passed undigested into the lower gut. Here, the oligosaccharides are fermented and metabolized by anaerobic bacteria. While this metabolic process does not result in digestive discomfort or adverse symptoms for all consumers, it has been associated with undesirable results such as abdominal cramping and excessive flatulence.

Despite the negative symptoms sometimes associated with the consumption of oligosaccharides, there is also evidence to suggest that they may be associated with some positive health outcomes similar to those thought to be linked with dietary fiber.⁴ Because of these potential health benefits, oligosaccharides have become popular food additives, with some manufactured oligosaccharides such as fructooligosaccharide and lactulose being added to food products that have been marketed as health foods.

Polysaccharides

Monosaccharide units may be bonded in chains that repeat many times over, far exceeding the 10-unit limit for a carbohydrate to be considered an oligosaccharide. When this occurs, the resulting compound is termed a polysaccharide. These polysaccharides represent the most commonly consumed form of carbohydrate in the diet and are often hydrolyzed into smaller subunits through digestive processes. Polysaccharides are commonly divided into two categories: starches and nonstarches.

Starches; Starches represent the most common form of digestible polysaccharides that can be found in plants and are primarily found in grains, legumes, and root vegetables. Starch is present in two forms, amylose and amylopectin, which are determined by the position of linkages that are found in the compound. Amylose, which contains only α -(1,4) glycosidic bonds, forms in linear, unbranched chains, while amylopectin contains both α -(1,4) and α -(1,6) glycosidic bonds resulting in branched chain polymers. While both types of starch are present in grains, legumes, and root vegetables, amylose represents only 15%–20% of total starch, while amylopectin represents between 80% and 85%. Starch is naturally present in the food supply in a number of sources such as root vegetables, cereal grains, seeds, and legumes. In addition to these natural sources, starch is also found in processed foods where it is used as a thickening agent.

Glycogen, which is structurally similar to amylopectin but more highly branched, is the preferred form of carbohydrate storage in both humans and in animals. Due to the high degree of branching of glycogen, it is readily broken down into glucose in order to meet the body's energy needs. Glycogen is primarily stored in

the liver and muscle tissues. Glycogen stored in the liver is accessible to meet the energy needs of the entire body and that stored in muscles being used for those tissues specifically.

In addition to amylose, amylopectin, and glycogen, which are all digestible starches, there are nondigestible, or resistant, starches as well. These starches are not digested in the small intestine and are instead fermented in the colon.

Nonstarches; Nonstarch polysaccharides are complex carbohydrates that are not easily digested like most other starches or other carbohydrates. Like resistant starches, they are instead fermented in the colon. A more common term for these carbohydrates is dietary fiber. These carbohydrates play a critical role in human health and are important for satiation, weight management, fecal bulking, and gut health.⁵ For more information on dietary fiber and its role in health and nutrition, please see Chapter 31 of this book, which is dedicated to dietary fiber.

Sugar alcohols

Sugar alcohols, also known as polyols, are derivatives of mono- and disaccharides that have been hydrogenated. The most common sugar alcohols include xylitol, erythritol, sorbitol, and maltitol. While these carbohydrate derivatives elicit a sweet taste like other sugars, they contribute less energy to consumers. Because of this, sugar alcohols are popular sweeteners in health foods and “sugar-free” candies. Overconsumption of products sweetened with sugar alcohols can result in gastric distress and discomfort for some consumers because these sugars are poorly digested. This poor digestibility and relationship to GI distress is one of the reasons that sugar alcohols are one of the food groups that are restricted on the low-FODMAP diet, discussed later in this chapter.⁶ Sugar alcohols are also noncariogenic and have been shown to provide some protective benefits against tooth decay because they are poorly metabolized by oral bacteria.⁷

II. NUTRIENT METABOLISM

A. Synthesis

Because the body needs a constant supply of glucose to meet its energy needs, there are number of metabolic pathways that can be triggered in order to maintain blood glucose levels. When an individual has gone several hours without consuming carbohydrates, or during brief periods of heightened stress, the body synthesizes glucose to keep the body supplied with energy. This process is termed gluconeogenesis.

As glucose levels in the body begin to decline, the pancreas begins to release the diabetogenic hormone glucagon. Glucagon serves a dual purpose when blood glucose levels are low: to signal liver cells to break down glycogen into glucose and release it into the blood stream and to stimulate the synthesis of new glucose via the gluconeogenic pathway. The gluconeogenic pathway is largely a reversal of the glycolytic pathway that is used by the body during the metabolism of glucose. That process is described in more detail in the “Metabolism” section of this chapter. While gluconeogenesis is essentially a reversal of the glycolytic pathway, it is important to note that three reactions that take place during glycolysis involve ATP and are, therefore, not reversible. This requires that these reactions be circumvented through other enzymatic mechanisms in order for gluconeogenesis to take place. Because of this requirement, the ability of an individual to successfully utilize the gluconeogenic pathway is dependent upon the presence of the enzymes and cofactors required to bypass those energy-requiring reactions.

Gluconeogenesis largely occurs in the liver, but during periods of extended fasting or starvation, the kidneys can also become a critical site for the production of glucose. Gluconeogenic precursors include pyruvate, lactate, glycerol, and some amino acids. The use of lactate as a precursor in gluconeogenesis is one of the ways that body is able to manage blood lactate levels following extended periods of strenuous activity.

B. Digestion

The digestion of carbohydrates occurs throughout the entire gastrointestinal (GI) tract, beginning in the mouth. In addition to the manual digestion that occurs when foods are chewed, the enzyme salivary amylase is present in the mouth and begins that process of breaking down starches into smaller polysaccharides. Salivary amylase is able to hydrolyze only the α -(1,4) glycosidic linkages of starch in the mouth, with both α -(1,6) and β -(1,4) bonds resistant to its effects (Fig. 3.2).

As carbohydrates are swallowed and move down the GI tract from the mouth to the stomach, stomach acids denature and deactivate salivary amylase, halting the digestion of carbohydrates. As food moves into the small intestine, pancreatic bicarbonate is released, creating an environmental pH that is more suitable to enzymatic digestion. Pancreatic amylase is then released and continues the process of cleaving α -(1,4) glycosidic, further breaking down starches into smaller subunits.

As starches are reduced in size to oligosaccharides and disaccharides, microvilli on the intestinal lumen utilize the digestive enzymes maltase, sucrase, and lactase and isomaltase. When the carbohydrates have been broken down to the monosaccharide level of glucose,

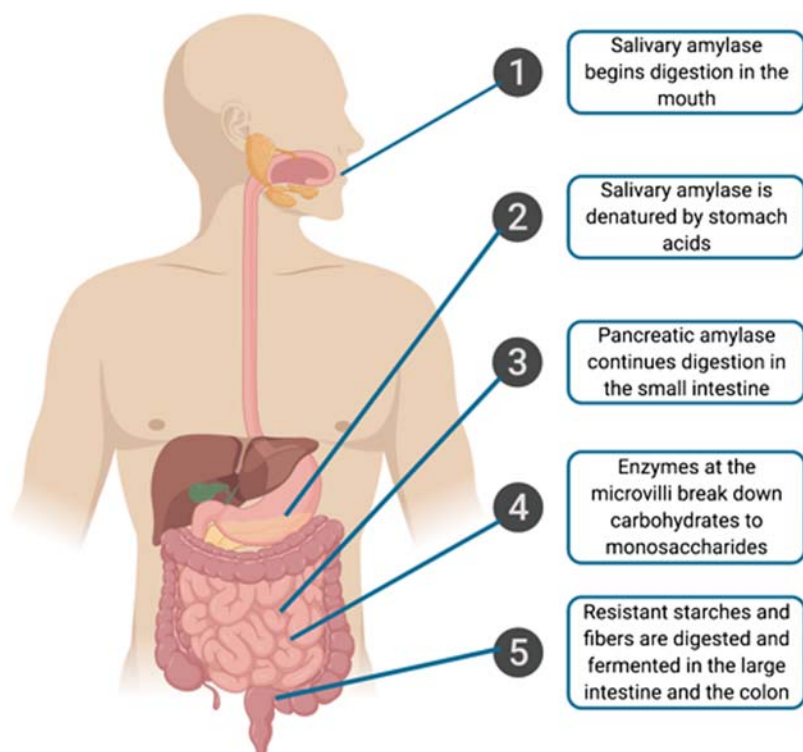


FIGURE 3.2 Carbohydrate digestion begins in the mouth and ends in the large bowel where resistant starches are fermented by colonic bacteria.

galactose, and fructose, they are absorbed by mucosal cells and transported to the liver via the hepatic portal vein. This process is described in more detail in the “Transport” section of this chapter.

Fibers, resistant starches, and other carbohydrates that are not digested in the small intestine are then moved to the large intestine and the colon where they are fermented by bacterial colonies. This fermentation process can result in the formation of short-chain fatty acids and, in some cases, large amounts of gas. The remaining carbohydrates are then excreted in the feces.

C. Transport

The primary site of transport for carbohydrates that have been broken down into monosaccharides is the mucosal cells lining the walls of the small intestine. Both glucose and galactose are absorbed through sodium-glucose transporter 1 (SGLT1) by active transport. SGLT1 is an Na^+/K^+ ATPase pump that uses energy to transport the monosaccharides through the mucosal cells. In order for transport to occur, the carrier must be primed with sodium. Then, one glucose and two sodium ions are transported into the mucosal cell. Glucose exists the mucosal cell at the basolateral surface following one of three routes: leaking back into the intestinal lumen,

diffusing into circulation, or being transported into the blood stream via the glucose transporter 2 (GLUT).

Unlike both glucose and galactose, fructose is transported passively using the GLUT5 transporter. While glucose absorption occurs much more slowly than that of glucose or galactose, its absorption is independent of glucose levels and will occur even when the concentration of glucose in the intestinal lumen is quite high. Once inside the mucosal cell, fructose is moved into circulation by the GLUT2 transporter.

Monosaccharides enter the blood stream and are then transported almost immediately to the liver via the hepatic portal vein. The liver serves as the major site of metabolism for both fructose and galactose, a process that will be described in more detail in the “Metabolism” section of this chapter.

Glucose serves as the preferred energy source for most of the cells functioning in the human body, and as such, the body needs a steady, well-regulated supply of glucose to various sites throughout the body depending on each site’s specific energy needs. As a highly polar molecule, glucose cannot simply diffuse across cellular membranes in order to reach its target site. Instead, a family of highly specific carrier proteins serves as glucose transporters throughout the body. There are 12 known GLUT isomers, and each isomer is responsible for the transportation of glucose in a distinct set of tissues.⁸

D. Metabolism

The metabolic response to individual monosaccharides at the time of digestion and absorption is largely dependent upon the energy needs of the body at that time. The main metabolic pathways of carbohydrates in the human body include glycogenesis, glycolysis, the tricarboxylic acid cycle (TCA cycle), and gluconeogenesis (Fig. 3.3).

Glycogenesis

Glycogenesis is the process by which glucose is converted into glycogen, the preferred form of carbohydrate storage for animals. While a vast majority of the glycogen in the body is stored in the skeletal muscle, the primary site of glycogenesis is actually the liver. Glycogen stored in the liver is primarily responsible for the homeostatic regulation of blood glucose levels, while glycogen stored in the skeletal muscle serves as a source of energy to be used by the respective muscle tissue when it is expending energy. In order for glycogenesis to begin, glucose must first be phosphorylated upon entering the cell, either through the action of glucokinase or by hexokinase. Hexokinase facilitates phosphorylation of glucose in muscle cells, while

glucokinase facilitates the phosphorylation in the liver. The product of this phosphorylation, glucose 6-phosphate, is what initiates the glycogenesis pathway and results in highly branched glycogen being stored in both the muscle and liver cells for later use.

Glycolysis and the TCA cycle

Glycolysis is the initial step in the metabolic process by which the body converts glucose molecules into energy, in the form of ATP. Glycolysis produces two pyruvate units, as well as a triose. Under normal, that is aerobic conditions, the generated pyruvate is transported into the mitochondria where it is decarboxylated to acetyl co-enzyme A (CoA). Acetyl CoA is then used to initiate the TCA cycle, during which it is oxidized to H_2O and CO_2 and generates large amount of energy in the form of ATP through the process of oxidative phosphorylation.

Gluconeogenesis

As described earlier in the chapter in the “Synthesis” section, gluconeogenesis is the process used to meet the body’s energy needs through the synthesis of glucose. For a fuller description of gluconeogenesis, refer to [Section II.A](#), synthesis of this chapter.

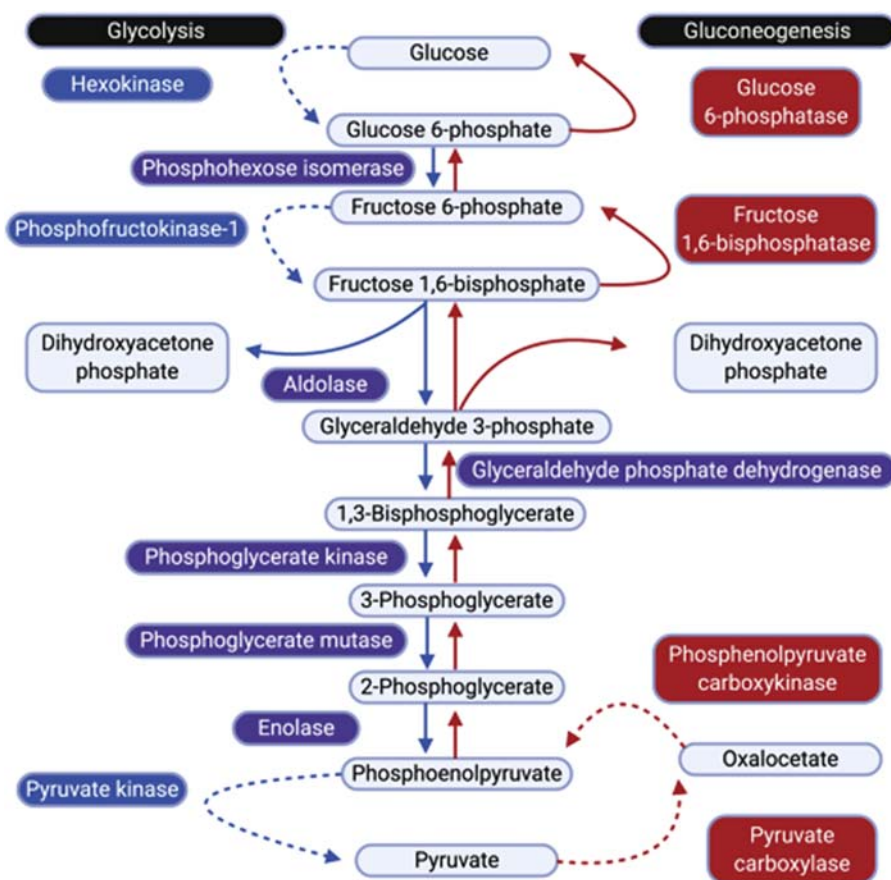


FIGURE 3.3 Glycolysis is represented on the left, and gluconeogenesis is represented on the right. Dashed lines represent steps that require energy.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Gene Regulation

Because glucose serves as the primary energy source for the body, it is critical that blood glucose levels and the amount of glucose available to the cells are tightly regulated. A number of hormones are required to ensure that blood glucose levels remain steady, including insulin, glucagon, epinephrine, and glucocorticoids (Fig. 3.4).

Following the consumption of a meal containing carbohydrate, the β cells of the pancreas release the anabolic hormone insulin, which is then carried to the liver sinusoids through the portal vein. In response to elevated insulin levels, glycogen storage and glycolysis are both increased resulting in reduced levels of glucose secretion from the liver. Elevated insulin levels also stimulate peripheral tissues such as adipose tissues and skeletal muscle tissues, to increase glucose uptake via the GLUT4 transporter. These metabolic changes result in a rapid decrease in blood glucose levels. Elevated insulin levels also result in gene regulation of key enzymes in the carbohydrate metabolic pathway. When insulin levels are high, glucokinase expression is activated, while the expression of both glucose 6-phosphate and phosphoenolpyruvate carboxykinase is inhibited.

When the body is in a state of prolonged fasting, or starvation, the body seeks to prevent blood glucose levels from dropping too low. When blood glucose levels begin to fall, pancreatic α cells release glucagon, a polypeptide hormone, into the blood stream. When glucagon reaches the liver, it results in increased hepatic production of glucose via glycogenolysis and gluconeogenesis in an effort to stabilize blood glucose levels. Glucagon is able to regulate a number of the rate-determining enzymes involved in glucose synthesis, but the most consequential is likely phosphoenolpyruvate carboxykinase, which is activated when glucagon levels are high.

Both epinephrine and glucocorticoids are produced in the adrenal gland, originating in the adrenal medulla and the adrenal cortex, respectively. And both hormones are released in response to decreased blood glucose levels, much like glucagon. The effect of epinephrine, a catecholamine, in the various tissues of the body, is dependent upon the amount and type of adrenergic receptors in that tissue. When epinephrine stimulates α -1 receptors, glycogenolysis is increased, stimulation of α -2 receptors results in increased lipolysis, and stimulation of β -receptors results in increased gluconeogenesis, glycogenolysis, and lipolysis. Glucocorticoids are steroid hormones that enhance the effects of glucagon.

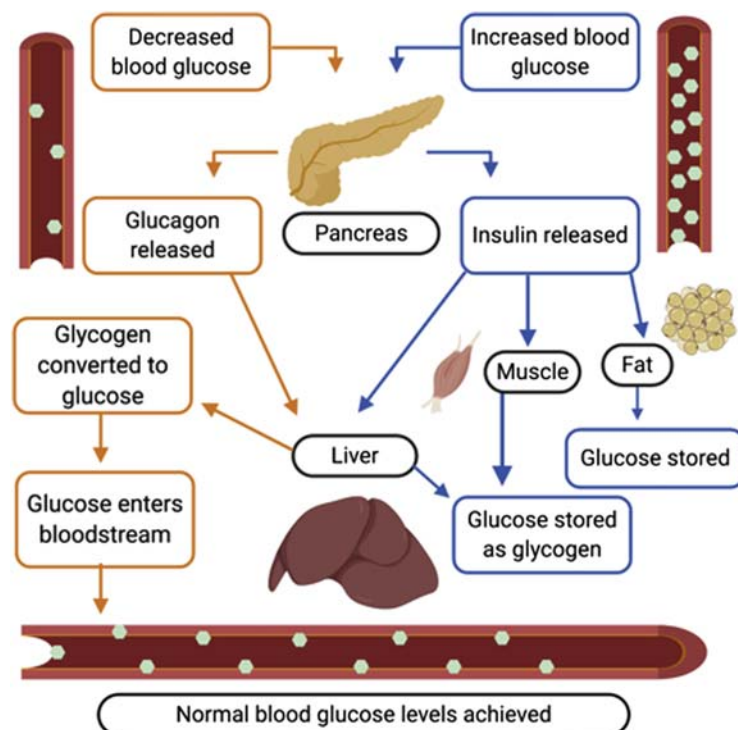


FIGURE 3.4 Glucagon and insulin are used to tightly regulate blood glucose levels, with glucagon resulting in more glucose entering the blood stream and insulin resulting in glucose storage in the tissue.

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Maintenance of adequate glucose for brain function

The human brain, like the brain of all mammals, depends upon a steady stream of glucose to meet its energy needs and maintain normal function. The brain is the largest consumer of glucose in the human body, and even though it accounts for only 2% of the body by weight, it is responsible for utilizing roughly 20% of energy derived from glucose.^{9,10} Because of the brain's need for glucose in order to maintain normal function, the needs of the brain are used to establish dietary recommendations for carbohydrate consumption in the US diet. Moreover, because of the brain's dependence upon glucose, the body will work to meet glucose needs even when the diet does not deliver adequate amounts. Those dietary recommendations, and the mechanisms by which the body meets the brain's energy needs when glucose is scarce, are detailed below.

Protein catabolism to provide three-carbon units from gluconeogenic amino acids

Because glucose serves as the primary energy source for the body, it is critical that blood glucose levels be maintained even during periods of long fasting, or when carbohydrates are not being frequently consumed as a part of the diet. To that end, even when carbohydrate intake is reduced, glucose can still be acquired through the breakdown of glycogen and the synthesis of glucose via gluconeogenesis.

Gluconeogenesis is an energy-consuming process that takes place largely in the cells of the liver and of the kidneys. Through gluconeogenesis, glucose is synthesized from three-carbon molecules, like amino acids, that are largely created through the breakdown of protein molecules in the body. These amino acids can be metabolized to form pyruvate or oxaloacetate, both of which can then be used in the production of glucose.

The proteins metabolized to provide the amino acids used in gluconeogenesis are proteins that could be otherwise used for key functions such as tissue maintenance or growth. This is not a problem when excess dietary protein is consumed, but, in low protein intake situations, protein deficiency signs, for example, low serum albumin levels, will occur, which is why diets with adequate supplies of carbohydrate are often recommended to consumers.

Ketosis

When carbohydrate consumption has been restricted to very low levels, ketone bodies are produced to

provide energy sources by conserving oxaloacetate when carbohydrate levels are low. When oxaloacetate is conserved, acetyl CoA does not take part in the citric acid cycle for the purpose of producing ATP. Instead, acetyl CoA is converted to the ketone bodies acetoacetate, β -hydroxybutyrate, and acetone in the liver and is released into the blood stream.

These ketones are then able to provide energy to tissues throughout the body, including muscle and heart tissue. If the body maintains elevated ketone levels, the brain, which normally requires only glucose as an energy source, will eventually also begin to use ketones in order to reduce its glucose needs and spare the protein that would otherwise be catabolized for gluconeogenesis.

Individuals who are fasting for prolonged periods, starving, or maintaining a very low carbohydrate diet may reach high levels of ketone bodies in the blood or enter "ketosis." While there is variation among individuals, it seems that most people who consume fewer than 50 g of carbohydrate per day will enter ketosis within two to four days.^{11,12} Mild ketosis has been associated with some adverse symptoms such as headaches and muscle soreness.¹³ Severe ketosis, which can also occur when an individual has untreated diabetes due to lack of insulin permitting entry of glucose into cells, can result in a condition known as diabetic ketoacidosis, and if left untreated can result in coma and death.

B. Reference Intakes

General carbohydrate recommendations

The Recommended Dietary Allowance (RDA) for carbohydrate put forth by the Food and Nutrition Board, Institute of Medicine in 2002 for Canada and the United States is 130 g/day.¹⁴ This RDA provides adequate carbohydrate for children and most adults, with infants requiring less carbohydrate in the form of glucose to meet their energy needs and pregnant and lactating females requiring greater amounts of carbohydrate, 175 and 210 g/day, respectively, to meet their energy needs for brain function. The Estimated Average Requirement (EAR) for adults and children ≥ 1 year of age of 100 g/day was based on estimates of the minimum average amount of glucose used by the brain when not depending on partial gluconeogenesis and was then increased by 30% for the RDA, to cover the needs of almost all individuals.¹⁴ For infants (≤ 1 year), an Adequate Intake (AI) was set at 60 g/day.

In addition to an RDA of 130 g/day, carbohydrates also have a recommended Acceptable Macronutrient Distribution Range (AMDR) aimed at helping individuals eat a healthful diet that does not exceed their energy needs.⁹ The AMDR for carbohydrate intake is 45%–65% of energy or 225–325 g of carbohydrate in a standard

2000 kcal per day diet.¹⁴ This AMDR allows for individuals to consume enough carbohydrate to meet recommendations for dietary fiber intake without failing to consume adequate amounts of protein and fat.

While data were reviewed regarding AIs of sugars or added sugars, there were insufficient data or evidence upon which to set daily intake goals of sugars or added sugars for individuals.¹⁴

Dietary fiber

The Food and Nutrition Board, IOM, established an AI of 14 g/1000 kcal consumed in 2002 for Canada and the United States.¹⁴ This translates to roughly 38 g/day for young adult men and of 25 g/day for young adult women. Because the AI for dietary fiber is based on daily energy intake, AIs are lower for young children and for individuals over the age of 50 and are higher for individuals consuming greater amounts of energy than normal such as women who are pregnant or lactating. These recommendations are based on the associations between dietary fiber consumption and risk of cardiovascular disease (CVD). For additional discussions about the role that dietary fiber plays in health, see Chapter 31 in this book on dietary fiber.

Whole grains

In 2015, the Dietary Guidelines for Americans (DGAs) established that healthy eating patterns should focus on increasing the consumption of whole grains, while limiting the intake of refined grains. While whole grains can serve as a good source of dietary fiber, not all whole grains are high in fiber. Whole grains are also good sources of key nutrients like iron, zinc, manganese, thiamin, niacin, copper, riboflavin, vitamin B6, vitamin A, phosphorous, and magnesium.¹⁵

Whole grains are those grains that are still fully intact and have not lost the bran, the endosperm, or the germ to processing. When grains are refined, much of the bran and the germ are stripped away leaving just the endosperm. The bran is the part of a whole grain that contains most of the grain's fiber, while the germ contains many of the key nutrients that are associated with whole grains. What is left after processing, the endosperm, is largely starch.

In order to recapture some of the nutrients that are lost during processing, some grains are enriched. During enrichment, iron, thiamin, riboflavin, and niacin are added back to grain and then they are fortified with folic acid. Because enrichment adds back these nutrients of concern, individuals who eat refined grains should choose enriched grains whenever possible. Additionally, those who consume only whole grains should consider adding some grains that have been fortified with folic acid into their diet, especially individuals who are pregnant or may become pregnant.¹⁶

Added sugar

In addition to its recommendations on whole grains, the 2015 DGAs advised that a healthy dietary pattern was one that restricted the consumption of added sugar to less than 10% of calories consumed in a single day. It is important to note that this recommendation is based on modeling for a healthful diet and that the recommendation to restrict added sugar consumption is not a Tolerable Upper Intake Level (UL). Instead, the recommendation is based on the understanding that as individuals obtain a greater amount of their calories from added sugars, a smaller amount of their energy comes from food sources that can deliver critical nutrients of concern.¹⁵

To help consumers understand the amount of added sugars that is present, the US Food and Drug Administration amended regulations for the Nutrition Facts panel in 2016. The new Nutrition Facts panel for packaged foods includes a number of changes to make nutrition information more accessible to customers, including a new value added sugars that is listed as a subsection to carbohydrates.¹⁷ All Nutrition Facts and Dietary Supplement panels are to be in compliance with the new labeling requirements by January 1, 2021.

Tolerable upper intake levels; Sugars such as sucrose, fructose, and dextrose that are present in foods have been associated with various adverse effects. The data available in 2002 on effects of sugars on behavior, dental caries, hypertriglyceridemia and cholesterol levels, cancer, and CVD as well as type 2 diabetes and obesity were all evaluated.⁹ Based on the review of the evidence, no ULs for carbohydrates or simple or added sugars were established.⁹ However, a maximal intake level of 25% or less of energy from added sugars was suggested, based on modeling which showed decreased intakes below the EARs of some micronutrients of American subpopulations when exceeding this level.⁹

The upper limit of the percent of carbohydrate in the AMDR was based on the minimal acceptable intakes of dietary protein and dietary lipids.⁹

C. Dietary Sources

Carbohydrates are present in many components of the diet, due in part to the wide variety of whole foods that contain carbohydrates and the vast number of processed foods that have been sweetened with them. Carbohydrates can be found in fruits, vegetables, whole grains, milk, and milk-based products and in many processed foods. Fats and oils, and meats, are not sources of carbohydrate in the diet.

Studies examining the National Health and Nutrition Examination Survey (NHANES) data set found that soft drinks, including diet sodas, were the leading source of

carbohydrate in the US diet for individuals aged 19 and older. Following soft drinks were (1) yeast breads and rolls; (2) cakes, cookies, quick breads, pastry, and pie; (3) candy, sugars, and sugary foods; and (4) fruits.¹⁸

US survey data from the NHANES What We Eat in America 2013–16 cohort reported that while all adults met the RDA for carbohydrate (130 g/day), energy intake from dietary carbohydrate was within the AMDR (45%–65%) in only 58% of adult men 19 and over and 67% of adult women 19 and over. The median intake of total sugars was 116 g/day for men and 90 g/day for women.¹⁹

V. ISSUES RELATED TO CARBOHYDRATES IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Dental caries

Perhaps the most well-documented negative health outcome associated with the consumption of carbohydrates is dental caries, also known as dental cavities. When sugars and certain types of fermentable carbohydrates are consumed, they are metabolized by bacteria found in the oral cavity. This process produces organic acids and results in the reduction of the pH of the dental plaque. In turn, tooth enamel becomes more susceptible to demineralization. If this process is repeated at a frequency that outpaces the remineralization of the enamel, the tooth with decay and dental caries will result.

A great number of variables influence the extent to which diet increases the likelihood of dental caries. Some variables include the frequency, form, and amount of carbohydrate consumed. For example, stickier carbohydrates are more likely to reside on the teeth for extended periods of time and are, therefore, more likely to contribute to dental caries.^{20,21} The fluoridation of many municipal water supplies and the frequent use of fluoridated toothpastes are believed to contribute to reduced incidence of dental caries in individuals.²²

Weight management and satiety

An examination of NHANES from 2001 to 2012 for both children and adults explored the relationship between whole grain consumption, body mass index (BMI), and waist circumference. The study, by Albertson et al., found that while a majority of US children and adults met daily recommendation for total grain intake, only an incredibly small proportion of consumers were meeting daily recommendations for whole grain intake.

Additionally, researchers found that consumption of whole grains was inversely related to both BMI and waist circumference.²³

Some clinical trials examining the relationship between whole grains and weight management have reached similar conclusions as the NHANES study mentioned above, but when the body of research is taken as a whole, the evidence is inconclusive. Much stronger evidence exists for the role of dietary fiber in promoting weight management, particularly through its satiating effect, but those relationships will be discussed in greater length in Chapter 31 of this book, which is focused on dietary fiber.²⁴

Heart health

While there is some evidence to suggest that a link between the consumption of whole grains and a reduced risk of coronary heart disease (CHD) and CVD may exist, it is certainly not conclusive. Epidemiological studies have found that increased consumption of whole grains is inversely associated with both CHD and CVD, but in randomized control trials, similar findings have largely not been replicated.²⁵ Instead, a systematic review of randomized control trials assessing the effect of whole grain consumption on CVD and its related risk factors found no difference in cardiovascular risk for those consuming whole grains versus control groups.²⁶

Insulin resistance and diabetes

The role of carbohydrates in the development and management of diabetes has long been an area of focus for nutrition and public health researchers due to the role carbohydrates play in altering blood glucose levels. Because type 1 diabetes, an autoimmune disorder that is often diagnosed when a child is very young, and type 2 diabetes, a disease that develops over a longer period of time in response to growing insulin resistance in the tissues of the body, result in poor blood glucose management, carbohydrates are of particular interest.

Whole grain diets have been shown to provide a host of benefits to individuals trying to manage an existing diabetes diagnosis and to those hoping to reduce their risk of developing type 2 diabetes. There is evidence suggesting that diets rich in whole grains confer some protective benefit against the development of type 2 diabetes, although the exact mechanism by which this benefit is conferred is unclear.²⁷ Additionally, randomized controlled trials have observed that individuals who consume diets rich in whole grains have an improved postprandial blood glucose tolerance and insulin sensitivity when compared to individuals eating a diet heavy in refined grains.²⁸

B. Known Interactions

Lactose intolerance

Of the known carbohydrate interactions, lactose intolerance is likely the most common and most well-known. Lactose intolerance is a condition that occurs in individuals who lack adequate amounts of the digestive enzyme lactase in the small intestine, making the digestion of the milk sugar lactose difficult. Individuals with lactose intolerance who consume dairy products often experience adverse symptoms such as cramping, flatulence, and diarrhea. These symptoms are caused by undigested lactose entering the colon and being rapidly fermented by colonic bacteria, resulting in the production of large amounts of acids and gas that cause GI distress.

Lactose intolerance is a condition that generally occurs as an individual ages because most infants are able to produce large enough quantities of the enzyme lactase to properly digest breast milk. As individuals age, lactase production and activity tend to decrease, which can result in lactose intolerance. Lactose intolerance is not uncommon, with an estimated 65% or more of the population experiencing a reduced ability to properly digest lactose. Prevalence of lactose intolerance varies among ethnic groups, with a higher proportion of East Asian and South African communities experiencing the condition than those located in Northern Europe and Australia.²⁹

The severity of GI symptoms, and the amount of dairy product required to trigger those symptoms, varies widely among individuals with some individuals requiring large amounts of dairy to experience adverse effects. Others experience symptoms after consuming even small amounts of dairy. Symptoms can be managed using over-the-counter enzyme supplements like the brand product Lactaid.

Celiac disease

While celiac disease is a condition in which individuals experience adverse GI symptoms in response to the consumption of gluten, a protein compound, the condition is worth noting in this chapter because gluten is found primarily in carbohydrates. Celiac disease is a genetic disorder in which in the consumption of gluten results in an autoimmune response where the villi of the small intestine are attacked, resulting in atrophy.

Gluten, which is a combination of both gliadin and glutenin proteins, is primarily found in foods that contain wheat grains, barley, rye, or triticale. Gluten has many uses in food production, particularly in the structure that it provides to baked goods, so it is quite ubiquitous in the food supply and can be difficult to avoid for those living with celiac disease. Cereals, baked goods, commercial soups, pastas, sauces, salad

dressings, and food colorings are all common sources of gluten. Even oats, which do not contain gluten, can commonly act as a source of gluten in the food supply because they are often processed in facilities where grains containing gluten are common.

While celiac disease has received increased attention from consumers, industry, and the research community in recent years, it is not an especially common condition. Epidemiological studies estimate a prevalence of roughly 1% among individuals living in what are traditionally thought of as Western nations, with a relatively low prevalence in non-Western nations. Some research has suggested that the prevalence of celiac disease in non-Western nations is either increasing or higher than once believed.³⁰ Celiac disease is most often diagnosed through the use of a serum antibody testing of the transglutaminase IgA antibody, while a small bowel biopsy is being used to confirm testing results. The only current treatment of celiac disease is a lifelong adherence to a gluten-free diet. Those with nonceliac gluten sensitivities, a condition with similar symptoms as celiac disease but lacking diagnostic tests, also appear to benefit from adherence to a gluten-free diet.³¹

C. As a Dietary Supplement

Prebiotics

Consumer interest in gut health and the role that both prebiotics and probiotics play in the management of the gut and its microflora have spiked in recent years. Because of this increased interest, many consumers have turned to the use of over the counter dietary supplements with the goal of improving their gut health. Some of these supplements are carbohydrate prebiotics, such as inulin and oligofructose, which have been shown to have some association with improving the diversity of colonic microflora and other markers of gut health.^{32,33} The role of dietary fiber and other complex carbohydrates on health, and their role as a prebiotic, is discussed at greater length in Chapter 31, which is focused on dietary fiber.

D. Dietary Patterns of Note

Ketogenic diet

The ketogenic diet, originally designed in the 1920s for the management and treatment of epilepsy, is a highly restrictive diet focused on drastically reducing the amount of carbohydrates present in the diet. The diet has been shown to be incredibly effective for this purpose, reducing the incidence of epileptic seizures in most patients and, in some cases, removing the need for medication entirely.³⁴ However, in the mid-to-late 1960s, the ketogenic diet, and other diets focused on

restricting carbohydrates, became a popular tool for weight loss with specific variations on carbohydrate restriction moving in and out of vogue in the decades that followed. Recently, over the course of the 2010s, the ketogenic diet has again become a popular dietary pattern for individuals who wish to quickly lose weight.

The ketogenic diet is characterized by severe reductions in the amount of carbohydrates an individual consumes accompanied by a corresponding increase in the amount of energy being derived from proteins and fats. The typical keto diet calls for a reduction of carbohydrate intake to less than 50 g/day, but many individuals adopt dietary patterns that are more restrictive of carbohydrates and limit their daily carbohydrate consumption to 20 g/day or less. This reduction in carbohydrate consumption then results in the body entering a ketotic state, which has been shown to facilitate weight loss.³⁵

After several days of carbohydrate restriction, glucose reserves in the body drop and reach levels that are insufficient for normal fat oxidation and are unable to meet the energy needs of the central nervous system (CNS). In turn, the CNS, no longer able to rely on glucose for energy, must instead utilize alternative energy sources. To meet these energy needs, acetyl CoA is overproduced, which results in elevated levels of ketone bodies such as acetone, β -hydroxybutyric acid, and acetoacetate. These ketone bodies are then utilized by the CNS and by other tissues throughout the body to meet energy needs that otherwise may have been met by glucose.

There is strong evidence to support the fact that utilization of a ketogenic diet is associated with weight loss, though there is little agreement on what mechanism leads to success in weight loss. Some studies have supported the belief that individuals adhering to a ketogenic diet, or a similarly low carbohydrate dietary pattern, are forced to “waste” calories when utilizing protein as their primary energy source and, therefore, are more likely to experience increased weight loss than individuals adhering to a “normal” dietary pattern. Others have suggested that there is no real metabolic advantage in the use of the ketogenic diet and instead suggest that users simply experience weight loss benefits when they adhere to any type of diet, with no specific weight loss diet being significantly more effective than another.

Low-FODMAP diet

Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (FODMAPs) and their role in gut health and in the management of GI disorders have recently seen heightened scientific interest. FODMAPs are carbohydrates and sugar alcohols that are not easily digested and absorbed in the intestine and instead are rapidly fermented in the colon. This

fermentation can result in unpleasant side effects like abdominal distention, stomach cramping and pain, bloating, and changes to motility. As such, recent research has examined the efficacy of dietary patterns that focus on the elimination of FODMAPs in the treatment and management of GI disorders such as irritable bowel syndrome, Crohn’s disease, and ulcerative colitis.³⁶

A low-FODMAP diet requires an individual to undertake a 2–6 week period of high restriction where many foods that contain any type of FODMAP are eliminated from the diet. After this restriction period, individuals are encouraged to slowly reintroduce FODMAP foods one by one to determine which foods might be responsible for triggering one’s GI symptoms. A high level of restriction is required for the diet to be successful, and as such, a wealth of online tools and publications have emerged to aim at helping individuals avoid foods that are considered to be high in FODMAPs. Researchers have even begun to determine which methods of dietary education would be the most effective in helping consumers adhere to the complex diet.³⁷ Adherence is further complicated by the fact that low-FODMAP food lists provided by different institutions often offer conflicting advice.³⁸

Evidence supporting the use of the low-FODMAP diet as a tool for the management of inflammatory bowel diseases is relatively strong. Clinical research focused on the efficacy of the low-FODMAP diet has found, with relative consistency, that individuals who adhere to the restrictive dietary pattern are less likely to report symptom flare ups.³⁹ While there is additional evidence to suggest that individuals who find early success with the low-FODMAP diet may continue to achieve desired results in the management of their GI symptoms over the long term, research should continue to examine the potential long-term impact such a diet may have on other aspects of health like the gut microbiome and the economic viability of sustaining the diet.

Glycemic index

The low glycemic index dietary pattern is not new and was first introduced in the early 1980s by Jenkins et al. The diet classifies carbohydrates based on their digestibility and their ability to contribute glucose to the body. The glycemic index is a ratio that explains the effect a specific food has on blood glucose levels compared to the response of a standard. Often, the standard used when calculating the glycemic index is 50 g of white bread or of pure glucose. A myriad of factors can influence the glycemic index of a food including the ratio of amylose to amylopectin, the processing techniques used to make the food, protein and fat content, and the amount of fiber present.⁴⁰

Because foods with a high glycemic index raise blood glucose levels more quickly than those with a low glycemic index, they also precipitate a more rapid and dramatic insulin response from the pancreas to regulate blood glucose levels. There is evidence suggesting that, over the long term, repeated exposure to these rapid fluctuations in blood glucose levels can have negative physiological effects on the body such as increased triglyceride levels, increased lipogenesis in the liver, and decreased satiety following meals.^{34,41} Because of these findings, some individuals, particularly those concerned about blood glucose levels such as individuals living with diabetes or those concerned with developing diabetes, may be interested in adopting a dietary pattern focused on consuming foods with a low glycemic index.⁴²

Use of the glycemic index as a means for establishing a healthful dietary pattern is certainly not without its shortcomings. Because the glycemic index is based on a 50 g carbohydrate serving, it fails to account for actual serving sizes of foods that an individual may consume. To account for this discrepancy, researchers have calculated the glycemic load for foods. The glycemic load takes into account both the calculated glycemic index for a food and the actual amount of carbohydrate that will be consumed. This calculation more accurately accounts for the glycemic response a food might have on an individual. Additionally, neither glycemic index nor glycemic load is a strong predictor of actual glycemic response when foods are consumed as part of a mixed meal and not on their own.

RESEARCH GAPS

- The role of whole grains in specific health outcomes such as cancer, heart disease, and obesity.
- The prebiotic effect of various carbohydrates and the role of the gut microbiome in human health.
- Whole grains as part of a sustainable global food system.
- Strategies for encouraging consumers to incorporate more whole grains into their diet.

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LIPIDS

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SUMMARY

Because of the hydrophobic nature of lipids, this class of macronutrients is handled differently than protein or carbohydrate with respect to digestion, absorption, and transport in circulation. Lipids are comprised mainly of nonesterified fatty acids, cell signaling molecules such as eicosanoids, and more complex lipids such as triacylglycerols, sterols, and phospholipids. Dietary complex lipids are broken down to varying degrees throughout the gastrointestinal system. Elegant systems operate to facilitate this process and ferry the resulting products from the gastrointestinal tract through the unstirred water layer into enterocytes. Through the coordination of numerous factors, including resynthesis and formation of transport particles (chylomicrons), lipids are released into the lymph system as chylomicrons and eventually bloodstream. During subsequent modification of chylomicrons by plasma enzymes and co-factors, cell surface receptors, and intracellular trafficking molecules, dietary lipids are delivered to their target sites. In peripheral tissue, lipids serve critical structural roles, are components of catabolic and anabolic pathways, and serve as regulatory molecules for a wide range of functions and energy reservoir. Given the diversity of fatty acids and other lipid compounds in edible fats and oils, as well as their metabolic products, their role in cellular structure–function relationships, and their implications for chronic disease prevention and treatment, they represent an active area of investigation. Some of these efforts have been directed at the development of functional foods intended to offer added health benefits, while others have been directed at decreasing dietary sources of some and replacing them with others.

Keywords: Catabolic and anabolic pathways; Cholesterol; Eicosanoids; Fatty acids; Lipids; Phospholipids; Sterols; Structural roles.

I. INTRODUCTION

Lipids have long been recognized as critical components of cell membranes, structural moieties, and core elements of lipoprotein particles as well as serve as carriers for fat-soluble compounds, precursors for bioactive compounds involved in a wide range of biological functions, a source of metabolic energy, and regulators of gene expression. The bioavailability of lipid-soluble compounds is dependent on the ability to absorb fat and deliver it to target tissues. With the exception of essential fatty acids, phytosterols, and fat-soluble vitamins, humans have the capacity to synthesize sterols and fatty acids from carbohydrate and protein. The

objective of this chapter is to provide an overview of the structure, digestion, absorption, and subsequent metabolism of principal lipid components, in particular fatty acids, including their dietary requirements and role in maintaining health and wellness.

II. NUTRIENT METABOLISM

A. Chemistry

Fatty acids

Fatty acids are hydrocarbon (acyl) chains with a methyl and carboxyl group at either end.¹ Most dietary

TABLE 4.1 Common fatty acids found in foods and in the body.

Code	Common name	Formula
Saturated		
4:0	Butyric acid	$\text{CH}_3(\text{CH}_2)_2\text{COOH}$
6:0	Caproic acid	$\text{CH}_3(\text{CH}_2)_4\text{COOH}$
8:0	Caprylic acid	$\text{CH}_3(\text{CH}_2)_6\text{COOH}$
10:0	Capric acid	$\text{CH}_3(\text{CH}_2)_8\text{COOH}$
12:0	Lauric acid	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$
14:0	Myristic acid	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$
16:0	Palmitic acid	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$
18:0	Stearic acid	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$
Monounsaturated		
16:1n-7 <i>cis</i>	Palmitoleic acid	$\text{CH}_3(\text{CH}_2)_5\text{CH}=(\text{c})\text{CH}(\text{CH}_2)_7\text{COOH}$
18:1n-9 <i>cis</i>	Oleic acid	$\text{CH}_3(\text{CH}_2)_7\text{CH}=(\text{c})\text{CH}(\text{CH}_2)_7\text{COOH}$
18:1n-9 <i>trans</i>	Elaidic acid	$\text{CH}_3(\text{CH}_2)_7\text{CH}=(\text{t})\text{CH}(\text{CH}_2)_7\text{COOH}$
Polyunsaturated		
18:2n-6,9 all <i>cis</i>	Linoleic acid	$\text{CH}_3(\text{CH}_2)_4\text{CH}=(\text{c})\text{CHCH}_2\text{CH}=(\text{c})\text{CH}(\text{CH}_2)_7\text{COOH}$
18:3n-3,6,9 all <i>cis</i>	α -Linolenic acid	$\text{CH}_3\text{CH}_2\text{CH}=(\text{c})\text{CHCH}_2\text{CH}=(\text{c})\text{CHCH}_2\text{CH}=(\text{c})\text{CH}(\text{CH}_2)_7\text{COOH}$
18:3n-6,9,12 all <i>cis</i>	γ -Linolenic acid	$\text{CH}_3(\text{CH}_2)_4\text{CH}=(\text{c})\text{CHCH}_2\text{CH}=(\text{c})\text{CHCH}_2\text{CH}=(\text{c})\text{CH}(\text{CH}_2)_4\text{COOH}$
20:4n-6,9,12,15 all <i>cis</i>	Arachidonic acid	$\text{CH}_3(\text{CH}_2)_4\text{CH}=(\text{c})\text{CHCH}_2\text{CH}=(\text{c})\text{CHCH}_2\text{CH}=(\text{c})\text{CHCH}_2\text{CH}=(\text{c})\text{CH}(\text{CH}_2)_3\text{COOH}$
20:5n-3,6,9,12,15 all <i>cis</i>	Eicosapentaenoic acid	$\text{CH}_3(\text{CH}_2)\text{CH}=(\text{c})\text{CH}(\text{CH}_2)_5(\text{CH}_2)_3\text{COOH}$
22:6n-3,6,9,12,15,18 all <i>cis</i>	Docosahexaenoic acid	$\text{CH}_3(\text{CH}_2)\text{CH}=(\text{c})\text{CH}(\text{CH}_2)_6(\text{CH}_2)_2\text{COOH}$

(c), *cis*; (t), *trans*.

fatty acids have an even number of carbon atoms that are aligned in a straight chain, which ranges in length from 4 to 22 carbons (Table 4.1). Fatty acids with 16 and 18 carbons comprise the bulk of those found in both the diet and the human body. Individual fatty acids are distinguished from each other by chain length, degree of saturation, double-bond number, double-bond conformation, and double-bond(s) location. A fatty acid with no double bond is termed saturated (Fig. 4.1A), with a single double bond is termed monounsaturated (Fig. 4.1B or D), and with two or more double bonds is termed polyunsaturated (Fig. 4.1C or E). Double bonds within fatty acids usually exist in the *cis* configuration, with hydrogen atoms on the same side of the carbon atoms (Fig. 4.1B or C). Less commonly, they exist in the *trans* configuration, with hydrogen atoms on opposite sides of the carbon atoms (Fig. 4.1D). Most double bonds occur in a nonconjugated sequence where a carbon atom with single carbon–carbon bonds separates the carbons forming the double bonds; however, some occur in the conjugated form where no intervening carbon separates the carbons forming the double bonds (Fig. 4.1E).

Differences in the position and conformation of double bonds within an acyl chain result in fatty acids having multiple isomers (i.e., same number of carbon, hydrogen, and oxygen atoms but a different configuration). Geometric fatty acid isomers result from differences in the conformation of a double bond. The presence of a *cis* relative to a *trans* double bond results in a greater bend or “knuckle” in the acyl chain (Fig. 4.1E). This bend impedes the fatty acids from aligning or closely packing with neighboring acyl chains, thereby influencing the physical properties of the fat such as increasing fluidity. The presence of a *trans* double bond reduces the internal rotational mobility of carbon atoms and makes them less reactive to chemical change than a *cis* double bond. The result is a more stable fat with a higher melting point.

Positional isomers of fatty acids are defined by differences in the location of double bonds along the acyl chain. These differences alter the physical properties of the fatty acid and, in some instances, may result in large changes in their metabolic fate and functions in the body. The most common distinction among positional

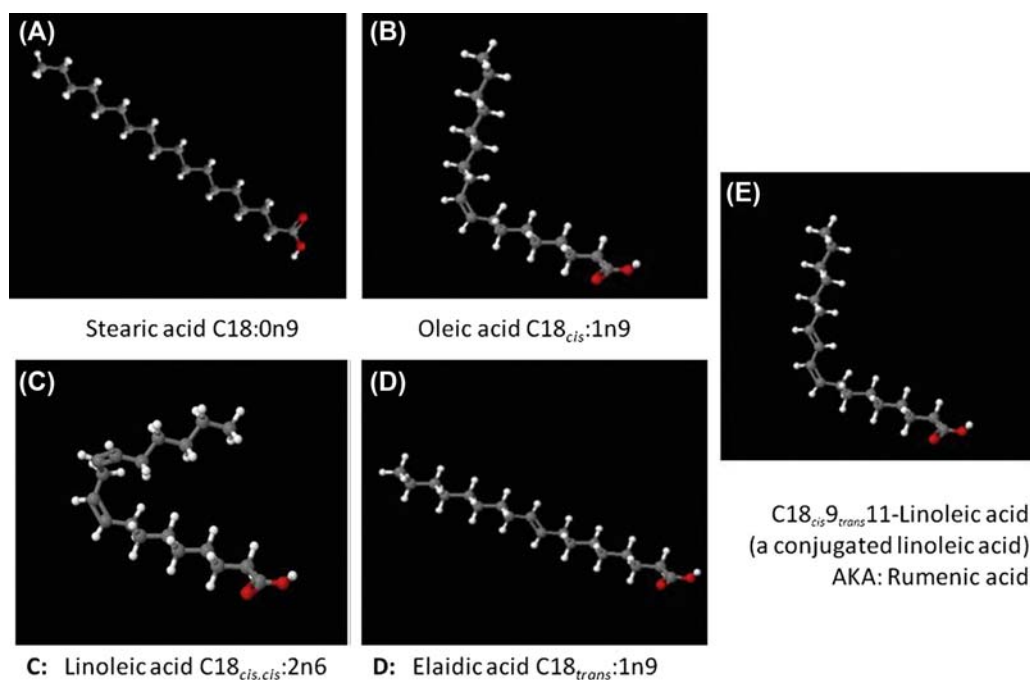


FIGURE 4.1 Examples of C18 fatty acids. Compare effect of saturation on conformation of fatty acids: a saturated fatty acid (A, stearic acid) with a monounsaturated fatty acid containing one *cis* double bond (B, oleic acid) and a polyunsaturated fatty acid containing two double bonds (C, linoleic acid). Compare conformational differences between a *cis* double bond (B, oleic acid) and a *trans* double bond (D, elaidic acid). Finally, compare the conformations of two isomers of linoleic acid: *c9,c12*-linoleic acid (C) and *c9,t11*-conjugated linoleic acid (E).

isomers is the location of the first double bond from the methyl end of the acyl chain. Fatty acids in which the first double bond occurs three carbons from the methyl group are designated as omega-3 fatty acids, denoted as ω -3 or *n*-3 fatty acids.² Fatty acids in which the first double bond occurs six carbons from the methyl end are termed ω -6 or *n*-6 fatty acids.

Triacylglycerols

Triacylglycerols, commonly referred to as triglycerides, are composed of three fatty acids individually esterified to each carbon of a glycerol molecule. This allows for the formation of stereochemically distinct fatty acid bond positions: *sn*-1, *sn*-2, and *sn*-3. The fatty acid moieties of the triacylglycerol molecule account for ~90% of its molecular weight, depending on the chain length and number of double bonds of the constituent fatty acids. Likewise, the physical form of a triacylglycerol is determined by its component fatty acids: chain length, number, position and conformation of the double bonds, and stereochemical position of each fatty acid on the triacylglycerol molecule.

Phospholipids

Phospholipids are composed of two fatty acids esterified to each of two carbon atoms of a glycerol molecule (*sn*-1 and *sn*-2) and one polar head group attached via a

phosphate linkage to the third carbon atom (*sn*-3). Phospholipid molecules are amphipathic; the fatty acids confer its hydrophobic properties and the polar head group confers its hydrophilic properties. The most predominant polar head groups are choline, serine, inositol, and ethanolamine. Phosphatidylcholine is also known as lecithin.

In vivo, the amphipathic properties of phospholipids allow them to serve as structural components of cellular membranes (bilayers) and lipoprotein particles (monolayers). The fluidity of cell membranes, which comprise approximately one-third of phospholipid, is primarily determined by their fatty acid profile. Due to the preferential incorporation of highly unsaturated fatty acids into the *sn*-2 position of phospholipids, they serve as a substrate reservoir for the synthesis of bioactive compounds (e.g., eicosanoids [see Section III.C]).

Cholesterol and cholesteryl ester

Cholesterol is an amphipathic molecule composed of a steroid nucleus and a branched hydrocarbon tail. Cholesterol is commonly found in two forms, free or esterified to a fatty acid at carbon number 3. Free cholesterol is a component of cell membranes, and, along with the phospholipid fatty acids, modifies fluidity.³ When taken up by a receptor mediated by feedback regulation, intracellular free cholesterol mediates cholesterol homeostasis in three ways⁴: (1) inhibits

3-hydroxy 3-methylglutaryl CoA reductase activity, the rate-limiting enzyme in de novo cholesterol biosynthesis; (2) increases acyl coenzyme A (CoA) cholesterol acyltransferase (ACAT) activity, the intracellular enzyme that esterifies free cholesterol thereby lowering intracellular concentrations; and (3) decreases the synthesis of low-density lipoprotein (LDL) cell surface receptors, thereby diminishing the uptake of additional cholesterol from blood. These three factors, if functioning optimally, protect against excess intracellular cholesterol accumulation, particularly important because a high concentration of intracellular free cholesterol is cytotoxic.⁵

Free cholesterol is more polar than esterified cholesterol. As a consequence, free cholesterol tends to be localized on the aqueous domains of cells and outer surface of lipoprotein particles, while cholesteryl ester is sequestered in the core of lipoprotein particles or intracellular lipid droplets. The majority of cholesterol in plasma (~two-thirds) is carried in LDL in the esterified form. In blood, cholesteryl ester is formed as a result of lecithin cholesterol acyltransferase (LCAT) activity, which catalyzes the formation of a bond between the number 3 carbon of free cholesterol and a fatty acid derived from the sn-2 position of lecithin. Intracellularly, cholesteryl ester is formed as a result of ACAT activity, with the fatty acid derived from a fatty acyl CoA, and because its nonpolar nature is sequestered in lipid droplets.⁶ Cholesteryl ester forms the core of atherosclerotic plaques. It is derived from the deposition of macrophages that infiltrate the arterial intima and take up cholesterol-laden LDL particles via receptors that are not feedback inhibited.⁷

Phytosterols (plant sterols)

Lipid fractions derived from plant materials contain phytosterols, compounds structurally related to cholesterol, and are sometimes referred to as plant sterols. Cholesterol and phytosterols differ with regard to their side-chain configuration and steroid-ring bonding patterns. The most common phytosterols are β -sitosterol, campesterol, and stigmasterol.⁸ Synthetic derivatives of phytosterols termed phytostanols are formed by hydrogenation. The bioavailability (percent absorbed) of phytosterols and phytostanols is considerably lower than cholesterol, and their concentrations in plasma tend to be low. Because they compete for and thus displace cholesterol from intestinal micelles, phytosterols and phytostanols reduce the absorption efficiency of cholesterol. And due to their interference with cholesterol absorption, hence increasing cholesterol excretion, consumption of 2 g/d of phytosterols or phytostanols represents an effective method of lowering circulating LDL concentrations.

B. Digestion of Fat

For the majority of people living in the North America, the average percent energy consumed as fat is 35% of kcalories, ranging from 26% at the 5th percentile to 46% at the 95th percentile. Total fat intakes are 77 g/d for adult women and 93 g/d for adult men (median), with an overall median intake for all age groups of 77 g/d.⁹ In contrast to dietary carbohydrate and protein, the apolar nature of fat prevents it from interfacing with the aqueous milieu of the gastrointestinal tract, so a specialized system exists to accommodate the unique challenges posed for its digestion and absorption (Fig. 4.2).

Triacylglycerols

Mouth, esophagus, and stomach; Digestion begins within the oral cavity where salivation and mastication occur. Lingual lipase, an enzyme released from the von Ebner (serous) glands of the tongue, along with saliva, causes the release of small amounts of fatty acids from dietary triacylglycerol. Lingual lipase principally cleaves fatty acids from the sn-3 position of the triacylglycerol molecule, with higher efficiency toward short-chain fatty acids, such as those under 14 carbon chain length. Lingual lipase activity continues as food travels through the esophagus and into the stomach. In the stomach, gastric lipase is released from the gastric mucosa.¹⁰ This enzyme cleaves triacylglycerol primarily at the sn-3 position. It has been estimated that 10%–30% of fat hydrolysis occurs prior to the masticated bolus of food entering the small intestine.¹⁰ This percent is relatively high in infants during the period when they primarily consume milk, composed of fat with a relatively high proportion of short-chain fatty acids. The increased pH of the intestine on entry of the food bolus inhibits the activity of the lingual and gastric lipases.

Intestinal fat digestion; The intestine is where most fat digestion occurs. Here, the process of triacylglycerol digestion is dependent on both pancreatic lipase and liver-derived bile acids. Bile acids are synthesized in the liver from cholesterol, stored in the gall bladder, and secreted in bile in response to the presence of fat as chyme reaches the duodenum. The rate-limiting enzyme in the synthesis of bile acids is 7α -hydroxylase.¹¹ Bile, per se, is composed of bile acids, phospholipid, and free cholesterol. The major function of bile acids is to emulsify the intestinal contents, also referred to as chyme. Cholesterol secreted as a component of bile is the primary mechanism humans have to excrete cholesterol. Emulsification of chyme serves to increase its surface area, thereby expediting its interaction with intestinal lipase. Bile acids have a steroid nucleus and an aliphatic side chain conjugated in an amide bond with taurine or glycine.¹² The hydroxyl and ionized sulfonate or carboxylate groups of the

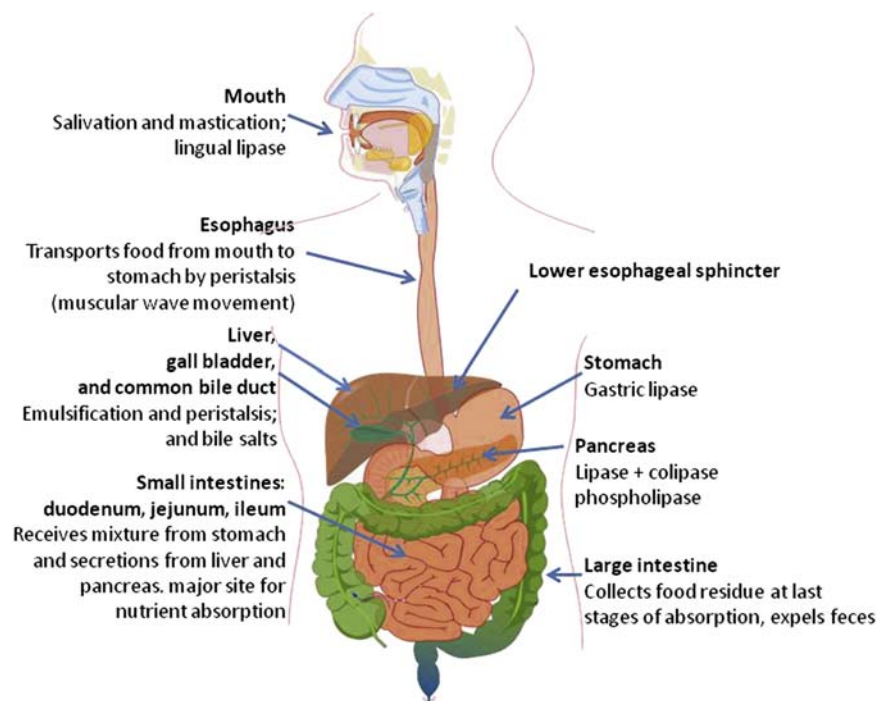


FIGURE 4.2 Diagram of digestion and absorption.

conjugate make bile acids water soluble. Primary bile salts are cholate and chenodeoxycholate (tri- and dihydroxy bile salts, respectively), which are synthesized in the liver directly from cholesterol. Secondary bile acids, deoxycholate and lithocholate, are converted from primary to the secondary bile acids cholate and chenodeoxycholate, respectively, by bacteria normally present in the gut microbiota. Secondary bile salts are recycled via resorption at a relatively high rate and can be further modified by hepatocytes or gut microbiota. These products include sulfated esters of lithocholate and ursodeoxycholate.

The entry of fat into the duodenum, in addition to stimulating the contraction of the gall bladder, causes cholecystokinin secretion and release of pancreatic lipase and colipase. In the intestine, pancreatic lipase is responsible for the majority of triacylglycerol hydrolysis.¹³ Pancreatic lipase hydrolyzes the sn-1 and sn-3 positions of triacylglycerol, while the sn-2 (middle) position of glycerol is resistant to hydrolysis by lipases. Colipase, synthesized by the pancreas, binds pancreatic lipase and facilitates the enzyme's adhesion to lipid in the gut.¹³ In the small intestine, the hydrolytic products of triacylglycerol (2-monoacylglycerol and free fatty acids), along with bile acids, cholesterol, and other fat-soluble substances, coalesce to form micelles for subsequent absorption. Deficits in the availability of an adequate amount of pancreatic lipase or bile acids can result in steatorrhea, the presence of undigested fat in the stool.

Phospholipid

The majority of the phospholipid in the small intestine is derived from bile with a smaller component coming from the diet. Phospholipase A₂, a pancreatic enzyme secreted in bile, mediates phospholipid digestion. This enzyme hydrolyzes the ester bond at the sn-2 position of the phospholipid, resulting in a free fatty acid and lysophosphoglyceride. These products are incorporated into micelles along with the other apolar compounds for subsequent absorption.

Cholesterol and cholesteryl ester

Bile, sloughed intestinal cells, and to a lesser and more variable extent diet, contribute cholesterol to the contents of the small intestine.¹⁴ Cholesterol originating from bile and intestinal cells is present in the free form. Cholesterol originating from the diet occurs as free cholesterol and cholesteryl ester. Prior to absorption cholesteryl ester hydrolase cleaves cholesteryl ester resulting in the formation of free cholesterol and a free fatty acid.

C. Absorption

Triacylglycerol

The efficiency of triacylglycerol absorption is high over a wide range of fat intakes. In the enterocyte, fatty acids with 12 or more carbon atoms are reesterified to glycerol to form triacylglycerol, incorporated into chylomicron particles, and secreted into the lymphatic

system. Free fatty acids with 10 or fewer carbon atoms, sometimes referred to as short- and medium-chain fatty acids, are absorbed directly into the portal circulation, along with amino acids and monosaccharides.

Transport of the micellar components into enterocytes is dependent on micelles traversing the unstirred water layer that separates the intestinal contents from the brush border of the small intestine^{14,15} (Fig. 4.3). Micelles traverse the unstirred water layer because of their relatively small size (30–100 Å) and apolar nature, attributed to the bile acids and phospholipids. Since the concentration of lipid-soluble compounds in the unstirred water layer and enterocytes is lower than in the intestinal lumen, the products of hydrolysis flow down this concentration gradient. Hence, the majority of lipid-soluble compounds are absorbed by passive diffusion. This net loss of micellar constituents results in a shift in the distribution of hydrolytic products down the concentration gradient from micelles in the intestinal lumen to micelles in the unstirred water layer and similarly from the unstirred water layer into enterocyte cells.

Bile acids are recycled with high efficiency. They are taken up by hepatocytes and resecreted into bile, with or without further modification, which is released back into the duodenum (termed enterohepatic circulation). Bile acids are absorbed by both passive and active mechanisms. Passive absorption of unconjugated bile salts occurs along the entire length of the small intestine and colon. Active bile acid absorption occurs in the ileum with approximately 95% efficiency.¹⁶ Interfering with the reabsorption of bile acids forces the liver to use

more cholesterol for their synthesis. This process has been exploited for the treatment of hypercholesterolemia with bile acid sequestrants and plant sterols specifically intended to interfere with reabsorption of both cholesterol and bile acids.

Short- and medium-chain fatty acids are absorbed directly into the portal circulation in the free form and transported to the liver bound to albumin. The majority of short-chain fatty acids are either oxidized rapidly as energy sources or converted to other metabolites within the hepatocyte.¹⁷

Phospholipid

Products of phospholipid digestion, free fatty acid, and lysophosphoglyceride are incorporated into intestinal micelles and are absorbed by a process similar to that described for the hydrolytic products of triacylglycerol.

Cholesterol and cholesteryl ester

The efficiency of free and esterified cholesterol absorption has been estimated to range from 40% to 60%. Free cholesterol and the free fatty acids resulting from the hydrolysis of cholesteryl esters are incorporated into intestinal micelles. The mechanism of free cholesterol absorption is complex, involving adenosine triphosphate (ATP)–binding cassette (ABC) G5 and G8 and Niemann-Pick C1-Like 1 (NPC1L1) transporters.^{18–21} NPC1L1 system draws cholesterol across the epithelial cell membrane from the lumen while the ABC proteins move the cholesterol back into the intestinal lumen,²² explaining why cholesterol absorption is less efficient than that for fatty acids.

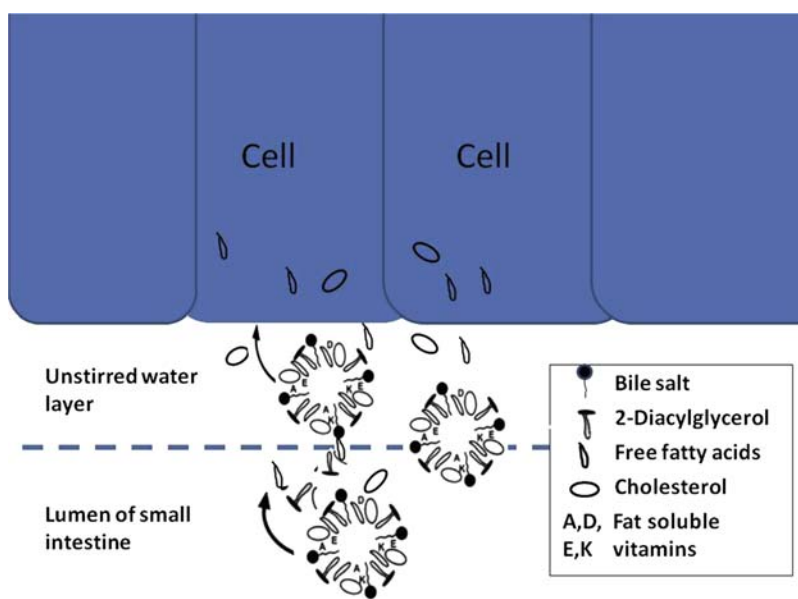


FIGURE 4.3 Fat absorption from the small intestine. Passive diffusion across a concentration gradient of most lipid components, including lipid-soluble vitamins, after micelles formed with bile acids in the lumen enter the unstirred layer and then diffuse into the enterocyte.

Plant sterols and sterol esters

The efficiency of plant sterol absorption is relatively low, approximately 1%–10% depending on the specific plant sterol, and somewhat lower for the saturated form of the compounds, stanols.⁸ The poor efficiency of absorption is attributed to the activity of the ABC proteins G5 and G8, which transport plant sterols, but less so cholesterol, out of the enterocyte,²³ low solubility of the sterols in micelles, and inability of the intestinal cell to reesterify the sterols once absorbed. Despite their poor solubility in micelles, plant sterols interfere with the incorporation of cholesterol into the intestinal particles, diminishing cholesterol absorption efficiency.^{24–26} Lower rates of cholesterol absorption result in, on average, a 10% reduction in plasma LDL cholesterol concentrations in most individuals consuming plant sterols.²⁶

D. Transport and Metabolism

Lipid-soluble compounds, with the exception of short- and medium-chain fatty acids, are transported out of the intestine and through the bloodstream in the form of lipoprotein particles. Lipoprotein particles have a broad range of sizes and compositions but share some common features. They are all spherical and composed of an apolar core containing triacylglycerol and cholesteryl ester, surrounded on the surface by a polar phospholipid monolayer also containing apolipoproteins (proteins that specifically associate with lipids) and free cholesterol.²⁷ Fat-soluble vitamins and other lipid-soluble compounds are sequestered in the core of lipoprotein particles. Lipoprotein particles are distinguished by the relative proportion of lipid constituents and type and amounts of apolipoproteins.

Lipoprotein particles

Chylomicrons; Chylomicrons are intestinally derived lipoprotein particles formed and secreted as a result of

the digestion and absorption of dietary fat. Their main function is to provide a mechanism, whereby dietary fat and fat-soluble compounds are transported from the intestine to peripheral tissues for subsequent uptake and utilization.^{28,29}

The first step in the formation of chylomicron particles is the resynthesis of triacylglycerol and phospholipids from fatty acids and glycerol and sn-2 monoacylglycerols or lysophospholipids, respectively. The fatty acid composition of the triacylglycerol portion of the chylomicron typically reflects that of the dietary fat.³⁰ A large proportion of the cholesterol that enters the enterocyte is reesterified, a reaction catalyzed by ACAT, prior to incorporation into chylomicron particles.⁶ Chylomicron particles have the largest diameter of all lipoprotein subclasses³¹ (Table 4.2). In humans, the distinguishing apolipoprotein of intestinally derived chylomicrons is apolipoprotein (apo) B-48, in contrast to the distinguishing apolipoprotein of hepatically derived particles, which is apoB-100. Apo B-48 is a relatively large hydrophobic protein synthesized on the rough endoplasmic reticulum and incorporated into chylomicron particles in the enterocyte. Once released into the lymph and then peripheral circulation, additional apolipoproteins are added to the surface. These include apo A-I, apo A-IV, apo A-II, apo C, and apo E.³² It has been suggested that the release of apo A-IV is stimulated by dietary fat and has a role in the regulation of upper gut function and satiety.³³

Chylomicrons are assembled in the enterocyte endoplasmic reticulum of the enterocyte from apo B-48, triacylglycerol, cholesteryl ester, phospholipid, and other lipid-soluble compounds. Microsomal triacylglycerol transfer protein (MTP) transports and facilitates the insertion of triacylglycerol into the nascent chylomicron core. The lipid-rich nascent particle is then transferred into the endoplasmic reticulum lumen and eventually to the Golgi apparatus for excretion. The particles are then released into the thoracic lymph

TABLE 4.2 Characteristics of lipoproteins.

Lipoprotein	Density (g/dL)	Molecular mass (daltons)	Diameter (nm)	Lipid (%) ^a		
				Triacylglycerol	Cholesterol	Phospholipid
Chylomicron	0.95	1400×10^6	75–1200	80–95	2–7	3–9
VLDL	0.95–1.006	$10–80 \times 10^6$	30–80	55–80	5–15	10–20
IDL	1.006–1.019	$5–10 \times 10^6$	25–35	20–50	20–40	15–25
LDL	1.019–1.063	2.3×10^6	18–25	5–15	40–50	20–25
HDL	1.063–1.21	$1.7–3.6 \times 10^5$	5–12	5–10	15–25	20–30

HDL, high-density lipoprotein; IDL, intermediate-density lipoprotein; LDL, low-density lipoprotein; VLDL, very-low-density lipoprotein.

^aPercentage composition of lipids; apolipoproteins make up the rest.

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duct, which feeds into the subclavian vein. Once in the circulation, the triacylglycerol component of chylomicron particles is hydrolyzed by lipoprotein lipase. Lipoprotein lipase is synthesized in adipose tissue and heart and skeletal muscle and migrates to the surface of the capillaries where it functions to hydrolyze triacylglycerol.^{34,35} Apo CII is a cofactor required for lipoprotein lipase activity. The hydrolysis of chylomicron triacylglycerol in the circulation provides a mechanism by which dietary lipid is transferred from the gastrointestinal system to peripheral tissue for subsequent metabolism or storage. Chylomicron particles depleted of the triacylglycerol component are termed chylomicron remnants and are taken up by the liver via either the LDL receptor or an LDL receptor–like protein receptor that recognizes apo E.^{36,37} The components of chylomicron particles are then either used by the liver directly or incorporated into a newly synthesized hepatically derived lipoprotein particle, very-low-density lipoprotein (VLDL).

VLDL and intermediate-density (IDL) lipoproteins; VLDL particles are hepatically derived particles that mediate the transport of lipid from the liver to peripheral tissue. The triacylglycerol in VLDL is derived from endogenously synthesized fatty acids and subsequent formation of triacylglycerol and the triacylglycerol harvested from the uptake of chylomicron remnants. The major apolipoprotein in VLDL is apoB-100.³⁸ ApoB-100 is synthesized on the rough endoplasmic reticulum and transferred to the Golgi apparatus where it is combined, with the involvement of MTP, triacylglycerol, phospholipid, and cholesterol, to form a nascent VLDL particle. Inadequate triacylglycerol or the absence of MTP results in internal degradation of apoB-100.³⁹ Once released into circulation, VLDL acquires apo E and apo C. The lipid components of VLDL particles are similar to chylomicrons; however, the proportion of triacylglycerol is less (Table 4.2), resulting in smaller, somewhat denser particles.

Once in circulation, the initial stages of VLDL metabolism are similar to those of chylomicron metabolism. Lipoprotein lipase hydrolyzes the triacylglycerol present in the core of the particle.^{40,41} The resulting fatty acids are then taken up by cells locally and oxidized for energy, used for the synthesis of structural components (e.g., phospholipid) or bioactive compounds (e.g., leukotrienes, thromboxanes), or stored as triacylglycerol. Triacylglycerol-depleted particles, VLDL remnants, can either be taken up directly by receptor-mediated mechanisms in the liver or remain in circulation. Delipidated VLDL particles are termed IDL particles and, eventually, as the particles continue to be delipidated, termed LDL.^{40,41} This process is

facilitated by both lipoprotein and hepatic lipase.^{40–42} Like lipoprotein lipase, hepatic lipase has the capacity to hydrolyze both triacylglycerols; however, it also has the capacity to hydrolyze phospholipid. As its name implies, it is localized to the liver. Ultimately, in addition to being depleted of triacylglycerol and some phospholipid, the conversion of VLDL to LDL also results in the loss of all apolipoproteins except apoB-100 and becomes enriched in cholesterol.

LDL; LDL particles can be taken up by receptor-mediated or scavenger receptors.⁴³ There are multiple LDL receptors, collectively referred to as LDL receptor family members. Receptor-mediated receptors include the apoB-100 receptor, LDL receptor-related protein (LRP), apo E receptor 2 protein, multiple epidermal growth factor–containing protein 7, VLDL receptor, LRP1B, megalin, LRP 5, and LRP 6.^{44,45} Receptor-mediated LDL receptors predominate in tissues such as liver, smooth muscle cells, fibroblasts, central nervous system neurons and astrocytes, epithelial cells of the gastrointestinal tract, testis Leydig cells, ovarian granulosa cells, and kidney dendritic interstitial cells. While the LDL receptor mediates the uptake of apoB-100 or apo E–containing lipoproteins, other members of the LDL receptor family recognize multiple apolipoproteins, proteases, and protease/inhibitor complexes and additional signaling molecules.⁴⁶

Once LDL particles are taken up by the receptor-mediated pathways, they disassociate from the receptor, releasing the receptor to be recycled. The LDL particle then fuses with a lysosome and is degraded into its component constituents. This step is critical for whole-body cholesterol homeostasis because the cholesterol taken up from LDL in circulation and released by the lysosome has three distinct effects, as discussed earlier in the “Cholesterol and cholesteryl ester” section. The process is self-limiting because elevated intercellular cholesterol concentrations downregulate further influx into the cell via a receptor-mediated process by decreasing the rate of LDL receptor synthesis. Alternatively, LDL can be taken up by a scavenger receptor on macrophages resident in various tissues. Unlike receptor-mediated uptake, this system is not feedback inhibited and has a high affinity for modified (e.g., oxidized) LDL particles.⁴³ Hence, scavenger-receptor uptake is proportional to circulating LDL cholesterol concentrations and is the predominate mechanism by which cholesteryl ester accumulates in atherosclerotic plaques.

Lipoprotein(a) (Lp(a)); Lp(a) is composed of an LDL-like particle and a single copy of apo(a), covalently bound to apoB-100. Blood Lp(a) concentrations are highly

heritable and affected by the apo(a) gene (LPA) located on chromosome 6q26-27. Apo(a) proteins comprise a family of proteins differing in size. The size of apo(a) is defined by the number of kringle IV repeats incorporated into the protein as a function of the LPA gene size polymorphism (KIV-2 VNTR). Although the precise function of Lp(a) still remains elusive, high Lp(a) concentrations are associated with high coronary heart disease and stroke risk.^{47,48}

High-density lipoprotein; Nascent high-density lipoprotein (HDL) particles are synthesized in the liver and intestine.⁴⁹ In addition, during delipidation of chylomicrons in blood, excess phospholipid, free cholesterol, and apolipoproteins from the surface of remnant particles form HDL. Critical for their formation is the availability of apo A-I. The primary role of HDL particles is to participate in “reverse cholesterol transport” by shuttling excess cholesterol from peripheral tissues to the liver for excretion, metabolism, or storage.⁵⁰ An integral part of this process is the scavenger receptor (SR-B1). This hepatic HDL receptor selectively takes up the cholesteryl ester component of HDL, thus enabling HDL to continually shuttle additional cholesterol from peripheral tissue to the liver.⁵¹

HDL is a heterogeneous group of particles that differ in both the apolipoprotein composition and size. All HDL particles contain apo A-I. Additional apolipoproteins can include apo A-II, A-IV, and Cs.³² The presence of apo A-I, paraoxonase, and platelet-activating factor acetyl hydrolase on HDL particles appears to protect other lipoproteins from oxidative modification.⁵² Plasma HDL concentrations are inversely related to triacylglycerol concentrations and risk of developing cardiovascular disease (CVD).

While in circulation, HDL undergoes modification through the interaction with other lipoprotein particles.⁵³ This includes exchange and modification of lipoprotein constituents and apoproteins. LCAT circulates on HDL and esterifies free cholesterol on its surface. Apo A-I serves as a cofactor for LCAT, and HDL-associated phosphatidylcholine serves as a source of fatty acids.⁵³ The formation of cholesteryl ester and its subsequent migration to the core of the HDL particle facilitates the subsequent uptake of additional free cholesterol from peripheral tissue and prevents the cholesterol on the HDL particle from transfer back to peripheral tissue. Cholesterol ester transfer protein facilitates the exchange of cholesteryl ester from HDL to VLDL or chylomicrons for triacylglycerol.⁵³ Phospholipid transfer protein activity results in HDL remodeling through the exchange of phospholipid. These processes enhance reverse cholesterol transport.

III. LIPIDS—NORMAL CELL AND ORGAN FUNCTION

A. Lipids and Cell Function

Lipid-soluble compounds are required for a wide range of cellular processes including structure, substrate, and energy-related. Polyenoic fatty acids provide the hydrophobic moiety of phospholipids, which are used for membrane structure and serve as precursors for eicosanoids and other compounds that regulate cellular activity. Oxidation of fatty acids contributes to providing fuel for the generation of ATP equivalents and body heat. Lipids contain more than twice the energy per gram (9 kcal/g) compared to carbohydrate or protein (4 kcal/g). Humans preferentially store lipid in the form of triacylglycerol as the primary energy reservoir. Dietary lipids are a source of lipid-soluble vitamins and sterols.

To maintain normal cellular function, an elaborate system of control regulates fatty acid biosynthesis, oxidation, and intracellular trafficking. This homeostatic regulatory system ensures that pathways of fatty acid anabolism and catabolism mesh with those of other macronutrients. In addition to providing a source of energy, fatty acids also modify membrane fluidity and serve as substrates for bioactive compounds.

B. Lipid Biosynthesis

Fatty acid synthesis

Fatty acids are synthesized *de novo* from acetyl CoA, initiated by the enzyme acetyl CoA carboxylase, which converts acetyl CoA to malonyl CoA. Malonyl CoA units are used to sequentially add 2 carbon units to the growing fatty acid chain culminating in the formation of palmitic acid (C16:0). From this point, palmitic acid can be modified by elongation and desaturation. One limitation is that humans do not possess enzymes necessary to insert a double bond below the n-7 carbon, with the result that n-6 and n-3 fatty acids are essential and must be consumed as part of the diet. Desaturase enzymes are membrane-bound and occur in the endoplasmic reticulum of several tissues. Desaturases are specific for the position of the double bond in the carbon chain and require electrons supplied by nicotinamide adenine dinucleotide or nicotinamide adenine dinucleotide phosphate.

Long-chain fatty acids are synthesized from C18:0, C18:2n-6, and C18:3n-3, the latter two coming from dietary sources, by alternating desaturation and elongation steps. The synthesis of C20:4n-6 from C18:2n-6 and of C20:5n-3 from C18:3n-3 is achieved through

D6 desaturation followed by elongation and D5 desaturation (Fig. 4.4). The synthesis of C22:6n-3 from C20:5n-3 occurs through elongation, desaturation, and partial α -oxidation.⁵⁴

Fatty acid elongation does not need to commence with 18-carbon precursors. To meet n-6 essential fatty acid requirements, it is possible to use C16:2n-6, derived from fats of plant origin in place of linoleic acid, thereby effectively reducing the requirement for the latter fatty acid.⁵⁵

The synthesis of long-chain fatty acids is regulated by the D6 desaturase enzyme, which in turn is influenced by hormones and dietary fatty acids. The D6 desaturase preferentially targets the most highly unsaturated fatty acid; the preferential desaturation order is therefore C18:3n-3 > C18:2n-6 > C18:1n-9. As such, the competitive nature of fatty acid desaturation and elongation among the three classes of fatty acids (Fig. 4.4) impact essential fatty acid requirements.⁵⁶ Consumption of diets rich in n-6 fatty acids can result in suppression of the elongation and desaturation of C18:3n-3 to C20:5n-3 and C22:6n-3.

Reversal of the elongation and desaturation process may also occur intracellularly. Very-long-chain ($C \geq 22$) and long-chain n-3 and n-6 fatty acids ($C \leq 20$) may undergo shortening and saturation through retroconversion to shorter fatty acids⁵⁷ in either the cytoplasm or peroxisome of the cell.

Cholesterol biosynthesis

Humans likely once relied extensively on endogenously produced cholesterol for cellular requirements, as ancestral dietary cholesterol intakes have been

estimated to be much lower than present levels of intake. Today, most people consume diets in which cholesterol intake is high enough such that cholesterologenesis is less relied on to meet cellular needs. The total body pool of cholesterol is estimated to be 75 g. Of the 1200 mg cholesterol turned over daily, 300–500 mg is absorbed from a typical North American diet and from bile, and de novo synthesis accounts for 700–900 mg.⁵⁸

As with fatty acids, the process of cholesterol biosynthesis begins with acetyl CoA generated through the oxidative decarboxylation of pyruvate or the oxidation of fatty acids. In the initial phase of the pathway, acetyl CoA molecules condense to form mevalonic acid (Fig. 4.5). The final enzyme of this initial phase, hydroxymethylglutaryl CoA reductase, is the rate-limiting enzyme in the overall cascade of cholesterologenesis. The later phase of cholesterol biosynthesis involves phosphorylation, isomerization, and conversion to geranyl-B and farnesyl-pyrophosphate, which in turn form squalene. From the squalene stage, loss of three methyl groups, side chain saturation, and bond rearrangement result in the formation of cholesterol.

Diet modulates cholesterol biosynthesis in several ways. For many years, it was thought that high levels of dietary cholesterol increased circulating levels of cholesterol; however, more recent research has shown that total and LDL cholesterol levels as well as cholesterol biosynthesis appear to be minimally affected by dietary cholesterol within current intake levels.⁵⁹ Conversely, it appears that the *type* of dietary fat more substantially perturbs the rate of cholesterol biosynthesis and circulating lipoprotein cholesterol concentrations. In particular,

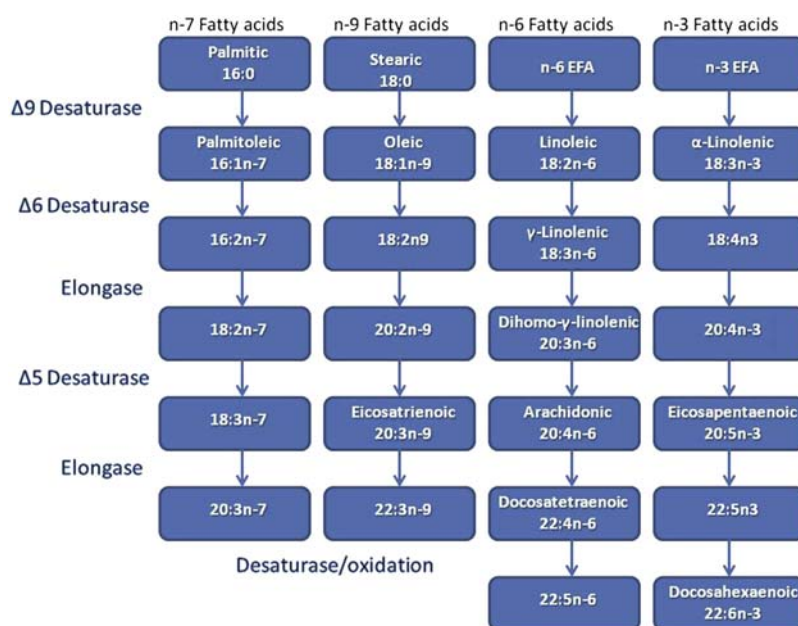


FIGURE 4.4 Interconversions of nonessential and essential fatty acids (EFAs).

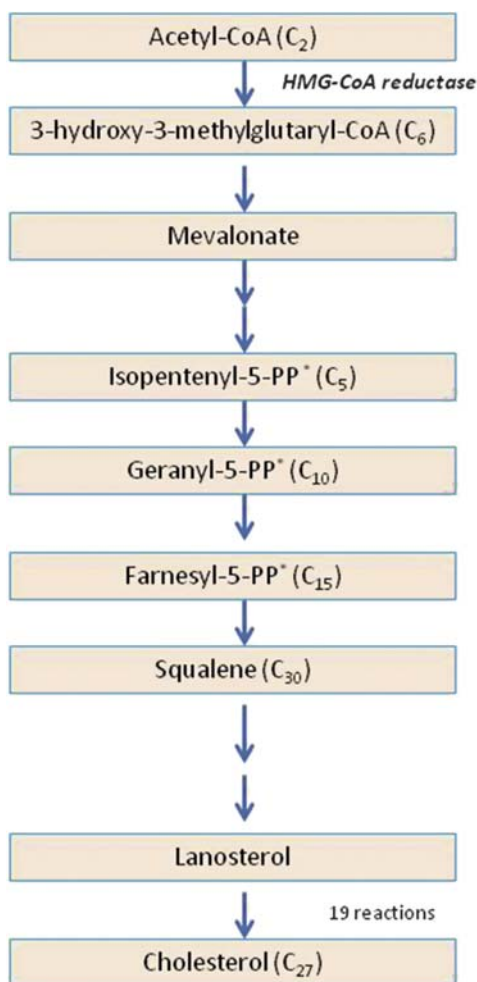


FIGURE 4.5 Steps in cholesterol biosynthesis.

dietary polyunsaturated fat intake enhances cholesterol biosynthesis, possibly due to the higher plant sterol content of those oils high in polyunsaturated fatty acids (PUFAs), interfering with cholesterol absorption.⁶⁰ Increasing the number of meals consumed per day, while holding total caloric intake constant, has also been shown to reduce cholesterol biosynthesis rates. Of the dietary factors capable of modifying cholesterol synthesis, energy restriction has the greatest effect. Humans who fast for 24 h exhibit a complete cessation of cholesterol biosynthesis.⁶⁰ Overall, within the context of current North American intake levels (329 mg/d = median intakes for men while for women they were 237 mg/d),⁶¹ dietary cholesterol per se is no longer considered a nutrient of public health concern.

Fatty acid oxidation

Mitochondrial fatty acid β -oxidation generates acetyl CoA. During this process, the fatty acyl chain undergoes

cyclical degradation through four stages, including dehydrogenation (removal of hydrogen), hydration (addition of water), dehydrogenation, and cleavage. Where double bonds exist within the acyl chain, the initial dehydration step does not occur. This four-stage process is repeated until the fatty acid is completely degraded to acetyl CoA, as shown by the detectable absence of chain-shortened n-3 or n-6 fatty acids within cells or in the bloodstream. Fatty acyl CoAs containing 18 carbons or fewer are transported into the mitochondria attached to carnitine. Short- and medium-chain fatty acids bypass this shuttle system and are transposed directly across the mitochondrial membrane. Beta-oxidation also occurs in peroxisomes through a similar, but not identical, process and is predominately limited to the incomplete oxidation of long-chain fatty acids (>18 carbons). The products are free fatty acids and short-chain carnitine acyl esters, the latter facilitating subsequent mitochondrial degradation.⁶²

The likelihood that fatty acids will be oxidized rather than stored is not identical across all chain lengths and degrees of unsaturation. A higher proportion of short- and medium-chain saturated fatty acids is oxidized than long-chain fatty acids. This is likely due to preferential transport of the former linked to albumin through the portal circulation and ability of short- and medium-chain fatty acids to bypass the carnitine transport system to enter the mitochondria.¹⁷

C. Eicosanoid Production and Regulation

Eicosanoids include prostaglandins, thromboxanes, leukotrienes, hydroxy acids, and lipoxins. Production of eicosanoids is governed by fast-acting and rapidly inactivated enzyme systems. While prostaglandins and thromboxanes are generated via cyclooxygenase (COX) enzymes, leukotrienes, hydroxy acids, and lipoxins are produced through the action of lipoxygenase (LOX) enzymes.⁶³ Major pathways for eicosanoid synthesis are depicted in Fig. 4.6. The process begins with the action of phospholipase A₂, hydrolyzing a fatty acid from the sn-2 position of the phospholipid molecule. Unsaturated fatty acids are preferentially incorporated into the sn-2 position of phospholipid species (e.g., phosphatidylcholine, phosphatidylethanolamine). The resultant free fatty acids serve as substrates for eicosanoid production via the COX and LOX enzyme cascades.

For the n-6 series, the major fatty acid substrate for eicosanoid synthesis, cleaved arachidonic acid is converted to prostaglandin H₂ (PGH₂) by prostaglandin H synthase-2 or its isoform, prostaglandin H synthase-1. These enzymes catalyze the conversion via a COX reaction and then reduction to PGH₂ by a peroxidase

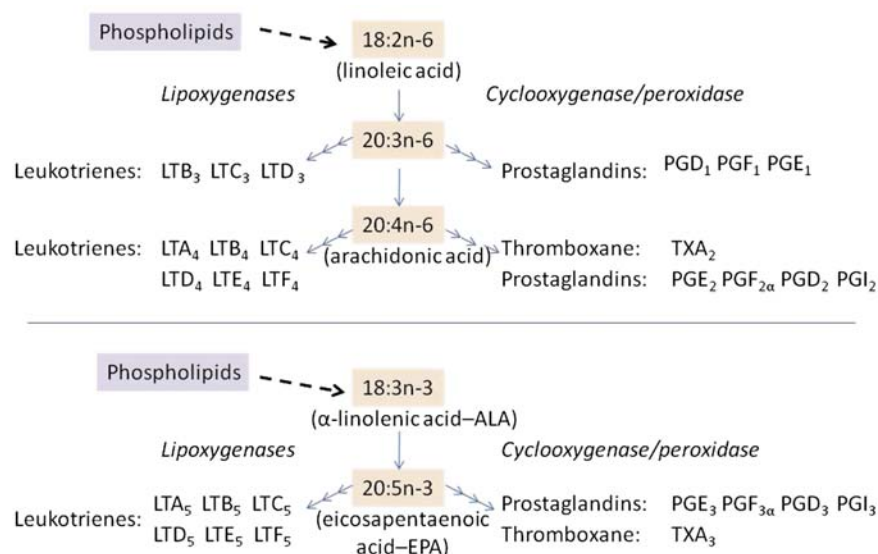


FIGURE 4.6 Synthesis of major eicosanoids from n-3 and n-6 fatty acids.

reaction. This latter intermediate then undergoes rapid conversion to form other bioactive forms of prostaglandins, thromboxanes, or prostacyclins. Via an alternative pathway, arachidonic acid also undergoes oxidation via a series of LOX enzymes to form other bioactive eicosanoids. The 5-LOX pathway generates leukotrienes B₄, C₄, and D₄ (LTB₄, LTC₄, and LTD₄) which are believed to serve as mediators in the immune response.⁶⁴

Eicosanoids are also derived from the n-3 series of fatty acids cleaved from membrane phospholipids. Eicosanoids derived from n-3 fatty acids evoke less-potent responses than do those of the corresponding n-6 eicosanoids. Thus, PGE₃ derived from eicosapentaenoic acid (C20:5n-3) (EPA) invokes a weaker inflammatory response than PGE₂, derived from arachidonic acid (C20:4n-6). Similarly, LTB₅ derived from EPA invokes a weaker proinflammatory response than does LTB₄, derived from arachidonic acid.

Eicosanoids are primarily synthesized from phospholipid fatty acids in the sn-2 position. Diets high in n-3 fatty acids are associated with less inflammation and prolonged bleeding times.⁶⁵ Dietary fatty acid composition has been shown to play an important role in eicosanoid composition, hence, activities. Diets rich in n-3-rich fats result in higher levels of n-3 fatty acids in membrane phospholipids. These fatty acids, when cleaved from phospholipids, compete with arachidonic acid for synthesis into eicosanoids. The enhanced bleeding times observed in Inuit populations who consume high levels of n-3 fatty acid-containing fish reflect high levels of n-3-based eicosanoid formation.⁶⁵

IV. DIETARY REQUIREMENTS

A. Fatty Acid Essentiality

The essentiality of dietary fat was recognized in the first part of the 20th century. In 1927, Evans and Burr demonstrated that animals fed semipurified, fat-free diets had impaired growth and reproductive failure.⁶⁶ These findings suggested the essentiality of fat, termed vitamin F. Subsequently, Burr and Burr⁶⁷ then documented the nutritional essentiality of a specific essential component of fat, linoleic acid (C18:2n-6). The absence of this nutrient in experimental animals results in scaly skin, impaired fertility, and growth retardation.⁶⁶⁻⁶⁸ Thus, the concept of “essential” fatty acids was introduced to represent fatty acids required by mammals that are not synthesized *in vivo*. The requirement is attributed to the absence of enzymes that can introduce double bonds above n-7. In the 1970s, dietary deficiency of n-3 fatty acids was linked to abnormal electroretinographic recordings in experimental animals.⁶⁹ Identification of essential fatty acids followed in humans. In 1958, studies in infants fed fat-free milk developed skin symptoms that were alleviated with the addition of linoleic acid.⁷⁰ In adults, the use of fat-free parenteral solutions containing only glucose, amino acids, and micronutrients resulted in clinically essential fatty acid deficiency, which was reversed by addition of linoleic acid. Subsequently, n-3 fatty acid deficiency symptoms in humans included neuropathy and were demonstrated to be ameliorated by γ-linolenic acid (C18:2n-3).⁷¹ Defining the dietary requirements for arachidonic acid (C20:4n-6),

docosahexaenoic acid (DHA) (C22:6n-3), and EPA (C20:5n-3) is currently under investigation.^{56,72} Dietary essential fatty acid requirements in infants are likewise under active investigation. There may be a need to provide not only linoleic acid but also arachidonic acid and DHA.

Experimental linoleate deficiency in rats, in contrast to general unsaturated fatty acid deficiency can be reversed with 2% of dietary energy fed as linoleic acid.⁵⁵ For α -linolenic acid, classification of deficiency symptoms in humans has been difficult, since the symptoms are more subtle than for linoleic acid. Also, pure deficiencies are hard to induce because symptoms may remain absent in the presence of DHA.

Because both brain and retina have high levels of DHA, it has been hypothesized that DHA in breast milk may confer a developmental advantage to both term and preterm infants, as has been reported with breastfeeding.⁷³ However, a recent review of the literature concluded that there was no beneficial effect (or harm) of n-3 fatty acid supplementation on neurodevelopmental outcomes of formula-fed full-term infants or visual acuity.⁷⁴

B. Reference Intakes

The most recent Dietary Reference Intakes (DRIs) from the Food and Nutrition Board of the Institute of Medicine, established in 2002 for the United States and Canada, have provided values for essential fatty acids, recommendations for dietary cholesterol, *trans* fatty acids, and saturated fatty acids, and ranges for total fat as part of a healthy diet. Based on the data available at that time, the recommendations for dietary cholesterol, saturated fat, and *trans* fatty acids, none of which are essential, were to limit consumption at levels as low as possible while consuming a nutritionally adequate diet.⁷⁵

Primarily for the prevention of CVD, recent dietary guidance relative to lipids in the diet has focused on replacing saturated fat with unsaturated fat.⁷⁶ The 2015 US Dietary Guidelines for Americans Advisory Committee recommended a limit of 10% of energy for saturated fat and that major sources of dietary saturated fat in the diet be replaced by liquid vegetable oils and not carbohydrate, particularly refined carbohydrate.⁷⁷ Additionally, there was no target or guidance related to total fat, unlike in earlier US Dietary Guidelines.⁷⁸

Recommended intakes for α -linolenic acid (n-3) and linoleic acid (n-6) fatty acids were 0.6%–1.0% and 5%–10% of total energy, respectively.⁷⁵ The median energy intake of men in the United States in 2013–16 was 2413 kcal/d and for women 1777 kcal/d.⁶¹ This is equivalent to median n-3 intakes of 2.1 g/d for men and 1.6 g/

d for women and n-6 fatty acid intakes of 20 g/d for men and 15 g/d for women.⁷⁵ Since approximately 10% of dietary fatty acids are long-chain PUFAs, 10% of these amounts can come from long-chain PUFA. These values were established based on the levels of fatty acids required to prevent or alleviate essential fatty acid deficiency. Deficiency symptoms were documented when patients were fed total parenteral nutrition solutions devoid of essential fatty acids in the 1960s. Skin irritations that developed during fat-free diets were alleviated when patients were fed solutions containing linoleic acid.⁷⁰

C. Dietary Sources of Lipids

Lipids are found in most foods (Food Data Central, USDA) (Table 4.3). Butter and full-fat dairy products are major sources of saturated fatty acids, particularly those with short chains. Coconut oil is a major source of saturated medium-chain fatty acids. Meat contains long-chain saturated and monounsaturated fatty acids. Marine animals are good sources of n-3 very long-chain polyunsaturated fatty acids, EPA, and DHA. Plant oils are major dietary sources of n-6 polyunsaturated essential fatty acids with 18 carbons in length. The fatty acid profile of plant oils varies widely, and therefore different oils possess various proportions of linoleic acid and α -linolenic acid. Safflower, sunflower, corn, and soybean oils are high in linoleic acid. Olive and canola oils, as well as newer varieties of safflower and soybean oils, are high in the monounsaturated fatty acid oleic acid. Soybean and canola oils, as well as flaxseed and linseed oils, provide significant amounts of α -linolenic acid. The longer-chain n-6 fatty acid, arachidonic acid, is found in foods of animal origin, primarily meat, similar to cholesterol, while phytosterols (or plant sterols) occur at relatively low concentration in plant oils.

Some functional foods that incorporate essential fatty acids and other lipids have been introduced into the food supply. Examples include bakery products made with flaxseed oil, eggs containing very long chain n-3 fatty acids, and spreads incorporating phytosterols, or their saturated derivatives, phytostanols.

V. ISSUES RELATED TO FAT IN HUMAN HEALTH

A. Relevant Health Outcomes and Issues of Concern

The relation between the dietary fatty acid profile and chronic disease risk has been of interest since the turn of the 20th century. For certain diseases common to developed countries, the evidence for reducing risk by

TABLE 4.3 Average triacylglycerol fatty acid composition of various foods and oils.

Food	Average fat %	Average fatty acid composition (%)						
		Saturated fatty acids			Monounsaturated fatty acids	Polyunsaturated fatty acids		
		16:0	18:0	Total	18:1n-9	18:2n-6 ^a	18:3n-3	20:4n-6
Almond oil	100	1	6	8	65	23	Trace	—
Beef tallow	100	20	29	53	42	2	Trace	—
Butter	81	10	22	53 ^b	20	3	0.3	—
Canola	100	2	4	7	62	19	9	—
Coconut oil	100	3	10	88 ^b	6	2	—	—
Corn oil	100	2	11	13	25	55	Trace	—
Cottonseed oil	100	3	25	30	18	51	Trace	—
Flaxseed oil	100	4	5	9	20	13	53	—
Grapeseed oil	100	4	7	11	20	68	Trace	—
Groundnut (peanut) oil	100	3	11	19 ^b	40–55 ^c	20–43 ^c	—	—
Herring ^d (menhaden)	16–25	4	19	30	13	1	1	—
Milk (cow's)	3.5	30	12	69 ^b	26	1–3	2	Trace
Olive oil	100	3	14	17	71	10	Trace	—
Palm kernel oil	100	2	7	80 ^b	14	1	—	—
Palm oil	100	5	45	52	38	10	—	—
Pork fat (lard)	100	13	28	42	46	6–8	2	2
Rapeseed oil ^e	100	2	5	7	53	22	10	—
Safflower seed oil	100	3	7	10	15	75	Trace	—
Salmon	13	0.5	2	3	3	0.2	Trace	Trace
Sesame oil	100	5	9	15	39	40	1	—
Soybean oil	100	4	11	15	23	51	7	—
Sunflower seed oil	100	4	6	12	24	60–70	Trace	—
Walnut	63	2	7	10	15	60	10	—

The percentages given are approximations. Trace = <1% detected; dash (—) = Nondetectable amount.

^aAlso contains C18:3n-6.

^bRemaining saturated fatty acids are from fatty acids C-12 (butter, 14%); C12 + C14 (milk, 14%; coconut and palm kernel 65%–70%); and C-20 + C22 (groundnuts, ~4%).

^cGroundnuts from Argentina have relatively low C18:1 and high C18:2 concentrations.

^dMenhaden herring oil varies depending on fishing grounds.

^eContrary to new rapeseed varieties like canola and low erucic acid rapeseed (LEAR), old varieties of rapeseed oil have 10% C20:1n-9 and 30%–50% C22:1n-9.

adopting certain fat intake profiles is compelling; for other diseases, the link remains speculative (Fig. 4.7).⁷⁶ This rapidly progressing field will undoubtedly see further development as research advances improved understanding.

Box 4.1 lists diseases linked to a relationship with dietary fat. By far, the most data exist for CVD. A recent comprehensive review concluded randomized controlled trials that replaced saturated fat with polyunsaturated plant oil reduced CVD by ≈30%.⁷⁶ It

further concluded that for prospective observational studies across a range of populations, there is a negative association between diets higher in polyunsaturated and monounsaturated fat and lower in saturated fat and CVD and all-cause mortality rates. Replacement of saturated fat with carbohydrate, primarily refined carbohydrates, was not associated with lower CVD risk. The biologic underpinning of these observations was attributed to lower LDL cholesterol concentrations.

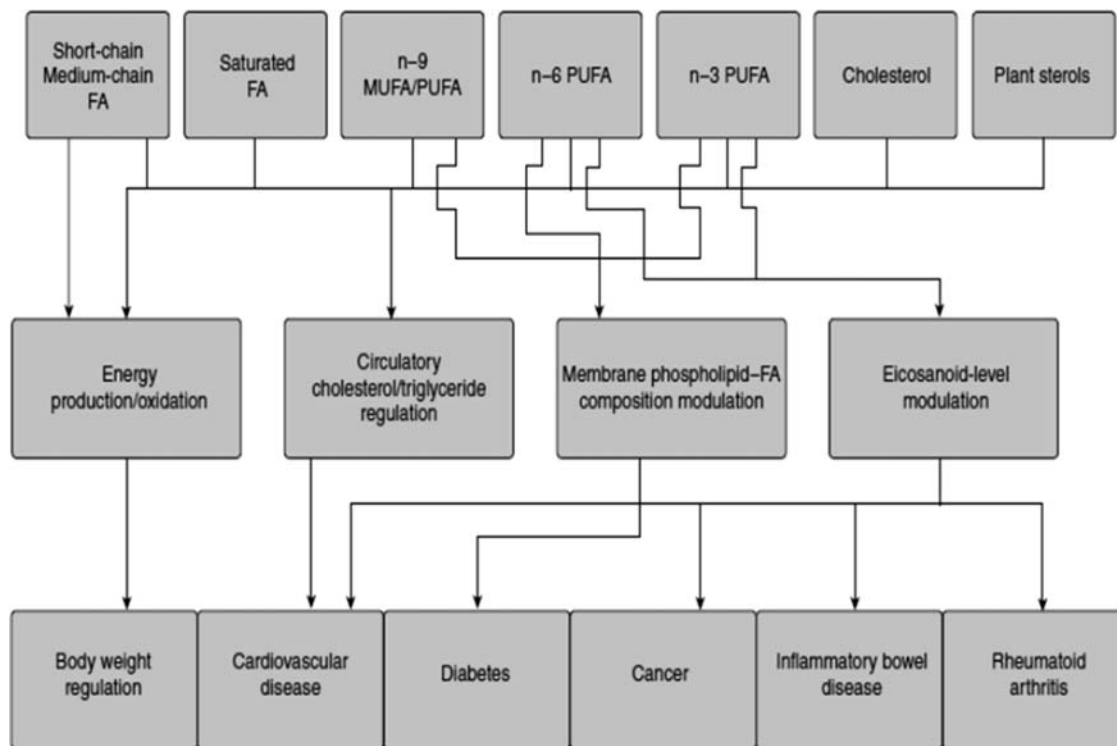


FIGURE 4.7 Interactions among dietary lipid components, metabolic systems, and disease/health outcomes.

BOX 4.1

Diseases influenced by the type of fat consumed in humans

Coronary heart disease and stroke
 Essential fatty acid deficiency during development
 Autoimmune disorders, including lupus and nephropathy
 Type 2 diabetes
 Inflammatory bowel disease
 Breast, colon, and prostate cancers
 Rheumatoid arthritis

The surprisingly low cardiovascular mortality rate in Native Alaskans consuming traditional diets high in EPA and DHA alerted researchers to the possibility that very-long-chain n-3 fatty acids may be effective in reducing CVD risk. However, a comprehensive meta-analysis concluded that there is high strength of evidence of no significant effects of marine oils on systolic or diastolic blood pressures.⁷⁹ Supplementation with n-3 fatty acids was associated with small and

statistically significant increases in HDL and LDL cholesterol concentrations. The clinical significance is thus unclear. Very long-chain n-3 PUFAs were associated with a dose-dependent decrease in triacylglycerol concentration in individuals with hypertriglyceridemia.

In observational studies, there is low strength of evidence that higher n-3 fatty acid intakes were associated with lower ischemic stroke risk.⁷⁹ Among randomized controlled trials and observational studies, the strength of evidence for an association with higher n-3 fatty acid intake and lower CVD event risk is weak. Evidence regarding the plant source of n-3 fatty acid, α -linolenic acid, is sparse. The evidence was judged as moderate for no significant effect on blood pressure or lipoprotein concentrations, and no significant association with coronary heart disease, atrial fibrillation, or congestive heart failure.

A recent supplementation trial with 1 g per day of fish oil (840 mg total n-3 fatty acids [460 mg of EPA and 380 mg of DHA], the vitamin D and omega-3 trial [VITAL]) for primary prevention and with a median follow-up of 5.3 years in over 25,000 subjects concluded there was no significant effect on the primary endpoint, a composite of myocardial infarction, stroke, or death from cardiovascular causes.⁸⁰ For secondary endpoints, significantly lower hazard ratios were observed for the

expanded composite endpoint of cardiovascular events, total myocardial infarction, total stroke, and death from cardiovascular causes.

Around the same time, individuals with elevated blood triacylglycerol concentrations and treated with statins were given 2 g twice a day (total 4 g/d) of a purified and stable form of EPA (ethyl ester, icosapent ethyl) for median of 4.9 years (Reduction of Cardiovascular Events with Icosapent Ethyl—Intervention Trial [REDUCE-IT]). This resulted in a significantly lower risk of ischemic events, including cardiovascular death.⁸¹ Some evidence also supports the notion that low omega-3 intakes associate with increased risk of overweight.⁸²

B. Future Directions

North American Mediterranean dietary intervention study

The PREDIMED (Prevención con Dieta Mediterránea) study, conducted in Spain, reported significant benefits of providing over 7000 individuals (>50 years of age) for a median period of 4.8 years with modest amounts of extra virgin olive oil or nuts compared to no additional food for a wide range of health outcomes related to both cardiovascular and cognitive health.⁸³ Whether the benefits resulting from giving individuals living in a Mediterranean country and consuming a Mediterranean-type diet, extra virgin olive oil and nuts would be similar to those resulting from giving virgin olive oil and nuts to individuals living in North America, and consuming a typical North American-style diet is uncertain. It is necessary to determine whether the beneficial health outcomes were due to the extra virgin olive oil and nuts, per se, or the combination of the background diet and lifestyle, and supplemental extra virgin olive oil or nuts.

Biomarkers of food intake

The assessment of food intake, particularly in large cohorts, is critical to our understanding of the relation between diet quality and health outcomes. The task of accurately assessing dietary patterns is challenging and usually relies on self-reported data. Inherent in the tools that rely on self-reported data are the subjective biases superimposed on the responses. Objective approaches to assess dietary patterns can be intrusive, disruptive, and extremely expensive, imposing their own biases and making them unsuitable for large cohort studies. Critically needed are validated, generally accepted biomarkers of food intake. Validation will depend on conducting well-controlled feeding trials and taking advantage of the rapidly evolving field of “omics” to identify a comprehensive battery of measures that will allow for the cost effective objective assessment of diet quality.

Interesterified dietary fats

Dietary fat forms an important part of the diet; however, consumption of *trans* and saturated “solid” fats has been associated with increased risk of CVD. Certain foods, such as spreads and bakery products, need solid fat in order to produce what is currently considered desirable mouthfeel, texture, and shelf life. However, health risks associated with *trans*, the major source of which is partially hydrogenated fat, and saturated fats mean that different and novel approaches to create solid fats are required. One such approach that has been proposed⁸⁴ is to incorporate interesterified (IE) fats. Interesterification is an industrial process that results in the reorganization of three fatty acid across the sn-1, sn-2, and sn-3 positions of the glycerol backbone of triacylglycerol molecules, with a focus on increasing the proportion of saturated FA in the sn-2 position. This change decreases the viscosity of the fat.

IE fats are now being increasingly utilized as hard fat replacements for partially hydrogenated in foods such as spreads, bakery products, and confectionary products. Currently, there is no legal requirement for food manufacturers to include the level of IE fats on food labels. Estimates from US data suggest that approximately 3%–5% of energy intake would come from IE fats, if they were the sole replacement of partially hydrogenated fat.⁸⁵

Despite the rapid uptick in the global use of IE fats in a wide range of foods, their health effects have not been carefully investigated.⁸⁴ As with molecular geometry, there is a possibility that positional composition may affect dietary fat metabolism, digestibility, and subsequent effects on cardiovascular health.

Influence of genetic polymorphism on response to dietary fat modifications

With recent advances in high throughput “omics” screening techniques, and the arrival of personalized medicine, an area of increasing research interest is defining how an individual’s genetic architecture alters the metabolic disposal of dietary fats in a manner that affects markers of disease risk. Over the past two decades, it has been shown that there is a substantial interindividual variability in the response to dietary perturbations on the basis of single-nucleotide polymorphisms (SNPs) for genes coding for enzymes involved with absorption, synthesis, and transport of fats.^{86,87} In these studies, individuals who possessed certain alleles for SNPs within these genes were found to respond differently in terms of their lipid-level responses to various dietary intakes, compared to individuals possessing other alleles.

Recent studies have examined this concept both from a qualitative and quantitative standpoint pertaining to fat intake. Moreover, a growing area of interest is

identification of not just SNPs, but clusters of SNPs, that contribute in predictable and complementary patterns to influence disease risk as a function of consumption of specific fat types. For example, three SNPs in distinct but related genes, working in concert, can predict whether consumption of dairy fat increases or decreases circulating LDL-cholesterol concentrations.⁸⁸

Oxylipins and health

Oxylipins, also previously referred to as eicosanoids due to their formation from twenty carbon essential fatty acids, function as metabolic regulators, having potent effects on many organ and cell systems. Certain eicosanoids, including leukotrienes B4 and C4, are pro-inflammatory, while others such as resolvins and protectins are antiinflammatory and are involved in the repair process which follows tissue injury.^{89,90} Oxylipins act both in an autocrine or paracrine manner, targeting peroxisome proliferator-activated receptors to modify adipocyte formation and function. Most oxylipins are formed from linoleic acid or α -linolenic acid.

Linoleic acid–derived oxylipins are usually present in blood and tissue in higher concentrations than any other PUFA oxylipin, despite the fact that α -linolenic acid is more readily converted to oxylipins. Thus, in animal tissues, polyunsaturated fatty acids from the n-6 and n-3 families are the biosynthetic precursors of oxylipins, including the eicosanoids (prostaglandins, leukotrienes, thromboxanes, and lipoxins) and docosanoids (protectins, resolvins, and maresins), while in plants, hormones such as the jasmonates are derived from α -linolenic acid.⁸⁹

Linoleic acid–derived oxylipins can be antiinflammatory, but are more often proinflammatory, associated with atherosclerosis, nonalcoholic fatty liver disease, and Alzheimer's disease. Lowering dietary linoleic acid results in fewer linoleic acid oxylipins. In contrast, levels of n-3 EPA-derived and DHA-derived oxylipins increase with an omega-3 fatty acid–enriched diet.⁹¹ An understanding of the biological roles of the myriad oxylipin species now being identified by newer analytical tools is forthcoming.

RESEARCH GAPS

- Assess whether the purported beneficial health outcomes of a Mediterranean diet are due to the extra virgin olive oil and/or nuts, per se, or the combination of the background diet, lifestyle, and supplemental extra virgin olive oil or nuts.
- Identify and validate biomarkers of food intake, taking advantage of the rapidly evolving “omics” approaches.
- Explore mechanisms determining the postprandial handling and trafficking of IE fats with different fatty acid profiles at relevant doses.
- Apply artificial intelligence–based learning algorithms to predict responsiveness of health-related biomarkers to dietary factors on the basis of an individual's genetic makeup.
- Utilize high-precision mass spectrometry approaches to define the roles of complex oxylipins in health outcomes.
- Standardize methodologies for identification and quantification of fatty acids in foods and biological tissues.
- Elucidate more fully the role of individual dietary n-3 polyunsaturated fatty acids in health promotion and prevention of disease.

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S E C T I O N B

Vitamins

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VITAMIN A AND PROVITAMIN A CAROTENOIDS

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SUMMARY

Vitamin A is an essential micronutrient required to maintain vision, the immune response, barrier function, growth and differentiation, male and female reproduction, and fetal development. It is acquired from the diet either as preformed vitamin A from animal foods or as provitamin A carotenoids acquired from plant foods. Vitamin A actions in the body involve two biochemical mechanisms. Its role in vision involves 11-*cis*-retinaldehyde, which is the chromophore for the visual pigment rhodopsin. The other involves all-*trans*-retinoic acid serving as a ligand for three distinct retinoic acid nuclear receptors, which regulate transcription of literally hundreds of genes. Vitamin A deficiency remains a major public health problem in the developing world. The vitamin and related factors are implicated in the causation/prevention of human metabolic disease, including obesity, diabetes, hepatic disease, and cardiovascular disease, as well as proliferative disorders, including cancers and skin disease.

Keywords: β -Carotene; Metabolic disease; Retinoic acid; Retinoid; Retinol; Transcription; Vision.

I. INTRODUCTION

A. Background

Vitamin A was first identified more than a century ago as a component present in the diets of growing chicks that, when removed from the diet, led to an impaired rate of growth.^{1,2} Since this essential component could be removed through extraction of the diet with lipid solvents, the factor was initially named fat-soluble A to distinguish it from essential water-soluble dietary components that were referred to as water-soluble B. Subsequently, fat-soluble A became vitamin A.

By definition, vitamin A is all-*trans*-retinol. However, in common modern usage, vitamin A usually refers collectively to the abundant/physiologically important vitamin A metabolites that are found within the body. These consist primarily of retinyl esters, retinol, and retinoic acid (Fig. 5.1). The preponderance of these metabolites is present in the all-*trans*-configuration. When

the geometric configuration of a metabolite is not specifically stated, it can be assumed that the all-*trans*-metabolite is being considered. As noted above, 11-*cis*-retinaldehyde acts centrally in vision, and as discussed below, both the 9-*cis*- and 13-*cis*-retinoic acid isomers may have roles in mediating vitamin A actions in the body. When *cis*-vitamin A isomers are being considered, the *cis*-configuration will be noted.

In the late 1970s, the term retinoid was coined to refer collectively to vitamin A and its metabolites, as well as to chemically synthesized vitamin A-like compounds that had been created in the laboratory as pharmacological agents to treat disease.³ At the time, a retinoid was defined as being any chemical that bears a structural resemblance to all-*trans*-retinol, with or without the biological activity of vitamin A. Thus, retinoids comprise both the naturally occurring vitamin A metabolites, as well as synthetic compounds. The terms vitamin A and retinoid are commonly used interchangeably in the literature.

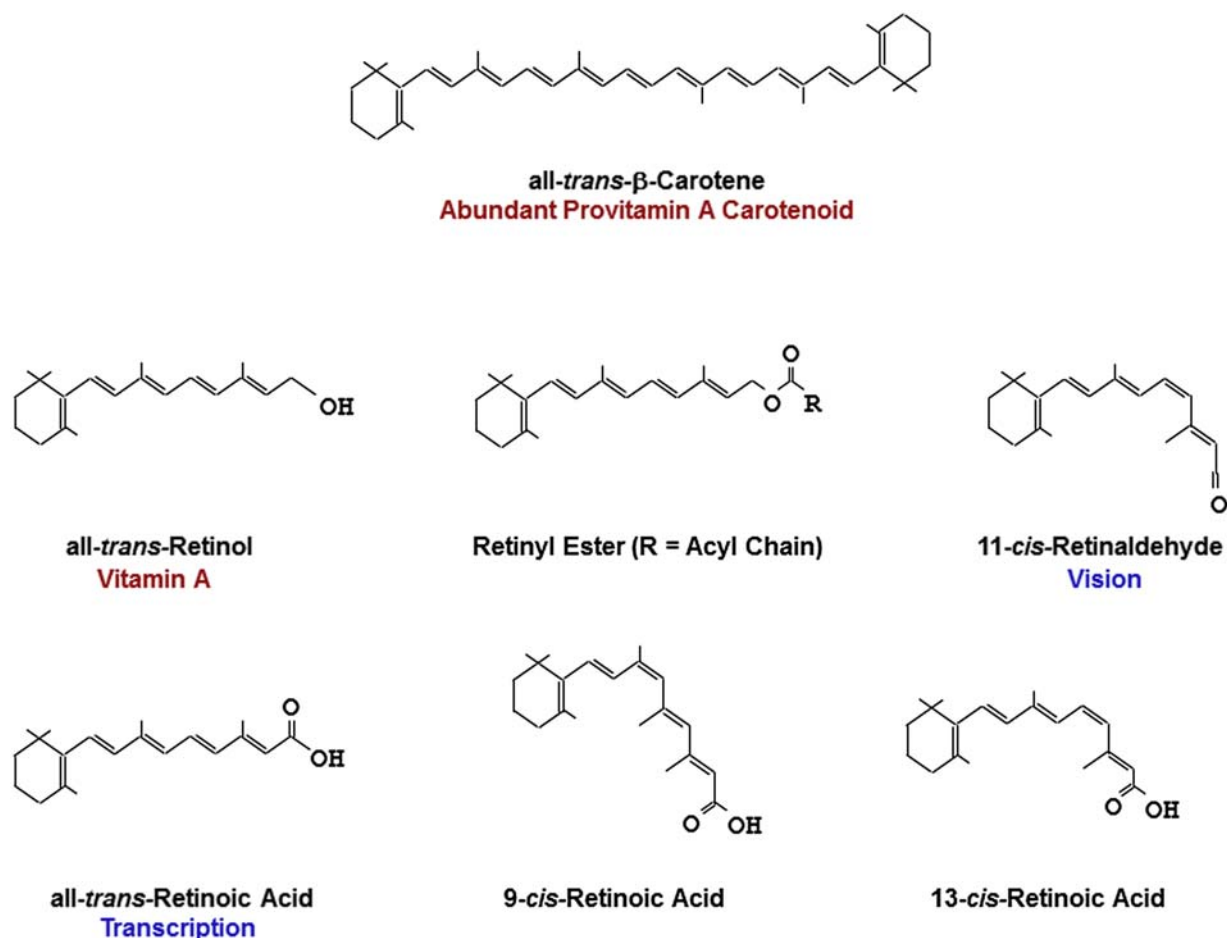


FIGURE 5.1 Chemical structures of β -carotene and major vitamin A forms.

B. Vitamin A Functions and Structure

Retinyl esters and retinol account for greater than 95%, and often as much as 99%, of all of the vitamin A that is present within the body.^{4–6} Depending on the tissue, retinyl esters will generally be more abundant than retinol. In addition, present within tissues are the vitamin A metabolites that are needed for mediating vitamin A's molecular actions. In the neural retina, this will include relatively high levels of 11-*cis*-retinaldehyde and its photoisomerization product all-*trans*-retinaldehyde.^{7,8} The transcriptional regulatory activities of vitamin A are mediated by all-*trans*-retinoic acid and as discussed below, possibly 9-*cis*-retinoic acid and/or its metabolites.^{9–12} Consequently, retinoic acid is present in most tissues and cells. In addition, oxidized and conjugated forms of both retinol and retinoic acid are present in tissues and blood.^{4–6,13–15} It is generally assumed that these are catabolic products destined for elimination from the body.

Retinol is the immediate precursor for the enzymatic synthesis of retinaldehyde and subsequently retinoic acid.^{4–6} It is also the predominant form of vitamin A

present in the fasting circulation. Retinol is readily generated through the enzymatic hydrolysis of retinyl esters. Retinyl esters have two roles within the body. They are storage forms of vitamin A found within intracellular lipid droplets where they are stored. They are also the vitamin A form that is packaged into nascent chylomicrons allowing for absorption of vitamin A following its uptake from the diet by the intestine. The physiologically relevant retinyl esters contain only long-chain fatty acyl groups, primarily the esters of palmitate, stearate, oleate, and linoleate, although other long-chain esters can be formed, albeit at a much lower level of abundance. Short chain retinyl esters, like retinyl acetate, are not synthesized by humans or animals but can be found in food and vitamin supplements. These short-chain esters are quickly hydrolyzed to retinol by gut hydrolases/esterases.

Since humans and other mammals cannot synthesize vitamin A *de novo*, all vitamin A that is present in the body was originally synthesized from provitamin A carotenoids that were produced by plants or microorganisms.¹⁶ The prototypical provitamin A carotenoid is β -carotene (see Fig. 5.1), which is enzymatically

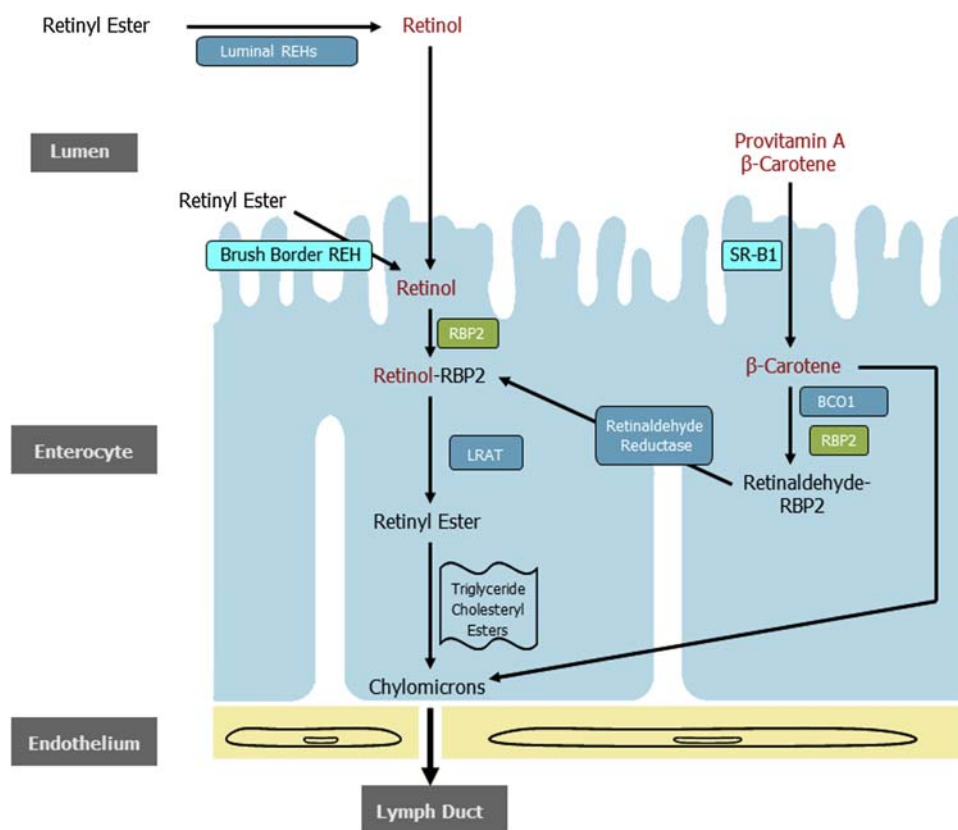


FIGURE 5.2 Schematic summary of the uptake and metabolic processing of dietary vitamin A and provitamin A carotenoid within the intestine.

cleaved at its central 15,15'-double bond to form retinaldehyde.^{16–18} Other common dietary carotenoids, including α -carotene and β -cryptoxanthin, have provitamin A activity, albeit less than β -carotene. At least one unmodified cyclohexyl ring at one end of the carotenoid molecule is required for the carotenoid to have provitamin A activity. A number of very abundant dietary carotenoids, including lycopene, zeaxanthin, and lutein, lack provitamin A activity. But each of these nonprovitamin A carotenoids is proposed to have other physiologically important activities (i.e., antioxidant or other activities) within the body that do not involve the formation of vitamin A.¹⁶ Recent research has suggested that some of these vitamin A-independent activities may involve the eccentric (asymmetrical) cleavage of the carotenoid to apocarotenoids that directly influence cellular processes.¹⁷

II. VITAMIN A METABOLISM

A. Vitamin A Synthesis

All vitamin A in the body must be acquired from the diet, since higher animals are incapable of its *de novo* synthesis.^{6,16} This could involve either the uptake and enzymatic conversion of dietary provitamin A

carotenoids into vitamin A or the direct dietary consumption of preformed vitamin A that had been formed from provitamin A carotenoids by animals lower on the food chain. The great preponderance, if not all of the vitamin A present in the body, has been formed through the actions of the enzyme β -carotene 15,15'-oxygenase (BCO1). This is the sole enzyme present in higher animals that are able to catalyze the central cleavage of provitamin A carotenoids to form retinaldehyde. The retinaldehyde formed through central carotenoid cleavage is immediately reduced to retinol (see Section II.B below and Fig. 5.2). BCO1 is expressed highly in enterocytes as well as in other tissues of the body. Since provitamin A carotenoids can be absorbed and taken up into the body without modification by intestinal enzymes, the presence of BCO1 within many tissues implies that provitamin A carotenoids can be converted to vitamin A throughout the body.

B. Vitamin A Uptake by and Metabolism in the Intestine

Dietary retinol is taken up without modification by enterocytes lining the proximal small intestine, dietary retinyl esters however are not absorbed intact and must first be acted upon by luminal and/or enterocyte brush border retinyl ester hydrolases (REHs) to form

free retinol (Fig. 5.2).^{4–6,19,20} The molecular identity or identities of REH(s) that are physiologically significant for retinol uptake is not established. Many intestinal hydrolases and esterases, when studied in vitro, possess REH activity. However, compelling in vivo evidence to support an important physiological role for many of the candidate enzymes possessing REH activity in the intestine is lacking.^{6,20}

There has been a long-standing interest in identifying transporters in the intestinal brush border membrane that facilitate retinol uptake into enterocytes. Although it is attractive to hypothesize that retinol absorption by enterocytes may involve a membrane transporter, there are at present no experimental data that support this hypothesis.^{6,21}

Unlike for retinol, there is strong evidence that dietary β -carotene is taken up into enterocytes by a membrane transporter. Both studies of mutant mouse models and in vitro cell culture experiments employing human enterocytes have established that scavenger receptor B1 (SR-B1) is a key mediator for uptake of β -carotene and other carotenoids from the intestinal lumen into the enterocyte.^{18,21–23} Carotenoid uptake by the intestine is regulated by nutritional vitamin A status.²² The intestine-specific transcription factor *Isx* (intestine-specific homeobox) plays a key role in regulating expression of both *SR-B1* and *Bco1* in the intestine.^{22,24} Severe vitamin A deficiency markedly decreases *Isx* expression, and this is accompanied by increased *SR-B1* and *Bco1* expression in both the duodenum and jejunum of the small intestine, allowing for increased carotenoid uptake and conversion to vitamin A. Retinoic acid, acting through the nuclear retinoic acid receptors, reduces *Isx* expression.^{22,24} This effect of retinoic acid on *Isx* expression results in a downregulation of both *BCO1* and *SR-B1*. Thus, dietary carotenoid uptake by enterocytes and its conversion to vitamin A are regulated by a diet-responsive regulatory network that controls *SR-B1* expression and β -carotene absorption, and vitamin A production by *BCO1* through negative feedback regulation of *Isx*.²²

Like other dietary lipids, dietary vitamin A and dietary carotenoids, not converted to vitamin A within the enterocyte, are taken up into the body as components of newly synthesized chylomicrons (Fig. 5.2). Within the enterocyte, newly absorbed or newly formed retinol must first be esterified to retinyl ester before it can be packaged into nascent chylomicrons.^{4–6} Cellular retinol-binding protein, type II (RBP2) is present at high concentrations in enterocytes where it binds both retinaldehyde and retinol.²⁵ Retinaldehyde, formed upon carotenoid cleavage by *BCO1*, also binds to RBP2, and this is the preferred substrate for reduction to retinol by an intestinal retinaldehyde reductase. Retinol bound to RBP2 is esterified to long-chain fatty acids through the action of lecithin:retinol

acyltransferase (LRAT), which utilizes retinol bound to RBP2 as a substrate for esterification.^{25,26} The resulting retinyl esters are packaged along with unmodified dietary carotenoid and other dietary lipids into chylomicrons that then are secreted into the lymphatic system in route to the general circulation.^{4–6}

LRAT is responsible for the preponderance of retinyl ester synthesis within the intestine, and elsewhere within the body.⁶ The older literature suggested a role for an acyl-CoA-dependent enzyme (an acyl-coA:retinol acyltransferase) in intestinal retinyl ester formation in times of normal dietary vitamin A intake. This has been shown to be incorrect.

C. Vitamin A Transport in the Circulation

A number of different vitamin A metabolites are present in the circulation, and these can differ quantitatively between the fasting and postprandial states (Fig. 5.3). These include retinol bound to retinol-binding protein 4 (RBP or RBP4) (The older literature refers to this protein as simply retinol-binding protein or RBP, whereas the recent literature uses the genetic nomenclature RBP4. Both RBP and RBP4 refer to the same gene/protein.); retinyl esters in chylomicrons, chylomicron remnants, very-low-density lipoproteins (VLDL), low-density lipoproteins (LDL), and high-density lipoproteins (HDL); retinoic acid bound to albumin; and the water-soluble β -glucuronides of retinol and retinoic acid. The delivery of vitamin A to tissues is complex, involving the different vitamin A forms, transport proteins, and membrane receptors. Quantitatively, the two most important pathways are those involving retinol bound to RBP4 and the postprandial delivery pathway involving chylomicron retinyl esters. However, the importance of retinoic acid delivery to tissues from the blood should not be discounted, since the accumulation of retinoic acid from the blood is tissue-dependent.²⁷ This notion is further substantiated by the effective use of orally administered retinoic acid in experimental studies and its pharmacological administration in humans. In this regard, the delivery of vitamin A to tissues, as either retinol or retinoic acid, is not different from the delivery of vitamin D or thyroid hormone, involving the presence of relatively large concentrations of the transcriptionally inactive precursor (retinol, 25-hydroxy-vitamin D, or T_4) and relatively low concentrations of the transcriptionally active metabolite (retinoic acid, 1,25-dihydroxy-vitamin D, or T_3). This implies that it may be correct to refer to retinoic acid as a hormone as it sometimes is in the literature.

Vitamin A transport in the fasting circulation

In the fasting circulation of humans, and many mammals, retinol bound to RBP4 is the predominant vitamin

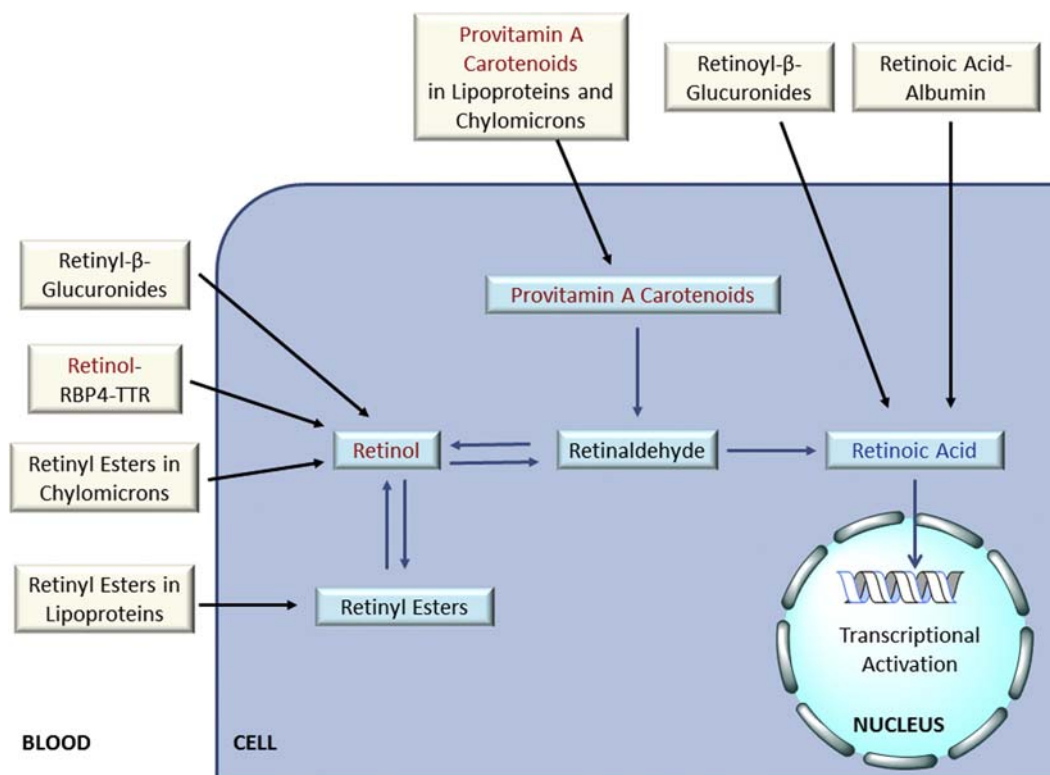


FIGURE 5.3 Many different vitamin A forms are present in the fasting and postprandial circulations.

A species, with normal concentrations ranging from 2 to 4 μM in healthy humans.^{6,28} This normally accounts for approximately 95% of the total vitamin A that is present in the fasting human circulation. Blood levels of retinol do not change significantly in the postprandial circulation when much greater concentrations of retinyl esters are present. Retinol bound to RBP4 cannot be considered to be an essential vitamin A transport pathway since a number of patients who completely lack RBP4 have been identified. Humans who lack RBP4 present with impaired vision (night blindness) or total blindness and other ocular defects, but not with severe systemic lesions.^{29–31} A similar phenotype is observed in mice where *Rbp4* expression had been genetically ablated. Thus, although retinol–RBP4 is very important for transport of vitamin A in the circulation, it is not essential since its absence is not lethal.

The retinol–RBP4 complex binds another plasma protein, transthyretin (TTR). This stabilizes the complex in the circulation, reducing renal filtration and the loss of the vitamin A in urine.²⁸ One molecule of RBP4 binds to one molecule of tetrameric TTR. The molar concentration of tetrameric TTR in the blood exceeds that of RBP4, thus holo-RBP4 is normally bound to TTR and some TTR is present not bound to RBP4.²⁸

How retinol is taken up by cells from the circulating retinol–RBP4–TTR complex has long been the subject

of research interest. Studies of intestinal cells imply that retinol can enter these cells by diffusion, and this is likely true for other cell types as well.²⁰ However, a cell surface receptor for RBP4 termed STRA6 (a product of the *stimulated by retinoic acid 6* gene) has been identified.^{32–34} STRA6 is expressed in a number of tissues/cells that have a high demand for vitamin A, especially the retinal pigmented epithelial (RPE) cells of the eye.^{32–34} But STRA6 is not expressed in many others that have important roles in vitamin A storage and metabolism, including liver cells.^{6,34} STRA6 interacts with RBP4 and increases cellular uptake of retinol.³² In addition, STRA6 reportedly facilitates retinol efflux from cells.^{35,36}

The human liver secretes some retinyl ester bound to nascent VLDL, and, upon metabolism of the VLDL, some of this retinyl ester will be found in LDL or transferred to HDL by cholesteryl ester transfer protein. Concentrations of retinyl esters in the fasting human circulation can vary considerably but are generally in the range of 100–200 nM for individuals who do not take vitamin supplements.^{37,38} Although it was proposed in the 1970s that retinyl esters in the fasting circulation are markers of vitamin A toxicity, especially hepatic toxicity, this hypothesis is not supported by later analysis of serum retinyl ester concentrations for 6547 adults who participated in the National Health and

Nutrition Examination Survey (NHANES III).^{38,39} Analysis of NHANES III data revealed that the prevalence of fasting serum retinyl ester concentrations that were greater than 10% of the serum total vitamin A concentration were not associated with abnormal liver function. Thus, the presence of retinyl esters in fasting human blood is a reflection of normal vitamin A transport in the circulation. The vitamin A present in VLDL, LDL, and HDL presumably is taken up by tissues along with the lipoprotein particles by their cell surface receptors, but this has not been definitively established.

The role of plasma lipoproteins in the delivery of vitamin A to tissues is not generally recognized and is underappreciated. Comparative biology supports the notion that vitamin A, as retinyl ester, is delivered through the circulation to tissues bound to VLDL, LDL, and HDL. In dogs, the delivery of retinyl esters to tissues through the fasting circulation accounts for 70% or more of the total vitamin A present in fasting blood.^{39–41} After exercise, the plasma retinyl ester concentrations of sled dogs nearly doubled, suggesting that exercise mobilizes retinyl esters, possibly from liver and/or adipose tissue.⁴² In domestic cats, retinyl esters are also the predominate vitamin A metabolite, accounting for approximately 70% of the total vitamin A present in the fasting circulation.⁴³ This is also true for ferrets where lipoprotein bound retinyl esters account for 80% or more of the total vitamin A present in the fasting circulation.^{44,45} The fasting circulations of chimpanzees and orangutans contain considerable retinyl esters, albeit at levels that account for only approximately 20% of total vitamin A present in the fasting circulation.⁴⁶

Retinoic acid is present in both the fasting and postprandial circulations where it is bound to albumin. Within the fasting circulation, blood levels of all-*trans*-retinoic acid range from 3 to 5 nM.⁴⁷ However, as early as 30 minutes after consumption of a vitamin A-rich meal consisting of 100 g of turkey liver, human blood levels of all-*trans*-retinoic acid are reported to reach 80–90 nM.⁴⁸ This level is quickly restored to fasting levels within several hours.^{48,49} This suggests that the intestine can contribute significantly to retinoic acid present in the postprandial circulation. Other tissues responsible for contributing retinoic acid to the fasting circulation remain to be established. At present, it is unclear whether only one or a few tissues contribute to the circulating retinoic acid pool or whether retinoic acid is simply “leaking” into the fasting circulation from most or all tissues.

Fasting concentrations of retinyl- and retinoyl- β -glucuronides are in the range of 5–15 nM with higher levels present postprandially.^{50,51} It has been suggested that retinyl- and retinoyl- β -glucuronides serve as sources of vitamin A for tissues.^{52,53} However, these fully water-soluble vitamin A species are generally thought to be catabolites destined for filtration by the kidney.

Vitamin A transport in the postprandial circulation

In the postprandial circulation, following consumption of a vitamin A-rich meal, retinyl ester concentrations can reach 5–10 μ M, although this will depend directly on the quantity of vitamin A consumed in the meal.⁵⁴ Some chylomicron retinyl ester is taken up by tissues as the chylomicron undergoes lipolysis and remodeling. In experimental model studies in rats, 66%–75% of chylomicron retinyl esters were found to be cleared by the liver and the remainder by peripheral tissues.⁵⁵ Thus, 25%–33% of dietary vitamin A is delivered to peripheral tissues without first entering liver stores. The uptake of chylomicron vitamin A by tissues allows humans lacking RBP4 and *Rbp4*-null mice to remain healthy, albeit with vision loss.^{29–31,56,57} It should be noted that this would not be possible without very regular intake of dietary vitamin A, a nutritional state that has become common only in the modern era. Nevertheless, chylomicron-bound vitamin A must be considered to be an important delivery pathway through which tissues acquire vitamin A.

Carotenoid transport in the circulation

Dietary carotenoids can be absorbed by the intestine unmodified and are found in the blood as a component of chylomicrons and their remnants, VLDL, LDL, and HDL.^{58,59} Fasting blood levels in humans of the canonical provitamin A carotenoid β -carotene can be as great as 5–8 μ M.⁵⁹ In the postprandial circulation, these levels are elevated in a manner that is directly proportional to dietary carotenoid consumption. There is no evidence as to whether circulating carotenoids are specifically taken up by cells and tissues or whether they are simply taken up along with the other neutral lipids that may be present in lipoprotein particles.

D. Vitamin A Storage in the Body

The ability of the body alternatively to store or to mobilize vitamin A in response to its dietary availability is unique among the vitamins (Fig. 5.4). Healthy, well-nourished individuals who have ingested vitamin A-sufficient diets and accumulated vitamin A stores throughout life can go many weeks or even months before they succumb to the adverse effects of consuming a vitamin A-insufficient diet. As noted above, vitamin A and several of its metabolites are present in both the fasting and postprandial circulations. However, only retinol can be mobilized from tissue retinyl ester stores in response to insufficient dietary vitamin A intake. That is to say, blood retinol levels are maintained at a constant level, or defended, in response to extended consumption of a vitamin A-insufficient diet. This is accomplished through the specific interaction of retinol

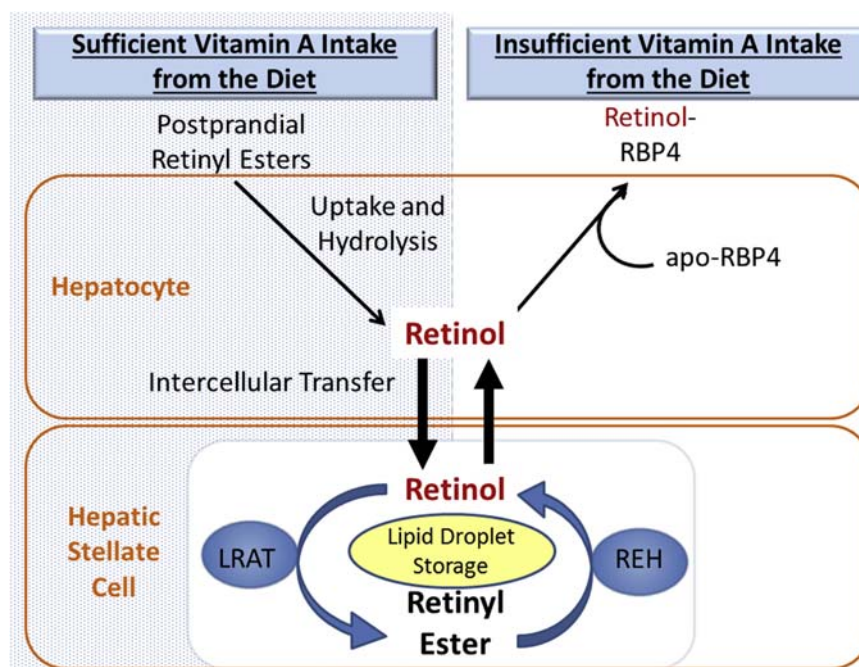


FIGURE 5.4 Depending on the vitamin A status of an individual, newly absorbed dietary vitamin A will either be stored in hepatic stellate cells or immediately resecreted from the hepatocyte bound to RBP4. In times of sufficient dietary vitamin A intake, newly absorbed vitamin A is transferred from the hepatocyte, the cellular site of dietary vitamin A uptake into the liver, to hepatic stellate cells where it is esterified by LRAT and incorporated into intracellular vitamin A-containing lipid droplets for storage. In times of insufficient dietary vitamin A intake, retinyl ester stores within hepatic stellate cells will undergo hydrolysis by a retinyl ester hydrolase (REH) allowing for mobilization of vitamin A stores from the liver bound to RBP4. It is important to note that a much greater percentage of newly absorbed vitamin A will be resecreted from hepatocytes bound to RBP4 without undergoing storage in hepatic stellate cells in times of insufficient dietary vitamin A intake.

with its serum transport protein, RBP4. RBP4 allows for the efficient mobilization of retinol from tissue (primarily hepatic) vitamin A stores. Hence, *RBP4*-deficient humans will be prone to developing vitamin A deficiency if they are unable to obtain sufficient dietary vitamin A on a regular basis.

The healthy human liver stores the majority of vitamin A that is present within the body. Two distinct hepatic cell types play central roles in the storage and metabolism of vitamin A within the liver, the parenchymal cells, or hepatocytes, and the nonparenchymal hepatic stellate cells (HSCs). Hepatocytes constitute approximately two-thirds of all cells present in the liver and approximately 90% of total hepatic protein.^{60,61} These large and relatively abundant cells are responsible for most of the metabolic activities associated with the liver. They are the cellular site of chylomicron remnant clearance in the liver, and consequently the uptake of dietary vitamin A by the liver.^{6,19,62,63} The hepatocyte is also the cellular site of RBP4 synthesis and secretion in the liver.^{6,28} Thus, hepatocytes are importantly involved in both the uptake of dietary vitamin A by the liver and its mobilization from the liver. However, hepatocytes account for only a relatively small proportion of the total vitamin A present within the liver. Estimates of the hepatic total vitamin A that is found within hepatocytes

range from 10% to 20%.⁵ The remainder, 80%–90%, is found in the lipid droplets of HSCs, which account for approximately 5%–8% of the cells in the liver and about 1% of total hepatic protein. It is thought that newly absorbed dietary vitamin A is retained in and secreted from the hepatocyte bound to RBP4 and that some is transferred to the HSC for storage. The relative proportion that is secreted bound to RBP4 depends on vitamin A nutritional status, with a greater percentage secreted in times of insufficient vitamin A intake. Thus, the relative distribution of hepatic total vitamin A between hepatocytes and HSCs changes in response to vitamin A nutritional status (Fig. 5.4).

Since many tissues, including liver, lungs, testes, and eyes, express BCO1, the enzyme responsible for β -carotene cleavage to vitamin A, intact provitamin A carotenoids delivered to these tissues may be converted in situ to vitamin A.¹⁶ Measurements of carotenoid levels in human tissues obtained postmortem establish that the liver contains significant concentrations of the common dietary carotenoids, β -carotene, α -carotene, β -cryptoxanthin, and lycopene (a nonprovitamin A carotenoid obtained from tomatoes) that ranged from undetectable to greater than 20 nmol/g tissue.⁶⁴ The kidney and lung also accumulate each of these dietary carotenoids, albeit at concentrations which were an order of magnitude less

than the liver.⁶⁴ It remains to be established whether tissue provitamin A carotenoid pools act as true “stores” that are used in a regulated manner for vitamin A synthesis. It is not known if these “stores” are drawn upon to defend tissue and/or blood vitamin A levels in times of insufficient dietary vitamin A intake. Thus, although tissues can accumulate significant concentrations of β -carotene and other provitamin A carotenoids, it remains unclear whether or how this is integrated into the whole-body economy of vitamin A.

E. Metabolic Activation of Vitamin A

The vitamin A metabolites that maintain vitamin A-dependent functions, retinaldehyde and retinoic acid, are formed through the enzymatic oxidation of retinol (Fig. 5.5).^{4–6,10,12,26} This involves two distinct enzymatic steps, the oxidation of retinol to retinaldehyde and the subsequent oxidation of retinaldehyde to retinoic acid. The first of these steps can be catalyzed by a number of distinct retinol dehydrogenases (RDHs) that have specific tissue/cellular localizations. This reaction is reversible owing to the actions of retinaldehyde reductases that reduce retinaldehyde to retinol. Three distinct retinaldehyde dehydrogenases have been identified, and each is expressed in tissue/cell type-specific manners. The formation of retinoic acid from retinaldehyde is irreversible. Thus, retinoic acid cannot be reconverted to retinaldehyde or retinol. To prevent excessive accumulation of retinoic acid, it must be oxidized and eliminated from the body. Three cytochromes (Cyp26A1,

Cyp26B1, and Cyp26C1) are specifically involved in catalyzing the oxidation of retinoic acid. The genes encoding each of these retinoic acid-metabolizing cytochromes are induced when retinoic acid levels are high. This provides an autoregulatory loop that prevents over accumulation of retinoic acid within tissues.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Gene Regulation

Vitamin A is required for maintaining normal immunity, barrier integrity, male and female reproduction, growth and development (by regulating cell proliferation and differentiation, and cell death), and vision.¹² Aside from photoperception in the eye, retinoic acid is responsible for maintaining other vitamin A-dependent processes in the body. Retinoic acid is a very potent transcriptional regulator acting through members of the nuclear hormone superfamily of ligand-dependent transcription factors.^{9–12} Canonically, the three retinoic acid receptors (RAR α , RAR β , and RAR γ) along with the three retinoid X receptors (RXR α , RXR β , and RXR γ) mediate the transcriptional regulatory activity of vitamin A. In excess of 500 diverse genes are transcriptionally responsive to vitamin A.⁶⁵ The RARs and RXRs are members of the steroid/thyroid/retinoid superfamily of nuclear hormone receptors. Each of the members of this nuclear receptor superfamily share similar structural properties

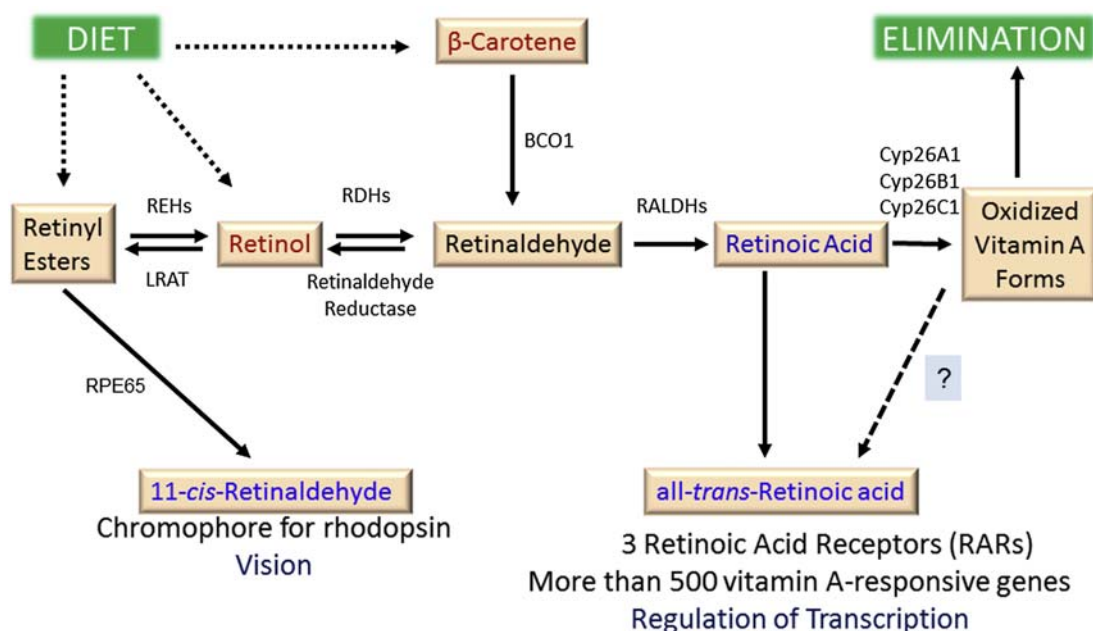


FIGURE 5.5 Summary scheme for the activating metabolism of vitamin A within the body.

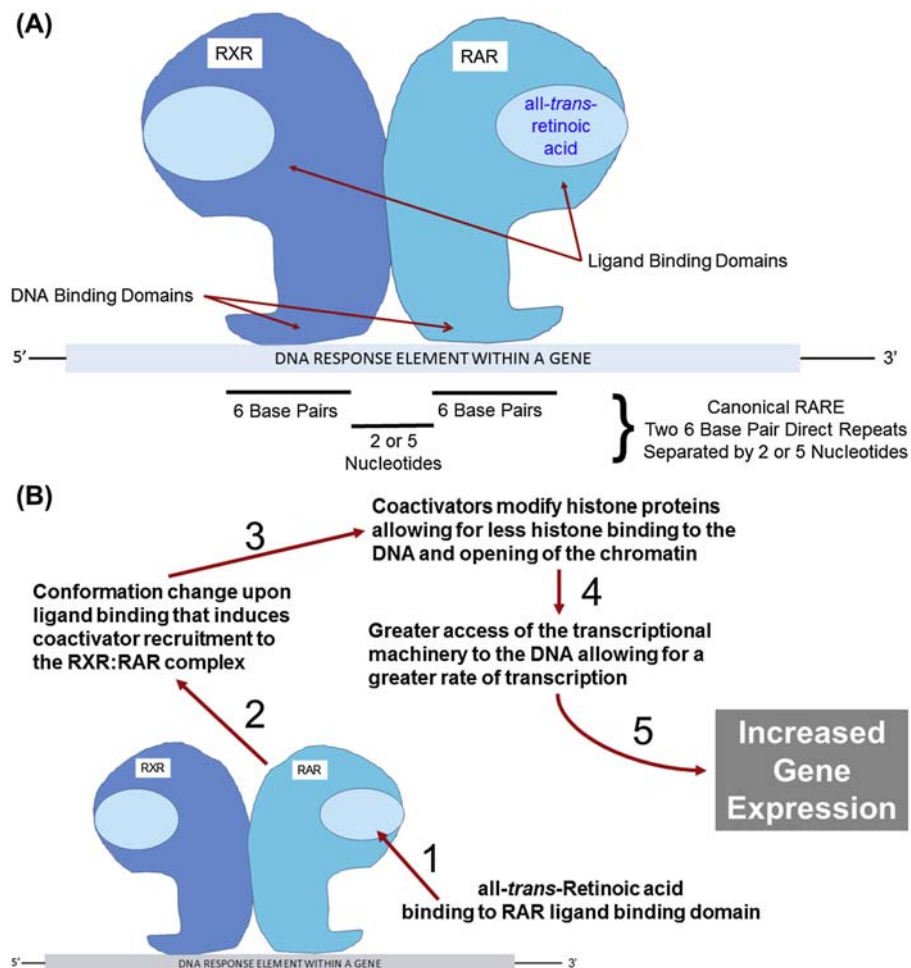


FIGURE 5.6 Structural features of the retinoic acid receptors (RARs) and their actions as transcriptional activators. Panel (A) To facilitate vitamin A–dependent transcriptional regulation, one retinoid X receptor (RXR) and one RAR heterodimerize forming a dimeric complex that binds to a retinoic acid–responsive DNA response element within the promoter of a gene (referred to as RARE). A canonical RARE consists of two 6 base pair direct repeats that are separated by either 2 or 5 nucleotides. The identities or sequences of these intervening nucleotides are not important for recognition of the response elements; only the sequence of the two 6 base pair repeated elements is needed for response element recognition. The portion of the nuclear receptor that recognizes the RARE is referred to as the nuclear receptor’s DNA-binding domain. The RAR ligand, all-*trans*-retinoic acid, binds to a portion of the nuclear receptor referred to as the ligand-binding domain. Panel (B) Upon binding of all-*trans*-retinoic acid to the ligand-binding domain of the RAR present on a RARE (step #1), a conformational change in the RXR:RAR complex is induced (step #2). This change in conformation induces the recruitment of coactivators to the complex and removal of any corepressors that may have been associated with the ligand-free complex (step #3). The coactivators possess enzymatic activities that modify the histone proteins sequestering the genomic DNA resulting in diminished histone binding to the DNA and an opening of the chromatin (highly compact DNA) (step #4). The transcriptional machinery now has greater access to the newly exposed DNA resulting in greater rates of transcription and increased levels of gene expression (step #5).

and similarities in how they act within the cell nucleus (Fig. 5.6).

All-*trans*-retinoic acid can bind to each of the RARs with high affinity and is universally considered to be the physiological ligand for these three nuclear receptors.^{9–12} The early literature regarding the properties of the RXRs proposed that 9-*cis*-retinoic acid is the physiological ligand for the RXRs.^{66,67} However, there is presently no consensus as to whether 9-*cis*-retinoic acid is truly a physiologically relevant ligand for the RXRs. This lack of consensus arises from differences in the analytical methods used to assess tissue and

cellular 9-*cis*-retinoic acid concentrations.⁶⁸ Owing to the lack of clear consensus on this issue, the current state of understanding has been conservatively summarized as “Indeed RXRs cannot bind all-*trans*-retinoic acid, and although its 9-*cis* isomer was initially considered as a bona fide RXR ligand, it is now controversial due to the inability to detect this compound in vivo.”¹¹

A number of naturally occurring molecules, some of them related to vitamin A, including β-apo 13-carotenone (formed from asymmetric cleavage of β-carotene), are reported to be able to bind to RXRs with relatively high affinity and able to influence

gene transcription *in vitro*.⁶⁹ Recently, 9-*cis*-13,14-dihydroretinoic acid, a natural metabolite of 9-*cis*-retinoic acid that is abundant in cells and tissues, has been proposed to be a natural endogenous RXR ligand.^{70,71} Presently though, there is no general agreement as to whether any of these is truly a significant natural RXR ligand acting *in vivo*.

Each member of the nuclear hormone receptor family possesses a ligand-binding domain (Fig. 5.6A).^{9-12,72-74} This is the site within the protein where the ligand specific to the receptor binds. For RAR α , RAR β , or RAR γ , this is the site on the receptor where all-*trans*-retinoic acid binds. Analogously, for the thyroid hormone nuclear receptor, thyroid hormone, but not all-*trans*-retinoic acid, will bind the thyroid hormone receptor's ligand-binding domain. Thus, the ligand-binding domain of each nuclear hormone receptor confers ligand specificity to the nuclear receptor. Each nuclear hormone receptor also possesses a defined DNA-binding domain.^{9-12,72-74} The DNA-binding domain recognizes a specific sequence within the promoter regions of genes to which the nuclear receptor will bind (Fig. 5.6A). The sequences of DNA within genes recognized by the DNA-binding domain are referred to as response elements. For the three RARs, these response elements are known as retinoic acid response elements or RAREs. The presence of a RARE within the promoter of a gene confers transcriptional responsiveness of the gene to the RAR and all-*trans*-retinoic acid. DNA response element sequences for each nuclear hormone receptor have been determined and cataloged. As would be expected, the response elements recognized by a specific hormone receptor are very similar, with similar sequence motifs. This confers DNA-binding specificity to each of the nuclear receptors. Most RAREs consist of a directly repeated six base pair sequence where the two repeats are separated by either 2 or 5 intervening nucleotides. An example of this is the RARE that is present in the human gene encoding Cyp26A1 where the RARE sequence is a direct six base pair repeat separated by five base pairs, AGTTCACcaagAGTTCA. To bind to its response elements, the RARs must either dimerize with another RAR to form a homodimer or heterodimerize with one RXR. It is the RAR-RAR homodimer or the RAR-RXR heterodimer that binds to RAREs to regulate transcription.

Bound to their respective response elements, nuclear hormone receptors interact with a number of other nuclear proteins.^{9-12,72-74} There are two very important functional classes of these, the coactivators and the corepressors. Nuclear receptor interactions with coactivators activate the basal transcription machinery, increasing the rate of transcription of the gene and hence gene expression. Interactions with corepressors slow the basal transcription machinery, decreasing transcription rates and gene expression. Through these interactions,

nuclear hormone receptors like the RARs and RXRs affect responsive gene transcription. Generally, when all-*trans*-retinoic acid is bound to an RAR, this will allow for interactions with coactivators and increased gene transcription (Fig. 5.6B). When all-*trans*-retinoic acid is unavailable to bind its ligand-binding site within the RAR, transcription rates are repressed. Thus, all-*trans*-retinoic acid usually increases gene expression. When all-*trans*-retinoic acid binds an RAR, this induces a conformational change in the nuclear receptor that allows for coactivator binding and displacement of bound corepressors. However, for some genes this is reversed and when the all-*trans*-retinoic acid/RAR complex binds to its response element, this represses transcription. Transcriptional repression by all-*trans*-retinoic acid is dependent on the context of the gene and other transcriptional regulators that may be involved in regulating expression of the specific gene.

The RARs and RXRs have been extensively studied at both the gene and protein levels and the molecular details of how these transcription factors activate/repress transcription are known in considerable detail.⁷²⁻⁷⁴

Retinoic acid regulates the transcription of diverse genes needed to support a variety of physiological, cellular, and biochemical responses within the body. Some of these are needed for maintaining and regulating vitamin A metabolism, transport, and actions within the body, examples of these include *LRAT*, *CYP26A1*, *CYP26B1*, *RBPI*, *STRA6*, and *RAR β* . A large number of vitamin A-responsive genes are involved in regulating normal cell proliferation and differentiation. Genes that are retinoic acid-responsive and involved in modulating these responses include many encoding proteins with homeobox domains, other transcription factors, and genes encoding proteins involved in regulating the cell cycle. Genes encoding either extracellular matrix proteins or enzymes involved in extracellular matrix remodeling contain RAREs and are responsive to all-*trans*-retinoic acid. Expression of a number of genes encoding protein hormones as well as a number of enzymes involved in the synthesis of nonpeptide hormones contain RAREs. Many genes encoding proteins important to lipid metabolism and transport require retinoic acid and RARs for their regulation. The list of genes that are regulated by vitamin A is extensive, as are the physiological responses that the genes support.^{10,12,65}

B. Vision

Since the seminal studies of Wald in the 1930s, it has been known that the photoreceptors in the retina (rods and cones) and RPE cells contain high concentrations of both the 11-*cis*- and all-*trans*-isomers of retinaldehyde,

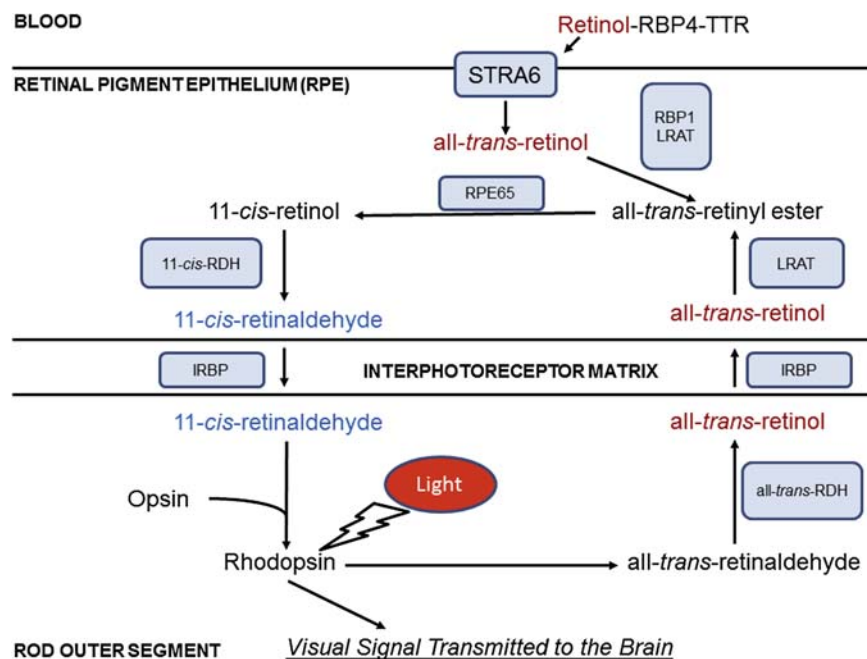


FIGURE 5.7 The visual cycle of vitamin A.

retinol, and retinyl ester.^{7,8} The rods and cones are the sites of phototransduction. RPE cells are the predominant site of vitamin A storage in the eye and are highly enriched in retinyl esters.^{8,75,76} The actions of vitamin A in vision are often referred to as the “Visual Cycle of Vitamin A” or simply the “Visual Cycle” (Fig. 5.7). Vitamin A (all-trans-retinol) uptake into the RPE from the circulation is mediated by STRA6 from the retinol-RBP4-TTR complex. Once within the RPE, all-trans-retinol binds to cellular retinol-binding protein, type 1 (RBP1), which metabolically channels the all-trans-retinol to LRAT for esterification. The all-trans-retinyl ester formed will undergo a coupled hydrolysis-isomerization catalyzed by the enzyme RPE65 to yield 11-cis-retinol. Thus, RPE65 is the isomerohydrolase that is solely expressed in the RPE and that is responsible for the formation of 11-cis-forms of vitamin A that are found within RPE cells and rods.^{8,75–77} A specific 11-cis-RDH then catalyzes the oxidative formation of 11-cis-retinaldehyde within the RPE. The 11-cis-retinaldehyde is transported from the RPE to the photoreceptors where light is perceived. This involves the actions of interphotoreceptor matrix retinol-binding protein (IRBP), which is found in the extracellular space between the RPE and the photoreceptors. Once taken up into the photoreceptors, the 11-cis-retinaldehyde complexes with opsin to form the visual pigment rhodopsin. When a photon of light excites the 11-cis-retinaldehyde bound to opsin, it immediately undergoes photoisomerization becoming opsin-bound all-trans-retinaldehyde. This photoisomerization is the primary visual event leading to G protein activation, membrane depolarization,

and the transmission of a signal to the brain. The all-trans-retinaldehyde dissociates from opsin and is reduced to all-trans-retinol by an all-trans-RDH that is specific to the photoreceptors. The all-trans-retinol is transported back to the RPE by IRBP, where it again binds RBP1 and the visual cycle begins anew.

It should be noted that the cycle described above is one of two that functions within the retina. This is the best-understood vitamin A cycle within the eye, functioning in the rod cells that were first identified and studied in the 1930s.^{7,8} However, considerable recent evidence indicates that a somewhat different second cycle functions in the retinal cone cells that are involved in color vision.^{75,76} The primary differences between the rod and cone cycles involve the individual proteins that catalyze the formation of 11-cis-forms of vitamin A and the metabolic fluxes of vitamin A in the cone cycle, and not qualitative differences in the forms of vitamin A present in the cones.

IV. DIETARY REQUIREMENTS

A. Indicators of Vitamin A Deficiency or Inadequacy

Xerophthalmia refers to the ocular manifestations arising from vitamin A deficiency.^{78–81} These manifestations are the first signs of vitamin A deficiency that are recognizable clinically without employing biochemical or histochemical analysis. Consequently, the term xerophthalmia is used synonymously with vitamin A

deficiency. Xerophthalmia can occur in any age group, but it is most commonly found in preschool-age children, adolescents, and pregnant women. Children are at higher risk of vitamin A deficiency and xerophthalmia owing to their greater need for vitamin A to support growth and development. Because vitamin A acts critically in the maintenance of the immune response, children experiencing vitamin A deficiency are also at higher risk of intestinal infestations and infections. These can impair vitamin A absorption and increase loss of the vitamin. It is this impairment to the immune response that contributes most significantly to the morbidity and mortality associated with vitamin A deficiency; it is the progressively worsening stages of xerophthalmia that allow for the clinical recognition and diagnosis of vitamin A deficiency. It should be noted that a number of studies have identified vitamin A-dependent impairments to the immune response occurring prior to xerophthalmia.

The current classification scheme for the stages of xerophthalmia considers night blindness to be the first stage of the disease (Stage XN).^{78,79} This is followed in order of appearance and severity by epithelial disruptions of the cornea and conjunctiva, including conjunctival xerosis (Stage X1A), Bitot spots (Stage X1B), corneal xerosis (Stage X2), corneal ulceration/keratomalacia (Stages X3A and X3B), corneal scar (Stage XS), and xerophthalmic fundus (Stage XF). Night blindness responds rapidly to vitamin A therapy, within 1–2 days. Prompt treatment with vitamin A generally results in the full preservation of eyesight up to the stage of corneal xerosis (Stage XS). If not promptly treated, the progressively worsening later stages of xerophthalmia will result in irreversible blindness.

Night blindness, the loss of vision in dim light arising from insufficient vitamin A levels within the RPE and retina to allow the visual cycle to run optimally, is the earliest readily observable manifestation of vitamin A deficiency. This is accompanied by very low circulating retinol–RBP4 levels.^{78–80} Impaired night vision can begin when circulating retinol–RBP4 concentrations fall below 1.0 μM , but is more frequently observed when they fall below 0.7 μM . The later, more severe stages of xerophthalmia, occur when circulating retinol–RBP4 levels fall below 0.35 μM . From the perspective of the biological origins of xerophthalmia, night blindness arises from an impairment in vitamin A actions in vision; however, the later stages of xerophthalmia all involve impairments to retinoic acid signaling and dysregulated vitamin A-dependent transcription. The corneal liquefaction resulting in keratomalacia and irreversible blindness arise through derepression of retinoic acid-dependent genes encoding collagenases and other enzymes that digest extracellular matrix proteins.

B. Assessment of Vitamin A Status and Criteria for Adequacy

To assess vitamin A adequacy in subjects, one needs to understand tissue vitamin A stores. This can be done functionally (i.e., assessing night blindness and xerophthalmia), biochemically (assessing blood or fluid vitamin A levels or responses to a vitamin A challenge), or histologically (assessing changes in cellularity brought on by vitamin A deficiency).

Because of the ease of obtaining blood samples, measurements of serum/plasma retinol concentrations are most commonly used to assess vitamin A status. Serum/plasma levels of retinol $\leq 0.70 \mu\text{M}$ indicate moderate vitamin A deficiency, and levels $\leq 0.35 \mu\text{M}$ reflect severe vitamin A deficiency.^{79,80} Although measurement of serum retinol levels is relatively noninvasive and technically simple, this approach has some drawbacks. This is because blood retinol levels are not always a good indicator of hepatic vitamin A stores.^{79–82} The liver secretes retinol mobilized bound to newly synthesized RBP4 in order to maintain blood retinol levels within a narrow range. Hepatic retinol mobilization will continue at a constant rate until hepatic vitamin A stores are nearly exhausted.^{79–82} Thus, an individual will maintain a normal blood vitamin A level even though hepatic stores are close to exhaustion, possibly reaching exhaustion within only a matter of days. This could incorrectly result in a subject being classified as having adequate vitamin A stores. In addition, since RBP4 is a negative acute-phase response protein, serum/plasma retinol concentrations are decreased by acute and chronic infections.^{79–82} In the case of underlying infection, serum retinol levels will mistakenly be taken to indicate insufficient hepatic vitamin A stores. Thus, serum/plasma retinol concentrations are not always a reliable indicator for diagnosing vitamin A deficiency, especially for individuals. However, for the purpose of clinical practice, plasma retinol levels alone have been recommended as being sufficient for documenting significant vitamin A deficiency.⁸³ Among populations, a frequency distribution of serum/plasma retinol concentrations is highly informative for assessing the vitamin A status of the population.

Breast milk retinol levels are sometimes measured for nursing mothers to assure that the infant is receiving sufficient vitamin A. Cutoffs for retinol levels in breast milk that reflect vitamin A insufficiency are $\leq 1.05 \mu\text{M}$ or $\leq 8 \mu\text{g/g}$ milk fat.⁸⁴

Since the liver is the predominant site for vitamin A storage in the body, hepatic vitamin A levels are the gold standard for understanding vitamin A nutritional adequacy. Although direct measurement of hepatic vitamin A levels is not technically difficult for human liver biopsies, given the morbidity and mortality

associated with obtaining a human liver biopsy, this is not feasible. A number of different indirect approaches have been developed, standardized, and validated for use in estimation of hepatic vitamin A reserves and/or predicting vitamin A status. These approaches include the isotope dilution method, the relative dose-response method, and the modified dose-response method.^{81,82} Each of these involves the use of mathematical models where vitamin A, either unlabeled, labeled with a stable isotope, or chemically modified in some manner, is administered to a subject, and the levels in the circulation of vitamin A and/or its tracer are measured over a number of days. This information is then considered in a mathematical model that allows for estimation of the concentration of vitamin A present in the liver and/or whether this is adequate or insufficient. Although these methods are less prone to misclassifying an individual's vitamin A status, they are also more technically sophisticated and hence more difficult to undertake.

Before the clinical onset of xerophthalmia, mild vitamin A deficiency leads to early keratinizing metaplasia and losses of mucin secreting goblet cells on the bulbar surface of the conjunctiva of the eye.⁸⁰ Functional changes on the ocular surface can be detected by microscopic examination of stained epithelial cells obtained by briefly applying a cellulose acetate strip or disc against the temporal conjunctivum. Referred to as conjunctival impression cytology, this method allows for classification of normal and differing degrees of abnormality based on the density and distribution of stained normal epithelial cells, goblet cells, and mucin spots that are present on the surface of the conjunctiva.

C. Reference Intakes

Intake recommendations for vitamin A in the United States and Canada were established as by the Dietary Reference Intakes (DRIs) in 2001 by the Food and Nutrition Board of the Institute of Medicine (IOM), the National Academies (Institute of Medicine, 2001).⁸⁵ DRI is the general term for a set of reference values used for planning and assessing nutrient intakes of healthy human beings. These values vary by age and gender. The Recommended Dietary Allowance (RDA) is the average daily level of intake that will be sufficient to meet the nutrient requirements of nearly all healthy individuals in the age and sex group to which they pertain. The RDAs for vitamin A in 2001 were given as micrograms (μg) of retinol activity equivalents (RAE) to account for the different bioactivities of retinol and provitamin A carotenoids. Since the body converts the major dietary sources of vitamin A into retinol in the intestine with differing degrees of overall efficiency, 1 μg retinol is equivalent to 12 μg β -carotene, 24 μg α -carotene, and

24 μg β -cryptoxanthin. Previously, vitamin A content listed on food and supplement labels in the United States has been as international units (IUs). By definition, 1 IU of retinol is 0.3 μg RAE; 1 IU of β -carotene from dietary supplements is 0.15 μg RAE; 1 IU of β -carotene from food is 0.05 μg RAE; and 1 IU of α -carotene or β -cryptoxanthin is 0.025 μg RAE. Because of this, an RAE cannot be directly converted into an IU without knowing the source of vitamin A. Consequently, new labeling requirements by the US Food and Drug Administration require that all labeled values appear as RAE as of January 1, 2020.⁸⁶

The RDAs for vitamin A for individuals 14 years and older are 900 μg RAE for males and 700 μg RAE for females.⁸⁵ These recommended intakes are based the Estimated Average Requirement (EAR) to maintain adequate liver stores, the criterion of adequacy chosen to establish the recommended intakes. Lower EAR values based on the amount needed to prevent night blindness were also provided,⁸⁵ but the criterion chosen by the IOM for recommended intakes for the United States and Canada is based on maintaining adequate liver storage. A consideration of the factors that led to these recommendations can be found at <http://www.nap.edu/openbook.php?isbn=0309072794>.

D. Vitamin A Toxicity (Hypervitaminosis A)

Vitamin A toxicity (hypervitaminosis A) is a disorder in which there is too much vitamin A in the body.^{83,87} It is rare for vitamin A toxicity to occur simply from consumption of vitamin A-rich foods. Usually, supplement consumption is involved. However, there are published reports from early last century indicating that Arctic explorers who had consumed polar bear or seal liver (two very rich sources of vitamin A) had experienced acute vitamin A toxicity.⁸⁸

Both acute and chronic vitamin A toxicities are associated with excessive vitamin A intake. Acute vitamin A toxicity occurs almost immediately, most often when an adult takes several hundred thousand IUs of vitamin A. Chronic vitamin A toxicity may occur over time in adults who regularly take more than 25,000 IU a day. Diagnosis of vitamin A toxicity is clinical. Blood vitamin A levels correlate poorly with toxicity.^{83,87} Differentiating vitamin A toxicity with other disorders may be difficult. Symptoms reported for vitamin A toxicity vary but include blurred vision; double vision; vomiting; bone pain or swelling; changes in alertness or consciousness; decreased appetite; dizziness; drowsiness; changes in skin including cracking in the corners of the mouth, peeling, and itching; loss of hair or oily hair; increased cerebrospinal fluid pressure, headache; irritability; liver damage; nausea; and poor weight gain and abnormal softening of the skull bone in infants and children.^{83,87}

Treatment involves simply stopping supplements (or changing food patterns) that contain vitamin A. Recovery usually occurs when vitamin A ingestion stops. Symptoms of chronic toxicity usually disappear with 1–4 weeks. Most people fully recover.

The most dangerous irreversible effect of excessive vitamin A intake is the teratogenicity associated with vitamin A and synthetic retinoids. This is not too surprising given the importance of vitamin A in regulating normal cell differentiation and proliferation. The teratogenicity of vitamin A/retinoids is seen in the many teratogenic outcomes experienced by pregnant women taking the drug Accutane (isotretinoin) as treatment for severe cystic acne.^{89,90} Accutane is 13-*cis*-retinoic acid, a natural metabolite of retinoic acid that is found normally at low levels (1–5 nM) in the human circulation.^{47–49} Owing to the teratogenicity of vitamin A, pregnant women, women intending to become pregnant, or breast-feeding women are advised not to consume more than 10,000 IU vitamin A per day to avoid possible damage to the fetus or infant.

Although provitamin A carotenoids can be converted to vitamin A in the body, excessive ingestion of carotenoids causes carotenemia, not vitamin A toxicity. Carotenemia is usually asymptomatic but may lead to carotenosis, in which the skin becomes yellow. However, as discussed in Section VI below, when taken as a high dose supplement, β -carotene has been associated with increased cancer risk. Importantly, cancer risk does not seem to increase when β -carotene is consumed in fruits and vegetables.

E. Basis for Individual Variability in Requirements for Vitamin A Intake

Vitamin A deficiency can result from inadequate vitamin A intake and is often found associated with protein–energy malnutrition. Insufficient dietary fat intake or fat malabsorption, intestinal infestations or infections and accompanying diarrhea, and liver disorders including later stage liver disease that affect hepatic metabolism and/or HSC storage of vitamin A all can contribute to development of vitamin A insufficiency.⁸³ Most people with cystic fibrosis have pancreatic insufficiency, increasing their risk of vitamin A deficiency due to difficulty absorbing fat. Alcoholic patients also have altered needs for vitamin A. Chronic excessive alcohol consumption results in the loss of hepatic vitamin A stores affecting the vitamin A needs and status for the individual.⁹¹

F. Tolerable Upper Intake Levels for Vitamin A

The Tolerable Upper Intake Level (UL) for vitamin A is the highest average daily intake of vitamin A that is

likely to pose no risk of adverse health effects to almost all individuals in the general population.⁸⁵ As intake increases above the UL, the potential risk of adverse effects may increase. This is based on evidence that this level of vitamin A intake is unlikely to cause adverse health outcomes when consumed either as food, as a supplement, or as a combination of both. The ULs for vitamin A apply to preformed vitamin A only, and take into account levels of preformed vitamin A intake associated with an increased risk of liver abnormalities in men and women, teratogenic effects, and a range of toxic effects in infants and children. ULs for β -carotene or other provitamin A carotenoids have not been established and are not included in the UL. A full listing of ULs for preformed vitamin A is available.⁸⁵

G. Dietary Sources of Vitamin A

There are many dietary sources of vitamin A. These include foods from animal sources that contain primarily preformed vitamin A and plant-based foods that contain primarily provitamin A carotenoids. Foods from animal sources that provide relatively high levels of vitamin A include liver, salmon, tuna, other fish, milk, milk products like cheese, and eggs. Vitamin A-rich plant-based foods include most leafy green vegetables, orange and yellow vegetables, tomato products, fruits, and some vegetable oils. The US Department of Agriculture's USDA National Nutrient Database available at <http://ndb.nal.usda.gov/ndb/> lists the nutrient content of common foods and provides a comprehensive listing of foods containing preformed vitamin A and provitamin A carotenoids.⁹²

US survey data from the NHANES What We Eat in America 2013–16 cohort reported the median vitamin A consumption for all adult men of 627 μg RAE, and women of 556 μg RAE. Using the EAR for men for vitamin A of 625 μg RAE (IOM, 2001)⁹², 50% of men had inadequate intakes of vitamin A, meaning that liver storage was inadequate. Applying the EAR for women of 500 μg RAE/day, 41% had inadequate intakes. Less than 3% of men or women consumed more than the UL of 3000 μg of preformed vitamin A/day.⁹³

Some foods are routinely supplemented with vitamin A. Milk, milk products, and ready-to-eat cereals are examples of these. Vitamin A also is available in multivitamins and as a stand-alone supplement. Usually supplemental vitamin A is provided in the form of retinyl acetate or retinyl palmitate. In some instances, a portion of the supplemental vitamin A is in the form of β -carotene. Some foods are supplemented with β -carotene alone. Supplement labels usually indicate the percentages of each form of the vitamin that are present in the product. Multivitamin supplements typically contain 750–3000 μg (2,500–10,000 IU)

preformed vitamin A. The amounts of vitamin A in stand-alone supplements vary widely. Given potential concerns regarding vitamin A toxicity, careful consideration is needed when choosing which stand-alone supplement to use.

V. ISSUES RELATED TO VITAMIN A IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Vitamin A was first identified over 100 years ago owing to its role as a dietary substance that is required for preventing vitamin A deficiency.^{1,2} In the present era, vitamin A deficiency remains a major public health problem throughout the developing world. Vitamin A toxicity is now increasingly seen in the developed world, although the magnitude of this problem remains very small compared to the magnitude of vitamin A deficiency encountered worldwide.

In the modern era, there is considerable interest in understanding the possible roles that vitamin A may have in the prevention and/or causation of disease states commonly encountered in the developed world. The actions of vitamin A in regulating transcription influence many essential cellular processes that are required for maintaining good health. Consequently, there has been considerable interest in using natural vitamin A metabolites and/or synthetic retinoids to prevent or to treat disease. In addition, there is growing recognition that genes/proteins involved in mediating vitamin A transport, metabolism, and actions are associated with disease causation especially proliferative disorders, hepatic disease, and other metabolic diseases including obesity, impaired insulin responsiveness, and cardiovascular disease. It needs to be noted that the actions of vitamin A in these disease states are not thought to directly involve systematic vitamin A deficiency or toxicity arising from dietary intake. Rather, this modern interest focuses on local lesions in vitamin A metabolism and/or actions, within specific cells in specific tissues that affect disease processes.

Vitamin A deficiency

According to the World Health Organization (WHO),⁹⁴ 190 million preschool-aged children and 19 million around the world have a serum retinol concentration $\leq 0.70 \mu\text{M}$. The WHO estimates that 250,000–500,000 vitamin A-deficient children become blind every year, half of them dying within 12 months of losing their sight. Thus, vitamin A deficiency remains a major public health problem in many developing

countries. There remains continuing worldwide efforts by governmental and nongovernmental agencies aimed at eliminating vitamin A deficiency. Although in the last 40 years there have been many notable public health successes for eradicating vitamin A deficiency, the long-standing goal of eliminating totally vitamin A deficiency is not yet achieved.

Vitamin A deficiency usually arises from prolonged dietary deprivation. Frank vitamin A deficiency is now relatively rare in the developed world. Vitamin A deficiency remains a problem in developing countries where residents have either limited access to foods containing preformed vitamin A from animal-based food sources due to poverty or traditional dietary patterns that do not include available foods that contain provitamin A carotenoids.⁸³

Proliferative disorders

Because of the role vitamin A plays in regulating cell growth and differentiation, there has been much research exploring the role of vitamin A in proliferative diseases. Most of this interest has focused on cancer. Many different types of studies, including retrospective, prospective, and large intervention trials where supplements were provided to human subjects at risk of cancer, have been undertaken. To date, this literature fails to provide reproducible and compelling evidence establishing a protective role for supplemental vitamin A in cancer prevention. However, it provides a cautionary lesson regarding supplement use.

In the 1980s and 1990s, several prospective and retrospective observational studies involving current and former cigarette smokers, as well as people who had never smoked, had found that higher dietary intakes of carotenoids, fruits and vegetables, or both were associated with a lower risk of lung cancer.⁸³ However, two large clinical trials designed to verify the existence of an inverse relationship between β -carotene intake and lung cancer were unable to do so. One, the Carotene and Retinol Efficacy Trial (CARET), enrolled 18,314 current and former smokers who took daily supplements containing 20 mg β -carotene and 25,000 IU retinyl palmitate for 4 years on average.⁹⁵ A second, the Alpha-Tocopherol, Beta-Carotene (ATBC) Cancer Prevention Study, enrolled 29,133 male smokers who took supplements of 50 mg/day α -tocopherol (vitamin E), 20 mg/day β -carotene, a combination of 50 mg/day α -tocopherol and 20 mg/day β -carotene, or a placebo for 5–8 years.⁹⁶ Neither of these studies was able to prove the hypothesis that high levels of β -carotene intake prevent lung cancer. Rather, the results of both the CARET and ATBC studies suggest that large supplemental doses of β -carotene, with or without retinyl palmitate, have detrimental effects resulting in an excess of both cancer and cardiovascular disease

morbidity and mortality in subjects receiving the supplement. This finding suggests that caution is needed when considering the use of high dose β -carotene supplements.

Metabolic disease

Vitamin A is required for regulating multiple genes and signal transduction pathways that have been linked to metabolic disease (obesity, insulin resistance, type 2 diabetes, hepatic disease, cardiovascular disease, and others). Consequently, it is not too surprising that the literature has implicated vitamin A and/or vitamin A-related parameters (vitamin A-binding proteins and enzymes involved in vitamin A metabolism) to metabolic disease development and progression.⁹⁷ However, this literature is not yet sufficiently mature to allow for definitive conclusions to be drawn regarding whether these effects arise through one common pathway involving retinoic acid signaling or through multiple different pathways.

The most extensive literature regarding human metabolic disease development and vitamin A-related parameters involves the actions for RBP4 in obesity, insulin resistance, type 2 diabetes, nonalcoholic fatty liver disease (NAFLD), cardiovascular disease, and other obesity-related metabolic diseases. However, definitive understanding of how RBP4 contributes to these disease states remains elusive. In 2005, high circulating RBP4 levels were found to be associated with impaired glucose clearance, with high RBP4 levels inducing insulin resistance.^{98,99} This finding led to the proposal that adipocyte-derived RBP4 is a signal that contributes to the pathogenesis of type 2 diabetes, linking obesity with type 2 diabetes, as well as other obesity-related metabolic diseases.^{98,99} An idea underlying this hypothesis was that obesity and increased adipocyte numbers result in increased RBP4 synthesis and secretion, and consequently increased blood RBP4 levels that signal diminished insulin responsiveness. Numerous cross-sectional studies, collectively involving thousands of subjects, have demonstrated that elevated circulating RBP4 levels are strongly associated with body mass index, waist circumference, glucose intolerance, insulin resistance, and hyperlipidemia.⁹⁷ Significant associations between circulating RBP4 levels and insulin sensitivity, percent trunk fat, and plasma triglyceride and LDL levels also were identified in 20–50 year-old subjects undergoing hyperinsulinemic-euglycemic clamp studies.⁹⁷ In addition, a number of different studies have identified single nucleotide polymorphisms present in the *RBP4* gene that are directly associated with obesity and insulin resistance. Studies carried out in animal models generally support findings obtained in humans. However, not all human or animal model studies have established a relationship between circulating RBP4 levels

and insulin resistance and obesity-related metabolic disease.⁹⁷ It has been proposed that this lack of full agreement involves methodological differences between studies.¹⁰⁰ Nevertheless, a consensus regarding the actions of RBP4 in human metabolic disease has not been reached.⁹⁷

Hepatic disease and hepatic stellate cell activation

The literature establishes that impairments to vitamin A signaling and metabolism can contribute to the development of NAFLD and steatohepatitis (NASH) as well as later stage liver disease including fibrosis, cirrhosis, and hepatocellular carcinoma (HCC). NAFLD is considered to be the first stage of liver disease development and is characterized by excessive accumulation of fat within hepatocytes. It is a major cause of later stage liver disease. NAFLD is present in approximately 90% of adults who are morbidly obese, where the disease can range from mild hepatic steatosis to the more severe NASH. NAFLD and NASH are reversible, but, if it is not resolved, disease can progressively worsen to fibrosis, cirrhosis, and HCC.¹⁰¹ This later stage liver disease involves activation of HSCs. Thus, both hepatocytes and HSC play central roles in both hepatic disease development and progression⁶¹ as well as hepatic vitamin A metabolism and storage.^{5,6}

Vitamin A and vitamin A-related proteins, especially $RAR\beta$, contribute to the development of liver disease. Hepatic retinol and retinyl ester levels as well as $RAR\beta$ mRNA levels in NAFLD patients have been reported to show a strong inverse correlation with the severity of the disease.¹⁰² Immunohistochemical study of liver tissue obtained from 22 patients with cirrhosis and 10 matched healthy individuals established that $RAR\beta$ protein levels are markedly reduced in the cirrhotic livers.¹⁰³ *In silico* analysis of gene expression data from 372 HCC patients and 50 controls similarly showed a reduction in $RAR\beta$ mRNA levels for the HCC patients.¹⁰³ Studies employing primary HSCs obtained from healthy human liver tissue established that treatment of these cells with all-*trans*-retinoic acid increases $RAR\beta$ mRNA levels and also reverses other changes in HSC gene expression that are associated with HSC activation.¹⁰³ Collectively, these findings from humans and human tissues implicate retinoic acid signaling in hepatic disease development. This conclusion is strongly supported by animal model studies where the disruption of RAR signaling in hepatocytes present within the mouse liver results in the spontaneous development of hepatic fibrosis, cirrhosis, and HCC.¹⁰⁴ Thus, normal vitamin A metabolism, storage, and actions are dysregulated in the development of hepatic disease and this likely contributes to the progression of disease.

B. Known Nutrient–Nutrient Interactions

Direct correlations between hemoglobin and blood retinol concentrations have been observed in both human and animal model studies.^{105,106} A cross-sectional study of children in Thailand found that serum retinol concentrations were positively correlated with serum iron and ferritin concentrations.¹⁰⁶ Interventions involving women in Indonesia demonstrated that combining vitamin A and iron supplementation was more effective in increasing hemoglobin concentrations

than giving iron alone.¹⁰⁷ It has been proposed that vitamin A deficiency impairs iron mobilization from stores and therefore vitamin A supplementation improves hemoglobin concentrations.^{108,109}

Although studies in animal models have suggested that blood retinol concentrations decline and rise with experimental zinc deficiency and repletion respectively, cross-sectional studies and supplementation trials in humans have failed to establish a consistent relationship between zinc and vitamin A status.¹¹⁰

RESEARCH GAPS

- What controls the whole-body economy of vitamin A?
- What role do apocarotenoids play in mediating vitamin A actions?
- How does vitamin A move within the liver between the hepatocyte and HSCs where it is stored?
- Why are HSCs the key cellular site for vitamin A storage in the liver?

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VITAMIN D

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SUMMARY

Vitamin D is a conditionally required nutrient that is biologically inactive but can be converted to a steroid hormone, 1,25(OH)₂D₃. This hormone regulates gene expression by interacting with the vitamin D receptor. The classical biological action of vitamin D is part of an endocrine hormone system that regulates calcium homeostasis. This system coordinates the renal metabolism of the vitamin D₃ prohormone 25(OH)D₃ into 1,25(OH)₂D₃, which is then released to act as an endocrine hormone on classical target tissues such as the intestine, kidney, and bone. However, it is now known that 1,25(OH)₂D₃ can regulate gene expression in other tissues where it influences a broad array of non-calcium metabolism-related processes, for example, development of cancer, regulation of the immune system, and brain and neural cells, as well as modifying risk of development of metabolic diseases. The expanded roles for vitamin D have led to considerable controversy regarding the level of vitamin D status that is optimal for human health.

Keywords: Bone; Calcium; Cancer; Diabetes; Inflammation; Muscle; Osteoporosis; Parathyroid hormone; Rickets.

I. INTRODUCTION

Vitamin D is a conditionally required nutrient that is essential for life in higher animals. While it is synthesized in the skin in response to ultraviolet (UV) light, in the absence of this stimulus, it must be obtained from the diet. The most important biological role for vitamin D is as a regulator of calcium homeostasis. These effects are achieved only after vitamin D is metabolized into the steroid hormone 1,25 dihydroxyvitamin D₃ (1,25(OH)₂D₃). This metabolite is produced in the kidney and acts on target tissues by interacting with the vitamin D receptor (VDR) to regulate gene transcription.

Although 1,25(OH)₂D₃ action is most closely linked to calcium metabolism, there is also evidence that it regulates cells and tissues not primarily related to mineral metabolism. This is because many cell types express the VDR and, therefore, are targets of 1,25(OH)₂D₃ action. As a result, vitamin D action may influence the

incidence or progression of nonbone diseases or conditions including cancer, inflammation, diabetes, and multiple sclerosis (MS).

The purpose of this chapter is to provide a succinct overview of current understanding of vitamin D. In addition to a discussion of the metabolism of vitamin D and the mechanisms by which its biologically active metabolite, 1,25(OH)₂D₃, regulates biology, the role that vitamin D action may play in several important human diseases is explored. Finally, nutritional aspects of vitamin D and other practical aspects of how current scientific understanding of vitamin D action is used to optimize human health are presented.

A. Background

The hallmark of vitamin D deficiency is rickets, a condition that was first described by Daniel Whistler in 1645 (see Ref. 1 for a detailed historical overview). However,

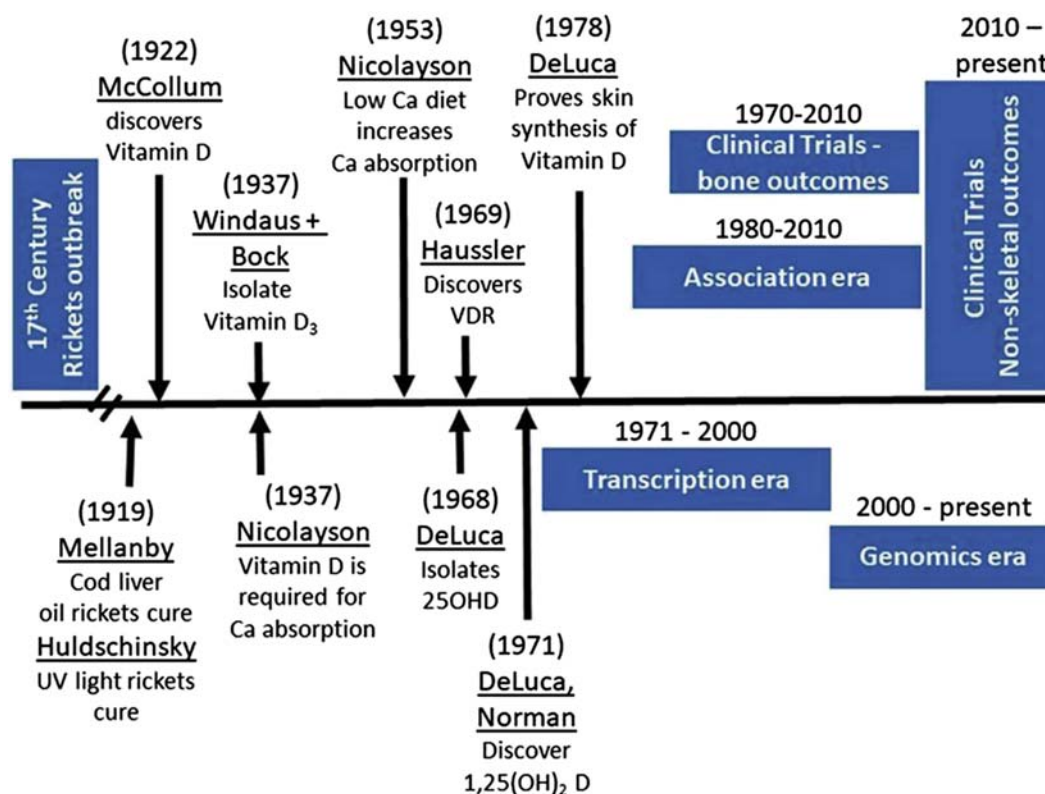


FIGURE 6.1 A timeline of major discoveries in vitamin D research. In the earliest days of vitamin D research (up to ~1940), the goal was to identify the factor to prevent rickets and link it to the physiology of calcium metabolism. The goal through the mid 1970's was to identify vitamin D metabolites and determine how these metabolites affected biology. Since then, mechanistic research has examined the primary action of the vitamin D hormone, 1,25-dihydroxyvitamin D, as a regulator of gene transcription through the vitamin D receptor (VDR). More recently, this mechanistic work has expanded to examine genome-wide actions of vitamin D in a wide variety of target tissues. Concurrent to the mechanistic work on vitamin D biology, clinical scientists and translational researchers have tried to associate vitamin D status with health outcomes. Initially, this was done by associating serum 25-hydroxyvitamin D levels to traditional outcomes (i.e., bone health and calcium metabolism) as well as nonskeletal outcomes (e.g., risk of chronic diseases such as cancer, diabetes, multiple sclerosis). The causal nature of these associations was tested in various subject populations. The first clinical studies were on bone outcomes while preclinical studies were establishing “proof of principle” for causal effects of vitamin D for non–bone-related outcomes. Since 2010, a number of large clinical intervention trials have been initiated to test the impact of vitamin D on many of these nonbone endpoints.

scientists did not identify vitamin D as a protective factor against rickets for almost three centuries. Fig. 6.1 provides a timeline of highlights in the history of vitamin D research. In 1919, Kurt Huldshinsky demonstrated that ultraviolet B rays (UVB light) were sufficient to cure rickets while Sir Edward Mellanby established that rickets could be caused by a deficiency of a trace component in the diet and that this component was in cod liver oil. E.V. McCollum separated the biological actions of fat-associated vitamins into vitamin D and vitamin A using animal studies and later coined the term “vitamin D.”

Despite the general understanding that a compound, “vitamin D”, was important for bone and calcium metabolism, the chemical structure of vitamin D was not determined until the 1930s by Adolf Windaus. His lab chemically characterized vitamin D₂ produced from UV irradiation of the plant/yeast steroid ergosterol and vitamin D₃ produced from UV irradiation of 7-dehydrocholesterol. Virtually simultaneously, the antirachitic component of cod liver oil was identified as

the newly characterized vitamin D₃. These results clearly established that the antirachitic substance vitamin D was chemically a steroid, more specifically a secosteroid (see below).

The molecular era of vitamin D research began in the late 1960's with the discovery and chemical characterization of 1,25(OH)₂D₃ by Hector DeLuca and Anthony Norman in 1971 and its nuclear receptor, the VDR, by Mark Haussler in 1973. Research on the molecular actions of vitamin D continued throughout the last century as scientists discovered how 1,25(OH)₂D₃ acts through the VDR to regulate gene transcription in multiple cell types. The reporting of the human (and other) genomes has extended this era further so that the regulation of genes by 1,25(OH)₂D₃ can be viewed as a system affecting genes across the genome.² In addition, as discoveries on vitamin D action in various tissues were occurring, others were conducting studies that associated vitamin D status with the risks of various diseases. The cause and effect relationships underlying these associations

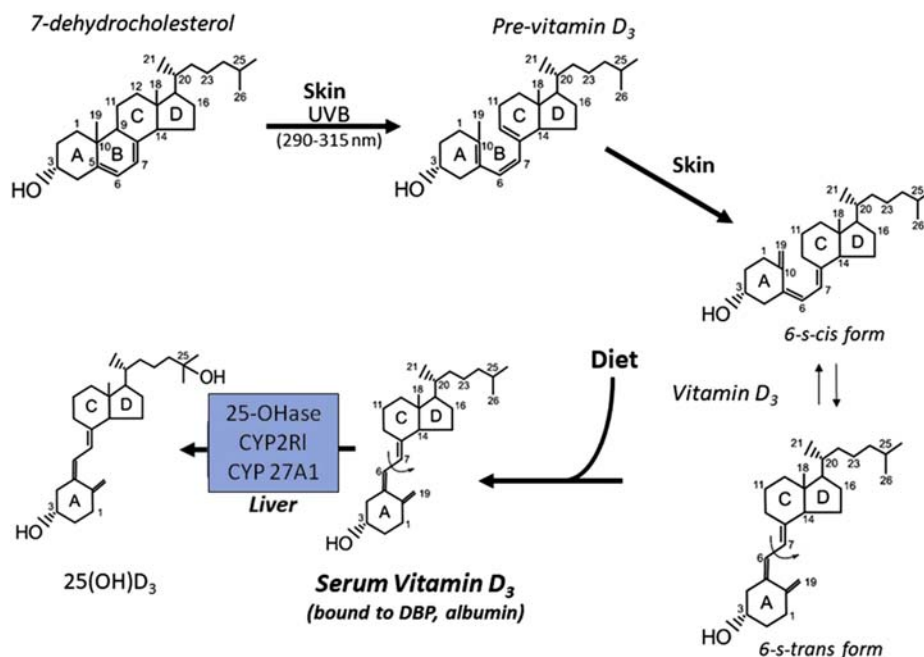


FIGURE 6.2 The basic chemistry, skin synthesis, and intestinal absorption related to vitamin D status. The precursor of vitamin D is the compound 7-dehydrocholesterol. Like other cholesterol metabolites, this comprises of a four-ring structure (labeled A, B, C, and D). When 7-dehydrocholesterol is exposed to ultraviolet B light (UVB, 190-315 nm), the B ring is broken at the 9,10 bond to become the seco B previtamin D₃, which then thermally isomerizes to become vitamin D₃ (cholecalciferol). The vitamin D₃ obtained in the diet and produced in the skin is bound to serum proteins (e.g., the vitamin D-binding protein (DBP), albumin) and delivered to the liver where they are rapidly converted to the prehormone metabolite, 25-hydroxyvitamin D₃ (25(OH)D₃), by the addition of a hydroxyl group to the 25 carbon on the vitamin D₃ side chain. This is the most stable metabolite of vitamin D₃ and is used as a vitamin D status indicator in humans.

have been explored in various preclinical studies, clinical studies, and large intervention trials. These explorations will be discussed later in the chapter.

B. Nutrient Function and Structure

The structures of the mammalian form of vitamin D, vitamin D₃ (cholecalciferol), and its precursor, 7-dehydrocholesterol, are presented in Fig. 6.2. There is also a plant-derived form of vitamin D, vitamin D₂ (ergosterol, structure not shown). Vitamin D is a generic term and indicates a molecule with the general ring structure from cholesterol (with four rings labeled A, B, C, and D) that has differing side chain structures. The ring structure is derived from the cyclopentanoperhydrophenanthrene ring structure common to cholesterol and other steroids, but with the C9,10 carbon-carbon bond of ring B broken.

The broken B ring gives the structure of vitamin D and all of its metabolites more flexibility than other steroids. It can rapidly flux positions at the C6 carbon to have either an open “6-s-trans” conformation or a more closed “6-2-cis” conformation that more closely resembles the parent cholesterol form.

Vitamin D was named according to the rules of the International Union of Pure and Applied Chemists. Because it is derived from a steroid, vitamin D retains its numbering from the parent compound cholesterol

(Fig. 6.2). Thus, the official name of vitamin D₃ is 9,10-seco(5Z,7E)-5,7,10(19)-cholestatriene-3β-ol. In contrast, because vitamin D₂ has a C22/23 double bond and a C24-methyl group in its side chain, the official name of vitamin D₂ is 9,10-seco(5Z,7E)-5,7,10(19),22-ergostatriene-3β-ol.

II. NUTRIENT METABOLISM

Vitamin D₃ itself has no intrinsic biological activity. To regulate biology, it must be metabolized to form the prohormone, 25(OH)D₃, and then the active metabolite, 1,25(OH)₂D₃, which is released into the circulation where it acts on other tissues as an endocrine hormone. These multiple steps are described below (more detailed information is available³).

A. Skin Synthesis of Vitamin D₃

In most higher animals, vitamin D₃ can be produced photochemically from 7-dehydrocholesterol when the epidermis is exposed to UV light with a wavelength of 290–315 nm (UVB).⁴ The conjugated double bond system in the B ring allows the absorption of UVB light to initiate transformation of the precursor into vitamin D₃ (Fig. 6.2). Thus, when a person is regularly exposed to

UVB light, vitamin D₃ is endogenously produced and there may be no need for a dietary vitamin D source. The vitamin D₃ produced in the skin is rapidly released into the circulation where it interacts with the vitamin D-binding protein (DBP) to be transported to the liver. However, low UVB exposure is common in countries at high latitudes, in people with deeply pigmented skin, in those who cover their skin with strong sunblock or for ethnic and religious reasons, and in the homebound. These groups are particularly at risk of low vitamin D status in wintertime, a period when UVB irradiation is markedly decreased.

B. Intestinal Absorption of Vitamin D

Vitamin D is present in the diet as cholecalciferol (D₃) from animal sources and ergocalciferol (D₂) from plant sources. Fat-soluble vitamins including vitamin D have traditionally been thought to “follow the fat” during intestinal absorption.⁵ This requires association of the vitamin with bile salts for incorporation into micelles, movement of the micelles into the body, and repackaging of the vitamin into chylomicrons for transport by the lymphatic system. This model is consistent with the observations that gastrointestinal and hepatobiliary diseases leading to fat malabsorption commonly cause vitamin D deficiency in humans. Studies in rats with experimental nephrotic syndrome show that despite a significant loss of vitamin D metabolites in urine and reduced serum 25(OH)D₃ levels, vitamin D absorption does not increase.⁶ This suggests that there are no homeostatic regulatory mechanisms to increase intestinal vitamin D absorption in times of need. Recent studies also show that intestinal absorption of vitamin D is not a simple passive diffusion process but instead requires the cholesterol transporters SR-BI and NPC1L1.⁷

C. Transport and Metabolism

Vitamin D within chylomicrons or bound to DBP is rapidly transported to the liver or into body fat. As a result, the serum half-life for vitamin D is only several hours long.⁸ The vitamin D in body fat is not selectively removed in times of physiologic need and is only released when fat mass is lost. As such, body fat is not a true storage depot. The metabolism of vitamin D begins in the liver (Fig. 6.2). The biochemical processes that convert vitamin D₃ to its prehormone, 25(OH)D₃, and then the endocrine hormone, 1,25(OH)₂D₃, are described below.³

Three enzymes are responsible for the conversion of vitamin D₃ into its two key metabolites.⁹ All three enzymes are members of the cytochrome P-450 mixed-function oxidase family. The first action is the vitamin D₃-25-hydroxylase (25-hydroxylase) enzyme in the liver that places a hydroxyl group on the C25 carbon on the

vitamin D₃ side chain (Fig. 6.2). Multiple genes, including CYP2R1 and CYP27A1, encode this enzyme. The conversion of vitamin D₃ to 25(OH)D₃ is rapid, driven primarily by the level of vitamin D₃ present, and is poorly inhibited by the product, 25(OH)D₃. After it is synthesized, 25(OH)D₃ is rapidly released from hepatocytes. 25(OH)D₃ has high affinity for DBP and is efficiently reabsorbed from the renal filtrate through a process dependent upon cubilin and megalin, two glycoprotein endocyte receptors. As a result, the serum half-life of 25(OH)D₃ is approximately 2 weeks and thus is used as a marker of vitamin D status (see discussion of markers of dietary adequacy below).

The amount of 1,25(OH)₂D₃ released from the kidney requires the coordinated effort of two kidney enzymes that are localized in mitochondria of the proximal tubules of the kidney. The synthesis of 1,25(OH)₂D₃ is mediated by the 25(OH)D₃-1 α -hydroxylase (1 α -hydroxylase) that is encoded by the CYP27B1 gene (Fig. 6.3). This enzyme places a hydroxyl group on the C1 carbon of the A ring in a forward or alpha configuration. In contrast, the hydroxyl group at the C3 carbon of vitamin D is in the back or beta position. This 1 alpha, 3 beta configuration of the hydroxyl groups on the A ring makes 1,25(OH)₂D₃ a high-affinity ligand for the VDR. Specific mutations in the CYP27B1 gene cause the rare human genetic disease vitamin D-dependent rickets type 1A.

The second enzyme affecting the amount of 1,25(OH)₂D₃ released from the kidney is the 25(OH)D₃-24R-hydroxylase (24-hydroxylase) encoded by the CYP24A1 gene (Fig. 6.4). This enzyme places a hydroxyl group on the C24 carbon in the vitamin D side chain. By doing so, this enzyme creates 1,24,25(OH)₃D₃, a short-lived metabolite that is a substrate for downstream enzymes that degrade the vitamin D hormone. CYP24A1 is also expressed in vitamin D target tissues where it inactivates the vitamin D hormone. The CYP24A1 gene is strongly transcriptionally upregulated by 1,25(OH)₂D₃. As such, it is a critical part of the negative feedback loop that limits both excessive 1,25(OH)₂D₃ production by the kidney and excessive vitamin D signaling at target tissues.

The renal 24-hydroxylase is also responsible for creating the second most abundant vitamin D metabolite in serum, 24R,25(OH)₂D₃ (i.e., where the -OH addition forms a chiral center with a “right-handed” stereocenter). There has been some debate as to whether there is a physiologic role for 24R,25(OH)₂D₃ or whether it is simply a degradation product of 25(OH)D₃.¹⁰ However, there are data to indicate that 24R,25(OH)₂D₃, either alone or in conjunction with 1,25(OH)₂D₃, is involved in growth plate development and fracture healing due to effects on resting zone chondrocytes.¹¹

A crucial feature of the vitamin D endocrine system is the stringent control of 1,25(OH)₂D₃ production that

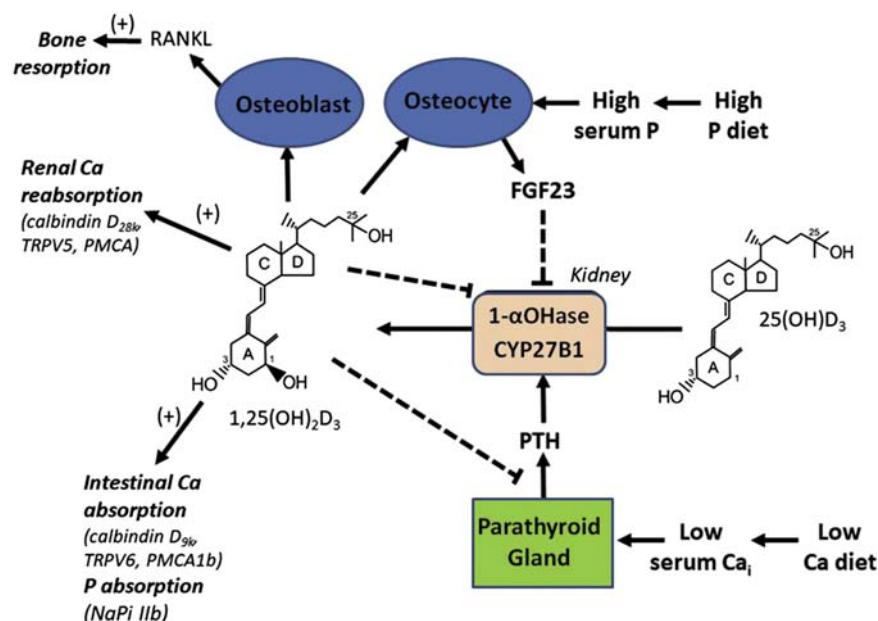


FIGURE 6.3 Formation and regulation of the endocrine hormone, 1,25-dihydroxyvitamin D₃. The vitamin D prehormone, 25-hydroxyvitamin D₃ (25(OH)D₃), is not biologically active but must be converted to the vitamin D hormone 1,25-dihydroxyvitamin D₃ (1,25(OH)₂D₃). This is done through the action of the enzyme, 25-hydroxyvitamin D₃ 1- α hydroxylase (1 α OHase) encoded by the CYP27B1 gene. 1 α OHase places a hydroxyl group on the 1 carbon of the A ring of 25(OH)D₃. (Because 1,25(OH)₂D₃ now has three hydroxyl groups, it is also called calcitriol.) Under most conditions, the bulk of 1,25(OH)₂D₃ is produced in the kidney. Renal 1 α OHase is regulated by a number of physiologic hormones whose levels are regulated by changes in dietary calcium (Ca) or phosphorus (P). When dietary Ca is low, changes in serum Ca lead to increased production of parathyroid hormone (PTH). PTH activates renal CYP27B1 gene expression leading to increased serum 1,25(OH)₂D₃ levels. Elevated serum 1,25(OH)₂D₃ levels upregulate the expression of genes that increase intestinal Ca absorption (i.e., TRPV6, calbindin D_{9k}, PMCA1b), renal Ca reabsorption (TRPV5, calbindin D_{28k}, PMCA), and bone resorption (i.e., osteoblast RANKL expression that promotes osteoclast differentiation). Elevated serum 1,25(OH)₂D₃ also causes a feedback inhibition of renal 1,25(OH)₂D₃ production by suppressing renal CYP27B1 expression and activating production of the 1,25(OH)₂D₃-inactivating enzyme 25-hydroxyvitamin D-24-hydroxylase, encoded by the CYP24A1 gene. When dietary P is high, this stimulates the production of fibroblast growth factor 23 (FGF23) by osteocytes in bone. FGF23 suppresses CYP27B1 gene expression leading to lower serum 1,25(OH)₂D₃ levels. Conversely, low dietary P will release the constraints on renal 1,25(OH)₂D₃ production imposed by FGF23. Elevated serum 1,25(OH)₂D₃ is also a stimulus for increased production of FGF23 by the osteocyte.

occurs through regulation of the renal 1 α -hydroxylase (Fig. 6.3). This regulation is best understood in the context of calcium deficiency.⁵ Whole body calcium metabolism is controlled by a multitissue axis of intestine, kidney, bone, and the parathyroid gland. When dietary calcium is habitually low, this triggers a physiologic adaptation that starts at the parathyroid gland. Low serum calcium is detected at the parathyroid gland by the calcium-sensing receptor. This initiates signaling that induces the synthesis and release of parathyroid hormone (PTH). This condition is called nutritional, secondary hyperparathyroidism. PTH travels to the kidney where it binds to the PTH receptor and stimulates protein kinase A-mediated phosphorylation of the CREB (cAMP response element-binding protein) transcription factor leading to the expression of the CYP27B1 gene. The elevated serum levels of 1,25(OH)₂D₃ stimulate gene expression at classical target tissues. Thus, 1,25(OH)₂D₃ increases intestinal calcium absorption due to increased expression of the apical membrane calcium channel TRPV6, the intracellular calcium-binding protein calbindin D_{9k}, and the basolateral membrane calcium ATPase

PMCA1b. In the kidney, 1,25(OH)₂D₃ increases renal calcium reabsorption (by increasing expression of TRPV5, calbindin D_{28k}, and PMCA [plasma membrane Ca²⁺ ATPase]).¹² Finally, in osteoblasts within bone, 1,25(OH)₂D₃ induces expression of RANKL (nuclear factor kappa-B ligand), a protein that stimulates osteoclast differentiation leading to more bone resorption.¹³ Collectively, this normalizes serum calcium and shuts down the production of PTH (and indirectly 1,25(OH)₂D₃).

CYP27B1 gene expression is also strongly downregulated by a negative feedback loop. Thus, when 1,25(OH)₂D₃ gets too high, the hormone interacts with the VDR to suppress CYP27B1 gene expression and to upregulate the renal expression of CYP24A1.¹⁴ This prevents vitamin D-induced hypercalcemia.

1,25(OH)₂D₃ production is also regulated in response to dietary phosphorus intake (Fig. 6.3). In this case, high dietary phosphorus stimulates osteocytes in bone to produce fibroblast growth factor 23 (FGF23).¹⁵ FGF23 is a negative regulator of CYP27B1 gene expression and therefore reduces 1 α -hydroxylase activity and 1,25(OH)₂D₃ production. Conversely, low dietary phosphorus releases

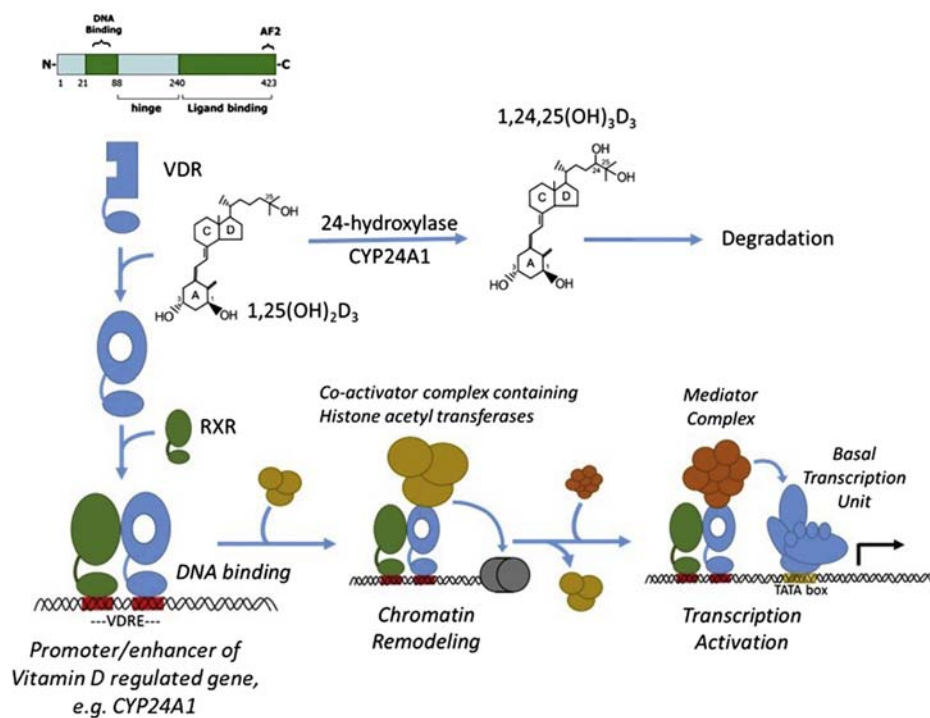


FIGURE 6.4 Model for how $1,25(\text{OH})_2\text{D}_3$ regulates biological responses by activating gene expression. The activation of gene expression by $1,25(\text{OH})_2\text{D}_3$ requires binding to the vitamin D receptor (VDR, see cartoons of the linear organization and folded VDR in the figure). The VDR is a steroid hormone receptor superfamily member that binds $1,25(\text{OH})_2\text{D}_3$ with high affinity (shown as an open oval in the VDR ligand-binding domain in the figure). Upon binding, VDR dimerizes with the retinoid X receptor (RXR). Together, the VDR-RXR heterodimer acts as a transcription factor that binds to specific DNA sequences called vitamin D response elements (VDREs). VDREs can be found close to a gene's transcriptional start site (i.e., proximal promoter elements) or far upstream or downstream from the start site as well as in introns (i.e., enhancer elements). Once the VDR-RXR heterodimer has bound to a VDRE, it will recruit other proteins that remodel chromatin to make it more accessible (i.e., coactivator proteins) or more closed (i.e., corepressor proteins). Most vitamin D-regulated genes are activated, so coactivator complexes are recruited. After chromatin has been opened, transcriptional activation continues when the coactivator complex is replaced with the mediator complex. This complex allows the VDR-RXR heterodimer to recruit the basal transcription unit (BTU) to the transcription start site of genes. The BTU includes RNA polymerase II and other proteins that activate RNA polymerase II; this causes the active production of mRNA from a gene. Chromatin Immunoprecipitation (ChIP) assays with an anti-VDR antibody coupled to next generation sequencing shows that there are many vitamin D-regulated genes in multiple vitamin D responsive tissues (e.g., intestine¹³⁷, osteoblasts and osteocytes,¹³⁸ prostate epithelial cells).¹¹⁵ As an example, the figure shows the most studied vitamin D responsive gene, CYP24A1, encoding the 24-hydroxylase. This enzyme adds a hydroxyl group to the 24 carbon on the $1,25(\text{OH})_2\text{D}_3$ side chain, a step that initiates the degradation of the vitamin D hormone.

the inhibition of CYP27B1 gene expression by FGF23 and leads to increased serum $1,25(\text{OH})_2\text{D}_3$ levels. These increases in $1,25(\text{OH})_2\text{D}_3$ are associated with increased intestinal phosphate absorption.

In addition to its central role in the vitamin D endocrine system, CYP27B1 is also expressed at low levels in a wide variety of tissues.¹⁶ This extrarenal expression is thought to supply $1,25(\text{OH})_2\text{D}_3$ locally for use as an autocrine- or paracrine-signaling factor. This may be why high serum $25(\text{OH})\text{D}_3$ levels are associated with reduced risk of various health outcomes (see discussion on nonskeletal actions of vitamin D below). Extrarenal CYP27B1 is not regulated by the hormones that control renal CYP27B1 expression (i.e., PTH, FGF23). However, a number of immune regulators (e.g., lipopolysaccharide, cytokines, interferons) stimulate $1,25(\text{OH})_2\text{D}_3$

production in activated macrophages, and this accounts for the elevated serum $1,25(\text{OH})_2\text{D}_3$ levels that have been seen during bacterial infections.¹⁶

Mechanisms of action of $1,25(\text{OH})_2\text{D}_3$

Regulation of gene transcription; $1,25(\text{OH})_2\text{D}_3$ is a steroid hormone that affects biology by regulating gene transcription (Fig. 6.4). This genomic response is due to a stereospecific interaction of $1,25(\text{OH})_2\text{D}_3$ with the VDR, a steroid hormone receptor that is a ligand-activated transcription factor. A detailed discussion of the VDR and its participation in the regulation of gene transcription is available elsewhere.¹⁷ VDR is a 50 kDa protein that binds $1,25(\text{OH})_2\text{D}_3$ with high affinity ($K_d \approx 0.5 \text{ nM}$). In contrast, $25(\text{OH})\text{D}_3$ binds with much

lower affinity (0.1%–0.3% compared to the binding of $1,25(\text{OH})_2\text{D}_3$), while the parent compound, vitamin D_3 , does not bind to the VDR at all. The central role for VDR in the biological actions of vitamin D was firmly established by several independent groups who created genetically modified mice with deletion of the VDR gene (VDR knockout or VDR-KO). These mice have severe disruption of calcium and phosphorus metabolism that results in stunted growth, severe rickets, and alopecia, i.e., they are a model for human vitamin D-resistant rickets, type II.¹⁸

The VDR is a member of the steroid hormone receptor superfamily that includes the classical members such as receptors for estrogen and testosterone, as well as nonclassical members such as the retinoic acid receptors and peroxisome proliferator-activated receptors. As with all nuclear steroid hormone receptors, the VDR has a number of functional domains (Fig. 6.4, top left): peptide sequences that drive nuclear localization, a zinc finger DNA-binding domain, a domain that controls both heterodimerization and ligand binding, and a transcriptional activation domain (AF2).¹⁹ The ligand-binding domain of the VDR consists of 12 α -helices arranged to create a pocket. In the absence of $1,25(\text{OH})_2\text{D}_3$, the VDR is transcriptionally inactive. However, when VDR binds its ligand, the ligand-binding domain undergoes a structural shift to wrap $1,25(\text{OH})_2\text{D}_3$ in a hydrophobic core.

The VDR can be found in both the cytoplasm and nucleus of vitamin D target cells. The structural changes resulting from binding of $1,25(\text{OH})_2\text{D}_3$ to the VDR exposes peptide sequences that allow VDR to form a heterodimer with the retinoid X receptor (RXR). This interaction is essential for VDR transcriptional activity, and some studies show that heterodimerization is required for migration of the RXR–VDR–ligand complex from the cytoplasm to the nucleus. Nonetheless, it is the $1,25(\text{OH})_2\text{D}_3$ –VDR–RXR complex that binds to specific sequences in DNA called vitamin D response elements.

Binding of the VDR–RXR dimer to DNA is through a zinc-finger structure that is a part of the DNA-binding domain of both VDR and RXR. Chromatin immunoprecipitation coupled to DNA sequencing (ChIP-seq) shows that the majority of DNA sites bound to VDR after $1,25(\text{OH})_2\text{D}_3$ treatment are far from the transcriptional start site of the genes they regulate, i.e., they are acting as enhancer elements. Regardless, once bound to DNA, the VDR–RXR heterodimer recruits protein complexes that alter chromatin structure.

For transcriptional activation, the proteins in the complex include those with histone acetyl transferase activity (e.g., CBP/p300, SRC1) as well as ATP-dependent remodeling activity (e.g., SWI/SNF), both of which are needed to release higher-order chromatin structure

that limits gene transcription. After chromosomal unwinding, the VDR–RXR dimer recruits the mediator complex to the promoter and utilizes it to activate the basal transcription unit containing RNA polymerase II.

While most VDR–RXR binding to DNA stimulates gene transcription, the transcription of several genes is suppressed by vitamin D signaling through the VDR (e.g., PTH, CYP27B1). For transcriptional repression, the VDR–RXR dimer recruits corepressors such as NCoR1, NCoR2/SMRT, and Alien that then recruit histone deacetylases and DNA methyltransferases that alter histone tails in ways that lead to a more compact chromatin structure.¹⁹ Additional details regarding the molecular activation of VDR and its role in gene transcription across the genome were recently reviewed elsewhere.²⁰

Rapid, nongenomic signaling; Although there is a strong consensus that $1,25(\text{OH})_2\text{D}_3$ regulates biology primarily through VDR-mediated regulation of gene transcription, there is evidence for another vitamin D signaling system. In this model, $1,25(\text{OH})_2\text{D}_3$ stimulates rapid activation of signal transduction pathways in a way that is typically associated with peptide hormone action, i.e., binding to a cell surface protein that transduces signals into the cell through protein phosphorylation cascades (a detailed discussion is available in Ref. 21). In contrast to genomic actions of $1,25(\text{OH})_2\text{D}_3$, which take hours to be observed, these rapid, nongenomic responses are seen within seconds or minutes. The rapid actions include opening of voltage-gated calcium channels that contribute to intracellular signaling, as well as a special type of intestinal calcium absorption called transcalcaltachia, and activation of phospholipase C, protein kinase C, tyrosine kinases, Src kinases, phosphatidylinositol 3'-kinase, and MAP kinases such as JNK and ERK. Rapid responses stimulated by $1,25(\text{OH})_2\text{D}_3$ are mediated either by a novel, nonnuclear role for the VDR or by a different protein called the membrane-associated, rapid response steroid-binding protein (MARRS).

While the genomic actions are mediated by $1,25(\text{OH})_2\text{D}_3$ that is in the open 6-trans position (i.e., where the broken B ring is in a more linear conformation), the rapid actions use $1,25(\text{OH})_2\text{D}_3$ in the closed 6-cis conformation (i.e., making vitamin D look more like a cholesterol structure). While there are many cell-based studies showing that $1,25(\text{OH})_2\text{D}_3$ induces rapid cellular responses, this area should be viewed with caution. There are few studies that have tested and shown that these rapid, nongenomic responses have an important role in physiology or in disease.

One area where nongenomic effects of vitamin D metabolites have been studied extensively is in the growth

plate chondrocyte. $1,25(\text{OH})_2\text{D}_3$ regulates the function of growth zone chondrocytes through both the VDR and MARRS receptors while $24\text{R},25(\text{OH})_2\text{D}_3$ works through a separate, membrane-associated receptor to regulate resting zone cells.²²

III. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Vitamin D status is linked to dietary or supplemental vitamin D_3 intake through a curvilinear relationship with the serum level of $25(\text{OH})\text{D}_3$ (Fig. 6.5). However, a fundamental challenge for defining vitamin D requirements is that it can be obtained from both diet and UVB exposure. In fact, substantial proportions of the US and Canadian populations are exposed to sub-optimal levels of sunlight, particularly during winter months.²³ Deprivation of sunlight through seasonal variation (winter),²⁴ or clothing habits due to religious practices,²⁵ can lead to the onset of clinical rickets or

osteomalacia, characterized by low serum $25(\text{OH})\text{D}_3$. Under condition where UVB exposure is limited, vitamin D becomes a true vitamin that must be supplied in the diet on a regular basis.

B. Criteria of Adequacy, Assessment, and Reference Intakes of Vitamin D

In 2011, the Dietary Reference Intakes (DRIs) for vitamin D for Canada and the United States were updated by a committee of the Food and Nutrition Board of the Institute of Medicine (IOM).^{25,26} To set the Estimated Average Requirement (EAR), the criteria of adequacy used were based on data where causality was established and where there was evidence of a dose response that could be used to support the reference value; in 2011, only bone endpoints met these criteria (See Fig. 6.6). Based on this assessment, a serum $25(\text{OH})\text{D}_3$ level of 20 ng/mL (50 nmol/L) was determined to be adequate to ensure good health (see Table 6.1 for serum $25(\text{OH})\text{D}_3$ levels defined as meeting human health needs¹⁵).

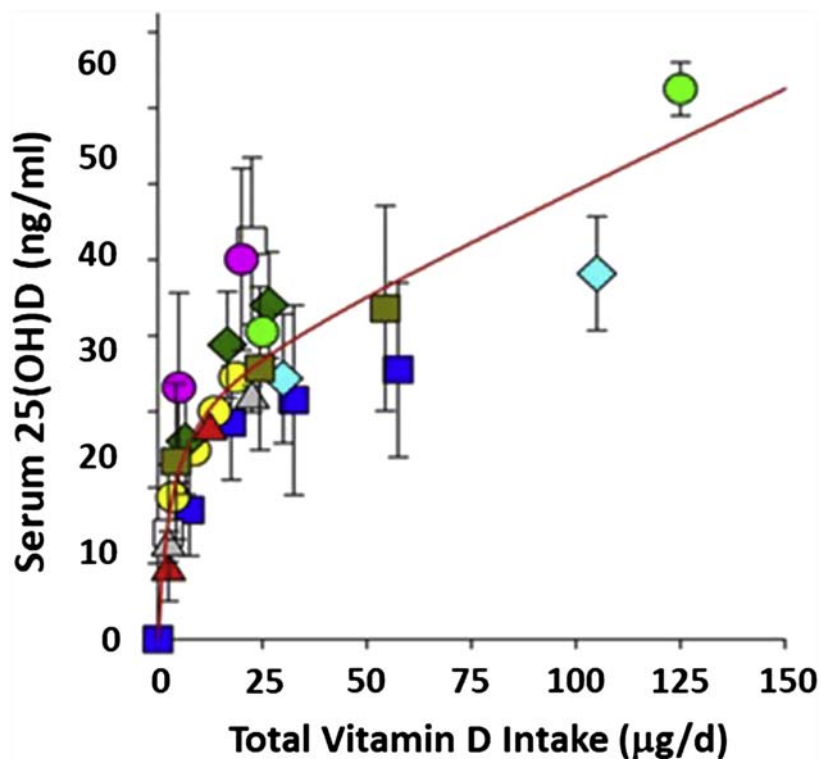


FIGURE 6.5 There is a curvilinear relationship between vitamin D intake and serum $25(\text{OH})\text{D}$ levels. The relationship between vitamin D intake levels and serum $25(\text{OH})\text{D}$ levels from nine supplementation trials reported in the IOM report on vitamin D requirements²⁶ plus an additional study using higher intakes.¹³⁹ These data show a curvilinear relationship between intake and vitamin D status. The curvilinear relationship means that people with low vitamin D status will be more responsive to a vitamin D supplement than those with high vitamin D status.

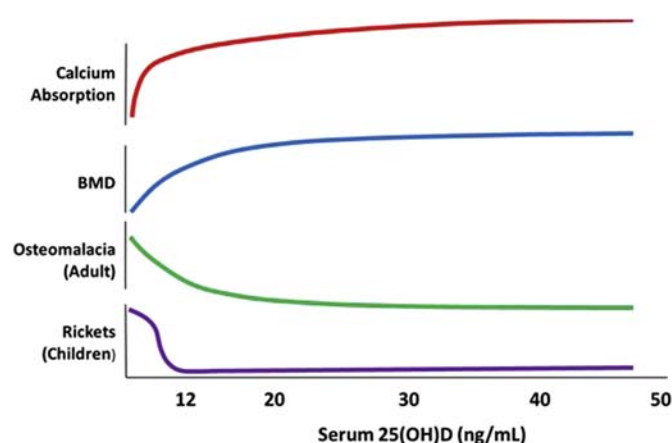


FIGURE 6.6 The relationship between serum 25(OH)D levels and indices of bone and calcium metabolism in humans. Nutritional reference values based on requirements for vitamin D were set by the US Institute of Medicine (IOM, now called the National Academy of Medicine) in 2011 for Canada and the United States based on clinical data from studies on bone endpoints. This figure was adapted from IOM report¹⁵; it shows the relationship between serum 25(OH)D levels and four classical bone/calcium metabolism outcomes (Fig. 6.6). These were the health outcomes used to set the 2011 vitamin D recommended intakes. All of these outcomes no longer changed when serum 25(OH)D levels had reached 15 ng/mL (~40 nmol/L).

Using dose–response relationships between dietary and supplement vitamin D₃ intake and serum 25(OH)D₃ levels, recommendations for the vitamin D₃ intake

level needed to reach serum levels of 20 ng/mL (50 nmol/L) were established (assuming no UVB exposure) (Table 6.2). Using this method, the Recommended Dietary Allowance (RDA) for vitamin D was set at 15 µg (600 IU)/day for ages 1–70 yr, regardless of pregnancy or lactation, and 20 µg (800 IU)/day for ages > 70 yr to take into account an expected increased variation in requirements in older individuals.¹⁵ For infants (0–12 months), an Adequate Intake based on amounts to reach the serum level cutoff was estimated at 10 µg (400 IU)/day. It was also assumed that infants who are breastfed receive vitamin D supplements of 10 µg (400 IU)/day.¹⁵ In addition to the recommended supplementation of the breastfed infant, the American Academy of Pediatrics recommends that infants consuming <32 oz of formula also receive vitamin D supplements.

Associations with nonskeletal outcomes

Although there was not enough evidence to make recommendations based on adequacy criteria for nonskeletal outcomes, there were strong associations between low serum 25(OH)D₃ levels (<20 ng/mL (50 nmol/L) and several compromised physiological systems (the immune system, the pancreas and metabolic homeostasis, heart and cardiovascular system, muscle, brain,

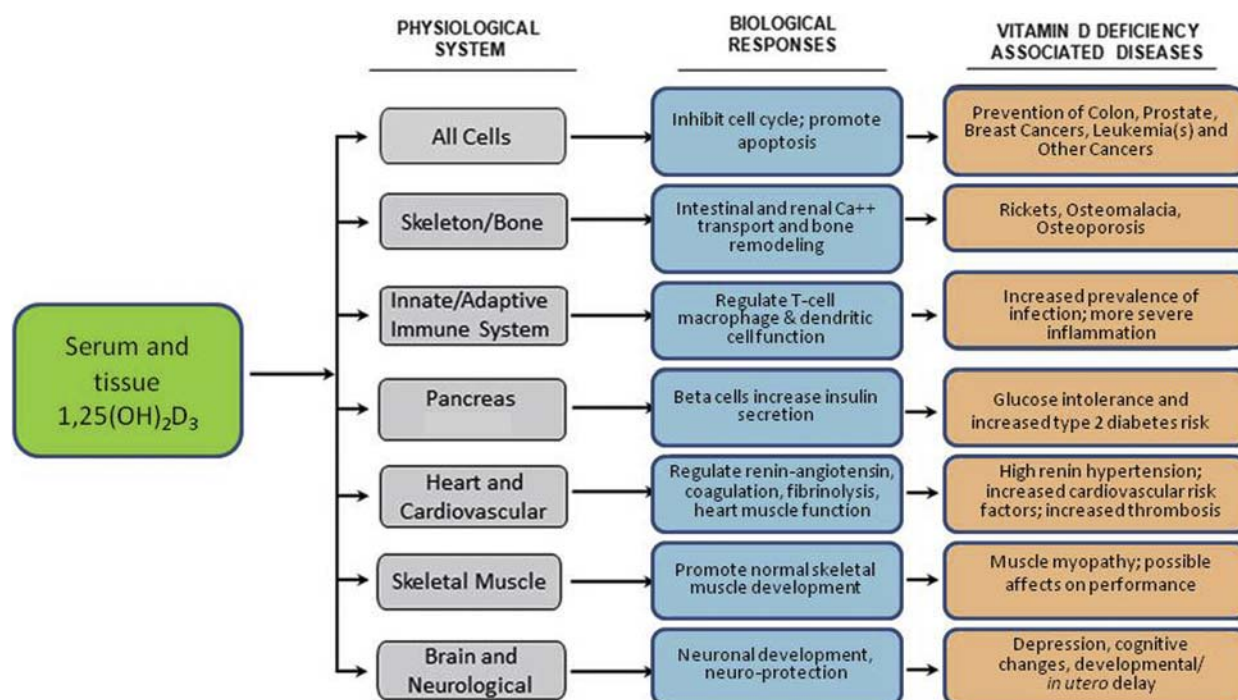


FIGURE 6.7 Vitamin D contributions to good health. Although the biology of vitamin D has been linked most closely to bone health and mineral metabolism, there is strong support for the existence of nonbone actions of vitamin D. This figure summarizes the breadth of vitamin D actions and their relation to various disease states. The three columns on the right-hand side summarize the physiologic systems that are dependent upon adequate vitamin D nutritional status, the biological responses that are induced by 1,25(OH)₂D₃-mediated activation of the VDR, and the disease states that have been evaluated for associations with low serum 25(OH)D₃ levels in humans. These systems are dependent upon the availability of vitamin D₃, its metabolism to 25(OH)D₃, and its activation to the steroid hormone, 1,25(OH)₂D₃.

TABLE 6.1 Serum levels of the vitamin D metabolite, 25-hydroxyvitamin D₃ (25(OH)D₃), and vitamin D status.

Serum 25(OH)D		Description
ng/mL	nmol/L	
<12	<30	Deficient
12 to <20	30 to <50	Low
20–50	50–75	Sufficient
>100	>250	Toxicity risk

Serum 25(OH)D levels used as cutoffs²⁶ to describe vitamin D status levels based on bone outcomes or indices of vitamin D intoxication.

TABLE 6.2 A summary of the current vitamin D recommended intakes for various age groups.

Age	Recommended Dietary Allowance (RDA) μg (IU)	Tolerable Upper Intake Level (UL) μg (IU)
0–6 months	10 (400) ^a	25 (1000)
7–12 months	10 (400) ^a	38 (1500)
1–3 years	15 (600)	63 (2500)
4–8 years	15 (600)	75 (3000)
9–18 years	15 (600)	100 (4000)
19–70 years	15 (600)	100 (4000)
Pregnancy or lactation 14+ years	15 (600)	100 (4000)
>70 years	20 (800)	100 (4000)

The levels for adequacy are based on intakes to maintain serum 25(OH)D₃ levels at or above 20 ng/mL (~50 nmol/L) in 97.5% of the population assuming no skin production of vitamin D₃.²⁶ The upper level of intake (UL) was based on preventing development of hypercalcemia or hypercalciuria.

^aAdequate Intake.

etc.)¹⁵ and this is discussed below. This suggests there is a positive association between serum 25(OH)D₃ levels and nonskeletal health outcomes (Fig. 6.6).

Because of these associations, some scientists believe that serum 25(OH)D₃ levels between 20 and 30 μg/mL indicate a state of marginal or insufficient vitamin D status.²⁷ However, most of the research needed to confirm that higher serum 25(OH)D₃ levels will benefit nonskeletal outcomes is still ongoing and the evidence from completed randomized control trials (RCTs) largely show negative findings^{28–32} (see Table 6.3 for a partial list of large intervention trials with this research goal).

Thus far, these trials have not indicated evidence of further clinical benefit of serum values above the range of 20–50 ng/mL (50–125 nmol/L), so researchers and clinicians should avoid applying the term “vitamin D insufficiency” to patients with serum 25(OH)D₃ concentrations

in the range of 20–30 ng/mL (50–72 nmol/L). Given the emerging data supporting a pandemic of low vitamin D status (serum 25(OH)D₃ < 20 ng/mL or < 50 nmol/L) in Europe and other countries,^{33,34} improving vitamin D status to protect health represents an enormous challenge to nutrition and health agencies around the world.

Vitamin D repletion in clinical populations

When vitamin D deficiency is encountered in the clinical setting (e.g., older or sick individuals; newborn infants), the physician usually prescribes supplemental vitamin D that can be in the form of either vitamin D₂ or D₃. For patients in the United States who have a poor vitamin D status, the only high-dose vitamin D formulations approved by the US Food and Drug Administration for clinical use are vitamin D₂ at 1.25 mg (50,000 IU)/week for 8 weeks; this is considered effective in ameliorating low vitamin D status.³⁵ Once vitamin D–deficient patients have normal serum levels of 25(OH)D₃ restored, they can be given daily vitamin D doses at or above the RDA for vitamin D (i.e., 15–20 μg (600–800 IU)/d) or at-risk patients may be prescribed a maintenance dose of 1.25 mg (50,000 IU) every other week. High doses of vitamin D₃ may be more effective than high doses of vitamin D₂ in raising 25(OH)D levels, but one well-designed study showed that at a dose of 25 μg (1000 IU)/day, vitamins D₃ and D₂ were equivalent.³⁶

C. Basis for Individual Variability in Requirements

There are many factors that can lead to variability in serum levels of 25(OH)D₃ and different requirements for dietary vitamin D.^{26,37} At the genetic level, there is evidence that serum 25(OH)D₃ levels are influenced by gene polymorphisms affecting skin production of vitamin D, serum DBPs, and vitamin D metabolism.^{38,39} There are also a number of demographic issues that place persons at risk for low levels of 25(OH)D₃. This includes highly pigmented skin (e.g., persons with African, South Asian, and Middle Eastern heritage), obesity (possibly due to greater deposition into adipose tissue or an attenuated hydroxylation to 25(OH)D₃ in the liver), and some diseases or medications. In addition, because UVB radiation penetrates uncovered skin and converts cutaneous 7-dehydrocholesterol to previtamin D₃, persons with almost no exposure to sunlight (e.g., institutionalized elderly persons such as those in nursing homes) or clothing traditions for ethnic or religious reasons that require limited skin exposure can also have lower 25(OH)D₃. UV energy–induced skin vitamin D synthesis is affected by season, length of day, time of day, cloud cover, pollution, skin melanin content, and use of sunscreen. It is suggested that at least 20% of the body’s

TABLE 6.3 Randomized controlled trials examining the impact of vitamin D supplementation on bone and nonbone outcomes.

Study name, location, & principal investigators	Subjects' age at entry (y) and number	D ₃ dose per day and other interventions	Duration (y)	Primary outcomes	First publication
VITAL, United States, Manson & Buring	50+ 25,871	Placebo, 2,000 IU, 840 mg Omega-3 fatty acids	5.3	CVD cancer	2019
ViDA, New Zealand, Scragg & Carmargo	50+ 5,110	Placebo, 1,00,000 IU/mo	3.3	CVD, respiratory disease, fractures	2016
D2d, USA, Pittas	25+ 2,423	Placebo, 4,000 IU	2.5	Diabetes	2019
FIND, Finland, Tomi-Pekka, Tuomainen & Voutilainen	60+ 2,500	Placebo, 1,600 IU, 3,200 IU	5	CVD cancer	2020*
DO-Health, 8 Cities in Europe, Bischoff-Ferrari	70+ 2,157	Placebo, 2,000 IU, 1 g omega-3 fatty acids, home exercise	3	Cognition, bone, immunity, blood pressure, functional decline (SPPB)	2021*

CVD, cardiovascular disease; D2d, Vitamin D and Type 2 Diabetes; DO-Health, VitaminD₃-Omega-3-Home Exercise HeALTHy Ageing and Longevity Tria; FIND, Finnish Vitamin D Trial; SPPB, short physical performance battery; ViDA, ViDA (Vitamin D Assessment) Trial; VITAL, VITamin D and Omega-3 Trial.

* Estimated.

surface must be exposed to the sun for 15–20 min for serum 25(OH)D₃ levels to increase.²⁶ While certain individuals may need higher intakes, as described above, the DRIs for vitamin D were determined assuming limited sun exposure due to public health concerns about skin cancer.¹⁵

D. Tolerable Upper Intake Levels of Vitamin D Intake

Excessive amounts of vitamin D are not normally available from usual dietary sources, and thus reports of vitamin D intoxication are uncommon. However, there is always the possibility that vitamin D intoxication may occur in individuals who are taking excessive amounts for various reasons. Typical symptoms of vitamin D intoxication include hypercalcemia, hypercalciuria, anorexia, nausea, vomiting, thirst, polyuria, muscular weakness, joint pains, diffuse demineralization of bones, and general disorientation.⁴⁰ If the condition is allowed to go unchecked, it can be fatal. The extent of vitamin D toxicity has been shown in some instances to be related to the level of dietary intake of calcium.⁴¹ The available evidence suggest that symptoms of vitamin D intoxication appear when serum 25(OH)D₃ levels rise above 100 ng/mL (250 nmol/L).⁴²

As part of the DRI process, the 2011 IOM committee set the Tolerable Upper Level (UL) for vitamin D₃ at 100 µg (4000 IU)/day for individuals 9 years or older. This reference value was supported by evidence that there are no observable adverse effects at this level of intake, while there was evidence of adverse effects,

including hypercalcemia, at intakes at or above 1.25 mg (50,000 IU)/day for several weeks or months.¹⁵ More recent studies have not reported any safety concerns with 100 µg (4000 IU)/day in healthy individuals or in pregnant women.^{28,43,44}

Case reports of vitamin D intoxication have occurred due to drinking milk that had been fortified with inappropriately high amounts of vitamin D.⁴⁵ This resulted in serum 25(OH)D₃ levels at about 300 ng/mL (750 nmol/L) in eight subjects, seven of whom had hypercalcemia levels. Also, a 73-year-old man who consumed 1.5 mg (60,000 IU) vitamin D/day was brought into the clinic with extreme delirium requiring sedation. His serum 25(OH)D₃ level was 220 ng/mL (550 nmol/L) with barely detectable PTH, and with hypercalcemia and creatinine at 200 µmol/L (normal 59–100). Treatment included a bisphosphonate and IV fluids.⁴⁶ These cases highlight the potential dangers of excessive vitamin D intake.

These cases highlight the potential dangers of excessive vitamin D intake. It has been proposed that because there is an absence of toxicity in clinical trials at 0.25 mg (10,000 IU)/day, it is safe to consume this level of the vitamin,⁴⁷ but this has not been sufficiently tested in long-term studies.

E. Measurement of 25(OH)D₃ and Assay Variability

A substantial body of literature exists demonstrating problems in assay comparability when measuring 25(OH)D₃.⁴⁸ To reduce variability in measurements, an

external quality assessment system, the Vitamin D External Quality Assessment Scheme, was developed based on comparability to standard samples. More recently, the Vitamin D Standardization Program within the US National Institute for Standards and Technology developed reference materials to include as an external standard to evaluate assay variation in an unbiased way.⁴⁹ Using external standards is strongly recommended to provide a more reliable assessment of vitamin D status and allow researchers to compare 25(OH)D₃ values with findings in other studies. The gold standard method to measure serum 25(OH)D₃ levels is the LC/MS/MS methodology.⁵⁰ However, while LC/MS/MS is considered the gold standard, other methods are also reliable when utilized with external standards and when the coefficient of variation of the assay is <10%.

F. Dietary Sources of Vitamin D

There are few naturally occurring food sources of vitamin D. These include fatty fish, fish liver oil, and egg yolk. In the United States, milk and plant-based milk alternatives can be voluntarily fortified with up to 5.1 µg (206 IU) vitamin D₃ per cup (245 g). Other foods may also be fortified with vitamin D, including certain orange juice formulations, yogurt, cheese, margarine, bread, and cereal. Canada and Finland have mandatory fortification of fluid milk and margarine with vitamin D. In the United States, there are also vitamin D-enhanced foods, such as UV-exposed mushrooms (which supply vitamin D₂).

Recent data in the United States indicate that few individuals have adequate dietary vitamin D intake in spite of voluntary food fortification. In the 2013–16 “What We Eat in America” cohort, 92% of men of 19+ years and >97% of women of 19+ years did not meet the EAR for vitamin D.⁵¹ The diets of children were similarly poor, in that 86% of 1- to 3-year-old boys and 93% of 1- to 3-year-old girls had inadequate dietary vitamin D intakes. It is important to note that this represents amounts from food and beverages only and thus does not include contributions from sunlight and supplement usage. For persons who consume a dairy-rich diet or fatty fish at least three times per week (e.g., salmon, tuna, mackerel), dietary vitamin D intake is generally going to meet the recommended intakes. However, for those who get limited sun exposure and do not eat these foods, supplemental vitamin D is needed.

IV. ISSUES RELATED TO VITAMIN D IN HUMAN HEALTH

Over the past several decades, at least 38 tissues that express the VDR⁵² have been identified, thus demonstrating

that vitamin D signaling is not restricted to the control of calcium homeostasis. The newer physiological systems known to be influenced by vitamin D include the immune system (both innate and adaptive), cardiovascular system, pancreas, muscle, and brain. Examples of documented biological responses are provided in Fig. 6.6 for each new system, as well as disease states that have been found to be associated with vitamin D deficiency.⁵² In this section, the evidence supporting a beneficial impact of high vitamin D status on a various health outcomes is summarized.

A. Relevant Health Outcomes and Conditions of Concern

Bone health

Osteomalacia and rickets; The bone diseases rickets (in children) or osteomalacia (in adults) are considered classic vitamin D deficiency states, and they are due either to the absence of vitamin D in the diet or inadequate sunlight exposure. The clinical features depend upon the age of onset. Rickets includes deformity of the bones in the knees, wrists, and ankles, as well as changes in the costochondral joint functions that is sometimes referred to as the rachitic rosary.⁵³ Only minor skeletal changes may occur if rickets develops in the first 6 months of life; however, younger infants may suffer from convulsions or tetany due to a low blood calcium level (usually <7 mg/100 mL). After 6 months, bone pain and tetany are likely to be present. Osteomalacia occurs in adulthood, after growth and development of the skeleton are complete, and its main symptoms are muscular weakness and bone pain with little bone deformity.⁵³ In general, rickets is more prevalent in developing countries and only appears when vitamin D levels are <12 ng/mL (30 nmol/L).⁵⁴ Because mothers of rachitic children are often vitamin D deficient, public health messages to mothers may reduce the incidence of rickets in young children.⁵⁵

The differences between osteomalacia and osteoporosis are subtle but important. Osteomalacia arises from a problem in the bone-building process, whereas osteoporosis arises from a weakening of mature bone via a general loss of bone calcium from the ongoing daily process of bone remodeling. While osteomalacia is not typically identified because a bone biopsy is needed for diagnosis, a study of adults with accidental deaths showed evidence of osteomalacia in adults in all ages and both sexes in those with lower levels of serum 25(OH)D₃.⁵⁶

Osteoporosis; Osteoporosis is the most common generalized disorder of bone; it is estimated that in the United States, over 54 million people, 80% of whom are female, have some form of osteoporosis.⁵⁷ In the United States,

24% of women over 65 years of age have osteoporosis while one in two women, and one in four men, over the age of 50 years will break a bone due to osteoporosis.⁵⁸ Chronic vitamin D deficiency in the elderly (>70 years) can result in an imbalance in the remodeling rate of bone and exacerbate the bone loss. One factor that leads to osteoporosis is inadequate vitamin D intake that reduces calcium absorption and can result in a negative calcium balance. When this occurs, bone resorption increases to remove calcium from bone to maintain normal blood levels of calcium.⁵³ Blood levels of calcium are tightly regulated by the intestine, kidney, and bone. Because of this tight regulation, blood levels of calcium cannot be used as a marker of low dietary calcium intake, and thus blood levels of calcium and vitamin D are not markers for osteoporosis. Osteoporosis is diagnosed with X-rays or fracture of vulnerable sites, including the hip, spine, and wrist. In most populations, it is a combination of low intakes of calcium and vitamin D that ultimately increase risk of osteoporosis. Importantly, higher doses of vitamin D intake (~83–100 µg (3300–4000 IU)/d for 1–4 years) do not show benefit over recommended intakes on bone mineral density or quality²⁸ or fracture risk.³²

Immunity and inflammation

VDR is expressed in immune cells (T and B cells and antigen-presenting cells), and these immunologic cells can also synthesize the active vitamin D metabolite, 1,25(OH)₂D₃ from 25(OH)D₃. Vitamin D can act in an autocrine manner in a local immunologic milieu and can impact the innate and autocrine adaptive immune responses. Briefly, innate immunity is the first defense against pathogens. The linings of the respiratory and GI tracts and skin are major components of the innate immune system, acting as physical and chemical barriers against pathogens. Macrophages also play a large role in innate immunity. The adaptive immune response refers to antigen-specific immune response, mediated by lymphocytes (B and T cells), and has a memory, permitting more efficient work against an antigen. Vitamin D deficiency is associated with an increased susceptibility to infection.⁵⁹ In addition, vitamin D deficiency has been associated with multiple autoimmune conditions including rheumatoid arthritis (RA), systemic lupus erythematosus (SLE), MS, type 1 diabetes (T1D), dermatomyositis, inflammatory bowel disease (IBD), hepatitis, psoriasis, and asthma, as discussed below.

Type 1 diabetes; Epidemiological and observational studies indicate a relationship between adequate vitamin D status and a decreased risk of developing T1D. In addition to vitamin D–modulating T-regulation, pancreatic beta cells have VDRs and express 1-α hydroxylase (encoded

by CYP27B1), and the human insulin gene promoter has a vitamin D–response element. In human studies, lower levels of maternal DBP in systemic and cord blood have been associated with increased risk of T1D in offspring,⁶⁰ but a large prospective study (Environmental Determinants of Diabetes in the Young) demonstrated no benefit of maternal vitamin D supplementation during pregnancy on T1D risk in the offspring.⁶¹ There are few randomized controlled studies investigating vitamin D replacement to prevent T1D or in treating it at the time of diagnosis as a means to prolong endogenous insulin secretion.

Multiple sclerosis; MS is a chronic, inflammatory, and neurodegenerative disorder of the central nervous system (CNS), which affects young adults and likely has an autoimmune origin. The pathogenesis and progression of MS is complex, yet there is evidence that it is associated with vitamin D deficiency.⁶² One mechanism of action of vitamin D in MS is immunomodulatory; it also has neurotrophic effects on the CNS.⁶³ Studies report an association between MS risk and lower sunlight exposure, vitamin D intake, and serum 25(OH)D₃ levels.^{62,64,65} The risk for MS for people living at high latitudes (>35 degrees) is about twofold higher, as compared to those who live at lower latitudes, and there is a higher MS incidence in children born in the summer rather than winter.⁶⁶ However, vitamin D supplementation has not consistently been shown to reduce symptoms,⁶⁷ and some studies suggest that high doses may result in deleterious side effects.³⁵ Overall, avoiding deficiency beginning in early life may prevent MS, and because some studies indicate that vitamin D supplementation may attenuate symptoms and MS relapses,⁶⁸ supplementation can be considered, but in moderate doses.

Rheumatoid arthritis, systemic lupus erythematosus, and juvenile dermatomyositis; RA, another chronic inflammatory disorder, largely affects cartilage tissues and results in swelling and deformity of joints and bone erosion. Studies indicate RA susceptibility and symptom severity are greater with low levels of vitamin D (<20 ng/mL or <50 nmol/L). In patients with diagnosed RA, vitamin D supplementation may reduce disease recurrence in those with vitamin D deficiency but does not consistently attenuate symptoms.^{69,70}

Studies have reported an inverse association between 25(OH)D₃ levels with the occurrence and severity of SLE and juvenile dermatomyositis, especially in those with 25(OH)D₃ levels <20 ng/mL (<50 nmol/L).^{71,72} Vitamin D supplementation may reduce symptoms,⁷³ but UV radiation exposure may worsen symptoms and therefore sun exposure is not recommended for SLE or dermatomyositis.⁷⁴

Psoriasis; Psoriasis is a chronic inflammatory autoimmune skin disorder with abnormal keratinocyte differentiation and hyperproliferation, resulting in dermatosis with red and scaly areas on the skin. The VDR is expressed in keratinocytes in the skin, and vitamin D contributes to the maintenance of the cutaneous barrier integrity. In addition to the use of topical corticosteroids, psoriasis is treated with topical vitamin D analogs with good evidence in a rodent model⁷⁵ that it reduces inflammation. There are studies examining the effect of oral supplementation on psoriasis,⁷⁶ and one study indicates a positive effect of higher dose vitamin D₂ supplementation and improvement in the psoriasis disease severity score, with no adverse events.⁷⁷ The Medical Board of the National Psoriasis Foundation only weakly recommends vitamin D supplementation in patients with psoriatic arthritis and emphasizes that any dietary intervention considered should only be used in conjunction with standard medical therapy.⁷⁸

Inflammatory bowel disease; Vitamin D deficiency may impact the disease course in patients with inflammatory bowel disorders and is commonly found at pediatric IBD diagnosis.⁷⁹ This is supported by cell and murine studies showing that the addition of vitamin D prevents epithelial barrier dysfunction and intestinal inflammation^{80,81} and observational findings that show use of vitamin D supplements are associated with a lower Crohn's disease activity index and markers of inflammation.⁸² Currently, supplementation trials are lacking, and thus it is not clear if higher doses of vitamin D will consistently attenuate symptoms in IBD.

Respiratory infections and asthma; Vitamin D supplementation (50 µg (2000 IU)/day) does not reduce the severity of upper respiratory tract infection symptoms compared with standard dose (10 µg (400 IU)/day) supplementation in preschool children.⁸³ Among black infants born preterm, sustained supplementation with vitamin D (10 µg (400 IU)/day), compared with diet-limited supplementation, resulted in a reduced risk of recurrent wheezing by 12 months of age. Others have found that asthma exacerbation increased with low 25(OH)D₃ and that it may be reduced with vitamin D supplementation in children and adults who had low serum 25(OH)D₃ levels.⁸⁴ This suggests that infants with vitamin D deficiency may benefit from vitamin D supplementation to ameliorate asthma symptoms.

Muscle mass and strength

VDRs are expressed in skeletal muscle cells and may explain associations between serum 25(OH)D₃ status and muscle strength in observational and interventional studies. In the Berlin Aging Study II, vitamin D

insufficiency (<20 ng/mL or < 50 nmol/L) was associated with a 1.5 higher odds ratio (OR) of being prefrail or frail.⁸⁵ The OR for persons with vitamin D insufficiency was 1.8 times higher OR for having low lean mass, and, interestingly, in a subanalysis, this was only significant in men.⁸⁶ The KORA-Age Study found that serum levels of 25(OH)D₃ < 10 versus ≥ 20 ng/mL (<25 vs. ≥ 50 nmol/L) were associated over 3 years with greater loss of muscle mass and physical performance in older adults.⁸⁷ In a prospective observational community cohort study in elderly women, serum 25(OH)D₃ < 20 ng/mL (50 nmol/L) was associated with higher frailty in all but the oldest old.⁸⁸ In men and women in the United States following hip fracture, serum 25(OH)D₃ concentrations <12 ng/mL (30 nmol/L) reduced the ability to walk independently afterward; the reduction was not seen in those with serum concentrations < 20 ng/mL (<50 nmol/L) (Hao, abstract 2019).

The effect of vitamin D supplementation on muscle outcomes has mixed findings. At a dose of 100 µg (4000 IU)/day over 4 months in elderly women, intramyonuclear VDR concentration and muscle fiber cross-sectional area increased.⁸⁹ In the D-Vitaal trial, 30 µg (1200 IU)/d for 12 months in older persons with low vitamin D status showed no effect on physical function,⁹⁰ while two metaanalyses examining the effect of vitamin D supplementation on muscle strength provided conflicting results.^{91,92} In community-dwelling older persons, one metaanalysis that included 15 studies found no improvement in muscle strength with vitamin D supplementation (with or without calcium).⁹¹ In another metaanalysis of 30 RCTs (of all ages), vitamin D supplementation had a small effect on improving muscle strength, but no effect on muscle mass or muscle power.⁹² However, in individuals with vitamin D deficiency (serum 25(OH)D₃ level < 12 ng/mL [<30 nmol/L]), the improvement in muscle strength was greater. In children and adolescents supplemented with 10, 50, or 100 µg (400, 2000, or 4000 IU)/day, increases seen in serum 25(OH)D did not influence muscle mass or strength in vitamin D-sufficient children and adolescents.⁹³

In the athlete, there is a strong belief in certain sports that vitamin D supplementation will improve performance. However, a metaanalysis, which included 13 RCTs (2005–16) with 532 athletes (vitamin D: n = 311, placebo: n = 221), found no significant improvement in physical performance with ≥ 75 µg (≥ 3000 IU) supplementation despite achieving vitamin D concentrations >30 ng/mL (75 nmol/L)/d.⁹⁴

In another metaanalysis using eight RCTs (involving 284 athletes), vitamin D supplementation had inconsistent effects, with positive effects on lower limb muscle strength but not upper limb muscle strength or muscle

power.⁹⁵ The very high doses used in other studies, the variability in study design, and protocols with a goal for serum 25(OH)D₃ levels to increase from the sufficient range (30–50 ng/mL (75–125 nmol/L)) to high levels (>50 ng/mL or 125 nmol/L) may be reasons why vitamin D supplementation has not shown consistent positive effects in the athlete.

Muscle strength is consistently reduced with vitamin D deficiency (25(OH)D₃ < 12 ng/mL or < 30 nmol/L), however, and in this case, supplementation is beneficial.^{91–95} Future studies using more physiological daily doses of vitamin D and conducted with participants with low baseline vitamin D status may demonstrate more beneficial effects on muscle and bone health.

Falls

Falls are common in the elderly, with serious consequences including major injury and fractures. A fall can occur due to lower muscle mass or sarcopenia or neurological reasons. Although vitamin D is beneficial for both these physiological systems, controlled clinical trials show that higher vitamin D intakes (e.g., 50–100 µg (2000–4000 IU)/day) either have no effect or even associated with increased falls risk.^{32,96–98} In one study, it was shown that a high daily dose of 100 µg (4000 IU) vitamin D/day slowed reaction time in older women.⁹⁹ Whether this or other neuropsychological effects of vitamin D supplementation can explain why falls are increased by those consuming additional vitamin D in these studies^{32,96–98} is not known. Together, findings from various studies^{32,96,98,100} indicate that vitamin D supplementation between levels of 15–40 µg (600–1600 IU)/day may reduce risk of falls.

Neuropsychological outcomes

Vitamin D has been considered a neurosteroid given that it may have an important role in neuroprotection, brain development, and immunomodulation, as well as in neurological diseases. Animal studies indicate that supplementation with vitamin D is protective against biological processes associated with cognitive decline and that it enhances learning and memory.¹⁰¹ The presence of 1- α -hydroxylase, the enzyme responsible for synthesis of active 1,25(OH)₂D₃, as well as the presence of the VDR in the brain, supports a role for vitamin D as a neuroprotective hormone.¹⁰² Serum 25(OH)D₃ deficiency of <12 ng/mL (30 nmol/L) is consistently associated with cognitive impairments,^{103,104} although other studies suggest a threshold of <20 ng/mL (<50 nmol/L)^{105,106} or no association at all.¹⁰⁷ In addition, there are a greater number of winter births for people with neurological disease, including epilepsy, MS, and cerebral palsy.^{66,108} Numerous studies suggest that hypovitaminosis D is associated with

increased risk of developing Alzheimer's disease (AD) and dementia.^{101,109}

Clinical intervention studies indicate small effects of supplementation (50 µg (2000 IU)/day) on cognitive domains,⁹⁹ while another trial failed to associate increased serum 25(OH)D₃ levels with improved cognitive outcomes.¹¹⁰ In this study, there was no superior cognitive benefit of 50 µg (2000 IU)/day compared with 20 µg (800 IU) vitamin D/day among relatively healthy older adults over 24 months.¹¹⁰

The D-Vitaal study investigated the effect of vitamin D supplementation (60 µg (1200 IU)/day compared to placebo) on depressive symptoms and found no effect on depression.⁹⁰ While some studies indicate benefits of 50 µg (2000 IU)/day on learning and memory, higher doses (100 µg (4000 IU)/day or more) may negatively affect other cognitive domains, such as reaction time.⁹⁹

Although there are no studies showing prevention of AD, vitamin D has been shown to affect physiology that is disturbed in AD, including neurotransmission, amyloid- β peptides and Tau accumulation, vascularization, oxidative stress, and inflammation; therefore, supplementation may have a positive effect.^{101,111}

Overall, maintaining serum 25(OH)D₃ levels >20 ng/mL (>50 nmol/L) is likely to slow, prevent, or improve neurocognitive decline in both healthy or cognitively impaired individuals. However, whether consuming the RDA or higher levels of vitamin D supplementation will have a beneficial effect has not yet been demonstrated in humans and possibly may have a negative effect.⁹⁹

Cancer

The relationship between vitamin D and cancer is a well studied nonskeletal health outcome and has been reviewed in depth by others.^{112,113} Early studies suggested that low serum 25(OH)D₃ levels increase cancer risk but many of these studies were underpowered and so their results are unreliable. To overcome this, researchers have implemented data pooling projects. One such study pooled data from 17 case-control cohorts comparing 5706 colorectal cancer cases with 7107 control subjects.¹¹⁴ Compared to the reference group (20–25 ng/mL (50–63 nmol/L)), there was a 31% increase in colorectal cancer risk for those with serum 25(OH)D₃ levels < 20 ng/mL (<50 nmol/L) and that risk was 27% lower in subjects with serum 25(OH)D₃ levels between 35 and 40 ng/mL (88–100 nmol/L).

Animal studies have been used to establish “proof of concept” that there is a causal link between low vitamin D status and increased health outcomes, including cancer. These studies show that severe vitamin D deficiency or VDR deletion can increase cancer incidence and severity in models for colon, prostate, and breast cancer and other cancers. Dietary vitamin D₃ intakes that

maintained serum 25(OH)D₃ levels at 22 ng/mL (55 nmol/L) were sufficient for normal bone mass in mice, but higher vitamin D₃ intakes (raising serum levels to 34 or 56 ng/mL (85 or 140 nmol/L) reduced early stage prostate cancer.¹¹⁵

A variety of molecular mechanisms might account for how signaling through the VDR might mediate the protective effect of vitamin D against cancer. Studies on the impact of 1,25(OH)₂D₃ on tumor cells reveal that 1,25(OH)₂D₃ regulates a large number of cancer-relevant biological processes, including the suppression of genes controlling cell proliferation and prooxidant stress as well as activating the genes controlling apoptosis and DNA repair.¹¹² Vitamin D signaling can also modulate the biology of other cells found within the tumor microenvironment including reducing inflammatory stress that promotes cancer.¹¹²

Surprisingly, given the evidence above, very few vitamin D intervention trials have been conducted. The results of two trials suggest that vitamin D intake can reduce cancer risk. In a 4 year, double-blind, placebo-controlled, RCT, healthy postmenopausal women were supplemented daily with either 100 µg (2000 IU) vitamin D and 1.5 g calcium (n = 1156) or placebo (n = 1147). Before the intervention, the study population had adequate serum 25(OH)D₃ levels (average of 32 ng/mL (80 nmol/L), and the supplement increased levels to an average of 44 ng/mL (110 nmol/L); this resulted in a trend toward reduced cancer incidence (P = .06).¹¹⁶

In the largest vitamin D intervention trial conducted to date, the VITamin D and OmegaA-3 Trial (VITAL), 50 µg (2000 IU) vitamin D₃/d or placebo was provided to more than 25,000 participants and invasive cancer incidence after a mean 5.3 years follow-up period was recorded.¹¹⁷ As with the first trial,¹¹⁶ the baseline vitamin D status of this group was adequate (31 ng/mL (78 nmol/L)).¹⁸ Although neither the incidence for invasive cancer nor cancer deaths were lowered by vitamin D supplementation during the 5.3 years period, when the first year of data were excluded (to eliminate subjects who probably had cancer at the start of the intervention), there was a significant reduction in cancer deaths in the vitamin D group (hazard ratio: 0.79). Collectively, these trials, coupled to the other types of data available, suggest that high vitamin D status is protective against cancer. However, data for specific cancers and subtypes of cancers are lacking.

Cardiovascular disease and hypertension

Vitamin D has various effects on the cardiovascular system, and this is likely because VDR and its 1- α -hydroxylase are found in a variety of tissues that are important to cardiovascular health. Vitamin D activates the nuclear receptor in cardiomyocytes and vascular endothelial cells. In addition, vitamin D is a negative

regulator of the renin–angiotensin–aldosterone system, an important regulator of blood pressure.¹¹⁸ There are numerous cross-sectional studies between vitamin D deficiency and elevated blood pressure and cardiovascular events suggesting that vitamin D metabolism affects the pathogenesis of hypertension (HTN) and cardiovascular disease (CVD). In the Multi-Ethnic Study of Atherosclerosis (MESA), 3002 persons (ages 45–84 years), lower serum 25(OH)D₃ concentrations (<20 vs. >30 ng/mL (<50 vs. > 75 nmol/L)) were not associated with a greater risk of incident hypertension. However, higher levels of PTH (>65 pg/mL), such as the levels that occur in response to low serum 25(OH)D₃ levels, increased the risk of HTN by 27%.¹¹⁹ In this same population (MESA), it was found that associations of 25(OH)D₃ with coronary heart disease were strengthened following adjustment for DBP levels among participants who were white or Chinese, but not black or Hispanic.¹²⁰

It has been proposed that free or bioavailable 25(OH)D₃ (i.e., that which is not bound to the serum DBP) may be a better indicator of vitamin D status than total 25(OH)D₃ levels because it is able to enter cells.¹²¹ One study indicated that bioavailable 25(OH)D₃ was associated with coronary heart disease events only among white participants.¹²⁰ In another study focusing on stroke, it was found to be associated with severe vitamin D deficiency.¹²² It is possible that lower serum 25(OH)D₃ levels do not lead to a higher risk of stroke or other CVD events but are instead a consequence of disease.

Vitamin D supplementation (50 µg (2000 IU)/day) or placebo was randomly assigned to over 25,000 older adults in the VITAL study for a median follow-up period of 5.3 years (previously discussed relative to cancer incidence); there was no reduced risk of major cardiovascular events.¹⁸ It is possible that this was partially due to baseline levels of serum 25(OH)D₃ levels >20 ng/mL (50 nmol/L) in most individuals. However, a metaanalysis,¹²³ which included more than 83,000 participants, also showed that vitamin D supplementation was not associated with reduced major adverse cardiovascular events, individual CVD endpoints (myocardial infarction, stroke, CVD mortality), or all-cause mortality.¹²³

While current findings indicate no cardiovascular protection with vitamin D supplementation, future studies could address whether specific vitamin D metabolites improve the prognosis for CVD outcomes in persons who are deficient, examine longer follow-up periods, and test higher vitamin D dosages.

Type 2 diabetes

For decades, evidence has demonstrated a role for vitamin D in pancreatic β -cell function and insulin

secretion.¹²⁴ In in vitro and in vivo studies, vitamin D deficiency reduces glucose-mediated insulin secretion in pancreatic β -cells,¹²⁵ while vitamin D supplementation restores insulin secretion. A few mechanisms may be responsible, such as a direct effect on β -cell function, or indirectly (possibly through the effect of vitamin D on osteocalcin). Observational studies suggest that improving vitamin D status may reduce the risk of type 2 diabetes (T2D). Accordingly, a number of studies have examined whether supplementation of vitamin D can slow the progression to T2D as another lifestyle change. This could have a high impact in the field since more than 84 million persons have prediabetes with glucose levels higher than normal range but below the threshold for diabetes.¹²⁶

A few large studies have now addressed this. In a study in Norway,³¹ 511 white adults with prediabetes were randomly assigned to 500 μg (20,000 IU)/week (approximately 72 μg (2900 IU)/day) of vitamin D or placebo for 5 years; there was no difference in the risk of diabetes between groups. In the VITamin D and Type 2 Diabetes Trial conducted at multiple sites in the United States, 2423 persons at high risk for T2DM were randomly assigned to vitamin D₃ supplementation (100 μg (4000 IU)/day) or a placebo for a median follow-up period of 2.5 years. Supplementation did not lower the risk of diabetes compared to placebo.³⁰ However, in subjects with vitamin D deficiency (serum 25(OH)D₃ levels < 12 ng/mL (30 nmol/L)) before supplementation, the risk of diabetes was reduced in those assigned to vitamin D.³⁰

Overall, the inability to consistently show evidence that vitamin D supplementation affects risk of T2D may be related to the study populations in which the majority of persons included had sufficient serum 25(OH)D₃ levels (>20 ng/mL (50 nmol/L)) where additional intake would not lower the progression to diabetes.

Obesity

Circulating 25(OH)D₃ levels are lower in the obese compared to lean individuals while PTH is higher in obesity. The lower 25(OH)D is attributed to volume dilution into adipose tissue but could also partially be due to altered hepatic metabolism caused by a fatty liver, greater clothing cover-up, or shade-seeking behavior in the obese population.²⁵ Alternatively, low serum 25(OH)D₃ levels may increase propensity to weight gain and obesity, as shown in animal models. Whether the lower 25(OH)D₃ level in obesity acts to increase risk of disease states associated with low 25(OH)D₃ levels, such as CVD and diabetes, is possible. Supplementation of vitamin D in obese individuals does not increase serum 25(OH)D₃ concentrations as much as in

leaner individuals, and therefore obese persons may need higher intakes of vitamin D to achieve serum levels >20 ng/mL (50 nmol/L),^{127,128} although effects on health outcomes, specifically in the obese, require further research. Interestingly, obese persons who lose weight respond by increasing serum 25(OH)D₃ levels.¹²⁹

The rise in serum 25(OH)D₃ levels with supplemental vitamin D (63 μg (2500 IU)/day) doubled in those who lost weight compared to those who received the supplement and did not lose weight.¹³⁰ In obese persons with low serum 25(OH)D₃ levels, both supplemental vitamin D and weight loss should be advised to achieve optimal increases in circulating levels.

Mortality

Many factors contribute to mortality and many attributes of vitamin D metabolism modify physiological systems. This can explain why a deficiency as demonstrated by low serum concentrations of 25(OH)D₃ could negatively affect multiple organs and systems and together increase the risk of death due to various causes. Many observational studies show that both low and high serum 25(OH)D₃ levels are associated with increased risk outcomes for mortality. In the VITAL trial, which included ~20% black participants, supplementation with 50 μg (2000 IU)/day vitamin D did not significantly affect mortality, including death from cancer (breast, prostate, and colorectal) and death from any cause.²⁹ In a metaanalysis, vitamin D supplementation also was not associated with all-cause mortality in adults compared with placebo or no treatment, but it was associated with a 16% reduced risk of cancer death.¹³¹

Summary of skeletal and nonskeletal outcomes

Observational studies of serum 25(OH)D suggest that the nadir of serum 25(OH)D₃ for disease risk and mortality is between 20 and 30 ng/mL (50–75 nmol/L). The new large randomized placebo-controlled trials examining whether nonskeletal health outcomes can improve with higher vitamin D intakes are shown in [Table 6.3](#). Overall, initial findings indicate that there are no adverse outcomes of vitamin D supplementation and resulting higher serum 25(OH)D₃ values, but thus far, most of the large trials show little or no measurable improved clinical benefits.

B. Known Interactions

There is no evidence that other nutrients negatively interact with vitamin D metabolism. However, the presence of fat in a meal (but not the type of fat) increases absorption of a vitamin D supplement.¹³² Medications that decrease vitamin D absorption (and other fat-soluble vitamins) include orlistat (a drug used for weight loss) and

cholestyramine (a drug used to decrease cholesterol absorption). Other medications that affect vitamin D absorption or metabolism are known to decrease serum 25(OH)D₃ concentrations and include anticonvulsants, antituberculosis types, cimetidine, some corticosteroids, and the antibiotic rifabutin.^{133,134} Persons with health conditions requiring any of the above medications should be monitored carefully. Additional medications interact with vitamin D.²⁶

C. Vitamin D as a Dietary Supplement

Supplementation with vitamin D is recommended for individuals with serum 25(OH)D₃ levels less than 20 ng/mL (50 nmol/L); the amount to supplement should be consistent with the DRIs²⁶ (Table 6.2). Testing for low levels in people at risk for vitamin D deficiency can be justified, but the US Preventive Task Force does not recommend routine testing for the general asymptomatic population.¹³⁵ In the absence of a serum 25(OH)D₃ test, the decision to take a vitamin D supple-

ment should consider risk factors for low vitamin D status (e.g., dark skin, obesity, low sunlight exposure).

Vitamin D supplementation is widespread due to self-selection by individuals, physician prescription, and recommendations by other health care practitioners. However, there is some evidence that there may be adverse health outcomes that result from routine, high dose vitamin D supplementation at levels above the UL of 100 ug/day (4000 IU/day).²⁶ This practice of high dose supplementation should thus be avoided in otherwise healthy persons unless it is being used to treat more severe vitamin D deficiency (e.g., in people with serum 25(OH)D₃ levels < 10 ng/mL).

For people with low vitamin D status, it is common for physicians to prescribe an 8 week course of ergocalciferol (vitamin D₂) supplementation at 1.25 mg (50,000 IU)/week or a high, daily over-the-counter supplement (such as 125 ug (5000 IU)/day) that is typically in the form of vitamin D₃. Under these conditions, retesting of serum 25(OH)D₃ levels should be performed after a 2 month period.

RESEARCH GAPS

- Mechanistic studies that identify the vitamin D-regulated genes responsible for the protective effects of vitamin D on disease risk.
- Studies to address the controversy^{27,136} whether there is a higher requirement for vitamin D to protect nonbone outcomes than the recommended intake to protect bone health.
- Ancillary studies on the large vitamin D supplementation trials that have recently been completed to address other nonbone outcomes that have previously been associated with serum 25(OH)D₃ levels.
- Studies to improve understanding of vitamin D metabolites (e.g., free circulating 25(OH)D₃) and their effects on health outcomes.
- Studies to define factors influencing the diversity of individual serum 25(OH)D₃ responses to increasing levels of vitamin D intake.
- Further research on sex- and race-specific factors affecting requirements for vitamin D.
- Studies to improve understanding of vitamin D requirements at various life stages.

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VITAMIN E

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SUMMARY

Discovered in 1922, the term “vitamin E” is used to describe eight structurally related lipid-soluble antioxidants classified as either tocotrienols or tocopherols, but only α -tocopherol has been shown to be essential for humans and is the focus of this chapter. Herein, the use of vitamin E describes general biological effects and those pertaining to α -tocopherol—specific effects shown in humans. For example, in the case of halting lipid peroxidation, all vitamin E forms exhibit this function, whereas only α -tocopherol prevents destruction of peripheral nerves that otherwise cause progressive, sensory neuropathy leading to the deficiency disorder of ataxia. While the diet of 76% of Americans is considered inadequate compared to the Estimated Average Requirement for α -tocopherol (12 mg/d), seemingly few people experience overt clinical signs of inadequacy—is the Estimated Average Requirement too high? And, is there a role for supplementing the diet despite inconsistent outcomes from large-scale randomized controlled trials? These questions, along with an overview of gut-to-liver trafficking and metabolism of α -tocopherol, its antioxidant function along with the adverse effects of oxidative stress, and the basis for determining dietary requirements of vitamin E, will be discussed in this chapter.

Keywords: α -Tocopherol transfer protein; Antioxidant; Dietary requirements; Fat-soluble vitamin; Lipid peroxidation; Pharmacokinetics.

I. INTRODUCTION

Vitamin E is unique among vitamins because it is not a cofactor or nuclear receptor ligand; its function is that of a fat-soluble antioxidant, specifically a peroxyl radical scavenger.¹ Consequently, α -tocopherol deficiency symptoms in target tissues are dependent not only upon its concentration but also on the degree of oxidative stress, presence of oxidizable compounds (e.g., polyunsaturated fatty acids [PUFAs]), and the level of other antioxidants and antioxidant enzymes.² This chapter describes vitamin E structures and antioxidant properties; absorption and lipoprotein transport; delivery to tissues and catabolism; distribution in food; safety; and its role in chronic disease prevention.

A. Background

Vitamin E was discovered in 1922 because it prevented fetal resorption in deficient pregnant rats consuming rancid fat.³ By the mid-1930s, not only α -tocopherols but also β - and γ -tocopherols were isolated from vegetable oils. It took until the mid-1960s to recognize that there are eight different forms of chromanols made by plants: four tocopherols and four tocotrienols (see [Section I B](#)). By the 1940s, vitamin E was recognized as a “chain-breaking” antioxidant based on its ability to prevent the chain reaction of free radicals during lipid peroxidation.⁴ All of the vitamin E forms have antioxidant activity, but the body preferentially retains α -tocopherol, which is the only form that reverses clinical deficiency symptoms, and therefore only α -tocopherol meets human vitamin E requirements (see [Section IV](#)).

The nutritional implications of vitamin E became apparent with the elucidation of lipid peroxidation of dietary fats.^{5,6} Importantly, vitamin E was effective in inhibiting lipid peroxidation, not only in homogeneous chemical solutions but also in heterogeneous systems such as micelles, membranes, and lipoproteins.⁷ Further, α -tocopherol partitions into membrane domains that are enriched in polyunsaturated phospholipids, amplifying its concentration in the location where its antioxidant activity is needed most.⁸ Rate constants of radical reactions with antioxidants showed synergistic interactions of vitamin E and vitamin C (ascorbate).^{9,10}

B. Nutrient Function and Structure

Naturally occurring forms

Vitamin E is colloquially known as the collective name for molecules synthesized by plants that exhibit the antioxidant activity of α -tocopherol¹¹ and includes eight different naturally occurring molecules (tocopherols and tocotrienols; Fig. 7.1).¹² These forms vary in the number of methyl groups on the chromanol ring: trimethyl (α), dimethyl (β or γ), and monomethyl (δ). The tocopherols have a chromanol ring with a phytyl side-chain, while the tocotrienols have an unsaturated side chain. Importantly for human nutrition, α -tocopherol has the highest antioxidant and biologic activities¹³ and is the form that meets human vitamin E requirements¹² (see Section IV).

Antioxidant activity

During lipid peroxidation, carbon-centered radicals in the presence of oxygen form peroxy radicals, which react 1000 times faster with vitamin E than with PUFAs, and thus vitamin E stops the lipid peroxidation chain reaction.¹⁰ In contrast, in the absence of vitamin E, peroxy radicals readily oxidize PUFAs to generate a hydroperoxide and a new carbon-centered radical that further propagates lipid peroxidation, thereby causing a chain reaction and depleting PUFAs (Fig. 7.2).¹⁰ It should be noted that vitamin E does not prevent lipid peroxidation by intercepting the initiating radical.¹⁴

During lipid peroxidation, the hydroxyl group of the vitamin E reacts with the peroxy radical to form the corresponding lipid hydroperoxide and the chromanoxyl radical.¹³ All of the vitamin E forms, as well as most phenols, are chain-breaking antioxidants.¹⁵ The free radical reaction is halted because the chromanoxyl radical of vitamin E then reacts with vitamin C (or other hydrogen donors), thereby oxidizing the latter and regenerating vitamin E in its reduced form.

Antioxidant interactions

The phenomenon by which vitamin E and other reducing agents interact with each other has led to the concept of “vitamin E recycling.”¹⁶ Oxidized vitamin E is usually not detected in vivo because the chromanoxyl radical is reduced rapidly to its unoxidized form by ascorbate or other reducing agents.^{10,17} This interaction of vitamin C with α -tocopherol has been demonstrated in cigarette smokers.^{18,19}

Although various vitamin E oxidation products have been demonstrated in vitro,²⁰ the biological relevance of these oxidized products and whether they are generated in vivo are unknown.²¹ α -Tocopheryl quinone has been detected in the plasma following high-intensity exercise²² and in the plasma of healthy persons and those with fatty liver.²³ It is unclear if these low α -tocopheryl quinone concentrations (up to 0.2 $\mu\text{mol/L}$) in persons with circulating α -tocopherol concentrations of $\sim 30 \mu\text{mol/L}$ are a result of in vivo α -tocopherol oxidation²³ or are contaminants absorbed from food.²⁴

The efficiency of the various tightly interrelated antioxidant systems is apparent when examining oxidation of PUFAs. For example, docosahexaenoic (DHA) and eicosapentaenoic acids (EPA) are much more resistant to lipid peroxidation in biological systems (e.g., in foods) containing various additional antioxidant mechanisms compared with isolated fish oils containing DHA and EPA.²⁵

Ferroptosis

All of the observations concerning the in vivo α -tocopherol mechanisms of action result from its role as a lipid-soluble antioxidant¹ and anti-ferroptotic agent.²⁶ Lipid peroxidation is a self-propagating cycle that generates radicals and cytotoxic products, such as reactive aldehydes. Increased lipid peroxidation changes bioenergetics²⁷ and causes mitochondrial dysfunction²⁸ that exacerbates pathology.²⁹

Vitamin E is a chain-breaking antioxidant that prevents lipid peroxidation propagation and generation of new radicals and is regarded as nature's most efficient anti-ferroptotic agent.³⁰ Ferroptosis^{31,32} is a non-apoptotic, non-caspase-dependent form of regulated cell death caused by lipid peroxidation.^{33,34} Vitamin E is anti-ferroptotic because it preferentially scavenges peroxy radicals thousands of times faster than they can react with PUFAs.^{35,36} The specific signaling and biochemical mechanisms^{37,38} that induce ferroptosis are not well-characterized, with the exception that glutathione (GSH) depletion increases ferroptosis,³⁴ which is prevented by vitamin E.^{26,36} Further, a large number of biochemical mechanisms have been

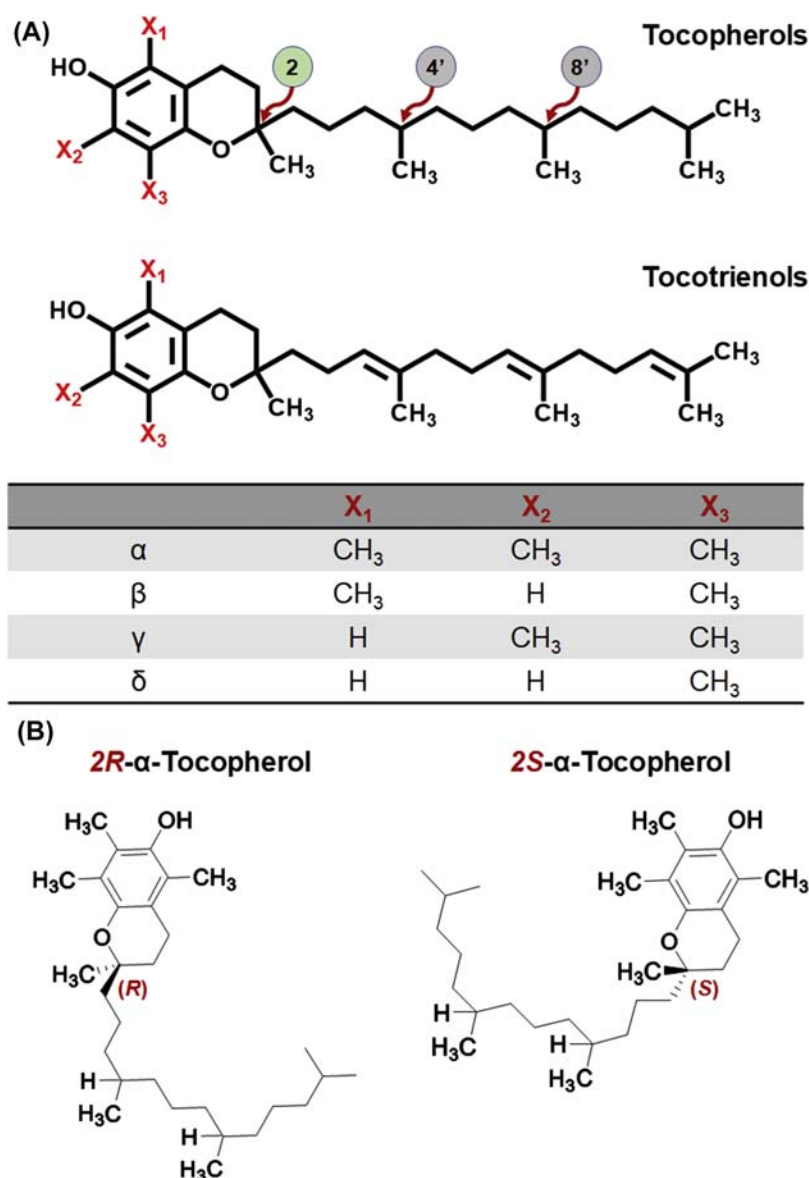


FIGURE 7.1 Vitamin E structures. (A) Structural differences between tocopherols and tocotrienols, as well as differences between alpha, beta, gamma, and delta forms are indicated. Note that tocotrienols contain three double bonds in the side chain. (B) Shown are the stereochemical differences between 2-*R*- and 2-*S*- α -tocopherols at the chiral carbon position 2. The *dotted wedge* indicates the group pointing away, and the *solid wedge* indicates the group pointing toward reader. The chiral carbons at 4' and 8' can be either *R* or *S*. Thus, synthetic, *all rac*- α -tocopherol is an equimolar stereoisomer mixture of four 2*R*- α -tocopherols (*RRR*-, *RRS*-, *RSR*-, *RSS*-) and four 2*S*- α -tocopherols (*SSS*-, *SSR*-, *SRS*-, *SRR*-). Only α -tocopherols in the 2*R* stereoisomer configuration contribute toward meeting human vitamin E requirements.¹²

implicated in cell death mechanisms, including cysteine deprivation,^{39,40} alterations in mitochondrial energy metabolism,²⁸ and enzymatically generated, toxic oxidized lipids.⁴¹ This is an emerging area of antioxidant and vitamin E research, particularly with regard to neurodegeneration.⁴²

C. As a Dietary Supplement or Food Fortificant

α -Tocopherol synthesized by plants has three chiral centers that are all in the *R*- conformation, *RRR*-

α -tocopherol.⁴³ Unlike most other vitamins, chemically synthesized α -tocopherol such as in most dietary supplements is not identical to the naturally occurring form. Synthetic α -tocopherol is called *all racemic*- α -tocopherol (*all rac* or *dl*) and contains an equal mixture of eight different stereoisomers (*RRR*, *RSR*, *RRS*, *RSS*, *SRR*, *SSR*, *SRS*, *SSS*), which differ in the stereochemistry of the phytyl side chain¹². The stereochemistry does not impact the antioxidant activity¹³; however, the 2 position of α -tocopherol (Fig. 7.1) is critical for α -tocopherol biologic activity (see Section II).

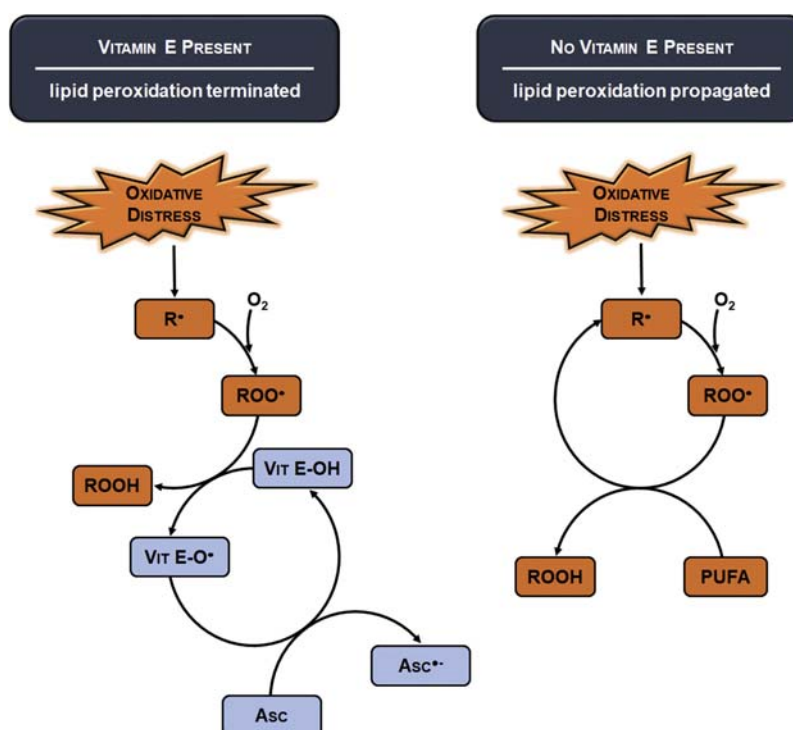


FIGURE 7.2 Antioxidant activity. When peroxy radicals (ROO•) are formed from carbon-centered radicals (R•), they react 1000 times faster with vitamin E (Vit E-OH) than with PUFA (RH); thus, membranes and lipoproteins are protected from the chain reaction of lipid peroxidation by vitamin E. The vitamin E hydroxyl group reacts with the peroxy radical to form the hydroperoxide (ROOH) and the vitamin E radical (Vit E-O•). The vitamin E radical (Vit E-O•) reacts with vitamin C (Asc, or other hydrogen donors), thereby oxidizing the latter to ascorbate free radical (Asc••) and returning vitamin E to its reduced state. It should be noted that free metals, such as iron or copper, can reinitiate lipid peroxidation by reaction with ROOH to form an alkoxyl radical (RO•).

The stereochemical and molecular differences between vitamin E forms are the basis for the differences in biological activity. Although all the various forms are peroxy radical scavengers,¹³ the body preferentially retains 2R- α -tocopherol. These structural differences are also the basis for the US Food and Drug Administration's definition of vitamin E for the Nutrition and Dietary Supplement Facts labels.⁴⁴ As of 2016, the vitamin E units on these labels have changed from International Units (IU) to mg α -tocopherol, with vitamin E defined as 1 mg α -tocopherol = 1 mg RRR- α -tocopherol = 2 mg *all rac*- α -tocopherol.⁴⁴ Thus, twice as much synthetic (*all rac*) compared with natural (RRR- α -tocopherol) is needed to meet the vitamin E requirement.

Vitamin E supplements are often sold as esterified forms such as α -tocopheryl acetate, succinate, or nicotinate and are typically racemic mixtures (*all rac*). Esterification with the hydroxyl group on the chromanol head of α -tocopherol prevents vitamin E from acting as an antioxidant and thus prolongs its shelf life. These esters are readily hydrolyzed in intestinal lumen to permit absorption of α -tocopherol.⁴⁵

II. NUTRIENT METABOLISM

A. Digestion and Intestinal Absorption

The role of fat in α -tocopherol absorption

Fat digestion and the formation of micelles are critical factors for vitamin E absorption, with all forms of natural and synthetic vitamin E equally well-absorbed (Fig. 7.3).^{46–48} Pancreatic and biliary insufficiency due to cystic fibrosis⁴⁹ or severe pancreatic exocrine insufficiency⁵⁰ can cause vitamin E malabsorption in humans. Similarly, defects in bile acid synthesis or secretion limit vitamin E absorption.⁵¹ Novel vitamin E formulations have demonstrated that micellarization of vitamin E is necessary for uptake into enterocytes.⁵² Moreover, vitamin E bioavailability is highly influenced by prandial status⁵³ in a process that takes several hours or longer to complete.⁵⁴

Depending on the meal fat content, vitamin E bioavailability derived from foods naturally containing vitamin E, present as a food fortificant, or consumed as a supplement is different. For example, dietary fat (40% vs. 0% fat)

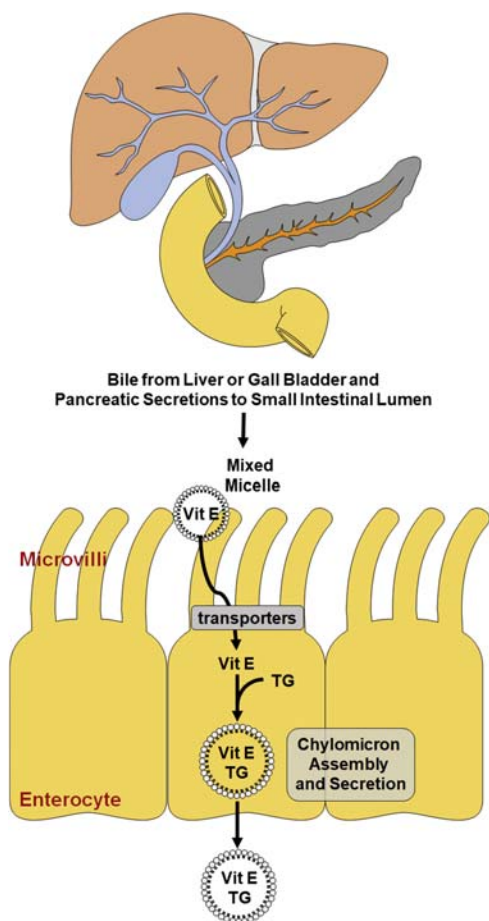


FIGURE 7.3 Vitamin E absorption. The various dietary forms of vitamin E (tocopherols and tocotrienols, see Fig. 7.1) are absorbed by the intestine due to the concerted actions of bile acids and pancreatic secretions that facilitate vitamin E micellization. Uptake of vitamin E at enterocytes is likely mediated by transporters (Niemann-Pick C1 like 1, scavenger receptor class B type I, cluster of differentiation 36, and ATP-binding cassette transporter A1) and then subsequently secreted from enterocytes as part of chylomicrons. During chylomicron catabolism, some vitamin E is transferred to circulating lipoproteins; the remainder is delivered to the liver.

consumed in a breakfast with deuterated α -tocopherol did not influence α -tocopherol absorption.⁵⁴ This “lack of fat effect” also occurred when α -tocopherol bioavailability was estimated in response to its coingestion with dairy milks containing different levels of milk fat (0.2–7.9 g).⁵⁵ Thus, unesterified α -tocopherol apparently does not require fat for absorption when a meal is consumed 4–12 h subsequent to dosing.⁵⁴

There was no effect of fasting for 24 h following α -tocopherol dosing with dinner in participants consuming a 15% fat diet for 5 weeks compared to a baseline 35% fat diet.⁵⁶ Apparently, 15% meal fat was sufficient for absorption. By contrast, when a 0% fat meal was followed by a 12 h fast, deuterated RRR - α -tocopherol absorption was delayed but was not significantly different from when meals were

eaten.⁵⁴ Apparently, the α -tocopherol accumulates in enterocytes until the next meal is consumed. During a subsequent meal, fat absorption hypothetically allows vitamin E that has been retained in enterocytes, likely in lipid droplets that coalesce with nascent chylomicrons, enriching their triglyceride contents with vitamin E, and thus facilitating vitamin E absorption.⁵⁴ A similar process has been shown for cholesterol⁵⁷ and for carotenoids.⁵⁸

The above studies used stable isotopes to trace RRR - α -tocopherol absorption. By contrast, the increase in α - and γ -tocopherol absorption from salad with 3 g canola oil in response to 0 g, 75 g, or 150 g cooked eggs was measured in the triglyceride-rich fraction of plasma in the subsequent 10 h.⁵⁹ The area under the curves for both tocopherols increased ~ 4 -fold with the high-egg meal compared with the other two meals. These data suggest that dietary components in addition to fat present in eggs (e.g., protein, cholesterol, and/or phospholipids) may promote vitamin E absorption.

The role of fat in α -tocopheryl acetate bioavailability

Dietary fat is required to stimulate mechanisms that hydrolyze α -tocopheryl acetate, promote enterocyte α -tocopherol uptake, and facilitate α -tocopherol secretion in chylomicrons.⁴⁶ When participants consumed either α -tocopherol or α -tocopheryl acetate with a fat-containing breakfast, there were no effects on the relative bioavailabilities.⁴⁵ However, when consumed without fat, α -tocopheryl acetate absorption was poor,⁶⁰ likely due to the necessity for biliary and pancreatic secretions for hydrolysis of the α -tocopheryl acetate.⁵⁸ Studies using apples fortified with deuterium-labeled RRR - α -tocopheryl acetate demonstrated that absorption increased linearly when meal fat content increased from 0 to 11 g.⁶¹ Thus, dietary supplements and fortificants containing α -tocopheryl acetate should be consumed with a meal containing fat.

Mechanisms of uptake into enterocytes

The major steps from micellar uptake to enterocyte trafficking and chylomicron incorporation are largely unknown. Vitamin E uptake into enterocytes can be facilitated by a number of transporters, including scavenger receptor class B type I (SR-B1), cluster of differentiation 36 (CD36), and ATP-binding cassette transporter A1 (ABCA1).^{58,62} α -Tocopherol binding at the N-terminal portion of the Niemann-Pick C1 like 1 (NPC1L1) transporter⁶³ also promotes its uptake into intestinal cells via the NPC1L1.⁶⁴ The steps that have been studied have evaluated α -tocopherol, but these are generally nonspecific mechanisms that facilitate fat or cholesterol absorption, so it is likely that all vitamin E forms are likely transported by these mechanisms.

Role of chylomicron secretion

Lipid droplets are rapidly formed inside enterocytes during fat consumption.⁶⁵ These lipid droplets transiently store triglyceride and thus likely function to limit extreme plasma triglyceride increases in response to a fat-rich meal.⁶⁶ The process of incorporating lipid droplets into chylomicrons is dependent upon Mfge8 (milk fat globule epidermal growth factor–like 8).⁶⁷ Thus, triglycerides are retained in enterocytes rather than being immediately secreted in chylomicrons.⁶⁷ In mice, apolipoprotein B48 deletion prevents chylomicron formation, allows large accumulations of lipid droplets in enterocytes, and limits α -tocopherol absorption,⁶⁸ thereby demonstrating the necessity for chylomicron formation for vitamin E absorption. Notably, persons with defective chylomicron secretion (e.g., abetalipoproteinemia or chylomicron retention disease⁶⁹) become severely vitamin E deficient with neurologic impairments if not given large vitamin E supplements⁷⁰ (see Section III A).

B. Hepatic α -Tocopherol Transfer Protein

Once vitamin E–containing chylomicron remnants reach the liver, the hepatic α -tocopherol transfer protein (α -TTP) mediates the preferential secretion of α -tocopherol into very low-density lipoproteins (VLDL)⁷¹ (Fig. 7.4), as reported in patients with genetic α -TTP defects^{72,73} and in α -TTP null mice.⁷⁴

α -TTP structure

Liver α -TTP has been isolated, its cDNA sequences reported and crystallized, and the α -tocopherol-binding pocket identified.^{75,76} α -TTP has high specificity for α -tocopherol compared with other tocopherols and tocotrienols,⁷⁷ which results from the specific α -tocopherol binding in α -TTP's hydrophobic pocket.⁷⁶ The pocket recognizes the three methyl groups on the chromanol head, the hydroxyl group, and the 2-R conformation of the natural α -tocopherol. The pocket causes the phytol tail of α -tocopherol to fold such that the 2-position on the chromanol ring is critical for the fit into α -TTP⁷⁶ and is the basis for only 2R- α -tocopherols (RRR, RSR, RSS, RRS, of which only RRR is naturally occurring in plants) contributing to meet the human requirement for vitamin E.¹²

α -TTP function

α -TTP belongs to a family of hydrophobic, ligand-binding proteins that have a *cis*-retinal binding motif sequence (CRAL_TRIO), all of which can bind α -tocopherol, but only α -TTP appears to have sufficient affinity to be a physiological mediator of its transport.⁷⁷ Thus, discrimination between vitamin E forms takes place in

the liver through the specificity of α -TTP,⁷⁸ but not in the intestine because nascent chylomicrons are formed from retained lipid droplets that contain various vitamin E forms,⁵⁴ as discussed above.

Studies to elucidate the mechanism by which α -TTP facilitates α -tocopherol secretion from the hepatocyte have used fluorescent α -tocopherol analogs to follow trafficking⁷⁹ and assess α -TTP ligand binding.⁸⁰ α -Tocopherol bound to α -TTP is exchanged for phosphatidylinositol phosphate in the membrane.^{78,81} Subsequently, ABCA1^{82–84} transfers α -tocopherol from the membrane to an acceptor lipoprotein, possibly either VLDL or HDL, outside the hepatocyte (Fig. 7.4). ABCA1 also mediates the α -tocopherol efflux from peripheral cells to HDL, similarly to “cholesterol reverse transport.”⁸² The precise mechanism of how α -TTP transfers α -tocopherol to acceptor lipoproteins, the specific acceptor, and the genetic abnormalities that result in abnormal transfer are under intense study.

C. Lipoprotein Transport

Transfer from VLDL to other lipoproteins

Since α -tocopherol is preferentially secreted from the liver in VLDL into the plasma, this is the mechanism by which the plasma becomes α -tocopherol–enriched relative to the other forms of vitamin E.^{72,73} Thus, the hepatic α -TTP is critical for plasma α -tocopherol enrichment (see Section III A, below).

During VLDL metabolism, α -tocopherol is distributed to all of the circulating lipoproteins^{85,86} (Fig. 7.4). Specifically, VLDL are delipidated to form LDL and the surface is transferred to HDL.⁵⁴ When deuterated α -tocopherol was administered intravenously (IV) in a lipid emulsion, the lipoprotein trafficking patterns differed between α -T_{IV} versus α -T_{oral}. The α -T_{IV} dose was rapidly cleared (<10 min) from plasma. In contrast, α -T_{oral} resulted in enrichment of deuterium-labeled α -tocopherol in the triglyceride-rich lipoprotein fraction that was more than three times greater than those of HDL or LDL. This enrichment was attributed to the oral α -tocopherol dose likely being taken up by enterocytes and retained until sufficient dietary fat in a subsequent meal was consumed to promote chylomicron secretion^{68,87} and subsequent VLDL secretion as described above.

Delivery to tissues

Unlike other fat-soluble vitamins that have specific plasma transport proteins, all lipoproteins transport vitamin E, and all mechanisms for delivery of lipids from lipoproteins to tissues (e.g., receptors and transporters) deliver vitamin E along with the other lipoprotein contents to tissues.⁴⁶

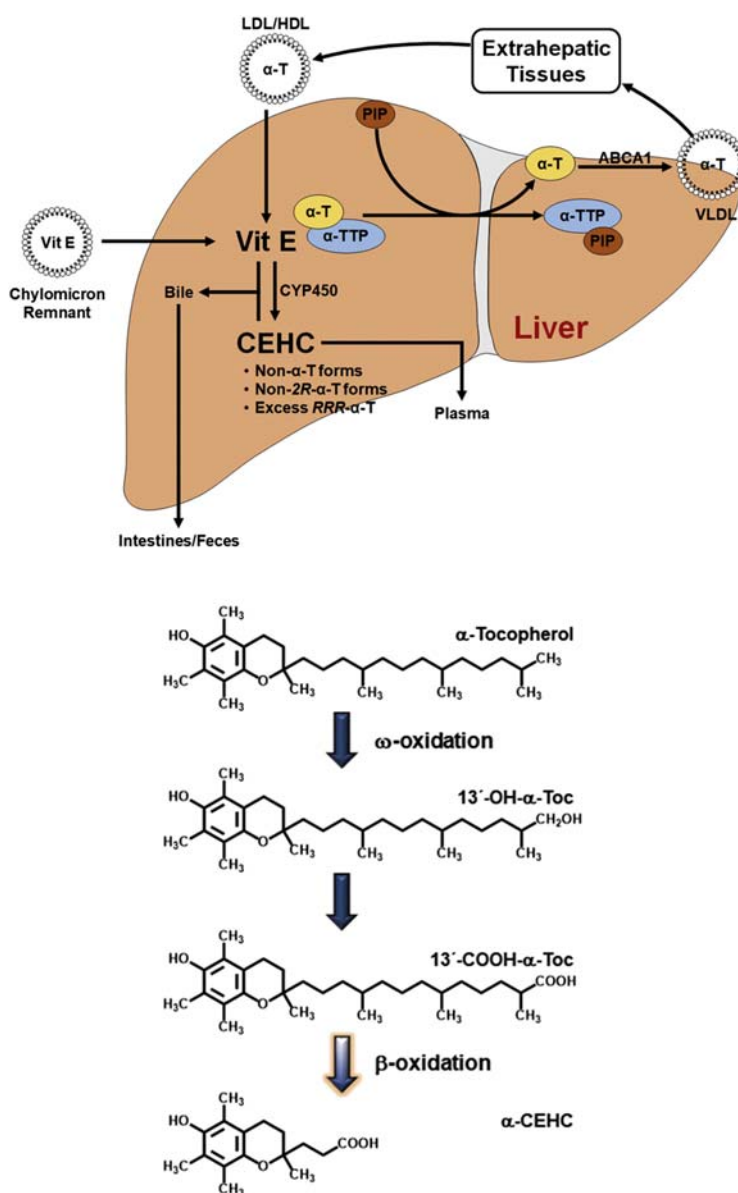


FIGURE 7.4 Hepatic vitamin E metabolism. Chylomicron remnants deliver vitamin E to the liver. The hepatic α -tocopherol transfer protein (α -TTP) discriminates among the various vitamin E forms to facilitate the preferential secretion of RRR- α -tocopherol into plasma as part of VLDL. α -TTP is thought to deliver α -tocopherol to the hepatocyte plasma membrane in exchange for membrane-derived phosphatidylinositol phosphate (PIP).⁷⁸ ATP-binding cassette transporter A1 (ABCA1) then facilitates α -tocopherol efflux into nascent VLDL that transports α -tocopherol to extrahepatic tissues. Because of their limited binding affinity to α -TTP, non- α -tocopherol forms along with non-2R-forms of α -tocopherol and excess RRR- α -tocopherol are directed for biliary elimination or cytochrome P450-mediated catabolism to corresponding carboxyethyl hydroxychromanol (CEHC) forms. In the lower panel, α -tocopherol is shown as an example to illustrate the initial steps in catabolism. α -Tocopherol is initially catabolized by ω -oxidation to form 13'-OH- α -tocopherol, and following five cycles of β -oxidation that successively truncate the phytol tail, the terminal catabolite, α -CEHC, is formed. The tocotrienol-derived intermediates differ subtly as a result of the double bonds in the tail (for additional structures, see Ref. 96).

Because all forms of vitamin E are absorbed, transported by chylomicrons, and delivered by lipoproteins to tissues, forms other than α -tocopherol are detectable in tissues.⁸⁸ However, hepatic catabolism and excretion of non-2R- α -tocopherol forms (such as γ -tocopherol and SRR- α -tocopherol) result in their relatively rapid elimination (see Section II E). This rapid excretion was

demonstrated in rats fed with α -tocotrienol⁸⁹ and in humans provided with deuterium-labeled γ -tocopherol⁴⁷ or SRR- α -tocopherol.⁸⁵ Moreover, sesamin in sesame oil, which interferes with the catabolism of γ -tocopherol,⁹⁰ slowed plasma γ -tocopherol disappearance.⁹¹ Thus, these data emphasize that the preferential catabolism of non-2R- α -tocopherol forms of vitamin E

is another mechanism that promotes preferential enrichment of the plasma with 2*R* forms of α -tocopherol.

D. Vitamin E Kinetics

Half-life of vitamin E isomers

Vitamin E kinetic models have been described.^{54,92,93} In normal subjects, the fractional disappearance rates of *RRR*- α -tocopherol (0.4 ± 0.1 pools per day) were significantly ($P < .01$) slower than for the synthetic form, *SRR*- α -tocopherol (1.2 ± 0.6). The apparent half-life of *RRR*- α -tocopherol was about 48 h, while *SRR*- α -tocopherol had a half-life of approximately 13 h.⁹² Using radioactive *RRR*- α -tocopherol (1.8 nmol) and complete urine and fecal collections for 21 d after dosing, with serial blood samples over 70 d after dosing in 12 healthy adults, Novotny et al.⁹³ estimated that adipose tissue was the slowest compartment for turning over *RRR*- α -tocopherol, with a residence time of 499 ± 702 d and a half-life of 184 ± 48 d. This study estimated that the entire body contains ~ 26 mmol α -tocopherol.⁹³

Kinetics of α - and γ -tocopherols have been compared.⁴⁷ Plasma *RRR*- γ -tocopherol exponential disappearance rates (1.39 ± 0.44 pools per day) were much faster than those of *RRR*- α -tocopherol (0.33 ± 0.11 , $P < .001$). Indeed, the γ -tocopherol half-life was 13 ± 4 h compared with 57 ± 19 h for *RRR*- α -tocopherol. Thus, *RRR*- α -tocopherol is retained in the plasma $\sim$ four times longer than *SRR*- α -tocopherol⁹² or *RRR*- γ -tocopherol.⁴⁷ The similarity in the disappearance rates for γ -tocopherol and *SRR*- α -tocopherol strongly supports the view that all the non-2*R*- α -tocopherol forms are excreted or catabolized.

Quantitation of α -tocopherol absorption

To assess α -tocopherol absorption in humans, two different deuterium-labeled α -tocopherols were used, in which one was administered IV and the other orally.⁵⁴ α -Tocopherol absorption by healthy women following a 40% fat breakfast and meal eating 4 and 8 h later was $55\% \pm 3\%$. These levels of α -tocopherol absorption are higher than those estimated when only oral deuterium-labeled α -tocopheryl acetate was administered⁶¹ and lower than when a very small dose of radiolabeled α -tocopherol was administered orally.⁹³

E. Vitamin E Catabolism

Structure and functions of catabolites

Vitamin E catabolites (α -CEHC and γ -CEHC [carboxyethyl hydroxychromanols]) are derived from α - and γ -tocopherol (as well as α - and γ -tocotrienols), respectively, and can be detected in urine, bile, and plasma,²¹ as well as liver homogenates⁹⁴ (Fig. 7.4). Intermediates of γ -tocopherol catabolism, including 9'-COOH-, 11'-

COOH, 13'-COOH, and 13'-OH- γ -tocopherol, have been identified⁹⁵; similar catabolites have been described for α -tocopherol.⁹⁶ Although CEHC is the terminal catabolite and is excreted in urine, the long-chain catabolites, such as 13'-carboxychromanol, are detectable in plasma and are excreted in feces.⁹⁷

These catabolites have some in vivo functions that are distinct from those of vitamin E. For example, γ -CEHC has natriuretic activity.^{98,99} The long-chain carboxychromanols, but not the chromanols, are inhibitors of cyclooxygenases.¹⁰⁰ 13'-Carboxychromanol inhibits lipooxygenase-5¹⁰¹ and alters sphingomyelin metabolism during inflammation.¹⁰² These reports have come from experimental animal models; further research is needed to evaluate these effects in humans.

Biochemistry of catabolism

Vitamin E catabolism is mediated initially by cytochrome P450s (CYPs). The tocopherols or tocotrienols are initially ω -hydroxylated, and CYP4F2 is the major P450 enzyme involved.^{103,104} However, there is cross talk between P450 enzymes, and CYP3A may also ω -hydroxylate when liver vitamin E concentrations are particularly high.¹⁰⁵ The gene for the mouse Cyp3a11 was induced in mice consuming diets with *RRR*- α -tocopheryl acetate (200–500 mg/kg diet).^{106,107} Additionally, when the mouse version of the CYP4F2 gene (Cyp4f14) was knocked down, other mechanisms partially took over catabolism.¹⁰⁸

Following ω -hydroxylation, β -oxidation takes place both in the mitochondria and the peroxisomes.¹⁰⁹

Conjugation of catabolites

The catabolites are conjugated with sulfate or glucuronide and excreted in urine or bile.²¹ Novel conjugates of α -CEHC in mouse and human urine including α -CEHC glycine, α -CEHC glycine glucuronide, and α -CEHC taurine have been reported.¹⁰⁶

Sulfation may be an important early step in intracellular trafficking to guide vitamin E catabolism, or at least γ -tocopherol catabolism. Sulfated- γ -CEHC is the major γ -CEHC conjugate, both in rats and in human cells in culture, and sulfated intermediates were found between 13'-OH- γ -tocopherol and γ -CEHC.^{110–112}

Kinetics of catabolism

Vitamin E catabolism is important in discriminating between various tocopherols and tocotrienols and thus also a key regulator of vitamin E bioavailability.¹⁰⁴ When equimolar amounts of deuterium-labeled tocopherols (~ 50 mg each d₆- α - and d₂- γ -tocopheryl acetates) were administered to humans, plasma disappearance rates of γ -CEHC did not differ from those of γ -tocopherol, but both were much faster than those of α -tocopherol.⁴⁷ Similarly, synthetic (*all rac*) as compared with

RRR- α -tocopherol was preferentially catabolized to α -CEHC by humans.¹¹³ Notably, tocotrienol turnover in human plasma is very rapid (<4 h),⁴⁸ and tocotrienols are rapidly catabolized to CEHCs.¹¹⁴

Because of their rapid catabolism, orally administered γ -tocopherol or tocotrienols do not accumulate appreciably in plasma and tissues.^{89,115} In humans, a 2-week daily supplementation of a γ -tocopherol-rich preparation (containing 623 mg γ -tocopherol, 61.1 mg α -tocopherol, 11.1 mg β -tocopherol, and 231 mg of δ -tocopherol per capsule) raised plasma γ -tocopherol concentrations to nearly those of α -tocopherol.¹¹⁶ However, circulating levels returned to baseline concentrations within 1 week of stopping the dosing. In addition, α -tocopherol supplementation decreased circulating γ -tocopherol by increasing γ -CEHC.¹¹⁷ Depletion of γ -tocopherol caused circulating levels of 5-nitro- γ -tocopherol that were otherwise elevated in smokers compared with nonsmokers to decrease. This decrease occurred in association with a tendency for plasma nitrite and nitrite concentrations to increase, suggesting that the availability of γ -tocopherol is needed to scavenge reactive nitrogen species.¹¹⁷

F. Liver Biliary Excretion

Vitamin E is not accumulated in the liver to “toxic” levels, suggesting that excretion and catabolism prevent adverse effects of vitamin E excess. α -Tocopherol is excreted into bile via multidrug resistance gene 2 (MDR2),¹¹⁸ an ATP-binding cassette transporter that also facilitates biliary phospholipid excretion.

HDL delivers α -tocopherol from the periphery to the liver via scavenger receptor-B type I (SR-BI).¹¹⁹ In SR-BI-null mice compared with wild-type mice, plasma α -tocopherol concentrations were increased, biliary α -tocopherol decreased, but liver α -tocopherol was unchanged¹²⁰; therefore, it appears that SR-BI-mediated hepatic α -tocopherol uptake is coupled to its biliary excretion.¹²⁰ Importantly, SR-BI protein is increased in vitamin E-deficient rats, suggesting that the liver increases SR-BI to increase hepatic α -tocopherol uptake.¹²¹ However, the regulation of hepatic vitamin E concentrations has not been extensively studied.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. α -Tocopherol Deficiency in Humans

α -TTP genetic defects

Although rare, severe α -tocopherol deficiency occurs in humans as a result of genetic abnormalities in α -TTP^{122–126} and is called “Ataxia with Vitamin E Deficiency” (AVED). Over 20 mutations in α -TTP resulting

in AVED have been reported.¹²² The most commonly encountered mutant is 744delA and affected persons are predominantly found in the Mediterranean region. The biochemical defect in AVED is attributed to the function of α -TTP that maintains plasma α -tocopherol concentrations via α -tocopherol secretion in VLDL.^{72,73} Extensive evaluation of phenotypic abnormalities and their relationship to specific α -TTP gene defects have been reported.¹²⁷ Ataxia observed in patients has also been mimicked in mice lacking α -TTP expression.¹²⁸

Persons with AVED have extraordinarily low plasma α -tocopherol concentrations (0–4 μ mol/L [normal, 20–30 μ mol/L plasma])¹²⁹ that can be restored within hours following the ingestion of α -tocopherol supplements.¹³⁰ However, circulating α -tocopherol concentrations revert to deficient levels within days of halting α -tocopherol supplementation. A dose of 800–1200 RRR- α -tocopherol mg/day is usually sufficient to prevent further deterioration of neurologic function, and in some cases improvements have been noted.^{131–133}

Postmortem analysis of a person with AVED demonstrated that oral administration of 300 IU(135 mg)/day of *all rac*- α -tocopheryl acetate for 11 y from age 62 y resulted in an increase in his serum α -tocopherol concentrations (15.5 μ g/mL or 35 μ mol/L) to normal ranges, allowed brain α -tocopherol accumulation, and prevention of Purkinje cell loss.¹³⁴

Visual system

Eyes are an extension of the nervous system, and α -tocopherol is particularly necessary for the maintenance of nerve function. Low serum α -tocopherol levels are associated with the development of retinitis pigmentosa and macular degeneration, as reported in a person with AVED.¹³⁵ Aged persons with macular degeneration were benefited by a supplement cocktail that included α -tocopherol,¹³⁶ and a Cochrane database metaanalysis concluded that antioxidants may slow, but not prevent, macular degeneration.¹³⁷

Genetic defects in lipoprotein metabolism and fat malabsorption syndromes

Genetic abnormalities in either lipoprotein synthesis (e.g., hypobetalipoproteinemia) or the microsomal triglyceride transfer protein (e.g., abetalipoproteinemia) result in vitamin E malabsorption or abnormally low plasma α -tocopherol, respectively.¹³⁸ α -Tocopherol deficiency can also occur secondary to fat malabsorption (e.g., cholestatic liver disease).¹³⁸

Cystic fibrosis can also cause complications in both biliary and pancreatic secretions resulting in nutritional deficiencies.¹³⁹ α -Tocopherol is particularly relevant^{140–142} because its deficiency has serious clinical consequences, such as progressive peripheral neuropathy. Importantly,

α -tocopherol supplementation to cystic fibrosis patients can normalize their α -tocopherol status and deficiency symptoms.¹⁴²

B. Pregnancy, Neural Tube Defects, and the Brain

Vitamin E was discovered because it prevented embryonic mortality in rats.³ α -Tocopherol in humans is important for pregnancy because increased early miscarriage is associated with low circulating α -tocopherol concentrations.¹⁴³ α -Tocopherol deficiency has been recognized since the 1960s to cause cerebellar ataxia in humans.¹⁴⁴ In the 1980s, α -tocopherol was shown to be necessary for normal neurologic function in humans,⁷⁰ as well as in mice.^{128,145–147}

Neural tube defects

α -Tocopherol deficiency in rat embryos is associated with neural tube defects (NTDs).^{148,149} Similarly, mouse embryos exhibit NTDs when α -tocopherol delivery to target tissues is impaired due to abnormal lipoprotein production^{150–152} or deletion of lipoprotein transporters, such as SR-B1.¹⁵³ In contrast, higher α -tocopherol intakes in humans are associated with lower NTD incidence.¹⁵⁴ These observations raise public health concerns since estimates of inadequate dietary α -tocopherol intakes (based on the adult Estimated Average Requirement [EAR] in the United States and Canada of 12 mg/d) exceed 80% of the global ≥ 14 -year-old population,¹⁵⁵ as well as 98% of American women.¹⁵⁶

In early embryo development in humans, α -TTP is expressed in the yolk sac.¹⁵⁷ In zebrafish, α -TTP mRNA is detectable in fertilized eggs and increases dramatically by 48 h postfertilization (hpf), showing its importance for transferring α -tocopherol from the yolk to the developing embryo.¹⁵⁸ α -Tocopherol is necessary to prevent NTDs in zebrafish and in rodent embryos at the analogous times that NTDs occur in human embryos with inadequate folate status (10–19 hpf for zebrafish,¹⁵⁹ 9–12 days for rats,¹⁶⁰ and 17–30 days humans^{161–163}). Thus, α -TTP and α -tocopherol are critical during embryonic development, especially during neurulation (neural plate and tube formation).¹⁵⁸

C. Omics

Various “omics” technologies, including proteomics, lipidomics, and genomics, have been used to understand α -tocopherol’s function and mechanism of action. For example, (1)H-NMR metabolomics has been used to investigate the biochemical effects in livers of rats in response to long-term (6 months) feeding of an α - and

γ -tocopherol–deficient diet; four polar hepatic metabolites were affected.¹⁶⁴ Specifically, glucose was lower and creatine, phosphocholine, and betaine were higher in deficient compared with sufficient rats. Further, transcription factors controlling energy metabolism (PGC1 α , PPAR α , and PPAR γ) were lower and that of fatty acid translocase CD36 higher in deficient compared with sufficient rats, suggesting altered hepatic energy metabolism.¹⁶⁴

Lipidomics and metabolomics

Lipidomic studies performed using α -tocopherol–sufficient and α -tocopherol–deficient zebrafish demonstrated the importance of α -tocopherol adequacy to maintain arachidonic and DHA¹⁶⁵ and protect against lipid peroxidation, especially in the brain^{166,167}. Studies in α -tocopherol–deficient zebrafish embryos also demonstrated altered phospholipid metabolism, which contributed to the observed functional impairments (e.g., locomotor function).¹⁶⁸ α -Tocopherol–deficient zebrafish embryos experienced depletion of choline, in association with increased lipid peroxidation and NADPH depletion.¹⁶⁹ Overall, these embryonic metabolic changes in response to α -tocopherol deficiency resulted in altered cellular energy homeostasis¹⁶⁹ similar to that described in the α -tocopherol–deficient rats.¹⁶⁴

Proteomics

Proteomics have been applied to the study of the sufficiency of α -tocopherol and other micronutrients in individuals from developing countries. In undernourished Nepalese children, the identification of 4705 unique proteins led to the development of a regression model showing that apolipoprotein C-III was well-correlated with α -tocopherol status.¹⁷⁰ Separate studies in school-aged Nepalese children established a plasma α -tocopherome where 119 proteins were correlated with α -tocopherol status.¹⁷¹ Most of the identified proteins had functions relating to lipid transport, blood coagulation, innate host defense, neuronal function and repair, and homeostasis. However, six specific proteins accounted for 71% of the plasma variability in α -tocopherol and could predict the prevalence of α -tocopherol deficiency ($<12 \mu\text{mol/L}$) at 51.4% compared with a prevalence rate of deficiency at 54.8% when measured directly. These studies show the potential for the quantification of protein biomarkers to effectively predict micronutrient deficiencies. However, further studies and the advent of technologies that better integrate various “omics” data sets within and between experimental models and human cohorts are needed.

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Anemia

Anemia is reported as an α -tocopherol deficiency symptom in humans¹⁷² and animals.¹⁷³ α -Tocopherol deficiency in premature infants has been associated with hemolytic anemia¹⁷⁴ and is responsive to α -tocopherol supplementation, particularly in those babies with a hematocrit <26% and a high reticulocyte count, and is receiving supplemental oxygen.¹⁷⁵ Children with α -tocopherol deficiency secondary to cystic fibrosis were reported to have a faster erythrocyte turnover due to hemolysis, which could be normalized with α -tocopherol supplements.¹⁷²

Erythrocyte hemolysis in response to added peroxide was used to assess development of vitamin E deficiency in a long-term trial in men.¹⁷⁶ These latter data were used for the setting the Recommended Dietary Allowance (RDA) for vitamin E in 1968¹⁷⁷ and the Estimated Average Requirement as part of the Dietary Reference Intakes (DRI) process for the United States and Canada in 2000.¹²

Plasma or serum concentrations

Deficiency levels have been defined as plasma α -tocopherol <12 $\mu\text{mol/L}$, the cutoff at which the response to known supplemental α -tocopherol intakes prevented in vitro peroxide-induced erythrocyte hemolysis.¹² In the National Health and Nutrition Examination Survey (1999–2000), serum α -tocopherol concentrations were $30.09 \pm 0.45 \mu\text{mol/L}$ (mean $\pm$ SEM), while γ -tocopherol concentrations were $5.74 \pm 0.22 \mu\text{mol/L}$.¹⁷⁸ Serum α -tocopherol concentrations in the global population were 22.1 $\mu\text{mol/L}$ (5.3; 42.2) (median [minimum, maximum]), as reported in a systematic review of α -tocopherol status, where 61% of estimated intakes were below 12 mg α -tocopherol/day.¹⁵⁵ By contrast, in France, the mean α -tocopherol was 29.8–33.4 $\mu\text{mol/L}$ with 5% CI ranging from 16–47, but average intakes were also estimated to be 12 mg α -tocopherol/day.¹⁷⁹

Normal circulating (either plasma or serum) α -tocopherol concentrations vary with serum lipids,¹⁸⁰ as well as with α -tocopherol intakes.¹⁸¹ Thus, as lipids increase with age, plasma α -tocopherol concentrations also increase.¹⁸²

Dietary intakes

Methods to determine dietary α -tocopherol intakes are so inaccurate that they vary little, while serum α -tocopherol concentrations can triple. For example, in 4087 adults aged >20 y who participated in the National Health and Nutrition Examination Survey (1999–2000), the mean dietary vitamin E intakes for healthy

TABLE 7.1 Serum α -tocopherol and dietary α -tocopherol intake in non-supplement user subgroup, NHANES 1999–2000.

Estimated dietary α -tocopherol (mg/d)	Serum α -tocopherol ($\mu\text{mol/L}$)
6.8 ± 1.3	<14
6.4 ± 0.6	14 to <16
6.5 ± 0.6	16 to <18
6.6 ± 0.3	18 to <20
6.9 ± 0.2	>20

Adapted from Ref. [178].

participants (nonusers of supplements) in relation to serum concentrations ranged from 6.4 to 6.9 mg/d while serum α -tocopherol values ranged from less than 14 $\mu\text{mol/L}$ to greater than 20 $\mu\text{mol/L}$ (Table 7.1).¹⁷⁸ Thus, dietary intakes are not useful for predicting serum α -tocopherol concentrations.¹⁸¹

Peripheral nervous system

The target tissue in severe α -tocopherol deficiency in humans is the large-caliber, myelinated axons in peripheral sensory nerves that causes a progressive peripheral neuropathy.¹³¹ Axonal degeneration rather than demyelination is the primary sensory nerve abnormality.^{130,183} Potentially, nerve degeneration caused by lipid peroxidation can be reversed by adequate α -tocopherol concentrations.¹⁸⁴ This marker of nerve degeneration has not been used for assessment of adequacy.

B. Criteria for Adequacy

The lack of a specific α -tocopherol metabolic function has made assessment of α -tocopherol adequacy difficult. Adipose tissue α -tocopherol concentrations, which are depleted in deficiency¹⁸³ and vary with intake,^{185,186} have been suggested as markers of long-term α -tocopherol status in humans.¹⁸⁷ Limited assessment of measured vitamin E intakes with adipose tissue α -tocopherol levels has been reported in healthy persons.⁹³ Adipose tissue α -tocopherol concentrations have also been studied in men with prostate cancer¹⁸⁸ and in severely burned pediatric patients.¹⁸⁹

C. Assessment of Status

The α -tocopherol requirement in humans is based on the observation that only supplements containing α -tocopherol have been shown to reverse α -tocopherol deficiency symptoms, for example, delay or even halt progression of peripheral neuropathy in patients with AVED.¹²⁹

As mentioned earlier, the dietary α -tocopherol requirement is based primarily on the amount necessary to correct abnormal erythrocyte hemolysis in healthy subjects who had consumed experimental α -tocopherol-deficient diets for 5–7 years.¹² Repletion intakes resulting in plasma α -tocopherol concentrations $>12\ \mu\text{mol/L}$, the cutoff at which the response to known supplemental α -tocopherol intakes prevented in vitro peroxide-induced erythrocyte hemolysis, were used to estimate average requirements.

D. Reference Intakes

α -Tocopherol intake recommendations and reference values for Americans and Canadians were established as part of the DRI process and last reviewed in 2000 by the Food and Nutrition Board of the Institute of Medicine.¹² EAR is based on prevention of erythrocyte hemolysis and is shown in Table 7.2; for adults, the amount is 12 mg/d. Using a standard deviation of 10%, RDA for the United States and Canada is set at a level of daily α -tocopherol intake that will meet or exceed the needs of almost all (97.5%) of the population; for adults, the RDA is $\text{EAR} + 2\ \text{S.D.} = 15\ \text{mg/d}$. Thus, the RDA is expected to exceed the needs of almost all within a group.¹² Table 7.2 includes the IOM DRIs¹² for various age and gender groups.

E. Basis for Individual Variability in Requirements

Oxidative distress increases plasma α -tocopherol disappearance in endurance exercise athletes¹⁹⁰ and in cigarette smokers.¹⁹ Thus, increased oxidative distress may increase disease risk; however, measures of oxidative distress are frequently not obtained in studies. α -Tocopherol supplements have decreased heart attack risk in persons with demonstrably inadequate antioxidant protection. For example, haptoglobin 2–2 genotype results in a dysfunctional haptoglobin protein and causes increased oxidation by free heme.¹⁹¹ In a placebo-controlled study carried out in diabetic patients who have the haptoglobin 2–2 genotype, daily supplementation of 400 IU (268 mg) of α -tocopherol reduced cardiovascular events.¹⁹² Additionally, haptoglobin 2–2 genotype subjects are more susceptible to vitamin C deficiency due to the impaired haptoglobin antioxidant function.¹⁹³ The lower vitamin C status may also increase α -tocopherol depletion.^{18,19}

F. Tolerable Upper Intake Levels

The Tolerable Upper Intake Level (UL) reference value established through the DRI process for the United States and Canada for vitamin E in 2000 is

1000 mg for adults¹² (Table 7.2). Reports of adverse effects of only vitamin E supplements in humans¹⁹⁴ were sufficiently rare that data from multiyear studies in rats fed high dietary α -tocopherol levels were used to set the UL.¹² No UL was set for infants as food was recommended as the only vitamin E source for infants. Since there was evidence of hemorrhaging and necrotizing enterocolitis in premature infants,^{195,196} a UL of 21 mg/day was suggested for premature infants, extrapolated from the adult UL on the basis of body weight.

The UL set for vitamin E applies to any form of supplementary α -tocopherol, which would primarily include supplements and synthetic vitamin E used in fortified foods. The UL was defined as including both 2R- and 2S- α -tocopherols because all of the stereoisomeric forms in *all rac*- α -tocopherol (derived synthetically) are absorbed and delivered to the liver. The appropriate conversion factors for *all rac*- α -tocopherol are different from those above when estimating dietary intakes relative to requirements. In the past, recommended intakes for vitamin E were in terms of IU. In 1989, the Food and Nutrition Board adopted the use of α -tocopherol equivalents (α -TEs), based on the view that part of the vitamin E requirement could be met by other naturally occurring forms of vitamin E (β - and γ -tocopherol, as well as α -tocotrienol).¹⁹⁷

Based on new data as discussed earlier, since these other natural forms are not recognized by α -TTP for transport, and since they are quickly excreted, the unit term mg α -TE was dropped with the DRI review in 2000.¹² Thus, the vitamin E reference value unit is now mg α -tocopherol. The US Food and Drug Administration issued new label values in 2016 (Daily Value, DV) which now require the use of mg, setting the DV for labeling at 15 mg.⁴⁴ However, many dietary supplements have used the term IU, as do some other countries, so it is important to know the conversion required. The factors to convert IU to mg for the UL are 0.9 times the IU for *all rac*- α -tocopherol (synthetic) and 0.67 times the IU for *RRR*- α -tocopherol. The IOM UL amounts (1000 mg) given in IU are 1100 IU for *all rac*- α -tocopherol and 1500 IU for *RRR*- α -tocopherol.¹²

An expert review on vitamin E safety has been published after the UL was established,¹⁹⁸ finding little evidence of concern with high supplement intakes. However, reports from three clinical trials have suggested adverse vitamin E effects in humans under special circumstances.¹⁹⁹ A 3 year, double-blind trial of antioxidants (vitamins E and C, β -carotene, and selenium) in 160 subjects on simvastatin-niacin therapy showed that there was less benefit of the drugs in raising HDL cholesterol than was expected.²⁰⁰ An increase in clinical end points (arteriographic evidence of coronary stenosis, or the occurrence of a first cardiovascular event — death, myocardial infarction, stroke, or revascularization) was

TABLE 7.2 Criteria and dietary reference intakes for vitamin E (α -tocopherol) by life stage group.^a

Life stage	Criterion	EAR ^b (mg/d)	RDA ^c (mg/d)	AI ^d (mg/d)	UL ^e (mg/d)
Premature infants				—	21
0–6 months	Average vitamin E intake from human milk			4	
7–12 months	Extrapolation from 0 to 6 months AI			5	
1–3 years	Extrapolation from adult EAR	5	6		200
4–8 years	Extrapolation from adult EAR	6	7		300
9–13 years	Extrapolation from adult EAR	9	11		600
14–18 years	Extrapolation from adult EAR	12	15		800
>18 years	Intakes sufficient to prevent hydrogen peroxide-induced erythrocyte hemolysis <i>in vitro</i> in a human study	12	15		1000
<i>Pregnancy</i>					
≤18 years	Same as adolescent	12	15		800
19–50 years	Same as adult	12	15		1000
<i>Lactation</i>					
≤18 years	Adolescent EAR plus average amount of vitamin E secreted in human milk	16	19		800
19–50 years	Adolescent EAR plus average amount of vitamin E secreted in human milk	16	19		1000

^aAdapted from Food and Nutrition Board, Institute of Medicine, 2000 [12].

^bEAR, Estimated Average Requirement. The intake that meets the estimated nutrient needs of half the individuals in a group.

^cRDA, Recommended Dietary Allowance. The intake that meets the nutrient needs of almost all (97%–98%) of individuals in a group.

^dAI, Adequate Intake. The observed average or experimentally determined intake by a defined population or subgroup that appears to sustain a defined nutritional status. For healthy infants receiving human milk, the AI is the mean intake.

^eUL, Tolerable Upper Intake Level. The highest level of daily nutrient intake that is likely to pose no risk of adverse health effects in almost all individuals.

also reported.²⁰⁰ The Women's Angiographic Vitamin and Estrogen Trial was a randomized, double-blind trial of 423 postmenopausal women with at least one coronary stenosis at baseline coronary angiography. In the postmenopausal women on hormone replacement therapy, all-cause mortality was increased in women assigned to antioxidant vitamins compared with placebo (HR, 2.8; 95% CI, 1.1–7.2; $P = 0.047$).²⁰¹ The extension of the Canadian Heart Outcomes Prevention Evaluation (HOPE) trial in patients >55 years with vascular disease or diabetes mellitus (Hope-TOO) suggested that patients at high risk for coronary heart disease taking α -tocopherol were at increased risk of left ventricular dysfunction.²⁰² Interestingly, none of these trials had the same adverse effect of α -tocopherol therapy. Moreover, a meta-analysis evaluating the relationship of α -tocopherol supplements

with all-cause mortality did not associate any adverse effects with an α -tocopherol-specific mechanism.²⁰³

Perhaps, these disparate outcomes could be attributed largely to the selection of persons with underlying health conditions (e.g., cardiovascular disease, high risk for cancer, neurological disorders) or the difficulty in conducting timely placebo-controlled clinical interventions with a micronutrient that cannot be fully omitted from the diet because of its essentiality.

Studies using the metaanalysis approach to evaluating clinical trials have made claims from dramatically increased mortality risk to virtually no risk from antioxidant supplements.^{199,203–205} In fact, the use of randomized clinical trials to assess nutrient benefits has been called into question, given the fact that the known adverse effect of a nutrient in humans results from its

deficiency and that subjects in the placebo group may not consume inadequate intakes.²⁰⁶ Others have suggested that a risk/benefit ratio be used to evaluate α -tocopherol effects.¹⁹⁹

G. Dietary Sources

γ -Tocopherol is the most abundant form of vitamin E in the US diet.²⁰⁷ However, α -tocopherol, not γ -tocopherol, and specifically only the 2R-forms of α -tocopherol, contributes to meeting human vitamin E requirements.¹² Based on the We Eat in America NHANES 2013–16 cohort, only 9.5% of males and 7.9% of females in the United States had adequate usual intakes of α -tocopherol.¹⁵⁶ This low level of adequacy may be due to most persons obtaining dietary vitamin E from high-energy, high-fat foods that are not particularly rich in α -tocopherol.¹⁵⁶ Some examples of vitamin E food sources are shown in Table 7.3. As shown in the table, excellent sources of natural α -tocopherol are many oils, tree nuts, some vegetables, and seafood; by contrast, French fries are rich in γ -tocopherol. The USDA Dietary Guidelines for

Americans (2015), while recommending reducing dietary fat intakes, did not provide a recommendation to optimize α -tocopherol intakes, despite it being a short-fall nutrient.²⁰⁸

V. ISSUES RELATED TO VITAMIN E IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

α -Tocopherol deficiency due to genetic defects in lipoprotein metabolism and fat malabsorption syndromes

Genetic abnormalities in either lipoprotein synthesis (e.g., homozygous hypolipoproteinemia) or the microsomal triglyceride transfer protein (i.e., abetalipoproteinemia) result in α -tocopherol malabsorption or abnormally low plasma α -tocopherol, respectively.¹³⁸ α -Tocopherol deficiency can also occur secondary to fat malabsorption (e.g., cholestatic liver disease).¹³⁸

Cystic fibrosis can also cause complications in both biliary and pancreatic secretions resulting in nutritional deficiencies.¹³⁹ α -Tocopherol is particularly relevant because its deficiency has serious clinical consequences such as hemolytic anemia, peripheral neuropathy, and decreased cognitive function,^{140–142} but α -tocopherol supplementation can normalize their status.¹⁴²

Metabolic syndrome and fatty liver disease

Metabolic syndrome (MetS) afflicts 35% of Americans and >50% of those over 60 y of age²⁰⁹ and is a known precursor to more serious cardiometabolic diseases.^{210–214} α -Tocopherol bioavailability in persons with MetS is impaired compared with healthy adults⁵⁵; possibly, α -tocopherol is “trapped” in steatotic enterocytes that limit α -tocopherol secretion in chylomicrons.⁵⁵ Such individuals also had decreased hepatic α -tocopherol secretion in VLDL, consistent with (1) limited hepatic α -tocopherol delivery from gut-derived chylomicrons, (2) increased oxidative depletion, or (3) potentially “trapping” of α -tocopherol in steatotic livers.²¹⁵

Nonalcoholic fatty liver disease (NAFLD) is the hepatic manifestation of MetS and afflicts 80–100 million Americans.^{216,217} NAFLD is a precursor to nonalcoholic steatohepatitis (NASH), in which fatty liver leads to cellular injury and inflammation with increased oxidative damage. Although circulating α -tocopherol concentrations in NASH patients are sometimes lower than in healthy persons,^{218,219} their levels by standard metrics are not considered deficient. However, dietary supplementation with pharmacologic α -tocopherol doses reduced NASH severity in children and adults.^{220–223}

TABLE 7.3 α - and γ -Tocopherol (T) contents of a serving of selected foods in decreasing order of α -tocopherol content per serving.¹

Food	Serving size	α -T (mg)	γ -T (mg)
Sunflower seeds, dry roasted	1 oz	10.30	0.13
Spinach, raw	1 cup	6.90	0.61
Almonds	1 oz	6.78	0.20
Oil, sunflower	1 tbsp	5.59	1.24
Hazelnuts	1 oz	4.26	0.00
Tomato sauce	1 cup	3.53	0.24
Asparagus, cooked	1 cup	2.70	0.38
Potato, French-fried	1 large	2.57	5.59
Oil, canola	1 tbsp	2.44	3.83
Oil, peanut	1 tbsp	2.12	2.15
Oil, olive	1 tbsp	1.94	0.11
Broccoli, chopped, raw	1 cup	1.90	0.42
Margarine	1 tbsp	1.82	4.09
Salmon (sockeye)	3 oz	1.60	NR ²
Oil, soybean	1 tbsp	1.11	8.74
Quinoa, cooked	1 cup	1.17	2.20
Pecans	1 oz	0.40	6.93

¹Source: USDA National Nutrient Database for Standard Reference Legacy Release, April 2018.

²NR, values not reported.

Chronic disease protection and public health implications

Is there benefit to α -tocopherol supplements? The questions would be easier to answer if there were specific metabolic pathways that required α -tocopherol such that a marginal deficiency could be defined. Certainly, signaling pathways and specific genes have been identified, which are altered in response to α -tocopherol concentrations,²²⁴ but these “molecular” effects are difficult to distinguish from changes in oxidative distress-dependent mechanisms.²²⁵ Further, while circulating α -tocopherol can be routinely measured, its assessment may be limited for evaluating vitamin E sufficiency.

Complementary biomarkers have been considered. In the Energetic Study, urinary α -CEHC was correlated with plasma α -tocopherol, and the inflection point in which α -CEHC increased occurred at an intake level of 12.8 mg of α -tocopherol,¹⁸¹ which is close to the current EAR of 12 mg/d.¹² Urinary α -CEHC was also shown to increase in response to 16-week daily consumption of α -tocopherol-rich hazelnuts, which occurred without affecting circulating α -tocopherol concentrations.²²⁶ In contrast, persons with MetS had less urinary excretion of deuterium-labeled α -CEHC following oral ingestion of deuterium-labeled α -tocopherol,²¹⁵ in agreement with the concept of impaired gut–liver trafficking of α -tocopherol in MetS.⁵⁵ Taken together, measures of urinary α -CEHC may be a useful biomarker to assess α -tocopherol adequacy, especially in populations having underlying hyperlipidemias that increase circulating α -tocopherol while masking the fact that α -tocopherol is insufficient at target tissues. Measures of α -CEHC may also be helpful to assess the efficacy of α -tocopherol administration in clinical trials, but no large-scale studies to date have utilized this putative biomarker.

B. Known Interactions

Vitamin C

An α -tocopherol and vitamin C interaction has been demonstrated in humans under oxidative distress. Cigarette smokers with the lowest plasma ascorbic acid concentrations had the fastest plasma α -tocopherol disappearance rates.¹⁹ Furthermore, when cigarette smokers were supplemented with vitamin C, their α -tocopherol E disappearance rates returned to levels seen in nonsmokers.¹⁸ This change is attributed to vitamin C recycling α -tocopherol (Fig. 7.2), as discussed in earlier (see Section I B). Additionally, a combined α -tocopherol and vitamin C deficiency causes devastating neurologic damage in guinea pigs²²⁷ and increases the severity of atherosclerosis in susceptible mice unable to synthesize vitamin C.²²⁸

Vitamin K

High α -tocopherol intakes are associated with an increased tendency to bleed.¹² Increased bleeding may be a result of decreased platelet aggregation caused by an inhibition of protein kinase C by α -tocopherol,²²⁹ some other platelet-related mechanisms,^{230,231} or decreased clotting due to a vitamin K and α -tocopherol interaction²³² causing abnormal blood clotting.¹² Importantly, in the Women’s Health Study, α -tocopherol supplements (600 IU, 402 mg) (every other day for 10 years) decreased the incidence of thromboembolism.²³² Thus, there may be a benefit to α -tocopherol supplements, as well as harm, by altering blood clotting.

Selenium

Hypothetically, the tocopheroxyl radical is reduced by ascorbic acid forming the ascorbyl radical (Fig. 7.2), which is reduced by ascorbyl reductase using glutathione (GSH) as the reductant.¹⁷ The subsequent reduction of GSSG (the GSH oxidation product) uses endogenous reducing equivalents, primarily NADP(H).¹⁶ This close association between GSH and vitamin E is further emphasized by studies of GPx4, a selenium-dependent peroxidase that uses GSH to reduce phospholipid hydroperoxide.²³³ Further, a combined α -tocopherol–deficient and selenium-deficient diet caused a fatal myopathy in guinea pigs, while deficiency of either nutrient was nonlethal.²³⁴

The SELECT trial investigated the efficacy of selenium and α -tocopherol on the risk for developing prostate cancer in more than 35,000 healthy men.^{235,236} However, neither nutrient, alone or combined, was found to prevent prostate cancer. The study was stopped prematurely and participants were instructed to discontinue their use of the supplements. Paradoxically, follow-up analysis revealed that the incidence of prostate cancer was higher among those previously taking α -tocopherol supplements but not those taking selenium alone or in combination with α -tocopherol.¹⁸⁸

VI. FUTURE NEEDS

Long-standing questions persist for α -tocopherol, focusing on whether any specific nonantioxidant function exists for α -tocopherol and leveraging technological advances for the accurate determination of α -tocopherol requirements that not only meet basal needs for humans but also protect against chronic diseases and pathological conditions impacted by oxidative distress.

Many of significant discoveries in the 21st century about α -tocopherol have been made possible by the use of stable-isotope labeled α -tocopherol. Advances have included

accurate measures of intestinal absorption, interactions of α -tocopherol with dietary components that potentiate bioavailability (e.g., fat) as well as nutrients that alleviate its oxidative turnover (e.g., vitamin C), and the adverse impact of oxidative distress and inflammation. Such adverse effects have been shown to not only accelerate α -tocopherol disappearance but also lead to a contemporary cardiometabolic disorder (i.e., MetS) that impairs gut-to-liver trafficking of α -tocopherol. Advances in the α -CEHC measures and proteomics-based approaches have also revealed novel biomarkers that can be potentially used to establish α -tocopherol nutriture. On the immediate horizon is to utilize new experimental findings to support improved estimates of dietary α -tocopherol requirements

previously established in large part on data generated from a few subjects who underwent multiyear α -tocopherol depletion–repletion investigations.

Critical questions remain including “how is adequacy defined?” and “for what function?”. These answers are further complicated by the fact that ~76% of Americans appear to have inadequate diets but do not experience any overt or easily measurable clinical or biochemical signs of α -tocopherol deficiency. Continued study is also needed to reconcile the largely equivocal outcomes of large-scale clinical trials involving α -tocopherol supplements. These questions along with growing consumer interest in personalized nutrition are expected to be at the forefront of research for the next several decades.

RESEARCH GAPS

- Identify specific nonantioxidant functions or roles as well as further elucidate transport mechanisms
- Establish novel technologies to assess vitamin E adequacy in relation to vitamin C and other nutrients in the antioxidant network based on these functions and roles
- Refine estimates of dietary requirements for healthy persons, and consider influences of oxidative stress, metabolic disorders, and interactions with other nutrients
- Identify benefits of α -tocopherol in chronic disease risk, especially fatty liver disease

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VITAMIN K

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SUMMARY

Historically discovered for its role in blood coagulation, vitamin K is emerging as a vitamin of wide-ranging physiological implications. Whether through the vitamin K–dependent protein family or specific K vitamers, vitamin K is now known to be involved in bone and cardiovascular metabolism, inflammation, and brain and endocrine function. This chapter discusses the most recent scientific literature relating to vitamin K and points to ongoing and future research efforts.

Keywords: Anticoagulant therapy; Bone health; Cardiovascular disease; Coagulation; Menaquinones; Phylloquinone; Vitamin K.

I. INTRODUCTION

A. Background

The history of vitamin K dates back to 1929 when, as part of his work on sterol metabolism, Danish nutritional biochemist Henrik Dam observed that chicks fed fat-free diets developed subcutaneous hemorrhages and anemia. Further work by Dam determined that the antihemorrhagic substance was fat soluble and occurred in extracts of liver and various plant tissues. In 1935, Dam named this new substance vitamin K in reference to the first letter of the word “koagulation,” as spelled in the German and Scandinavian languages. By 1939, the two naturally occurring forms of the vitamin—vitamin K₁ and vitamin K₂—had been isolated from alfalfa and putrefied fish meal, respectively. In 1941, the first vitamin K antagonist (VKA) was discovered when a new substance identified in spoiled sweet clover hay had been reported to cause hemorrhagic disease in cattle in the United States and Western Canada in the 1920s. This substance, 3,3'-methyl-bis-(4-hydroxycoumarin), later became known as dicoumarol. In the period after the discovery of dicoumarol, several derivatives of coumarin were synthesized for use as clinical agents in anticoagulant therapy. One of these, warfarin (3-[*a*-acetylbenzyl]-4-hydroxycoumarin), has been used successfully as a clinical agent since

1941. Access to VKAs eventually helped specify the role of vitamin K in blood coagulation and proved to be invaluable to vitamin K research in general. A detailed account of the discovery of vitamin K has recently been published.¹

Although the bleeding condition reported by Dam was originally associated with decreased prothrombin (factor II) activity, it was later established that three other coagulation proteins (factors VII, IX, and X) were also depressed in vitamin K deficiency states. For many years, this participation in blood coagulation was assumed to be the sole physiological role for vitamin K. However, the discovery in the early 1970s of γ -carboxyglutamic acid (Gla), a new amino acid found in all vitamin K–requiring proteins, later led to the discovery of additional vitamin K–dependent proteins (VKDPs) not involved in hemostasis and greatly contributed to present understanding of the action of vitamin K at the molecular level. To this day, the participation of vitamin K in Gla synthesis remains the only well-defined function of this vitamin.²

B. Structure

Compounds with vitamin K activity have a common 2-methyl-1,4-naphthoquinone ring but differ in the structure at the 3-position. Vitamin K occurs naturally in two forms. Phylloquinone (2-methyl-3-phytyl-1,

4-naphthoquinone; also referred to as vitamin K₁) is synthesized in plants and represents the main source of dietary vitamin K in Western countries.³ Menaquinones (2-methyl-1,4-naphthoquinones; also referred to as vitamin K₂), are produced by bacteria and form a family of compounds with unsaturated isoprenyl side chains of varying length at the 3-position. The predominant forms of the menaquinone series contain 6–10 isoprenoid units, but menaquinones containing up to 13 units have been isolated.⁴ One of the menaquinones, MK-4, is not a common product of bacterial synthesis but is synthesized from phyloquinone with menadione as an intermediate (see later in this chapter).

The parent structure of all K vitamins, 2-methyl-1,4-naphthoquinone, also called menadione or vitamin K₃, does not occur in nature but can be alkylated to MK-4 in avian and mammalian tissues. This synthetic form has been used as a source of vitamin K in a wide range of animal feeds.⁴ The different K vitamers are illustrated in Fig. 8.1.

Current methods to assess vitamin K in plasma, biological tissues, and foods are based on high-performance liquid chromatography (HPLC). Vitamin K is first extracted from matrices with organic solvents and then purified by solid-phase chromatography, before being selectively isolated by HPLC. Quantitation is usually achieved using fluorescence detection, and procedures enabling simultaneous determination of phyloquinone and the menaquinones have been developed and applied reliably. Methods combining chromatographic and mass spectrometry techniques have also been developed and used in studies involving labeled phyloquinone. A review describing methods for sample pretreatment and quantitation of the K vitamers in various biomatrices has recently been published.⁵

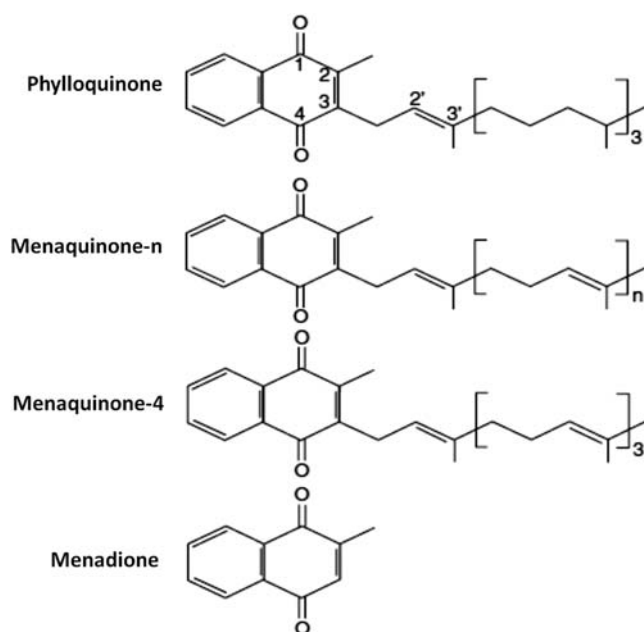


FIGURE 8.1 Chemical structures of phyloquinone, menaquinones, menaquinone-4 (MK-4), and menadione.

In 2009, a vitamin K external quality assurance scheme was established to assist in the harmonization of methods for vitamin K analyses among laboratories worldwide and to improve the comparability of clinical and nutritional studies.⁶ Finally, as part of metabolic studies, methods were developed to assess urinary menadione⁷ and the various phyloquinone metabolites.⁸

C. Biochemical and Physiological Function

Vitamin K–dependent carboxylation

Over 40 years after its discovery, vitamin K was shown to act as a cofactor in the posttranslational synthesis of Gla from glutamic acid (Glu) residues contained in precursor proteins. γ -Carboxyglutamic acid is common to all VKDPs and increases the affinity of these proteins for calcium. As illustrated in Fig. 8.2, the γ -carboxylation of glutamate residues is catalyzed by a microsomal enzyme called γ -glutamyl carboxylase (GGCX) located at the luminal surface of the endoplasmic reticulum in a reaction that requires the reduced form of vitamin K, hydroquinone, and carbon dioxide and oxygen. It is currently believed that the vitamin K hydroquinone and oxygen react to form a strong base capable of abstracting a proton from the γ -carbon of the Glu residue to form a carbanion intermediate, which then undergoes carboxylation to yield a Gla residue.⁹ The carboxylase hence uses the energy of vitamin K hydroquinone oxygenation to convert Glu residues to Gla residues in VKDPs. Carboxylation of the Glu residues is accomplished by a processive mechanism (defined

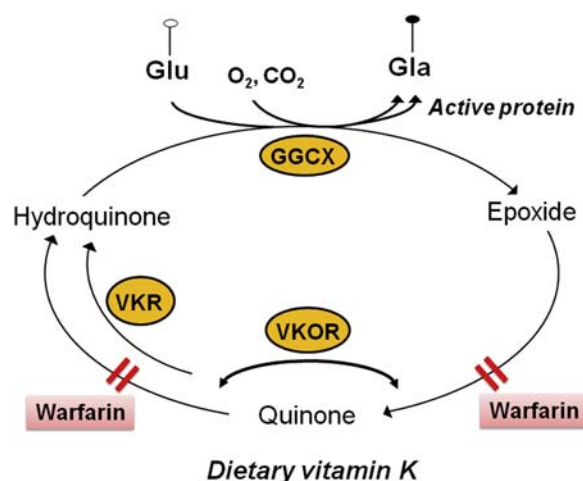


FIGURE 8.2 The vitamin K cycle. Carbon dioxide is incorporated into glutamic acid residues on inactive vitamin K–dependent proteins (VKDPs) in the presence of oxygen with the reduced form of vitamin K, hydroquinone, acting as a cofactor to the enzyme γ -glutamyl carboxylase (GGCX) to form γ -carboxyglutamyl (Gla) residues, thus activating the proteins. The resulting vitamin K 2,3-epoxide metabolite is then reduced back to vitamin K quinone by VKOR, the vitamin K oxidoreductase enzyme; its action is inhibited by coumarin-type drugs such as warfarin. VKR refers to the vitamin K reductase enzyme, which reduces vitamin K to hydroquinone; it is also inhibited by drugs such as warfarin.

as an enzyme's ability to catalyze consecutive reactions without releasing its substrate) and is facilitated by a carboxylase recognition signal propeptide that tethers the Glu residues to the enzyme (reviewed in Ref. 10).

The γ -carboxylation reaction results in the generation of the vitamin K 2,3-epoxide metabolite, which is reduced to vitamin K quinone by the enzyme vitamin K oxidoreductase (VKOR). The quinone is then reduced to its bioactive hydroquinone form by VKOR, but more likely by a single (or multiple) yet unidentified vitamin K reductase enzyme(s) (VKR).

Activity of VKOR is known to be inhibited by 4-hydroxycoumarin derivatives such as warfarin.⁴ The blocking action of these drugs (also referred to as VKAs) forms the basis of their pharmacological action as anticoagulants. The identity of VKR is not yet known; however, it is generally admitted that, at least in the liver, there is a reductase that is insensitive to warfarin. This comes from the observation that in cases of warfarin intoxication, the effect of the drug can be overcome by administering large doses of vitamin K. This enzyme operates at high tissue concentrations of vitamin K and can support carboxylation of the hepatic vitamin K-dependent coagulation factors (as reviewed by Tie and Stafford¹¹). Collectively, these reactions make up the vitamin K cycle (Fig. 8.2).

In the absence of vitamin K or in the presence of VKAs, carboxylation of precursor proteins is incomplete and proteins are secreted in plasma in various undercarboxylated forms. These proteins, referred to as PIVKAs (protein induced by vitamin K absence or antagonism), lack biological activity.

Vitamin K-dependent proteins

Although γ -carboxyglutamic acid residues have been detected in both vertebrate and invertebrate species, only proteins of mammalian origin will be discussed here.

Blood coagulation proteins; There are seven VKDPs involved in blood coagulation, namely prothrombin (factor II), factors VII, IX, and X, protein C (PC), protein S (PS), and protein Z. All the vitamin K-dependent coagulation proteins (46,000–80,000 Da) are synthesized in the liver and contain between 10 and 12 Gla residues. The Gla residues enable Ca^{2+} -mediated binding of the proteins to the negatively charged phospholipid surfaces provided by blood platelets and endothelial cells at the site of injury. With the exception of proteins S and Z, Gla-containing blood proteins are zymogen forms of serine proteases and have considerable structural homology.

Prothrombin was among the proteins associated with the coagulation cascade as described in the middle of the 19th century. The identification of factors VII, IX, and X came much later (1950s) through the study of patients with hereditary bleeding tendencies, and proteins C, S,

and Z were discovered in the mid-1970s. Prothrombin and factors VII, IX, and X represent the classical vitamin K-dependent plasma clotting factors and participate in the cascade resulting in the formation of a fibrin clot.

A key element in the formation of fibrin is the generation of thrombin from prothrombin by activated factor X. Vitamin K-dependent factors VII and IX activate factor X by the extrinsic and intrinsic pathways, respectively. The extrinsic pathway is activated by external trauma that causes blood to escape from the vascular system. This pathway is quicker than the intrinsic pathway. It involves tissue factor and factors VII and IX. The intrinsic pathway is activated by trauma inside the vascular system triggered by internal damage to the wall of a vessel and is activated by platelets, exposed endothelium, chemicals, or collagen. This pathway is slower than the extrinsic pathway, but it is more important. It involves factors VIII, IX, XI, and XII. Both pathways meet and finish the pathway of clot production in what is called the common pathway. The common pathway involves factors I (fibrinogen), II, V, and X.

In contrast, proteins C, S, and Z are inhibitors of the procoagulant system. PC exerts its inhibitory activity by inactivating factors Va and VIIIa and enhances fibrinolysis with PS as a cofactor, and protein Z serves as a cofactor for the inhibition of factor Xa by protein Z-dependent protease inhibitor¹ (see review by Davie¹² and O'Donnell et al.).¹³ An abbreviated diagram of the coagulation cascade is presented in Fig. 8.3.

In addition to their hemostatic action at the cell surface, many of the coagulation proteins are now known to also possess cell signaling activity and influence a wide range of cellular events. Hence, predominantly through its interaction with protease-activated receptors (PARs), thrombin has been involved in physiologic processes such as embryogenic development, angiogenesis, wound healing, and inflammation, as well as in pathological conditions such as atherosclerosis, sepsis, cancer, and neurodegeneration.^{14,15} Similarly, factor Xa interacts with PARs and has been shown to increase the expression of a wide range of inflammatory cytokines associated with conditions such as fibrosis and atherosclerosis.¹⁶ In contrast, PC and its activated form (APC) are cytoprotective and have been associated with antiapoptotic and antiinflammatory activities, favorable alterations of gene expression, and stabilization of endothelial barriers. Furthermore, APC-initiated signaling has been shown to support cell proliferation and differentiation, as well as angiogenesis, actions which are an integral part of tissue regeneration and wound healing.^{17,18}

In contrast to the other VK-dependent blood factors, which are principally associated with the liver, PS is synthesized by different cell types (e.g., megakaryocytes, endothelial and vascular smooth muscle cells [VSMCs], and osteoblasts) and has been found in many extrahepatic

The second vitamin K–dependent bone protein to be discovered, MGP, has a molecular weight of ~9.6 kDa and contains five Gla residues. In contrast to OC, which is exclusively associated with mineralized tissues, MGP is expressed in many soft tissues as well as bone, and in cell types including VSMCs, although the protein itself only accumulates in calcified tissues such as bone and cartilage²³ and calcified arterial vessel walls.²⁴ The physiological function of MGP as an inhibitor of calcification was established when mice that lacked MGP showed calcification of their arteries, which led to hemorrhagic death due to vessel rupture within 2 months. These animals also presented inappropriate calcification of various cartilages, which resulted in short stature, osteopenia, and fractures.²⁵

The third Gla protein present in bone, PS, is synthesized and secreted by osteoblasts²⁶ and has been shown to enhance the bone-resorbing activity of mature osteoclasts, an action deemed related to its interaction with TAM receptors.²⁷

Gas6

Discovered in 1993, Gas6, thus named after it was found to be the product of the growth arrest–specific gene 6, is a high molecular weight protein (75 kDa) that contains 11–12 Gla residues. It has a structure similar to that of PS (44% amino acid homology) and is expressed in many cell types, including neurons, VSMCs, mesangial cells, endothelial cells, bone marrow stromal cells, and macrophages, as well as in human platelets and plasma.^{28,29} Gas6 binds and activates the receptor tyrosine kinases of the TAM family (Tyro3, Axl, and Mer), a property that is dependent on its Gla residues.

Since its discovery, Gas6 has been associated with a wide range of cellular processes that include cell differentiation, proliferation and activation, cell adhesion and chemotaxis, phagocytosis, and protection against apoptosis. Physiologically, it has been involved in the nervous system and the retina, in platelet metabolism and hemostasis, in vascular smooth muscle homeostasis, in spermatogenesis, and in osteogenic differentiation and osteoclast function. In light of these actions, Gas6 has been linked to conditions such as inflammation, thrombotic events, atherosclerosis, glomerular kidney disease, and cancer. The multiple actions of Gas6 have recently been reviewed.^{30,31}

Gla-rich protein

In 2008, a new Gla-rich protein (GRP) was isolated from calcified cartilage. This 10.2-kDa protein contains 16 Gla residues, which is the highest proportion of Glas reported among the VKDPs.³² It is widely distributed but its expression is highest in cartilage with chondrocytes, chondroblasts, osteoblasts, and osteocytes expressing the protein. GRP has been shown to act as

an inhibitor of calcification in the cardiovascular and articular systems through a mechanism that could involve the inhibition of the calcium-induced signaling pathways and direct mineral binding to inhibit crystal formation/maturation. Recently, GRP was also shown to decrease proinflammatory responses in articular and immune cell systems. A review of the GRP functional activities has recently been published.³³

Transmembrane Gla proteins

Among the other VKDPs that have been identified are those of the transmembrane Gla family (TMG), which comprises four proteins: proline-rich Gla proteins 1 and 2 (PRPG1, PRPG2)³⁴ and transmembrane Gla proteins 3 and 4 (TMGP3 and TMGP4).³⁵ In contrast to the previously discussed VKDPs, these are not secreted but rather are single-pass integral membrane proteins. Like Gas6, however, the TGMs have a wide tissue distribution.^{34,35} Their functions *in vivo* are presently unclear but their chemical conformations suggest that they could be involved in cell transduction and have broad physiological activities.

Periostin

In 2008, periostin, a protein associated with the extracellular matrix known to promote cell migration and angiogenesis, was identified as a VKDP by virtue of its γ -carboxylation modification.³⁶ However, a recent study conducted with human periostin extracted from fibrotic lung or secreted from a cell line engineered to optimize γ -carboxylation did not confirm the initial finding that the protein is subject to vitamin K–dependent γ -carboxylation.³⁷ In light of this, it remains to be determined whether periostin is a member of the VKDPs. A summary of the VKDPs and their functions is given in [Table 8.1](#).

II. NUTRIENT METABOLISM

A. Absorption and Bioavailability

Vitamin K is absorbed from the proximal intestine into the lymphatic system by a process that requires the presence of bile and pancreatic juices. Consequently, conditions interfering with these functions or associated with fat malabsorption will impair vitamin K absorption. In healthy adults, absorption of phylloquinone has been estimated to be ~80% when administered in its free form but decreases significantly when it is absorbed from foods.³⁸ When investigated using the area under an absorption curve, absorption of phylloquinone from spinach was found to be 4%–17% that of phylloquinone from a suspension or a tablet.^{39,40} A similar estimate was also reported for phylloquinone from kale using compartmental modeling.⁴¹

TABLE 8.1 Summary of the vitamin K–dependent proteins and their functions.

Protein category	Physiological functions	Cell signaling–related actions
<i>Blood coagulation</i>	<i>Hemostasis</i>	
Prothrombin	Procoagulant	<i>Thrombin</i> Embryonic development Angiogenesis Proinflammatory action Wound healing
Factor VII	Procoagulant	
Factor IX	Procoagulant	
Factor X	Procoagulant	<i>FXa</i> Proinflammatory action
Protein C	Anticoagulant	<i>Protein C</i> Antiinflammatory action Antiapoptotic actions Cytoprotective action
Protein S	Anticoagulant	<i>Protein S</i> Modulator of C4BP system Phagocytosis of apoptotic cells Cell proliferation and survival Vessel integrity and angiogenesis
Protein Z	Anticoagulant	
<i>Bone</i>		
Osteocalcin	Negative regulator of bone formation Endocrine function	
Matrix Gla protein	Inhibitor of calcification ^a	
Protein S	Undetermined	
<i>Others</i>		
Gas6	Cell differentiation, proliferation, adhesion, and chemotaxis Phagocytosis and protection from apoptosis	
TMGs	Undetermined	
Gla-rich protein	Undetermined	

TMG, transmembrane Gla proteins.

^aThis role as calcification inhibitor applies to soft tissues as well as bone.

Phylloquinone absorption from vegetables can be improved when consumed with fat³⁹ but absorption efficiency remains lower than when phylloquinone is being consumed in an oil form.⁴² Moreover, a study that used stable isotope methodology suggests that absorption of phylloquinone is influenced by meal components.⁴³

Data for the menaquinones are more limited but when MK-4 was administered with butter³⁹ and MK-7 as part of the Japanese food natto,⁴⁴ the bioavailability of these two vitamers was higher than that of phylloquinone from spinach.

Recently, two independent groups identified at least three intestinal protein transporters that are deemed to

be implicated in the uptake of phyloquinone by the intestine. In 2014, a French group⁴⁵ found evidence that two members of the class B scavenger receptor family, namely scavenger receptor class B type I and cluster of differentiation 36, are potentially involved in the intestinal absorption of phyloquinone. Both these receptor proteins have a broad specificity and are known to be transporters for dietary cholesterol and vitamin E.⁴⁵ In 2015, a group in Japan reported that Niemann-Pick C1-Like 1 (NPC1L1), previously known to be a transporter for dietary cholesterol and α -tocopherol, is also a key regulator of intestinal phyloquinone absorption.⁴⁶ Of the three transporter proteins, NPC1L1 is associated with the strongest evidence, as its action was documented using different experimental models. Results from the studies of these two groups were recently reviewed.⁴⁷

B. Transport and Clearance

Absorbed phyloquinone is incorporated into chylomicrons and transported to the liver, where it is cleared from chylomicron remnants through an apolipoprotein E (apoE) receptor. Unlike other fat-soluble vitamins, vitamin K has no known carrier protein. In the circulation, phyloquinone is principally carried in triacylglycerol-rich lipoproteins (TRLPs) (>50%) with each of the low-density lipoprotein (LDL) and high-density lipoprotein (HDL) fractions accounting for ~15% of the circulating vitamin.³⁸ This likely explains the strong positive correlation observed between circulating phyloquinone and triacylglycerols in various studies.⁴⁸

Data on the lipoprotein distribution of ingested menaquinones is more limited. In a comparative study that involved the administration of equimolar amounts of phyloquinone, MK-4, and MK-9, all molecular forms were predominately associated with TRLPs during the first 4 h after intake. Thereafter, MK-4 appeared in LDLs and HDLs, and by 8 h, it was largely (>80%) associated with the LDL fraction. In contrast, MK-9 was not found in HDLs at any time point and appeared in LDLs after 8 h. By 24 and 48 h, ~50% and >90% of MK-9 was present in the LDL fraction, respectively.⁴⁹

These differences in lipoprotein profile probably explain the different clearance rates of the K vitamins from the circulation. When consumed in physiological amounts, postprandial plasma phyloquinone concentration peaks after about 6 h, returning to baseline by 24 h. Clearance rate of unlabeled MK-4 was also found to be similarly rapid. This is in contrast to the long-chain menaquinones, which remain in the circulation for much longer periods; terminal plasma half-lives for MK-9 and MK-7 are estimated at 60 and 72 h, respectively (see review by Shearer and Newman⁵⁰).

Compared to the other fat-soluble vitamins, phyloquinone circulates in blood in very small concentrations, and values ranging between 0.25 and 2.7 nmol/L have been reported in healthy young and old adults (Sadowski et al.⁵¹). Fasting plasma phyloquinone has been linked to the genetic polymorphism of apoE; however, results have been inconsistent among the groups investigated.³⁸ Population reference values for circulating menaquinones are not yet available.

C. Cellular Uptake and Catabolism

Uptake of phyloquinone into tissues including bone is deemed to be via clearance of chylomicron remnants by apoE. Furthermore, internalization of phyloquinone in osteoblasts appears to vary according to apoE genotype.⁵² Cellular uptake of menaquinones has not been investigated specifically, but it has been suggested that long-chain menaquinones could enter osteoblasts via the LDL receptor in light of their close association with this lipoprotein fraction.³⁸

Older studies that used pharmacological dosage protocols showed phyloquinone to be rapidly metabolized with about 20% being excreted in the urine and 40%–50% excreted in the feces via the bile.⁵³ Urinary vitamin K metabolites consist mainly of glucuronide conjugates of derivatives in which the phytyl side chain has been oxidized and cleaved. Biliary metabolites have not been clearly identified. A method for measuring two major urinary metabolites of phyloquinone, 5C- and 7C-aglycones, by HPLC with electrochemical detection in the redox mode, has been published⁸ and used successfully in metabolic studies.⁵⁴

D. Tissue Stores and Biosynthesis of MK-4

As the site of synthesis of the coagulation proteins, liver has traditionally been considered the main storage organ for vitamin K and consists of around 90% menaquinones, with MK-10 and MK-11 being particularly abundant, and 10% phyloquinone.⁴ However, phyloquinone and the menaquinones are also present in extrahepatic tissues. In postmortem specimens, phyloquinone concentrations in heart and pancreas were found to be comparable to that in liver while those in lung, kidney, and brain were much lower.⁵⁵ Menaquinone-4 is also widely distributed and is present in tissues in variable amounts. In brain and kidney, for instance, MK-4 concentrations largely exceed those of phyloquinone, whereas in the pancreas, MK-4 and phyloquinone are present in comparable amounts.⁵⁶ Vitamin K is also present in bone, with phyloquinone predominating over the menaquinones (MK-4 through MK-8).³⁸

The presence of MK-4 in tissues has been the object of investigation for over 50 years; however, it is only in recent years that the origin of this menaquinone has been elucidated. In a series of studies, it was recently established that during intestinal absorption, a fraction of dietary phyloquinone is cleaved by an unknown enzyme(s) to release menadione, which then enters the blood circulation via the thoracic lymphatic system. Menadione is delivered to target tissues and is then transformed into MK-4 through a prenylation reaction with geranylgeranyl diphosphate in a reaction catalyzed by the enzyme UBIAD1 (UbiA prenyltransferase domain-containing protein 1) (see review by Shearer and Okano.²

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Vitamin K and Bone Health

The discovery of OC and the evidence that the glutamic acid residues it contained needed to be carboxylated for its function in bone soon fostered interest in the role of vitamin K nutriture in bone health. Numerous observational studies have been conducted over the years, opening the way to randomized controlled trials of vitamin K supplementation, the results of which have recently become available. In older observational studies, high phyloquinone intakes were associated with lower risk of hip fracture, in contrast to the relationship between phyloquinone intake and bone mineral density (BMD), which has been more variable.⁵⁷

In a recent metaanalysis comprising four cohort studies and one nested case–control study published between 1999 and 2012, and including 80,982 participants and a total of 1114 fracture cases, subjects with the highest intake of dietary phyloquinone were found to have a 22% reduction in the risk of fractures (95% CI: 0.56–0.99), when compared with those with the lowest intake. However, in subgroup analyses, the statistically significant inverse association between dietary phyloquinone intake and risk of total fractures was seen in studies including both men and women, but not in studies that included women only. Moreover, the association was observed only in studies with more than 10 years of follow-up.⁵⁸

Observational studies focusing on menaquinones are more limited. In Japanese studies, intake of natto, a native fermented soybean dish high in phyloquinone and MK-7, was associated with higher BMD in elderly men and reduced bone loss in elderly women. Conversely, total menaquinone intake was not found to be protective in Norwegian men and women aged 71–75 when assessed over a 10-year period.⁵⁹

When investigated with respect to vitamin K nutritional status, high plasma phyloquinone and lower ucOC concentrations, reflective of a high status, have been associated with positive outcome on BMD in some studies but not all⁵⁷ and have not necessarily correlated with bone turnover markers.⁶⁰ It should be stressed that the methodology for the estimation of ucOC is subject to numerous sources of variation, and, without proper standardization of the assay, ucOC values may not be reliable. Indeed, as ucOC is strongly correlated to total OC, it should always be expressed as a percentage of total OC or as a ratio of carboxylated OC to account for bone activity, a precaution not always respected, and which may partly explain the lack of consensus among studies.⁶¹

Despite their value, a major limitation of dietary association studies is that they make it difficult to discern the specific role of vitamin K from those of other nutrients in the diet. Diets rich in vitamin K also tend to be rich in fruits and vegetables, which in themselves are beneficial to bone health.⁶² Recent access to randomized controlled trials of vitamin K supplementation has been helpful in this regard. A metaanalysis of RCTs assessing the effect of vitamin K on BMD and fractures in adult populations was recently published.⁶³ Thirty-six studies were included totaling 11,112 participants with follow-up ranging from 6 to 48 months. Fracture data were available for 12 of the trials. Of the trials reporting clinical fractures, six used menaquinones (five with 45 mg of MK-4; one with 360 µg of MK-7), and three used phyloquinone (ranging from 200 µg to 10 mg). Of the trials reporting vertebral fractures, six used menaquinones (five with 45 mg of MK-4; one with 360 µg of MK-7) and one used phyloquinone (5 mg).

For postmenopausal or osteoporotic patients, the odds ratio (OR) of any clinical fractures was statistically lower for the vitamin K versus control group; however, when restricting the analysis to trials with low risk of bias, the OR was no longer statistically significant. The metaanalyses showed no difference in vertebral fractures between the groups.⁶³

BMD data (hip, lumbar spine, femur, and radius) for postmenopausal or osteoporotic women were available for 22 trials. Trials used phyloquinone and menaquinones in doses ranging from 100 µg to 5 mg and from 180 µg to 45 mg, respectively. At 1 year, BMD was not statistically significantly different between the vitamin K and control groups. At 2 years, the mean difference in BMD was statistically significantly higher in the vitamin K group at the lumbar spine and the femur but not at other sites.⁶³ However, when sensitivity analyses were conducted and included only studies with a low risk of bias, the percentage change in lumbar spine at 1 and 2 years were not statistically significant. Based on their analyses, authors concluded that in postmenopausal or

osteoporotic patients, vitamin K supplementation does not appear to significantly impact BMD and/or offer protection against fracture.⁶³

Given that individuals on warfarin therapy are by definition in states of chronic vitamin K deficiency, bone status in these patients has been investigated. A metaanalysis investigating fractures and BMD and comparing patients treated with VKAs, with healthy controls and medical controls, was recently published. Twenty-one studies including 79,663 individuals treated with VKAs and 597,348 controls were included. Compared with healthy controls, VKA-treated individuals showed significantly higher fracture risk in cross-sectional and longitudinal studies and more incident hip fractures.^{63a} However, when limiting the analysis to studies where VKA participants and healthy controls were matched on at least one variable (e.g., age, sex), both these findings became nonsignificant. Likewise, BMD of VKA-treated patients and medical controls were similar at all investigated sites. Based on their results, the authors concluded that VKAs neither increase the risk of fracture when matching eliminated confounding factors nor reduce BMD beyond the effects of medical illness.^{63a}

Results from these most recent clinical trials do not appear to support a protective role for vitamin K in age-associated bone loss and fracture risk. However, the role of vitamin K in bone is complex, and research needs to be pursued. Future studies could target patients at higher risk of fracture or individuals with long-standing suboptimal nutritional status. Likewise, investigation of the alternative actions of the K vitamers should be encouraged. MK-4 has been shown to be a potent agonist of the steroid and xenobiotic nuclear receptor and to regulate the transcription of various bone-related genes including those of the extracellular matrix involved in collagen accumulation, as reviewed recently.⁶⁴ Whether these actions are modulated by nutritional status remains to be determined.

B. Vitamin K and Cardiovascular Health

In addition to the animal work described earlier, evidence for a role of vitamin K in calcification has come from reports derived from genetic diseases. Individuals suffering from Keutel syndrome, a disorder characterized by abnormal cartilage calcification, have been shown to possess mutations of the MGP gene that result in nonfunctional MGP.⁶⁵ Similarly, patients presenting with pseudoxanthoma elasticum, a multisystem disorder characterized by dystrophic mineralization of soft connective tissues, have been shown to have lower

concentrations of total MGP when compared to healthy controls.² Finally, immunohistochemical studies comparing normal and atherosclerotic human tissues showed MGP to be constitutively expressed in the vessel wall during calcification and in atherosclerotic plaque.⁶⁶

These initial findings prompted subsequent epidemiological studies addressing the association between vitamin K status and cardiovascular disease (CVD). Results from these studies have been summarized in recent review articles^{67,68} and have recently been included in metaanalyses. In one report totaling 222,592 participants, the authors assessed the prospective association between vitamin K intake and CVD events and all-cause mortality.⁶⁹ Specifically, higher dietary phylloquinone and higher dietary menaquinone were associated with lower coronary heart disease. No significant associations were observed between dietary vitamin K and all-cause mortality, CVD mortality, or stroke. Elevated plasma levels of inactive MGP (dp-ucMGP) were associated with an increased risk of all-cause mortality and CVD mortality. In the dietary studies, the intakes of vitamin K were all self-reported using FFQs. As the accuracy of FFQ data depends on memory and the precise estimation of the frequency and portion size of the foods consumed over a specific period of time, conclusions about the role of dietary vitamin K on CVD events and all-cause mortality are subject to some limitations.

A second recent systematic review and metaanalysis partly addressed these limitations by assessing the effect of vitamin K status and supplementation on vascular calcification and stiffness, two well-established surrogate measures of vascular health. It also assessed the association between the inactive forms of MGP (dp-ucMGP) and OC (ucOC) and incident CVD and mortality.⁷⁰ This metaanalysis included 13 controlled clinical trials (n = 2162) and 14 longitudinal studies (n = 10,726). When compared to control, vitamin K supplementation was associated with significant reduction in vascular calcification (−9.1%), dp-ucMGP (−44.7%), and ucOC (−12.0%). An improvement in vascular stiffness was also observed but did not reach statistical significance.⁷⁰ In longitudinal studies with median follow-up of 7.8 years, VKDP levels were associated with a combined endpoint of CVD or mortality (hazard ratio = 0.45). The authors stressed the significant heterogeneity of the longitudinal studies with respect to the populations assessed, thresholds used to define levels of inactive VKDP, and the duration of follow-up.⁷⁰ Likewise, the clinical trials were generally small with variable duration. While these results provide evidence of vitamin K supplementation on vascular health, larger trials with strong endpoints are needed before further recommendations can be made.

C. Emerging Functions for Vitamin K

Inflammation

Menaquinone-4 has been shown to limit the production of interleukin-6 (IL-6)⁷¹ and of prostaglandins⁷² in vitro and to limit inflammation in models of encephalomyelitis.⁷³ Similarly, phylloquinone has been reported to suppress lipopolysaccharide-induced inflammation in the rat⁷⁴ and recently reviewed.⁶⁷ Furthermore, recent epidemiological cohort studies point to a protective role for vitamin K in inflammation. In a cross-sectional study involving participants from the Framingham Offspring Study ($n = 1381$; mean age = 59 ± 8 y), higher vitamin K intake and plasma phylloquinone were associated with lower concentrations of several proinflammatory biomarkers (i.e., IL-6, intracellular adhesion molecule-1, tumor necrosis factor receptor 2, and C-reactive protein).⁷⁵ Similar positive effects of vitamin K on inflammation burden have also been reported in the PREDIMED trial⁷⁶ and the Multi-Ethnic Study of Atherosclerosis study.⁷⁷

To gain insight on causality, the role of vitamin K in inflammation has been investigated as secondary outcomes in two randomized trials conducted in older adults. In both cases, supplementation with phylloquinone and menaquinone-7 failed to reduce circulating inflammatory biomarkers in older men and women over a 3 year period.⁷⁸ Noteworthy, both these trials included healthy older adults, which may have limited the ability to see an effect of vitamin K in reducing inflammation.

Osteoarthritis, a condition associated with a proinflammatory state, has offered additional opportunities to investigate the role of vitamin K in the context of inflammation. Epidemiological studies conducted thus far suggest that low vitamin K status is a risk factor for developing the condition.^{79,80} Intervention studies of vitamin K supplementation are very limited and have thus far been inconclusive. Whether the action of vitamin K in the inflammatory process involves the VKDP or is independent of its function as an enzymatic cofactor in the carboxylation reaction remains to be specified. The role of vitamin K in inflammation has recently been reviewed.^{67,78}

Brain function and cognition

Vitamin K is widely distributed in cerebral structures where it is predominantly present as MK-4. In rodents, MK-4 has been shown to represent >98% of total vitamin K in brain, with concentrations increasing as a function of phylloquinone intake.^{81,82} Vitamin K participates in brain function through the VKDPs Gas 6 and PS, both ligands of the TAM receptors. In brain, Gas 6 has been involved in a wide range of actions including cell survival (neurons, glia), chemotaxis, mitogenesis, cell growth, and the myelination process. Similarly, PS has

been shown to offer neuronal protection during ischemic/hypoxic injury through its antithrombotic functions and TAM-related signaling actions (as reviewed by Ferland³¹). Vitamin K also participates in the synthesis of sphingolipids, a family of lipids largely present in brain cell membranes which, in addition to their structural functions, have key cell actions (i.e., proliferation, differentiation, and senescence).⁸³ Associations between brain tissue MK-4 and various sphingolipids have been reported in rodents.⁸¹

Furthermore, evidence has been mounting for a role of vitamin K in cognition in both animal models and humans. In rodents, lifetime consumption of a low phylloquinone-containing diet has been shown to result in cognitive deficits in old age, a finding associated with alteration of sphingolipids in the hippocampus, a key brain region for memory consolidation, and one that is largely affected in Alzheimer's disease.⁸⁴ In a group of older adults (≥ 65 years) participating in the CLIP (Cognition and LIPophilic vitamins) study, participants in the highest third of dietary phylloquinone intake showed better global cognition and better behavior when compared to those in the lowest third of intake.⁸⁵ Similarly, when investigated in older adults presenting subjective memory complaints, those with more serious complaints had lower mean dietary phylloquinone intakes.⁸⁶ In a study involving cognitively healthy adults (67–84 years), better verbal episodic memory was observed in subjects with higher vitamin K status as assessed by plasma phylloquinone.⁸⁷ Finally, patients treated with coumarin anticoagulants have provided evidence for potential detrimental effects of their low vitamin K status on cognitive function^{88,89} and brain volume.^{90, 90a} The role of vitamin K in brain and cognition has recently been reviewed.⁹¹

Endocrine function

The pancreas has long been known to contain significant amounts of vitamin K per unit weight,⁵⁶ and older reports suggested that rats fed a low phylloquinone-containing diet had a delayed insulin response to glucose infusion and increased plasma glucose levels.⁹² In 2007, a series of studies using transgenic mice reported that OC increased insulin secretion and insulin sensitivity and decreased the severity of type 2 diabetes.^{93,94} Notably, these studies suggested a specific role for the uncarboxylated form of OC, a finding then at odds with what was known of the role of vitamin K in the biological activation of the VKDP through carboxylation.⁹⁵

Since the publication of these reports, a number of observational studies have been conducted, and, in contrast to what was observed in the mice model, studies tended to report beneficial effects of increased phylloquinone intake on glucose homeostasis and insulin resistance (see review by Bourron and Phan⁹⁶). In a recent

report, circulating phylloquinone was significantly and independently lower in type 2 diabetes patients compared with age-matched control patients and was significantly and inversely associated with insulin resistance.⁹⁷ Mechanisms underlying the beneficial effect of vitamin K against insulin resistance remain to be determined but could involve its action in inflammation.

The role of vitamin K in insulin resistance has also been investigated in a limited number of supplementation studies that used phylloquinone or the menaquinones (MK-4 and MK-7). Results have generally been inconclusive; however, as discussed in Bouron and Phan,⁹⁶ these studies had limitations as they were largely ancillary studies. As a result, they presented great heterogeneity in their design, e.g., study duration, vitamin K form and posology, participants' characteristics and medical condition, etc. Future studies should be planned to specifically address the role of vitamin K (both phylloquinone and the menaquinones) on glucose metabolism and progression to diabetes, using optimal experimental designs.

Anticancer effect

Numerous *in vitro* and a few *in vivo* studies have suggested anticarcinogenic effects for vitamin K, notably the menaquinones. This protective role of vitamin K has been linked to its ability to influence cell differentiation and induces apoptosis.⁹⁸ In epidemiological studies, intakes of menaquinones, but not of phylloquinone, have been associated with reduced risk of cancer and mortality in the large EPIC-Heidelberg study.^{99,100} As for the other actions of vitamin K, its anticancer potential will have to be investigated in rigorous intervention trials.

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency

A number of indicators have been used to assess vitamin K status; however, only prothrombin time has been linked to adverse clinical outcome. Prolonged prothrombin time expressed as the normalized international ratio is the indicator of reference for vitamin K deficiency in the clinical setting. Plasma or serum phylloquinone, as an indicator of vitamin K status, reflects recent intakes and responds to dietary changes. As discussed earlier, vitamin K insufficiency leads to the release of undercarboxylated forms of the VKDPs. Antibodies enabling the detection of the inactive forms of prothrombin, OC, and MGP are now commercially available. Although these forms have been shown to

respond to alterations in dietary vitamin K, their clinical value remains a subject of debate.³

B. Criteria of Adequacy

A "normal" range for phylloquinone is not yet defined clinically; however, a concentration ~ 1.0 nmol/L has been used as the minimum level expected when Adequate Intakes (AIs) are met.³

C. Assessment of Status

Clinically significant vitamin K deficiency is associated with an increase in prothrombin time and, in severe cases, bleeding. However, overt vitamin K deficiency is uncommon in adults and has mainly been associated with gastrointestinal disorders associated with fat malabsorption (e.g., due to bile-duct obstruction, inflammatory bowel disease, chronic pancreatitis, cystic fibrosis) and liver disease.¹⁰¹ Hospitalized patients with low food intake or poor nutritional status may also be at increased risk of vitamin K deficiency, especially if treated with antibiotics or other drugs interfering with vitamin K metabolism.¹⁰²

D. Reference Intakes

As part of the Dietary Reference Intakes process, the Food and Nutrition Board of the Institute of Medicine updated dietary recommendations for vitamin K in 2001 for the United States and Canada.³ The recommended AI for vitamin K is 120 $\mu\text{g}/\text{day}$ for men and 90 $\mu\text{g}/\text{day}$ for women based on the amount needed to maintain serum phylloquinone levels at or above 1 nmol/L. The AI for children during the first 6 months of life is 2 $\mu\text{g}/\text{day}$ and from 7 to 12 months is 2.5 $\mu\text{g}/\text{day}$. AIs for children are 30 $\mu\text{g}/\text{day}$ for ages 1–3 years, 55 $\mu\text{g}/\text{day}$ for ages 4–8 years, 60 $\mu\text{g}/\text{day}$ for ages 9–13 years, and 75 $\mu\text{g}/\text{day}$ for ages 14–18 years. The recommendations for pregnancy and lactation are the same as for nonpregnant females of similar age (90 $\mu\text{g}/\text{day}$).

The European Union Scientific Committee on Nutrition,¹⁰³ the United Kingdom,¹⁰⁴ and the WHO/FAO¹⁰⁵ have established 1 μg of phylloquinone per kilogram body weight per day as their official guideline for vitamin K. As part of its review process in 2017, the EFSA panel assessed the evidence available since 1993 regarding the menaquinones (e.g., food composition databases, intake, absorption, function, body/organ contents, etc.) and deemed it insufficient for consideration in setting the AI.

Considering the respective reference body weights of men and women, the EFSA proposed an AI of 70 µg phyloquinone/day for adults of both sex, irrespective of age. This amount was also deemed adequate to meet the requirements of pregnant and lactating women.¹⁰³

E. Tolerable Upper Intake Levels

As part of the DRI process in 2001, no evidence of adverse health effects or toxicity for the natural form of phyloquinone or the menaquinones was found, even when large amounts are administered,³ hence no Tolerable Upper Intake Levels for vitamin K have been established. However, synthetic menadione has been shown to produce hemolytic anemia, hyperbilirubinemia, and kernicterus when administered in amounts of more than 5 mg/day to infants.¹⁰⁶ Consequently, menadione is no longer used as a therapeutic agent.

F. Dietary Sources and Usual Intakes

The application of high-performance liquid chromatography analysis to vitamin K and the development of relatively straightforward assay procedures have enabled analysis of a large number of foods for vitamin K in the last decade in both the United States and Europe. Vitamin K is found in highest amounts in green vegetables and certain oils; a recent analysis based on the 2011–12 NHANES survey showed that vegetables contributed 35%–60% of total vitamin K intake, followed by mixed dishes representing 15%–30% of total intake by virtue of the oils added to the foods.¹⁰⁷

Vegetables such as collards, swiss chard, and spinach, contain in excess of 300 µg phyloquinone/100 g, while broccoli, Brussels sprouts, and cabbage contain between 100 and 200 µg phyloquinone/100 g. The phyloquinone content of oils is variable, soybean and canola oils being the richest sources (100–200 µg/100 g), followed by olive oil with 50–100 µg/100 g. Oils derived from corn and sunflower seeds are not good sources of phyloquinone, with a content of <10 µg/100 g.³

Hydrogenation of plant oils to form solid shortenings results in some conversion of phyloquinone to 2',3'-dihydrophyloquinone. This form of the vitamin is most prevalent in margarines and processed foods and can represent a substantial portion of total vitamin K in some diets. Dihydrophyloquinone has lower bioavailability and biological activity compared to phyloquinone.¹⁰⁸

In contrast to phyloquinone, the menaquinones are found in animal-based foods, notably liver products, meat, and poultry, and cheese and other dairy. Animal liver and foods that undergo fermentation such as cheeses typically contain long-chain menaquinones (i.e., MK-8 and MK-9).¹⁰⁹ Natto, a traditional Japanese dish made of fermented soybeans, is an excellent source of

MK-7.¹¹⁰ The relative contribution of the menaquinones to total vitamin K intake remains to be specified.¹⁰³

Access to the vitamin K content of foods has made it possible to obtain reasonable estimates of the dietary phyloquinone intake of the North American, European, and Asian populations. In the United States, recent analyses of the NHANES surveys point to mean intakes that are in line with the AI. In the study by Harshman et al.,¹⁰⁷ mean phyloquinone intakes were found to be ~120 µg/day for both sex, 43.0% of men and 62.5% of women meeting the AIs of 120 and 90 µg, respectively. In the latest release of the 2013–16 cohort of the *What We Eat in America—NHANES* survey, mean phyloquinone intakes were 118 and 124 µg phyloquinone/day for men and women, respectively, which indicated that 40% of men and 60% of women 19+ years had intakes greater than their AI.¹¹¹

Similarly, as part of its initiative in 2017, the EFSA panel reviewed vitamin K consumption data from 13 dietary surveys from nine European countries (Finland, France, Germany, Ireland, Italy, Latvia, the Netherlands, Sweden, and the United Kingdom). Notwithstanding limitations of the databases available, including uncertainties about the contribution of the different K vitamers, “total” vitamin K estimates were obtained for all age groups from infants to adults. Mean estimates of intake met the AIs for phyloquinone in all age strata with values for adults ≥18 years ranging between 72 and 196 µg/day. Leafy vegetables and vegetables from the *Brassica* family were the most important contributors to total vitamin K intake in all age strata except infants for whom specialty food for “young population” was the main source of vitamin K. Composite dishes were also significant contributors to total vitamin K intake in part due to the vegetable-based ingredients in the dishes, the contribution of animal and vegetable fats and oils, and the group comprising legumes, nuts, oilseeds, and spices.¹⁰³

V. ISSUES RELATED TO VITAMIN K IN HUMAN HEALTH

A. Relevant Health Outcomes and Human Health

Newborn infants

Newborn infants are a clinical population with a well-established risk for vitamin K deficiency. Factors contributing to the condition are poor placental transfer, low concentrations of plasma clotting factors due to hepatic immaturity, and low vitamin K content of breast milk. Individually or in concert, these factors increase the risk of bleeding in infants in the first weeks of life, a condition known as vitamin K deficiency bleeding

(VKDB).¹¹² Given that VKDB can be effectively prevented by administration of vitamin K, the American Academy of Pediatrics¹¹³ and the Canadian Paediatric Society¹¹⁴ recommend that phyloquinone be given to all newborns as a single, intramuscular (IM) dose of 0.5–1 mg within 6 h of birth, a recommendation also supported by the World Health Organization¹¹⁵ and the European Society for Pediatric Gastroenterology, Hepatology, and Nutrition.¹¹⁶

When the clinical context precludes IM treatment or when parents decline this route of administration, vitamin K should be administered orally at a dose of 2.0 mg at birth and again at 2–4 and 6–8 weeks of age. Reported concerns from parents who decline IM administration include aspects such as perceived risk to their newborn, preference for more “natural” approaches (e.g., oral vitamin K or increasing the mother’s own pre/postnatal dietary vitamin K intake), and lack of trust toward health care providers and pharmaceutical companies.¹¹⁷ Incidentally, concerns in the early 1990s of a possible association between IM administration of vitamin K and childhood cancer have not been substantiated.¹¹³

Finally, preterm infants are at higher risk of VKDB; however, there is currently insufficient evidence to support a different clinical practice for them. The IM administration hence applies to this situation as does the oral route, although the latter could be limited by the infant’s health status.¹¹⁶

Possible effect of antibiotics

Although bleeding episodes in antibiotic-treated patients have often been attributed to an acquired vitamin K deficiency resulting from a suppression of menaquinone-synthesizing organisms in the gut, data to substantiate this hypothesis are lacking. Furthermore, interpretation of these reports has generally been complicated by the possibility of general malnutrition in the patients under investigation.³

B. Known Interactions

Vitamin E

High doses of vitamin E have been shown to interfere with vitamin K and precipitate deficiency states, especially in subjects with low vitamin K status. The underlying mechanisms for this nutrient–nutrient interaction are not yet known but likely involve metabolic pathways^{118,119}.

Anticoagulation therapy

Patients treated with coumarin drugs undergo a particular drug–nutrient interaction as warfarin blocks the vitamin K cycle by inhibiting vitamin K epoxide

reductase activity. In light of this tight coupling, alteration in vitamin K intake will influence warfarin efficacy, as has been reported on numerous occasions. For this special population, it was recommended in 2001 that once the dose of warfarin has been established, individuals avoid any complications regarding coagulation resulting from variations in vitamin K intake by continuing to follow their normal dietary patterns, meaning that they should maintain a continued similar intake of vitamin K from foods and supplements.³ However, recent studies have shown that anticoagulated patients who have usual low dietary intake of vitamin K are more likely to present unstable anticoagulation while on the drug.¹²⁰ In fact, a growing number of reports indicate beneficial effects of high intakes of vitamin K with respect to anticoagulation control.¹²¹ In one study, high usual vitamin K intake was associated with more stable anticoagulation and a decreased need for control tests,¹²² high intake being notably associated with lower relative variability in vitamin K intake.¹²³

While avoidance of vitamin K–rich foods, particularly green vegetables, is common among warfarin users, this translates into lower usual vitamin K intakes. This points to the need for better nutrition counseling in these patients as well as for continuing education activities for health care professionals involved in the management of warfarin therapy.¹²⁴

C. As a Dietary Supplement

Phylloquinone, MK-4, and MK-7 are commercially available as dietary supplements. As discussed in earlier sections, whether higher vitamin K intakes, including through use of supplements, can help alleviate chronic conditions remains a subject of investigation. Irrespective of this issue, in recent years, a growing trend in the supplement industry has been to use the general term “vitamin K2” to refer indiscriminately to MK-4 and MK-7, which, as mentioned earlier, have significantly different bioavailabilities and half-lives. This practice should be discouraged for, in addition to misleading the consumer, it could carry potential detrimental health consequences, such with warfarin-based anticoagulation therapy, where half-lives of the menaquinones have been shown to influence therapeutic control.¹²⁵ Finally, consumers should be aware that most over-the-counter vitamin K oral supplements are not subjected to stringent quality control. Although product quality of currently available supplements is difficult to ascertain, one report did raise concerns about content accuracy and uniformity between tablets.¹²⁶

RESEARCH GAPS

- Further characterize the cellular and physiological functions of all known VKDPs as they relate to human physiology and health and vitamin K status;
- Further characterize the actions of the K vitamers independent of their role in the carboxylation reaction;
- Investigate the newly identified vitamin K intestinal protein transporters in normal and pathological conditions and as a function of vitamin K status;
- Conduct large-scale clinical studies of vitamin K supplementation to specify the role of vitamin K in cardiovascular health using well-established clinical endpoints; these studies should be designed so as to also assess the clinical value of surrogate measures of vitamin K status;
- Conduct basic and clinical research to further understand the role of vitamin K in emerging functions, i.e., inflammation, brain, and endocrine; and
- Conduct vitamin K intervention studies aimed at optimizing anticoagulation therapy.

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VITAMIN C

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SUMMARY

Vitamin C is a potent reducing agent in mammals and plants. Humans rely on vitamin C for the activity of numerous enzymes, and vitamin C status impacts physiological health, including risk for infections, cardiovascular disease, and cancer. The recommended intake for vitamin C in adults varies by gender and by country (from 45 to 110 mg/d). The Tolerable Upper Intake Level is 2000 mg/d. Supplemental vitamin C should not replace high intakes of fruits and vegetables, but it may offer health benefits under certain circumstances. Low intake of fruits and vegetables increases risk for scurvy, a concern for isolated populations, refugees, the critically ill, and the elderly. Smokers and patient populations are also at risk for suboptimal vitamin C status. Current vitamin C deficiency rates in developed countries range from 8% to 19%. Individuals with a history of renal stone formation or conditions associated with iron overload should use caution when supplementing with vitamin C.

Keywords: Antioxidant; Immunoenhancement; Injection therapy; Scurvy; Vascular health.

I. INTRODUCTION

A. Background

Vitamin C was first isolated from the adrenal glands of cows in 1928 and termed “hexuronic acid” by the Hungarian scientist Albert Szent-Györgyi. By 1932, the antiscorbutic property of hexuronic acid was established independently by Szent-Györgyi and C.G. King of the University of Pittsburgh, and the compound was renamed “ascorbic acid.” Chemist Norman Haworth, at Birmingham, England, working in collaboration with Szent-Györgyi, elucidated the structure of ascorbic acid in 1933. Both men received a Nobel Prize in 1937 for their work leading to the discovery and structure of ascorbic acid.

B. Nutrient Function and Structure

Ascorbic acid is a six-carbon, water-soluble (33 g/100 cm³ at 20°C) γ -lactone (MW 176.14). Its acidic nature results from the ionization of the enolic hydroxyl (OH) on C-3 ($pK_a = 4.25$) at neutral pH. Vitamin C is a redox-active system comprising ascorbic acid, the free

radical monodehydro-L-ascorbic acid (AFR), and oxidized ascorbate, or dehydro-L-ascorbic acid (DHA) (Fig. 9.1). The ascorbate radical/ascorbate thermodynamic couple, the one-electron reduction potential, is below that of most physiological redox systems. Hence, ascorbic acid is a primary nonenzymatic antioxidant for many reactive compounds in vivo, including the α -tocopherol free radical, the glutathione radical, peroxy radicals, the hydroxyl free radical, the superoxide radical, and the urate free radical.¹ AFR is detected directly by electron spin resonance spectroscopy, and new methods are under development utilizing this characteristic to precisely pinpoint the location of oxidative stress in vivo to facilitate treatment modalities.²

Liquid chromatography electrochemical detection methods are considered the most sensitive and precise systems for measuring ascorbic acid in biological samples. As mentioned above, DHA cannot be detected directly in these systems and is measured indirectly by subtraction after its pre- or postcolumn reduction to ascorbic acid.³ Ascorbic acid degrades rapidly in extracted plasma; hence, biological samples must be carefully handled and rapidly preserved and frozen. Venous blood is commonly collected on K₂EDTA, chilled, and

processed within 1 h. Plasma is deproteinized using metaphosphoric acid and stored at -80°C until analysis.⁴ Since much of the DHA in the sample (typically $<10\%$ of total vitamin C) is a consequence of inadvertent oxidation during blood processing, reduction of DHA prior to analyses is recommended, and the total vitamin C content of the sample is then reported.

Assay development using fluorescent and colorimetric optical sensing techniques for detecting ascorbic acid in food and biological samples is rapidly advancing and may permit rapid and sensitive quantification of ascorbic acid in the near future.^{5,6}

As a cofactor for mixed function oxidases, vitamin C participates in the synthesis of various macromolecules including collagen, carnitine, and norepinephrine (Table 9.1). In these reactions, vitamin C promotes enzyme activity by maintaining metal ions in the reduced form. In collagen and carnitine syntheses, α -ketoglutarate-dependent dioxygenases incorporate one atom of oxygen into succinate and one into the oxidized product of the substrate in the presence of ferrous iron (Fig. 9.2). Three ascorbate-dependent dioxygenases are required for the synthesis of collagen (prolyl 4-hydroxylase, prolyl 3-hydroxylase, and lysyl-hydroxyl

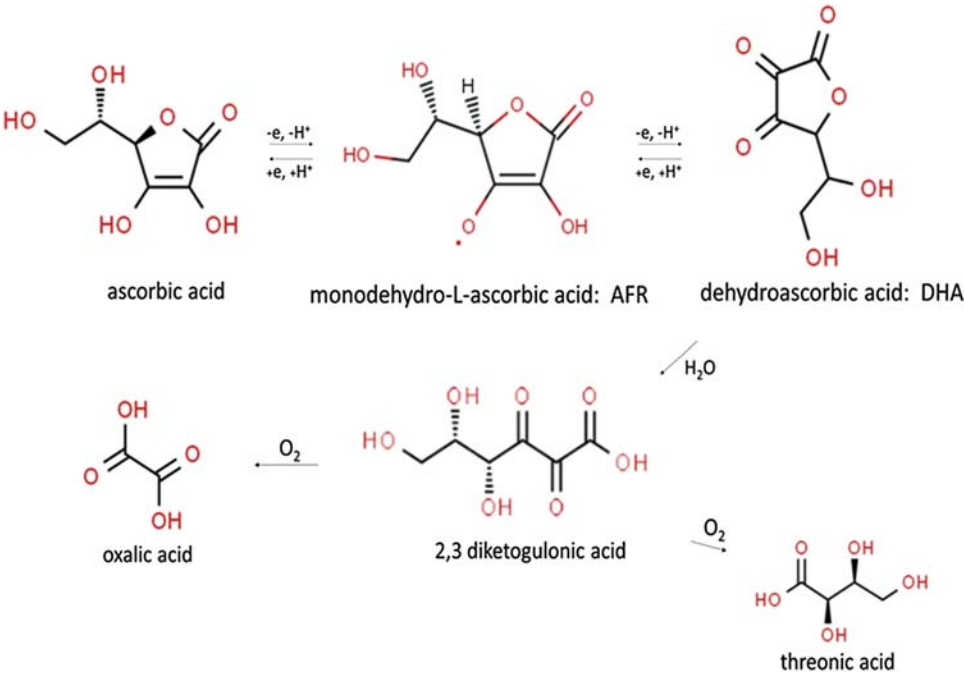


FIGURE 9.1 Structure and metabolism of ascorbic acid.

TABLE 9.1 Roles of vitamin C in maintaining enzyme activity and the tissue of enzyme action.

Vitamin C–dependent enzymes	Role	Tissue
Prolyl 4-hydroxylase Prolyl 3-hydroxylase Lysyl-hydroxylase	Collagen synthesis	Fibroblasts
6-N-Trimethyl-L-lysine hydroxylase α -Butyrobetaine hydroxylase	Carnitine synthesis	Liver, kidney
Dopamine β -hydroxylase	Norepinephrine synthesis	Adrenal medulla
Peptidylglycine α -amidating monooxygenase	Increased neuropeptide activity	Neurons
RNA/DNA demethylases	Stem cell pluripotency	Mesenchymal stromal cells

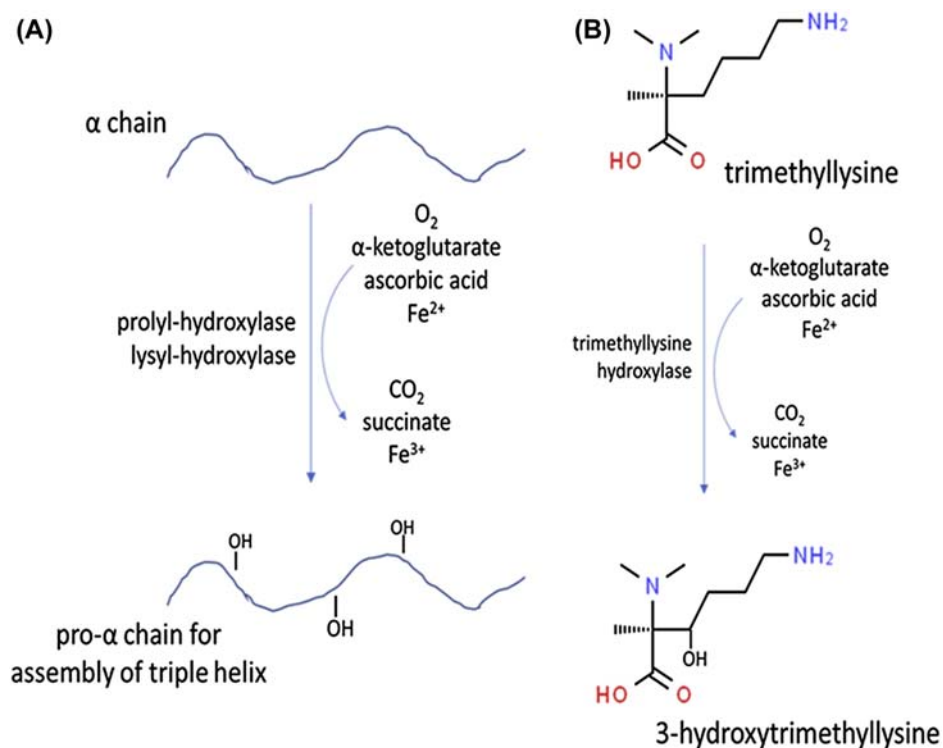


FIGURE 9.2 The vitamin C—dependent biosynthesis of collagen and carnitine. α -Ketoglutarate—dependent dioxygenases incorporate one atom of oxygen into succinate and one into the oxidized product of the substrate in the presence of ferrous iron. Ascorbate-dependent dioxygenases are required for the synthesis of (A) collagen (prolyl-hydroxylases and lysyl-hydroxylase) and (B) carnitine (trimethyl-L-lysine hydroxylase and α -butyrobetaine hydroxylase [latter pathway not shown]).

ase), and two are involved in carnitine biosynthesis (6-N-trimethyl-L-lysine hydroxylase and γ -butyrobetaine hydroxylase). Many of the symptoms of the vitamin C deficiency disease scurvy are attributed to weakened collagenous structures (thus bleeding and gum deterioration, impaired wound healing, and epiphyseal and metaphyseal plate lesions in bone) and insufficient tissue carnitine (perhaps resulting in fatigue and lassitude). As a cofactor for the synthesis of collagen, vitamin C promotes wound healing, and the vitamin C requirement is increased in surgical patients.⁷ In a randomized, controlled clinical trial, patients undergoing procedures for dental implants displayed improved wound healing at 14 days postsurgery when given vitamin C (300 mg/d) versus a placebo.⁸

There is an extensive literature linking impaired collagen synthesis with the connective tissue insufficiency in scurvy; however, there is little empirical evidence linking the fatigue of scurvy with reduced tissue carnitine in humans. In scorbutic guinea pigs, tissue carnitine concentrations are depressed >50%. In humans, primary carnitine deficiency is associated with fatigue, which is reversed with carnitine supplementation.⁹ Case reports demonstrate that the tiredness associated with latent scurvy is alleviated only by vitamin C supplementation.¹⁰

Evidence linking tiredness and tissue carnitine in vitamin C deficiency is needed to verify this premise.

Vitamin C is considered the predominant reductant for the copper-dependent enzyme dopamine β -hydroxylase (used in the synthesis of norepinephrine from dopamine in the adrenal medulla) (Table 9.1). Although clinical reports have documented cardiovascular failure in vitamin C—deficient patients linked to an impaired vasomotor tone that was reversed by vitamin C administration,¹¹ clinical investigations examining the role of vitamin C in sympathetic noradrenergic function are lacking. There is mounting interest in high-dose ascorbic acid therapy to treat the severe refractory hypotension characterizing septic shock.¹² In a double-blinded randomized clinical trial conducted in patients with septic shock, ascorbic acid administration (25 mg/kg intravenous ascorbic acid every 6 h for 72 h) reduced the requirement for epinephrine administration by nearly 50% and reduced patient mortality compared to control (14% and 64%, respectively).¹³

Vitamin C is also the required cofactor for the copper-dependent enzyme peptidylglycine α -amidating monooxygenase responsible for the activation of various hormones and neurotransmitters^{14,15} (Table 9.1). This monooxygenase controls the production of more than

one-half of all neuropeptides, and reduced enzyme activity is believed to contribute to the debilitating neurological symptoms observed with copper deficiency.¹⁶ The involvement of vitamin C in this synthetic pathway suggests an important yet largely unexplored role of this nutrient in neuroendocrine biology.

II. NUTRIENT METABOLISM

A. Synthesis

Long before vitamin C was utilized as a cofactor for enzymes in mammalian systems, it was a ubiquitous constituent of plants. The presence of duplicative biosynthetic pathways for vitamin C in algae and land plants suggests an important role for vitamin C in the evolutionary success of higher plants.¹⁷ In the animal kingdom, invertebrates and teleost fish, the dominant fishes in oceans and fresh water, cannot manufacture vitamin C; however, primitive fish, amphibians, and reptiles produce vitamin C as do most mammals. The extremely stressful conditions associated with terrestrial life, including high oxygen tension, desiccation by dry air, and hot sun, may have necessitated selection pressure for species capable of vitamin C synthesis.¹⁸ Several divergent lines of mammals (bats, guinea pigs, nonhuman primates, and man) lost the ability to synthesize vitamin C; however, these species possess marked increases in copper and zinc superoxide dismutases and circulating concentrations of uric acid (due to the loss of the uric acid oxidase gene) compared with the vitamin C-synthesizing amphibians, reptiles, and mammals and thereby maintain a strong defense system against oxygen toxicity.^{19,20}

In all plant species tested, ascorbic acid is synthesized from GDP-D-mannose in the L-galactose pathway; in mammals, the vitamin is synthesized from D-glucose in the hexuronic acid pathway of the liver or kidney (Fig. 9.3). In the vitamin C-dependent mammals, a genetic lesion inhibits the synthesis of L-gulonolactone oxidase, the terminal enzyme for ascorbic acid synthesis, and L-gulonic acid is processed via the pentose phosphate pathway to reenter carbohydrate metabolism.²¹

Although incapable of manufacturing vitamin C, humans are particularly adept at conserving the vitamin since adequate plasma concentrations are maintained at lower intakes of vitamin C (~1 mg/kg per day) as compared to other vitamin C-dependent species (3–6 mg/kg per day).²² Chatterjee¹⁸ calculated that the rate of vitamin C synthesis in small mammals ranged from 150 mg/kg body weight/d in rats to nearly 275 mg/kg body weight/d in rabbits and mice. In these species, the body pool of vitamin C ranged from 30 to 100 mg/kg, and blood vitamin C concentrations varied from 0.5 to 1.0 mg/dL (28–57 $\mu\text{mol/L}$).²³ In contrast,

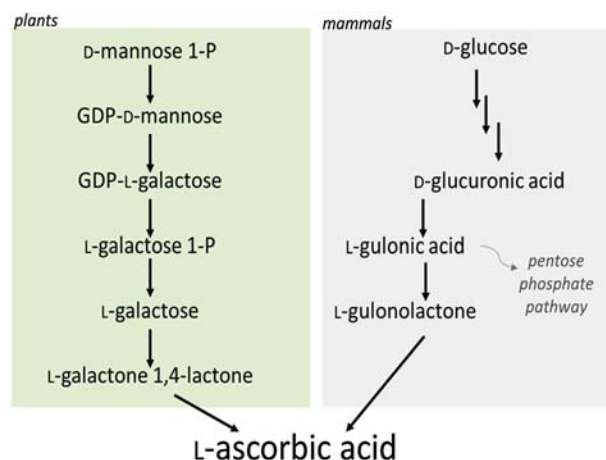


FIGURE 9.3 De novo synthesis of vitamin C. In plants, ascorbic acid is synthesized from GDP-D-mannose in the L-galactose pathway; in mammals, the vitamin is synthesized from D-glucose in the hexuronic acid pathway of the liver or kidney. In vitamin C-dependent mammals, L-gulonolactone oxidase synthesis is inhibited, and L-gulonic acid is processed via the pentose phosphate pathway.

vitamin C-dependent humans maintain a body pool of vitamin C near 20 mg/kg and a plasma vitamin C concentration near 0.5–0.7 mg/dL (28–40 $\mu\text{mol/L}$).²⁴ Also, the half-life for vitamin C in humans (which ranges from 14 to 40 d) appears to be an order of magnitude above that for rats or guinea pigs (of 3–6 d).²⁵ However, the half-life for vitamin C falls dramatically to about 2 h when high-dose ascorbic acid is administered intravenously in humans, reflecting saturation of the renal reabsorption capacity and rapid elimination of excessive vitamin C.²⁶

B. Digestion and Transport

Vitamin C does not require digestion prior to absorption into intestinal cells. Ascorbic acid is transported into cells via the sodium-dependent transporter SVCT1, located on epithelial cells; the sodium- and $\text{Ca}^{2+}/\text{Mg}^{2+}$ -dependent transporter SVCT2 transports ascorbic acid into metabolically active and specialized tissue cells.²⁷ Hence, SVCT2 likely promotes the antioxidant activities of ascorbic acid in tissues, while SVCT1 regulates whole body homeostasis of the vitamin. SVCT1 expression has been shown to increase and SVCT2 expression decrease with aging in murine models.^{28,29}

In cultured human-derived liver epithelial cells, SVCT1 was upregulated under ascorbic acid-deficient conditions but downregulated with vitamin supplementation, whereas SVCT2 expression was unaffected by change in substrate level. Although similar results were observed in liver extracts from vitamin C-deficient guinea pigs, there was no change in transporter expression in kidney extracts (SVCT1 and SVCT2) or brain extracts (SVCT2) from these animals.³⁰ Recent

research demonstrating SVCT2 expression in mucin-containing goblet cells of bronchial epithelium and subcellularly in mitochondrial membranes suggests a role for vitamin C at these sites.

In cell culture, oxidant stressors increased SVCT2 mRNA expression via multiple signaling pathways, including mitogen-activated protein kinases and NF- κ B, and elevated SVCT2 expression was associated with reduced cellular oxidative stress.^{31,32} In healthy, vitamin C-sufficient adults, vitamin C supplementation (1000 mg/d for 6 weeks) increased SVCT2 protein expression in skeletal muscle, but did not alter the cellular antioxidant potential.³³ However, in individuals with type 2 diabetes managed with oral antiglycemic agents, vitamin C supplementation (1000 mg/d for 4 months) was associated with increased SVCT2 protein levels in skeletal muscle, increased intracellular ascorbic acid, and reduced oxidative stress in insulin-stimulated, but not rested, muscle.³⁴ Hence, ascorbate-associated reductions in cellular oxidative stress may require more SVCT2 transporters as well as elevated oxidative stress.

Genetic variation in the gene that codes for the ascorbic acid transporter SVCT1, *SLC23A1*, has been associated with variations in circulating ascorbic acid concentrations in large population studies.^{35,36} This same allelic variation was also linked to a 20%–35% increased risk for aggressive periodontitis, a condition commonly observed in vitamin C deficiency.³⁷ These results demonstrate the important role that this transport system plays in vitamin C homeostasis. Genetic variations in the SVCT2 transporter are associated with risk

for cancer³⁸ and preterm delivery,³⁹ indicating an important role for this transporter in vitamin C functionality.

C. Metabolism

In its metabolic roles, ascorbic acid is converted to AFR in blood and tissues. This radical does not contribute to free radical chain reactions in vivo since it is rapidly recycled to ascorbic acid via NADPH-dependent reduction by thioredoxin reductases or by microsomal or mitochondrial oxidoreductases, which are present in all cell types⁴⁰ (Fig. 9.1). At sites of marked oxidant stress, the dismutation of two molecules of AFR to DHA and ascorbic acid likely occurs. Although DHA undergoes irreversible hydrolysis to 2,3-diketogulonic acid with a half-life of less than 2 min, nearby cells can bind DHA via specific facilitative glucose transporter isoforms (GLUT1, ubiquitous distribution; GLUT3, largely confined to neuronal tissues; and/or GLUT4, largely confined to muscle and adipose tissues).⁴¹ This uptake of DHA via glucose transporters by many cell types, followed by the immediate two-electron reduction of DHA to ascorbic acid by several mechanisms (glutathione-dependent systems or NADPH-dependent systems mediated by thioredoxin reductase or by 3 α -hydroxysteroid dehydrogenase), suggests an efficient system for the recycling and salvaging of vitamin C at sites of high oxidative stress.⁴² (Fig. 9.4)

However, research in SVCT2 knockout mice has clearly demonstrated that a recycling pathway is not

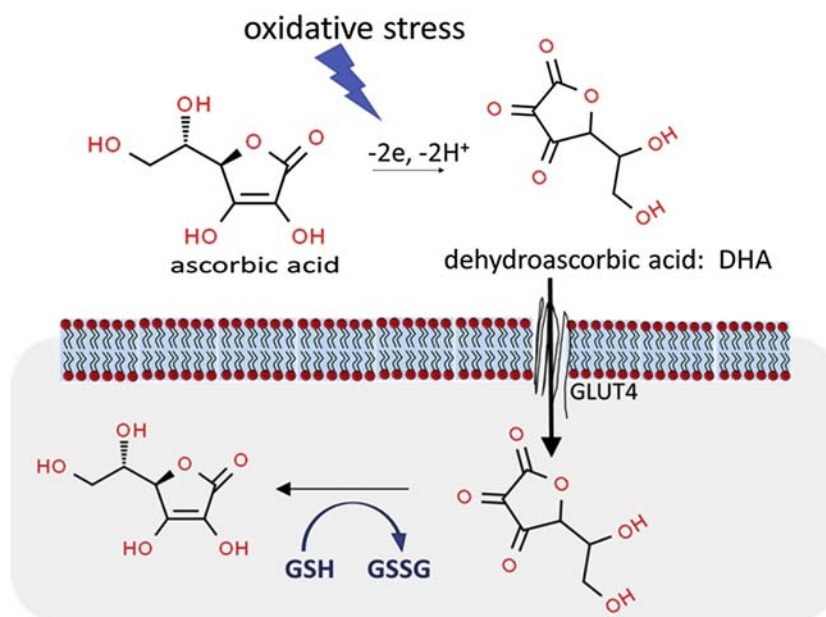


FIGURE 9.4 Role of vitamin C in oxidative stress. At sites of marked oxidant stress, ascorbic acid is oxidized to dehydroascorbic acid (DHA). Nearby cells can bind DHA via specific facilitative glucose transporter isoforms (GLUT1, GLUT3, and/or GLUT4). The uptake of DHA via glucose transporters by many cell types is followed by the immediate two-electron reduction of DHA to ascorbic acid by several mechanisms, including the glutathione-dependent systems ($GSH \rightleftharpoons GSSG$).

sufficient to compensate for the lack of direct ascorbic acid transfer into cells, as these mice die within minutes of birth.⁴³ Yet, in erythrocytes (cells that do not express SVCT2), DHA uptake and reduction to ascorbic acid appears to have physiologic relevance, since red blood cell ascorbate directly protects red blood cell integrity and stabilizes plasma ascorbate concentrations via transmembrane electron transfer.⁴⁴

If not recycled to ascorbate, DHA is irreversibly delactonized to 2,3-diketogulonic acid, and its further degradation to oxalic acid likely represents the major catabolic pathway of vitamin C in nonsupplemented individuals. Other possible catabolic products of vitamin C include L-threonic acid, L-xylonic acid, L-lyxonic acid, and L-xylose (Fig. 9.1). In a carefully conducted vitamin C depletion–repletion pharmacokinetic study in seven healthy inpatient volunteers, the bioavailability of an orally administered dose of ascorbic acid ranged from 100% at 200 mg, 73% at 500 mg, and 49% at 1250 mg.⁴⁵ At these same doses, the fraction of ascorbic acid excreted unmetabolized after 24-h was approximately 50%, and at steady state (once plasma ascorbic acid reached a plateau), the 24-h oxalate excretion was approximately 30, 35, and 40 mg.

In subjects administered intravenous ascorbic acid (1.5 g/kg or approximately 100 g), within 6 h, about 80% of the administered dose was excreted unmetabolized, while <0.5% of the administered dose was excreted as oxalic acid (80 mg oxalic acid).^{46,47} In healthy adults, oxalate excretion averages about 25 mg/d, and normal oxalate excretion is considered to be less than 40 mg/d. Thus, practitioners utilizing high-dose intravenous vitamin C therapy, such as with cancer patients,⁴⁷ need to recognize the possibility of hyperoxaluria.

In guinea pigs and rats, as well as in primates, there is considerable conversion of vitamin C to carbon dioxide. Over 65% of an injected dose was excreted as carbon dioxide within 10 days in guinea pigs, with 20% of the dose catabolized to carbon dioxide within the first 24 h.⁴⁸ In humans, this respiratory pathway seems insignificant, since less than 5% of an injected dose appeared as carbon dioxide after a 10-day period.

The principal route of excretion of vitamin C and its metabolites in humans is through the urine, mainly as oxalate and unmetabolized ascorbic acid. Utilizing C¹⁴-labeled ascorbic acid to trace excretion routes, it was determined that, after 10 days, about 40% of an injected dose of ascorbic acid was excreted in urine in humans compared with only 10% in guinea pigs.⁴⁹

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Gene Regulation and Metabolomics

Recent research suggests a novel molecular role of vitamin C in stem cell biology. Mesenchymal stem cells (MSCs) harvested from various sources, including the oral cavity, bone marrow, adipose tissue, and umbilical cord, are used therapeutically for the regeneration of bone, muscle, brain, tendon, and gum tissues. Trials show that ascorbic acid is an effective proliferating agent for the expansion of MSC in vitro by promoting culture stability, cell proliferation, and maintenance of stemness.^{50,51} Other work demonstrates the ability of vitamin C to facilitate the gene expression and epigenetic modifications necessary for stem cell pluripotency, functions explained in part by vitamin C's role as a cofactor for histone and DNA/RNA demethylases.⁵²

A genome-wide transcriptome analysis in THP-1 macrophages revealed a wide spectrum of biological processes either up- or downregulated by vitamin C treatment (~100 μ M for 96 h).⁵³ These functional responses included “signal transduction,” “cAMP metabolic process,” “cholesterol transport,” and “immune system.” Other work has delineated the role of vitamin C in cAMP regulation via its inhibitory action on adenylyl cyclase,⁵⁴ and since cAMP is a key second messenger and signaling agent for gene expression, this function of vitamin C should be explored.⁵⁵

For example, in vitro, cAMP signaling initiates the process of adipogenesis from multipotent MSCs. In two separate cell lines with adipogenic capacity, vitamin C (IC₅₀, 150 μ M/L) inhibited the differentiation of mesenchymal cells into mature adipocytes, an effect attributed to vitamin C inhibition of cAMP production.⁵⁶ Moreover, the expression of the gene encoding SVCT2 decreased during the adipocyte differentiation process and was absent in mature adipocytes; however, vitamin C treatment increased SVCT2 expression and was associated with reductions in lipid accumulation. This reversal of lipid accumulation by vitamin C in vitro is noteworthy, given the reported inverse association between adiposity and plasma vitamin C concentrations in adults.^{57–59} Alternatively, obesity-induced inflammation may accelerate the metabolism of vitamin C in vivo reducing plasma concentrations of the vitamin, and some evidence has shown that vitamin C promoted locomotive behaviors and lipolytic responses in cafeteria diet-fed rats⁶⁰ and inhibited gene expression linked to obesity onset.⁶¹

There is evidence demonstrating cofactor roles for vitamin C in several transcription and enzymatic reactions that regulate osteoblast differentiation and bone maintenance.⁶² High intakes of vitamin C are related to higher bone mineral density and a lower risk of osteoporosis.⁶³ There is also evidence that vitamin C treatment altered gene expression in cultured skin fibroblasts from burn patients: 10 of 84 genes related to oxidative stress were significantly upregulated, including six genes associated with antioxidant proteins.⁶⁴

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Subcutaneous and intramuscular hemorrhages, leg edema, neuropathy, and cerebral hemorrhage characterize scurvy, the vitamin C deficiency disease, and these symptoms are generally attributed to weakened collagenous structures. If untreated, the condition is ultimately fatal. Throughout the course of civilization, scurvy has plagued whole populations without access to fresh fruits and vegetables. At present, scurvy is reported in malnourished and refugee populations across the globe⁶⁵ and also observed in developed nations, particularly among alcoholics, institutionalized elderly, men who live alone, and children or adults who consume restrictive diets.^{66,67} Patients complain of lassitude, weakness, and general malaise and often seek medical advice following the appearance of a skin rash, bruising, lower extremity edema, and bone pain.^{66,67}

Data from the second National Health and Nutrition Examination Survey (NHANES II) conducted in the United States from 1976 to 1980 indicated that the prevalence of vitamin C deficiency defined as serum vitamin C < 0.2 mg/dL (<11 $\mu\text{mol/L}$) ranged from 0.1% in children (3–5 y) to 3% in females (25–44 y) and 7% in males (45–64 y).⁶⁸ The prevalence of marginal serum vitamin C status (0.2–0.5 mg/dL or 11–28 $\mu\text{mol/L}$) in the NHANES 1976–80 cohort ranged from 17% in adult females to 24% in adult males. In the most recent NHANES data for blood vitamin C analyses (2003–04), vitamin C deficiency rates ranged from 8% to 11% for women and men (20–39 y and 20–59 y respectively).⁶⁹ The marginal vitamin C prevalence rate for US adults in this 2003–04 cohort remained fairly stable at about 20% of the population.⁷⁰

Vitamin C deficiency rates for a sample of young Canadian adults (serum vitamin C < 0.2 mg/dL [11 $\mu\text{mol/L}$]) and for a representative sample of low income adults in the United Kingdom (plasma vitamin C < 0.2 mg/dL [11 $\mu\text{mol/L}$]) were reported at 14% and 19%, respectively.^{71,72} In a small cohort of Australian patients attending an urban surgical practice (n = 306), 21.4%

were vitamin C deficient (plasma vitamin C $\leq$ 11.4 $\mu\text{mol/L}$).⁷³ At a single urban Canadian hematology clinic, a retrospective chart review identified 22 patients presenting with scurvy over an 8-year period.⁷⁴ Of 47 adults entering an alcohol detoxification program in Saint-Louis, France, less than one-third had adequate vitamin C status and 57% had vitamin C deficiency (plasma vitamin C < 11 $\mu\text{mol/L}$).⁷⁵ Hence, subclinical scurvy is surprisingly evident in various affluent populations.

B. Assessment of Status

Functional measures of vitamin C status are not currently available, so vitamin C concentrations in plasma or leukocytes are used to determine vitamin C status. Leukocyte concentrations are difficult to measure and interpret; hence, the measurement of plasma vitamin C is the most widely applied test for vitamin C status. Plasma concentrations <0.2 mg/dL (11 $\mu\text{mol/L}$) indicate vitamin C deficiency, and concentrations between 0.2 and 0.5 mg/dL (11 and 28 $\mu\text{mol/L}$) represent marginal vitamin C status, defined as moderate risk for developing vitamin C deficiency due to inadequate tissue stores. Intakes at adult recommended levels, 75–90 mg/d, are associated with plasma vitamin C concentrations near 0.8 mg/dL (45 $\mu\text{mol/L}$), and tissue saturation is achieved at intakes from 100 to 200 mg/d, corresponding to plasma concentrations near 1.0 mg/dL (about 60 $\mu\text{mol/L}$).⁴⁵

C. Reference Intakes

In 2000, the Panel on Dietary Antioxidants and Related Compounds of the Institute of Medicine, a part of the United States National Academies, released the Dietary Reference Intakes (DRIs) for vitamin C (Table 9.2), a joint effort undertaken for Canada and the United States.⁷⁶ For adult men, the Estimated Average Requirement (EAR) for vitamin C, the level of intake that is estimated to meet the requirements of half the healthy individuals in a life stage and gender group, was set at 75 mg/d to maintain near-maximal tissue concentrations of vitamin C and provide antioxidant protection. EAR for adult women, 60 mg/d, was extrapolated based on body weight differences from the EAR for men. The EAR is used to estimate the rate of inadequacy in a similar population group and is the basis for the RDA.

The RDAs were calculated as 120% of the EAR and are believed to cover the needs of 97%–98% of individuals (90 and 75 mg/d for adult men and women, respectively) (Table 9.2). Older age groups (>50 years) have decreased lean body mass compared with their younger counterparts; however, the EARs and RDAs for older

TABLE 9.2 Daily vitamin C recommendations for age and gender groups by various health agencies^a.

Age	mg vitamin C/day			
	RDA IOM	RNI WHO/FAO	PRI EU	RDA NIHN
0–6 mo	40 ^b	25	—	40 ^b
7–12 mo	50 ^b	30	20	40 ^b
1–3 yr	15	30	20	35–40
4–8 yr	25	30–35	30–45	40–60
9–13 yr (boys)	45	35–40	45–70	60–95
14–17 yr (boys)	75	40	70–100	95–100
9–13 yr (girls)	45	35–40	45–70	60–95
14–17 yr (girls)	65	40	70–90	95–100
18+ yr (men)	90	45	110	100
18+ yr (women)	75	45	95	100
14–18 yr (pregnant)	80	55	105	110
19–50 yr (pregnant)	85	55	105	110
14–18 yr (lactating)	115	70	155	145
19–50 yr (lactating)	120	70	155	145

^aAbbreviations: EU, European Union; IOM, Institute of Medicine (for Canada and the United States); NIHN, National Institute of Health and Nutrition, Japan; PRI, Population Reference Intake; RDA, Recommended Dietary Allowance; RNI, Reference Nutrient Intake; WHO/FAO, World Health Organization/Food and Agricultural Organization of the United Nations

^bAdequate Intake.

men and women were not decreased from that for younger adults because oxidative stress, which causes higher vitamin C needs, increases with age.⁷⁶ The IOM report recommended that smokers should consume an additional 35 mg of vitamin C daily, totaling 125 mg/d for adult men and 110 mg/d for adult women, due to increased oxidative stress. Data from the NHANES cohort 2013–16 reported that 23% of children and 48% adults had inadequate intakes of vitamin C (e.g., daily intake fell below the EAR).⁷⁷ The daily average intakes for vitamin C in this cohort ranged from 75 to 85 mg for adult females and males, respectively.⁷⁸

Higher vitamin C intakes were recommended for pregnant and lactating women to offset losses from maternal body pools (Table 9.2). RDAs for children ranged from 15 to 25 mg/d for younger children and from 45 to 75 mg/d for preadolescents and adolescents. Since there were no functional criteria for vitamin C status in infants, the recommended intakes of vitamin C at this age level were based on Adequate Intakes reflecting the vitamin C ingestion of mainly breastfed infants (Table 9.2).

D. Variation in Requirements

The World Health Organization and Food and Agriculture Organization of the United Nations⁷⁹ set the recommended nutrient intake for vitamin C at 45 mg/d for all adults and elderly individuals including smokers, a value estimated to achieve 50% tissue saturation in 97.5% of this population. Pregnant and lactating women require an additional 10 and 25 mg daily; recommended intakes were arbitrarily set at 25 mg/d for infants gradually rising to 40 mg/d for adolescents (Table 9.2). The European Food Safety Authority recommended vitamin C intakes of 95–110 mg/d for adults and 105 and 155 mg/d for pregnant and lactating women, respectively.⁸⁰ The Ministry of Health, Labour and Welfare of Japan⁸¹ set the DRIs for vitamin C at 40 mg/d for infants and children up to 1 year of age; the recommendation gradually rises to 75 mg/d for children 10–11 years of age and to 100 mg daily for all age categories 15 years and older. For pregnant and lactating women, the recommendations are 110 and 145 mg daily (Table 9.2).

E. Tolerable Upper Intake Levels

High doses of vitamin C (as high as 2 g/d) are well tolerated by healthy individuals. Based strictly on reports of gastrointestinal disturbances, the Tolerable Upper Intake Level (UL) for vitamin C was set at 2000 mg/d for adults by the Institute of Medicine.⁷⁶ The UL represents a dosage level that is with high probability tolerated biologically and provides guidance to individuals using dietary supplements. Claims that supplemental vitamin C may lead to rebound scurvy, the iron-catalyzed oxidation of ascorbic acid, red blood cell hemolysis, and vitamin B12 deficiency did not withstand scientific scrutiny.⁷⁶ Based on the NHANES cohort 2013–16, <3% of US adults have daily vitamin C intakes over the UL.

F. Dietary Sources

Vitamin C is found exclusively in fruits and vegetables with the exception of the liver, which may contain as much as 28 mg/100 g (cooked). Excellent sources of vitamin C (per 100 g raw product) include guava (228 mg), sweet yellow pepper (184 mg), kiwifruit (93 mg), kale (93 mg), broccoli (89 mg), and Brussel sprouts (85 mg). Papaya, strawberries, oranges, orange juice, and cauliflower contain lesser amounts (50–60 mg/100 g). The commonly consumed fruits and vegetables in the United States (apples, bananas, grapes, potatoes, and tomatoes) contain only 5–16 mg vitamin C/100 g raw. Since vitamin C is easily oxidized by exposure to light, air, and heat, processing methods can significantly reduce the vitamin C content of fruits

and vegetables. For example, vitamin C losses in cooked spinach reached nearly 60%.⁸² Hence, consuming a vitamin C-rich raw fruit or vegetable daily is advisable to meet dietary recommendations.

V. ISSUES RELATED TO VITAMIN C IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Vascular health

Current evidence suggests possible roles of vitamin C in the promotion of vascular health beyond its role as a general antioxidant.⁸³ In coronary heart disease, oxidative stress reduces intracellular concentrations of the vasodilator nitric oxide (NO) and elevates vascular cell adhesion molecules and proinflammatory substances. Vitamin C stimulates endothelial NO synthesis by attenuating the oxidative degradation of tetrahydrobiopterin, an essential cofactor for endothelial NO synthase.⁸⁴ Vitamin C also functions in the synthesis and deposition of type IV collagen in the vascular endothelial basement membrane and the stimulation of endothelial cell proliferation.⁸⁵

In healthy adults, vitamin C ingestion (1500 mg) rapidly reversed the endothelial dysfunction that occurred acutely during a 3-h period of uninterrupted sitting, an effect attributed to the restoration of NO activity.⁸⁶ Vitamin C also reversed the endothelial dysfunction produced by induced hypoglycemia and by recovery hyperglycemia in individuals with type 1 diabetes.^{87,88} Moreover, data from 44 clinical trials supported a role for vitamin C in reducing endothelial dysfunction, particularly in diabetic and heart failure patients.⁸⁹

Although several large randomized clinical trials did not show clear benefits of supplemental vitamin C for reducing risk of coronary heart disease or for improving clinical symptoms in patient populations,^{90,91} other evidence suggests that vitamin C regimens have clinical value. In men with hypercholesterolemia (serum cholesterol ≥ 5.0 mmol/L), vitamin C supplementation (250 mg slow release daily for 3 years) displayed a 56% lower risk for atherosclerotic progression as compared to placebo.⁹² A dose-response analysis of five prospective trials measuring circulating vitamin C and incidence of stroke ($n = 25,433$) suggested that a 20 $\mu\text{mol/L}$ increase in plasma vitamin C was associated with a 19% reduction in risk of stroke ($\text{RR} = 0.81$, 95% CI: 0.75–0.88).⁹³ Interestingly, short telomere lengths, which indicate premature vascular aging, are linked to low plasma vitamin C concentrations.⁹⁴ Moreover, vitamin C supplementation reduced collar-induced intima-media thickening, an independent predictor of stroke, in rabbits after 14 days.⁹⁵

Given these potential mechanisms, vitamin C nutriture may need to be ample over the course of decades to favorably impact risk for stroke.

Cancer

Much recent epidemiologic evidence suggests that vitamin C may protect against cancer, including breast, pancreatic, and upper gastrointestinal tract cancers,^{96–102} and one report demonstrated a benefit of postdiagnosis vitamin C supplementation for reducing breast cancer mortality risk.¹⁰³ Yet, there are many reports suggesting neutral or even promotive relationships between vitamin C status and cancer risk^{104–107}; hence, the efficacy of vitamin C supplementation for lowering cancer risk in the general population is equivocal.

There are many plausible theoretical mechanisms for the anticancer effects of vitamin C including reductions in cellular DNA and protein oxidative damage; blockage of the Warburg effect; epigenetic reprogramming of stem cells; inhibition of N-nitroso compound formation from nitrite; and inhibition of the tumor promoting hypoxia-inducible factors (see¹⁰⁸). However, much of the current interest is focused on the efficacy of high-dose vitamin C injection therapy as a cancer treatment.^{109,110} This therapy delivers 0.1–1.5 g/kg ascorbic acid intravenously achieving transient peak plasma vitamin C concentrations as high as 15 mmol/L (physiological reference ranges for plasma vitamin, 50–70 $\mu\text{mol/L}$). At mM concentrations in biological systems, vitamin C acts as a prooxidant generating H_2O_2 that directly destroys cancer cells but not normal cells. The discriminating factor between cell types may be the high concentration of labile iron in cancer cells due to the specific downregulation of ferritin and the upregulation of hepcidin in tumor cells.¹¹¹ The reduction of this tumor-specific elevation in labile iron by vitamin C then drives the formation of superoxide anion from O_2 that then dismutates to H_2O_2 .¹¹²

Many in vitro reports delineate this anticancer effect of pharmacologic concentrations of vitamin C; moreover, vitamin C injection therapy is reasonably safe,^{113,114} and phase II studies are encouraging. A recent metaanalysis identified 23 phase I/II trials in 401 cancer patients receiving intravenous vitamin C (variable dosages: 1–1.5 g/kg injection; 3–7 days per week; 5–24 weeks). The authors concluded that intravenous vitamin C appeared useful as a single agent or combined with a primary therapy and that randomized controlled trials are now necessary to establish efficacy of treatment.¹¹⁵ There is one randomized controlled trial in patients with stage III/IV ovarian cancer ($n = 25$) that examined the efficacy of infusion vitamin C therapy (target peak plasma vitamin C of 20–23 mM) in combination with standard therapy. The results demonstrated decreases in most categories of toxicity and improved overall survival (median

time for disease progression/relapse was nearly 9 months longer in the vitamin C arm).¹¹⁶

Immunoenhancement

Supplemental vitamin C has been demonstrated to enhance immune parameters in certain patient populations, including neutrophil activation in sepsis and Chediak-Higashi syndrome, natural killer cell activity in thalassemia, and improved neutrophil motility in furunculosis.^{117–120} However, trials conducted in healthy individuals show no benefit of vitamin C supplementation,¹²¹ and a definitive role for vitamin C in immunoenhancement has not been established.

In vitro vitamin C directly inhibits NF- κ B activation and selectively inhibits inflammatory cytokine production, suggesting a possible antioxidant-independent effect of vitamin C in controlling inflammatory responses, which would impact both acute and chronic disease.^{122–125} Inhalation of lipopolysaccharide, a product of Gram-negative bacteria, stimulates alveolar macrophages to secrete the inflammatory mediators (IL-1, TNF- α , and IL-6) via NF- κ B activation, which, if unchecked, can lead to severe inflammation and pneumonia.¹²⁶ Although well-designed trials are scarce, a systematic review reported that vitamin C treatment favorably influenced the duration of pneumonia in several trials and that the prophylactic use of vitamin C may reduce the incidence of pneumonia in high-risk populations.¹²⁷

Serum vitamin C was inversely related to seroprevalence of *Helicobacter pylori* in a large, population-based US sample.¹²⁸ In a randomized, controlled trial (n = 312 patients with *H. pylori* infection), the addition of vitamin C (500 mg/d) to the standard *H. pylori* treatment regimen significantly improved the *H. pylori* eradication rate by nearly 60% (from 49% to 78%).¹²⁹ Others have also noted beneficial effects of supplemental vitamin C in combination with *H. pylori* eradication regimens in randomized clinical trials.^{130,131} Since the virulence of *H. pylori* is related to NF- κ B signaling,¹³² the benefit of vitamin C therapy may be related to its ability to inhibit NF- κ B activation.

Beneficial effects of vitamin C have been reported for several viral infections including Epstein–Barr and herpes zoster. A retrospective chart review of 218 patients with elevated Epstein–Barr antibodies linked high-dose intravenous vitamin C therapy to reductions in viral antibodies and symptom duration.¹³³ In several randomized clinical trials, intravenous administration of ascorbic acid (5 g on days 1, 3, and 5) reduced spontaneous pain in patients with herpes zoster as compared to the saline control, and the incidence of postherpetic neuralgia was reduced 46% in one of these patient populations.^{134,135} The broad antioxidant effect of ascorbic acid is the likely mechanism for the relief observed with vitamin C treatment since reactive oxygen species are implicated in the peripheral neuropathic pain

associated with these infections. In patients recovering from wrist fractures, oral vitamin C dosing (>500 mg/d for 50 d) reduced the occurrence of complex regional pain syndrome for up to 1 year.¹³⁶ These reports suggest a possible role for vitamin C therapy in pain management. Indeed, an analysis of a national data set (n = 4742 adults aged ≥ 20 y) demonstrated strong associations between suboptimal vitamin C status and spinal pain.⁷⁰

The role of supplemental vitamin C in combating the common cold has long been debated. A metaanalysis of 29 controlled trials concluded that vitamin C supplementation reduced the incidence of colds in people under stressful conditions such as severe physical activity (RR = 0.48; 95%CI 0.35 to 0.64) but not in normal, unstressed people (RR = 0.97; 95%CI 0.94 to 1.00).¹³⁷ Since the incidence of the common cold varies across the year (highest incidence November to March) and vitamin C status is variable among individuals, these factors should be considered. Trials conducted during the winter months or over several years that recruit vitamin C–depleted individuals are necessary to demonstrate the efficacy of vitamin C in cold prevention. A 5-y randomized, double-blinded, placebo-controlled trial demonstrated that vitamin C supplementation (500 mg/d) reduced the frequency of colds 20% and reduced the risk of suffering more than 2 cold episodes over a 3-year period by 66%, compared to the placebo treatment (50 mg vitamin C/d).¹³⁸ In vitamin C–depleted male college students, vitamin C supplementation (1000 mg/d for 8 weeks during the winter months) reduced the incidence of colds by 45% in comparison to placebo.¹³⁹

B. Known Interactions

Vitamin C regenerates vitamin E as well as antioxidant polyphenols and acts in synergy with these antioxidants, and with glutathione and selenium, to promote maximal antioxidant protection. For the many pathological conditions associated with oxidative stress and reactive radicals, including cataract, neurodegenerative diseases, and arthritis, vitamin C is linked to favorable disease outcomes.^{140–142} Emerging evidence suggests a role for vitamin C in pain management, possibly via oxidative stress reduction or the biosynthesis of analgesic neurotransmitters and neuropeptide hormones.^{100,101,143}

Vitamin C plays a well-documented role in nonheme iron absorption in the gut, and recent research is expanding the roles of vitamin C in iron metabolism and homeostasis.¹⁴⁴ Dietary vitamin C is linked to improved iron status in compromised patient populations^{145,146} and populations consuming plant-based diets.¹⁴⁷

C. As a Dietary Supplement

In the United States, 43% of adults over the age of 20 y reported taking vitamin C supplements, with about 75%

of these individuals taking <500 mg/d.⁷⁰ Supplemental vitamin C is generally well tolerated, yet a small portion of individuals report diarrhea, nausea, and other gastrointestinal disturbances when supplementing vitamin C. This is due to the osmotic effect of unabsorbed vitamin C in the gastrointestinal tract and represents the premise for setting the UL for vitamin C at 2000 mg/d. However, because dietary vitamin C may enhance mealtime iron absorption, high-dose vitamin C regimens may aggravate conditions associated with increased iron absorption and storage, notably hemochromatosis. About 0.5% and 10% of the US population are homozygous and heterozygous, respectively, for the hemochromatosis HFE gene mutation. The addition of vitamin C–fortified orange juice to a test meal did increase nonheme iron absorption in homozygous patients compared to controls,¹⁴⁸ but increased iron absorption was not observed in patients heterozygous for the HFE gene mutation.¹⁴⁹ Analyses of NHANES data from 1976 to 1980 showed that women with the highest serum vitamin C concentrations (>2.1 mg/dL or 116 $\mu\text{mol/L}$) had a twofold

increased prevalence of elevated serum ferritin levels.¹⁵⁰

In thalassemia major, an iron-overload disease characterized by impaired globin chain synthesis, excessive vitamin C supplementation by patients may mobilize iron, increasing oxidative stress and exacerbating tissue damage. However, modest vitamin C supplementation (100 mg/d) concomitant with chelation therapy may benefit patients by enhancing urinary iron excretion and reducing oxidative stress.¹⁵¹

High-dose vitamin C regimens increase urinary excretion of oxalic acid and uric acid, constituents of renal calculi, and, theoretically, may promote the formation of kidney stones. Epidemiologic investigations have reported mixed results regarding a direct association between vitamin C supplementation and stone incidence. In one prospective cohort analysis (156,735 women and 40,536 men), dietary vitamin C intake was not associated with the incidence of kidney stones; however, in men only, supplemental intakes of vitamin C > 700 mg/d were associated with a significant increased risk for stones.¹⁵²

RESEARCH GAPS

Research into the contribution of vitamin C deficiency to morbidity and mortality is needed since 8%–19% of North American and European citizens fall into this category. Diet–gene interactions are important considerations for trials examining associations between vitamin C intake and disease risk. Also, because it is inexpensive and relatively nontoxic, continued investigation of the potential benefits of supplemental vitamin C should be pursued.

Note: Additional sources of information in the published literature regarding Vitamin C in disease states¹⁵³ and epigenomics¹⁵⁴ are available.

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THIAMINE

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SUMMARY

Thiamine (vitamin B1) was the first vitamin characterized, and its discovery was at the origin of the concept of vitamin. The main form of thiamine is thiamine diphosphate, an essential coenzyme in cell metabolism. Thiamine deficiency mainly affects the nervous and cardiovascular systems and causes two classical diseases, beriberi and Wernicke–Korsakoff syndrome (anterograde amnesia resulting from brain lesions in alcoholics). Considering the large spectrum of manifestations of beriberi (wet, dry, infantile) and Wernicke’s encephalopathy, it was recently proposed to replace the archaic nomenclature by the umbrella term thiamine deficiency disorders. In this chapter, we discuss the role of thiamine derivatives, thiamine dietary requirements, thiamine transport mechanisms, and metabolism. Current research interests are focused on the metabolism and role of thiamine derivatives and the biochemical and pathophysiological mechanisms by which thiamine deficiency induces specific brain lesions and may be involved in other disorders such as Alzheimer’s disease and diabetes.

Keywords: Alzheimer’s disease; Beriberi; Coenzyme; Diabetes; Korsakoff’s syndrome; Thiamine deficiency disorders; Thiamine diphosphate; Vitamin B1; Wernicke’s encephalopathy.

I. INTRODUCTION

A. Background: Thiamine and The Concept of Vitamins

Thiamine (thiamin, [Fig. 10.1](#)) is a water-soluble vitamin of the B group (vitamin B1, aneurin). Most bacteria, fungi, and plants are able to synthesize thiamine *de novo*, while animals solely rely on exogenous sources. Nutritional thiamine deficiency in the adult commonly leads to a polyneuritic syndrome called (dry) beriberi. This used to be a major public health problem in East Asia, especially in the 19th century, because the population in these countries heavily relied on polished rice (poor in thiamine) as staple food. Another, relatively common form of thiamine deficiency called Wernicke–Korsakoff syndrome, which affects the central nervous system and is generally associated with chronic alcoholism.

The discovery of thiamine is a famed story at the origin of the concept of vitamins.¹ Beriberi was described in Chinese medical texts as early as 4000 years

ago. The first description of beriberi in Japan appeared in the 9th century, but its relationship with diet was not suspected until the end of the 19th century. In 1890, the Dutch physician Christiaan Eijkman, working at the Medical Laboratory of Weltevreden (Batavia, now Djakarta in Indonesia), made the significant observation that fowl, when fed on polished rice, soon died from paralysis. The early symptoms could be reversed by switching to whole rice. Gerrit Grijns, Eijkman’s successor in Batavia, recognized that polyneuritis was caused by the absence of an essential substance, the anti-beriberi factor, that is virtually absent in the endosperm but is present in the bran of rice. At the same time, the concept of vitamin started to be developed by the English biochemist Frederick Hopkins, who first postulated that food, in addition to the well-known fundamental components, i.e., proteins, sugars, and fats, must also contain “accessory factors” that, although present only in low amounts, are nonetheless indispensable for a normal functioning of the organism. The demonstration of an anti-beriberi factor was a major

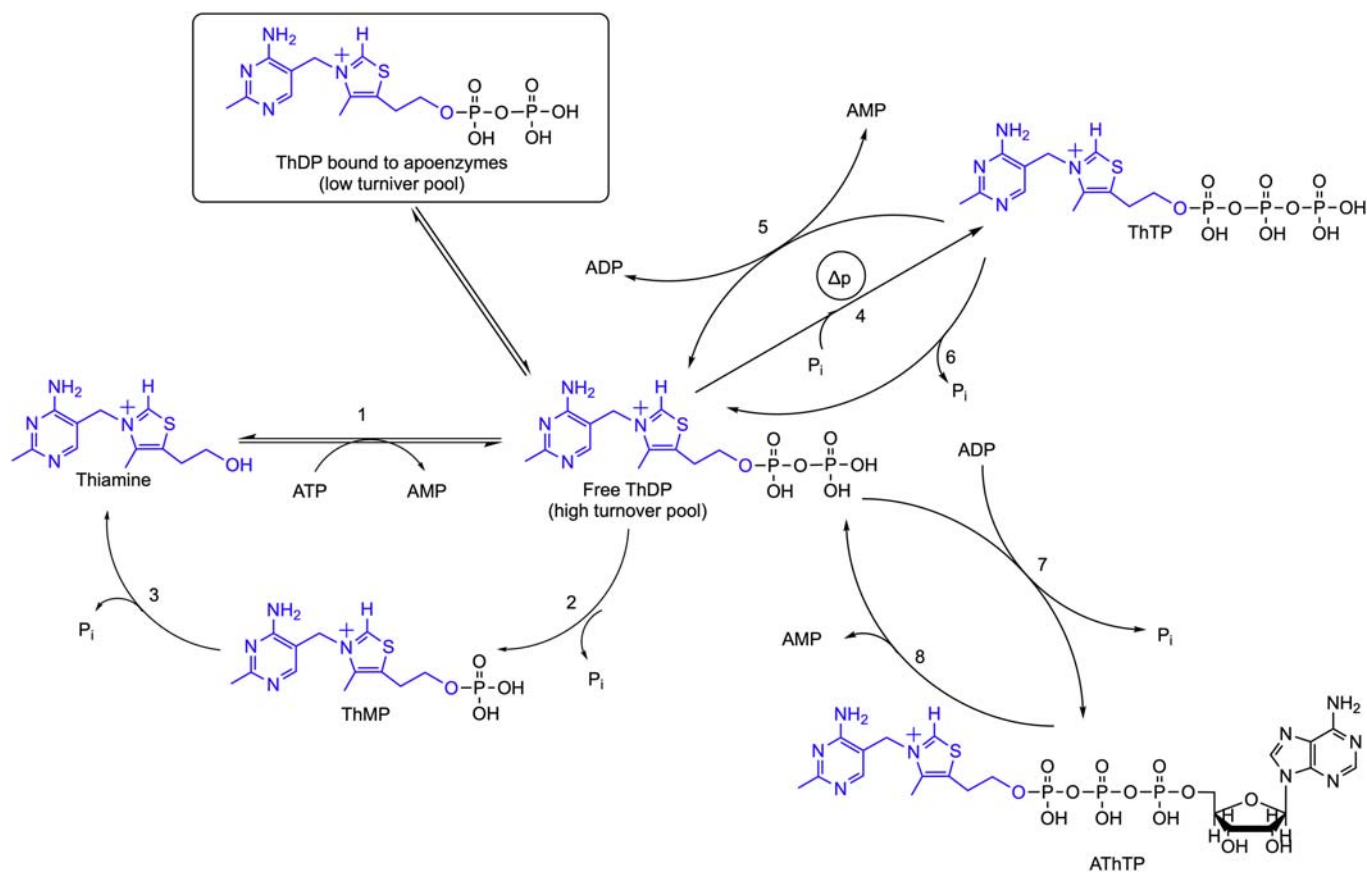


FIGURE 10.1 Structure and metabolism of thiamine derivatives. ThDP is synthesized from thiamine and ATP by thiamine pyrophosphokinase (1). This free, high turnover pool can either be hydrolyzed to ThMP by thiamine pyrophosphatases (2), bind to apoenzymes (low turnover pool), or transformed to ThTP or AThTP. ThMP can be hydrolyzed to thiamine by thiamine monophosphatases (3). ThDP can be phosphorylated to ThTP by two mechanisms: mitochondrial F₀F₁-ATP synthase (4) and cytosolic adenylate kinase (5). ThTP can be hydrolyzed to ThDP by a very specific cytosolic 25-kDa thiamine triphosphatase (6). ThDP can also be converted to AThTP by a ThDP adenylyl transferase (7). AThTP can be hydrolyzed to ThDP and AMP by a putative AThTP hydrolase (8).

step toward the acceptance of the concept of vitamins. It took many years to isolate the antiberiberi factor, and it was finally obtained in pure crystals in 1926 by Eijkman's countrymen Barend Jansen and Willem Donath (Weltevreden, Batavia). The American chemist R.R. Williams determined its structure in 1934 and synthesized the vitamin for the first time. It was first called "aneurin" for antineuritic vitamin, and it was later internationally agreed to call it "thiamine" for "sulfur-containing vitamin," but its exact properties and role remained unknown.

B. Thiamine Structure and Metabolic Function

The first clues concerning the fundamental role of thiamine in carbohydrate metabolism were provided by the classical studies carried out from 1929 to 1940 by Peters and associates, using the thiamine-deficient pigeon as a model.² The deficient birds (fed on polished rice) show a typical head retraction behavior called opisthotonos.

Remarkably, thiamine administration at an early stage of the deficiency allows a complete recovery within 30 min. As no morphological brain lesions were observed before or after thiamine treatment, these observations led to the concept of "biochemical lesion."

In 1937, Lohman and Schuster demonstrated that thiamine diphosphate (ThDP) was an indispensable cofactor for the oxidative decarboxylation of pyruvate, hence the now obsolete name of cocarboxylase.³ Between 1950 and 1960, Breslow and coworkers investigated the mechanism by which the thiamine moiety of ThDP catalyzes the decarboxylation of pyruvate in the reaction catalyzed by yeast pyruvate decarboxylase.⁴ The same mechanism is involved in the first step of oxidative decarboxylation of 2-oxoacids by mitochondrial pyruvate and 2-oxoglutarate dehydrogenase (OGDH) complexes. These studies revealed that ThDP exerts its function by proton substitution on the carbon in position 2 of the thiazolium ring.

In addition to ThDP, two other phosphorylated derivatives (Fig. 10.1), thiamine monophosphate (ThMP)

and thiamine triphosphate (ThTP), have been known for many years.⁵ More recently, an adenylated thiamine derivative, adenosine thiamine triphosphate (AThTP), has been described.^{6,7} The biological roles of derivatives other than ThDP are still unclear.

II. NUTRIENT METABOLISM

A. Thiamine Biosynthesis

Most prokaryotes, protozoans, fungi, and plants are able to synthesize thiamine. The thiamine biosynthesis pathways are highly complicated and vary between bacteria and eukaryotes. The phosphorylated thiazole and the pyrimidine moieties are synthesized separately and then assembled to form ThMP.⁸ In plants and yeast, ThMP is hydrolyzed to thiamine and then pyrophosphorylated by thiamine di(pyro)phosphokinase 1 (TPK1, EC 2.7.6.2) to the coenzyme ThDP. Bacteria, which lack TPK1, directly phosphorylate ThMP to ThDP by a thiamine (mono)phosphate kinase (EC 2.7.4.16, Fig. 10.1). Thiamine biosynthesis is regulated by riboswitches found in bacteria, fungi, and plants. Riboswitches are specific sequences located either in the terminal untranslated regions or in the introns of pre-mRNAs that bind small ligands in order to regulate the expression of vitamins or amino acids. Binding of ThDP to the sensory domain of the ThDP riboswitch induces a change in conformation of the mRNA halting its transcription or translation.⁹

B. Intestinal Absorption, Excretion, and Homeostasis

Human plasma thiamine concentrations are very low, approximately 10 nM, with free concentrations probably even lower.^{10,11} Thanks to the existence of high-affinity thiamine transporters, such a low concentration is sufficient to maintain ThDP-dependent apoenzymes at near coenzyme saturation. The half-life of coenzyme-bound ThDP, the so-called low-turnover pool, is approximately 17 h.¹² However, insufficient dietary intake possibly in combination with higher metabolic needs (lactation, carbon-rich diet, hyperemesis gravidarum, excessive alcohol consumption) can rapidly lead to a state of thiamine deficiency.

Global thiamine homeostasis is maintained by balancing intestinal uptake and renal loss. However, different mechanisms seem to prevail in different organs, shielding the brain from important variations in thiamine content. Hence, during thiamine deficiency, thiamine levels decrease less steeply in the brain than in the liver.^{13,14} On the other hand, administration of a high dose of thiamine (or a thiamine precursor with a higher bioavailability) leads to an important increase in total thiamine (mainly in the form of ThDP) in

erythrocytes and in the liver, but not in the brain (see below). There are no long-term thiamine body stores, and once the plasma thiamine concentration tends to normalize following increased excretion, erythrocytes and liver will release thiamine (after dephosphorylation of excess ThDP), which will be rapidly eliminated.^{15,16} Approximately 20–24 h after a bolus administration, thiamine levels have returned to normal.

Total thiamine concentration in human milk is approximately 200 µg/L (~0.5 µM), mostly in the form of ThMP.¹⁷

Intestinal thiamine transport

Thiamine phosphate esters taken up from the food are hydrolyzed to thiamine or ThMP by intestinal alkaline phosphatase and transported across the brush border membrane to the portal system. At low concentrations (<1 µM), thiamine is absorbed through an active process, while at high intraluminal concentration, a passive process seems to prevail. As a result, a larger proportion of a small dose of thiamine is absorbed when its concentration is low.

The high-affinity transport process ($K_m = 0.01$ – 1 µM) across the brush border membrane of the enterocyte involves an electroneutral thiamine/ H^+ antiport and is thus facilitated by an outwardly directed H^+ gradient,¹⁸ which is probably the driving force for intestinal thiamine transport in the duodenum.

Thiamine degradation and excretion

Degradation and elimination of thiamine have not been extensively studied. After administration of high doses, excess thiamine and some of its metabolites (2-methyl-4-amino-5-pyrimidine carboxylic acid and 4-methyl-thiazole-5-acetic acid) are excreted in the urine. Urinary thiamine excretion is decreased during fasting.¹⁹ Human urine thiamine concentrations are ~0.5 mg/L or 1.5 µM.²⁰ In the rat, after intraperitoneal administration of ^{14}C -pyrimidine-labeled thiamine, ~30% of the vitamin taken up is eliminated by urinary excretion and 13% in the feces.²¹ However, only ~6% of the excreted radioactivity was genuine thiamine, the rest being found in at least 22 degradation products.²² Probably most of these are products of detoxification enzymes present in the liver and kidney. None of these enzymes is known to be specific for thiamine.

Thiamine homeostasis

There is no doubt that the administration of radioactively labeled thiamine in rats leads to a much more rapid and higher accumulation of radioactivity in the blood and the liver than in the brain.²³ This is in agreement with the observation that thiamine is not actively transported across the blood–brain barrier, except maybe in choroid plexus.²⁴

After oral administration of large doses of thiamine or synthetic thiamine precursors such as benfotiamine (see [Section V.C](#)) in mice, blood thiamine concentrations strongly increase, followed by an increase in liver thiamine content. However, brain content hardly increases.¹⁶ Those results can be explained if the rate of thiamine transport across the blood–brain barrier is low compared to the rate of uptake by hepatocytes.²⁵

Net thiamine accumulation by the brain is also influenced by efflux of thiamine through the blood–brain barrier, an efflux considerably increased by transstimulation.²⁶ Thiamine efflux was also stimulated by extracellular thiamine in cultured rat neuroblastoma cells, possibly by a self-exchange mechanism.²⁷ Such a transstimulation of thiamine efflux would limit excess thiamine uptake by the brain.

In conclusion, the emerging picture is that thiamine, after entering the bloodstream, is mostly captured and phosphorylated by the liver (which is the main short-term thiamine storage organ) and, to a lesser extent, by erythrocytes acting as a thiamine buffer. In contrast, thiamine levels in the brain are highly regulated to minimize variations in thiamine and thiamine phosphate levels: mostly unknown regulatory mechanisms control both the influx and efflux of thiamine.

C. Thiamine Transporters

High-affinity thiamine transport across plasma membranes is achieved by two homologous transporters (*SLC19A2* and *SLC19A3*) belonging to the *SLC19* solute (folate/thiamine) carrier family.²⁸ The transport most probably involves a thiamine/H⁺ antiport according to the reaction: Thiamine (out) + nH⁺ (in) $\rightleftharpoons$ Thiamine (in) + nH⁺ (out). Intracellular thiamine pyrophosphorylation, subsequent uptake of ThDP by mitochondria and binding to apoenzymes are additional factors favoring the intracellular accumulation of thiamine derivatives.

THTR1

SLC19A2 codes for a high-affinity thiamine transporter (THTR1) with a K_m of $2.5 \pm 0.6 \mu\text{M}$.²⁹ In humans, it is highly expressed in the skeletal muscle, heart, and placenta, but in mice, the highest expression is observed in the liver, kidney, and brain, with practically no expression in the skeletal muscle.^{30,31} Mutations in *SLC19A2* lead to thiamine-responsive megaloblastic anemia (TRMA) accompanied by diabetes mellitus and sensorineural deafness.

THTR2

THTR2, the *SLC19A3* gene product, is a thiamine transporter with a very high apparent affinity for thiamine (K_m of $27 \pm 8 \text{ nM}$).^{32,33} THTR2 is rather ubiquitously

expressed, with the highest levels in the small intestine, placenta, liver, kidney and heart, which might explain why these tissues are spared in TRMA. Furthermore, its expression is low in the bone marrow. THTR2 is expressed along the whole length of the human digestive tract, with the highest expression in the proximal part of the intestine (duodenal and jejunal areas), which has the highest capacity to absorb thiamine. THTR2 is also present on colonocytes where it might play a role in the absorption of thiamine produced by the microflora of the large intestine.³⁴ Expression of THTR2 seems to be low in human brain. Its gene product is scattered throughout blood vessels, mainly in the basement membrane and pericytes, suggesting that it is important for thiamine transport through the blood–brain barrier.³⁵ Indeed, mutations in *SLC19A3* lead to biotin–thiamine-responsive basal ganglia disease as well as Wernicke-like encephalopathies.^{36,37} Though THTR1 is widely expressed in brain, circulating thiamine concentrations are only about 10 nM and loss of the high-affinity THTR2 will impact brain thiamine uptake.

Reduced folate carrier protein (RFC-1)

SLC19A1 codes for RFC-1, which is able to transport ThMP and ThDP, but not thiamine.^{38,39} While ThDP is not physiologically encountered in extracellular media, ThMP is present in plasma and cerebrospinal fluid, and therefore RFC-1 may be an alternate route for cells to capture the vitamin. The choroid plexus, which plays an important role in the brain thiamine homeostasis, accumulates the vitamin by an active process and releases thiamine and ThMP on the apical (CSF) side.²⁴ RFC-1 is highly expressed on the apical side of the choroid plexus and may thus be responsible for the release of ThMP into the CSF.

Organic cation transporters

The nonspecific organic cation transporters (OCT) 1 and 2 seem to be important for the secretion of thiamine into the milk of lactating mice.⁴⁰ They might also be involved in high-capacity intestinal and liver thiamine transport, but probably with a low affinity ($K_m \approx 30 \mu\text{M}$). These transporters could be involved in thiamine uptake by liver cells when plasma concentrations are high. In most cell types, there are high-affinity (THTR1 and 2) and low-affinity thiamine uptake mechanisms.⁴¹ It is however not clear whether the low affinity is mediated by a transporter of the OCT family in cells other than intestinal epithelium and hepatocytes.

Mitochondrial ThDP transporter

A mitochondrial ThDP transporter (*SLC25A19*) was characterized in yeast and in animals.⁴² Mutations in *SLC25A19* cause Amish lethal microcephaly, an autosomal recessive disease characterized by extreme

microcephaly, severe 2-oxoglutarate aciduria, and fatal outcome within the first year. SLC25A19 seems to act as an antiporter, exchanging extramitochondrial ThDP against intramitochondrial nucleotides.

Colonic high-affinity ThDP transporter

In ruminants and other grazing animals, thiamine and ThDP synthesized by bacteria of the digestive tract are important sources of the vitamin. Recently, a human colonic high-affinity ThDP transporter, product of the gene SCL44A4, was identified.⁴³ The transport may be involved in the absorption of microbiota-generated ThDP and thus contribute to host thiamine homeostasis.

D. Metabolism of Thiamine and its Phosphate Derivatives

Thiamine diphosphate synthesis

In animal cells, thiamine transported into the cells is rapidly phosphorylated to ThDP according to the reaction: thiamine + ATP $\rightleftharpoons$ ThDP + AMP, catalyzed by TPK1. An isoform of this enzyme (TPK2) is present in chloroplasts.

Tissue content and metabolism of thiamine phosphate derivatives

Thiamine phosphate derivatives are found in all organisms.^{11,44} In most cases, the cofactor ThDP is the major thiamine compound and represents 70%–90% of total thiamine. In the brain, most of it is bound to apoenzymes, but a higher proportion of free ThDP is found in hepatocytes, enterocytes, and skeletal muscle fibers. ThDP is synthesized from thiamine and ATP in the cytosol (Fig. 10.1) and is hydrolyzed by thiamine diphosphatases yielding ThMP, which can be hydrolyzed to thiamine. Several enzymes have been reported to hydrolyze ThDP, but all thiamine diphosphatases studied so far also hydrolyze nucleoside diphosphates. Likewise, no hydrolase has been found to be specific for ThMP, which may be hydrolyzed by alkaline phosphatases, acid phosphatases, or an ecto-5'-nucleotidase. ThMP and free thiamine generally amount to less than 15% of total thiamine in animal cells. The triphosphorylated derivative, ThTP, is a minor compound (less than 1% of total thiamine), but it is found in most organisms, including humans.^{1,11} In *Escherichia coli*, ThTP is transiently synthesized when the bacteria are switched from an amino acid-rich to an amino acid-poor medium, suggesting that it might be a signal involved in the adaptation to amino acid starvation.⁴⁵ In animal tissues, ThTP is consistently found in nervous tissue, skeletal muscle, liver, and other tissues. At least two kinds of enzymes have been shown to hydrolyze ThTP. A soluble 25-kDa thiamine triphosphatase (THTPA, EC

3.6.1.28) has been characterized at the molecular level.⁴⁶ It is very specific for ThTP and relatively ubiquitously expressed in mammalian tissues. In the brain, it seems mainly localized in the neurons.⁴⁷ The mechanism of ThTP synthesis is controversial. Adenylate kinase 1 (EC 2.7.4.3) synthesizes ThTP according to the reaction ThDP + ADP $\rightleftharpoons$ ThTP + AMP,^{48,49} but the rate of this reaction is 6–7 orders of magnitude lower than the natural reaction 2 ADP $\rightleftharpoons$ ATP + AMP. Nevertheless, this enzyme may be responsible for the accumulation of cytosolic ThTP in skeletal muscle of birds and pigs, where THTPA is either absent or inactive.¹¹ In the brain, ThTP is localized in mitochondria and might be synthesized by a chemiosmotic mechanism resembling that of ATP synthesis.⁵⁰ Such a mechanism, involving F₀F₁-ATP synthase, also seems to prevail in *E. coli*.⁵¹

AThTP was recently discovered in *E. coli*, where it accumulates during acute energy stress.⁵² It might be synthesized by a ThDP adenyllyl transferase.⁵³ It is found in many animal cells, but we have no indication as to its potential roles.

III. ROLE IN NORMAL CELLS AND ORGANS

ThDP participates as a coenzyme in many reactions, most of them in prokaryotes. All ThDP-dependent enzymes share common pyrophosphate and pyrimidine-binding domains separated by a variable catalytic domain. They have evolved from a common, possibly transketolase-like, ancestor.^{54,55}

The coenzyme ThDP is tightly bound to its apoenzymes. ThDP bound to apoenzymes is considered the “slow turnover ThDP pool,”^{12,56} though the stability of the different coenzyme–apoenzyme complexes varies: ThDP dissociates faster from pyruvate dehydrogenase than from OGDH complex and transketolase. It can be estimated that in the brain cells or neuroblastoma cells, the slow ThDP pool represents 90%–95% of the total cellular ThDP and has a turnover of 6–20 h. For the smaller free ThDP pool, the turnover is of the order of 1–3 h. In other cell types such as hepatocytes or red blood cells, the free cytosolic ThDP pool may be much more important, in particular after thiamine administration.

A. Mammalian ThDP-Dependent Enzymes

In this chapter, we will only discuss mammalian ThDP-dependent enzymes (Fig. 10.2⁵⁷), but it must be kept in mind that many other ThDP-dependent enzymes exist in prokaryotes, not to mention pyruvate decarboxylase in yeast and some bacteria.

In mammalian cells, ThDP is a coenzyme for the cytosolic transketolase (EC 2.2.1.1) and possibly other

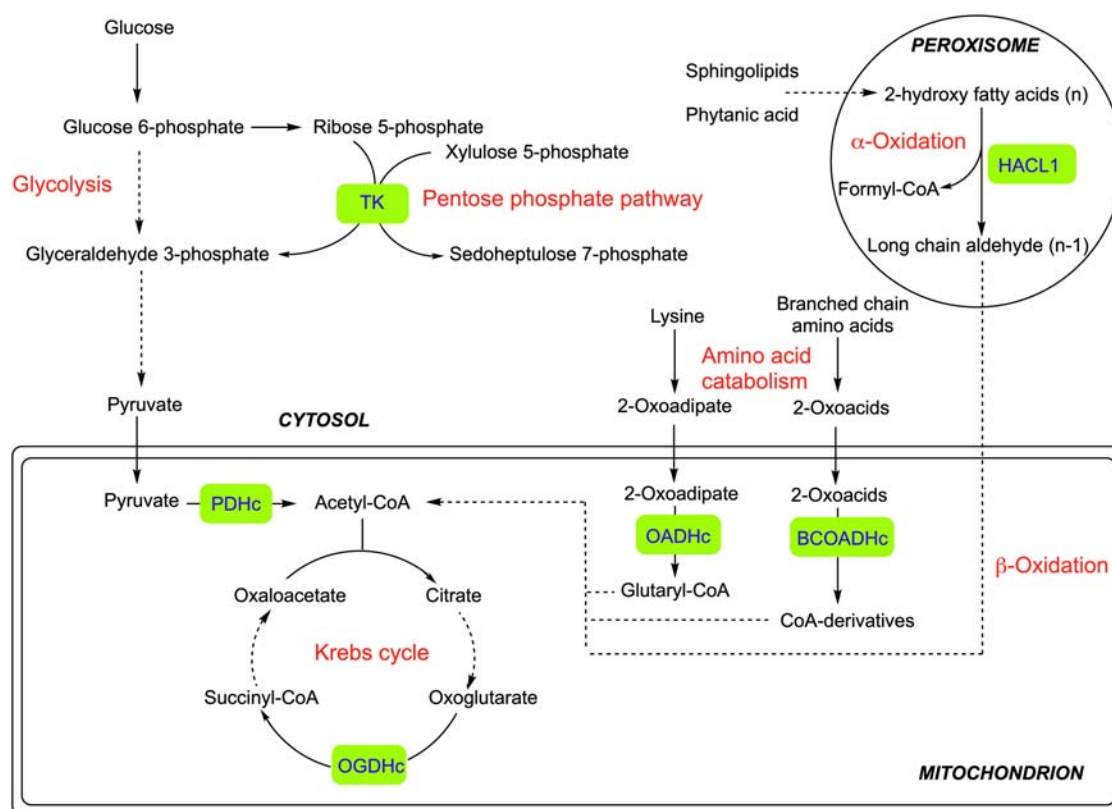
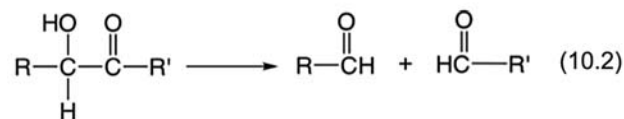
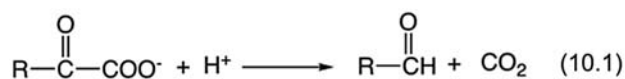


FIGURE 10.2 Role of ThDP-dependent enzymes in cellular metabolism. Thiamine-dependent enzyme is indicated in a green box. BCOADHc, branched-chain 2-oxoacid dehydrogenase complex; HACl1, 2-hydroxyacyl-CoA lyase 1; OADHc, 2-oxoadipate dehydrogenase complex; OGDHc, 2-oxoglutarate dehydrogenase complex; PDHc, pyruvate dehydrogenase complex; TK, transketolase. Figure updated from Bettendorff (2012).⁵⁷

transketolase-like enzymes. In mitochondria, ThDP is the cofactor of E1 subunits of pyruvate dehydrogenase (PDH, EC 1.2.4.1), OGDH (or α -ketoglutarate) (EC 1.2.4.2), and branched-chain 2-oxoacid dehydrogenase (BCOADH, EC 1.2.4.4) complexes. More recently, another member of the E1 dehydrogenase family has been described, 2-oxoadipate dehydrogenase complex (OADH, no EC number assigned), that seems to be involved in the oxidative decarboxylation of 2-oxoadipate to glutaryl-CoA in the L-lysine degradation pathway.^{58–60} Finally, 2-hydroxyacyl-CoA lyase (HACl1, no EC number assigned) has been described in peroxisomes where it is involved in the degradation of 3-methyl-branched-chain and of 2-hydroxy long-chain fatty acids.⁶¹

B. Mechanism of ThDP-Dependent Catalysis

In all ThDP-catalyzed enzymatic reactions, the coenzyme catalyzes the cleavage or the formation of a C–C bond immediately adjacent to a carbonyl group (α cleavage or α condensation) such as in the decarboxylation of 2-oxoacids (Eq. 10.1, 2-oxoacid dehydrogenases, pyruvate decarboxylase) or the cleavage of α -hydroxyketones (Eq. 10.2, transketolase and 2-hydroxyacyl-CoA lyase):



As the cleavage of the C–C bond directly adjacent to the carbonyl group (α -cleavage or α -fission) would yield an unstabilized carbanion, there is no simple acid-base-catalyzed mechanism for such reactions and a special catalyst is required that may act as an electron sink and stabilize the carbanion. Breslow was the first to propose a mechanism for thiazolium-catalyzed reactions that is considered to be similar to the one involved in ThDP-catalyzed enzyme reactions.⁴ According to this mechanism, the catalytic action of ThDP is linked to the ability of its thiazolium ring to form a ylide (carbanion) intermediate by deprotonation of the thiazolium C-2 and attack of the resulting carbanion on a substrate carbonyl. The catalytic mechanism of the thiazolium ring of ThDP in 2-oxoacids dehydrogenases is depicted in Fig. 10.3A. The positive charge on the quaternary

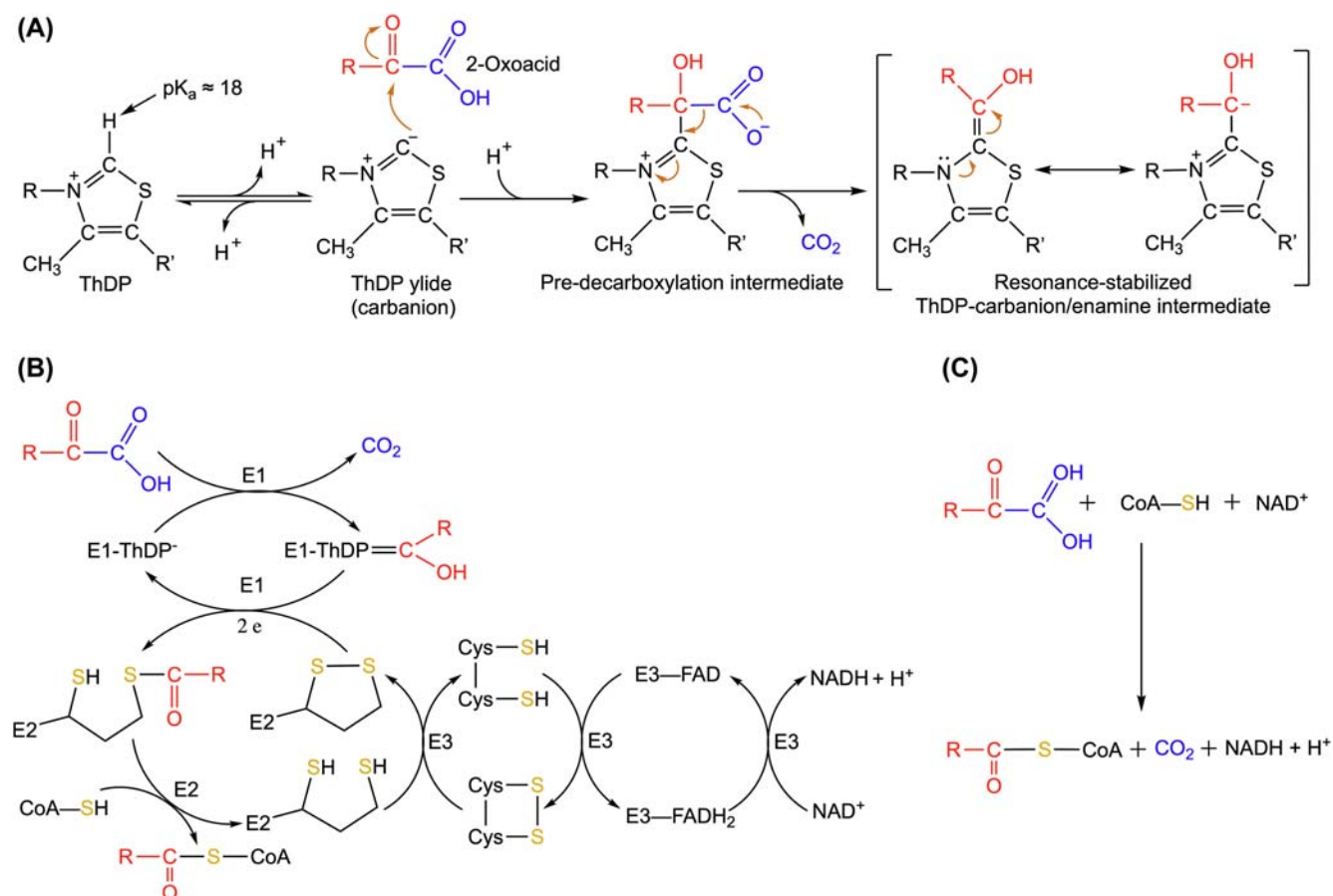


FIGURE 10.3 Role of ThDP in oxidative decarboxylation in 2-oxoacid dehydrogenases. (A) Role of the thiamine thiazole ring in catalysis. The catalytic cycle starts with the formation of a ThDP ylide with a negatively charged carbanion that will act as a nucleophile attacking the carbonyl carbon of pyruvate. This results in formation of an unstable covalent predecarboxylation intermediate that is rapidly decarboxylated in a process during which electrons are displaced toward the positively charged quaternary nitrogen of the thiazolium ring. This leads to a resonance-stabilized ThDP-carbanion/enamine intermediate, carrying the activated aldehyde. (B) General mechanism of 2-oxoacid dehydrogenases. After ThDP-dependent decarboxylation, the activated aldehyde is transferred to lipoamide, which is reduced while the activated aldehyde is oxidized before transfer to CoA and formation of the CoA-thioester. Finally, dihydrolipoamide is oxidized via a Cys-Cys disulfide group and E3-bound FAD, and NADH is released. The transferred acyl group is in red. R: CH₃ for PDHc, -OOC-CH₂-CH₂- for OGDHc, -OOC-CH₂-CH₂-CH₂- for OADHc, and leucine, isoleucine, and valine residues for BCOADHc. (C) Global reaction for the oxidative decarboxylation of a 2-oxoacid by ThDP-dependent dehydrogenase complexes.

ammonium acts as an electron drain thereby weakening the C-H bond on the adjacent carbon increasing the acidic character of the proton. In this mechanism, the role of the aromatic thiazolium ring of ThDP is the resonance stabilization of the carbanion.

It is interesting to note that this mechanism resembles cyanide catalyzed reactions. We could thus imagine that thiamine is a mechanistic analog of cyanide in which the cyanide part is buried in such a way as to be non-toxic. This may have been an important factor in the selection of this molecule in the early days of evolution.⁶²

Though the pyrimidine ring of ThDP is not essential for catalysis, it has been shown that in most ThDP-dependent enzymes, ThDP is held in the so-called V conformation, making an angle between the thiazolium and pyrimidine rings in such a way as to bring close the

aminopyrimidine group (as an imino tautomer) and the thiazolium C-2 proton in order to favor deprotonation.⁶³ The diphosphate group plays no role in catalysis itself and is only required for tight binding of ThDP to the apoenzymes.

C. ThDP-Dependent Enzymes and Their Roles

ThDP-dependent enzymes exist in most mammalian cells because of their aerobic metabolism.^{64,65} Brain and heart are most sensitive to a decrease in the activity of these enzymes, in particular PDH and OGDH.

All four 2-oxoacid dehydrogenase complexes (PDHc, OGDHc, OADHc and BCOADHc) are large multi-enzyme complexes composed of multiple copies of three

core enzymes, a decarboxylating dehydrogenase referred to as the E1 component, a dihydrolipoyl acyl-transferase (E2), and a dihydrolipoyl dehydrogenase (E3) (Fig. 10.3B). E1 and E2 are specific for each complex (except for OGDHc and OADHc that share the same E2 component), while E3 is common to the four complexes.

E1 catalyzes the oxidative decarboxylation of the 2-oxoacid according to the ThDP-mechanism shown in Fig. 10.3A, followed by transfer of the n-1-activated aldehyde from ThDP to a lipoamide residue belonging to E2. During the latter reaction, the activated aldehyde is oxidized to an acyl residue while the lipoamide disulfide is concomitantly reduced to a substituted dihydrolipoamide residue. Hence, the E1 component catalyzes two reactions: ThDP-dependent decarboxylation and reduction of lipoamide disulfide.

E2 catalyzes the transfer of the acyl moiety from acyl-dihydrolipoamide to coenzyme A. The lipoamide, which is covalently linked to a lysine residue via an amide bond, extends out of the enzyme and, behaving like a swinging arm, transfers the lipoamide amide group from E1 to E3 while catalyzing the formation of a thioester bond between the acyl moiety and CoA.

E3 is a flavoprotein catalyzing the regeneration of lipoamide disulfide by reduction of an E3 Cys-Cys disulfide. The two electrons are then transferred to the releasable cosubstrate NAD^+ via a tightly bound FAD prosthetic group.

PDHc and OGDHc complexes are essential for mitochondrial oxidative metabolism that, especially in neurons, is crucial for survival.

OADHc was only recently identified as a separate enzyme from OGDHc and recognized as the missing link in L-lysine degradation.⁵⁹ Like the other 2-oxoacid dehydrogenases, it contains a specific E1 subunit (E1a) encoded by the human DHTKD1 gene, while the E2 and E3 subunits are shared with OGDH.

BCOADHc complex plays a central role in the degradation of the branched-chain amino acids leucine, isoleucine, and valine.

Transketolase is cytosolic and is a key enzyme in the pentose phosphate pathway, a major source of NADPH for reductive biosyntheses (fatty acids for instance) and production of ribose for nucleic acid synthesis. It presents as a homodimer (each with a molecular mass of 75 kDa). Each protomer carries an active site located at the contact surface; hence, dimerization is essential for enzyme activity. In humans, three isoforms of the enzyme have been described: transketolase, transketolase-like 1, and transketolase-like 2.⁶⁶ All three isoforms are located on different chromosomes and probably arose by gene duplication. The answer as to the enzymatic competence of transketolase-like 1 protein still awaits a carefully done study on the purified recombinant protein.⁶⁷

Peroxisomal 2-hydroxyacyl-CoA lyase was the first ThDP-dependent enzyme to be discovered in peroxisomes.⁶¹ In the liver, it is involved in the degradation of 3-methyl-branched fatty acids, essentially phytanic acid (3,7,11,15-tetramethyl hexadecanoic acid) present in human diet as a product of chlorophyll degradation.

D. Gene Regulation

To our knowledge, no direct effects of thiamine in the regulation of gene expression have been described (we will not discuss riboswitches here as they are not relevant in animals).

IV. DIETARY REQUIREMENTS

There are no long-term body stores for thiamine and any excess is rapidly eliminated by the kidneys. Thiamine deficiency will affect mainly the nervous (peripheral or central), the cardiovascular, and to a lesser extent the gastrointestinal systems.

A. Indicators of Deficiency or Inadequacy

Early symptoms of thiamine deficiency are rather unspecific and include fatigue, irritability, poor memory, sleep disturbances, precordial pain, anorexia, and abdominal discomfort; hence, TDDs are difficult to recognize. Overt thiamine deficiency may present with a wide range of pathologies, mainly involving neurological (peripheral and central), psychiatric, and cardiovascular syndromes.⁵²

Classically, pure nutritional thiamine deficiency in the adult is called beriberi, which itself can adopt a variety of forms. The “dry” form is a polyneuritic syndrome that may result in paralysis of the lower limbs. The “wet” form is characterized by cardiac insufficiency and edema. A rare variant of fulminant wet beriberi with severe lactic acidosis and multiorgan failure is referred to as Shoshin beriberi. The improvement can be rapid and spectacular after treatment with thiamine. Just as the nervous system, the heart muscle heavily relies on oxidative metabolism, explaining its high sensitivity to thiamine deficiency.

Infantile beriberi is mainly characterized by heart failure, but its early clinical features are highly unspecific and it is therefore often missed except possibly in its aphonic form.⁶⁸ Shoshin beriberi is the predominant form in breastfed babies (1–3 months) and often misdiagnosed as a septic shock.⁶⁸

Wernicke’s encephalopathy is another neurological form often, but not always, associated with alcoholism. This syndrome, involving confusion and ataxia, is rapidly reversed by thiamine treatment but, when untreated deficiency becomes chronic and irreversible

lesions appear in the diencephalon, resulting in Korsakoff's psychosis (see below).

The different thiamine deficiency conditions have a large clinical spectrum with overlapping symptoms and sometime age-specific forms making the terminology archaic. For these reasons, the Gates Foundation-Sackler Institute Thiamine Deficiency Consultation group suggested replacing the previous terms by the more general umbrella terminology of thiamine deficiency disorders (TDDs).⁶⁸ TDD can be further subdivided in three categories: (overt) thiamine deficiency, thiamine insufficiency, and at risk of thiamine deficiency (i.e., belonging to a population with thiamine intakes associated with clinical insufficiency or deficiency).⁵²

B. Criteria of Adequacy

Biochemical changes precede clinical symptoms of thiamine deficiency. Essentially, three methods have been used to determine the preclinical thiamine status: urinary thiamine excretion, erythrocyte transketolase activity, and determination of the concentration of thiamine and its phosphate esters in plasma or whole blood.⁶⁹ However, none of these indicators by itself provides an adequate basis on which to estimate the thiamine requirement (Table 10.1^{52,68}).

Urinary thiamine excretion has been used mainly in metabolic studies and has been supplanted by the measurement of transketolase activity and blood thiamine levels.

The transketolase assay is based on the fact that in a person with a normal thiamine status, this enzyme, which is highly abundant in erythrocytes, will be close

to saturation with the coenzyme ThDP. Hence, after addition of external ThDP to lysed erythrocytes, we should observe at best a very small increment of enzyme activity. The so-called erythrocyte transketolase activity coefficient (ETKac, the fractional ratio of stimulated over unstimulated activity) should be larger (typically > 1.20) in patients suffering from thiamine deficiency as basal ETK activity is low (transketolase far from saturation), and the ETKac should be proportional to the severity of thiamine deficiency. Hence, this method is a measure of the biological activity of the coenzyme ThDP, whose lack is thought to be responsible for the symptoms observed in thiamine deficiency.

The ETK assay can be complemented by a direct measurement of blood ThDP concentrations using a sensitive high-performance liquid chromatographic method (HPLC). In blood, the main thiamine vitamer is ThDP, which is almost totally located in erythrocytes, where it is formed by pyrophosphorylation of thiamine by TPK1. There is indeed a high-affinity thiamine transporter (THTR1) but no ThDP transporter in the erythrocyte plasma membrane. ThDP is more stable in frozen erythrocytes than transketolase. Human plasma contains small concentrations of thiamine and ThMP, typically in the 10 nmol/L concentration range.¹⁰ Plasma thiamine concentrations vary with nutrient intake.

Presently, we lack enough data to correlate infantile, juvenile, and adult TDDs with biochemical markers of low thiamine status.⁵² Indeed, among people with the same blood thiamine concentrations or transketolase effect, some can show symptoms of thiamine deficiency while others do not. Also, as far as the ETKac is concerned, there seem to be large interindividual variations,

TABLE 10.1 Potential cutoff values for the evaluation of thiamine status.

Evaluation	Cutoff	Status	Advantages	Disadvantages
Plasma thiamine			Indicates recent intake	Not an indicator of thiamine status
Plasma ThMP			Indicates recent intake	Not an indicator of thiamine status
Blood ThDP	<80 nmol/L 70–90 nmol/L	Deficiency limit	Biologically active vitamer and indicator of thiamine status	Unstable, requires freezing for long-term storage and chromatographic equipment with fluorescence detector
Erythrocyte ETKac	$\alpha < 1.2$	Deficiency	Functional assay of biological activity	Assay not readily available, requires rapid separation of erythrocytes

Adapted from Refs. 52,68.

possibly linked to biochemically distinct forms of transketolase.⁷⁰

Hence, it is difficult to determine exact cutoff value to discriminate between the notions of thiamine deficiency, insufficiency, and low status, and one should not totally rely on experimental indicators. This issue has been extensively discussed recently.^{52,68} Fortunately, most TDDs rapidly (often within hours) respond to thiamine administration, which is presently the ultimate criterion to assess a TDD.⁷¹

C. Assessment of Status

As discussed above, HPLC is the method of choice for the determination of thiamine in body fluids. The most direct measure is the determination of ThDP in blood samples.⁷² This method is rapid, suitable for pediatric diagnostics, highly sensitive, and easy to standardize. However, such methods are generally not routinely available in most medical laboratories. The ETKac is a widely used method to assess thiamine status in humans (Table 10.1).^{52,68} An abnormally large ThDP effect can be interpreted as a low saturation of transketolase by the cofactor, as a consequence of thiamine deficiency. A good correlation is obtained between direct erythrocyte ThDP determination and the ThDP effect. The latter is however an indirect determination of the thiamine status that can be influenced by factors other than thiamine deficiency and it is less sensitive.

TABLE 10.2 Recommended dietary allowances (RDAs, mg/day) for thiamine in humans.

	Males	Females
<i>Infants and children</i>		
0–6 months	0.2	0.2
7–12 months	0.3	0.3
1–3 years	0.5	0.5
4–6 years	0.6	0.6
7–9 years	0.9	0.9
<i>Adolescents</i>		
10–18 years	1.2	1.1
<i>Adults</i>		
19+ years	1.2	1.1
Pregnant woman	—	1.4
Lactating woman	—	1.5

D. Reference Intakes

Reference intakes are listed in Table 10.2 adapted from World Health Organization and Food and Agriculture Organization of the United Nations report released in 2004.⁷³ They mirror closely those established by the Institute of Medicine for Americans and Canadians in 1998.⁶⁷ The Recommended Dietary Allowance (RDA) is defined as the average daily level of intake sufficient to meet the nutrient requirements of nearly all (>97%) healthy individuals in a specific population group, while the Estimated Average Requirement (EAR) is the amount that is sufficient to meet the nutrient requirements of half of a population group and is used to evaluate adequacy of nutrient intakes.⁶⁷ The indicators used to establish adequacy were urinary excretion of thiamin in depletion/repletion experiments, erythrocyte transketolase activity, and development of clinical signs and symptoms of deficiency such as irritability and inability to concentrate.⁶⁷

Lactating women transfer about 0.2 mg of thiamine to their infants through the milk, and an additional 0.1 mg is needed to compensate for the increased energy cost of lactation.⁷³ For that reason, lactating women, especially when already living in regions where a low thiamine status is common, are particularly at risk of thiamine deficiency.

E. Basis for Individual Variability in Requirements

Thiamine requirement may depend on various factors. It is known that a diet rich in carbohydrates will increase thiamine requirements. Several studies have pointed out a low thiamine status in the elderly.^{74,75} The reason for this could be decreased intestinal absorption of the vitamin with age, and it could mean that an RDA of 1.2 mg/day is not adequate in these people and should be revised. Blood ThDP levels were reported to be weakly correlated with Mini-Mental State Examination score.⁷⁶ Blood total thiamine levels are decreased in diabetic patients, suggesting that those people would require a higher dietary thiamine intake.⁷⁷ Furthermore, it is well known that people with alcohol dependence often have a low thiamine status, but other groups (people with HIV/AIDS or who have undergone bariatric surgery) may be in a similar situation.

F. Tolerable Upper Intake Levels

Reference values for upper levels of thiamine have not been established. Oral and intramuscular administrations are considered safe, while high-dose intravenous injections (100 mg IV bolus) can, in rare cases

(1%), induce hypersensitivity and other adverse effects.⁷⁸ Anaphylactic shock has been observed but is extremely rare. A 67-year-old woman developed ventricular fibrillation following parenteral administration of a mixture of B1, B2, B6, C, and nicotinamide, which may have been caused by the thiamine component.⁷⁹

Toxic effects of thiamine are negligible (with doses up to 500 mg), the renal clearance is fast, and there is a rapid decline in absorption following high doses. This does not affect. In animals, the LD₅₀ is 3710 mg/kg (<https://www.drugbank.ca/drugs/DB00152>)⁸⁰ for oral administration and 118 mg/kg (<http://www.t3db.ca/toxins/T3D2696#reference-A308>)⁸¹ for intravenous injection.

G. Dietary Sources

Main sources of dietary thiamine are meat, in particular pork, fish, and cereals. Plants and fruits are less rich in thiamine, except probably plant germs that contain thiamine stores.⁵ Whole rice is a rich source of thiamine, but, as already mentioned, it is only present in the bran. For that reason, diets mainly based on polished rice can lead to thiamine deficiency. Therefore, in many countries, including the United States and European countries, white rice is enriched with thiamin, riboflavin, niacin, and iron. The US Department of Agriculture maintains a database with the nutrient content of most common foods (<https://ndb.nal.usda.gov/ndb/>). Cooking and pasteurization can lead to the loss of up to 20%–30% of the original thiamine content.⁵²

The US Food and Drug Administration's Daily Value of 1.2 mg/day for thiamine, used for food labeling, is based on the RDAs for adults on a 2000 kcal diet.⁸² Dietary intakes of thiamine in the United States are adequate for most people, with the exception of teenage girls (11% < EAR) and younger (9% < EAR) and older women (13% < EAR).⁸³

V. ISSUES RELATED TO THIAMINE IN HUMAN HEALTH

The importance of thiamine in human health is not only limited to the role of ThDP as coenzyme, but recently several pharmacological benefits of thiamine and some of its synthetic precursors have been highlighted in neurodegenerative diseases and diabetes. In this respect, we will also discuss the benefits of thiamine supplementation as well as some of its pharmacological effects.

A. Relevant Health Outcomes and Conditions of Concern

Thiamine deficiency induces changes in the brain, the most prominent being lactic acidosis, oxidative stress, neuroinflammation, and excitotoxicity.⁸⁴ These changes

are most probably the consequences of a decrease in ThDP-dependent enzyme activities as well as possibly the decrease of other thiamine derivatives. Thiamine deficiency mainly results in metabolic changes (the "biochemical lesion") as evidenced by the rapid reversal of the symptoms after thiamine administration.

In addition to purely nutritional TDDs (the various manifestations of beriberi), discussed above, subclinical and infantile beriberi can be relatively common in some populations. Thiamine deficiency can also result from external causes such as chronic alcohol consumption, antithiamines, general malnutrition, gastrectomy,⁸⁵ bariatric surgery, hyperemesis gravidarum,⁸⁶ fast-growing hematologic malignant tumors,⁸⁷ drug abuse, and AIDS.⁸⁸ Finally, thiamine deficiency can be inborn, due to mutations in genes involved in thiamine metabolism. All these disorders share the characteristic of being at least partially reversible after thiamine administration.

In this section, we will also focus on recent research involving the possible involvement of thiamine in diabetes and neurodegenerative diseases with emphasis on Alzheimer's disease. It should be stressed that in these conditions, thiamine deficiency is not the main cause of the disease but a possible consequence.

Subclinical thiamine deficiency (thiamine insufficiency)

Subclinical thiamine deficiency in humans is probably more widespread in developed countries than initially thought, especially in malnourished and elderly people^{89,90} as well as in some risk groups such as infants and pregnant and lactating women.⁹¹ Maternal thiamine deficiency may be a problem in displaced populations, either as a result of insufficient thiamine intake or the consumption of thiaminase-containing food (see below).⁹² It is important to emphasize that circulating and brain thiamine levels are much lower in humans than in rodents.^{11,75} This might be one reason why humans are more prone to thiamine deficiency than rodents and most other animals.

Infantile thiamine deficiency

TDDs are endemic in many low- and middle-income countries, especially among children and in displaced populations. Exclusively breastfed children are particularly at risk of thiamine deficiency.^{52,68} Because infantile thiamine deficiency, especially in newborns, will very often lead to mental retardation, the health burden of the disease can be very high for the society.

Considering the multiple forms of thiamine deficiency diseases and, particularly in children, the lack of specific symptoms, a true assessment of the prevalence is difficult. Several studies, mainly carried out in Asia, suggest that infantile thiamine deficiency is one of the leading causes of death among children aged

less than 1 year. In Myanmar, it was reported that infantile beriberi accounts for 17% of infant deaths on a national level. In Laos, the infant mortality was estimated to about 50 deaths among 467 live births.⁵² In low- and middle-income countries, the risk of thiamine deficiency is associated with the consumption of staple food with a low thiamine content such as polished rice in many areas throughout Asia or processed cassava root in West Africa.⁹³ A change in cultural habits is often difficult to implement. Parboiling, a process during which rice is partially boiled within the husk allowing diffusion of nutrients into the endosperm, is practiced in India for instance and can be a solution.

Rare cases of accidental thiamine deficiency may also occur, such as an episode of encephalopathy in infants in Israel in 2003, caused by a defective soy-based formula.⁹⁴

Thiamine status in alcoholic patients

The most common factor for thiamine deficiency in developed countries is alcohol consumption. It seems that alcohol interferes with intestinal thiamine absorption, in particular by reducing the expression of THTR1 in the intestine.⁹⁵ An impairment of phosphorylation of thiamine to ThDP has also been suggested.^{96,97} Thiamine deficiency related to chronic alcohol abuse leads to a severe impairment of brain function, known as Wernicke–Korsakoff syndrome. This remains the third most common cause of dementia after Alzheimer’s disease and vascular dementia, contributing to between 10% and 24% of all cases of dementia.^{98,99} However, not all alcoholics develop Wernicke–Korsakoff syndrome, and environmental and maybe genetic factors might also play a role.

The clinical features of Wernicke’s encephalopathy (which is the first acute stage of the syndrome) include ophthalmoplegia, global confusional state, and ataxia. Korsakoff’s psychosis appears as a consequence of chronic thiamine deficiency in the alcoholic patient, often after several episodes of Wernicke’s encephalopathy. Administration of thiamine at an early stage of Wernicke’s encephalopathy leads to rapid improvement (biochemical lesion).^{84,100,101} Otherwise, there are irreversible sequelae such as residual ataxia and horizontal nystagmus, deficits in learning, memory loss, and personality changes, which are hallmarks of Korsakoff’s psychosis. This is linked to irreversible brain lesions that are rather specific, affecting the thalamus, the mammillary bodies, and sometimes the cerebellum, the cerebral cortex being largely spared. The lesions possibly arise from decreased OGDHc activity impairing energy metabolism in endothelial cells and neurons, but it remains unknown why OGDHc activity is more sensitive to thiamine deficiency in vulnerable brain

regions.¹⁰⁰ Hence, the elucidation of the mechanisms involved in the selective vulnerability of certain brain regions to thiamine deficiency is an important subject of research.

Thiamine status in diabetic patients

A poor thiamine status has been reported in diabetic patients, with blood thiamine concentrations significantly lower than in control patients, possibly as a result of higher renal excretion.⁷⁷ Interestingly, benfotiamine, a thiamine precursor with high bioavailability (see [Section V.C](#)), seems to protect against diabetic complications such as retinopathy, neuropathy, and nephropathy.¹⁰² Diabetes is also a consequence of inborn errors of SLC19A2 probably inducing thiamine deficiency in pancreatic islets.

Inborn disorders of thiamine metabolism and ThDP-dependent enzymes

Inborn disorders are due to direct inherited impairment of ThDP-dependent enzymes or indirectly to impairment of thiamine transport function or metabolism (mainly TPK1), leading to decreased ThDP coenzyme function. These diseases that start to manifest in the earliest childhood have a wide spectrum of mostly neurological symptoms summarized in [Table 10.3](#).^{42,59,101,104–109} In many cases, the outcome is fatal if not treated. Hence, it is essential that the involvement of thiamine (in the widest sense) be recognized as early as possible in order to start a treatment with thiamine supplementation that may help overcome the metabolic impairment.

Thiamine in neurodegenerative diseases

As pointed out by Gibson and colleagues, TDD and neurodegenerative diseases share common features, including impairment of brain oxidative metabolism.¹¹⁰ Moreover, it has been shown that postmortem brains from patients with Alzheimer’s disease have decreased 2-oxoglutarate activities¹¹¹ and decreased brain ThDP levels.¹¹² More recently, the group of Zhong showed that blood ThDP reduction correlates with glucose hypometabolism in the brain.¹¹³ They suggested that blood ThDP levels might have a diagnostic value in patients with Alzheimer’s disease.^{114,115}

B. Known Interactions

Thiamine is nontoxic, and there are no known examples of interference with any medication. However, some drugs, food, and food additives can interfere with thiamine metabolism or stability and induce thiamine deficiency disease.

TABLE 10.3 Inherited human diseases due to mutations in thiamine transporters, TPK1, or ThDP-dependent enzymes.

Component involved	Disease	Phenotype
Thiamine transporter 1 (SLC19A2)	Thiamine-responsive megaloblastic anemia	Diabetes mellitus and sensorineural deafness
Thiamine transporter 2 (SLC19A3)	Biotin–thiamine-responsive basal ganglia disease ³⁷ Wernicke-like encephalopathy	Subacute encephalopathy, fatal ^a Confusion, ophthalmoplegia, ataxia, fatal ^a
Mitochondrial ThDP transporter (SLC25A19)	Amish lethal microcephaly ^{42,103}	Congenital microcephaly, 2-oxoglutaric aciduria, fatal ^b
Thiamine pyrophosphokinase 1	(Leigh-like) encephalopathy ¹⁰⁴	Encephalopathies with various degrees, α -ketoglutaric aciduria, and lactic acidosis, fatal ^a
Pyruvate dehydrogenase (thiamine-responding form)	Leigh's disease ¹⁰⁵ (subacute necrotizing encephalomyelopathy)	Encephalopathy, lactic acidosis, fatal ^b
Branched-chain 2-oxoacid dehydrogenase	Maple syrup urine disease ¹⁰⁶ (branched-chain ketoaciduria)	Organic acidemia, sweet-smelling urine, fatal ^c
Transketolase variants	Predisposition to Wernicke–Korsakoff syndrome ¹⁰⁷	Confusion, ophthalmoplegia, ataxia
2-Hydroxyacyl-CoA lyase	Refsum-like disease ¹⁰⁸	Metabolic impairment with liver damage
2-Oxoadipate dehydrogenase (DHTKD1 gene)	α -Aminoadipic and α -ketoadipic aciduria ⁵⁹ Charcot–Marie–Tooth disease	Metabolic disorder with neurological abnormalities

^aFatal outcome only if not treated.^bDoes not generally respond to thiamine therapy.^cDoes not generally respond to thiamine therapy but requires a diet poor in branched-chain amino acids.

Fluorouracil-based chemotherapy

Thiamine deficiency was found to be high in cancer patients, in particular those with fluorouracil-based chemotherapy,¹¹⁶ raising the possibility that fluorouracil may interfere with thiamine metabolism. This deficiency can be overcome with thiamine administration.

Furosemide

Loop diuretics such as furosemide increase thiamine excretion and may induce thiamine deficiency in patients already having a low thiamine status.¹¹⁷

Sulfites and caffeic acid

Sulfites are highly used as food preservatives. Sulfites can degrade thiamine by cleaving the methylene bridge between the thiazole and methylene rings. Cases of thiamine deficiency in dogs due to feeding on sulfite-preserved meat have been described.¹¹⁸ Earlier studies had suggested that 3,4-dihydroxycinnamic acid (caffeic acid), tannic acid, and similar substances had an antithiamine activity, but this was later disproved.¹¹⁹

Antimalarial drugs

Some antimalarials such as chloroquine¹²⁰ and the antitrypanosomal drug melarsoprol¹²¹ were shown to inhibit thiamine transport.

Thiaminases

Some foodstuff may contain thiaminases (thiamine-destroying) enzymes. Some fish (carp, eel, Baltic herring, or catfish, for instance), shellfish, the African silkworm (*Anaphe venata*) larvae, and raw cassava may contain thermolabile thiaminase I, a pyrimidine transferase (EC 2.5.1.2). This is also the case of some ferns (*Pteris aquilina*) that, when consumed by grazing cattle or horses, may result in severe thiamine deficiency. Cooking destroys thiaminases, but consumption of raw food containing thiaminase may cause thiamine deficiency in humans.

C. Thiamine as a Dietary Supplement

The human body contains approximately 25–30 mg of thiamine, with about 80% being under the form of ThDP. About 4%–5% of this amount is replaced each

day, in agreement with and RDA of 1.2 mg for a healthy adult. Unlike other B vitamins, intestinal thiamine absorption is a relatively slow process, especially in humans; hence, supplementation of thiamine may be justified in many cases.¹²²

Thiamine supplementation

The most common thiamine supplements are thiamine mononitrate and thiamine hydrochloride, which are stable and highly water soluble. Many dietary supplements containing either thiamine alone or in a multivitamin complex are available. Typical daily supplement doses are less than 10 mg. In order to overcome thiamine deficiency, therapeutic doses of up to 200 mg can be administered. In order to overcome the relatively slow intestinal absorption of thiamine, several synthetic thiamine precursors with higher bioavailability were developed. The most used are the thiamine disulfides sulbutiamine (*o*-isobutrylthiamine disulfide) and fursultiamine (thiamine tetrahydrofurfuryl disulfide) and the thioester benfotiamine (*S*-benzoylthiamine *O*-monophosphate).

Food fortification

Food fortification could be a successful means to eradicate TDD. Recently, Southeast Asia has seen a significant decrease in TDDs upon food fortification programs.¹²³ However, food fortification can only be successful when implemented on a large scale after identification of a food category suitable for delivering thiamine to the at-risk group. As in Asia, the most consumed food is rice, and it would be a good target under the condition that it can be done at a low cost and on a large scale. In Myanmar, where there are about 16,000 rice millers, fortification would require a very decentralized approach. In Laos, fortification of the widely consumed fish sauce is probably a good alternative.¹²³

Food fortification requires that the micronutrient is affordable, has no side effects even when taken in high doses, and is chemically stable. This seems to be the case with thiamine, at least for the first two conditions. However, thiamine is heat labile, light sensitive, and sensitive to cleavage by sulfites. As sulfites are largely used food preservatives, this requires some precautions.⁶⁴ Stable thiamine precursors could be useful alternatives to thiamine.

Finally, the possibility of crop biofortification, i.e., fortification through breeding or genetic engineering, has been raised.¹²⁴ It is however not clear how such crops could be used in low-income countries at a low cost. Indeed, such genetically modified organisms would probably be protected by patents and require the continuous purchase of new seeds.

Pharmacological effects of thiamine and its synthetic precursor benfotiamine

In a preliminary study, Pan et al. showed that oral administration of benfotiamine, a thiamine precursor with better bioavailability, led to a long-term cognitive improvement in five patients with mild-to-moderate Alzheimer's disease.¹²⁵ However, this study, besides the fact that the number of patients was very small, was not placebo-controlled. Hence, a definite answer as to the utility of thiamine or thiamine precursors as therapeutic agents in Alzheimer's disease requires further clinical studies.

Two different studies also suggested improvement of thiamine administration in patients with Parkinson's disease.^{126,127} However, both studies suffer from the same flaws as the previously mentioned Alzheimer study.

Encouragingly, two in-depth studies carried out in transgenic mouse models of Alzheimer's disease¹²⁸ and tauopathies¹²⁹ showed that long-term oral treatment with high doses of benfotiamine (200 mg/kg) improved cognitive function and decreased amyloid deposition and tauopathy. Surprisingly, none of the two studies showed an increase in brain ThDP levels after treatment. This observation that benfotiamine treatment does not lead to increased brain ThDP levels is shared by other studies,^{16,130} suggesting that benfotiamine may act by different more pharmacological mechanisms, possibly involving activation of kinases (GSK3 α and β)^{131–133} or the Nrf2/ARE¹²⁹ antioxidant stress response.

Recently, powerful antidepressant and antiaggressive effects of thiamine and benfotiamine have been reported, but again the molecular mechanisms involved remains unknown.^{130,131}

RESEARCH GAPS

- Biological roles of all phosphorylated thiamine derivatives other than ThDP
- Mechanisms leading to selective vulnerability of certain brain regions to thiamine deficiency
- Regulatory mechanisms involved in thiamine brain homeostasis
- Mechanism of pharmacological effects of thiamine and precursors in diabetes and Alzheimer's disease
- Mechanism of antioxidant properties of thiamine
- Need for rapid diagnostic methods in TDDs

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RIBOFLAVIN

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SUMMARY

Riboflavin (vitamin B₂) is biosynthesized by bacteria, yeasts, and higher plants, but most animals must obtain it from the diet. It is widely distributed in food, with kidney, liver, cheese, eggs, and milk containing especially high levels. Upon consumption of the free vitamin or flavoenzymes that are hydrolyzed in the gut, riboflavin is absorbed by facilitated transport, circulates in blood, and is taken up by (and effluxes from) cells via several transporters. After intracellular conversion to flavin mononucleotide and flavin adenine dinucleotide, the coenzymatic forms are bound noncovalently (or in several cases, covalently) to enzymes that catalyze oxidation–reduction reactions or, in rare instances, perform nonredox functions. Riboflavin and derived flavin products are secreted in milk and excreted in urine. Nutritional deficiencies arise from inadequate intake and, less frequently, defects in transport or formation of functional flavoproteins. Some of these diseases benefit from supplementation, but normal individuals receive ample vitamin by eating a variety of foods, some already fortified with riboflavin.

Keywords: Coenzymatic forms of riboflavin; Deficiencies due to genetic defects; Flavoprotein functions; Metabolism and excretion; Riboflavin structure; Riboflavin transport.

I. INTRODUCTION

A. Background

Riboflavin is so named because it has a ribose-like part of its structure and is yellow (Latin, *flavus*); it was historically designated as the second vitamin in the B complex of water-soluble vitamins (the first being thiamin), so it is also called vitamin B₂. Much of the early history of this vitamin has been detailed in the series edited by Sebrell and Harris,¹ subsequent book chapters,^{2,3} and articles accessible online.^{4,5} The early studies were complicated by the experimental diets being deficient in multiple water-soluble vitamins, but the confusion was sorted out in the 1930s when the yellow compounds that had

been isolated from milk (lactoflavin), eggs (ovoflavin), and other sources were structurally defined as “riboflavin” by chemical synthesis by the labs of Kuhn and Karrer and established to restore the growth of animals fed deficient diets (notables for these investigations include Birch, György, Harris, Kuhn and Wagner-Jauregg). The first characterization of deficiency symptoms in humans was published by Sebrell and Butler.⁶ Over this same timeframe, the function of riboflavin (phosphate) as a coenzyme (i.e., a factor that could be dissociated from the holoenzyme then added back to the apoenzyme to restore the activity) was demonstrated by Theorell using the “old yellow enzyme” discovered by Warburg and Christian. The history of characterization of riboflavin metabolism has also been described.^{7,8}

B. Nutrient Function and Structure

Riboflavin and major coenzymatic forms

Riboflavin (7,8-dimethyl-10-[1'-D-ribyl]isoalloxazine) is a yellow fluorescent compound that is widely distributed throughout the microbial, plant, and animal kingdoms. The “flavin” portion is called an isoalloxazine because all but one subcategory (the 5-deazaflavins) are 10-substituted derivatives of isoalloxazine, the parent tricyclic ring system, with nitrogens in positions 1, 3, and 5 (Fig. 11.1). The major coenzyme forms of riboflavin (Fig. 11.1) are flavin mononucleotide (FMN, also named riboflavin 5'-phosphate), which may account for as much as 10% of the total flavin in diverse cells, and the more frequently encountered flavin adenine dinucleotide (FAD), which comprises nearly 90%. The coenzyme forms of riboflavin and its structural variants (Fig. 11.2) function as prosthetic groups of numerous flavoenzymes (in the ballpark of 70–80 for humans,^{9,10} based on proteomic analysis, and estimated to be ~400 for all organisms based on “flavogenomic” analysis¹¹). Flavoenzymes catalyze diverse¹² and often essential one- and two-electron redox reactions that capitalize on the special capacity of flavins to be interconverted among the oxidized (quinoid) form, half-reduced (semiquinoid or radical) and fully reduced (hydroquinoid) forms, as summarized in Fig. 11.3. These and other physiologically relevant redox forms have been structurally detailed in earlier reviews.^{2,13} They are also now known to participate in

some nonredox functions, such as light sensing¹⁴ and circadian cycling regulation.¹⁵

Additional flavins and coenzymatic forms

Approximately 40 different flavoquinones have been isolated from natural sources, and Fig. 11.2 shows the major as well as some of the minor and novel subspecies. Most mammalian flavoproteins bind FMN and FAD, with a few enzymes utilizing flavins altered at positions 6 or 8 α (i.e., for humans, 8 α -N³-histidyl-FAD for succinate dehydrogenase, sarcosine dehydrogenase, and dimethylglycine dehydrogenase; 8 α -cysteinyl-FAD for monoamine oxidase [both A and B types] and L-pipecolate oxidase; and 6-hydroxy-FAD for apoptosis inducing protein).⁹ Additional structural variants are found in mammalian plasma or urine from flavin degradation, such as lumichrome and lumiflavin from photodegradation and microbial action¹⁶ and 6-hydroxy derivatives that may arise from oxidative turnover of FMN and FAD during function (as discussed below and shown in Fig. 11.6).

A large number of modifications have been found in other organisms (Fig. 11.2); for example, 6-S-cysteinyl-FMN, 8-hydroxy flavins, and 8 α -O-tyrosyl-FAD have been found in certain bacterial systems and 8 α -S-cysteinyl- and both 8 α -N¹- and N³-histidyl-FAD are found in several lower and higher organisms, including some mammals. Interestingly, an enzyme with the N¹-histidyl linkage, L-gulonolactone oxidase, accounts for the ability of some organisms (e.g., rats), but not humans, to biosynthesize L-ascorbic acid. One flavoenzyme, the berberine bridge enzyme, which is involved in alkaloid biosynthesis, contains FAD linked in the 8 α -position to a histidyl residue and in the 6-position to a cysteinyl thiol.¹⁷ Coenzyme F₄₂₀ has a 5-carba-5-deaza nucleus and was originally characterized from purification from *Methanobacteria* but subsequently found in a number of other organisms, including *Mycobacterium tuberculosis*.¹⁸ Prenylated FMN is a relatively recently discovered flavin from reversible decarboxylase enzymes (ubiD and related genes that are widely distributed in archeal, bacterial, and fungal genomes).^{19,20}

These structural variations usually affect the nature of protein binding (e.g., tightness and orientation in the active site) and adjust the redox chemistry of the flavin to match the reaction(s) it assists. For example, the 6-cysteinyl-, 8 α -histidyl-FAD of berberine bridge enzyme appears to correctly position the isoalloxazine ring in proximity to the substrate and has the highest redox potential so far known for any flavoprotein, which strongly predisposes the enzyme for a hydride ion transfer from its substrate.¹⁷ In some cases, the flavins serve other biologic functions, such as the detection of light by plants²¹ and the tapetum of the cat's eye and other animals

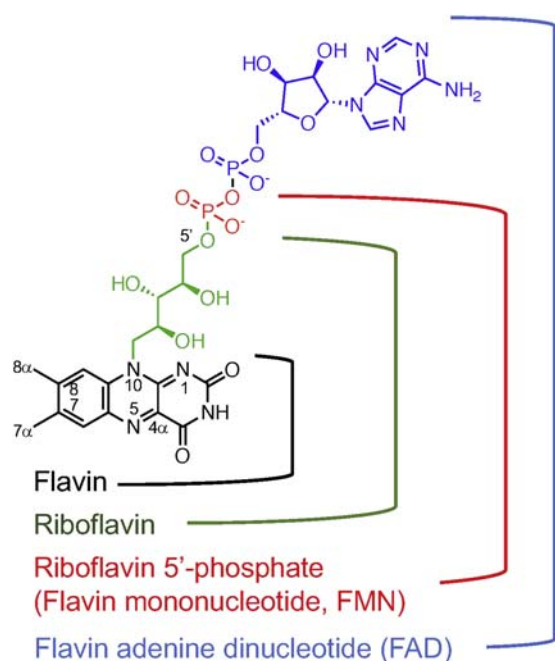
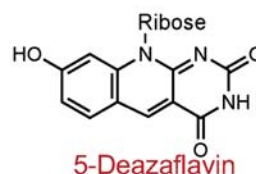


FIGURE 11.1 Riboflavin and coenzymatic forms.

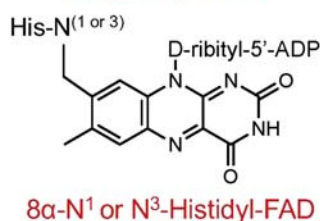
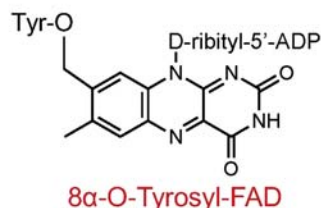
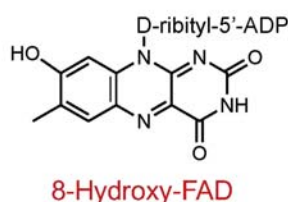
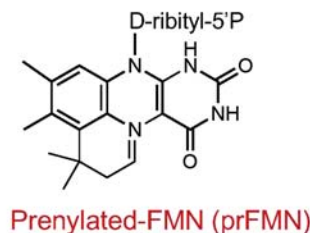
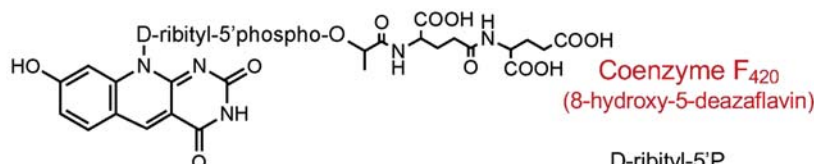
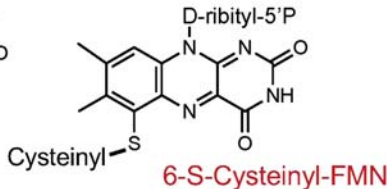
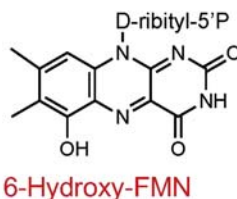
Naturally occurring flavins

Riboflavin
(see Fig. 1)



Flavocoenzymes found with flavoenzymes

FMN & FAD
(see Fig. 1)



[and 6-S-Cysteiny-8α-N³-HistidylFAD, not shown]

FIGURE 11.2 Flavin structural variants.

(where neko flavin was first found),²² and as antimicrobial agents (roseoflavin and 8-demethyl-8-amino-riboflavin).^{23,24}

Biological processes utilizing flavins

Flavoproteins can generate and react with substrate hydride equivalents, radical equivalents, and carbanion

equivalents. There are hundreds of biological processes in prokaryotes and eukaryotes that utilize flavins for enzymatic catalyzes of dehydrogenase, oxidase, monooxygenase, and reductase reactions in bioenergetics pathways, redox homeostasis, protein folding, DNA repair, chromatin remodeling, apoptosis regulation, photochemistry, and bioluminescence.^{9,12} Flavoenzymes are also involved in the formation and/or function of

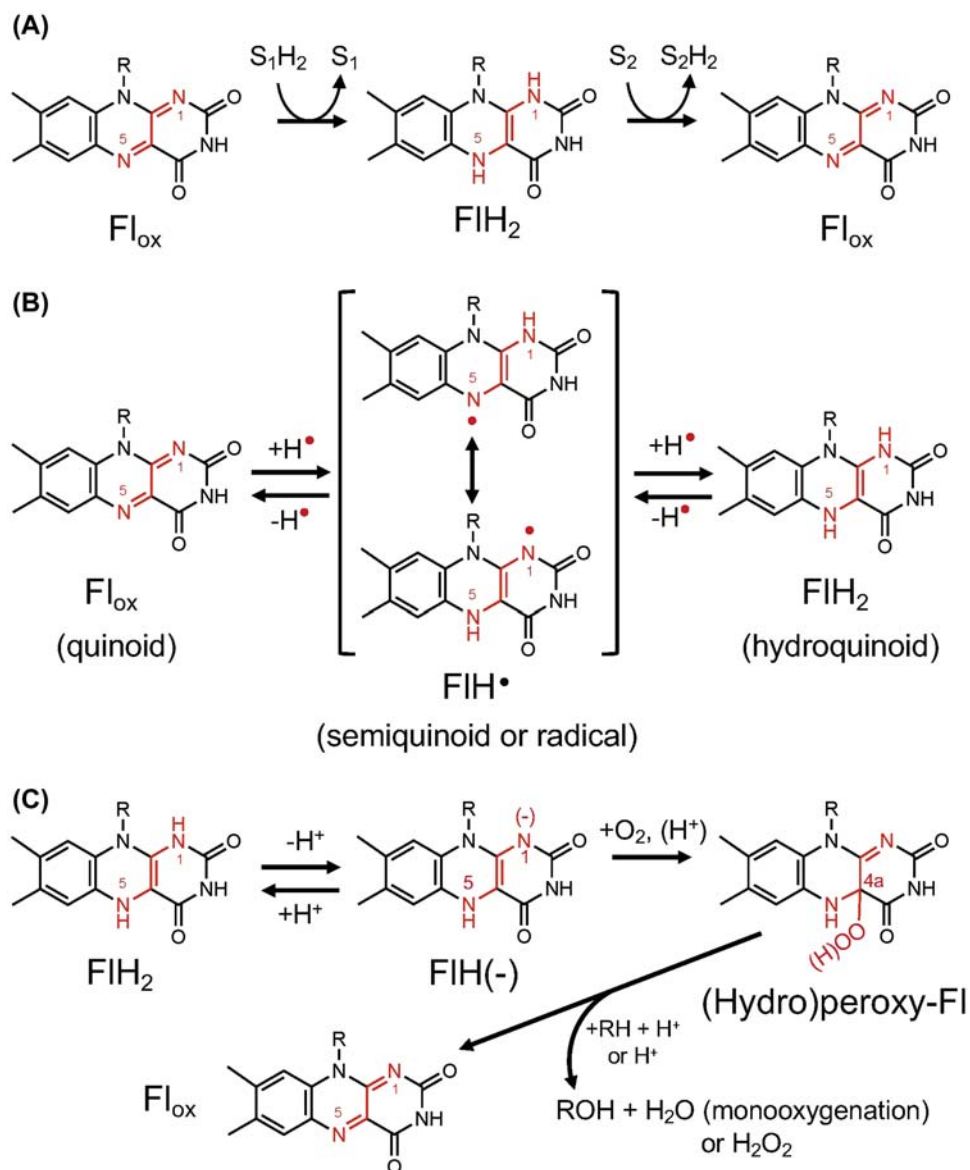


FIGURE 11.3 Flavoenzyme reactions.

other vitamins, namely, in the formation of the aldehyde form of vitamin B6 (pyridoxal 5'-phosphate) by pyridoxamine (pyridoxine) 5'-phosphate oxidase, for some of the redox reactions involving niacin, and the activity of methylenetetrahydrofolate reductase (MTHFR) in the folate cycle (and therefore participates in homocysteine metabolism). A relatively recently characterized (non-mammalian) enzyme, UDP-galactopyranose mutase, represents a new class of flavoenzyme in which the flavin functions as a catalytic center in a nonredox reaction.²⁵

For humans, the ones that appear to be most impacted by dietary deficiencies are involved in energy generation via the mitochondrial electron transport chain, lipid metabolism (flavoenzymes that participate in fatty acid oxidation, desaturation of essential fatty

acids, and synthesis of other important lipid categories), and oxidant defense mechanisms that depend on reduced glutathione, since the high intracellular ratio of reduced (GSH) to oxidized glutathione (GSH vs. GSSG) is maintained by the FAD-dependent glutathione reductase. The basic principles for how flavins are involved in such reactions are briefly summarized in the next subsection and more information can be found in an in-depth review.¹³

Flavoenzyme-catalyzed reactions

Flavoenzymes operate in catalytic cycles that involve (a) transfer of one or two electrons from a substrate to the flavin moiety (Fl_{ox}), thereby oxidizing the substrate and reducing the flavin (FlH_2), followed by (b) transfer

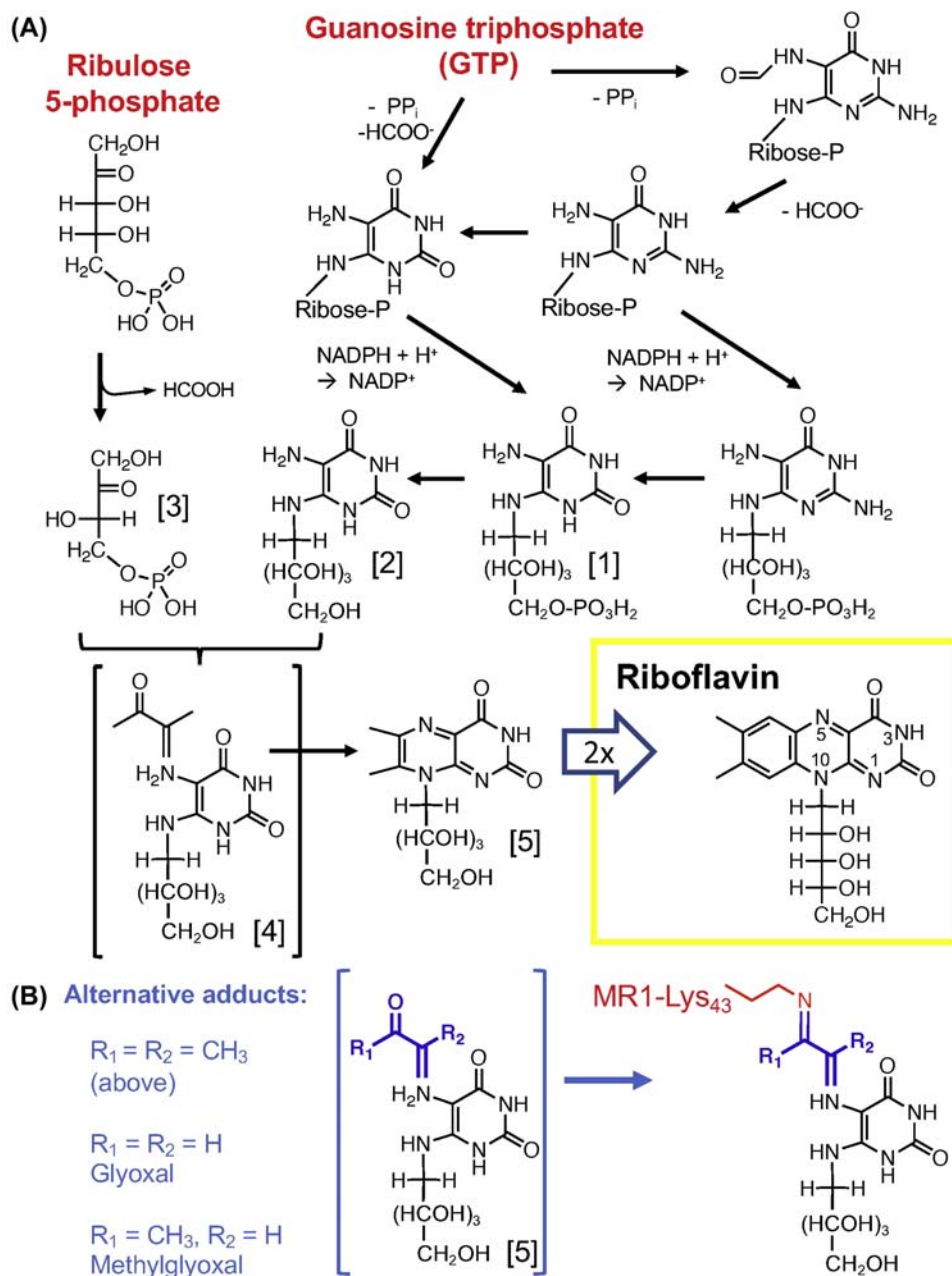


FIGURE 11.4 Riboflavin biosynthesis and origin of MAIT antigen.

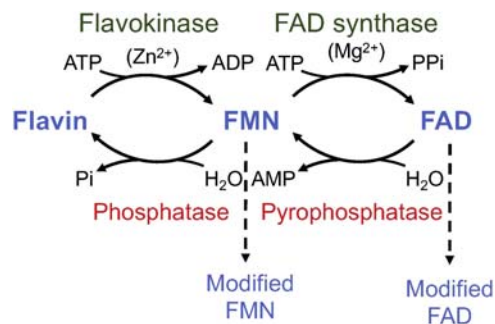


FIGURE 11.5 Interconversion of flavin coenzymes.

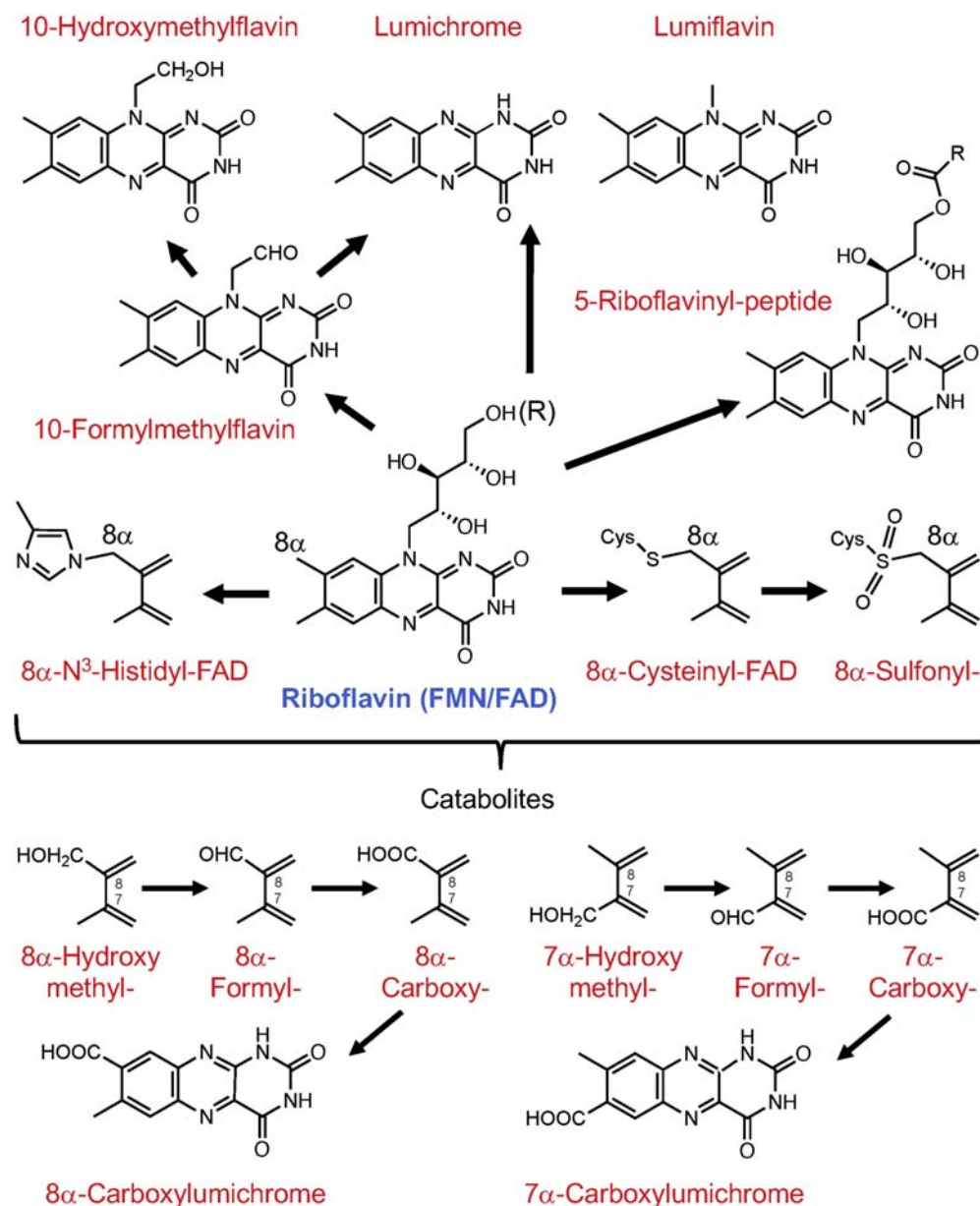


FIGURE 11.6 Catabolism and photodegradation of riboflavin reflected by urinary products from mammals.

of electrons from the reduced flavin to a second substrate to return the flavin to its original oxidized state and reducing the substrate. A general example of this type of reaction for two electron transfer is shown in Fig. 11.3A. An example of a biochemical reaction that uses these steps is FAD-dependent glutathione reductase which catalyzes the reaction $\text{NADPH} + \text{H}^+ + \text{GSSG} \rightarrow \text{NADP}^+ + 2 \text{GSH}$.

The chemistry that is involved in a one electron redox reaction is shown in Fig. 11.3B, wherein transfer of one electron to the flavin (in this diagram from a hydrogen radical, $\text{H}\cdot$) results in the one electron reduced semiquinone ($\text{FlH}\cdot$) that is stabilized by the two resonance forms shown. Addition of another electron can form the two-electron reduced dihydroflavin (FlH_2), or the

semiquinone can transfer the $\text{H}\cdot$ to a single electron acceptor (likewise, FlH_2 can transfer one electron to an acceptor to form $\text{FlH}\cdot$). This has been described as “an example of a biochemical reaction that uses these steps as ‘step down transformer’”¹³ between a substrate that only functions to transfer two electrons, such as $\text{NAD(P)H} + \text{H}^+$, and acceptors that only accept one electron, such as $\text{Fe}^{\text{III}}/\text{Fe}^{\text{II}}$.

Covalent adducts form in some flavoenzyme-catalyzed reactions,²⁶ such as the addition of oxygen to the C4a position to yield peroxides (Fig. 11.3C). C4a is also the site for connection of thiols, whereas N5 is favored for hydride and carbanion addition. Reduced flavins are highly reactive with oxygen and often

undergo rapid autoxidation with one-electron transfer to produce $O_2^{\bullet-}$ or two-electron transfer to produce H_2O_2 (this outcome—transfer of electrons to the flavin from a substrate followed by oxidation of the reduced flavin to H_2O_2 —is termed a flavoprotein oxidase reaction). If the O—O bond in the flavin-4a-OOH intermediate is cleaved and an oxygen atom is transferred to another substrate, the flavoprotein has served as a monooxygenase.

The wide diversity in the types of reactions conducted by flavoproteins is achieved by the creation of environments around the flavin, which is often tightly bound, and cosubstrates that favor the intended reaction. In some flavoproteins, covalent attachment of the flavin to the protein via 8α -S-cysteinyl-, 8α -N¹- or N³-histidyl-, or other modifications of the dimethylisoalloxazine ring play important roles in the tight binding and enhanced (and controlled) reactivity of the flavin.

Useful sources for additional information

Reviews that briefly cover the history of riboflavin and coenzymes, flavin and flavoprotein chemistry and properties, and related topics are found in two books dealing with all the vitamins required by humans, *Modern Nutrition in Health and Disease* (with a useful chapter on riboflavin therein)²⁷ and *Handbook of Vitamins*,²⁸ and more recent editions in these series, when available. Highly useful sources for chemical and biochemical methods for flavin and flavoprotein isolation, synthesis, characterization, and quantitation are the numerous volumes (Parts A through L) of *Methods in Enzymology, Vitamins and Coenzymes* that were edited by McCormick et al. over nearly 40 years (with the most recent volume regarding riboflavin being Part J in 1997).²⁹ In addition, ongoing discoveries about new flavins, the proteins they associate with, and the biochemical processes in which they participate are enlarged upon in volumes entitled *Flavins and Flavoproteins* resulting from periodic symposia (the 16th volume of such proceedings was published in 2008)³⁰ and a special edition of *The FEBS Journal* published in 2015.³¹ One of the most comprehensive reviews⁹ lists 77 human flavoproteins with FMN, FAD or covalently attached flavin cofactors, the structure clan (family), gene symbol, and gene location and includes a listing (Table 2 of reference 9) of 50 disease-related human flavoproteins.

II. NUTRIENT METABOLISM

A. Synthesis

Commercial synthesis and bioproduction

As described in thorough reviews,^{32,33} the development of a commercially viable method for synthesis of riboflavin began at Merck soon after the discovery of vitamin B₂, and chemical synthesis processes dominated

for about 50 years. This was followed by a combined chemoenzymatic process and then by development of an overproducing strain of the fungus *Ashbya gossypii* by classical mutagenesis approaches and other engineering methods that supplanted chemical synthesis by the beginning of this century. Fermentation methods continue to be improved to optimize costs and yield.³³

Biosynthesis

Studies conducted primarily with bacteria and fungi^{5,34} (sometimes to enhance industrial production, as described for *A. gossypii* in the preceding section)^{32,33} have provided an in-depth metabolic map for riboflavin biosynthesis (Fig. 11.4A). Guanosine triphosphate (GTP) serves as the precursor for both the ribityl side chain and half of the 7,8-dimethylisoalloxazine ring, and ribulose 5-phosphate is the source of the carbons for the dimethylbenzene ring portion. There are some variations in the initial reactions, but they generally proceed as shown: GTP (from purine biosynthesis) undergoes the hydrolytic release of one of the guanine ring carbons as formate, hydrolysis of the triphosphate to release pyrophosphate, and ribose ring opening and reduction to converge on a common intermediate [1] (5-amino-6-ribitylamino-2,4(1H,3H) pyrimidinedione 5'-phosphate) that is dephosphorylated to [2] (5-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione). The other precursor, ribulose phosphate (from the pentose phosphate pathway), undergoes rearrangement and formate release to form 3,4-dihydroxy-2-butanone 4-phosphate [3] which reacts with [2] (catalyzed by lumazine synthase) to form the chemically unstable intermediate in brackets [4] that converts to 6,7-dimethyl-8-ribityllumazine [5]. Dismutation of two 6,7-dimethyl-8-ribityllumazines by riboflavin synthase produces riboflavin and replenishes [2] that is available for condensation with [3] for another synthetic cycle.

Fig. 11.4B illustrates how the intermediate [4] and alternative condensation products from glyoxal or methylglyoxal [5] can be trapped by reaction with other nucleophiles, which recently—and provocatively—has been found to include the side chain amino group of Lys₄₃ of MR1 (MHC-related protein 1), and thereby serve as antigens for mucosal associated invariant T (MAIT) cells.^{35,36} MAIT cells are a T-cell subset present in blood and mucosal tissues, including the lungs, liver, intestines, and genitourinary tract of humans and other mammals that have been suggested to offer protective immunity to pathogens (noting that riboflavin biosynthesis does not occur in most mammals but is widespread in microbes) and also might play a role in human autoimmune and immunologic disease.³⁷

Most of the flavins shown in Figs. 11.1–11.3 and 11.6 derive from metabolic processes to convert the vitamin into its coenzymatic forms or degradation products and will be described in subsequent sections of this review.

Two exceptions are roseoflavin produced by *Streptomyces davawensis* and 5-carba-5-deazaflavins, including coenzyme F₄₂₀ formed by *Methanobacterium* sp. and other microorganisms. The biosynthesis pathway for roseoflavin has been recently elucidated to proceed from FMN to 8-demethyl-8-formyl-FMN (via oxidation of the 8-methyl-group to 8-formyl-), conversion of the 8-formyl-group to 8-carboxy- in a thiamine-dependent reaction (oxygen is not required), followed by decarboxylation and transamination at position 8 to produce 8-demethyl-8-amino-FMN.³⁸ After removal of the 5'-phosphate by an uncharacterized phosphatase, 8-demethyl-8-amino-riboflavin undergoes two rounds of methylation of the 8-amino-group with each using S-adenosyl-methionine as the donor,^{38,39} resulting in roseoflavin (Fig. 11.2).

For coenzyme F₄₂₀ biosynthesis, intermediate [2] of riboflavin biosynthesis, 5-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione (Fig. 11.4), is condensed with tyrosine to generate 7,8-didemethyl-8-hydroxy-5-deazariboflavin, the 5-deazabackbones shown in Fig. 11.2.^{40,41} This compound was originally found in methanogens but is widely synthesized by aerobic soil bacteria.⁴¹ The remainder of the side chain was originally thought to arise from transfer of the lactyl-group from L-Lactyl-2-diphospho-5'-guanosine followed by the addition of the γ -glutamates by γ -L-glutamyl ligase but now appears to involve (at least for prokaryotes) transfer of enolpyruvyl-phosphate from enolpyruvyl-diphospho-5'-guanosine then reduction and polyglutamylation.⁴²

The regulation of riboflavin biosynthesis depends on the specific organism because some rely entirely on de novo biosynthesis, some on both biosynthesis and uptake from exogenous sources, and others (humans and most mammals, but also a few bacteria and fungi) must obtain all of their riboflavin from exogenous sources.³⁴ In general, Gram-negative bacteria are exclusively dependent on de novo synthesis, whereas Gram-positive bacteria either possess systems for both de novo synthesis and uptake or rely on uptake only.⁴³ For those organisms that biosynthesize riboflavin, there is considerable variability in the organization and regulation of the genes for the pathway enzymes. One of the most widely distributed regulatory mechanisms for bacteria is the so-called "riboswitch," which is a sequence in the 5'untranslated region of messenger RNA that adopts alternative secondary structures that allow or repress the expression of downstream genes. Riboswitches offer considerable promise as targets of antibacterial agents and two agents, ribocil-C and roseoflavin (Fig. 11.2), have been found to inhibit dual FMN riboswitches that separately control riboflavin biosynthesis and uptake processes that are essential for the growth and pathogenesis of medically significant Gram-positive bacteria, including methicillin-resistant *Staphylococcus aureus*.⁴⁴

The transport systems for riboflavin are numerous and complicated, with some bacterial transporters

facilitating uptake and others the efflux of flavins. The latter, when it occurs, can play important roles in utilization of extracellular nutrients, for example, by participating in the reduction of extracellular insoluble ferric (Fe³⁺) to soluble ferrous (Fe²⁺) iron and in the settlement of symbiotic interactions.³⁴ Mammalian transport systems will be described in the next sections.

B. Digestion and Absorption

Since a large fraction of the vitamin B2 in food occurs as FAD and FMN, plus small amounts of flavinyl peptides, the first steps of digestion are hydrolysis by nonspecific phosphodiesterases, phosphatases, and hydrolases in the stomach and GI tract to release free riboflavin and flavinyl peptides. Riboflavin is absorbed by humans in the proximal small intestine by a saturable uptake process that is rapid and proportional to dose before leveling off in adults at about 27 mg.⁴⁵ Bile salts appear to facilitate uptake, and a modest amount of the vitamin circulates via the enterohepatic system.

Using isolated enterocytes,⁴⁶ everted jejunal segments,⁴⁷ and human intestinal brush border membrane vesicles,⁴⁸ transport was found to be carrier-mediated, Na⁺-dependent, and saturable at low, physiologically relevant, concentrations (e.g., micromolar), whereas simple diffusion was the prevailing transport mechanism at high concentrations. Lumiflavin inhibited riboflavin transport via the carrier-mediated saturable component.

Three riboflavin transporters have subsequently been characterized⁴⁹: RFVT1/*SLC52A1* (formerly named RFT1),⁵⁰ which is mostly expressed in small intestine and placenta; RFVT2/*SLC52A2* (formerly RFT3), which is highly expressed in brain and salivary gland and at lower levels in many other tissues; and RFVT3/*SLC52A3* (formerly RFT2), which is most highly expressed in testis, small intestine, prostate, and some other tissues. Kinetic analysis of these transporters has found sub- to low-micromolar Michaelis–Menten constants for riboflavin, and transport is inhibited by lumiflavin.^{50,51} In contrast to the earlier findings with intestinal transport models,^{46–48} riboflavin transport by these proteins is not dependent on Na⁺, membrane potential, or pH (except RFVT3/*SLC52A3* is more active at pH 6 than 7.4).⁴⁹ RFVT3 has been proposed to be the major intestinal transporter for riboflavin,⁵¹ and mice with conditional, intestinal-specific, knockout of RFVT3/*SLC52A3* display riboflavin deficiency, growth and development retardation, premature death, and severe reductions in carrier-mediated riboflavin uptake, and supplementation with pharmacological doses of vitamin B2 restored growth.⁵²

Intestinal uptake is adaptively upregulated in riboflavin deficiency,⁵³ and increased expression of RFVT3 mRNA and protein at the cell surface accounts for this adaptation to increase uptake.⁵⁴ Other factors that have been noted to affect intestinal absorption are aging,

uremic conditions, antacids, and fiber intake.⁵⁵ The latter effect has been attributed to RFVT3, which is thought to be important for colonic uptake of riboflavin and is induced by butyrate, a short-chain fatty acid produced in the large intestinal lumen by bacterial fermentation of fiber.⁵⁶

Physiologically, riboflavin uptake has been characterized by pharmacokinetics⁴⁵ and kinetic modeling using stable-isotope labels.⁵⁷ Care must be taken in interpreting human uptake by appearance in plasma because a large fraction of newly absorbed riboflavin is removed by the liver on “first pass,” so that appearance in plasma underestimates bioavailability from foodstuffs.

C. Transport

Circulatory transport

In circulation, riboflavin is loosely associated with albumin and somewhat more tightly bound by immunoglobulins in plasma from healthy individuals^{58,59} and patients with certain types of cancer.^{60,61} By use of flavinyl affinity chromatography, riboflavin-binding IgG, IgM, and IgA were isolated and papain cleavage yielded Fab fragments that still bound riboflavin⁶²; hence, the antigenic binding site is likely to be involved. The ability of flavins to serve as antigens has also been shown by haptenic challenge⁶³ and might be related to the antibodies generated by riboflavin biosynthesis intermediates, discussed in [Section II.A](#).^{35–37}

Riboflavin-binding proteins (RBP) play important roles in development, as exemplified by estrogen-induced avian proteins that are incorporated into eggs.^{55,64–66} These are related to RBPs found in other egg-laying species (fish, amphibians, reptiles, and monotremes),⁶⁶ and there is evidence that pregnancy-specific RBPs are produced by mammals (pregnant cows, rats, bonnet monkeys, and humans),^{67,68} perhaps to account for differences in riboflavin concentrations in maternal and cord blood in providing riboflavin for embryonic development.^{27,55,65} The genes and mechanisms for pregnancy-specific mammalian RBP have not been identified, but there might be structural similarity to the avian proteins because immunization of animals (rats, mice, and the bonnet monkey) with the avian RBP or injection of antibodies against the avian RBP can affect pregnancy.⁶⁸ Another possibility is that they are related to, or the same as, the recently discovered mammalian protein retbindin, which has 27% sequence homology with the avian RBP⁶⁹ and is a key player in the acquisition and retention of flavins in the neural retina.⁷⁰

In addition to having roles in transport and storage, RBPs might facilitate metabolism because an interaction between chick liver riboflavin kinase and RBP has been demonstrated.^{27,55,64,65}

Transport into cells

Cellular uptake of riboflavin from circulation has been noted to have similarities (and differences) from enteric transport mechanisms.^{27,71} Hepatocytes exhibit an initial rapid and specific intake that is relatively insensitive to Na^+ , followed by slower passive diffusion of the vitamin; once inside the cell, riboflavin becomes metabolically trapped by phosphorylation.⁷² In contrast, studies with proximal tubular epithelial cells from rat kidney found that the faster, facilitated phase of flavin uptake exhibits Na^+ dependence.⁷³ The RFVT1 and RFVT2 transporters have been suggested to play roles in the transport of riboflavin from cells to circulation,⁴⁹ and their properties are consistent with the findings with hepatocytes. There are also instances where the RFVT family transporters are elevated in cancer cells.⁷⁴ More studies are needed to characterize which of these, and possibly additional, transporters function in the uptake of riboflavin by different cells.

D. Metabolism

Riboflavin metabolism includes its conversion to FMN and FAD, as summarized in [Fig. 11.5](#), plus the additional processes whereby these flavocoenzymes are covalently attached to the polypeptide backbone of some of the flavoproteins, such as mammalian mitochondrial monoamine oxidase (as 8α -S-cysteinyl-FAD) and succinate and sarcosine dehydrogenases (as 8α -N³-histidyl-FAD). Turnover of FMN and FAD involves simple hydrolytic reactions ([Fig. 11.5](#)), but the full complement of catabolic reactions is more complicated, as will be described in [Section ii](#).

Coenzyme formations and interconversions

FMN and FAD biosynthesis; Riboflavin kinase (flavokinase) has been purified from many sources, including rat liver,⁷⁵ plants (mung bean),⁷⁶ bacteria (*Brevibacterium ammoniagenes*),⁷⁷ and subsequently others. They share some molecular properties (including the requirement of Zn^{2+} for kinase activity), but a substantial distinction is that mammalian and plant kinases form just FMN, whereas the enzymes from *B. ammoniagenes* and many other prokaryotes are bifunctional and perform both riboflavin kinase and FAD synthetase reactions.^{77,78} Conversely, the mammalian FAD synthetase (FMN: ATP adenylyl transferase) ([Fig. 11.5](#)) cannot serve as a flavokinase,⁷⁹ but the human enzyme has been found to be able to serve both as an FAD synthase when Mg^{2+} is the cofactor and as an FAD hydrolase when Co^{2+} is added.⁸⁰ Riboflavin kinase is cytosolic and possibly in mitochondria⁸¹; FAD synthase is located in the cytosol, mitochondria, and the nucleus.⁸²

The kinetics of substrate and product interactions with the kinase/synthetase system of rat livers has been discerned,⁸³ and interestingly (and now known to be “importantly”), FAD was found to be an inhibitory endproduct that might regulate its own formation. What has been subsequently found is that FAD is tightly bound by FAD synthase to enable the synthase to operate as a “FAD chaperone” to deliver it to apoenzymes via physical interaction of the synthase and recipient, as has been seen with nuclear lysine-specific demethylase 1, mitochondrial dimethylglycine dehydrogenase.⁸⁴

The regulation of riboflavin kinase and FAD synthase warrants further study in light of the interesting findings about these enzymes and to explain intriguing observations about the factors that had already been seen to affect their activity, such as the thyroid hormone-induced increase in a more active form of liver flavokinase and a concomitant decrease in a less active form⁸⁵ and activation by tumor necrosis factor to enhance the incorporation of FAD into NADPH oxidases.⁸⁶

FMN and FAD hydrolysis; Mammalian phosphatases that can hydrolyze FMN to riboflavin and inorganic phosphate not only include those with somewhat acidic pH optima,⁸⁷ such as those located in lysosomes, but also include those with alkaline optima in the intestinal brush border.⁸⁸ FAD pyrophosphatases from both liver and intestine are optimal at alkaline pH⁸⁹ and are influenced by age but not riboflavin status. As noted above, FAD synthase can also have FAD hydrolase activity under special conditions.⁸⁰

Coenzyme addition to apo-flavoenzymes (noncovalent); Flavoenzymes typically bind their cofactor tightly and are usually found primarily in the holoenzyme (i.e., cofactor bound) form. For at least some flavoenzymes, acquisition of the coenzyme is facilitated by interaction with the proximal synthase, as discussed above for the chaperone function of FAD synthase.⁸⁴

Another reason that apoenzymes are present in low amounts is that they are typically unstable, as initially shown for the mitochondrial flavoproteins isovaleryl-CoA dehydrogenase and short-chain acyl-CoA dehydrogenase,⁹⁰ but have been expanded to many more by a recent proteomic analysis.¹⁰ That study examined the proteome of B16 cells cultured for 24 h in riboflavin-free versus riboflavin-sufficient medium and found that the median abundance of the identified flavoenzymes (68) decreased by ca 13%, whereas the median abundance of the remaining proteome (6578 proteins) was not different. Degradation of the apo-flavoenzymes was linked to the protein quality control machinery.

Some apo-flavoenzymes are present at sufficient levels to be useful indicators of deficiency status by assaying them with and without added coenzyme (i.e., erythrocyte pyridoxamine (pyridoxine) 5′phosphate oxidase for FMN⁹¹ and glutathione reductase for FAD^{92,93}).

Coenzyme addition to apo-flavoenzymes (covalent); The means by which the covalent attachment of flavocoenzymes to the apo-flavoenzymes occur are still being elucidated. Formation of the 8 α -substituted riboflavins (e.g., S-cysteinyl or N3-histidyl derivatives) prior to coenzyme biosynthesis has been argued against because they are not substrates for riboflavin kinase or FAD synthetase.⁹⁴ Instead, intact coenzyme (in this case, FAD) was shown to be incorporated during translation of the nascent polypeptide chains of 6-hydroxy-D-nicotine oxidase from *Arthrobacter oxidans*, resulting in an 8 α -N₃-histidyl-FAD.⁹⁵ That flavinylation occurs co- or posttranslation has been extended to other flavoenzymes with 8 α -modified flavins, with the thinking being that the covalent attachments may be autocatalytic, capitalizing on the reactive properties of the isoalloxazine ring itself,⁹⁶ and in some cases utilize additional proteins.⁹⁷

An additional type of covalent attachment has been found for the bacterial Na⁺-translocating NADH:quinone oxidoreductase, the formation of a phosphodiester of the 5′-phosphate group of FMN with a threonine residue, and this is produced by a specific flavin transferase enzyme.⁹⁸

Catabolism, excretion, and secretion

Mammals are relatively limited in their abilities to catabolize riboflavin, and the diversity of flavin-derived products in mammalian urine reflects turnover events in somatic cells (producing mainly the free vitamin and cleavage products from flavinyl polypeptides⁹⁹), degradation by gastrointestinal microbes, and photochemical events that occur at the dermal level.^{16,100–103} These diverse compounds and their interconnections are summarized in Fig. 11.6.

Gastrointestinal microbes can cleave the ribityl side chain to 10-formylmethylflavin (found in urine from ruminants), which can be reduced to 10-hydroxyethylflavin¹⁰⁴ or oxidized by alimentary bacteria of the ruminant and human to 10-carboxymethylflavin.¹⁰⁵ The side chain catabolites 10-hydroxymethylflavin and lumiflavin/lumichrome-level compounds are found in urine from rats¹⁰² and humans,¹⁰¹ and the latter may also result from the action of light on flavin within the dermal tissue (lumichrome has also been recovered from rat feces).¹⁰⁶ That at least a portion of the side chain catabolites are produced by microbial degradation has been confirmed by antibiotic administration.^{102,106} Other ribityl chain metabolites in urine (probably from tissue metabolism) are a peptide ester of riboflavin¹⁰³ and an α -D-glucoside, which has also been shown to be utilizable as a vitamin source because it can be taken up and metabolized.¹⁰⁷ Terminal ribityl side chain oxidation of riboflavin by oxidative enzymes produced by the food fungus *Schizophyllum commune* leads to vitaminically inactive schizoflavins and thereby reduces the riboflavin available to other competing organisms.^{108,109}

Catabolites derived primarily from oxidations within tissue include 7- and 8-hydroxymethylriboflavins found in human urine.^{101,110} The 7 α -hydroxyriboflavin has been found in human blood.¹¹¹ 7- and 8-carboxylumichromes have been found in rat urine.^{102,112} These methyl oxidized products reflect microsomal mixed-function oxidase activity.¹¹³ Other 8 α -flavin catabolites arise from 8 α -flavinyl-peptides⁹⁹ and include an 8 α -sulfonylriboflavin found in human urine that may derive from the 8 α -cysteinyl-FAD of monoamine oxidase.¹⁰¹ And one of the most surprising findings regarding modification at the 8-position,¹¹⁴ the human electron-transferring flavoprotein, which serves as a central electron carrier in the mitochondrial matrix, undergoes irreversible and pH-dependent oxidation of the 8 α -methyl group to 8-formyl-FAD, which increased the affinity for dimethylglycine dehydrogenase, one of its electron transport partners.

All together, we can account for >95% of flavins excreted in human urine. For normal adults eating varied diets, riboflavin comprises about two-thirds; 7-hydroxymethylriboflavin, 10%–15%; a few to several percent of the 8-hydroxymethyl- and 8 α -sulfonylriboflavins occur; there are lesser amounts of riboflavinyl peptide ester and such side chain–altered flavins as the 10-hydroxyethyl and 10-formylmethyl compounds; and only traces of carboxymethylflavins and lumiflavin.

Additional metabolic derivatives of flavins are known to occur naturally as products and metabolites by other organisms, such as certain bacteria of the genus *Pseudomonas* and other bacterial, fungal, or plant organisms but are not normally of consequence to humans.¹¹⁵

Secretion into milk; The transport pump for riboflavin has been identified as the multidrug transporter breast cancer resistance protein (BCRP/ABCG2), a member of the ATP-binding cassette family of transmembrane drug transporters.¹¹⁶ In milk from cows¹¹⁷ and from humans,¹¹⁸ the flavin in highest concentration other than the free vitamin is FAD, much of which is hydrolyzed during pasteurization. The presence of 10-(2'-hydroxyethyl)flavin is notable because this catabolite is an inhibitor of riboflavin uptake⁷² and phosphorylation¹¹⁹ and subtracts modestly from the biological activity of the milk. Also present are 7- and 8-hydroxymethylriboflavins and other catabolites, including the 10-formylmethylflavin and lumichrome.^{117,118}

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Gene regulation

Early studies of riboflavin deficiency and gene expression^{120,121} noted that some pathways were more

sensitive to the reduced availability of the flavocoenzymes—namely, the mitochondrial electron transfer chain was preserved while enzymes for fatty acid β -oxidation were diminished. They speculated that this might be due to alterations in gene expression and/or protein stability. Similar observations have been made in recent proteomic studies^{10,122} and have been attributed to the instability of the apoenzymes due to activation of the protein quality control machinery¹⁰ that recognizes faulty proteins, and possibly also the activation of ER stress and apoptosis processes in riboflavin-deficient cells.¹²² In addition to elevated stress markers, in HepG2 cells, riboflavin deficiency increased the abundance of damaged (carbonylated) proteins and increased the incidence of DNA strand breaks.¹²³

As one of the nutrients involved in one-carbon metabolism, riboflavin can also affect gene expression by epigenetic mechanisms.¹²⁴ From an opposite direction, supplementation with vitamin B2 (in combination with B1 and niacin) has been reported to help normalize the gene expression profile of mice exposed to acute hypoxia.¹²⁵

B. Metabolomics

Riboflavin has often been included in studies that have analyzed the major biomolecules in blood or urine and it and its metabolites have surfaced as potentially important biomarkers in more recent “metabolomics” investigations of diseases such as cancer.^{126,127} However, its full potential has not been fully exploited, in part because flavoproteins are involved in so many processes. Knowing the metabolic pathways that have enzymes that require FMN or FAD for activity, it should be possible to identify deficiencies (and defects in particular flavoenzymes) by metabolomic analysis and a genome-scale metabolic model with a curated and extended flavoproteome has been developed for this purpose.¹²⁸ It predicted the metabolic impact of flavin depletion and multiple-acyl-CoA-dehydrogenase deficiency,¹²⁸ as proof of concept; therefore, it might be able to facilitate the characterization of the flavometabolome for other deficiencies and disease.

IV. DIETARY REQUIREMENTS

Human riboflavin requirements, biochemical indicators for these, and factors affecting the requirements are conveniently summarized on a web site maintained by the National Institutes of Health Office of Dietary Supplements (<https://ods.od.nih.gov/factsheets/Riboflavin-HealthProfessional/>).¹²⁹ The text box summarizes some of the information from that web site and sources that are cited.^{130–133} These topics are discussed in the following subsections (Box 11.1).

BOX 11.1

Summary of Dietary Requirements and Deficiencies for Vitamin B₂

Adequate Intakes (AI),* Recommended Dietary Allowances (RDA)** & Dietary Guidelines***

Age	Male	Female	Pregnancy	Lactation
Birth to 6 months	0.3 mg*	0.3 mg*		
7–12 months	0.4 mg*	0.4 mg*		
1–3 years	0.5 mg**,***	0.5 mg**,***		
4–8 years	0.6 mg**,***	0.6 mg**,***		
9–13 years	0.9 mg**,***	0.9 mg**,***		
14–18 years	1.3 mg**,***	1.0 mg**,***	1.4 mg**	1.6 mg**
19–50 years	1.3 mg**,***	1.1 mg**,***	1.4 mg**	1.6 mg**
51+ years	1.3 mg**,***	1.1 mg**,***		

Daily Values (DV)

The U.S. Food and Drug Administration has developed Daily Values to help consumers compare the nutrient contents of products within the context of a total diet.¹³²

Adults and children ≥ 4 years	1.3 mg
Infants 1 through 12 months	0.4 mg
Children 1 through 3 years	0.5 mg
Pregnant women and lactating women	1.6 mg

Tolerable Upper Intake Level (UL)

The UL for riboflavin could not be established,¹³⁰ but this “does not signify that people can tolerate chronic intakes of these vitamins at levels exceeding the Recommended Dietary Allowance (RDA) or Adequate Intake (AI).” See section V. F. for more information.

Indicators of Deficiency or Inadequacy¹²⁹

The signs and symptoms of riboflavin deficiency (ariboflavinosis) include:

- skin disorders (seborrheic dermatitis)
- cheilosis (swollen, cracked lips)
- angular stomatitis (mouth lesions)
- sore throat
- itchy and red eyes
- edema of the mouth and throat
- reproductive problems
- hair loss
- glossitis (magenta tongue)
- hyperemia and edema of the pharyngeal and oral mucous membranes
- degeneration of the liver and nervous system
- Anemia and cataracts can develop if riboflavin deficiency is severe and prolonged.

Sources¹²⁹

Foods especially rich in bioavailable riboflavin are yeast, kidney, liver, cheese, eggs, and milk. Green vegetables are reasonably good sources. Natural grains and cereals are relatively poor sources, but fortification and enrichment have led to a considerable increase in intake of riboflavin from these food items.

A. Indicators of Deficiency or Inadequacy

In all animals, the symptom of severe riboflavin deficiency (also known as ariboflavinosis) is retardation of growth, and additional effects include loss of hair, disturbances of the skin, etc. The symptoms for humans are listed in the text box.

B. Criteria of Adequacy

The text box has several types of reference values for differences in age, sex, and special needs (e.g., pregnancy and lactation) that have been based by the Food and Nutrition Board of the Institute of Medicine¹³⁰ using the following criteria of adequacy: the Adequate Intake (AI) for infants from birth to 12 months was assumed to be equivalent to the mean intake of riboflavin in healthy, breastfed infants and the Recommended Dietary Allowance (RDA) for all other ages is the average daily level of intake sufficient to meet the nutrient requirements of nearly all (97%–98%) healthy individuals, and this has been estimated by a number of criteria, principle among which were the erythrocyte glutathione reductase assay, the erythrocyte flavin concentration, and urinary excretion of the vitamin in fasting, random, or 24-h specimens or by load tests. The RDA for riboflavin¹³⁰ was used as the basis for the Dietary Guidelines for Americans, 2010,¹³¹ and the 2015–20 Dietary Guidelines for Americans, Eighth Edition, US Department of Health and Human Services and US Department of Agriculture.¹³³

C. Assessment of Status

Biochemical indicators for estimating riboflavin requirements and determining status include assays of red cell and urinary flavin¹⁰⁰ (both enhanced by HPLC and fluorometry) and the erythrocyte assays of glutathione reductase ($\pm$ FAD)^{92,93} and pyridoxine (pyridoxamine) phosphate oxidase ($\pm$ FMN).⁹¹ Some groups have concluded that the inflection point in the urinary excretion curve represents the primary biomarker of riboflavin requirement.¹³⁴ Although widely used, the erythrocyte glutathione reductase assay^{92,93} has known limitations, such as that it cannot be used in persons with glucose-6-phosphate dehydrogenase deficiency.¹³⁵ Therefore, for assessment by enzymatic assay, there are advantages to the use of the FMN-dependent pyridoxine (pyridoxamine) phosphate oxidase assay, which has been validated in rats¹³⁶ and humans.¹³⁷

D. Reference Intakes

The Dietary Reference Intakes defined by the Institute of Medicine in their 1998 report¹³⁰ still stand for

Americans and Canadians for riboflavin. In the cited sources can be found the RDA (the estimate to meet the requirement of 97%–98% of healthy individuals), the Estimated Average Requirement (EAR) (the estimate to meet the requirement of half of the healthy individuals in a group and used to evaluate the adequacy of group intakes), the AI (based on approximations of nutrient intake by healthy people, therefore, assumed to be adequate), and the Tolerable Upper Intake Level (the highest daily nutrient intake level that is likely to pose no risk of adverse health effects to almost all individuals). The US Food and Drug Administration's Daily Values (DV) are based on the RDAs for adults, but with fewer subdivisions to help consumers compare the nutrient contents of products within the context of a total diet.¹³² These are summarized in the text box or, for EAR, found in the Report.¹³⁰

E. Basis for Individual Variability in Requirements¹²⁹

The following groups are among those most likely to have inadequate riboflavin status: pregnant and lactating women and their infants, especially those who rarely consume meats or dairy products (such as those living in developing countries and some vegetarians in the United States); vegetarian athletes, due to the stress that exercise puts on metabolic pathways that use riboflavin; people who are vegan and/or consume little milk; and people with genetic defects or other diseases that adversely affect vitamin B2 absorption or utilization (see below).

F. Tolerable Upper Intake Levels

An upper level of intake was not set for riboflavin by the IOM because no toxicity has been reported when riboflavin intake from food has exceeded the RDA by several fold nor when supplements of 400 mg/day were taken with meals for at least 3 months.¹³⁰ A follow-up of the latter study¹³⁸ reported that there were no significant changes in blood pressure or body weight, but one woman in the riboflavin group withdrew from the study with diarrhea and another patient complained of polyuria but completed the study. The paucity of adverse effects is thought to be due to riboflavin's low aqueous solubility and limited capacity to be absorbed in the gastrointestinal tract.¹³⁰ Riboflavin absorption from the gastrointestinal tract is limited to about 27 mg at any one time from an oral dose given to an adult, which is well below the oral LD₅₀ in rats (>10,000 mg/kg).¹³⁹ Excess riboflavin can cause yellow-colored urine.¹⁴⁰

Riboflavin and derivatives have been used in a number of phototherapies that could produce potentially toxic reactive oxygen species, oxidized amino acid residues,¹⁴¹ and lumichrome from cleavage of the ribityl side chain (which is the major photodegradation product of riboflavin in milk exposed to sunlight¹⁴² and found in urine¹⁶).

G. Dietary Sources

Riboflavin is widely distributed in food in the form of the free vitamin (usually the minor form unless there has been supplementation), noncovalently bound FMN and FAD, and smaller amounts of the flavin coenzymes covalently bound by protein (which are not significantly reutilized)¹⁴³ (however, in general, vitamin B2 bioavailability appears high with urinary monitoring suggesting that riboflavin from spinach is as bioavailable as it is from milk).⁵⁷

Foods that are particularly rich in vitamin B2 are organ meats (kidney, liver), dairy products (milk and cheese) and eggs, whereas ungerminated grains, especially polished rice, fruits and vegetables contain much lower amounts. However, grains and cereals are fortified with riboflavin in the United States and many other countries. A useful online resource for the riboflavin content of specific food is FoodData Central (US Department of Agriculture, Agricultural Research Service, FoodData Central, 2019, <https://fdc.nal.usda.gov>).¹⁴⁴ FoodData Central is an integrated data system that provides expanded nutrient profile data and links to related agricultural and experimental research.

V. ISSUES RELATED TO RIBOFLAVIN IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Dietary intakes of riboflavin in the United States are adequate. An analysis of data from the 2013–16 National Health and Nutrition Examination Survey (NHANES) “What We Eat in America” showed that less than 3% of the US population has an intake of riboflavin from foods and supplements below the EAR with the exception of teenage girls (7%), young women (4%), and men and women over 70 yrs of age (4%).¹⁴⁵ For those who have an inadequate intake of riboflavin, there is an obvious need to increase their vitamin B₂ intake (preferably via the diet, but possibly also via supplementation). Care should also be taken by those with increased physiologic needs (e.g., in pregnancy, lactation, or high physical activity) (Section IV.E.) or defects in utilization of riboflavin or its coenzymatic forms.

There has been an expansion in our knowledge of inborn errors of metabolism that affect riboflavin transport, metabolism, or utilization by functional flavoproteins, and with the many ongoing investigations to identify functionally significant variants in the human genome, it is certain that more will be found. Three well-defined and one emerging new example of riboflavin-responsive variants are briefly summarized below.

Riboflavin-responsive multiple acyl-CoA-dehydrogenase deficiency (RR-MADD), also known as glutaric aciduria type II

MADD are inborn errors of metabolism have been known for decades¹⁴⁶ to be due to defects in the FAD-dependent electron transfer flavoproteins (ETF) or ETF dehydrogenase, which link mitochondrial fatty acid oxidation and amino acid metabolism (mostly that of sarcosine and dimethylglycine) to the mitochondrial respiratory chain. A subset of MADD patients respond to riboflavin supplementation (RR-MADD), and these have been attributed to multiple causes, i.e., defects in (a) the flavoprotein¹⁴⁷; (b) the cellular riboflavin transporters (RFVT1/SLC52A1, RFVT2/SLC52A2, RFVT3/SLC52A3)^{148,149}; (c) the mitochondrial FAD transporter (SLC25A32)¹⁵⁰; or (d) FAD synthase (FLAD1).¹⁵¹

Brown-Vialetto-Van Laere syndrome (also known as Fazio–Londe disease)

This is a rare neurological disorder that has some features that mimic MADD and was found to be improved by supplementation with high doses of riboflavin and attributed to defects in riboflavin transporters RFVT2/SLC52A2 and RFVT3/SLC52A3.^{152,153}

MTHFR C677T polymorphism

The C677T variant of MTHFR has been associated with risk of high blood pressure by many studies, and there is also evidence that riboflavin supplementation can modify the effect. The mechanism is not yet known, but might be due to increasing coenzyme binding and/or stabilization of the variant MTHFR enzyme to restore more normal activity.^{154,155} A relationship between this polymorphism and riboflavin status has also been associated with schizophrenia and schizoaffective psychosis.¹⁵⁶

NAD(P)H quinone oxidoreductase 1 (NQO1, DT-diaphorase) variant p.P187S

NAD(P)H quinone oxidoreductase 1 is involved in antioxidant defense and protection from cancer, and the p.P187S variant has been associated with increased risk of smoking-related lung cancer and other cancer risk. One of the consequences of this structural variation is a lower binding affinity for the FAD and increased proteolytic degradation of the protein, and addition of FAD restores the stability to wild-type levels.¹⁵⁷

Therefore, individuals with this variant might benefit from riboflavin supplementation to stabilize the enzyme.

Other areas of concern regarding riboflavin status

Include interactions with other components of the diet that can affect absorption, such as the polysaccharide chitosan that is used as antiobesity supplement¹⁵⁸; losses in chronic renal failure and dialysis that can necessitate supplementation¹⁵⁹; and losses during phototherapy for hyperbilirubinemia where no,¹⁶⁰ or some,^{161,162} reduction in the saturation of erythrocyte glutathione reductase by FAD has been reported.

In addition to these well-defined cases, there are a number of conditions where intake of elevated amounts of vitamin B₂ has been suggested to have benefit (such as for migraine headaches and cancer prevention) but with inconclusive data for or against supplementation. These are discussed on the “Riboflavin and Health” link on the “Fact Sheet for Health Professionals: Riboflavin” maintained by the National Institutes of Health Office of Dietary Supplements.¹²⁹

B. Known Interactions

Certain antimalarial, antimicrobial, anticancer, tricyclic antidepressant, and antipsychotic agents have been observed to alter riboflavin utilization.¹⁶³ For example, boric acid complexes with the polyhydroxyl ribitol side chain of riboflavin, and individuals who accidentally consume boric acid or one of its derivatives excrete high levels of riboflavin; the phenothiazine ring of chlorpromazine has structural similarity to flavins, and chlorpromazine increases urinary levels of riboflavin. Studies of riboflavin reuptake by renal tubules

suggest that agents such as probenecid, trifluoperazine, and chlorpromazine can affect transport.¹⁶⁴ Since ethanol consumption together with riboflavin deficiency reduces hepatic GSH and the activities of enzymes controlling GSH metabolism, there is concern that this combination might enhance alcohol-induced liver injury.¹⁶⁵

In addition, flavoproteins are involved in the metabolism of at least four other vitamins (with already established effects of riboflavin deficiency on the first two): conversion of the pyridoxamine- and pyridoxine-5'-phosphates to pyridoxal 5'-phosphate by FMN-dependent pyridoxamine(pyridoxine) 5'-phosphate oxidase, an enzyme that has been highly characterized biochemically^{166,167} and for its role in vitamin B₆ metabolism¹⁶⁸ under normal and riboflavin-deficiency conditions^{91,168}; reduction of 5,10-methylenetetrahydrofolate to 5-methyltetrahydrofolate by the FAD-dependent MTHFR, which has also been shown to be sensitive to riboflavin deficiency¹⁶⁹; utilization of the alkylcobalamins (forms of vitamin B12) by CblC, which utilizes FMN or FAD¹⁷⁰; and multiple enzymes in vitamin D metabolism.¹⁷¹

C. As a Dietary Supplement

Since a low percentage of the US population has an intake of riboflavin below the EAR,¹⁷² there is relatively little need for additional supplementation, except for the individuals with identified needs discussed in [Sections IV.E](#) and [V.A.](#)¹³⁹ When supplements are taken, it is worth bearing in mind that riboflavin absorption by humans levels off in adults at about 27 mg⁴⁵; therefore, little above this amount is likely to be utilized.

RESEARCH GAPS

Most of the new discoveries that have surfaced about the biochemical, genetic, and (patho)physiologic aspects of riboflavin utilization and function since the last edition of *Present Knowledge in Nutrition* have shifted the focus of research from establishing an average dietary requirement for everyone in an age, gender, and activity category to the identification of those, often relatively few, individuals with particular needs due to their genetic and lifestyle profiles. Accomplishing this will require more basic research and application of the tools of genomics, metabolomics, and possibly other “omics” to uncover when (and why) an individual’s needs differ from the general population and to monitor responses to the selected interventions. It is fitting that this new perspective has begun to emerge as those who study flavins and flavoproteins are beginning to celebrate the centennial anniversary of the discovery of riboflavin.

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NIACIN

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SUMMARY

Human niacin deficiency, known as pellagra, was the most devastating nutritional disease in modern American history. Even today, after enrichment of refined foods, niacin deficiencies may occur through poor food habits and disease processes that actively cause depletion. Vitamin B3 is the precursor to the various essential pyridine nucleotide structures, including oxidized and reduced forms of nicotinamide adenine dinucleotide and its phosphorylated derivatives (in total, referred to as NAD[P]), which are used in more reactions (>470) than any other vitamin-derived molecule. Nicotinic acid (niacin), nicotinamide (niacinamide), nicotinamide riboside, tryptophan, and nicotinamide mononucleotide will all prevent pellagra but possess distinguishable characteristics as NAD precursors. Nicotinic acid also activates the GPR109 receptors to mediate many of the niacin-dyslipidemia-correcting activities. NAD working through sirtuins provides benefits in preventing many disease processes. ADP-ribose polymerase inhibitors simultaneously preserve NAD levels while decreasing genomic stability. Tryptophan, glutamine, thiamin, ascorbate, riboflavin, and pyridoxal phosphate are required for various aspects of formation and interconversion of NAD[P], leading to complex nutritional interactions. Continued basic and clinical study of NAD metabolism has great potential to improve human health.

Keywords: Niacin; Nicotinamide; Nicotinic acid; Pellagra.

I. INTRODUCTION

There are three forms of vitamin B3: nicotinic acid (also known as niacin), nicotinamide (also known as niacinamide), and nicotinamide riboside. Humans cannot specifically synthesize vitamin B3 endogenously. Humans depend on plants and microorganisms to introduce these chemicals into the food chain. Technically, niacin refers to just one of the forms of vitamin B3, but it is often used to denote the total B3 content. Vitamin B3 is by definition the essential dietary precursor for the endogenous formation of nicotinamide adenine dinucleotide (NAD). NAD is essential to all forms of life, and humans convert these various sources of vitamin B3 to NAD by distinct metabolic pathways. Ultimately, NAD functions as a cofactor, substrate, or ligand in reactions involving over 400 different gene products. NAD is involved in more reactions than any

other known vitamin-derived molecule. Today, NAD research continues to expand with the increased focus on its role in life span, and potential therapeutic benefits seen working through sirtuins, NAD-depleting poly(-ADP-ribose) polymerase (PARP) enzymes in cancer, and many more aspects.

Vitamin B3 is essential to life, plays roles in longevity and various chronic disease processes, and is actively depleted in many clinical conditions. Current and future research will be key to understanding how we can focus on these pathways for preventing the progression of chronic, infectious, and other diseases of aging.

A. Background

Niacin deficiency was first described by Casal in 1762 in Europe as a condition characterized by extreme weakness and crusty skin.¹ Accordingly, Casal coined the

disease “pellagra,” meaning angry skin. Pellagra spread throughout Europe following the dependence on corn, which had been imported from the Americas. Nicotinic acid in corn is tightly bound, requiring alkaline treatment for bioavailability. This was accomplished in traditional preparations, but the knowledge was lost as corn moved between continents and cultures. Just over 100 years later, pellagra would reach epidemic proportions in the Southern United States, due in part to the invention and distribution of industrial-scale milling devices in the 1870s. Suddenly, the first micronutrient-depleted refined foods, white flour, and rice became available to the American masses. Thereafter, pellagra and beriberi became pervasive, depending on the variety of individual diets. Pellagra exerted many effects on physical and psychiatric health and was the leading cause of death in mental hospitals in 1907. By the 1920s, pellagra had become the most severe deficiency disease in the history of the United States. Over 120,000 people died from pellagra epidemics in the United States during the first two decades of the 20th century.^{1,2}

Pellagra was finally understood in the 1930s, thanks in large part to the work of the epidemiologist Dr. Joseph Goldberger. Initially, Dr. Goldberger’s hypothesis that pellagra was a dietary deficiency disease was rejected in favor of corn toxins or poor sanitation. Elvjem identified the pellagra-preventing factor in 1937 as the molecule nicotinic acid. A canine model, in which niacin deficiency caused tongue discoloration, was used to isolate a curative rice-polishing component that was named the pellagra-preventing factor, later identified as niacin. The government established legally mandated standards of niacin enrichment of flour in 1942. Technically, niacin refers to the chemical nicotinic acid, while niacinamide refers to the chemical nicotinamide. However, niacin is commonly used to refer to either form. This is unfortunate because there are significant physiological differences between niacin and niacinamide.

Doctors who witnessed pellagrous dementia noticed many similarities to schizophrenia and theorized that schizophrenics might have a genetic disorder that required far greater levels of niacin. Experiments performed in the 1950s revealed that administration of gram quantities of niacin to schizophrenic patients frequently exerted therapeutic benefits with no adverse reaction.³ However, this approach remains controversial today. This may be due, in part, to the fact that schizophrenia could enter a stage of irreversible change that is no longer responsive to chemical approaches. These same researchers discovered that high-dose nicotinic acid, but not nicotinamide, effectively lowered blood cholesterol.⁴ Today, high-dose niacin is a more effective elevator of the high-density lipoprotein (“good” cholesterol) than any pharmaceutical.⁵ This approach also lowers triglycerides and very-low-density lipoprotein

(“bad” cholesterol). Interestingly, high-dose niacin will also raise cholesterol for individuals with abnormally low cholesterol. Administration of high doses of the various alternative NAD precursors nicotinic acid/niacin, nicotinamide/niacinamide, and/or nicotinamide riboside remains an ongoing area of clinical research.

B. Nutrient Function and Structure

Vitamin B3 is defined as the dietary precursors to NAD, other than the amino acid tryptophan. These include niacin (also known as nicotinic acid or pyridine-3-carboxylic acid), niacinamide (also known as nicotinamide or pyridine-3-carboxamide), and added more recently, a third form of vitamin B3, nicotinamide riboside.⁶ Structures for the three molecules are shown in Fig. 12.1.

Ultimately, these precursors function following conversion to one of the forms of NAD[P], which are used in more reactions than any other vitamin-derived cofactor. Over 400 proteins use NAD for chemical reactions. Ultimately, amino acid polymorphisms residing within cofactor-binding domains result in decreased binding affinity with associated decreased enzyme kinetics. There are known examples where administration of vitamin cofactor supplements can rescue the symptoms arising from decreased enzyme efficiency due to these mutations.⁷ Known niacin-responsive genetic diseases include mutations in either the neutral amino acid transporter encoded by the SLC6A19 gene, the aldehyde dehydrogenase encoded by ALDH2, or the glucose-6-phosphate-1-dehydrogenase gene. Mutations causing a loss of SLC6A19 function result in Hartnup disease, which resembles pellagra and leads to death unless treated early. This occurs due to an inability to transport tryptophan into cells. Symptoms are rescued via the administration of niacin, thus supporting the potential importance of tryptophan as a NAD precursor in diets low in vitamin B3.⁸ Up to one-third of all known enzyme-encoding gene mutations may result in decreased enzyme efficiency due to altered binding affinity for the cofactor.⁷ Given that there are more NAD-binding proteins than any other cofactor and most have never been examined from this perspective, it is expected that many more undiscovered mutations are in fact responsive to administration of high doses of NAD precursors.

Good niacin status is required to prevent birth defects.⁹ Loss of function in the kynurenine pathway, connecting tryptophan to the synthesis of NAD, can result in congenital malformations with lethality. These are referred to as congenital NAD deficiency disorders. Studies performed in mice reveal that postpartum mothers also increase mammary circulation of their NAD metabolites by at least 20-fold for the benefit of offspring as part of a process known as homeorhesis.¹⁰ Hepatic and mammary NAD systems are altered during

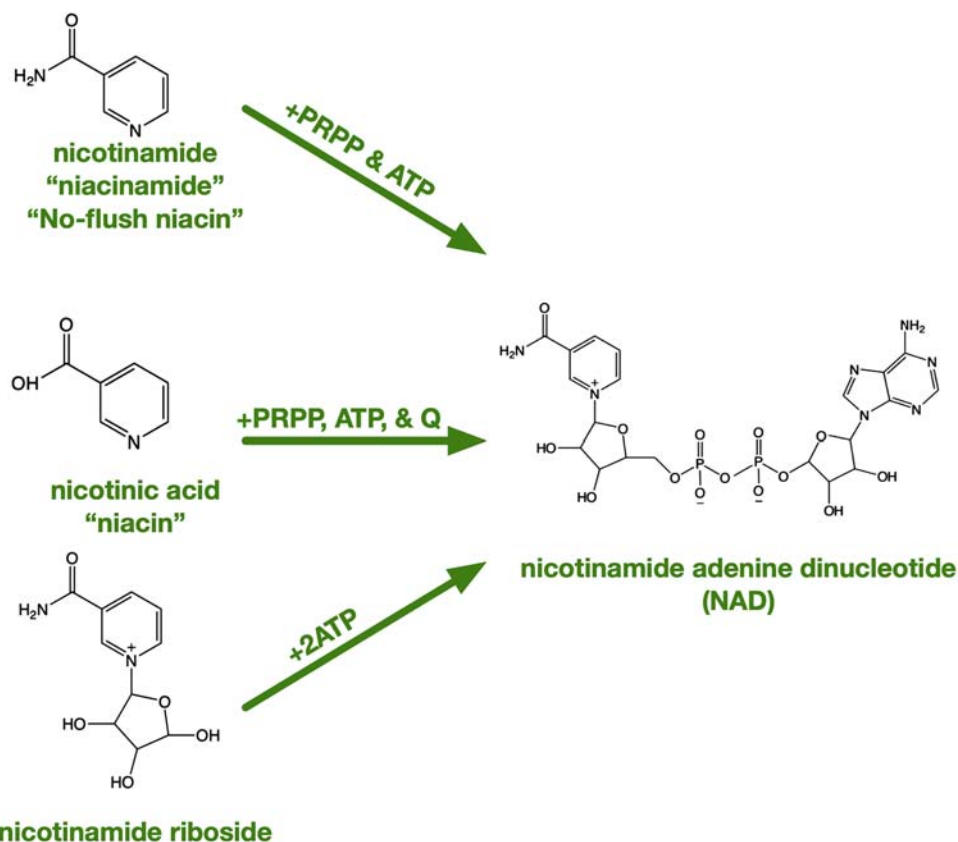


FIGURE 12.1 Forms of vitamin B3. Various dietary precursors can support the formation of the essential molecule, nicotinamide adenine dinucleotide (NAD). The cofactors required to convert each vitamin B3 precursor to NAD are shown. *PRPP*, 5-phosphorylribose-1-pyrophosphate; *Q*, glutamine.

this postpartum phase. Supplementation with nicotinamide riboside to mothers increases their lactation, nursing behaviors, and general transmission of nutrients to offspring. Mouse pups from nicotinamide riboside-supplemented mothers retain their benefits even into adulthood, manifested as increased physical performance, spatial memory, and hippocampal neurogenesis. A similar response is expected for nicotinamide or nicotinamide mononucleotide-supplemented pregnant mothers based on the pathway.

II. NUTRIENT METABOLISM

A. Synthesis

Humans cannot specifically synthesize nicotinic acid or nicotinamide endogenously, rather we depend on plants and microorganisms to introduce these chemicals into the food chain. NAD is essential to all forms of life, and humans convert various sources of niacin to NAD by distinct metabolic pathways. Plant products in the diet are primarily sources of nicotinic acid, while animal products provide nicotinamide through the breakdown of NAD[P] during the digestive process.

Pathways to and from NAD

The NAD precursors (nicotinic acid/niacin, nicotinamide/niacinamide, nicotinamide riboside, or tryptophan) follow separate pathways to generate NAD within the cell, but all of these pathways ultimately require the one enzyme, nicotinamide mononucleotide phosphoribosyltransferase (NMNAT). There are three different NMNAT isoforms (encoded by NMNAT1, NMNAT2, and NMNAT3) expressed in distinguishable subcellular locations. Overexpression of any of these enzymes increases cell viability in response to various challenges that threaten cellular NAD concentrations.^{11–15} In this section, we review the NAD-producing pathways as shown in Fig. 12.2.

Nicotinic acid, obtained from the diet, is converted to NAD via the Preiss-Handler pathway, starting with nicotinic acid phosphoribosyltransferase (encoded by the NPT1 gene). This enzyme is not inhibited by nicotinamide, whereas the equivalent mononucleotide-generating enzyme for nicotinamide (encoded by the NAMPT gene) is inhibited. Accordingly, high doses of nicotinic acid are known to increase the amount of NAD within cells to a greater level than does nicotinamide. Whether this greater

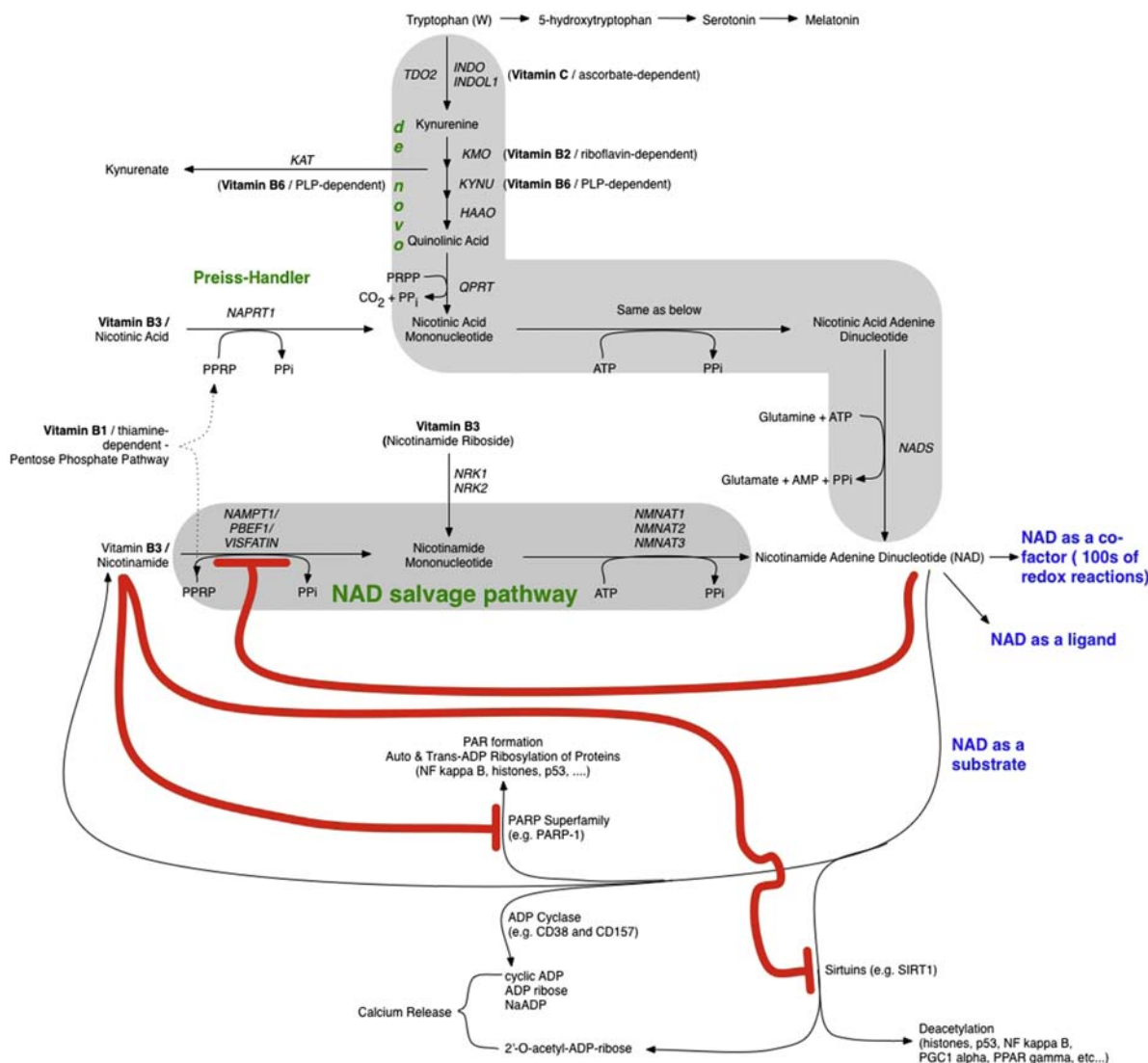


FIGURE 12.2 Pathways forming and utilizing NAD. The three biosynthetic pathways, including the de novo, Preiss-Handler, and salvage pathways, are followed with gray shading. Vitamin B2 (riboflavin), vitamin B6 (pyridoxal phosphate), and vitamin C (ascorbate) are all required for completion of the NAD de novo pathway starting from tryptophan. Otherwise, vitamin B1 (thiamin) is required for synthesis of phosphoribosyl pyrophosphate (PRPP) in the pentose phosphate pathway. The uses of NAD as a cofactor, ligand, or substrate are shown to the right. Gene names are in italics.

increase in intracellular NAD is part of the reason for nicotinic acid's superior performance in preventing atherosclerosis is not very clear. Next, nicotinic acid mononucleotide is converted to nicotinic acid adenine dinucleotide via NMNAT, which requires ATP. Lastly, glutamine is used by NAD synthase to replace the acid with the amide to produce NAD. While glutamine is technically not an essential amino acid, it is essential in the niacin-originating synthesis of NAD, and glutamine status could impact this pathway.¹⁶

Nicotinamide, obtained from the diet, or produced by NAD glycohydrolases or ADP-ribosyltransferases, is re-cycled back to NAD via the NAD salvage pathway (Fig. 12.2). The first enzyme, nicotinamide phosphoribosyltransferase, is rate-limiting in controlling NAD synthesis.¹⁷ Changes in NMNAT levels become more important in controlling NAD production during times of stress, which are known to induce expression of NAMPT.¹⁸ Increases in the efficiency of the NAD salvage pathway have been shown to provide greater protection against otherwise lethal shocks in a wide variety of cell types.

Lastly, nicotinamide riboside was discovered in 2004 to be a third form of vitamin B3.⁶ By contrast, nicotinic acid and nicotinamide were discovered and have been studied for over 60 years to date. Nicotinamide riboside is converted to NAD by a simple two-step pathway starting with a reaction catalyzed by nicotinamide riboside kinase (Fig. 12.2). Less is known about nicotinamide

yltransferase, is rate-limiting in controlling NAD synthesis.¹⁷ Changes in NMNAT levels become more important in controlling NAD production during times of stress, which are known to induce expression of NAMPT.¹⁸ Increases in the efficiency of the NAD salvage pathway have been shown to provide greater protection against otherwise lethal shocks in a wide variety of cell types.

riboside, but it is present in milk and examination of nicotinamide riboside supplementation (500 mg, twice a day) revealed no toxicity, and increased NAD levels.¹⁹ Experiments to date indicate that nicotinamide riboside boosts NAD levels, activates Sir2, and extends replicative life span in yeast, suggesting potential for therapeutic benefits in humans.²⁰ Nicotinamide riboside is distinguished in vertebrate development for serving essential pathways in muscular development.²¹ Nicotinamide riboside has been shown to prevent loss of skeletal muscle stem cells, which happens normally with aging.²² Whether other NAD precursors such as niacin can do this also remains to be determined.

The *de novo* pathway of NAD synthesis starts with tryptophan. Requiring over seven chemical reactions, this is the longest NAD biosynthetic pathway. One intermediate, 2-amino-3-carboxymuconate-6-semialdehyde (ACMS), has to accumulate and degrade nonenzymatically to allow the pathway to be completed. The enzyme that degrades ACMS toward acetyl CoA lowers the efficiency of tryptophan to niacin conversion,²³ likely explaining the individual and interspecies variation that is observed. Tryptophan has various roles in metabolism, including support of protein metabolism and synthesis of serotonin and melatonin, in addition to *de novo* biosynthesis of NAD. Tryptophan is found at the lowest concentrations of any amino acid under basal conditions, and regulation of tryptophan metabolism is not thought to be closely linked to niacin status. However, the importance of tryptophan as a precursor to NAD is evident in Hartnup disease, which arises from a mutation in a tryptophan transporter. Hartnup disease presents with many symptoms that resemble pellagra, most of which are rescuable by administration of preformed niacin or niacinamide.^{24–26} This rescue illustrates the lack of requirement for tryptophan as a niacin precursor if other sources of B3 are adequate. Conversion of tryptophan to NAD also requires riboflavin (vitamin B2), pyridoxal phosphate (vitamin B6), ascorbate (vitamin C), and glutamine in order to go to completion to NAD (Fig. 12.2), suggesting that this pathway will be impaired in malnourished patients. Much of the tryptophan to NAD conversion occurs via the hepatic tryptophan dioxygenase-catalyzed pathway. However, the *de novo* pathway is highly regulated by the immune system in specific cell types as well. Interferon gamma activates the rate-limiting enzyme indoleamine-2,3-dioxygenase (IDO), which is required for interferon's biological activities. The intermediates in this NAD synthetic pathway such as kynurenine and kynurenate have significant physiological roles. IDO is also expressed in endothelial cells, which are vulnerable to excessive immune cell depletion of plasma

tryptophan.²⁷ Persistent activation of IDO is seen in many autoimmune diseases and cancer, where decreased serum tryptophan is a common diagnostic indicator of poor prognosis. Complementary administration of high doses of NAD precursors has been shown to rescue these pathogenic processes in animal models as well as in examined clinical cases.²⁸

B. Digestion

Some of the absorption of nicotinic acid occurs via passive diffusion in the stomach, while both nicotinic acid and nicotinamide are absorbed in the intestine by a sodium-dependent saturable transporter as well as by a passive diffusion.²⁹ This transport increases at higher pharmacological doses.

C. Transport

Low intakes of niacin result in little direct excretion in the urine because both vitamins are actively reabsorbed from glomerular filtrates.³⁰ Several N-methylnicotinate metabolites are primarily formed enzymatically in the liver and do appear in urine.

D. Metabolism

NAD participates in more reactions than any other vitamin-derived molecule.³¹ At least 470 proteins use NAD(P(H)), see supplementary Table in Ref. 32 NAD functions in hundreds of redox reactions as a cofactor, tens of reactions as a substrate, and single-digit numbers of proteins use NAD as a ligand. More and more proteins are being discovered by biochemists to be directly involved in NAD(P(H))-connected activities at the biochemical level. The importance of NAD concentrations in regulating these various processes is a perennial area of vibrant clinical research interest.

NAD as a cofactor in redox reactions

In general, redox enzymes use NAD(H) for catabolic reactions and NADP(H) more commonly for anabolic (biosynthetic) reactions.³¹ For example, NADPH serves as an important reducing agent for the synthesis of lipids and steroids, while NAD⁺ is required to oxidize substrates in the Krebs cycle.

Ninety percent of the NAD pool is maintained in a highly oxidized state (NAD⁺) primarily by the action of the mitochondrial electron transport chain.³³ By contrast, the NAD(P) pool is maintained predominantly in a reduced state, mainly by the action of the pentose phosphate pathway. Consequently, other enzymes linked to the NAD redox couple are strongly oxidizing,

whereas those linked to the NAD(P) redox couple are strongly reducing.

Both NAD and NADP serve essential bioenergetic functions in the intracellular respiratory chain of all cells. NAD assists in the stepwise transfer of electrons from various energy substrates to the cytochromes. NADH usually donates its electrons to a flavin coenzyme in the mitochondrial electron transport chain responsible for ATP production. Reducing equivalents are carried to the mitochondria by the malate–aspartate or the glycerol phosphate–dihydroxyacetone shuttles. Approximately 57 different cytochrome P450 monooxygenase proteins are used by humans for a wide variety of metabolic processes including synthesis and degradation of drugs, xenobiotics, prostaglandins, leukotrienes, retinoic acid, vitamin D, cholesterol, bile, and steroids.³⁴ Pyridine nucleotides and flavocoenzymes funnel reducing equivalents to the mitochondrial respiratory chain.

NAD as a substrate and NAD depletion

Four categorical enzymes play dominant roles in controlling NAD levels. These enzymes respond to DNA damage, immune activation, and other stimuli. This section covers aspects of these NAD concentration controlling enzymes: PARPs 1–18, NAD-dependent deacetylases (sirtuins 1–7), ADP cyclases (CD38 and CD157), and IDOs. The first three categorical enzymes produce nicotinamide as a side product that is recycled back to NAD via the salvage pathway as shown in Fig. 12.2.

PARP1 and PARP2 NAD-consuming activities are highly activated by any kind of DNA damage.³⁵ These enzymes use NAD to produce the anionic polymer poly(ADP)ribose. They also transfer ADP-ribose directly to histones, p53, NF kappa B, and other important proteins. The degree of PARP activation is directly proportional to the degree of DNA damage. This response serves a critical function in stopping cell division after the DNA is damaged, enabling DNA repair, and encouraging apoptosis, thus minimizing cancer cell expansion. However, hyperactivation of PARP1 leads to severe depletion of intracellular NAD and ATP. This results in uncontrolled necrotic cell death, and this mechanism has been shown to contribute to injury in many models of acute and chronic disease. In fact, PARP inhibition prevents a wide variety of otherwise lethal shocks in mammals, but it may come with an increased likelihood for tumor formation.³⁶

We cannot change our genetic makeup significantly during our lifetime, but epigenetics is very responsive to nutritional status. Increased NAD levels can activate SIRT1, which alters chromatin structure through histone deacetylation. SIRT1 activation has been attracting interest as it is thought to be the mechanism underlying the

life span extension associated with caloric restriction. While other histone deacetylases only possess constitutive activity, the sirtuins are activated by increases in NAD concentrations. This NAD-dependent activation of sirtuin HDAC activity is thus quite unique because we can alter our NAD levels to ultimately alter chromatin structure. However, NAD-dependent PARP1 activity also tends to relax chromatin structure, and the combination of these two influences of niacin status on chromatin structure has not been properly investigated.³⁷

Activated NAD-dependent sirtuins deacetylate histones, p53, NF kappa B, and many other important proteins. Sirtuins can negatively regulate PARP activity by deacetylation.³⁸ A tremendous amount of investment has been devoted to sirtuin research owing in large part to the excitement surrounding SIRT1-dependent caloric restriction-mediated life span extension.³⁹ Animal models have revealed that sirtuin activation is beneficial for treatment of diabetes⁴⁰ and neurodegeneration.^{41,42} Most models use resveratrol as a sirtuin activator, and there is a lack of research looking at the possible synergy between resveratrol treatment and niacin supplementation, to maximize deacetylation activity.

The ADP-ribosyl cyclase enzyme, CD38, uses NAD as a substrate to generate known mediators of intracellular calcium release.⁴³ This mechanism is required for chemotaxis of many types of immune cells. The CD38-deficient mouse has persistent elevated NAD levels and increased energy and does not become obese when fed a high-fat diet.⁴⁴ However, CD38-deficient mice are more susceptible to infections.⁴⁵ CD38 is highly expressed in the brain, where its potential neurological role is an active area of research.⁴⁶ It is likely that NAD-dependent calcium signaling in neurons or astrocytes plays a role in the dementia observed in pellagra patients. Other NAD-dependent ADP-ribosyl cyclase enzymes, such as CD157, likely have important and less studied functions.

IDO is highly regulated by immune responses.⁴⁷ IDO plays significant roles by decreasing available plasma tryptophan concentrations and potentially controlling NAD concentrations, similar to Hartnup disease. Tryptophan is present at the lowest concentrations of all amino acids under basal conditions, and it is used for the NAD biosynthesis *de novo* pathway. During pregnancy, IDO is highly expressed in the trophoblasts surrounding the developing fetus. This starves local immune cells, thus conferring fetal immunotolerance and preventing rejection of the foreign fetus by the immune system. During infections and in autoimmune disease, IDO is highly activated by interferons in specific professional antigen-presenting immune cells (dendritic cells, macrophages, and B cells). Activation of IDO also

exerts antibacterial and antiviral activities, particularly intracellularly. IDO is frequently persistently activated in autoimmune disease and cancer.⁴⁸ Persistent or acute excessive activation of IDO during autoimmune disease or cancer weakens neighboring cells due to exceedingly low serum tryptophan levels. Accordingly, high doses of vitamin B3 have been shown to complement many effects of this deficiency of de novo NAD precursor tryptophan, thus relieving many of the pellagra-like symptoms similar to the effects seen in Hartnup disease.²⁸

The α -amino- β -carboxymuconate- ϵ -semialdehyde decarboxylase (ACMSD) enzyme, which functions within de novo NAD biosynthetic pathway starting from tryptophan, can also alter NAD levels by effectively siphoning tryptophan away from NAD synthesis.⁴⁹ By contrast, inhibitors of ACMSD can lead to more de novo NAD synthesis.

NAD as a ligand

NAD binds to the receptors P2Y1 (colonic cells) or P2Y11 (in monocytes and granulocytes) to function as a neurotransmitter.^{50,51} NAD functions as an inhibitory neurotransmitter inhibiting colonic muscle contraction. This causes an increase in intracellular calcium. Some of these NAD effects are ultimately due to associations between the P2Y receptors and other G-protein coupled receptors.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

Molecular biologists use gene deletion to determine gene function. We can consider similar loss of function analysis for determining nutrient function. More significantly, however, we are able to readily perform pharmacological complementary rescue of loss of function symptoms by administering the vitamin molecule. Experiments involving vitamin deficiencies can only be performed on humans for brief periods⁵²; however, we can still learn from the history of pellagra and through experimentally controllable animal models of niacin deficiency.

A. Gene Regulation

As a precursor to NAD, niacin can change the concentration of NAD. Changing levels of NAD concentrations can influence the gene expression of thousands of genes. This occurs in part through epigenetic pathways involving NAD-dependent Sirtuin family of histone deacetylase genes as well as the poly-ADP-ribose family of proteins. There are many genes that are regulated. It is

complex and an area of active ongoing research investigation.⁵³

IV. DIETARY REQUIREMENTS

Humans can meet their vitamin B3 requirement through ingestion of nicotinic acid, nicotinamide, nicotinamide riboside, and/or tryptophan.^{31,119} Plant products deliver mainly nicotinic acid, while animal products release nicotinamide from its nucleotide forms during food processing and digestion. "Niacin" is a term that refers to direct vitamin precursors for NAD formation, whereas "niacin equivalent" (NE) takes into consideration the contribution of tryptophan. The conversion factor for the endogenous transformation of tryptophan to NAD is approximately 60 mg of tryptophan per 1 mg of NAD. Thus, ingestion of 60 mg of tryptophan counts as 1 NE.⁵⁴ The tryptophan content of protein is approximately 1%. Thus, a diet of approximately 100 g of protein would provide approximately 16 NE, thus meeting the recommended daily allowance for NAD without inclusion of the vitamin forms of niacin.

A. Indicators of Deficiency or Inadequacy

The three classic "Ds" of pellagra include diarrhea, dermatitis, and dementia, which occur in unpredictable orders and combinations, depending perhaps on genetics, lifestyle, or infectious conditions.^{2,55,56} These are followed ultimately by death of the patient if the diet is not improved. Pellagrous dermatitis has a characteristic appearance on parts of the body, such as the face, neck, hands, feet, and elbows, and is triggered by exposure to sunlight, heat, or mild trauma. These lesions are typically bilaterally symmetric about the head. Accordingly, they were first termed "Casal's necklace." Mental symptoms include irritability, headaches, sleeplessness, loss of memory, hallucinations, and emotional instability. Women were more suicidal, while men were more violent during the pellagra epidemics. Pellagra-like symptoms are also caused by chronic alcoholism, carcinoid syndrome,⁵⁷ and some drugs such as the anti-tuberculosis medicine isoniazid or chemotherapeutics.⁵⁸

B. Criteria of Adequacy and Assessment of Status

Traditionally, measures of the urinary metabolites N'-methylnicotinamide and 2-pyridone have been used as indirect indicators of niacin status.⁵⁹ Biochemical assessment of niacin status is ideally determined by measuring tissue NAD and NADP levels.^{57,60} A frequently used index is the niacin number, which is a ratio of NAD to NADP in whole

blood.^{52,61} As NADP levels are relatively stable during niacin deficiency, whereas NAD declines,⁶² this ratio is a dependable measure of niacin status. Restricting niacin intake to 6.1 or 10.1 NE decreased NAD by 70% within 5 weeks, while nicotinamide adenine dinucleotide phosphate (NADP) levels remained constant.⁵² These decreased NAD levels precede pellagra symptoms and are also frequently seen in carcinoid syndrome patients.⁵⁷ Analysis of over 1000 women in Sweden in the 1980s using this technique surprisingly revealed that 15%–20% of modern-day populations in developed countries may have significant niacin deficiency.⁶³

C. Reference Intakes

Recommended Dietary Allowances (RDAs) originated during World War II as minimal standards for food relief and have continued to evolve. The RDA represents the amount of a vitamin necessary to meet the needs of 97.5% of the population. Since tryptophan can also be converted to nicotinic acid mononucleotide (Fig. 12.2), but with 1/60th the efficiency of nicotinic acid, 60 mg of tryptophan is considered 1 NE.

As part of the Dietary Reference Intakes process, in 1998 the Food and Nutrition Board of the Institute of Medicine established Estimated Average Requirements (EARs), RDAs, Adequate Intakes (AIs), and Tolerable Upper Intakes (ULs) for Canadians and Americans for niacin⁶⁴. The current RDA for niacin for adult men is 16 mg NE, while for women, it is 14 mg NE. It was based on deriving an NE intake that would result in excretion of 1.0 mg N¹-methylnicotinamide/day. A weighted average of 11.6 mg NE/day was obtained from four studies in humans that correlated NE intake with excretion of N¹-methylnicotinamide at a level of 1.0 mg/day.⁶⁴ This was used to establish the EAR of 12 mg/adult men and 11 mg/day for adult women. Because of the wide variation in response noted in the studies, a CV of 15% was used to establish the RDA at 2 standard deviations above the EAR. The RDA for men was then set at 16 mg NE/day, while that for women was set at 14 mg NE/day.

For infants, AI values are based on the average breast milk composition of niacin from unsupplemented mothers of 1.8 mg/L. Because the tryptophan content of human milk was about 210 mg/L, and the expected high rate of protein turnover and net protein retention associated with infancy, the use of the ratio of 1:60 for NE to tryptophan was considered an overestimate of NE contribution. Thus, for infants 0–6 mo, the AI is 2 mg/day of preformed niacin. For older infants 6–12 mo, the AI of 4 mg/d was based on extrapolating from the adult EAR value based on metabolic body size and adding a factor for growth. The EAR and RDA for pregnancy and lactation are derived from the

TABLE 12.1 Recommended dietary allowances (RDAs) and adequate intakes (AIs) for Canada and the United States for niacin.⁶⁴

Population group	NE ^a /day
Infants	
0–0.5 y ^b	2
0.5–1.0 y ^b	4
Children	
1–3 y	6
4–8 y	8
9–13 y	12
Males	
≥14 y	16
Females	
≥14 y	14
Pregnant	18
Lactating	17

Niacin equivalents (NE) refers to the amounts of niacin & tryptophan, while adequate intake (AI) refers to the amount required to avoid overt deficiency. So NE is a description of the units and AI is a description of the criteria for activity - no overt deficiency.

^aNiacin equivalents except where noted for infants.

^bNiacin only.

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EAR for nonpregnant or lactating women and then increased to account for increased niacin to support the increased energy expenditure during these periods (Table 12.1).^{64,65}

D. Basis for Individual Variability in Requirements

There is a wide range of interindividual variability in the dependence on niacin supplementation depending on the NAD-binding domain genotype of individuals.⁷ Alterations in the NAD-binding domain for any of the more than 400 NAD-utilizing genes can result in decreased binding affinity and therefore a greater dependence on niacin for achieving healthy and normal gene function. This is an area of ongoing research investigation requiring genetic and biochemical resources. Examples where the mutations in the NAD have been identified include the ALDH2, G6PD, and UUR gene.⁷

The mitochondrial aldehyde dehydrogenase (ALDH2) catalyzes the NAD-dependent oxidation of toxic acetaldehyde to generate acetic acid. Glu487 → Lys substitution

in the ALDH2 is the single genetic factor most strongly correlated with reduced alcohol consumption and greater incidence of alcoholism. This allele is found in 50% of the Asian population and is associated with a phenotypic loss of ALDH2 activity in both heterozygotes and homozygotes. ALDH2-deficient individuals exhibit an adverse response to ethanol consumption with increased risk for Alzheimer's disease (AD), and cancer. The Glu487→Lys variant causes a 150-fold increase in K_m for its NAD cofactor.⁶⁶ Accordingly, data suggest that higher NAD levels achieved through higher niacin supplementation may decrease the effects of this polymorphism.

There are likely to be more genetic polymorphisms in the NAD-binding domain of genes discovered with time and analysis. These polymorphisms have the potential to cause a greater individual variability in the requirements for niacin.

E. Tolerable Upper Intake Levels

As part of the Dietary Reference Intakes process, a UL for niacin was established based on the adverse effect of skin flushing noted following a dose of nicotinic acid. For adults, this was conservatively set at 35 mg/day of preformed niacin obtained from synthetic sources, including supplements and fortification. It does not apply to NE derived from food. While other adverse effects were evaluated resulting from high doses of niacin and nicotinic acid, they were at levels above that noted with vasodilation and thus the UL based on skin flushing will protect against these more serious adverse effects.⁶⁴

The UL is not meant to apply to individuals taking high-dose niacin for the treatment of dyslipidemia under medical supervision.⁶⁴ While the flush can be uncomfortable, it is linked to the beneficial effect of correcting dyslipidemia. By contrast, nicotinamide does not cause the flush response and it also has not exerted the therapeutically beneficial effects on lipid profiles.

F. Dietary Sources

Nicotinic acid and pyridine molecules are first made in plants and thereafter distributed in the food chain (Table 12.2).⁶⁷ For example, niacin can be synthesized in grasses, which are ingested by cows. Then rumen bacteria can synthesize more niacin. Ultimately, the vitamin can be consumed in plant products or transmitted through milk or meat consumption to people. Among plant foods, nuts, legumes, and some whole grains are good sources of nicotinic acid. Coffee can also be a major source of niacin in the diet, with the darker roasts having more niacin content.⁶⁸ Corn (without alkaline treatment)

TABLE 12.2 Food sources of preformed niacin.^a

Food source	Amount of niacin
Dairy	0.2 mg per 1 cup whole milk 0.05 mg per 1 egg
Meats (per 3oz, 85 g)	14 mg beef liver 10–11 mg tuna, halibut, swordfish 7 mg rainbow trout 2–6 mg beef, lamb, pork, poultry, other fish
Cereals and grain products	2–10 mg per 1 cup ready-to-eat cereal 3 mg per cup barley or rice, cooked 3–4 mg per 4-in (3-oz/85-g) bagel 2 mg per 2-oz/56-g hard roll 2 mg per cup noodles or pasta
Vegetables (per cup)	4 mg canned tomato product 3 mg mushrooms 2 mg corn 2 mg potatoes 0.5 mg of mustard greens
Other	4 mg per oz (28 g) peanuts

^aThe contribution of tryptophan is not included.

Data from USDA/ARS (2005) USDA National Nutrient Standard Database, Release 18. USDA/ARS, Washington, DC.

and refined grains are poor sources, and dependence on these is a risk factor for niacin deficiency. Fortified flour and cereal products are a good source of nicotinic acid. Meat products are also a good source of nicotinamide, obtained through the digestion of pyridine nucleotides. Lastly, high-protein foods provide tryptophan, which contributes to meeting vitamin B3 requirements.⁶⁷ Data from the National Health and Nutrition Examination Survey indicate that most people in the United States obtain niacin levels similar to the EARs.⁶⁹

V. ISSUES RELATED TO NIACIN IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

The questions as to which is the most effective NAD precursor in pharmacological use and which diseases are most responsive to these treatments are among the most exciting in molecular therapeutic clinical research today. We therefore briefly review here what happens

to concentrations of NAD, tryptophan, poly(ADP-ribose) (PAR), and other relevant molecules over the course of pathogenesis for several diseases.

Pharmacological niacin

Gram-level quantities of standard/immediate release or prescription sustained release niacin have been used with large populations of hyperlipidemic patients with only rare adverse events. Sustained release niacin, however, has resulted in chemical hepatitis.^{70,71} Other rare side effects include cystoid macular edema. Inositol hexanicotinate has also been prescribed as a form of niacin ("flush free"); however, there is evidence that the nicotinic acid is poorly released from the inositol and that correction of dyslipidemia is less effective.

Cardiovascular disease

More people ultimately die due to cardiovascular events than as a result of any other disease. High-dose niacin is well known as one of the most effective agents preventing atherosclerosis-related strokes and heart attacks.⁷⁰ Niacin increases HDL (the good cholesterol) levels more than any other pharmaceutical, while simultaneously decreasing cholesterol, VLDL, and triglycerides in otherwise dyslipidemic individuals. A 15-year follow-up to a cohort receiving 3 g niacin per day showed a 15% reduction in mortality.⁷² Similar reductions in mortality were seen in patients with previous myocardial infarctions.⁷³ The book "The Eight-Week Cure for Cholesterol" written by a medical journalist described his struggle to treat severe dyslipidemia using niacin.⁷⁴

The mechanism of action of niacin in correcting dyslipidemia is a perennial cardiovascular research question. The high-affinity nicotinic acid G-protein-coupled receptors GPR109a and GPR109b serve primary roles in the niacin dyslipidemia-correcting mechanism of action.⁷⁵ These are highly expressed in adipose tissue, but also in neutrophils, professional antigen-presenting cells (dendritic cells, macrophages, and B cells), intestine, retina, and other areas within the central nervous system.^{76–78} In adipocytes, nicotinic acid signaling through GPR109 causes a decrease in cyclic AMP levels, which decreases hormone-sensitive lipase.⁷⁹ In other cells, a wide variety of prostaglandins are produced at high levels, mediating potentially beneficial effects including increased peripheral circulation. The endogenously produced prostaglandin PGJ2 most potently activates PPAR gamma and other PPARs similar to the diabetes and dyslipidemia drugs, thiazolidinediones and clofibrate, respectively. NAD working through sirtuin-PPAR pathways is also potentially involved in positive lipid-correcting activities. Still other nicotinic acid associated pathways, including regulation of membrane-bound ATP synthases and diacylglycerol

acyltransferase, are potentially involved in the myriad of high-dose niacin/nicotinic acid benefits⁸⁰.

Cancer

While the death rate for cardiovascular disease has dropped by over 60% over the past 50 years, that for cancer has been disappointingly unchanged by comparison, having decreased by merely 5% adjusted for age and population.⁸¹ Cancer is initiated by unrepaired DNA damage. Under healthy conditions, cells with excessive DNA damage undergo apoptosis. NAD plays significant roles in controlling both genomic stability and the cell death process. Moreover, later in the progression of cancer disease, NAD can become a dominant theme in metabolic chemotherapy.

The rate of NAD turnover in cancerous transformed cells is much higher than that of normal cells. This is due to an increased demand for NAD in glycolysis and also the increased PARP1/2 activity associated with genomic instability. Cancer cells have a far greater essential requirement for glycolysis than for oxidative phosphorylation as a means of generating ATP. Tumor cells typically perform glycolysis at 30 times greater rate than that of normal cells.⁸² Moreover, the rate of entry of glucose into cancer cells is also 20 to 30 times greater than into untransformed cells in order to supply essential levels of glucose. Oxygen-independent glycolytic ATP production requires regeneration of NAD by lactate dehydrogenase. Interestingly, clinical pellagra may be observed during chemotherapy.⁵⁸

Chemotherapeutics have been developed to decrease NAD via inhibition of NAMPT and/or IDO.⁸³ Other interesting areas of niacin cancer research include the observation that GPR109a can function as a tumor suppressor in the colon.⁸⁴ Whether administration of niacin can help in colon cancer is unknown. However, niacin is known to provide therapeutic benefit in the related ectodermal skin cancer models.^{60,85,86}

Neurodegenerative disease

As excitable energy-demanding tissues, neural cells (neurons and glia) have distinct metabolic NAD-dependent requirements.⁸⁷ Similar to tumor cells, neurons are very dependent on glycolysis as a source of ATP generation. Oxidizing glucose in glycolysis and the pentose phosphate pathway, with low mitochondrial activity, decreases oxidant stress in these sensitive tissues. Glial cells function as support cells to neurons, synthesizing cholesterol and lactate, which are both delivered to neurons. Increases in baseline NAD or NAD salvage pathway efficiency are well known to prevent otherwise lethal stress-induced neurodegeneration in cells and animal models.^{41,88–91} Thus, we must understand what happens to NAD levels in the diseases of neurodegeneration: dementia/AD, Parkinson's disease

(PD), Huntington's disease (HD), amyotrophic lateral sclerosis, multiple sclerosis (MS), and even schizophrenia. AD is the most common form of dementia recognized today. This is followed by vascular dementia. The NAD deficiency disease pellagra is characterized by dementia, where niacin is one of the rare molecules known to prevent a form of dementia while also providing many therapeutic vascular benefits.

Diseases defined by the formation of insoluble protein plaques include AD (amyloid protein; tauopathies), HD (huntingtin protein), and PD (α -synuclein; Lewy bodies). While surgery and stem cells may hold promise for the future, currently these latter diseases are irreversible after a certain point in disease progression. Primary emphasis is placed on prevention or stay of progression. An increase in dietary niacin from 15 to 40 mg/day has been correlated with a 70% decrease in the likelihood of AD, in a study examining 6158 patients over 65 years of age.⁹² NAD, potentially through activation of SIRT1, can reduce amyloid precursor protein processing to plaques in transgenic mouse models of AD.⁴² Studies show that high apolipoprotein A-1 (apoA-1) decreases AD risk,⁹³ and high-dose niacin is effective in raising apoA-1 containing HDL, suggesting that this might be effective in delaying the progression of brain injury. Meanwhile, high ApoE is associated with increased risk of AD^{94,95} and is found in the chylomicrons, VLDL, and LDL particles. ApoE mutations can cause familial AD and dyslipidemia. ApoE-deficient mice are used as a model of dyslipidemia, where administration of niacin is able to correct the elevations in cholesterol and triglycerides but not the decrease in HDL unless cholesteryl ester transfer protein is overexpressed.⁵ More research is needed to understand how niacin nutrition and pharmacological use relate to the frequency of AD, ApoE, and vascular dementia as well as how best to incorporate niacin into treatments.

NAD levels in the hippocampus decrease with age.⁹⁶ The hippocampus functions in long-term memory. The calcium calmodulin-dependent serine protein kinase Cask, which functions in memory, is highly regulated by NAD levels and is decreased with aging. Nicotinamide mononucleotide can ameliorate much of the cognitive age-related cognitive decline in mice.

PD occurs due to a deficiency of dopamine-producing neurons in the substantia nigra part of the midbrain. The mechanisms of PD are poorly understood; however, excessive methylnicotinamide (MN) has been proposed to be part of the cause of PD pathogenesis.⁹⁷ Urinary MN is elevated in PD. Since this is the metabolic by-product of niacin metabolism, this suggests that NAD metabolism may be involved. Administration of NADH has been tested on PD patients and found to

stimulate increased dopamine production.⁹⁸ Future work will be required to understand the role in brain metabolism of various NAD precursors and metabolites.

MS is the most commonly diagnosed disease of the central nervous system. An examination of the levels of serum antibody levels in MS patients indicates that they have a 50% reduction in the levels of NAD as compared to controls.⁹⁹ Neurons are exquisitely sensitive to degeneration when NAD levels are low. Animal models of MS have revealed that NAD precursors can prevent the onset and damage of MS pathogenesis.^{14,41} PAR formation increases at sites of pathogenesis, where preservation of NAD levels is key to maintaining health.⁸⁷

Other ongoing areas of neurobiology research involve the slow Wallerian degeneration (Wlds) mouse, which is remarkably resilient to stress-induced neuronal cell death. Razor excision of wild-type neurons leads to neurodegeneration in less than 24 h. Remarkably, Wlds mouse neurons survive for up to 2 weeks and are still capable of depolarization.¹⁰⁰ The gene mediating this effect includes the NAD biosynthetic enzyme, nicotinamide adenine mononucleotide adenylyl transferase-1 (Ube4b/NMNAT1).¹⁰¹ Follow-up research has determined that increased expression of NMNAT1, NMNAT2, or NMNAT3 is extremely neuroprotective and that NAD itself can exert a similar survival effect.^{12,15,41,102} NAD-mediated neuroprotection may be mediated by increased sirtuin activity,⁴¹ but NAD-chaperone survival activities have also been observed for NMNAT.¹⁰³

Diabetes

NAD levels decrease with aging and with many pathogenic processes including diabetes. In diabetes, NAMPT-mediated NAD biosynthesis is compromised. Adipose tissue regulates NAD⁺ biosynthesis at a systemic level via secretion of NAMPT.¹⁰⁴ NAMPT is present in the cell (iNAMPT) and outside the cell (extracellular; eNAMPT). SIRT1 can deacetylate iNAMPT to cause NAMPT to leave adipose tissue and circulate (eNAMPT), where it can lead to increased NAD biosynthesis in the hypothalamus, to affect feeding, physical, and neural activity. Elimination of adipocyte NAMPT in mice leads to insulin resistance, while supplementation with niacin or NMN normalizes this diabetes-related metabolic dysfunction.¹⁰⁵ Supplementation with nicotinamide mononucleotide, the product of NAMPT catalyzed reaction, can significantly correct glucose intolerance and lipid profiles in age-induced type 2 diabetic mice.¹⁰⁶ The efficacy of NMN in treating diabetes in patients remains to be determined.

The availability of NAD systemically decreases with age and in certain diseases. Increasing NAD synthesis by supplementation with nicotinamide mononucleotide can effectively mitigate age-associated physiological decline in mice including improving insulin sensitivity, eye function, body weight, immune function, and others, with no apparent toxicity.¹⁰⁷ Diabetic retinopathy is the leading cause of vision impairment and blindness among working-age adults. NAD depletion in photoreceptor cells may be a major part of the most common forms of blindness, where NMN supplementation may be useful in preventing blindness.¹⁰⁸

Nicotinic acid is distinguished as the only NAD precursor that also independently activates the GPR109a high-affinity nicotinic acid receptor by virtue of the monocarboxylate functional group on the nicotinate molecule.⁷⁵ This occurs at levels of nicotinic acid that are greater than that found in normal foods. In nature, GPR109a is activated by the ketone beta-hydroxybutyrate under starvation conditions, where it functions to inhibit lipolysis and conserve precious stored fats by inhibiting lipolysis.¹⁰⁹ The GPR109a receptor is downregulated in type II diabetes, and studies in mice suggest this interaction may prevent high-fat diet induced obesity in a GPR109a-dependent fashion, thus supporting that pharmacological use of niacin may be useful in preventing obesity-related diabetes.^{110,111} Elimination of GPR109a in mice leads to progressive weight gain with hepatic fat accumulation.¹¹¹ By contrast, niacin treatment protects against hepatic de novo lipogenesis and increased brown fat thermogenic activity in a GPR109a-dependent fashion. Collectively, niacin-GPR109a signaling may be a useful target in preventing obesity and diabetes.

Autoimmune disease

Some autoimmune diseases, listed from more to less common, include hypothyroiditis, rheumatoid arthritis, hyperthyroiditis, type 1 diabetes, systemic lupus erythematosus, myocarditis, MS, and inflammatory bowel disease. Autoimmune diseases commonly arise from an undesirable clonal expansion of autoreactive T cells. One of the best-understood mechanisms for controlling this expansion involves starving the T cells of tryptophan by interaction with tryptophan-hoarding professional antigen-presenting cells (dendritic cells, macrophages, B cells, and microglia).⁴⁷ IDO is frequently activated in autoimmune diseases as the immune system works to control this undesirable expansion. This ultimately contributes to depletion of extracellular

tryptophan where low tryptophan levels are characteristic and correlate highly with a poor prognosis.¹¹² This activated immune system may also activate PARP1, which additively contributes to NAD depletion.¹¹³ Ultimately, the complementary administration of NAD precursors to this situation makes sense and has been shown to prevent onset of pathogenesis of disease as well as in some clinical examples.²⁸

B. Known Interactions

NAD is involved in phase 1 drug metabolism.¹¹⁴ Accordingly, high-dose vitamin B3 supplementation has the potential to increase the metabolism and clearance of pharmaceuticals and xenobiotics. For example, when a hepatocyte is exposed to DDT, then cyp1A1 can be induced to such a strong extent that it can constitute up to 10% of the total protein of the cell. At this point, the enzyme and substrate are at elevated levels and so the level of NAD⁺ cofactor can become limiting in the metabolism toward clearance of the xenobiotic.²⁸

Claims have been made that niacin may exacerbate statin-associated adverse events. However, no major clinical trial has suggested a potential drug interaction between statins and niacin, and there is no proposed theoretical mechanistic reason to expect such a drug interaction.⁷¹

C. As a Dietary Supplement

Nicotinic acid supplementation has been used at pharmacological doses for over 50 years to treat dyslipidemia and reduce the risk for atherosclerosis.⁷⁰ Typically, 3–5 g of nicotinic acid daily has been used without serious adverse events.⁷¹ Nicotinic acid has a long-standing safety record. Nicotinic acid boosts NAD concentrations and corrects dyslipidemia in many individuals. Triglycerides and VLDL are lowered if they are too high, and HDL is elevated more than any known pharmaceutical.^{115,116} This is not achievable using nicotinamide or nicotinamide riboside.

Experimentation using rats demonstrated that nicotinamide riboside has a similar safety profile to that of nicotinamide.¹¹⁷ No adverse reactions were observed at 300 mg/kg/d or less. This corresponds to over 23 g for an individual that is 170 pounds. The safety and metabolism of nicotinamide riboside were recently examined in randomized, placebo controlled, double-blind trials involving healthy overweight adults.¹¹⁸ A safety profile similar to nicotinamide was observed.

RESEARCH GAPS

There is much that remains to be determined in niacin research. The impact of low and high niacin status on basic energy metabolism is poorly appreciated and is a logical area in which to apply cutting-edge techniques in metabolomics. The use of NAD as a substrate in the formation of poly, mono, and cyclic ADP-ribose, as well as in support of sirtuin activities, suggests that there are complex interactions between intermediary metabolism and control through ADP-ribosylation reactions. It is clear that ADP-ribosyltransferases, sirtuins, and metabolic families of enzymes all play important roles in cancer and in autoimmune and neurodegenerative diseases, with much still to be learned. With the benefits of inhibiting ADP-ribose metabolism under certain conditions (ischemia-reperfusion, treatment of certain cancers), and the risks of doing so at other times, clinical management of ADP-ribose metabolism is in its infancy. Given the extremely large number of niacin-dependent proteins in metabolism, cataloging polymorphisms and individual variation in niacin requirements is another long-term goal.

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VITAMIN B6

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SUMMARY

Vitamin B6 exists as a family of chemical species that ultimately can be converted metabolically to the coenzyme pyridoxal 5'-phosphate (PLP). PLP serves as a coenzyme in almost all aspects of human metabolism, including amino acid metabolism, one-carbon metabolism, glycogenolysis, and gluconeogenesis. Recent evidence indicates that inflammation suppresses PLP in the plasma and liver. Inflammation also increases vitamin B6 catabolism. However, the mechanistic connection between vitamin B6 insufficiency and chronic disease remains unclear.

Keywords: Amino acid metabolism; One-carbon metabolism; Pyridoxal; Pyridoxal 5'-phosphate; Pyridoxamine; Pyridoxine; Vitamin B6.

I. INTRODUCTION

A. Background

Evidence of the existence of an unknown water-soluble factor later termed vitamin B6 first emerged in the late 1920s, and its purification, identification, and synthesis followed quickly thereafter in the 1930s (for historical review, see Ref. 1). Identification of pyridoxal (PL) and pyridoxamine (PM) as natural forms of the vitamin, and the recognition of a phosphorylated form of PL, eventually identified as pyridoxal 5'-phosphate (PLP), followed in the 1940s (for review, see Ref. 2). Much of this work evolved from bioassays involving the growth of rats and the prevention of a characteristic skin condition termed acrydynia^{1,3} and the use of vitamin B6-dependent bacteria.² We are indebted to these early researchers for their biochemical ingenuity, insight, and experimental skills in an era when modern analytical tools were lacking.

B. Nutrient Function and Structure

Vitamin B6 is the term for the family of water-soluble vitamin species with the basic structure of 2-methylpyridine,

3-hydroxypyridine, 5-hydroxymethyl pyridine (Fig. 13.1). Through its coenzymatic function in processes including amino acid metabolism, one-carbon metabolism and nucleotide synthesis, neurotransmitter metabolism, heme synthesis, gluconeogenesis, and glycogenolysis, vitamin B6 affects nearly all aspects of metabolic function and cellular homeostasis. The various forms differ with respect to the nature of the C-4 substituent, which defines pyridoxal (PL, aldehyde, -CHO), pyridoxine (PN, alcohol, -CH₂OH), and pyridoxamine (PM, amine, -CH₂NH₂). Further variation among the constituents of this family exists through the ability of PN, PM, and PL to exist as a phosphate ester (designated PLP, PNP, and PMP for PLP, pyridoxine 5'-phosphate, and pyridoxamine 5'-phosphate, respectively) or with PN as a β -D-glucoside derivative (PNG). 4-Pyridoxic acid (4-PA) is the primary catabolic product. PN, as the HCl salt, is the synthetic form used in food fortification and dietary supplements mainly for reasons of its superior stability.

The ability of PLP (and PL to a lesser extent) to condense with the uncharged amino group of amines and amino acids to form Schiff base (aldimine) and related complexes is well known and, indeed, serves

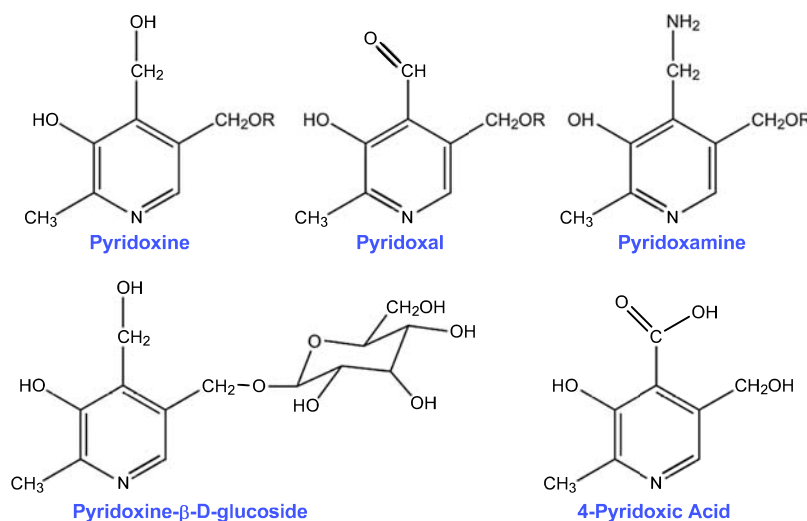


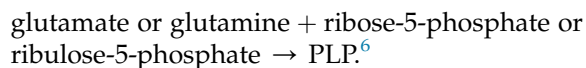
FIGURE 13.1 Chemical forms of vitamin B6. R = -H for nonphosphorylated pyridoxine, pyridoxal, and pyridoxamine and R = -PO₃H for their 5'-phosphates. Copyright held by ILSI.

as the basis for the coenzyme function of vitamin B6. This reactivity with amines and aldehydes also accounts for the observed nonenzymatic interconversions between PLP and/or PL with PMP and/or PM during food processing.⁴ In addition, PLP is very sensitive to photochemical oxidation forming 4-pyridoxic acid phosphate, while other vitamin B6 species also exhibit some photochemical instability. Consequently, care must be taken to minimize light exposure in vitamin B6 analysis and other research procedures. The greater reactivity of PLP toward amines is due the presence of the phosphate group that prevents the formation of an internal hemiacetal bridge between the C-5 hydroxymethyl group and the C-4 aldehyde, as occurs readily in PL. PL and PLP also can undergo reductive binding to the ε-amino groups of protein lysyl side chains to form ε-pyridoxyl-lysine during the thermal processing of food. This complexed form of the vitamin is enigmatic as it is not detected in common vitamin B6 assays, and it exhibits partial vitamin B6 activity and also competes with other vitamin B6 species for cellular uptake.⁵

II. NUTRIENT METABOLISM

A. Synthesis

Mammalian species lack the ability to conduct de novo synthesis of vitamin B6 and, thus, must obtain it from the diet. Plants, which can synthesize vitamin B6, constitute a major source of vitamin B6 in most mammalian diet. In plants, vitamin B6 is synthesized directly as PLP by the PLP synthase-catalyzed reaction:



PN HCl is the major form of vitamin B6 used in vitamin supplements and the fortification of certain foods. Approaches to the commercial synthesis of PN HCl have been reviewed.^{7,8}

Vitamin B6 disposition and metabolic function

The in vivo disposition of vitamin B6 is summarized in Fig. 13.2.

B. Digestion

Intestinal absorption of vitamin B6 from dietary sources occurs primarily in the small intestine and also from bacterial production in the large intestine. Uptake of dietary vitamin B6 occurs predominantly in the jejunum, with rate of absorption varying according to the B6 species present.^{9–11} Glycosylated forms of vitamin B6 found in plant foods are hydrolyzed partially by brush border lactase-phlorizin hydrolase¹² prior to absorption. Some absorption of intact PNG also occurs, after which it may be partially hydrolyzed by β-glucosidase apparently in the kidney or excreted in the urine.^{13–15} PMP and PLP are major vitamin B6 species in animal products and also exist in lesser proportions in plant-derived foods. All such phosphorylated vitamers require dephosphorylation by alkaline phosphatase and other phosphatases prior to absorption. Kinetics reported in many studies are consistent with intestinal absorption of PL, PM, and PN by simple, nonsaturable diffusion. However, evidence exists for carrier-mediated intestinal absorption of the vitamin, with rate of transport both pH dependent and saturable.^{16,17} A specific mucosal carrier that is not sodium dependent and has an acidic pH optimum also has been detected, but its molecular identity is not determined.¹⁸ Vitamin

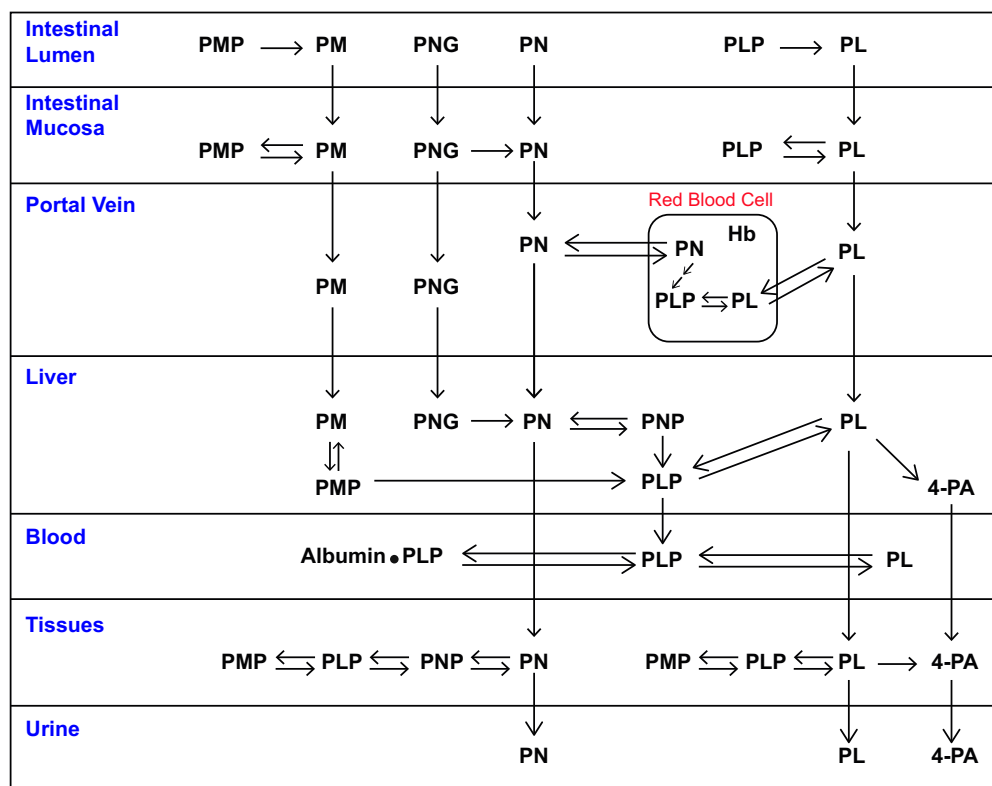


FIGURE 13.2 In vivo disposition of the forms of vitamin B6. 4-PA, 4-pyridoxic acid; Hb, hemoglobin. PL, pyridoxal; PLP, PMP, and PNP, their respective phosphorylated forms; PM, pyridoxamine; PN, pyridoxine; PNG, pyridoxine β-D-glucoside. Copyright held by ILSI.

B6 synthesized by the microflora in the human colon may be a secondary exogenous source available for absorption. Studies with mouse and human colonic cells have provided evidence of colonic absorption of vitamin B6.¹⁷ While the contribution of bacterially produced vitamin B6 to overall nutritional status warrants further investigation, dietary intake certainly constitutes the major source of vitamin B6 nutrition.

Upon initial absorption, phosphorylation catalyzed by pyridoxal kinase in the enterocytes provides metabolic trapping of the newly absorbed PL, PM, and PN.¹¹ The negative charge provided by the phosphate moiety hinders diffusion of the vitamin across the cell membrane and promotes protein binding. PNP, PMP, and PLP are dephosphorylated before crossing the basolateral membrane and entering portal circulation.¹¹ Uptake by the liver is also facilitated by metabolic trapping.

The liver is at the core of vitamin B6 metabolism. While most tissues can phosphorylate PL, PN, and PM, the liver is the major site where the various species of vitamin B6 undergo interconversion to form PLP. PLP serves as the primary coenzymatic form of vitamin B6 and is the major form of the vitamin in the body.^{19–21} Upon absorption to the liver, the nonphosphorylated forms of vitamin B6 are first phosphorylated by pyridoxal kinase, which uses zinc and ATP as cofactor and phosphoryl donor.²¹ Hepatic activity of pyridoxal kinase

is 10-fold that of the phosphatase, therefore phosphorylated species of vitamin B6 predominate.²¹ Binding of PLP to high-abundance PLP-dependent enzymes such as serine hydroxymethyltransferase, aminotransferases, and glycogen phosphorylase also renders PLP less accessible to phosphatases for dephosphorylation. The formation of PLP by oxidation of PNP and PMP occurs by the action of the flavin mononucleotide-dependent PNP–PMP oxidase.^{22,23}

C. Transport

Vitamin B6 is carried to organs and tissues primarily in plasma. In plasma, the main form is PLP and to a lesser extent PL, which are bound to albumin.²⁴ In erythrocytes, PLP and PL are bound to hemoglobin.^{25,26} Tissue uptake of circulating plasma PLP requires prior removal of the 5'-phosphate group, whereas plasma PL is immediately available. The importance of alkaline phosphatase in vitamin B6 metabolism is evident in the transport as well as in the initial absorption of the vitamin. Prior to uptake by tissue or transfer between cell compartments, membrane phosphatase converts PLP to PL.²⁷ Metabolic trapping by phosphorylation, as seen in the enterocyte, is repeated in tissues in order to retain the vitamin.

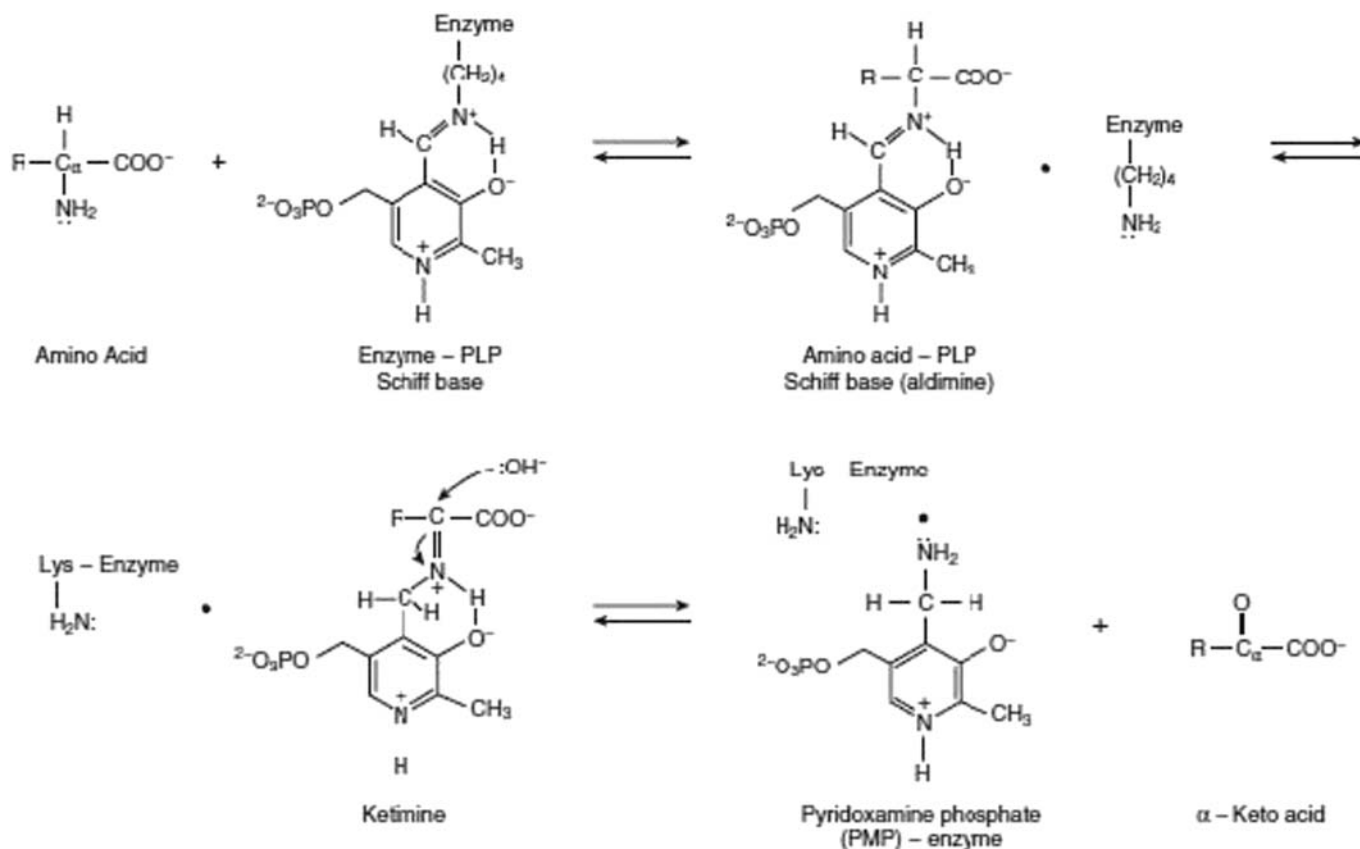


FIGURE 13.3 Mechanism of bond labilization via Schiff-base formation in PLP-dependent transamination reactions. Copyright held by ILSI.

D. Vitamin B6 Metabolism and Role as a Coenzyme

The largest body pool of vitamin B6 is in skeletal muscle, which constitutes 70%–80% of the whole-body vitamin B6 mainly as PLP associated with glycogen phosphorylase.^{28,29} An analysis of the vitamin B6 content of human muscle via biopsy estimated the total body pool in adult humans to be $\sim 1000 \mu\text{mol}$ ($\sim 170 \text{ mg}$).¹⁹ Muscle concentration of vitamin B6 has been found to be quite resilient to effects of dietary restriction and even supplementation, whereas plasma vitamin B6 showed marked changes under the same conditions.³⁰ Studies in rats have shown that muscle PLP concentration and glycogen phosphorylase activity decreased in response to a caloric deficit but not to dietary depletion of vitamin B6,³¹ but functional consequences of these findings are unclear.

The PLP molecule readily condenses with free amino groups of proteins and small molecules to form a Schiff base. As the coenzyme for aminotransferases and most other PLP-dependent enzymes, the pyridine ring and the C-4 carbonyl group enable PLP to facilitate many reactions involving amino acid substrates. In the absence of substrate, PLP is bound to an ϵ -amino group of a

specific lysine residue. In the case of aminotransferases, the binding of the incoming substrate amino acid to the enzyme–PLP complex allows migration of the PLP to yield a new Schiff base with the amino acid. Subsequent shifting of charge in this complex allows bond labilization at the α carbon, with hydrolysis of the complex to release the keto acid product and concurrent generation of PMP. Schiff base formation of an alternate keto acid leads to an analogous process of charge transfer, bond labilization, and transfer of the NH_2 group from PMP (i.e., originally from the first substrate amino acid) to the keto acid to generate a new amino acid. See Fig. 13.3 for a diagram of the mechanism by which the Schiff base formation labilizes the α -carbon bond on an amino acid.

While this reactivity of PLP with lysyl residues of PLP-dependent enzymes and the α -amino groups of their substrates accounts for the primary coenzymatic mechanism of PLP, its ability to form such Schiff bases indiscriminately when present at higher concentrations probably explains why an overaccumulation of cellular PLP would have adverse effects. The body minimizes the risk of detrimental levels of PLP when vitamin B6 intake is high by the combined action of phosphatases and oxidases. The strong inhibition of PN (PM)

phosphate oxidase by its product, PLP, retards accumulation of the latter.³² Dephosphorylation of the phosphorylated forms also contributes to cellular homeostasis of PLP by promoting vitamin B6 turnover, through enhanced conversion of PL to a biologically inactive form, 4-PA.³³ This irreversible conversion is known to occur in the liver, and it has not been well examined in other tissues. Perfusion studies with isolated rat kidney have suggested that the kidney may also be a site of 4-PA production.³⁴ 4-PA, the major vitamin B6 catabolite, is measurable in plasma and in the urine.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Gene Regulation

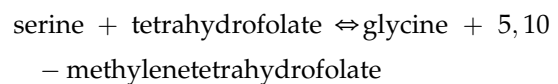
Experimental studies have shown that intracellular PLP concentration affects the production of certain proteins. Whether PLP or other form of vitamin B6 influences gene expression directly or by an indirect mechanism often is unclear in such studies. For example, the level of PLP could influence gene expression indirectly by processes such as contributing to epigenetic regulation of DNA methylation or by direct modification of transcription factors or nuclear receptors. Higher intracellular PLP concentration has been shown to depress the transcriptional activation of androgen and glucocorticoid receptors, which may serve as a regulatory mechanism for steroid-regulated genes.^{35,36} However, the regulatory mechanism is unclear. PLP also appears to act as a chaperone for alanine:glyoxylate aminotransferase by enhancing proper folding of the newly synthesized enzyme.³⁷ Recently, transcriptome analysis indicated that 525 genes were differentially expressed in pancreatic acinar cells cultured in vitamin B6-supplemented (1 mM PN in culture medium) versus vitamin B6-deficient conditions.³⁸ This finding suggests that certain levels of vitamin B6 adequacy versus deficiency can affect many aspects of gene expression. Whether less extreme variation in levels of vitamin B6 status influences gene expression is unclear.

B. Cellular and Physiological Function

PLP-dependent processes are involved either directly or indirectly in almost all phases of metabolism. Many processes in amino acid interconversion and catabolism involve PLP-dependent enzymes. Aside from playing important roles in amino acid metabolism, the formation of α -keto acids plays a role as metabolic intermediates. An indirect role of PLP in the metabolism of n-3 and

n-6 fatty acids can be inferred from reported changes in fatty acid profiles during vitamin B6 deficiency^{39,40}; however, the biochemical mechanism is unknown.

Another important role of PLP-dependent enzymes is in the area of one-carbon metabolism. One-carbon metabolism constitutes a collection of reactions and cycles that generate methyl groups (as 5-methyltetrahydrofolate and S-adenosylmethionine), facilitate methionine recycling and regulate homocysteine concentration, and serve in the synthesis of purine nucleotides and thymidylate, which are essential for DNA synthesis and repair. Fig. 13.4 illustrates the PLP-dependent reactions in the one-carbon pathway. Serine hydroxymethyltransferase (SHMT) isoforms exist in cytoplasm, mitochondria, and nuclei. SHMT catalyzes the reversible reaction:



These reactions facilitate cellular one-carbon metabolism by allowing the interconversion of serine and glycine as well as providing a means of supplying the folate one-carbon pool by generating 5,10-methylene tetrahydrofolate. The mitochondrial glycine cleavage system is a 4-enzyme complex containing a PLP-dependent glycine decarboxylase. This glycine cleavage process has been shown to yield at least 20 times as much

One Carbon Metabolism

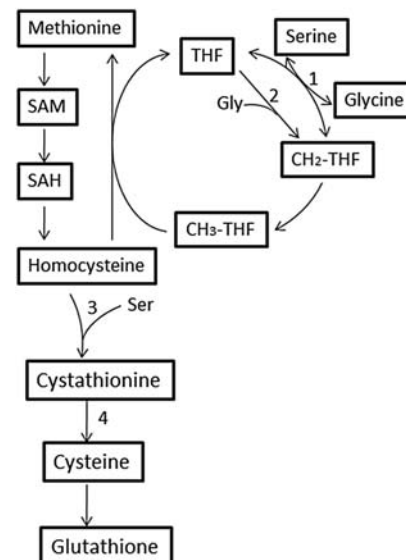


FIGURE 13.4 PLP-dependent reactions in one-carbon metabolism and transsulfuration reactions. 1. Serine hydroxymethyltransferase; 2. Glycine cleavage system; 3. Cystathionine β -synthase; 4. Cystathionine γ -lyase. SAH, S-adenosylhomocysteine; SAM, S-adenosylmethionine; THF, tetrahydrofolate. PLP-dependent metabolism of serine, glycine, and cysteine are not shown. Copyright held by ILSI.

5,10-methylenetetrahydrofolate as needed for methylation demands and to maintain glycine cleavage flux that far exceeds the rate of SHMT-catalyzed processes.⁴¹ Stable isotopic tracer studies conducted before and after vitamin B6 restriction in humans showed the resiliency of these critical processes in maintaining one-carbon metabolism.^{41–43}

PLP also serves as the coenzyme for both components of the transsulfuration pathway, which is responsible for the catabolism of homocysteine and the synthesis of cysteine. In the first step, cystathionine β -synthase (CBS) exhibits allosteric stimulation by SAM and exhibits little change in activity as a result of moderate vitamin B6 deficiency. The second enzyme, cystathionine γ -lyase (CGL), catalyzes the cleavage of cystathionine to form cysteine and α -ketobutyrate plus ammonia. CGL exhibits much greater susceptibility to loss of its PLP coenzyme during vitamin B6 deficiency, which leads to cystathionine elevation in plasma and tissues in vitamin B6 insufficiency. Unexpectedly, cysteine production and cysteine concentration in tissue and plasma are maintained over a wide range of suboptimal levels of vitamin B6 nutriture^{42,44,45} apparently because the greatly elevated cystathionine concentration tends to maintain CGL flux to maintain cysteine production.⁴²

Side reactions of CBS and CGL are important sources for the physiological production of hydrogen sulfide.⁴⁶ Vitamin B6 deficiency has the potential to reduce the production of hydrogen sulfide by CGL, whereas that from CBS is less affected by B6 deficiency.⁴⁷

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Vitamin B6 deficiency can contribute to adverse health effects because of the many metabolic roles played by PLP. Overt vitamin B6 deficiency is rare in developed countries but includes clinical manifestations such as microcytic anemia, convulsions, and dermatitis. PLP is a cofactor for δ -aminolevulinate synthase, the rate-limiting enzyme in heme biosynthesis. In vitamin B6 deficiency, heme synthesis is depressed, which limits hemoglobin synthesis and leads to the production of smaller-than-normal red blood cells and a condition known as microcytic anemia.⁴⁸ PLP also is involved in the synthesis of neurotransmitters, such as δ -aminobutyric acid and serotonin. Depressed neurotransmitter levels have been observed in vitamin B6-deficient animals.⁴⁹ PLP's role in neurotransmitter production may explain the seizures observed in infants fed vitamin B6-deficient formula; administration of PN alleviated the problem.⁵⁰ The risk of all such disorders associated

with vitamin B6 deficiency would undoubtedly be affected by vitamin B6 intake, the rate of depletion if associated with an abrupt change in intake, and the duration and extent of a chronic deficiency.

Vitamin B6 deficiency can be caused by inadequate intake, impaired absorption, or inefficient conversion of absorbed forms of the vitamin to PLP.²¹ Certain drugs, such as isoniazid,⁵¹ phenelzine,⁵² dopamine,⁵³ and gentamicin,⁵⁴ bind to the carbonyl group of PLP or PL and can induce functional vitamin B6 deficiency. Theophylline, an asthma drug, inhibits PL kinase, which interferes with PLP synthesis.⁵⁵ The effect of oral contraceptives on vitamin B6 status has been debated,^{56–58} but several studies report significantly lower plasma PLP concentration in oral contraceptive users compared to nonusers.^{59–61} Results from recent studies suggest that the current formulation of oral contraceptives has little effect on vitamin B6 status or function in healthy young women.^{62,63}

More relevant in developed countries is the issue of marginal vitamin B6 status, which often is viewed as that associated with the range of plasma PLP concentration of 20–30 nmol/L. Though not accompanied by clinical symptoms of deficiency, marginal status may increase risk for chronic diseases such as cancer and cardiovascular disease as described in [Section V](#) below. Studies involving controlled depletion of healthy adults frequently show changes in plasma amino acids, specifically mild elevation of glycine and a doubling of cystathionine concentration,^{41,42} and alterations in glutathione synthesis in certain individuals.⁶⁴ While this level of deficiency yields little or no elevation in fasting plasma homocysteine, a transient postprandial rise in homocysteine could occur. The relationship between these metabolic changes and pathophysiological processes remains unclear.

B. Criteria of Adequacy

Abnormal electroencephalographic patterns were observed in women consuming a controlled diet providing less than 0.05 mg of vitamin B6 per day having mean plasma PLP of 9.8 nmol/L.⁶⁵ Therefore, plasma PLP concentrations of at least 10 nmol/L are suggested as a lower limit of nutritional adequacy based on this physiological criterion.

As reviewed previously,⁶⁶ many studies have shown that plasma PLP values in the range of 20–30 nmol/L are common and have not been associated with obvious health effects. While it was recommended in 1990 that 30 nmol/L plasma PLP be viewed as indicative of the lower end of the normal range of vitamin B6 status,⁶⁷ the range of 20–30 nmol/L has been considered as marginal deficiency.^{68,69}

C. Assessment of Vitamin B6 Nutritional Status

An in-depth discussion of direct and functional indicators of vitamin B6 status has recently been published.⁷⁰ Vitamin B6 nutritional status can be measured by either direct or indirect methods. The most commonly used direct criterion is plasma PLP concentration, which is typically measured by the tyrosine decarboxylase assay or by HPLC. Plasma PLP has been shown to correlate well with tissue stores in rats⁷¹ and serves as a reasonable indicator of long-term vitamin B6 nutritional status.

Despite its widespread utility, plasma PLP may be influenced by factors other than vitamin B6 intake that may complicate its interpretation as a status indicator. These factors include pregnancy,⁵⁹ age,⁷² exercise,⁷³ sex,^{61,74} inflammation,⁷⁰ smoking,⁷⁵ and alcohol intake.⁷⁶ Thus, the potential for these various factors to influence B6 metabolism should be recognized when evaluating plasma PLP data.

Other direct measures of vitamin B6 status include total plasma vitamin B6, erythrocyte PLP, urinary 4-PA, and total urinary vitamin B6 excretion, all of which have been used previously for nutritional evaluations. Approximately 50% of total vitamin B6 intake is excreted as urinary 4-PA. The measurement of urinary 4-PA as a nutritional status indicator is limited because it is influenced by both vitamin B6 tissue stores as well as short-term dietary intake. While assessments using urinary 4-PA generally involve burdensome complete 24-h urine collections, the validity of assays based on untimed urine collections has been reported and may alleviate this issue.⁷⁷ The value of plasma 4-PA measurements in vitamin B6 status assessment requires further evaluation.

Indirect methods of vitamin B6 status evaluate concentrations of metabolites or the activity of PLP-dependent enzymes. The tryptophan load test has been used as a functional indicator of vitamin B6 status.⁶⁷ The activity of kynureninase, a PLP-dependent enzyme involved in the catabolism of tryptophan, is decreased in vitamin B6 deficiency, altering the flux of the pathway.⁷⁸ A more contemporary method of vitamin B6 status assessment that does not require a tryptophan dose is based on the measurement of constituents of the tryptophan catabolic pathway in plasma using liquid chromatography coupled with tandem mass spectrometry.⁷⁹ An examination of the ratios of various kynurenes led to the observation that the ratio of plasma concentrations to hydroxykynurenine to xanthurenic acid was a very sensitive and specific assessment tool for vitamin B6 status.⁸⁰ More recently, the hydroxykynurenine ratio ($\text{HKr} = \text{HK} / (\text{kynurenine acid} + \text{xanthurenic acid} + 3'\text{-hydroxyanthranilic acid} + \text{anthranilic acid})$) has been proposed as a specific and sensitive improved vitamin B6 biomarker that reflects the PLP-dependent steps of the kynurenine pathway.⁸¹

The methionine load test, which involves a 3-g oral methionine load,⁸² is similar to the tryptophan load test in that it measures the functioning of a vitamin B6-dependent pathway. In effect, the methionine load test evaluates the balance of homocysteine production from the bolus of methionine versus the capacity for its metabolic clearance.⁸³ Under the conditions of this procedure, homocysteine remethylation is suppressed while homocysteine production is extensive. Thus, the methionine load test as applied to vitamin B6 status assessment is a functional evaluation of the transsulfuration pathway of homocysteine catabolism. As stated above, both CBS and CGL are PLP dependent, with the latter highly subject to effects of inadequate vitamin B6 status.^{42,44} In contrast to the utility of postmethionine-load plasma homocysteine as a diagnostic criterion of vitamin B6 status, fasting plasma homocysteine is only weakly related to vitamin B6 status.⁸³ Fasting plasma cystathionine without a methionine load also can be used as a functional biomarker of vitamin B6 nutritional insufficiency. In vitamin B6 deficiency, rat liver cystathionine and human plasma cystathionine exhibit elevation over a wide range of vitamin B6 deficiency states ranging from marginal to severe.^{42,44,84} Deficiencies of folate and vitamin B12 also cause elevated cystathionine, so these should be ruled out if using cystathionine as a vitamin B6 biomarker.

Erythrocyte alanine aminotransferase (EALT) and aspartic aminotransferase (EAST) activity and stimulation also can be used to measure vitamin B6 status indirectly. Coenzyme stimulation is the increase in enzyme activity following the addition of PLP *in vitro*, as vitamin B6 deficiency increases the proportion of apoenzyme to holoenzyme in the erythrocyte. EALT and EAST are viewed as indicators of long-term vitamin B6 status.⁶⁷

Many controlled dietary studies and observational investigations have concluded that vitamin B6 intake and dietary protein intake^{85,86} are the most significant determinants of plasma PLP concentration. Plasma PLP concentrations appear to increase 12 nmol/L for every 1 mg/day increase in vitamin B6 intake,⁶¹ while high protein intake lowers plasma PLP concentrations.^{85,87} Extensive literature exists regarding the relationship between vitamin B6 intake and various other indices of vitamin B6 status such as urinary total vitamin B6, urinary 4-PA, EALT and EAST activity and PLP stimulation, plasma total vitamin B6, plasma 4-PA, plasma PLP, and the methionine and tryptophan load tests. However, none of these provide the truly functional information from a biochemical or physiological basis on which to base the absolute requirement.

TABLE 13.1 Recommended Dietary Allowances (RDAs) for Canada and the United States for vitamin B6⁶⁶.

Age	Male (mg)	Female (mg)	Pregnancy (mg)	Lactation (mg)
Birth to 6 months	0.1 ^a	0.1 ^a		
7–12 months	0.3 ^a	0.3 ^a		
1–3 years	0.5	0.5		
4–8 years	0.6	0.6		
9–13 years	1.0	1.0		
14–18 years	1.3	1.2	2.6	2.8
19–50 years	1.3	1.3	2.6	2.8
51 + years	1.7	1.5		

^aAdequate Intake (AI).

D. Reference Intakes

As mentioned earlier, the observation of altered electroencephalographic patterns in a vitamin B6 depletion study that induced plasma PLP levels to ~10 nmol/L provides evidence of a functional inadequacy in this very low and undoubtedly inadequate range.⁶⁵ In 1998, the Food and Nutrition Board of the Institute of Medicine developed Dietary Reference Intakes for the United States and Canada for vitamin B6. The Estimated Average Requirement (EAR) was set at an intake level that should maintain plasma PLP at 20 nmol/L and was increased by twice the assumed 10% CV to establish Recommended Dietary Allowances (RDAs) that should meet the needs of almost all in each subgroup of the population.⁶⁶

The current RDAs, which are 20% above the EARs for vitamin B6 across the life span, are given in Table 13.1.

In establishing these EARs, it was commented that even this conservative criterion may overestimate the actual requirement for half of the population on the basis of health maintenance.⁶⁶

Despite this conservative view on which the current RDA is based, the dilemma remains on how to define the requirement in the absence of obvious causal relationships between vitamin B6 and specific health outcomes.

An assessment of vitamin B6 biomarkers in a subsequent controlled dietary intake study suggested that the current RDA for young women (1.3 mg/d) may be inadequate.⁸⁸ However, an evaluation of vitamin B6 intakes from NHANES 2003–06 from foods and supplements indicated that only 8% of individuals had insufficient intakes.⁸⁹ More recent analysis from NHANES 2013–16 provided distribution of percentiles of vitamin B6 intake across age groups and sexes.⁹⁰ This analysis showed evidence of inadequate vitamin B6 intakes among older Americans, with 15% of men 71 yr and above having vitamin B6 intake less than their

EAR of 1.3 mg/day and 31% of women 71 yr and above with intakes less than their EAR of 1.1 mg/day EAR. Intakes of younger men (19–50 yr) were adequate when compared to their EARs, while 9% of younger women 19–50 y had inadequate intakes.

E. Basis for Individual Variability in Requirements

Little is known about the genetic and physiological variables that are responsible for individual variability in requirements.

Food composition has the potential to affect bioavailability, which indirectly influences requirements. The bioavailability of total vitamin B6 in a mixed diet is approximately 61–81%.⁹¹ In plant-derived foods, pyridoxine β-D-glucoside can represent 5%–75% of total vitamin B6.⁹² The glucoside derivative exhibits bioavailability as a source of available vitamin B6 of approximately 50%. The incomplete utilization of dietary pyridoxine β-D-glucoside has been estimated to be equivalent to a loss of 15%–18% of total vitamin B6 consumption in diet with high glucoside content.⁹³

The intake of dietary protein has long been recognized as a determinant of vitamin B6 requirements in humans and animals. The vitamin B6 requirement increases with increasing protein intake.^{85,94} Leklem summarized a number of studies and concluded that vitamin B6 intake >0.020 mg vitamin B6/g dietary protein would provide adequacy vitamin B6 nutriture.⁶⁷ The current RDA for vitamin B6 does not account for the level of protein intake.

F. Tolerable Upper Intake Levels

PN supplements of over 500 mg/day have been reported to cause sensory neuropathy.⁹⁵ In order to minimize this risk, the Tolerable Upper Intake Level for

vitamin B6 was set at 100 mg/day for adults.⁶⁶ This range far exceeds the formulation of typical multivitamin supplements that provide 2 mg/daily dose.

Several types of genetic disorders are vitamin B6 responsive, and therapeutic vitamin B6 supplementation is used in their management. These conditions include inborn errors that affect enzymes involved in the interconversion of vitamin B6 isoforms, reduce absorption of vitamin B6, or affect PLP-dependent enzymes; for example, PN-responsive anemia, cystathionuria, and homocysteinuria.⁹⁶ Furthermore, certain inborn errors result in the generation of metabolites that inactivate PLP. High doses of PN have been used therapeutically in PN-responsive disorders with few reports of toxic effects, although health care professionals should be cognizant of risks of toxicity.⁹⁶

G. Dietary Sources

Vitamin B6 is widely distributed in foods of plant and animal origin. Muscle-derived foods including beef, pork, chicken, and fish are good sources, while organ meats are particularly rich in vitamin B6. Whole grains are good sources of vitamin B6, as are bananas and, to a lesser extent, potatoes. While whole wheat flour is a good source of the vitamin, white flour is not (which reflects the relative lack of vitamin B6 in the endosperm).

The presence of glycosylated forms of PN in plants, particularly as PNG, complicates their nutritional evaluation. This is because dietary PNG exhibits only about 50% bioavailability in humans, and it also exerts weak antagonistic effects on the utilization of nonglycosylated PN.^{15,97} The content of PNG in plant foods varies widely, for example, ranging from 5% in bananas to 75% in carrots as a fraction of total vitamin B6. The overall bioavailability of vitamin B6 in a mixed diet has been reported to be approximately 75%.⁹¹

In spite of the potential for lower bioavailability of vitamin B6 in certain plant foods due to the presence of PNG and potential effects of fiber and cellular structure, there is little evidence that vegetarians with sufficiently varied diet and good total intake of the vitamin would be particularly at risk of insufficient vitamin B6 status.⁹⁸ However, individuals with marginal intake of total vitamin B6 would be more prone to reduced nutritional status due to this incomplete bioavailability.

While sources of vitamin B6 in the US diet are distributed over a wide range of food, breakfast cereals fortified with PN constitute an important source. Various fortified sports bars, meal replacement products, and beverages also constitute major sources of vitamin B6 intake in many individuals. Unlike thiamin, niacin, riboflavin, and folic acid, PN is not added to enriched flour and other enriched grain products.

V. ISSUES RELATED TO VITAMIN B6 IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

The lower ranges of vitamin B6 intake, especially among women, are well below recommended levels, and this marginal range is associated with increased risk of cardiovascular disease, stroke, and venous thrombosis. These lower ranges also have been associated recently with incidence of certain cancers. At present, the key word is association—the mechanistic connections (if any) between low vitamin B6 status and chronic disease remain to be determined.

Chronically inadequate vitamin B6 nutritional status has been associated with aberrant one-carbon metabolism and health. Suboptimal vitamin B6 nutriture was first associated with vascular disease in monkeys,⁹⁹ a relationship confirmed in human epidemiological studies. Verhoef et al.¹⁰⁰ reported that the frequency of myocardial infarction was negatively associated with both folate and vitamin B6 nutriture, and these findings were confirmed in the Nurses' Health Study.¹⁰¹ Women in the highest quintile of B6, folate, or folate plus B6 intake had less coronary heart disease than those in the lowest quintile of each intake category. An Italian study¹⁰² also reported an independent association of low plasma PLP concentration with coronary artery disease risk. Dalery et al.¹⁰³ described the results of large trials examining plasma PLP concentration and risk of vascular disease. Even in healthy control subjects, low folate or B6 status increased the risk of vascular disease, and risk of vascular disease in low vitamin B6 status was independent of plasma homocysteine concentration. Kelly et al. observed that low vitamin B6 status (plasma PLP < 30 nmol/L) was associated with more than a twofold increase in risk of stroke or transient ischemic attack^{104,105} independent of folate status or plasma total homocysteine (tHcy). Further, a prospective study has shown that people with plasma PLP < 23.3 nmol/L had 1.8-fold higher recurrent risk of venous thromboembolism than those with PLP > 23.3 nmol/L.¹⁰⁶ In spite of the strength of these associations, acute supplementation trials with mixed folic acid, vitamin B12 and vitamin B6 (e.g.,^{107,108}), or PN alone compared to mixed vitamins (e.g.,^{109,110}) for *secondary* prevention (i.e., preventing *recurrence* of cardiovascular disease) have yielded largely negative results.

The mechanism by which low vitamin B6 intake is associated with risk of vascular disease is not known. Since vitamin B6 deficiency has little tendency to raise fasting plasma tHcy but yields an elevated tHcy response following a methionine load, low B6 nutriture

may lead to repeated transient mild hyperhomocysteinemia following meal consumption.

Several reports of associations between elevated plasma C-reactive protein (CRP) and low vitamin B6 status have raised the hypothesis that systemic inflammation is prone to occur during vitamin B6 deficiency or contributes to low B6 status.^{111–113} The mechanism behind this association is unclear because controlled dietary restriction of vitamin B6 in healthy adults does not cause a change in CRP.⁴² In light of the reports of a relationship between CRP, vascular disease, and low vitamin B6 status in observational studies such as NHANES, it appears that vitamin B6 deficiency alone is not an inflammatory condition, but the deficiency may predispose an individual to the effects of other inflammatory processes. Alternatively, inflammation may lead to tissue-specific deficiency¹¹⁴ by altering vitamin B6 cellular metabolism, turnover, or transport. Altered glutathione metabolism^{44,64} may be one such metabolic connection between impaired vitamin B6 status and inflammatory response. The mechanisms involved in inflammation and their effect on vitamin B6 status have been reviewed.^{115,116}

A linkage between vitamin B6 status and certain aspects of cancer risk also has been reported. For example, a metaanalysis evaluated prospective studies and found a significant inverse relationship between plasma PLP concentration and colorectal cancer risk.¹¹⁷ In this study, the evidence was less conclusive regarding vitamin B6 intake and risk, possibly as a result of greater uncertainty in assessing vitamin B6 consumption patterns. Greater risk of lung cancer also has been shown to be

inversely associated with plasma PLP concentration.¹¹⁸ Little consistent evidence of other connections between vitamin B6 and cancer has been reported, and further investigation is needed.

B. Known Interactions

Theophylline doses within the range used in asthma treatment lower plasma PLP because theophylline is a noncompetitive inhibitor of pyridoxal kinase.¹¹⁹ These authors reported that 10 mg/day PN HCl overcomes this antagonistic effect. Isoniazid (isonicotinylhydrazine), the drug of choice used for tuberculosis prevention and therapy, acts as a vitamin B6 antagonist by forming complexes with PL and/or PLP to form a hydrazine derivative. This hydrazone is a potent inhibitor of pyridoxal kinase. Commercially available combination doses of isoniazid and PN HCl are available to lessen the antivitamin B6 effect. However, many case reports of pellagra occurring during isoniazid therapy even with PN HCl suggest that the doses of PN were not capable of overcoming the induced vitamin B6 deficiency. Pellagra occurrence then occurred from resulting suppression of nicotinamide production from the PLP-dependent kynurenine pathway. Penicillamine and L-DOPA also have the ability to form complexes with PLP, but these complexes do not inhibit pyridoxal kinase.¹²⁰

C. As a Dietary Supplement

PN HCl is the most common form of vitamin B6 in supplements, typically at doses providing 1.5–3 mg of PN.

RESEARCH GAPS

1. Vitamin B6 is a family of related compounds consisting of PN, PL, PM, and their 5' -phosphates. Fruits and vegetables also contain pyridoxine- β -D-glucoside in variable and often substantial amounts. The development of more complete food composition tables would improve dietary assessment.
2. The development of a better understanding of the linkage between vitamin B6 nutritional status and disease susceptibility is an important need. Small-scale controlled mechanistic studies examining connections of metabolomic patterns, dietary intake, genetics, and in vivo kinetics may help in the development of diagnostic tools and biomarkers for use in larger population studies.
3. Understanding vitamin B6 nutrition on a personalized level will be important in assessing optimal intakes and consequences of deficiency. For example, how do genetic and nutritional factors interact to define an individual's health and susceptibility to chronic disease?
4. Further characterization of the interaction of inflammatory processes and vitamin B6 status may be an important key in understanding optimal nutrition and in interpreting biomarkers such as plasma PLP and inflammatory markers.
5. What is the role of vitamin supplementation and food fortification in human health, specifically with respect to primary prevention of cardiovascular and other forms of chronic disease?

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FOLATE

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SUMMARY

Folate is a water-soluble B vitamin present in diets as naturally occurring polyglutamyl food folates or as monoglutamyl synthetic folic acid (in fortified foods and supplements). Folate coenzymes function in the acceptance and donation of one-carbon entities critical to nucleotide synthesis, cellular methylation reactions, and amino acid metabolism. Genetic variants in folate-metabolizing genes may lead to metabolic abnormalities that have been linked in some cases with disease and birth defect risk particularly under conditions of suboptimal folate status. Folate intake recommendations consider differences in the bioavailability of folate, which varies depending on the source (naturally occurring food folate is less bioavailable than folic acid in fortified foods or supplements). A separate folic acid intake recommendation along with folic acid fortification of the food supply in the United States, and numerous other countries, is specifically designed to reduce the risk of neural tube birth defects.

Keywords: Dietary folate equivalent; Folate; Folic acid; Folic acid fortification; Neural tube defects; One-carbon metabolism.

I. INTRODUCTION

A. Background

The pioneering work of Lucy Wills in the late 1920s revealed that a substance in yeast extract, later identified as folate, could cure the megaloblastic anemia of pregnancy.¹ Folate received its name in 1941 when an “acid nutritive” with “vitamin-like properties” was isolated from 4 tons of spinach (folium is latin for leaf).² The pure crystalline form of folate (i.e., folic acid) was then synthesized in 1943. The generic term “folate” is used to denote all the different chemical forms of the vitamin including naturally occurring reduced folates from whole foods and synthetic oxidized folic acid used in fortified foods and dietary supplements. Folate coenzymes function in the acceptance and transfer of one-carbon (1-C) moieties involved in the synthesis,

metabolism, and modification of nucleotides, amino acids, and other cellular components.

B. Nutrient Structure and Function

Common structural features of the folate compounds include a pteridine bicyclic ring system, para-aminobenzoic acid, and one or more glutamic acid residues (Fig. 14.1). Naturally occurring food folates are present in the reduced tetrahydrofolate (THF) form and have side chains composed of five to eight glutamate residues joined in γ -peptide linkages. Folic acid, the synthetic fully oxidized monoglutamate form of the vitamin, rarely occurs in nature but is the form added to fortified food products and used in dietary supplements. Reduction of the pteridine moiety to dihydrofolate (DHF) and THF, elongation of the glutamate chain, and acquisition of single

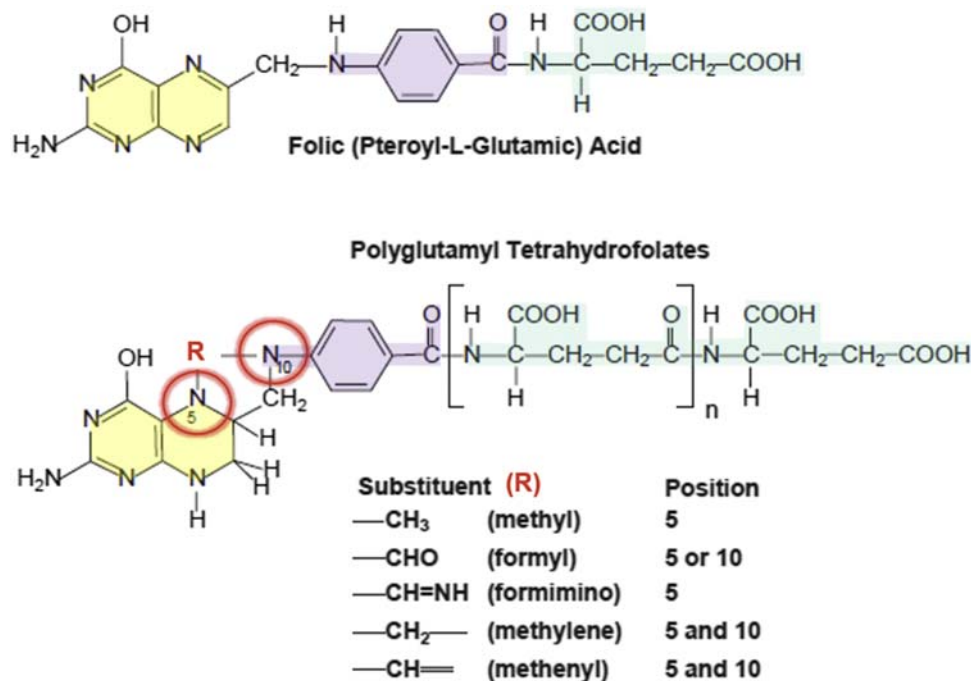


FIGURE 14.1 Folate structures: Folic acid consists of a *p*-aminobenzoic acid molecule (shown in purple) linked at one end to a pteridine ring (shown in yellow) and at the other end to a molecule of glutamic acid (shown in green). Food folates exist in various forms, containing from 2 to 10 additional glutamate residues joined to the first glutamate. The folate/folic acid structure can vary by reduction of the pteridine moiety to form dihydrofolate and tetrahydrofolate (THF), elongation of the glutamate chain, and substitution of 1-C units at the N5, N10, or both positions. Folate coenzymes are formed by the attachment of 1-C units including methyl (CH₃), methylene (—CH₂—), methenyl (—CH=), formyl (—CH=O), or formimino (—CH=NH) groups to the polyglutamyl form of the THF molecule.

carbons on the N-5, N-10, or both positions are required for folate to participate in cellular 1-C metabolism.³ The 1-C-substituted folate vitamers function with methyl (CH₃), methylene (—CH₂—), methenyl (—CH=), formyl (—CH=O), or formimino (—CH=NH) groups mainly as the polyglutamate form of the THF molecule.

C. Analytical Methodology to Measure Folates

There are three main analytical methods used to measure folate, with the choice of method dependent in part on the purpose of the assessment.⁴ Advantages and disadvantages of each method are discussed briefly below and described in further detail in a 2015 expert panel review.⁴

Microbiologic assay (MBA)

Often considered the “gold standard” procedure for measuring serum and red blood cell (RBC) folate, this procedure is based on the growth of a folate-dependent microorganism (*Lactobacillus rhamnosus*) in response to all metabolically active forms of folate present in the sample. Although the assay is somewhat labor-intensive and lengthy, it requires only a small sample volume and simple instrumentation, making it ideal for low-resource settings.⁴ Further, it can be “harmonized” through the use of standard techniques and common critical reagents to produce comparable results across laboratories and over time (see Section IV.C).⁵

Protein-binding assays

Protein-binding assays were developed primarily for use in diagnosing folate deficiency in clinical settings, are available in commercial kit form, have a quick turnaround time, and are relatively inexpensive.⁴ As they rely on highly specific folate binding proteins to “extract” folate from the sample, assay results can be influenced by different binding affinities of the various folate forms and by lot-to-lot variability of assay reagents.⁴

Chromatography-based assays

Chromatography-based assays are highly selective and specific, provide information of individual folate forms, and offer good precision but require expensive instrumentation, an experienced operator, and complex sample extraction and data interpretation procedures.⁴ They are most suited for the research setting.

II. NUTRIENT METABOLISM

A. Digestion

Naturally occurring food folate is present in the polyglutamate form and must be converted to the monoglutamate form prior to absorption. This conversion occurs primarily in the proximal small intestine, the main site of folate absorption, and involves the brush border enzyme

glutamate carboxypeptidase II (GCP II). GCP II is an exopeptidase that hydrolyzes terminal γ -linked glutamate residues from the polyglutamate tail, ultimately producing a folate monoglutamate.⁶ Maintenance of the optimal pH of 6.5 is critical for the complete deconjugation of polyglutamate folates by GCP II.^{6,7} Folic acid is consumed as a monoglutamate and thus does not require deconjugation for intestinal uptake.

B. Absorption and Transport

Intestinal uptake and efflux

The primary mediator of intestinal folate absorption is the proton-coupled folate transporter (PCFT) encoded by the *SLC46A1* gene.⁸ The proton coupling action of PCFT allows the concentrative transport of folates with optimal activity around pH 5.5, which corresponds to the acidic environment of the proximal small intestine brush border. PCFT exhibits similar affinity for reduced folates and folic acid (K_m values 0.5–0.8 μ M at pH 5.5).⁹

A second folate transporter, the reduced folate carrier (RFC) encoded by the *SLC19A1* gene, is also widely expressed in the small intestine. In contrast to PCFT, RFC operates optimally at a pH of 7.4 and has low affinity for folic acid ($K_m = 200$ –400 μ M). While RFC was long considered the primary route of folate absorption, studies of hereditary folate malabsorption and PCFT-null mice confirm that the principal route of folate absorption occurs via PCFT in the duodenum and jejunum.⁸

While the main site of folate absorption is the small intestine, folate can also be taken up in the colon.¹⁰ PCFT and RFC are both expressed in the large intestine; however, colonic pH does not favor PCFT activity, and the primary mechanism of colonic folate absorption remains unclear.⁸ When pharmacological doses (>10 μ M) of folate are consumed, absorption also takes place by a nonsaturable diffusion-like process and most of the vitamin appears unchanged in portal circulation.³

Within the intestinal cells, reduced folate vitamers are converted to 5-methyltetrahydrofolate (5-methylTHF). A small portion of folic acid may also be converted to 5-methyl-THF by enterocytes. Folate efflux of monoglutamate 5-methylTHF and unmetabolized folic acid from the enterocyte into portal blood is likely mediated by the multidrug resistance-associated proteins (MRPs), specifically MRP3, which is expressed at the basolateral intestinal membrane.⁸

Hepatic uptake and efflux

The predominant forms of folate in portal blood are monoglutamate 5-methylTHF and to a lesser extent folic acid that escaped metabolism to 5-methylTHF within the intestinal cell. Hepatic uptake of 5-methylTHF and folic acid is mostly mediated by the PCFT.¹¹ The bidirectional RFC is also expressed by the

liver, but its role in hepatic folate transport has not been established.⁸ Folate monoglutamates are exported from hepatocytes via MRPs 3 and 4.⁸ 5-MethylTHF is the predominate form of folate exported to plasma and is thus the major circulatory form. Folic acid that has not been converted to a coenzyme form can also enter circulation via these same transporters.

Systemic transport and tissue uptake

Low- and high-affinity proteins bind folate in circulation with the largest percentage ($\sim 50\%$) bound loosely to albumin.¹² Cellular uptake of folate by peripheral tissues utilizes mainly RFC, although PCFT is also widely expressed and prominent in the kidney, liver, and brain.¹¹ In the central nervous system, folate uptake is mediated largely by high affinity folate receptors (e.g., FR α) that utilize an endocytotic process for the cellular internalization of folate.¹³ PCFT plays a role in FR α -mediated endocytosis by serving as a route of export of folates from acidified endosomes.¹¹ Folate receptors are also expressed in kidney proximal tubes, erythropoietic cells, and placenta.¹¹

C. Cellular Storage

Tissues are limited in their ability to store folate beyond that required for metabolic function. The total body content of folate in humans is estimated to be approximately 15–30 mg based on measurements of liver tissue concentration,¹⁴ assumptions of liver mass (~ 1400 g), and relative folate content (i.e., $\sim 50\%$ of total body folate).¹⁵ Calculations based on in vivo kinetic studies yield similar estimates of body folate pools,¹⁶ but conflicting data have been reported.¹⁷

D. Excretion

Intact folates and folate cleavage products are excreted in urine and feces. The fraction of plasma folate not associated with protein is freely filtered at the glomerulus and efficiently reabsorbed in the proximal renal tubules.³ In unfortified populations, urinary excretion occurs mainly as folate cleavage products, and the urinary excretion of intact folates represents only a small percentage of dietary folate.^{3,18} However, in folic acid-fortified populations, the consumption of folic acid supplements may lead to the urinary excretion of intact folates representing a substantial proportion of dietary folate intake.¹⁹ Biliary secretion of folate via hepatic MRP2 has been estimated to be as high as 100 μ g/day providing for enterohepatic circulation of folate.³ Fecal folate losses are difficult to assess due to the contribution of microbial folate synthesis in the large intestine; however, fecal folate excretion has been estimated to be similar to that of total urinary excretion of intact folate and cleavage products.²⁰

E. Metabolism

Upon cellular uptake, 5-methylTHF is demethylated to THF by methionine synthase, while folic acid requires reduction of its pteridine ring to THF by dihydrofolate reductase (DHFR). Unmetabolized 5-methylTHF and folic acid are rapidly effluxed from the tissue to plasma. THF generated from the metabolism of 5-methylTHF and folic acid is subsequently polyglutamated through the actions of folypolyglutamate synthetase. More than 95% of tissue folates are polyglutamate species, which usually exhibit higher affinities for folate-dependent enzymes than monoglutamates. Due to their charge and being bound to folate enzymes, folate polyglutamates do not cross cell membranes, thus polyglutamylation serves as a mechanism to sequester folate within the cell. Prior to release from tissues, folate polyglutamates require conversion to the monoglutamate form by γ -glutamyl hydrolase. Within the cell, THF can be metabolized to other coenzymatic forms that function as 1-C acceptors or donors in nucleotide biosynthesis, methylation reactions, amino acid metabolism, and 1-C unit generation (see Section III).

F. Bioavailability

Bioavailability broadly refers to efficiency of nutrient utilization, including the physiological and biochemical processes involved in intestinal absorption, transport, metabolism, and excretion.²¹ Folate bioavailability depends upon source and chemical form. Naturally occurring food folate exhibits variable and often incomplete bioavailability related primarily to dietary variables and physiological conditions. These include (i) entrapment of naturally occurring folates in the cellular structure or insoluble matrix of certain foods; (ii) instability of labile THF during passage through the stomach; (iii) inhibition of the intestinal deconjugation of polyglutamate folates by food constituents; and (iv) indirect impairment of folate deconjugation and absorption by alteration of jejunal pH. These factors contribute to the wide range of reported bioavailability of naturally occurring food folate with an estimated average bioavailability of 50%.^{22,23} In contrast, folic acid in supplements and fortified foods does not require deconjugation prior to absorption and is more chemically stable than naturally occurring food folates and thus exhibits more complete bioavailability. It is estimated that folic acid consumed in fortified foods is ~85% bioavailable.²⁴ The greater bioavailability of folic acid provided the rationale for the calculation of Dietary Folate Equivalents (DFEs) currently used for folate intake recommendations as

described in more detail later in this chapter (see Section IV).

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

Folate coenzymes function in the acceptance and donation of 1-C entities critical to the synthesis of purines and thymidylate, the methionine cycle and methylation reactions, and the interconversion and catabolism of certain amino acids. Folate-dependent pathways are localized to three interconnected cellular compartments: cytoplasm, nucleus, and mitochondrion. Each compartment is associated with a particular metabolic function.²⁵

A. Purine Biosynthesis

The required folate coenzyme for the de novo synthesis of the purine nucleotides, adenine, and guanine in the cytosol is 10-formylTHF (Fig. 14.2 and Table 14.1). The formyl group of 10-formylTHF is incorporated at the C-2 and C-8 positions of the purine ring by reactions catalyzed by glycinamide ribonucleotide transformylase (GARFTase) and aminoimidazolecarboxamide ribonucleotide transformylase (Fig. 14.2, reactions 1 and 2). 10-formyl is generated by the coupling of formate (from the mitochondria) to THF in a reaction catalyzed by the activity of formylTHF synthetase (S) of the MTHFD1 trifunctional enzyme (Fig. 14.2, reaction 3). Trifunctional MTHFD1 also facilitates the interconversions of 10-formylTHF, 5,10-methenylTHF, and 5,10-methyleneTHF in the reversible reactions catalyzed the activities of methenylTHF cyclohydrolase (C) and methyleneTHF dehydrogenase (D) (Fig. 14.2, reactions 4 and 5, respectively). In addition to the mitochondria, 1-Cs for 10-formylTHF production can be generated in the cytosol from serine via the action of cytosolic serine hydroxymethyltransferase 1 (SHMT1), which produces 5,10-methyleneTHF (Fig. 14.2, reaction 6), as well as from the catabolism of histidine and purines, which enter the cytoplasmic pool as 5,10-methenylTHF.²⁶ Nonetheless, these alternative routes for 10-formylTHF production are not major contributors since the reductive environment in the cytoplasm favors the conversion of 10-formylTHF to 5,10-methyleneTHF in the reactions catalyzed by the cyclohydrolase and dehydrogenase activities of the MTHFD1 trifunctional enzyme.²⁷

B. Thymidylate Biosynthesis

5,10-MethyleneTHF is the required folate coenzyme for the de novo biosynthesis of the pyrimidine nucleotide, thymidylate, also known as deoxythymidine

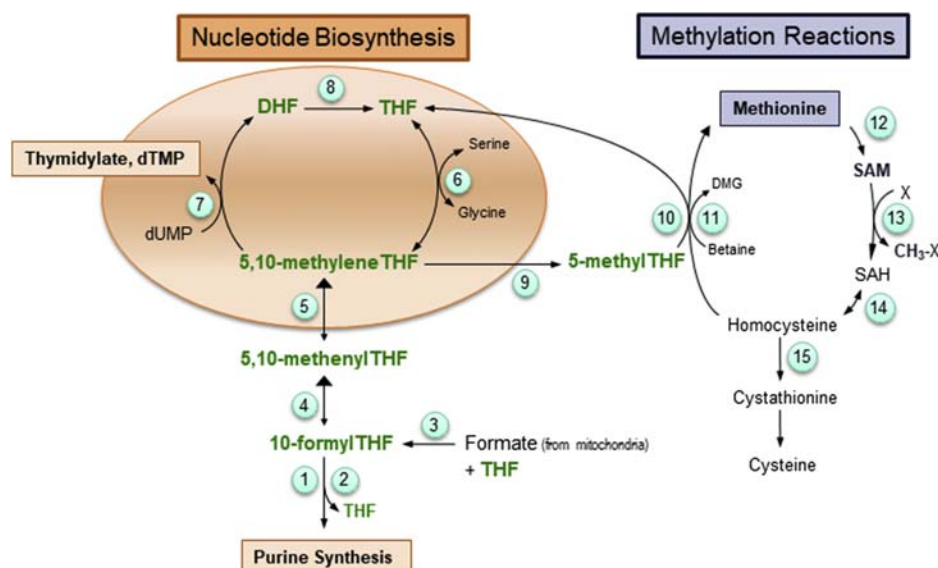


FIGURE 14.2 Folate-mediated biosynthetic reactions and interconversions of folates (polyglutamates) emphasizing the mostly unidirectional flow of 1-C units through reactions 4 and 5. The thymidylate cycle denoted by the orange background also occurs in the nucleus. Each reaction shows the product plus THF, which is recycled to accept additional 1-C units (mostly as formate from the mitochondria). The encircled numbers in this diagram denote enzymatic reactions that are listed in Table 14.1. *DHF*, dihydrofolate; *DMG*, dimethylglycine; *SAH*, S-adenosylhomocysteine; *SAM*, S-adenosylmethionine; *THF*, tetrahydrofolate; *X*, methyl (CH_3) group acceptor.

TABLE 14.1 Major cytoplasmic and nuclear metabolic enzymes and reactions for folate (as shown in Fig. 14.2).

Reaction*	Enzyme	Abbreviations
1	Glycinamide ribonucleotide transformylase	GARTase
2	Aminoimidazolecarboxamide ribonucleotide transformylase	AICARTase
3	Methylenetetrahydrofolate dehydrogenase 1 with formylTHF synthetase activity	MTHFD1 (S)
4	Methylenetetrahydrofolate dehydrogenase 1 with methenylTHF cyclohydrolase activity	MTHFD1 (C)
5	Methylenetetrahydrofolate dehydrogenase 1 with methyleneTHF dehydrogenase activity	MTHFD1 (D)
6	Serine hydroxymethyltransferase 1	SHMT1
7	Thymidylate synthase	TYMS
8	Dihydrofolate reductase	DHFR
9	Methylenetetrahydrofolate reductase	MTHFR
10	Methionine synthase	MTR
11	Betaine homocysteine S-methyltransferase	BHMT
12	Methionine adenosyltransferase	MAT
13	Various methyltransferases	MTs
14	S-adenosylhomocysteine hydrolase	SAHH
15	Cystathionine β -synthase	CBS

* Enzymes 3, 4, and 5 represent the three activities associated with the trifunctional MTHFD1.

monophosphate (dTMP) (Fig. 14.2 and Table 14.1). Thymidylate synthase (TYMS) catalyzes the transfer of a 1-C from 5,10-methylene THF to deoxyuridine monophosphate to form dTMP (Fig. 14.2, reaction 7); this is the rate-limiting step during the cell cycle that enables DNA replication to proceed. This reaction is unique in

folate 1-C transfers in that the THF carrier is oxidized to DHF as the 1-C transferred to dTMP is reduced from a methylene to a methyl group. In order to play a further role in 1-C metabolism, DHF must be subsequently reduced to THF in a reaction catalyzed by DHFR (Fig. 14.2, reaction 8).

5,10-Methylene-THF is generated from 10-formyl-THF through the cyclohydrolase and dehydrogenase activities of trifunctional MTHFD1 using formate as the 1-C source (Fig. 14.2, reactions 4 and 5). Alternatively, 5,10-methylene-THF can be obtained by SHMT1, a cytosolic enzyme that catalyzes the reversible transfer of formaldehyde from C3 of serine to THF to generate 5,10-methylene-THF and glycine (Fig. 14.2, reaction 6). Formate is estimated to contribute about 70% of 1-C groups used by TYMS, while serine contributes the other 30%.

De novo thymidylate biosynthesis can occur in the cytosol and nucleus. Nuclear synthesis of thymidylate enables the generation of dTMP at the site of DNA replication.²⁸ During cell division, the cytosolic enzymes, constituting the thymidylate cycle (SHMT1, TYMS, DHFR, and MTHD1), are translocated to the nucleus following their modification (sumoylation) by small ubiquitin-like modifier proteins.²⁹

C. Methionine Cycle and Methylation Reactions

5-Methyl THF is the required folate coenzyme for the remethylation of homocysteine to methionine in the methionine cycle (Fig. 14.2 and Table 14.1). The folate-dependent remethylation process is catalyzed by methionine synthase (MTR) and requires 5-methylTHF and vitamin B12 (Fig. 14.2, reaction 10). 5-MethylTHF is produced from 5,10-methyleneTHF in the unidirectional reaction catalyzed by methylenetetrahydrofolate reductase (MTHFR) (Fig. 14.2, reaction 9). The production of 5-methylTHF by MTHFR is an important functional and regulatory step in folate-mediated 1-C metabolism because it commits 1-C units for methionine regeneration at the expense of thymidylate biosynthesis.

Methionine, either from dietary protein or produced from homocysteine, can be activated to S-adenosylmethionine (SAM), the principal methylating agent in all cells (Fig. 14.2, reaction 12). SAM is the methyl donor in over 100 transmethylation reactions, including methylation of DNA, RNA, histones, neurotransmitters, and phospholipids. The main consumers of SAM include phosphatidylethanolamine *N*-methyltransferase (PEMT), guanidino acetate methyltransferase, and glycine *N*-methyltransferase (GNMT), which synthesize phosphatidylcholine, creatine, and sarcosine, respectively. Sarcosine is a source of 1-Cs following its degradation in the mitochondria as described in Part D. As a result of methyl group transfer by a variety of methyltransferases, SAM is converted to S-adenosylhomocysteine (SAH) (Fig. 14.2, reaction 13), which is subsequently hydrolyzed to homocysteine and adenosine by SAH hydrolase (Fig. 14.2, reaction 14). Although this reaction is reversible, utilization of adenosine and the remethylation

of homocysteine provide the kinetic impetus that drives this reaction in the direction of homocysteine formation.

Homocysteine can be removed permanently from the methionine cycle via the transsulfuration pathway where homocysteine is ultimately converted to cysteine. Cystathionine β -synthase (CBS) (Fig. 14.2, reaction 15) catalyzes the committing step in transsulfuration in which homocysteine and serine are coupled to form cystathionine in a pyridoxal phosphate-dependent reaction that is activated by SAM. This promotes CBS activation and favors the transsulfuration pathway under conditions of ample methionine.

D. Generation of 1-C Units

THF participates in the mitochondrial generation of 1-C units in the form of formate for cytosolic and nuclear folate-mediated 1-C metabolism (Fig. 14.3 and Table 14.2). The C3 of serine, derived from the diet, glycolytic intermediates, and serine produced by cytosolic SHMT1, is a primary source of 1-C units from which 5,10-methyleneTHF is formed from THF through the action of mitochondrial SHMT2 (Fig. 14.3, reaction 1). 5,10-MethyleneTHF is converted to 10-formylTHF by a bifunctional enzyme encoded by *MTHFD2* (embryos) or *MTHFD2L* (adults) with dehydrogenase and cyclohydrolase activities^{25,30} (Fig. 14.3, reactions 2 and 3, respectively). In the final step, 10-formylTHF is hydrolyzed to formate and THF by the monofunctional mitochondrial enzyme, *MTHFD1L*³¹ (Fig. 14.3, reaction 4). Redox and equilibrium conditions support the flow of 1-C units through the mitochondria in the oxidative direction of 5,10-methyleneTHF to 10-formylTHF to formate²⁵ as shown in Fig. 14.3. The formate is then delivered to the cytosol where it is converted to 10-formylTHF for purine synthesis (Fig. 14.2, reaction 3) or is reduced to 5,10-methyleneTHF, which can be diverted to thymidylate biosynthesis (Fig. 14.2, reaction 7) or methylation reactions (Fig. 14.2, reaction 9). Alternatively, the 1-C unit in the form of 10-formylTHF may be oxidized to CO₂ by the enzyme, formyltetrahydrofolate dehydrogenase, which functions to remove excess 1-C groups (Fig. 14.3, reaction 5).

Additional sources of 1-C units in select tissues (i.e., liver) include the mitochondrial glycine cleavage system and the oxidative catabolism of choline. The glycine cleavage system is a multienzyme complex that cleaves glycine to produce CO₂, ammonia, and 5,10-methyleneTHF (Fig. 14.3, reaction 6), the latter of which can be oxidized to 10-formylTHF to recover 1-C units. Choline catabolism involves its oxidation to betaine, which, like 5-methylTHF, is used in the remethylation of homocysteine to methionine in a reaction catalyzed by cytosolic betaine-homocysteine

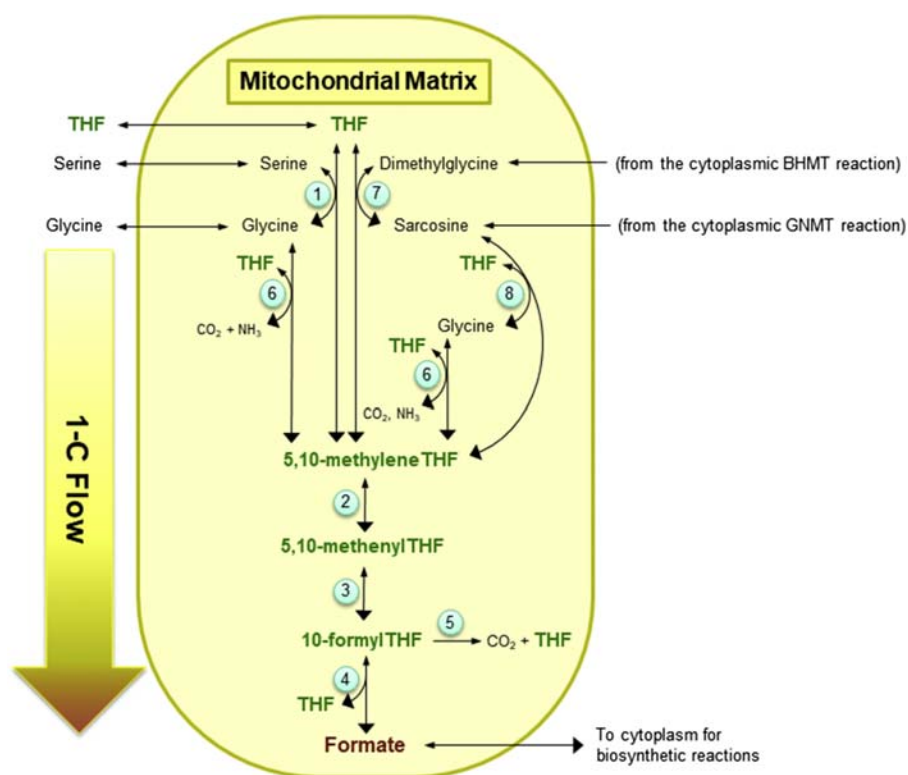


FIGURE 14.3 Mitochondrial folate-mediated one-carbon metabolism emphasizing the generation of formate from the catabolism of serine, glycine, and choline. Serine, glycine, and dimethylglycine (from oxidative choline catabolism) can traverse the mitochondrial inner membrane enabling their entry from the cytoplasmic compartment and their conversion to formate. The encircled numbers in this diagram denote enzymatic reactions that are listed in Table 14.2.

TABLE 14.2 Major mitochondrial enzymes and reactions (as shown in Fig. 14.3).

Reaction*	Enzyme	Abbreviations
1	Serine hydroxymethyltransferase 2	SHMT2
2	Methylenetetrahydrofolate dehydrogenase 2 (like) with methyleneTHF dehydrogenase activity	MTHFD2/MTHFD2L
3	Methylenetetrahydrofolate dehydrogenase 2 (like) with methenylTHF cyclohydrolase activity	MTHFD2/MTHFD2L
4	Methylenetetrahydrofolate dehydrogenase 1 (like)	MTHD1L
5	FormylTHF dehydrogenase	FTHFD
6	Glycine cleavage system	GCS
7	Dimethylglycine dehydrogenase	DMGD
8	Sarcosine dehydrogenase	SDH

* Enzymes 2 and 3 represent the two activities associated with the bifunctional MTHFD2 (embryo) or MTHFD2L (adult).

S-methyltransferase (BHMT) (Fig. 14.2, reaction 11). The other product of this reaction, dimethylglycine, undergoes two sequential mitochondrial oxidations catalyzed by dimethylglycine dehydrogenase and sarcosine dehydrogenase to form glycine and two molecules of 5,10-methyleneTHF (Fig. 14.3, reactions 7 and 8, respectively).

Only a limited exchange of free folate occurs between the cytoplasmic and mitochondrial compartments. However, these compartments are metabolically connected by transport of 1-C donors (serine, glycine, and formate) across the mitochondrial membranes, supporting a mostly unidirectional flow from serine to formate (mitochondria) and onto methionine (cytoplasm).²⁵

E. Amino Acid Interconversions and Catabolism

Folate coenzymes are used in amino acid metabolism as alluded to in previous sections. The interconversion of glycine and serine is catalyzed by the folate-dependent enzyme, SHMT (cytosol and mitochondria; [Figs. 14.2 and 14.3](#)), while methionine regeneration from homocysteine is catalyzed by the folate and vitamin B12-dependent enzyme, methionine synthase ([Fig. 14.2](#), reaction 10). Folate coenzymes also participate in the catabolism of select amino acids. Serine and glycine are degraded by the folate-dependent glycine cleavage system ([Fig. 14.2](#), reaction 6). Histidine catabolism involves a folate-dependent reaction in the final step, whereby glutamate formimidoyltransferase catalyzes the transfer of the formyl group from formiminoglutamic acid to THF to yield 5-formiminoTHF and glutamate. 5-FormiminoTHF can be converted to 5,10-methenylTHF and used as a folate coenzyme in cytosolic biosynthetic pathways ([Fig. 14.2](#)).

F. Regulation of the Biosynthetic Pathways

The folate-mediated biosynthetic cycles are interconnected, and factors that regulate any one cycle of 1-C metabolism will influence folate availability for the other cycles or pathways. Studies in animals and human cells suggest that folate cofactors are preferentially partitioned toward the nuclear compartment (at the expense of homocysteine remethylation) under conditions of folate deficiency, thereby ensuring proper functioning of the *de novo* thymidylate synthesis pathway.³² The partitioning of MTHFD1 from the cytosol to the nucleus appears to mediate this provision of 1-C units for thymidylate biosynthesis.²⁹ Additional enzymes with regulatory functions include formylTHF dehydrogenase that regulates the supply of formate for folate-mediated nucleotide biosynthesis by depleting 10-formylTHF pools³³ ([Fig. 14.3](#), reaction 5); MTHFR that mediates the flow of 1-C units between thymidylate biosynthesis and methylation reactions ([Fig. 14.2](#), reaction 9); and cSHMT that preferentially partitions cSHMT-derived 5,10-methyleneTHF toward thymidylate biosynthesis at the expense of methylation reactions ([Fig. 14.2](#), reaction 6).³⁴

Several important metabolic regulators also exist within the biosynthetic cycles. For example, SAM and SAH are the major regulators of the use of methyl groups in the methionine cycle and exert control at several regulatory points. Surplus SAM binds to the regulatory domain of MTHFR and inhibits enzyme activity thereby reducing the production of 5-methylTHF, which is required for the recycling of homocysteine through the methionine synthase reaction. Diminished production

of 5-methylTHF by MTHFR also yields an active GNMT, which is normally inhibited by 5-methylTHF. GNMT catalyzes the SAM-dependent methylation of glycine to sarcosine thereby metabolizing excess SAM. Sarcosine is converted back to glycine and 5,10-methyleneTHF via sarcosine dehydrogenase following its transport into the mitochondria ([Fig. 14.3](#), reaction 8), which allows for the conservation of 1-C units. SAM also activates CBS ([Fig. 14.2](#), reaction 15), the first step in the transsulfuration pathway that initiates homocysteine catabolism and its permanent removal from the methionine cycle. Under conditions of inadequate SAM concentrations, methionine conservation is preserved by recycling homocysteine to methionine through activation of MTHFR and inhibition of CBS. Further, as a result of enhanced MTHFR activity and greater production of 5-methylTHF, GNMT is inhibited sparing SAM for essential methylation reactions.

SAH is a potent product inhibitor of most methyltransferases.³⁵ Accumulation of SAH and the associated inhibition of cellular methyltransferases will occur under metabolic conditions that interfere with product removal of homocysteine or adenosine. Increased cytosolic SAH concentrations have also been reported to upregulate CBS activity, decrease BHMT activity, and decrease MTHFR activity in certain tissues.

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

The initial manifestations of folate deficiency or inadequacy are decreasing blood folate concentrations and increasing plasma homocysteine. Over time, severe folate deficiency leads to decreased DNA synthesis and impaired maturation of erythropoietic precursors resulting in a gradual increase in RBC mean cell volume and a clinical diagnosis of megaloblastic anemia. With folate deficiency, impaired cell division also leads to decreased numbers of white cells and platelets. Because the intestinal mucosa undergoes continuous regeneration, with replacement of epithelial cells every 3 days, its folate requirements are greater than other tissues. Gastrointestinal symptoms frequently result from severe folate deficiency and are often associated with impaired absorption.^{36,37}

B. Assessment of Status

Folate status is most often assessed using blood folate concentrations ([Table 14.3](#)). Functional indicators, including plasma homocysteine and genomic biomarkers, can also offer insight into the function and capacity of folate-dependent metabolic pathways.⁴

TABLE 14.3 Serum and RBC folate concentrations for folate status assessment.^{a,36,39}

	nmol/L	ng/mL	Interpretation
IOM criteria for dietary inadequacy			
Serum	< 7	< 3	Deficiency based on increased risk of megaloblastic anemia
RBC	< 305	< 140	
WHO criteria for insufficiency			
RBC ^b	< 906	< 400	Insufficiency based on increased risk of NTDs with population-level status below cutoff
RBC ^c	< 748	< 330	

^aHomocysteine concentrations were also considered in determining the adequacy of folate intake.

^bAssay specific cutoff for folate analysis by microbiologic assay with chloramphenicol-resistant strain and folic acid calibrator.⁴⁴

^cAssay specific cutoff for folate analysis by microbiologic assay with chloramphenicol-resistant strain and 5-methyl-THF calibrator.⁴⁴

Serum folate

Serum or plasma folate is a sensitive indicator of recent dietary folate intake and may also be used to assess long-term status when recurrent measures in the same individual are conducted overtime.

RBC folate

In contrast to serum folate, RBC folate is responsive to changes in status over the long term. Folate is taken up by developing reticulocytes in the bone marrow and not by mature RBCs in circulation, thus RBC folate represents the amount accumulated early in the ~120 day RBC life span. In addition, RBC folate concentration is associated with liver folate content and is considered representative of tissue folate stores.³⁶

Functional indicators of folate status

Plasma homocysteine concentration; Total plasma homocysteine can be a sensitive functional indicator of folate status. 5-MethylTHF is used in the remethylation of homocysteine to methionine (Fig. 14.2), and it is well established that plasma homocysteine rises under conditions of folate inadequacy.³⁶ However, elevated total plasma homocysteine concentration is not specific to folate deficiency as homocysteine levels may also be influenced by several other nutrient deficiencies (e.g., vitamin B12), genetic abnormalities, or renal insufficiency.³⁶ In folic acid–fortified populations, elevated plasma homocysteine is more likely to be associated with inadequate vitamin B12 than folate deficiency.³⁸

Genomic biomarkers; Because of the essential role folate plays in nucleotide biosynthesis and methylation reactions, folate inadequacy has the potential to impact indices reflecting DNA synthesis, repair, modification, and stability (e.g., the misincorporation of uracil into DNA, DNA methylation, and micronuclei formation).⁴ However, like plasma homocysteine, these genomic biomarkers can be affected by other nutritional and

environmental factors and cannot be used in isolation to assess folate status. Nevertheless, used in the proper context, these indicators can provide valuable information about the functional effects of folate status at the genomic level, as well as folate-mediated 1-C metabolism at the cellular level.⁴

C. Criteria of Adequacy

Criteria used to establish Dietary Reference Intakes (DRIs)

Blood folate concentration cutoffs used by the Institute of Medicine (IOM) in 1998 to evaluate folate adequacy represent the most recently established reference values for the United States and Canada (Table 14.3).³⁶ IOM deficiency cutoffs for serum and RBC folate concentrations are linked to increased megaloblastic anemia risk and were used as the basis of the determining dietary inadequacy.³⁶ Homocysteine concentrations were also considered in determining the adequacy of folate intake.³⁶

The IOM DRI process defined folate deficiency based on serum or RBC folate cutoffs of <7 nmol/L (<3 ng/mL) and <305 nmol/L (<140 ng/mL), respectively, when establishing the Estimated Average Requirement (EAR).³⁶

World Health Organization (WHO) criteria for insufficiency

In 2015, WHO defined an RBC folate cutoff for folate “insufficiency” in women of reproductive age (WRA) as an indicator of NTD risk within a population.^{39,40} In contrast to the IOM cutoffs for deficiency, the WHO cutoff was defined as the “optimal” blood folate concentration for WRA required to support rapidly dividing cells during early embryonic development and associated with the lowest NTD risk.⁴¹ WHO guidelines defined the “optimal” RBC folate concentration to achieve the greatest reduction in folate sensitive NTDs as

TABLE 14.4 Folate dietary reference intakes.³⁶

Age and/or life stage	AI (μg DFE/day)	EAR (μg DFE/day)	RDA (μg DFE/day)	UL (μg folic acid/day)
Infants				
0–6 months	65	ND	ND	ND
7–12 months	80	ND	ND	ND
Children and adolescents				
1–3 years	—	120	150	300
4–8 years	—	160	200	400
9–13	—	250	300	600
14–18	—	330	400	800
Adults*				
≥ 19 years	—	320	400	1000
Pregnant women*				
≤ 19 years	—	520	600	800
≥ 19 years	—	520	600	1000
Lactating women*				
≤ 19 years	—	450	500	800
≥ 19 years	—	450	500	1000

* Women capable of becoming pregnant are recommended to consume 400 μg folic acid/day for the prevention of NTDs in addition to intake of food folate from a varied diet. It is assumed that women will continue consuming 400 μg from supplements or fortified food until their pregnancy is confirmed and they enter prenatal care, which ordinarily occurs after the end of the periconceptional period. AI- Adequate Intake; EAR- Estimated Adequate Requirement; DFE- Dietary Folate Equivalent; ND- Not determined; RDA- Recommended Daily Allowance; UL- Tolerable Upper Intake Level .

906 nmol/L (400 ng/mL; [Table 14.3](#)).³⁹ This value, as established, should be used only to assess folate insufficiency in population groups of WRA and not to assess NTD risk of individuals and is thus a threshold value, not a mean or median value. The threshold represents the lowest end of RBC folate concentration that all WRA should achieve for optimal NTD risk reduction in the population.⁴¹

Harmonization of assays to measure blood folate

WHO recommends that RBC folate should be assessed using an MBA to obtain comparable results across countries.³⁹ In addition, as a comparison study found that different laboratories reported dissimilar MBA results depending on the microorganism or folate calibrator used,⁴² correct interpretation of folate status may require the use of adjusted assay-specific cutoffs ([Table 14.3](#)).^{43,44} To produce comparable results across laboratories and over time so that the same cutoff values can be applied, it is necessary to “harmonize” the MBA using common critical reagents and standard technique.⁵ Harmonization of the MBA for folate status assessment of populations is a key component of an ongoing international project designed to improve maternal folate status to prevent

infant NTDs in low- and middle-income countries and regions.^{4,45}

D. Reference Intakes

DRIs

IOM established DRIs for folate in 1998 ([Table 14.4](#)).³⁶ As reviewed in [Section II.B](#), folic acid in fortified foods is more bioavailable than naturally occurring food folate. To account for this difference, the folate DRI values present intake recommendations in DFEs ([Table 14.4](#)). DFEs are calculated using the following equation: μg DFE = μg naturally occurring food folate + (1.7 × μg folic acid).³⁶

The folate EAR and Recommended Daily Allowance (RDA) for adult men and women is 320 and 400 μg DFE per day, respectively ([Table 14.4](#)). By definition, the EAR is the estimated dietary folate intake at which 50% of healthy individuals in a given population maintain folate adequacy. The folate RDA is calculated as 120% of the EAR in order to provide an estimated dietary folate intake that accounts for individual variability in folate requirement and at which ~98% of healthy individuals in a given population will maintain adequacy.

The folate RDAs for pregnant and lactating women are increased to 600 and 500 μg DFE per day, respectively, in order to account for greater 1-C metabolism and DNA synthesis related to rapid cell division during pregnancy and the amount of folate excreted in breast milk.³⁶

The folate EARs and RDAs for children and adolescents ages 1–18 years were derived from adult values, while the folate Adequate Intake (AI) for infants is based on breast milk folate content (Table 14.4). The DRIs defined an AI when there is insufficient data to set an EAR and thus an RDA. All infant DRI intake values were based upon observational mean nutrient intakes of exclusively breastfed infants; thus, the infant folate AI represents the average daily folate content of breast milk provided to an exclusively breastfed infant.

Folic acid intake recommendation for neural tube defect risk reduction

Maternal supplementation with folic acid during the periconceptional period reduces the risk of an NTD-affected pregnancy⁴⁶ as discussed in Section V.A. Because unplanned pregnancy is relatively common and the NTD-protective effect of maternal folic acid supplementation is observed early in gestation (i.e., first 8 weeks), the IOM additionally recommended that all women capable of becoming pregnant consume 400 μg folic acid/d from supplements or fortified foods in addition to folate from a varied diet.³⁶ This recommendation is consistent with guidelines from other public health bodies, including the US Public Health Service, March of Dimes, and numerous professional organizations (e.g., American Academy of Pediatrics),^{47–50} and should not be confused with the folate RDA. For women at increased risk of an NTD-affected pregnancy due to a prior history of an NTD-affected pregnancy, the Centers for Disease Control recommended intake is 4000 μg folic acid per day.⁵¹

E. Basis for Individual Variability in Requirements

Genetic variation

Normal folate metabolism is dependent on genes encoding ~ 150 proteins directly or indirectly involved in folate metabolism and transport. Hundreds of single nucleotide polymorphisms (SNPs) have been identified in these genes.⁵² Genetic variants or polymorphisms in folate-metabolizing genes can alter enzyme activity and regulation as well as influence gene expression, which may result in altered concentrations of critical metabolites.⁵³

The *MTHFR* 677C $\rightarrow$ T SNP has the most robust effect on folate metabolism and requirement and is also the most extensively studied. The *MTHFR* enzyme

catalyzes the reduction of 5,10-methyleneTHF to 5-methylTHF, the methyl donor for homocysteine remethylation to methionine (Fig. 14.2). The C to T substitution at base pair 677 in the *MTHFR* gene causes the substitution of a valine for an alanine amino acid that yields reduced enzyme stability and activity and results in decreased 5-methylTHF and increased homocysteine concentrations.^{54,55} When compared with the 677CC genotype, the 677TT genotype has been associated with a 10%–35% reduction in circulating folate^{56–58} and elevated plasma homocysteine^{59,60}; however, greater intracellular folate concentrations stabilize the *MTHFR* 677T protein so that the effects of the *MTHFR* 677TT genotype are most evident when folate status is low.^{55,56} It is estimated that 11% of the US population has two copies of the 677C $\rightarrow$ T SNP (i.e., homozygous *MTHFR* 677TT genotype); however, prevalence varies widely by race/ethnicity and geography.⁵⁷

The recommended intakes for folate account for individual variability in nutrient requirement, including that due to genetic variation, by adding two estimated coefficients of variation (each 10%) to the folate EAR.³⁶ In clinical trials conducted since the folate DRIs were published, women with the *MTHFR* 677TT genotype consuming the folate RDA of 400 μg DFE per day maintain adequate folate status,^{61,62} while men with the 677TT genotype do not.⁶³ This research indicates that the coefficient of variation used to calculate the current folate RDA may not fully account for the effect of genetic variation on individual folate metabolism and requirement.

F. Tolerable Upper Intake Level

The folate Tolerable Intake Level (UL) refers specifically to folic acid consumed from fortified foods and/or dietary supplements, not naturally occurring food folate, and is expressed as $\mu\text{g}/\text{d}$ folic acid rather than DFEs of total folate intake (Table 14.4). The folic acid UL was established by the IOM in 1998³⁶ based on case reports primarily from the 1940–50s, a time in which folate deficiency could not yet be distinguished from vitamin B12 deficiency which both result in macrocytic anemia that is responsive to either B12 or folic acid.^{64,65} However, only B12 deficiency can lead to neurological consequences, which, if untreated with B12, are irreversible. In the studies reviewed in establishing the folate UL, many were single case reports of patients with macrocytic anemia who appeared to be successfully treated with folic acid (some at levels of 5–10 + mg/d) but later developed neurological disease due to failure to diagnose and treat the vitamin B12 deficiency. This led to the conclusion that an excess intake of folic acid could “mask” the diagnosis of B12 deficiency and lead to irreversible neurological damage; a concept

that was recently challenged.^{64–66} Clinical laboratory testing and practice have advanced considerably over the past 70 years with specific methods for folate and B12 status assessment. In addition, vitamin B12 therapy is now readily available, making the probability of delaying or “masking” the diagnosis of vitamin B12 deficiency with high intake levels of folic acid extremely unlikely.^{64,65}

G. Food and Supplement Sources

Humans lack the ability to synthesize folate, making exogenous sources essential. Plants and select microorganisms produce reduced folates via similar biosynthetic pathways that include formation of the pteridine ring from guanosine triphosphate, the condensation of the pteridine ring with para-aminobenzoic acid, and the addition of glutamate.^{67,68} Naturally occurring food folate is concentrated in select foods, including dark green leafy vegetables, orange juice, asparagus, strawberries, and legumes such as peanuts, black beans, and kidney beans.⁶⁹

Another source of folate is that synthesized by microorganisms in the small and large intestines.^{70,71} This source of folate was not considered in the development of reference values for the DRIs for the United States and Canada. How folate synthesis and absorption in the gut may impact folate status, intake requirements, and human health remain active areas of research.^{72,73}

Folic acid is chemically manufactured and is the form used in fortified foods and dietary supplements. Since January 1, 1998, all cereal grain products labeled as “enriched” (e.g., flour, bread, pasta, breakfast cereal, and rice) have been required by the FDA to be fortified in the United States with folic acid.⁷⁴ FDA also approved voluntary folic acid fortification of corn masa flour, a staple ingredient of Mexican cuisine not included in the original regulation, in 2016.^{75,76} Outside the United States, ~80 other countries have fortification of wheat flour, maize flour, or rice.⁷⁷ In addition to fortified foods, folic acid-containing over-the-counter dietary supplements, including those with folic acid only and those containing multiple vitamins with folic acid, are available. Prenatal supplements and therapeutic doses of folic acid may be prescribed.

H. Folate Intake and Status in the United States and in Women of Reproductive Age in Low and Middle-Income Countries

Folate and folic acid intake estimates

Total dietary folate intake includes naturally occurring food folate plus folic acid from fortified grain products (140 µg folic acid/100 g flour) and ready-to-eat (RTE) cereals, which can contain up to 400 µg folic acid/serving.⁷⁴ Using the USDA Food and Nutrient

Database for Dietary Studies, estimates of dietary folate intake are categorized as (i) naturally occurring food folate; (ii) folic acid; (iii) total folate as µg/day; (iv) and total folate as µg/day DFEs (folic acid and folate intake).⁷⁸ Folate from dietary supplements, usually in the form of folic acid, is an additional source of the vitamin.

Results from recent National Health and Nutrition Examination Surveys (NHANES) indicate most Americans achieve recommended dietary folate intakes. Analysis of NHANES 2013–16 data estimates an average daily total folate intake among adult men and women of 564 and 433 µg DFE, which includes naturally occurring food folate and folic acid from fortified foods.⁷⁸ Earlier analyses of NHANES 2003–04 and 2005–06 data examining sources of folic acid found the usual median intakes of folic acid for adults consuming enriched cereal grain products (ECGP) only, ECGP plus RTE, ECGP plus supplements (SUPP), and ECGP + RTE + SUPP were 138, 274, 479, and 635 µg/day, respectively.⁷⁹ Those most likely to meet or exceed the folic acid UL of 1000 µg/day do so by consuming dietary supplements rather than folic acid-fortified products.^{80,81}

Importantly, nonpregnant WRA are estimated to consume only 128 µg folic acid/day through diet alone, highlighting the role dietary supplements play for most women in achieving the daily 400 µg folic acid intake guideline for the prevention of NTDs.⁸²

Folate status in the United States

Since fortification of the food supply with folic acid in 1998, folate status in the United States has improved population-wide. A comprehensive evaluation of NHANES blood folate data from 1988 to 2010 demonstrates that serum and RBC folate concentration postfortification (1999–2010) increased 2.5 times and 1.5 times, respectively, compared to the prefortification period (1988–94).⁸³ Analyses of NHANES 2011–12 data found that blood folate concentrations continued to be similar to earlier postfortification analyses with <1% of the population falling below serum and RBC folate deficiency cutoffs.⁴³

Folate status of WRA in low- and middle-income countries

Primary prevention of birth defects, including NTDs, is an important strategy for meeting the health outcomes of the United Nations Sustainable Development Goals year 2030.⁸⁴ To this end, a recent technical consultation sought to determine the global prevalence of folate insufficiency in WRA as an indicator of NTD risk.⁴⁴ Current evidence on the folate status of WRA was evaluated in a systematic review that accounted for methodological issues and included results from 45 relevant studies conducted in 39 countries between 2000 and 2014.⁴⁴ The

prevalence of folate deficiency was >20% in countries with low-income economies. Only 11 of the 45 surveys reported the prevalence of folate insufficiency reflecting NTD risk, which was >40%. To better inform global efforts to prevent NTDs, additional surveys using the harmonized MBA to assess folate status are needed.

V. ISSUES RELATED TO FOLATE IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Neural tube defects (NTDs)

Congenital malformations of the central nervous system, commonly referred to as NTDs, result from a failure of fusion of the neural folds in the developing embryo during the first weeks after conception.⁸⁵ NTDs often cause permanent disability due to damage in the brain, spinal cord, and/or spinal nerves and can also be fatal. A recent systematic review reported an estimated 260,100 annual NTD affected birth outcomes worldwide, which is likely an underestimation due to lack of surveillance systems and underreporting from the regions of the world most affected.⁸⁶

Folic acid taken periconceptionally significantly reduces the risk of NTDs, a conclusion based on definitive evidence from randomized controlled intervention trials and supported by observational studies.⁴⁶ In the United States, the FDA mandated fortification of enriched grains with folic acid (as described in [Section IV](#)) in order to increase intake among WRA and reduce NTD incidence. The implementation of mandatory folic acid fortification programs in the United States and many other countries around the world has resulted in convincing reductions in NTDs during the postfortification era, providing further evidence of the efficacy of this vitamin for NTD risk reduction.⁷⁷

There are a variety of risk factors for NTDs, including environmental exposures and genetic variation. Adequate maternal folate intake and status reduce, but do not eliminate, the risk of NTD development as a certain percentage of NTDs are considered nonfolate responsive. The mechanism(s) by which maternal folate intake facilitates the reduction in NTD risk is not fully elucidated and is likely multifactorial. In some cases, maternal/fetal folate metabolism is disordered, and this metabolic error may affect uptake and/or metabolism of folate, which can in turn adversely affect gene expression critical to normal embryogenesis.⁸⁷ For example, the *MTHFR* 677T variant, a well-known modifier of folate metabolism and requirement (see [Section IV.E](#)), has been associated with an increased risk for NTD,^{88–90} and it is hypothesized that sufficient maternal

folate intake/status reduces the risk of NTDs by overcoming metabolic inefficiencies linked to genetic abnormalities.⁸⁷

De novo thymidylate synthesis may also be an important factor in the etiology of folate-responsive NTDs.²⁹ In a mouse model with impaired SHMT1 function, both folic acid and deoxyuridine (a thymidylate precursor) supplementation are protective of NTD development.^{91,92} SHMT1 is an enzyme that generates 5,10-methyleneTHF, which is used for thymidylate biosynthesis and is particularly important for proper DNA replication during cell division (see [Section III.B](#)). Impaired SHMT1 activity leads to increased uracil incorporation in DNA (vs. thymidylate), resulting in reduced DNA integrity and cell death. Taken together, this body of research suggests that dysfunctional DNA replication, related to impaired thymidylate synthesis and/or uracil accumulation, plays a role in the development of some NTDs.

Pregnancy and fetal outcomes

Pregnant women are at increased risk of developing a folate deficiency due to the accelerated demands placed on the supply of folate for the synthesis of DNA and other 1-C transfer reactions.⁹³ Beyond the folate responsive megaloblastic anemia first characterized in the 1930–40s, a number of pregnancy complications including placental abruption, preeclampsia, spontaneous abortion, stillbirths, and fetal growth restriction have been inversely associated with impaired folate status and linked to hyperhomocysteinemia; however, data are inconclusive.⁹³

Cancer

In epidemiological studies, low folate status has been associated with increased risk for leukemia, lymphoma, colorectal, breast, and other cancers.^{94–98} The mechanism by which inadequate folate status initiates carcinogenesis may involve enhanced uracil “misincorporation,” increasing the risk of mutagenic DNA strand breaks and genome instability as well as abnormalities in DNA methylation (hypo and hyper) leading to altered expression of oncogenes and tumor suppressor genes.⁴ Conversely, it has been proposed that once foci of precancerous cells are established, an elevated folate intake may accelerate the growth of established tumors.⁹⁸ However, a recent review,⁹⁹ including several expert scientific panel reports,^{66,100} failed to find evidence of a dose-response relationship between folate status or intake of folate/folic acid and increased rates of tumor growth. Notably, a metaanalysis of RCTs that included over 50,000 individuals found no significant effect of folic acid supplementation on the incidence of site-specific cancer (large intestines, prostate, lung, breast, or other specific sites) during the first 5 years of treatment.¹⁰¹

Cognitive outcomes

Associations between high folate and low vitamin B12 blood concentrations and impaired cognition in older adults taking multivitamins have been attributed to malabsorption of vitamin B12. The observed cognitive decline may be attributed to low B12 status with its associated neurological and cognitive impairments.⁹⁹ Importantly, there is considerable heterogeneity in the observational studies, with either no influence or even a protection against cognitive impairment associated with high serum folate in individuals with normal vitamin B12 status.^{99,102,103}

Vascular disease

Folic acid supplementation, either alone or in combination with other B vitamins, reduces circulating concentrations of homocysteine, a risk factor for cardiovascular disease, but has no effect on the risk of recurrent myocardial infarction.^{104–107} However, the potential for a benefit of folic acid in stroke prevention has been shown¹⁰⁸ with the greatest risk reduction observed in countries without mandatory folic acid fortification^{109,110} and among trials with a lower percent use of statins.¹⁰⁹ A benefit of folic acid supplementation in combination with an antihypertension medication on the primary reduction of stroke has also been reported among adults with hypertension in China¹¹¹ and in a metaanalysis of patients with hypertension and hyperhomocysteinemia.¹¹²

B. Select Key Nutrient Interactions

Vitamin B12

egaloblastic anemia (enlarged RBCs due to impaired DNA synthesis and cell division) can arise from a deficiency of either folate or vitamin B12. The biochemical explanation of why a single deficiency of either vitamin leads to the same megaloblastic changes in the bone marrow arises from the dependence of methionine synthase on both folate and vitamin B12. The remethylation of homocysteine to methionine involves the sequential

transfer of a methyl group from 5-methylTHF to cobalamin, and then to homocysteine, thus forming methionine. In a vitamin B12 deficiency, folate is “trapped” in the 5-methylTHF form and THF is not regenerated for the formation of 5,10-methyleneTHF, which is required for thymidylate biosynthesis and thus DNA synthesis and cell division.

Choline

The metabolism of folate and choline intersects at multiple points: (i) both 5-methylTHF and choline are methyl donors in the remethylation of homocysteine to methionine; (ii) 5-methylTHF is a source of methyl groups for SAM-dependent de novo biosynthesis of phosphatidylcholine through PEMT pathway; and (iii) choline is a source of 1-C units for folate-mediated 1-C metabolism.¹¹³ Studies in animals have reported depletion of liver choline metabolites in response to folate deficiency.¹¹⁴ In humans, dietary folate depletion decreases circulating concentrations of the choline metabolites, phosphatidylcholine, and sphingomyelin.¹¹⁵ These folate-induced changes in choline status could be due to an increased use of choline as a methyl donor or impaired de novo biosynthesis of phosphatidylcholine via the PEMT pathway.

C. Safety of Folic Acid

The safety of folic acid has been recently reviewed.⁹⁹ Folic acid is consumed via fortified foods and/or dietary supplements, with high folic acid intakes achieved nearly exclusively by supplement use (see [Section IV.H](#)). The “totality of evidence” reviewed by two recent expert panels including the British Scientific Advisory Committee on Nutrition⁶⁶ and the US National Toxicology Program,¹⁰⁰ and several recent reviews^{64,65,99} have failed to demonstrate risks for adverse consequences of folic acid intake. The data available in 1998 upon which the UL was based (see [Section IV.F](#)) were limited, and the recent expert panels^{66,100} have outlined the types of research needed to determine if adverse consequences of high folic acid intake related to vitamin B12 are of concern.

RESEARCH GAPS

- Global folate status of WRA using harmonized assays as a basis of NTD risk assessment and folic acid intervention programs
- Effectiveness of alternative folic acid fortification vehicles for NTD risk reduction
- Impact of drugs commonly used to treat diseases on folate metabolism and status
- The development of effective global in-country surveillance systems to assess efficacy of folic acid intervention programs on birth defect prevalence
- Role of RFC in intestinal and hepatic uptake of folate
- Use of computational models to enhance current understanding of folate-mediated one-carbon metabolism in the pathology of disease

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VITAMIN B12

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SUMMARY

Vitamin B12 also known as cobalamin is a cobalt-containing corrin ring cofactor required for two enzymes in higher animals, methionine synthase and L-methylmalonyl-Co-A mutase. It is a product of microbial synthesis; therefore, elaborate mechanisms of uptake and transport are necessary to insure availability of this rare nutrient. Vitamin B12 ingested in the diet binds to intrinsic factor (IF) and is internalized in the distal small intestine by the cubam receptor. Human B12 deficiency is caused mainly by lack of animal source food or lack of IF. Human deficiency causes megaloblastic anemia and/or demyelinating neurologic syndromes with deficiency, and methylmalonic acid and homocysteine increase in body fluids. Treatment with parenteral or high dose oral vitamin B12 is effective in malabsorption syndrome and corrects megaloblastic anemia. The development of inexpensive B12-containing foods acceptable for populations unwilling to eat or unable to afford animal source foods would be of benefit.

Keywords: Cobalamin; Demyelination; Homocysteine; Megaloblastic anemia; Methylmalonic acid; Vitamin B12.

I. INTRODUCTION

A. Background

Vitamin B12 (cobalamin, Cbl) holds a unique position in the science of human nutrition since there is a specific disease (pernicious anemia), which results in the isolated malabsorption of vitamin B12, and was ultimately fatal prior to the development of vitamin therapy. This unique syndrome of megaloblastic anemia and demyelinating lesions of the central nervous system was recognized and described by Thomas Addison as early as 1855.¹ Pernicious anemia has such specific manifestations and such a dramatic response to the replacement of vitamin B12 that a patient response could actually be used as a bioassay. Minot and Murphy showed that a diet containing large amounts of liver would stimulate red blood cell production in patients with pernicious anemia in 1926.² Their early work eventually culminated in the purification of vitamin B12 from liver by Folkers at Merck, Sharp and Dohme in 1948, and Smith and Parker at Glaxo in the same year.³ During the next 50 years, the structure, X-ray crystallography, synthesis of coenzyme forms, and chemistry and biochemistry of the

cobamides have been elucidated.⁴ Much of what we know about the role of vitamin B12 in metabolism and the pathophysiology of the deficient state has been determined by studying patients with pernicious anemia since this accident of nature results in a very selective model of a deficiency disease not complicated by protein-calorie malnutrition or multiple vitamin and mineral deficiencies.

B. Vitamin B12 Function and Structure

Structure of vitamin B12

The structure of vitamin B12 (OH-Cbl) is shown in Fig. 15.1. It is a very complex molecule containing a corrin ring, (which coordinates a cobalt molecule), a base (5,6-dimethylbenzimidazole), a sugar, and an aminopropanol group. An upper axial ligand is coordinated to the cobalt and can be hydroxy, cyano, glutathione or the coenzyme forms, methyl (CH₃), and adenosyl.⁵ The chemistry of the carbon-cobalt bond found in the coenzyme forms is unique and has been studied extensively.^{5,6} Only microorganisms retain the ability to synthesize Cbls, and microbial synthesis pathways have

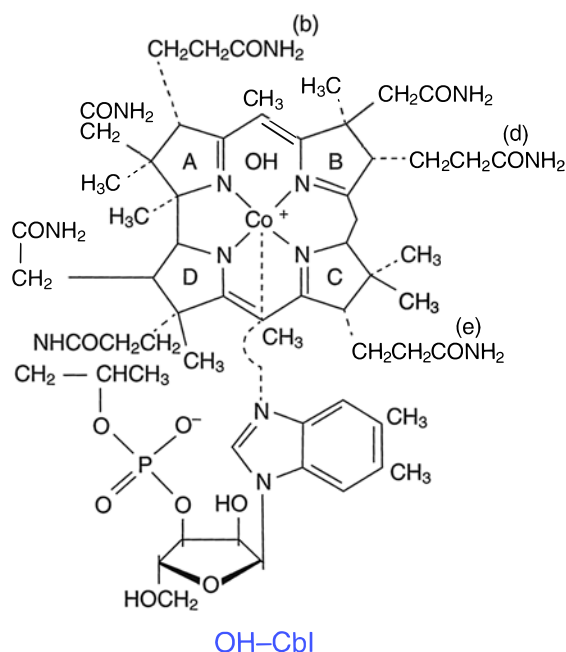


FIGURE 15.1 The structure of OH-cobalamin is shown.

been elucidated in a series of elegant studies.⁷ Plants do not use Cbl; therefore, the source of Cbl in all higher animals is the product of microbial synthesis. Analogues of Cbl with bases other than 5,6-dimethylbenzimidazole are widely found in nature including in the human intestinal tract⁸ but do not support coenzyme activity in higher animals, although they are utilized by various microorganisms.

Function of vitamin B12

Microorganisms use different forms of Cbl in many reactions, including methionine synthesis, carbon skeleton mutation, elimination reactions, aminomutations, and acetate and methane synthesis.⁷ Higher animals require Cbl as a cofactor for only two enzymes, L-methylmalonyl-CoA mutase and methionine synthase.⁵ The binding of the coenzyme forms to these two enzymes has been studied by X-ray crystallography.^{7,9–11} It appears that the dimethylbenzimidazole is displaced from the cobalt, and a histidine residue from the enzyme coordinates at that position. Although both of these enzymes use Cbl, they perform very different types of chemical reactions. During methionine synthesis and other methyl transfer reactions, the methyl–cobalt bond undergoes heterolytic cleavage, whereas the mutase performs a homolytic cleavage of adenosyl-Cbl, which results in the formation of a radical. Methionine synthase must be reactivated since the Cbl cofactor is occasionally oxidized (every 2,000 cycles).¹² Recent reviews describe structural–functional relationships of the Co–carbon bonds in the coenzymes and the role bond cleavage plays in the specific reactions.^{7,12}

II. NUTRIENT METABOLISM

A. Synthesis

Vitamin B12 is only synthesized by microorganisms including bacteria and archaea. There are two biosynthesis routes, aerobic and anaerobic.^{13,14} Some microorganisms can use a salvage pathway wherein the corrin ring is combined with bases other than 5,6-dimethylbenzimidazole to form alternative corrinoids that are preferred by some bacteria.¹⁵ There are more than 25 different steps in the synthesis. The commercial production of vitamin B12 has utilized the classic organisms *Pseudomonas denitrificans*, *Propionibacterium shermanii*, or *Sinorhizobium meliloti*. However, *Escherichia coli* is increasingly used as an organism to produce useful chemicals, including Cbl after gene transfers from other organisms.¹⁴ An ATP-binding cassette transport system is used to transport corrinoids into Gram-negative bacteria.¹⁵ Large oral doses of vitamin B12 in human subjects increase the fecal content of multiple analogues, which is proof of the importance of the salvage pathway conserving the corrin ring for bacterial metabolism.⁸

B. Digestion

Vitamin B12 is generally bound to one of the two enzymes or other carrier proteins in food; therefore, it must be released prior to absorption. This process starts when food is chewed and mixed with saliva, which contains a Cbl-binding protein, haptocorrin (R-protein). The bound Cbl is further released in the acid environment of the stomach where peptic digestion of proteins begins. The released vitamin B12 is bound to haptocorrin and carried to the duodenum. The specific Cbl-binding protein gastric intrinsic factor (IF) is released by the gastric parietal cells but does not bind Cbl until stomach acid is neutralized in the duodenum, and digestive enzymes remove the haptocorrin from the vitamin.^{16,17} The binding of IF is very specific for Cbl and thus may prevent the binding and absorption of Cbl analogues also found in foods. The IF-Cbl is carried to the ileum where it is taken up by the cubam receptor (formed from cubilin and amnionless, a transmembrane protein)^{18,19} in the brush border. The endocytosed complex is degraded in lysosomes and is eventually released from the cell bound to transcobalamin (TC). Inborn errors of IF, cubilin, or amnionless cause juvenile megaloblastic anemia and Imerslund-Grasbeck syndrome.^{20,21} Two interacting transporters LMBD1 and ABCD4 are necessary for releasing Cbl from the lysosome into cytoplasm.²² The CblIF defect has now been shown to be defective LRMB01²³ and CblJ a defect in ABCD4.²⁴

It is not known exactly how the absorbed Cbl is then processed so that a complex of transcobalamin-Cbl (TC-

Cbl) is secreted into the portal blood and thus delivered to the liver and ultimately all tissues.

C. Transport

TC is the primary serum transport protein for vitamin B12. The TC-Cbl complex is taken up by receptor-mediated endocytosis by the recently purified TCblR¹⁶ and fused to lysosomes in which the Cbl is released. Recent evidence suggests that Cbl export from cells may be mediated by the ATP-binding cassette drug transporter, ABCC1.²⁵ Cbl in the circulation is bound to haptocorrin (R-binder, TCI, and III) 70%–90%, and TC, 10%–30%. The function of haptocorrin is unclear but could be protective since it also binds corrinoids other than Cbl and can deliver them to the liver.²⁶ Defects of TC cause severe megaloblastic anemia, elevations of methylmalonic acid (MMA) and homocysteine, neurologic disease, and immune deficiency in the first few months of life.^{9,27} Deficient haptocorrin appears to be relatively common, a cause of the “false positive” low serum B12 level and is a benign condition.

D. Metabolism

After the Cbl is released from the lysosomes, the coenzyme forms must be synthesized. A number of defects of intracellular Cbl metabolism and transport have been described, which form eight complementation groups (CblA–CblI, CblF, CblG, CblJ) with three generally similar clinical syndromes.^{17,27} If there is a defect in

methionine synthase or in regenerating $\text{CH}_3\text{-Cbl}$ (MS, CblE, CblG), the patients have only hyperhomocysteinemia. Other complementation groups have defects in L-methylmalonyl-CoA mutase or in the formation of adenosyl-Cbl (mut^0 , mut^- , CblA, CblB), and these patients have only methylmalonicacidemia. However, there are also patients who have difficulty with the lysosomal release of Cbl (CblF, CblJ), described above, or with a reduction step common to the synthesis of both coenzymes (CblC, CblD) who have combined hyperhomocysteinemia and methylmalonic aciduria.^{27,28} Recently CblD has been shown to be associated also with isolated hyperhomocysteinemia or methylmalonic acidemia.²⁹

The pathway of propionyl-CoA metabolism is shown in Fig. 15.2. The metabolism of various amino acids such as valine, isoleucine, and odd chain fatty acids results in the formation of propionyl-CoA, which is carboxylated to form D-methylmalonyl-CoA. A racemase interconverts the two isomers, and L-methylmalonyl-CoA is a substrate for the adenosyl-Cbl-dependent enzyme L-methylmalonyl-CoA mutase.³⁰ D-methylmalonyl-CoA can be hydrolyzed to MMA. The mutase is a mitochondrial matrix enzyme. It has been purified from human³¹ and other sources, and the genes have been cloned and sequenced also from human³² and bacterial sources. The enzyme is a homodimer that binds 2 mol of adenosyl-Cbl per dimer. During the last decades, the reaction mechanisms have been intensively studied.⁷

The mutase is not completely saturated with adenosyl-Cbl in vivo.³³ OH-Cbl infusions in rats over 14 days increased holo L-methylmalonyl-CoA mutase activity

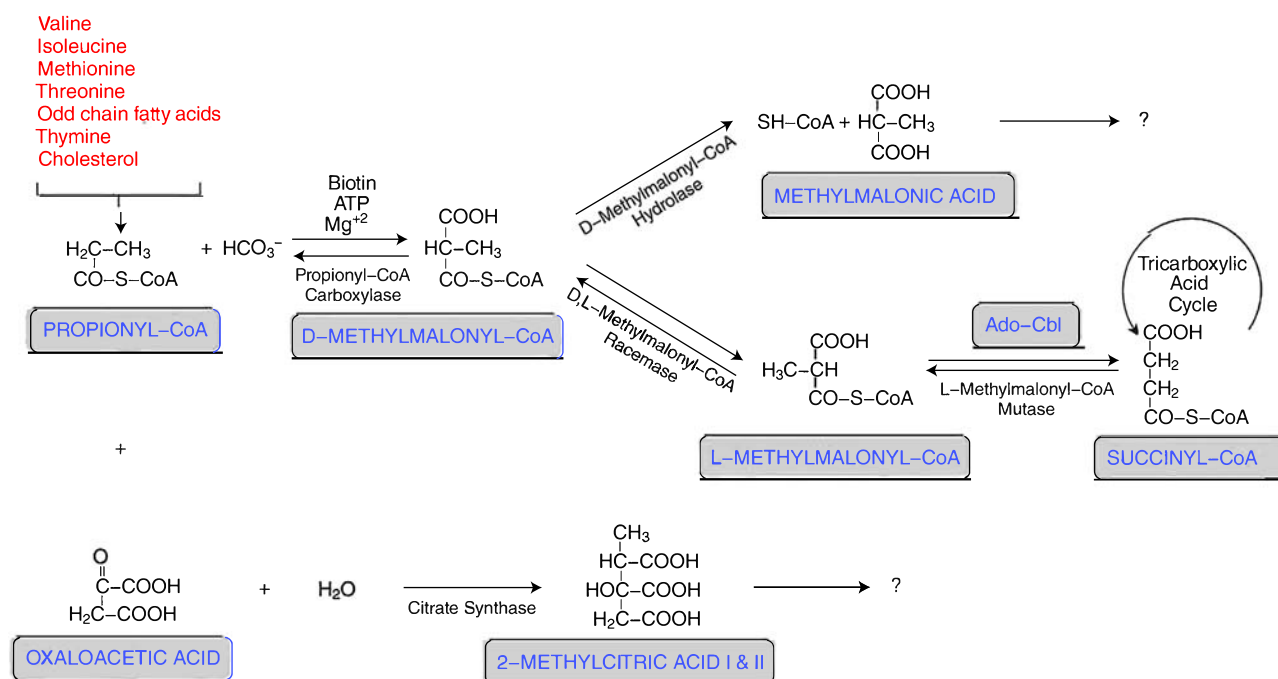


FIGURE 15.2 The pathways of propionyl-CoA and methylmalonyl-CoA metabolism are shown.

and decreased serum MMA.³⁴ High-dose oral vitamin B12 lowered mean MMA concentrations in normal subjects who had baseline concentrations >240 nmol/L.³⁵

This pathway is an important source of energy in ruminants since the bacterial fermentation in the rumen produces large quantities of propionic acid. However, this pathway is also important in humans since congenital defects of the mutase or of the ability to synthesize adenosyl-Cbl result in life-threatening methylmalonic aciduria, which is complicated by severe metabolic ketoacidosis.^{27,30} Serum, urine, and CSF MMA concentrations are always increased in conditions of B12 deficiency,^{36,37} human or animal nitrous oxide–induced Cbl inactivation^{34,38} experimental animal deficiency³⁴ ruminants with cobalt deficiency³⁹ and Cbl deprived tissue culture cells.⁴⁰

Methionine synthase

The pathway of methionine metabolism is shown in Fig. 15.3.⁴¹ Methionine is an essential amino acid that is necessary for protein synthesis and also is a crucial methyl donor after activation to S-adenosylmethionine (SAM). SAM is a source of methyl groups for the synthesis of creatine and phospholipids (quantitatively the most important together) and also for synthesis of neurotransmitters, DNA, RNA, and protein methylation.⁴² After donating the methyl group, S-adenosylhomocysteine (SAH) is

formed, which is cleaved to homocysteine and adenosine by SAH-hydrolase. The homocysteine that is generated can either be remethylated to form methionine or condensed with serine to form cystathionine by a vitamin B6–dependent enzyme, cystathionine beta-synthase. Cystathionine is further metabolized to cysteine and alpha ketobutyrate by another vitamin B6–dependent enzyme, gamma-cystathionase.⁴²

Methionine can be synthesized by two different enzymes, Cbl-dependent methionine synthase and betaine-homocysteine methyltransferase.^{43,44} Methionine synthase transfers a methyl group from 5-methyltetrahydrofolate to homocysteine. The methyl-Cbl bound to the methionine synthase is demethylated in the reaction with the homocysteine and then remethylated by the reaction with 5-methyltetrahydrofolate. SAM is also required because occasionally the active Cbl form, cob(1)alamin, is oxidized and must be reduced by obtaining a methyl group from SAM.¹² Recent investigations show that there is also a human methionine synthase reductase which is a flavoprotein.⁴⁵ Methionine synthase is a cytoplasmic enzyme. It contains 1 mol of Cbl per mol of protein. Mechanisms of methionine synthase have been studied extensively.⁴³

It is readily appreciated from Fig. 15.3 that the balance of methionine metabolism is dependent on three vitamins, vitamin B12, folate, and vitamin B6. Congenital

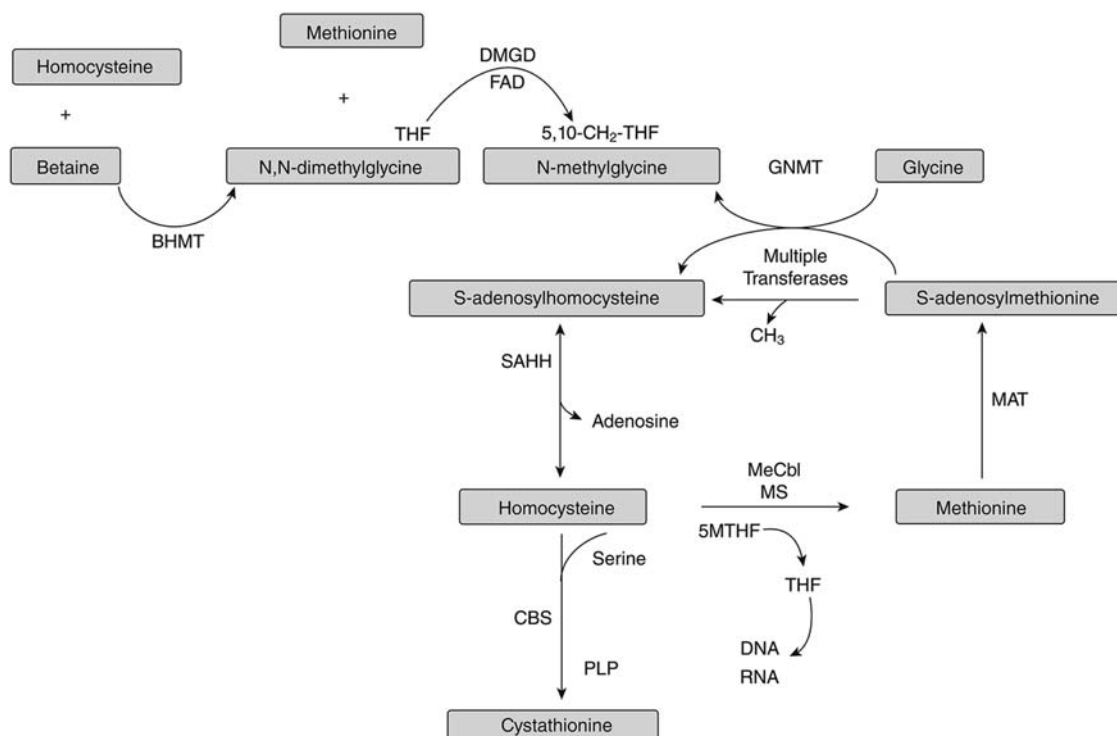


FIGURE 15.3 The pathways of methionine metabolism are shown. The enzymes shown are as follows: *BHMT*, betaine-homocysteine methyltransferase; *CBS*, cystathionine beta-synthase; *DMGD*, dimethylglycine dehydrogenase; *GNMT*, glycine N-methyltransferase; *SAHH*, S-adenosylhomocysteine hydrolase; *MAT*, methionine adenosyltransferase; *MS*, methionine synthase.

defects of the methionine synthase (CblG), methionine synthase reductase (CblE), or the synthesis of methyl-Cbl (CblC/D) have been described, and they cause severe hyperhomocysteinemia.^{17,27} Deficiencies of either vitamin B12 or folate also result in hyperhomocysteinemia.^{46,47} Elevations of serum total homocysteine (tHcy) are associated with accelerated vascular disease and thrombosis, thus this field of congenital and acquired defects has generated much interest.⁴⁸

Homocysteine is at a branch point between the remethylation to methionine or the reaction to form cystathionine, which is termed transsulfuration. Transsulfuration is not reversible and eventually results in removal of the sulfhydryl group. Cystathionine beta-synthase is activated by SAM; thus, in conditions of high methionine intake, transsulfuration is stimulated and homocysteine is removed.⁴⁹ The concentrations of SAM are regulated by the methylation of glycine by glycine N-methyltransferase an enzyme that is found in high concentrations in the liver.⁵⁰ This reaction is inhibited by 5-methyltetrahydrofolate, a form of folate which may accumulate in B12 deficiency because methionine synthase is impaired.

Another area of interest is in the regulation of the concentrations of cystathionine since both vitamin B12 and folate deficiency cause accumulation of serum cystathionine, despite the fact that there is experimental evidence and theoretical reasons to believe that the deficiency of SAM in these conditions should result in a decrease in transsulfuration and the formation of cystathionine.⁵¹

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Gene Regulation and DNA Synthesis

Gene regulation

Several patients who appeared to have a CblC mutation but when studied had no mutations in the respective gene MMACHC were discovered. Instead, there were mutations in a transcriptional coregulator host cell factor C1.⁵² This factor forms a complex with transcription factor THAP domain containing protein 11 and zinc finger protein 143, and mutations in these proteins have also been shown to cause metabolic disease resembling CblC deficiency.⁵³ An epigenetic silencing of MMACHC has also been described.²⁷ No regulatory functions have been described for the other complementation groups reviewed in Ref. 9.

DNA synthesis

Fig. 15.3 demonstrates that there are interactions between vitamin B12 and folate metabolism, which are important for hematopoiesis.⁵⁴ The irreversible reaction

forming 5-methyltetrahydrofolate results in a metabolically inactive form of folate unless it is demethylated by methionine synthase. 5-Methyltetrahydrofolate is a poor substrate for folyl polyglutamate synthase and thus is not retained intracellularly and can be lost in the urine. Thus, there is “trapping” of folate as 5-methyltetrahydrofolate in severe vitamin B12 deficiency, and this may result in secondary folate deficiency.⁵⁵ N-Methylglycine is elevated in folate-deficient but not B12-deficient patients suggesting that “trapped” methylfolate may be bound to and inhibiting glycine N-methyltransferase⁴¹ further sparing SAM but not SAH concentrations in B12 deficiency.^{50,56} Since other forms of folate are necessary for thymidine and purine synthesis, there are resulting impairments in DNA synthesis likely causing megaloblastosis. There are also abnormalities in one carbon metabolism in vitamin B12 deficiency, probably due to the secondary folate deficiency.⁵⁷

B. Pathophysiology of Vitamin B12 Deficiency

Megaloblastic anemia

In humans, vitamin B12 and folate deficiency cause identical forms of megaloblastic anemia.^{58,59} Thus, the secondary block in folate metabolism in vitamin B12 deficiency must be the underlying cause. This constellation of morphologic changes and clinical and laboratory abnormalities is due to an imbalance of decreased DNA synthesis with adequate RNA synthesis. Impaired methylation due to an increase in SAH, which inhibits methyltransferases, may also be important.⁵⁶ The nuclei of the developing hematopoietic precursor cells in the bone marrow remain immature as compared to the cytoplasm. The morphologic result is a macrocytic red blood cell precursor (high mean cell volume), with an open chromatin pattern. The white blood cells are enlarged with hypersegmentation of granulocyte nuclei. Many cells die in the bone marrow (ineffective erythropoiesis) possibly by apoptosis that leads to the cellular release of bilirubin and lactate dehydrogenase. The megaloblastic nuclei demonstrate decreased DNA synthesis, increased precursors in S phase, misincorporation of uracil into DNA, little evidence for decreased DNA methylation, chromosomal breakage, and rescue by both purines and thymidine.⁵⁹ The cellular bone marrow packed with very immature appearing erythroblasts has led to the mistaken diagnosis of acute leukemia.⁵⁸ The fragmentation of the red blood cells may also lead to erroneous diagnoses such as hemolysis or thrombotic microangiopathes.⁶⁰ Megaloblastic anemia can be completely cured with vitamin B12 treatment and can be partially corrected with folate treatment which leads to diagnostic errors, as has been thoroughly reviewed by Savage and Lindenbaum.⁵⁴ Ultimately, many of such

incorrectly treated patients succumbed to relapse of anemia and/or neurologic disease.

Neurologic abnormalities

Vitamin B12 deficiency, whether naturally occurring or induced by nitrous oxide (which inactivates methionine synthase), leads to a demyelinating disorder of the central nervous system in humans,^{38,61,62} nonhuman primates,⁶³ fruit bats,⁶⁴ and swine^{65,66} but not other animal species. This lesion was described long before the underlying vitamin B12 dependence was recognized. It has been termed “subacute combined degeneration” or “combined systems disease.” The pathology includes swelling of myelin sheaths and patchy vacuolation of myelin with spongy degeneration of the spinal cord, starting in the thoracic and cervical dorsal columns and progressing to the lateral columns. Lesions are also seen in the brain and optic nerves and possibly peripheral nerves.^{58,61} Signs and symptoms vary considerably; however, the most common symptom seen in a large review of well-documented cases was painful paresthesia of the extremities.⁶¹ The most common sign was loss of proprioception and vibration sense in the toes or ankles. One of the most intriguing discoveries has been the strong inverse relationship between the severity of the hematologic and the neurologic abnormalities.^{61,67} Only about one-third of patients with megaloblastic anemia will develop the neurologic abnormalities, and 25% of these will have almost completely normal hematologic parameters.⁶⁷ Serum MMA and homocysteine and all of the other metabolic changes are similar in the subjects with involvement of the nervous system as compared to those with just hematologic involvement.⁶⁸ Vitamin B12 treatment will partially or completely correct the lesions if begun promptly.⁶¹

The underlying biochemical abnormalities leading to demyelination of the central nervous system are not understood. There is intriguing negative data from observation of individuals with methylmalonic aciduria due to mutase defects who do not develop the demyelinating disease of the spinal cord seen in vitamin B12 deficiency.³⁰ Likewise, patients with severe folate deficiency and elevations of homocysteine similar to the levels seen in vitamin B12 deficiency do not develop myelopathy.⁵⁴ However, the combined defects (CblC, CblD, CblF) do cause central nervous system disease very similar to that seen with B12 deficiency.²⁷ It appears that the activity of both Cbl-dependent enzymes needs to be impaired in order to cause the demyelinating disease, although defects of methionine synthase might be an exception. There is also no explanation for why most animal species are resistant to the development of spinal cord defects. A gastrectomized rat model has shown abnormalities of cytokines and growth factors⁶⁹ in the nervous system. Similar changes were found in humans with B12 deficiency.⁷⁰ Magnetic resonance

imaging techniques of the brain and spinal column demonstrate the areas of demyelination and surprisingly often show improvement long before signs and symptoms abate.⁶² Functional studies have shown brain choline depletion, especially in infants.^{71,72}

There are alterations in the total coenzyme-A pools and in carnitine metabolism as a result of the buildup of methylmalonyl-CoA in animal models of vitamin B12 deficiency.^{73,74} The increase in propionyl-CoA appears to lead to increased formation of branched chain and odd chain fatty acids in both inborn errors of the mutase and severe vitamin B12 deficiency.^{75,76}

Box 15.1 shows the metabolites; MMA, homocysteine, SAH, cystathionine, 2-methylcitric acid, and N, N-dimethylglycine that are found in elevated concentrations in body fluids (serum, urine, cerebrospinal fluid) and tissues in both humans with pernicious anemia and other forms of vitamin B12 deficiency as well as experimental models of vitamin B12 deficiency. Cox and White in 1962 demonstrated that 95% of symptomatic vitamin B12-deficient humans had elevated concentrations of urine MMA that responded to therapy.³⁶ Their work has been repeatedly confirmed, and it is now well accepted that virtually every patient who has clinical abnormalities that will respond to vitamin B12 replacement has elevated MMA.^{47,77–79} Elevations of MMA are highly specific to Cbl metabolism with the exception of modest elevations seen in chronic renal insufficiency.^{80,81} Thus, monitoring MMA is extremely useful in diagnosis, studying response to treatment, and in documenting vitamin B12 deficiency in animal or cellular models.

Fig. 15.2 shows that propionyl-CoA can be condensed with oxaloacetic acid to form 2-methylcitric acid. Elevated concentrations of 2-methylcitric acid are found in patients with inborn errors of mutase and in severely affected vitamin B12-deficient subjects.⁸¹

tHcy is frequently elevated in the serum in humans with pernicious anemia^{46,47} and other forms of vitamin B12 deficiency as well as animal models of vitamin B12 deficiency.³⁴ Elevated homocysteine is not specific to vitamin B12 deficiency, in contrast to MMA, but is found in folate deficiency, defects of cystathionine beta-synthase, methionine adenosyltransferase, 5-10-methylenetetrahydrofolate reductase, methionine synthase, or in the methyl-Cbl reductive pathway.⁴⁸ Homocysteine concentration is highly correlated with the serum creatinine, at least partially because of the role of SAM in the formation of creatine, the precursor of creatinine.⁸² The mean homocysteine value of a population is highly dependent on the dietary and supplement intake of folate and vitamin B12. These additional influences make elevated tHcy less specific as a diagnostic tool for vitamin B12 deficiency. However, it has been shown that elevated homocysteine due to vitamin B12 deficiency will not correct with folic acid treatment.⁸³

BOX 15.1**Laboratory Abnormalities in Vitamin B12 Deficiency**

<i>Metabolites</i>		
	High	Methylmalonic acid Total homocysteine Cystathionine 2-Methylcitric acid N, N-dimethylglycine S-Adenosylhomocysteine
<i>Megaloblastic anemia</i>		
Peripheral blood	Low	Red blood cell count Hemoglobin and hemocrit White blood cell count Platelet count
	High	Mean cell volume
	High	Red cell distribution width Hypersegmented granulocytes Hypercellularity Nuclear-cytoplasmic dysynchrony Open immature nuclear chromatin pattern Giant metamyelocytes and bands Intramedullary cell death
Bone marrow		
<i>Blood chemistry</i>		
		Increased unconjugated bilirubin Increased lactate dehydrogenase Decreased haptoglobin

The serum cystathionine concentration is elevated in most subjects with severe vitamin B12 deficiency.⁵¹ Like the tHcy, it is not specific to vitamin B12 deficiency since folate deficiency and especially vitamin B6 deficiency also cause elevated concentrations.

The clinical manifestations of vitamin B12 deficiency have been best studied in the human with pernicious anemia. Although there are spontaneously occurring animal models of vitamin B12 deficiency such as ruminants pastured on cobalt deficient soil³⁹ and dogs⁸⁴ with selective malabsorption of vitamin B12, these animals do not display similar syndromes. In one investigation, Cbl analogue treatment of minipigs caused spinal lesions and mild anemia and leukopenia although the rise in mean red cell volume and morphologic changes in the bone marrow were very subtle as compared to what would be expected in a human.⁶⁶ The lack of convenient animal models has certainly been a detriment to the study of the pathophysiology of vitamin B12.

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

The gold standard indicator of B12 deficiency is elevated MMA in body fluids or tissues (animal models) that normalizes with B12 supplementation.^{17,34,37,40,46,47} The serum Cbl level has poor sensitivity and specificity although it has often been used to assign status.⁵⁸ Elevated homocysteine that falls with sole B12 treatment is also a sensitive and specific indicator.⁴⁶

B. Criteria of Adequacy

Adequate Cbl status is that which provides normal growth and development of children and also prevents myelopathy and megaloblastic anemia at all ages. There may be no clinical effects in mild metabolic deficiency since MMA is a very sensitive indicator of mutase

function and thus adequate Cbl coenzyme. Improving B12 nutrition alone in resource poor settings is often disappointing since iron, folate, zinc, and protein deficiency often coexists.^{17,85} In such situations, it is difficult to determine what role the inadequate B12 status is playing in adverse health outcomes.

C. Assessment of Status

The mainstay of assessment of vitamin B12 deficiency has historically been the demonstration of a low serum Cbl level,^{86,87} although the specificity and sensitivity vary according to lab reference ranges. Many methods have been used to assay total Cbl in serum, starting with microbiologic assays with *Lactobacillus leichmannii*. The original IF-competitive binding assays that utilized radioisotopes have been replaced with chemiluminescent microparticle immunoassays and chromatographic methods.⁸⁸ Standardization between assays is problematic.^{58,89,90} The new assays usually report higher values than the historical assays and technical problems with samples with large amounts of blocking IF antibodies exist. Surveys from accredited labs show marked variability of results on the same samples. The use of serum assays to assign status is also problematic because only about 20% of total serum Cbl is carried by TC (the cellular delivery protein). Also, the known roles for Cbl are all intracellular, so tissue levels may not necessarily reflect those found in the plasma. In general, there is a rough correlation of status with serum Cbl values such that extremely low concentrations (<100 pg/ml) often do indicate clinical deficiency and values above the mean for a population generally can be interpreted as adequate.^{37,58} Values between 100 and 300 pg/ml include a mix of patients with elevated MMA and clinical deficiency and false positive low values, the proportions depending on the pretest probability of deficiency in the cohort.⁵⁸ There is a theoretical advantage to measuring Cbl on TC, known as holo-TC, and tests are commercially available in Europe and Canada, but not yet the United States. However, diagnostic accuracy for elevated MMA is about equal between total Cbl and holo-TC.^{91,92}

Both the serum MMA and homocysteine concentrations can reach extremely high levels in symptomatic vitamin B12 deficiency, as shown in Fig. 15.4. Only 3 out of 313 episodes of well-documented vitamin B12-deficient megaloblastic anemia had normal concentrations of both metabolites.⁴⁷ The MMA and homocysteine were elevated in 121 subjects who had neurologic syndromes without anemia in the same report, and as can be seen in Fig. 15.4, patients with normal hematocrit are found at all levels of metabolites.

Elevated values of MMA and/or homocysteine due to inadequate Cbl status will fall immediately on replacement of the vitamin and can be used to document

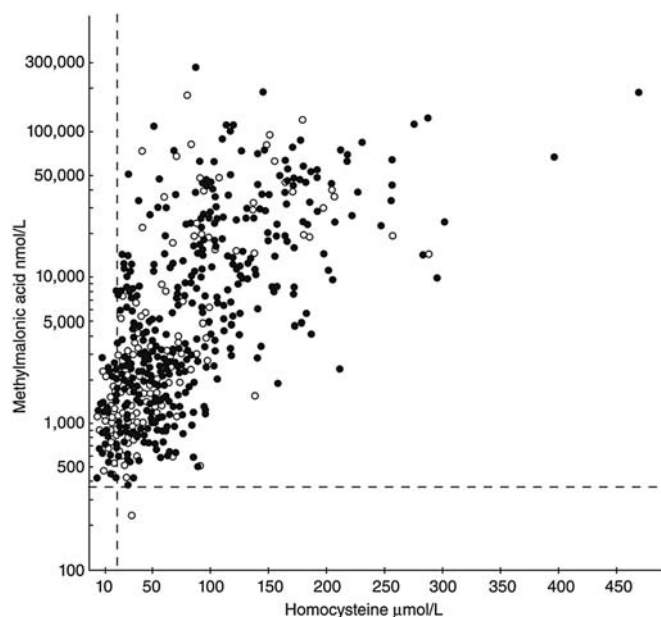


FIGURE 15.4 Serum methylmalonic acid and total homocysteine concentrations are shown in 491 episodes of B12 deficiency. Closed circles represent patients with hematocrit less than 38% and open circles greater than or equal to 38%. Patient values are collated from the following refs.^{37,38,46,47,56,60,67,79,83,106,110,112,113,120,123} The hatched lines represent three standard deviations above the mean for normal controls, MMA; 376 nmol/L and total homocysteine, 21.3 μmol/L.⁴⁷

inadequate intake or absorption in individuals or populations.^{46,79} Gradual correction of megaloblastic anemia and/or myelopathy will follow adequate replacement. If MMA is not elevated in the untreated patient, despite seemingly inadequate Cbl level, there will be no change in clinical abnormalities.⁷⁶

D. Reference Intakes

The minimum quantity of B12 required to sustain life is not known, but is likely less than 0.5 mcg per day, but would not maintain normal biochemical values. Although the most recent RDA for Canada and the United States was set in 1998 at 2.4 mcg for adults,⁹³ MMA is higher in those with intakes less than 6 mcg indicating inadequacy, thus dietary supplements should contain more than 2 μg.⁹⁴ Recommended intakes for pregnant and lactating women were increased to 2.6 and 2.8 μg, respectively. For children, the RDAs increase from 0.4 in infants to 1.8 μg in adolescents. Given the high proportion of older individuals unable to absorb naturally occurring vitamin B12 due to atrophic gastritis and/or achlorhydria, it was recommended in 1998 that those older than 50 years meet their RDA mainly through fortified foods or dietary supplements although the amount was not specified.⁹³

The IF-mediated absorption is saturable and inversely proportional to the dose given, with 50% of 1 mcg and 20% of a 5 mcg dose absorbed.^{95a,96} Doses separated by

4–6 h are absorbed as initially. The bioavailability is almost 50% from food but again is dependent on the dose taken with less from the foods with the highest content. The total body content of vitamin B12 is estimated between 1,000 and 4,000 mcg, although there is a large range as reviewed in Ref. ^{95a}. Daily losses were about 0.13% per day or less than 1 mcg/day. Vitamin B12 is excreted into bile, which if not bound to IF in the duodenum will be lost to the body. The estimated reabsorption is 67% in normal individuals.^{95a}

US survey data from the NHANES What We Eat in America 2013–16 cohort reported the median vitamin B12 consumption for all adult men of 5.1 mcg and women of 3.5 mcg.^{95b} Using the Estimated Average Requirement (EAR) for adults for Vitamin B12 of 2 mcg,⁹³ less than 3% of men and 8% of women in the United States had inadequate diets using this comparator. However, 11% of girls 14–18 years had intakes less than their EAR of 2.0 mcg.

E. Basis for Individual Variability in Requirements

It is not known if there are differences in requirements for absorbed vitamin B12 even though research has uncovered a number of polymorphisms in TC.⁹⁷ However, there is variability in the ability to absorb B12 from the diet since a healthy and intact gastrointestinal tract is necessary with normal peptic activity, gastric acid, IF excretion, and normal ileal mucosa. The

common conditions causing malabsorption of B12 are described in [Section V](#) and in [Box 15.2](#).

F. Tolerable Upper Intake Levels

The Institute of Medicine report on dietary reference intakes stated that there would be no upper limit for vitamin B12 because no adverse effects had been associated with it in healthy individuals.⁹³ Hydroxy-Cbl in gram quantities is used as an intravenous treatment for cyanide toxicity without any side effects. Patients with CblC mutations and other inborn errors are treated with high dose hydroxy-Cbl injections 0.1 mg/per kg daily.⁹⁸ Concerns about the cyanide in the common supplement, cyanocobalamin, are unwarranted as the quantities are much smaller than allowable cyanide in drinking water.⁹⁹

Very rarely injected Cbl has been associated with an anaphylactic-like reaction. A few patients have reported an acneiform rash after frequent Cbl injections.¹⁰⁰ It has been reported that *Propionibacterium acnes* in the skin converts the corrin ring from the Cbl into porphyrin that can then cause inflammation.¹⁰⁰

G. Dietary Sources

All animals require vitamin B12 and obtain it ultimately from the products of microbial synthesis. Ruminant animals actually carry bacteria that synthesize Cbls in their rumen.¹⁰¹ Humans obtain vitamin B12

BOX 15.2

Causes of Vitamin B12 Deficiency

Animals

Cobalt-deficient soil—ruminants only
Congenital ileal malabsorption—dogs and cats

Human

Dietary

Lack of foods of animal origin
Vitamin B12—deficient breast milk

Malabsorption

Pernicious anemia
Protein-bound B12 malabsorption
Pancreatic insufficiency
Jejunal bacterial overgrowth
Fish tapeworm

Tropical sprue

Total or partial gastric resection or bypass
Ileal disease or resection
Ileal urinary conduit

Drugs

Nitrous oxide
Metformin
Stomach acid blockers

Congenital

Intrinsic factor defect
Transcobalamin defect
Immerslund—Gräsbeck syndrome
CblA—CblJ defects

from the products of animal origin including meats, fish, shellfish, etc., dairy products, and eggs.^{85,102,103} Organ meats and shellfish have the highest amounts. Improved methods of corrinoid analysis have shown that fermented vegetable foods, some algae, and other plant products may contain B12.¹⁰² Synthetic vitamin B12 has been added to many prepared foods such as cereal, sport nutrition products, and other supplemented foods, especially those targeted to vegetarians in the United States.

V. ISSUES RELATED TO VITAMIN B12 IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Acquired vitamin B12 deficiency

The major causes of acquired vitamin B12 deficiency are shown in Box 15.2. By far, the most common causes of vitamin B12 deficiency relate to acquired malabsorption.

The autoimmune disease, pernicious anemia, is the most common cause of severe malabsorption of vitamin B12.^{17,58} Chronic autoimmune atrophic gastritis (type A atrophic gastritis) develops with antibodies to gastric parietal cells and in 50% of the individuals also to IF.^{104,105} The gastric parietal cell H^+/K^+ -ATPase has been shown to be the antigen against which parietal cell antibodies are directed. There is a complete loss of IF so vitamin B12 malabsorption ensues. The disease is found in all races and ethnic groups throughout the world^{85,106,107} and the prevalence increases with age and female sex.^{87,108} Though rare, pernicious anemia occurs even in young people and those of African origin may be at higher risk for young onset.¹⁰⁸ One study found that the prevalence in persons over age 65 years was 1.9%.¹⁰⁹ In pernicious anemia, there is impaired enterohepatic circulation of Cbl since the Cbl secreted in the bile cannot be bound to IF and is lost in the stool leading to more rapid depletion of Cbl when treatment is discontinued.³⁷

Surgical manipulations of the gastrointestinal tract including total and partial gastrectomy, bariatric surgery such as gastric bypass and gastric sleeve operations, ileal resections, or ileal urinary conduit constructions frequently lead to vitamin B12 malabsorption.^{105,110–112} Parasitic infection with the fish tapeworm and jejunal bacterial overgrowth, chronic inflammatory diseases of the ileum such as Crohn's disease, and tropical sprue have been shown to cause vitamin B12 deficiency. *Helicobacter pylori* may be a cause of atrophic gastritis also.¹⁰⁵

Up to 15% of elderly subjects are found to have mild-to-moderate vitamin B12 deficiency documented by elevated MMA and homocysteine concentrations and decreased serum Cbl levels.^{112–114} Only a minority of these subjects discovered in screening studies have true pernicious anemia, and the others may have abnormalities in the release of vitamin B12 from food and varying degrees of atrophic gastritis and achlorhydria.¹⁷

Recent reports suggest that if vitamin B12–deficient patients are treated with nitrous oxide anesthesia, they can develop a rapid onset of the demyelinating syndrome in the weeks to months postoperatively.⁶⁰ The widespread use of stomach acid blockers such as the histamine blockers or the proton pump inhibitors may lead to a syndrome similar to the malabsorption of protein bound B12. Metformin (widely used in treatment of diabetes) may induce vitamin B12 deficiency.¹¹⁵

Dietary vitamin B12 deficiency is found in resource poor settings where populations do not have access to animal source food, especially in regions of Africa, Asia, and South America.⁸⁵ It is also very common in populations who avoid most animal products such as in India.⁸⁵ The rise in vegetarianism in North America and Europe has caused vitamin B12 deficiency in otherwise well-nourished individuals.^{85,94}

The vitamin B12–deficient mother may have no symptoms, but the infant is born deficient and if breastfed exclusively will develop severe deficiency since the breast milk becomes depleted of B12. These infants have failure of brain growth and myelination and permanent disability.^{27,72}

The megaloblastic anemia described above is the classic picture of severe vitamin B12–deficient anemia. More recently it has been noted that there are many patients who have only mild abnormalities of blood cells and bone marrow but who may have severe abnormalities of the nervous system.⁶⁷ Fig. 15.4 demonstrates that many of the patients with normal or near-normal hematocrit (open circles) had profound elevation of MMA and/or homocysteine. The reasons for susceptibility to megaloblastic anemia versus neurologic abnormalities are unexplained. The use of MMA and homocysteine as screening tests has also led to the realization that there is a significant fraction of individuals with biochemically severe vitamin B12 deficiency who appear to have no signs of megaloblastic anemia or neurologic abnormalities.

Vitamin B12 deficiency has a very wide spectrum of severity and manifestations.⁵⁸ In addition to anemia and/or neurologic disease, some subjects may have additional symptoms such as glossitis of the tongue, megaloblastic gut, weight loss, mental changes, and

even infertility. The clinical spectrum in infants is also markedly different than in adults,^{27,72} with failure to thrive, irritability, movement disorders, and poor brain growth leading to permanent developmental delay in some cases.¹¹⁶

B. Known Interactions

The interaction of Cbl and folates has been described in Section III. A comparison of the serum Cbl concentration versus serum or red cell folate concentration has been frequently used to distinguish between the megaloblastic anemia resulting from deficiencies of either Cbl or folate. This is problematic because red cell folate can be low in both vitamin B12 and folate deficient subjects and serum folate can occasionally be low in vitamin B12 deficiency.⁵⁴ However, since MMA concentrations are elevated only due to vitamin B12 deficiency, it is possible to determine that a patient with megaloblastic anemia has at least vitamin B12 deficiency, although in some cases there may also be coexisting folate deficiency. Treatment with folic acid doses greater than 800 mcg may partially rescue megaloblastic anemia in the Cbl-deficient patient yet allow the neurologic syndrome to progress which is why a specific diagnosis of Cbl deficiency should be made.⁵⁴

Folic acid has been added to fortified grain products in the United States since January 1998 (and in many other countries). Since then, low serum folate concentrations have virtually been eliminated in the United States,¹¹⁷ whereas Cbl status has been largely unchanged. Patients with pernicious anemia or other forms of severe Cbl deficiency may have been put at risk by masking of megaloblastic anemia. New concerns that B12-deficient seniors with cognitive disorders may have been harmed by high folate intake have surfaced.¹¹⁸ A diet deficient in vitamin B12 is due to lack of animal source food, thus usually also iron, zinc, and protein deficient which means there may be interactions on optimum growth and development of children from multiple nutrients.¹¹⁹

C. As a Dietary Supplement

Proper supplementation of vitamin B12 depends on the condition targeted. Dietary deficiency requires 2–6 mcg added in fortified food, multivitamins, or single vitamin supplements if not already deficient. However, most symptomatic deficiency is due to malabsorption.

If a vegetarian diet lacking in animal source foods has been followed for many years, there may be a total body deficit of 1,000–2,000 mcg at the time supplementation

is begun.^{95a,96} Since the IF-mediated absorption is limited to 2–6 mcg, it will take 6–9 months to repair the deficit if small supplements (<500 mcg) are given. This is particularly important during pregnancy since the fetus will be growing in a deficient environment for months. Deficient vegetarians should be loaded with high dose oral or parenteral B12 until repleted. Most symptomatic B12 deficient patients have malabsorption due to lack of IF rather than deficient diets. Without IF to bind Cbl, replacement must be either parenteral or high dose oral, which bypasses the need for IF. A commonly used regimen is to start with weekly intramuscular injections of 1,000 mcg of cyano-Cbl or hydroxy-Cbl for 4–8 weeks until the clinical response is clearly evident followed by monthly injections for life.⁵⁸ About 150 mcg is retained after each injection of 1,000 mcg. An occasional patient will have serum MMA levels above the normal range on monthly injections and will require more frequent treatment.¹²⁰ Hydroxy-Cbl is often used in Europe and likely requires less frequent dosing.¹²¹ There are no advantages to using the expensive and light-sensitive coenzyme forms methyl or adenosyl-Cbl for treatment.¹²¹ High dose daily oral therapy may be a good alternative to intermittent parenteral treatment.

Investigations from 50 years ago showed subjects with or without pernicious anemia will absorb about 1% of a radioactively labeled oral dose.¹²² A randomized trial of 2,000 mcg oral daily resulted in much higher serum Cbl concentrations and lower MMA concentrations than monthly injections of cyano-Cbl.¹²⁰ Patients should be offered a choice of high dose oral versus parenteral therapy. The commonly available multivitamin preparations in the United States contain much lower quantities of vitamin B12, usually 6–9 mcg although some marketed to seniors contain up to 100 mcg. Although serum Cbl concentrations are higher in seniors who take multivitamins, the serum MMA concentrations are not corrected in many individuals¹²³ and high oral doses are needed. In most cases, vitamin B12 therapy must be continued for life, thus it is imperative to have fully documented the deficiency prior to treatment and to convince the patient that this treatment is an ongoing necessity. An alternative to supplements would be to fortify the food supply with B12 as well as folic acid^{17,124} for persons ingesting deficient diets. Bread fortified with folic acid and vitamin B12 did increase serum vitamin B12 values and lowers MMA values by 10% in a cohort of older adults.¹²⁴ These data are encouraging as a means to improve B12 nutrition; however, questions about adequate dose, stability of B12 after cooking and storage (since B12 analogues could be harmful), and the intended target population

all remain.¹²⁵ The recognition that B12 deficiency may be related to birth defects and poor pregnancy outcomes is part of the impetus to implement population wide supplementation.¹²⁶ Evaluating inexpensive microbial or fermented vegetarian food sources of B12 may have a

large impact for the world populations who either cannot afford or chose not to consume animal source foods. Newer methods of analysis that distinguish B12 analogues from true B12 will be necessary for monitoring this endeavor (Research Gaps).

RESEARCH GAPS

1. The cause of demyelination of the spinal cord and brain in vitamin B12 deficiency is largely unknown.
2. The risks of high folate status in vitamin B12 deficiency should be further investigated.
3. Methods to screen for and treat vitamin B12-deficient women of childbearing age and their infants need to be developed.
4. The benefits of treatment of mild vitamin B12 deficiency (especially in seniors) should be investigated.
5. Vitamin B12-supplemented vegetarian foods should be developed, which are inexpensive and acceptable to populations that will not or are unable to obtain animal source food.

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PANTOTHENIC ACID

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SUMMARY

Identified over 80 years ago, pantothenic acid is an essential vitamin, which serves as the metabolic precursor for coenzyme A (CoA). In the form of CoA and as a component of acyl carrier protein, pantothenic acid is a participant in myriad metabolic reactions involving lipids, proteins, and carbohydrates. Though essential, pantothenic acid deficiency in humans is rare due to its ubiquitous distribution in foods of both animal and plant origin. Supplementation with pantothenic acid or its derivatives may have some health benefits, but further investigation into various health claims is necessary before any specific recommendations may be given.

Keywords: Acyl carrier protein; Coenzyme A; Dexpanthenol; Hallervorden-Spatz syndrome; Pantetheinase; Pantetheine; Pantothenate kinase 2; Pantothenic acid; Sodium-dependent multivitamin transporter.

I. INTRODUCTION

A. Background

The discovery and identification of pantothenic acid as an essential dietary factor was mainly due to the efforts of research groups associated with R. J. Williams, C. A. Elvehjem, and T. H. Jukes. Williams et al. established that pantothenic acid was required for the growth of certain bacteria and yeast.¹ Next, Elvehjem and associates² and Jukes and associates^{3,4} demonstrated that pantothenic acid was a growth and antidermatitis factor for chickens. The eventual characterization of pantothenic acid by Williams took advantage of observations that the antidermatitis factor in acid extracts of various food sources, i.e., pantothenic acid, did not bind to fuller's earth under acidic conditions. Using chromatographic and fractionation procedures (solvent-dependent chemical partitioning), Williams isolated several grams of pantothenic acid for structural determination from 250 kg of liver as starting material.⁵ With this information, a number of research groups contributed to the chemical synthesis and commercial preparation of pantothenic acid.

In the 1950s, one of the functional forms of pantothenic acid, coenzyme A (CoA), was discovered as the cofactor essential for the acetylation of sulfonamides and choline.⁶ In the mid-1960s, pantothenic acid was next identified as a component of acyl carrier protein (ACP) in the fatty acid synthesis complex.⁷ These developments, in addition to a steady series of observations throughout this period on the effects of pantothenic acid deficiency in humans and other animals, provide the foundation for our current understanding of this vitamin.

B. Structure and Function

Pantothenic acid is ubiquitous in nature and distributed widely in foodstuffs. Its name, coined by Williams,^{1,5} is derived from the Greek term, *pántothen*, which means "from all quarters." It consists of pantoic acid and β -alanine linked by an amide bond (Fig. 16.1A). In plants, fungi, and some bacteria, pantothenic acid is synthesized by the condensation of pantoic acid with β -alanine. Animals, in contrast, require a dietary source because of the inability to synthesize pantoic acid, an α -hydroxy acid derived

$$\text{HO}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{CH}_2-\text{NH}-\overset{\text{O}}{\parallel}{\text{C}}-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\underset{|}{\text{C}}}}-\text{CH}_2-\text{OH}$$
$$\text{HS}-\text{CH}_2-\text{CH}_2-\text{NH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{CH}_2-\text{NH}-\overset{\text{O}}{\parallel}{\text{C}}-\underset{\text{OH}}{\underset{\text{CH}_3}{\text{CH}}}-\underset{\text{CH}_3}{\text{C}}-\text{CH}_2-\text{OPO}_3^{2-}$$
$$\text{HS}-\text{CH}_2-\text{CH}_2-\text{NH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{CH}_2-\text{NH}-\overset{\text{O}}{\parallel}{\text{C}}-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\underset{|}{\text{C}}}}-\text{CH}_2-\text{O}-\text{PO}_2^- - \text{O}-\text{PO}_2^- - \text{O}-\text{CH}_2-\underset{\text{NH}}{\underset{|}{\text{CH}}}-\overset{\text{C}=\text{O}}{\underset{|}{\text{C}}}$$
C[C@H](O)C(COP(=O)([O-])OP(=O)([O-])OC[C@@H]1O[C@H](n2cnc3c(N)ncnc32)[C@H](O)[C@@H]1O)COCCNC(=O)CCCNC(=O)CCS

from 2-oxoisovalerate.^{8,9} Pantothenic acid is the vitamin component of 4'-phosphopantetheine (Fig. 16.1B). 4'-Phosphopantetheine includes a mercaptoethylamine moiety bound in amide linkage to the terminal carboxyl group of pantothenic acid. The mercaptoethylamine moiety of 4'-phosphopantetheine is essential for the activity of ACP (Fig. 16.1C) and CoA (Fig. 16.1D).

pure state is water soluble, viscous, yellow in color, and stable at neutral pH with a molecular mass of 219.23 g/mol. It is present in nature only as the d- or (R)-enantiomer.¹⁰ Exposure to acid, alkali, and heat causes loss of vitamin activity. Calcium pantothenate, which is white, odorless, and crystalline, is more stable than pantothenic acid¹¹ and therefore is the common dietary supplement form of the vitamin.

Section B. Vitamins

TABLE 16.1 Selected Functions of Coenzyme A (CoA) and Acyl Carrier Protein (ACP).

Function	Importance
<i>Carbohydrate-related</i>	
Citric acid cycle transfer reactions	Oxidative metabolism; energy production (ATP production)
Acetylation of sugars (e.g., N-acetylglucosamine)	Production of carbohydrates essential to cell structure
<i>Lipid-related</i>	
Phospholipid biosynthesis	Cell membrane formation and structure
Isoprenoid biosynthesis	Cholesterol and bile salt production
Steroid biosynthesis	Steroid hormone production
Fatty acid elongation, acyl and triacylglyceride synthesis	Ability to modulate cell membrane fluidity and energy sequestration or storage
β -oxidation	Oxidative metabolism; energy production (ATP production)
<i>Protein- and amino acid-related</i>	
Protein acetylation	Altered protein conformation and allosteric interactions Activation of certain hormones, cell signaling molecules, and enzymes Transcriptional regulation (e.g., acetylation of histones and transcription factors)
Protein acylation (e.g., myristic and palmitic acid additions and prenylation)	Compartmentalization of proteins and activation hormones and transcription factors
Amino acid synthesis and catabolism reactions	Nitrogen metabolism
Protein CoAthiolation	Protection from reactive oxygen species; facilitates the modulation of targeted proteins to new subcellular locations and regulation of certain functional activities

pyruvate in the citric acid cycle and the transport and transfer of carbon atoms within cells. Accordingly, the production of CoA derivatives and ACP is highly regulated.

II. NUTRIENT METABOLISM

A. Synthesis

About 6000 tons of calcium D-pantothenate are produced annually for use in animal feeds, pharmaceuticals, and supplements. It is prepared by chemical synthesis from D-pantolactone and calcium β -alanine. However, racemic separation of L- and D-forms of the final product is often necessary (e.g., separation by chromatography or recrystallization). Other approaches to synthesis use microorganisms in which only the D form of pantothenic acid is synthesized, and thus the separation of racemates is not necessary.¹²

B. Digestion

Pantothenic acid in food is usually present as CoASH or as protein- or peptide-bound 4'-phosphopantetheine. In order to be absorbed, a near quantitative release of pantothenic acid or 4'-phosphopantetheine from associated CoA derivatives, proteins, and peptides is achieved by the sequential actions of two intestinal hydrolases with pyrophosphatase and phosphatase activity. Another intestinal hydrolase, pantetheinase, further hydrolyzes 4'-phosphopantetheine to pantothenic acid.¹³

C. Transport

The uptake of luminal pantothenic acid occurs by a sodium-dependent multivitamin transporter (SMVT) found at the surface of intestinal and colonic enterocytes. The human isoform of SMVT is one of the 12 members of the solute carrier gene family 5 (SLC5A) transporters.^{14,15} As a multivitamin transporter, SMVT is also a transporter for biotin, lipoic acid, various sugars, myo-inositol, iodide, short-chain fatty acids, and choline. For pantothenic acid, SMVT cellular uptake is saturable with an apparent K_m of 15–20 μ M. Cell signaling associated with protein kinase C and calmodulin-dependent processes is also essential to the regulation of pantothenic acid uptake.^{16,17} For example, cellular energy and growth inhibitors such as ouabain, gramicidin D, cyanide, azide, and 2,4-dinitrophenol inhibit pantothenic acid uptake. Replacing sodium by potassium or reducing the external sodium concentration (below 40 mM) also reduces pantothenic acid uptake. In contrast, once an individual cell meets its need for pantothenic acid, the cellular efflux of the vitamin is not affected by the addition of pantothenic acid, sodium, ouabain, gramicidin D, or 2,4-dinitrophenol.^{14–17}

In blood plasma, circulating pantothenic acid is not bound tightly to a specific protein. Rychlik¹⁸ estimated the free and bound pantothenic acid in human plasma; over 90% is in the free state at concentrations ranging from 0.5 to 1.0 μ M. Accordingly, SMVT-active transport is essential for the transfer of free pantothenic acid from blood into cells. This is underscored by the differences in cellular versus plasma concentrations. For

example, the cellular concentrations of free pantothenic acid in the liver and heart, respectively, are 10–15 and $\sim 100 \mu\text{M}$, i.e., 10–100 times that of plasma. Also, the concentration of CoA derivatives is 100–400 μM , and the concentration of ACP is 10 μM or more in the cytosol of typical cells. In mitochondria, the CoA concentration can be as much as 10- to 20-fold higher than in cytosol. Mitochondria account for 70%–90% of the total cellular CoA content. The unidirectional influx of pantothenic acid across cerebral capillaries (the blood–brain barrier) occurs by a low-capacity, saturable transport system with a half-saturation concentration approximately 10 times the plasma pantothenic acid concentration.^{19–21}

Concerning whole body retention of pantothenic acid, tubular reabsorption by the kidney appears to be the primary mechanism for conserving the vitamin; amounts above the pantothenic acid renal threshold are excreted. Other than 4'-phosphopantetheinate, no other compounds or products derived from pantothenic acid have been identified as excretory products. Daily renal excretion of pantothenic acid is also closely linked to the daily intake of the vitamin.^{22,23}

D. Metabolism

Cellular regulation and synthesis of CoA

The first step in the synthesis of CoA is the phosphorylation of pantothenic acid to 4'-phosphopantetheine (Fig. 16.2), a reaction catalyzed by pantothenic acid

kinase.^{8,9,24–27} Pantothenic acid kinase possesses a broad pH optimum (between pH 6 and 9) with a K_m for pantothenic acid of $\sim 20 \mu\text{M}$. Mg-ATP is the nucleotide substrate for the phosphorylation reaction with a K_m of $\sim 0.6 \text{ mM}$. The pantothenic acid kinase reaction also serves as the primary control point in the pathway. For example, feedback inhibition of pantothenic acid kinase by CoA or CoA derivatives represents a significant control on the flux of substrate through the subsequent steps in the pathway and defines the upper threshold for intracellular CoA cofactor levels. Inhibition by acetyl-CoA is slightly higher than that of free CoA. The inhibition by free CoA is uncompetitive with respect to the pantothenate concentration, with a K_i for inhibition of 0.2 μM .²⁷

Other inhibitors and activators for CoA synthesis include L-carnitine, which is essential for the transport of fatty acids into mitochondria. Carnitine does not affect pantothenic kinase activity by itself but reverses the inhibition of pantothenic acid kinase by CoA. Such properties provide potential mechanisms for the control of CoA synthesis and regulation of cellular pantothenic acid content, i.e., feedback inhibition by CoA and its acyl esters, which is reversed by changes in the concentration of free carnitine. In this regard, it is important to note that the free concentration of acyl CoA in cells can be low and variable because the bulk of acyl derivatives are protein bound. Moreover, similar to CoA, carnitine exists in both free and acylated forms, and reversal of kinase inhibition by CoA does not occur when carnitine is acylated.²⁵ The ratio of free to acylated carnitine also varies considerably depending on feeding and

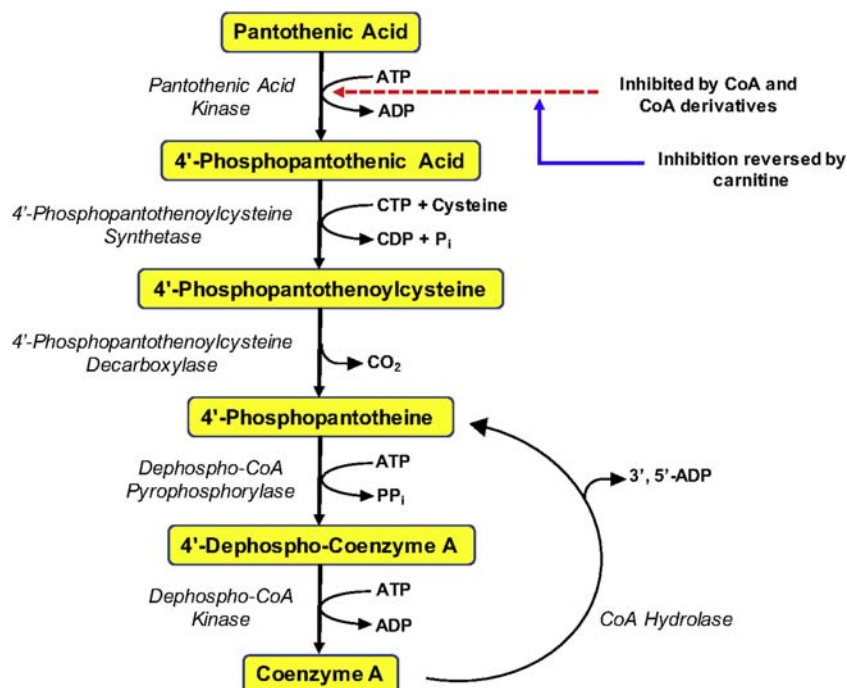


FIGURE 16.2 Metabolic conversion of pantothenic acid to coenzyme A.

hormonal influences. Insulin status is particularly important; fasting and diabetes (states of low insulin) increase pantothenic acid kinase activity and the total content of CoA.^{25–28} The perfusion of heart preparations or incubation of liver cells with glucose, pyruvate, or palmitate also inhibits pantothenic acid phosphorylation, owing to a reduction in free carnitine, and increases in the free and acylated forms of CoA.

There are four distinct human isoforms of pantothenic acid kinase, each with different tissue locations and K_m/K_i values for activation or inhibition.^{24–28} These differences allow for cellular regulation of CoA over a wide range of concentrations in keeping with the observation that depending on location and metabolic activity, cellular CoA levels may span more than a 10-fold range.^{24–28} Moreover, in addition to regulation at the substrate level, transcriptional regulation of the pantothenic acid kinase isoforms involves factors, such as the tumor suppressor protein, p53, which regulates cell cycling, and peroxisome proliferator-activated receptor agonists, which can play key roles in the regulation of energy homeostasis and immunity. For example, using metabolomic approaches for assessments in humans (e.g., high-resolution NMR and mass spectrometry), significant reductions in both urinary pantothenic acid and acetyl carnitine occur in response to PPAR agonists, such as fenofibrate²⁴; the reduction in urinary excretion of pantothenic acid and acetyl carnitine is taken as an indirect measure of CoA synthesis. Phosphatidylethanolamine-derived metabolites, often referred to as endocannabinoids, are also activators of pantothenic acid kinase isoforms.^{27,28}

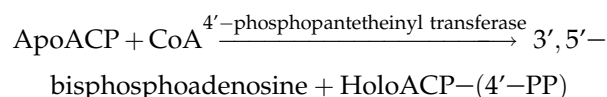
Following 4'-phosphopantetheine acid formation, the subsequent steps in CoA synthesis are carried out on a protein complex (~400,000 Da) with multifunctional catalytic sites. Important enzymatic features of this complex include dephospho-CoA-pyrophosphorylase activity, which catalyzes the reaction between 4'-phosphopantetheine and ATP to form 4'-dephospho-CoA; dephospho-CoA kinase activity, which catalyzes the ATP-dependent final step in CoA synthesis; and a CoA hydrolase site that catalyzes the hydrolysis of CoA to 3', 5'-ADP and 4'-phosphopantetheine. This sequence of reactions, the CoA/4'-phosphopantetheine cycle, provides a mechanism by which the 4'-phosphopantetheine can be recycled to form CoA (Fig. 16.2). Each turn of the cycle utilizes two molecules of ATP and produces one molecule of ADP, one molecule of pyrophosphate, and one molecule of 3', 5'-ADP.^{25–28}

Cellular regulation and synthesis of ACP

Fatty acid synthesis in plants, many forms of bacteria, and yeast is catalyzed by multiple enzymes,

designated as a type II fatty acid synthesis or FAS system, i.e., distinct and individual enzymes perform each of the functional activities needed for fatty acid synthesis.^{29,30} In this regard, ACP is sometimes referred to as a “macrocofactor” because, as a small polypeptide chain (M.W. ~8500–8700 Da), it may be isolated and separated from the enzymes for which it serves as a cofactor.

In contrast, in mammalian cells, ACP is associated with fatty acid synthase,³¹ an enzyme complex composed of two large protein subunits (each with M.W. ~250,000 Da). The system in mammals is designated as a type I FAS system. In type I FAS systems, the 4'-phosphopantetheine moiety, found in the ACP functional domain of fatty acid synthetase, acts as an arm or extension to facilitate the sequential movement of substrates and respective products from each unique catalytic domain to the next during synthesis and elongation of a fatty acid (cf. **PKN 11 chapter 4**). In both prokaryotes and eukaryotes, CoA serves as the substrate for 4'-phosphopantetheine. Phosphopantetheinyl transferase catalyzes the reaction^{29–31} depicted here:



Although there are few data specifically related to the regulation of the holo-ACP peptide or the fatty acid synthesis ACP domain, it is reasonable to assume overall production of ACP activity is linked closely to CoA production.

Features of CoA and 4'-Phosphopantetheine related to mechanisms of action

The intermediates arising from the transfer reactions catalyzed by CoA and the 4'-phosphopantetheine prosthetic group act as “high-energy” compounds. CoA and ACP react with acetyl or acyl groups to form thioesters. Thioesters (-S-CO-R) are thermodynamically less stable than typical esters (-O-CO-R) or amides (-N-CO-R). The double bond character of the C–O bond in -S-CO-R does not extend significantly into the C–S bond, i.e., S-CO-R is not stabilized by resonance. Accordingly, thioester bonds have a more negative standard free energy of hydrolysis (–7.5 kcal/mol compared with most oxygen ester bonds, which are about –5 kcal/mol). Thioesters are also more reactive than oxygen esters in nucleophilic displacement reactions involving acyl groups, i.e., for most of the reactions involving CoA or ACP, no additional energy from an energy source, such as ATP, is required for transfer of an acetyl or acyl group. For reference, the $-\Delta G$ of hydrolysis is ~–7.5 kcal/mol for acetyl-CoA compared to 7–8 kcal/mol for the hydrolysis of ATP to AMP and pyrophosphate or ADP and phosphate.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

Essential functions of CoA and ACP are listed in Table 16.1 and Fig. 16.3. Acetylated, prenylated, or acylated derivatives of CoA are involved in transfer reactions and processes that range from fatty acid and neurotransmitter synthesis to protein acetylations, prenylations, and acetylations. As a succinylated or propionylated derivative, CoA is essential as a cofactor in amino acid metabolic pathways. As a β -hydroxy β -methylglutaryl derivative, CoA is an intermediate in the mevalonate and ketogenesis pathway important to cholesterol and steroid hormone synthesis.^{24–28} Regarding ACP, in addition to fatty acid synthesis,^{29–31} in bacteria and plants, ACP performs a wide range of functions that include amino acid synthesis and the formation of polyketides. Polyketides are a diverse group of secondary metabolites that include antibiotics, such as erythromycin; cholesterol-lowering drugs, such as lovastatin; and putative antiaging compounds, such as resveratrol.³²

Recently, CoA has also been identified as a cofactor in redox regulation, acting in effect as a reactive oxygen species (ROS) inhibitor.^{33,34} The covalent attachment of CoA to proteins in response to oxidative and metabolic stress is termed “S-thiolation” or “CoA-thiolation” and prevents irreversible sulfhydryl oxidation. CoA-thiolation can also alter the charge and activity of modified proteins. Protein cysteines are targets for numerous posttranslational modifications and are often targets for ROS (e.g., free radicals, peroxides in various forms, or heavy metals). Protein CoA-thiolation offers protection, analogous to the protection provided by glutathione. For example, glutathione also reduces disulfide bonds by serving as an electron donor and prevents potential damage to components, wherein the availability of sulfide groups is essential for activity. Also, acetyl-CoA is essential to the acetylation of histones, specific transcription factors, and enzymes where their acetylation is essential to activity regulation. Such acetylations are in turn essential for cell growth and mitosis.

For all of the aforementioned functions, it is also important to appreciate that each is inextricably linked

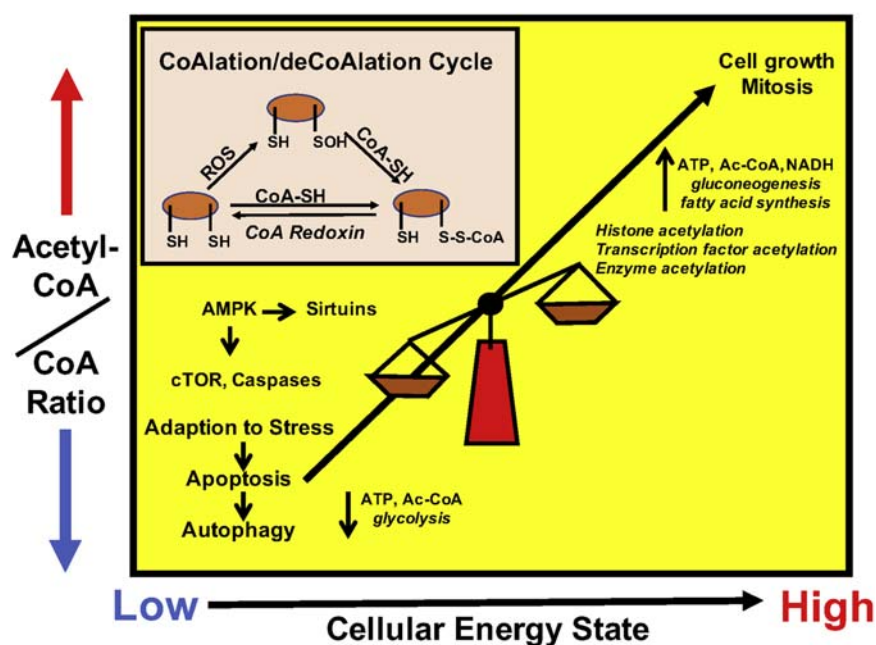


FIGURE 16.3 Coenzyme A (CoA) and the energy state of cells. The relative abundance of cellular acetyl coenzyme A (acetyl-CoA) and coenzyme A (CoA) is linked to the energy state of all cells whether of prokaryote or eukaryote origin. When the acetyl-CoA to CoA ratio is elevated, the activation of certain histones, transcription factors, and enzymes related to cell growth can occur more easily along with increased gluconeogenesis and fatty acid synthesis.^{33,34} In contrast, when the balance shifts and results in an elevated CoA to acetyl-CoA ratio, proapoptotic signaling factors and enzymes are activated, such as mTOR, AMPK, and the sirtuin and caspase family of enzymes.^{33–38} The accumulation of CoA at the expense of acetyl-CoA favors activation of sirtuins and promotes the activation of AMPK, a master regulator of cellular energy homeostasis; mTOR, a target of AMPK, is an intracellular nutrient sensor involved in the control of protein synthesis and cell growth.³⁶ Sirtuins influence a wide range of cellular processes, such as aging, stress resistance, and mitochondriogenesis. Reversible acetylation catalyzed by sirtuins activates acetyl-CoA synthetase.³⁷ Further, at low levels of acetyl-CoA, the enzymes in the caspase family of proteinases are activated. Caspases are proteases that play important roles in cell remodeling when the cell energy state is low and when cells are damaged or stressed.³⁸ The CoAlation/deCoAlation cycle is also included to reflect that when cellular stress results from chemically reactive species containing oxygen (ROS: peroxides, superoxide, hydroxyl radicals, singlet oxygen, or alpha-oxygen), CoA can play a protective role.^{33,34}

to the energy charge of cells (Fig. 16.3). For example, the need for CoA and its derivatives is primarily controlled by the cell's ability to make and utilize ATP. Metabolic pathways that use energy are activated when the cellular energy charge is high and inhibited when cellular energy charge is low. A relative increase in cellular ATP has the potential to downregulate energy-generating processes (glycolysis, citric acid cycle) and modulate positively energy-consuming pathways (amino acid synthesis, gluconeogenesis). Likewise, a fall in acetyl-CoA leads to a decrease in mitochondrial energy generation via the citric acid cycle and causes a decrease in the amount of 4-carbon organic acids participating in the cycle (Fig. 16.3).

At normal stages of cellular energy flux, a complex homeostatic system involving signaling pathways and nutrient sensors dictates the balance between anabolic and catabolic processes. Cell signaling pathways that are particularly sensitive to changes in acetyl-CoA and CoA cellular levels are those involving mTOR,³⁵ AMPK,³⁶ and sirtuin signaling.³⁷ Mammalian target of rapamycin (mTOR) is a member of the phosphatidylinositol 3-kinase-related family of signaling molecules and is involved in cell growth, motility, and survival.³⁵ AMPK (5'-AMP-activated protein kinase) is considered as the master regulator of cellular energy homeostasis and is activated when the ATP pool in cells is diminished.³⁶ The sirtuins are a family of enzymes with either mono-ADP-ribosyltransferase or a wide range of deacetylase and deacylase activities. Sirtuins influence a wide range of cellular processes, such as aging, stress resistance, and mitochondriogenesis.³⁷ In addition, CoA and CoA derivatives influence the activity of caspases (cysteine-aspartic proteases).³⁸ For example, at low levels of acetyl-CoA, the caspase proteases play roles in programmed cell death and inflammation.³⁸ In many cases, the control of the activities mentioned above is dependent upon the degree of acetylation or acylation as an activation signal.^{35–38} Indeed, taken together, the full range of functions attributed to pantothenic acid-derived cofactors and their importance are profound.

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

The essentiality of pantothenic acid has been documented in a wide variety of animal species. The classical signs of deficiency, first recognized by Elvehjem, Jukes, and colleagues in chickens, include growth retardation and dermatitis.^{2–4} Many other physiological systems are affected by pantothenic acid deficiency, owing to the diversity of metabolic functions in which CoA and ACP participate. Neurological, immunological, hematological, reproductive, and gastrointestinal pathologies

TABLE 16.2 Effects of Pantothenic Acid Deficiency in Selected Species.^{2–4,39–53}

Chicken	Dermatitis around beak, feet, and eyes; poor feathering; spinal cord myelin degeneration; involution of the thymus; fatty degeneration of the liver.
Rat	Dermatitis; loss of hair color; loss of hair around the eyes; hemorrhagic necrosis of the adrenals; duodenal ulcer; spastic gait; anemia; leukopenia; impaired antibody production; gonadal atrophy with infertility.
Dog	Anorexia; diarrhea; acute encephalopathy; coma; hypoglycemia; leukocytosis; hyperammonemia; hyperlactemia; hepatic steatosis; mitochondrial enlargement.
Pig	Dermatitis; hair loss; diarrhea with impaired sodium, potassium, and glucose absorption; lachrymation; ulcerative colitis; spinal cord and peripheral nerve lesions with spastic gait.
Human	Numbness and burning of feet and hands; headache; fatigue; insomnia; anorexia with gastric disturbances; increased sensitivity to insulin; decreased eosinopenic response to adrenocorticotrophic hormone (ACTH); impaired antibody production.

have been reported. The effects of pantothenic acid deficiency in different species are summarized in Table 16.2.^{2–4,39–53} In humans, clinical pantothenic acid deficiency (plasma values <1 μ M; urinary excretion <1 mg pantothenic acid per day) has been reported to occur after 6 weeks.^{52–55}

Because pantothenic acid is such a ubiquitous component of foods, both animal and vegetable, deficiency of this vitamin in humans is very rare. If present, pantothenic acid deficiency is usually associated with multiple nutrient deficiencies, thus making it difficult to discern effects specific to a lack of pantothenic acid. What is known about pantothenic acid deficiency in humans comes primarily from two sources of information. During World War II, malnourished prisoners of war in Japan, Burma, and the Philippines experienced numbness and burning sensations in their feet. While these individuals suffered multiple deficiencies, this specific syndrome was only reversed upon pantothenic acid supplementation.⁵¹ Experimental pantothenic acid deficiency has also been induced in both animals and humans by administration of the pantothenic acid kinase inhibitor, ω -methylpantothenate, in combination with a diet low in pantothenic acid.^{52,53,56} Observed symptoms in humans included numbness and burning of the hands and feet similar to that experienced by the World War II prisoners of war, as well as myriad other symptoms listed in Table 16.2. Some of the same symptoms are produced when individuals are fed a semisynthetic diet from which pantothenic acid has been essentially eliminated, but without addition of ω -

methylpantothenate.⁵⁴ Another pantothenic acid antagonist, calcium hopantenate, has been shown to induce encephalopathy with hepatic steatosis and a Reye-like syndrome in both dogs and humans.^{48,57}

Further evidence of the essentiality of pantothenic acid metabolism is revealed by an autosomal recessive disorder, initially called Hallervorden-Spatz syndrome and subsequently referred to as pantothenate kinase-associated neurodegenerative disease or neurodegeneration with brain iron accumulation 1.^{58–61} Mutations in the gene encoding pantothenate kinase 2 (gene map locus: 20p13-p12.3) underlie this disorder, which is characterized by iron accumulation in the brain, progressive neurodegeneration, and early death. The pathogenesis of the disorder is believed to be related to mitochondrial CoA deficiency, inhibition of fatty acid β -oxidation, and oxidative stress. In addition, lack of phosphopantothenic acid synthesis may lead to accumulation of cysteine, the substrate required for step 2 of pantothenic acid metabolism, i.e., synthesis of phosphopantotheynoylcysteine (Fig. 16.2). It has been proposed that accumulated cysteine rapidly autooxidizes in the presence of free iron leading to the generation of free radicals and additional oxidative stress.^{58,59}

B. Criteria of Adequacy and Assessment of Status

Pantothenic acid status is reflected by both whole blood concentration and urinary excretion. Whole blood concentrations typically range from 1.6 to 2.7 $\mu\text{mol/L}$,⁶² and a value of $<1 \mu\text{mol/L}$ is considered low.⁶³ It is important to note, however, that plasma levels of the vitamin are considered less reliable as an indicator of pantothenic acid status than urinary excretion because they are often not well correlated with changes in intake.^{63,64} Urinary excretion is considered a more reliable indicator of status because it is closely related to dietary intake.^{52–54,65,66} Excretion of $<1 \text{ mg}$ pantothenic acid per day in urine is considered low.^{63,64,67}

Pantothenic acid concentrations in whole blood, plasma, and urine are measured by microbiological assay employing *Lactobacillus plantarum*. For whole blood, enzyme pretreatment is required to convert CoA to free pantothenic acid since *L. plantarum* does not respond to CoA. Other methods that have been employed to assess pantothenic acid status include radioimmunoassay, ELISA, and gas chromatography. The topic of pantothenic acid status assessment has been reviewed.⁶³

C. Reference Intakes and Basis for Individual Variability in Requirements

An Estimated Average Requirement (EAR) and a Recommended Dietary Allowance (RDA) were not established

in 1998 for the United States and Canada as part of the last Dietary Reference Intake evaluation of pantothenic acid by the Institute of Medicine, Food and Nutrition Board. However, Adequate Intakes (AIs) for men and women throughout the life cycle were established based on observed mean intakes and estimates of basal excretion in urine (Table 16.3).⁶⁷ An EAR and an RDA are difficult to establish because CoA and ACP are involved in so many aspects of metabolism that biochemical markers or clinical deficiency criteria *specific* for pantothenic acid deficiency in humans have been challenging to define and because there is a lack of data on dietary insufficiency that could demonstrate a level of intake that would be inadequate for a group of individuals in order to determine EAR values. In this regard, more information is needed related to the extent to which dietary pantothenic acid supplementation can stimulate or influence CoA or ACP synthesis.

Urinary excretion of pantothenic acid only exceeds basal levels when intakes are greater than 4 mg/d in young adult males. Thus, an intake of 4 mg/d likely reflects the level at which saturation of the body pool occurs.⁶⁵ Estimates of dietary intake in healthy adults have ranged from 4 to 7 mg/d.^{65,68–70} There is no evidence to suggest that this range of intake is inadequate, and 5 mg/d has been set as the AI for adults. For adults older than 51 years, the AI remains the same (5 mg/d) as there is currently no basis for expecting an increased requirement in elderly individuals. During pregnancy, the AI is increased to 6 mg/d based on usual intakes of 5.3 mg/d⁷² with rounding up. During lactation, the AI is increased further to 7 mg/d, accounting for additional secretion of the vitamin in human milk (1.7 mg/d) and the lower maternal blood concentrations reported when intakes are about 5–6 mg/d.^{64,71,72} This is likely the result of efficient sequestering of the

TABLE 16.3 Adequate Intakes (AI) for Pantothenic Acid.⁶⁷

	mg/day
Infants	
0–5 months	1.7
6–12 months	1.8
Children	
1–3 years	2.0
4–8 years	3.0
9–13 years	4.0
Adolescents	
14–18 years	5.0
Adults	
19–50 years	5.0
> 50 years	5.0
Pregnancy	6.0
Lactation	7.0

vitamin in human milk, estimated to be 0.4 mg for every 1 mg pantothenic acid consumed during active lactation.⁷³

The AI for infants reflects the mean intake of infants fed principally with human milk. Human milk contains about 2.5 mg/L.⁷³ Values for children and adolescents have largely been extrapolated from adult values. These values are supported by studies comparing intake and urinary excretion of the vitamin in preschool children.⁷⁴ Dietary intake of pantothenic acid was 3.8 and 5 mg/d in children of high and low socioeconomic status, respectively, and urinary excretion was 3.36 and 1.74 mg/d, respectively. In a separate study, 35 healthy girls, 7–9 years of age, were fed defined diets, and urinary excretion was measured.⁷⁵ The average daily excretion was 1.3 mg/d when intake was 2.79 mg/d and 2.7 mg/d when intake was 4.45 mg/d. Therefore, intakes of 2.8–4.5 mg/d exceed urinary excretion of the vitamin. In healthy adolescents (13–19 years of age), 4-day diet records indicated that the average pantothenic acid intake was 6.3 mg/d for males and 4.1 mg/d for females.⁶⁶ Average urinary excretion in this latter study was 3.3 and 4.5 mg/d for males and females, respectively, while whole blood pantothenic acid concentrations averaged 1.86 $\mu\text{mol/L}$ and 1.57 $\mu\text{mol/L}$, respectively. Normal blood concentrations of the vitamin in healthy individuals have been reported to range from 1.6 to 2.7 $\mu\text{mol/L}$.⁶² Taken together, these data indicate that intake of 4 mg/d is sufficient to maintain normal blood concentrations in adolescents.

It is possible to meet the AI for adults with a mixed diet containing as little as 100–200 g of solid food using the estimate of 20–50 μg pantothenic acid per gram typically found in edible animal and plant tissues, i.e., the equivalent of a mixed diet corresponding to 600–1200 kcal or 2.4–4.8 MJ. The typical Western diet contains 6 mg or more of available pantothenic acid.⁶⁵ A detailed review on estimates of intake and of the AI for pantothenic acid is available.⁶⁷

D. Tolerable Upper Intake Levels

Oral pantothenic acid, even in doses as high as 10–20 g/d, is well tolerated.^{76,77} Occasional mild diarrhea may occur, particularly when the supplement is in the form of calcium D-pantothenate.

E. Dietary Sources

Pantothenic acid is found in a wide variety of foods of both plant and animal origin at levels ranging from 20 to 50 μg per gram. Particularly rich sources of pantothenic acid include poultry, beef, seafood, liver and

other organ meats, eggs, milk, whole grains, mushrooms, avocados, potatoes, broccoli, peanuts, sunflower seeds, chickpeas, and tomato products.^{78,79} Royal bee jelly and ovaries of tuna and cod also have high levels of the vitamin.⁸⁰ Because of its thermal lability and susceptibility to oxidation, significant amounts of pantothenic acid are lost from highly processed foods, including refined grains and cooked or canned meats and vegetables. Processing and refining whole grains result in a 37%–47% loss of pantothenic acid, while canning of meats, fish, and dairy products leads to losses of 20%–35%.⁸¹ Greater losses of the vitamin occur during canning (46%–78%) and freezing (37%–57%) of vegetables.⁸¹ Pantothenic acid is also synthesized by intestinal microorganisms,^{82–84} though the potential maximal contribution of the vitamin from this source with respect to the human dietary requirement has been estimated to be very low (0.078% of the AI).^{83,84}

The primary source of pantothenic acid in food is CoA. Intestinal phosphatases and nucleosidases are capable of very efficient hydrolysis of CoA so that near quantitative release of pantothenic acid occurs as a normal part of digestion. Further, the overall K_m for pantothenic acid intestinal uptake is 10–20 μM . At an intake of ~ 10 –15 mg of CoA, the amount of CoA in a typical meal, the pantothenic acid concentration in luminal fluid would be ~ 1 –2 μM . At this concentration, pantothenic acid does not saturate the transport system, and as a consequence, it is efficiently and actively absorbed.⁸⁵

F. Issues Related to Pantothenic acid in Human Health

Although overt deficiency of pantothenic acid is extremely rare in humans, an internet search for “pantothenic acid” reveals numerous websites providing background information, health claims, and of course an opportunity to buy the vitamin for oral consumption. Many of the claims made on these websites are completely unwarranted. For example, the use of pantothenic acid to prevent and treat graying hair was based on the observation that pantothenic acid deficiency in rodents causes their fur to turn gray.⁴³ No association between graying of hair in humans and pantothenic acid status has ever been demonstrated. Moreover, although other claims for pantothenic acid have a more credible scientific basis and are summarized below, it should be noted that many such claims are based on studies that were conducted in the 1940s, 50s, and 60s and still await validation.

G. Cholesterol Lowering

Pantothenic acid is not particularly effective in lowering serum cholesterol levels. Rather, oral doses of its metabolite, pantetheine, or more specifically the dimer, pantethine, induce favorable effects on serum cholesterol concentrations. Several studies indicated that pantethine, in doses typically ranging from 500 to 1200 mg/d, can lower total serum cholesterol, low-density lipoprotein cholesterol, and triacylglycerols and raise high-density lipoprotein cholesterol in individuals with dyslipidemia, hypercholesterolemia, and hyperlipoproteinemia associated with diabetes.^{86–91} The effects are favorable when compared with those of the more conventional lipid-lowering drugs, such as lovastatin, which may be associated with side effects and liver toxicity. There appear to be no adverse side effects associated with high-dose pantethine therapy. Furthermore, evidence exists that pantethine therapy is more effective than dietary modification in reducing serum cholesterol and lipid concentrations.⁸⁶ The mechanism by which pantethine exerts its hypolipemic effects is unclear. A hypothesized site of action is in the regulation of liver sterol biosynthesis. Because pantethine is a coenzyme precursor, it may shunt active acetate from sterol synthesis to mitochondrial oxidative and respiratory pathways.⁹² Additionally, pantethine may promote improved triacylglycerol and low-density lipoprotein cholesterol catabolism, as well as reduced cholesterol synthesis via inhibition of the enzyme hydroxymethylglutaryl-CoA reductase.^{93–95}

H. Enhancement of Athletic Performance

Scientific support for an effect of pantothenic acid supplements on athletic performance is limited. Until recently, most of the potential benefit has been inferred from animal studies. More than 70 years ago, frog muscles soaked in pantothenic acid solution were shown to do twice as much work as control muscles before exhaustion,⁹⁶ and more than 50 years ago, rats supplemented with high doses of pantothenic acid were shown to withstand exposure to cold water longer than unsupplemented mice.⁹⁷ Moreover, rats deficient in pantothenic acid became exhausted more rapidly during exercise than did pantothenic acid replete controls.⁹⁸ In this latter study, deficiency was associated with lower tissue CoA concentrations and greater depletion of glycogen reserves during exercise.

Studies assessing the influence of pantothenic acid on human performance are mixed. In one study, well-trained distance runners were supplemented with 2 g/d of pantothenic acid for 2 weeks.⁹⁹ These athletes outperformed other equally well-trained distance runners

who received placebo. Those who received the supplements also used 8% less oxygen to perform equivalent work and had ~17% less lactic acid accumulation. However, in a separate study, no effect on performance was observed in highly conditioned distance runners after receiving 1 g/d of pantothenic acid for 2 weeks.¹⁰⁰ Additionally, no difference in performance was observed among highly trained cyclists given a combination of thiamin (1 g) and pantethine/pantothenic acid (1.9 g) or placebo. The supplement or placebo was given for 7 days before each exercise test. The investigators found no effect on any physiological or performance parameters during steady-state or high-intensity exercise.¹⁰¹

I. Rheumatoid Arthritis

Almost 70 years ago, researchers noted that young rats made acutely deficient in pantothenic acid suffered defects in growth and development of bone and cartilage, which were reversed by repletion of the vitamin.¹⁰² Subsequently, blood levels of pantothenic acid in humans with rheumatoid arthritis were found to be lower than in healthy controls. On the basis of this finding, a clinical trial in which 20 patients with rheumatoid arthritis were injected daily with 50 mg calcium pantothenate was conducted.¹⁰³ Blood levels of pantothenic acid increased to normal, and relief from rheumatoid symptoms was achieved in most cases. Symptoms recurred when supplementation was discontinued. Similar results were obtained in arthritic vegetarians.¹⁰³ Later, it was found in a double-blind, placebo trial that oral doses of calcium pantothenate (≤ 2 g/d) reduced the duration of morning stiffness, degree of disability, and severity of pain in patients with rheumatoid arthritis.¹⁰⁴ Individuals with other forms of arthritis were not helped by the supplements, indicating that the therapeutic effect of pantothenic acid is specific for rheumatoid arthritis.

J. Wound Healing and Skin Care

In cultured human skin cells given an artificial wound, adding calcium D-pantothenate increased the number of cells and the distance that they migrated across the edge of the wound.¹⁰⁵ These effects are likely to accelerate wound healing. Little *in vivo* human data, however, exist to support the findings of accelerated wound healing in cell culture and animal studies in response to oral pantothenic acid. A small randomized double-blind, placebo-controlled study of adults with mild to moderate acne vulgaris reported a reduced number of lesions in the group receiving a daily oral pantothenic acid–based dietary supplement (Pantothén) for 12 weeks compared to

those receiving a placebo.¹⁰⁶ In contrast, a randomized, double-blind study examining the effect of supplementing patients undergoing surgery for tattoo removal with 1000 mg of vitamin C and 200 mg of pantothenic acid did not demonstrate any significant improvement in the wound healing process in those that received the supplements.¹⁰⁷ Furthermore, no benefits were observed when the doses were increased to 3000 mg of ascorbic acid and 900 mg of pantothenic acid.¹⁰⁸

More effective in wound healing is direct, topical skin application of dexpantenol, a stable alcohol form of pantothenic acid.¹⁰⁹ These effects of pantothenic acid are likely related to upregulation of genes involved in wound healing, such as IL-6, IL-1 β , CYP1B1, CXCL1, CCL18, and KAP4-2,^{110,111} which may promote fibroblast proliferation and accelerated epithelialization. Dexpantenol also appears to play a role in the management of minor skin disorders. This is likely related to hygroscopic properties of the drug that promote moisturization and regeneration of the stratum corneum skin barrier.¹⁰⁹ Conditions that respond to topical dexpantenol treatment include atopic dermatitis¹¹² and diaper rash.¹¹³ Dexpantenol may also help maintain skin hydration in cases of radiation dermatitis¹¹⁴ and may reduce skin irritation caused by experimental sodium lauryl sulfate exposure.¹¹⁵ Dexpantenol has also been recommended to treat cheilitis and dry nasal mucosa associated with treatment with the acne drug, isotretinoin.¹¹⁶

K. Lupus Erythematosus

It has been hypothesized that lupus erythematosus, a systemic autoimmune disorder that affects the skin, joints, and various internal organ systems, may be the result of pantothenic acid deficiency.¹¹⁷ The hypothesis is based on the supposition that pantothenic acid deficiency may be induced by three drugs—procainamide, hydralazine, and isoniazid—that are also known to cause drug-induced lupus erythematosus. These drugs are metabolized via CoA-dependent acetylation, and the increased demand for CoA may cause pantothenic acid deficiency. It must be noted, however, that no data have been generated on the effect of these drugs on cellular CoA or pantothenic acid concentrations. It is further postulated that non-drug-induced systemic lupus erythematosus may be the consequence of an

increased need for pantothenic acid in susceptible individuals with genetic polymorphisms in CoA-dependent enzymes. Such polymorphisms remain to be identified. Nonetheless, it is recommended that lupus erythematosus be treated with a combination of vitamins and minerals, including 10 g/day of pantothenic acid.¹¹⁷ Support for such pharmacological doses comes from studies carried out in the 1950s. Some symptoms of lupus erythematosus, but not all, were alleviated with high doses (8–15 g/day) of pantothenic acid derivatives (calcium pantothenate, panthenol, or sodium pantothenate) alone¹¹⁸ or in combination with vitamin E supplements.^{119,120} No improvements in disease symptoms were observed with lower doses (400–600 mg) of calcium pantothenate.¹²¹ With modern technology available to probe genes for polymorphic variability, supplementation trials in lupus erythematosus patients should be repeated to test the hypothesis that a genetic-based increased requirement of pantothenic acid underlies the pathogenesis of this disease.

L. Antimalarial Drugs

More recently, *inhibitors* of the pantothenic acid pathway have been investigated for use as antimalarial drugs, most specifically related to the malaria parasite *Plasmodium falciparum*, which is the most virulent of the *Plasmodium* species in humans. *P. falciparum* requires an external supply of pantothenic acid to support its intracellular growth.¹²² Erythrocytes infected with *P. falciparum* rapidly take up pantothenate via the “new permeability pathways” (NPPs) induced in the erythrocyte membrane by the maturing parasite.¹²³ This increased permeability of the membrane via the NPP is not present in uninfected erythrocytes. The *P. falciparum* parasite relies on the endogenous synthesis of CoA from pantothenate and does not require exogenous CoA.¹²⁴ Thus, the CoA biosynthesis pathway has been a target for antimicrobial drug discovery. Pantothenamides, amide analogs of pantothenic acid, and other derivatives have shown promise in killing the parasite. The antiplasmodial activity of the many already available pantothenic acid analogs now needs to be proven in mammalian models. As the spread of the resistance to antimalarial agents widens, the development of new therapies for the treatment and prevention of malaria is essential.

RESEARCH GAPS

- Requirements: An RDA has yet to be established for pantothenic acid.
- Drug interactions: Pantothenic acid and cofactors such as biotin share common receptors.¹²⁵ Oral contraceptives containing estrogen and progestin may increase the requirement for pantothenic acid.¹²⁶ Though pantetheine has lipid-lowering effects,^{86–91} little is known if combining pantetheine with other lipid-lowering drugs will produce additive effects on blood lipid profiles. Pantothenic acid inhibitors as antimalarial agents are also of interest for future investigation.^{122–124}
- Inflammation and oxidative stress: In animal models of various disorders, including lung injury,^{127,128} endometriosis,¹²⁹ necrotizing enterocolitis,¹³⁰ urethral trauma,¹³¹ hypospadias,¹³² and renal complications in diabetes mellitus,¹³³ dextranpanthol supplementation reduces inflammation, oxidative stress, and related pathology. Moreover, dietary pantothenic acid intake is inversely correlated with C-reactive protein in humans.¹³⁴ Paradoxically, the activity of pantetheinases (products of the vanin genes, *VNN1*, *VNN2*, and possibly *VNN3*), which convert pantetheine to pantothenic acid and cysteamine, is proinflammatory.¹³⁵ Knockout of the *Vnn1* gene in mice and inhibition of pantetheinase by RR6, a pantothenone similar in structure to pantothenic acid and dextranpanthol, is associated with reduced inflammation and protection against related pathologies.^{136,137} Future investigations are required to resolve this paradox, including the possibility that pantothenic acid and dextranpanthol may be feedback inhibitors of the pantetheinases.

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BIOTIN

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SUMMARY

Biotin is a water-soluble vitamin and serves as a coenzyme for five carboxylases, which catalyze key steps in the metabolism of fatty acids, glucose, and amino acids. Holocarboxylase synthetase (HCS), biotinidase, sodium-dependent multivitamin transporter, and the biotin transporters SMVT and MCT1 play crucial roles in biotin homeostasis in mammals. Biotin also regulates gene expression by biotinylation of lysine residues of histones H2A, H3, and H4. Human biotin requirements are unknown, and recommendations for dietary intake are based on estimates of biotin intake in healthy populations (“adequate intake”). Individuals carrying mutations in genes coding for HCS or biotinidase require lifelong supplementation with pharmacological doses of biotin. Reliable markers for biotin status include the activity of propionyl-CoA carboxylase and methylcrotonyl-CoA carboxylase, but urinary metabolite markers are unreliable measures of biotin deficiency. Anticonvulsants and lipoic acid may interfere with biotin metabolism, thereby increasing biotin requirements. Severe biotin deficiency has been linked to birth defects, multiple sclerosis misdiagnosis, and impaired immune function.

Keywords: Biotin; Holocarboxylase; Multiple carboxylase deficiency; Multiple sclerosis.

I. INTRODUCTION

Humans and other metazoans cannot synthesize the water-soluble vitamin biotin and depend on dietary biotin that originated in plants and microorganisms.¹ Biotin is now known to play key roles in the metabolism of macronutrients, histone modification, and various signal transduction pathways. Importantly, clinical evidence now exists that life stages (e.g., pregnancy) and lifestyle factors (e.g., smoking) affect biotin requirements. This chapter reviews the current state of knowledge in biotin nutrition in humans and, occasionally, other organisms.

A. Background

Biotin was first discovered in nutritional experiments demonstrating a requirement for the water-soluble vitamin biotin in mammals over 80 years ago.² Boas observed that

rats fed uncooked egg whites as the protein source lost hair and developed scaly dermatitis and neurological presentations. Later, it was revealed that uncooked egg whites contain the protein avidin, which binds biotin tightly, thereby preventing its absorption.

B. Nutrient Function and Structure

Holocarboxylase synthetase (HCS; also known as biotin–protein ligase) covalently attaches biotin to five human carboxylases, each of which serves biological functions.¹ In addition, biotin also plays unique roles in gene regulation, mediated by both biotinylation of histones and “classical” pathways of cell signaling. The chemical structure of biotin was determined by du Vigneaud et al.³ (Fig. 17.1). Biotin was first chemically synthesized by Harris et al. (1943).⁴

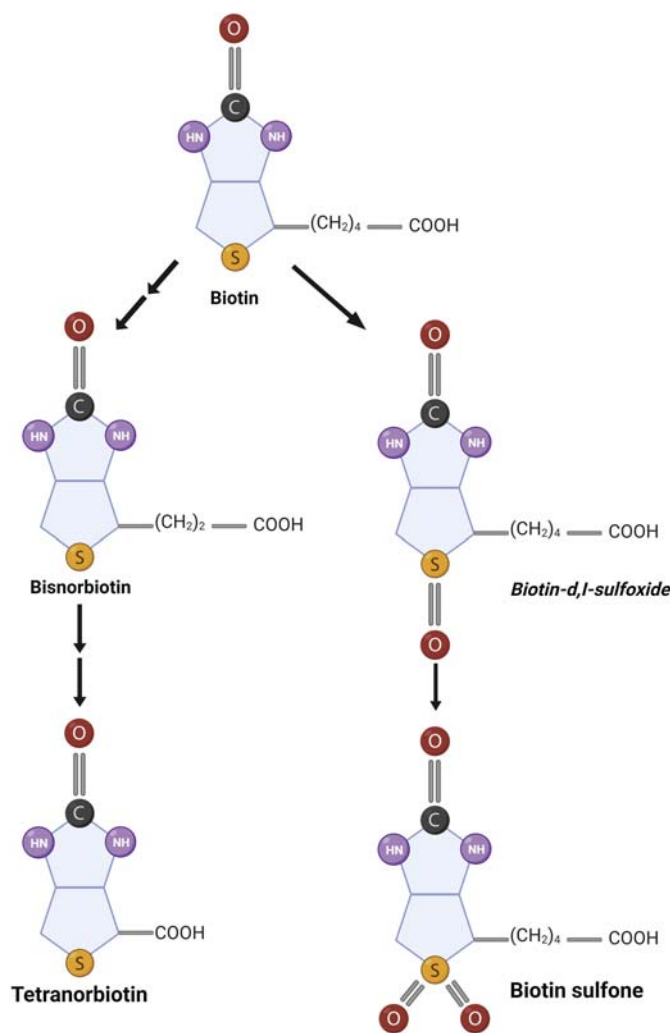


FIGURE 17.1 Pathways of biotin catabolism. Data from Ref. 5.

Biotin-dependent carboxylases

In mammals, biotin serves as a covalently bound coenzyme for acetyl-CoA carboxylases (ACC) 1 and 2, pyruvate carboxylase (PC), propionyl-CoA carboxylase (PCC), and 3-methylcrotonyl-CoA carboxylase (MCC). The attachment of biotin to the ϵ -amino group of a specific lysine residue in holocarboxylases is catalyzed by HCS; biotinylation of carboxylases requires ATP and proceeds in the following two steps.

1. $ATP + \text{biotin} + HCS \rightarrow \text{Biotinyl-5'}-AMP-HCS + \text{pyrophosphate}$
2. $\text{Biotinyl-5'}-AMP-HCS + \text{apocarboxylase} \rightarrow \text{holocarboxylase} + AMP + HCS$
3. (Net) $ATP + \text{biotin} + \text{apocarboxylase} \rightarrow \text{holocarboxylase} + AMP + \text{pyrophosphate}$

Both the N- and C-termini in HCS are important for recognition of carboxylases.⁶ These regions contain highly conserved sequences of amino acids (e.g., Met-Lys-Met at the attachment site). Holocarboxylases mediate the covalent binding of bicarbonate (not carbon dioxide), using 1'-N-carboxybiotinyl as an HCS-bound carboxyl donor.⁷

Both the cytoplasmic ACC1 and the mitochondrial ACC2 catalyze the binding of bicarbonate to acetyl-CoA to generate malonyl-CoA, but the two isoforms play distinct roles in intermediary metabolism.⁸ ACC1 produces malonyl-CoA for the synthesis of fatty acid synthesis in the cytoplasm; ACC2 is an important regulator of fatty acid oxidation in mitochondria. The malonyl-CoA produced by ACC2 inhibits mitochondrial uptake of fatty acids for β -oxidation.

PC, PCC, and MCC localize in mitochondria. PC is a key enzyme in gluconeogenesis. PCC catalyzes an essential step in the metabolism of propionyl-CoA, which is produced in the metabolism of some amino acids, the cholesterol side chain, and odd-chain fatty acids. MCC catalyzes an essential step in leucine metabolism (Fig. 17.2). Both PCC and MCC are composed of nonidentical subunits, i.e., biotinylated α subunits and nonbiotinylated β subunits, which are encoded by distinct genes.

Proteolytic degradation of the five holocarboxylases leads to the formation of biotinyl peptides and biocytin (biotin- ϵ -lysine). These compounds are further degraded by biotinidase to release biotin, which is then recycled in holocarboxylase synthesis.⁹

II. NUTRIENT METABOLISM

A. Synthesis

Mammals and other metazoans cannot synthesize biotin and depend on dietary biotin originating in microbial and plant biosynthetic pathways. The route of biosynthesis of biotin was largely elaborated by Eisenberg et al. working with *Escherichia coli*.^{10,11} In this pathway, dethiobiotin is formed from pimelyl-CoA (which can be synthesized from oleic acid) and carbamyl phosphate.¹² Sulfur is incorporated into dethiobiotin in a synthase-dependent step, generating biotin.¹³

B. Digestion

Biotin in foods is largely protein bound.⁹ Several gastrointestinal proteases may hydrolyze biotin-containing proteins to generate biotinyl peptides. Biotinyl peptides are further hydrolyzed by intestinal biotinidase to release biotin. Intestinal biotinidase is found in pancreatic juice, secretions of the intestinal glands, bacterial flora, and the brush border membranes. Biotinidase activities are similar in mucosa from duodenum, jejunum, and ileum. The primary site or sites for hydrolysis of biotinyl peptides are unknown. Small quantities of biotinyl peptides may be absorbed without prior hydrolysis.¹⁴

C. Transport

Early investigations of biotin transport in rat jejunum suggested that intestinal biotin uptake is mediated by both saturable and nonsaturable components.¹⁵ At biotin

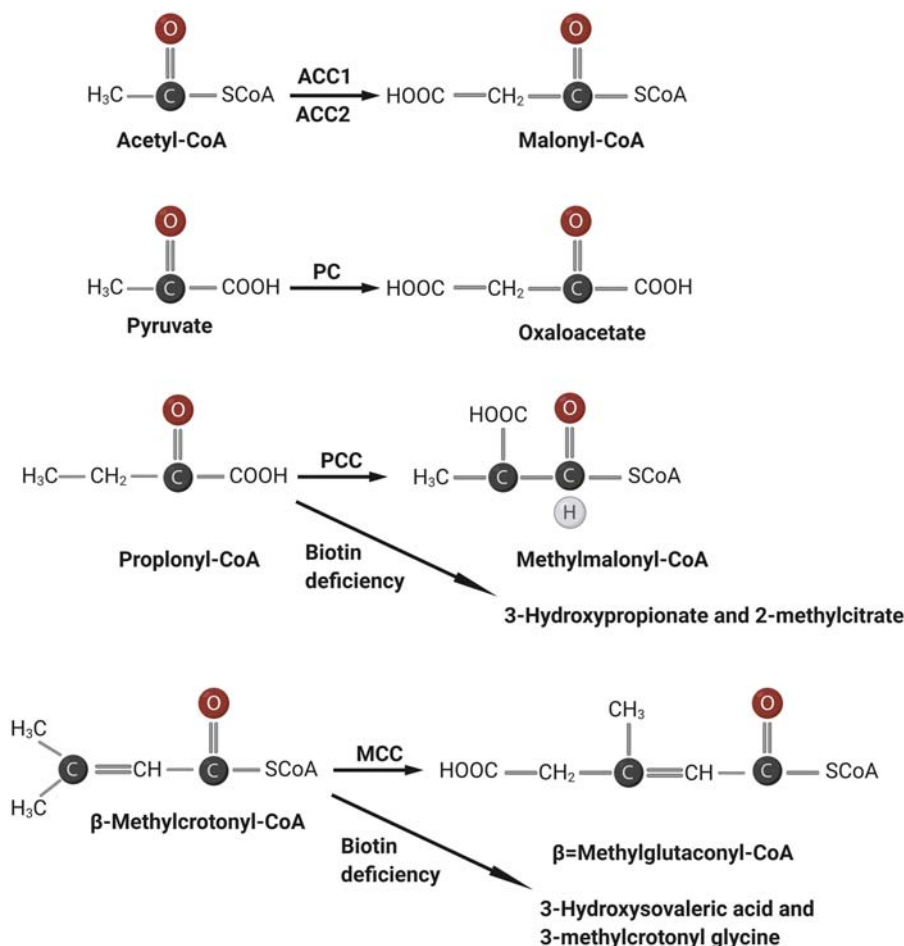


FIGURE 17.2 Biotin-dependent carboxylases. ACC, acetyl-CoA carboxylase; MCC, 3-methylcrotonyl-CoA carboxylase; PC, pyruvate carboxylase; PCC, propionyl-CoA carboxylase. Data from Ref. 1.

concentrations less than 5 $\mu\text{mol/L}$, biotin absorption proceeds largely by the saturable process, whereas, at concentrations above 25 $\mu\text{mol/L}$, nonsaturable uptake predominates. The saturable mechanism of biotin transport is sodium dependent.¹⁶ Transport of biotin is faster in the jejunum than in the ileum and is minimal in the colon (85 vs. 36 vs. 2.8 pmol/g tissue wet weight/25 min, respectively); the apparent K_m (Michaelis–Menten constant) of the transporter for biotin in rat jejunum is 3.7 $\mu\text{mol/L}$. The biotin transporter was cloned and has comparable affinities for biotin, pantothenic acid, and lipoic acid, if overexpressed in mammalian cells.¹⁷ Consequently, this transporter was named the sodium-dependent multivitamin transporter (SMVT). Four splicing variants of SMVT transcripts have been identified in rats.¹⁸ The expression of SMVT is regulated by protein kinase C and histone biotinylation.¹⁹

Human albumin, α -globulin, and β -globulin bind biotin rather unspecifically.²⁰ Biotinidase has one high-affinity (dissociation constant $K_d = 0.5 \text{ nmol/L}$) and one low-affinity ($K_d = 50 \text{ nmol/L}$) binding site for biotin and might serve as a biotin carrier protein in the plasma of normal adults.²¹ A biotin-binding glycoprotein (molecular weight 66,000) is present in serum of

pregnant and estrogenized female rats and appears to be essential for embryonic survival.²² The protein binding of biotin in plasma is somewhat controversial; evidence has been provided for the importance of a low-affinity, high-capacity system such as albumin.²³

The SMVT is the most important transporter for mediating biotin uptake into the liver and most peripheral tissues.¹⁸ In contrast, monocarboxylate transporter 1 contributes importantly toward biotin uptake into lymphoid cells and perhaps keratinocytes.²⁴ Studies indicate that biotin requirements are not uniform across tissues, but some tissues might be more susceptible to biotin deficiency than others.²⁵

D. Metabolism

McCormick and coworkers identified two pathways of biotin catabolism in microorganisms and mammals (Fig. 17.1). In one pathway, biotin is catabolized by β -oxidation of the valeric acid side chain.⁵ The repeated cleavage of two-carbon units leads to the formation of bisnorbiotin, tetranorbiotin, and related catabolites that are known to result from β -oxidation of fatty acids (i.e., α , β -dehydro, β -hydroxy, and β -keto intermediates).

β -Ketobiotin and β -ketobisnorbiotin are unstable and may decarboxylate spontaneously to form bisnorbiotin methyl ketone and tetranorbiotin methyl ketone.⁵ After degradation of the valeric acid side chain to one carbon (tetranorbiotin), microorganisms cleave and degrade the heterocyclic ring⁵; degradation of the heterocyclic ring is quantitatively minor in mammals.²⁶ In a second pathway of biotin catabolism, the sulfur in the heterocyclic ring is oxidized to produce biotin-l-sulfoxide, biotin-d-sulfoxide, and biotin sulfone.⁵ It is likely that sulfur oxidation in the biotin molecule occurs in the smooth endoplasmic reticulum in a reaction that depends on nicotinamide adenine dinucleotide phosphate.²⁷ Finally, biotin is also catabolized by a combination of both β -oxidation and sulfur oxidation, producing compounds such as bisnorbiotin sulfone.

[3H]Biotin efflux from human lymphocytes is best described by using a three-compartment model; the half-lives of the three elimination phases are approximately 0.2 h, 1.2 h, and 22 h.²⁸ Likely, the half-life of [3H]biotin during the terminal, slow phase of efflux is determined by the breakdown of biotin-dependent carboxylases to release free [3H]biotin, followed by efflux of [3H]biotin from lymphocytes. The half-life of [3H]biotin during the terminal phase of efflux (22 h) resembles the half-lives of PC (28 h) and ACC (4.6 days). Studies in pigs also generated valuable insights into biotin pharmacokinetics.²⁹

Biotin is distributed unequally across cellular compartments.³⁰ For example, the vast majority of biotin in rat liver localizes in mitochondria and cytoplasm, whereas only a small fraction localizes in microsomes. The relative enrichment of biotin in mitochondria and cytoplasm is consistent with the role of biotin as a coenzyme for carboxylases in these compartments. A quantitatively small but qualitatively important fraction of biotin localizes in the cell nucleus, i.e., about 0.7% of total biotin in human lymphoid cells can be recovered from the nuclear fraction.³¹ The relative abundance of nuclear biotin increases to about 1% of total biotin in response to proliferation, consistent with a possible role for biotinylated histones in cell proliferation.

A relatively large fraction of intravenously administered biotin accumulates in rat liver, consistent with a role for this organ in biotin storage.³⁰ Depletion and repletion experiments of biotin-dependent carboxylases in rat liver suggest that ACC2 serves as a reservoir for biotin,³² but studies in the Zempleni laboratory are not consistent with a role of any holocarboxylase as a biotin reservoir.²⁵

Healthy adults excrete approximately 100 nmol per day of biotin and catabolites into urine.³³ Biotin accounts for approximately half of the total; bisnorbiotin, biotin-d,l-sulfoxides, bisnorbiotin methyl ketone, biotin sulfone, and tetranorbiotin-l-sulfoxide account for most of the balance. If physiological or pharmacological doses of biotin are administered parenterally to humans,

rats, or pigs, 43%–75% of the dose is excreted into urine^{26,34}; the remainder is unaccounted for and might be due to excretion of metabolites not detectable by avidin assays; incorporation into biotin sinks such as histones or carboxylases; or excretion via nonrenal routes. Renal epithelia reclaim biotin that is filtered in the glomeruli in an SMVT-mediated process.³⁵

The biliary excretion of biotin and catabolites is quantitatively minor, i.e., <2% of an intravenous dose of [14C] biotin.³⁶ The concentration ratios of biotin, bisnorbiotin, and biotin-d,l-sulfoxide (bile vs. serum) in pigs are at least one order of magnitude smaller than the ratio measured for the cholephilic compound, bilirubin.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Biotin Deficiency

Biotinidase deficiency

Low activities of the enzyme biotinidase cause a failure to recycle biotin from degraded carboxylases, i.e., to release biotin from biocytin.³⁷ Substantial amounts of biocytin are excreted into urine,³⁸ eventually leading to biotin deficiency. Thus, clinical and biochemical features in children with biotinidase deficiency are similar to those described for biotin deficiency.

Typically, symptoms of biotinidase deficiency appear at age 1 week to >1 year in humans.³⁷ Wolf proposed to distinguish between patients with profound biotinidase deficiency (<10% of normal serum biotinidase activity) and patients with partial deficiency (10%–30% of normal biotinidase activity). Biotin deficiency in humans is rare because biotin is continuously recycled by the action of biotinidase. The estimated incidence of profound biotinidase deficiency is 1 in 112,000, and the incidence of partial biotinidase deficiency is 1 in 129,000. The combined incidence of profound and partial deficiency is 1 in 60,000 live births; an estimated 1 in 123 individuals is heterozygous for the disorder. Most of the children in whom biotinidase deficiency has been diagnosed are Caucasian. Mutations of the biotinidase gene have been well characterized at the molecular level.³⁹

Biotinidase deficiency leads to multiple carboxylase deficiency in humans with ATP deficit.⁴⁰ The mouse model of biotinidase deficiency dramatically reveals loss of myelin with neurodegeneration, which is in part rescuable with repletion with biotin.⁴¹ Studies using biotinidase-deficient transgenic knockout mice revealed that the animals become symptomatic by day 8 of life and exhibit failure to thrive.⁴² Symptomatic animals exhibit demyelination, axonal degeneration, and corpus callosum compression in the brain. Injection of biotin into these mice prevents the development of the symptoms.

Today, biotinidase deficiency is distinguished as one of the few metabolic diseases for which we can identify

patients and successfully rescue them.⁴³ The first child in the world identified to have profound biotinidase deficiency was identified by newborn screening in 1985 after screening 37,000 newborns, where newborn screening is now incorporated into all US states and most countries.⁴⁴ Children with profound biotinidase deficiency are treated with 5–20 mg of biotin per day to compensate for urinary loss of biotin.³⁷ If identified early, symptomatic patients improve rapidly after life-long biotin supplementation is initiated.

Procedures for prenatal diagnosis of biotinidase activity in cultured amniotic fluid cells and for neonatal screening using blood samples have been proposed.³⁷ Biotinidase activity is measured by quantifying the release of p-aminobenzoic acid from N-biotinyl-p-aminobenzoate.⁴⁵ The mean ($\pm$ SD) normal activity of biotinidase is 5.8 ± 0.9 nmol p-aminobenzoate liberated/min/mL serum. Synthetic inhibitors are available for studies of biotinidase in vitro.⁴⁶

The first case of adult-onset biotinidase deficiency with adult-onset was presented in 2015.⁴⁷ It described a 22-year-old man with extensive myelopathy and bilateral optic neuropathy that was initially diagnosed with neuromyelitis optica. Just a small amount of biotin is needed in the diet, so some individuals who have mutations in their biotinidase gene do not overtly present until later in age. In one example, a 36-year-old woman who had progressive optic atrophy from the age of 13 also had stroke-like episodes and spastic diplegia in her 20s before it was determined that she had a mutation in her biotinidase gene.⁴⁸ Unfortunately, some of the damage was irreversible and could not be resolved with biotin treatment. Untreated vision and hearing problems are usually irreversible unless treatment is done within a critical time window.⁴⁹ All states in the United States screen for the disorder, but some countries such as France do not.

Most urgently, biotin can be administered to rescue some patients previously diagnosed with multiple sclerosis, but in fact have biotinidase deficiency. Multiple sclerosis is the most common inflammatory demyelinating disease of the central nervous system, and the cause of MS is unknown. To some extent, it is a default diagnosis with loss of myelin as one of the key signatures of diagnosis. Many of the symptoms of multiple sclerosis are indistinguishable from biotin deficiency. It is now highly recommended that individuals presenting with myelopathy, neuropathy, and vision loss be tested for by biotinidase deficiency because the biotinidase enzyme test is so easy and just 10 mg per day of biotin can prevent irreversible nerve damage, reversing vision loss, or spastic paraparesis, but it can take weeks to months for complete resolution.⁵⁰

The symptoms of biotinidase deficiency can be wide-ranging. Multisymptomatic children have presented

with tetraplegia due to myelopathy with and without vision loss.⁵¹ For these reasons, it is highly recommended that individuals presenting with myelopathy should be screened for by biotinidase deficiency. Many children have gone weeks to months without being correctly diagnosed with biotinidase deficiency. Rapid recognition of biotinidase is critical to preventing permanent vision loss neurological damage.

Carboxylase deficiencies

Early investigations of biotin transport in rat jejunum suggested that intestinal biotin uptake is mediated by both saturable and nonsaturable components.¹⁵ At biotin concentrations less than 5 μ mol/L, biotin absorption proceeds largely by the saturable process, whereas, at concentrations above 25 μ mol/L, nonsaturable uptake predominates. The saturable mechanism of biotin transport is sodium dependent.¹⁶ Transport of biotin is faster in the jejunum than in the ileum and is minimal in the colon (85 vs. 36 vs. 2.8 pmol/g tissue wet weight/25 min, respectively); the apparent K_m (Michaelis–Menten constant) of the transporter for biotin in rat jejunum is 3.7 μ mol/L. The biotin transporter was cloned and has comparable affinities for biotin, pantothenic acid, and lipoic acid, if overexpressed in mammalian cells.¹⁷ Consequently, this transporter was named SMVT. Four splicing variants of SMVT transcripts have been identified in rats.¹⁸ The expression of SMVT is regulated by protein kinase C and histone biotinylation.¹⁹

Chemically pure biotin is about 100% bioavailable, even if the dose administered is 600 times the normal dietary intake.³⁴ Biotin exits enterocytes at the basolateral membrane by a carrier-mediated, sodium-independent process that does not accumulate biotin against a concentration gradient (Said et al., 1988).⁵²

Biotin transporter deficiency

A patient with biotin transporter deficiency did not have any of the cutaneous symptoms of biotin deficiency.⁵³ Rather, central nervous system signs and profound metabolic disturbances consistent with deficiencies of multiple carboxylases dominated that patient's clinical picture. All resolved with biotin supplementation. Evidence was provided that biotin transporter deficiency was not caused by abnormal SMVT, but the identity of the afflicted transporter remained unknown.

B. Gene Regulation

Biotinylation of histones

Histones are quantitatively the most important proteins in chromatin and play essential roles in processes such as DNA packaging and repair, replication, and gene regulation.⁹ Mammals express five major classes

of histones: linker histone H1 and core histones H2A, H2B, H3, and H4. Histones consist of a globular domain and a flexible N-terminal tail. Chromatin is made of repetitive nucleoprotein complexes, the nucleosomes. Each nucleosome ("nucleosomal core particle") consists of 146 base pairs of DNA, wrapped around an octamer of core histones (one H3/H3/H4/H4 tetramer, two H2A/H2B dimers). The N-terminal tails of core histones protrude from the nucleosomal surface; covalent modifications of these tails affect the structure of chromatin and play critical roles in gene regulation. Some modifications also exist in the globular domain and C-terminal regions^{1,54} (Fig. 17.3).

Biotinylation of distinct lysine (K) residues is among the recognized modification marks in histones.^{31,55} Biotinylated chromatin causes a condensation to a smaller nucleosome structure with less DNA being wrapped around a nucleosome.⁵⁶ Biotinylated histones have been repeatedly detected in repressed loci.^{57,58} HCS can be detected in both nuclear and extranuclear compartments^{59,60}; nuclear HCS is a chromatin protein that interacts directly with histone 3 that can biotinylate lysine 9 and 18.⁶¹ Phenotypes of HCS knockdown include decreased life span and heat survival in *Drosophila melanogaster*. Knockout of HCS in drosophila as well as human and mammalian cells in culture results in derepression of

long terminal repeats and chromosomal instability due to effects on gene expression rather than reduced activities of the biotin-dependent carboxylases. MicroRNAs (miR) are very small, just 22 nucleotides in length. They are increasingly being appreciated as sensors of nutrients involved in regulation of associated genes. The miR-539 promotor is a biotin sensor in human cells. It is also involved in the regulation of HCS enzyme expression, making more HCS when more biotin is present.⁶²

Evidence is emerging for the biological functions of K12-biotinylated histone H4 (H4K12bio), while information regarding other biotinylation marks is scarce. The H4K12bio mark is enriched in heterochromatin repeats (telomeres, long terminal repeats, and pericentromeric alpha satellite repeats) and transcriptionally repressed genes.¹ H4K12bio plays a role in the regulation of biotin transporter expression and the repression of long-terminal repeats.^{19,63} Importantly, the abundance of histone biotinylation marks depends on biotin supply. Low levels of histone biotinylation in biotin-deficient cells and model organisms have been linked to increased frequency of retrotransposition events, suggesting a role for histone biotinylation in chromosomal stability. Biotinylation of lysines other than K12 in histone H4 appears to have biological functions similar to that described for H4K12bio.⁵⁸

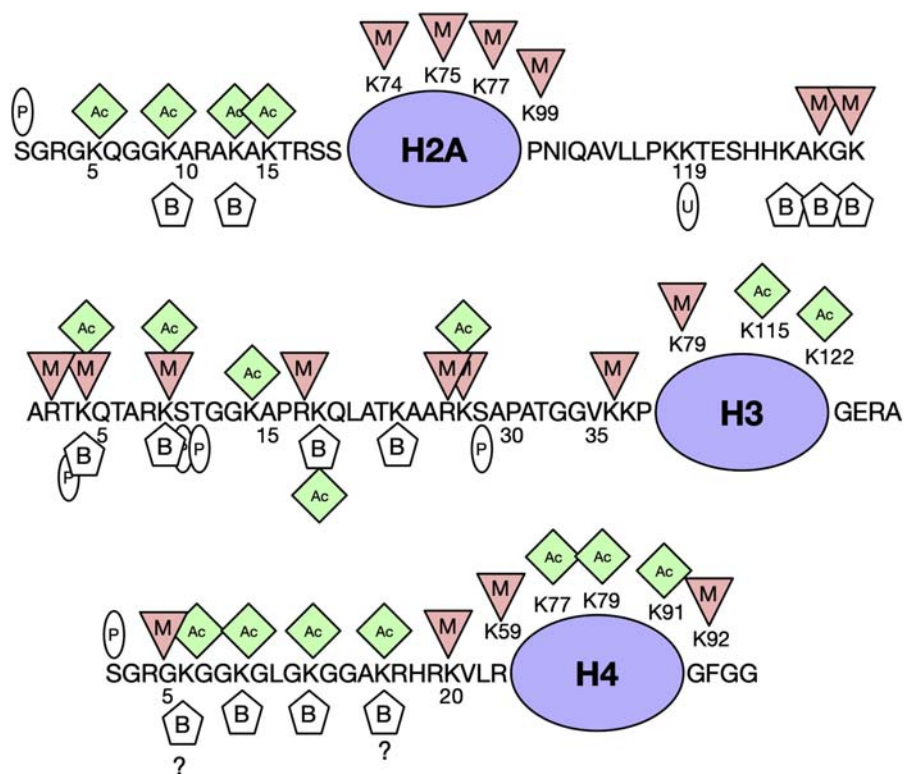


FIGURE 17.3 Modification sites in histones H2A, H3, and H4. Ac, acetate; B, biotin; M, methyl; P, phosphate; U, ubiquitin. Data from Refs. 1 and 54.

C. Gene Expression

Pharmacological doses of biotin have been shown to increase the expression of genes with critical roles in glucose homeostasis, including hepatic glucokinase⁶⁴ and insulin receptor in diabetic rats.⁶⁵ In addition, biotin supplementation produced in vivo changes in the pancreatic islets including increased insulin secretion, islet cell relative volume, and glucose tolerance.^{65,66} Biotin supplementation also demonstrably ameliorated hyperglycemia in animal models of diabetes.⁶⁵ However, pharmacological doses of biotin in the clinic have resulted in mixed data.^{65,67} Thus, whether supplemental biotin could be beneficial for the management of hyperglycemia in patients with impaired glucose tolerance remains to be proven.

IV. DIETARY REQUIREMENTS

Healthy adults excrete approximately 100 nmol per day of biotin and catabolites into urine.³³ Biotin accounts for approximately half of the total; bisnorbiotin, biotin-*d,l*-sulfoxides, bisnorbiotin methyl ketone, biotin sulfone, and tetranorbiotin-*l*-sulfoxide account for most of the balance. If physiological or pharmacological doses of biotin are administered parenterally to humans, rats, or pigs, 43%–75% of the dose is excreted into urine^{26,34}; the remainder is unaccounted for and might be due to excretion of metabolites not detectable by avidin assays; incorporation into biotin sinks such as histones or carboxylases; or excretion via nonrenal routes. Renal epithelia reclaim biotin that is filtered in the glomeruli in an SMVT-mediated process.³⁵

A. Indicators of Deficiency or Inadequacy

Blood biotin concentration is not an early or sensitive indicator of impaired biotin status.⁶⁸ Indicators of biotin deficiency have been experimentally validated by causing asymptomatic biotin deficiency by consuming raw egg whites. Urinary biotin excretion declines to abnormal values in roughly 90% of subjects after 3 weeks of raw egg white consumption.^{69–71} Detectable increases in both 3-hydroxyisovaleric acid and 3-hydroxyisovalerylcarnitine are observable in the urine in 90% of subjects after 2 weeks of eating raw egg whites, thus indicating that biotin depletion decreases the activity of the biotin-dependent enzyme MCC.

The activity of the biotin-dependent enzyme PCC in lymphocytes isolated from venous blood samples is the best indicator of biotin status.⁷² However, this assay is technically demanding, and so the abundance of holo-PCC and holo-MCC determined by gel densitometry

and fluorescent-labeled streptavidin is the best practically useful indicator of marginal biotin deficiency.⁷³

B. Criteria of Adequacy

Data for the requirements for biotin are lacking. For this reason, the Food and Nutrition Board of the Institute of Medicine has offered only Adequate Intakes (AIs) for biotin.⁷⁴ Biotin AI for an adult is 30 µg and for an infant is 5 µg. The AIs for older infants (6 µg/day), children (8–12 µg/day), adolescents (25 µg/day), and adults (30 µg/day) were all extrapolated from the values for younger infants (Fig. 17.2).

C. Assessment of Status

Microbial growth assays 4'-hydroxyazobenzene-2-carboxylic acid

Biotin auxotroph microorganisms have been used for biotin analysis, but these microbes lack chemical specificity and show variable growth responses to biotin and the various biotin precursors, catabolites, and analogs (Zempleni, 2000).⁷⁵ In addition, binding of biotin to proteins can be a meaningful confounder in microbial growth assays.

(Strept)Avidin-binding assays

The proteins avidin and streptavidin are widely used in biotin analysis because they bind biotin with extraordinary strength and specificity; the dissociation constant of the avidin–biotin complex is 1.3×10^{-15} M.⁷⁶ Avidin and streptavidin are purified from egg white and *Streptomyces* sp., respectively. Streptavidin has greater specificity for biotin compared with avidin and is the probe of choice in avidin-binding assays. The Zempleni laboratory relies on an assay developed by Kuroishi et al.⁷⁷

Self-administration of pharmacological doses biotin supplementation can interfere with laboratory assays employing streptavidin technology.^{78–80} This is likely to continue with the increasing interference with clinical tests that depend on a biotin-streptavidin technology. Nutritionists, endocrinologists, physicians, nurses, and other health care providers should determine the biotin supplementation patient nutritional history before administering these tests and be made aware of this concern for potential false positives.

Biotin catabolites and compounds that are structurally similar to biotin also bind to avidin, causing the following potential pitfalls in biotin analysis. First, avidin binds biotin catabolites less tightly compared with biotin (Zempleni, 2000).⁷⁵ Hence, avidin-binding assays may underestimate the true concentration of biotin plus catabolites if calibrated by using biotin. Second, compounds other

than biotin or catabolites may bind to avidin and cause artificially large readings for “apparent biotin.” Therefore, avidin-binding compounds in biological samples need to be resolved by chromatography prior to analysis of individual chromatographic fractions against authentic standards, as reviewed elsewhere.⁷⁵

4'-Hydroxyazobenzene-2-carboxylic acid

4'-Hydroxyazobenzene-2-carboxylic acid is a robust probe to quantify chemically pure biotin at concentrations that exceed those typically found in biological samples, e.g., in pharmaceutical preparations.⁷⁵

Biotin analogs

Synthetic biotin analogs (e.g., iminobiotin) and some naturally occurring biotin metabolites (e.g., biocytin) are available commercially.⁷⁵ Likewise, a large number of reagents are available for chemical biotinylation of amines, sulfhydryl groups, carboxyl groups, nucleic acids, and others.

Direct measures

The serum concentrations and the urinary excretions of biotin and catabolites are potential markers of biotin status. There are no currently used reliable markers of biotin status in standard clinical laboratory settings. Analysis from multiple angles reveals that the most reliable markers of biotin status are levels of MCC and PCC.⁷³ Urinary analysis of biotin status does not work for detecting deficiency. Urinary tests are only useful for the detection of biotin supplementation.

Biotin accounts for about half of the total of all biotinyl compounds in human urine and serum¹ (Table 17.1). Tetranorbiotin-l-sulfoxide was detected in human urine,

but its avidin affinity is too low to allow meaningful quantitation.

The urinary excretion of biotin and biotin catabolites decreases rapidly and substantially in biotin-deficient individuals.⁶⁸ However, serum concentrations of biotin, bisnorbiotin, and biotin-d,l-sulfoxide do not decrease in biotin-depleted individuals and in patients on biotin-free total parenteral nutrition⁷⁸ during reasonable periods of observation. Hence, serum concentrations are not good indicators of marginal biotin deficiency.

Indirect measures

Lymphocyte PCC and its activation by biotin *ex vivo* are reliable markers for human biotin status.⁷⁹ The carboxylase activation index equals the ratio of PCC activities in lymphocytes incubated with and without excess biotin.⁷⁸ High values for the activation index suggest that a substantial fraction of total PCC is apo-PCC, indicating biotin deficiency.

Reduced carboxylase activities in biotin deficiency cause metabolic blocks in intermediary metabolism (Fig. 17.2). Reduced activity of MCC impairs leucine catabolism. As a consequence, 3-methylcrotonyl-CoA is shunted to alternative pathways, leading to an increased formation of 3-hydroxyisovaleric acid and 3-methylcrotonyl glycine. The urinary excretion of 3-hydroxyisovaleric acid is an early and sensitive indicator of biotin status.⁸⁰ The derivative, carnitine-3-hydroxyisovaleric acid, also appears to be a robust indicator of biotin status.⁷¹

Reduced activity of PCC causes a metabolic block in propionic acid metabolism. Consequently, propionic acid is shunted to alternative metabolic pathways. In these pathways, 3-hydroxypropionic acid and 2-methylcitric acid are formed. The urinary excretion of 3-hydroxypropionic acid and 2-methylcitric acid is not a good indicator of marginal biotin deficiency.⁸⁴ The accumulation of propionyl-CoA likely leads to occasional incorporation of a three-carbon fragment (instead of the usual acetyl group) in fatty acid elongation and consequently to significant accumulation of odd-chain fatty acids in plasma phospholipids, plasma triglycerides, and RBC membrane phospholipids.

ACC and PC activities might also be reduced in biotin deficiency, but the responses of these pathways to biotin depletion have not yet been thoroughly investigated. Apparently, lactate levels increase in biotin-depleted *Drosophila melanogaster* compared with biotin-sufficient controls, presumably due to decreased PC activity.⁸⁵ This observation is consistent with the biotin responsiveness of lactate and pyruvate levels observed in an individual with inborn biotin transporter deficiency.⁵³

TABLE 17.1 Serum concentrations and urinary excretions of biotin and catabolites.

Compound	Serum (pmol/L)	Urine (nmol/24 h)
Biotin	244 ± 61	35 ± 14
Bisnorbiotin	189 ± 135	68 ± 48
BSO ^a	15 ± 33	5 ± 6
BNBMK ^a	ND ^a	9 ± 9
Biotin sulfone	ND ^a	5 ± 5
Total biotin	464 ± 178 ^b	122 ± 66

Means ± SD are reported (*n* = 15 for serum; *n* = 6 for urine).

^aBSO, Biotin-d,l-sulfoxides; BNBMK, bisnorbiotin methyl ketone; ND, not determined (BNBMK and biotin sulfone had not been identified at the time when this study of serum was conducted, and, hence, quantification of these “unknowns” was based on using biotin as a standard.

^bIncluding three unidentified biotin catabolites.

Data from Ref. 1.

D. Reference Intakes

Adequate intakes

The Food and Nutrition Board of the Institute of Medicine released recommendations for the United States and Canada for AIs of biotin in 1998 (Table 17.2).⁷⁴ These reference values are based on biotin intakes of infants exclusively fed human milk and showing no signs of inadequacy. These values were then extrapolated on the basis of body weight to the 0.75 power for adults and various age groups of children and should be adequate to maintain normal excretion of 3-hydroxyisovaleric acid, based on the data of Mock et al., 1997.⁸⁶ AIs may serve as goals for the nutrient intake of individuals. Biotin supplements may contribute substantially to biotin intake. For example, 15%–20% of individuals in the United States report consuming biotin-containing dietary supplements.⁷⁴

Life stages and lifestyle factors that affect biotin requirements

Pregnancy and use of drugs may be associated with an increased requirement for biotin as described above. Smoking might further accelerate biotin catabolism in women,⁸⁷ and alcohol impairs human placental biotin transport.⁸⁸

The AI (Table 17.2) for pregnant and lactating women may be insufficient.⁸⁹ Lactation may generate an increased demand for biotin. At 8 days postpartum, biotin in human milk is approximately 8 nmol/L and accounts for 44% of biotin plus catabolites; bisnorbiotin and biotin-d,l-sulfoxide account for 48% and 8%, respectively.⁸⁶ By 6 weeks postpartum, the biotin concentration

increases to approximately 30 nmol/L and accounts for about 70% of biotin plus catabolites; bisnorbiotin and biotin-d,l-sulfoxides account for about 20% and less than 10%, respectively. Marginal biotin deficiency occurs spontaneously in a substantial proportion of women during normal human pregnancy,⁹⁰ suggesting either these changes may be normal for pregnancy or that biotin intakes at least two to three times the AI may be needed during pregnancy and lactation.⁷⁴

E. Basis for Individual Variability in Requirements

Variations in individual requirements for biotin are rarely observed since biotin is required in such small amounts and the human body has mechanisms in place for recycling endogenous biotin. However, genetic polymorphisms in HCS, biotinidase, or biotin transporters are predicted to potentially cause variations in an individual's requirement for biotin.

F. Tolerable Upper Intake Levels

No Tolerable Upper Intake Level for biotin has been established. Biotin has no reported toxicity in humans. Daily use of 100 mg oral up to 20 mg intravenous for the treatment of subjects with biotin-responsive inborn errors of metabolism does not produce reports of toxicity. High-dose biotin administration (300 mg/day) leads to high biotin blood concentrations with increased unchanged urinary excretion.^{91,92}

G. Dietary Sources

The majority of biotin in meats and cereals appears to be protein bound.⁹³ Most studies of biotin content in foods have depended on using bioassays. Despite potential analytical limitations due to interfering endogenous compounds, protein binding, and lack of chemical specificity for biotin, there is reasonably good agreement among published reports. Biotin is widely distributed in natural foodstuffs. Foods relatively rich in biotin include egg yolk, liver, and some vegetables. Due to the lack of data on biotin contents of foods, there are little data available on the usual food and dietary supplement intakes in the United States. There is one report of the biotin content in 87 selected foods.⁹⁴ The dietary biotin intake in Western populations is about 35–70 µg per day (143–287 nmol per day). Infants consuming 800 mL of mature breast milk per day ingest approximately 6 µg (24 nmol) of biotin.⁹⁵ It remains unclear whether biotin synthesis by gut microorganisms contributes importantly to the total biotin absorbed.⁹³

TABLE 17.2 Adequate intakes of biotin.

Life-stage group	Adequate intake (µg per day)
<i>Infants</i>	
0–6 months	5
7–12 months	6
<i>Children</i>	
1–3 years	8
4–8 years	12
<i>Adults</i>	
9–13 years	20
14–18 years	25
19 years and older	30
Pregnant women	30
Lactating women	35

Data from Ref. 74.

V. ISSUES RELATED TO BIOTIN IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Frank biotin deficiency

Signs of frank biotin deficiency have been described in patients receiving parenteral nutrition without biotin supplementation⁹³ and in patients with biotinidase deficiency.³⁷ Frank biotin deficiency may also be observed in individuals consuming large amounts of raw egg white, which contains nondenatured avidin. Binding of biotin to avidin in the gastrointestinal tract prevents absorption of biotin.⁹⁶

Clinical findings of frank biotin deficiency include periorificial dermatitis, conjunctivitis, alopecia, ataxia, hypotonia, ketolactic acidosis/organic aciduria, seizures, skin infection, and developmental delay in infants and children.³⁷ In addition, the following symptoms were observed in adults and adolescents on egg white diet:

- thinning hair, often with loss of hair color
- skin rash described as scaly (seborrheic) and red (eczematous); in several cases, the rash was distributed around the eyes, nose, and mouth
- depression, lethargy, hallucinations, and paresthesias of the extremities.

Neurologic system

Disturbances in biotin homeostasis in the CNS cause encephalopathies.⁹⁷ Factors leading to biotin imbalances in CNS include deficiencies of biotinidase, HCS, and perhaps biotin transporters as described below. Afflicted patients typically respond to the administration of large doses of biotin by maintaining normal neurologic function. The reader should note that the biotin-responsive defect observed by Ozand et al.⁹⁷ turned out to be a thiamin transporter defect⁹⁸; the link between biotin and thiamin transport remains to be elucidated.

Moderate dietary biotin deficiency is typically not associated with neurologic symptoms because the CNS maintains normal concentrations of biotin at the expense of other tissues. Evidence has been provided that biotin deficiency causes a >90% decrease in holocarboxylases in rat liver, whereas brain carboxylases remain unchanged.⁹⁹ Apparently, biotin deficiency decreases expression of SMVT in rat liver while maintaining normal expression of SMVT in brain.

Multiple sclerosis is the most common inflammatory demyelinating disease of the central nervous system, and the cause of multiple sclerosis is unknown, but there is an autoimmune etiology. High-dose biotin (300 mg per day) has been shown in a singular placebo-controlled clinical trial of multiple sclerosis to provide

benefit based on expanded disability status scale scores.^{100,101} High doses of biotin (100–300 mg per day) were tested in 23 patients with primary or secondary progressive MS. Improved clinical outcomes were reported in nearly all subjects. In four patients with prominent visual impairment related to optic nerve injury, visual acuity improved significantly and visual evoked potentials improved in two. One patient with left-sided blindness for both eyes steadily improved after several months of biotin. Sixteen patients out of the 18 with prominent spinal cord involvement improved. While biotin deficiency is well known to cause loss of myelin characteristic of multiple sclerosis, the explanation for why multiple sclerosis patients sometimes responded to higher doses of biotin is completely unknown other than for the rare examples of biotinidase deficiency misdiagnosed as multiple sclerosis as described in [Section IV A](#) of this chapter.

Immune system

Biotin deficiency has adverse effects on cellular and humoral immune functions. For example, children with hereditary abnormalities of biotin metabolism developed candida dermatitis; these children had absent delayed hypersensitivity skin test responses, IgA deficiency, and subnormal percentages of T lymphocytes in peripheral blood.¹⁰² In biotin-deficient rats, the synthesis of antibodies is reduced.¹⁰³ Biotin deficiency in mice decreases the number of spleen cells and the percentage of B lymphocytes in spleen¹⁰⁴; inhibits thymocyte maturation¹⁰⁵; and increases the production of proinflammatory cytokines.⁷⁷ Studies indicate that biotin deficiency has a profound effect on increasing inflammatory markers in dendritic cells and that this may be one of the reasons for the increased inflammation and biotin-deficient individuals.¹⁰⁶ Studies of salmonella food poisoning indicate that this causes biotin deficiency not due to direct interruption of absorption but rather due to increased inflammation.¹⁰⁷

Cell proliferation

Biotin deficiency causes decreased rates of cell proliferation and cell cycle arrest in HeLa cells and choriocarcinoma cells.¹ Human lymphoid cells respond to mitogen-induced proliferation with a 200%–600% increase in biotin uptake compared with quiescent controls. This effect is mediated by an increased expression of biotin transporters rather than by an increased affinity of transporters for biotin.^{108,109} The increased biotin uptake in proliferating cells is paralleled by increased activity.

Cell stress and survival

HCS deficiency mimics biotin deficiency. HCS knock-down flies exhibited a decreased survival in response to heat stress in comparison with wild-type controls.¹¹⁰ In

contrast, biotin deficiency increased survival of human lymphoid cell cultures treated with cell death signals and antineoplastic agents, possibly due to increased nuclear translocation of the transcription factor NF- κ B.¹¹¹ *Drosophila melanogaster* exhibits greater survival rates in response to stress compared with biotin-sufficient controls, if fed biotin-defined diets for 11 days.¹¹² The reason for the differences in HCS- and biotin-deficient organisms is unknown.

Lipid metabolism

Three biotin-dependent carboxylases are directly linked to lipid metabolism: ACC1, ACC2 (fatty acid synthesis and β -oxidation), and PCC (metabolism of odd-chain fatty acids). Biotin deficiency causes alterations of the fatty acid profile in liver, skin, and serum of several animal species, which was paralleled by a >80% decrease in carboxylase activities.¹¹³ The observed increase in the abundance of odd-chain fatty acids suggests that odd-chain fatty acid accumulation might be a marker for reduced PCC activity in biotin-deficient individuals. In patients who developed biotin deficiency during parenteral alimentation, the percentage of odd-chain fatty acids (15:0, 17:0) in serum increased for each of the four major lipid classes, i.e., cholesterol esters, phospholipids, triglycerides, and free fatty acids, but the relative changes have not always been consistent among studies. Some tissues might be more susceptible than others, e.g., biotin deficiency affected the fatty acid composition in liver to a much greater extent than in brain tissue.

Teratogenic effects of biotin deficiency

Severe biotin deficiency is teratogenic in several animal species, including cleft palate, micrognathia, and micromelia in rats and mice.¹¹⁴ About half of the pregnant women in the United States are marginally biotin deficient despite a normal dietary biotin intake,¹¹⁵ judged by increased urinary 3-hydroxyisovaleric acid, and decreased urinary biotin and PCC activity in lymphocytes. Marginal biotin deficiency in pregnant women might be caused by accelerated biotin catabolism and fetal accumulation of biotin.^{68,116} The fetal-to-maternal concentration ratio of biotin plus metabolites is about 6:1, suggesting that fetal biotin accumulation might contribute to maternal biotin depletion.¹¹⁷ Yet, the activity of PCC is reduced by approximately 90% in the fetus at term in response to feeding an egg white diet that causes only a 50% reduction in maternal hepatic PCC activity in mice, suggesting that the fetus may not really be an efficient biotin parasite.¹¹⁸ If conclusive evidence was provided that marginal biotin deficiency causes birth defects in a quantitatively important fraction of pregnant women, this line of research would have major implications for health care professionals and policy makers.

B. Known Interactions

Anticonvulsants

Biotin requirements may be increased during anticonvulsant therapy. The anticonvulsants primidone and carbamazepine inhibit biotin uptake into brush border membrane vesicles from human intestine.⁹³ Long-term therapy with anticonvulsants increases both biotin catabolism and urinary excretion of 3-hydroxyisovaleric acid. Phenobarbital, phenytoin, and carbamazepine displace biotin from biotinidase, conceivably affecting plasma transport, renal handling, or cellular uptake of biotin. This is associated with decreased plasma concentrations of biotin. Accordingly, it is advisable to increase biotin supplementation when taking anticonvulsant regularly to avoid the serious health problems associated with anticonvulsant-induced biotin deficiency.

Lipoic acid (thioctic acid)

Lipoic acid may be administered to treat heavy metal intoxication, reduce signs of diabetic neuropathy, and enhance glucose disposal in patients with non-insulin-dependent diabetes mellitus.¹¹⁹ Lipoic acid competes with biotin for binding to SMVT,¹⁷ potentially decreasing the cellular uptake of biotin. Indeed, chronic administration of pharmacological doses of lipoic acid decreased the activities of PC and MCC in rat liver to 64%–72% of controls.¹¹⁹ Biotin supplementation should be considered when taking high-dose lipoic acid to avoid the health problems associated with biotin deficiency.

C. As a Dietary Supplement

Empirically, ingestion of pharmacological doses of biotin has been considered safe. For example, lifelong treatment of biotinidase deficiency patients with biotin doses that exceed the normal dietary intake by 300 times does not produce frank signs of toxicity.³⁷ Likewise, no signs of biotin overdose were reported after acute oral and intravenous administration of doses that exceeded the dietary biotin intake by up to 600 times.³⁴ Note, however, that biotin supplementation is associated with alterations of gene expression in healthy adults and human cell culture models as described above.¹²⁰ Some of these changes might have undesired effects in cell biology. For example, biotin supplementation decreases expression of the gene coding for sarco/endoplasmic reticulum ATPase 3 in human lymphoid cells; decreased expression of this calcium transporter is associated with impaired protein folding in the endoplasmic reticulum, causing cell stress.¹²¹ Moreover, there is no evidence that the pharmacological doses of biotin work any more effectively than standard doses. Further, as noted above in [Section V C](#), pharmacological doses of biotin supplementation can interfere with laboratory assays employing streptavidin technology.^{78–80}

RESEARCH GAPS

In the foreseeable future, biotin research likely will focus on the following priority areas:

1. Decipher the complex role of histone biotinylation in gene regulation and genome stability.
2. Determine what may cause adult-onset biotinidase deficiency.
3. Use transgenic animal models to characterize the roles of HCS, biotinidase, and biotin transporters in biotin homeostasis and histone biotinylation.
4. Determine the biological effects and whether marginal biotin deficiency causes birth defects in humans.
5. Quantify human biotin requirements and determine the true prevalence of biotin deficiency in vulnerable subgroups of the population.
6. Determine whether polymorphisms in genes coding for HCS, biotinidase, and biotin transporters affect biotin requirements.

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CHOLINE

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SUMMARY

Choline is an essential nutrient important for metabolism and numerous biological functions. Dietary requirements for choline are highly individualized and are based on several factors including genetics and gender; this nutrigenetic approach serves as an example of a new way of thinking about heterogeneity in individuals' nutritional requirements for optimal health. Low choline intake leads to multiple physiological changes, resulting in a number of disease states and altered development. In this chapter, fundamental aspects of choline function and its role in metabolism in health and disease are described.

Keywords: Brain development; Cardiovascular disease; Choline; Liver cancer; Methylation; Single-nucleotide polymorphisms.

I. INTRODUCTION

Choline is an essential dietary nutrient and is important for normal function of cells and organs. Specifically, choline is needed for cholinergic neurotransmission, making cell membranes, and lipid transport, and it is a source of methyl groups for the universal methyl donor *S*-adenosylmethionine (AdoMet). Choline deficiency is associated with fatty liver and liver damage, muscle damage, and abnormal fetal development.¹

A. Background

At one time choline was not considered an essential nutrient for humans because there was an endogenous pathway for *de novo* biosynthesis of the choline moiety as part of forming phosphatidylcholine (PtdCho), also known as lecithin.² Despite this capacity to synthesize some choline in liver, healthy people with normal folate and vitamin B₁₂ status fed a low-choline diet (about 50 mg choline/d) developed fatty liver, liver damage (elevated plasma alanine and/or aspartate transaminases: ALT), or developed muscle damage (elevated plasma creatine phosphokinase) that resolved when choline was restored to the diet.^{3,4} Metabolic abnormalities (decreased clearance of homocysteine [Hcy]),⁵ elevations in markers of DNA damage,⁶ and increased lymphocyte gene

apoptosis⁷ also were observed in people fed the low-choline diet. In addition, individuals fed intravenously with solutions low in choline developed liver dysfunction that resolved when choline was added back to the solutions.⁸ The human requirement for the nutrient choline was officially recognized in 1998 with the establishment of Adequate Intake (AI) recommendations as part of the Dietary Reference Intake process for the United States and Canada by the Food and Nutrition Board of the Institute of Medicine of the US National Academies.⁹ Subsequently, in 2016, the US Food and Drug Administration set reference daily intake values for choline (<https://www.fda.gov/media/99069/download>) for nutrition labeling, and the European Food Safety Authority has set dietary reference values for choline for the European Union (<https://www.efsa.europa.eu/en/press/news/160817>).

B. Structure and Function of Choline and its Metabolites

Choline is metabolized to form betaine (trimethylglycine), acetylcholine (ACh), phosphocholine (PCho), glycerophosphocholine (GPCho), PtdCho, and sphingomyelin (SM) (some of these compounds are depicted in Fig. 18.1). Betaine is a methyl-group donor relevant for epigenetic regulation (discussed further in this chapter) and is a major cell osmolyte. It primarily

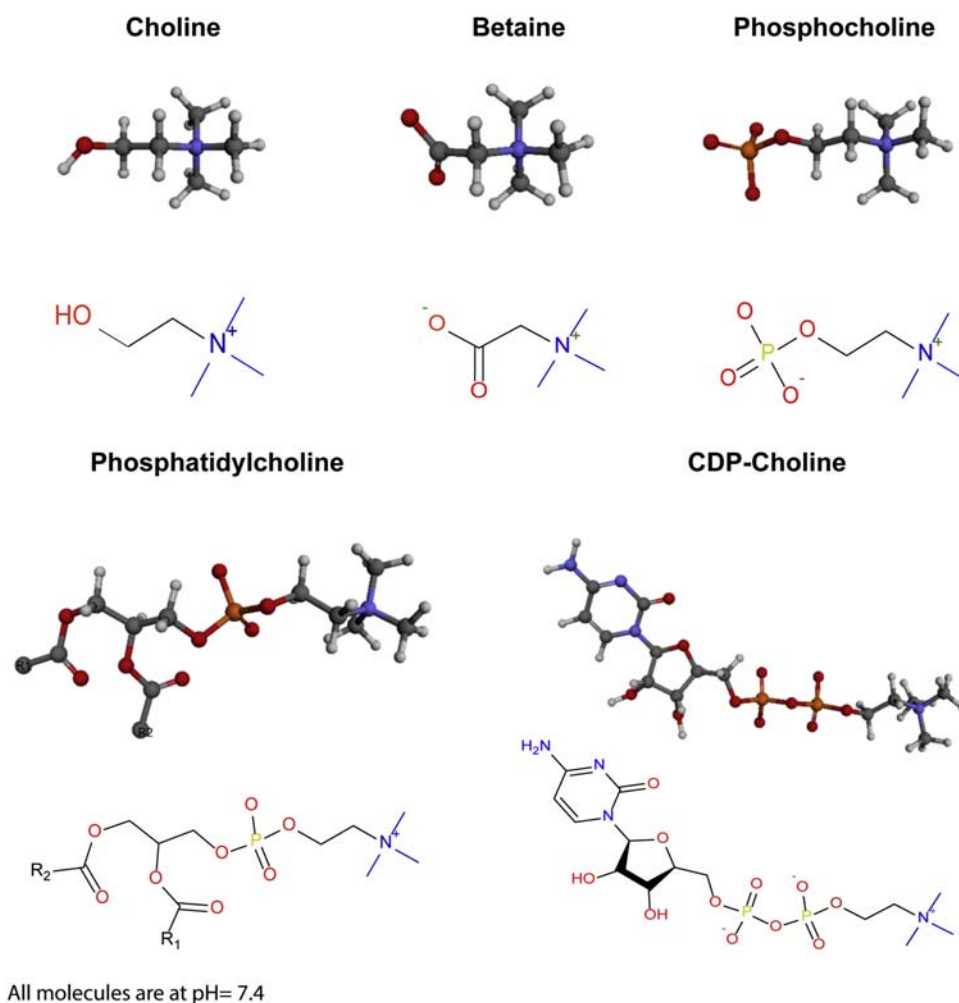


FIGURE 18.1 Structures of choline and major choline-related molecules. Choline and its metabolites are depicted in this figure. The 2D structures were drawn using ChemBioDraw and the 3D renderings were created with Discovery Studio Visualizer. Colors of atoms are as follows: carbon is denoted in gray, oxygen in red, hydrogen in white, nitrogen in blue, and phosphorus in yellow.

accumulates in the liver and kidney. Betaine transporters have been identified in the liver, renal medulla, and on the brain's surface.^{10,11} ACh is a neurotransmitter important for brain functions such as memory and mood.^{12,13} Neurons interfacing between the brain and the periphery use ACh to control skeletal muscles, heart rate, breathing, sweating, and salivation.¹³ Choline concentrations can be limiting for ACh synthesis in the brain and peripheral nerves.¹⁴ PtdCho is the most abundant phospholipid of all mammalian membranes, comprising 40%–50% of total cellular phospholipids¹⁵; it also serves as a major storage pool for fatty acids and contributes to the formation of second messengers such as diacylglycerol, arachidonic acid, and docosahexaenoic acid (DHA).^{16,17} PtdCho is also important for packaging hepatic lipids for export as lipoproteins.¹³ SM is an important constituent of myelin (insulating nerve axons); it is a component of membranes that modify the formation of lipid rafts; it is a precursor for formation of ceramides (important intracellular signals involved in cell migration, apoptosis, and cell survival); and it is a breast milk component.^{18–21} Overall, choline and its metabolites are fundamental for cellular functioning.

II. CHOLINE METABOLISM

A. Choline Synthesis

Choline in the body is derived from two sources: dietary intake and choline derived via *de novo* PtdCho synthesis catalyzed by phosphatidylethanolamine-N-methyltransferase (PEMT); this pathway is most active in the liver.²² PEMT catalyzes three sequential methylations of phosphatidylethanolamine to generate PtdCho, which can then be hydrolyzed to make choline available. These reactions are AdoMet-dependent, and at least two AdoMet/*S*-adenosyl-L-homocysteine (AdoHcy)–binding motifs are identified in eukaryotic PEMT proteins.²³

B. Digestion, Absorption, and Transport

Choline in foods is mostly in the form of PtdCho, SM, PCho, GPCho, and choline, with betaine, a metabolite of choline, also present in some foods. After ingestion and subsequent digestion by pancreatic and mucosal enzymes, the water-soluble compounds choline, betaine, PCho, and GPCho enter the portal circulation of the liver. They rapidly accumulate in the liver where most

are phosphorylated and used to make membranes and lipoproteins.²⁴ Approximately half of the two lipid-soluble choline compounds in foods, PtdCho and SM, persist intact after digestion. They enter via the lymphatic system in chylomicrons, miss first-pass metabolism by the liver, and thus can reach other organs.²⁵

Choline enters tissues by diffusion and mediated transport. The plasma membrane and the mitochondrial membrane have a choline transporter encoded by the gene *SLC44A1*.^{26–29} *SLC44A1* is expressed in the brain, spinal cord, colon, and many other tissues.^{3,29,30} Cholinergic neurons have a special high-affinity choline transporter (encoded by *SLC5A7*) that helps them accumulate choline³¹; under most conditions, this is the rate-limiting step for ACh synthesis.^{32,33}

C. Methyl Group Metabolism

Choline (via betaine) is a major dietary source of labile methyl groups that are needed for many biochemical reactions.³⁴ Methyltransferases use AdoMet as their methyl donor, and AdoMet synthesis depends on the methylation of Hcy to form the dietary essential amino acid methionine (a precursor for AdoMet) (Fig. 18.2). This

methylation can be accomplished through either of two parallel pathways. The first uses betaine as its methyl donor and is catalyzed by betaine-homocysteine-methyltransferase; the second uses 5-methyltetrahydrofolate as its methyl donor and is catalyzed by methionine synthase (with vitamin B₁₂ as an essential cofactor).^{35,36}

S-Adenosyl-L-homocysteine (AdoHcy) is a product of all AdoMet-dependent methylations, and it is further metabolized to form Hcy, creating a methylation cycle.³⁷ AdoHcy binds to the AdoMet-binding site on methyltransferases and competitively inhibits methylation.³⁸ Thus, the ratio of AdoMet:AdoHcy determines methyltransferase activity and is often referred to as the methylation potential. Because there are two parallel pathways, one using choline-derived methyl groups and one using folate-associated methyl groups, there is an important interaction that allows one nutrient to spare some of the requirements for the other nutrient. If animals or humans consume a diet deficient in choline, dietary folate requirements increase due to increased 5-methyltetrahydrofolate being used to remethylate Hcy in the liver.^{39,40} Alternatively, a diet deficient in folate increases dietary choline requirements since choline-derived betaine becomes the primary methyl donor.^{41–43}

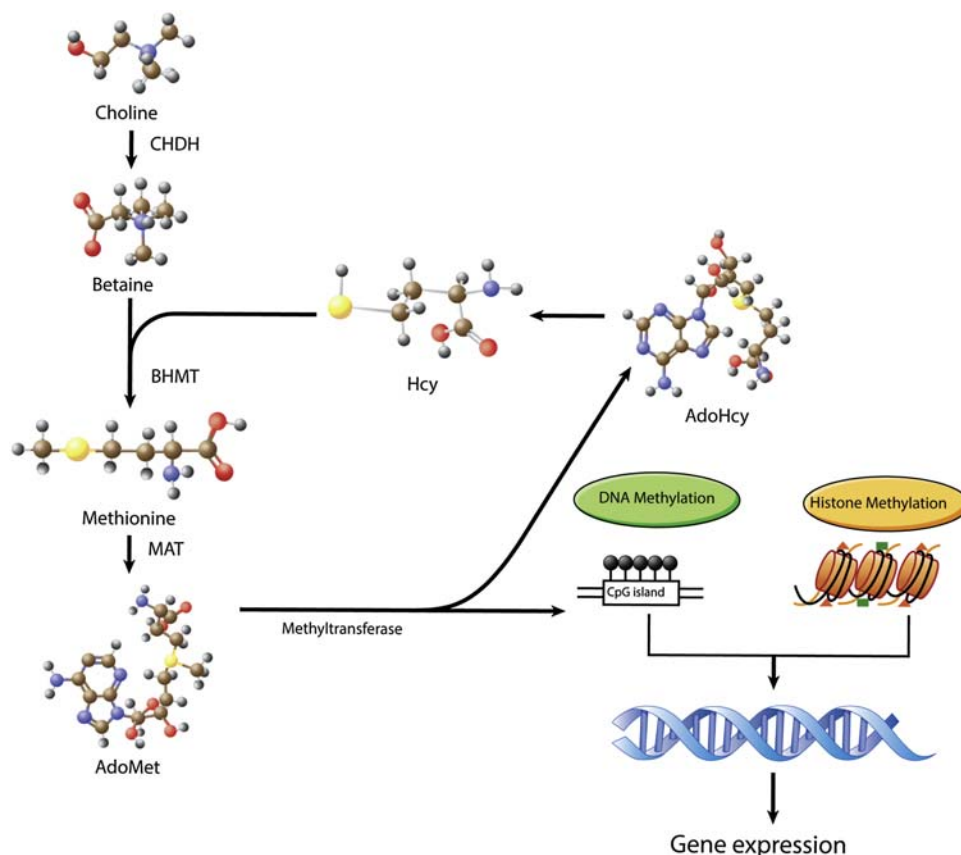


FIGURE 18.2 Choline and control of gene expression by methylation. Choline is oxidized to betaine (trimethylglycine) and is converted to methionine by betaine-homocysteine methyltransferase. The 1-carbon unit of AdoMet can be donated to a number of methyltransferases. These methyltransferases can attach methyl groups to many substrates, including DNA, histones, RNA, and proteins. Specifically, DNA and histone methyltransferases can alter genome-wide methylation status, resulting in altered gene expression. Colors of atoms are as follows: carbon is denoted in brown, oxygen in red, hydrogen in white, nitrogen in blue, and sulfur in yellow.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Gene Regulation

Methyltransferase-dependent epigenetic modifications in DNA and chromatin are influenced by choline availability and are associated with transcriptional regulation.⁴⁴ DNA methylation usually occurs where a cytosine nucleotide is followed by a guanine nucleotide in the 5' → 3' direction (CpG), forming 5-methylcytosine. Covalent posttranslational histone methylation occurs at the amino-terminal “tails” of histones (addition of one, two, or 3 methyl groups) (Fig. 18.2). DNA methylation usually silences genes, while the effects of histone methylations are site-specific (histone3 lysine4 [H3K4], H3K36, and H3K79 methylation acts as “active” marks and H3K9, H3K27, and H4K20 methylation as “silencing” marks).⁴⁵ Active histone modifications often coincide with active DNA methylation marks, and “cross talk” between the two modifications occurs.

Choline deficiency during pregnancy induces changes in DNA and histone methylation and, as a consequence, changes gene expression in multiple tissues. In rodents, maternal diet low in choline alters global histone methylation and DNA methylation at specific genes in the fetal brain.^{44,46–48}

B. Metabolomics

In people, choline deficiency is associated with fatty liver and liver cell death, as well as with rhabdomyolysis.^{4,49,50} Metabolomic profiling of individuals may predict which individuals would develop organ dysfunction when fed a choline-deficient diet. Metabolites in several pathways, including choline, lipid, carnitine, and products of amino acid metabolism, have predicted which individuals would develop choline deficiency–related fatty liver.⁵¹ Choline deficiency affects mitochondrial membranes, resulting in loss of mitochondrial membrane potential, increased leakage of reactive oxygen species, abnormal morphology, and abnormal bioenergetics.^{52–54} Perhaps in the future, mitochondrial function biomarkers can be used to assess choline status.

IV. DIETARY REQUIREMENTS FOR CHOLINE

A. Assessment of Choline Nutrient Status

Consequences of inadequate choline

Liver; The majority of choline metabolism occurs in the liver and its roles in this organ have been studied extensively. As discussed earlier, one important function of the

liver is the maintenance of whole-body lipid homeostasis. Choline, especially via its metabolite PtdCho, is essential for the export of lipids from the liver^{55,56} as it is needed for the membranes used in exporting triglycerides as very–low-density lipoproteins.⁵⁷ In animal models and in people, choline deficiency is associated with fat accumulation in the liver (steatosis). Such nonalcoholic fatty liver disease (NAFLD) is the most common chronic liver disorder, occurring in 30% of the general population and 65%–90% of overweight or obese individuals.⁵⁸

Although choline deficiency in people is associated with hepatic steatosis, not all individuals are affected equally. Thus far, genetics and gender have been implicated as factors that mediate risk.^{4,59–61} Some of the PtdCho in the liver (about 30%) is produced by PEMT in the liver. Rodent models lacking *Pemt* develop liver inflammation, nonalcoholic liver disease (NAFLD), and liver failure when fed a choline-deficient diet.⁶²

People with certain variants of the *PEMT* gene are at greater risk for developing NAFLD.¹⁵ As discussed earlier, many metabolic pathways contribute to choline metabolism, and in the liver, transport of PtdCho into bile becomes an important sink for choline. Polymorphisms in genes for transporters in this process, as well polymorphisms in the genes of choline and folate metabolism, can be used to predict which people will develop NAFLD as their body mass index increases.⁶³

Muscle; Choline availability has an impact on muscle membrane composition and is needed for proper muscle function.^{64,65} People with genetic variants in choline transporter *SLC44A1* and choline kinase beta (*CHKB*) may be more susceptible to muscle damage when they exercise while eating a diet low in choline.⁶⁵ The mechanism probably involves a reduction in membrane PtdCho, which makes myocytes fragile and susceptible to rupture.³ It is interesting that deletion of *CHKB* results in muscular dystrophy in mice,⁶⁶ due to increased mitophagy.⁶⁷

Brain; Choline is essential for the brain and spinal cord development. Increased total choline intake during pregnancy is highly associated with a decrease in neural tube and cleft palate defects.^{68,69} One of the great successes of nutrition science has been the identification of the role folate plays in normal neural tube closure; adequate dietary folate intake by the mother during pregnancy can prevent 50% or more of neural tube defects (NTDs) in babies.^{70,71} As discussed earlier, choline and folate metabolism are highly interrelated. Inhibition of choline uptake and metabolism is associated with the development of NTDs in mice.^{72,73} A retrospective case–control study (400 cases and 400 controls) of periconceptional dietary intakes of choline in women in California found that women in the lowest quartile for daily dietary choline intake had double the risk of having a baby with an NTD

compared to women in the highest quartile for intake.⁷⁴ Additional birth defects have been associated with low maternal dietary choline intake, including cleft lip,⁷⁵ hypospadias (defect in male urethra),⁷⁶ heart defects,⁷⁷ and congenital diaphragmatic hernia.⁷⁸

In rodent studies, if choline availability is increased above what is considered an adequate level during a critical period of embryonic development, between days 11 and 18 of gestation, the offspring performs better on memory-related tasks, such as Morris water maze and radial arm maze, throughout life.^{79–83} In people, higher intakes of choline during pregnancy are associated with enhanced information processing speeds in the mothers' children at 4, 7, 10, and 13 mo of age⁸⁴ and with better performance on memory tests at 7 years of age.⁸⁵

The importance of adequate choline supply for brain development has been well established.¹ The development of the brain is a process in which stem cells commit to the various lineages of cells that form the brain by forming progenitor cells. These progenitor cells first replicate and then differentiate. In general, choline regulates cell proliferation, migration, differentiation, and apoptosis in neural progenitor cells (NPCs).^{86–91} The likely mechanism for choline-mediated modulation of NPC's proliferation and differentiation is via regulation of the synthesis of the epidermal growth factor receptor (EGFR)—an essential signal involved in cortical neurogenesis.^{92–97} A specific miR, miR-129–5p, which is upregulated in low-choline NPCs, binds the 3' untranslated region of *Egfr* mRNA and inhibits its translation into protein.⁴⁸ Inhibition of miR-129–5p function in the developing cortex in vivo is sufficient to restore NPC proliferation and differentiation in low choline embryos. Thus, brain development is controlled by choline availability through the regulation of miR expression.⁴⁸

Neurological disorders; In addition to the role of choline in brain development, behavior, and memory, choline may modulate several neurological disorders.⁴⁷ In rodent models, choline and DHA promote neuronal membrane and synapse formation and diminish the accumulation of amyloid plaque in the brain, especially when used in combination.⁹⁸ These mechanisms are of particular importance in ameliorating the cognitive derangements associated with Alzheimer's disease.⁹⁸ Citicoline (cytidine5'-diphosphocholine), a choline-containing molecule, is being used to treat stroke and traumatic brain injury,^{99,100} as well as to treat hypoglycemia-induced brain injury.¹⁰¹

Choline administered in the perinatal period may also modify the cognitive deficits seen in a mouse model of Rett syndrome (a form of autism),^{47,102–104} the BTRB T + *l*trp3tf/J mouse model of autism,¹⁰⁵ rodent models of iron deficiency,¹⁰⁶ mouse models of fetal alcohol spectrum disorder (FASD),¹⁰⁷ and possibly people with FASD,^{108–110} and in a mouse model of Down

syndrome.^{111,112} Women with genetic polymorphisms in genes involved in choline metabolism (particularly *BHMT*) have an increased risk (up to fivefold) for having a baby with Down syndrome.¹¹³

Eye; The retina and brain neurons share a common embryological precursor,¹¹⁴ and thus it is not surprising that choline availability influences retinal development and vision. In a mouse model, low choline during pregnancy led to disruptions in embryonic retinal structure and changes in electroretinographic function. These alterations persisted into adulthood.¹¹⁵

Cancer; Choline is the only nutrient for which dietary deficiency causes development of hepatocarcinomas in rats without exposure to any known carcinogen.^{116,117} Interestingly, choline-deficient rats not only have a higher incidence of spontaneous hepatocarcinomas, but they are markedly sensitized to the effects of administered carcinogens.¹¹⁸ Choline deficiency is therefore considered to stimulate both cancer-initiating and cancer-promoting activities.¹¹⁷ Mice with a deletion of a gene in choline metabolism (*BHMT*, see Fig. 18.2) also develop hepatocarcinomas at 1 year of age.^{36,119}

In a metaanalysis of 11 studies reported since 2016, higher dietary intake of choline and betaine reduced cancer incidence in people.¹²⁰ Risk for breast cancer was modified by dietary choline intake: risk was reduced by 24% among women having high dietary intakes of choline and was increased in women with the *PEMT* rs12325817 SNP (reduces estrogen-induced *PEMT* activity, see below).¹²¹ Importantly, a subsequent analysis determined that breast cancer mortality was reduced by high intakes of choline and betaine.^{122,123}

Higher choline and betaine intakes also decreased the risk of colorectal cancer.¹²⁴ Several mechanisms have been proposed for the carcinogenic effects of choline deficiency, including enhanced apoptosis, decreased cell proliferation, increased DNA damage, and altered gene expression.¹ It appears that sufficient choline is important for protecting against liver, colon, and breast cancers; however, higher choline intakes have been associated with an increased risk for lethal prostate cancer.¹²⁵

Supply of choline to the fetus and infant

During pregnancy, choline is available to the fetus through facilitated placental transport.^{126–128} After birth, breast milk is a rich source of choline.^{20,129–131} This results in much higher concentrations of choline in tissues of the fetus and infant than in the adult.^{21,132} These high choline levels are needed by the developing fetus for cellular membrane genesis and for normal brain development. Mouse models have shown that maternal choline supplementation resulted in offspring with higher cognitive function compared with no maternal supplementation.⁴⁷

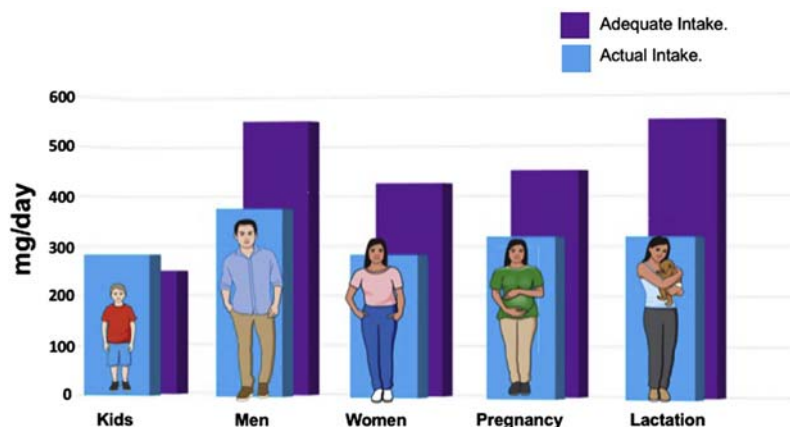


FIGURE 18.3 Dietary choline intake and recommendations in the United States. Adequate intake recommendations for choline per day are 200 mg for children 1–3 years, 250 mg for 4–8 years, 375 mg for 9–13 years, 550 mg for males ≥ 14 years, 400 mg for girls 14–18 years, 425 mg for women ≥ 19 years, 450 mg for pregnant women, and 550 mg for lactating women⁹ (purple bars). However, actual average daily dietary intake for choline as reported by What We Eat in America, NHANES 2013–16 cohort was less than the recommended intakes for much of the population, as were intakes for pregnant and lactating women (blue bars).¹³⁹

As mentioned in the previous section, during brain development, choline regulates the proliferation and differentiation of NPCs and is essential for NPC migration and subsequent cellular layering of the cerebral cortex.⁹² After birth, infants receive their choline from breast milk, typically ~ 1.3 – 1.5 mM.^{20,130,133,134} Due to the importance of choline, in the United States, all commercially available infant formulas were reformulated in 2007 to contain a comparable amount of choline as in breast milk (Code of Federal Regulations, 21CFR107.100). Choline supplementation during pregnancy and lactation increases the choline content in breast milk.^{135,136}

B. Reference Intakes for Choline

Choline is an essential nutrient

The current recommendations for choline intake for the United States and Canada were established in 1998 as part of the Dietary Reference Intakes process of the Institute of Medicine. The recommended dietary intake, termed an AI, for choline for adult men is 550 mg/day and for adult women 425 mg/day.⁹ The AI is based on the amount to prevent elevated plasma aminotransferase (ALT) values in normal subjects noted in one controlled trial available at the time⁵⁰ and was supported by studies providing choline as PtdCho to patients receiving total parenteral nutrition.^{8,137} Data were not available at the time to establish an AI based on intakes from apparently healthy populations nor were additional human studies available upon which to establish an Estimated Average Requirement (EAR) and thus a Recommended Dietary Allowance (RDA).⁹

Pregnancy creates increased demands for dietary choline as shown by pregnant female rats that deplete their hepatic stores of choline.¹³⁸ Thus, the AI for choline during pregnancy is 450 mg/day,⁹ greater than for nonpregnant women. Because lactation further

consumes maternal choline stores, the AI for lactating women is estimated to be 550 mg/day of choline.⁹ Though no experimental data exist, AIs for infants were extrapolated from amounts that would be consumed in breast milk, while AIs for children were extrapolated based on adult requirements per kg body mass, resulting in estimates of the AI that vary from 125 mg/day for infants to 375 mg/day for 9- to 13-year-old children (Fig. 18.3).⁹

Currently, studies are underway to refine these estimates of human choline requirements. Some reports indicate that a subset of adult humans become depleted of choline at intakes of 750 mg/day,⁴ and recent studies suggest that maternal intake of choline at twice the AI levels has beneficial effects on fetal brain development,⁸⁴ suggesting that the estimates for the AI may be too low for many adults.

C. Tolerable Upper Intake Levels

Excess dietary choline intake has been associated with fishy body odor due to the excretion of large amounts of trimethylamine (TMA; see later discussion of the potential toxic effects of trimethylamine *N*-oxide, TMAO).^{9,140} Additional side effects observed due to excess dietary choline include vomiting, sweating, gastrointestinal side effects, and hypotension. A Tolerable Upper Intake Level was set by the Institute of Medicine in 1998 for choline for adults at 3.5 g/day based, on these cholinergic effects and fishy body odor.⁹

D. Individual Variation in Choline Requirements

Dietary choline requirements are governed by an individual's ability to make choline *de novo*, and on genes of

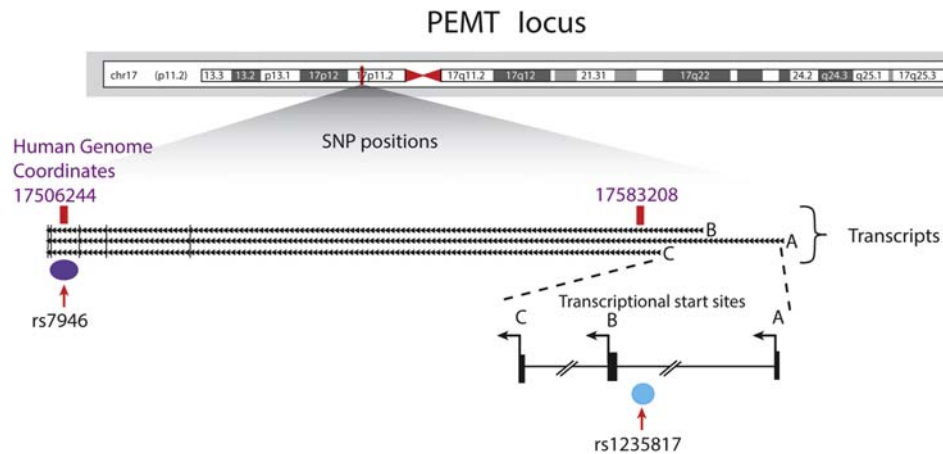


FIGURE 18.4 Schematic representation of the human *PEMT* locus and location of common genetic single-nucleotide polymorphisms (SNPs). The *PEMT* gene is located on chromosome 17 (in the human genome), is 40 kb long, and comprises of nine exons. The rs7946 polymorphism (in purple) leads to a nonfunctional *PEMT* protein. Additionally, the promoter region of the *PEMT* gene contains an estrogen receptor–binding site, increasing its expression. The rs12315817 polymorphism (in blue) is located near the estrogen receptor binding site, reducing estrogen-induced *PEMT* gene expression.

choline and folate metabolism that are intertwined in the metabolism of choline. Several very common single-nucleotide polymorphisms (SNPs) in these genes have a profound impact on choline needs.^{59,60} As previously discussed, when eating a low-choline diet, 77% of men and 80% of postmenopausal women developed fatty liver, liver damage, or muscle damage, while only 44% of premenopausal women develop such signs of organ dysfunction.^{4,59,60} The promoter for the *PEMT* gene is estrogen-responsive,^{59,141} and this likely explains why premenopausal women are more resistant to developing organ dysfunction when fed a low choline diet.¹⁴¹ The *PEMT* rs12325817 SNP prevents estrogen induction of the gene's expression and has an obvious effect in premenopausal women where having one or two variant alleles is sufficient to increase the risk of developing choline deficiency–induced organ dysfunction by 25-fold.^{4,59,60} Another SNP in the *PEMT* gene (rs7946) is present more often in people with fatty liver.⁶¹ (see Fig. 18.4).

Premenopausal women who are carriers of a common 5,10-methylenetetrahydrofolate dehydrogenase SNP (*MTHFD1*; rs2236225) were more than 15 times as likely as noncarriers to develop choline deficiency–induced organ dysfunction.⁶⁰

A choline intake exceeding current dietary recommendations also preserves markers of cellular methylation and attenuates DNA damage in a genetic subgroup of folate-compromized men with an SNP in the methylenetetrahydrofolate reductase (*MTHFR*) gene.¹⁴²

E. Dietary Sources of Choline

Choline in various forms is available from many foods. Foods high in choline per serving include milk,

eggs, liver, meats, poultry, peanuts, fish, dairy foods, and egg-based dishes.¹⁴³ As mentioned earlier, average dietary intake of choline in the United States is below the recommended guidelines for almost every age group¹³⁹ (see Fig. 18.3). Betaine, a choline metabolite that cannot be converted to choline but can be used as a methyl donor, can spare some of the choline requirement.^{144,145} Spinach, beets, wheat, and shellfish are also good sources of betaine.¹⁴³

V. ISSUES RELATED TO CHOLINE IN HUMAN HEALTH

A. Impact of Choline on Gut Microbiome

Cardiovascular disease

In people, gut microbiota can metabolize choline to form TMA, which the liver converts to TMAO. This metabolite has been associated with increased risk of myocardial infarction and stroke.^{146–150}

In the human intestine, choline cleavage is performed by bacteria that express the *cut* cluster of genes,^{151–153} which allows them to hydrolyze choline, forming dimethylamine and TMA. Crucial genes in the cluster are *CutC* and *CutD*, encoding for choline TMA-lyase and its activating protein, respectively (Fig. 18.5).¹⁵⁵ Gnotobiotic mice do not produce TMA,^{156,157} and antibiotic treatment of standard mice decreases TMA formation, confirming that intestinal microbiota are required for choline cleavage.¹⁵⁷ Colonization of gnotobiotic mice with choline-converting bacteria increases TMA production.¹⁵⁶ Though most of the production of TMA occurs in

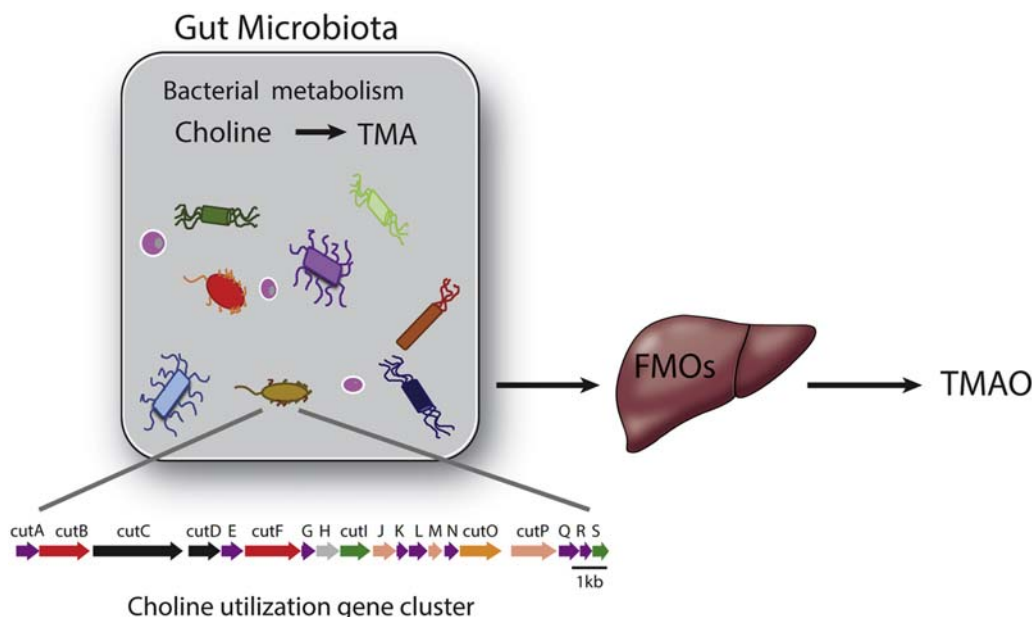


FIGURE 18.5 Gut microbial choline metabolism and trimethylamine N-oxide formation in the host. Choline is cleaved by microbes that express the genes *CutC* and *CutD*, resulting in TMA generation.^{151–153} TMA is absorbed by the intestine and is oxidized in the liver to form TMAO by flavin-containing monooxygenase enzymes (FMOs).¹⁵⁴ The genetic organization of the *Cut* cluster is shown.^{151–153} Structural proteins are represented in purple, glycyl radical enzymes and activating proteins in black, aldehyde oxidoreductase in red, aldehyde dehydrogenase in orange, phosphotransacetylase in gray, poorly conserved genes in black, and enzymes with unknown function in pink.

the cecum and colon, oral flora (e.g., *Streptococcus sanguis*; a facultative anaerobe) also can form TMA from choline.¹⁵⁸

TMA is absorbed from the intestine and then oxidized in the liver to produce TMAO.¹⁵⁴ In people, TMAO is excreted through urine, sweat, and breath.¹⁵⁹ TMAO modulates multiple aspects of cholesterol and steroid metabolism, including reverse cholesterol transport.^{160,161}

Choline, betaine, and TMAO were identified as molecules in plasma that were significantly associated with increased atherosclerotic plaque burden in subjects.¹⁵⁰ During 3 years of follow-up of patients who had undergone elective coronary angiography, increased plasma concentrations of TMAO were associated with an increased risk of a major adverse cardiovascular event.¹⁴⁹

Higher dietary intake of PtdCho, another precursor for TMAO formation, was associated with increased risk mortality in the Nurses' Health Study (1980–2012) and the Health Professionals Follow-Up Study (1986–2012).¹⁶² In a cohort of patients with stable coronary artery disease, elevated plasma TMAO concentrations were associated with greater long-term mortality risk.¹⁶³

Elevated concentrations of plasma choline and betaine, assessed in stable cardiac patients, were associated with incident mortality, myocardial infarction, and

stroke risk, independent of other traditional risk factors. However, these metabolites were predictive only when a simultaneous increase in plasma TMAO concentrations was observed.¹⁶⁴

A number of other clinical studies observed an association between TMAO and cardiovascular disease (CVD) risk; plasma TMAO concentrations were strongly associated with long-term mortality in heart failure patients, advanced left ventricular diastolic dysfunction, and atherosclerosis burden in patients with atherosclerotic coronary artery disease.^{147,148,163–165} However, this association between TMAO and atherosclerosis was not seen in the Coronary Artery Risk Development in Young Adults Study (CARDIA) where coronary artery atherosclerosis was the outcome measure.¹⁶⁶ Similarly, no association was found between dietary choline or betaine intake and incident CVD in the Atherosclerosis Risk in Communities study.¹⁶⁷ In the Jackson Heart Study, of African Americans, higher dietary choline was associated with reduced risk of incident ischemic stroke, while higher betaine intake was associated with a nonlinear higher risk of incident heart disease.¹⁶⁸ Conversely, dietary intakes of betaine and choline were not associated with CVD risk in postmenopausal Dutch women in the PROSPECT-EPIC cohort.¹⁶⁹

Renal disease

Patients with poor kidney function have elevated plasma TMAO concentrations.^{148,170,171} In patients with chronic kidney disease, glomerular filtration rate and impaired renal function were the main variables affecting plasma TMAO concentrations.¹⁷² In patients who underwent coronary angiography, plasma concentrations of TMAO or choline were significantly increased with decreasing renal function but were not associated with incidence or symptoms of coronary heart disease. The study concluded that plasma concentrations of TMAO were confounded by impaired kidney function since it was increased with declining glomerular filtration rate.¹⁷³ Renal transplantation corrects the high plasma TMAO concentrations seen in renal failure.^{172,174} Thus, atherosclerosis affects both heart and kidney, and when it is severe enough to cause cardiac events, it also likely affects renal function, thereby elevating TMAO. This may explain why the association between TMAO and CVD is mainly observed in people who are older and have more severe CVD and is not seen in populations who are younger and have less severe atherosclerosis.¹⁷⁵

B. Need for Improved Biomarkers of Choline Status

The last few decades of research invested in understanding the biological functions of choline in humans have resulted in an impressive body of knowledge about the health implications of choline insufficiency and supplementation. However, better biomarkers of choline status need to be identified. Self-reported dietary intake is very hard to measure accurately, and for choline self-report can significantly underestimate actual intake.¹⁷⁶ The largest stores of choline are located in the liver, and mass resonance spectroscopy of liver has been used in the past to assess choline status in

humans.^{4,59,60} This method is not practical for use as a biomarker in clinical or public health practice as it is expensive and the availability of the instrumentation is limited. Liver biopsy is risky and not practical, making measurement of hepatic choline and choline metabolite concentrations a poor choice for assessing choline status. Measurement of choline concentrations in plasma alone is not an adequate marker for choline status because choline concentrations drop quickly to 70% of normal when a person is fed a low-choline diet but do not decrease further with more severe depletion,^{4,50} even in people starved for 7 days.¹⁷⁷ The concentrations of PtdCho and SM also decrease in people fed a low-choline diet,^{4,50} but these phospholipids alone are not adequate biomarkers for choline status because they are carried as part of plasma lipoproteins and vary in concentration with changes in fat and cholesterol transport by these particles. Thus, such measurements are useful but not sufficient as biomarkers for choline status.

The understanding of the wide-reaching effects of choline in systems biology as a direct result of its role as a methyl donor is only beginning to be understood. The implications of epigenetics in health and disease are becoming increasingly evident with the availability of epidemiological studies linking low dietary choline intakes to higher concentrations of proinflammatory markers.^{178,179} The interesting links between genetics and metabolomics in determining risk for diseases associated with choline deficiency are initial examples of the complexities of the signatures that will need to be defined to distinguish those who are at risk of disease from those who are not. This is true not just for choline, but for all nutrients. The future of choline and nutrition research in general is exciting and complex. The hope is to ultimately understand how all nutrients act in concert with systems biology to determine the health outcomes of individuals.

RESEARCH GAPS

There are several areas of research that warrant further attention to enhance understanding of the choline needs of individuals and the potential to use this nutrient, and its derivatives, as therapeutic agents:

- Development of better biomarkers of choline status.
- Investigation of the role(s) of choline in inflammation.
- Assessment of the prevalence of choline-relevant gene variants in large populations around the world.
- Assessment of the impact of ethnicity and environment on the prevalence and consequences of choline-related SNPs. This will greatly enhance ability to evaluate the role of choline in disease.
- Elucidation of the specific mechanisms by which choline is involved in epigenetics.
- Confirmation with human studies of the data observed with rodent models of the influences of dietary choline on brain development that can perhaps moderate the severity of several neurological diseases.

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S E C T I O N C

Minerals

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CALCIUM

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SUMMARY

Calcium is the most abundant mineral in the body, existing primarily in the skeleton where it serves a structure–function role and also as a reserve when dietary calcium is inadequate. All homeostatic mechanisms are evolved to maintain constant serum calcium levels to ensure cell signal transmission. Dietary calcium intakes around the world are below recommended intakes. The prevalence of US adults below the Estimated Average Requirement for calcium from food only is $37.77 \pm 1.4\%$, but supplement use decreases that to $19.6 \pm 1.3\%$ (4).

Keywords: Bone; Calcium; Dairy; Kidney stones; Secondary messenger; Vitamin D.

I. INTRODUCTION

A. Background

Calcium is the most abundant mineral in the body, with $>99\%$ in the skeleton. The skeleton is a functional reserve that supports our mobility and can be drawn upon during periods of dietary inadequacy of calcium. Because essentially all body processes require calcium, finely tuned homeostatic control mechanisms to maintain constant blood levels of calcium have evolved, as have complex cellular mechanisms to control movement of intracellular calcium. Inadequate dietary calcium is associated with increased risk of a number of diseases.

B. Nutrient Function and Structure

Calcium is a divalent cation. Because calcium is of intermediate solubility, it exists in both solid form (bones) and in solution (plasma). Calcium binds to proteins through the oxygen atoms of glutamic and aspartic acid residues that fix the tertiary structures of proteins, thereby governing their activity and stability. In this capacity, calcium serves as the most common signal transmitter in cell biology. Calcium has only one oxidation state, so it is not prone to be toxic at high concentrations.

Calcium is unusual in that the storage form is also functional. As part of hydroxyapatite, $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$,

calcium is a building block of bone, which is organized in a hierarchical manner from nanometer to millimeter-sized structures (Fig. 19.1). This structure is strong enough to support our bodies for many decades but light enough to allow mobility. Lifelong cumulative calcium status can be assessed by measuring bone mineral content by bone densitometry with dual X-ray absorptiometry (DXA) because $>99\%$ of the body's calcium is in the skeleton and it is present in a constant ratio (36% of total body bone mineral content).

Bone remodeling occurs throughout life except in teeth. Bone resorption initiated by osteoclasts (bone-resorbing cells) results in microscopic pits on bone surfaces. This is necessary for modeling changes in bone size during growth, repairing microarchitectural damage, and maintaining serum calcium levels as well as serving as a source of other minerals. Bone formation under the control of osteoblasts fills in the pits. It exceeds resorption during growth and often lags behind resorption later in life, resulting in age-related bone loss.

II. CALCIUM METABOLISM

A. Transport

Calcium concentrations in blood and extracellular fluids are maintained under tight regulation at ~ 2.5 mM.

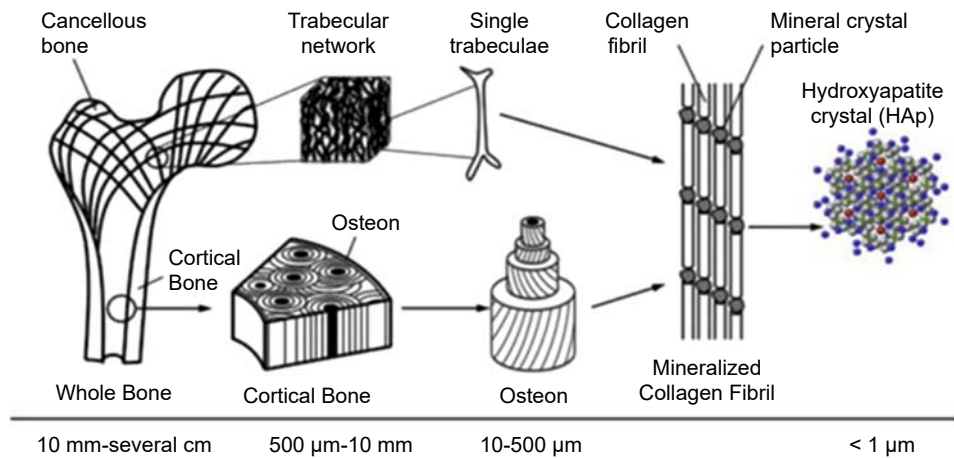


FIGURE 19.1 Calcium as part of hydroxyapatite constitutes the mineral crystals that make up the highly organized structure of bone. Lattices of crystals are interspersed among collagen fibers. These are organized into lamellae that are organized into a dense outer shell for cortical bone and a looser spongy structure for trabecular bone within the marrow cavity.¹

Thus, serum calcium concentrations do not reflect nutritional status. About half of the calcium in plasma exists in free ionized form and is functionally available. Most of the rest is bound to albumin; some to globulin; and $<10\%$ is complexed to phosphate, citrate, and other anions.

Extracellular calcium levels can be perceived by a surface Ca^{2+} -sensing receptor (CaR), a member of the superfamily of G-protein–coupled receptors in the parathyroid, kidney, intestine, lung, brain, skin, bone marrow, osteoblasts, breast, and other cells.² The CaR permits Ca^{2+} to act as an extracellular first messenger in the manner of a calcitropic hormone. For example, parathyroid glands detect small changes in extracellular concentrations of Ca^{2+} , which regulate the release of parathyroid hormone (PTH) into the circulation. Similarly, CaR allows minute-to-minute regulation of renal tubular calcium reabsorption. The role of these actions in calcium homeostasis is described below.

Extracellular calcium serves mainly as a source of Ca^{2+} to the skeleton and cells but fulfills essential functions in its own right including participating in blood clotting and intercellular adhesion.

B. Homeostatic Regulation

The homeostatic regulation of serum calcium to ensure a constant supply to tissues is complex and incompletely understood. When calcium levels fall even slightly, levels are returned to normal by a PTH–vitamin D–controlled increase in calcium absorption, increase in renal tubular reabsorption, and bone resorption (Fig. 19.2). Vitamin D produced in the skin or ingested in the diet is hydroxylated in the liver and converted to calcitriol (1,25-hydroxy vitamin D) in the kidney by $1\alpha[\text{OH}]\text{ase}$. FGF23 (fibroblast growth factor 23) is a potent counterregulatory

hormone to $1\alpha[\text{OH}]\text{ase}$, which is blunted when serum calcium levels fall.³ Osteoclast-mediated bone resorption is normally responsible for the majority of the supply of calcium to the blood. However, it was not required when bone resorption was completely blocked by osteoprotegerin, a RANKL (receptor activator of nuclear factor kappa-B ligand) inhibitor, and serum calcium levels were maintained by the other physiological adaptations.⁴

Elevated levels of extracellular Ca^{2+} inhibit secretion of PTH and production of calcitriol as well as stimulate secretion of calcitonin, a peptide hormone produced in the thyroid gland. This results in decreased calcium absorption, increased urinary calcium excretion, and decreased bone resorption. Extracellular Ca^{2+} binds to CaR on the surface of parathyroid cells, which stimulates a conformational change in the receptor leading to an inhibition of PTH secretion from the parathyroid.⁵ CaR also has been located on the calcitonin-secreting c-cells of the thyroid gland, in the calcitriol-producing cells of the renal proximal tubule and the intestine, and in osteoblast cell lines,² suggesting that CaR plays a similar role in monitoring Ca^{2+} levels at these sites in order to normalize serum calcium levels. In the kidney, CaR regulates serum calcium concentration independent of PTH.⁶

C. Absorption

Calcium is not efficiently absorbed or retained in the body. The intestine is the dominant site of adaptation to dietary calcium deficiency. Calcium absorption efficiency (fractional absorption) is influenced by calcium status and physiological state (higher in adolescence, pregnancy, and lactation and lower in aging). However, prolonged calcium deficiency is not completely corrected by an increase in absorption efficiency, so bone loss ensues to maintain serum calcium concentration.

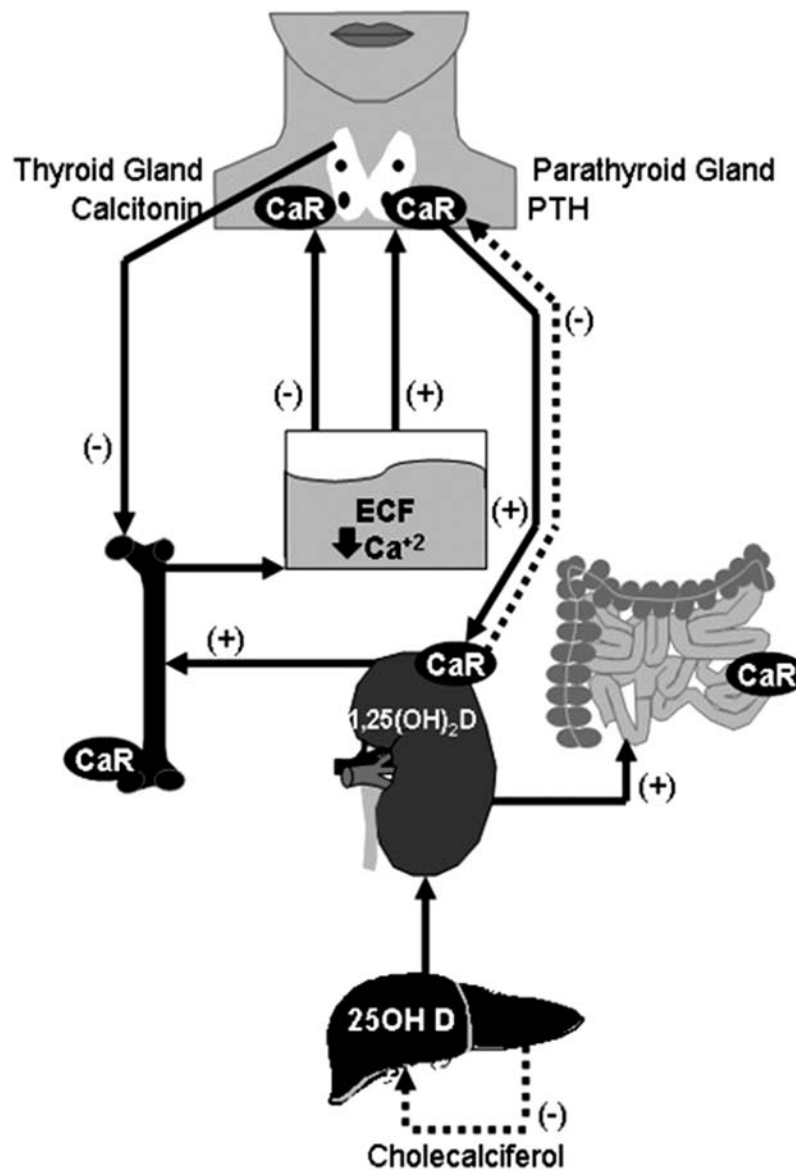


FIGURE 19.2 Homeostatic regulation of calcium. The + and – signs indicate the stimulatory and inhibitory responses to a fall in serum or extracellular fluid (ECF) Ca^{2+} below 2.5 mM. The Ca^{2+} -sensing receptor, CaR, has been localized in each of the cells depicted, but its direct role in calcium homeostasis in the intestine has not been demonstrated.

Calcium absorption occurs by two routes. The vitamin D–PTH-dependent transcellular route is saturable and subject to homeostatic regulation. Calcium absorption by this route is upregulated with increased calcitriol due to reduced levels of serum calcium, as shown in Fig. 19.2, which interacts with the vitamin D receptor in the enterocyte and stimulates the synthesis of calcium-binding proteins. Epithelial calcium channels, i.e., TRPV6, mediate calcium entry into the cell, and calcium transporters, i.e., calbindin 9K, shuttle Ca^{2+} across the intestinal epithelial cell.⁷ Calcium absorption by the paracellular route (passive diffusion that occurs between cells) is largely

dependent upon intestinal luminal calcium concentration. Some regulation occurs when calcitriol increases claudins 2 and 12, paracellular calcium channels.

On low calcium intakes and with increase age, net calcium absorption can decrease to below 200 mg per day and calcium balance becomes negative. For calcium intakes approaching 800–1000 mg/day, the saturable component of calcium absorption may account for half of total calcium absorption.⁸ Calcium absorption is also influenced by the calcium content of the meal. As the calcium load increases, absorption efficiency from the saturable pathway decreases,^{9,10} although net calcium

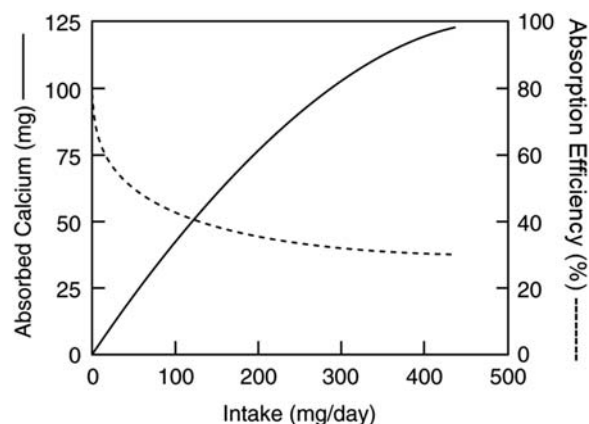


FIGURE 19.3 Theoretical relationship between calcium intake and net calcium absorbed (solid line) and absorption efficiency (dashed line). To convert grams to moles, multiply by 0.023.

absorption continues to increase, but the nonsaturable component becomes increasingly dominant (Fig. 19.3). Practically, calcium absorption is more efficient if consumed in divided doses throughout the day.

D. Calcium Excretion

Approximately 10 g calcium is filtered through the kidney daily and most is reabsorbed. Urinary calcium excretion accounts for ~50% of the variability in calcium retention. Calbindin-D 28k controls the apical calcium channel TRPV5-mediated calcium influx following increase in calcitriol when serum levels of calcium fall.¹¹ However, calcium intake explains only 1%–6% of the variation in urinary calcium.¹² Urinary excretion cannot reduce to zero during low calcium intakes as for some other minerals, i.e., magnesium, because of obligatory losses.

Some dietary constituents have an important role in calcium retention. Dietary sodium is the key dietary factor that influences urinary calcium loss. Every additional 43 mmol (1 g) sodium results in an additional loss of ~0.66 mmol (26.3 mg) calcium in adults.¹³ The effect of sodium on urinary calcium loss and net calcium retention is greater in white than black children.¹⁴ A longitudinal study in postmenopausal women illustrated the negative consequences of high salt intakes on bone density.¹⁵ In this study, no bone loss occurred on calcium intakes as high as 44.2 mmol/day (1768 mg/day) or on urinary sodium excretion (reflecting dietary intakes) as low as 92 mmol/day (2115 mg/day). Dietary protein also increases urinary calcium loss but does not decrease net calcium retention due to protein's role in offsetting changes in endogenous secretion or calcium absorption.¹⁶ In fact, bone loss and fractures in the elderly are reduced by higher protein intakes.¹⁷

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Second Messenger

Intracellular calcium concentrations, at 100 nmol/M, are >10,000-fold less than extracellular calcium concentrations. In response to a chemical, electrical, or physical stimulus interaction with a cell surface receptor, intracellular calcium concentrations rise from an influx of extracellular calcium or from intracellular calcium stores such as the endoplasmic or sarcoplasmic reticulum. The rise in intracellular calcium triggers a specific cellular response, usually through activating one or more kinases to phosphorylate one or more proteins. Thus, calcium acts as a second messenger to activate a wide range of physiological responses including muscle contractions, hormone release, neurotransmitter release, visual phototransduction, glycogen metabolism, coagulation, and cellular differentiation, proliferation, and motility.

A number of enzymes are activated or stabilized by Ca^{2+} , a function unrelated to changes in intracellular calcium concentration. These include several proteases and dehydrogenases.

B. Genetic Regulation

Genetic regulation of serum calcium and urinary calcium excretion as well as genetic regulation of their hormone systems, PTH and vitamin D, was reviewed by Bonny et al.¹⁸ Rare monogenic diseases have led to the identification of some genes involved in regulating serum calcium, and genome-wide association studies approaches have helped identify loci involved in calcium homeostasis. Specifically, vitamin D polymorphisms have been associated with fractional calcium absorption.¹⁹ In other work, use of recombinant inbred mouse lines²⁰ identified gene clusters that regulate calcium absorption in response to low-calcium diets.

IV. CALCIUM REQUIREMENTS

A. Indicators of Calcium Deficiency

Low calcium intake has been associated with a multitude of disorders. When the functional reserve (skeleton) is depleted chronically to maintain normal serum calcium levels, low bone mass ensues and can lead to osteoporosis. Low amounts of calcium reaching the lower bowel (unabsorbed calcium) can increase vulnerability to colon cancer and kidney stones. Failure to maintain extracellular calcium concentrations may increase risk of hypertension, preeclampsia, premenstrual syndrome, obesity, polycystic ovary syndrome, and hyperparathyroidism.

B. Criteria of Adequacy

The only criteria for selecting an indicator of adequacy of calcium with sufficient strength and quality of evidence according to the Institute of Medicine²¹ were indicators of bone health. These indicators include measures of bone accretion, bone maintenance, and bone loss as appropriate for different life stages. Dose–response studies to determine intakes for average or maximal retention at different life stages are strongest. During growth, calcium retention is positive. During early adulthood, neutral calcium retention is the goal. During aging, bone loss may occur even when calcium intakes are adequate. Determining calcium retention by metabolic balance studies has the advantage of being able to maintain a controlled diet with a constant level of calcium for a sufficient period to come into steady state. Determining calcium retention by measuring bone mineral content over time requires using a strong bone outcome measure as calcium is a constant fraction of bone mineral. Therefore, an imaging measure is more precise than retention derived from metabolic balance studies. However, it is practically too difficult to maintain a controlled diet with a known calcium intake sufficiently long to measure bone changes prospectively by DXA.

C. Assessment of Status

There is no good objective measure of calcium status. Total body calcium levels can be calculated

from bone mineral content measured by DXA, but that is an indicator of cumulative calcium retention and is influenced by many factors including genetics and physical activity.

Dietary assessment of calcium intake is the current approach to assessing calcium status. Dietary assessment is usually done by self-report but can take many forms, none of which is ideal.²²

Calcium intakes of most populations globally, especially adolescent females, are below the calcium requirements of their country (Fig. 19.4). Generally, calcium intakes are highest in Scandinavian countries and lowest in Asian countries. Adolescent and elderly males and most females over age 9 years are the age groups of most concern for inadequacy. In the United States, supplement use decreased the prevalence of adults (≥ 20 years) not meeting the Estimate Average Requirement from $37.7 \pm 1.4\%$ to $19.6 \pm 1.3\%$, but the amount was dependent on race/ethnicity, i.e., from $35.9 \pm 1.0\%$ to $24.2 \pm 0.7\%$ in white, $56.3 \pm 1.4\%$ to $48.2 \pm 1.4\%$ in black, 36.4 ± 0.1 to $30.1 \pm 2.0\%$ in Hispanic, and $61.4 \pm 2.5\%$ to $47.0 \pm 2.6\%$ in Asian adults.²³

D. Reference Intakes

The 2010 Dietary Reference Intakes (DRIs) for calcium in the United States and Canada relied on bone health as the indicator in setting calcium recommendations, as with previous recommendations²¹ (Table 19.1).

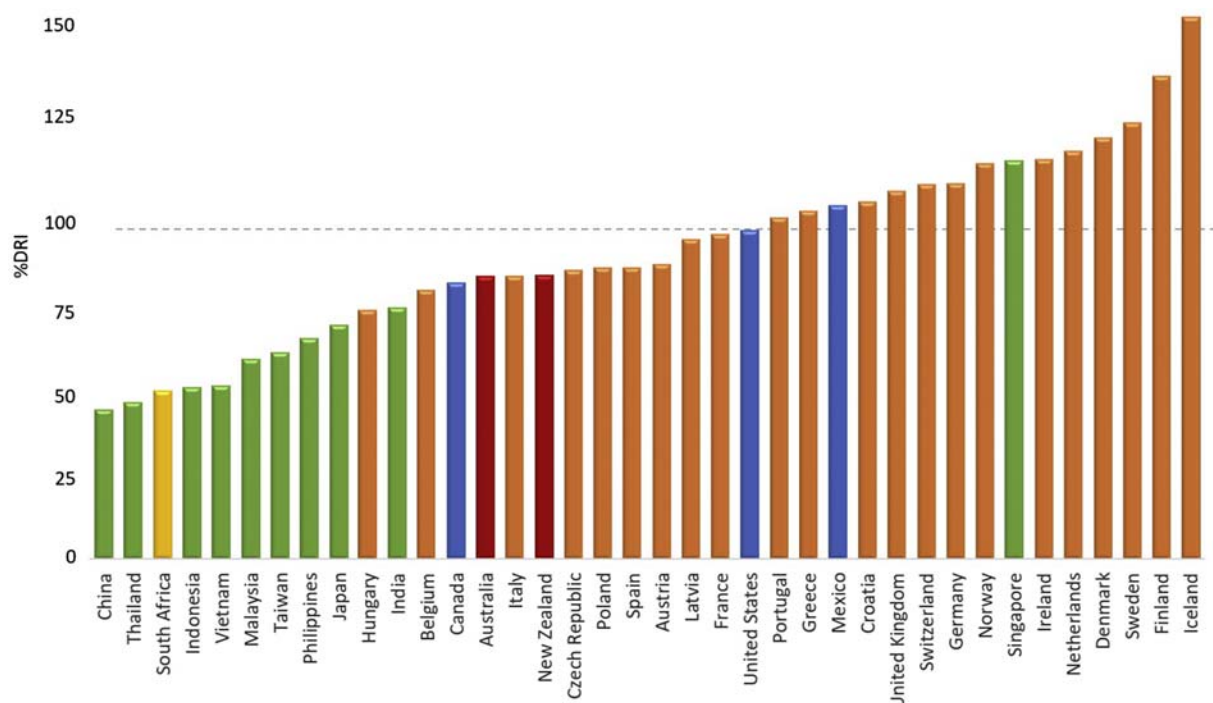


FIGURE 19.4 Calcium intakes in adults over age 20 years as a percentage of country-specific Dietary Recommended Intakes (DRIs). Colors are coded by continent. Data are from Refs. (24,25,21,26,27,28,29,30,31,32,33,34,35,36,37).

TABLE 19.1 Dietary Reference Intakes for calcium by life-stage group for the United States and Canada.

Life-stage group	AI (mg)	EAR (mg)	RDA (mg)	UL (mg)
0–6 months	200	—	—	1000
7–12 months	260	—	—	1500
1–3 years	—	500	700	2500
4–8 years	—	800	1000	2500
9–13 years	—	1100	1300	3000
14–18 years	—	1100	1300	3000
19–30 years	—	800	1000	2500
31–50 years	—	800	1000	2500
51–70 years	—	800	1000	2000
71+ years	—	1000	1200	2000
14–18 years pregnant	—	1100	1300	3000
19–30 years pregnant	—	800	1000	2500
31–50 years pregnant	—	800	1000	2500
14–18 years lactating	—	1100	1300	3000
19–30 years lactating	—	800	1000	2500
31–50 years lactating	—	800	1000	2500

AI, Adequate Intake; EAR, Estimated Average Requirement; RDA, Recommended Dietary Allowance; UL, Tolerable Upper Intake Level.

Data from Ref. (21).

In Europe, Dietary Reference Values were determined by the factorial approach and balance studies³¹, which could be construed as bone health since almost all the body's calcium is in the skeleton.

The range in calcium recommendations around the world is large; for example, recommended calcium intakes for adolescents range between 500 and 1300 mg/day. In addition to assumed differences among people, differences in calcium requirements occur through aiming for the prevention of deficiencies versus optimal health and preventing chronic disease, with different end points from maximal retention to replacing losses, and using different assumptions for absorption efficiency, losses, and safety margins. Many countries simply adopt the requirements determined by North America, the United Kingdom, or authoritative bodies such as the FAO or WHO.³⁸ Perhaps 1 day nutrient reference values will be harmonized globally.³⁹

Calcium intake needs may be higher during growth as in puberty or later in life when utilization is diminished. However, recommended calcium intakes are not higher for pregnancy and lactation in the North America or Europe. Calcium supplementation during lactation does not prevent lactation-induced bone loss, which amounts

to about 5% per year of lactation; however, bone calcium is regained upon weaning.⁴⁰ Calcium absorption efficiency is enhanced by the second trimester of pregnancy and greatest during the third trimester, when fetal demand for calcium is greatest.⁴¹ Calcium supplementation during pregnancy may improve bone gain postpartum since 1 g/d improved bone mineral density (BMD) of women 1 year postpartum compared to baseline measures,⁴² although more evidence is needed.⁴³ Generally, fetal skeletons are protected by homeostatic upregulation at all but exceptionally low calcium intakes of mothers, and no association has been identified between pregnancy, lactation, and risk of fracture.⁴⁴ A longitudinal study showed some compromise of pelvic and spine BMD despite an increase in arm and leg BMD during pregnancy.⁴⁵ However, low dairy intake was associated with decreased fetal femur length in pregnant African-American adolescents.⁴⁶ The need for calcium in breast milk to support neonatal development is met by maternal adaptation of increased bone resorption.⁴⁷

E. Basis for Individual Variability in Requirement

Calcium requirements depend on life stage, which define whether bone is in a state of accrual, maintenance, or loss as discussed earlier. Within a life stage, there is some evidence that requirements vary with body size, although this has not been considered to date. Larger bone mass either requires more calcium to achieve or greater efficiency or both. One of the largest determinants of calcium absorption efficiency is height, an indicator of gut length, and increased absorptive surface.⁴⁸ Calcium absorption efficiency is also related to body size measures, i.e., height and Z-score height in children⁴⁹, and race^{50,51}. Adolescent boys are more efficient at absorbing and retaining calcium on a given calcium intake than girls.⁵² Serum IGF-1 and IGF-BP3 were the strongest predictors of calcium retention in adolescent boys after calcium intake.⁵³

F. Tolerable Upper Levels

Potential problems of excessive calcium intake used to determine Tolerable Upper Intake Levels (ULs) by the Food and Nutrition Board for North America were established for the first time in 1997 at 2500 mg/day. They were revised in 2010 (Table 19.1) to adjust for increased requirements for adequacy at various life stages. An upper limit of 2500 mg/day for calcium has also been adopted by the European Community, Japan, the Nordic countries, and Taiwan. The prevalence of intakes above the UL, even when supplements are considered, is small (<2% of adults from 2009 to 2012 NHANES) except for 4.8 + 0.3% for white adults.²³

The hallmark concern of excessive calcium intake is hypercalcemia, where serum calcium levels exceed 10.5 mg/dL or 2.63 mmol/L, and hypercalciuria, where urinary calcium exceeds 250 mg/day in women or 275–300 mg/day in men. This is rare in healthy people and is usually caused by conditions such as malignancy or urinary hyperparathyroidism but has been increasing, especially in postmenopausal and pregnant women with ingestion of large quantities of supplements (>3 g/day) together with alkali.⁵⁴ This calcium alkali syndrome may be accompanied by renal insufficiency and is associated with hypophosphatemia.⁵⁴ Symptoms of hypercalcemia include lax muscle tone, constipation, large urine volumes, anorexia, and, ultimately, confusion, coma, and death.

The primary basis for the UL for calcium in adults is risk of kidney stones. There was a 17% increase in incidence of kidney stones in the Women's Health Initiative trial.⁵⁵ However, obtaining calcium from food or by taking supplements with food to bind any oxalates minimizes risk of stones as will be discussed later under kidney stones. Individuals should avoid exceeding the ULs. However, at present, inadequate calcium intake is still the primary concern for optimizing calcium nutrition for most individuals.

G. Dietary Sources of Calcium

Calcium intakes declined when cultivated cereal grains became the staple plants in the diet of most of the world, as the calcium content of grains and fruits is typically quite low. Consequently, the major source of concentrated calcium from foods in most parts of the world is dairy products. It is estimated that in the United States, 64% of food calcium comes from dairy products and mixed foods containing dairy ingredients.²¹

Sources of calcium should be evaluated for their calcium bioavailability as well as content. A summary of calcium bioavailability from a variety of foods and a comparison of the number of servings necessary to provide an amount of absorbable calcium equivalent to that from a glass of milk is given in Table 19.2.

Typically, bioavailability of calcium is dictated by the ability of the cation to become free of its ligands or dissociated from the salt and then to remain soluble. However, calcium does not have to be dissociated from low molecular weight salts such as calcium carbonate or calcium oxalate to be absorbed, presumably by the paracellular route or by pinocytosis.⁵⁶ That low molecular weight calcium salts can be absorbed without the aid of a vitamin D-inducible calcium transporter, as is required in active calcium absorption, suggests possible treatments for individuals with impaired kidney function who cannot synthesize calcitriol. Solubility of calcium salts in water over the wide range of 0.1–10.1 mmol/L

did not influence calcium absorption.^{9,10} The condition of achlorhydria also does not reduce calcium absorption as long as the salt is consumed with food.⁵⁷ Not considered by Recker⁵⁷ is the possibility that appreciable quantities of calcium carbonate were absorbed intact in patients with achlorhydria. Thus, either the pH of the intestinal milieu is an important factor in calcium bioavailability or we need to revise our understanding of calcium absorption because it appears that solubility of a calcium source may be less important than we have traditionally thought.

Calcium absorption can be inhibited by some ligands and enhanced by others. The strongest known inhibitor of calcium absorption, oxalate, forms a salt with calcium that has a solubility (0.04 mmol/L) considerably below the solubility range discussed above.^{9,10} Other modest inhibitors to calcium absorption form salts that cannot be completely dissociated in the gut and are too large to be absorbed intact by the paracellular route. Phytic acid is such an inhibitor. A threefold difference in phytic acid content in soybeans reduced calcium absorption by 25%.⁵⁸ Although phytate is a more modest inhibitor of calcium absorption than oxalate, it is consumed in higher amounts. However, in developed countries, phytate consumption is less detrimental where bread is leavened and phytate complexes are hydrolyzed by enzymes in yeast during fermentation. Fiber was once thought to inhibit calcium absorption, but it is now considered to be more likely that the phytate associated with fibers in seeds is the inhibitor. Purified fibers have little effect on 5-hour calcium absorption.⁵⁹ Enhanced calcium absorption can result from extremely soluble salts. Enhancers may work by preventing precipitation of calcium by phosphates in the gut, or they may alter the calcium absorption capacity of the intestinal epithelium.

Calcium citrate malate, with a solubility of 80 mmol/L, is the best-studied example of a salt with superior calcium absorption.^{9,10} Some casein and whey peptides prevent precipitation of calcium by phosphates.⁶⁰ Several prebiotic fibers including inulin, fructooligosaccharides, galactooligosaccharides, soluble corn fiber, and others have been shown to increase calcium absorption, suppress bone resorption, improve accretion during growth, and inhibit bone resorption later in life.^{61,62,63,64,65} As a rule, calcium absorption determined on the same calcium intake or load is similar for most dairy foods, salts used to fortify foods, and supplements that provide the major contributors of calcium to the diet. Consequently, when assessing dietary calcium, it is appropriate to ensure adequate total intake without undue focus on bioavailability. However, if alternatives to dairy products are selected as the primary source of calcium, it is important to ensure that nutrients other than calcium provided by dairy products including magnesium, vitamin D (fluid milk), riboflavin, and vitamin B12 are also adequate.

TABLE 19.2 Comparing food sources for absorbable calcium.

Food	Serving Size (g)	Calcium Content (mg)	Fractional Absorption (%)	Estimated Absorbable Ca/ serving (mg)	Servings needed to equal one cup milk
Milk	240	300	32.1	96.3	1.0
Beans, pinto	86	44.7	26.7	11.9	8.1
Beans, red	172	40.5	24.4	9.9	9.7
Beans, white	110	113	21.8	24.7	3.9
Bok choy	85	79	53.8	42.5	2.3
Broccoli	71	35	61.3	21.5	1.0
Chinese mustard green	85	212	40.2	85.3	1.1
Juices, fortified with most calcium salts	240	300	32.1	96.3	1.0
Juices, fortified with calcium citrate malate	240	300	52.0	15.6	0.62
Kale	85	61	49.3	30.1	3.2
Soymilk, fortified with calcium carbonate	240	300	23.7	71.1	1.35
Spinach	85	115	5.	5.9	16.3
Sweet potatoes	164	44	22.2	9.8	9.8
Rhubarb	120	174	8.54	10.1	9.5
Yogurt	240	300	32.1	96.3	1.0

Adapted from Ref. (13).

The US Food and Drug Administration has developed Daily Values (DVs) for use on food and dietary supplement labels to assist the US population in comparing products. The DVs are based on the DRIs for adults on a 2000 kcal diet. The DV for calcium is mandated to be listed on these labels. The DV for calcium originally was established as 1000 mg for adults and children aged 4 years and older.⁶⁶ This DV has recently changed to 1300 mg through implementation of new labeling laws that must appear on food products and dietary supplements beginning in January 2020.⁶⁶

V. ISSUES RELATED TO CALCIUM IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Osteoporosis

Of the relationships between dietary calcium and disease prevention, osteoporosis is the most studied (see Chapter 29). Connecting calcium intake to skeletal health is obvious given that 99% of the body's calcium resides in the skeleton and there are endogenous obligatory losses and continual bone turnover. Osteoporosis is characterized by reduced bone mass resulting in increased skeletal fragility and susceptibility to fractures.

Most guidelines recommend adequate calcium intake as an integral part of the prevention or treatment of osteoporosis.^{67,68} Calcium enhances skeletal strength, both as the principal constituent of mineralization and by lowering bone turnover rate through reduction of serum PTH.⁶⁹ Calcium is not the only nutrient important to bone health, but it is the one most likely to be deficient. Good nutrition is not the only requisite for good bone health; also important are physical activity, especially weight-bearing exercise, and not smoking. Hormonal sufficiency promotes bone maintenance, but estrogen therapy after menopause has also been associated with increased risk of breast cancer and stroke.⁷⁰ Adequate calcium intake can potentiate the advantage to the bone of physical activity in growth⁷¹ and adulthood⁷² and estrogen in postmenopausal women.⁷³ The primary strategies for reducing the risk of osteoporosis are to maximize development of peak bone mass during growth and to reduce bone loss later in life. Adequate calcium intakes are important for both of these aims.

A National Osteoporosis Foundation (NOF) position paper reported the results of 19 systematic reviews of factors influencing the development of peak bone mass. Only dietary calcium and physical activity received an A grade for strength of evidence for the relationship.^{74,75} A trial level metaanalysis found that calcium and vitamin D supplement reduced total fractures in adults by 14% and hip fractures by 39%.^{74,75}

Maximizing peak bone density protects against fracture at any age. Children who avoid milk had a fracture rate 1.75 times higher than expected from their birth cohort.⁷⁶ Milk may have skeletal benefits beyond adequate calcium. Rats provided calcium from nonfat dry milk solids during growth grew stronger bones than rats fed calcium from calcium carbonate, and the advantages were largely retained after switching to a low-calcium diet where all calcium came from calcium carbonate during maturity.⁷⁷ Whether the advantage of milk is due to bioactive constituents or simply to providing one or more nutrients such as phosphorus at higher levels requires further investigation. Stronger study designs are needed to determine the effect of dairy intake on fracture.⁷⁸

Response to calcium as a supplement (usually combined with vitamin D) depends in part on the nutrient status at the start of the intervention and the magnitude of change with the intervention. An 18-month intervention of three servings of dairy daily did not affect bone accrual in early pubertal children habitually consuming two servings daily.⁷⁹ The lack of benefit in this study likely reflected too little increase in a population consuming only marginally inadequate calcium at baseline on top of a heterogeneous growth spurt during the intervention.

On cessation of calcium supplementation, the skeletal advantage gained during treatment in randomized, controlled trials disappears with follow-up in most trials⁸⁰ unless study participants were on low habitual calcium intakes when the trial began.⁸¹ In children habitually consuming approximately 850 mg/day, skeletal advantages in the radius and total body BMD with calcium supplementation compared with the placebo group gained during puberty disappeared on average in spite of continued supplementation during postpuberty.⁸² However, the ability for girls to exhibit "catch-up" skeletal growth was size dependent as those who were taller at peak bone mass had reduced bone accretion at the forearm if assigned to the placebo group compared with those in the calcium treatment group from age 10 to 18. Furthermore, other skeletal sites such as the hip failed to exhibit complete catch-up growth.⁸³ At this time, the best recommendation is to maintain adequate calcium intakes throughout life to optimize development of peak bone mass during growth and to minimize bone loss later in life.

Hypertension and cardiovascular disease

The ability of calcium supplementation to control hypertension has been controversial. High dietary calcium intake may reduce cardiovascular risk through decreasing blood lipid and cholesterol levels.^{84,85} Lipids are less well absorbed and are therefore excreted in the stools when calcium binds fatty acids and forms insoluble soaps. Ensuring adequate calcium intake, particularly through

dairy products, is a primary nonpharmacologic strategy for reduction of high blood pressure recommended by the Joint Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure.⁸⁶ In 85,764 women participating in the Nurses' Health Study, those who had calcium intakes in the lowest quintile of calcium intake had a significantly greater incidence of ischemic stroke than those in the highest quintile of calcium intake.⁸⁷ Metaanalyses of randomized-controlled calcium supplement trials show that blood pressure reduction is greater in persons with low habitual calcium intakes⁸⁸ and during pregnancy.⁸⁹ However, a metaanalysis of RCTs of calcium supplementation with or without vitamin D found no relation to CVD.⁹⁰

The influence of dietary patterns rather than individual nutrients is more closely linked to disease prevention. The Dietary Approaches to Stop Hypertension (DASH) study showed an impressive and quick response of lowered blood pressure to a diet rich in fruits and vegetables and an even greater reduction if three servings of low-fat dairy foods were consumed daily.⁹¹ Approximately 70% of subjects who would have required antihypertension drug therapy experienced normalization of blood pressure on the DASH diet. Calcium may influence blood pressure by reducing sympathetic nervous system activity or through its role in intracellular signaling.

Cancer

Overall, there is no consistent relationship between calcium intake and different kinds of cancers as some show a protective effect, some show increased risk, and some show no significant effect or association.^{92,93} A 4-year, double-blind, placebo-controlled RCT of 1500 mg calcium and 2000 IU vitamin D3 daily in 2303 postmenopausal women did not lower risk of all types of cancer.⁹⁴ Increasing calcium and dairy food intakes appears to reduce the risk of colon cancer, particularly of the distal colon.^{95,96} Calcium may lower fecal-free bile acid and free fatty acid concentrations, thereby lowering cytotoxicity, and/or it may have direct effects on the colorectal epithelium.⁹⁶ Recurrent colorectal adenomas were decreased ~20% with calcium supplementation.⁹⁷

A metaanalysis resulted in a 19% reduction in relative risk (RR 0.81, 95% + 0.72–0.90) of breast cancer in the highest compared to lowest quartile of calcium intake.⁹² However, there was no effect of calcium supplementation at 1000 mg/day on breast cancer incidence or risk in the Woman's Health Initiative trial.⁹⁸

Kidney stones

Observational studies suggest that adequate dietary calcium decreases incidence of kidney stones.⁹⁹ A controlled trial demonstrated that increased calcium

intake reduced kidney stone incidence by half and was associated with a decreased urinary oxalate excretion.¹⁰⁰ In the gut, calcium combines with oxalate from dietary sources. The rather insoluble calcium oxalate salt is poorly absorbed,⁵⁶ and oxalic acid is less available for stone formation. However, when calcium supplements are not consumed with meals, they have no opportunity to bind with dietary oxalate and may increase urinary calcium levels that are often already high in people who form stones, further aggravating stone formation.¹⁰¹ This is the likely explanation for the increase in kidney stones in the Women's Health Initiative discussed earlier. People who form kidney stones are often characterized by a renal leak of calcium that reduces skeletal mass.¹⁰² Lowering calcium intake in such individuals would not likely correct their kidney stone problem but would undoubtedly further compromise their skeletal health.

Other disorders

Adequate calcium intake may be associated with protection against several other disorders. One multicenter trial showed that calcium supplementation at 30 mmol/day (1200 mg/day) of calcium for 3 months significantly reduced symptoms of premenstrual syndrome by 48% compared with 30% in the placebo group.¹⁰³ An observational study showed a protective effect of calcium and vitamin D against polycystic ovarian syndrome, a common cause of menstrual dysfunction and infertility.¹⁰⁴ All of these relationships may work through regulation of PTH levels by dietary calcium.

B. Known Interactions

Adequate vitamin D status is necessary for calcium absorption by the active pathway as discussed earlier. The 2010 IOM DRI committee considered the interdependence of calcium and vitamin D and concluded that "calcium appears to be the more critical nutrient in the case of bone health" and "calcium, or lack thereof, 'drives' the need for vitamin D."²¹ In Western youth populations, vitamin D status appears to be sufficiently high so that supplementation has little benefit for calcium absorption.¹⁰⁵

Two dietary constituents that were once associated with negative calcium balance and, consequently, were thought to be risk factors for fracture are protein and caffeine. In recent studies, dietary protein has been shown to be beneficial to the bone in some individuals and caffeine intake has a modest effect. The ability of dietary protein to increase BMD or reduce fracture risk appears to depend on adequate calcium intake.¹⁰⁶ The ability of increased calcium absorption to offset increased urinary

calcium losses discussed earlier in the [Section II.D](#) necessitates adequate calcium intake. Authors of a systematic review on caffeine safety concluded that up to 400 mg caffeine per day did not have an adverse effect on calcium metabolism or bone end points with a moderate to low level of confidence with the current state of evidence.¹⁰⁷

Some concern about calcium supplementation and reduced absorption of trace elements has been raised, especially regarding iron. However, Minihaan and Fairweather-Tate¹⁰⁸ showed that although calcium at 10 mmol (400 mg) can significantly reduce iron absorption in a single meal, chronic calcium supplementation at 30 mmol/day (1200 mg/day) does not decrease iron status. Presumably, iron absorption is upregulated as stores decrease. Neither iron status nor zinc metabolism or retention was affected by 1 g calcium supplementation in girls.^{109,110}

C. As a Dietary Supplement

Dietary supplements are popular in some populations. From an analysis of NHANES 2009–12 data, 61.1 ± 0.8% of US adult whites, 40.7 ± 1.2% of blacks, 36.6 ± 1.3% of Hispanics, and 53.6 ± 3.2% of Asians used dietary

supplements.²³ A cost–benefit analysis estimated that use of calcium and vitamin D supplements would reduce fractures by 544,687 per year in Europe at a savings of £6.9 billion and 323,566 fractures per year in the United States at a savings of \$3.9 billion.¹¹¹

Concerns over prolonged calcium supplementation have been raised for risk of kidney stones as already discussed and prostate cancer, myocardial infarction, and vascular calcification. Risk of prostate cancer has been associated with calcium intake in observational studies. A twofold increased risk in prostate cancer was reported for milk consumption but not for other dairy products or dietary calcium.¹¹² Generally, risk for prostate cancer has been observed at calcium intakes associated with supplement use, i.e., >2000 mg/day, and the association lacks strong biological plausibility.

Concern over calcium supplementation and soft tissue calcification led to the development of a systematic review to inform a position paper by the NOF and American Society for Preventative Cardiology. These groups concluded that a lack of evidence existed for such a relationship with a moderate level of strength.¹¹³

Additional sources of information are available in the published literature (Refs. [114–116](#)).

RESEARCH GAPS

More is known about the relationship between calcium and bone health than for any other disorder, yet several questions remain. The US Surgeon General's Report on Bone Health and Osteoporosis⁶⁷ emphasized the role of good nutrition, and calcium and vitamin D in particular, for normal bone growth and osteoporosis prevention and treatment. Important areas for future research include strategies to maximize peak bone mass in boys and girls in ethnically diverse populations. Should calcium recommendations be based on body size? The prevention and reversibility of calcium and vitamin D deficiency and disease states need to be better understood. Can calcium protect the skeleton against low vitamin D status except for very low serum levels, i.e., <10 nmol/L? Future research should focus on more complex interactions that consider not only calcium and vitamin D but also phosphorus.

For each of the other calcium nutrition–related disorders outlined above, additional research will clarify benefits of adequate calcium and risks of excess. We need to better understand the importance of calcium intake in late-stage cancers. An area of active research stemming from increased incidence of obesity is the consequence of bone loss accompanying weight loss. The Surgeon General's report identified finding mechanisms and interventions to preserve bone during weight loss as an important area for future research. Adequate dietary calcium has been shown to be a useful countermeasure.¹¹⁷ Epidemiological studies would become more useful in identifying the relationship of calcium intake and disease once more accurate means for assessing calcium intake are developed. For each calcium-related disease and metabolic parameter, the interaction between diet and genetics will be a dominant area of investigation in the coming decades. The first genetic mapping of gene–dietary calcium interactions has been reported. Further insights into genetics and nutrition are likely to help explain racial and sex differences in calcium absorption as well as identify individuals who can benefit by rigorous interventions. Concern over excesses is a modern phenomenon as more people turn to supplements to meet their calcium needs. Still, we should understand the risks of this popular alternative to meeting calcium recommendations through food.

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PHOSPHORUS

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SUMMARY

Phosphorus is an essential micronutrient involved in a number of critical biological processes including bone metabolism and intracellular signaling. Systemic phosphorus homeostasis is tightly regulated through an intricate system of endocrine feedback loops. Parathyroid hormone and fibroblast growth factor 23 are the primary effector hormones in this system that coordinates phosphorus trafficking between dietary phosphorus absorption, urinary phosphorus excretion, and exchanges with bone and soft tissue stores. Nutritional phosphorus deficiency is rare in the developed world since most individuals consume more phosphorus than is required for daily needs. Excess dietary phosphorus intake is more of a concern in Western diets, in part due to the widespread use of phosphorus-based food additives. Evidence that excess phosphorus intake is linked to kidney, cardiovascular, and bone disease requires careful consideration of how to properly balance safety, quality, and convenience of the food supply with chronic disease risk in the general population.

Keywords: Calcium; Diet; Fibroblast growth factor 23; Hyperphosphatemia; Osteomalacia; Parathyroid hormone; Phosphorus; Rickets.

I. INTRODUCTION

A. Background

The word phosphorus derives from the ancient Greek words $\phi\omega\varsigma$ (phôs) meaning “light” and $\φορ\upsilon\varsigma$ (phorus) meaning “bearer.”¹ The origin of this name comes from the initial discovery of phosphorus in the 17th century, when a small sample of phosphorus purified from urine was found to spontaneously emit pale-green light in the dark without producing any heat. Further investigation of this new element paved the way for its eventual recognition as an essential constituent of the fabric of all life.

Phosphorus in the biosphere most commonly occurs in its pentavalent form, as phosphate (PO_4^{3-}),¹ and it is in this form that it is most often found in living organisms. Phosphorus is a limiting nutrient in the sense that the mass of the biota in any given environment is determined strictly by phosphorus availability in that environment.² As a reflection of that dependence, and with

the exception of guano deposits, virtually all of the phosphorus in an environment will be tied up in its biota.^{2,3}

B. Nutrient Function and Structure

In addition to its role in the mineral component of the endoskeleton of vertebrates, phosphorus is involved in cell membrane structure as phospholipids, in information coding as DNA and RNA, in energy metabolism as ATP and GTP, and in enzymatic activation by phosphorylation of catalytic proteins. Additionally, inorganic phosphate (P_i) concentration in the extracellular fluid (ECF) of higher vertebrates plays a vital role in supporting both mineralization of bone and the intermediary metabolism of all body tissues.⁴

Phosphorus is the sixth most abundant element in the human body, comprising in adults about 1.0%–1.4% of fat-free mass or ~ 12 g (0.4 mol) per kilogram. Of this total, 85% is in the mineral of bones and teeth, with 15% in the blood and soft tissues (especially myelin).⁴ Thus, in a

70 kg adult with 25% fat mass, total body phosphorus is ~630 g (~21 mol).

II. PHOSPHORUS METABOLISM

Extracellular inorganic phosphate (P_i) makes up <0.1% of total body phosphorus.⁴ Phosphorus from digestion of food in the gut and resorption of mineral from bone are both transferred into this compartment, and it is from this compartment that bone mineral is formed and urinary phosphorus is derived. At physiological pH (7.4), ECF P_i exists as a mixture of $H_2PO_4^-$ and HPO_4^{2-} , with an effective valence of -1.8.¹ While serum P_i is not always a faithful reflection of tissue phosphate levels, many situations of phosphate deficiency or excess express themselves as excessively low or high values of serum P_i .

All procedures used to measure phosphorus in clinical laboratories actually are measuring P_i . In the United States, these values are typically expressed in mg/dL phosphorus instead of P_i , in part because the valence of P_i changes depending on the pH of the medium being sampled.¹ With an atomic weight of ~31 g/mol, a serum phosphorus concentration of 4 mg/dL actually equates to the serum phosphorus concentration and not the serum P_i concentration, which would be ~1.3 mmol/L at a physiological pH of 7.4.

A. Phosphorus Digestion/Absorption/Transport

Bioavailability

With the exception of seeds and unleavened breads, most food phosphorus is in the form of readily hydrolyzable organic phosphate esters.⁵ Seed foods contain, in addition to such digestible phosphorus, an organic storage form of phosphorus—phytic acid (hexainositol phosphate)—which can contribute to available phosphorus in soils low in inorganic phosphate which is required for plant embryos to grow. The principal significance of phytic acid for human nutrition is that the human intestine lacks the enzyme phytase and is hence unable to utilize most of the phytate phosphorus present in foods. Colonic bacteria, which do possess phytase, are able to release some of that phosphorus for absorption. Additionally, yeasts used in food preparation hydrolyze phytic acid, and hence leavened cereal-grain foods (e.g., many breads) exhibit good phosphorus availability, whereas unleavened products such as matzah or flour tortilla do not. Diets high in unleavened phytate-containing foods can lead to decreased iron, zinc, and calcium absorption due to their binding to phytate, and hence result in deficiencies of these minerals, because the ingested phytate—mineral complexes are not absorbed.

Aside from phytate, the principal factor influencing phosphorus bioavailability is not the food itself but coingested calcium, which binds not just phytate phosphorus but other forms of dietary phosphorus; the resulting calcium:phosphate complex cannot be absorbed.⁵

Intestinal absorption of phosphorus

Most organisms get the phosphorus they need by consuming the tissues of other organisms (plant and animal); such ingested phosphorus is absorbed with high efficiency. In adult humans, for example, net phosphorus absorption typically ranges from 55% to 80% of ingested intake^{6,7} and in infants from 65% to 90%.^{8–10} The most active intestinal absorptive site is the jejunum.

Most cell organelle phosphorus in ingested foods is quickly hydrolyzed by intestinal phosphatases and hence most absorbed phosphorus is in the form of P_i .¹¹ The principal exception is phosphorus from phytic acid. Otherwise, intrinsic bioavailability of most food phosphorus sources is high.

Absorption of phosphorus is by a combination of active transport and passive diffusion, with the former being the regulated component.¹² Active phosphorus absorption is influenced by vitamin D status. Vitamin D—stimulated active absorption of phosphorus occurs in the intestine,¹³ where the primary function of vitamin D is to promote absorption of calcium and phosphorus. Studies have shown that dietary phosphorus loading in healthy volunteers induces a decrease in serum 1,25-dihydroxyvitamin D ($1,25(OH)_2D$) concentrations, demonstrating endocrine feedback regulation of phosphorus absorption.^{14,15} Using an intestinal washout method, large daily doses of calcitriol ($1,25(OH)_2D$) enhanced dietary phosphorus absorption in five patients on chronic hemodialysis, producing essentially normal phosphorus absorption efficiency and suggesting that the loss of renal synthesis of $1,25(OH)_2D$ reduced phosphorus absorption in patients with end-stage kidney disease (ESKD).¹⁶ Nevertheless, despite low $1,25(OH)_2D$ production, dietary phosphorus is still readily absorbed in ESKD patients, frequently resulting in the development of hyperphosphatemia in the absence of dietary phosphorus restriction—which is the rationale for use of intestinal phosphorus binders in such patients.¹⁷

Despite the foregoing evidence of an effect of vitamin D on absorption, the net result in normal individuals is probably small. It is instructive to note that net phosphorus absorption in adults is approximately 65%, leaving relatively little room for vitamin D—mediated absorption to increase. Such vitamin D effects on phosphorus absorption found in humans may be more indirect than direct, as the following data suggest.

Heaney and Nordin, pooling two large data sets of human metabolic balance studies, showed that over a wide range of ingested calcium:phosphorus ratios, the principal determinant of fecal phosphorus (and

therefore inversely of absorbed phosphorus) was fecal calcium, with phosphorus intake itself exerting a significant but weaker effect.¹⁸ Together, fecal calcium and diet phosphorus explained about 80% of the observed variance of phosphorus absorption. Each 10 mmol of ingested calcium, by complexing phosphorus in the intestinal lumen, blocked the absorption of ~ 4 mmol of dietary phosphorus. This is the basis for the use of calcium salts as phosphorus binders in patients with ESKD. This very high correlation leaves little room for an effect of vitamin D status. However, as calcium absorption rises in response to vitamin D, less calcium is left behind in the intestinal lumen to bind the still unabsorbed phosphorus, and hence phosphorus absorption would predictably rise under conditions of high vitamin D status. This does not necessarily mean that vitamin D directly stimulated phosphorus absorption. Thus experiments in which calcitriol increased absorption of both calcium and phosphorus in rats cannot be interpreted unambiguously.¹⁹ In brief, the actual effect of vitamin D on adult phosphorus absorption under usual conditions and in health remains unclear.²⁰

Endogenous fecal phosphorus

The emphasis in the foregoing section is on transfer of phosphorus from the gut lumen into the blood, but it should be noted that there is a substantial transfer in the opposite direction, i.e., from the mucosa into the chyme. The magnitude of this outward-directed flux has been estimated from studies involving systemic administration of a phosphorus isotope, with measurement of its appearance in the feces.^{21,22} Estimates for the magnitude of this flux lie in the range of 8–10 mmol (250–300 mg)/day. Probably much of that total is in the form of sloughed-off gastrointestinal mucosal cells (which typically turn over approximately every 5 days). This phosphorus source is digested just as is food phosphorus, and thereby becomes available for reabsorption. Using 8.8 mmol (275 mg)/day for intestinally secreted phosphorus, and 65% for net phosphorus absorption, and taking the median phosphorus intake for women ages 51–70 from NHANES 2013–2016² of 37.3 mmol (1167 mg)/day, one can compute that the total phosphorus load presented to absorptive sites amounts to 46.1 mmol (1429 mg)/day, with fecal phosphorus amounting to ~ 13 mmol (400 mg)/day. This means that gross absorption of dietary phosphorus was 72%. This is but one indication of the avidity of the organism for ingested phosphorus.

Urinary phosphorus excretion

Normally, most phosphorus loss from the body is through the kidneys. Under steady-state conditions (neither building nor reducing tissue mass), 24 h urinary phosphorus excretion is equal to net intestinal

absorption of phosphorus. The kidney capacity for excretion of phosphorus is very great.²³ Across a doubling of phosphorus intake, ECF P_i rises very little, with essentially all the absorbed phosphorus being excreted by the kidney.²⁴ Very little elevation of the filtered load (as reflected in the serum P_i) is needed to clear the body of unneeded phosphorus.

B. Phosphorus Homeostasis

Phosphorus, along with calcium, magnesium, sodium, and potassium, are the mineral elements found in largest amounts in humans. The principal factors regulating phosphorus homeostasis differ substantially from those regulating calcium homeostasis. For phosphorus, the regulatory apparatus is optimized to deal with environmental scarcity, whereas, for calcium, the organism must deal with environmental excess, likely due to evolutionary adaptations to primitive diets rich in calcium but poor in phosphorus.²⁵ Increase in body mass, as in the generation and growth of new organelles or tissues, creates an obligatory requirement for phosphorus. In addition to the structural role of phosphorus in cell function, phosphorus as inorganic phosphate buffers pH changes in the ECF and is involved in the storage of intracellular energy (i.e., glycogen synthesis).

In complex vertebrate organisms, phosphorus availability is determined by the concentration of P_i in the ECF bathing all of the tissues and organs. Maintenance of an adequate concentration of P_i in the ECF is critical for organisms. At physiological pH and pCO_2 , the form of calcium phosphate salt that would likely precipitate from the ECF is $CaHPO_4$.⁴ Normally, ECF is only about half saturated with respect to this product. However, ECF calcium and phosphorus concentrations are supersaturated with respect to another crystal form, hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$), which readily precipitates in the presence of a suitable crystal nucleus. Hence, the ECF supports mineralization when and where the body creates a suitable crystal nucleus but is indefinitely stable as it circulates and bathes the other tissues.

The concentrations of ionized calcium and phosphorus in serum (and ECF) of adults are typically 1.1–1.3 mmol/L for calcium and 0.9–1.4 mmol/L (2.8–4.3 mg/dL) for phosphorus.¹ These values produce a $Ca \times P$ ion product $\cong 1.3 \text{ mmol}^2/\text{L}^2$. Since serum calcium is typically held much more constant than is serum phosphorus, most of the clinically encountered variation in the $Ca \times P$ product is produced by variations in serum P_i .

There is a common misconception that ECF concentrations of calcium and phosphorus are reciprocally related by solubility product considerations. In other words, when one goes up, the other must go down. The mechanisms underlying these changes will be described later. While sometimes the concentrations of

these two ions do seem to vary inversely, that is through physiological regulation of their respective levels, not because of a physical chemical equilibrium.

Much has been made of the utility of the ECF $\text{Ca} \times \text{P}$ product for assessing bone health and the propensity for spontaneous calcification of nonosseous tissues.²⁶ One of the original reports of the $\text{Ca} \times \text{P}$ product suggested that rickets in children occurred when the $\text{Ca} \times \text{P}$ product was less than $35 \text{ mg}^2/\text{dL}^2$.²⁷ That an elevated $\text{Ca} \times \text{P}$ is the main factor causing or precipitating calcification of nonosseous tissues is less clear, as evidenced by studies showing that a very high (and nonphysiologic) $\text{Ca} \times \text{P}$ product alone fails to cause calcification of vascular tissue in vitro.^{28,29} Further, vascular calcification can occur even in the absence of elevations of either ionized calcium or P_i or both,^{30,31} indicating that an elevated $\text{Ca} \times \text{P}$ is not essential for spontaneous calcification of nonosseous tissue. Instead, calcification is likely driven by a derangement in calcium and P_i homeostasis combined with the loss of calcification inhibitors in the serum (e.g., pyrophosphate).²⁹

Very high phosphorus consumption can increase serum P_i concentrations,³² which, if chronic, can lead to metastatic calcification, particularly of the kidney and the coronary arteries, and, in some species, increased porosity of the skeleton. The kidney effects have been studied mainly in rats and mice,^{33–35} have required very high phosphorus loads in addition to the animals' already high basal phosphorus intakes, and in several

reports have required partial reduction of renal tissue mass for their expression. The bony lesions have been described in rabbits³⁶ and bulls.³⁷ As with kidney toxicity, the bony lesions required extremely large phosphorus intakes—in rabbits approximately 40 times the equivalent of human intake (on a per kilogram basis) and in bulls providing feed with phosphorus at levels typically given to cows to support milk production. These situations may not be relevant to human nutrition or to usual human dietary intakes of phosphorus.

C. Regulation of ECF P_i Concentration

The relatively small mass of P_i in the ECF turns over many times each day. Phosphate traffic in and out of the ECF consists mainly, on the input side, of intestinal absorption of ingested phosphorus and release of bone mineral phosphorus in the resorptive phase of bone remodeling (Fig. 20.1).³⁸ On the output side, phosphate leaves the ECF mainly in the urine, in gastrointestinal secretions, and in the mineralization of newly forming bone (in the reconstruction phase of bone growth and remodeling).³⁸

While ECF P_i is inevitably affected by this ion traffic, the actual regulation of this critical concentration, as is also true for calcium, is mainly through adjusting the kidney threshold for excretion. Approximately 200 mmol (6200 mg) of phosphorus is filtered each day at the kidney, the vast majority of which is reabsorbed from the

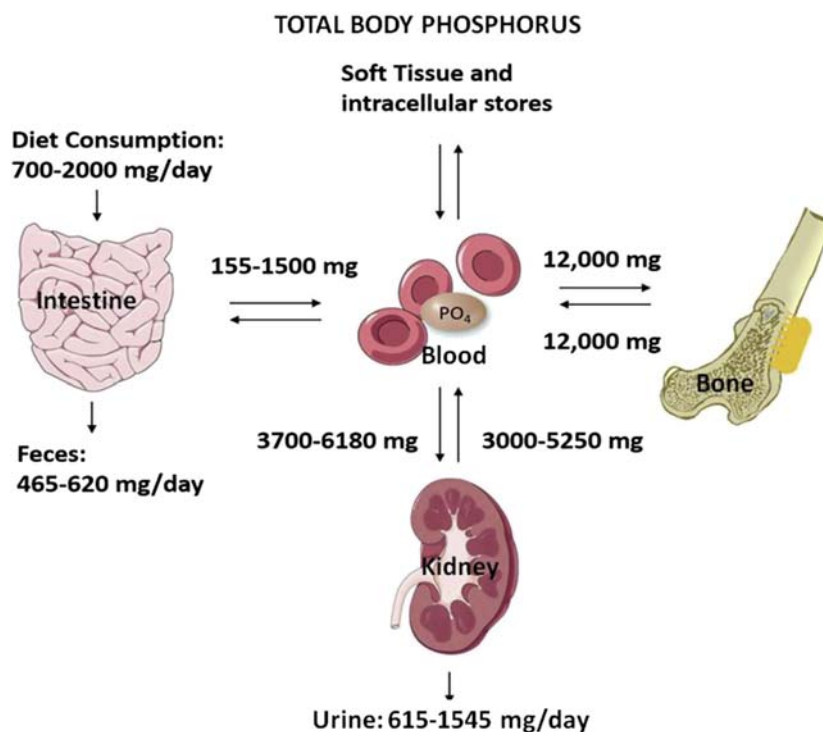


FIGURE 20.1 Phosphorus handling between body compartments. Adapted from Lomashvili et al.²⁹ and Blaine et al.³⁸

glomerular filtrate in the proximal tubule.⁴ This tubular reabsorption is mediated by sodium-dependent phosphate cotransporters in the brush border of the proximal tubular epithelium. This reabsorption has a limited capacity (termed the “tubular maximum” for phosphate [TmP]), which in turn is regulated up or down to adjust serum P_i concentration. The two factors most involved in this regulation are parathyroid hormone (PTH) secreted by the parathyroid glands and fibroblast growth factor 23 (FGF23) secreted by bone cells. Both FGF23 and PTH lower the TmP and hence increase renal phosphorus clearance (urinary excretion), i.e., the virtual volume of serum effectively cleared of its P_i per unit time. When the TmP decreases (i.e., serum phosphorus clearance rises), serum P_i falls, and vice versa.

Role of FGF23

As a negative feedback control loop, the principal regulator of systemic phosphorus balance is FGF23, produced in bone by cells of the osteoblast lineage.³⁹ FGF23 was initially discovered as the causative factor for several rare forms of osteomalacia/rickets that are either inherited (X-linked hypophosphatemic rickets [XLH] or autosomal dominant hypophosphatemic rickets [ADHR]) or acquired (tumor-induced osteomalacia [TIO]). All of these conditions are characterized by hypophosphatemia due to urinary phosphorus wasting and low or inappropriately normal circulating 1,25-(OH)₂D concentrations.^{40,41} Investigations to determine the genetic basis for ADHR eventually led to the discovery that mutations in the gene encoding FGF23 cause ADHR.⁴² Experimental data later showed that these mutations resulted in the synthesis and secretion of an FGF23 peptide that was resistant to proteolytic cleavage, resulting in the accumulation of excess quantities of biologically active FGF23 in the circulation.^{43,44} Mechanistic studies later showed that the *fgf23* gene encodes a protein of 251 amino acids and is most highly expressed in osteocytes and osteoblasts.⁴⁵ The N-terminal domain of FGF23 contains the FGF homology domain, whereas the C-terminal domain contains the unique 71 amino acid sequence that distinguishes it from other known members of the FGF family.⁴⁶

Interestingly, FGF23 belongs to a subfamily of FGFs (known as the FGF19 subfamily) that lack several amino acid residues in their C-terminal domain allowing them to be more soluble in circulation due to their low binding affinity to heparan sulfate, and thus able to exert systemic actions.⁴⁶ Binding of FGF23 to its cognate receptor is mediated by Klotho, a transmembrane protein that is most highly expressed in the kidney and parathyroid glands.⁴⁷

The principal stimulus to FGF23 production is unclear. The factors that most strongly induce FGF23 secretion appear to be 1,25(OH)₂D, dietary phosphorus intake, and, potentially, PTH.^{14,48–54} Iron deficiency,

hypoxia, and inflammation also powerfully stimulate FGF23 secretion,⁵⁵ but this is usually coupled with increased proteolytic cleavage of the FGF23 peptide, resulting in high concentrations of C-terminal FGF23 fragments of unclear physiologic significance, and stable circulating concentrations of the physiologically active intact peptide. Interestingly, in vitro and in vivo studies have failed to convincingly show that inorganic phosphate can directly stimulate FGF23 secretion itself, suggesting that FGF23 may regulate net phosphorus balance instead of ECF P_i per se.⁵⁶

Primarily secreted by bone cells, the main sites of action of FGF23 are the kidneys and parathyroid glands (Fig. 20.2). In the kidneys, FGF23 stimulates the endocytosis of sodium–phosphate cotransporters (NPT2a and NPT2c) from the apical membrane of renal proximal tubular cells, resulting in diminished reabsorption of inorganic phosphate filtered at the glomerulus.⁵⁷ In addition, FGF23 inhibits the synthesis of renal 25-hydroxyvitamin D-1 α -hydroxylase,^{51,57–59} the enzyme required to convert 25(OH)D to its more active metabolite, 1,25(OH)₂D, and upregulates the activity of 24-hydroxylase,⁶⁰ the enzyme that represents the major catabolic pathway for both 25(OH)D and 1,25(OH)₂D. In the parathyroid glands, FGF23 directly inhibits the synthesis and secretion of PTH,^{61,62} perhaps representing a secondary, indirect mechanism for inhibiting 1,25(OH)₂D synthesis. By lowering the TmP, FGF23 increases phosphorus clearance, leading to a fall in ECF P_i in states of phosphorus excess, such as high dietary phosphorus intake or kidney failure. Not surprisingly, FGF23 concentrations are typically quite high in patients with ESKD⁶³; whether sustained high levels of FGF23 are themselves responsible, through “off-loop” effects, for some of the morbidity of this condition is uncertain.⁶⁴

Role of parathyroid hormone

The actions of FGF23 on phosphorus handling in the kidney mirror those of PTH, which, as noted, also lowers the TmP and reduces serum P_i , but this effect is not a direct part of the P_i feedback control loop. The actions of FGF23 and PTH diverge with respect to vitamin D metabolism. As mentioned, FGF23 regulates renal synthesis of 1,25(OH)₂D, but in a direction opposite to the effect of PTH. FGF23 downregulates and PTH upregulates the expression of the renal 25-hydroxyvitamin D-1 α -hydroxylase. This discordance is a direct reflection of the differing control systems for which these hormones are the effector molecules. PTH is most responsive to ECF Ca^{2+} , and its upregulation of 1,25(OH)₂D synthesis leads to increased calcium entry from the gut and decreased renal calcium clearance, thus raising ECF Ca^{2+} . FGF23, for its part, is more responsive to phosphorus balance and adjusts renal phosphorus clearance to keep phosphorus balance normal.¹⁴ PTH, on the other hand, by

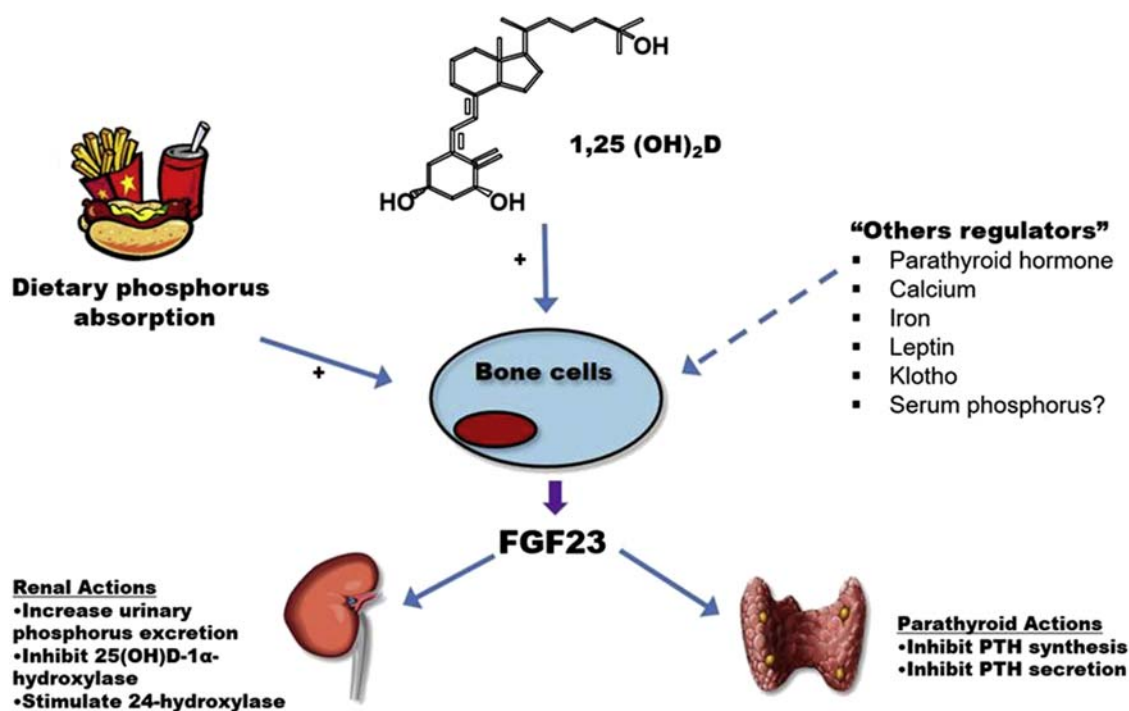


FIGURE 20.2 Overview of fibroblast growth factor 23 (FGF23) regulation of phosphorus. High levels of the active form of vitamin D, 1,25-(OH)₂D₃, or high dietary intake of phosphorus stimulate production of FGF23 in osteoblasts; its main sites of action are on kidney cells where it increases urinary phosphorus excretion and inhibits the activity of 25(OH)D-1α-hydroxylase and stimulates 24-hydroxylase, both of which act to reduce the production of 1,25(OH)₂D, and on parathyroid gland, to decrease production and transport of parathyroid hormone (PTH), thus increasing kidney excretion of phosphorus.

lowering P_i , enhances osteoclastic release of bone mineral at preexisting resorption loci,⁶⁵ thereby raising ECF Ca^{2+} . Where the two control systems intersect is in their similar effect on the TmP and, accordingly, P_i .

Serum P_i is itself active in this regulation, with low P_i leading to independent upregulation of renal 1,25(OH)₂D synthesis. By contrast, FGF23, although it lowers serum P_i as a consequence of increasing renal phosphorus clearance, does not lead to the expected increase in 1-α-hydroxylation of 25(OH)D.⁴⁸ This is because, as noted above, FGF23 itself downregulates 25-hydroxyvitamin D-1α-hydroxylase, thus preventing the increase in 1,25(OH)₂D synthesis that would otherwise have accompanied the drop in serum P_i .

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Bone Mineralization

There are three main features of the relationship of phosphorus to bone: (1) bone mineral consists of not just calcium but specifically a complex calcium phosphate salt (hydroxyapatite) comprising 85% of total body phosphorus; (2) adequate quantities of ingested phosphorus, expressed physiologically as a serum phosphate concentration of 1.5–2.0 mmol/L (4.7–6.2 mg/

dL), are essential for bone building during growth; and (3) hypophosphatemia from whatever cause limits mineralization at new bone-forming sites at all ages, impairs osteoblast function, and enhances osteoclastic bone resorption. The amorphous calcium phosphate formed in the first stages of mineralization exhibits a Ca:P molar ratio of about 1.33:1 or very close to the molar ratio of Ca:P in adult ECF.⁴ At active bone-forming sites, ECF is depleted of both its calcium and its phosphate. Osteoblast function seems not to be appreciably affected by ECF Ca^{2+} , but, like other tissues, the osteoblast needs a critical level of P_i in its bathing fluid for fully normal cellular functioning. Local P_i depletion both impairs osteoblast function and limits mineral deposition in previously deposited matrix.

B. Bone Resorption

P_i also affects the responsiveness of the osteoclast to PTH: for any given PTH level, resorption is higher when P_i is low,⁶⁵ while high P_i values reduce bony responsiveness to PTH, leading to increased PTH secretion in order to maintain calcium homeostasis; this lowers ECF P_i . Similarly, high plasma P_i suppresses renal synthesis of 1,25(OH)₂D,¹⁵ thereby slightly reducing net phosphorus absorption from the diet. Both mechanisms reduce phosphorus input into the ECF when P_i is high and augment it when P_i is low. However, in neither circumstance is the effect on plasma P_i more than modest.

IV. DIETARY REQUIREMENTS

A. Phosphorus Deficiency

Phosphorus deficiency is generally manifested as hypophosphatemia. This only rarely occurs because of inadequate dietary phosphorus intake and is almost always due to nondietary metabolic disorders.⁶⁶

Only limited quantities of phosphate are stored within cells, and most tissues depend upon ECF P_i for their metabolic phosphate. When ECF P_i is low, the result is cellular dysfunction in all tissues. The resulting effects of hypophosphatemia may include bone pain, rickets, anemia, muscle weakness, general disability, parathesias, ataxia, anorexia, decreased resistance to infection, confusion, and at times, death.⁶⁶ The observed muscle weakness involves especially proximal muscle groups and when prolonged or severe can lead to muscle fiber degeneration. Depending on the age of the individual, either osteomalacia or rickets are evident. In either of these conditions, mineralization of forming growth plate cartilage or bone matrix is the disorder that is represented by impaired function of chondroblasts and osteoblasts.⁶⁷ This functional disturbance both slows osteoid deposition and disturbs the normal maturation process in the hypertrophic zone of the growth cartilage. These acute disorders typically occur when ECF P_i falls below ~ 0.3 mmol/L (~ 1 mg/dL). However, for reasons discussed above, when growth is rapid, beginning bone lesions may be found at P_i values in the peripheral blood as high as 0.5–0.6 mmol/L (1.5–1.8 mg/dL).⁶⁶

B. Criteria for Adequacy

An adequate intake for any nutrient is one such that additional inputs of the same nutrient confer no further benefit. This concept is expressed in Fig. 20.3, which is a schematic plot of effect (or benefit) against intake of different individuals in a similar population subgroup. Intakes at levels above the average group requirement are judged “adequate” in the sense that no further benefit accrues to that individual’s health if intake of the individual is increased. Not everyone requires the same amount of the nutrient due to differences in absorption, metabolism, genetics, etc., but the median intake of the group (the Estimated Average Requirement or EAR) can be used as the average requirement for that subgroup, or as a cutoff, based on statistical assumptions.¹¹ Since the RDA is set at a value to meet the needs of almost all in the subgroup, when not much is known about an individual’s requirement, consuming an intake at or above the RDA provides some assurance that an individual’s needs are met.

The intake required to sustain ECF P_i at or above the lower limit of normal (0.87 mmol/L [2.7 mg/dL]⁴) is a

possible functional indicator to use as a cutoff for determining the dietary phosphorus requirements. However, whether maintaining serum P_i concentrations within a normal range is the best functional indicator of dietary phosphorus adequacy is not clear, since functional regulators of total body phosphorus balance (like PTH and FGF23) can markedly vary across a wide spectrum of dietary phosphorus intakes despite little or no change in serum P_i concentrations.⁶⁸ It might be possible to develop normal ranges for PTH and/or FGF23 plasma concentrations as possible integrated measures of dietary phosphorus adequacy (particularly when the ECF P_i is within the normal range), but at the moment, no such ranges are established.

An often-cited metric used to determine the general relationship between absorbed phosphorus intake and plasma P_i in adults comes from infusion studies.^{11,24} In these studies, hyperphosphatemia was sustained at a steady state following the infusion of a neutral phosphate solution. If one were to assume the same linear relationship between dietary phosphorus consumption as with intravenous phosphorus infusion, the achieved plasma P_i could thus be directly related to actual intakes. When intake is low, the filtered load will be below the TmP, and urinary loss of the infused (considered to be equivalent to absorbed) phosphorus will be minimal. At higher amounts consumed, urinary excretion increases when the TmP is surpassed, thus matching the amount absorbed and resulting in a slow elevation of serum phosphorus concentrations. It should be emphasized, however, that the relationship between infused phosphorus and absorbed phosphorus in these studies was not derived from directly measured values, but instead was based on the estimated relationship between infused and absorbed phosphorus, and has not been validated in other studies or other populations.

C. Dietary Reference Intakes

Recommended intakes for phosphorus

Reference intake values for phosphorus were last established as part of the Dietary Reference Intakes (DRI) process by the Food and Nutrition Board of the Institute of Medicine (IOM) for the United States and Canada in 1997 (Table 20.1).¹¹ The adult value for the EAR in Table 20.1 is based on the average intake required to sustain ECF P_i at or above the lower limit of normal: 0.87 mmol/L (2.7 mg/dL). To derive the Recommended Dietary Allowance (RDA), the EAR was increased by 2 standard deviations (the coefficient of variation was assumed to be 10% as the variation in requirements in the various age groups was unknown) to meet the needs for phosphorus of almost all (97%) of the group. The EAR and RDA for

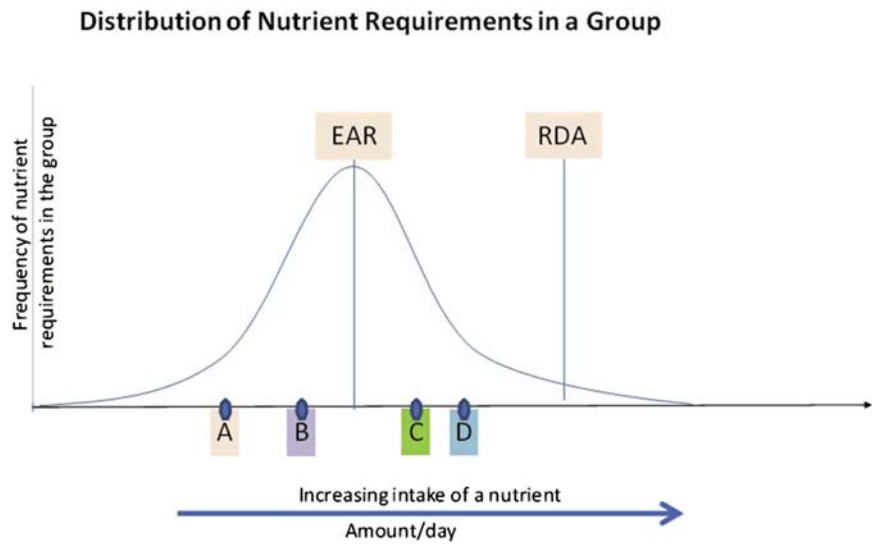


FIGURE 20.3 Schematic of Gaussian or normal distribution of nutrient requirements in a group. This figure depicts the relationship of the requirements of four individuals (persons A, B, C, and D) to the intake level at which half of the group (persons A and B) have their needs met for the nutrient (defined as the estimated average requirement, or EAR¹¹), while half do not (persons C and D). Thus C and D need additional amounts of the nutrient above the median requirement to obtain full benefit of the nutrient. The RDA has typically been set at a level 2 s.d. above the EAR (median requirement) to cover the needs of almost all (97.5%) of the group.¹¹

pregnancy and lactation are the same as for women of the same age.

The EARs and RDAs in Table 20.1 reflect the obvious need for higher intakes as body size is increasing and particularly for periods of the greatest tissue accumulation (e.g., adolescence). However, the recommendations for children and adolescents, while reasonable, are not as firmly based as for the adult. Of concern is that mean serum P_i may rise by less than 0.2 mmol/L (0.6 mg/dL) as intake doubles—from 20 to 40 mmol/day (620–1240 mg/day).^{11,60} Serum P_i in this range is generally considered “normal,” and whether the higher end of that range would be more or less beneficial than the lower end is simply unknown.

TABLE 20.1 Dietary reference intakes for phosphorus.^a

Ages (year)	EAR ^b	RDA ^b	UL ^b
1–3	380 (12.3)	460 (14.8)	3000 (96.8)
4–8	405 (13.1)	500 (16.1)	3000 (96.8)
9–18	1055 (34)	1250 (40.3)	4000 (130)
19–50	580 (18.7)	700 (22.6)	4000 (130)
51–70	580 (18.7)	700 (22.6)	4000 (130)
71+	580 (18.7)	700 (22.6)	4000 (130)

^aValues given as mg/day (mmol/day) and apply to both male and female.
^bEAR, Estimated Average Requirement; RDA, Recommended Dietary Allowance; UL, Tolerable Upper Intake Level.
Data from Institute of Medicine.¹¹

Tolerable Upper Intake Levels

The DRI reference value established by the IOM in 1997 to provide guidance regarding potential for adverse effects of overconsumption of a nutrient is the Tolerable Upper Intake Level (UL). The UL established in 1997 for adults in Canada and the United States is 4000 mg (129 mmol)/day.¹¹ Few individuals ever approach that level of intake. The highest 95th percentile intake of phosphorus from the NHANES 2013–2016 cohort is 2504 mg for men 19–30 years of age.⁶⁹ The UL chosen in 1997 is based on the quantity of dietary phosphorus intake that was estimated to raise serum P_i above the normal range as extrapolated from the infusion studies of Bijovet.²⁴

While this was the adverse effect used to determine the UL, other more sensitive indicators identified since 1997 may be more appropriate upon which to base the UL. For example, a study in healthy adults showed that increasing dietary phosphorus consumption from an average of 1100 mg/day to an average of 1700 mg/day, a level far lower than 4000 mg/day, is sufficient to increase circulating FGF23 and osteopontin, a structural protein normally found in bone, which may have adverse effects on bone and cardiovascular health.⁷⁰

Kidney function; The relationship demonstrated in the infusion studies⁶⁰ holds only in adult individuals with adequate renal function, that is, the slow rise of plasma P_i seen with rising phosphorus intake over most of the intake range applies only so long as excess absorbed phosphate can be spilled into the urine. Phosphorus clearance changes little with decreased renal function in adults until

the glomerular filtration rate (GFR) is less than 20% of normal values; at lower levels, plasma P_i rises to higher levels in order to excrete phosphorus levels equal to the amount absorbed.⁷¹ Hence, one of the major features of ESKD is hyperphosphatemia.

Concern has also been expressed about an apparent rise in phosphorus intake in recent years because of population-level increases in phosphorus intake through such sources as cola beverages and phosphorus-based food additives.^{72,73} A high phosphorus diet can produce a slightly higher level of plasma P_i , especially during the absorptive phase after eating.³² Adverse effects on the skeleton may occur when serum P_i is elevated, even when still within the normal range, as well as by elevated PTH levels.^{72,74,75} In the animal studies cited above, increased bony porosity develops.^{36,37} It is known that phosphorus loads in humans lead rapidly to very small decreases in ECF Ca^{2+} as well as increased levels of PTH.⁷² However, these changes revert to normal within 5 days while still on the high-phosphorus intake.⁷⁶ This acute, mild hypocalcemic effect has frequently been ascribed to the formation of calcium phosphate complexes in plasma with a resultant reduction in Ca^{2+} . This in turn would lead to enhanced PTH release. A more plausible scenario is that the fall in plasma Ca^{2+} occurs as the result of inhibition of calcium release from bone due to inhibition of PTH as previously mentioned.⁶⁵

Calcium:phosphorus ratio; Diets low in calcium and high in phosphorus produce increases in PTH.^{74,75} This has been presumed to be harmful for bone, and thus some have advocated a specific dietary calcium:phosphorus ratio. However, diets lower in phosphorus but also low in calcium result in a similar increase in PTH,⁷⁶ so a diet high in phosphorus is not the main culprit of the sustained rise in PTH, but more so the low calcium intake. More to the point, a diet containing 2000 mg (~65 mmol) phosphorus per day consumed by men for 8 weeks resulted in little effect on either calcium balance or calcium absorption compared to a diet of 800 mg (26 mmol) phosphorus.^{77,78}

Feeding diets that were low or high in calcium did not alter the lack of effect on calcium balance or absorption, thus questioning usefulness of prescribing a specific dietary Ca:P ratio in adults. Other studies evaluating the effect on bone turnover using calcium tracers in women where phosphorus intakes were doubled from an intake of 1147–2300 mg (37–74.5 mmol/day, an intake well above the 95th percentile for women in NHANES⁶⁹) on a constant intake of calcium showed little effect on the bone turnover processes measured following 4 months on the of treatment.⁷⁸ This argues against the concern that high phosphorus intakes, within the range currently experienced by the US population, adversely affect bone health in women.

D. Dietary Sources

Because phosphorus is intimately involved in virtually all of the functions and structures of living organisms, phosphorus content of most animal tissues varies little, ranging from 7.75 to 20.5 mg (0.25–0.65 mmol) per gram of protein. Good sources are dairy, meats, grains, legumes, and nuts. The resulting ubiquitous distribution of phosphorus in all natural foods makes it all but impossible to construct a diet that is low in phosphorus and at the same time nutritionally adequate for individuals who need a low phosphorus diet, such as patients with ESKD.

Estimated phosphorus intakes for US adults from the NHANES 2013–2016 cohort by age and sex are shown in Table 20.2.⁶⁹ Median dietary phosphorus intake decreased from 1179 mg/day (38 mmol) for women ages 19–30 years to 1058 mg/day (34 mmol) for those 71 years and older. In men, median dietary phosphorus intake decreased from 1645 mg/day (53 mmol) for men

TABLE 20.2 Usual phosphorus intakes (mg/day) for men and women in the United States (NHANES 2013–2016).⁶⁹

	Percentiles						
Age (y)	5	10	25	50	75	90	95
<i>Boys</i>							
1–3	696	759	878	1025	1185	1342	1448
4–8	849	921	1060	1224	1402	1579	1690
9–13	820 ^a	925 ^a	1120	1364	1639	1921	2100
14–18	933 ^a	1045 ^a	1259	1520	1814	2111	2297
<i>Girls</i>							
1–3	659	721	837	977	1132	1287	1385
4–8	735	802	925	1075	1241	1402	1508
9–13	702 ^a	797 ^a	980 ^a	1207	1465	1722	1891
14–18	614 ^a	707 ^a	876 ^a	1090	1333	1578	1735
<i>Men</i>							
19–30	984	1111	1352	1645	1972	2298	2504
31–50	961	1086	1322	1612	1939	2265	2474
51–70	907	1033	1260	1541	1858	2172	2374
71+	777	887	1092	1360	1650	1948	2143
<i>Women</i>							
19–30	727	816	980	1179	1397	1606	1742
31–50	759	849	1013	1214	1435	1653	1794
51–70	713	800	961	1157	1372	1588	1723
71+	641	722	873	1058	1264	1474	1602

^aBelow their EAR.¹¹

19–30 years to 1360 mg/day (44 mmol) for individuals 71 years and older. For any given age group (e.g., 51–70 years), phosphorus intake more than doubled, from 713 mg/day (23 mmol) at the 5th percentile to 1723 mg (56 mmol) at the 95th. Compared to the EAR for phosphorus for adult men and women of 580 mg (18.7 mmol)/day,¹¹ all adult age groups had fewer than 5% with intakes below the EAR, indicative of adequate dietary intakes in adult age groups. However, due to the much higher EAR values for adolescence, 33% of girls and 11% of boys 14–18 years had intakes less than the EAR.⁶⁹

V. ISSUES RELATED TO PHOSPHORUS IN HUMAN HEALTH

A. Hypophosphatemia

As noted above, hypophosphatemia almost never occurs spontaneously for dietary reasons alone. However, with respiratory alkalosis or diabetic ketoacidosis, or after major catabolic episodes such as following alcoholic bouts or other similar situations, the use of calorie-rich feeding with inattention to phosphorus intake may result in serious hypophosphatemia that at times may be fatal.^{79–81}

The so-called “refeeding syndrome” or “refeeding hypophosphatemia” is a well-recognized problem following gut shut down for any cause.^{82–89} It is generally considered to be due to cellular uptake of P_i as glucose moves into cells replenishing exhausted energy stores and can occur following as little as 48 h without oral feeding, particularly in previously malnourished individuals. Finally, aluminum-containing antacids, by binding dietary phosphorus in the gut, can also produce hypophosphatemia in their own right, as well as aggravating phosphate deficiency related to these other problems.⁹⁰

In infants, severe hypophosphatemia is likely to occur when given parenteral nutrition containing inadequate amounts of phosphorus, when rapid oral refeeding occurs following a period of severe restriction, or when renal phosphorus excretion losses are great due to improper fluid and electrolyte therapy.^{91,92} Severely malnourished infants, often with diarrhea, frequently present with hypokalemia, hypotonia, and hypophosphatemia.⁹³

The spectrum of hypophosphatemia dysfunction is common to other situations unrelated to dietary phosphorus intake.⁶⁶ They include severe vitamin D deficiency rickets, in which the hypophosphatemia is due not so much to insufficient absorption of dietary phosphorus as to high levels of production of PTH (which lowers the TmP and hence plasma P_i). The same is true for renal tubular acidosis, familial hypophosphatemia, Fanconi’s syndrome, or any condition in which

tubular P_i reabsorption is reduced (directly or indirectly) and hence renal clearance of P_i is high.⁶⁶

Examples mentioned earlier of genetic mutations which result in hypophosphatemia include XLH, ADHR, and, in some cases, fibrous dysplasia.⁹⁴ In XLH, the defect is due to an inactivating mutation in an endopeptidase (PHEX), leading to increased synthesis of FGF23 (and hence renal phosphate wasting).⁹⁵ In ADHR, an activating mutation of the FGF23 gene renders FGF23 resistant to proteolytic change.⁹⁵ TIO comes about because of inappropriate secretion of FGF23 (and other phosphatonins) by the neoplastic tissue.⁹⁶

B. Excess Dietary Intake Due to Food Processing and a Western Diet

The already high intake of phosphorus in modern diets has raised concerns about the potential adverse public health impact of phosphorus added to food in the course of food processing. A few decades ago, regulatory agencies—such as the Joint Food and Agriculture Organization/World Health Organization Expert Committee on Food Additives (JECFA) and the Select Committee on Generally Recognized as Safe Substances (SCOGS) of the US Food and Drug Administration (FDA)—commissioned separate studies to assess the safety of phosphorus additives in processed foods. The JECFA report, released in 1964, and the SCOGS report, released in 1975, essentially came to the same conclusion, namely that phosphorus-based food additives posed little threat to consumer safety when used in quantities that “are now current or might reasonably be expected in the future.”⁹⁷ Such approved uses as food additives (affirmed in April 2018) include as an acidulant, a sequestrant, and as a nutrient. As summarized by the 1975 FDA report, “None of the Generally Recognized as Safe phosphates is intrinsically harmful and their use in foods does not present a hazard when the total amount of phosphorus ingested and the intakes of calcium, magnesium, vitamin D and other nutrients are satisfactory.”⁹⁷

Estimates of intake from phosphate additives have varied from 140 to 1000 mg/day.⁵ While it is possible that phosphorus-based food additives are safe for public consumption when used under these conditions, critical limitations in the literature used to support this conclusion should prompt caution.

First, the vast majority of animal studies cited by the reports mentioned above were conducted in the 1960s and 1970s, 20–30 years before the biological basis for a direct link between excess phosphorus and cardiovascular disease (i.e., vascular calcification) was first reported.⁹⁸ As a result, while renal and bone toxicities were carefully evaluated in these studies, the impact of excess phosphorus intake on cardiovascular health was examined in less detail. Moreover, critical hormones

involved in phosphorus homeostasis, most notably FGF23, were unknown in that era. High FGF23 concentrations have been strongly associated with cardiovascular disease, particularly left ventricular hypertrophy.^{99–102} Since FGF23 was not discovered until the beginning of this century,⁴² none of these older studies examined the potential adverse effects of phosphorus additives on FGF23 secretion. Finally, very few of these studies were conducted in humans. This is a critical gap in the literature given that phosphorus toxicology research in animals rarely accounts for food processing conditions such as cooking, which may modify the biochemical properties of food additives.¹⁰³

High dietary phosphorus intake has been associated with increased all-cause mortality. In a study of NHANES participants, higher dietary phosphorus consumption was associated with higher risk of all-cause mortality

independently of potential confounders.¹⁰⁴ Similarly, higher dietary phosphorus consumption has been independently associated with higher left ventricular mass in adult participants of the Multi-Ethnic Study of Atherosclerosis.¹⁰⁵ Finally, higher dietary phosphorus intake has been linked to increased risk of cancer, particularly prostate cancer in men.¹⁰⁶ Notably, however, other studies have shown no association of dietary phosphorus consumption with adverse health outcomes.^{107,108} For all these reasons, the full public health implications of high use of phosphorus additives by the food industry, and possible adverse effects on vascular health, of excess phosphorus intake at levels less than the UL, remain to be further evaluated; it is thus premature to reach the conclusion that the current quantity of phosphorus additives consumed in the modern Westernized diet has negligible effects on human health.

RESEARCH GAPS

Given that the fundamental importance of phosphorus for life is well established, four important areas of the understanding of phosphorus metabolism need elucidation to improve the knowledge of phosphorus in health and disease processes:

- Better understanding of the role of the phosphatonins (and especially FGF23) to include quantitative details of the endocrine feedback loop regulating serum P_i .
- Elucidation of the consequences (if any) of variations in serum P_i ; specifically, is there an optimal value within the nominal “normal” range, and, if so, does this optimum vary with age?
- Development of more effective means of controlling hyperphosphatemia in patients with ESKD.
- Determination of whether high FGF 23 concentrations, produced by high dietary phosphorus intake, exert adverse off-loop effects on various body systems.

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MAGNESIUM

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SUMMARY

Magnesium is an essential nutrient that is involved in practically every major metabolic and biochemical process within the cell. The human body contains about 25 g of magnesium, with about 50%–60% in the bone and the remainder in soft tissue. Magnesium (Mg^{2+}) is a divalent metal ion, the fourth most abundant cation in the body after calcium, potassium, and sodium. Food sources high in magnesium include green, leafy vegetables, unpolished grains, and nuts. Strong evidence suggests magnesium deficiency contributes to cardiovascular disease, the metabolic syndrome, type 2 diabetes, and osteoporosis. Magnesium status is difficult to measure but best defined by dietary intake coupled with serum magnesium concentrations and urinary magnesium excretion.

Keywords: Ca:Mg ratio; Cardiovascular disease; Depression; Diabetes; Magnesium status magnesium transporters; Magnesium supplementation; Osteoporosis.

I. INTRODUCTION

A. Background

Magnesium (Mg^{2+}) is an essential nutrient that is involved in virtually every major cellular metabolic and biochemical process within the cell. It is part of most cellular ATP interactions¹ and is a co-factor for over 600 enzymatic reactions,² including those involving energy production (Krebs cycle, glycolysis), and nucleic acid and protein synthesis. Magnesium is required for cell growth as well as DNA repair, oxidative, immune, and neuromuscular functions, and bone development. It is also necessary for electrolyte balance, and calcium, sodium, and potassium homeostasis, which are all essential for stabilizing excitable membranes.^{3,4} Magnesium acts as a calcium channel antagonist and affects blood vessel dilation/contraction by stimulating production of vasodilator prostacyclins and nitric oxide; it also alters vascular responses to vasoactive agonists.⁵

Throughout the first half of the 20th century, magnesium intakes in Western countries were less than those in Eastern countries.⁶ However, with the spread of processed food, dietary magnesium intakes have declined worldwide.⁷

Almost half (48%) of the U.S. population consumes less than the recommended amount of magnesium from foods, especially teens and those over 70 years of age.⁸ The Dietary Guidelines for Americans has identified magnesium as an under consumed nutrient.⁹

Overt signs of clinical magnesium deficiency have not been routinely recognized in healthy individuals, however, relatively low magnesium intake and/or status has been strongly associated with critical health issues, including hypertension,^{10,11} cardiovascular

disease (CVD),^{12,13} metabolic syndrome^{14–16}; type 2 diabetes (T2DM), and depression.^{17–20} Magnesium deficit has also been associated with poor musculoskeletal health^{21–25} and several neurological and central nervous system conditions.²⁶ Low magnesium intakes coupled with high calcium intakes and high calcium-to-magnesium (Ca:Mg) ratios can increase risk factors for CVD and metabolic syndrome²⁷ by inducing disturbed intracellular calcium ion signaling or toxicity.^{2,28,29}

B. Nutrient Function and Structure

A divalent metal ion, Mg^{2+} is the fourth most abundant cation in the body after calcium, potassium, and sodium³⁰; the most prevalent intracellular divalent cation; and the second most abundant intracellular cation after potassium.^{30,31} Usually bound to ligands, Mg^{2+} forms comparatively stable complexes.^{32,33} Mg^{2+} (55%–70%) represents the physiologically active form of the mineral; however, protein-bound (20%–30%) and chelated magnesium (5%–15%) buffer the ionized pool.³⁴ Magnesium's principal conformational roles are to complex highly charged anions such as adenosine triphosphate (ATP) and nucleic acids, and support enzyme–substrate interactions, and stabilize polymer conformation.^{1,35,36} Magnesium is also needed for cell growth.³⁷

The human body contains about 25 g of magnesium, of which approximately 50%–60% is in the skeleton; two-thirds of this is within the hydration shell and one-third is on the bones' crystal surfaces.^{4,38} Muscle pools comprise 27% while and soft tissues contain 19% with serum contributing less than 1% of total body magnesium.⁴ Approximately 30% of serum magnesium is bound to protein, whereas the majority of the residual portion is ionized and filtered through the kidneys.

Magnesium is required for a number of metabolic reactions including aerobic and anaerobic metabolism, both directly as an enzyme activator and indirectly as part of the magnesium–ATP complex¹; protein synthesis³⁷; and oxidative phosphorylation.^{4,31,39} Magnesium is required for the adequate supply of purines and pyrimidines for RNA and DNA synthesis. In the nucleus, magnesium is involved in DNA stability/repair and regulates the activity of DNA and RNA polymerases.² Within the cell, magnesium regulates ATP binding, enzymatic activity of more than 600 enzymes, cell proliferation, and tRNA and mRNA stability.² Magnesium may directly augment adenylate cyclase activity as well as sodium-potassium-ATPase activity,⁴⁰ which is required for active transport of potassium.^{41,42} Magnesium also plays a significant role in protein kinase activity, particularly enzymes involved in glycolysis, and it plays an important role in bone and mineral homeostasis.

II. NUTRIENT METABOLISM

The body tightly regulates magnesium levels via three main mechanisms: absorption through the gastrointestinal tract; renal excretion after filtration–reabsorption; and exchange from the large pool in bone and muscle² (see Fig. 21.1).⁴³

A. Digestion/Absorption

Magnesium is absorbed throughout the intestinal tract via active transport (transcellular) and passive diffusion (paracellular). Active transport accounts for a greater fractional absorption of magnesium at low dietary intakes, whereas passive diffusion occurs at higher dietary

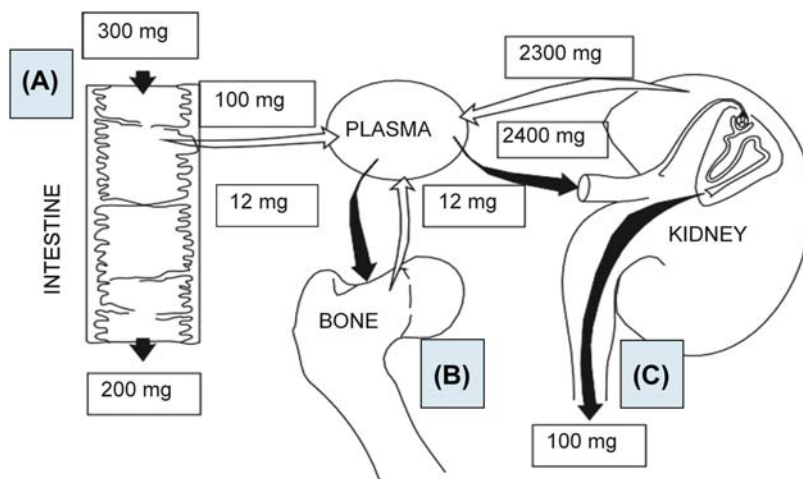


FIGURE 21.1 Magnesium homeostasis in humans. A schematic representation of magnesium metabolism indicating (A) its absorption from the alimentary tract, (B) its distribution to and from bone, and (C) its dependence on the kidney for excretion. Homeostasis depends upon the integrity of intestinal and renal absorptive processes and translocation to and from body stores (mostly bone). Adapted from Rude.⁴³

intakes, resulting in lower fractional absorption.^{4,44} Active transport occurs mostly in the ileum or lower part of the small intestine whereas passive diffusion occurs in the rest of the small intestine. Passive diffusion, which accounts for approximately 90% of intestinal magnesium absorption, depends on a high concentration of magnesium within the jejunum and ileum along with a transcellular potential difference generated by sodium transport.^{4,45} Magnesium absorption in the cecum and colon is thought to be transcellular, mediated by the magnesium transporters transient receptor potential (TRP) melastatin type 6 TRPM6 and TRPM7.²

Magnesium is relatively well absorbed by the gut; oral bioavailability varies from 35% to 70%³ with about a 40%–60% net absorption from food.⁴² In contrast to other minerals, intestinal magnesium absorption is poorly regulated and depends mainly on magnesium intake.^{2,46} When intakes are low, dietary magnesium absorption increases, urinary excretion decreases, and bone reserves are used. When intakes are adequate, the opposite occurs.

A number of factors can decrease magnesium absorption, including protein, fiber, phytates and oxalates. Phytate in high-fiber foods appears to bind magnesium to its phosphate groups, decreasing magnesium absorption.^{47–51} Foods containing non-fermentable fiber such as cellulose and lignins also bind magnesium. In addition, phosphorus, calcium, and protein can decrease magnesium absorption.

Vitamin D and its active metabolites increase intestinal magnesium absorption.⁴ Calcium intakes above normal requirements can decrease magnesium balance with low dietary magnesium intakes.⁵² High dietary phosphorus can decrease magnesium absorption^{4,52} but it can also decrease magnesium excretion,^{50,52,53} leaving magnesium balance unchanged.

Natural bacterial fermentation of carbohydrates generates short-chain fatty acids, producing an acidic environment that promotes magnesium solubility and absorption.^{54,55} In addition, the ingestion of oligosaccharides, such as oligofructose-enriched inulin, increases magnesium absorption, as does a high fructose diet (20% vs. 4% of energy).⁵²

B. Excretion

Approximate 2400 mg of magnesium is filtered by the glomeruli on a daily basis with the nephron recovering 95%–99% of this by coordinated transport processes.² Only 10%–25% of magnesium is reabsorbed via passive diffusion in the proximal tubule, which is associated with calcium, sodium and water transport,^{2,30} with hormonal effects on sodium reabsorption affecting magnesium reabsorption. Unlike most other electrolytes that are absorbed in the proximal convoluted tubule,

approximately 50%–70% of magnesium reabsorption occurs at the thick ascending limb of the loop of Henle via passive diffusion.² Flow through the distal convoluted tubule determines final urinary magnesium concentrations because no absorption takes place beyond this segment. TRPM6 is specifically expressed in this segment and its activity is regulated by intracellular magnesium.^{2,56} When magnesium intakes are excessive, the kidneys excrete high amounts of magnesium, but during magnesium deficiency, the kidneys conserve magnesium by excreting less than 12–24 mg/day.³⁰ Diets high in sodium, calcium, and protein may increase renal magnesium excretion, as does caffeine and alcohol consumption.^{30,57–61}

C. Translocation of Magnesium From Body Stores

Intracellular magnesium concentrations are tightly regulated as they are vital to intracellular functions. When magnesium intakes are low, intracellular magnesium concentrations can be maintained by releasing magnesium from phospholipids in cell membranes.^{2,62}

If dietary magnesium intake remains so low that the body's physiological mechanisms to increase intestinal magnesium absorption and decrease renal excretion are exceeded, resorption of surface bone magnesium stores in the hydroxyapatite form occurs.

D. Transport

Much has been learned in recent years about the molecular identities of the magnesium transporting proteins through genetic screening and microarray-based expression studies. In particular, several proteins are critical to magnesium homeostasis. These include the channel kinase TRPM7, which is ubiquitously expressed and is responsible for much of the magnesium absorption in the colon as well as magnesium flux into the cell.^{63,64} This passive magnesium transport process is further supported by TRPM7's sister channel kinase TRPM6.^{65,66} The TRPM6 channel is the gatekeeper controlling the body's magnesium balance.⁶⁷ Indeed, the two proteins can form heteromeric proteins when co-expressed in a cell,⁶⁸ and the tight up-or down-regulation of TRPM6 expression determines TRPM7 channel activity, thereby influencing the saturable active transport movement of divalent cations from the

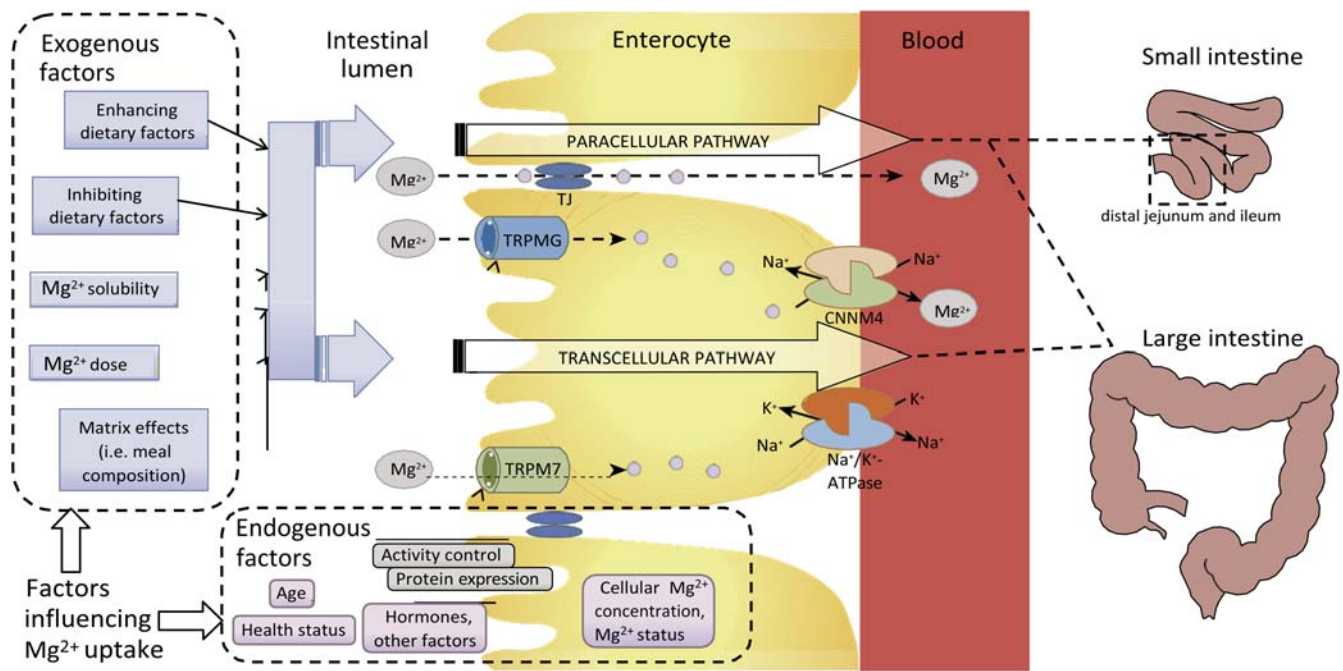


FIGURE 21.2 Intestinal magnesium absorption and influencing factors. The mechanism of Mg^{2+} absorption through the enterocytes into the bloodstream shows a dual kinetic process that involves two mechanisms: a saturable (transcellular) active pathway and a non-saturable (paracellular) passive pathway. The intestinal absorption occurs predominantly in the small intestine—mainly in the distal jejunum and ileum via the paracellular pathway, which is regulated by the paracellular Tight Junctions (TJ). The fine-tuning of Mg^{2+} uptake occurs in the caecum and colon of the large intestine via the transcellular pathway mediated by membrane TRPM6/7 channel proteins and the paracellular pathway. It has been suggested that the basolateral Mg^{2+} extrusion mechanism of the enterocyte is performed via CNNM4, a Na^+/Mg^{2+} -Antiporter. The driving force is a Na^+ -gradient that is established via the Na^+/K^+ -ATPase. The bioavailability and intestinal absorption efficacy of orally ingested Mg^{2+} are influenced by various endogenous and exogenous factors. CNNM4, cyclin M4; TRPM6, transient receptor potential melastatin type 6; TRPM7, transient receptor potential melastatin type 7. Reproduced from Schuchardt and Hahn.⁷⁰

intestinal lumen into the cells^{69,70} (see Fig. 21.2).⁷⁰ For example, magnesium deficiency leads to an increased expression of TRPM6, which in turn may lead to higher expression of heteromeric channel kinase expression in native tissue. Heteromeric TRPM7/TRPM6 channel kinases exhibit a reduced sensitivity to intracellular magnesium, which serves to boost magnesium absorption in the intestine and to reduce excretion through urine in the kidney.⁶⁸ Other proteins that are likely involved in magnesium transport have been described in detail by Kolisek et al.⁷¹ and de Baaij et al.²

E. Physiology: Hormonal Influences on Magnesium Homeostasis

Magnesium absorption, excretion and translocation are influenced by various hormones. Vitamin D and its active metabolites increase intestinal magnesium absorption. Magnesium is also a cofactor for the binding of vitamin D to its transport protein, vitamin D binding protein. Moreover, conversion of vitamin D into the active, hormonal form ($1,25(OH)_2D$) is magnesium-dependent. Both in vitro and in vivo studies indicate that magnesium deficiency affects enzymes that synthesize and

metabolize $25(OH)D$ and $1,25(OH)_2D$ ^{72–74} and alters bone specific binding sites for $1,25(OH)_2D_3$ in magnesium deficiency.⁷⁵

Magnesium deficiency, which leads to reduced $1,25(OH)_2D$ and PTH response^{76,77}; has been implicated in magnesium-dependent vitamin-D-resistant rickets which was reversed by magnesium supplementation.⁷⁸

Estrogen and PTH are involved in renal magnesium excretion.⁷⁹ Estrogen stimulates TRPM6 channel expression,⁷⁹ and TRPM6 expression appears to be regulated by serum magnesium levels (but not by $1,25(OH)_2D$ or PTH).⁸⁰ Hormonal effects on sodium reabsorption in the proximal tubule will also affect magnesium reabsorption because sufficient sodium transport is required to drive water transport, which is a prerequisite for magnesium reabsorption.² PTH can have profound effects on magnesium homeostasis by enhancing magnesium reabsorption in the kidney, absorption in the gut, and release from the bone.⁸¹

Insulin administration causes magnesium to shift from extracellular to intracellular spaces, as indicated by a decline in plasma magnesium levels and a rise in red cell magnesium levels, with minimal alteration in renal magnesium reabsorption.⁸²

F. Insights From Gene Regulation

Serum magnesium concentrations have a heritable component, with heritability estimates of approximately 30%. The Cohorts for Heart and Aging Research in Genomic Epidemiology (CHARGE) consortium evaluated single nucleotide polymorphisms (SNPs) across the genome in association with serum magnesium levels in 15,366 participants of European descent.⁸³ Six different genomic regions were identified, containing variants associated with serum magnesium concentrations. The identified SNPs (including TRPM6) were also associated with clinically defined hypomagnesemia and some with traits linked to serum magnesium levels, including kidney function, fasting glucose, and bone mineral density.

Data suggest that common SNPs in the TRPM6 gene are associated with an increased risk of developing T2DM⁸⁴ because TRPM6 cannot be activated by insulin when these SNPs are present.² In a large prospective study of postmenopausal women, genetic associations between magnesium-related ion channel genes and risk of T2DM were modified by total magnesium intake and ethnicity.⁸⁵ Several SNPs (CLDN19, SLC41A1, and MGMT1) were associated with an increased risk of developing T2DM in African American and Hispanic American women when they consumed less than 250 mg/day of magnesium. However, among African American women with high magnesium intake, the CNNM1 gene was found to have a protective association with T2DM.

III. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

The current United States (US) magnesium Estimated Average Requirements (EARs)³ are based on magnesium balance studies, as are the United Kingdom (UK) EARs,⁸⁶ whereas those established by the WHO/FAO⁸⁷ are based on magnesium balance studies plus usual intakes. The European Food Safety Authority (EFSA) used usual intakes to generate an Adequate Intake (AI) rather than an EAR.⁸⁸ Because high magnesium intake from foods has not been shown to cause adverse effects, the tolerable upper intake level (UL) for magnesium applies only to magnesium from supplemental sources.

Table 21.1 shows the Magnesium Dietary Reference Intakes (DRIs)³ published by the Institute of Medicine (IOM). Recommended Dietary Allowances (RDAs) for magnesium in the US are slightly higher than those in some countries (e.g., The UK has an adult Reference Nutrient Intake (RNI) of 300 mg/day for men and 270 mg/day for women), but in general, healthy adult magnesium requirements range from 220 to 450 mg/day throughout the world.

None of these recommended intakes for healthy humans take into account translocation of body magnesium stores nor the global trend of rising human body weights, so magnesium requirements may need reevaluation to prevent depletion of magnesium stores which can lead to development of chronic latent magnesium deficiency (CLMD).⁸⁹

Recent analysis of USDA metabolic unit magnesium balance studies,⁹⁰ which included mainly white adults, estimated that magnesium requirements may be as low as 250 mg/day for a healthy 70-kg adult. This assessment of magnesium requirement is lower than the IOM's adult RDAs. Compared with other countries' magnesium requirements, the assessment is again lower than adult UK RNIs and EFSA AIs, but comparable to the WHO/FAO RNIs for adults.

B. Dietary Sources

Magnesium is a ubiquitous mineral, widely found in foods. Food sources high in magnesium include green, leafy vegetables, seeds, unpolished grains, legumes, dark chocolate and nuts. Intermediate sources of magnesium include meats, hard cheeses, some fresh and dried fruits, and most seafoods. Refined grains are poor sources of magnesium after removal of the mineral-dense bran and germ.⁹¹

C. Assessment of Status

A simple, rapid, and reliable single measurement to determine the presence of magnesium deficiency is lacking. Each of the three most commonly used indicators of magnesium status (intake, circulating level, urinary excretion) has shortcomings.^{92,93}

Measurement of magnesium body stores via the magnesium load retention test⁹⁴ also has limitations. This test determines the percent of a magnesium load that is retained by the body by measuring the percent of the load that is excreted in the urine within 24–48 h of administration. It is currently the only tool to assess adequacy of body magnesium stores, but it is cumbersome and relies heavily on the reliability and standardization of the 24-h urine measurement. Additionally, this test is costly and impractical for most clinical settings; it is more suitable for research

TABLE 21.1 US Dietary reference intakes for magnesium.

Age and sex	EAR	AI (mg/day)	RDA (mg/day)	UL ^a (mg/day)
<i>Infants (boys and girls)</i>				
0–6 months	NA	30	NA	Unable to establish for magnesium supplementation
7–12 months	NA	75	NA	
<i>Children (boys and girls)</i>				
1–3 years	65	NA	80	65
4–8 years	110	NA	130	110
9–13 years	200	NA	240	350
<i>Males</i>				
14–18 years	340	NA	410	350
19–30 years	330	NA	400	350
31–>70 years	350	NA	420	350
<i>Females</i>				
14–18 years	300	NA	360	350
19–30 years	255	NA	310	350
31–>70 years	265	NA	320	350
<i>Pregnancy</i>				
14–18 years	335	NA	400	350
19–30 years	290	NA	350	350
31–50 years	300	NA	360	350
<i>Lactation</i>				
14–18 years	300	NA	360	350
19–30 years	255	NA	310	350
31–50 years	265	NA	320	350

^aUL only includes intake from supplements, not food and water. AI, adequate intake; EAR, estimated average requirement; NA, not applicable; RDA, recommended dietary allowance; UL, tolerable upper intake levels.

Adapted from Institute of Medicine.³

units.⁸⁹ However, it has been found to be applicable for outpatients as well as reliable in hip replacement and renal transplantation patients.⁹⁴ Table 21.2 summarizes the current methodologies to assess magnesium status.⁸⁹

D. Serum Magnesium

Serum magnesium does not necessarily reflect true total body magnesium content and can remain in the “normal” range when body stores have been depleted. This conclusion is supported by controlled metabolic unit studies in postmenopausal women that found serum magnesium concentrations responded slowly (several weeks) with changes from deficiency to adequacy, or vice versa.^{95,96} Nonetheless, total serum

magnesium concentration is the predominant clinical test used to assess magnesium status. The reference interval used for total serum magnesium concentration was determined by measuring magnesium in 15,820 “healthy individuals” aged 18–74 years in the 1974 in NHANES 1. The central 95th percentile was established as the “normal” range for magnesium, which resulted in a reference interval of 0.75–0.95 mmol/L.⁹⁷ This reference interval likely did not consider CLMD and does not reflect current magnesium content of foods and eating habits. Interpretation of the serum reference range is questionable in individuals with magnesium concentrations between 0.75 and 0.85 mmol/L, who might have CLMD^{39,98} (see Fig. 21.3). The recent Canadian Health Measure Survey Cycle 3, conducted in 2012–13, measured serum magnesium in subjects

TABLE 21.2 Magnesium (Mg) assessment measures and their utility¹

Biomarker	Type	Use	Utility	Factors to be considered	Assessment methods
Dietary Mg	Exposure	Individual Population	Inadequate intake of dietary Mg contributes to etiology of Mg deficiency	Impacted by diets high in phytates, calcium, and phosphorus.	Requires data on intake of phytates, calcium, and possibly phosphorus as well as food preparation methods and extent of consumption of refined grains. Diet history and FFQ and 24-hr recall should be used to align with individuals; 24-hr recall and weighed or estimated diet records should be aligned with populations.
Serum/plasma Mg concentration	Exposure status (deficiency or excess)	Individual Population	Responds to Mg restriction, wasting, or supplementation. Generally not influenced by age or sex	Feasible and inexpensive. Not reflective of total body stores. Impacted by time of day (circadian rhythm), inflammation, poor or compromised renal function, poorly controlled diabetes, drugs (diuretics, cancer therapies; PPIs). Hypoalbuminemia may lead to spuriously low Mg values.	Assay method of choice: Atomic absorption spectrophotometry (AAS). Acceptable methods: Colorimetric is used by the majority of clinical laboratories; enzymatic also acceptable. NIST standard available. Reference range: 0.75–0.95 mmol/L.
24-hr urinary Mg excretion	Exposure status (deficiency or excess)	Individual Population	Responds to Mg restriction, wasting, or supplementation	Considerable subject burden. Susceptible to incomplete collection impacted by time of day (circadian rhythm), inflammation, poor or compromised renal function, and poorly controlled diabetes, drugs (diuretics). In adults, it appears to be an effective biomarker.	Assay method of choice: AAS. In presence of hypomagnesemia, urinary Mg excretion <12 mg (0.5 mmol) is consistent with intact renal response and urinary Mg >24 mg (1 mmol) excretion indicates abnormal renal wasting. Reference range: 3.0–5.0 mmol/24 hr
Fractional excretion of Mg (FEMg)	Exposure status for deficiency	Individual	Used to distinguish between GI and renal Mg losses	Impacted by time of day (circadian rhythm), inflammation, poor or compromised renal function, poorly controlled diabetes, and drugs (diuretics, cancer therapies).	$FEMg = (uMg \times sCr / sMg \times uCr \times 0.7) \times 100$. u=urine, s=serum, Cr = creatinine. FEMg >2% indicates inappropriate renal wasting. FEMg <2% indicates appropriate renal handling.
Red cell Mg concentration	Exposure status (deficiency or excess)	Individual	Responds to Mg restriction, wasting, or supplementation. May be more accurate than serum measures	Care must be taken not to hemolyze blood sample. Unknown if reflective of body stores.	Assay method of choice: AAS. Reference range: 1.65–2.65 mmol/L. Test is not commercially available.
Blood mononuclear cell Mg concentration	Exposure status for deficiency	Individual	Has been used in experimental human Mg deficiency and in patient populations	May be more accurate than serum measures.	Test is not commercially available.

Continued

TABLE 21.2 Magnesium (Mg) assessment measures and their utility¹—cont'd

Biomarker	Type	Use	Utility	Factors to be considered	Assessment methods
Serum-ionized Mg concentration	Exposure (deficiency or excess)	Individual	Responds to severe Mg restriction, wasting, or supplementation	Ion-selective electrode probes susceptible to interference by calcium, lipophilic cations, and thiocyanate in the blood. Unknown if reflective of body stores.	Ion-selective electrodes can measure ionized Mg (70% of total Mg) in serum, plasma, and whole blood. Reference range: 0.55–0.75 mmol/L.
Buccal cell Mg concentration	Exposure (deficiency or excess)	Individual	Relatively new. Few data from randomized trials	Unknown if reflective of body stores.	Assay method yet to be validated. Commercial kits are available.
Mg load retention test	Exposure (deficiency)	Individual	Based on 24-hr urine collection and ability to administer IV Mg sulfate	Invasive and cumbersome. Values correlate well with bone Mg but have yet to be shown to correlate with other Mg measures—such as serum or plasma blood levels.	Standard protocols exist. Based on amount of Mg retained postinfusion of 15 mmol over 4-hour period. <20% retention —normal Mg status. >20 and <50% —borderline deficient Mg status. >50%—deficient Mg status.

¹Table compiled from Witkowski, Hubert, and Mazur, 2011.

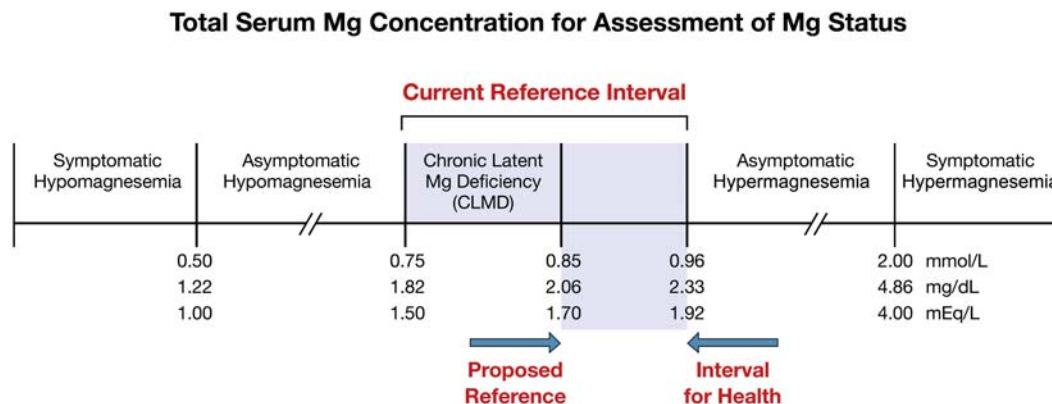


FIGURE 21.3 Total serum Mg concentration for assessment of Mg status. Conversion factor: mg/dL to mmol/L multiply by 0.4114; mmol/L to mg/dL multiply by 2.43; mmol/L to mEq/L divide by 0.5. Reproduced from Costello et al.⁹⁸

aged 3–79 years and found that substantial proportions of the adults (9.5%–16.6%) and adolescents 12–19 years (15.8%–21.8%) had serum magnesium concentrations <0.75 mmol/L,⁹⁹ a level currently accepted as an indication of magnesium deficiency.

E. Dietary Magnesium

Many Americans have mineral intakes that are below amounts recommended by the IOM, and European adult mean and median intakes are lower than bodyweight-corrected RDA values.⁸⁸ Moving forward, it may be prudent to examine the combined effect of several minerals in relation to chronic diseases, such as

the Ca:Mg ratio in cancer,¹⁰⁰ or sodium, potassium, calcium, and magnesium in CVD.¹⁰¹

F. Urinary Magnesium

Measuring 24-h urinary magnesium excretion is useful when combined with other magnesium status indicators, or when evaluating the magnesium status of population groups. The clinical urinary magnesium excretion reference interval is 3.00–5.00 mmol/24 h.¹⁰² Controlled metabolic experiments indicate that 40–80 mg (1.65–3.29 mmol) is the low range of daily magnesium excretion when magnesium intakes are <250 mg/day, and 80–160 mg (3.29–6.58 mmol) is

the daily range when intakes are >250 mg/day.⁹⁶ Similar excretion rates were noted for healthy adults consuming a self-selected diet.¹⁰³ However, these studies found that urinary magnesium excretion can change rapidly (within a few days) when magnesium intakes increase from below to equal or above the current RDA, or vice versa. Analysis of datasets from the Nurses' Health Study I and II and the Health Professionals Follow-up Study found that the population's 24-h urinary excretion of magnesium was stable (intra-class correlation coefficient >0.70) across repeated measurements within 1 year.¹⁰⁴

A circadian effect on magnesium cellular concentration¹⁰⁵ is reflected in magnesium blood levels and has been demonstrated for urinary,^{106,107} serum¹⁰⁶ and plasma ionized magnesium.¹⁰⁸ Plasma levels peak around 10 a.m. and 8 p.m. and vary by about 6% throughout the day. The circadian rhythm is also present after supplementation.¹⁰⁹ This significant within-day variability should be taken into account in clinical and research settings.

G. Basis for Individual Variability in Requirements

In fact, average adult body weights are substantially higher than the reference values used to set intake recommendations.¹¹⁰ For example, *body weight corrected* recalculation of DRIs for adults aged 31–50 years, using average U.S. adult body weights,¹¹⁰ would result in EARs of 336 mg/day (women) and 414 mg/day (men), and RDAs of 406 mg/day (women) and 496 mg/day (men). However, translocation of magnesium body stores and prevention of their depletion is not included in these body-weight corrected calculations.

True magnesium requirements depend on total body weight,⁵² but because most official magnesium intake recommendations are set using healthy reference body weights, magnesium requirements for obese and overweight individuals are actually higher than established values.⁸⁹

Older individuals were once thought to have a greater rate of renal magnesium excretion and a lower rate of intestinal magnesium absorption, exacerbating the consequences of low intakes.³ However, an analysis of human magnesium balance studies indicates that magnesium balance does not decrease with age.⁹⁰ Another consideration for older adults is that many

take medications, such as proton pump inhibitors, that can interact with magnesium and reduce its absorption.³

According to early studies in animals, increased calcium intakes exacerbate magnesium deficiency.⁶ Limited studies in humans also indicate that high calcium intakes decrease magnesium balance or retention,²⁷ especially in individuals with low magnesium intakes and/or status. However, individuals ingesting adequate magnesium may be able to handle high calcium intakes without affecting magnesium retention or balance.^{6,52} Because many calcium channels rely on magnesium, intracellular calcium levels increase with magnesium deficiency.³⁰

High dietary phosphorus stimulates PTH secretion which might increase magnesium intake needs.¹¹¹ Excessive or deficient protein intakes might also increase the need for magnesium, but results are not consistent.⁴² Additional acute and chronic conditions that affect magnesium needs are shown in Fig. 21.4.¹¹²

H. Criteria of Adequacy

This approach circumvents the difficulties of a magnesium load test to determine body magnesium stores. It has been suggested that magnesium deficiency may be defined as a combined serum magnesium <0.85 mmol/L; 24-h urinary magnesium excretion <80.0 mg/day; and magnesium intakes <250 mg/day or <3.57 mg magnesium per kg body weight/day.⁸⁹

Currently, a combined determination of serum magnesium concentration and urinary magnesium excretion along with a dietary magnesium history adjusted for body weight, seems to be the propitious choice for assessing magnesium status.

I. Effects of Excessive Magnesium Intake

High magnesium intake from food shows no evidence of harm; however, excessive supplemental magnesium can have adverse effects.³ The IOM established a UL of 350 mg for supplemental magnesium (1997). EFSA and the German Federal Institute for Risk Assessment (BfR) recommend a maximum daily intake of supplemental magnesium of no more than 250 mg.^{113,114}

The main acute effect of excess magnesium is diarrhea; magnesium has a cathartic effect because of the osmotic activity of unabsorbed magnesium salts.⁴² Nausea and abdominal cramping may also occur.⁴²

Health Risks of Magnesium Depletion

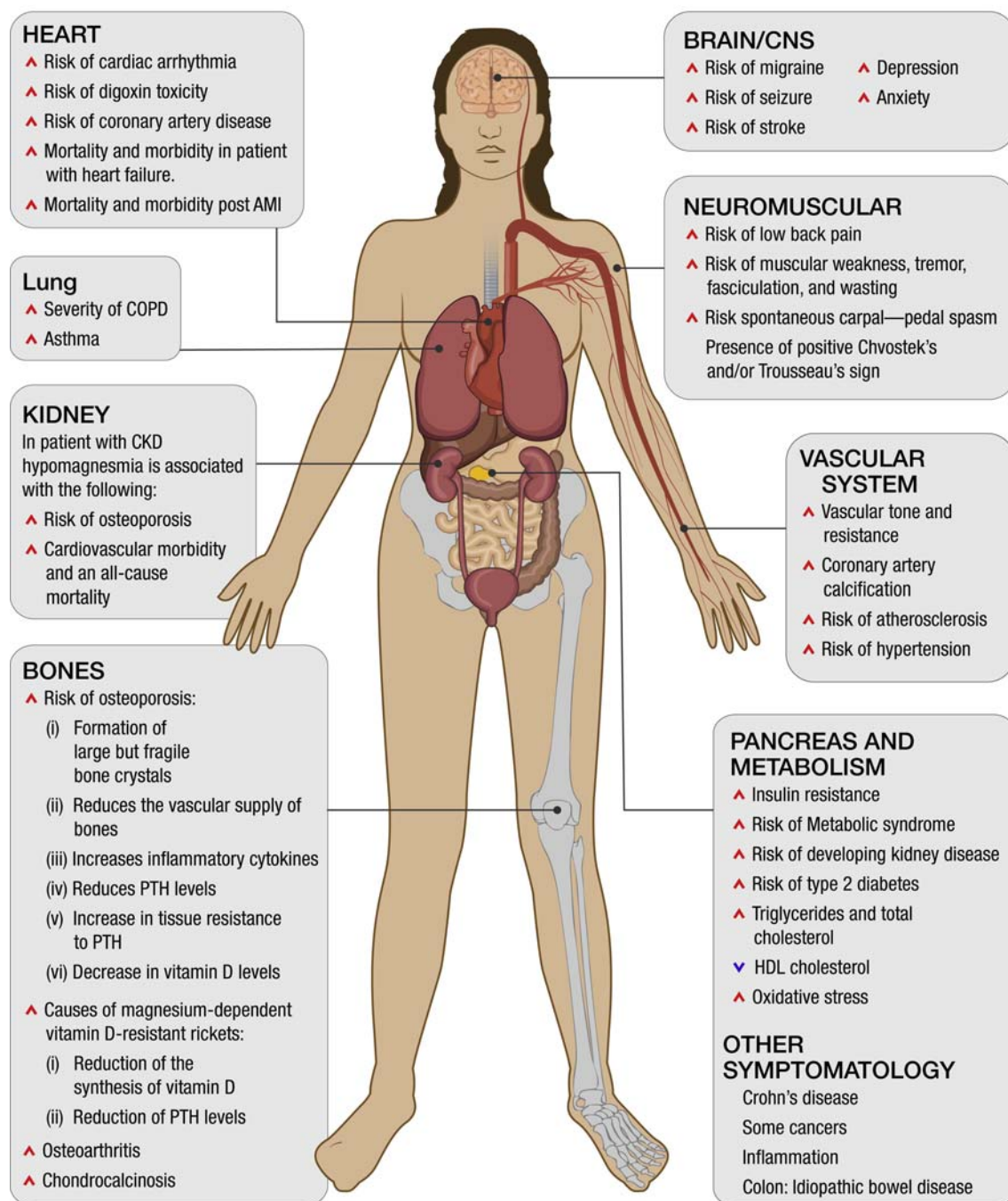


FIGURE 21.4 Health risks of magnesium depletion. AMI, acute myocardial infarction; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; HDL, high density lipoprotein; PTH, parathyroid hormone. Adapted from Al Alawi et al.¹¹²

High serum magnesium concentrations from high supplemental magnesium intake are very rare but can cause renal failure.^{115,116} Very large doses of magnesium-containing laxatives and antacids (typically providing more than 5000 mg/day magnesium) have

been associated with symptoms of magnesium toxicity.¹¹⁷ This toxicity, which can be similar to signs of magnesium deficiency, usually develop after serum concentrations exceed 1.74–2.61 mmol/L, and can include hypotension, nausea, vomiting, facial flushing,

retention of urine, ileus (gastrointestinal atony), and depression. At very high serum magnesium concentrations, above approximately 7.50 mmol/L lethargy can occur before progressing to muscle weakness, difficulty breathing, extreme hypotension, irregular heartbeat, and cardiac arrest. The risk of magnesium toxicity increases with impaired renal function or kidney failure because the body's ability to remove excess magnesium is reduced or lost.^{3,4,118}

IV. ISSUES RELATED TO MAGNESIUM AND HUMAN HEALTH

A. Calcium-to-Magnesium (Ca:Mg) Ratio

Magnesium ion is a physiological antagonist of calcium ion (Ca^{2+}) within cells.¹¹⁹ This makes the cellular Ca:Mg ratio of major importance for Ca^{2+} -dependent signaling events which, when induced, result in a range of downstream cascades. It is possible that cytosolic calcium activation results from a magnesium deficit. Even small changes in magnesium ion concentrations within the cell may cause disturbed Ca^{2+} signaling or Ca^{2+} toxicity.^{2,27,28}

A strong physiological/cellular link exists between rising intracellular Ca:Mg and risk factors for CVD and metabolic syndrome including hypertension, hyperinsulinemia, insulin resistance, and left ventricular cardiac hypertrophy^{29,120} (see Fig. 21.5). High cellular Ca:Mg can also cause inflammation syndrome,¹²¹ and the impact of soft tissue calcification is a growing area of research in the pathogenesis of disease.^{122,123} One theory to explain how excess calcium might antagonize magnesium is that magnesium and calcium share a homeostatic regulating system that involves the calcium sensing receptor. Thus, high intakes of calcium could adversely impact individuals with diets exceedingly low or high in magnesium.¹²⁴

Raised blood Ca:Mg levels have been associated with cancers,^{125,126} and dietary Ca:Mg ratios above 2.7 wt:wt (1.6:1 mol:mol) have been associated with increased risk of colon cancer in humans with certain genetic profiles.^{100,127} High dietary Ca:Mg intake ratios has been noted in cancer studies,¹²⁸ lung cancer in smokers,¹²⁹ and in factors contributing to all-cause mortality.¹³⁰

Traditional recommendations for a healthy dietary Ca:Mg ratio are 2.0 wt:wt (1.2:1 mol:mol),¹³¹ and recent evidence points to a healthy range of 1.7–2.6 wt:wt (1:1 to 1.6:1 mol:mol).^{61,124} Dietary Ca:Mg ratios have increased in the United States over the past 35 years, rising to 3.0 wt:wt in 2000.¹³² There may be a rising global trend in dietary Ca:Mg as human populations move from traditional to modern processed food diets.²⁹

B. Relevant Health Outcomes and Conditions of Concern

Inflammation

Magnesium insufficiency has consistently been shown to increase chronic low-grade inflammation, which is thought to play a major role in chronic disease etiology.¹¹² Many pathological conditions involving low serum and/or dietary magnesium status have been associated with an increased inflammatory response and increased oxidative stress in cell culture, animal models and human studies.^{121,133–136} Magnesium deficiency often occurs in obesity because of an increased need to counteract the inflammatory stress induced by adipose tissue dysfunction.⁵²

Cardiovascular disease (CVD)

A large body of evidence from epidemiological studies, randomized controlled trials (RCTs), and meta-analyses supports inverse associations between magnesium intake and CVD.^{12,137}

Studies have shown that water with high magnesium content (10–40 ppm) benefits heart health.¹³⁸ A recent meta-analysis including 10 of these studies concluded that the concentration of magnesium in drinking water was significantly and inversely associated with coronary heart disease mortality, particularly in the European populations that were studied.¹³⁹

Magnesium is believed to be linked to CVD risk through a broad range of physiological roles that include low circulating levels of magnesium associated with impaired glucose homeostasis and insulin action, dyslipidemia, elevated blood pressure, chronic inflammation, impaired vasomotor tone and increased carotid intima-media thickness, peripheral blood flow, and electrocardiogram abnormalities.^{11,140–143}

Prospective studies in generally healthy populations link low circulating magnesium with higher risk of CVD morbidity and mortality.⁹⁸ Many of these observational studies were included in one or both of two meta-analyses published in 2013.^{12,144} Del Gobbo et al. found that a 0.2 mmol/L increment in circulating magnesium lowered total CVD risk by 30%, risk of ischemic heart disease (IHD) by 17% and risk of fatal IHD (e.g., fatal myocardial infarction) by 39%.¹² Results of a meta-analysis by Qu et al.¹⁴⁴ were similar, although much

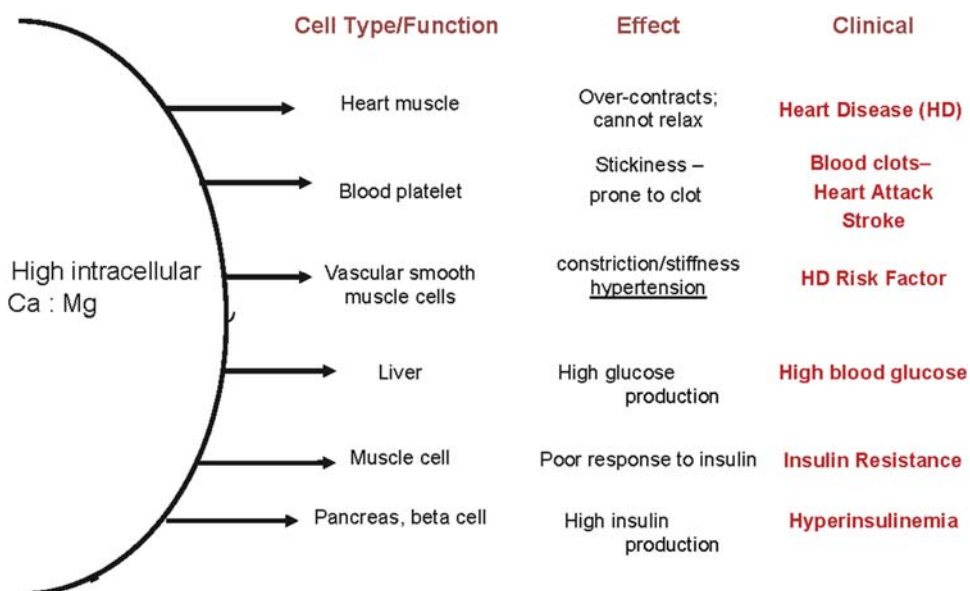


FIGURE 21.5 Cellular effects in humans of high Ca:Mg ratio. Reproduced from Rosanoff et al.²⁹

weaker, with only a 9% reduction in risk of CVD events. Collectively across prospective studies, higher risk of CVD morbidity and mortality tended to begin at circulating magnesium concentrations <0.75–0.85 mmol/L.

The Atherosclerosis Risk in Communities Study (ARIC) is particularly informative when evaluating the optimal serum magnesium concentration for CVD health, because the study includes a large (~16,000) U.S. population-based sample of blacks and whites, and participants have experienced many clinical CVD events over more than 25 years of follow-up. Within ARIC, low serum magnesium has been linked to greater risk of incident CVD-related risk factors, such as hypertension,¹⁴⁵ diabetes,¹⁴⁶ and chronic kidney disease¹⁴⁷ which is a risk factor for CVD.¹⁴⁸ It has also been linked to numerous CVD outcomes, including ischemic heart disease,¹⁴⁹ sudden cardiac death,¹⁵⁰ heart failure,¹⁵¹ atrial fibrillation,¹⁵² and ischemic stroke.¹⁵³ Levels of serum magnesium >0.75 mmol/L appear to be associated with lower CVD risk.

Coronary artery calcification

The Framingham Heart Study has provided data on associations between magnesium intake and coronary calcification. Using multi-Detector Computer Tomography of the heart and abdomen, the investigators found that adjusted self-reported total (dietary and supplemental) magnesium intake was inversely associated with arterial calcification.¹⁵⁴ Dietary magnesium intakes >380 mg/day were also shown to help prevent coronary calcification.¹⁵⁵ Serum magnesium concentrations have also been shown to be negatively associated with coronary artery calcification density in chronic kidney

disease patients.¹²³ Yet to be determined is the precise mechanisms involving magnesium's role in vascular calcification and cardiovascular protection.

Hypertension

In addition to directly affecting vascular turgidity,¹⁵⁶ magnesium is needed to maintain a healthy endothelium which is critical for the regulation of blood pressure through vasoconstriction and vasodilation.¹⁵⁷ Indirectly, magnesium affects blood pressure via interactions with sodium, potassium, and calcium.¹⁵⁸

In the vast majority of prospective and cross-sectional studies, serum, urinary, and dietary magnesium have been inversely associated with hypertension and blood pressure, and numerous clinical trials and their meta-analyses show a clear effect of magnesium status on this important measure of cardiovascular and overall health.^{11,159}

Prospective cohort studies provide the strongest support for a beneficial effect of dietary magnesium on blood pressure. A recent review and meta-analysis of seven prospective cohort studies found a 8% risk reduction of hypertension comparing the lowest intake group with the highest across magnesium intakes ranging from 96 to 425 mg/day.¹⁵⁹ Two studies^{160,161} were supportive, whereas three studies^{162–164} provided suggestive evidence consistent with the premise that dietary magnesium helps reduce the risk of high blood pressure. Only two studies^{145,165} provided data that did not support an effect of dietary magnesium on blood pressure.

Over 40 RCTs examining the effects of magnesium supplements on blood pressure in normotensive, hypertensive, and/or individuals with metabolic syndrome

have been published since 1983 and have been included in six meta-analyses.^{10,11,166–169} The latest meta-analysis, including 34 trials ranging from 3 weeks to 6 months duration,¹¹ shows a significant lowering of both systolic (–2.00 mm Hg) and diastolic (–1.78 mm Hg) blood pressure with either organic or inorganic magnesium supplementation (median dose of 368 mg/day). Two quantitative reviews of all available RCTs^{170,171} have shown a substantial difference in blood pressure response to magnesium supplementation between non-medicated hypertensive subjects and subjects who are taking anti-hypertensive medications: 600 mg/day magnesium is required to elicit a significant reduction in both systolic and diastolic blood pressure in non-medicated hypertensive subjects, whereas only 240 mg/day is needed in medicated hypertensive subjects.

Lastly, although intravenous magnesium sulfate is effective for pre-eclampsia and eclampsia during pregnancy, data from intervention studies are mixed on whether oral magnesium supplementation is beneficial for pregnant women who have normal blood pressure.^{172,173} A recent RCT in pregnant women showed that magnesium supplementation of 300 mg/day starting at week 25 significantly blunted the rise in diastolic blood pressure at 37 weeks compared with placebo.¹⁷⁴

Stroke

High intakes of both magnesium and potassium, but not calcium, significantly reduced the risk of stroke in women in the Nurses Health Study¹⁷⁵ as well as in men in the Health Professionals Follow-up Study.¹⁷⁶ In addition, DASH style diets, which are high in both potassium and magnesium, have been shown to reduce the risk of stroke,^{177,178} and a recent RCT in 291 Taiwanese individuals showed that using a potassium and magnesium-enriched salt for 6 months after hospital discharge improved neurologic performance measures in stroke patients.¹⁷⁹ A meta-analysis¹⁸⁰ of prospective cohort studies found an association between high magnesium intake and reduced risk of both total and ischemic stroke. In addition, Bain et al.,¹⁸¹ found a 41% reduced risk of stroke in men in the highest 30% versus lowest 10% of magnesium intakes in the European Prospective Investigation into Cancer and Nutrition (EPIC) study.

Low serum magnesium has also been linked with acute ischemic stroke. In the ARIC Study¹⁵² and the Framingham Heart Offspring Study,¹⁸² each with 18–20 years of follow-up, lower circulating magnesium (<0.76 and <0.73 mmol/L, respectively) was associated with approximately 30% and 50% higher risk of incident atrial fibrillation which has long been associated with a high risk of stroke.¹⁸³ In another study, stroke patients with lower serum magnesium upon hospital admission showed higher mortality.¹⁸⁴

Heart failure (HF)

Other than sodium reduction, data are sparse on nutritional factors, such as magnesium intake, that may impact the incidence or recurrence of HF. Nonetheless, heart failure has been associated both with low dietary magnesium intakes^{137,185} and low circulating magnesium levels.^{151,186–188}

A longitudinal analysis of incident HF in the Jackson Heart Study cohort of 4916 African Americans free of HF but with other co-morbidities found the median time from baseline to initiation of HF was 1837 days with 270 incident cases reported. Participants in the lowest quartile of magnesium intake (0.52–2.30 mg/day/kg body wt [approx.181 mg/day]) had a 34% increased risk of HF hospitalization compared with those in the remaining quartiles (>2.30 mg/day/kg body wt).¹⁸⁵ A similar analysis of 58,615 healthy Japanese adults aged 40–79 years found that dietary magnesium was inversely associated with HF mortality over a 14.7 year follow-up period.¹³⁷

In the ARIC Study, subjects with low serum magnesium (<0.70 mmol/L) had a 70% higher risk of incident HF than those with higher magnesium levels.¹⁵¹ In the Finish Kuopio Ischemic Heart Study,¹⁸⁷ men with baseline serum magnesium concentrations >0.81 mmol/L had a 13% lower incidence of HF compared with those who had lower concentrations. In the British Regional Heart Study in older men, serum magnesium was inversely associated with the risk of incident HF events after adjustment for cardiovascular risk factors. Risk was shown to decline at levels >0.85 mmol/L.¹⁸⁸

In contrast to these findings, a 2016 meta-analysis of the association between serum magnesium and cardiovascular and all-cause mortality outcomes pooled data from seven prospective studies enrolling 5172 patients with chronic HF.¹⁸⁹ Patients with hypermagnesemia (6 studies, serum magnesium >0.89 mmol/L) were approximately 38% and 35% more likely to die from CVD and all causes, respectively, than those with normal serum magnesium. However, no significant effects of hypomagnesemia (5 studies, serum magnesium <0.74 mmol/L) were observed. The risk of cardiovascular mortality almost doubled when hypermagnesemia ranged between 1.05 and 1.09 mmol/L.

Metabolic syndrome and type 2 diabetes (T2DM)

Magnesium's central role in carbohydrate metabolism has been well described.¹⁹⁰ Insulin receptors, which are part of the tyrosine kinase receptor family, are dependent on the binding of magnesium which allows the transfer of phosphate from ATP to protein. In magnesium-deficient animals, the levels of insulin receptor phosphorylation are decreased in skeletal muscles, thus impairing glucose accumulation in these

tissues.⁵ Inflammatory mediators such as IL-1, IL-6, IL-8, TNF- α , as well as norepinephrine, epinephrine, and reactive oxygen species are also thought to contribute to insulin resistance in magnesium deficiency.

Several recent studies have shown that increased magnesium intakes decrease the risk of developing metabolic syndrome. These include studies using magnesium supplements,^{191–193} a high magnesium diet such as DASH,¹⁹⁴ or a diet supplying magnesium at levels above the EAR from either food or food plus supplements.¹⁹⁵ Meta-analyses also indicate that adequate magnesium intake reduces the risk of metabolic syndrome.^{14–16,196}

Adequate magnesium intakes also reduce the risk of T2DM.^{197–199} An analysis of dietary magnesium intake data from 40 prospective cohort studies across nine countries involving 637,922 subjects with 26,828 incident cases of T2DM (22 studies) found that after multivariate adjustment, the risk of developing diabetes was reduced by 26% per 100 mg/day increment in dietary magnesium intake, comparing lowest to highest intake levels. Moderate heterogeneity was associated with this finding.¹⁹⁹

Like dietary intakes of magnesium, serum and/or plasma magnesium levels also support an association between magnesium and metabolic syndrome,^{16,192,200–203} as well as T2DM.^{204,205}

Serum magnesium levels also appear to correlate with disease progression, with lower levels in subjects with higher numbers of metabolic syndrome measures.²⁰⁶ In addition, patients with the lowest serum magnesium levels have more complications from diabetes than those with higher levels.^{200,204,207} The Canadian Health Measures Survey from 2012 to 2013 provides the most recent support of this relationship.⁹⁹ It measured serum magnesium concentrations in subjects aged 3–79 years and determined that between 9.5% and 16.6% of adult sex-age groups were hypomagnesemic (serum magnesium <0.75 mmol/L). Serum magnesium concentrations were 0.04–0.07 mmol/L lower in adults ($n = 217$) with diabetes than in those without diabetes. In addition, serum magnesium concentration was negatively associated with BMI, serum glucose, serum insulin, HbA1c, and the homeostasis model assessment of insulin resistance (HOMA-IR).

The American Diabetes Association states that there is insufficient evidence to support the routine use of magnesium to improve glycemic control in people

with diabetes and prediabetes.²⁰⁸ It further notes that there is no clear scientific evidence that vitamin and mineral supplementation benefits people with diabetes who do not have underlying nutritional deficiencies. However, two recent systematic reviews and meta-analyses evaluating the effects of magnesium supplementation on insulin sensitivity and glucose control^{196,209} in diabetic and pre-diabetic subjects of varying age and disease status both showed that magnesium supplementation significantly improved the HOMA-IR index as well as either fasting plasma glucose levels or 2 h oral glucose tolerance test levels. These data suggest that oral magnesium supplementation may exert a beneficial effect on insulin sensitivity regardless of metabolic glucose status.

Bone health, fractures, osteoporosis

The degree of mineralization of bone (bone density measured at tissue level) and the characteristics of the mineral deposited (apatite crystals) are major determinants of bone strength.

Magnesium availability can influence the structure of apatite crystals as magnesium ions bind at the surface of the hydroxyapatite crystals, which increases the solubility of phosphorus and calcium hydroxyapatite and affects crystal size and formation.² Crystals deposited in magnesium deficient bone are larger, which can promote bones to be brittle and more prone to fractures (de Baaij et al.² as above) magnesium deficiency is also associated with reductions of PTH and 1,25(OH)₂D levels, and increased inflammation, all of which can lead to bone loss.⁴

Evidence suggests that magnesium is an important factor in bone turnover. Low serum magnesium concentrations inhibit the activity of osteoblasts while promoting that of osteoclasts. Low extracellular magnesium inhibits osteoblast growth by increasing the release of nitric oxide through the upregulation of inducible nitric oxide synthase.²¹⁰ At the same time, it increases the number of osteoclasts generated from bone marrow precursors.²¹¹ Indirectly, magnesium affects the regulation of calcium homeostasis through PTH, 1,25(OH)₂D, and magnesium-dependent calcium channels²¹² and alters inflammatory cytokine release which amplifies osteoclast function while inhibiting osteoblast function.²¹³

Serum magnesium concentrations are closely related to bone metabolism because bone surface magnesium is continuously exchanged with blood magnesium.

Dietary magnesium intakes have been shown to be positively associated with bone mineral density (BMD) in cross-sectional studies assessing dietary magnesium intakes^{24,214–216} as well as in clinical studies of magnesium supplementation.^{217–220} However, this may not translate into lower risk of fracture.

Data on whether magnesium intakes affect fracture risk with age have been mixed. A recent analysis of data from the Women's Health Initiative suggested that lower dietary magnesium intake was not associated with an increased risk of fractures.²² In addition, a recent meta-analysis of 12 cohort and cross-sectional studies found that high magnesium intakes were also not associated with risk of hip or total fracture.²¹ In contrast, an analysis of data from 3765 participants of the U.S. Osteoarthritis Initiative Study over a follow-up period of 8 years showed a significant reduction in fracture risk for individuals with the highest baseline magnesium intakes.²²¹ The effect was greater for women than men, and women meeting the magnesium RDA had a 27% decreased risk for future fractures compared with those who didn't. This study population, however, had a very high prevalence of low magnesium intakes (161 ± 34 mg/day for men and 144 ± 34 mg/day for women in the lowest quintile).

Findings from the UK Biobank Cohort, a cross-sectional study designed to evaluate skeletal muscle and bone health in 156,575 men and women aged 39–72 years, found that higher intakes of dietary magnesium were positively associated with greater grip strength, indices of skeletal muscle mass (fat free mass), and BMD.²⁴ This was seen in both men and women in middle and older age groups across the distribution of magnesium intakes which ranged from a mean of 238 mg/day in the lowest quintile to 532 mg/day in the highest quintile. These results suggest that dietary magnesium has a role in musculoskeletal health and could have relevance for population prevention strategies not only for fractures but also sarcopenia, frailty, and falls.

Data from the Kuopio Ischemic Heart Disease prospective cohort study demonstrated that low serum magnesium levels (<0.74 mmol/L) were independently associated with an increased risk of total and femoral fracture over a median follow-up of 25.6 years. The investigators also observed a continuous decrease in risk of femoral fractures with increasing levels of serum magnesium. However, they found no evidence of an association between dietary magnesium and the risk of any fractures.²³

Male, but not female, participants enrolled in the EPIC-Norfolk study showed significant inverse trends in fracture risk across serum magnesium concentration groups for spine fractures, and the composite of total hip, spine, and wrist fractures over a mean follow-up

of 13.4 years. At baseline, mean serum magnesium concentration for males was 0.81 ± 0.12 mmol/L.²²²

Red blood cell magnesium levels have been shown to be lower in postmenopausal women with osteoporosis versus healthy controls.²²³ In addition, many postmenopausal women with osteoporosis who are vitamin D deficient and have low PTH levels are also magnesium deficient, and magnesium supplementation has been shown to correct these biochemical abnormalities.²²⁴

Osteoarthritis (OA)

Dietary magnesium intake has been shown to be inversely associated with radiographic knee OA and joint space narrowing, but not with osteophytes (bony growths or deposits) in 1626 healthy Chinese adults.²²⁵ Among 2112 participants from the Johnston County Osteoarthritis Project, the prevalence of knee OA was 36% and the relationship between dietary magnesium intake and radiographic knee OA was modified by race.²²⁶ An inverse association starting at 186 mg/day of dietary magnesium was observed among whites but not among African Americans. In another study of 2548 participants in the Osteoarthritis Initiative cohort, lower magnesium intakes were associated with worse knee OA pain and function scores throughout the 48 months of study among participants with baseline radiographic knee OA and magnesium intakes below the EAR (68% of the men, 44% of the women) compared to those with higher intakes.²²⁷

To evaluate the relationship between serum magnesium concentrations and radiographic knee OA investigators studied 2855 participants from Hunan Province, China with a prevalence of OA was 30%.²²⁸ A significant inverse association between serum magnesium concentration and radiographic knee OA was observed after adjustment for age, sex, and BMI. Knee OA was reduced by 28% in the highest serum magnesium quartile (median 1.00 mmol/L) compared with the lowest quartile (median 0.84 mmol/L).

Chondrocalcinosis

Chondrocalcinosis, which is characterized by a deposition of calcium pyrophosphate crystals in joints, is a known complication of a hereditary disorder called Gitelman syndrome.²²⁹ Hypomagnesemia may promote the formation of calcium pyrophosphate crystals in joints because magnesium ions increase the solubility of calcium pyrophosphate crystals and are important activators for alkaline phosphatase, which hydrolyzes pyrophosphates into inorganic phosphate.²³⁰ Therefore, magnesium supplementation is the cornerstone for the prevention of chondrocalcinosis in individuals with Gitelman syndrome.

The association between knee chondrocalcinosis and serum magnesium was evaluated by pooling the data from two Chinese population-based cross-sectional studies in 13,947 adults living in Hunan province, China.²⁵ The results showed that subjects with lower levels of serum magnesium, even within the normal range, had a higher prevalence of knee chondrocalcinosis as measured by bilateral knee radiographs, in a dose-response manner. While this study may not be generalizable to other population groups outside of China and is limited by the absence of dietary intake data, it does suggest future research in this area is warranted.

Depression

Magnesium affects the function of the hypothalamic-pituitary-adrenal axis,²³¹ and dysregulation of this axis increases levels of catecholamines and cortisol, the stress hormones associated with depressive symptoms. Magnesium can also moderate the *N*-methyl-D-aspartate (NMDA) receptor whose dysfunction is a major cause of depression pathology.^{2,232,233} Inhibitory gamma-amino butyric acid receptors further amplify this process and are also regulated by magnesium.²

Animal studies show that a magnesium-deficient diet enhances depression and anxiety-related behavior in mice.²³⁴ Magnesium supplementation induces antidepressant and anxiolytic-like effects in mice and has been shown to improve post-traumatic depression and anxiety in rats.^{20,233,235}

Dietary magnesium intake and the risk of depression was evaluated in the cross-sectional analysis of data from 17,730 adults who participated in the 2007–14 NHANES surveys.²³⁶ Dietary magnesium (diet alone and diet plus supplemental intake) was inversely associated with the risk of depression, even after adjustment for important confounding factors including energy intake and BMI. The multivariate adjusted odds ratio for depression for the highest quartile (>347 mg/day) versus lowest quartile (<200 mg/day) of dietary magnesium intake was 0.47. In subgroup analysis, this statistically significant inverse association was observed among women but not men. These NHANES data are in agreement with data from 5708 community dwelling healthy adults participating in the Hordaland Health Study in Australia²³⁷ that showed a significant inverse relationship between

magnesium intake and depressive symptoms. Miki et al.,²³⁸ reported data from the Furukawa Nutrition and Heart study and also found that higher dietary magnesium intake among 2006 adults aged 19–69 years from Japan was associated with a 37% decreased prevalence of depression.

Derom et al.,²³⁹ however, found no association between magnesium intake and depression in a cohort of 12,939 Spanish university graduates who were followed for a mean of 5.8 years, possibly due to the high magnesium intake of the study population (median intake, 404 mg/day). Similarly, no conclusive evidence for an association between generally high dietary magnesium intake and the incidence of depression was found in 15,836 participants in the SUN Mediterranean cohort after 9.3 years of follow-up (mean magnesium intake of highest quintile, 597 mg/day).²⁴⁰

Li et al.¹⁷ conducted a meta-analysis that pooled data from 11 cohort and cross-sectional studies from various countries, some of which were described above. Comparing highest versus lowest magnesium intakes showed a reduction in risk of depression of 19% (with moderate heterogeneity), and dose response analysis found a non-linear association with the largest risk reductions observed at intakes of 320 mg/day. A prior systematic review on the topic was inconclusive.²⁰ Phelan et al.²⁴¹ pooled data from 62 effect-size estimates and found higher magnesium levels in bodily fluids (serum, red blood cell, plasma, whole blood, cerebrospinal fluid, and/or urine) in treated patients with a mood disorder compared with untreated patients or healthy controls. They also found that adherence to a magnesium-rich diet in cross-sectional (but not prospective) studies was associated with a 34% decrease in risk of depression, although study design flaws were present. In contrast to this review, a pooled analysis of 11 studies by You et al. found decreased levels of serum magnesium in patients with depression (many of whom were inpatients) compared with controls.²⁴²

In a cross-sectional study conducted in Iran, which was not included in the review by You et al.,²⁴² the prevalence of hypomagnesemia (<1.80 mg/dL; 0.74 mmol/L) in 650 subjects diagnosed with depression was 13.7%, significantly higher than the prevalence of hypomagnesemia in non-depressed individuals. In addition, the authors found a significant relationship between hypomagnesemia and intensity of depression.²⁴³ In a subsequent RCT by the same group of investigators,²⁴⁴ 60 depressed subjects with hypomagnesemia were randomized to receive 500 mg magnesium oxide or placebo for 8 weeks. Along with the repletion of serum magnesium in most subjects receiving magnesium, the mean Beck Depression Inventory score declined significantly in these subjects compared

Magnesium affects several biological pathways associated with neuromuscular and psychiatric symptoms.

with those receiving the placebo, i.e., -15.65 ± 8.92 versus -10.40 ± 7.90 , respectively.

Stress and anxiety

In a recent RCT, 268 healthy adults with low serum magnesium (mean 0.80 mmol/L), were given either 300 mg magnesium or 300 mg magnesium plus 30 mg of vitamin B6 daily for 8 weeks for stress reduction.²⁴⁵ Both magnesium alone and magnesium plus vitamin B6 reduced stress, as determined by the Depression Anxiety Stress Scales (DASS-42), by approximately 40% among all subjects. In those with severe or extremely severe stress and low serum magnesium concentrations, the magnesium-vitamin B6 combination provided a 24% greater reduction in stress than magnesium alone at week eight.

Migraine headaches

Magnesium deficiency is related to factors that promote headaches, including neurotransmitter release and vasoconstriction.²⁴⁶

In addition, people who experience migraine headaches have lower levels of serum and tissue magnesium than those who do not.^{247–249}

Intravenous magnesium sulfate is often used to treat migraine headaches and the Canadian Headache Society gives a weak recommendation against its use to treat migraine pain in emergency settings based on moderate quality of evidence.²⁵⁰ Measuring serum ionized Mg^{2+} levels may be useful in identifying patients with migraine headaches who may respond to an intravenous infusion of magnesium sulfate.²⁵¹

Oral magnesium preparations have been deemed “probably effective” for migraine prevention by the American Academy of Neurology and the American Headache Society.²⁵² Chiu et al.²⁵³ pooled eight studies with 789 participants to determine the effects of oral magnesium on the frequency of migraine. The doses and formulations varied widely. Oral magnesium significantly alleviated the frequency and intensity of migraine, but with high heterogeneity across studies. In addition, it was difficult to determine whether magnesium was responsible for the beneficial effects as some of the included studies did not use adequate randomization methods and because some of the oral magnesium preparations were administered in combination with other therapies.

In a meta-analysis by Okoli et al. of three RCTs in 226 adults, children, and adolescents, oral magnesium

therapy did not significantly decrease migraine severity compared with placebo.²⁵⁴ Although the data on migraine frequency (4 trials, 226 participants) could not be pooled due to high heterogeneity, the investigators found a reduction in migraine frequency with magnesium therapy. Overall, the authors concluded the evidence was insufficient to determine whether magnesium is efficacious for the treatment of migraine headaches.

A systematic review of five RCTs with 253 participants that evaluated oral magnesium for migraine prophylaxis in adults 18–65 years of age concluded that magnesium was “possibly effective” (Grade C evidence level) for the prevention of migraine.²⁵⁵ Lastly, an umbrella review of meta-analyses for multiple health outcomes found a 62% reduction in the frequency of migraine events (3 RCTs, 186 participants) and a 59% reduction in the intensity of migraine (2 RCTs; 146 participants) with magnesium supplementation.²⁵⁶

C. Interactions With Medications

A number of prescription medications can affect magnesium status either by increasing or decreasing magnesium absorption, excretion, or translocation.^{2,82,257,258} Among these drug categories are diuretics (loop, thiazide and potassium sparing), hypoglycemics including insulin, anti-epidermal growth factor receptor modulators, antimicrobials, aminoglycoside antibiotics, cardiac glycosides, proton-pump inhibitors, bisphosphonates and estrogen. Monitoring of serum or urinary magnesium concentrations may be warranted as well as increased consumption of dietary magnesium.

D. Magnesium Supplementation Practices

In the United States, dietary supplement use is widespread, with over half of adults reporting use, and older adults reporting the highest use.^{259,260} Chen et al.²⁶¹ pooled data from 30,899 adults aged 20 years or older who participated in six cycles of NHANES (1999–2000 to 2009–2010) and provided information on dietary supplement use. Of the 24,763 dietary supplement users, 33.3% used magnesium supplements, with a mean dose of 146.8 mg/day. Although the use of magnesium supplements decreased the number of adults with intakes below the magnesium EAR, 49.7% of those surveyed still remained below the EAR. Another NHANES analysis for survey years 2009–12 showed a 10%–19% decrease in the prevalence of magnesium inadequacy (below the EAR) with supplement use.²⁶²

Magnesium supplements are available in a variety of salt forms, including magnesium oxide, citrate, chloride, and others.²⁶³ The Supplement Facts panel on a dietary

supplement label declares the amount of *elemental* magnesium in the product, not the weight of the entire magnesium-containing compound.

The water solubility of a magnesium salt affects its bioavailability. A number of small studies evaluating the bioavailability of magnesium supplements indicate that organic salts, such as magnesium aspartate, citrate, and lactate have a higher solubility than inorganic salts such as magnesium oxide, and sulfate.²⁶⁴ However, a meta-analysis by Zhang et al.¹¹ found that studies using

inorganic magnesium forms lowered high blood pressure more effectively than organic forms.

Magnesium is a primary ingredient in some laxatives. Phillips' Milk of Magnesia, for example, provides 500 mg elemental magnesium (as magnesium hydroxide) per tablespoon. Magnesium is also included in some remedies for heartburn and upset stomach due to acid indigestion. Extra-strength Rolaids, for example, provides 55 mg elemental magnesium (as magnesium hydroxide) per tablet.

RESEARCH GAPS

- The IOM recommended intake levels for healthy humans do not take into account translocation of body magnesium stores. Reevaluation may be needed to set magnesium requirements at levels that prevent depletion of magnesium stores and reduce the risk of developing chronic latent magnesium deficiency (CLMD).
- A revision of the serum reference interval reflective of health and subclinical disease is needed. The lack of a standardized laboratory test that accurately reflects magnesium status remains one of the most vexing challenges associated with the magnesium field.
- Updated food composition tables that reflect analytically-derived magnesium content of the current food supply are needed.
- Moving forward, it may be more prudent to examine the combined effect of several minerals in relation to chronic diseases, such as the Ca:Mg ratio in relation to cancer, or sodium, potassium, calcium, and magnesium in cardiovascular disease (CVD).

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IRON

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SUMMARY

Iron is essential for acquiring, distributing, storing, and using oxygen. Over 80% of systemic iron is involved in acquiring and delivering oxygen (hemoglobin) or storing it (myoglobin), and the rest is in cellular depots (ferritin) or is used to support and control the systemic use of oxygen for energy creation and expenditure; synthesis of tissues, hormones, and active vitamins; substrate metabolism; the breakdown of xenobiotics; innate immunity; and antioxidant activity. Since iron has such diverse and fundamental roles, the effects of iron deficiency are protean, subtle, and difficult to distinguish from other nutrient deficiencies; however, anemia and compromised physical performance, psychomotor development and function, and immune function are important consequences. However, the human health effects of preanaemic or nonanemic iron deficiency, particularly in early life, are not well characterized. The systemic trafficking and use of iron exploit its interchange between ferrous and ferric states and protect tissues from the oxidative damage that ionic iron can cause. This protection involves minimizing the entry of iron into the body by downregulating its absorption. The body has no means for excreting systemic iron, and it maintains its iron content by recycling endogenous iron and compensating basal iron losses by mobilizing from its ferritin depots and by derepressing intestinal absorption of iron. The systemic response to hypoxia involves, among other things, adaptation in the cellular and systemic turnover of iron involving upregulation of its absorption. In contrast, in response to inflammation, iron turnover is immobilized, iron absorption is further down regulated, and the metal is retained and sequestered in tissues. This results in a functional deficiency known as an anemia of inflammation. Such adaptations of iron turnover and those that occur in growth and reproduction make it difficult to assess iron deficiency and the efficiency of iron absorption from foods and to set dietary reference values. The causes of iron deficiency and the population groups at risk are the same as for other nutrients. Iron deficiency is likely to be accompanied by other deficiencies, and some defects may have been inappropriately attributed to iron deficiency. In a population context, anemia has often been seen solely as an iron deficiency anemia, and interventions have been introduced to increase iron intakes. However, these anemias may be due to inflammation, the causes of which, e.g., systemic and intestinal parasitic and bacterial infections, may also be causing systemic plasma and blood loss. International agencies' reports have emphasized the need for preventive as well as therapeutic and strategies to reduce the prevalence of iron deficiency and to eradicate its socioeconomic sequelae, while warning that iron supplementation risks precipitating or exacerbating infections.

Keywords: Anemia; Ferritin; Hemoglobin; Iron; Iron deficiency; Myoglobin.

I. INTRODUCTION

A. Background

Iron (atomic weight 55.85, atomic number 26) has four naturally occurring isotopes. These are, with their

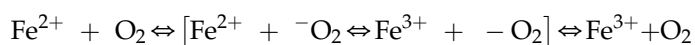
abundances, ^{56}Fe (92%) and ^{54}Fe (5.8%) and, two low abundance stable isotopes, ^{57}Fe (2.12%) and ^{58}Fe (0.28%). The low abundance isotopes along with the radioisotope ^{59}Fe (beta decay with a half-life of 44.5 days) are used as tracers to enable biological kinetic and functional studies of the metal. There are 24 radioactive iron isotopes, but these

are of most interest to geo and cosmic sciences. Iron comprises 6.2% of the Earth's crust where it is the fourth most abundant element, after oxygen, silicon, and aluminum. Iron has oxidation states ranging from -2 to $+6$. The feature underpinning iron's biological relevance is its ability to alternate between the ferrous (Fe^{2+}) and ferric (Fe^{3+}), respectively, by losing and gaining an electron. Iron's biological role further depends on the nature of the complexes it forms with sulfur, nitrogen, and oxygen. With sulfur, it forms four coordinated complexes with four or fewer sulfur atoms to form iron–sulfur clusters (ISCs), and with N it forms six-coordinated complexes within a porphyrin ring thereby forming heme.^{1,2}

ISCs were the predominant iron compounds supporting metabolism in early life forms 3,000 million years ago (mya) when the atmosphere was acidic, sulfurous, and anoxic. At that time, oxygen was bound in carbon dioxide and water in the planet's crust and water. In such conditions, iron was in the ferrous state (Fe^{2+}); it was water soluble and easily acquired by the early anaerobic life forms that used sulfur in ISCs as an H acceptor for electron transfer in generating energy. About 2500 mya, photosynthesis emerged; this metabolized CO_2 and water and released oxygen into the atmosphere, which became less acidic resulting, around 2000 mya, in crustal ferrous iron being oxidized to the less water-soluble ferric state. This is evidenced by the appearance at that time of red geochemical beds of ferric iron salts and the primary sources of the metal, namely magnetite (Fe_3O_4), hematite (Fe_2O_3), limonite ($\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$), and siderite (FeCO_3). Some sulfur deposits remain, for example, in iron pyrites (FeS_2).

Thus, while early life forms were able to acquire ferrous iron to support their metabolism, their evolutionary descendants had to adapt to the lower availability of environmental iron. Consequently, anaerobic microbes and fungi, and possibly some plants, developed low molecular weight, high-affinity iron-chelating compounds (siderophores) to acquire ferric iron, solubilize it, and translocate it intracellularly, where it was reduced to ferrous iron for metabolic use. Thus, siderophores were and still are fundamental to enabling iron to enter the food chain, and their iron chelating activity is an important virulence factor contributing to the pathogenicity of microorganisms.^{3,4}

The present 20% oxygen in the atmosphere came 580 million years ago. Aerobic life forms have developed systems to acquire and facilitate the metabolic use of oxygen for oxidation and energy generation, while simultaneously avoiding damaging exposure to oxidative species (i.e., hydroxyl $\cdot\text{OH}$ and OH^- and superoxide O_2^-) created during the oxidation of ferrous to ferric iron in reactions such as the Haber–Weiss and Fenton reactions:



The oxidative radicals, many of which are produced by iron-dependent enzyme activities, damage proteins, nucleic acids, lipids, and carbohydrates and generate other reactive species involving sulfur and nitrogen. Organisms have evolved extracellular transporters and intracellular carriers (metallochaperones) that bind iron, thereby minimizing the oxidative risk, and deliver it to its functional sites. Simultaneously, these overcome problems arising from the poor solubility of iron, particularly ferric iron, at physiological pH. Overall, these mechanisms achieve an intracellular concentration of iron at 10^{-4} M, which is higher and more metabolically facilitative than the aqueous solubility of ferric iron (10^{-18} M); simultaneously, the intracellular free iron is restricted to 10^{-24} M, thereby reducing the hazard of systemic and cellular oxidative damage.

A general background for this chapter is provided by a review of the early understanding of the metabolism of iron⁵ and by reviews by national⁶ and international committees^{7,8} on how current insights can be applied in public health.

B. Iron Function and Structure

The biological functions of iron are achieved via four categories of metalloproteins listed in [Table 22.1](#).

ISCs operate in cellular regulatory assemblies in the mitochondria, cytoplasm, and nucleus.^{9,10} Their roles in the fundamental control of the homeostasis of iron and oxygen are being increasingly recognized. They function as catalysts and participate in electron transfer. They also participate in the regulation of gene expression, DNA synthesis, repair and configuration, and in ribosomal assembly protein translation. Dysfunction of ISCs is associated with hematological and neurological defects.

Heme is synthesized in the mitochondria and cytoplasm and is completed in mitochondria where ferrochelatase inserts Fe^{2+} iron into protoporphyrin IX. Since heme has limited water solubility and is an oxidant, it needs chaperones to export it from the mitochondria to the cytosol where most of it is incorporated with a globin molecule to form a globin–heme such as hemoglobin, or the heme is conveyed to other cytosolic sites to produce heme enzymes. The protein moieties in these complexes control iron's potential toxicity and its interactions in electron transfer and with oxygen.²

There are many globin–hemes in vertebrates and invertebrates. They have diverse functions including transporting oxygen, carbon dioxide, and carbon monoxide; decomposing nitric oxide (e.g., hemoglobin and neuroglobin), storing oxygen (e.g., myoglobin and neuroglobin), and scavenging free radicals (cytoglobin). However, the precise physiological roles of many heme–globin complexes are uncertain.^{11,12} This might even apply to hemoglobin whose major function is discussed below, but whose presence in tissues other than erythrocytes excites interest in what else it might be doing.

TABLE 22.1 Iron-dependent activities.

Iron-dependent activities	Type of function and comment
Nonheme non-iron-sulfur oxygenases	Substrate attacked by oxygen
Phenylalanine hydroxylase	Phenylalanine to tyrosine
Tyrosine hydroxylase	Tyrosine to dopaminergic acid
Prolyl and lysyl hydroxylase	Regulation of hypoxia-inducible factors, Procollagen and collagen synthesis
Ribonucleotide reductase	Ribo- to deoxyribonucleotide; DNA metabolism
Cysteine oxygenase	Cysteine metabolism
Prostaglandin synthetase	Arachidonic acid to prostaglandin
Iron-sulfur proteins	Iron-sulfur proteins in iron-sulfur clusters involved with mitochondrial substrate metabolism TCA cycle. Metallochaperones, conformational influencers of proteins, and DNA in cells and mitochondria, and in initial responses to iron depletion and hypoxia
Dehydrogenases: e.g., isocitrate, NADPH, succinate	Citric acid cycle in mitochondria, serial oxidation producing NADPH, ATP, GTP, and substrates for synthesis of amino acids, lipogenesis, and glycogenesis
Aconitase and iron-sulfur-sensing cluster	Aconitase is also involved with regulation of iron kinetics, cell signaling, and heme synthesis
Xanthine oxidase	Purine metabolism. hypoxanthine to xanthine and xanthine to uric acid. Produces reactive oxygen and nitrogen species
Heme proteins	Several variants of heme involved
Cytochromes a, b, c Cytochrome c oxidase	Electron transfer chain in mitochondria and its inner membrane; oxidizing NADPH from citric acid cycle to drive synthesis of ATP (i.e., oxidative phosphorylation).

Continued

TABLE 22.1 Iron-dependent activities.—cont'd

Iron-dependent activities	Type of function and comment
Cytochrome P450 monooxygenases	Microsomal. Cytochrome P450 oxidases have over 11,000 activities, including the metabolism of endogenous substrates including organic acids, fatty acids, prostaglandins, steroids, and sterols including cholesterol, and vitamins A, D, and K and hormones. Synthesis of steroid hormones, phase I metabolism of xenobiotics, involving a diverse range of metabolic pathways and substrates
Fatty acid desaturases	Long-chain fatty acid synthesis: myelination by oligodendrocytes
Peroxidases	Numerous substrates (e.g., iodide to iodate)
Myeloperoxidase	Neutrophil bactericide
Catalase	$2\text{H}_2\text{O}_2$ to $2\text{H}_2\text{O} + \text{O}_2$
Tryptophan-2,3-dioxygenase	Neurotransmitters, serotonin synthesis, 5-hydroxyindoleacetic acid, niacin, and nicotinamide metabolism
Reductases sulfite), nitrite metabolism (sulfite oxidase)	Sulfite and nitrate/nitrite metabolism
Duodenal cytochrome b ferric reductase	Fe^{3+} to Fe^{2+} on enterocytes brush border
Heme and globin	Heme is embedded in a globin, the nature and conformation of which enables iron to bind dioxygen reversibly
Hemoglobin	Transport O_2 , CO_2 , NO. Hemoglobin is also found in the lung alveolar cells, substantia nigra, macrophages, retina pigment cells, hepatocytes, and other epithelial cells, possible antioxidant One gram of hemoglobin contains 3.47 mg of iron
Myoglobin	A monomeric store of oxygen in muscle. It functions analogously to hemoglobin, but the heme-globin complex is adapted to function in the ambient pO_2 , pH, and pCO_2 of muscle

Continued

TABLE 22.1 Iron-dependent activities.—cont'd

Iron-dependent activities	Type of function and comment
Neuroglobin	Exact function is unknown 20%–25% sequence homology with Hb. Stores dioxygen in CNS but has poor affinity for oxygen and Hb; 94% homology with murine ngb and possibly neuroprotective.

5HIAA, 5 hydroxyindoleacetic acid; ATP, adenosine triphosphate; CNS, central nervous system; Dcytb, duodenal cytochrome b; GTP, guanosine prep triphosphate; NADPH, nicotinamide adenine dinucleotide phosphate.

Hemoglobin is a tetramer comprising two pairs of α - and β -globin–heme units, each of which binds a molecule of oxygen.¹¹ A molecule of hemoglobin transports four oxygen molecules (amounting to 1.34 mL of oxygen per gram of hemoglobin), and as a result, blood carries 50–70 times more oxygen than would plasma alone. Hemoglobin binds and releases oxygen, forming oxyhemoglobin and deoxyhemoglobin respectively: the binding depends on the conformation of the globin molecule and on a link between the heme and a histidine in the globin chain. This interaction controls the spin state of the ferrous iron in the heme and enables it to bind oxygen without becoming oxidized (in contrast to what happens in heme enzymes).² The iron is in a low-spin state at the pulmonary alveolar–capillary interface, where, relative to the peripheral tissues, the pO_2 is high, the pCO_2 is low, and the pH is less acidic. In peripheral tissues, the higher pCO_2 , lower pO_2 ,

and more acidic milieu protonate the histidine, which switches the iron to a high-spin state that reduces its affinity for oxygen, thereby releasing it. This phenomenon, including its enhancement by cooperativity between the hemoglobin subunits, is known as the Bohr effect. The release of oxygen is also facilitated by phosphate anions, higher temperature, and by 2,3-diphosphoglycerate, a glycolytic product, which displaces oxygen from heme. Deoxyhemoglobin transports CO_2 to the lungs, but since CO_2 is highly soluble in plasma, this mechanism accounts for only 15%–20% of carbon dioxide excretion.

II. IRON METABOLISM

A. Digestion

The efficiency with which dietary iron is absorbed from the diet and utilized in the body is influenced by (1) the physicochemical form of iron and the factors that influence the release of iron from the food and its solubilization in the intestinal lumen and therefore the amount available for intestinal mucosal uptake and (2) the homeostatic settings of the intestinal mucosa to take up and transfer iron to the body.¹³ (Table 22.2).

The principal determinants of the bioavailability of iron are usually the systemic need and homeostatic setting for the mucosal uptake and transfer of iron. The dietary influences of bioavailability are of more importance when intestinal adaptation is insufficient to acquire enough of the iron released by digestion iron to meet systemic needs for the metal.

TABLE 22.2 Factors influencing intestinal luminal solubilization, mucosal uptake, and transfer of iron.

Enhancers	Inhibitors
<i>Dietary factors affecting solubilization and availability of iron for mucosal adsorption and uptake</i>	
Amino acids (lysine, cysteine, histidine) peptides Organic acids: Citric, ascorbic, lactic, malic, tartaric acid, and citric acid, which occur naturally in fruit and vegetables and in fermented products Fructose, sorbitol, ethanol Additives (e.g., NaFeEDTA) using chelates	Phosphates, e.g., phytates found in whole-grain cereals, legumes, nuts, and seeds Casein phosphatides and other phosphates (e.g., caseins) whose effects may be exacerbated by calcium and magnesium Antacids, hydroxides, complexes Acid secretion blockers, gastric surgery Tannins Polyphenols such as those in tea, particularly black tea, and coffee are more powerful inhibitors than those from herb teas, cocoa or red wine, perhaps because they have higher levels of galloyl esters and tannates vegetables (spinach, aubergine) Soy and legumes Competition at mucosal uptake sites: Ca, mg, zn, cd
<i>Systemic</i>	
Systemic need for iron, iron deficiency, new tissue synthesis (e.g., increased erythropoiesis) Fasting Pregnancy	Anemia of inflammation Systemic iron adequacy or excess Achlorhydria Copper deficiency

The effects of inhibitors and enhancers on bioavailability are often estimated in single-meal studies, but these estimates are unable to allow for adaptation to the diet, which might take 3–7 days. Fortunately, in the case of iron, it is possible to use isotopes as tracers differentially to correct for the subjects' systemic need for iron as manifested by intestinal adaptation.

The forms of dietary iron available for absorption are nonheme iron and heme iron (mainly from meat). Lactoferrin and other ferrins are not regarded as significant dietary sources of iron. In contrast, ferritins and phyto-ferritins might be an appreciable source of iron.¹⁴

Both heme and nonheme iron are released from food by gastric and proximal intestinal proteolysis. The nonheme iron in the proximal small intestine is sustained in solution by the acidity and by binding to the coincidentally released amino acids and peptides, which facilitate mucosal adsorption. Heme iron is less soluble in an acidic milieu than it is in alkaline conditions.

Soaking, germination, or fermentation of vegetable sources before cooking may enable intrinsic phytase to break down phytate. Cooking may improve the solubilization of iron by degrading phytate; this, for example, is beneficial for oats, which have little intrinsic phytase activity. The effect of processing grain on its value as a source of iron is unpredictable; for instance, milling grain removes its husk, which is rich in phytate, but the husk is also the principal source of the grain's iron. Thus, although the processing may have improved the release of iron, there may be insufficient iron left after milling for it to be of any benefit. Adding milk to black tea or coffee does not improve the availability of iron. Heat processing may destroy some of factors enhancing solubilization. Meat improves the uptake of nonheme iron, but it is not clear how its constituent peptides achieve this.^{5–8}

Studies of solubilization enable dietary sources of iron to be ranked as sources of iron; however, quantitative values of availability cannot be extrapolated reliably to a food or diet universally, partly because of uncertainties about the dietary characteristics and importantly also because of systemic adaptation. The use of dietary characteristics to derive equations and models to predict the bioavailability of iron are of limited value.¹⁵ However, broad characterizations of diets on the bioavailability of their iron content have been used in deriving reference intake values from estimated physiological requirements. In general, the bioavailability values applied have been around 15%–18%.^{7,8}

B. Iron Absorption: Gastrointestinal Mucosal Uptake and Transfer

Gastroduodenal acidity and proteolysis in the proximal intestine release nonheme iron from its matrix, but more extensive digestion is needed to release heme and ferritin. Ferric and ferrous irons are most soluble in acid conditions,

and nonheme iron is predominantly absorbed as ferrous iron in the proximal duodenum, i.e., before hepatopancreatic secretions increase the luminal pH. Mucosal uptake of inorganic iron involves its aggregation by the mucus and glycocalyx of the epithelial brush border. Ferric iron is converted by a brush border ferrireductase, duodenal cytochrome B reductase, to ferrous iron which is then transported across the brush border membrane, down a proton gradient, by an enterocytic proton symporter divalent transporter (DMT1). Nonheme iron also can be absorbed in the distal intestine, and it is possible that intestinal uptake of iron may occur by other mechanisms; DMT1-deficient mice are able to absorb iron, but humans with an inherited defect in DMT1 are iron deficient. Other enterocytic membrane translocators for the uptake of iron, including Zip 8 and Zip 14, have been proposed.¹⁶

Since heme is more soluble at neutral or alkaline pH, it is feasible that the sites for its mucosal uptake are distributed more distally in the gut than those of nonheme iron. A definitive carrier for heme's uptake has not been characterized. Ferritin in an intact or partly degraded form is taken up by carrier-mediated endocytosis in the small intestine.¹⁴

In the enterocyte, heme is degraded by hemoxygenase, and the released iron is thought to join a common transit pool of ferrous iron. The ferrous iron is transported to ferroportin (FPN) in the basolateral membrane. The metallochaperones conducting this transit may be members of the poly(rC)-binding protein (PCBP) family, which, *in vitro*, can transfer ferrous iron between DMT 1 and FPN. It is noteworthy that cell membrane DMT1 and FPN are the principal, if not the only, carriers for the cellular ingress and egress of iron. At FPN, the ferrous iron is oxidized by hephaestin, a copper-dependent oxidase which is collocated with FPN, and exported, as ferric iron, to the portal ECF and circulation where it binds to apotransferrin-forming transferrin (TFN) for distribution to the portal and systemic circulations. Ceruloplasmin, another copper-dependent oxidase, in the portal ECF may also be involved with this process.^{16–18}

C. Transport

TFN is the main carrier of iron in the circulation and extracellular space; overall, TFN iron turnover approximates 30 mg/24 h; it is a sialylated glycoprotein synthesized in the liver as apotransferrin. TFN binds one or two ferric molecules, which it delivers to a cell surface TFN receptor (TFR1), which is on all cells apart from the lens and mature erythrocytes. TFR1 membrane surface density reflects the cellular need for iron. Usually, they are densest on erythroblasts, lymphoid cells, and the neurepithelium. TFN's degree of sialylation affects its function; in pregnancy, it is highly sialylated, favoring iron uptake by the placenta, whereas with inflammation

or eclampsia, it is less sialylated and binds TFRs less effectively. The binding of iron by TFN is saturable and after large doses of oral iron, like those used for supplements, iron complexes with organic acids such as citrate can be found in the circulation. The half-life of newly absorbed iron in plasma is about 75 min, and some 80% percent of TFN iron goes to erythroid cells in the bone marrow. A second TFN receptor TFR2 is found on the hepatocytic basolateral membrane and is pivotal in iron homeostasis.

The cellular uptake of iron is by endocytosis of the TFN–TFR1 complexes. The endosome is acidified to pH 5.5, which releases the ferric iron to be reduced by a membrane related metalloreductase (STEAP) and exported by DMT1 into the cytoplasm. As with other cell types (e.g., enterocytes), there seem to be separate DMT1-independent means of translocating non–TFN-bound iron across cellular and endosomal membranes.

The residual endosomal complex returns to the cell surface, and the apotransferrin is recycled. It is not clear how iron is trafficked in the cytosol, but here again, it has been proposed that PCBP2 family metallochaperones delivers ferrous iron to ferritin. Iron is also passed to sites for the synthesis of ISCs and enzymes or is conveyed to mitochondria where it binds to a mitochondrial carrier, mitoferrin, and is distributed to mitochondrial ferritin; used to synthesize ISCs; or transferred to ferrochelatase for heme production.^{16,17}

D. Metabolism and Turnover

Adult women and men contain 3.5–4 g iron, which represents about 4.3×10^{22} atoms of iron to underpin its functions. The compartmentation of iron (see Fig. 22.1) comprises hemoglobin: 3000–3500 mg; myoglobin: 400–500 mg; heme and nonheme enzymes:

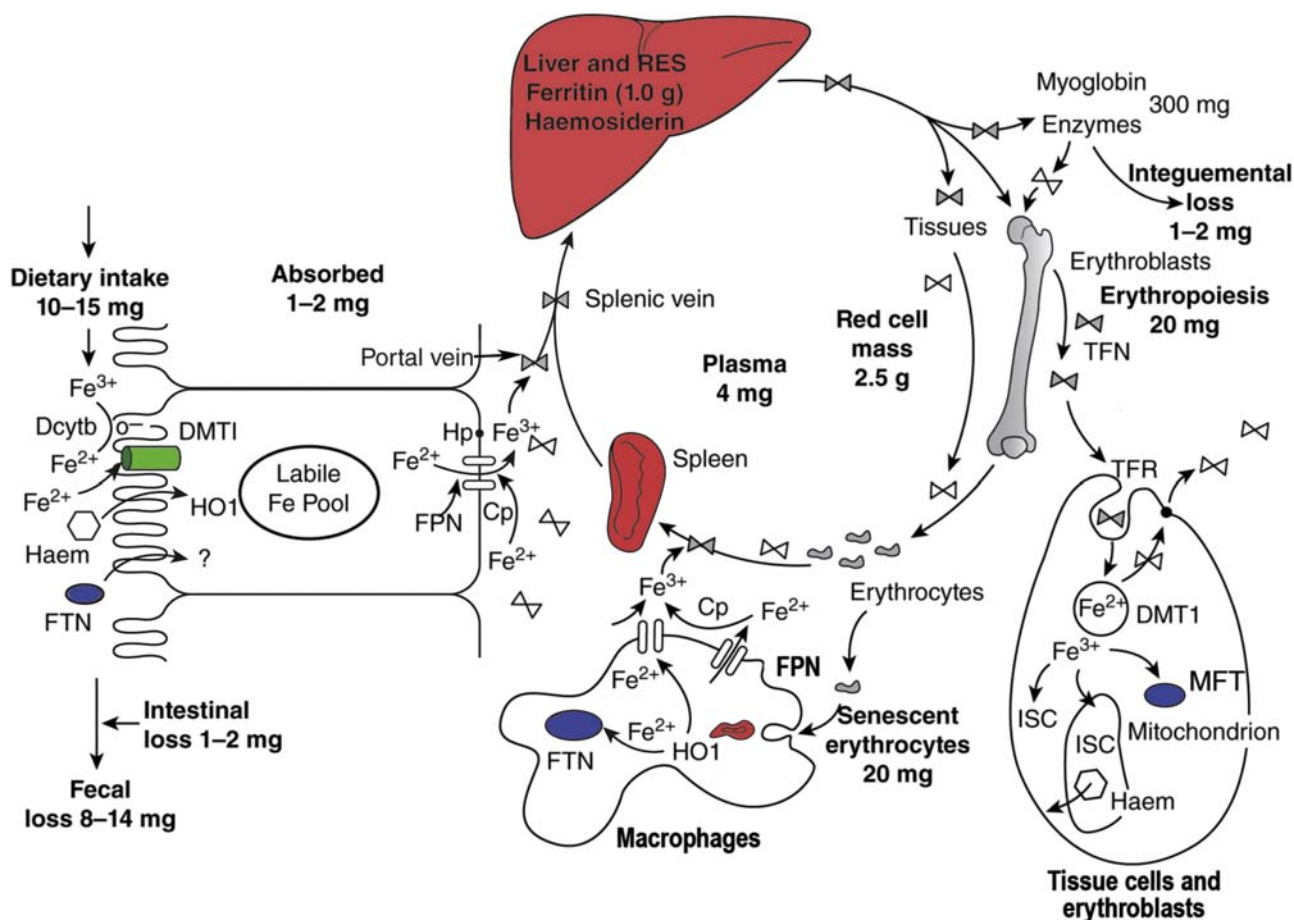


FIGURE 22.1 Diagrammatic representation of the daily systemic turnover of iron. There is a cycle centered principally on the use and reuse of endogenous iron to sustain hemoglobin concentrations in the erythrocytes, alongside a smaller pool in other tissues that turns over as the tissues are renewed. The major depots of iron are in the reticuloendothelial system (RES) as ferritin and in the liver as ferritin and hemosiderin, the red cell hemoglobin as well as smaller pools in muscle and other organs. Iron is lost from the gut in shed enterocytes and minor blood loss, and from integumental loss which totals 1–2 mg/day. These losses, and other needs to replace menstrual loss or to support the synthesis of new tissue, are replenished from ferritin and upregulated intestinal absorption. Blood from the spleen mixes with that from the intestine in the portal circulation and enters the liver where the hepatocytes can sense the TFN iron load and adjust the production of hepcidin secretion to control the absorption of iron. Cp, Ceruloplasmin; Dcytb, duodenal cytochrome B reductase; DMT1, divalent metal transporter 1; FPN, ferroportin; FTN, ferritin; HO1, hemoxygenase 1; Hp, hephaestin; ISC, iron–sulfur cluster; MFT, mitochondrial ferritin; TFR, transferrin receptor; ∇ , apotransferrin; ∇ , transferrin.

100 mg; ferritin and hemosiderin: 1000 mg; TFN: 3 mg; and an intracellular or transit pool: 7 mg. The body cannot excrete iron and the systemic iron pool is efficiently conserved. The recycling of endogenous iron, involving about 30 mg of iron daily, is at least 90% efficient; hence, the body only needs to absorb enough iron to replace basal integumental losses (i.e., hair nails and epithelia) approximating 14 $\mu\text{g/kg}$ body weight per day, which is about 1–2 mg/day. This loss and those from menstruation and the requirements for the synthesis of new tissue are balanced by mobilizing iron from ferritin and by adjustments in gastrointestinal absorption.

The systemic turnover of iron recycles iron from all tissues; the major focus is the salvage and recycling of heme iron from the erythron. About 10^{11} senescent red cells are phagocytosed daily, mostly by macrophages in the spleen. The heme is degraded by macrophage hemoxygenase to biliverdin, carbon monoxide, and iron, which is then either deposited in the macrophages' ferritin or exported via FPN to apotransferrin for utilization elsewhere, which is usually in the erythron. Free heme is an active oxidant, but there is a protein, hemo-pepin, which scavenges any free heme in the plasma and delivers it to hepatocytes for degradation.

he major systemic depot of iron is ferritin, which is found in the cytosol and mitochondria of all cells. It is a highly conserved protein and is found in all species, pro- and eukaryotes alike. It is a hollow sphere of 24 apoferritin subunits with 6 channels through which iron can enter and leave the sphere. There are heavy (H) and light (L) types of apoferritin subunits; their relative amounts vary between organs and somehow influence the mobility of the iron. H chains predominate in the heart and brain and L chains in the liver and spleen. H chains have ferroxidase activity that facilitates storage of iron as a ferric hydroxyphosphate and prevents oxidative damage. Each ferritin molecule can reversibly bind up to 4,500 atoms of iron, but usually only 20%–50% of this capacity is occupied. Mitochondrial ferritin has a 75% homology to H ferritin and a smaller homology to L ferritin. Hepatocytic ferritin is a depot of iron complementing the upregulation of intestinal iron absorption as a reservoir of systemic iron in response to iron loss or hypoxia. It is the primary repository of excess iron, and it contributes to innate immunity in that it sequesters iron in response to inflammation, infections, and the risk of oxidative damage. The other large ferritin pool, in the RES, is part of the endogenous recycling pool of iron supporting the erythron and this, not hepatic ferritin, is the source of ferritin in plasma where ferritin may be an iron transporter.

Normally, during cell turnover, ferritin is degraded by proteasomal lysis and the iron is recycled. However, with chronic iron excess or inflammation, ferritin might be inadequately degraded forming hemosiderin, which is a deposit of immobilizable iron for metabolism, but which, none the less, can cause oxidative hepatocellular damage.

E. Homeostasis of Absorption and Systemic Turnover of Iron

It was once thought that the reduced absorption of iron in response to low iron needs or high intakes was attributable to the induction in the enterocytes of a “mucosal block” arising from the local induction of ferritin. It is now realized that enterocytic uptake and transfer of iron is usually downregulated by the hormone hepcidin and that any “mucosal block” might only occur at high exposures to iron.

The intestinal mucosal uptake and transfer of dietary iron is inversely related to serum ferritin concentrations below 60 $\mu\text{g/L}$. This is indicative of an adaptive response relative to systemic, or more accurately RES, depots of iron. This adaptation may involve increased uptake and transfer of iron by enterocytes that can be achieved by programming enteroblasts in the epithelial crypts as part of an intestinal adaptation, but, more immediately, it reflects the derepression of DMT1 and FPN imposed by the regulatory hormone hepcidin.

Hepcidin is a hormone of 25 amino acids synthesized by hepatocytes. Its expression is regulated by a sensory assembly on hepatocytic membranes. This assembly involves an interaction of HFE (a membrane-bound protein that has long been known to be linked with hereditary hemochromatosis), with TFR1, and TFR2: usually HFE is associated with TFR1, but it can be displaced by iron-saturated TFN. When this happens, HFE binds instead to TFR2, and the HFE–TFR2 complex activates a membrane complex of a bone morphogenic protein (BMP6) and hemojuvelin (HJV), which costimulate phosphorylation and activation of a transcription factor for the expression of hepcidin. Hepcidin is also synthesized by monocytes, macrophages, pancreatic beta, and renal cells, and adipocytes.^{17,18}

Hepcidin blocks the export of iron from enterocytes and macrophages by inducing the internalization and degradation of FPN and reducing DMT1 activity. The iron retained in the cells is sequestered in ferritin and, in the case of enterocytes, it is subsequently lost into the gut lumen when the cells are shed. Hepcidin is downregulated by increased iron needs and by hypoxia. It is upregulated by inflammation, and this is an element of the dysregulation of iron metabolism that occurs in the pathogenesis of the anemia of inflammation (AI).¹⁹

The homeostatic control of systemic and cellular iron metabolism involves an intricate reciprocal interaction between mechanisms ensuring adequacy of iron supply with those which mediate innate immunity and others which maintain the supply of oxygen to cells. In normoxic, noninflammatory conditions cellular iron homeostasis is managed by two iron-responsive proteins (IRP1 and IRP2). These bind to iron-responsive elements (IREs) on untranslated regions (UTRs) of the mRNAs for proteins

involved in iron kinetics. When iron supply is limited, IRP1 binds the 5'UTRs and represses the translation of the mRNAs for the two apoferritin chains, FPN, and α -levulinic acid synthetase, which is the rate-limiting enzyme in heme synthesis. These steps stop ferritin sequestration of iron and its export by FPN and reduces its use by inhibiting synthesis of erythropoietin and heme; it acts to retain iron in the cell. Simultaneously, IRP binding to the 3'UTR of the mRNAs for TFR1, DMT1, and an organizing molecule necessary for endocytosis upregulates the cellular apparatus for the acquisition of iron.^{17,18}

If cells have an adequate supply of iron, the IRPs dissociate from the IREs, and their production is reduced. IRP1 binds a cytosolic ISC forming a complex that has aconitase activity. When there is a cellular lack of iron, the aconitase disassembles and IRP1 can bind to its IREs as is described above. Both IRP1 and IRP2 may be controlled by the same mechanism involving F-box/LRR-repeat protein 5 (FBXL5). This is a highly conserved protein analogous to an oxygen-binding nonheme protein, hemerythrin, which is found in invertebrates and bacteria. In iron-replete cells, FBXL5 enables the proteasomal lysis of IRPs, but in the absence of iron, FBXL5 disassembles and IRP2 and IRP1 are not proteolyzed and are able to bind to the IREs, thereby activating systems that acquire iron.

The intracellular and systemic homeostasis of iron is intricately linked (IRP1, for example) and has some control of an hypoxia-inducible factor (HIF) that is an initiator of the responses described next with, but subordinate to, the (i) systemic reactions to hypoxia and (ii) to the innate responses designed to protect the host from inflammatory damage, and from infection, by denying pathogens access to endogenous iron.

Cells sense and maintain oxygen homeostasis to avoid hyperoxia and oxidative damage as well as the metabolic sequelae of hypoxia. The relevant adaptive changes need to be rapid and sensitive to support the narrow range of normoxia. The mediators of this adaptation are HIFs. HIFs are dimers comprising HIF- α and HIF- β and some control units. These complexes can activate transcription by binding to hypoxia response elements present on a wide range of genes involved in processes including angiogenesis and vascular regulation, extracellular matrix metabolism, growth and apoptosis, transcriptional regulation (which is of relevance to carcinogenesis), metabolism, and erythropoiesis and iron metabolism.²⁰

HIF is being produced constantly, but it is almost simultaneously being degraded by a system of oxygen activated prolyl hydroxylase domain enzymes (PHDs), which are essentially oxygen sensors in that adequate intracellular pO_2 induces them to degrade proline links in the HIF subunits thereby preventing the many HIF-mediated reactions described above. However, if hypoxia develops, HIF can initiate an immediate response; this would include both the repression of hepcidin

expression and thereby increased absorption of iron and stimulation of erythropoietin production and erythropoiesis, thus ensuring the iron-dependent arm of the systemic response to hypoxia.²⁰

F. Influence of Inflammation on Iron Turnover: Anemia of Inflammation

The HIF/PHD mechanism is also activated by inflammation, even in normoxia.^{19,20} Chronic and acute infections, malignancy, inflammatory conditions in the bowel, joints, and kidneys, and even low-grade inflammatory states such as obesity are associated with dysregulated iron turnover. This is an adaptive state called the anemia of inflammation (AI) previously known as the anemia of chronic disease.¹⁹ AI is manifest as a mild iron resistant anemia (hemoglobin 80–95 g/L) with normal red cell indices and reduced erythropoiesis; a normal or elevated serum ferritin; hypoferrremia; a normal or reduced serum TFN, which has a low iron saturation; within reference values for TFR1; and a ratio of serum TFN receptors/log ferritin concentrations <1 .¹⁹ These changes are mediated by inflammatory cytokines (interferon- γ , tumor necrosis factor- α , and interleukins 1 and 6) that upregulate hepcidin and by the cellular responses to reactive species. Collectively, these increase the uptake and sequestration of iron by cells, especially macrophages, and block its export by FPN. Thus, increased output of hepcidin reduces the absorption of iron and augments the signals for the retention of iron by macrophages. The inflammatory mediators stimulate phagocytosis of senescent red cells and reduce erythropoietin production and its effectiveness by direct actions on the peritubular renal cells and erythroblasts. The adaptation may have evolved as a protection against damage caused by “free iron” liberated by tissue damage and to deny pathogens, such as Gram-negative bacteria, access to iron.^{3,19,20} AI can become sustained and lead to a significant paradoxical functional iron deficiency syndrome despite there being appreciable amounts of iron in tissue depots.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

Iron metabolism is essentially all about controlled energy production and exploitation of oxygen to support systemic metabolism. Table 22.1 illustrates the ubiquity of iron-dependent activities in metabolism. It is obviously vital for oxidative phosphorylation. The citric acid cycle and respiratory chain involve six different heme proteins and six iron–sulfur assemblies, as well as copper-based electron transfer from cytochrome c to cytochrome c oxidase at the ultimate transfer of electrons to molecular oxygen. In addition to this role in energy creation and expenditure, iron facilitates the synthesis of connective tissue and bone matrix, hormones, and active vitamins,

nucleic acid replication, substrate metabolism, phase 1 breakdown of xenobiotics and some endogenous metabolites, innate immunity, and antioxidant activity.

The elicitation of the pathogenesis of syndromes arising from inborn errors of iron metabolism has provided insight on its metabolic regulation and physiological functions as well as alerting us to a better understanding and awareness of the clinical features that might arise from iron deficiency both with and without anemia.^{9,10,21–23}

The Online Mendelian Inheritance in Man²¹ database has 427 entries relating to genes affecting iron metabolism. There are at least six inherited iron overload syndromes, the hemochromatoses, arising from unregulated absorption of iron caused by defects in HFE, HJV, hepcidin, TFR2, and FPN.^{21–23} Another syndrome, African Iron Overload Syndrome, previously called Bantu Siderosis, is an ecogenetic disorder arising from a combination of a genetic defect and increased exposure to iron from food and beer that have been prepared or stored in iron utensils. In contrast to the hemochromatoses, the abnormal iron overload is distributed in the Kupffer (RES) cells as well as in hepatocytes.²¹

Dysregulation of iron turnover also results from defects of DMT1, hephaestin, ceruloplasmin, TFN and its sialylation, ferritin, matriptase, heme oxygenase, IRP, and ferrochelatase. Defects in ISCs affect their interaction with DNA and their function as iron chaperones. These might account for why ISC defects are associated with altered mitochondrial function, abnormalities of the heart, abnormal iron accumulation and distribution in the brain (e.g., Friedrich's ataxia), developmental delay or regression, epilepsy, myopathies, ineffective erythropoiesis, and anemia (sideroblastic anemias).^{9,21,22}

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Circumstances in which iron deficiency might occur are shown in Table 22.3. These are similar to those that apply to all nutrients. Pure iron deficiency is arguably rare. It may be encountered in infants and toddlers who are inappropriately fed and in others on “fad diets.” Otherwise conditions causing iron deficiency would be associated with other conditions that carry a risk of causing other nutrient deficiencies arising from endogenous losses and malabsorption.

In such instances, iron deficiency should be regarded as a marker of potential multinutritional deficits, particularly in communities where anemia is endemic. Vegetarians have been reported to have lower serum ferritin values than omnivores, but even so, these values are usually within the reference range.^{6,8} The hematological and biochemical markers used to assess iron status would need to be applied and analyzed intelligently in the clinical and public health context in which they are being

TABLE 22.3 Causes of iron deficiency.

Low iron intake relative to systemic needs

Periods of rapid tissue synthesis, i.e., growth and rehabilitation; pubertal growth spurt, menarche, delayed and inappropriate diversification of diet in infancy, e.g., too much cows' milk displacing other foods in infants and toddlers; inappropriate diets in general

Maldigestion and malabsorption

Reduced gastroduodenal acidity caused by gastrectomy gastric bypass surgery, proton pump inhibitors, antacids, intestinal resection enteropathies

Systemic loss

Genitourinary system, menorrhagia, childbirth, trauma gastrointestinal blood loss, protein losing gastro enteropathies, enterocolitis intestinal parasitism (*necator americanus*, *Ancylostoma duodenale*, *Trichuris trichiura*, schistosomiasis, *helicobacter pylori*)

used and the influences of physiological events such as growth, adolescence and pregnancy, and of inflammatory states as in AI would need to be considered (see below).

Features of iron deficiency

Studies in animals can model severe, marginal, chronic, and acute conditions of iron deficiency both with and without anemia. Importantly, it is evident from such models that many of the subsequent defects are evident in tissues, particularly in those with a faster turnover than erythrocytes, before an iron-deficient anemia (IDA) develops. These models demonstrate inefficient energy metabolism, with altered glucose and lactate metabolism, reduced muscle myoglobin content, and impaired muscle strength and endurance. Reduced cytochrome oxidase activity has been found in muscle and the intestinal mucosa. Impaired collagen synthesis and associated osteoporosis have been noted, and the latter may also in part be secondary to interrupted hydroxylation of vitamin D. Vitamin A metabolism is also disturbed. Other defects include altered prostaglandin metabolism; impaired dopaminergic and serotonin neurotransmission, particularly in dopaminergic areas such as the substantia nigra, globus pallidus, cerebellar nuclei, and hippocampus; defective neuro-myelination and synapse and dendrite development; and altered brain membrane fatty acid profiles with, for example, less docosahexaenoic acid.^{24,25} Associated functional defects include impaired memory and spatial navigation and delayed responses to auditory and visual stimuli. There are few data giving a dose–response relationship for these effects and many accompany iron deficiencies that are more severe than are usually encountered in humans. Nonetheless these phenomena, together with information about the functional defects associated with all inborn errors of iron metabolism, provide plausible mechanistic but not definitive bases for inferring that iron deficiency, both

with and without anemia, would have similar effects in humans. These also show that susceptible organs are those with a rapid turnover, a specialized function or high energy dependency, such as immunocytes, enterocytes, brain, and muscle.

Iron deficiency in humans

Since iron deficiency might be part of a variable multi-nutritional deficit in which secondary disruption of the use of other nutrients and substrates might occur, it is difficult to ascribe specific features to iron deficiency or, as has been said, even to expect that pure isolated iron deficiency per se exists in clinical and public health contexts. Thus, classic features ascribed to iron deficiency such as koilonychia, soft nails, glossitis, cheilitis, mood changes, muscle weakness, and impaired immunity may be secondary, in part at least, to other coincident deficiencies for example of zinc, vitamin D, vitamin A, and iodine. Many studies use anemia as a surrogate indicator of iron deficiency and do not characterize the “deficiency” sufficiently to identify thresholds of anemia or of iron deficiency at which adverse outcomes occur. This is important because of the implications for psychomotor and growth impairments found in infants and young children with anemia and in whom focus should be on anticipating these outcomes. Characterization is also important to get evidence on preanemic features thereby enabling thought to be given to providing effective interventions both to correct the causes and to treat iron and nutritional deficiencies before any anemia develops.

Iron deficiency and cognitive, motor, and behavioral development

Iron is transported across the blood–brain barrier (BBB) by mechanisms analogous to those in other tissues.²⁴ The BBB is an effective boundary, and it protects the CNS from systemic iron overload such as in the hemosideroses. Ferric iron is delivered by TFN to neurons and oligodendrocytes, and the CNS accumulates iron until early adulthood. Undoubtedly, iron deficiency in animal models compromises neurological and psychomotor development particularly at critical periods in early development, when the defects are irredeemable. Furthermore, models suggest that in young animals, iron is preferentially distributed to erythroid tissues rather than the brain and other organs, implying that deficiencies of iron might occur in the brain unaccompanied by anemia.²⁵

Iron-deficient and anemic infants and children have impaired attention, poor recognition memory, and reduced reward-seeking behaviors. They are withdrawn and have poor social interactions. There is evidence that adolescent girls who were anemic as toddlers have altered memory and spatial awareness, but the vulnerable periods for the effects of early life deficiencies of iron have not been clearly identified. There is concern that, based on outcomes in models, defects might persist

and are irredeemable by subsequent iron supplementation, thereby setting affected children on a life course of poor achievement, and disadvantage. There are emerging data alerting us that this may be the case.²⁵ However, research in this area is confounded by concomitant nutrient deficiencies, socioeconomic factors, infections, and difficulties in standardizing outcomes and characterizing the causes and degree of iron deficiency. Many studies only consider the degree of anemia and assume that it is caused by iron deficiency.^{6,8}

Although data are inconclusive, there is a risk that iron deficiency, or even mild anemia, or both, reduces voluntary activity and work productivity. Iron deficiency reduces the proportion of circulating T lymphocytes and impairs T-cell lymphocyte proliferative responses. Impairments also affect neutrophil activity, phagocytosis, and microbicidal activity with a weakened neutrophil respiratory burst, free radical production and myeloperoxidase activity, and secretion of interleukin 2. However, B-cell functions are little affected.

In adults, low hemoglobin concentrations have been associated with lethargy, fatigability and impaired concentration, reduced voluntary activity, physical work capacity, and productivity. Iron-responsive fatigability and defects in endurance capacity of muscle during aerobic exercise training have been seen in groups without anemia in whom serum ferritin concentrations were less than 16 mcg/L.²⁶

Although many studies poorly reported outcomes and characterization of iron deficiency, there is an implication from these data overall that iron-responsive defects occur at hemoglobin concentrations below 80, 95, and 100 g/L. However, most of these thresholds were derived from stratified data rather than from anemia as a continuous variable. A possible exception is that a decline in muscle endurance and energetic efficiency has been observed as hemoglobin concentrations drop below 130 g/L, with the effect becoming greater with every further 10 g/L fall. Another study using phlebotomy to reduce hemoglobin values to 100–110 g/L reduced aerobic capacity by 16%–18%.⁸

There are other conditions that have a variable and unpredictable response to iron therapy. These include sleep disturbances, disturbed thermal regulation, breath holding attacks in early childhood,²⁷ and restless leg syndrome in adults.²⁸

B. Criteria of Adequacy

Adequate iron status (i.e., sufficiency) implies having normal erythropoiesis and iron functions that are not limited by iron supply, as well evidence of a reserve.

C. Assessment of Iron Status

Markers used in the assessment of iron status are shown in [Table 22.4](#). There is no single unequivocal

TABLE 22.4 Markers of Iron Adequacy, Deficiency, and Excess.

Marker Representative reference range (adults)	Diagnostic use confounding factors
<i>Homeostasis</i>	
Serum hepcidin Not yet fully validated.	Immunochemical and mass spectrometric assays are not cross standardized yet to enable inter laboratory usage. Intra-laboratory studies show values correlate with serum ferritin. Values are higher in men than in women and demonstrate physiological responses. It has good potential to differentiate AI and IDA and to assess iron status. Potentially very useful but is not yet sufficiently standardized for routine clinical or public health use.
<i>Tissue iron need</i>	
Serum transferrin receptor 2.8–8.5 mg/L	Inversely related to iron supply to cells and increased with new tissue synthesis (thus increased in growing children) and erythropoiesis. Less perturbed by AI than is serum ferritin but its value for children in regions where malaria and infection are endemic is less certain. Expensive and not well-standardized methodology.
<i>Tissue iron supply and turnover</i>	
Serum transferrin/iron 10–30 $\mu\text{mol/L}$	Most extracellular iron is bound to transferrin. Subject to diurnal and prandial fluctuations. Reduced with AI. Raised with iron overload.
Transferrin saturation 15%–60%	
Serum total iron binding capacity	Inversely related to transferrin saturation
<i>Iron in tissues</i>	
Serum ferritin 12–300 $\mu\text{g/L}$ females 12–200 $\mu\text{g/L}$ males	Important marker. 1 $\mu\text{g/L}$ of serum ferritin equates to 7–10 mg of stored iron. Serum ferritin originates from the RES macrophages and not from hepatocytes; the value therefore underestimates total systemic ferritin iron. Since hepatocytes are the first tissue to accumulate excess iron, serum ferritin is a poor marker of early iron overload.
Bone marrow	Stainable iron in aspirate of marrow. Invasive, expensive, and assessment is subjective. Not suitable for population surveys.

Continued

TABLE 22.4 Markers of Iron Adequacy, Deficiency, and Excess.—cont’d

Marker Representative reference range (adults)	Diagnostic use confounding factors
<i>Iron-dependent functions</i>	
Hemoglobin (see Table 22.5) Females 120–160 g/L Males 130–180 g/L	Commonly used but it is a late marker of iron deficiency; poor specificity and selectivity. Good monitor for therapeutic responses.
Mean corpuscular hemoglobin 27–32 pg	Useful for monitoring therapeutic response. Probably not useful for surveys. Other measures more practical.
Mean cell volume 84–99 fL	
Reticulocyte hemoglobin <25 pg	Suggestive of iron deficiency
Zinc protoporphyrin <800 µg/L	Marker of defective heme synthesis. Good marker of iron deficiency

marker of iron deficiency or large excess, except, perhaps, for serum ferritin at extreme deficiency or excess. Assessment of deficiency and excess relies on the use and contextual interpretation of markers of (1) homeostatic control, (2) functional iron, (3) the adequacy of iron supply to tissues, and (4) tissue depots.²⁹ Each of these markers is subject to confounding from other nutritional deficiencies and from the AI, and consequently, they should be used critically.²⁹

WHO Reference Values for hemoglobin values are given in Table 22.5.

Attempts have been made to use a battery of analyses to identify IDA. An approach was developed for NHANES II using serum ferritin, TFN saturation, and erythrocyte protoporphyrin or a combination of MCV, TFN saturation, and erythrocyte protoporphyrin. Individuals with two or more abnormal values in either trio could be considered iron deficient. Other approaches

use hemoglobin, serum ferritin values, and serum TFN receptors coupled with C-reactive protein and alpha-1-acid glycoprotein as indicators of acute and chronic inflammatory disease.³⁰ The ferritin index (serum TFR/log ferritin ratio) may be useful as an indicator of iron reserve to discriminate between AI (<1) and AI/IDA (>2).¹⁹ Diagnosing iron deficiency based on the response to an iron supplement is unreliable but might have to be used if a possible cause of AI cannot be found. In the future, hepcidin values might be of use to identify AI.

D. Reference Intakes

Table 22.6 gives the sets of reference dietary intake values for iron in North America⁷ and the European Union;⁸ other reference values are summarized in the recent EFSA opinion.⁸ The advisory groups involved

TABLE 22.5 Hemoglobin thresholds used to define anemia at sea level (WHO).

Group	Hemoglobin (g/L) ^a
Children	
0.5–4.99 years	110
5–11.99 years	115
12–14.99 years	120
Nonpregnant women >15 years	120
Pregnant women	110
Men (>15 years)	130

^aValues for those of African heritage are 4–10 g/L lower than those of Caucasian origin. A further amendment used by the WHO classifies anemia as mild, moderate, or severe, according to whether the values are above 80%, 80%–60%, or below 60% of the appropriate population threshold.

TABLE 22.6 Dietary reference values for iron developed for the United States and Canada (IOM 2001) and for the European Union (European Food Safety Agency, 2016).

The United States and Canada Dietary Reference Intakes (IOM 2001)			European Union Dietary Reference Values (EFSA 2016)		
Age	Average Requirement (mg/d)		Age	Average Requirement (mg/d)	Population Reference Intake (mg/d)
0–6 months	0.27		0–6 months		—
7–12 months	11.0		7–11 months	8	11
1–3 years	7.0		1–6 years	5	7
4–8 years	10.0		7–11 years		11
	Male	Female			
9–13 years	8.0	8.0			
14–18 years	11.0	15.0	12–17 years (M)	8	11
19–50 years	8.0	18.0	12–17 years (F)	7	13
50 + years	8.0	8.0	18 + years (M)	6	11
Pregnancy	—	27.0	18 + years (F)	7	16
Lactation	—		Postmenopausal	6	11
14–18 years		10.0	Pregnancy	As for nonpregnant, premenopausal women	
18 + years		9.0	Lactation		

explain the bases for their values and describe the limitations of the data that they had available. The values support public health measures and are cautious. Although they support assessments of dietary intakes, they are not diagnostic of iron deficiency or excess. The differences between the two sets of recommendations arise in the areas of most uncertainty, which are the needs of children, particularly at puberty which happens at a wide age range that is difficult to accommodate in a simple range of values, and of menstruating, pregnant, and lactating women, and assumptions about the impact of homeostatic adaptation on the bioavailability of iron. The uncertainties in the reference values, composition tables, and status determination are reflected in the lack of congruence between population surveillance data on iron intakes, the apparent prevalence of iron deficiency, and in dietary reference values.⁶

Iron and women in their reproductive years

The demands of menstruation, pregnancy, lactation, and, in adolescence, growth on iron homeostasis and iron requirements are uncertain. Women may be at risk of inadequate iron supply because contemporary lifestyle and diets favor reduced energy expenditure and the consumption of diets with a lower iron density than those on which humans evolved.⁸

Menstrual blood loss of adolescent girls is marginally less than that of older women, and in nutritional risk assessments of iron nutriture, it is assumed that adolescent

menstrual losses match those in adult women. There is little intraindividual variation in menstrual blood loss but considerable interindividual variation. Estimates of menstrual loss in different countries are consistent in showing that 95% of women lose 118 mL or less per cycle with median and mean losses of 30 and 44 mL, respectively: these equate to iron losses of 0.49 and 0.7 mg/day.⁸ Intrauterine devices increase menstrual blood loss two- to fourfold, whereas oral contraceptive users have smaller median menstrual blood and iron losses (e.g., 18 mL/cycle and 0.26 mg/day) with mean losses of 26 mL/cycle and 0.43 mg/day. There is an inverse correlation between menstrual blood loss and serum ferritin concentrations; on average, an increase in menstrual iron loss of 1 mg/day is associated with a reduction in serum ferritin of 7 µg/L. It should be noted that in adolescent girls and boys serum ferritin levels may fall with their growth spurt; therefore, in girls, this would happen before menarche.

Iron in pregnancy and lactation

Red cell mass and oxygen carrier capacity increases until 32–34 weeks' gestation in response to a two- to four fold increase in erythropoietin production. The usual inverse relationship between erythropoietin and red cell mass is lost, but it reappears later in pregnancy if the hemoglobin falls below 90 g/L. TFN production also increases with a denser sialylation that favors its binding to placental receptors. However, the hematocrit, hemoglobin, and serum iron, ferritin, and TFN fall

because from early gestation the plasma volume expands by about 1.3 L. This increase is independent of the preceding nonpregnant plasma volume, which is about 2.7 L. The increase is larger for multiple pregnancies and multigravida. Thus, the declining values reflect the physiological hemodilution of pregnancy and are not necessarily indicative of iron deficiency or anemia. The plasma volume expansion stops at the end of the second trimester and decreases near term.

The efficiency of nonheme iron absorption increases during pregnancy and is sustained during early lactation. Iron bioavailability has been shown to increase from 7% at 12 weeks' gestation to 66% at 30 weeks. This of course contradicts ideas that women who become pregnant need to increase their iron intakes and is consistent with observations that women do not increase their intakes during pregnancy.

The risk of adverse outcomes such as low birth weight and premature delivery is increased at hemoglobin values below 90 g/L or above 130 g/L.³¹ This range reflects the healthy range of hemoglobin in pregnancy: lower values may indicate a degree of anemia, whereas values above the range indicate failure of normal adaptation, possibly secondary to other causes of failed pregnancy. The limits are not well defined. The lower limits from various studies vary between 86 and 105 g/L, at least in part because the values were not taken at the same stage of pregnancy. The most predictive values are those determined in early pregnancy. Later values, particularly those from the last trimester, correlate poorly with outcomes.

Interpreting such data is complicated by the possible effects of AI. Infections, especially parasitism, are associated with impaired placental perfusion and function, intrauterine growth retardation, and early onset of labor,³¹ but it is not clear if these effects are due to dysregulation of iron supply. Iron supplements may improve women's hematological parameters in pregnancy but have little impact in reducing adverse outcomes in women with normal levels of hemoglobin. Some studies suggest beneficial effects of iron supplementation on reducing the incidence of low birth weight if it is initiated early in pregnancy. There are few data to enable a confident risk assessment of the influence of birth spacing on iron requirements in pregnancy. Although it is accepted that physiological requirement for iron is increased to support pregnancy, there is no agreement on a need for increased reference intakes for pregnancy and lactation. Some advisory bodies, for example, EFSA⁸ and SACN,⁶ think that extra iron can be supplied by systemic adaptation and increased absorption. Others such as the Institute of Medicine advised an increase in reference values.⁷ None the less, many agencies no longer recommend regular iron supplements during pregnancy, recommending them only for women with

hemoglobin values less than 110 g/L in the first trimester and 105 g/L at 28 weeks' gestation.

Iron and the infant

Embryos and fetuses have forms of hemoglobin that have a higher affinity for oxygen than adult hemoglobin. This enables them to acquire oxygen at the low placental pO₂ from maternal hemoglobin. Neonates have 150–250 mg of iron, much of which they have gained during the last trimester of gestation and hemoglobin levels of 160–180 g/L. They respond to the higher pO₂ ex utero, by reducing synthesis of fetal hemoglobin and hemoglobin values fall to 90–110 g/L by about 3–4 months. The degraded hemoglobin releases 50–60 mg of iron, which is deposited in reticuloendothelial ferritin. In early infancy, serum ferritin concentrations are as much as 400 µg/L. These can fall to about 30 µg/L as the endogenous iron depot is used for new tissue synthesis and for producing adult hemoglobin from 4 to 6 months of age. This coincides with the relocation of erythropoietin production from the liver to the renal peritubular cells. There is a risk of iron deficiency at 6–9 months. However, the iron reserve can be increased by 30–50 mg of iron if, at birth, the cord is not clamped until it has stopped pulsing, thereby letting these late pulsations transfer the 32% of the neonate's blood volume, which is in the placental component of the fetal circulation back to the baby.³² Thus, until 6 months of age, healthy babies have little if any need for iron from external sources.³³ Compared with that available from its systemic depots, relatively little iron is received by babies from breast milk, the iron content (0.2–0.4 mg/L) of which is not altered by either maternal iron deficiency or supplementation. Breast milk, however, contains the iron-binding protein lactoferrin, which is thought to protect the infant against pathogenic bacteria by limiting the iron available to them and favoring a protective intestinal microflora of bifidobacteria and lactobacilli. Intestinal iron absorption is downregulated until 6 months of age when infants become more reliant on exogenous iron.³⁴

E. Tolerable Upper Intake Levels Iron Overload

The risk of iron overload from dietary sources is negligible with normal intestinal function.^{6,35} Chronic iron overload with hemochromatosis has an increased incidence of systemic cancer, cardiovascular disease, neurological disease, arthropathies, and diabetes mellitus, but these associations do not occur in the general population, including heterozygotes for hemochromatosis. Although there is an association between the incidences of colorectal cancer with meat consumption, this relates to red meats and to some processed meat products and is not related to their iron content.^{6,36}

Acute intakes of 20 mg or more of elemental iron per kg bwt/day, particularly without food, cause corrosive hemorrhagic necrosis of the intestine causing diarrhea which might lead to systemic fluid and blood loss, shock, systemic oxidative tissue, and organ damage and failure. The onset of gastritis, nausea, abdominal pain, and vomiting after acute doses of iron has been used to gauge Upper Levels. However, there are insufficient data to inform a risk assessment to set safe upper limits for iron consumption. In North America, a guidance value of iron intake from all sources of 45 mg/day has been set based on interactions with other transition elements,⁷ and in the United Kingdom, a guidance level for supplemental iron intake has been set at 17 mg a day for adults based on the above gastrointestinal effects. The European Union has not set an Upper Level.³⁵

F. Dietary Sources

Good food sources of iron include red meat, liver and kidneys, fish, cereals, beans and nuts, egg yolks, dark green vegetables, dried apricots, figs and prunes, potatoes, and fortified food products. The iron content of dairy products and many fruits and vegetables is lower. In Western dietary patterns, such as those in the European Union, grains and grain products provide 20%–49% of iron intake. Typically, 70%–90% of dietary iron is in the nonheme form. Assays of iron in food products provide total iron values. The heme content of meat cannot be measured directly. It is estimated by first measuring the meat's apparent total inorganic nonheme iron content, then treating the meat to degrade its heme content to release its iron, and then remeasuring the total iron. The difference between the two values is assumed to be the heme iron content of the meat. The proportion of heme in the iron content of meat is 64%–78% for beef; 39% for pork, ham, bacon, pork-based luncheon meats, and veal; 26% for chicken and fish; and 21% of the iron in liver.^{6–8}

It is noteworthy, looking at compositional data,³⁷ that the iron content of fish (clams, mackerel, trout, and bass) is similar to that of chicken, pork, beef, and lamb: all provide 1–2 mg of iron per 75 g serving. Oysters and octopus can provide three to four times this amount.³⁷ Arguably, popular evaluation of dietary sources of iron has been conditioned by concepts that the iron in meat is all easily absorbed heme iron and meat is therefore better for consumers. However, if the body does not need the iron, it will not absorb it. Cooking reduces the water and fat content of meat, thereby increasing on a weight for weight basis its iron content. Cooking may denature heme and, additionally, cause loss of nonheme iron.

Cereal sources may contribute about 40% of iron intake, and although about half of that may come from iron fortification of breakfast cereals, plant products can be good sources of iron. The iron content of plant

cultivars varies appreciably, but generally servings (75 g) of beans, potatoes, and peas provide as much iron as do meats, and standard servings of cooked or fermented tofu, brown rice, soybeans, and other lentils are also good sources. Additionally, nonheme iron is acquired as fortification or restoration of iron lost during the processing of foods and from contamination arising from extrusion processes and from cast iron cooking utensils, particularly if citrate or similarly acidic ingredients are used in cooking. The assessment of dietary quality concerning the availability of its iron content has been aided by the compilation of data on the phytate content of foods, but these data might be of more qualitative than quantitative value.¹⁵

Advice on diversification of infants from breastfeeding or formula feeding has been derived from information like that above. Authorities recommend that diets based on beans, chickpeas, lentils, dark green vegetables (okra, broccoli, spring greens, watercress), whole grains, brown rice, mashed tofu, dried fruits such as apricots, figs, and prunes, and fortified cereals can provide enough iron for infants and young children. The foods need to be prepared appropriately for the child's age, i.e., well-cooked and mashed or pureed. This applies particularly to seeds and nuts that are also recommended sources of iron.

The use of food compositional data to assess dietary iron intakes is offset by an appreciation that, on the basis of variability in survey data, the number of days needed to gain some idea of an individual's true mean daily intake of iron would require around 50 days of observation to be within 10% of the true mean intake.³⁸

V. ISSUES RELATED TO IRON IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern, Iron Supplements, and Fortification

In most situations, apart from severe malnutrition and in areas where malaria and other infectious diseases are endemic, iron and micronutrient supplements containing iron in balanced amounts have proven safe and efficacious as a population intervention alongside other measures to counter the causes of anaemia and its associated sequelae. However an increased incidence of diarrheal disease, septicemia, and activation of infections (e.g., malaria) has been observed in communities where iron therapy has been used as a general measure. The World Health Organization recommends that in areas of high risk of malarial transmission supplementation should be targeted at children who are anemic and at risk of iron deficiency and that they

should also receive concurrent protection from malaria and other infectious diseases.”^{39–41} The basis of this problem is that most if not all bacteria have powerful siderophores, which are significant virulence factors because they can strip iron from heme, ferritin, despite its avidity for iron, TFN, and other iron proteins. If iron is available from hemolysis as with malaria, from hemolytic anemias, or from oral supplementation, bacterial proliferation, commonly with enteric organisms, may arise. A similar problem might apply to interactions between iron plasmodia, HIV, and tuberculosis. An analogous relationship with nondiarrhoeal and respiratory disease is not so clear. Iron supplements, both alone or with other micronutrient supplements, are associated with a decrease in the amount of bifidobacteria and lactobacilli and an increase in enteropathogenic bacteria in the gut microbiome,⁴¹ the implications of which are important for infant, childhood, and population feeding and dietary strategies in both clinical and public health. Iron supplementation and fortification and its use in food additives have not been fully explored to determine how this phenomenon should be approached for risk assessment and for risk management in policy and practice.^{40,41}

B. Treatment and Prevention of Iron Deficiency

For both populations and individuals, it is important to treat the cause of the iron deficiency and any coexisting nutritional deficiencies. Often, as in nutritional rehabilitation from pan-malnutrition, it is best to do this before giving any iron.

This applies particularly to clinical management. Iron salts used to treat iron deficiency include for oral use ferrous sulfate (60 mg of iron/300 mg), ferrous fumarate (65 mg of iron/200 mg), and ferrous gluconate (35 mg of iron/300 mg). Other preparations are available. For parenteral use ferric carboxymaltose, iron dextran, iron isomaltoside preparations are available.

Additional treatments would be needed in circumstances when other nutritional deficiencies or infections are present to ensure that the iron supplements would be effective. The response to iron should be monitored, and if this is not possible or practical, then dosage should be cautious. In any event, the above shows that it is not wise to use large doses of iron. Strategies are evolving in which hepcidin levels are being used to guide iron therapies.⁴⁰ Guidelines drawing on all these issues and uncertainties in the use of iron supplements for the young population have been published by the WHO.⁴²

At a population level, iron fortification of foods can be undertaken and has been well considered.⁴³ Strategies for the selection of fortificants must consider the

oxidative effects of inorganic iron salts on the organoleptic properties and shelf-life of the products. Preferred fortificants are ferrous sulfate or ferrous fumarate, either as simple salts or encapsulated within shells of vegetable oils or stearates, electrolytic iron, and sodium iron EDTA. Recent innovations include microencapsulated ferrous fumarate “sprinkles” that can be distributed over food and a micronutrient powder containing a phytase.⁴⁴ The latter is used to fortify complementary feeds and to optimize the availability of the intrinsic iron, thereby minimizing the risks of high exposure to iron. Strategies involving multimicronutrient supplements are increasingly being applied to address coincident deficiencies and to assuage uncertainty about the identity of the limiting nutrients other than iron.⁴⁵

C. Public Health and Preventive Measures to Combat Iron Deficiency

The data on the effects of iron deficiency are not as sound as those on the association of defects with anemia, which suggest that, irrespective of their pathogenesis, hemoglobin values of less than 110 g/L have adverse effects. Thus, it is understandable that public health measures should address anemia and its several causes, rather than iron deficiency per se.

The elements of preventing iron deficiency are similar in both developed and developing countries, although appropriate cultural and locale-specific emphases need to be applied. The key components at all ages involve measures to reduce the environmental causes of blood loss and iron loss from intestinal and other parasites, ensuring access to adequate diets, and being aware that multinutrient supplements are probably needed to ensure adequate nutrition in general and effective use of dietary and supplemental iron.

In infants, delayed clamping of the cord at birth and sustaining breastfeeding are additionally important.⁶ It needs governmental interventions to support the infrastructural interventions to achieve these measures, which would have a broader benefit because they would break the infection–malnutrition cycle. National, regional, and local governments need to appreciate the socioeconomic benefits of reducing the prevalence of iron deficiency.⁴⁶ The political will to act on this could be generated by an increased engagement and empowerment of the citizenry who themselves could undertake some of the basic public health and sanitary measures that would control some of the causes of iron deficiency. The potential benefits are exemplified in a strategic costing exercise in Pakistan, which concluded that micronutrient deficiencies in young children cost Pakistan 1.44% of its GDP and 4.45% of disability adjusted life years; these are appreciable obstacles to national development.⁴⁷

D. Known Interactions

There are unresolved concerns about the significance of interactions between iron, zinc, and copper along their kinetic pathways. For example, they all may use DMT1 and all bind to TFN. Iron can inhibit the utilization of zinc and copper thereby

compromising in early life, linear growth,^{6,8} immune function, cupro-enzyme activity in low birth weight infants, and ceruloplasmin concentrations. However, the circumstances in which these phenomena occur, and their significance are not fully characterized, and it is not clear which components of bioavailability are involved.

RESEARCH GAPS

Uncertainties persist about how much iron we need, how to determine iron status, and how to ensure adequate iron status in individuals and populations. The assessment and management of iron deficiency and anemia need to be differentiated. The challenge lies in how to use new knowledge in cell biology to address these core issues. Part of this strategy will entail refining our current approaches by using improved understanding of iron trafficking and to use markers, such as hepcidin, of the control of iron kinetics to characterize iron status. It is unlikely that new markers of iron status will emerge until new markers of adaptive mechanisms to iron deprivation and excess have been identified and validated. It should be possible using systems biology better to validate existing markers or to evolve algorithms to assess the adequacy of dietary intakes, systemic iron supply, and function.

It is particularly important to appreciate the difference between iron deficiency and anemia and to be able to consider, in the context of risk assessment and management, whether the differentiation matters or not. Law makers and public health strategists should consider how measures to provide increased intakes and measures, such as improved sanitation and disease control, to prevent iron and other deficiencies should be prioritized. Assessment of requirements and of status will benefit from applying improved awareness of how iron homeostasis interacts with that of other nutrients and possibly more importantly with other systemic conditions such as any degree of inflammation. This should also influence the use of iron supplementation and fortification programs, bearing in mind concerns about trace element interactions causing other deficiencies, impaired growth, and activation of infections, in which regard the role of enteric pathogens is important. Overall, it is important that iron should be considered in a broad nutritional and health context. The socioeconomic implications of iron deficiency in early life on adult health and achievements should be a motivator for research, practice, and policy.

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ZINC

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SUMMARY

Zinc is an essential micronutrient for all forms of life, with diverse and indispensable catalytic, structural, and regulatory functions. Dietary and acquired zinc deficiencies are common health problems in both developing and industrialized countries. Zinc deficiency results in growth failure, with the epidermal, gastrointestinal, central nervous, immune, skeletal, and reproductive systems most affected. Due to its ubiquitous distribution and involvement in diverse metabolic processes, the biochemical and physiological underpinnings of zinc deficiency syndrome have not been fully established. The discovery of transporter-mediated cellular and subcellular zinc trafficking and zinc's emerging functions as a signaling ion have raised interest in biochemical and clinical zinc nutrition research.

Keywords: Absorption; Bioavailability; Catalytic; Deficiencyhomeostasis; Metabolism; Regulatory; Signaling; Structural role; Transporter.

I. INTRODUCTION

A. Background

Although zinc is an essential micronutrient for all forms of life, its essentiality and importance in the human diet was slow to be recognized. In 1934, zinc was discovered to be essential in experimental animals.¹ Endogenous zinc concentrations in human tissues were determined in 1926,^{2,3} suggesting that zinc might have biological functions in humans, but this was not confirmed until 1961. In the 1960s, growth retardation, impaired sexual maturation, severe anemia, and cognitive deficits were reported in parts of the Middle East, where the diet consisted of unrefined flatbread, potatoes, beans, and milk.⁴ When affected individuals were provided with a hospital diet, their growth improved, and their hypogonadism was corrected. Subsequent dietary supplementation studies proved that the primary reason for this syndrome was the lack of zinc in the diet. In the United States, dietary zinc supplementation studies confirmed the occurrence of a growth-limiting zinc deficiency in otherwise healthy

infants and young children.^{5,6} In 2002, the World Health Organization (WHO) recognized zinc deficiency as a major global risk factor for disease mortality and morbidity.⁷ Currently, 1.2 billion people in developing countries are estimated to have a dietary zinc deficiency. In both developing and industrialized countries, acquired zinc deficiency is even more common and has been observed in patients with malabsorption syndromes, liver disease, chronic renal disease, sickle cell disease, and other chronic illnesses, such as diabetes and obesity.⁸ Dietary and acquired zinc deficiencies present with similar symptoms.⁹

B. Biochemistry and Function

Understanding its biological functions requires consideration of zinc's chemical features. Zinc²⁺ (Zn²⁺) is a strong Lewis acid (electron acceptor; copper [Cu²⁺] > Zn²⁺ > iron [Fe³⁺]). Thus, in biological reactions, Zn²⁺ strongly binds to thiolate and amine electron donors.¹⁰ Although Zn²⁺ is redox-neutral under physiological conditions, it plays a central role in controlling redox biology. Zn²⁺ release from Zn–thiolate

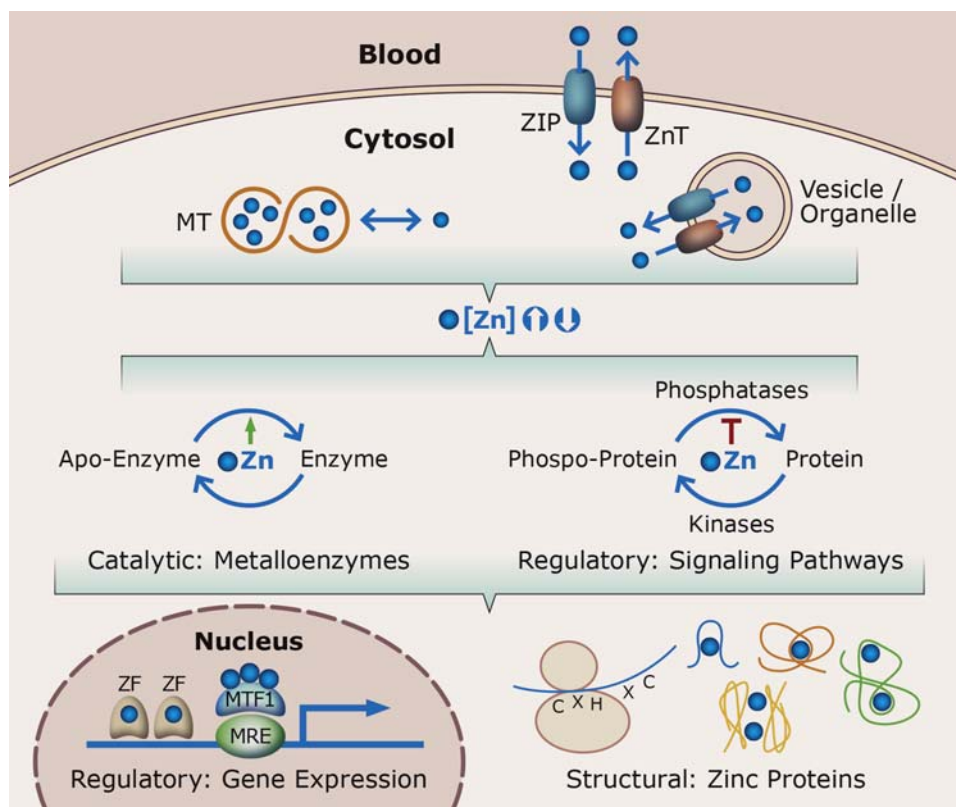


FIGURE 23.1 General scheme of catalytic, structural, and regulatory biological functions of zinc.^{25,44–48,80,88,89} MRE, metal-response element; MT, metallothionein; MTF-1, metal-response element-binding transcription factor-1; ZIP, Zrt/Irt-like proteins; ZnT, zinc transporter; ZF, zinc finger.

clusters by an oxidant produces disulfide bonds, indicating that Zn–sulfur (S) thiolate bonds are sensitive to cellular redox status.¹¹ Among the redox-inert divalent metal ions (magnesium [Mg^{2+}], calcium [Ca^{2+}], and Zn^{2+}), Zn^{2+} interacts the most strongly with ligand donor atoms. Consequently, zinc is a potent effector at correspondingly lower concentrations.

To determine its specific functions, it is necessary to assess zinc concentrations throughout the body in both cellular and subcellular compartments. Analytical procedures include conventional atomic absorption spectrophotometry, microwave plasma atomic emission spectrometry, and inductively coupled plasma emission. Some early zinc kinetics studies used radiolabeled zinc (^{65}Zn), which allowed measurement of intestinal zinc transport without interference from endogenous zinc,¹² but radiolabeled zinc is no longer used in humans due to its long radioactive and biological half-lives. Zinc has five natural, stable isotopes¹³ that have been used in humans to understand whole-body zinc homeostasis and physiology. New, emerging techniques for imaging zinc include fluorescence resonance energy transfer using probes designed to sense zinc ions, X-ray fluorescence microscopy, and the use of highly specific fluorescent probes for zinc, such as FluoZin-3.^{14–20}

The first known biological function of zinc is its catalytic role in erythrocyte carbonic anhydrase (CA), discovered in 1939.²¹ Shortly thereafter, hundreds of zinc enzymes in all six enzyme classes were defined.²² In the 1980s, zinc finger proteins (ZNFs), in which zinc has primarily a structural role, were discovered.²³ Using computational approaches, the zinc proteome was estimated to include 9% of eukaryotic proteins and 10% of human proteins.²⁴ The most recent discovery in zinc biology is the recognition that zinc ions function as signaling molecules in a manner similar to that of calcium ions. These diverse biological roles of zinc are grouped into three general functional classes: in catalysis; as a structural element; and as a regulatory ion (Fig. 23.1).²⁵

II. METABOLISM

A. Body Zinc

Tissue zinc distribution

After iron, zinc is the second most abundant metal in the body. Total body zinc in an adult human ranges from an average of 1.5 g in women to an average of 2.5 g in

men. Approximately 60% of zinc is stored in skeletal muscle, 30% in bone, 5% in liver and skin, and the remaining 5% in other tissues.²⁶ Plasma zinc accounts for only 0.1% of the body's zinc and represents 20%–30% of the zinc in whole blood. In plasma, 80% of zinc is loosely bound to albumin, and 20% is tightly bound to 2-macroglobulin.²⁷ The dynamic state of zinc in the body depends upon both dietary zinc status and metabolic/physiological state. Kinetic data with both radioactive and stable isotopes revealed the presence of two metabolic zinc pools in humans:^{12,28} a rapid turnover pool (~12.5 days) found in liver, pancreas, kidney, and spleen and a slow turnover (~300 days) pool in muscle, red blood cells, bone, and the nervous system. The exchangeable zinc pool can be identified using stable zinc isotopes.²⁸ Approximately 10% of total body zinc is exchanged with isotope within 2 days. The size of the exchangeable zinc pool decreases by approximately one-third when dietary zinc is severely restricted, suggesting redistribution from the exchangeable pool to other tissues.²⁹

Intracellular zinc distribution

Approximately 95% of body zinc is intracellular, distributed in the cytoplasm (50%), nucleus (30%–40%), and membranes (10%).³⁰ The total cellular zinc concentration is thought to range between 100 and 500 μM .^{31–33} Cytosolic zinc is bound with proteins and sequestered into organelles and vesicles. Thus, the cytosolic labile ("free") zinc ion concentration is very low, ranging from picomolar to low nanomolar.^{17,34,35} Labile zinc concentrations in intracellular organelles have been measured as 0.14 to 300 pM in mitochondria, 0.9 to 5 nM in the endoplasmic reticulum (ER), and 0.2 pM in the Golgi complex.^{36–39} High zinc concentrations are found in brain synaptic vesicles (1–5 mM)⁴⁰ and in pancreatic beta cell insulin granules (10–20 mM).⁴¹ Zinc is also found in low concentrations in extracellular spaces (1–10 nM in the brain) and modulates intracellular events through binding to zinc-sensing receptors⁴² and ion channel receptors.⁴³

B. Zinc Transporters

Zinc mobilization requires specific zinc transport systems that carry it across biological membranes. In the body, starting from intestinal dietary zinc uptake, zinc transporters facilitate the cellular and subcellular distribution of zinc to and within tissues. Cellular zinc homeostasis involves the opposing actions of two families of metal transporters, SLC39A/ZIP (Zrt/Irt-like proteins) and SLC30A/ZnT (zinc transporter). To date, 14 ZIP (1–14) and 10 ZnT (1–10) family members have been identified. ZIP family transporters traffic metals into the cytosol from the extracellular space

and within the lumen of intracellular compartments. In contrast, ZnT family transporters traffic metals from the cytosol to the extracellular space or to intracellular compartments. Both metal transporter families display tissue-specific expression. Within cells, the subcellular localization and expression levels of zinc transporters differ depending on physiological conditions and body zinc status (reviewed in Ref. 44–48).

ZIP4 localizes to the apical membrane of enterocytes and functions in zinc absorption.⁴⁹ Mutations in human ZIP4 cause acrodermatitis enteropathica (AE) due to zinc malabsorption that results in severe zinc deficiency. Mutations in human ZIP13 cause Ehlers-Danlos syndrome (a connective tissue disorder).⁵⁰ ZnT8 localizes to pancreatic beta cells and facilitates zinc transport into insulin granules, where it is needed for insulin hexamer formation.⁵¹ Human ZnT8 mutations are associated with type 1 and type 2 diabetes. ZnT2 in the mammary gland transports zinc into breast milk, and a mutation in human ZnT2 causes lethal zinc syndrome.⁵²

Transient receptor potential channels (TRPs)⁵³ and metal-permeable channels, including N-methyl-D-aspartate receptors and α -amino-3-hydroxy-5-methyl-4-isoxazole propionic acid receptors, also transport zinc.⁵⁴ Fig. 23.2 summarizes current data on the cellular distributions of zinc-transporting proteins.

Some zinc transporters also transport other metals. The exact mechanism for ligand specificity of mammalian zinc transporters is unknown due to the lack of crystal structures. Models generated using crystal structures of their bacterial homologs suggest that ligand specificity might be determined by ligand abundance and by conformational changes due to the physiological state of the cell.^{55,56}

C. Intestinal Uptake and Absorption

Dietary zinc absorption occurs along the entire intestinal tract.⁵⁷ From perfusion studies, the jejunum appears to have the highest rate of zinc absorption.⁵⁸ However, as the duodenum is the first site of bolus zinc exposure during the postprandial period, it also likely contributes to zinc absorption.⁵² Kinetic measurements from studies of zinc uptake in the small intestine revealed the presence of both nonsaturable and saturable transport systems with high (>1.8 mM) and low to normal (luminal concentration 0.1–1.8 mM) zinc intakes, respectively.^{58,59} The saturable components may represent zinc transporter activity.

Dietary zinc enters the polarized enterocytes through the apical membrane.⁶⁰ Zip4 mRNA is abundantly expressed throughout the small intestine in both humans and mice.⁶¹ Expression of apically localized intestinal ZIP4 is upregulated under conditions of zinc

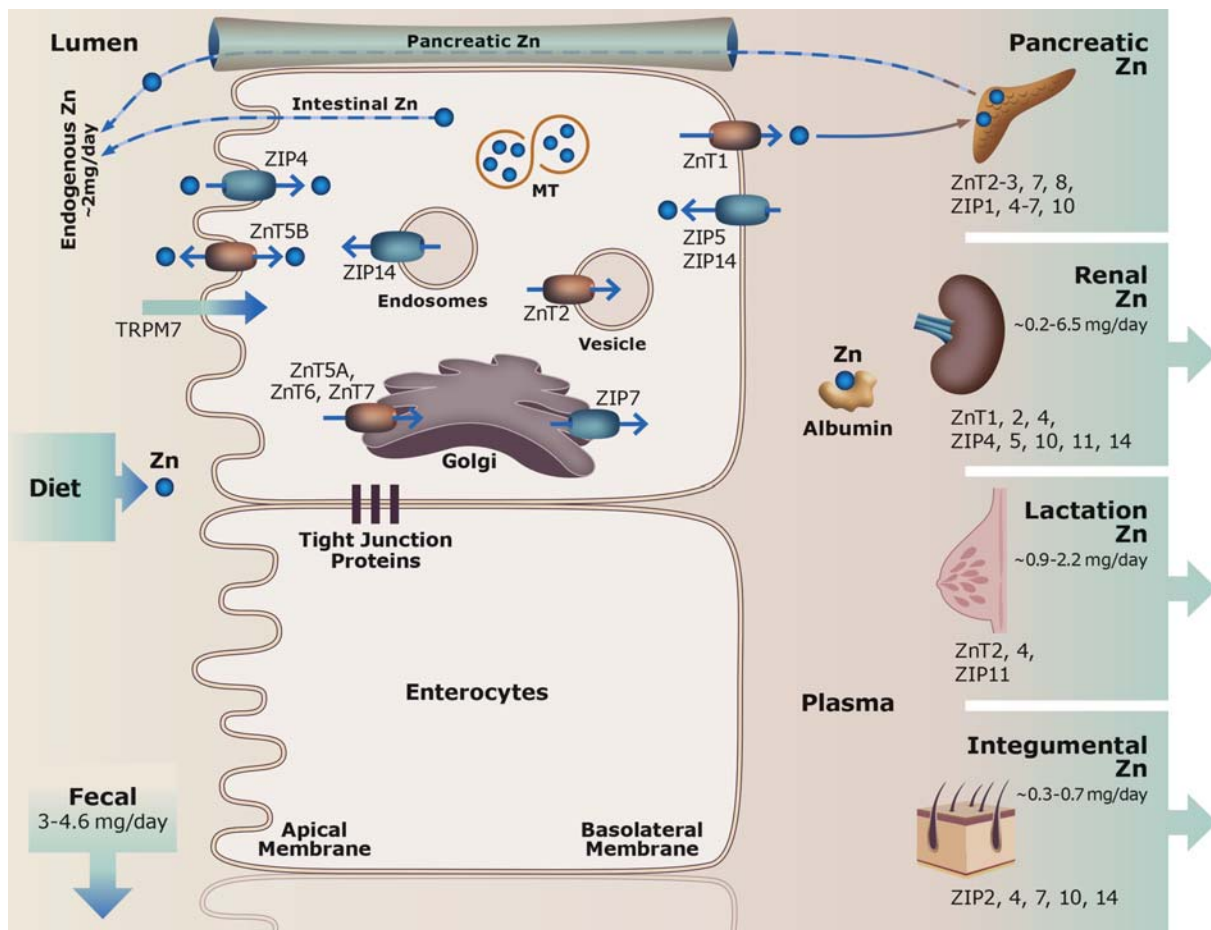


FIGURE 23.3 General pathways of transporter-mediated intestinal zinc absorption and major routes of zinc loss.^{61–63,66–69,71–76}. MT, metallothionein; TRP, transient receptor potential channel; ZIP, Zrt/Irt-like proteins; ZnT, zinc transporter.

sources and are highly dependent upon dietary composition and zinc content. The amount of zinc lost in semen was 0.63 mg per ejaculum, and 5–6 µg/day of zinc are lost during menstruation on a baseline diet containing 15.7 mg Zn/day.⁷¹ Zinc transporters are present in tissues involved with zinc loss (Fig. 23.3).^{70,77}

E. Homeostasis

At the organismal level, zinc homeostasis is dependent on the absorption of exogenous (dietary) zinc, reabsorption of endogenous zinc secreted into the gastrointestinal lumen, and excretion.⁷⁸ Both intestinal zinc absorption and endogenous excretion increase with high dietary zinc intake, whereas endogenous excretion decreases with low dietary zinc intake.

Cellular zinc homeostasis is controlled by cytosolic zinc-binding proteins, such as MTs and zinc-sensing metal-response element (MRE)-binding transcription factor-1 (MTF-1), and by zinc mobilization via transporters. MTs are low molecular weight (6–7 kD), highly

conserved, cysteine-rich proteins that bind metals through thiol clusters.⁷⁹ MTs modulate intracellular zinc through sequestration, mobilization, or release. Excess zinc induces MT expression via the zinc-sensing transcription factor MTF-1, whereas Zn deficiency causes its release from MTs to maintain the size of the intracellular free zinc pool.⁸⁰

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

Biological roles of zinc are grouped into three general functional classes: catalytic; structural; and regulatory. These are not mutually exclusive and can coexist within multistep cellular processes.

A. Catalytic and Structural

Zinc is an integral component of the catalytic sites of enzymes and of structural proteins. The zinc

coordination sphere in proteins is composed primarily of sulfur, oxygen, and nitrogen atoms,⁸¹ and the main zinc coordinating ligands are Cys, His, Asp, Glu, and water. Structural studies of crystallized zinc metalloproteins revealed that His residues are more prevalent in the zinc coordination sphere of catalytic sites, whereas Cys residues are found more frequently in structural proteins.⁸²

Zinc plays a catalytic role in enzymes from all six enzyme classes (oxidoreductase, transferase, hydrolase, lyase, isomerase, ligase).²² In zinc metalloenzymes, zinc can be catalytic or cocatalytic.⁸³ In a catalytic zinc site, the zinc directly participates in the bond-making or bond-breaking step (e.g., CA and carboxypeptidase A). In a cocatalytic zinc site, several metal ions are bound in proximity to one another, with one playing a catalytic role and the others enhancing the catalytic activity (e.g., alkaline phosphatase [AP], phospholipase C, nuclease P1, and leucine aminopeptidase). Zinc binding is a posttranslational modification, and removal of the bound zinc leads to a loss of enzymatic activity. Although how zinc is donated to apometalloenzymes has not been established, new findings suggest the involvement of zinc transporter activity. ZnT5/ZnT7 heterodimers and ZnT7 homooligomers provide zinc ions to activate AP activity.⁸⁴ Similarly, ZIP14-mediated zinc transport was required for activation of the insulin-degrading enzyme (IDE) zinc metallopeptidase to deactivate the insulin receptor (IR).⁸⁵ This is an example of the hybrid catalytic and regulatory function of zinc, as its catalytic activity in IDE modulates the downstream insulin receptor signaling pathway.

Zinc in structural sites participates in the tertiary, quaternary, and quinary structures of proteins, in protein aggregation, and protein–protein, protein–DNA/RNA, and protein–lipid interactions.⁸⁶ ZNFs are the first group of proteins in which a structural role of zinc was described.²³ Currently, 30 types of ZNFs have been identified based on their zinc finger domain structure,⁸⁷ which is maintained by the zinc ion coordinating different combinations of cysteines and histidines. ZNFs have broad cellular distributions and are best known for functioning as transcription factors (e.g., retinoic acid, calcitriol nuclear receptor, krüppel-like factor 4). ZNFs also play roles in numerous physiological processes, including cell proliferation (e.g., ZFP36 RNA-binding protein) and differentiation, (e.g., ZNF750 interaction with chromatin modifiers). Zinc is required for protein–protein interactions.⁸⁶ For example, ZnT8-mediated zinc transport into insulin granules is required for insulin hexamer assembly in pancreatic β -cells, suggesting that intracellular zinc concentrations may control protein complex formation.⁵¹ However, this has not been established for all zinc proteins.

B. Regulatory

Gene expression

In mammals, the best-characterized zinc-regulated gene expression system is binding of the transcription factor MTF-1 to the MRE consensus sequence in the promoter region of the MT1,2 and ZnT1,2 genes.^{88,89} At higher levels of zinc availability, following zinc binding, MTF-1 translocates to the nucleus and activates transcription of the MT1,2 and ZnT1,2 genes. In contrast, MTF-1 binding to MREs located downstream of the translation start site represses ZIP10 expression.⁹⁰ There are ~27 MTF-1 target genes, 13 of which involved in ion metabolism were identified.⁸⁸

Cell signaling

Transient cellular or extracellular changes in zinc concentrations initiate zinc signaling and occur following its release from zinc-binding MTs, its transport across membranes by transporters, or release from vesicles (Fig. 23.4). Extracellular and intracellular zinc ions act as first and second messengers, respectively (reviewed in Ref. 91). Zinc's role as a second messenger was first shown in mast cells.⁹² In response to high-affinity immunoglobulin E receptor activation, the cytosolic zinc concentration increases within minutes, either from loosely bound zinc from MTs or the perinuclear region including the ER and induces NF- κ B activation and cytokine expression (early response). In contrast, inflammation-induced increases in the intracellular zinc concentration provide negative feedback by inhibiting toll-like receptor 4 signaling for NF- κ B activation in adipocytes.⁹³ Zinc transporter-mediated changes in the cytosolic zinc concentration in response to extracellular stimuli (late response) also regulate signaling pathways. Inhibition by phosphatases is the primary mode of action for zinc ions. Zinc transported by ZIP14 regulates proliferation and apoptosis in the liver via inhibition of protein tyrosine phosphatase PTP1B during regeneration and ER stress response, respectively.^{94,95} Similarly, ZIP8-mediated zinc transport from lysosomes regulates T-cell activation via inhibition of calcineurin phosphatase activity.⁹⁶

Zinc may control signaling events via transient interactions with proteins that are not zinc metalloproteins. By transiently binding to allosteric sites or to the ligand-binding site, zinc can modulate the activity of ion channel receptors and subsequent ion-mediated signaling events that control hormone secretion, neurotransmitter release, muscle contraction, volume regulation, growth, motility, and apoptosis.⁴³ Zinc's effect on ion channel receptors could be influenced by both the extracellular and intracellular zinc pools, and regulation by zinc may result in inhibition or activation.

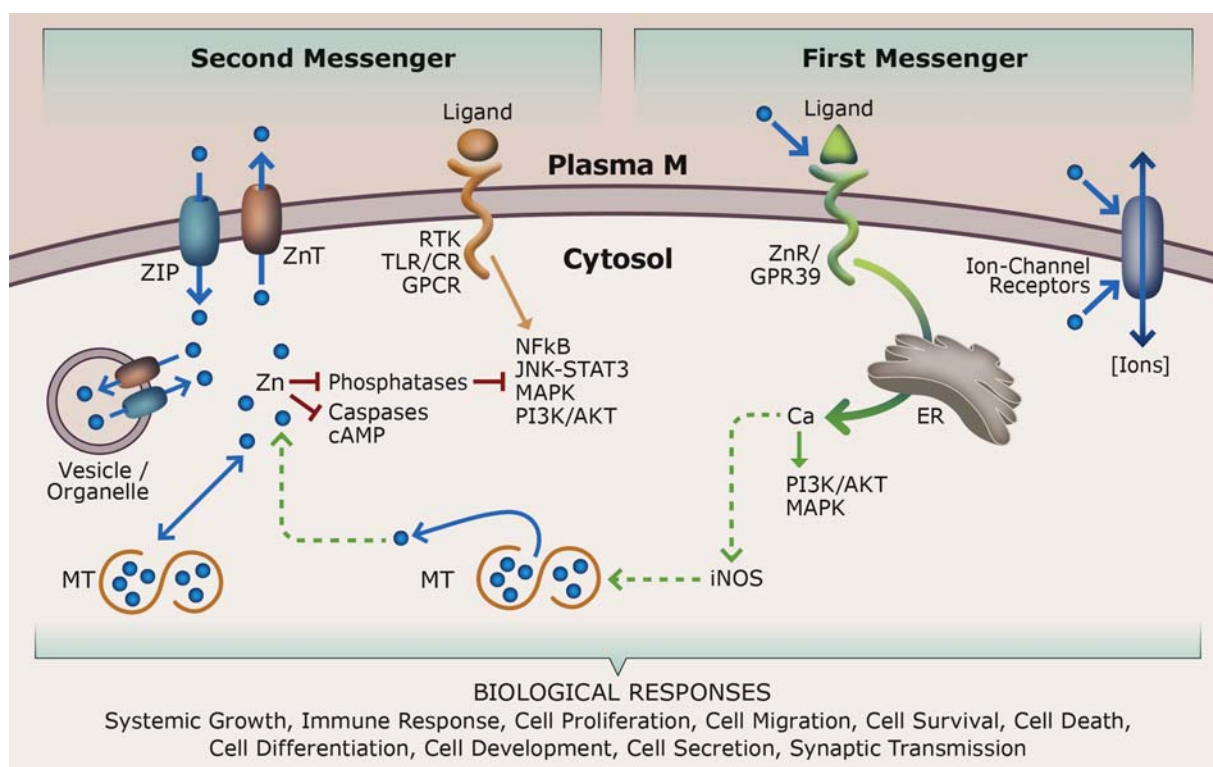


FIGURE 23.4 Summary of zinc-regulated cellular signaling pathways.^{42,91–99} *AKT*, protein kinase B; *cAMP*, cyclic adenosine monophosphate; *CR*, cytokine receptor; *ER*, endoplasmic reticulum; *GPCR*, G protein–coupled receptor; *iNOS*, inducible nitric oxide synthase; *JNK*, c-Jun N-terminal kinase; *MAPK*, mitogen-activated protein kinase; *NF-κB*, nuclear factor kappa-light-chain-enhancer of activated B cells; *PI3K*, phosphoinositide 3-kinase; *Plasma M*, plasma membrane; *RTK*, receptor tyrosine kinase; *STAT3*, signal transducer and activator of transcription 3; *TLR*, toll-like receptor; *ZnR/GPR39*, zinc-sensing receptor/G protein–coupled receptor 39.

Ion channel receptors have a central pore through which ions (potassium [K^+], sodium [Na^+], Ca^{2+} ; hydrogen [H^+]; chloride [Cl^-]) cross biological membranes. Zinc could act as a competitive inhibitor for Ca^{2+} permeability in L-, N-, P/Q-, T-type, and store-operated Ca^{2+} channels in the presence of sufficient Ca^{2+} levels.⁴³ In contrast, both intracellular and extracellular zinc potentiate ATP-sensitive potassium (K_{ATP}) channels to provide negative feedback on glucose-induced insulin secretion in pancreatic cells and protect against ischemic damage in cardiac and neural cells.^{97,98}

ZnR/GPR39 is a zinc-sensing, orphan, plasma membrane G protein–coupled receptor.⁴² Its ligands and zinc-sensing mechanism are not fully established. Ligand-screening studies suggest that exogenous, allosteric zinc binding to ZnR/GPR39 causes calcium release from thapsigargin-sensitive ER stores.⁹⁹ Changes in calcium concentration modulate downstream PI3K/AKT signaling, which controls cell growth and survival. ZnR/GPR39 activity is found in tissues relevant to zinc signaling, including neurons, colon epithelial cells, skin epidermal cells, pancreatic cells, prostate cancer cells, salivary gland cells, and bones.

IV. DIETARY RECOMMENDATIONS

A. Requirements and Considerations

There is no functional reserve or storage site that serves as an alternative source of zinc for systemic zinc homeostasis when the exogenous supply is restricted. Thus, adequate uniform dietary intake of zinc is essential to maintain the size of the whole-body pool within an appropriate range.

Daily intake

Current Daily Reference Intakes (DRIs) for zinc are based upon factorial analyses of the amount of zinc needed to replace endogenous zinc losses and to meet the additional requirements for development, growth, pregnancy, and lactation (Table 23.1).¹⁰⁰ The recommendation for infants aged 0–6 months was determined from the observed mean zinc levels provided through exclusive breastfeeding. For subsequent life stages up to 18 years of age, the primary considerations relate to the amount of zinc consumed for tissue accretion and growth. The Recommended Dietary Allowance (RDA) for adult women and men is 8 and 11 mg/day, respectively. Higher

TABLE 23.1 Dietary Reference Intakes of zinc by life stages.

Life stage	EAR, mg/day		RDA, mg/day		AI, mg/day	UL, mg/day
	Females	Males	Females	Males		
0-6 months	–	–	–	–	2	4
7-12 months	2.5	2.5	3	3	–	5
1-3 years	2.5	2.5	3	3	–	7
4-8 years	4.0	4.0	5	5	–	12
9-13 years	7.0	7.0	8	8	–	23
14-18 years	7.3	8.5	9	11	–	34
≥19 years	6.8	9.4	8	11	–	40
Pregnancy						
14-18 years	10.5	–	12	–	–	34
19-50 years	9.5	–	11	–	–	40
Lactation						
14-18 years	10.9	–	13	–	–	34
19-50 years	10.4	–	12	–	–	40

AI, adequate intake; EAR, estimated average requirement; UL, tolerable upper intake level.

recommendations for pregnant and breastfeeding women are to support developing maternal, embryonic, and fetal tissues and to prevent losses in human milk, respectively. The Institute of Medicine report on DRI provides further details of the rationale, scientific premises, and methodology for these recommendations.¹⁰⁰

Bioavailability

The current DRI of zinc is based upon factorial analyses with a defined fractional absorption for each life stage (e.g., 40% for adult men and 30% for preadolescent children).¹⁰⁰ However, the amount of dietary zinc that is absorbed can vary greatly, from 5% to more than 50% of intake, depending on the presence of other dietary constituents affecting its bioavailability.¹⁰¹ Of particular relevance is phytate (myoinositol hexaphosphate), which forms a tenacious complex with zinc in the intestinal lumen, rendering it insoluble and biologically unavailable.¹⁰² Accordingly, the phytate:zinc molar ratio has been used as an estimate of the quality of a mixed diet as a source of zinc and can be calculated as follows:

$$\text{Phytate: zinc molar ratio} = \frac{\text{Grams of phytate in foods}/660}{\text{Grams of zinc in foods}/65.4}$$

where 660 and 65.4 represent the molecular weights of phytate and zinc, respectively.¹⁰³

A phytate:zinc ratio less than five indicates a good dietary source of zinc, whereas a ratio between 5 and 15

reflects moderate zinc bioavailability.¹⁰⁴ A ratio above 15 indicates poor zinc bioavailability, and consumption of such diet can produce a negative zinc balance.¹⁰⁵ Many plant foods, especially whole-grain cereals and legumes, have phytate:zinc ratios above 15. A mathematical model of the effects of dietary phytate on zinc absorption estimates a daily absorption of ~4.5 mg zinc from intake at the current RDA without phytate.¹⁰⁶ With a gram of phytate, the estimated physiological requirement for adequate absorbable zinc becomes twofold that of a phytate-free diet. Thus, very strict vegetarianism or limited access to low-phytate zinc-rich animal products, such as meat and seafood, is likely to increase the levels of zinc intake that is physiologically adequate.¹⁰⁷

B. Deficiency and Supplementation

Zinc is needed for the normal functioning of essentially all cells in the human body. Due to the wide range of biological processes in which zinc is involved, there is no sign of zinc deficiency that distinguishes it from other malnutrition disorders. Symptoms are rather generalized clinical or metabolic changes that include slow growth velocity, stunting, wasting, impaired immunity, and poor sexual maturation and pregnancy outcomes. Zinc therapy has been effective in improving specific clinical conditions, particularly when targeting populations with high risk of malnutrition. Relevant health outcomes include growth rate, pregnancy outcome, and immune response. In trials with children in developing countries in which diarrhea- and pneumonia-associated morbidity and mortality are leading causes of death, supplemental zinc was observed to have therapeutic and preventive effects.¹⁰⁸

C. Assessment of Status

Serum (or plasma) zinc concentrations

About 17% of the global population is estimated to be at risk of inadequate zinc intake.¹⁰⁹ The prevalence of this risk is as high as 30% in low-income regions such as South Asia. Despite our understanding of the essentiality of zinc and of the biological components for maintaining organismal zinc balance, translation of these findings to public health benefits has been hindered due to the lack of a reliable diagnostic means of zinc deficiency. Serum (or plasma) zinc concentration is the only biomolecular index currently recommended by WHO for estimates of individual zinc status in population-based studies.¹¹⁰ With a normal range of 80–120 µg/dL, current lower cutoffs for zinc adequacy in healthy adult women and men at the morning fasting state are at 70 and 74 µg/dL of serum zinc, respectively.¹¹¹ Other

recommended indices for zinc intake at the population level are percentage of zinc intake below the Estimated Average Requirement and prevalence of stunting in children less than 5 years of age, both of which increase with higher risk of zinc deficiency.¹¹⁰

Plasma zinc concentration has been successfully employed as a biomarker of zinc intake or status under highly controlled experimental settings and remains as the most frequently measured laboratory marker of zinc status in trials and national surveys, e.g., the United States National Health and Nutrition Examination Study. A systematic review of 32 potential zinc indices from 46 studies identified plasma zinc concentrations as the only means to reflect dietary zinc intake in a dose-dependent manner.¹¹² Moreover, plasma zinc correlated best with total body zinc among putative zinc indices, of which others were plasma AP activity and plasma retinol-binding protein levels.¹¹³ Yet, the power of plasma zinc as a valid diagnostic tool for zinc deficiency becomes weak in field settings primarily due to the lack of specificity in its response. Plasma zinc concentrations can drop as much as 19% postprandially¹¹⁴ and fluctuate by ~14% over a day.¹¹⁵ Moreover, acute physiological and psychological stress is sufficient to produce a 33% drop in plasma zinc concentrations, even when daily zinc intakes are sustained above 20 mg.¹¹⁶ Such magnitudes of responses are comparable to those produced by experimental dietary zinc restriction (e.g., ~20% reduction by a nearly complete depletion of dietary zinc over a 10- to 11-day period).^{113,117}

Hypozincemia is a part of acute phase response¹¹⁸ that is intended to limit zinc availability to invasive pathogens in which zinc is also essential. Thus, administration of zinc to correct hypozincemia by active infection may produce a zinc-rich microenvironment more favorable for pathogen growth. Notably, the risks of infection and other inflammatory conditions are generally higher in regions where zinc deficiency is a major public health concern.¹¹⁹ Thus, there is a critical need for an alternative index that provides a zinc-specific mode of response, discriminating zinc deficiency from acute phase reactions.

Hemolysis may serve as a major artifact leading to a misinterpretation from plasma zinc concentrations. Plasma contains 20–30% of the zinc in circulation, whereas its volume constitutes about one-half of whole blood. Typically, the level of plasma zinc is within the order of 1 µg/mL. Erythrocytic zinc concentration is ~1 ng/10⁶ cells.¹²⁰ With a volume of one million red cells at about 0.1 µL, the erythrocytic concentration of zinc is estimated to be ~10 µg/mL. Thus, even a small fraction of erythrocytic zinc leaked into the plasma is expected to produce a relatively large impact on the plasma. Zinc deficiency can increase osmotic fragility of erythrocytes,¹²¹ leading to increased hemolysis, and

thus contamination of the plasma zinc pool by erythrocytic zinc. Such an amount of zinc added to the plasma would potentially negate any losses produced by a negative zinc balance.

Collectively, plasma or serum zinc concentration is a valid indicator of changes in dietary zinc intake under controlled experimental or clinical settings and considered the typical laboratory index for the assessment of zinc status. Discovery of an alternative biomarker for zinc that is refractory to other physiological cues (e.g., inflammation and food intake) and produces amplified responses to changes in zinc remains a major task for the field of zinc nutrition.

Blood cell zinc concentrations

Attempts to characterize cellular zinc contents as potential assessment tools for zinc status have been made, particularly with blood cells due to the noninvasive nature of the sampling procedures. Repression of erythrocytic zinc has been produced by acute experimental zinc depletion in healthy men (0.55 mg zinc/day; 12 days).¹²² Lower erythrocyte zinc content and its correction by dietary zinc supplementation have been measured in elderly subjects with low zinc scores determined by dietary records.¹²³ Positive association of zinc contents in isolated lymphocytes and granulocytes have been observed in zinc-supplemented patients with sickle cell disease¹²⁴ and in healthy men with experimental zinc restriction.¹²⁵ However, the responses of blood cellular zinc levels to dietary intake have not been consistent. Erythrocyte, neutrophil, or platelet zinc contents were spared by extensive dietary zinc restriction protocols despite the repression in plasma zinc concentrations.¹²⁶ No changes in the zinc contents of mononuclear white cells and neutrophils of postmenopausal women were produced by a 6-month dietary restriction regimen.¹²⁷ Similar to the case of plasma or serum zinc, the mixed or moderate responses of cellular zinc concentrations in blood cells are attributable to the efficient regulation of zinc homeostasis at both the cellular and systemic levels. Moreover, cellular zinc contents of immune cells are likely to be affected by responses of zinc transporters to inflammatory cues.^{96,128,129} Thus, as with plasma zinc levels, any changes in blood cellular zinc concentrations with zinc status may be confounded by the presence of inflammation or disease, which would largely limit their value as a diagnostic tool for zinc deficiency.

Functional indicators of zinc

Considerable data from both animal and human studies demonstrate the differential expression of genes related to zinc metabolism (e.g., MT and zinc transporters) in response to fluctuations in dietary zinc intake. Elevated cellular zinc causes MTF-1 to be translocated from the cytosol to the nucleus, where it

activates MT genes via binding to upstream MRE.¹³⁰ Reduced and supplemental zinc intake have been shown to decrease and increase blood MT transcript levels in humans, respectively.^{117,131} Repression of MT by experimental zinc depletion has been measured in buccal RNA samples isolated from oral swabs.¹¹⁷ MT rapidly responds to both acute dietary zinc restriction and supplementation at the protein level in human erythrocytes,^{122,132} which lack transcription machinery. This implies activation of a yet unknown posttranslational mechanism promoting the turnover of MT when zinc becomes limited. Notably, proteolysis has been proposed as a mechanism for the release of MT-bound zinc.¹³³ Conversely, the lack of zinc on MT may cause conformational changes making these proteins more susceptible to erythrocytic proteases.¹³²

Differential expression of zinc transporter genes is fundamental to cellular and systemic zinc homeostasis. Of particular relevance is the ubiquitously expressed zinc exporter, ZnT1, which harbors a functional MRE in its promoter region.¹³⁴ Elevation in ZnT1 transcripts by supplemental zinc in humans has been measured with dried blood spots.¹³¹ This inverse relationship of whole blood ZnT1 mRNA abundance with dietary zinc intake levels was confirmed with a subsequent dietary restriction study with male subjects.¹¹⁷ Other zinc transporter gene responses may potentially serve as biomarkers of zinc status. Repressed whole blood ZIP3 and leukocyte ZIP1 mRNA have been reported as consequences of dietary zinc supplementation in healthy men and women, respectively.^{131,135} Although zinc homeostatic genes are promising diagnostic targets for zinc status, the application of relevant assays to the field requires further establishment of factors including zinc specificity, standards for quantitation, reference values for adequacy, and field-friendly means for sample collection, preservation, and processing.

Other potential zinc indices

Because zinc is essential for the activity of various enzymes, the potential effects of dietary zinc restriction on metalloenzymes have been tested. The enzymatic activity of serum AP, which requires zinc as a catalytic component, is affected by dietary zinc deficiency in animals.^{136,137} The association of AP activity with plasma zinc has been measured in humans,^{138,139} however with a lack of response to marginal dietary zinc restriction.¹²⁶ The validity of other metalloenzymes, for example, α -D-mannosidase and superoxide dismutase (SOD), as indicators of dietary zinc deficiency has been investigated with animal and human models.^{126,140,141} Due to inconsistencies in the results, there is no consensus for any zinc-dependent enzyme as a biomarker for zinc assessment.

Zinc has regulatory roles in immune and inflammatory responses. The recently discovered essentiality of

zinc transporters for the activation of immune cells, such as ZIP8 and ZIP6 for T lymphocytes and dendritic cells, respectively, implies a critical role of zinc in immunity.^{96,128} Zinc induces releases of interleukin-1 β and tumor necrosis factor- α from by peripheral blood mononuclear cells in vitro.^{142,143} Immunoregulatory properties of zinc have also been demonstrated by shifts in cytokine production and T-cell subpopulations resulting from dietary zinc restriction in humans,¹⁴⁴ indicating the potential of cytokines to reflect zinc status. The potency of immune responses to ex vivo immunochallenges, for example, by phytohemagglutinin or lipopolysaccharide, has been positively associated with dietary zinc intakes at both supplemental and inadequate levels.^{117,131} Thus, functional analyses focusing on immune responses of blood cells may serve as a means for identifying dietary zinc depletion, particularly when responses to low zinc are expected to be the opposite of those produced by infection or inflammation.

D. Toxicity and Tolerable Upper Intake Levels

Naturally occurring zinc from food does not appear to cause zinc toxicity. Yet, high zinc intake in supplement forms can produce adverse health effects, including gastric distress, nausea, dizziness, headaches, and loss of appetite. Large doses of zinc above 200 mg can be emetic, and gastrointestinal distress has been reported with doses as low as 50 mg. Accidental intravenous dosage of >7 g of zinc over a 60-hour period has caused death.¹⁴⁵ Zinc toxicity by chronic occupational exposure to zinc (e.g., by galvanization workers) manifested by elevated serum zinc and by lower serum copper and whole-blood hemoglobin levels has been reported.¹⁴⁶ A few cases of severe zinc toxicity from ingestion of nonfood zinc-containing items have been reported. Massive and prolonged ingestion of coins consisting primarily of zinc may cause copper deficiency anemia by zinc toxicosis in patients with metal pica.^{147,148} Denture adhesives with high zinc content, up to 17–34 mg/g of product, have produced hematological and neurological abnormalities attributable to zinc-induced copper deficiency.¹⁴⁹ Notably, hematological and neurological manifestations of zinc toxicity from ingestion of coins or denture cream were improved by removal of the sources of excess zinc and copper supplementation.^{148,149}

The Tolerable Upper Intake Level (UL) is the highest daily intake level expected to pose no risk of any adverse health effects to the majority of the population.¹⁰⁰ The UL of zinc is 40 mg/day for healthy adults. The UL of zinc concerns the levels of ingestion at which zinc adversely affects copper status, reflected by reduced erythrocyte copper–zinc SOD activities.¹⁵⁰ Copper–zinc SOD is a metalloenzyme, which requires zinc as a structural component. Its enzymatic activity is sensitive to copper

status, as copper serves as the catalytic metal cofactor for the protein. Pharmacological doses of oral zinc can be effective in treatment of certain inherited and acquired disorders. Thus, patients can easily exceed the current UL. Implementation of such zinc therapy to patients with higher zinc requirements should not be discouraged due to the UL, which is set for the general population with recommended intakes at the current RDA.

V. ISSUES RELATED TO ZINC IN HUMAN HEALTH

A. Genetic Variants of Metal Dyshomeostasis

Genetic mutations that impair zinc handling can lead to inherited zinc deficiency disorders requiring a substantially elevated daily zinc intake. AE, a hereditary, autosomal-recessive, metabolic disorder is attributed to abnormal zinc absorption due to mutations in the *SLC39A4* gene.¹⁵¹ Manifestations of severe zinc deficiency due to loss of ZIP4 activity include dermatological lesions, diarrhea, lack of weight gain, and impairments in the immune and reproductive systems.¹⁵² AE symptoms can be corrected by daily oral administration of zinc doses that may be an order of magnitude higher than recommended for healthy individuals.¹⁵² This suggests the presence of ZIP4-independent paracellular or low-affinity transport mechanisms for zinc absorption. TRPM7, a potential channel for transport of divalent cations, including zinc and magnesium, from the intestinal lumen into absorptive cells may be particularly relevant.⁶³

Patients with inherited systemic copper buildup, known as Wilson's disease, benefit from higher zinc intake levels. Wilson's disease mutations impair the activity of ATP7B, a copper transporter integral to the biliary excretion of excess body copper.¹⁵³ The US Food and Drug Administration has approved multiple oral administrations of high levels of zinc acetate as maintenance therapy for patients with Wilson's disease.¹⁵⁴ Although this is similar to zinc therapy for patients with AE, the mode of action in Wilson's disease is to correct the inherited copper overload by harnessing the inhibitory effects of high zinc on copper absorption.

B. Malabsorption Disorders

Endogenous factors affecting zinc absorption or excretion may shift the zinc balance. Of relevance are malabsorption disorders or pathological conditions of hyperzincurea, either of which lead to a negative zinc balance. Diarrhea may impair absorption and promote loss of intestinal zinc, thus precipitating secondary zinc deficiency.¹⁵⁵ Other gastrointestinal disorders, including short bowel syndrome and inflammatory bowel diseases

(e.g., Crohn's and celiac diseases), have been associated with low serum zinc levels.¹⁵⁵ Gastric and intestinal bypass surgeries promoting malabsorption are increasingly common for bariatric or weight-loss purposes.¹⁵⁶ Persistent decreases in plasma zinc have been reported after 12 and 24 months of duodenal switch and Roux-en-Y gastric bypass surgeries.¹⁵⁷ Furthermore, oral zinc tolerance tests indicate impaired dietary zinc assimilation following Roux-en-Y-gastric bypass surgery.¹⁵⁸ Notably, lower plasma zinc concentrations have been noted in children and adolescents with obesity,¹⁵⁹ implying risks of preexisting zinc deficiency. Oral zinc supplementation in forms of zinc gluconate, sulfate, or acetate, providing 8–15 mg of elemental zinc, has been recommended for treatment of postoperative zinc deficiency in patients of bariatric surgery.¹⁶⁰

C. Alcoholism and Diabetes

Hyperzincuria has been associated with hypozincemia under pathologic conditions such as alcoholic liver disease and type 1 and type 2 diabetes.^{161,162} Albumin produced by the liver is the primary carrier protein for plasma zinc. Thus, hypoalbuminemia by alcoholic cirrhosis may be related to the accompanying excess zincuria. Hyperzincuria in diabetics tends to worsen with the severity of the disease,¹⁶² indicating a potential involvement of zinc depletion in the progression of diabetic symptoms. A recent systematic metaanalysis of 25 zinc trials indicates the potential benefits of zinc supplementation on glycemic control and lipid parameters in diabetics.¹⁶³ The exact molecular mechanisms of the excess zinc excretion and the roles of supplemental zinc in diabetes remain uncertain. The latter could relate to the insulin-like effect of zinc¹⁶⁴ and its regulatory functions on insulin receptor signaling.¹⁶⁵ Notably, a pancreatic β cell-specific zinc transporter, ZnT8, was implicated in insulin processing and secretion and the pathogenesis of diabetes.¹⁶⁶ The ZnT8 gene, *SLC30A8*, was identified in genome-wide association studies as a risk locus for type 2 diabetes¹⁶⁷ and was also characterized as an autoantigen gene for type 1 diabetes.¹⁶⁸

D. Cancer

Changes in intracellular zinc levels and zinc transporter activities have been associated with cancer pathogenesis and progression.⁷⁷ ZIP4 overexpression, which causes hyperaccumulation of tissue zinc and enhances cancer progression in animal models, is characteristic of human pancreatic tissues.¹⁶⁹ Directly targeting ZIP4 expression by RNA interference represses pancreatic cancer growth and improves survival of mice carrying pancreatic xenografts.¹⁷⁰ Aberrant ZIP6 and ZIP10 expression has been identified in breast cancer and renal

carcinoma cells, respectively.^{171,172} A *ZIP11* mutation (rs11077654) was significantly associated with survival in bladder cancer in a candidate gene association study focused on variants of all 24 zinc transporter genes.¹⁷³

Biochemically, zinc protects against oxidative stress, which may impair DNA integrity and lead to carcinogenic mutations.¹⁷⁴ Peripheral blood mononuclear cells exhibited DNA damage resulting from mild zinc deficiency, which was reversed by dietary zinc repletion.¹⁷⁵ Dietary zinc modulates the exosomal contents of cancer-related miRNA.¹¹⁷ Among the most zinc-responsive miRNA, miR-204 and miR-296-5p have been characterized as regulatory components of oncogenes associated with the prognosis of gastric, colorectal, and prostate cancer.^{176–178} Collectively, dysregulated zinc homeostasis by zinc malnutrition or aberrant zinc transporter expression may have pathogenic implications for cancer development and progression. Zinc metabolism and related genes may, therefore, be potential therapeutic targets for cancer treatment.

VI. CONCLUSION

Zinc, in different forms such as ionic or protein-bound, serves as a fundamental component for diverse biological processes essential to human life. While dietary zinc deficiency remains a major public health problem particularly prevalent in low-income nations, a reliable diagnostic means for zinc status is lacking. Moreover, many human diseases have been linked to dysfunctional zinc transporters and consequent zinc dyshomeostasis (acquired zinc deficiency) in both developing and industrialized countries. Manifestations of dietary and acquired zinc deficiencies are similar; however, efficacy of zinc therapy can differ by the etiology of deficiency. Thus, further research for the discovery of biomarkers for these zinc deficiency disorders, and development of therapeutic approaches effectively correcting zinc dyshomeostasis and its associated symptoms, due to either inadequate intake or genetic variation, is warranted.

RESEARCH GAPS

- Microbiome: Changes to the proximal intestine in environmental enteropathy influence the nature and amount of zinc delivered into the colon. Microbiota-generated phytases might improve colonic zinc absorption. In rodents, colonic zinc absorption is enhanced when microbially fermentable carbohydrates (specifically resistant starch) are consumed; it is unclear whether this occurs in humans.
- Metallomics: Using metallomics, accurate reference values for normal distributions and concentrations in tissues and cells of metal ions, including zinc, could be established and be used to understand changes relevant to disease.
- Human Genetics: Hypozincemia due to differential expression of zinc transporter genes has been implicated in innate host defense against pathogenic microorganisms. Likewise, genetic variants of zinc homeostatic genes could confer metabolic advantages (e.g., by limiting zinc flux toward the invading pathogen's microenvironment). Development and phenotyping of experimental models harboring polymorphic zinc transporter gene variants introduced by gene-editing techniques would be particularly relevant.

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COPPER

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SUMMARY

Intensive research over the past several years has demonstrated (or predicted) that pathological perturbations in copper homeostasis may contribute to the development of ischemic heart disease and neurodegenerative and connective tissue disorders in humans. Copper was also shown to facilitate the host response to bacterial infection. Emerging evidence links copper to common lipid-related pathologies such as nonalcoholic fatty liver disease. Also, research on copper chaperones continues to reveal novel roles for these intracellular proteins. Despite these major achievements, important issues remain unresolved. Sensitive and specific copper nutrition–related biomarkers need to be identified, to accelerate research on copper metabolism in humans. Moreover, recent experimental evidence suggests that frank copper deficiency in the United States is more common than traditionally thought and that the copper RDA for adults should be increased. These and other unresolved issues related to copper nutrition will undoubtedly be a focus for future investigation.

Keywords: Anemia; ATOX1; ATP7A; ATP7B; Ceruloplasmin; CTR1; Menkes disease; Metallothionein; Superoxide dismutase; Wilson's disease.

I. INTRODUCTION

A. Background

Almost two centuries ago, it was recognized that copper was a normal constituent of marine invertebrates and plants, but the presence of copper in animal tissues was not firmly established until 1912 when Bodansky documented that the human brain contains copper.¹ Also around this time, it was established that copper was required (in liver extracts), along with iron, to correct experimental anemia in rats.² A pathological role for copper in humans was initially recognized when the etiology of Wilson's disease³ (WD) (a copper-overload disorder) was first described in the 1940s.^{4,5} Interestingly, however, it was not appreciated until over 45 years later⁶ that WD was caused by mutations in the gene encoding ATPase copper-transporting beta (ATP7B),^{7,8} a copper transporter that facilitates biliary excretion of excess copper. Also, as early as the 1930s, an association between copper depletion and development of anemia in humans was suspected. Experimental proof of this association came 40 years later when Menkes disease (MD)⁹ was linked to

defective intestinal copper absorption and systemic copper deficiency,^{10,11} exemplifying the importance of copper in human health and development. The causative gene in MD, which was discovered two decades later,^{12–14} encodes ATPase copper-transporting alpha (ATP7A), which is required for intestinal copper absorption. Based at least in part on the discovery and characterization of WD and MD, it is now universally recognized that copper is an essential nutrient for humans and that it can be toxic when in excess. It is present in body fluids and tissues at $\mu\text{g/g}$ (ppm) to ng/g (ppb) concentrations. Given that copper depletion and overload cause significant physiological perturbations, tightly controlled biological systems for regulating copper absorption, transport, storage, utilization, and excretion have evolved in humans (and other mammals). These molecular regulatory systems and other pertinent aspects of copper nutrition will be considered in detail in this chapter.

B. Copper Structure and Function

Copper (atomic number, 29; atomic mass, ~ 63.5 Da) exists in two predominant oxidation states in biological

systems, Cu^{2+} (cupric) and Cu^+ (cuprous). Copper is redox active, and it shifts between these oxidation states during copper-mediated enzyme catalysis. Protein-bound copper (in cuproenzymes) facilitates single-electron transfer reactions involving oxygen. Seven radioisotopes of copper exist in nature. The two with the longest half-lives (^{67}Cu , ~ 70 h and ^{64}Cu , ~ 13 h) are utilized to track copper utilization in experimental animals; however, their usefulness in humans is limited by the short half-lives and dangerously high energy emissions. Instead, two stable isotopes (^{63}Cu and ^{65}Cu) are most frequently utilized to track absorption, utilization, excretion, and turnover of copper in humans.¹⁵ Copper isotope ratios in experimental subjects are then assessed by inductively coupled plasma mass spectrometry or thermal ionization mass spectrometry. Cuprous copper (Cu^+) is insoluble in aqueous solutions (e.g., biological fluids) and is thus frequently complexed with other molecules.¹⁶ Excess (unbound) copper produces potentially damaging oxygen free radicals via Fenton- and Haber–Weiss-type reactions; fortunately, then, the reactivity of intra- and extracellular copper at physiological levels is mitigated via nonspecific (electrostatic) interactions

with amino acid side chains in various proteins and possibly other biological molecules.

II. COPPER METABOLISM

A. Copper Digestion, Transport, Storage, and Excretion

Overview

Typical copper intake from a normal mixed diet is ~ 1.2 – 1.3 mg/d (Fig. 24.1), of which ~ 0.8 mg/d is absorbed by the proximal small intestine. Excretion of excess copper occurs mainly via the bile (~ 0.4 mg/d), after transport (export) of copper from hepatocytes into bile canaliculi by the ATP7B copper transporter. Total fecal losses from unabsorbed dietary copper plus excreted biliary copper equate to ~ 1 mg/d. In the liver, copper is incorporated into ceruloplasmin (CP) (a secreted protein involved in iron metabolism) and other cuproenzymes. Body stores of copper are quite limited, although small amounts of copper may be stored within cells bound to metallothionein I and II (MTI/II). MTs are small, cysteine-rich proteins that bind to various heavy

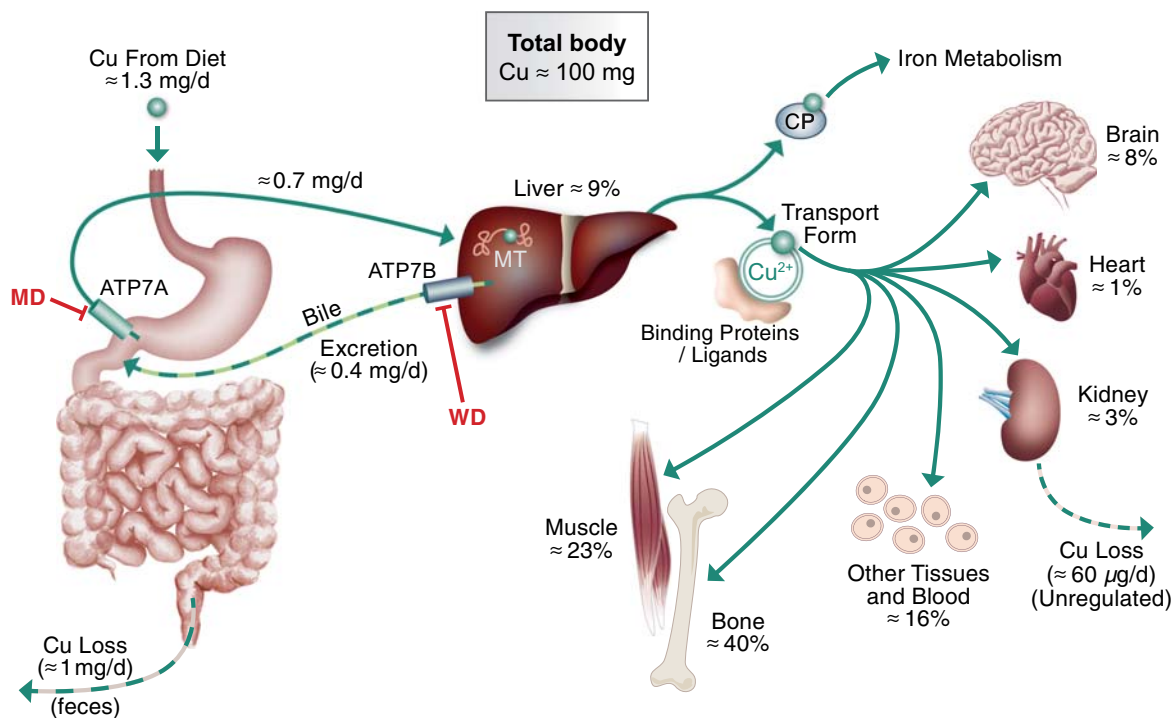


FIGURE 24.1 Overall body copper homeostasis in humans. The major regulatory mechanisms that control body copper levels are depicted. Copper-transporting ATPases play critical roles in intestinal absorption (ATPase copper-transporting alpha; ATP7A) and biliary excretion (ATPase copper-transporting beta; ATP7B), and their dysfunctionality underlies Menkes disease (MD) and Wilson's disease (WD), respectively (indicated in red). After absorption, most dietary copper enters the liver where it may be stored in metallothionein, incorporated into ceruloplasmin (CP) or distributed in the blood to various organs. Circulating CP participates in iron metabolism, while atomic cupric (Cu^{2+}) copper bound to various serum ligands functions as transport copper. Numbers beside or under various organs indicate the approximate percent of body copper that is present in that organ/tissue.

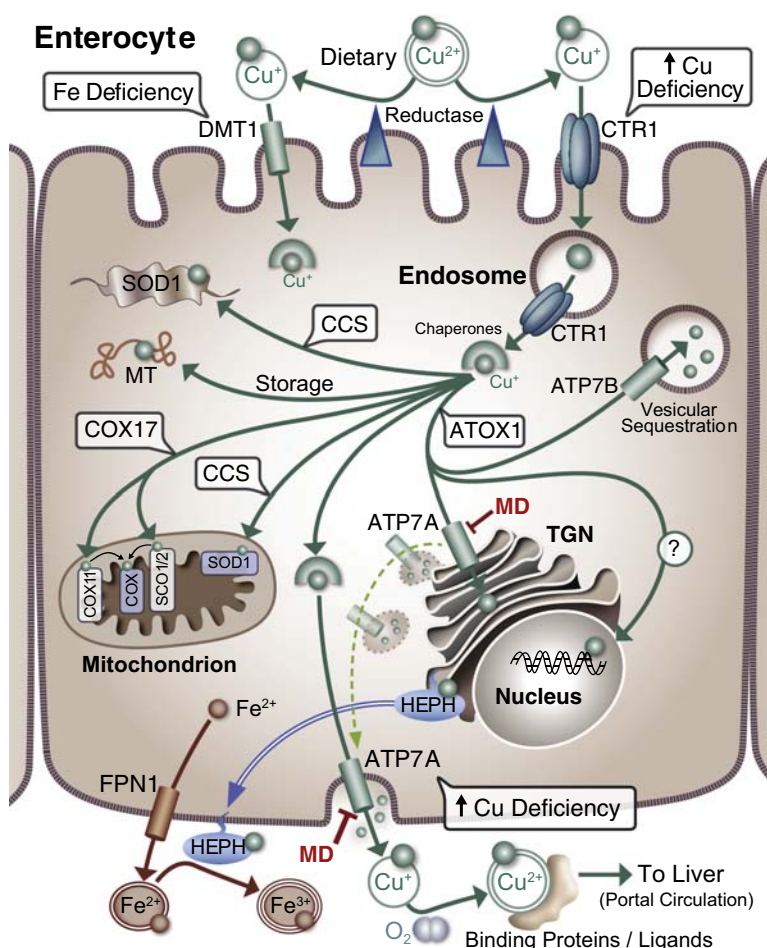


FIGURE 24.2 Copper homeostasis in enterocytes. An enterocyte is depicted showing the processes that are involved in enteral copper absorption. Dietary copper is reduced and then transported across the apical surface by copper transporter 1 (CTR1). Divalent metal-ion transporter 1 (DMT1) may import copper during iron deprivation. Once in cells, Cu^+ is bound to chaperones for distribution: copper chaperone for SOD (CCS) delivers copper to cytosolic and mitochondrial superoxide dismutase 1 (SOD1); antioxidant protein 1 (ATOX1) delivers copper to ATPase copper-transporting alpha (ATP7A) in the *trans*-Golgi network (TGN) and also possibly into the nucleus (where ATOX1 may also act as a transcription factor); and cytochrome *c* oxidase copper chaperone 17 (COX17) delivers copper to the mitochondrion. Excess copper can be bound to metallothionein (MT). Intracellular copper may be transported into vesicles by ATP7B for possible storage. Copper is distributed within mitochondria by COX11 and SCO cytochrome C oxidase assembly proteins 1 and 2 (SCO1/2) to support biosynthesis of cytochrome C oxidase. In the TGN, copper is incorporated into cuproenzymes, including the multicopper ferroxidase hephaestin (HEPH), which oxidizes ferrous iron that exits cells. When copper is in excess, ATP7A-containing vesicles traffic to the basolateral membrane, thus facilitating copper efflux. Cu^+ that exits cells is oxidized (probably) by O_2 and then bound by albumin and α -2 macroglobulin for transport to the liver. CTR1 and ATP7A expression/activity may be increased during copper deficiency. Dysfunctional ATP7A (shown in red) underlies Menkes disease (MD) in humans.

metal ions such as copper (or zinc) with high affinity (as recently reviewed¹⁷). These proteins protect cells against free copper-induced oxidative stress when copper efflux systems are inadequate (e.g., in neonates or in cholestasis) or after acute (accidental) ingestion of excess copper. Although most copper in the blood is associated with CP, atomic copper, which exits the liver and binds to serum proteins and other ligands, functions as “transport copper” to supply body tissues and cells (Fig. 24.1). Homeostatic control of copper levels occurs via coupling of intestinal absorption with hepatic copper excretion in bile. So, when dietary copper intake is high and intestinal copper absorption is elevated, biliary copper excretion is enhanced, and conversely, low intake and lower intestinal absorption leads to a repression of copper excretion.¹⁸

Transport of enteral copper into enterocytes

Dietary (and supplemental) copper, along with endogenous copper present in gastric, intestinal, and pancreatic secretions, constitutes the intestinal copper pool. Biliary copper secreted into the gut lumen is complexed with bile salts, which may limit (re)absorption. Prior to transport across the brush-border membrane (BBM) into enterocytes, dietary copper must be reduced from the cupric (Cu^{2+}) to the cuprous (Cu^+) state (Fig. 24.2). Several cupric reductases have been identified in experimental animal models, including cytochrome *b* [558] ferric/cupric reductase,¹⁹ six transmembrane epithelial antigen of the prostate (STEAP) metalloreductases,^{20,21} and cytochrome B reductase 1 (CYBRD1)²²; however, the relative functional contribution of each in the intestine

remains unclear.^{23,24} Additional reducing equivalents could be provided by dietary factors (e.g., ascorbic acid) or endogenous GI secretions (e.g., gastric acid), as with iron.²⁴ In any case, as revealed by studies in laboratory mice, reduced (cuprous) copper is likely transported into enterocytes via copper transporter 1 (CTR1 or SLC31A1).²⁵ The relative contribution of CTR1 to human intestinal copper absorption has, however, not been clearly defined. Moreover, given that copper is trapped within enterocytes in an unusable form when intestinal CTR1 is ablated²⁶ and since CTR1 has been localized on the basolateral membrane (BLM) by at least one group,²⁷ other non-CTR1-dependent copper intake mechanisms have been recently proposed.²⁸ Divalent metal ion transporter 1 (DMT1 or SLC11A2) may also import enteral copper,²⁹ but this is probably only physiologically relevant during states of iron deficiency, when DMT1 protein levels are very high.^{30–32}

Intracellular copper homeostasis within enterocytes

After import across the BBM, copper interacts with chaperone (or binding) proteins that deliver it to various intracellular locations for incorporation into apocuproproteins. These copper–ligand interactions prevent reaction of copper with oxygen, thus mitigating the production of superoxide radicals. It was recently suggested that the initial copper-interacting molecule (after import via CTR1) is intracellular glutathione (GSH) and that the Cu:GSH complex then passes copper along to protein chaperones.^{33–35} One caveat to this hypothesis, however, is that most supporting evidence comes from in vitro cell culture models. Other, perhaps better-established copper chaperones have been described, including cytochrome *c* oxidase copper chaperone 17 (COX17), which delivers the metal to mitochondria to support cytochrome C oxidase (COX) synthesis. Antioxidant 1 copper chaperone (ATOX1) delivers copper to membrane-embedded ATP7A in the *trans*-Golgi network (TGN) to support cuproenzyme synthesis. ATOX1 has also been shown to act as a copper-dependent transcription factor that functionally *trans*-activates the SOD3 gene.^{36,37} It was further shown to interact with several proteins capable of direct interaction with RNA and DNA, so its role in regulation of gene expression may be broader.³⁸ Logically, ATOX1 may also function in nuclear copper delivery, as has been demonstrated in vitro.³⁹ Copper is an essential component of chromatin and is known to accumulate preferentially in heterochromatic regions,⁴⁰ where it may be involved in DNA folding and repair.⁴¹ Given its redox potential, some have suggested that nuclear copper accumulation increases DNA damage and that copper may thus play a pathological role in tumorigenesis and even perhaps in neurodegenerative disorders.⁴² Another copper-binding

protein is copper chaperone for SOD (CCS), which delivers copper to support production of Cu/Zn-superoxide dismutase 1 (SOD1) in the cytosol and mitochondria (Fig. 24.2). The necessity of these proteins to maintain copper homeostasis has been established mainly by studies in knockout (KO) mice.⁴³ Additional chaperone proteins deliver copper to components of the electron transport chain machinery within mitochondria, including cytochrome *c* oxidase copper chaperone 11 (COX11) and SCO cytochrome C oxidase assembly proteins 1 and 2 (SCO1/2) (Fig. 24.2). Interestingly, SCO2 mutations in humans are associated with encephalopathy and cardiovascular dysfunction due to impaired respiratory chain function.⁴⁴ More details about these copper chaperones can be found in recent reviews on this topic.^{45,46} Intracellular (excess) copper may also be bound in cells by MT (as described above).

Copper efflux from enterocytes

Excess copper may be transported out of enterocytes by the copper efflux transporter ATP7A, which translocates within TGN-derived vesicles to the BLM when intracellular copper is elevated. Export is probably mainly by exocytosis, but small amounts of ATP7A on the BLM could also function in the transport of intracellular (cytosolic) copper (probably delivered by a chaperone) out of enterocytes (Fig. 24.2). Once copper exits enterocytes, the oxidizing environment of the interstitial fluids presumably converts cuprous copper (Cu^+) to cupric copper (Cu^{2+}), which binds to albumin (ABM) or α_2 -macroglobulin (and possibly other minor ligands) for delivery in the portal blood to the liver.^{24,47}

Transport and transfer

Most, but not all, dietary copper derived from the gut is taken up by the liver. Copper (Cu^{2+}) has to again be reduced (possibly by STEAP proteins) prior to import by CTR1 into the liver (Fig. 24.3).⁴⁸ In liver cells, as in enterocytes, copper binds to (probably the same) chaperones and is distributed to various intracellular compartments to support cuproenzyme synthesis. ATP7B pumps copper into the TGN where it is incorporated into CP and other cuproproteins. Copper exits the liver as an integral component of the CP ferroxidase protein (constituting 40%–70% of serum copper)⁴⁹ and as atomic copper (by an unknown transporter). It is the latter form, after binding to various serum ligands, that functions as transport copper to supply body tissues and cells. Interestingly, copper contained within *holo*-CP can be exchanged with cell surface cupric reductases and transporters (resulting in the formation of *apo*-CP),⁵⁰ but the physiological significance of this observation is currently unclear.

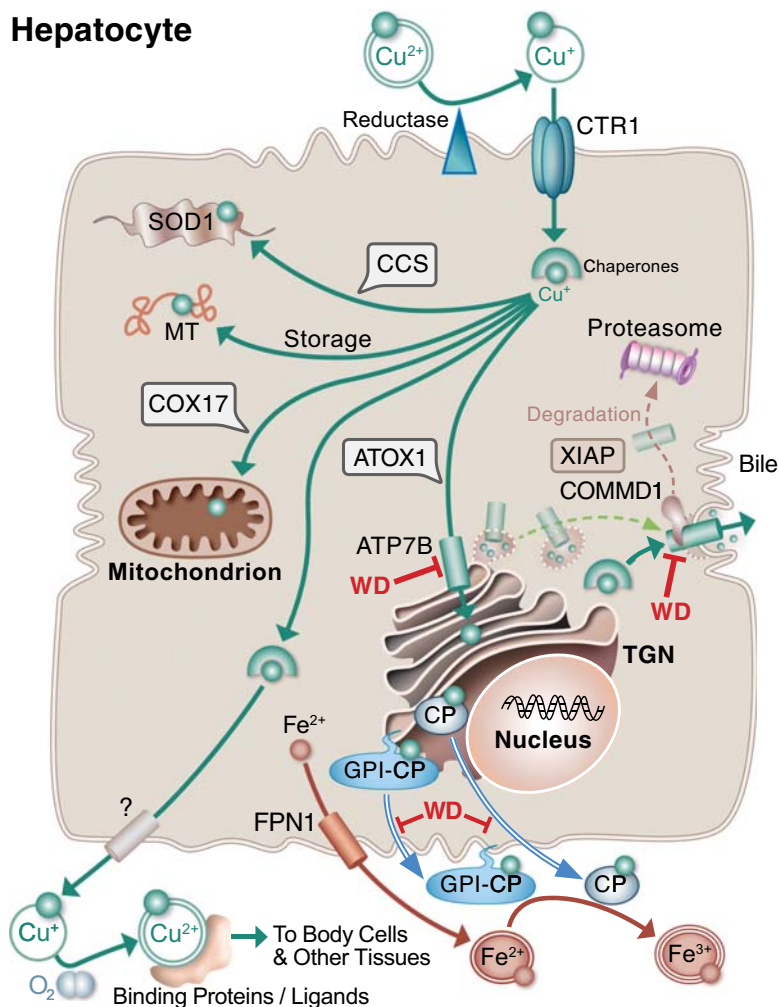


FIGURE 24.3 Copper homeostasis in hepatocytes. A hepatocyte is depicted with the major proteins involved in copper homeostasis in the liver. Copper in portal blood is reduced before being transported into the cell by CTR1. As in enterocytes, chaperones distribute copper within the cell, and excess copper may be stored bound to metallothionein (MT). ATPase copper-transporting beta (ATP7B) pumps copper into the *trans*-Golgi network (TGN) for incorporation into ceruloplasmin (CP) and other cuproenzymes. Soluble CP, and the membrane-anchored form (GPI-CP), oxidizes ferrous iron released by some tissues and cells, including hepatocytes. When copper is in excess, ATP7B-containing vesicles translocate to the canalicular membrane, thus facilitating copper excretion in the bile. ATP7B protein levels (and copper excretion) are controlled by copper metabolism domain containing 1 (COMMD1) and X-linked inhibitor of apoptosis (XIAP) via the proteasome pathway. Atomic copper also presumably exits hepatocytes, via an unknown transport process, and is then (spontaneously) oxidized and bound to various serum ligands for distribution to body cells and tissues. Dysfunctional ATP7B (shown in red) underlies Wilson's disease (WD) in humans.

Excretion

Excess copper in hepatocytes stimulates the translocation of ATP7B-containing vesicles from the TGN to the canalicular membrane (Fig. 24.3), facilitating exocytotic biliary excretion.⁵¹ Conceivably, ATP7B present on the canalicular membrane (after vesicular fusion) could also export excess cytosolic copper. Some biliary copper may be reabsorbed, but excess biliary and unabsorbed dietary and endogenous copper are lost via the feces. Biliary copper secretion is less efficient during the fetal and neonatal periods, increasing the risk of copper toxicity. Moreover, pathologic hepatic copper

accumulation was documented in intrahepatic cholestasis of childhood; chelation therapy was recommended if copper levels exceed 250 µg/g.⁵² Cholestasis in the elderly may also increase liver copper levels, elevating the likelihood of toxicity.

Storage

Total body copper in adults ranges from 50 to 120 mg. Humans do not store appreciable amounts of copper; thus, frequent dietary consumption is a necessity. Tissue copper levels likely reflect quantities of cuproenzymes.

B. Metabolic Regulation

Adaptive mechanisms have evolved in mammals that protect against copper depletion and copper excess. For example, absorption of dietary copper adapts to body requirements. Under physiological conditions, ~10% of dietary copper is absorbed,⁵³ but percent absorption increases when intake levels are low⁵⁴ (and vice versa). Also, when intake is high, copper may be sequestered in MT in enterocytes (equating to a mucosal block) and biliary excretion increases. These physiological adaptations are ineffective when daily copper intake is less than ~0.4 mg/d¹⁸; fortunately, this is not a concern for most individuals since the average daily copper intake in the United States is much higher (*i.e.*, ~1.2 mg/d). Moreover, when intake is excessive, regulatory mechanisms are slower to respond, so short-term copper imbalance is possible.

III. COPPER: ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Posttranslational Control of Copper Homeostasis

The dietary requirement for copper is undoubtedly influenced by individual genetic variation that influences gene expression levels and protein functionality. Molecular mechanisms governing copper homeostasis, however, probably do not involve modulation of gene transcription. To date, only one copper-responsive *trans*-acting factor has been identified, ATOX1, which translocated to the nucleus when intracellular copper levels increased.³⁶ Rather, copper homeostasis seems to be mostly regulated posttranscriptionally.⁵⁵ For example, steady-state mRNA levels for many copper transporters (*e.g.*, CTR1, ATP7A, ATP7B) do not change with variations in dietary copper intake; instead, regulation of copper intake and efflux seems to be regulated predominantly by copper-responsive protein trafficking.⁵⁶

B. Cuproenzymes and Copper-Binding Proteins

Overview

Copper has no known structural roles in proteins (as, for example, zinc may play in the Cu/Zn SOD1/3 proteins). Copper does, however, interact with certain proteins that are involved with its transport across biological membranes, intracellular disposition, and storage. Collectively, these are referred to as copper-binding proteins, as described below and discussed throughout this chapter. From a functional perspective, copper plays a predominant role in mammalian

physiology as an enzymatic cofactor (*i.e.*, a prosthetic group) for a host of cuproenzymes (Table 24.1). These enzymes are mostly oxidases that utilize oxidized (Cu^{2+}) or reduced (Cu^{+}) copper to facilitate single-electron transfer reactions between a substrate and molecular oxygen. The copper centers bind oxygen and release water, superoxide, hydrogen peroxide, or other organic reaction by-products. These cuproproteins are involved in fundamental biological processes such as ATP production, stabilization of collagen and bone, iron metabolism, dismutation of the free radical superoxide, activation of biogenic peptides, and neurotransmitter synthesis. Detailed descriptions of these proteins have been recently published elsewhere.^{57,58}

Mono- and diamine oxidases

Copper-dependent mono- and diamine oxidases catalyze the oxidative conversion of amines to aldehydes in the presence of (redox-active) copper and a quinone cofactor. These proteins function in the oxidative deamination of biogenic amines. For example, amine oxidases catabolize histamine, tyramine, and polyamines in plasma. Inactivation of polyamines may limit cell growth and has possible therapeutic relevance to cancer.⁵⁹ They also function in intracellular signaling via production of hydrogen peroxide.⁶⁰ Amine oxidase copper containing 3 (or vascular adhesion protein 1), a single-pass, type II membrane protein, may influence leukocyte trafficking^{61,62} and thus play a role in the immune response. Other examples are the monoamine oxidases A and B (MAOA/B), which are copper-dependent, flavocoenzymes anchored in the outer mitochondrial membrane that function in the metabolism of neuroactive and vasoactive amines in the central nervous system (CNS) and peripheral tissues. MAOs, for example, catalyze the deamination of the (monoamine) neurotransmitters serotonin, norepinephrine, tyramine, and dopamine in mitochondria. Dysfunction of MAOs has been linked to attention deficit disorder, depression, substance abuse, and delayed sexual maturation.⁶³ Moreover, selective MAOB inhibitors (rasagiline, safinamide) are currently used to treat Parkinson's disease.⁶⁴ Another amine oxidase, AOC1, which is secreted into the extracellular space, inactivates histamine, putrescine, and spermidine, which are biomolecules that function in allergic and immune responses, cell proliferation, tissue differentiation, tumor formation, and possibly apoptosis.

Lysyl oxidase

An additional (secreted) copper-dependent amine oxidase, lysyl oxidase (LOX), functions in the oxidative deamination of specific lysine residues in extracellular matrix proteins. This reaction initiates cross-linking (and thus stabilization) of elastin and collagen fibers

and is thus important for the formation of bone, blood vessels, skin, lungs, and teeth. Recently, LOX has been linked to human cancers since it cross-links the extracellular matrix in local tumor microenvironments. One recent investigation found that LOX traps EGF receptors at the cell surface, thus promoting tumor progression.⁶⁵

Peptidylglycine α -amidating monooxygenase

Peptidylglycine α -amidating monooxygenase (PAM) is an evolutionarily conserved cuproenzyme that converts precursor peptides to their bioactive forms. PAM is a membrane-associated protein found in cytoplasmic or secretory vesicles, with highest expression in cells of the nervous and endocrine (e.g., in the pituitary) systems. Copper and ascorbate are required as cofactors by a monooxygenase domain that converts C-terminal glycine residues to hydroxyglycyl residues, which are then hydrolyzed by a lyase domain to produce α -amidated (bioactivated) peptides.⁶⁶ Target peptides include calcitonin, adrenomedullin, vasopressin, vasoactive intestinal peptide, α -melanocyte-stimulating hormone, cholecystokinin, gastrin, neuropeptide Y, and substance P.⁴⁷ Mice lacking PAM show no detectable peptide amidation activity and die in utero,⁶⁷ underscoring the importance of this protein in mammalian physiology. Interestingly, PAM was also recently linked to defects in ciliogenesis (assembly of cilia).⁶⁸

Multicopper ferroxidases

Multicopper ferroxidases (MCFs) CP and hephaestin (HEPH) are additional example cuproenzymes. CP, literally meaning sky-blue plasma protein, is a copper-containing glycoprotein produced by hepatocytes and secreted into the blood. CP oxidizes ferrous iron (Fe^{2+}) that is released from some tissues to ferric iron (Fe^{3+}), which then binds to transferrin (i.e., the iron-transport protein) in the blood. In the human genetic disease aceruloplasminemia (caused by mutations in the *CP* gene) and in *Cp* KO mice, iron metabolism is perturbed, yet copper homeostasis is normal, demonstrating that CP does not have a specific copper-related function in mammals (such as copper transport/distribution in the blood). A GPI (plasma membrane)-anchored form of CP, which also functions as a ferroxidase, has been described in hepatocytes, spleen, brain, macrophages, and other cells/tissues.⁶⁹ GPI-CP expression was decreased in a mouse model of WD, which was associated with iron accumulation in hepatocytes and liver macrophages (i.e., Kupffer cells), resulting in a functional iron deficiency.⁷⁰ Pathologic iron accumulation in these liver cells was presumably due to decreased GPI-CP-mediated ferroxidase activity.

HEPH, a MCF with 50% homology to CP, was originally identified in the sex-linked anemia (*sla*) mouse.⁷¹ HEPH was initially described as a membrane-anchored,

intestinal ferroxidase; however, recent evidence has documented expression in the antrum of the stomach, the enteric nervous system, and pancreatic β -cells.⁷² HEPH is required for optimal intestinal iron absorption in weanling, adult, and pregnant mice under physiological conditions, but it was not required to upregulate iron absorption during hemolytic and iron-deficiency anemia.⁷³ HEPH expression decreases during copper deprivation, perhaps impairing absorption of dietary iron and thus contributing to the anemia associated with copper deficiency.⁷⁴

A third identified MCF, HEPH-like 1 (HEPHL1) (or zyklopen), was proposed to play a role in placental iron efflux in mice.⁷⁵ A recent case report described a young boy with compound heterozygous mutations in HEPHL1 that was linked to an abnormal hair phenotype.⁷⁶ The authors suggested that HEPHL1-mediated ferroxidase activity (possibly in hair follicles) is required for normal hair growth and development.

Cytochrome C oxidase

COX, the terminal electron acceptor complex of the mitochondrial respiratory chain, is a large complex consisting of multiple protein subunits and two heme groups, plus binding sites for three copper ions. COX mediates the reduction of molecular oxygen to form water and generates a proton gradient that is required for ATP production. COX activity may be impaired during copper deficiency, potentially disrupting energy balance. COX dependency upon copper for normal function is exemplified by recent work that links the activity of a mitochondrial copper transporter (SLC25A3) to COX biogenesis.⁷⁷

Dopamine β -hydroxylase

Dopamine β -hydroxylase (DBH) is a soluble cuproenzyme that is found within cytoplasmic and secretory vesicles, and chromaffin granules. It contains a copper atom in each of its four subunits and requires ascorbate as a cofactor. DBH catalyzes the conversion of dopamine to norepinephrine in the neurotransmitter synthesis pathway. Expression of DBH is highest in sympathetic neurons of the peripheral nervous system and in noradrenergic and adrenergic neurons in the brain, and in the adrenal medulla.⁴⁷ Mutations in the *DBH* gene in humans are associated with deficits in autonomic and cardiovascular function, and polymorphisms in *DBH* may increase risk for a variety of psychiatric disorders. In mice, *Dbh* gene inactivation leads to embryonic lethality, exemplifying its key role during development.⁷⁸

Superoxide dismutase

Superoxide dismutase (SOD) proteins are soluble, antioxidant proteins that mediate the conversion of

naturally occurring, highly reactive superoxide radicals to hydrogen peroxide and oxygen. Two of these proteins, SOD1 and SOD3, require copper for catalysis and utilize zinc as a structural component; a third (mitochondrial) isoform (SOD2) requires Mn for function. SOD1 exists as a 16 kDa homodimer and is found in the cytoplasm and mitochondria. SOD3 is a ~135 kDa tetramer that is secreted into various extracellular fluids, including plasma, synovial fluid, and lymph.⁷⁹ Importantly, mutations in SOD1 have been recently implicated in familial amyotrophic lateral sclerosis,⁸⁰ and recent evidence supports a role for SOD3 in the development of chronic obstructive pulmonary disease in humans.⁸¹ Moreover, low SOD3 levels have been linked to increased cancer incidence and poor treatment outcomes.⁸²

Tyrosinase

Tyrosinase (TYR) is a cuproprotein located in the melanosome membrane that possesses tyrosine hydroxylase and dopa oxidase activities. It functions in the melanin biosynthetic pathway where it catalyzes two copper-dependent steps: 1) conversion of tyrosine to dopamine and 2) oxidation of dopamine to dopaquinone (which is then converted to melanin in subsequent steps). TYR is thus required for normal pigmentation; logically then, mutations in *TYR* cause hypopigmentation or albinism. Nonpathological genetic polymorphisms in *TYR* have been associated with variations in skin pigmentation. The copper dependency of melanin synthesis is perhaps best exemplified by the achromotrichia observed in copper-deficient domestic and laboratory animals.⁴⁷

Copper-binding proteins

The proteins described above contain copper within the 3D architecture of the protein, at a presumed active site where the copper atom is involved in catalysis. Other proteins, that are not enzymes, bind copper, either transiently as in the case of a copper transporter (e.g., ATP7A) or a chaperone (e.g., CCS), or more long term as in the case of a copper storage protein (e.g., MT) (Table 24.2). Other similar examples are mentioned throughout the text. Besides binding copper, recent research has established that many of these proteins have additional biological functions.⁴³ Another copper-binding protein (i.e., transporter) of potential interest is a CTR1 homolog, copper transporter 2 (CTR2 or SLC31A2).⁸³

An additional example is prion protein (PRNP), which is a GPI-anchored, membrane-associated glycoprotein that is a disease-causing entity in Creutzfeldt–Jakob disease and other rare neurodegenerative disorders in humans. PRNP can bind up to eight copper atoms in an N-terminal region composed of octamer repeats. The primary physiological function(s) of PRNP

has remained elusive. It has been suggested that PRNP functions in copper transport into the CNS where it may deliver copper to antioxidant proteins (or function itself as an antioxidant).⁴³ Recently, it was proposed that the copper-binding sites in PRNP are required for neurotogenesis and that loss of the ability to bind copper may initiate pathological neurodegenerative processes.⁸⁴ Copper binding has also been suggested to promote PRNP oligomerization, which can lead to the production of pathological amyloid fibrils.

Amyloid beta precursor protein (APP) is another copper-binding protein with proposed functions related to neurite growth, neuronal adhesion, and axogenesis. APP is a cell-surface receptor and transmembrane precursor protein that is cleaved by secretases to produce several bioactive peptides, which function in transcriptional activation or may have antimicrobial and antifungal properties. These peptides can also form the protein component of amyloid plaques found in the brains of patients with Alzheimer's disease (AD). Copper binding to APP may lead to neurotoxic outcomes due to generation of reactive oxygen species.⁸⁵

Cellular copper homeostasis, copper-transporting ATPases (ATP7A, ATP7B)

These proteins play key physiological roles in copper transport and distribution within cells and between tissues. Both proteins are expressed in multiple tissues, with ATP7A having a more ubiquitous distribution and ATP7B being more restricted (e.g., in liver, certain neurons, cornea, and mammary tissue). Their essentiality is exemplified by their dysfunction in human MD (ATP7A) and WD (ATP7B) (as detailed below), which results in severe and often irreversible morbidities. Both proteins have been implicated in intestinal copper homeostasis²⁸ (see Fig. 24.2). ATP7A is required for enteral copper absorption, as it mediates copper efflux from enterocytes, the proposed rate-limiting step in intestinal copper absorption.^{86,87} It also functions intracellularly in copper import into the TGN, to support cuproenzyme synthesis. ATP7A is clustered in the TGN when copper is low, but it redistributes to vesicles and traffics toward the plasma membrane when intracellular copper levels increase (to facilitate copper efflux probably by vesicular exocytosis). ATP7A expression and activity may increase when body copper levels are low. Recently, ATP7B was also found to be expressed in enterocytes, where it may facilitate copper storage in intracellular vesicles.⁸⁸ The distribution of the copper-transporting ATPases, however, differed, with ATP7A mainly expressed in absorptive cells of the villus tip while ATP7B expression was mainly detected in intestinal crypts (in Paneth and stem cells). Unexpectedly, enteric ATP7B was proposed to be involved in lipid homeostasis in the absorptive epithelium.⁸⁸

Emerging roles of copper in normal and disturbed lipid homeostasis, for example, in nonalcoholic fatty liver disease (NAFLD), obesity, and diabetes were recently reviewed.²⁸ ATP7B (but not ATP7A) is also expressed in hepatocytes, where it pumps copper into the TGN and facilitates copper efflux into the biliary tree (probably also by vesicular exocytosis) (see Fig. 24.3). Importantly, ATP7A also plays a particularly important role in copper export from the placental trophoblast, thus facilitating copper distribution from mother to developing fetus.

C. Copper: Physiological Functions

Predictably, the known functions of the cuproenzymes outlined above define how copper influences mammalian physiology (and pathophysiology). As with any essential nutrient, including copper, inadequacy alters the activities of some, but not all, proteins that are dependent upon that nutrient for function; that is, when the nutrient is limiting, there is a hierarchy of influences on the functions mediated by these proteins. So, for example, a mild or moderate copper deficiency may depress the activity of some cuproenzymes, while others maintain normal activity. Moreover, it may be that, in some cases, we only know how deficiency alters the expression/activity of some cuproproteins in experimental animal models or livestock (which may be more severe than what naturally occurs in humans). These considerations are thus important for assessing the essentiality of any nutrient important in human health since functions that are mediated by that nutrient may be impacted by dietary inadequacy/deprivation.

Adenosine triphosphate generation

The copper-dependent protein, COX, located in the inner membrane of mitochondria, is the terminal step in the electron transport chain. COX is a multisubunit enzyme that catalyzes the reduction of molecular oxygen (O₂) to H₂O. COX facilitates electron transfer reactions, thus creating a proton electrical gradient that is used to generate the energy currency of cells, adenosine triphosphate (ATP). Severe copper deficiency may impair COX function and thus cause ATP depletion.

Connective and bone tissue formation

LOX facilitates formation of connective tissue, ensuring integrity of the heart and vasculature, and also bone tissue. Copper deficiency blunts LOX activity, thus resulting in the development of connective tissue disorders and bone defects, possibly including osteoporosis. Copper-deficient neonates may develop skeletal abnormalities very similar to what occurs in

vitamin C deficiency (i.e., scurvy).⁸⁹ Long-term copper supplementation may preserve bone health in adults,⁹⁰ but not all investigations on this topic have supported this contention.⁹¹ A recent metaanalysis, however, concluded that low serum copper was a risk factor for development of osteoporosis (along with low serum Fe and Zn).⁹²

Iron absorption and release from stores

Given the similar atomic radii, positive electrical charges, and redox potentials of copper and iron ions, it is a logical prediction that these two metals interact in biological systems, as is indeed the case.⁹³ An obvious connection between copper and iron metabolism is the MCFs CP and HEPH, which are cuproenzymes involved in iron metabolism. The expression/activity of CP^{94,95} and HEPH⁷⁴ decrease during copper depletion. Lack of ferroxidase activity impairs iron efflux from some tissues, perhaps most prominently affecting the liver, which is an iron storage organ. Impaired hepatic iron efflux during acquired copper deficiency may lead to iron accumulation to the extent of possible pathological consequence in humans.⁹⁶ Low copper intake may also lead to the development of an “iron deficiency–like” microcytic, hypochromic anemia. Reduced activity of HEPH in the intestine and low CP-mediated ferroxidase activity may impair iron absorption and iron release from stores, respectively, thus decreasing circulating iron levels and restricting iron delivery to the bone marrow to support erythropoiesis. Copper depletion can also lead to an inability of erythroid precursors to utilize iron for hemoglobin synthesis, but a pathological mechanism for this has not yet been elucidated.

Central nervous system development and function

Copper is required for normal development and function of the CNS, including the brain. Perhaps the best example of the essentiality of copper in brain development is the severe neuropathological phenotype (including tremors and ataxia) of infants suffering from the copper deficiency disease MD.⁹⁷ Moreover, inadequate copper intake during pregnancy and/or lactation may result in profound, long-lasting negative physiological outcomes for neonates since the late gestational and perinatal periods are key times for brain development. Associated pathological changes likely result from the reduced activity of various cuproenzymes in the CNS.⁴⁷ For example, during copper depletion, demyelination of nerve fibers may occur due to reduced COX activity (which is required the biosynthesis of sphingolipids), and neurotransmitter synthesis may be impaired due to low MAOA and DBH activity.

Pigmentation

Three critical steps in the biochemical pathway from tyrosine to melanin are catalyzed by the cuproenzyme TYR. Not surprisingly then, depigmentation of skin and hair is commonly observed in copper-deprived humans and animals.

Antioxidant functions

SOD1 and SOD3 are copper- and zinc-dependent antioxidant proteins that catalyze the conversion of superoxide radicals to H_2O_2 , which can subsequently be reduced to H_2O by other antioxidant enzymes.⁹⁸ Copper is involved in catalysis (*i.e.*, electron transfer), while zinc plays a structural role in these proteins. SOD1 is found in the cytoplasm of most body cells (including red blood cells), while SOD3 is an extracellular protein that is most abundant in the lung but low in plasma. CP is another cuproenzyme with antioxidant properties. First, by binding copper ions, CP mitigates copper-dependent production of reactive oxygen species. Second, CP facilitates oxidation of ferrous iron released from tissues, so it can be bound by transferrin in the blood, and by doing so, it prevents free iron from participating in deleterious free radical-generating reactions (*e.g.*, via Fenton chemistry).⁶³ And lastly, copper may indirectly regulate gene expression via its influence on antioxidant status of cells. Oxidative stress, possibly mediated by excess copper, activates a number of intracellular signal transduction pathways that *trans*-activate genes encoding proteins involved in detoxification of reactive oxygen species.⁹⁹

IV. DIETARY REQUIREMENTS

A. Copper: Indicators of Inadequacy and Assessment Status

Determining copper status of humans has proven to be challenging.¹⁰⁰ Recent efforts have thus sought to identify sensitive and specific biomarkers of copper nutritional status. This topic has been recently reviewed.¹⁰¹ Commonly used assessment methods have historically included quantification of plasma copper levels and serum CP activity, and the activity of cuproenzymes in various cells in blood (*e.g.*, erythrocyte Cu/Zn SOD).⁴⁷ Inductively coupled plasma mass spectroscopy and atomic absorption spectroscopy are commonly used techniques to quantify plasma (and tissue) copper content.⁶³ Plasma copper is lower and CP activity is blunted in human experimental trials only when consumption of a low-copper diet (≤ 0.6 mg/d) continues for at least 6 weeks.⁴³ Moreover, these measurements are confounded by the fact that blood copper content and CP activity are increased during, for example, the acute-phase response to infection and

inflammation, pregnancy, and cancer.⁴⁷ Newer methods have assessed serum AOC1 and skin LOX activity, copper in hair, and COX activity and copper levels in platelets and mononuclear cells. These assays have, however, proved to be of limited effectiveness since they do not consistently respond to a wide variety of copper intakes (0.38 mg/d to 6.0 mg/d).¹⁰¹ Other biomarkers have been recently considered. For example, in rats, serum and tissue PAM activity correlated with dietary copper intake,¹⁰² and similar observations were made in humans.¹⁰³ Also, the ratio of CCS to SOD1 protein levels in rodent erythrocytes has been proposed as a new biomarker, but as with the other assays described above, the utility of this approach to assess copper nutritional status in humans is unknown.⁴³

B. Copper: Dietary Reference Intakes

Dietary Reference Intakes for copper were established almost a decade ago¹⁰⁴ and are listed in Table 24.3. Based on a lack of experimental data, Adequate Intake levels for copper have been established for infants 0–6 months of age and for those between 7 and 12 months. The RDA increases throughout childhood and adolescence and is increased from adult levels during pregnancy and lactation. Upper Tolerable Intake Levels (ULs) have also been established for copper (see details below).

C. Copper: Upper Tolerable Intake Levels

Humans and other mammals have developed physiological mechanisms that control copper homeostasis, including regulated intestinal absorption and hepatic excretory systems. Ingestion of excess copper may, however, bypass homeostatic checkpoints that normally limit excessive intestinal absorption, potentially leading to hepatic toxicity. Due to these possible adverse consequences of high copper ingestion, a UL of 10 mg/d has been established (see more details below in the [Section V.D.I](#)).

D. Copper: Dietary Sources

The RDA for copper intake is 0.9 mg/d¹⁰⁴, but the average adult diet in the United States contains 1.0–1.6 mg of copper/d. Shellfish, wheat bran cereal, nuts, seeds, organ meats, whole grains, and chocolate are the richest dietary sources of copper. Determining the amount of copper in foods is difficult, and food composition tables may be inaccurate. Although copper absorption is lower from plant foods, consumption of a vegan diet does not necessarily equate to an increased risk for copper deficiency. Copper may also be obtained from dietary supplements; however, cupric oxide, which

has low bioavailability, is commonly used in these products.¹⁰⁵ Some have suggested that the RDA for copper intake in the United States is inadequate. For example, a recent National Health and Nutrition Examination Survey suggested that up to 40% of the US population had low serum copper reflecting a functional copper deficit,¹⁰⁶ most likely due to dietary insufficiency. These and other recent experimental analyses support a higher recommended intake of copper at 2.6 mg/d for adults.^{107,108}

V. ISSUES RELATED TO COPPER IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern Related to Copper

Physiological outcomes of copper deficiency

Severe copper deficiency in animals results in abnormalities in the immune, skeletal, and cardiovascular systems. Further consequences of copper deprivation are hypochromic anemia (which does not respond to iron supplementation), hypopigmentation, thrombocytopenia, and neutropenia.⁸⁹ More specific features include skeletal abnormalities, fractures, and spinal deformities; neonatal ataxia; depigmentation of hair and wool; abnormal keratinization of hair, wool, and fur; reproductive failure; cardiovascular abnormalities; and impaired immune function. Importantly, some of these symptoms have only been observed in one or two species, and some manifestations of copper deficiency in experimental and domesticated animals are more severe than what is typically observed in humans.

B. Groups at Risk of Copper Inadequacy

Overt copper deficiency in humans is rare, but some have suggested that mild or “subclinical” copper deficiency may be more common than generally presumed.¹⁰⁹ Proving this possibility is, however, quite challenging since specific, sensitive biomarkers of copper nutritional status have not been identified.¹¹⁰ Logically, dysregulation of copper homeostasis could result from inefficient intestinal copper absorption and/or excessive copper loss through the endogenous biliary excretion system (in, for example, patients with cholestasis). Individuals at highest risk for copper depletion include those receiving total parenteral nutrition lacking copper or those on restricted diets; premature infants consuming milk-based formulas (which may lack adequate copper); malnourished infants or those with chronic diarrhea; hospitalized patients on long-term peritoneal dialysis; severe burn and renal dialysis

patients; and those consuming supplemental zinc or copper chelators (as a treatment for WD) or aluminum-containing antacids.⁴⁷ Malabsorptive disorders, including celiac disease, sprue, and short bowel syndrome, may also result in copper deficiency. Moreover, surgical bowel resection for the management of morbid obesity may increase risk for acquired copper deficiency.^{101,111} Furthermore, a **neurologic** syndrome in adults with acquired copper deficiency has been recently described.¹¹² Symptoms include **CNS nerve fiber** demyelination, polyneuropathy, myelopathy, and **inflammation** of the optic nerve. The etiology is unknown, but dysfunction of intestinal ATP7A (due to genetic mutations) and concomitant copper malabsorption may be the underlying cause.¹¹³

Inherited copper deficiency

MD results from mutations in *ATP7A*, which encodes a protein that mediates copper transport within and export from most cell types (except, most notably, hepatocytes). As a result, in MD, intestinal copper absorption is blocked, and systemic copper depletion ensues (although paradoxically, copper accumulates in some tissues). This depletion impairs the activity of various cuproenzymes, with severe pathological outcomes. For example, copper transport into the brain is repressed, and cuproenzyme activity in neurons is impaired, resulting in progressive nerve degeneration.¹¹⁴ Additional disease manifestations include seizures, connective tissue disorders, subdural hemorrhage, and hair abnormalities. Treatments for affected individuals include subcutaneous injection with copper histidine, effectively bypassing the intestinal absorptive defect.⁹⁷ Nonetheless, copper transport into the brain is still impaired.

C. Copper and Health

Cardiovascular disease

Several pathological disturbances of the cardiovascular system have been documented in experimental animals deprived of copper, including biochemical, anatomical, mechanical, and electrical abnormalities.⁴³ For example, development of ischemic heart disease and hypertrophic cardiomyopathy is associated with copper depletion in rodents.^{115,116} Also, hypercholesterolemia and hypertriglyceridemia have been observed in severely copper deficient rats.¹¹⁷ CVS-related perturbations observed in copper-depleted animals, however, differ from the pathophysiological changes associated with atherosclerotic heart disease that most commonly afflicts humans. Nonetheless, the extensive literature describing these phenomena in experimental animals has provided the impetus to consider the role of copper

in cardiovascular system function/dysfunction in humans. Copper deficiency in humans may predispose individuals to developing CVD. For example, emerging evidence has linked low copper status with increased risk for developing ischemic heart disease^{109,118} and cardiac arrhythmias.⁴⁷ Copper depletion trials in humans have, however, not provided consistent results.^{119,120} Furthermore, the possibility that high copper could increase CVD risk has also been considered, since copper in excess may promote oxidative stress. Supporting this contention, several observational studies in humans have made correlative connections between high serum copper and increased incidence of coronary heart disease.^{121,122} However, serum copper largely reflects CP (which contains the largest pool of blood copper), which is not a sensitive biomarker of copper nutritional status. Moreover, CP is an acute-phase response protein that can be induced by infection and inflammation. Increases in serum copper could thus be secondary to the inflammation associated with atherosclerotic heart disease; thus, elevated copper in the blood may not have any direct relevance to disease pathogenesis. In conclusion, whether copper depletion or excess alters risk for developing CVD remains uncertain, since the copper nutritional status of those who were surveyed in epidemiological studies and those who participated in metabolic trials was largely unknown due to the lack of sensitive and specific copper-related biomarkers in humans.

Immune function

Copper is required for normal development and function of the immune system, but the exact role of copper in this process has not been elucidated. For example, copper deficiency has been frequently associated with an increased risk of infection,¹²³ suggesting that cellular and/or humoral immune system factors may be less effective when copper is low. Predictably, adverse effects of insufficient copper on immune system function are most deleterious in neonates. Severely copper-deficient infants with MD suffer from frequent and severe infections.^{124,125} Also, copper supplementation of copper-depleted, malnourished infants increased the ability of immune cells to phagocytize pathogens.¹²⁶ Neutropenia is another hallmark of copper deficiency in humans, and the function of some immune cells (e.g., macrophages, lymphocytes) may be impaired by even marginal copper deficiency. For example, proliferation of peripheral blood mononuclear cells isolated from adult males fed a low-copper diet (0.38 mg/d) for 42 days was suppressed after an *in vitro* immune challenge.¹²⁷ Study participants also had low plasma copper levels, and the activity of some cuproenzymes was repressed. Copper may also be required for the

innate immune response against bacterial infection.¹²⁸ Collectively, these investigations (and others not cited here) exemplify important roles for copper in immune system function, but definitive proof is lacking in part due to the inability to detect marginal, or subclinical, copper deficiencies in human participants in intervention trials.

Bone health

Connective tissue disorders, osteoporosis, and bone defects have been reported in dietary and inherited copper deficiency in humans and other mammals. This is perhaps not surprising given that a key enzyme in collagen formation, LOX, is a cuproenzyme. Infants with severe copper deficiency develop bone abnormalities resembling vitamin C deficiency⁸⁹ (which impairs collagen cross-linking and thus stabilization). Whether marginal copper deficiency in adults increases risk for developing osteoporosis is unclear, and human intervention trials are few. In one recent study, bone resorption occurred in healthy adult males with a marginal copper intake (0.7 mg/d) for 8 weeks.⁹¹ Also, supplementation with copper (3–6 mg/d for 6 weeks) had no effect on biomarkers of bone health in healthy adult men and women.^{129,130} In conclusion, since no consensus has been reached in the scientific literature, additional research is required to assess the impact of moderate copper deficiency on bone health and copper supplementation on the development of age-related osteoporosis in humans.

Neurodegenerative disorders

A possible relationship between dysregulation of copper homeostasis and the onset of cognitive decline in patients with AD has been recently considered. Confusingly, both low and high copper have been implicated in the pathogenesis of the disease. In AD, highly reactive, “free” copper, i.e., that fraction in the blood not associated with CP, increases.¹³¹ It has been hypothesized that this excess, non-CP copper interacts with (copper-binding) β -amyloid peptides and promotes the formation of pathological senile plaques.¹³² It was also suggested that polymorphisms in *ATP7B* may influence AD risk, since reduced *ATP7B* activity diminishes hepatic copper excretion and impairs CP production, thus increasing “free” copper in the blood and resulting in copper accumulation in the liver and brain. It is also conceivable that genetic variations (in *ATP7B* and/or other genes) could influence individual susceptibility to environmental exposure to excess copper (e.g., in drinking water), perhaps modulating risk for developing AD (or other neurodegenerative disorders).

It may also be important to consider the possibility that disturbed copper metabolism is not causative of

AD, but rather symptomatic of the disease. A recent investigation provided some clarity on this issue. Sixty adult AD patients participated in a randomized intervention trial, whereby they were given a zinc supplement (which impairs intestinal copper absorption) or placebo daily for 6 months. Notably, serum “free” copper declined and cognitive function stabilized¹³³ in the zinc-supplemented group, supporting a role for copper in the primary pathogenesis of the disease.

Quite contrary to the hypothesis that increased brain copper increases risk for AD, others have argued that AD is associated with copper depletion. Evidence supporting this contention include the observations that low copper levels and suppression of cuproenzyme activity have been reported in the brains of patients with AD,¹³⁴ and limited evidence showing that people with higher copper levels have a lower risk of developing AD.¹³⁵ Overall though and in summary, additional investigation is required to determine whether high or low blood copper levels are primarily associated with the pathogenesis of AD and whether copper intake from food, supplements, and water affects AD risk or severity of symptoms.

Lipid metabolism

Dietary copper levels and body copper status seem to influence lipid homeostasis.¹³⁶ Emerging evidence demonstrates that Cu deficiency is associated with increases in fat storage and alterations in lipid metabolism.^{106,137} For example, copper deprivation of rats led to accumulation of long-chain fatty acid-rich triglycerides in several tissues¹³⁸ and increased body weight and fat deposition.¹³⁹ It has also been shown that rats deprived of dietary copper develop hepatic steatosis, inflammation, and fibrosis, closely modeling NAFLD in humans.¹⁴⁰ This is consistent with the observations that hepatic copper is low in NAFLD patients, and hepatic copper was inversely associated with the degree of steatosis.¹⁴⁰ Consistent with this, serum copper and CP levels are also low in NAFLD patients.¹⁴¹ Hepatic copper deficiency is also noted in mouse models of NAFLD.^{142,143} Conversely, fat stores are mobilized, and storage decreased when body copper levels are elevated.^{106,137} Dietary copper supplementation studies in animals that support this contention include the following observations: decreases in adipose tissue mass in rabbits¹⁴⁴; decreased subcutaneous fat in steers¹⁴⁵; and increased lipolysis in adipose tissue in sheep.¹⁴⁶ Collectively, these observations support the possibility that copper depletion and excess both disrupt lipid homeostasis. The copper-dependent mechanisms by which these changes occur have not, however, been elucidated to date.

D. Health Risks From Excess Copper

Copper toxicity

Copper excess, or overload, has infrequently been documented in humans, although copper poisoning has occurred from ingestion of acidic beverages stored in copper-lined containers and from contaminated household water supplies.¹⁴⁷ As such, limits have been established for copper levels in drinking water by the US Environmental Protection Agency (1.3 mg/L) and by the World Health Organization (2 mg/L).¹⁴⁸ Acute ingestion of excess copper may cause abdominal pain, nausea, vomiting, and diarrhea, while more chronic exposure can result in severe liver damage, kidney failure, coma, and even death. Fortunately, in generally healthy adults, liver damage has not been observed with copper intakes up to 10 mg/d. This was the basis for establishing the UL for copper at that level. Some scientists, however, believe that the UL for copper may be too high. One study supporting this possibility involved giving men 7.8 mg of copper daily for 147 days. Body copper levels increased during the study period, and biomarker analysis suggested impairment of immune function and increased oxidative stress with high copper intake.^{149,150} In contrast, a similar investigation did not report any negative outcomes associated with consuming 8 mg of copper/day for half a year.¹⁵¹ One caveat is that the latter study was done in AD patients, while the former studies were done in healthy men, so direct comparison between these studies is challenging. Finally, some genetic disorders may increase risk for copper overload with copper intakes well below the UL, including WD (as discussed below), Indian childhood cirrhosis, and idiopathic copper toxicosis.

Wilson's disease

WD is an autosomal recessive genetic disorder that is typified by abnormal tissue copper distribution and storage.¹⁵² Mutations in the *ATP7B* gene cause dysfunction of the ATP7B copper transporter, thus disrupting copper homeostasis. WD occurs in 1:30,000 individuals worldwide.¹⁵³ In WD, excess copper accumulates in the liver (due to impaired excretion), brain, and cornea (forming Kayser–Fleisher rings), causing oxidative stress and eventual tissue and organ damage. If untreated, WD patients can develop hepatitis, cirrhosis, fibrosis, hemolytic crisis, and eventual hepatic failure. High copper in the brain can cause neurological damage, and copper accumulation in the eye may result in abnormal eye movements. Urinary copper excretion may also be elevated, and blood CP activity is characteristically low. Therapeutic intervention early in life may prevent some of these

pathological outcomes. Treatments include copper chelation therapy with penicillamine and/or trientine¹⁵⁴ or high zinc dosing (which interferes with absorption of dietary copper).

E. Known Interactions

Copper is not known to have clinically relevant interactions with any medications. The relative amount of copper in the diet seems to be the major predictor of absorption levels; however, several dietary factors, including certain amino acids and proteins, iron, zinc, molybdenum, vitamin C, and carbohydrates, have been reported to exert adverse effects on copper bioavailability.¹⁵⁵ Dietary components may influence copper absorption more profoundly in neonates, since digestive and absorptive functions are immature and homeostatic regulation of biliary copper excretion may be inefficient. The effects of iron, zinc, and vitamin C on copper metabolism are considered in more detail below.

Iron

Copper and iron interact in numerous ways,^{23,24} as recently reviewed.⁹³ These interactions have been appreciated for many decades,¹⁵⁶ but it has only been recently that molecular details have emerged. In the gut, HEPH activity is decreased by copper depletion, while *ATP7A* transcription is increased by iron depletion.^{30,157} DMT1 may transport iron and copper, CYBRD1 may reduce both metals, and FPN1 expression/activity may be influenced by copper. Moreover, hepatocytes produce and secrete the MCF CP, which is principally involved in iron metabolism. Liver copper levels increase during iron depletion, while hepatic iron levels increase during copper depletion.¹⁵⁸ This latter effect is presumably due to decreased CP activity, which impairs hepatic iron release.¹⁵⁹ Interestingly, bone marrow iron utilization seems to be copper dependent, since hemoglobin production is impaired by copper depletion despite serum iron levels being normal.^{23,24}

It was also suggested that excess iron can interfere with copper utilization.¹⁰⁹ For example, high iron intake interfered with copper absorption in infants and adults,¹⁰⁹ and high iron intake increased the dietary requirement for copper in rats.¹⁶⁰ Also, iron overload may perturb copper utilization.^{161,162} CP protein expression was decreased in hereditary hemochromatosis, a genetic iron-loading disorder.¹⁶³ Studies in rats⁹⁵ and mice¹⁶⁴ demonstrated that high iron intake impairs dietary copper utilization, resulting in the development of copper deficiency-related pathologies. Notably, moderate to severe copper deficiency was noted in growing rats when dietary iron was increased to ≈ 9.5 -fold over requirements.¹⁶⁵ This raises concerns about the possible

influence of iron supplementation on copper homeostasis in humans, especially since many Americans may have marginal copper intakes (thus putting them at increased risk).^{109,166} Copper depletion associated with iron supplementation could be most detrimental during pregnancy (when iron supplementation is widely recommended^{167–170}) due to the severe effects of copper deficiency on the developing fetus^{171,172} and in children and adolescents during periods of rapid growth. It has thus been suggested that iron supplements should contain extra copper.¹⁰⁹ Additional experimentation on this topic is clearly warranted.

Zinc

High dietary zinc intake impairs copper absorption, possibly due to the induction of MT (which sequesters zinc) in enterocytes. For example, copper depletion has been observed in humans consuming supplements containing 50 mg of zinc daily for extended periods of time. This observation provided the rationale for setting the UL for zinc at 40 mg/d for adults.¹⁷³ Also, severe copper deficiency may result from the ingestion of excessive amounts of zinc containing denture cream.¹⁷⁴

Ascorbic acid

Ascorbic acid supplementation may induce copper deficiency in experimental animals and could have a similar effect in humans. In premature infants, plasma vitamin C levels were negatively correlated with serum CP and antioxidant activity.¹⁷⁵ Other human studies have also suggested that ascorbic acid supplementation may perturb serum ferroxidase activity.

F. Copper as a Dietary Supplement

Identifying individuals at increased risk for copper deficiency is challenging, given the lack of sensitive and specific, copper nutrition-related biomarkers. Nonetheless, supplements containing only copper are available, and copper is also typically included in multivitamins, micronutrient supplements, and in fortified breakfast cereals. In supplements, copper may be included as amino acid chelates, copper gluconate, cupric oxide, or cupric sulfate. The relative bioavailability of these different forms of copper is uncertain.¹⁷⁶ Amounts of copper included in supplements can vary from a few μg up to 15 mg (which is 1.5-fold higher than the UL for adults).¹⁷⁷ Supplementation of adults with 10 mg cupric gluconate/d for 12 weeks did not cause gastrointestinal distress or liver damage.¹⁷⁸ Supplemental copper should be administered to neonates and infants with caution, since toxicity risks are increased due to enhanced intestinal absorption and inefficient biliary excretion. Copper loading is observed

clinically today in WD patients and in those with biliary cirrhosis and biliary atresia. Individuals with these conditions should avoid consuming supplemental copper.

VI. CONCLUDING REMARKS

Continuing research on copper biology aims to further define roles for this essential nutrient in various pathologies in humans, including neurodegenerative and connective tissue disorders, cardiovascular-related diseases, and conditions involving perturbations in lipid metabolism (e.g., NAFLD). Moreover, predicted relationships between functional copper depletion and ischemic heart disease should be tested in appropriate experimental settings. Recent investigative efforts have also focused on redefining dietary intake recommendations for copper, in part, by developing novel (and possibly more informative) copper nutrition–related

biomarkers of moderate (i.e., subclinical) copper deficiency. Being able to identify individuals with marginal copper status (but who are not overtly deficient) will likely provide the impetus for additional metabolic studies aimed at clarifying copper requirements in healthy adults and in various physiological (e.g., pregnancy and lactation, periods of rapid growth) and pathological (e.g., chronic inflammation) conditions. Additionally, the role of copper in regulating iron utilization by developing erythrocytes and possible influences of copper on intestinal lipid metabolism should be further investigated. These and other areas of copper-related research will undoubtedly increase knowledge of copper biology over the next 5–10 years (hopefully), thus permitting the further development of therapeutic modalities to treat human conditions in which disturbances in copper homeostasis may play a primary (or secondary) pathological role (e.g., various neurodegenerative disorders).

RESEARCH GAPS

- Revised RDAs for copper—some have suggested an RDA of 2.6 mg/d for adults
- Identification of sensitive and specific copper nutritional biomarkers
- Copper in neurodegenerative disorders, possible copper toxicity from drinking water
- Copper in lipid metabolism, role of intestinal ATP7B
- Role of copper in hemoglobin production
- Relationship between copper deficiency and ischemic heart disease

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IODINE AND THE IODINE DEFICIENCY DISORDERS

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SUMMARY

Iodine is an essential component of hormones produced by the thyroid gland. Thyroid hormones, and therefore iodine, are essential for mammalian life. Optimal dietary iodine intakes for healthy adults are 150–250 µg/d. In regions where iodine in soils and drinking water is low, humans and animals may become iodine deficient. Iodine deficiency has multiple adverse effects in humans due to inadequate thyroid hormone production that are termed the iodine deficiency disorders. Iodine deficiency during pregnancy and infancy may impair growth and neurodevelopment of the offspring and increase infant mortality. Deficiency during childhood reduces somatic growth and cognitive and motor function. Assessment methods include urinary iodine concentration, presence of goiter, newborn serum thyroid-stimulating hormone concentration, and serum thyroglobulin concentration. In most countries, the best strategy to control iodine deficiency in populations is carefully monitored iodization of salt.

Keywords: Cretinism; Goiter; Iodine; Iodine deficiency; Iodine excess; Iodized salt; Thyroglobulin; Urinary iodine.

I. INTRODUCTION

A. Background

Bernard Courtois, a French chemist, accidentally discovered iodine (atomic weight 126.9) in 1811 while producing gunpowder for Napoleon's army. He was extracting sodium carbonate from seaweed ash to use for manufacture of saltpeter (nitrate). Applying sulfuric acid to clean out the metal vats used to burn the seaweed, he produced an intense violet-colored vapor that cooled into dark violet crystals. He gave the crystalline material to Joseph Gay-Lussac, a French scientist, who identified it as a new element and named it "iode," from the Greek *ιώδης* (i.e., violet).¹ In 1819, Jean-François Coindet, in Geneva, successfully treated goiter with tincture of iodine. In 1896, Baumann and Roos discovered iodine in the thyroid. In the first two decades of the 20th century, pioneering studies by Swiss and American physicians demonstrated the efficacy of iodine prophylaxis

in the prevention of goiter and cretinism in populations. The first use of salt iodization as a general public health measure was by the surgeon H. Eggenberger in Northern Switzerland in the early 1920s.¹

B. Nutrient Function and Structure

Iodine is a required component of the hormones thyroxine and triiodothyronine; severe dietary iodine deficiency results in goiter and hypothyroidism in adults and cretinism in children. Iodine is widely but unevenly distributed in the earth's environment. The atomic number of iodine is 53, and it has a molecular weight of 126.9 amu, making it the heaviest of stable halogen elements. It is purple when in its elemental form I₂ and thus is ~254 g/mol. It mainly exists in nature as iodide, an anion (I⁻) with an oxidation state of -1; this form of iodine is often used to produce iodine supplements and iodized table salt. It also is found as iodate, IO₃⁻, another form used frequently as an additive to table salt.

In many regions, leaching from glaciation, flooding, and erosion have depleted surface soils of iodide, and most iodide is found in the oceans. The concentration of iodide in seawater is $\approx 50 \mu\text{g/L}$. Iodide ions in seawater are oxidized to elemental iodine, which volatilizes into the atmosphere and is returned to the soil by rain, completing the cycle.² However, iodine cycling in many regions is slow and incomplete, leaving soils and drinking water iodine-depleted. Crops grown in these soils will be low in iodine, and humans and animals consuming food grown in these soils become iodine deficient. In plant foods grown in deficient soils, iodine concentration may be as low as $10 \mu\text{g/kg}$ dry weight, compared to $\approx 1 \text{ mg/kg}$ in plants from iodine-sufficient soils.³

II. NUTRIENT METABOLISM

A. Digestion

Iodine is ingested in several chemical forms. Iodide is rapidly and nearly completely absorbed in the stomach and duodenum. The sodium/iodide symporter (NIS) and the sodium/multivitamin transporter on the apical surface of enterocytes mediate active iodine absorption.^{4,5} Iodate, widely used in salt iodization, is reduced in the gut and absorbed as iodide. In healthy adults, the absorption of iodide is $>90\%$.⁶ Organically bound iodine is typically digested and the released iodide absorbed, but some forms may be absorbed intact. For example, $\approx 75\%$ of an oral dose of thyroxine, the thyroid hormone, is absorbed intact.

B. Metabolism

The distribution space of absorbed iodine is nearly equal to the extracellular fluid volume. Iodine is cleared from the circulation mainly by the thyroid and kidney (Fig. 25.1), and while renal iodine clearance is fairly constant, thyroid iodine clearance varies with iodine intake. In conditions of adequate iodine supply, $\leq 10\%$ of absorbed iodine is taken up by the thyroid. In chronic iodine deficiency, this fraction can exceed 80% .³ During lactation, the mammary gland concentrates iodine and secretes it into breast milk to provide for the newborn.³ The salivary glands, gastric mucosa, and choroid plexus also take up small amounts of iodine.³ Iodine in the blood is turned over rapidly; under normal circumstances, plasma iodine has a half-life of $\approx 10 \text{ h}$, but this is shortened if the thyroid is overactive, as in iodine deficiency or hyperthyroidism.³

The body of a healthy adult contains $15\text{--}20 \text{ mg}$ of iodine, of which $70\%\text{--}80\%$ is in the thyroid.³ In chronic iodine deficiency, the iodine content of the thyroid may fall to $<20 \mu\text{g}$.³ In iodine-sufficient areas, the adult thyroid traps $50\text{--}60 \mu\text{g}$ of iodine/day to balance losses and maintain thyroid hormone synthesis.³ An NIS, a plasma membrane protein in the basolateral membrane,

transfers iodide into the thyroid at a concentration gradient 20 to 50 times that of plasma.⁷ The NIS concentrates iodine by an active transport process that couples the energy released by the inward translocation of sodium down its electrochemical gradient to the simultaneous inward translocation of iodine against its electrochemical gradient.

Thyroglobulin (Tg), a large glycoprotein (molecular weight $660,000$), is the storage protein for iodine in the thyroid (Fig. 25.2). At the apical surface of the thyrocyte, thyroperoxidase (TPO) and hydrogen peroxide oxidize iodide and attach it to tyrosyl residues on Tg, to produce monoiodotyrosine (MIT) and diiodotyrosine (DIT), the precursors of thyroid hormone. Thyroid hormone refers to both thyroxine (T_4) and triiodothyronine (T_3).³ TPO then catalyzes the coupling of the phenyl groups of the iodotyrosines through a diether bridge to form the thyroid hormones. Linkage of two DIT molecules produces T_4 , and linkage of an MIT and DIT produces T_3 (Fig. 25.3). Thus, T_3 is structurally identical to T_4 but has one less iodine (at the $5'$ position on the outer ring). Iodine comprises 65% and 59% of the weights of T_4 and T_3 , respectively. In the thyroid, mature Tg, containing $0.1\%\text{--}1.0\%$ of its weight as iodine, is stored extracellularly in the luminal colloid of the thyroid follicle.³ After endocytosis, endosomal and lysosomal proteases digest Tg and release T_4 and T_3 into the circulation. MIT and DIT are not normally released into the blood. Iodine is removed from these tyrosines by a selenium-dependent deiodinase within the thyroid and is then recycled for use within the thyroid, conserving iodine.³

In the circulation, thyroid hormone is bound noncovalently to carrier proteins, mainly thyroxine-binding globulin, and also to transthyretin and albumin. In target tissues—liver, kidney, heart, muscle, pituitary, and the developing brain— T_4 is deiodinated to T_3 . T_3 is the main physiologically active form of thyroid hormone and binds to nuclear receptors.³ The thyroid hormone receptors have been cloned and regulatory DNA elements identified in thyroid hormone-responsive genes.⁸ Hormone-receptor interactions stimulate several pathways, including the adenosine triphosphate and inositol phosphate- Ca^{2+} cascades, which in turn stimulate or inhibit protein synthesis.

Both T_4 and T_3 are degraded through a complex series of pathways, and their turnover is relatively slow: the half-life of T_4 is ≈ 5 days and for T_3 $1.5\text{--}3$ days.⁹ The released iodine enters the plasma iodine pool and can be taken up again by the thyroid or excreted by the kidney. More than 90% of ingested iodine is ultimately excreted in the urine, with only a small amount appearing in the feces.

The principal regulator of thyroid hormone metabolism is thyroid-stimulating hormone (TSH), a protein hormone (molecular weight $\approx 28,000$) secreted by the

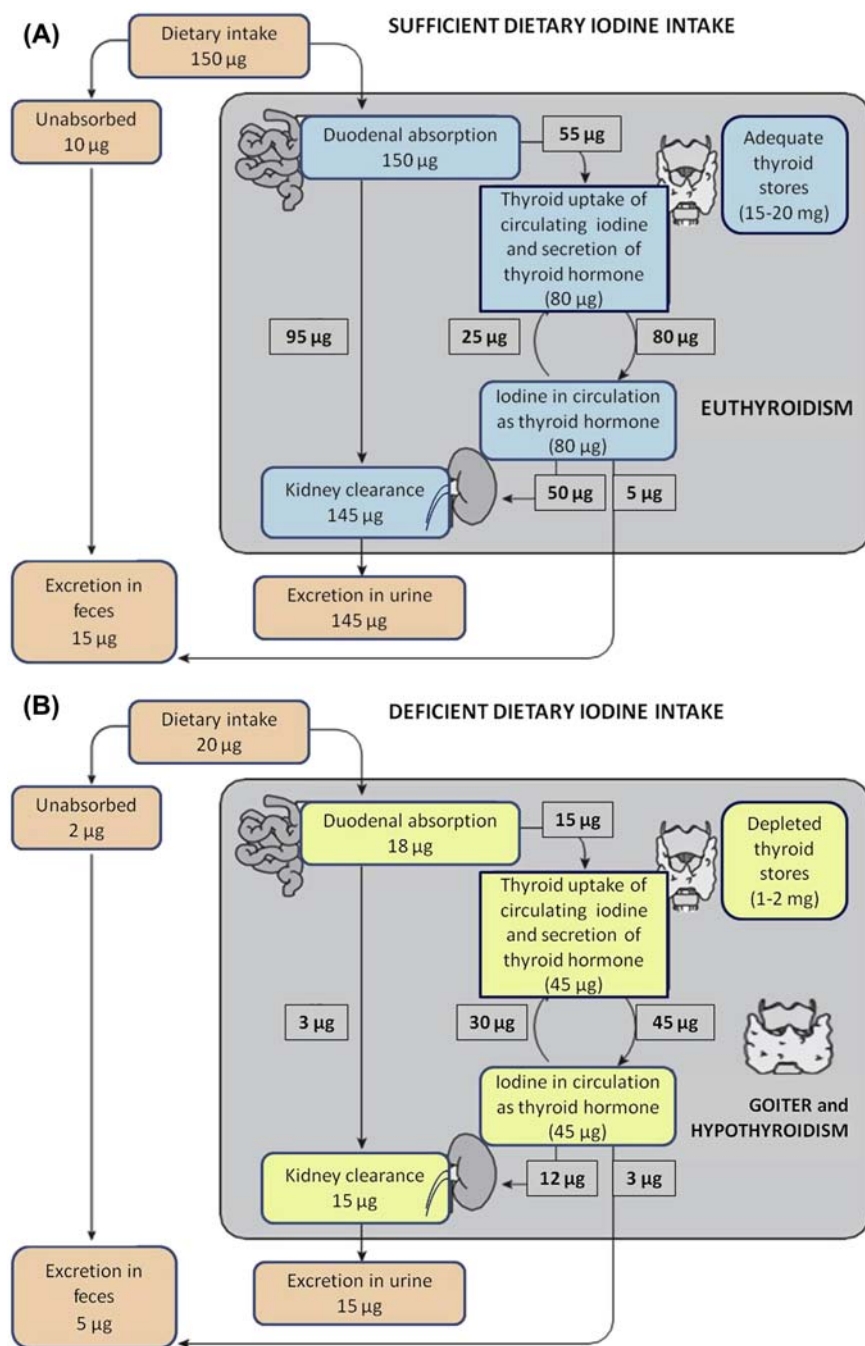


FIGURE 25.1 Over 90% of dietary iodine is absorbed in the duodenum. Iodine (as iodide) is cleared from the circulation mainly by the thyroid and kidney. Thyroid clearance varies with iodine intake. (A) (top): In iodine-sufficient individuals with adequate thyroid iodine stores, about 10%–35% of circulating iodide is taken up by the thyroid to balance losses and maintain thyroid hormone synthesis. (B) (bottom): In chronic iodine deficiency, the fraction of circulating iodide cleared by the thyroid increases to about 65%, but the iodine content of the thyroid is depleted and hypothyroidism develops.

pituitary. TSH secretion is controlled through negative feedback by the level of circulating thyroid hormone, modulated by TSH-releasing hormone from the hypothalamus.³ In the thyroid, TSH increases iodine uptake through stimulation of NIS expression. TSH exerts its action at the transcription level of the NIS gene through a thyroid-specific enhancer that contains binding sites for the transcription factor Pax8 and a cAMP-response

element-like sequence.¹⁰ TSH also stimulates breakdown of Tg and release of thyroid hormone into the blood but does not influence iodine absorption from the gut or renal iodide excretion. Because the primary stimulus to TSH secretion is circulating thyroid hormone, an elevated serum TSH concentration generally indicates primary hypothyroidism, while a low concentration indicates primary hyperthyroidism.³

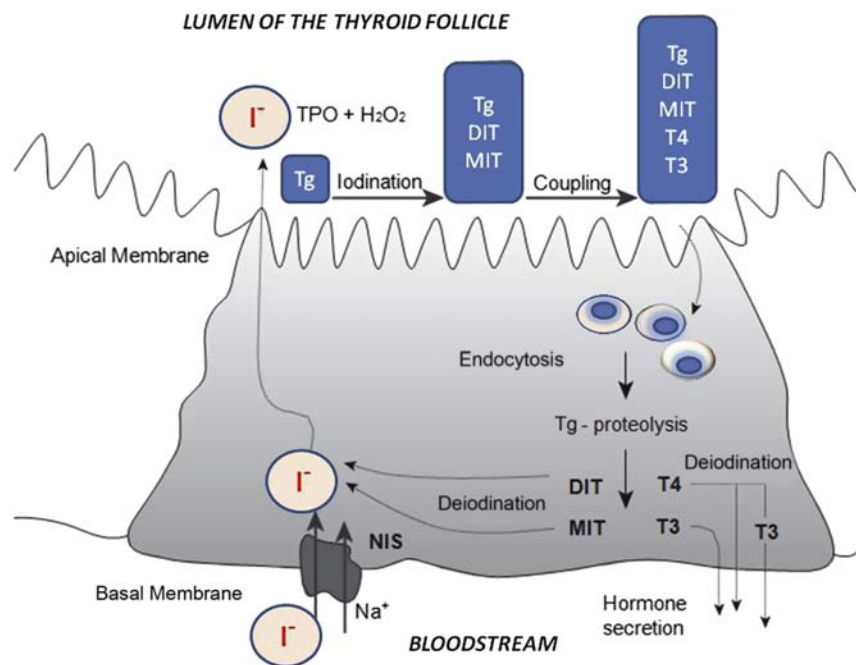


FIGURE 25.2 Iodine pathway in the thyroid cell. Iodide (I^-) is transported into the thyrocyte by the sodium iodide symporter (NIS) at the basal membrane and migrates to the apical membrane. The I^- is oxidized by thyroperoxidase (TPO) and hydrogen peroxide and attached to tyrosyl residues in thyroglobulin (Tg) to produce the hormone precursors iodotyrosine (MIT) and diiodotyrosine (DIT). The residues then couple to form thyroxine (T4) and triiodothyronine (T3) within the Tg molecule in the follicular lumen. Tg enters the cell by endocytosis and is digested. T4 and T3 are released into the circulation, and nonhormonal iodine on MIT and DIT is recycled within the thyrocyte.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

The only well-established function of iodine in humans is as an essential component of the thyroid hormones, which regulate a variety of physiologic processes, including reproductive function, growth, and development. During pregnancy, thyroid hormone crosses the placenta to the fetus early in the first trimester, before the fetal thyroid is functioning. In the developing brain, it influences cell growth and migration.¹¹ It also promotes growth and maturation of peripheral tissues and the skeleton.³ Thyroid hormone increases energy metabolism in most tissues and raises the basal metabolic rate.⁶ Other physiologic functions of iodine outside of thyroid hormones are uncertain: iodine may influence fibrocystic breast disease, may play a role in immune response or as an antioxidant, and may modify risk for gastric cancer.⁶

IV. DIETARY REQUIREMENTS

A. Indicators of Adequacy and Assessment of Status

Several methods are available for assessment of iodine nutrition. The most commonly used are measurement of

thyroid size and concentration of urinary iodine (UI).¹¹ Additional indicators include newborn thyrotropin (another name for TSH) and blood concentrations of thyroglobulin, thyroxine (T4), or triiodothyronine (T3). As discussed below, UI is a sensitive indicator of recent iodine intake (days) and serum Tg shows an intermediate response (weeks to months), whereas changes in the goiter rate reflect long-term iodine nutrition (months to years).³

B. Goiter

Two methods are available for measuring goiter: neck inspection and palpation and thyroid ultrasonography. Goiter surveys are usually done in school-age children.¹¹ By palpation, a thyroid is considered goitrous when each lateral lobe has a volume greater than the terminal phalanx of the thumbs of the subject being examined. In the classification system of WHO,¹¹ grade 0 is defined as a thyroid that is not palpable or visible, grade 1 is a goiter that is palpable but not visible when the neck is in the normal position (i.e., the thyroid is not visibly enlarged), and grade 2 goiter is a thyroid that is clearly visible when the neck is in a normal position.

In areas of mild-to-moderate iodine deficiency, where goiters are small, measurement of thyroid size by ultrasonography is a more objective and precise method, and is preferable to palpation. Portable

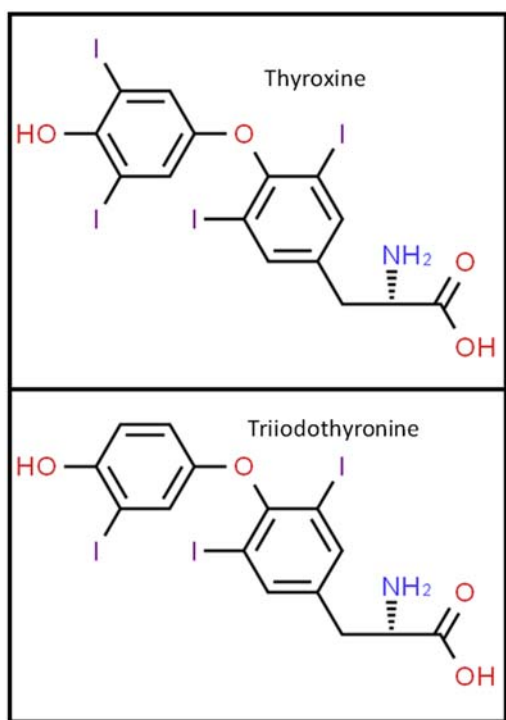


FIGURE 25.3 Iodine is an essential component of the thyroid hormones thyroxine (T4) and triiodothyronine (T3).

ultrasound equipment can be used in the field, and goiter can be classified according to international reference criteria for iodine-sufficient children by age, gender, and body surface area.¹¹ The total goiter rate of a group is used to define severity using the following criteria: <5%, iodine sufficiency; 5.0%–19.9%, mild deficiency; 20.0%–29.9%, moderate deficiency; and >30%, severe deficiency.¹¹

In areas of iodine deficiency, although thyroid size predictably decreases in response to increases in iodine intake, thyroid size may not return to normal for months or years after correction of iodine deficiency.¹² During this transition period, the goiter rate in areas of endemic goiter is difficult to interpret because it reflects both a population's history of iodine nutrition and its present status. Despite this lag period, a sustained salt iodization program will decrease the goiter rate to <5% in school-age children, and this indicates disappearance of iodine deficiency as a significant public health problem.¹¹

C. Urinary Iodine

Because >90% of ingested iodine is excreted in the urine, UI is an excellent indicator of recent iodine intake.^{6,11} Most methods of measuring UI are based on the Sandell-Kolthoff reaction, in which iodide catalyzes the reduction of yellow ceric ammonium sulfate to the colorless cerous form, in the presence of arsenious acid.³ UI can be expressed as a concentration ($\mu\text{g/L}$),

in relationship to creatinine excretion (μg iodine/ g creatinine), or as 24-h excretion ($\mu\text{g/day}$). To estimate iodine intakes in individuals, 24-h UI collections are preferable. For populations, because it is impractical to collect 24-h samples in field studies, UI can be measured in spot urine specimens from a representative sample of the target group, and expressed as the median, in $\mu\text{g/L}$ ¹¹ (Table 25.1). Variations in hydration among individuals generally even out in a large number of samples, so that the median UI in spot samples correlates well with that from 24-h samples. Creatinine may be unreliable for estimating daily iodine excretion from spot samples, especially in malnourished subjects where creatinine concentration is low.

Spot UI measurements from populations are often misinterpreted; it is a common mistake to assume that all subjects with a spot UI < 100 $\mu\text{g/L}$ are iodine deficient, but even in iodine-sufficient regions, individual spot UI concentrations are highly variable from day-to-day.

D. TSH (Thyrotropin)

Because serum TSH is determined mainly by the level of circulating thyroid hormone, which in turn reflects iodine intake, TSH can be used as an indicator of iodine nutrition. However, in older children and adults, although serum TSH may be slightly increased by iodine deficiency, values often remain within the normal range.¹¹ TSH is therefore a relatively insensitive indicator of iodine nutrition in adults. In contrast, TSH is a sensitive indicator of iodine status in the newborn period. Compared to the adult, the newborn thyroid contains less iodine but has higher rates of iodine turnover. Particularly when iodine supply is low, maintaining high iodine turnover requires increased TSH stimulation. Serum TSH concentrations are therefore increased in iodine-deficient infants for the first few weeks of life, a condition termed transient newborn hypothyroidism.

In areas of iodine deficiency, an increase in transient newborn hypothyroidism, indicated by >3% of newborn TSH values above the threshold of 5 mU/L whole blood, suggests iodine deficiency in the population.¹³ TSH is used in many countries for routine newborn screening to detect congenital hypothyroidism. If already in place, such screening offers a sensitive indicator of iodine nutrition. Newborn TSH is an important measure because it reflects iodine status during a period when the developing brain is particularly sensitive to iodine deficiency.

E. Thyroglobulin

Thyroglobulin (Tg) is synthesized only in the thyroid and is the most abundant intrathyroidal protein. In iodine sufficiency, small amounts of Tg are secreted into the circulation, and serum Tg is normally <10 $\mu\text{g/L}$. In areas of endemic goiter, serum Tg increases due

TABLE 25.1 Updated epidemiological criteria from the World Health Organization,¹¹ with modification,¹⁷ for assessment of iodine nutrition in a population, based on median or range of urinary iodine concentrations.

Urinary iodine concentration of a population group (µg/L)	Iodine intake	Iodine nutrition
<i>School-aged children</i>		
<20	Insufficient	Severe iodine deficiency
20–49	Insufficient	Moderate iodine deficiency
50–99	Insufficient	Mild iodine deficiency
100–299	Adequate	Optimum
>300	Excessive	Risk of adverse health consequences (iodine-induced hyperthyroidism, autoimmune thyroid disease)
<i>Pregnant women</i>		
<150	Insufficient	
150–249	Adequate	
250–499	More than adequate	
≥500 ^a	Excessive	
<i>Lactating women^b</i>		
<100	Insufficient	
≥100	Adequate	
<i>Children less than 2 years of age</i>		
<100	Insufficient	
≥100	Adequate	

^aThe term excessive means in excess of the amount needed to prevent and control iodine deficiency.

^bIn lactating women, the values for median urinary iodine excretion are less than the iodine requirements to account for the iodine excreted in breast milk.

to greater thyroid cell mass and TSH stimulation. Serum Tg is well correlated with the severity of iodine deficiency as measured by UI and is a sensitive indicator of iodine repletion.¹⁴ Tg can also be assayed on dried blood spots taken by a finger prick, simplifying collection and transport. Reference ranges for Tg on dried blood spots in school-age children and pregnancy are available,^{15,16} and Tg is recommended to assess iodine status in children.¹¹ Tg may also be a biomarker of iodine excess in populations.¹⁷

F. Thyroid Hormone Concentrations

In contrast, thyroid hormone concentrations are poor indicators of iodine status. In iodine-deficient populations,

serum T3 increases or remains unchanged, and serum T4 usually decreases. However, these changes are often within the normal range, and the overlap with iodine-sufficient populations is large enough to make thyroid hormone levels an insensitive measure of iodine nutrition.¹¹

G. Reference Intakes

The US Food and Nutrition Board of the Institute of Medicine (IOM) set an Adequate Intake for iodine in infancy and Recommended Dietary Allowances (RDAs) for children, adults, and pregnant and lactating women in 2002 for the United States and Canada⁶ (Table 25.2). The basis of the recommended intake for infants is the

TABLE 25.2 Recommendations for iodine intake ($\mu\text{g}/\text{day}$) by age or population group.

Age group	Institute of Medicine ⁶ For the United States and Canada				World Health Organization ¹¹	
	EAR	AI	RDA	UL	Age group	RNI
Infants 0–6 mo 7–12 mo		110 130		a	Infants 0–12 mo	90
Children 1–3 y 4–8 y 9–13 y 14–18 y	65 65 73 95		90 90 120 150	200 300 600 900	Children 1–5 y 6–12 y	90 120
Adults 19 y +	95		150	1100	Adults 12 y +	150
Pregnancy 14–18 y 19+ y	160 160		220 220	900 1100	Pregnancy	250
Lactation 14–18 y 19+ y	209 209		290 290	900 1100	Lactation	250

AI, Adequate Intake; EAR, Estimated Average Requirement; RDA, Recommended Dietary Allowance; RNI, Recommended Nutrient Intake; UL, Tolerable Upper Intake Level.

^aUL for infants not possible to establish, source of intake should be from food and formula only.

observed mean iodine intake of infants exclusively fed human milk, while the RDAs for children were based on UI excretion and relationship to prevalence of goiter in school children. Adult RDAs were based on thyroidal radioiodine accumulation in euthyroid adults⁶ and, for pregnancy and lactation, amounts needed for the growing fetus and the amount that is secreted in breast milk. The WHO established recommended nutrient intakes for iodine were most recently reported in 2007¹¹ (Table 25.2).

H. Tolerable Upper Intake Levels

The US Food and Nutrition Board, IOM, set a Tolerable Upper Intake Level (UL) for iodine.⁶ The UL is the highest level of daily intake that is likely to pose no risk of adverse health effects in almost all apparently healthy individuals. The UL is 200 $\mu\text{g}/\text{day}$ for ages 1–3 years, 300 $\mu\text{g}/\text{day}$ for ages 4–8 years, 600 $\mu\text{g}/\text{day}$ for ages 9–13 years, 900 $\mu\text{g}/\text{day}$ for ages 14–18 years, and 1100 $\mu\text{g}/\text{day}$ thereafter (Table 25.2). The adverse effects identified and upon which the ULs were based were two studies in which elevated TSH concentrations were seen in men and women receiving iodine supplements for 2 weeks or longer and who had normal UI excretion prior to the initiation of supplementation.⁶ It should be noted that individuals with autoimmune thyroid disease or chronic iodine deficiency may respond adversely to intakes lower than these.¹⁸

I. Dietary Sources

Iodine-deficient soils are common in mountainous areas (e.g., the Alps, Andes, Atlas, and Himalaya ranges) and areas of frequent flooding, especially in South and Southeast Asia (e.g., the Ganges River plain of Northern India). Many inland areas, including those of Central Asia and Africa and Central and Eastern Europe, are iodine deficient. Iodine deficiency in populations residing in these areas will persist until iodine enters the food chain through addition of iodine to foods (e.g., iodization of salt) or dietary diversification that introduces foods produced outside the iodine-deficient area.

J. Foods

The native iodine content of most foods and beverages is low. In general, commonly consumed foods provide 3–80 μg per serving.^{19,20} Foods of marine origin have higher iodine content because marine plants and animals concentrate iodine from seawater. Iodine in organic form occurs in high amounts in certain seaweeds. Inhabitants of the coastal regions of Japan, whose diets contain large amounts of seaweed, have remarkably high iodine intakes, amounting to 50–80 mg/day.³ In the US Total Diet Study, between 2008 and 2012, the estimated mean iodine intake for the total US population was 216 $\mu\text{g}/\text{person}/\text{day}$, with all age/sex groups exceeding their respective EARs.²⁰

Major dietary sources of iodine in the United States are bread and milk,²⁰ and iodine bioavailability from

milk is high.²¹ In Switzerland, based on direct food analysis, mean intake of dietary iodine is $\approx 140\text{ }\mu\text{g/day}$, mainly from bread and dairy products.¹⁹

In many countries, use of iodized salt in households for cooking and at the table provides additional iodine. Boiling, baking, and canning of foods containing iodated salt cause only small losses ($\leq 10\%$) of iodine content.²² Cooking potatoes, pasta, or rice in water containing iodized salt sharply increases their iodine content.²³ Iodine content in foods is also influenced by iodine-containing compounds used in irrigation, fertilizers, and livestock feed.²⁴ Iodophors used for cleaning milk cans and teats can increase the native iodine content of dairy products.²⁴ Traditionally, iodate was used in bread making as a dough conditioner,³ but it is being replaced by non-iodine-containing conditioners.

K. Iodized Salt

There are two methods commonly used to correct iodine deficiency in a population: iodized oil supplements and iodized salt. In nearly all regions affected by iodine deficiency, the most effective way to control iodine deficiency is through salt iodization.¹¹ Salt iodization is recommended because

- Salt is one of few foodstuffs consumed by virtually everyone.
- Salt intake is fairly consistent through the year.
- Iodization technology is simple and relatively inexpensive to implement.
- The addition of iodine to salt does not affect its color or taste.
- The quantity of iodine in salt can be simply and inexpensively monitored.

All salt for human consumption, including salt used in the food industry, should be continuously iodized.¹¹ In Switzerland, previously effected by endemic goiter and cretinism, a national iodized salt program, in place for over half a century, has effectively eliminated iodine deficiency. Iodine can be added to salt in the form of potassium iodide (KI) or potassium iodate (KIO₃). Because KIO₃ has higher stability in the presence of salt impurities, humidity, and porous packaging,¹¹ it is the recommended form. Iodine is usually added at a level of 20–40 mg iodine/kg salt, depending on local salt intake.¹¹ However in industrialized countries, because typically $\sim 80\text{--}90\%$ of salt consumption is from purchased processed foods, if only household salt is iodized, it will not supply adequate iodine. Thus, it is critical that the food industry use iodized salt. The current push to reduce salt consumption to prevent chronic diseases and the policy of salt iodization to eliminate

iodine deficiency do not conflict: iodization methods can fortify salt to provide recommended iodine intakes even if per capita salt intakes are reduced to $<5\text{ g/day}$.²⁵

V. ISSUES RELATED TO IODINE IN HUMAN HEALTH

A. Iodine Deficiency Disorders

Iodine deficiency has multiple adverse effects on growth and development in animals and (Table 25.3) humans.

B. Goiter

Thyroid enlargement (goiter) is the classic sign of iodine deficiency. It is a physiologic adaptation to chronic iodine deficiency. As iodine intake falls, secretion of TSH increases in an effort to maximize uptake of available iodine, and TSH stimulates thyroid hypertrophy and hyperplasia.³ Initially, goiters are characterized by diffuse, homogeneous enlargement, but over time, thyroid follicles may fuse and become encapsulated, a condition termed nodular goiter. Large goiters may be cosmetically unattractive, can obstruct the trachea and esophagus, and may damage the recurrent laryngeal nerves and cause hoarseness³ (Fig. 25.4A). Surgery to reduce goiter has significant risks, including bleeding and nerve damage, and hypothyroidism may develop after removal of thyroid tissue.

TABLE 25.3 Clinical signs and outcomes seen in the iodine deficiency disorders, by age group.^{11,18}

Age groups	Health consequences of iodine deficiency
All ages	Goiter Increased susceptibility of the thyroid gland to nuclear radiation
Fetus	Abortion Stillbirth Congenital anomalies Perinatal mortality
Neonate	Infant mortality Endemic cretinism
Child and adolescent	Impaired mental function Delayed physical development
Adults	Impaired mental function Reduced work productivity Toxic nodular goiter; iodine-induced hyperthyroidism Hypothyroidism in moderate-to-severe iodine deficiency



FIGURE 25.4 (A) Large nodular goiter in a 14-year-old boy in northern Morocco, with tracheal and esophageal compression and hoarseness, likely due to damage to the recurrent laryngeal nerves. (B) This 9-year-old girl from Western China demonstrates the three characteristic features of neurological cretinism: severe mental deficiency together with squint, deaf mutism, and motor spasticity of the arms and legs. *Reproduced with permission from Michael Zimmermann.*

C. Cretinism

Although goiter is the most visible effect of iodine deficiency, the most serious adverse effect is damage to reproduction. Severe iodine deficiency during pregnancy is associated with a greater incidence of stillbirths, abortions, and congenital abnormalities.³ Iodine prophylaxis with iodized oil in pregnant women in areas of severe deficiency reduces fetal and perinatal mortality.²⁶ The fetal brain is particularly vulnerable to iodine deficiency. Normal levels of thyroid hormones are required for neuronal migration and myelination of the central nervous system.²⁷ The most severe form of neurological damage from fetal hypothyroidism is termed cretinism (Fig. 25.4B). It is characterized by gross mental retardation along with varying degrees of short stature, deaf mutism, and spasticity.³

Two distinct types of cretinism have been described. The more common, neurologic cretinism, has specific neurologic deficits that include spastic quadriplegia with sparing of the distal extremities. The second type, the myxedematous form, is seen most frequently in central Africa and has the predominant finding of profound hypothyroidism, with thyroid atrophy and fibrosis.³ Up to 10% of populations with severe iodine deficiency may be cretinous. Iodine prophylaxis has completely eliminated the appearance of new cases of cretinism in previously iodine-deficient Switzerland.¹

D. Cognitive Impairment

Although new cases of cretinism are now rare, mild-to-moderate iodine deficiency still affects many countries

(see below) and can impair cognitive development. Two recent systematic reviews^{28,29} have confirmed the benefits of correcting iodine deficiency. The first systematic review looked at 89 studies that provided iodized salt to pediatric and adult populations and reported a significant 72%–76% reduction in risk for low intelligence (defined as IQ < 70) and an 8.2–10.5 point overall increase in IQ.²⁸ The second systematic review similarly concluded that iodine-sufficient children have a 6.9–10.2 point higher IQ than iodine-deficient children.²⁹ Iodine deficiency is thus considered one of the most common causes of preventable mental impairment worldwide.¹¹ Even in areas of mild-to-moderate iodine deficiency, cognitive impairment in school-age children is at least partially reversible by administration of iodine.^{30,31} Overall, iodine deficiency produces subtle but widespread adverse effects in a population, including decreased educability, apathy, and reduced work productivity, resulting in impaired social and economic development.³

E. Thyroid Disorders in Adults

Iodine status of a population is an important determinant of thyroid diseases in adults.¹⁸ In adults, iodine deficiency increases risk for diffuse goiter and thyroid nodules. As an adult population moves from severe iodine deficiency to mild iodine deficiency and then to iodine sufficiency, there is a shift from excess hypothyroidism to excess hyperthyroidism, which is transient, and then a small shift back toward excess mild subclinical hypothyroidism.¹⁸ Severe iodine deficiency causes hypothyroidism because, despite an increase in thyroid activity to maximize iodine uptake and recycling, iodine

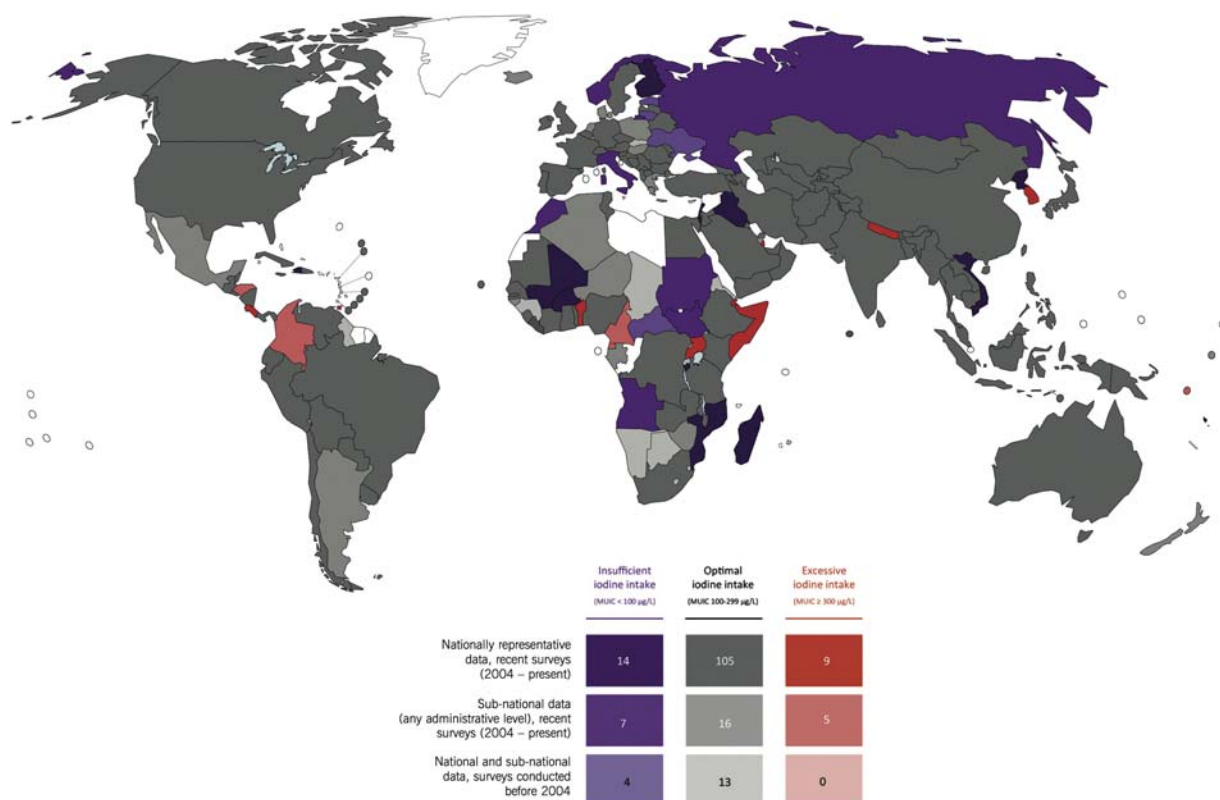


FIGURE 25.5 Global status of iodine nutrition in 2019, based on the median urinary iodine concentration (MUIC) for each country.³⁵ Reproduced with permission from the Iodine Global Network.

concentrations are simply not high enough to maintain thyroid hormone production. In mild-to-moderate iodine deficiency, the thyroid gland is able to compensate for deficient dietary intake by increasing thyroid activity, which maintains thyroid hormone production, but at a price: in some individuals, chronic stimulation of the thyroid leads to thyroid nodularity and autonomy.¹⁸ This increase in nodularity subsequently increases risk of hyperthyroidism if iodine intakes are raised by supplementation or fortification. However, this increased risk of hyperthyroidism in the population is transient, as iodine sufficiency normalizes thyroid activity, resulting, in the long term, in reduced nodularity and autonomy.³²

The small increase in mild subclinical hypothyroidism that occurs with a shift from deficient to optimum or excessive iodine intakes may be linked to an increased risk of thyroid autoimmunity and may also be transient, but more long-term studies are needed.³² The overall incidence of thyroid cancer in populations does not seem to be affected by the usual range of normal to high iodine intakes from dietary sources.³³ However, iodine deficiency is a risk factor for thyroid cancer, particularly for follicular thyroid cancer, and possibly for anaplastic thyroid cancer.³⁴

F. Epidemiology

Only a few countries—including Switzerland, the Scandinavian countries, Australia, the United States, and Canada—were completely iodine sufficient before 1990. Since then, widespread introduction of iodized salt has produced dramatic reductions in iodine deficiency. Over the past 15 years, the number of iodine-sufficient countries has increased from 67 to 120, showing the major progress achieved through increased salt iodization.³⁵ National or large subnational iodine surveys have been done in the past decade in over 150 countries, representing 98% of the world's population.³⁵ In 2019, iodine intake was found to be adequate in 121 countries, deficient in 21 countries, and excessive in 14 countries.³⁵ Of the 21 deficient countries, 19 were mildly iodine deficient, 2 were classified as moderately iodine deficient, and none was severely deficient, based on the mUIC (Fig. 25.5).

G. Iodine Excess

Acute iodine poisoning caused by ingestion of many times the UL causes gastrointestinal irritation, abdominal pain, nausea, vomiting, and diarrhea, as

well as cardiovascular symptoms, coma, and cyanosis.³ Excess iodine intake may very rarely precipitate iododerma, a skin disorder consisting of acneiform eruptions, pruritic rash, and urticaria. However, most people are remarkably tolerant to high dietary intakes of iodine.^{3,6}

Subclinical hypothyroidism

In iodine-sufficient individuals, the earliest effect of high iodine intakes is typically an increase in serum TSH without a decrease in serum T4 or T3, a condition termed subclinical hypothyroidism.¹⁸ Large excesses of iodine inhibit thyroid hormone production, leading to increased TSH stimulation, thyroid growth, and goiter. A clinical trial in healthy adults found TSH concentrations were increased by total iodine intakes of ≥ 750 $\mu\text{g/day}$.³⁶ In children, chronic intakes of ≥ 500 $\mu\text{g/day}$ are associated with increased thyroid volume, an early sign of thyroid dysfunction.³⁷ Iodine-induced goiter and hypothyroidism can occur in newborns due to high maternal intakes³⁸ or through exposure to excess iodine at delivery from the use of antiseptics containing beta-iodine.

Iodine-induced hyperthyroidism

A rapid increase in iodine intake of populations with chronic iodine deficiency may precipitate iodine-induced hyperthyroidism (IIH).³⁹ This is more likely to occur if the iodine is given in excess, for example, if the iodine content of iodized salt is too high, or when iodine-containing medication is given. IIH occurs mainly in older people with nodular goiter. Thyrocytes in nodules often become insensitive to TSH control, and if iodine supply is suddenly increased, these autonomous nodules may overproduce thyroid hormone.⁴⁰ Symptoms of IIH include weight loss, tachycardia, muscle weakness, and skin warmth, without the ophthalmopathy of Graves' disease. IIH is dangerous when superimposed on underlying heart disease and may be lethal.³⁹ Introduction of iodine prophylaxis has been associated with increased hospitalizations for IIH in Europe, the United States, and several African countries.³⁹ The incidence gradually abates but may rise again when the level of iodine in salt is increased. Its occurrence should not be an argument against salt iodization, as the underlying cause of most autonomous nodules and IIH is chronic iodine deficiency. To reduce risk for IIH, the iodine level in salt should be monitored and reduced if too high.

H. Known Interactions

Goitrogens

Iodine deficiency is the main cause of endemic goiter (see below), but other dietary substances that interfere with thyroid metabolism can aggravate the effect, and they are termed goitrogens.⁴¹ A well-known example is linamarin, a thioglycoside found in cassava, a staple

food in many developing countries. If cassava is not adequately soaked or cooked to remove the linamarin, it is hydrolyzed in the gut to release cyanide, which is metabolized to thiocyanate. Thiocyanate blocks thyroidal uptake of iodine.

Other goitrogenic substances are found in millet, sweet potato, beans, and cruciferous vegetables (e.g., cabbage). Soy-based flour can inhibit iodine absorption, and use of soy-based formula without added iodine can produce goiter and hypothyroidism in infants.⁴² Most of these substances do not have a major clinical effect, particularly if they are cooked before consumption, unless there is coexisting iodine deficiency.

Industrial pollutants, including resorcinol, perchlorate, and phthalic acid, may be goitrogenic, and cigarette smoking has been shown to impair iodine metabolism in women.⁴³

Nutrient deficiencies

Deficiencies of selenium, iron, and vitamin A exacerbate the effects of iodine deficiency. Glutathione peroxidase and the deiodinases are selenium-dependent enzymes. In selenium deficiency, accumulated peroxides may damage the thyroid, and impairment of deiodinase activity reduces thyroid hormone synthesis.⁴⁴ Iron deficiency reduces heme-dependent TPO activity in the thyroid and thus impairs production of thyroid hormone. In goitrous children, iron deficiency anemia blunts the efficacy of iodine prophylaxis while iron supplementation improves the efficacy of iodized oil and iodized salt.⁴⁵ Poor iron status during pregnancy may increase risk for maternal hypothyroidism.⁴⁶ Vitamin A deficiency in iodine-deficient children increases TSH stimulation and risk of goiter, probably through decreased vitamin A-mediated suppression of the pituitary TSH β gene.⁴⁷

I. As a Dietary Supplement

Iodine supplements

Dietary supplements often contain iodine. Based on data from the Third National Health and Nutrition Examination Survey (NHANES III), 12% of men and 15% of nonpregnant women took a supplement that contained iodine, and the median intake of iodine from supplements was ≈ 140 $\mu\text{g/day}$ for adults.⁶ Iodine supplements (≈ 150 $\mu\text{g/day}$) are recommended for pregnant and lactating women residing in areas of mild-to-moderate iodine deficiency.⁴⁸ In the United States, because it is uncertain if dietary iodine intakes are adequate in pregnancy, expert groups have recently called for iodine supplementation of this group.⁴⁹ A recent controlled study has demonstrated the safety of daily oral iron supplementation in mildly iodine-deficient pregnant women.⁵⁰ Other sources of iodine include water purification tablets, radiographic contrast media, medicines (e.g.,

amiodarone, an antiarrhythmic drug, contains 75 mg/tablet), and skin disinfectants (e.g., povidone-iodine contains ≈ 10 mg/mL).

Iodized oil and other forms of supplementation

On the population level, in some regions, iodization of salt may not be practical for control of iodine deficiency, at least in the short term. This may occur in remote areas where communications are poor or where there are numerous very small-scale salt producers. In these areas, other options for correction of iodine deficiency should be considered, such as iodized oil supplements.¹¹ Iodized oil is prepared by esterification of the unsaturated fatty acids in seed or vegetable oils and addition of iodine to the

double bonds. It can be given orally or by intramuscular injection. The intramuscular route has a longer duration of action (up to 2 years), but oral administration is more common because it is simpler.

Iodized oil is recommended for populations with moderate-to-severe iodine deficiency that does not have access to iodized salt and may be targeted to women of child-bearing age, pregnant women, and children.⁴⁸ Lactating women who receive one dose of 400 mg iodine as oral iodized oil soon after delivery can provide adequate iodine to their infants through breast milk for at least 6 months.⁴⁰ Iodine can also be given as potassium iodide or iodate as drops or tablets⁵¹ and in drinking or irrigation water.⁵²

RESEARCH GAPS

Research gaps in iodine nutrition include

- the associations between population iodine intake and autoimmune thyroid disease and papillary thyroid cancer,
- links between iodine nutrition and the immune response, and
- the effects of mild iodine deficiency during pregnancy on offspring development.

In public health, research gaps include

- how best to ensure adequate iodine during pregnancy and infancy;
- identification of the best monitoring indicators for pregnancy and infancy; and
- how to ensure quality control and sustainability of existing iodized salt programs.

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SELENIUM

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SUMMARY

Selenium is an essential trace element present as a component of many foods and marketed as a dietary supplement. Extreme deficiency is associated with several diseases in endemic areas while excess can lead to selenosis. Selenium has been heavily investigated with regard to its ability to prevent cancer based on animal data and human epidemiology, but attempts to establish this by conducting human trials assessing the benefits of selenium supplements have been disappointing. Many of the known biological roles for selenium can be attributed to being present in a class of proteins referred to as selenoproteins. There are 25 human selenoproteins that include selenium in the form of selenocysteine, the amino acid encoded by the UGA triplet in selenoprotein messenger ribonucleic acids. Many selenoproteins function in redox reactions with the biological functions of about one-third of which is being established.

Keywords: Cancer prevention; Natural supplements; Selenium; Selenoproteins; Selenosis.

I. INTRODUCTION

A. Background

Selenium (Se) is a mineral and essential micronutrient to human life, which was first discovered by a Swedish chemist, Jons Jacob Berzelius, in 1817. The interest in this element was sparked by an illness among sulfuric acid-manufacturing workers. After analysis of a red deposit on lead walls, Berzelius suspected tellurium was the causative agent, but little to none was found. Instead, a new element having similar properties to tellurium was isolated and was named after the moon goddess, Selene.¹ This work was later published in 1818, and many chemical characteristics initially described by Berzelius are still held true today. Throughout the late 1930 and 1940s, selenium was considered toxic to farm animals. Acute poisoning of horses was characterized by blindness, excessive perspiration, abdominal pain, colic, diarrhea, increased heart and respiration rates, and lethargy, a condition described as “blind staggers.” Chronic

toxicity, referred to as “alkali disease” included additional symptoms, including alopecia and cracked hooves. In contrast, inadequate intake of selenium can result in a variety of symptoms including muscle weakness referred to as “Nutritional Muscular Dystrophy,” weight loss, diarrhea, and decreased fertility.²

The essentiality of selenium to health was first established in 1957 by Schwarz and Foltz following experiments of liver necrosis when rats were fed different yeast-based diets. Vitamin E, mistakenly L-cysteine, and a mysterious factor 3 were reported to help alleviate liver necrosis in these animals. Factor 3 was later discovered to contain selenium, and selenium supplementation in vitamin E-deficient rats showed improvement of liver necrosis.³ A major breakthrough in the biology of selenium occurred in 1973 when Flohe et al. and Rotruck et al. revealed that selenium could be specifically incorporated into proteins.^{4,5} After these initial reports, an entire family of proteins in which selenium was present in the form of selenocysteine would later be discovered and many of these proteins still have unknown functions.

B. Nutrient Function and Structure

Selenium is a metalloid element of atomic number 34 with the same number of valence electrons as the other elements below it and above it on the periodic table, including sulfur and tellurium. Inorganic forms of selenium include selenite, selenate, and selenide, while the more common organic forms are selenocysteine and selenomethionine, amino acids where the sulfur atom of cysteine and methionine are replaced with selenium. The main function of selenium is to serve as a component of selenoproteins, most of which participate in redox reactions.^{6,7} In these proteins, selenium, in the form of selenocysteine, typically is positioned within the active site of the protein where it is well suited to support the reactions in which it participates due to its low pK_a (5.2), which assures that the amino acid is deprotonated at physiological pH.⁸

II. NUTRIENT METABOLISM

A. Synthesis

The best-understood biological role of selenium is its serving as an essential component of a family of selenium-containing proteins or selenoproteins. Although the translation of many proteins may involve the nonspecific incorporation of selenium instead of sulfur in sulfur-containing amino acids, there is a class of proteins that specifically contain selenium as the amino acid selenocysteine. In most of these proteins, selenocysteine exists at the active site of the enzyme and participates in redox reactions, readily donating hydrogen atoms to the enzyme's substrates. In humans, there are 25 separately encoded selenoproteins that contain one or more selenocysteines.⁹ Specific incorporation of selenocysteine is directed in response to an in-frame UGA codon in the selenoprotein messenger ribonucleic acid (mRNA) that would otherwise signal the termination of translation. Whether or not selenocysteine is incorporated in the response to the UGA codon during protein synthesis is dependent on the presence of a signal sequence in the 3'-untranslated region of the mRNA, referred to as the selenocysteine incorporation sequence or SECIS element.¹⁰

Biosynthesis of selenocysteine begins with the aminoacylation of a dedicated transfer ribonucleic acid (tRNA) with a serine, followed by the conversion of the serine to phosphoserine and then the replacement of the phosphate group with a selenium atom derived from hydrogen selenide.⁷ Thus, this tRNA serves the unique dual function of being both the site for selenocysteine synthesis from serine and as the adaptor molecule that recognizes the UGA triplet as the codon for selenocysteine.¹¹ Since the UGA triplet encodes the amino acid selenocysteine among representative species across all

of the life kingdoms,¹² selenocysteine is now considered the 21st biologically relevant amino acid, and this designation remains the only expansion of the genetic code since the 20 amino acid code was established.¹³

Nomenclature of human selenocysteine-containing proteins is divided into two categories.¹⁴ One category consists of those selenoproteins for which there is a determined function and the function is reflected in the name. For example, the selenoprotein that reduces methionine sulfoxides, methionine sulfoxide reductase, is designated MSR1. The other group is made up of selenocysteine-containing proteins whose functions are yet to be clearly defined. These proteins are named using the SELENO root followed by a letter, such as SELENOF or SELENOK.

B. Digestion

The bioavailability of selenium depends on the ingested forms (organic or inorganic) as well as other dietary components, such as heavy metals, which could inhibit uptake.¹⁵ Inorganic forms of selenium include selenite and selenate salts. Organic selenium is predominately found as selenomethionine. Selenocysteine and methylated forms of selenomethionine are also present to a lesser degree.¹⁶ Selenomethionine is the most readily absorbed and is metabolized following active transport from the small intestine by a sodium-dependent transporter.¹⁷ After absorption, selenomethionine is bound to hemoglobin and then accumulated in the liver and muscles.^{18,19} Selenocysteine can also be absorbed by the intestine, but at a lower efficiency than selenomethionine.²⁰ Red blood cell uptake of selenium is by a yet unknown transporter.²¹ Inorganic selenium forms are passively absorbed but are stored less efficiently than the organic forms.²² Selenite is more bioavailable than selenate. Experimentally, selenite can be found in erythrocytes within minutes after intravenous administration in rats.²³ Selenite is metabolized following the reduction selenide by glutathione and then is bound to albumin or hemoglobin where further processing will occur in the liver.^{24,25} Unmetabolized selenate is taken up in plasma and then transported to the liver for processing or excreted in urine.²⁶ Selenide is then used for conversion to selenocysteine for selenoprotein incorporation.²⁷ The major steps of selenium metabolism are summarized in Fig. 26.1.

C. Transport

The liver is the major organ responsible for controlling selenium levels in the body, where it is either designated for excretion or for further processing in the synthesis of selenoproteins. Selenoprotein P (SELENOP, previous known as SEPP1) is an extracellular protein that functions in selenium transport and may have antioxidant functions as well.²⁸ SELENOP has up to 10 selenocysteines

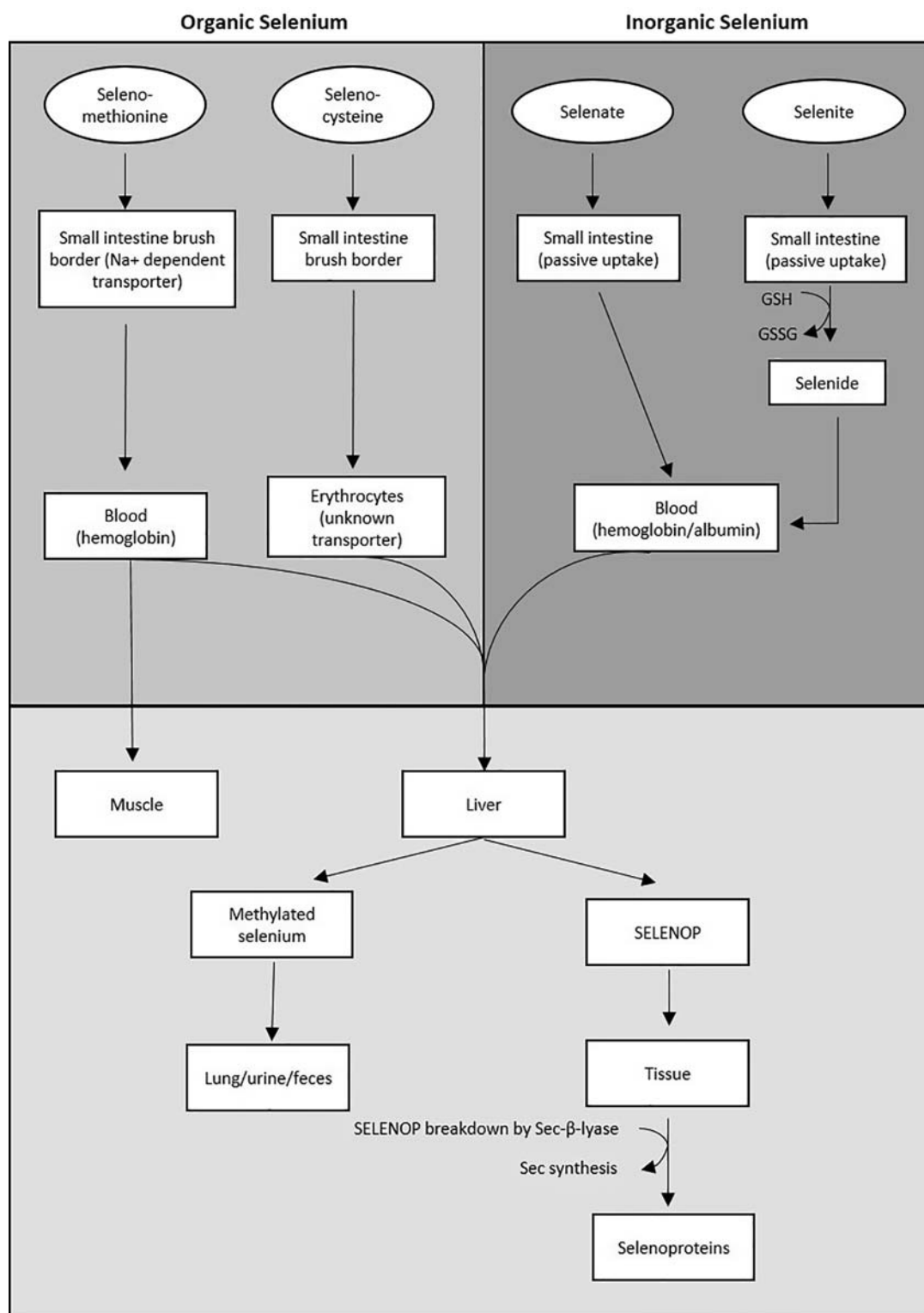


FIGURE 26.1 The major steps of selenium metabolism ranging from ingestion to excretion are presented. *GSH*, glutathione; *GSSG*, reduced glutathione; *Sec*, selenocysteine.

and comprises the major form of selenium in the plasma. Approximately 44% of the selenium in the plasma is SELENOP and another selenoprotein, glutathione peroxidase 3 (GPX3), accounting for 30% of the plasma selenium content. The remaining 37% is associated with albumin.²⁹ Selenium is supplied to the brain and testes by uptake of SELENOP via the apolipoprotein E receptor 2³⁰ and via megalin, another member of the lipoprotein receptor family, in the kidney proximal tubule epithelial cells and perhaps other organs as well.³¹ SELENOP enters the tissue where the protein is catabolized and selenocysteine is converted to selenide for use in selenoprotein synthesis by sec- β -lyase, and the products are funneled into selenoprotein synthesis.³²

D. Metabolism

Excretion of selenium begins in the liver where it is converted to trimethylselenonium by S-adenosylmethionine methylation or to selenosugars, with selenosugars being the likely predominant form excreted.³³ Metabolized selenium is primarily excreted via the kidneys in urine either as selenosugars or as methylated selenide, which, together with absorption, function to regulate homeostasis. In cases of excess selenium intake, selenium does not undergo trimethylation and is excreted from the lung as dimethyl selenide.³⁴ Rats injected with high doses of selenium (2 mg/kg) eliminate up to 26% of the initial dose of selenium via the lungs after 24 h, as compared to 14% excreted via the kidneys as trimethylselenonium.³⁵ Intestinal excretion of selenium was thought to be a minor method of excretion with most selenium being excreted by this route unabsorbed, however evidence points to intestinal excretion being a form of homeostatic control.³⁶ In men, given supranutritional levels of selenium, approximately half of the detected selenium was excreted in the feces.³⁷

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Gene Regulation

There is little evidence that selenium can directly regulate the transcription of specific genes. However, since many of the proteins that require selenium for synthesis and function are antioxidants, genes whose expression are regulated by reactive oxygen-sensitive transcription factors, such as nuclear factor erythroid 2-related factor 2 or nuclear factor kappa-light-chain-enhancer of activated B cells, are likely to be impacted by selenium levels. Several studies have examined the consequences of selenium status on the levels of both coding and noncoding RNAs without determining the biological impact of the observed changes.³⁸ Selenium

levels have a direct impact on the stability and translation of selenoprotein mRNAs. At very low selenium levels, selenoprotein-encoding mRNAs are degraded due to presence of in-frame UGA codons by a cellular surveillance system designed to recognize cellular mRNAs that mistakenly contain premature stop codons, referred to as nonsense-mediated decay.³⁹ Selenoprotein levels are also regulated at the level of protein synthesis. The stability and maturation of the tRNA essential for the synthesis of selenocysteine and the incorporation of that amino acid to the elongating peptide chain are regulated by selenium.^{40,41} Selective regulation of distinct subsets of selenoproteins by selenium is also mediated by the eukaryotic initiation factor 4A-III translation factor that blocks the synthesis of some selenoproteins by binding the elongation factor specific for selenoprotein synthesis, SECIS-binding protein 2 (SBP2).⁴² These mechanisms result in the so-called hierarchy of selenoprotein translation that retains "house keeping" selenoproteins at the expense of those whose functions are considered "stress-related."⁴³

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Myxedematous cretinism

Myxedematous cretinism is a disease found in regions of both selenium and iodine deficiency. This disease presents with aberrant skeletal growth, intellectual disability, hypothyroidism, and atrophy of the thyroid. The classic presentation includes the excessive accumulation of mucin, hence where the syndrome derives its name. Other symptoms include dry, scaly skin, infantile facial features, long bone growth distortion leading to short stature, lack of fusion of the cranial plates, irregular tooth implantation, delayed sexual maturation, and hypothyroidism associated with thyroid atrophy (involution).⁴⁴ "Stippled epiphysis" is the description given to the skeletal abnormalities that have been found due to the mineralization of multiple scattered locations in the epiphyseal cartilage.⁴⁴ There are two main forms of this disease, neurological cretinism and endemic myxedematous cretinism. Neurological cretinism is prevalent in areas with very severe iodine deficiency with or without selenium deficiency in the geographic region.⁴⁵ This disorder predominates in Latin America, Asia, and New Guinea.⁴⁶ In contrast, endemic myxedematous cretinism is prevalent in areas with very low selenium levels, approximately 7 times lower than areas with adequate selenium.⁴⁷ Differences between neurological and the endemic type are the appearance of widespread goiters in the endemic type.⁴⁷ This disorder predominates in Central Africa. In a population in the Northern Democratic Republic of Congo,

goiter was present in 65%–85% of its citizens. Upon further investigation, it was determined that myxedematous cretinism of this type was found in 2%–6% of this population.⁴⁷

Both forms of cretinism share insufficient thyroid hormone as a contributory factor, but differences in the disease presentation may be attributed to the time of exposure to the insufficiency, whether occurring during the expression of key developmental genes of the relevant organs.⁴⁸ Insufficient selenium status during fetal neurological development may contribute the neurological form, while insufficiencies during gestation or early childhood possibly result in myxedematous cretinism.⁴⁸ Selenium plays a critical role in thyroid hormone metabolism as the conversion of the prohormone 3,5,3',5'-tetraiodothyronine (T₄, thyroxine) to the active 3,5,3'-triiodothyronine (T₃) by thyronine deiodinases, which are all selenoproteins. Types I and II thyronine deiodinases convert T₄ to T₃ in tissues and plasma. Type III converts T₄ into an inactive isomer 3,3',5'-triiodothyronine (rT₃) and also inactivates T₃ by further deiodination.⁴⁵ Deiodinase type II is the least affected by selenium deficiency and during periods of selenium deficiency, being maintained in critical organs such as the brain by the preferred retention of selenium in that organ.^{49,50} Clinically, patients with myxedematous cretinism present with very low plasma T₃ and T₄ levels with an extremely high TSH level (about 20 times higher than patients without cretinism), indicating that the thyroid gland is less responsive to TSH stimulation. Noncretinous iodine with a selenium deficiency leads to goiter formation with elevated T₄ levels and normal levels of TSH.⁵¹

During fetal neurological development, the disease appears to develop due to the interactions of the maternal iodine and selenium. The fetal thyroid gland does not fully function until about 20 weeks of gestation. Although the gland is not active, fetal tissue deiodinases are active and dependent on maternal T₄.⁴⁶ When the mother is deficient in selenium, low deiodinase activity leads to a greater accumulation of circulating T₄ available to the fetus. During times when only iodine is deficient or combined deficiency in iodine and selenium in the mother, abnormal fetal neurological development occurs due to the low levels of maternal T₄.⁴⁶ For this reason, iodine supplementation to the populations in geographic areas of deficiency will be beneficial in reducing fetal neurological damage.⁵²

Therapy for myxedematous cretinism involves iodine supplementation of the pediatric population with hypothyroid cretinism under the age of five, restoring thyroid function to many individuals. One problem that arises with iodine supplementation is that the thyroid gland could lose responsiveness during childhood. The gland becomes involuted, highly fibrotic, necrotic, and incapable of

supporting growth and sexual maturation.⁴⁶ For individuals with both selenium and iodine deficiencies, both supplements may be administered at the same time. Supplementing with only selenium can be hazardous as only deiodinase activity will be enhanced. Individuals with both deficiencies have decreased deiodinase activity with an accumulation of T₄.⁵³ When selenium alone is provided, the levels of T₄ dramatically decrease after deiodinase activity is restored and T₄ is converted to T₃.⁵² The involuted or atrophied thyroid gland is unable to compensate for the increased metabolism and consumption of T₄.^{46,52} A clinical trial in Northern Zaire of selenium supplementation for 2 months restored recipients to selenium adequacy, defined by the saturation of GPX activity.⁵² Many endemic areas affected by this disease are nations of poverty, and this situation, as well as ongoing political instability, has hindered selenium supplementation and this preventable disease remains prevalent in such areas.⁵⁴

Keshan disease

Keshan disease was first identified in 1935 in Keshan County, China.⁵⁵ This region in China is a remote and hilly area that has low levels of selenium and iodine in the soil, making the food grown on this soil deficient in selenium and iodine.⁵⁶ Cardiomyopathy is the classic presentation, occurring mostly in women of childbearing age and preschool children. Other symptoms include congestive heart failure, stroke from diffuse cardiac thrombosis, and sudden death.⁵⁷ When autopsies of patients with chronic disease were performed, the hearts had dilations of heart chambers, cellular edema, mitochondrial swelling, overlapping fibrotic tissue, and cardiac necrosis.⁵⁸ The estimated average intake of selenium in endemic areas was 10 µg/day as compared to other areas with low selenium where populations do not exhibit cardiomyopathy and the average selenium intake was estimated at 20 µg/day.⁵⁹ The annual prevalence rates of Keshan disease can range from fewer than 10 to several 100 cases per 100,000 people.⁵⁶ Seasonal variation exists, with residents of northern China being more frequently afflicted in winter, while in southern China, this disease was more prevalent in the summer months indicating other possible etiological factors.⁵⁶

Diet was implicated in Keshan disease etiology in areas where the population mainly consumed the same or similar meals consisting of home-grown corn cultivated in low-selenium soil, while the disease was much less prevalent among people who also lived in regions of low selenium content in the soil, but whose diet included a greater variety of food, e.g., different grains and soybeans.⁵⁶ Selenium deficiency was suspected as contributing to Keshan disease, given the overlapping geographical distribution of the disease and low-

selenium areas.⁶⁰ What followed were several large interventional trials in which selenium was provided to residents of the afflicted areas in the form of an oral sodium selenite tablet. After supplementation, the incidence of Keshan disease was dramatically reduced, validating the causal relationship between selenium deficiency and incidence.⁵⁶ More recently, a meta-analysis of 35 studies from 1935–2017 indicated that selenium supplementation provided 80% protection from Keshan disease, affording evidence that the deficiency was deemed a causation by modern epidemiological standards.⁶¹ However, oversupplementation to prevent the disease may also be detrimental as a recent prospective cohort study indicated the possibility of a “u-shaped” association with selenium intake and incidence of disease in children.⁶² As the selenium concentrations increased, the signs and symptoms associated with Keshan disease started to increase.⁶²

In addition to selenium deficiency, environmental factors may impact Keshan disease incidence. An association between patients with Keshan disease and the presence of enteroviral genomic DNA with the closest homology to coxsackie B virus or coxsackie B–like virus has been reported.⁶³ Coxsackie B virus has been associated with cardiomyopathies.⁶⁴ Perhaps, relevant to these observations are animal studies where a nonvirulent strain of coxsackie viruses was injected into selenium-deficient mice and virulent virus emerged that acquired multiple point mutations accounting for the phenotypic change.^{65,66} Whether similar events occur in Keshan disease remains to be determined.

Kashin–Beck disease

Kashin–Beck disease, characterized by osteoarthropathy, is also endemic to China as well as Tibet, Siberia, and North Korea where selenium and iodine are deficient.^{67–69} In 1992, the number cases with Kashin–Beck disease were estimated at 2.2–3 million individuals with the prevalence being almost 80% in endemic areas.⁷⁰ The disease clinically presents with painful arthritis joint pain from enlarged or deformed joints that have limited motion, enlarged or shortened fingers, and morning stiffness.^{70,71} Symptom severity can vary from individual to individual with the common features presenting during childhood or pubertal growth spurt. Mild cases exhibit enlargement or crepitus of small joints, and more serious cases display distorted long bone growth with enlarged joints leading to shorter stature.⁷⁰ Recently, the estimated prevalence of this condition was reduced dramatically from 2.2–3 million to 640,000 individuals after prevention programs were implemented providing selenium-enriched table salt to the population.⁵⁶

Selenium concentrations in foods from endemic areas have been shown to be lower compared to areas not affected,⁷² which is an interesting association with this

disease in areas where both selenium and iodine were deficient, not one mineral or the other. Myxedematous cretinism, described previously, was found in areas where selenium and iodine were deficient, but Kashin–Beck disease is not found in Central Africa. This difference in disease prevalence indicates different and/or multiple etiologies and pathogenesis of each disease.⁷³ Some studies have implicated environmental factors such as mycotoxins and soil decomposition products (fulvic acids) that interact with selenium deficiency to contribute to the development of the disease. Fulvic acid is a water-soluble fraction of plant and animal matter from microbial and chemical degradation. Concentrations were highest in areas where Kashin–Beck disease was endemic compared to areas without the disease,⁷³ and research using animal models with selenium deficiency with fulvic acid supplementation induced fibrosis of joints.⁷⁴ Mycotoxins from contaminated stored grains were also found to be higher in these endemic areas.⁷⁵ Fungal genera included *Alternaria*, *Cladosporium*, *Drechslera*, and *Trichothecium* and the presence of all of these fungi were associated with elevated risk of the disease.⁷⁵ Although no familial trends have been associated with Kashin–Beck disease, one study implied associations between *GPX1* polymorphisms with disease incidence.⁷⁶ More recently, zinc deficiency in addition to selenium deficiency was considered to be a suspect in the pathogenesis of the disease, but more research is required to address the significance of this possibility.⁷⁷

Treatment for Kashin–Beck disease involves the correction or removal of any of the etiological factors discussed: iodine deficiency, selenium deficiency, fungal contamination, and fulvic acid.^{78,79} If corrected early before prolonged exposure, normal bone growth could be restored.⁷³ However, once the bone damage has progressed, there are few options for treatment other than the management of pain and mobility. Surgical methods may be another option to reshape the bones of the ankle and knee, but many cannot afford the cost of the procedure.⁸⁰ More recently, sodium hyaluronate injection into the knees of patients has been recommended for pain.⁸¹ Viscosupplementation of synovial fluid has been used for patients with osteoarthritis with good efficacy and safety, and similar results may be achievable in Kashin–Beck disease patients.^{81,82} With knowledge of the etiological factors, incidence of this disease has dropped dramatically in the present day.⁸³

B. Criteria of Adequacy

The Recommended Daily Allowance (RDA) for selenium indicated by the Food and Nutrition Board of the Institute of Medicine is 55 µg/day for healthy adults over age 13, with higher intake being suggested during pregnancy (60 µg/day) and lactation (70 µg/day).⁸⁴

These values were determined based on the expected amount of selenium required to maximize the activity of a GPX3, the second most abundant selenoprotein in the plasma after SELENOP. GPXs are very responsive to selenium, and the recommended RDA amount is below the levels required to maximize the levels of all selenoproteins. For example, a population in a region in China showed maximal activity of GPX3 when supplemented with 37 µg/day of selenomethionine and twice that amount of selenite.⁸³ When SELENOP levels were examined, the levels did not reach maximal activity at either of those doses or even at the highest dose administered at 61 µg/day of selenomethionine or 66 µg/day of selenite. There is a continuous debate as to what an appropriate selenium RDA should be on and how the appropriate level should be determined. Clearly, higher intake than the RDA would be required to maximize the levels of all selenoproteins.

C. Assessment of Status

Currently, there are multiple methods used to assess selenium status. Selenium levels can be directly quantified by gas chromatography/mass spectrometry⁸⁵ or by neutron activation analysis⁸⁶ depending on the source used. Different tissues and body fluids can be utilized, but the most common sources being blood, plasma, serum, erythrocytes, urine, hair, and nails.^{87,88} The biological source needs to be considered when assessing selenium status. Although plasma selenium measurements are most often cited, they may not be the most reliable.⁸⁹ Blood and urine provide an indication of short-term selenium status, while hair and nails are more typically used as a measure of long-term (months and years) selenium status. Plasma and serum samples may provide the best result for immediate changes in selenium status, whereas whole blood can provide long-term status.⁹⁰ For recent intake of selenium, urine is often used and provides information about intake that occurred in recent days.⁹⁰

Blood and urine are commonly used because of the relatively noninvasive collection methods and ease of processing and storage. Selenium levels in the urine are closely correlated with predicted intake as assessed through food frequency questionnaires and duplicate food samples.⁹¹ Hair and toenails are even more convenient due to the ease of collection and are typically a reliable assessment of long-term status.⁹¹ However, hair can be a problematic source because common grooming and cleansing products often contain selenium that can contaminate the samples and result in inaccurate estimated levels.⁹² Furthermore, hair growth rates vary among individuals, which may influence selenium accumulation. Nails, on the other hand, have a more consistent growth rate and are less prone to contamination.⁸⁶

Selenium status can be measured indirectly through the quantification of selenium-containing proteins. One approach is to measure the activity of glutathione peroxidase (GPX) using a coupled spectrophotometric assay either in plasma or erythrocytes. This approach may be best in determining selenium deficiency because maximum GPX activity can be achieved with adequate levels of selenium.⁹³ Therefore, assessing higher intake or toxicity can be problematic with this approach. Measurement of SELENOP, the major selenoprotein in plasma, has been considered as a candidate for selenium status.⁸³ SELENOP levels were shown to be optimized at 75 µg/day of selenium intake as compared to GPX in the serum, which expressed maximal activity at 35 mg/day.⁸³

D. Reference Intakes

As mentioned previously, the RDA for selenium intake is 55 µg/day for healthy adults over age 13, with higher intake being suggested during pregnancy (60 µg/day) and lactation (70 µg/day) based on the amount of selenium required to maximize GPX3 activity.⁸⁴ This intake did not take into account the differences in body weight. Taking into consideration lean body mass and historical selenium intake, it has been suggested that men require 80 µg/day and women require 57 µg/day.⁵⁴ More recently, it has been suggested that the reference intake should be adjusted to 70 µg/day for men and 60 µg/day for women based on SELENOP levels in the plasma rather than GPX3 activity.⁹⁴

E. Basis for Individual Variability in Requirements

The RDA for selenium increases as age increases, and selenium intake is recommended when a woman is pregnant and/or lactating.⁸⁴ Specific RDA requirements are presented in Table 26.1.

F. Tolerable Upper Intake Levels

According to the Food and Nutrition Board of the Institute of Medicine, the upper limit of selenium intake is suggested to be 400 µg/day. High levels of selenium intake can lead to selenosis or selenium toxicity. Most cases of selenosis were reported in areas where the plant foods consumed were grown in soils where the selenium levels were high.⁹⁵ Some common symptoms of selenium poisoning are alopecia, brittle and discolored nails, dermatitis if topical exposure was used, peripheral neuropathy, nausea, diarrhea, fatigue, irritability, and breath odor resembling garlic.⁹⁶ Many of the symptoms typically occur within about 10 days following exposure with the fingernail discoloration and loss and alopecia persisting for longer than 90 days.⁹⁷ Although selenosis

TABLE 26.1 RDA values for dietary reference intakes.

Estimated average requirement ⁹⁴			Recommended dietary allowance ⁸⁴		
Age	Male (µg/day)	Female (µg/day)	Age	Male (µg/day)	Female (µg/day)
0–4 months	10	10	0–6 months	15	15
4–12 months	15	15	7–12 months	20	20
1–4 years	15	15	1–3 years	20	20
4–7 years	20	20	4–8 years	30	30
7–10 years	30	30	9–13 years	40	40
10–13 years	45	45	14–18 years	55	55
13–15 years	60	60	Adults	55	55
15–19 years	70	60			
Adults	70	60			
Pregnant women	—	60	Pregnant women	—	60
Lactating women	—	75	Lactating women	—	70

Adapted from Kipp et al., *The Institute of Medicine, Food and Nutrition Board*.

presents with multiple symptoms, treatment involves stopping the source of the increased intake and treating the symptoms. Usually no hospitalization is required for selenium poisoning.

The exact causes of selenium toxicity symptoms are unknown, but most likely are multifactorial. One likely hypothesis involves the erroneous substitution of selenium for sulfur in sulfur-containing amino acids cysteine and methionine as demonstrated in yeast studies.⁹⁸ Furthermore, redox balance of cells by selenium-induced oxidative stress could be another contributing factor. Experimentally, when cells in culture or animals are exposed to high levels of selenium elevated mutagenesis, chromosomal abnormalities, micronuclei formation, and the induction of cell cycle arrest and apoptosis may occur.⁹⁹

An outbreak of selenosis occurred in the in the United States with 201 patients across 10 states reporting similar symptoms, with only one person being hospitalized.¹⁰⁰ During this outbreak, the median estimated dose of selenium that was consumed was 41,749 µg/day. As interviews of patients began, the use of a common liquid dietary supplement was found among the cases. This supplement claimed to provide a balance of nutrients that would “maintain energy and sustain health”.⁹⁷ According to the product’s nutritional fact sheet, it contained a blend of 16 vitamins, 12 elements, 58 “trace elements,” amino acids, essential fatty acids, spirulina, coenzyme Q10, and other antioxidants. The FDA analyzed the product and found that the selenium concentration was 1360 µg/mL, which was approximately 200 times higher than the advertised 6.6 µg/mL and was more than 700 times the RDA.⁹⁷ While examples of selenosis continue to be reported, they are rare and

the greatest concerns are the adverse consequences from selenium supplementation above the RDA.

G. Dietary Sources

Water

The levels of selenium in surface and ground water range from 0.06 µg/L to 400 µg/L depending on geographical location.¹⁰¹ Even in the oceans, selenium is detectable at approximately 0.09 µg/L to 0.11 µg/L.¹⁰² Selenium is also present in drinking water and only slightly contributes to the daily intake.¹⁰³ The major forms found in drinking water are inorganic selenate and selenite ions. The Environmental Protection Agency (EPA) and the National Primary Drinking Water Regulations put forth legally enforceable standards for the levels of contaminants. In the case of selenium, the maximum allowable amount is currently 50 µg/L.¹⁹ The World Health Organization (WHO) recommends a lower level in the water. In the 1950s, WHO recommended levels were similar to EPA standards, but in the 1960s, this limit was reduced to 10 µg/L and this recommendation still holds today with certain exceptions where areas with low selenium are known. Selenium levels in the water are important to consider because many aquatic life forms, such as marine invertebrate and vertebrates, take up the nutrient and become part of human consumption through the food chain.

Plant products

Plant selenium concentrations vary depending on the amount of selenium found in the soil. The selenium found in the soil comes from the earth’s crust unless added as fertilizer. Absorption of selenium by plants is

dependent on selenium concentration as well as the pH of the soil. Selenite is predominately found in acidic soil while the oxidized form, selenate, is found in alkaline soils.¹⁰⁴ Selenate is considered the more soluble form and is readily taken up by plants.¹⁰⁴ Wheat is one of the primary dietary sources of selenium, with the selenium concentration varying depending on the amount present in the soil. In the United States, wheat products can range from 3 to 100 µg/100g depending on the food.¹⁰⁵ The predominant species of selenium is selenomethionine (about 55%) with selenocysteine and inorganic selenium content contributing only slightly.¹⁰⁵ Typically grown naturally without selenium supplementation, most fruits and vegetables contain small amounts of selenium. There are the exceptions such as onions, wild leeks, garlic, and broccoli that can be grown under selenium-enriched environments and will accumulate selenium.¹⁰⁶ Again, the predominant form of selenium is selenomethionine. Brazil nuts have reported some of the highest plant-based selenium content, ranging from 1470–1917 µg/100g, with the majority being in the form of selenomethionine.¹⁰⁷

Animal products

For nonvegetarians and nonvegans, animal products comprise a relatively large amount of dietary selenium and are considered to be a good source of selenium. The bioavailability of selenium in meat product depends on the selenium that was consumed by the animal. In the United States and internationally, selenium supplementation of cattle, hogs, and chicken is a common practice.¹⁰⁸ Different cuts and sources of meat have varying selenium concentration with the most selenium content in organ meats. In the United States, meats such as beef, chicken, pork, and lamb have an average selenium concentration ranging from 10 to 40 µg/100g.¹⁰⁸ The varying amounts can be attributed to differences in soil selenium concentrations of the grazed land. Selenomethionine (about 60%) and selenocysteine (about 50%) are the predominant form of selenium when animals are not supplemented with selenium. Eggs contain approximately 26 µg/100g, and fish on average contains about 27 µg/100g.^{109,110} The selenium content in fish can range from 12 to over 70 µg/100g depending on the species and amount of selenium in the water environment.¹⁰⁹ When animals are supplemented with selenocysteine, the primary form of selenium for human consumption will be both selenocysteine and selenomethionine. On the other hand, when inorganic selenite or selenate is used in the feed, the primary form of human consumption will only be selenocysteine.¹¹¹

The US Food and Drug Administration (FDA) has developed Daily Values (DVs) for use on food and dietary supplement labels to assist the US population in comparing products. The DVs are based on the DRIs

for adults and children age 4 years and older and for pregnant and lactating women based on a 2000 kcal diet. Originally, no DVs were established for selenium. New DVs for selenium have been established as adults and children ≥4 years: 55mcg; infants through 12 months: 20 mcg; children 1 through 3 years: 20 mcg; and pregnant and lactating women: 70 mcg. These DVs were established through implementation of new labeling laws that require updated DVs on food products and dietary supplements beginning in January 2020.¹¹⁰

V. ISSUES RELATED TO SELENIUM IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Genetic diseases associated with selenium biology: The essentiality of selenoproteins for human health is apparent by the association of defects in selenoprotein synthesis and clinical syndromes. A mutation in the selenocysteine tRNA that prevents its maturation and ability to contribute to the translation of a subset of selenoproteins results in variety of symptoms including pain, fatigue, and muscle weakness.¹¹² While this syndrome is very rare, several families of Iranian, Moroccan, and Japanese ethnicity presented with a neurological illness that affects children, progressive cerebellocerebral atrophy, caused by mutations in the gene for selenocysteine synthase (*SEPSECS*), the protein responsible for the final step of selenocysteine synthesis from phosphoserine.¹¹³ Mutations in the gene for selenocysteine insertion sequence-binding protein 2 (*SCISBP2*), the specific translational elongation factor required for selenoprotein synthesis, result in a host of symptoms, including growth retardation and delayed bone development.¹¹⁴ The disease, referred to as SECISBP2 syndrome, is caused in large part by the impaired production of deiodinases that are selenoproteins and function in thyroid hormone metabolism.¹¹⁴ In addition to each of these syndromes that are the result of mutations that affect different aspects of selenoprotein synthesis, there are several diseases caused by mutation in individual selenoproteins. Mutations in the *SELENON* gene results in a neuromuscular disorder;¹¹⁵ mutations in the gene for thioredoxin reductase 2 (*TXNRD2*) is associated with cardiomyopathy and other symptoms.¹¹⁵ Some individuals diagnosed with sedaghatian-type spondylometaphyseal dysplasia, a disease exhibiting skeletal, heart, and brain pathology, were shown to carry mutations in the gene for glutathione peroxidase 4. A comprehensive review of the diseases attributed to mutations in selenoprotein biosynthesis or individual selenoproteins has recently been published.¹¹⁶

In addition to diseases attributed to mutations in the genes described above, polymorphisms in selenoprotein genes have been associated with increased risk of a number of diseases including Crohn's disease, infection, glaucoma, and diabetes, although the majority of associations have been to cancer risk or outcome.¹¹⁷ Many of these polymorphisms have been experimentally determined to be functional, affecting either the structure of the encoded protein or the regulation of the gene. For example, a single-nucleotide polymorphism (SNP) in the gene for GPX1 results in either a leucine or proline and may result in altering the distribution of the protein between the cytoplasm and mitochondria with distinct phenotypic consequences.^{118,119} An SNP in the gene for another selenoprotein, SELENOF in the 3'-untranslated regulatory region of the mRNA, affects the translational efficiency during protein synthesis.^{120,121} Both SNPs have been associated with increased risk of cancer mortality.¹²²

Selenium, selenoproteins, and cancer

Of the diseases potentially impacted by selenium intake, cancer has received the most attention. The interest in selenium as a chemopreventive agent initiated when Shrauzer et al. published an epidemiological study indicating inverse association between dietary intake of selenium and cancer incidence.¹²³ Many studies have addressed this issue, and a recent meta-analysis consisting of 15 observational cohort studies has indicated reduced cancer incidence and mortality with selenium status, with the decreases in cancer of the stomach, colorectal, lung, breast, bladder, and prostate being evident.¹²⁴ The authors of this publication have described the weaknesses of the studies used in their analysis.

In addition to observational studies, there have been a number of randomized controlled trials (RCTs) evaluating the benefits of selenium supplementation to reduce cancer risk. These trials were motivated by both encouraging epidemiological data and a large number of animal studies, which indicated that providing rodents a nontoxic dose of selenium could prevent tumor formation induced by a wide variety of carcinogens. The largest of these was the Selenium and Vitamin E Cancer Prevention Trial (SELECT), which provided selenium in the form of selenomethionine at 200 mg/day, vitamin E at 400 IU/day, both supplements or placebo to 32,000 men and assessed prostate cancer incidence.¹²⁵ The results of this trial indicated no benefit to the selenium supplementation group as compared to the placebo arm.¹²⁶ This study was included among 10 RCTs in a meta-analysis that indicated that there was no evidence for a chemopreventive benefit for selenium supplementation and that there may be an increased cancer risk at some sites among individuals who have a higher baseline selenium status.¹²⁷

Currently, there is little evidence that supplementation with selenium would be a useful chemopreventive strategy. However, human genetic data have indicated statistically significant associations between SNPs in several selenoproteins and cancer risk and/or mortality.¹²² Human genetic variation at some of these SNPs may be a confounding variable in determining which individuals could benefit from selenium supplementation. SNPs in two selenoproteins whose levels are responsive to selenium availability and potentially function as tumor suppressors, GPX1 and SELENOF, have been associated with cancer risk and mortality, respectively, and are likely to determine the translational efficiency of the corresponding mRNA.^{116–118} Therefore, certain individuals with specific genotypes may benefit from selenium supplements.

B. Known Interactions

Selenium has been purposed as a possible treatment option for heavy metal toxicity such as from high intake of mercury, arsenic, and cadmium. All three of these metals have been shown to form compounds with selenium.^{119–122} High levels of exposure to mercury results in a variety of symptoms as well as significantly decreasing or blocking the activity of the antioxidant protein GPX.¹²³ When mercury and selenium, in the form of selenite, were administered in equal molar ratios, a protective action against the toxic effects (kidney and cardiac damage) of mercury was found in rodent studies.^{124,125} Chloralkali plant workers exposed to significantly high levels of mercury were studied, and levels of GPX activity in erythrocytes were approximately half that found in the control group.¹²⁶ Arsenic and cadmium toxicity interacts with selenium in similar mechanisms as mercury toxicity through the inhibition of various enzymes including glutathione production.^{128,129} Experimentally, selenium significantly reduced lipid peroxidation and restored GPX activity in the liver of rats caused by arsenic resulting from arsenic toxicity.¹³⁰ Other examples of selenium's protective effects against arsenite-induced cardiovascular damage¹³¹ and immunotoxicity¹³² have been reported. In studies using animal models of cadmium toxicity, selenium was protective of the kidney,¹³³ the liver,¹³⁴ reproduction,¹³⁵ and the brain.¹³⁶

C. As a Dietary Supplement

As described above in the case of Keshan disease, selenium supplementation of the diets of the population of Keshan County, China, who were selenium deficient resulted in a dramatic reduction in the cardiomyopathy that characterized the disease. While low levels of selenium have been implicated in increased risk of several diseases in human epidemiological studies, much less

evidence supports the benefits of selenium supplements to improve health among the general population. In spite of this, selenium consisting of varying forms of the element is abundant on the market, both as an individual supplement and as a component of multivitamins/mineral supplements. Suppliers of these products indicate that taking a selenium supplement can reduce the risk of diseases such as cancer, heart disease, and asthma as well as improve the function of the immune system, the liver, and thyroid, energy levels, and mental health. Randomized controlled clinical studies have yet to report a clear benefit of selenium supplementation. Meta-analyses failed to reveal any benefits of selenium supplements for the prevention of prostate cancer¹²⁷ or cardiovascular disease.¹³⁷ The large selenium supplementation study SELECT that failed to see a benefit of

selenium supplementation in the reduction of the risk of prostate cancer¹³⁸ provided the opportunity to assess the benefits of selenium supplementation on diseases other than prostate cancer. As in the case of prostate cancer, these ancillary analyses from SELECT failed to indicate a benefit of selenium supplementation for the prevention of bladder cancer,¹³⁹ cataracts,¹⁴⁰ colorectal adenomas,¹⁴¹ and dementia.¹⁴² It should be noted that as a dietary supplement, the US FDA does not require selenium supplements to be effective in improving any health outcome and products must include the disclaimer indicating that “this product is not intended to diagnose, treat, cure, or prevent a disease.” Moreover, there have been some suggestions that selenium supplementation may increase the risk of type 2 diabetes¹⁴³ as well as increase the risk of certain skin cancers.¹⁴⁴

RESEARCH GAPS

- Determining the best method to assess selenium adequacy.
- The biological functions of many selenoproteins have yet to be resolved.
- Understanding why there is an inverse association between selenium in the diet and cancer risk when supplementation trials have failed to detect a benefit of supplementation.
- Are there health risks associated with selenium supplements to individuals with high baseline levels?

VI. REFERENCES

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CHROMIUM

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SUMMARY

Chromium, as the trivalent ion, has been postulated to be involved regulating carbohydrate and lipid (and potentially also protein) metabolism by enhancing insulin's efficacy. Yet, no convincing evidence that chromium is an essential element has been identified. Dietary chromium apparently is absorbed via passive diffusion, and the extent of absorption is low (~1%). Chromium is maintained in the bloodstream bound to the protein transferrin; chromium is generally believed to be delivered to tissues by transferrin via endocytosis. No unambiguous animal model of chromium deficiency has been established. One limitation in potentially characterizing chromium deficiency in humans is the lack of an accepted biomarker of chromium nutritional status. Attempts to identify a "glucose tolerance factor" have not provided a chemically defined functional compound that conforms with the proposed physiological role of chromium as a facilitator of insulin action *in vivo*.

Keywords: Chromium; Insulin resistance; Low molecular weight chromium-binding substance; Total parenteral nutrition; Transferrin.

I. INTRODUCTION

A. Background

The element chromium possesses two oxidation states that are considered to be biologically and environmentally relevant based on their stability in the presence of water and dioxygen. Compounds containing hexavalent chromium (chromium(6+)) when inhaled and also potentially when ingested orally in large quantity are mutagenic and carcinogenic. Chromium as the trivalent ion or chromic ion (chromium(3+)) will be the focus of this work as it was proposed to be an essential trace element for mammals about six decades ago. However, in the last 20 years, the essential status of chromium has been questioned, and evidence for its essentiality has been found lacking. Chromium is probably best known for its use as a dietary supplement for body mass loss and body composition change and increasing insulin sensitivity, although the former have been thoroughly refuted and the latter lacks evidence for its use to be recommended. Yet, chromium-containing supplements generate hundreds of million dollars in sales

annually as the supplement is available over the counter in numerous forms including pills, chewing gums, sports drinks, and nutrition bars. The history of chromium from its proposed status as an essential element to its declassification as such will be outlined along with a summary of what is known about its intake and transport, potential toxicity, and potential use as an agent to treat insulin resistance and related conditions.

II. NUTRIENT METABOLISM

A. Digestion

Understanding potential biological effects from Cr^{3+} requires establishing how chromium is transported and distributed in the body. Chromium absorption is low; only ~1% of chromium ingested orally is absorbed, with the rest lost through the feces. Chromium is believed to be absorbed from the gastrointestinal tract by passive diffusion. Chromium supplements, such as chromium picolinate, are believed to break down in the stomach, explaining the similar degree of absorption

to that of chromium from inorganic chromium salts such as CrCl_3 .¹ Intestinal perfusate studies suggest chromium is actively transported out of the intestinal cells, as ~94% of the chromium entering the cells was cleared from the cells (leaving only ~6% behind to be stored).² However, no transporter is known for chromium.

B. Transport

Chromium is generally believed to be transported from the bloodstream to tissues by the iron transport protein transferrin and transported from the tissues back to the bloodstream and ultimately to the urine for elimination by the peptide low molecular weight chromium-binding substance (LMWCr). However, each of these steps has been questioned, and data conflicting with this scheme have appeared. Transferrin and LMWCr have been proposed to function in chromium detoxification and to be associated with the potential pharmacological action of chromium at high doses.³

Over 80% of the chromium in the bloodstream is bound to the iron transport protein transferrin.⁴ Under physiological conditions, transferrin selectively binds Fe^{3+} ions, as the metal-binding sites have evolved to bind metal ions with large charge-to-size ratios. The transferrin molecule is comprised of two lobes, the N-terminal and C-terminal lobes, with a high degree of sequence homology. Each lobe possesses a ferric ion-binding site. In humans, the protein is normally about 30% saturated with iron, so that in vivo it has the potential to bind other metal ions. The binding of manganese and chromium, as the trivalent ions, may be physiologically relevant.⁵

The most detailed studies of Cr-transferrin movement have used $^{51}\text{Cr}_2$ -labeled transferrin administered intravenously to rats. Injection of $^{51}\text{Cr}_2$ -labeled transferrin into the bloodstream results in a rapid and insulin-sensitive movement of chromium into the tissues as Cr-transferrin; greater than 50% of the chromium is transported to the tissues within 30 min (plasma membrane recycling of transferrin receptors is sensitive to insulin as increases in insulin result in a stimulation of the movement of transferrin receptors from vesicles to the plasma membrane). Tissue levels of chromium were maximal 30 min after injection; decreases in tissue chromium with time were mirrored by increases in urine chromium. Thus, transferrin, in an insulin-dependent fashion, can transfer chromium to tissues from which chromium is excreted in the urine. The half-life of the ^{51}Cr in the bloodstream in the presence and absence of insulin was estimated to be 8.26 and 6.79 min, respectively; the specific loss of ^{51}Cr -transferrin loss from the blood has a half-life of 8.77 min in the absence of added insulin.⁶ The removal of ^{51}Cr from the blood is faster than the appearance of ^{51}Cr in the urine; the lag in time indicates that the Cr-transferrin in the blood and chromium in the urine are

not in direct equilibrium and that intermediates in the transport of chromium must be involved. One intermediate, LMWCr, has been identified.

However, questions have arisen about the transport of chromium by transferrin. The rate at which transferrin can bind chromium was proposed to be too slow to be physiologically relevant; however, recent studies mimicking physiological conditions in the bloodstream have found that Cr^{3+} can bind sufficiently rapidly.⁷ Transferrin binds chromium concomitantly with an equivalent of (bi)carbonate, making the binding of chromium first order in the concentration of bicarbonate. In the presence of 25 mM bicarbonate (as in the bloodstream), Cr^{3+} binds tightly to transferrin in less than 20 min. A more serious question has arisen as to whether Cr^{3+} loss from transferrin is fast enough to be physiologically relevant.⁸ Uptake of Cr_2 -transferrin has been reported to be significantly different in cultured skeletal muscle cells⁹ and hepatocarcinoma cells,⁸ with the former readily uptaking Cr_2 -transferrin by endocytosis and the latter not. Recent studies have shown that Cr^{3+} loss from one of the two metal-binding sites of transferrin is rapid (c. 6 min half-time, less than the 15 min of the endocytosis cycle) while loss from the other is slower such that only ~10% of the Cr^{3+} would be lost in 15 min (J.B. Vincent and J.F. Rasco, unpublished results). The situation may be more complicated than this as the binding of chromium by transferrin is accompanied by conformation changes in the metal-transferrin complex that may greatly affect the rate of chromium loss.

The existence of LMWCr was first reported by Wada and coworkers in 1981.¹⁰ A low molecular weight chromium compound was identified by size exclusion chromatography of the cytosol of liver cells of male mice injected with a single dose of potassium chromate. A similar low molecular weight compound was found in the feces and urine 2 h after injection in the plasma. These researchers suggested that an LMWCr was formed in the liver that participates in retention and excretion of chromium in the body. LMWCr was subsequently found to occur in urine normally, although the amounts were greatly increased after rats were injected with chromate. Wada and coworkers proposed the urine LMWCr was similar to that isolated from the liver and other organs and to be involved in removing excess chromium from the body.¹¹

LMWCr appears to be a naturally occurring oligopeptide composed of glycine, cysteine, aspartate, and glutamate with the carboxylates comprising more than half of the total amino acid residues. The complete amino acid sequence has not appeared, despite attempts at sequencing by Edman degradation, NMR, and mass spectrometry. The lack of additional characterization of the organic components of the materials was a matter of considerable concern, reminiscent of the situation with

glucose tolerance factor (GTF). However, treatment of LMWCr from bovine, alligator, and chicken livers and human urine with trifluoroacetic acid allowed for the isolation of a heptapeptide component of LMWCr; in mass spectral studies, the heptapeptide from both the livers sources and urine were found to possess a negatively charged molecular ion with a molar mass of 804. Subsequent mass spectrometry studies revealed that the peptide had the sequence pEEEEGDD, where pE is pyroglutamate¹²; the pyroglutamate is formed from glutamate during the isolation of the heptapeptide as LMWCr has an N-terminus that is free to react with chemicals that react with primary amines. Despite its small size, LMWCr from rabbit and bovine liver tightly binds four equivalents of chromic ions. The binding is quite tight and highly cooperative,¹³ such that only apoLMWCr and holoLMWCr (Cr₄-LMWCr) should coexist in solution. Spectroscopic studies suggest that the chromic ions comprise an anion-bridged multinuclear assembly supported by carboxylates from the oligopeptide.¹⁴

Intravenous injection of rats with ⁵¹Cr₂-transferrin results in the appearance of LMWCr loaded with ⁵¹Cr. Approximately 50% of the ⁵¹Cr appeared in the urine within 6 h of injection of Cr-transferrin into the tail vein of rats in the absence of added insulin; insulin treatment concurrent with injection of ⁵¹Cr-labeled transferrin results in approximately 80% of the label appearing in the urine within 180 min, an appreciable enhancement in the movement of Cr from the blood to the tissues to the urine. Separation of the urine components by G-25 size exclusion chromatography revealed that Cr occurred in the urine as a single low molecular weight species, which comigrates with LMWCr when the urine was spiked with LMWCr.⁶

The appearance of ⁵¹Cr in the urine could be fit to a biphasic model. In the presence or absence of insulin, the rate constants were statistically equivalent, 0.0486 and 0.0005 min⁻¹. Thus, insulin did not affect these rates (although the fit is not as good as desired for the data in the absence of insulin suggesting a more complex behavior). However, the relative amounts of Cr that appeared in each pool were affected by insulin. Proportionally more Cr is found in the fast-appearing pool when insulin was coadministered with the ⁵¹Cr-transferrin.⁶

III. LACK OF A ROLE IN NORMAL AND ORGAN FUNCTION

A. Glucose Tolerance Factor

When Cr³⁺ was initially proposed to be an essential trace element for mammals, Cr³⁺ was equated to the term "glucose tolerance factor". In the late 1950s and early 1960s, Schwarz and Mertz examined the effects

of a Torula yeast diet on rats. The diet led to a liver disorder (resulting from the low selenium content of the diet) and an apparent inability to efficiently restore blood glucose levels after an intravenous glucose challenge. The authors postulated the existence of a dietary requirement that was absent from the diet leading to the glucose intolerance, GTF. The addition of inorganic compounds containing over three dozen different elements to the diet (at 200–500 µg/kg body mass) failed to restore glucose tolerance, while several Cr(III) complexes (200 µg Cr/kg body mass) apparently did, leading to the postulate that Cr³⁺ was the dietary factor, GTF.¹⁵ However, the chromium content of the diet was not examined, and the rats were kept in cages with metal components (possibly including stainless steel with its large chromium content); thus, the chromium intake of the rats cannot be ascertained. Additionally, questions about data handling and the significance of the effect observed have been raised, while recently, studies with rats generally fail to observe effects on glucose levels in glucose tolerance tests. Additionally, the amount of chromium supplementation is about 10³ times the typical daily content of a commercial rat chow; hence, the dose of chromium was supranutritional, leaving the possibility that the effect, if real, was pharmacologically relevant rather than nutritionally relevant.

Brewer's yeast, which accumulates chromium, was also shown to apparently restore glucose tolerance when added to the Torula yeast diet. Thus, attempts were made to isolate the chromium-containing species from Brewer's yeast. The product of this endeavor was also termed GTF by Mertz, leading to confusion whether the term relates to the inorganic Cr³⁺ ion or a complex or Cr³⁺ with a biomolecule(s). Since the late 1960s, the latter has become the most common usage of the term GTF. In the late 1960s, GTF was reported to be a complex of Cr³⁺ with nicotinate and the amino acids aspartate, glutamate, and cysteine; however, the full details of the isolation and characterization were not reported until 1977.¹⁶ Subsequently, a three-dimensional structure was even proposed for GTF. Thus, for nearly 40 years, chromium was believed to be an essential trace element as part of the biological molecule GTF. Unfortunately, as has been reviewed previously, the research behind the existence of this GTF complex is extremely flawed. Briefly, the isolation procedure involved refluxing an extract of Brewer's yeast for 18 h in 5 M HCl, conditions that would have destroyed any protein or peptides, nucleic acids, complex carbohydrate, or many other biomolecules that could have potentially bound Cr³⁺. The amounts of chromium, nicotinate, and the amino acids in the material were not reported so that the ratios of the components to one other, if they are even stoichiometric, cannot be determined. Worse still is that the

characterization was performed on a material that was impure, so that none of the components can definitely be said to be associated with the component of the mixture that was active in bioassays (for the potentiation of insulin-dependent glucose metabolism by isolated rat adipocytes). Subsequent investigations have shown that the bioactive component of Brewer's yeast in these assays could be separated from Cr^{3+} , so that chromium is not even a component of the species in the yeast responsible for the increased glucose metabolism.¹⁷ Amazingly, a recent attempt to isolate GTF has been reported but fell into the same pitfall of reporting the composition of an impure material.^{18,19} The use of the term GTF should be terminated.

B. Pharmacological Effects in Rodents

Several rodent models of diabetes and insulin resistance have been utilized to examine the effects of chromium supplementation.²⁰ Dividing these studies by the type of model utilized is important. Three of the models examined to date have symptoms arising from related mutations; the JCR:LA-cp, Zucker obese, and Zucker diabetic fatty rats all have mutations of the leptin receptor generating peripheral insulin resistance. When chromium is administered at a young age to these models, it generally has no effect on body mass and food intake. Chromium administration appears to have no or little effect on fasting blood glucose levels but lowers glycated hemoglobin levels. It also appears generally to be beneficial to lipid metabolism, lowering total cholesterol levels; however, effects on other lipid variables are inconsistent. In contrast, in Goto-Kakizaki rats, which have hepatic insulin resistance rather than peripheral insulin resistance, no effects from chromium supplementation have been found. These conflicting results between the two types of genetic models may suggest that the effects of chromium supplementation are primarily derived from the skeletal muscle.²⁰

For chemical models of insulin resistance, most notably rats treated with streptozotocin or alloxan, the results of chromium supplementation are too inconsistent to allow firm conclusions to be drawn.²⁰ Dietary models have also been examined. Cr administration of rats on high-sugar diets have been shown to have increased insulin sensitivity. Similarly rats on high-fat diets have also been shown to have increased insulin sensitivity, although for both diets conflicting results have been observed. Several studies have looked at the effects of chromium supplementation on rats on high-fat diet that were also treated with streptozotocin. While these studies are similar in observing reduced cholesterol and triglycerides levels, other results, most notably fasting glucose and insulin levels and body mass changes, are not consistent.²⁰

For all the rodent studies, no great dependence appears on chromium dose, length of time of chromium administration, or form of chromium. The origin of the diabetes appears to make a significant difference on the potential benefits of Cr administration. The number of studies with the different models is approaching the point where metaanalysis could prove worthwhile. The influence of commercial concerns needs to be addressed in such studies.²⁰ Yet, overall, rat studies observing beneficial effects from chromium supplementation generally provided rats between 80 and 1000 μg (or even more) Cr/kg body mass daily for periods of weeks or months. Based only on body mass, these doses would correspond to 5.2–65 mg Cr daily for an average 65 kg human. Even when corrected for the metabolic rate, surface area, etc., of rats compared to humans, this range corresponds to ~1–13 mg Cr daily for a human. Thus, human clinical trials may have only started to approach the dose necessary to see a beneficial effect in humans (*vide infra*). The amount of Cr used in clinical trials needs to be increased before ruling out that Cr supplementation has no effect on individuals with type 2 diabetes, insulin resistance, or related conditions. However, one also cannot rule out that beneficial effects might be unique to rodents and perhaps certain other animals.

All of these studies have utilized pharmacologically, rather than nutritionally, relevant doses of chromium. That these doses are all supranutrition has been demonstrated in a seminal study utilizing "low-chromium" diets.²¹ Rats were provided the AIN-93G—purified diet with no added Cr in the mineral mix (<20 μg Cr/kg diet, the lowest quantity of Cr in a rat diet examined to date) and were kept in cages with no access to metal for 6 months. This Cr content is similar to that of a human consuming 30 μg Cr daily (the average of the Dietary Reference Intake (DRI) Adequate Intake value for men and women). No differences in body mass, insulin sensitivity, or response to a glucose challenge were observed compared to rats on the complete AIN-93G diet (with 1000 μg Cr/kg). Adding additional Cr^{3+} to the diet (200 $\mu\text{g}/\text{kg}$ or 1000 $\mu\text{g}/\text{kg}$) also had no effects on body mass but resulted in increased insulin sensitivity as measured by reduced areas under the curve for insulin concentrations in glucose tolerance tests. Insulin sensitivity was shown to increase as a function of added Cr (Fig. 27.1). The key in this study is that beneficial effects from Cr supplementation required supranutritional doses.²¹ In fact, this study suggests that rats on the AIN-93G are already obtaining enough Cr to have beneficial effects from supplementation. Thus, the basal Cr levels in rodent diets must be considered when designing experiments, and effects from supplemental Cr could be difficult to detect on top of beneficial effects from high Cr content in the basal diets. In fact, studies that have misinterpreted the beneficial effect from the

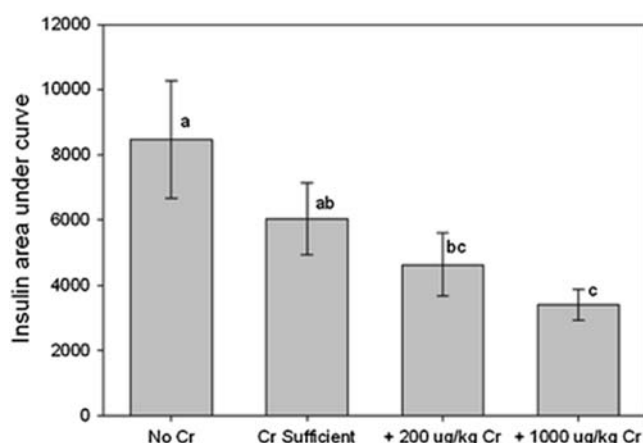


FIGURE 27.1 Area under the curve for plasma insulin concentrations in glucose tolerance tests for Zucker lean rats on the AIN-93G diets. Different letters indicate significant differences between groups. No Cr, rats on the AIN-93G diet, with Cr not included in the mineral mix; Cr sufficient, rats on a purified AIN-93G Cr-sufficient diet (Cr as $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$); +200 $\mu\text{g}/\text{kg}$ Cr, rats on the AIN-93G Cr-sufficient diet with an additional 200 μg Cr/kg (Cr as $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$); +1000 $\mu\text{g}/\text{kg}$ Cr, rats on the AIN-93G Cr-sufficient diet with an additional 1000 μg Cr/kg (Cr as $\text{KCr}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$). Reproduced from Di Bona et al.¹² with the permission of the copyright holder.

supranutritional dose of Cr in this diet have appeared (see, for example, Refs. 22, 23). The Cr content of the AIN-93 diets needs to be reexamined, and Cr probably should be removed from the mineral mix.

C. Molecular Mechanism(s) of Action

Numerous pathways by which a chromium-containing biomolecule could manifest effects on insulin sensitivity have been proposed.²⁴ However, research results in in vitro and in vivo systems are contradictory, such that the state of the field is not immediately clear.¹⁷ Attention has been focused on the insulin signaling cascade, most specifically insulin receptor and Akt (protein kinase B), and on cholesterol synthesis and metabolism as the potential sites of chromium action, although effects on oxidative stress and cytokine generation have also been proposed. A major issue with the in vitro studies is the forms of chromium used. Most commercial chromium supplements are believed to degrade in the gastrointestinal tract, so that the cells of the body never encounter the chromium-containing molecule comprising the supplement. For example, the best-selling supplement, chromium picolinate, is toxic to cells in vitro but harmless at nutritional doses when given orally (vide infra). Thus, the effects of adding the commercial supplements directly to cultured cells could have drastically different effects on insulin sensitivity, and the mechanisms thereof, than those of chromium in vivo, as presumably cells would normally be

exposed to chromium as a complex with transferrin. Such studies using chromium-containing transferrin are lacking. Also control experiments using the free ligands, e.g., picolinic acid, are not often performed.

In vivo studies have suggested several potential targets for the effects of chromium; however, results are not consistent, perhaps in part because different chromium compounds in different doses in different animals and models are being utilized. For example, some studies suggest a role of chromium in the insulin-signaling pathway. Cefalu and coworkers²⁵ observed increases in insulin receptor substrate 1 (IRS-1) tyrosine phosphorylation and IRS-1-associated phosphoinositol-3 (PI-3) kinase activity in skeletal muscle after insulin stimulation in obese JCR:LA-cp rats administered chromium as chromium picolinate compared with obese controls. The enhanced signaling occurred without changes in the concentration of IRS-1, Akt, or PI-3 kinase; however, levels of phosphotyrosine phosphatase 1B (PTP 1B) that dephorylates insulin receptor were reduced. Sreejayan and coworkers²⁶ have found that chromium supplementation as chromium tris(phenylalanine) significantly enhanced insulin-stimulated Akt phosphorylation and membrane-associated glucose transporter 4 (Glut-4) in skeletal muscles of sucrose-fed mice. Increased insulin-stimulated membrane-associated Glut-4 has also been observed in JCR:LA-cp rats administered chromium as chromium picolinate.²⁷ Unfortunately, not all studies are consistent with these results, including failure to observe changes in PTP 1B levels or activity, changes in protein phosphorylation without insulin stimulation, and changes in protein concentration; however, a consensus does seem to appear that chromium supplementation augments insulin-dependent Glut-4 translocation under insulin-resistant conditions.²⁸

IV. DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Currently, no symptoms of chromium deficiency have been definitively established.²⁹ The evidence most commonly cited for chromium being an essential element is case studies of chromium supplementation of patients who developed insulin resistance and received total parenteral nutrition (TPN). However, less than 10 case studies with beneficial effects have been reported.³⁰ Only one study attempting to apply similar conditions to multiple adult trauma subjects receiving TPN has appeared; the study observed no statistically significant effects of chromium supplementation on serum glucose or insulin levels.³¹ Subjects on TPN with known chromium content ($\sim 5\text{--}16 \mu\text{g}$ Cr/day) received oral

equivalent quantities of chromium in excess of the Adequate Intake before the TPN solutions were supplemented with chromium. Given the average Adequate Intake of 30 μg Cr/day and assuming 1% of chromium is absorbed, an individual on a typical American diet would have ~ 0.3 μg of chromium enter the bloodstream daily; as all the chromium in TPN is introduced into the bloodstream, the content of the TPN not supplemented with chromium provided greater than 15 times the AI. Subjects in the TPN studies treated with chromium-supplemented TPN received an additional 12, 40, or 150–250 μg Cr/day, another 40- to 800-fold increase above the AI. These doses clearly represent pharmacologically relevant, rather than nutritionally relevant doses.

Thus, no relationship can be established between the results of these case studies and chromium deficiency,³⁰ although the studies could suggest a pharmacological effect in humans with altered glucose or carbohydrate metabolism. Additional care must be taken in interpreting these case studies as the conditions of the subjects before entering the studies varied considerably as did their time on TPN, the amount of chromium received, and several other variables. The European Food Safety Authority determined that “it is unclear on the basis of these case reports whether deficiency of chromium could be considered the only cause of glucose intolerance in these patients, whether deficiency of chromium has occurred in these patients, and whether chromium deficiency occurs in healthy populations.”²⁹

B. Reference Intakes

In 2001, the Institute of Medicine determined that insufficient evidence existed to set an Estimated Average Requirement for chromium. Consequently, Adequate Intakes of 35 and 25 μg /day for young men and women, respectively, were set based on estimated mean intakes.³² A Recommended Daily Allowance, based on the Estimated Average Requirement, is defined to be the required daily dose of the nutrient that results in $>98\%$ of Americans not showing signs of deficiency. The AI is more conservative, meaning that more than greater than 98% of adult Americans would be Cr sufficient at ~ 30 μg Cr/day; thus, essentially all Americans are not chromium deficient. More recently in 2014, the Panel on Dietetic Products, Nutrition and Allergies of the European Food Safety Authority determined that “there is no evidence of beneficial effects associated with chromium intake in healthy subjects” and that “the setting of an Adequate Intake for chromium is also not appropriate”.²⁹

The Federal Trade Commission of the United States ordered entities associated with the nutritional supplement Cr picolinate to stop making several representations in 1997 because of the lack of “competent and

reliable scientific evidence.”³³ These representations include that Cr picolinate reduces body fat; causes weight loss; causes weight loss without dieting or exercise; causes long-term or permanent weight loss; increases lean body mass or builds muscle; increases human metabolism; controls appetite or craving for sugar; reduces serum cholesterol; lowers elevated blood sugar levels; is effective in the treatment or prevention of diabetes; or that 90% of US adults do not consume diets with sufficient chromium to support normal insulin function, resulting in increased risk of obesity, heart disease, elevated blood fat, high blood pressure, diabetes, or some other adverse effect on health.

C. Tolerable Upper Intake Levels and Toxicity

As no adverse effects have been convincingly associated with excess intake of chromium from food or supplements, no Upper Tolerable Limit was established for chromium as the trivalent ion in 2001 by the Institute of Medicine.³² In 2005, the US Food and Drug Administration (FDA) found that use of chromium supplements was safe up to 1 mg Cr/day, the highest amount used in clinical trials.³⁴ In 2003, the Scientific Committee on Food found insignificant evidence to set a Tolerable Upper Intake Level. From the limited number of studies, no evidence of adverse effects was found to be associated with supplemental chromium intake up to a dose of 1 mg/day; their recommendation was for sources of Cr^{3+} other than Cr picolinate.³⁵ The EFSA Panel on Contaminants in the Food Chain in 2014 derived a Tolerable Daily Intake of 300 μg Cr/kg body mass daily from the lowest no-observed-adverse-effect level identified in a chronic oral toxicity study in rats.³⁶ The Food Standard Agency (United Kingdom) has determined that doses up to 10 mg of chromium daily should be safe for humans.³⁷ Concerns recently have arisen over the use of supplemental chromium in TPN solutions as subjects on TPN solutions that provide 10 or more μg Cr/day accumulate chromium. The American Society for Parenteral and Enteral Nutrition (ASPEN) has made examining chromium requirements in parenteral nutrition their number one urgent priority.³⁸ This concern is based on TPN solutions currently providing 10–15 μg Cr/day in the United States. Assuming the average AI for men and women of 30 μg Cr/day and an $\sim 1\%$ absorption efficiency for chromium, humans would have ~ 0.3 μg chromium entering their bloodstream daily; thus, TPN solutions would provide to the bloodstream ~ 33 –50 times more chromium than a standard diet. While no reported cases of chromium toxicity in patients on long-term TPN are known, the ASPEN has recommended that TPN without added chromium should be made available.

This does not mean that no toxic effects might be associated with high intakes of trivalent chromium. No

significant health concerns have been found for current commercial chromium supplements at current doses, although questions about the use of chromium picolinate have been raised. Cell culture study and in vivo studies using mammals where the compound was administered intravenously or using fruit flies suggest it is toxic and clastogenic, mutagenic, and possibly carcinogenic.¹⁷ However, a study commissioned by the National Toxicology Program (National Institutes of Health [NIH]) has found that providing chromium picolinate up to 5% of the diet of male and female rats and mice for 2 years had no conclusive deleterious health effects.³⁹ This is probably the result of chromium picolinate dissociating in the gastrointestinal tract when provided orally. In the cell culture study and in vivo studies, the chromium picolinate molecule probably reached cells intact.¹⁷ Hence, the supplement should not be used intravenously, for example, in TPN. The unique potential for deleterious effects from chromium picolinate probably arises from its redox potential, which could possibly allow the chromic center to undergo redox chemistry in vivo.

D. Dietary Sources

Chromium is ubiquitous in foods but at very low concentrations. During the processing of food, particularly in stainless steel (~10%–40% chromium) equipment, the concentration of chromium appears to increase; thus, most of the chromium in some foods in fact may come from processing. Foods particularly rich in chromium (i.e., >100 ppb) include broccoli and black pepper and certain beers; however, values for vegetables must be considered carefully because of the variable amount of chromium that comes from soil contamination. The low concentrations of chromium in food, the ease of contamination, and the low suggested dietary requirement for chromium make preparation of a low-chromium (or chromium-deficient) diet difficult if even possible (as this assumes that chromium is actually an essential trace element). Given the amount of chromium in the diet that comes from modern processing, the chromium intake of early humans must have been extremely limited; thus, humans probably evolved on a diet containing significantly less chromium than the current diets of people of developed nations.

The US FDA has developed Daily Values (DVs) for use on food and dietary supplement labels to assist the US population in comparing products. The DVs are based on the DRIs for adults and children age 4 years and older and for pregnant and lactating women based on a 2000 kcal diet. Originally, no DVs were established for chromium. New DVs for chromium have been established as adults and children ≥ 4 years: 35 mcg; infants through

12 months: 5.5 mcg; children 1 through 3 years: 11 mcg; and pregnant and lactating women: 45 mcg. These DVs were established through implementation of new labeling laws that require updated DVs on food products and dietary supplements beginning in January 2020.⁴⁰

V. ISSUES RELATED TO CHROMIUM IN HUMAN HEALTH

No consistent dose–response relationships between chromium and a beneficial health outcome in humans have been established. Thus, currently chromium is not recommended for any clinical uses. Clinical studies have focused primarily on type 2 diabetic subjects, tend to have small subject pools, and tend not be well designed. One study has dominated attention; decreases in the concentration of fasting serum glucose, insulin, hemoglobin A1C, and total cholesterol as well as decreased glucose and insulin levels in response to glucose challenges were observed as a result of Cr supplementation in a dose-responsive manner.⁴¹ One hundred and eighty-eight individuals in China participated in the study. Unfortunately, attempts to reproduce these results with other populations, including Western populations, have been unsuccessful (for example, Refs. 42, 43). Agencies of the United States government have commissioned two metaanalyses that have generated inconclusive results on whether Cr affects symptoms of type 2 diabetes. In the first, published in 2002 with funding from the Office of Dietary Supplements of NIH, only four quality studies were identified for analysis, leading to the results being inconclusive as the combined results except for those of the Chinese study found no effects.⁴⁴ In the second, published in 2007 with funding from the Department of Health and Human Services, 18 studies were identified. The authors concluded Cr supplementation “may have a modest effect” on glucose metabolism in type 2 diabetics but that “large heterogeneity and the overall poor quality limit the strength of our conclusions.”⁴⁵ Unfortunately, the positive effects of greatest magnitude used in the analysis came from the 12 studies ranked lowest in quality, including the Chinese study. Additionally, a trend was observed that commercial industry-sponsored studies were more likely to observe beneficial effects. A recent metaanalysis that benefited from inclusion of reports of clinical studies since 2007 (so that only high quality studies could be included) and improved methodology found no significant effect of Cr supplementation.⁴⁶

In December 2003, the nutraceutical company Nutrition 21 petitioned the US FDA for eight qualified health claims. They were as follows: Cr picolinate may reduce the risk of insulin resistance; Cr picolinate may reduce the risk of cardiovascular disease when caused by insulin

resistance; Cr picolinate may reduce abnormally elevated blood sugar levels; Cr picolinate may reduce the risk of cardiovascular disease when caused by abnormally elevated blood sugar levels; Cr picolinate may reduce the risk of type 2 diabetes; Cr picolinate may reduce the risk of cardiovascular disease when caused by type 2 diabetes; Cr picolinate may reduce the risk of retinopathy when caused by abnormally high blood sugar levels; and Cr picolinate may reduce the risk of kidney disease when caused by abnormally high blood sugar levels. After extensive review, the FDA issued a letter of enforcement discretion allowing only one qualified health claim for the labeling of dietary supplements. The claim states "One small study suggests that chromium picolinate may reduce the risk of type 2 diabetes. FDA concludes that the existence of such a relationship between chromium picolinate and either insulin resistance or type 2 diabetes is highly uncertain."³⁴

The American Diabetes Association in recent years has been consistent in its position that "There is insufficient evidence to support the routine use of micronutrients,

such as chromium...to improve glycemic control in people with diabetes."⁴⁷

Recent case studies examining intravenous infusions of chromium (generally 3 µg/h) as a treatment for glucose intolerance have found that chromium reduced insulin requirements for subjects with hyperglycemia. These studies are suggestive of a pharmacological effect of chromium as supranutritional quantities of chromium were utilized, but only a small number of subjects have been examined to date.³⁰

These case study results in combination with the rodent studies showing that supranutritional doses of chromium are required before any effects are observed on insulin sensitivity may suggest that beneficial effects from chromium supplementation in humans are possible at higher doses than have been used to date in clinical trials. However, clinical trials using probably ~10 mg Cr/day (or perhaps greater) would be required before any definitive conclusions regarding potential beneficial effects of pharmacological doses of chromium in humans could be reached.

RESEARCH GAPS

The most important recent studies of chromium supplementation are rodent studies, which have clearly indicated that the beneficial effects of chromium supplementation on insulin sensitivity are pharmacological effects and not nutritionally relevant. This potentially explains the lack of effects of chromium in clinical trials as the daily doses of chromium utilized when adjusted for body mass have not approached those utilized to generate beneficial effects in the rodent studies. Clinical trials testing for beneficial effects on insulin sensitivity and cholesterol levels from using 10 mg Cr/day or greater would be required before any definitive conclusions regarding beneficial effects of pharmacological doses of chromium in humans could be reached. These would need, of course, to have careful observation for potential deleterious effects from the larger doses of chromium.

Studies probing the mechanism of action of chromium at a molecular level are also needed. Perhaps better designed studies utilizing cultured cells that involve physiologically relevant concentrations of chromium in physiologically relevant forms could lead to consistent results. These results would then need to be verified by in vitro studies.

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SODIUM, CHLORIDE, AND POTASSIUM

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SUMMARY

Many health professionals and authoritative groups today recommend decreasing dietary intake of sodium from high-sodium content foods and table salt while increasing potassium because such maneuvers should enhance overall health. The sodium content of the body correlates directly with fluid volume. However, many throughout the world unfortunately ingest far more sodium than generally recommended, leading to volume expansion, edema, and elevate blood pressure. The large INTERMAP and Dietary Approach to Stop Hypertension clinical investigations corroborate findings from earlier studies like INTERSALT by concluding that higher sodium intakes cause elevated blood pressure leading to many cardiovascular perturbations. On the other hand, too small a circulating volume emanating from low sodium intake can create symptoms/signs ranging from tiredness/low blood pressure to outright disorientation/shock. Conversely, higher potassium intakes are characteristically associated with lower blood pressures and better cardiovascular performance. Nevertheless, high and low circulating potassium can also have profound adverse effects—primarily on depolarization/contraction of the heart. With diminished renal function, even moderate potassium intake can cause serious cardiovascular complications.

Keywords: Blood pressure; Fluid volume; Potassium; Sodium.

I. INTRODUCTION

A. Life and the Ancient Seas

Many scientists believe that existence of ancient seas on Earth was essential for development of life.¹ Life is estimated to have begun 2 billion years ago in the Precambrian seas that provided mobility, dissolved substances such as sodium, potassium, and chloride, and stable physicochemical conditions.^{2–5} The presence of seas on Earth is unique in the solar system.^{6,7}

Although life probably started in the sea billions of years past, terrestrial forms are believed to have emerged only 360 million years ago.⁵ Strengthening the contention that life began in seas is knowledge that terrestrial life forms had to retain their own internal sea as extracellular and intracellular body fluids containing electrolytes in order to live on land.^{4,8} Man probably came upon the scene 2 million years ago.⁹ The proportions of sodium, chloride, and potassium in ancient seas have been closely linked with life. Cells in the body can survive, grow, and

function only by retaining a friendly environment via tight regulation of an internal sea.

B. Maintaining the Sea Within

Kidneys played a major role in the emergence of terrestrial life. Homer Smith^{3,8} and later Robert Pitts⁹ discussed the crucial role of kidneys in regulating ions and fluid in the body that allowed vertebrates to leave the sea. The renal regulation of sodium, chloride, and potassium in body spaces is especially crucial because the presence of these cations and anions dictates the volume of the body fluid compartments via effects on osmolality, and they play a significant role in acid–base balance. A major difference between plasma and both the past and current seas is the presence of proteins in the former. Proteins occupy about 6% of the blood plasma volume. Healthy kidneys protect the circulating concentrations of proteins by preventing renal losses.⁹

In addition to renal regulation, oral intake of these micronutrients is also of paramount importance.¹⁰

Dietary intake has changed over the years, and these changes play a prominent role in the overall health of numerous modern countries. Nutritional intake can strongly influence development of many prevalent current health perturbations.^{11,12} With the decline in disease and death due to infections at an early age and the emergence of the modern Western diet, cardiovascular and metabolic disorders have become relatively more common.^{11,12} Cardiovascular diseases (CVDs) account for approximately one out of three deaths in the United States. There are seven crucial areas in modern diets that may relate to the prevalence of modern diseases.¹³ Prominent among the listings is the sodium–potassium ratio. To strengthen the reasoning behind consuming less sodium and more potassium, these two cations will be discussed in detail separately.

II. SODIUM

A. Background

To gain better understanding of sodium balance, it is necessary to differentiate between sodium and salt (sodium chloride). Sodium (Na^+), molecular weight = 23 g/mol or Eq, is an essential micronutrient vital in the balance of body fluids. Dietary sodium is usually measured in grams (g), and sometimes in milliequivalents or millimoles. The most common source of dietary sodium is table salt, sodium chloride (NaCl), molecular weight = 58.4 g/mol. Because table salt is only 40% sodium, the remaining 60% being chloride, it is necessary to discern whether any reference to mass is discussing grams of sodium or salt. In the United States, one teaspoonful of table salt contains approximately 5.75 g of salt, equivalent to about 2300 mg of sodium.

In the past, most interested parties have attempted to set an upper limit of normal for daily sodium intake, primarily to control blood pressure.¹⁴ The most recent recommendations for adults are 2300 mg/day¹⁵ (see below, Reference Intakes). The American Heart Association continues to recommend that an intake of 2300 mg per day not be exceeded.¹⁶ To make matters even more complex, many experts in the field have supported individualized recommendations for certain groups. People with hypertension as suggested above, African Americans, and older adults tend to have a greater blood pressure response to a sodium challenge and should be advised appropriately.

B. Sodium Function

Claude Bernard first drew attention to the meticulous regulation of the internal sea within the human organism, the so-called *milieu intérieur*, via distribution of sodium, chloride, and potassium.^{17,18} Further, the regulation of

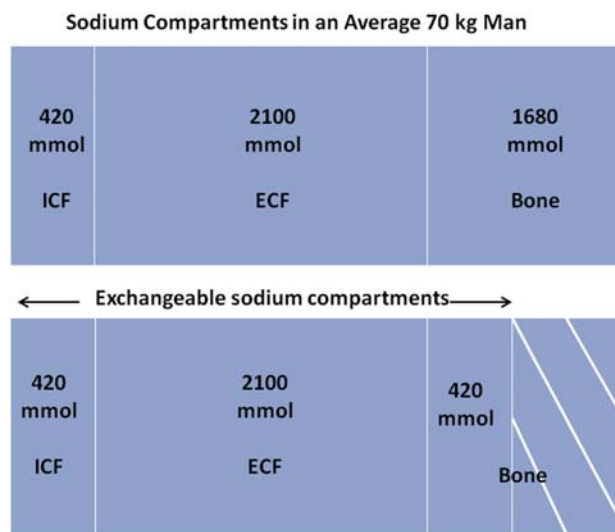


FIGURE 28.1 Sodium compartments. ECF, extracellular fluid; ICF, intracellular fluid.

water and electrolytes for optimal health that was to be maintained within narrow limits was designated *homeostasis*.

Total body sodium of a normal adult averages about 60 mmol/kg body weight.¹⁸ For the typical 70 kg person, this is approximately 4200 mmol—almost 100 g of sodium as depicted in Fig. 28.1. Bone contains roughly 1680 mmol, that is, about 40% of the total body sodium. Approximately 2520 mmol of the remaining sodium resides in the extracellular and intracellular fluid. In addition to that in bone, roughly 50% of total sodium is extracellular and 10% is intracellular.

Sodium stores are classified as exchangeable and non-exchangeable.¹⁸ Through measurements using radioactive isotopes, exchangeable sodium has been estimated at 42 mmol/kg body weight (Fig. 28.1). Exchangeable sodium consists of all extracellular sodium, all intracellular sodium, and less than half the bone sodium. Essentially all nonexchangeable sodium is in bone, where it is buried within the bone structure. Exchangeable sodium is important; because when sodium is lost from the blood plasma into the urine or feces, it can be replaced rapidly from other compartments via diffusion. In turn, sodium retained in edema is also spread out among these various compartments.

C. Sodium Metabolism

Sodium digestion, absorption, transport, excretion, and metabolism

The amount of sodium consumed (intake) must equal the amount of sodium lost (output) to maintain balance. Calculating intake is fairly simple. Sodium is absorbed by passive diffusion virtually completely in

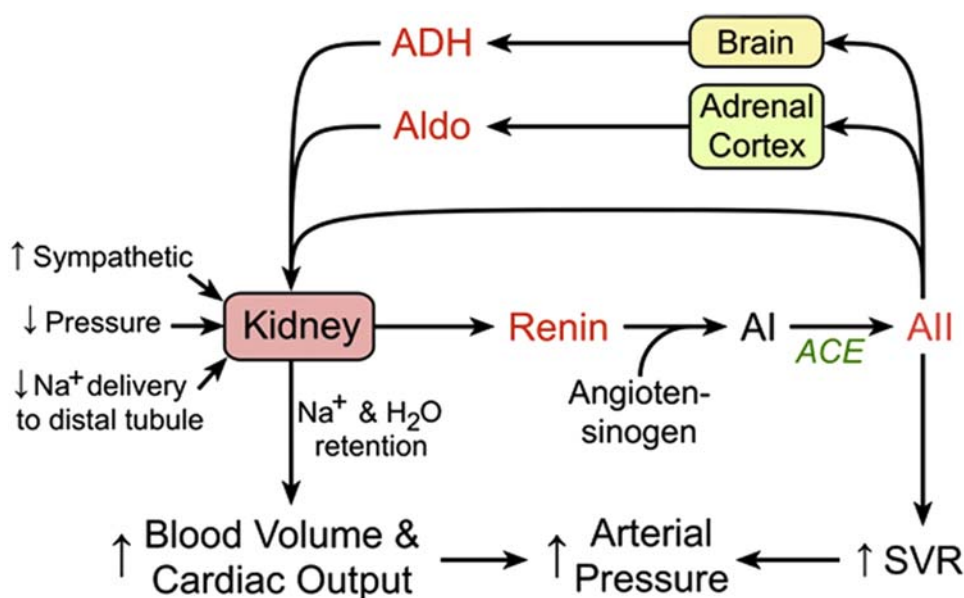


FIGURE 28.2 Brain/renin–angiotensin–aldosterone system. ADH is released by the pituitary gland in the brain in response to increased plasma osmolality; it acts on kidney collecting ducts to reabsorb water and thus Na^+ from glomerular filtrate by exchanging K^+ for Na^+ . Kidney secretes the hormone renin, which stimulates the secretion of angiotensinogen by the liver and converts angiotensin I from circulating angiotensinogen; angiotensin I is converted to active angiotensin II by the action of ACE. Angiotensin II then increases systemic vascular resistance (increases blood pressure) and stimulates production of aldosterone from adrenal gland, which then acts on renal tubule to reabsorb water and Na^+ . ACE, angiotensin-converting enzyme; ADH, antidiuretic hormone; Aldo: aldosterone; SVR, systemic vascular resistance. From: <https://www.cophysiology.com/Blood%20Pressure/BP015>.

the small intestines. Therefore, if 4 grams sodium is ingested, 4 grams is delivered to the body.

Salt intake can be influenced by salt appetite, and the brain/renin–angiotensin system is important in this respect (See Fig. 28.2). Intracerebral injection of angiotensin II stimulates salt appetite, whereas blocking formation of angiotensin II via central administration of captopril, an angiotensin-converting enzyme inhibitor, diminishes salt appetite.¹⁹ Thirst also plays a significant role in the regulation of volume and body space size. Changes in osmolality related to sodium and chloride homeostasis influence the thirst mechanism and the release of antidiuretic hormone (ADH) by the pituitary gland, which affects the handling of water in the renal collecting duct.²⁰

Loss via skin; In contrast to calculating input, calculating output can be difficult because more than one route must be considered in the estimate. Normal losses of sodium occur through the skin, feces, and urine. In the absence of considerable physical effort or heat stress, only small amounts are lost via the skin, mainly by sweating with lesser losses through skin sloughing. In subjects who were ambulatory but not actively working or sweating, daily restriction to 100–150 mg sodium led to average daily losses of <25 mg sodium via skin. The investigators attributed this small loss to desquamation of epithelial cells, sebaceous secretion unnoticed sweat, and possibly some insensible perspiration.²¹

Normal sodium concentration of sweat is typically 46–90 mg/day but is dependent on the rate of sweat,

sodium intake, and acclimatization to environmental heat.²² Changing production of salt-retaining hormones such as aldosterone can regulate losses of sodium in sweat. Conn²³ demonstrated that healthy individuals sweating as much as 5–9 L/day could decrease sodium concentrations in sweat to as little as 0.1 g/L after acclimatization. Under normal conditions, skin losses are small, but substantial amounts of sodium can be removed through excess sweating.

Fecal sodium loss; Loss of sodium via feces is small, even when sodium intake is high.²⁴ On a daily sodium intake ranging widely from 0.05 to 4.1 g/day, only 10–125 mg appeared in stools.²⁵ However, cases of severe diarrhea may result in substantial losses. Potential losses via hair, nails, saliva, semen, and menstruation are too negligible in the overall picture to be considered.

Urinary sodium losses; It is safe to state that more than 90% of sodium output is by means of renal excretion. When sodium intake is acutely reduced to an extremely low value, urinary sodium excretion falls exponentially over 4–5 days. In contrast, increased sodium intake above a certain level results in the excretion of an increased amount after a few days.²⁶

The intake of sodium is accurately estimated by determining the amount present in urine in the absence of gross sweating. Pragmatically, the small amounts in the sweat and feces can be ignored in the overall calculations of balance, since the amount of sodium in the modern Western diet is relatively large.

Sodium homeostasis

Healthy kidneys are amazing in their ability to compensate adequately under most circumstances for either too little or too much sodium intake. This is important to the average individual with normal renal function who from day to day pays little attention to sodium balance even though his intake varies markedly. An example of a usual balance of sodium chloride might be the following: 10.5 g/day in foods and a dietary output of 10.5 g/day (in urine [10.0 g/day], sweat [0.25 g/day], and feces [0.25 g/day]). In the case of too little intake, the healthy kidney conserves sodium at the expense of hydrogen and potassium.⁹ The renal conservation of sodium is so efficient that, if necessary, a daily intake of only a few millimoles is required to balance the small nonurinary losses of sodium⁹ (See Fig. 28.2).

The overall control of body sodium homeostasis through the interplay of various factors on kidney function is incompletely understood. Both intrarenal and extrarenal factors are involved. Associated with sodium homeostasis, extrarenal mechanisms working interdependently are plasma renin activity²⁷; plasma angiotensin II²⁸; aldosterone production²⁸; atrial natriuretic peptide²⁹; catecholamines such as adrenaline, noradrenaline, and dopamine³⁰; hormones such as vasoactive intestinal peptide³¹; and possible Na⁺/K⁺-ATPase inhibitors.^{32,33}

To reemphasize, the major control for sodium output is by means of renal excretion.³⁴ Fig. 28.3 provides an example of what generally happens when sodium intake suddenly increases. In this example, the initial usual daily consumption of 4 grams sodium is balanced

by 4 grams excreted (day 0). The body is in the so-called homeostasis, since output matches input. On day 1, intake is doubled to 8 grams, and the increased intake is continued over the ensuing period of time. On day 1, excretion rises more than the previous day when intake was less but falls short of equaling actual intake that day. On days 2 and 3, sodium excretion continues to rise with the higher sodium challenge and eventually matches the daily consumption. Balance is eventually reached with the increased intake as mentioned previously in this example at 4–5 days but at what cost?

The crucial point here is that excretion does not usually exceed intake to overcome the earlier retention with continuation of this higher intake. Accordingly, it is apparent from the depiction that sodium consumed earlier is retained such that total body sodium has risen a percentage point or two. This in turn leads to a similar rise in the extracellular fluid (ECF) volume. On the doubled sodium intake, individuals have once again reached a homeostatic point where output matches input but with a higher total body sodium and balance. Many physicians believe that this can be harmful over the long run and must be removed by lessening intake or increasing sodium excretion, perhaps via the use of diuretics. The lesson to be learned here is that cutting down intake of sodium should lead to lower total body sodium, less fluid volume, and in general a healthier status—especially in terms of blood pressure and the cardiovascular system.

Maintenance of body fluid compartments

Total body water content of an individual varies roughly between 45% and 70% of body weight, a range of 50%–60% being most representative of normal adults.^{9,18} (Fig. 28.4).

Importance of chloride to ECF: Early on, the chloride moiety was assumed to be responsible for the blood pressure disorders caused by salt—possibly because the measurement of chloride was much more accurate than that of sodium and afforded better evaluation.³⁵ Later, the development of the flame photometer led to more emphasis on the sodium moiety as the major perturbation in the significant elevation of blood pressure by salt. Nevertheless, the two ions seem to work together to control extracellular volume and blood pressure. It is worth repeating that sodium coupled to chloride—not so much to other anions such as bicarbonate—affects fluid distribution in the extracellular compartment.³⁶ Therefore, although most references here are to sodium, it is important to remember that both sodium and chloride are coupled most of the time in the actions, both good and bad.

Because of the virtual absence of protein with its overall negative charge on one side of the capillary wall

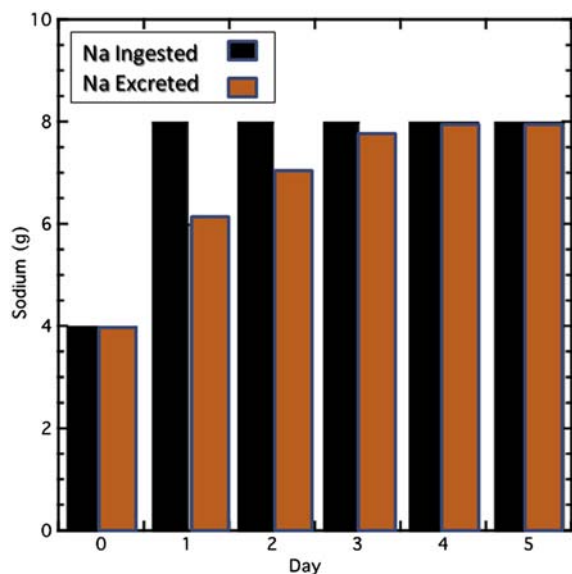


FIGURE 28.3 Results of increasing dietary sodium intake. Daily sodium intake doubled from 4 to 8 g on day 1. Sodium excretion initially does not equal the higher intake, but over 3–4 days, once more equals intake, never to exceed it.

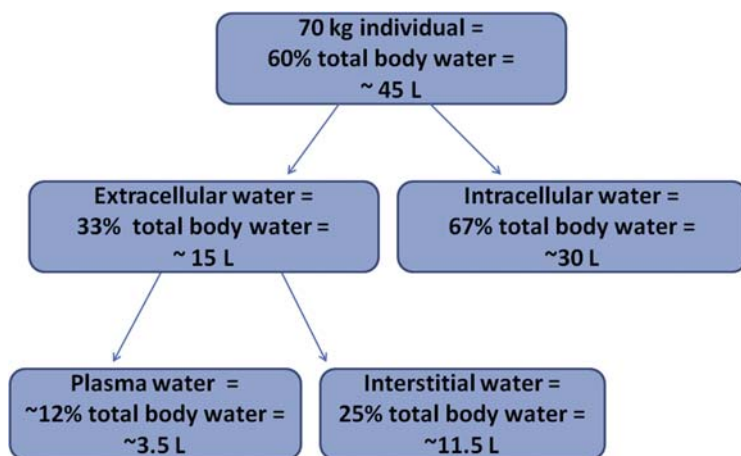


FIGURE 28.4 Body fluid compartments. Total body water is divided into the extracellular and intracellular compartments. Extracellular water includes plasma and interstitial water. Sodium and chloride determine the extracellular fluid volume (ECF) to a great extent.¹⁸ Circulating proteins also influence the relationship between the plasma and interstitial fluid volumes. The endothelium of the capillaries allows rapid distribution of diffusible ions and water but restricts the passage of protein between the plasma and interstitial fluid. Interstitial fluid, therefore, is an ultrafiltrate of plasma.

(protein content is 1% intracellular vs. 6% extracellular), diffusible ions distribute themselves according to the Gibbs–Donnan rule, i.e., more anions such as chloride and bicarbonate will be on the relatively protein-free side of the membrane (interstitial space).⁹ Nevertheless, the sum of the cations and anions must be equal on each side of the membrane. The presence of the nonpenetrable protein also causes a slight increase in oncotic pressure on the plasma side that is balanced by the hydrostatic pressure developed by the heart. Sodium is maintained in the extracellular compartment and potassium in the intracellular compartment mainly through the actions of the Na^+/K^+ -ATPase exchange pump (see

Fig. 28.5). Accordingly, sodium is the major cation in the ECF and the major osmotic particle outside the borders of the cell, whereas potassium is concentrated within the cells due to the action of this pump mechanism.

In the above discussion, sodium and chloride have been considered together. Total amounts of both sodium and chloride determine the size of the extracellular space. Restriction of dietary chloride without restriction of sodium prevents expansion, whereas it is well recognized that administration of sodium coupled to other anions such as bicarbonate has a negligible effect on expansion of the extracellular body space.³⁶ The average plasma concentration of sodium, estimated by specific

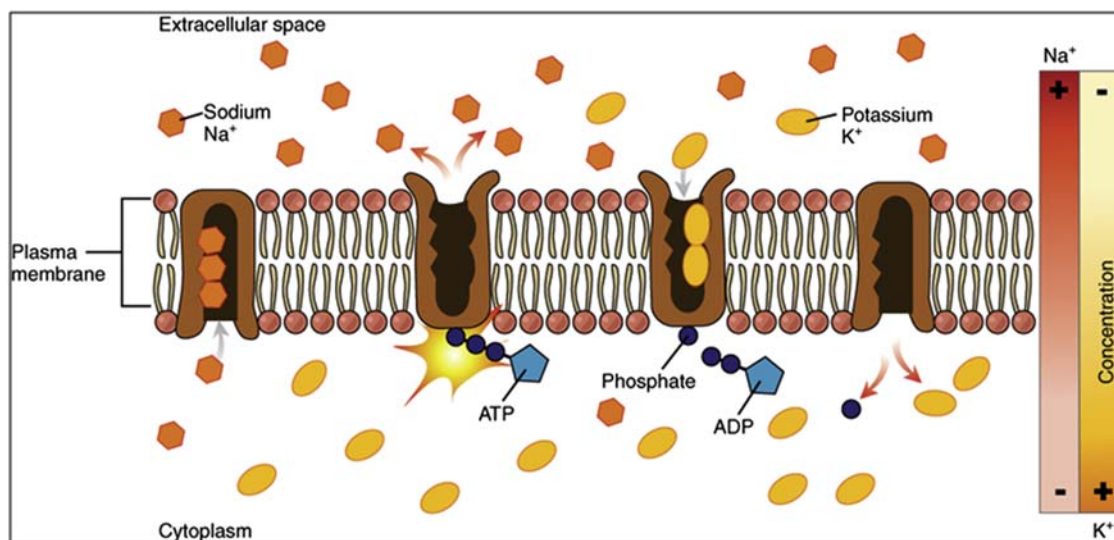


FIGURE 28.5 The sodium–potassium pump is found in many cell (plasma) membranes. Powered by ATP, the pump moves sodium and potassium ions in opposite directions, each against its concentration gradient. In a single cycle of the pump, three sodium ions are extruded from and two potassium ions are imported into the cell. URL: <https://cnx.org/contents/FPtK1zmh@8.108:q2X995E3@12/The-Cell-Membrane> Version 8.25 from the Textbook OpenStax Anatomy and Physiology Published May 18, 2016.

electrodes or flame photometry, is 135–145 mmol/L and of chloride is 98–108 mmol/L. Because of various forces maintaining high sodium concentration in the ECF, infusions of normal saline (0.9% sodium chloride) result predominantly in expansion of the extracellular space.^{18,37} Infusions of dextrose solutions result in fluid distribution throughout the total body water, distributed between intracellular and extracellular water compartments as the glucose is rapidly taken up by cells.

Kidney function

A detailed description of renal tubular handling of sodium is beyond the scope of this chapter. Briefly, like so many areas vital for existence of life, the ability of the kidneys to manage sodium balance depends on a number of checks and balances (Fig. 28.2). Tubular reabsorption of sodium is controlled by renal sympathetic nerves, various hormones, and the pressure and composition of the blood perfusing the kidney.³⁸ These elements include in addition to glomerular filtration rate, factors such as aldosterone, angiotensin II, arteriolar blood pressure, atrial natriuretic factor, and ADH.

D. Dietary Requirements for Sodium

Sodium deficiency

In contrast to sodium overload (see below), deficiency is less common. It is hard to exclude salt from the modern diet, and, as mentioned above, normal kidneys are primed to prevent excessive urinary losses. Harmful losses may nevertheless occur due to excess sweating during heavy exertion and/or may be associated with high external temperatures and require replacement.³⁹ Too severe restriction of dietary sodium can also lead to harmful body depletion.²¹ Physicians treating renal patients with hypertension and congestive heart failure by reducing the body content of sodium via reduced intake and/or diuresis are not infrequently surprised by an elevation of blood urea (BUN), indicating too much removal. This may be a sign of too little blood flow to the kidneys and suggests that other organs are being poorly perfused as well. Subjective signs of sodium deficiency include thirst, nausea, weakness, fatigue, disorientation, and cramps.⁴⁰

Assessment of status

Clinical conditions associated with sodium; Edema, the accumulation of excess fluid in the body, can lead to symptoms such as swelling of a part or all of the body.⁴¹ Accumulation of fluid in the lungs leads to difficulty in breathing. Causes of generalized edema are listed in Table 28.1: the major causes relate to cardiac, renal, or hepatic perturbations. Sodium retention may indicate a physiological response of the kidneys to what is perceived as inadequate arterial blood volume

or may reflect an abnormal renal response due to internal damage or hormonal imbalances.³⁷

In addition to treating the major cause of the edema, such excess fluid is often treated by limiting the intake of salt and/or removing excess sodium by using diuretics.⁴² Too small a circulating volume can cause symptoms and signs ranging from tiredness and low blood pressure to outright disorientation and shock. Judicious replacement of salt and water often overcomes this status.³⁷ Protein malnutrition may cause edema in a number of poor countries and can be overcome by including more protein in the diet to increase plasma protein to help maintain oncotic pressure.⁴³

Hyponatremia and hypernatremia; More recently, it has been postulated that low levels of intake may be connected with various adverse health effects including chronic disease risk^{44–46} (see below). An in-depth discussion of hyponatremia (low serum sodium) and hypernatremia (high serum sodium) is beyond the scope of this chapter. Suffice it to say, hyponatremia can be associated with an excess of total body sodium as well as with an excess of water because the overall water volume is important in the disturbance.⁴⁷ Thus, a patient with hyponatremia may not be depleted in total body sodium but instead may have an excess of intravascular water (e.g., if an increase in water volume exceeds an increase in sodium such as with a hypotonic solution), and hyponatremia will develop despite the increase in total body sodium.³⁷ Similarly, hypernatremia

TABLE 28.1 Causes of generalized edema.

Organ/Class	Disease/symptom
Cardiac	Congestive heart failure Pericardial disease
Renal	Nephrotic syndrome Renal insufficiency
Hepatic	Cirrhosis Hepatic venous disease
Endocrine	Myxedema Hyperaldosteronism
Pharmaceutically induced	Nonsteroidal antiinflammatory drugs Chlorpropamide Tolbutamide Steroids Estrogen
Protein malnutrition	Decreased oncotic pressure due to low albumin/globulin fractions
Idiopathic edema	Unknown

can occur in the face of low or normal body sodium if dehydration (i.e., water depletion) is present. In hospitalized patients, the prevalence of hyponatremia (defined as serum sodium concentration <135 mEq/L) is commonly as high as 15%–22%, but more severe dilutions of <130 mEq/L are seen in only 1%–4% of patients.⁴⁸ Symptoms may vary depending on the acuteness of the hyponatremia; slow decreases in sodium content result in nausea, lassitude, and muscle cramps and are usually less severe than rapid decreases in which confusion, coma, or convulsions occur³⁷.

Reference intakes for sodium

In 2004, the Food and Nutrition Board of the US Institute of Medicine (IOM) established recommendations for Adequate Intakes (AIs) of sodium for the United States and Canada as part of the Dietary Reference Intakes (DRIs) review; these were subsequently published in 2005 (IOM, 2005). AIs were established rather than Recommended Dietary Allowances (RDAs), as at the time, there were inadequate data regarding dose responses at lower levels of intake that could be used to establish an Estimated Average Requirement and then calculate a RDA (IOM, 2005).

Usually, an AI is established for a nutrient based on the average intake of a population group that is apparently healthy. However, for sodium, intakes of most people in the United States and Canada are in excess and thus average intakes were not useful in establishing recommended intakes. Based on an estimated median intake of 2150 kcal/day for adults in the United States (IOM, 2005), consuming food at the level of 1500 mg sodium/day, an intake level evaluated in the Dietary Approach to Stop Hypertension (DASH)–Sodium trial,⁴⁹ would meet three criteria: (1) the intake profiles of selected other nutrients met or exceeded the RDA or AI when consuming at this level, (2) it was an intake that was greater than the levels of sodium intake associated with adverse effects of low sodium intake noted related to insulin resistance and blood lipid concentrations, and (3) it would allow for excess sodium loss in unacclimatized individuals exposed to high temperatures (IOM, 2005).

Interestingly, it has been postulated subsequently that low levels of intake may be connected with various adverse health effects, including chronic disease risk.^{44–46} Most recently in 2019, a new review by the Food and Nutrition Board of sodium DRIs established an AI for sodium based on criteria other than that used to establish the 2005 values; however, the AI was set at approximately the same level, 1500 mg (66 mmol)/day, based on (1) the lowest levels of sodium intake evaluated in randomized trials and (2) evidence from the a well-designed balance study where no deficiency symptoms had been reported.¹⁵ The AIs for children 1–18 y were extrapolated

from the adult AI based on estimated energy requirements, while the sodium AI for infants was based on the sodium content of breast milk.¹⁵

Excess sodium intake

Historical perspective; Primal herbivorous humans probably consumed ≤ 10 mmol/day (0.2–0.3 g/day) of sodium.^{10,50} On a successful hunting day, carnivorous humans might even have ingested as much as 60 mmol (1.4 g) sodium. Over the ensuing years, however, humans developed a significant salt appetite that brought about an estimated increase in sodium intake of 87–260 mmol/day (2.0–6.0 g/day). No doubt this increase has occurred because most individuals experience enhanced flavor and improved texture of foods with added salt. Also, salt serves as a preservative and thickener. Unfortunately, the ubiquitous nature of salt in foods has allowed the general public to become accustomed to high levels.

Many observations and clinical studies over the past century have implicated salt (sodium and chloride) intake in hypertension.⁵¹ Around 1900, French workers proposed that hypertension arose from failure of kidneys to adapt to excess dietary salt.⁵² In support, Allen in 1925 showed that severe restriction of dietary salt in many hypertensive patients reduced blood pressure, which he believed was due to an unknown renal defect in sodium excretion.⁵³

The beneficial effects of the famous Kempner rice diet, an early low-sodium diet prescribed by many physicians, were attributed to low sodium content.⁵⁴ The most recent available US data on hypertension from the 2015–16 NHANES reported that for all US adults, the prevalence of hypertension among non-Hispanic blacks was 40.3%, higher than for non-Hispanic whites (27.3%), non-Hispanic Asians (25.0%), or Hispanic adults (27.8%).⁵⁵

Hypertension in the elderly and in blacks is generally characterized by low plasma renin activity, suggesting volume expansion and less sodium excretion when consuming sodium.⁵¹ Furthermore, the elevated blood pressure of blacks responds especially well to treatment with diuretics.⁵⁶ Accordingly, most clinicians accept dietary sodium as playing an important role in the development of hypertension.⁵⁷ Not all studies have found a conclusive relationship between blood pressure and sodium excretion,^{44,58–60} although recent systematic reviews have demonstrated moderate to strong dose–response relationships.^{43,61}

Salt sensitivity; A plausible explanation for the difficulty in associating sodium consumption with blood pressure may be that the connection pertains only to those individuals who are “salt sensitive.” Some individuals can eat large amounts of sodium without affecting blood pressure markedly because they excrete excess sodium adequately. However, more than half of those with

hypertension have a marked increase in blood pressure in response to a sodium challenge.⁶² These individuals tend to retain sodium because they excrete the challenge sodium load more slowly. Favoring a genetic connection, normotensive relatives of hypertensive subjects, especially older individuals, show a blunted natriuretic effect. Further support for the contention that sodium retention is important in the pathogenesis of hypertension is that individuals on a modern, Western diet had an ECF volume 15% greater than those on a more primitive diet who exhibited less hypertension.⁶³

Many believe that blood pressure rises to compensate for sodium and water retention because elevated blood pressure allows the kidney to excrete more sodium and water to maintain balance.⁶⁴ As mentioned, hypertension is more prevalent in the black population,⁶⁵ suggesting genetic predisposition or poor lifestyle choices.

Inappropriate renal sodium handling can derive from many initiating causes, for example, inheritance of inborn renal transport disorders, presence of circulating factors influencing renal reabsorption such as aldosterone and steroids, and/or decreasing renal mass. Other factors associated with salt sensitivity include female gender, aging, obesity, insulin resistance, and a positive history of hypertension.¹¹

Perhaps, the greatest ability of salt intake to elevate blood pressure occurs in those individuals said to have "resistant hypertension."^{66,67} Resistant hypertension is defined as being present in patients with uncontrolled blood pressure despite therapeutic use of three or more medications generally accepted as being beneficial for pressure reduction.⁶⁷ In a randomized crossover study, blood pressure in 12 patients with resistant hypertension was examined following 1 week on a low-sodium diet (1150 mg (50 mmol)) per day or on a high-sodium diet (5750 mg (250 mmol)/day. Comparing pressures, when on the lower level of sodium, the group showed a reduced average systolic blood pressure of 22.7 mm Hg and a diastolic of 9.1 mm Hg compared with when on the high sodium level.⁶⁷

Epidemiological studies; Two large international epidemiological studies examined the role of salt in hypertension. In the INTERSALT study, data on sodium excretion from 52 centers throughout the world were reported in 1988.⁶⁸ The ranges of intake, adjusted for body mass index and alcohol intake, were roughly from 1200 to 5800 mg (50–250 mmol) sodium per day. The INTERSALT Cooperative Research Group reported that the prevalence of hypertension was 1.7% in nonobese subjects consuming a low-sodium diet compared with 11.9% for nonobese subjects consuming a high-sodium diet. Interestingly, the four centers with the lowest sodium excretion also had the lowest average systolic and diastolic blood pressures. Although the INTERSALT study showed a positive correlation between

blood pressure and sodium ingestion from the 52 participating centers, removal of data from the four centers reporting the lowest sodium intake lessened the significance of the correlation.⁶⁹ One assumption based on these and other data is that a sodium intake of <2300 mg (<100 mmol)/day would lead to a healthful reduction in blood pressure even in individuals without hypertension.

Approximately 10 years later (1996–99), the INTERMAP study was conducted.⁷⁰ INTERMAP followed measurements of dietary sodium intake and urinary sodium excretion in men and women aged 40–59 years from 17 population samplings. Multiple samples were obtained from some countries: Japan (4), China (3), the United Kingdom (2), and the United States (8). Food recalls and timed urine specimens were used to estimate salt intake. In general, the INTERMAP study corroborated the findings of INTERSALT and most other epidemiologic studies in finding higher sodium intakes associated with higher blood pressures.^{70–72}

Intervention studies; In further corroboration, the well-recognized intervention studies with the DASH diet revealed that the lowest blood pressures were found with the lowest sodium intake: 500 mg/day.⁷³

Tolerable Upper Intake Levels for sodium

Previous DRI guidelines for the United States and Canada released in 2004 provided Tolerable Upper Intake Levels (ULs) for sodium that were based on decreasing blood pressure to decrease risk of CVD (IOM, 2005). Subsequently an expanded methodology to evaluate evidence relating nutrient intake to chronic disease risk recommended for nutrients, where excess intake may result in chronic disease risk, a new type of reference value be developed, and the risk assessment methodology used to derive a UL be used for other non-chronic disease–related adverse effects.⁷⁴ To that end, a recent application of this expanded approach was applied to sodium and potassium.

The DRIs, a relatively new approach to review intake recommendations, are a set of reference values for an apparently healthy population. The revised reference intake category for sodium, now named a Chronic Disease Risk Reduction level (CDRR), is based on application of the GRADE system in evaluating the available scientific literature using systematic reviews. The evidence that decreasing high sodium intakes reduces the risk of CVD was found to be moderate in strength but was judged higher in strength regarding lowering of blood pressure.⁶¹ The sodium CDRR for adults was set at 2300 mg (100 mmol)/day, the lowest level of sodium intake for which there was moderate strength of evidence of chronic disease risk reduction.¹⁵ This fresh evaluation keeps chronic disease end points in mind in order to make "estimates of dietary intake of sodium

and potassium *that* will be compatible with optimal health throughout the lifespan and may decrease risk of chronic disease where data indicate they play a role.”¹⁵ In terms of the review of the role of excess sodium resulting in nonchronic disease conditions using the DRI UL risk assessment methodology, data were inadequate upon which to base a UL (see below). This is work in progress.

Dietary sources of sodium

Over 60 years ago, Dahl¹⁰ remarked “Widespread use of salt has received little attention from nutritionists.” This certainly is not the present case because it is now generally recognized that a substance in such common use as salt might be noxious when consumed in amounts determined by dietary customs and tastes. See Table 28.2 for sodium content of selected foods.⁷⁵

Brown and colleagues reviewed salt intake throughout the world.⁷⁶ In the 21st century, sodium intake around the world remains well in excess of physiological needs, i.e., intake may be 5 to 20 times above need. In part, multiple sources of sodium contribute to the amounts of sodium ingested worldwide. In North America and Europe, 75% of sodium intake is via manufactured foods, with cereals and baked goods being the largest contributors. In the Far East, salt added at home and in soy sauce proved to be the largest contributors. In the United States, more than 70% of sodium comes from manufactured foods.⁷⁶

As has been reported, sodium intake varies greatly among countries.⁷⁷ In the INTERSALT study, daily intake was ≥ 6900 mg (300 mmol) by some in Japan; 5400 mg (235 mmol) in Finland; 3500–3900 mg (150–170 mmol) in the United States, Thailand, and New Zealand; 1400 mg (62 mmol) in Polynesia; and < 690 mg (30 mmol) in the Amazon Jungle, New Guinea Highlands, and the Kalahari Desert.⁷⁸ A more recent worldwide analysis (Fig. 28.6) indicates that age-adjusted mean intakes of sodium changed little between 1990 and 2010.⁷⁷

Recently compiled US survey data from the NHANES What We Eat in America 2013–16 cohort reported the median sodium intake for all adult men (19+ years) of 4000 mg (175 mmol) and women of 2900 mg (127 mmol).⁷⁹ Less than 3% of either gender had intakes less than the DRI AI for sodium of 1500 mg (65 mmol), while 97% of men and 79% of women had intakes above the CDRR level of 2300 mg (100 mmol)/day. Fewer older men and women had intakes above the CDRR reference level.^{15,79} Recent guidelines from the IOM urged the FDA to set government standards for amount of salt added by manufacturers, restaurants, and food service companies because analysts estimate that population-wide reductions in sodium intake could prevent more than 100,000 deaths annually.⁸⁰

TABLE 28.2 Approximate values for sodium and potassium in some common foods.

Food	SR nutrient database number ^a	Weight (g)	Sodium		Potassium	
			(mg)	(mmol)	(mg)	(mmol)
Apple, raw, with skin, 2–3/4" dia.	09003	149	1.5	0.1	159	4.1
Avocado, raw, large	09037	201	14	0.6	975	24.9
Banana, raw one medium, 7"	09040	118	1	0	442	11.3
Beef patty, 93% lean, broiled	23473	85	56	2.4	291	7.4
Beer, regular, one 12 oz can	14003	356	14	0.6	96	2.5
Bread, white, one slice	18069	25	122	5.3	32	0.8
Chicken, breast meat only, grilled, 3 oz	05747	85	44	1.9	332	8.5
Corn flakes, 1 c	08506	28	160	7.0	30	0.8
Egg, scrambled, one large	01132	61	88	3.8	80	2.0
Milk, 2%, 1 c	01079	244	115	5.0	342	8.7
Pancake, frozen, 6"	18288	40	184	8.0	36	0.9
Pizza, pepperoni, one slice, 1/8th 14" regular crust	21302	111	760	33.0	216	5.5
Red wine, 5 oz	14096	147	6	0.3	187	4.8
Spaghetti, cooked, enriched, no salt, 1 c	20121	124	1	0	55	1.4
Sweet potato, baked in skin, one medium, 5"	11508	114	41	1.8	542	13.9
Tomato juice, canned, salt added, 1 c	11540	243	615	26.7	527	13.5
Tuna, light, canned in oil, drained, 3 oz	15119	85	354	5.4	176	4.5

^aSR, USDA Standard Reference 28, National Nutrient Database (2019).⁷⁵

E. Issues Related to Sodium in Human Health

Tackling the problem of excess sodium

Japan in the late 1950s had the dubious distinction of having the highest death rate from strokes of any nation.⁸¹ In 1960, the Japanese government initiated a program to reduce salt intake from 13.5 to 12.1 g/day over 10 years. Paralleling this decrement were falls in blood pressure and an 80% reduction in stroke mortality despite increases in fat intake, cigarette smoking, alcohol consumption,

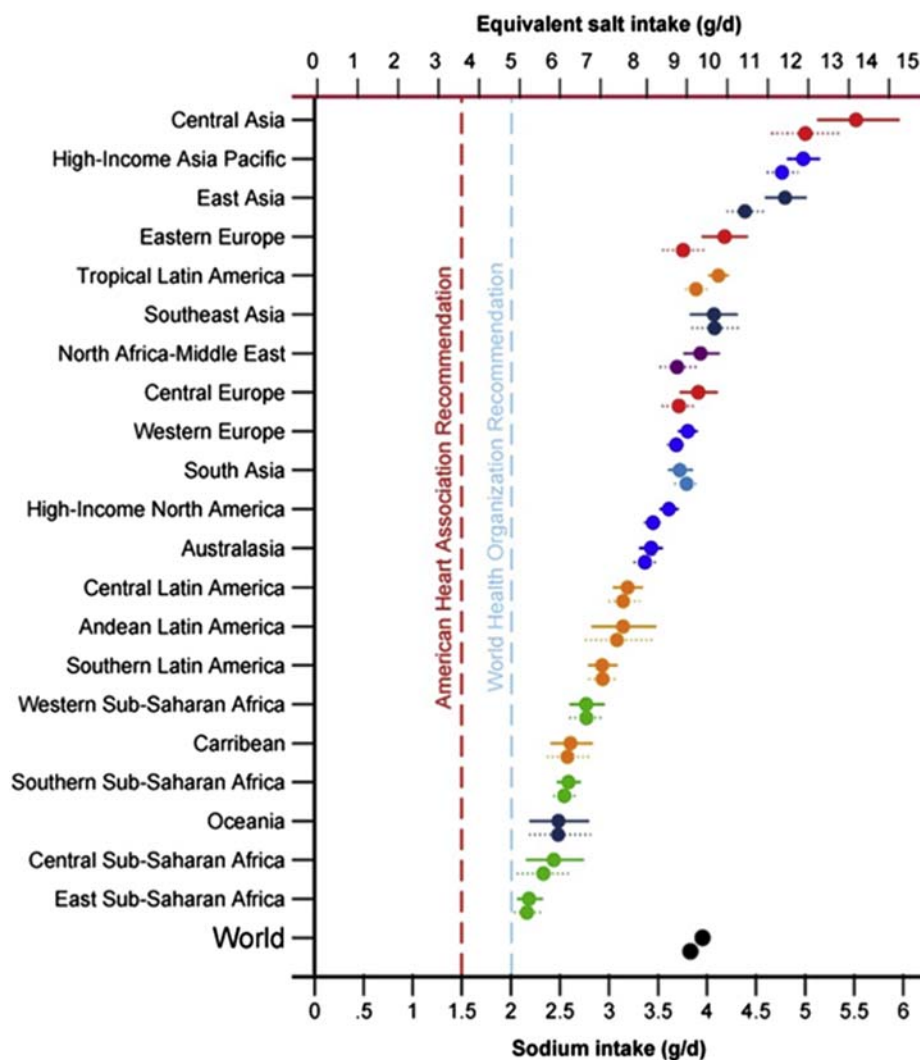


FIGURE 28.6 Mean (95% uncertainty interval) age-standardized sodium intakes (g/day) in 1990 and 2010 in 21 Global Burden of Disease regions. Note: the upper symbol for each pair is for 2010. Regions are ranked by levels in both sexes combined, ages 20+.⁷⁷

and obesity. Starting in the 1970s, the Finnish Ministry of Social Affairs and Health worked closely with food companies to decrease salt content of products and educate physicians to lower sodium intake.⁸² Sodium intake decreased at least 40% during this period,⁸³ and prevention of CVDs was noted.⁸⁴ Public Health England, the English government's public health authority, established goals in 2014 to be reached by 2017 to decrease sodium in the diet.⁸⁵ In the United States, Healthy People 2020 has a sodium reduction objective (NWS-19), to reduce sodium intake from the baseline of 3640 mg/day in 2010–2300 mg/day by 2020.⁸⁶

Why is this necessary? Lifetime probability of hypertension approaches 90%. Worldwide more than 26% of adults have hypertension. Virtually all reports on clinical trials examining the lessening of sodium in the diet found reduced blood pressure, a lowered incidence of strokes, and, to a lesser extent, fewer coronary events.⁸⁷

Although recommendations for the exact amount of salt intake to optimize blood pressure are different among a variety of sources, the general advice to lower it continues.^{88–90}

Although the problem usually lies on the intake side, recent changes have started to take place because the public is becoming alarmingly aware of the consequences of “too much salt.” The major effort to decrease sodium intake between 1980 and 1990 was via salt shakers (0.06 versus 0.02 mmol [1.4 vs. 0.5 g]). Unfortunately, sodium intake emanating directly from foods, by far the major source, was not affected.⁸³ Thus, overall sodium intake had not decreased enough to satisfy most health experts.^{91,92} A decrease in sodium can often be accomplished successfully without affecting consumer enjoyment of food products if it is done in a stepwise process that systematically and gradually lowers sodium levels across the food supply.⁸⁰ In addition to

hypertension, heavy consumption of salt-laden foods such as breads, grains, and cereal products increases the risk of strokes, osteoporosis, obesity, stomach cancer, and kidney stones, left ventricular hypertrophy, congestive heart failure, and severity of asthma.^{93–96}

Known interactions of sodium with other nutrients

It is further recognized that potential adverse effects from salt consumption are also influenced by genetic factors and the simultaneous intake of other nutrients, for example, potassium, magnesium, and calcium.⁹⁷ There is also increasing awareness that corresponding consumption of carbohydrates can greatly potentiate the adverse effects of sodium.^{98–100} The ability of consumed refined carbohydrates to amplify the blood pressure effects of sodium, i.e., enhance salt sensitivity, probably relates to the fact that these macronutrients are associated with salt retention.^{98–100}

For decades now, it has been noted that most people living in the Western world ingest far more sodium and less potassium than necessary for a healthy existence.¹⁰¹ This is unfortunate because, based on previous observations from large-scale trials, the assumption arises that overloading dietary sodium and insufficient potassium consumption alone or combined eventually can adversely influence long-term health status—particularly adverse cardiovascular events produced primarily through effects on blood pressure.¹⁰²

III. CHLORIDE

Total body chloride averages 33 mmol/kg body weight—that is, a 70 kg person has 2310 mmol chloride.¹⁸ Roughly 70% of chloride (molecular weight = 35.4 g/mol) is distributed in the ECF. Much of the remaining chloride is localized in the collagen of connective tissue and is largely exchangeable. Since chloride tends to follow sodium homeostasis, it has thus been discussed in the previous section on sodium.

IV. POTASSIUM

A. Background

Potassium is the major intracellular cation and plays a significant role in several physiological processes.^{103–106} Circulating potassium enters all tissues and has a profound effect on the function of several organs, particularly depolarization and contraction of the heart. With a molecular weight of 39.1 g/mol, it is typically found in the diet in plants and animal sources.

B. Potassium Function

A number of organ systems are involved in maintaining potassium balance. As indicated above (see Fig. 28.5), a major function of potassium is membrane polarization. Membrane polarization depends on the concentrations of internal and external potassium (on either side of the membrane). The major clinical features of disordered potassium homeostasis relate to perturbations in membrane function, especially in the neuromuscular and cardiac conduction systems. Accordingly, both deficient and excessive circulating potassium concentrations can lead to disorders in cardiac, muscle, and neurological functions.

C. Composition of Potassium in the Body

The total body potassium of a normal adult male averages 45 mmol/kg body weight,¹⁸ i.e., the body of a 70 kg person contains 3150 mmol (120 g) potassium. Only 60 mmol or about 2% of potassium is distributed in the ECF. Virtually all the body's potassium is labile and exchangeable. A typical balance of potassium for an average 70 kg man is depicted in Fig. 28.7. The circulating potassium concentrations are low, approximately 3.5–5.0 mmol/L, and the plasma concentration of potassium is often a poor indicator of tissue potassium stores. Intracellular potassium is maintained at 140–150 mmol/L.¹⁰⁷ Potassium status not only depends on the intake and output of potassium but also on its distribution as well.¹⁰⁴ Distribution depends on the energy-consuming processes at the cell membranes in which sodium extrusion is coupled to entry of potassium (see Fig. 28.5). Although the kidneys can adapt to high and low potassium intakes over a period of 2–3 weeks, the minimum rate of excretion is 5 mmol/day. Taken together with obligatory extrarenal losses, this implies that potassium balance cannot be achieved on intakes <10–20 mmol/day (<400–800 mg/day).¹⁰⁴

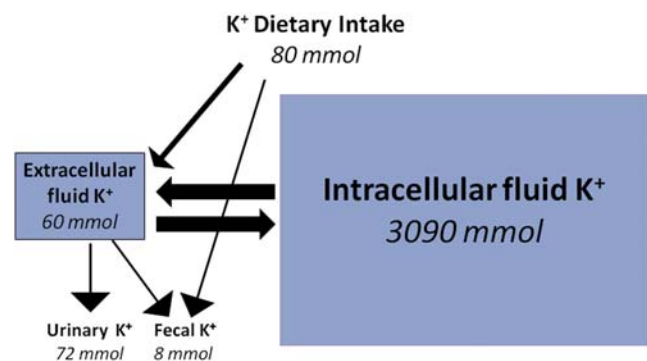


FIGURE 28.7 Potassium distribution balance in a normal 70 kg man. ECF, extracellular fluid; ICF, intracellular fluid.

D. Potassium Metabolism

Potassium digestion, absorption, and excretion

Potassium is readily absorbed in the small intestine in proportion to the presented load. Under usual circumstances, roughly 90% of ingested potassium is absorbed via the gastrointestinal tract for use in the body, leaving the remaining 10% to be excreted in feces (Fig. 28.7). Circulating concentrations are relatively stable because cells immediately take up most potassium entering the body via facilitated diffusion.

Cellular uptake is facilitated by the hormones insulin, catecholamines (epinephrin, norepinephrin), and aldosterone. Since the kidney is the principal excretory organ and decreased renal function from any cause may result in excess potassium retention, it usually takes from several days to as much as 3 weeks for the kidney to adapt to excess potassium intake.¹³ When kidneys cannot readjust adequately, the gastrointestinal tract may reestablish balance, at least in part, by eliminating increased amounts of potassium (i.e., 30%–40% of daily intake) via feces.¹⁰³

Potassium transport and exchange

Only a small portion of oral potassium absorbed remains in the extracellular compartment. Normal rates of dietary intake cause only negligible changes in plasma levels. At high intakes of 8–12 g (200–300 mmol)/day, patients unaccustomed to large loads may develop a significant elevation in circulating potassium despite normal renal function.¹⁰⁴ Such potassium loads augment secretions of insulin, catecholamines, and aldosterone, which help maintain balance.

E. Dietary Requirements for Potassium

Indicators of adequacy

Potassium and blood pressure; In contrast to deleterious effects of high sodium intake that frequently increase blood pressure, ingestion of more potassium may influence blood pressure favorably by lowering elevations.^{108,109} Many studies have indicated that potassium has a good blood pressure–lowering effect, especially in salt-sensitive individuals.¹¹⁰ The beneficial effects of potassium (even magnesium and calcium) work, at least in part, through an effect on sodium balance: a greater sodium excretion despite no change in intake.^{104,105}

A high intake of potassium has been reported to protect against increased blood pressure and other cardiovascular risks.^{108,110,111} Analysis of results from the INTERSALT study showed a lowering of systolic blood pressure at higher potassium excretory rates, presumably reflecting higher intakes.⁶⁸ Various clinical trials have demonstrated a lowering of blood pressure in persons taking potassium supplements.¹¹¹ One metaanalysis reviewed

19 clinical trials involving a total of 586 participants.¹⁰⁸ Results showed that oral potassium supplements significantly lowered systolic blood pressure (–5.9 mm Hg) and diastolic blood pressure (–3.4 mm Hg).

In mildly hypertensive individuals, a low-potassium diet may augment the already elevated systolic and diastolic pressures and also cause sodium retention.¹¹² In normal people, dietary potassium depletion can also cause sodium retention and augmented blood pressure.¹¹³ Low intake of potassium rather than excess sodium consumption might be more important resulting in the greater incidence of severe hypertension among blacks.^{114,115}

Over and above the natriuretic effects of potassium, increased urinary kallikrein¹¹⁶ and stimulation of the sodium–potassium ATPase in vascular smooth muscle cells and adrenergic nerve terminals^{117,118} may be important in lowering blood pressure.

The well-known DASH diet to ameliorate elevated blood pressure is higher in potassium (~105 mEq (4.1 g)/2100 kcal) than the control (typical American) diet (43 mEq (~1.7 g)/2100 kcal).¹¹⁹ The DASH diet seems particularly effective in lowering blood pressure as well as decreasing risk of heart failure.¹²⁰ Results from a number of DASH studies have been published.^{73,121,122} The basic diet resulted in a marked reduction of blood pressure. Although many nutrients are responsible for the overall effect, potassium is probably most important in producing the beneficial effects. In the second DASH-Sodium trial, salt intake was monitored at three different levels, with the greatest effects on blood pressure being seen with the lowest sodium intake. A 12 month follow-up study was instituted to determine the effects of the DASH diet with sodium reduction after discontinuation of the feeding intervention.¹²³ Compared with control participants, DASH diet participants ate more fruit and vegetables and maintained reduced blood pressure even though their sodium intake increased. As a final point, organic forms of potassium, such as typically found in food in which organic acids are the anions associated with potassium, may be more influential than inorganic forms such as potassium chloride found in salt substitutes in lowering blood pressure.¹²⁴

Sodium:potassium ratio; A recent report indicated that a higher sodium-to-potassium excretion ratio is associated with the increased risk of CVDs.^{125–127} This association is stronger for the ratio than for sodium and potassium alone.^{51,128,129} Therefore, diets that lower the overall intake of sodium while increasing the intake of potassium are recommended for the best cardiovascular health.

Hypokalemia; Total body potassium can be decreased and lead to diminished serum concentration of potassium, i.e., hypokalemia.^{103,107} However, hypokalemia can also

result from shifts of potassium out of the blood and into the cells even though total body potassium is normal. Common causes of such hypokalemia include increased renal excretion (poor renal tubular function, diuretic drugs), adrenal disorders (hyperaldosteronism), increased gastrointestinal losses (vomiting, diarrhea), increased uptake by cells (due to excess insulin, β -adrenergic agonists), and decreased food intake (chronic alcoholism, anorexia nervosa). Mild hypokalemia may be asymptomatic or present with muscle weakness, constipation, fatigue, and malaise.¹⁰⁵ Patients with underlying cardiac diseases are prone to arrhythmias.¹⁰⁵ Moderate hypokalemia may result in more severe constipation, an inability to concentrate urine accompanied by polyuria, and a tendency for the development of encephalopathy in patients with concomitant renal disease.¹⁰⁵ Severe hypokalemia can result in muscular paralysis and even in poor respiration because of immobilization of the diaphragm and decreased blood pressure.¹⁰⁵

Reference intakes for potassium

Our ancestors consumed a diet relatively high in potassium in contrast to that consumed in the modern societies. The lowering of intake is associated with increasing consumption of processed foods low in potassium and a lessened intake of fruits and vegetables.

In 2004, as part of the DRI process, the Food and Nutrition Board of the IOM set the recommended intake for potassium for Canada and the United States as an AI at 4700 mg (120 mmol), based on studies in which potassium blunted hypertension due to severe salt sensitivity in African American men and decrease risk of kidney stones, along with reviews of other supportive studies (IOM, 2005). At the time, there was inadequate data to determine an EAR, the intake at which half of the group would show indications of inadequacy based on these criteria and the other half would have their needs met. Without an EAR, RDA could not be established (IOM, 2005).

The recent update of sodium and potassium recommendations and requirements as part of the DRI process conducted by the Food and Nutrition Board focused on chronic disease risk reduction as well as other end points to determine potassium requirements. In reviewing the available data using the new methodology of basing CDRR on both a dose–response relationship as well as at least moderate strength of evidence from systematic reviews,⁷⁴ the decision to not set recommended intake values for potassium based on chronic disease risk due to a lack of data on dose response in the ability of potassium to lower blood pressure.¹⁵ Thus, a CDRR was not established based on the data available at this time. However, an AI was

established, based on estimated average intakes of apparently healthy individuals.¹⁵ For infants, it is based on potassium content of human breast milk. The 2019 DRIs established potassium AIs of 3400 mg (87 mmol)/day for men and 2600 mg (66 mmol)/day for women.¹⁵

Dietary sources

The potassium content of the average American diet ranges between 2000 mg (50 mmol) and 4000 mg (100 mmol)/day. The NHANES What We Eat in America 2013–16 cohort study reported median potassium intakes in the United States for adult men (19 + years) of 2925 mg (74 mmol) and women of 2325 mg (59 mmol)/day.⁷⁹ Over 30% of adult men had intakes greater than the recently revised AI for potassium of 3400 mg (87 mmol), while 31% of women had intakes above their AI of 2600 mg (66 mmol)/day.^{41, 91}

Because potassium is the principal cation of the intracellular fluid, the major source of dietary concentrations of potassium (>than 400 mg (100 mmol)/serving) include avocado, apricots, cantaloupes, lima beans, oranges, and all meats, poultry, and fish (See Table 28.2). Milk and yogurt, as well as nuts, are recognized as being excellent sources of potassium (IOM, 2005). Apple juice, asparagus beets, carrots, peas, and lettuce are considered to have moderate levels. Those often listed as low in potassium are blueberries, cabbage, beans, mushrooms, pineapples, and plums (IOM, 2005). Consequently, a potassium-free diet is virtually impossible to devise. Since most foods contain some potassium, mineral imbalances do not as a rule occur through dietary inadequacies but rather through excessive fluid losses from sweating, diarrhea, or the use of some diuretics that alter potassium excretion.

Tolerable Upper Intake Levels for potassium

An UL for potassium was not established in the 2004 DRI process because the available data related to adverse effects due to high levels of intake in apparently healthy people with normal kidney function were not conclusive (IOM, 2005). Hyperkalemia appears primarily associated in normal individuals with taking large amounts of potassium supplements. While GI discomfort has been reported with high potassium supplements, it was not used as an adverse effect upon which to establish a UL (IOM, 2005). The 2019 review and update of potassium DRIs did not establish a UL either, due to lack of an outcome that could be directly attributed to excess intakes of potassium in individuals with normal kidney function and not under therapy related to hypertension.¹⁵

F. Issues Related to Potassium in Human Health

Severe disturbances in potassium balance

Diuretic usage is the most common cause of hypokalemia in the United States.¹³⁰ Elderly women using long-acting thiazides or loop diuretics are at increased risk.¹³⁰ A low-sodium/high-potassium diet, lower doses of diuretic agents, and substitution of different antihypertensive drugs may ameliorate the problem. Patients with normal renal function who receive diuretics- and digitalis-type compounds may need potassium supplementation or an agent that reduces potassium loss via the kidneys, for example, aldactone, triamterene, or amiloride. Laxative abuse often leads to hypokalemia and can be overcome by ceasing to use the causative agent.¹⁰⁵

When high circulating concentrations of potassium are present (hyperkalemia), it is important to differentiate between pseudohyperkalemia and/or laboratory error and the actual increases in total body potassium. Pseudohyperkalemia is not a real increase in circulating potassium but rather a response to a condition that should be recognized. Pseudohyperkalemia may arise from hemolysis of red blood cells with potassium release into the serum or potassium release into serum from massive leukocytosis or thrombocytosis.¹⁰⁵

In the case of true hyperkalemia, the most important clinical manifestation is cardiac arrest caused by perturbations in electrical (membrane) conduction.¹⁰⁵ Various characteristic changes in the electrocardiogram assessment aid in the diagnosis. Neuromuscular symptoms of too much potassium include tingling, paresthesia, weakness, and flaccid paralysis.¹⁰⁵ In addition to spurious hyperkalemia caused by hemolysis or too many blood cells in the specimen, other common causes of true hyperkalemia are decreased renal excretion; adrenal disorders; and the use of medications such as spironolactone, triamterene, amiloride, angiotensin-converting enzyme inhibitors, nonsteroidal antiinflammatory drugs, and heparin.¹⁰⁵

Treatment for severe potassium perturbations

Hypokalemia; Oral replacement in deficiency states is generally preferable to intravenous administration, and the amount given depends on the body deficit. A rough rule is that a 1 mmol/L decrease in circulating levels is equivalent to 200–300 mmol of body potassium stores. Oral or intravenous potassium at a dose of 40–120 mmol/day generally improves all symptoms of hypokalemia. Any intravenous fluid should contain ≤ 40 mmol/L, and administration should be < 10 mmol/h. If more rapid administration is necessary, it should be performed with electrocardiogram monitoring.¹⁰⁴

Hypokalemia is often associated with metabolic alkalosis. Therefore, replacement of potassium in the form of

potassium chloride may ameliorate the alkalosis more effectively. Hypomagnesemia is somehow related to hypokalemia. The latter may prove resistant to replacement until the magnesium deficiency is corrected. Drugs causing hypokalemia include diuretics, antibiotics (carbenicillin, penicillin, polymyxin B, gentamicin), drugs causing magnesium wasting, and drugs causing metabolic alkalosis.^{104,105}

Hyperkalemia; Hyperkalemia is treated by eliminating excess intake and reversing the other causes mentioned above. When the problem is severe and immediate action must be taken, several therapeutic choices exist. First, one can antagonize the membrane effects by giving calcium gluconate or hypertonic saline. Second, cellular uptake of potassium can be stimulated by giving sodium bicarbonate, glucose, or insulin. Finally, potassium can be removed from the body by using certain diuretics (thiazides), cation exchange resins (Kayexalate), and dialysis (peritoneal or hemodialysis).¹⁰⁴

Chronic hyperkalemia is treated by eliminating excess potassium from the extracellular space and reversing the primary cause of the disturbance and any cofactors that would increase potassium retention. Many drugs and conditions accentuate potassium retention. Drugs causing hyperkalemia include β -adrenergic blocking agents, digitalis, arginine succinylcholine, potassium penicillin, potassium salts, angiotensin-converting enzyme inhibitors, and potassium-retaining diuretics (aldactone, triamterene, amiloride). Acidosis, hypoinsulinemia, low levels of catecholamines, and hypoaldosteronism may contribute to high levels of circulating potassium.^{104,105}

V. ISSUES OF CONCERN WITH SODIUM, POTASSIUM, AND CHLORIDE

A. Potassium Supplements in Patients with Renal Failure

Potassium supplementation must be considered carefully in the face of evident renal failure because the kidneys are the major regulator of potassium homeostasis. Because some potassium salts can be irritating to the gut lining, patients with severe gastrointestinal stress such as previous history of ulcers and bleeding should be considered carefully for oral potassium replacement.^{104,105}

B. General Lifestyle Recommendations Today

The effects of changing from the modern diet to a Paleolithic one containing more potassium and fiber have been evaluated.¹³¹ The Paleolithic diet includes lean meat, fruits, vegetables, and nuts and excludes

cereal grains, dairy, and legumes. Even short-term consumption (10 days) of a Paleolithic diet relative to the modern diet improves blood pressure and glucose tolerance, lessens insulin secretion, augments insulin sensitivity, and better lipid profiles.¹³²

Diets from earlier times and those from primitive areas of the world today compared with the modern Western diet contained higher proportions of potassium relative to sodium.¹³¹ This reversal of potassium and sodium concentrations in the food supply may play a significant role in the development of many chronic

diseases today.^{13,76,131} Too much sodium consumption has the proclivity to raise blood pressure and create cardiovascular perturbations.⁷⁶ Americans also consume too little potassium, roughly one-half the recommended AI. In contrast to sodium, potassium tends to benefit the cardiovascular system.¹³¹ Although some limitations exist in interpreting earlier studies,¹³³ lowering sodium intake and raising potassium intake is commonly recommended by most experts.^{134–136} Modest reduction of sodium from table salt consumption could save lives and reduce health costs.^{137,138}

RESEARCH GAPS

- Ascertain the influence on blood pressure of the link between refined carbohydrates (particularly dietary sugars like sucrose, high fructose corn syrup, fructose) and table salt (sodium chloride).
- Determine the precise importance of the interplay between many nutritional factors in addition to refined carbohydrates with sodium and/or potassium on blood pressure regulation.
- Understand the specific influence of sodium and potassium in the spectrum of insulin resistance and the development of the metabolic syndrome.
- Comprehend the effects of dietary sodium/potassium along with elevated blood pressure, even slight, on the aging process.

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MANGANESE, MOLYBDENUM, BORON, SILICON, AND OTHER TRACE ELEMENTS

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SUMMARY

This chapter presents current knowledge about the essentiality, biochemical function, beneficial actions, deficiency signs, absorption, transport, retention, excretion, beneficial intakes, and nutritional importance of manganese, molybdenum, boron, silicon, nickel, and vanadium in human and animal nutrition. Emerging evidence indicates that these elements have beneficial bioactivity in nutritional or reasonable supranutritional amounts. This bioactivity has been found to have ameliorative or preventive effects on chronic diseases such cardiovascular disease, cancer, diabetes, and osteoporosis. Thus, these elements, especially manganese, boron, and silicon, should receive attention for intake recommendations for promoting health.

Keywords: Boron; Manganese; Molybdenum; Nickel; Silicon; Trace elements; Vanadium.

I. GENERAL INTRODUCTION

Mineral elements reviewed in this chapter are those whose nutritional importance generally have been considered of relatively limited public health concern. Manganese and molybdenum are well-established essential elements because they have been identified as enzyme cofactors. However, nutritional manganese and molybdenum deficiencies have been difficult to induce or identify in humans. Boron and silicon have been shown to have beneficial effects in humans, but their acceptance of being of nutritional importance is hampered by not having a definitive biochemical function to establish essentiality. These four elements will receive the most attention in this update. Numerous other elements have been suggested to possibly be beneficial in nutritional amounts because they have been shown to affect some physiological or biochemical function mostly in animal models. Since the 10th edition of *Present Knowledge in Nutrition* was published in 2011, some limited new information about beneficial nutritional or pharmacological effects of nickel and vanadium has been reported, and thus these elements will be briefly discussed. Essentially no new information about any nutritional benefits

of arsenic, fluoride, and strontium has appeared recently, so these elements will not be addressed here. Possible nutritional or pharmacological importance of a number of other elements in addition to those mentioned above has been presented in past issues of *Present Knowledge of Nutrition*. No major new findings supporting their beneficial effects have appeared since then. Instead of repeating that information here, readers are invited to see past issues that present findings about these elements and arsenic, fluoride, and strontium.^{1,2}

II. MANGANESE INTRODUCTION

A. Background

Manganese is an established essential nutrient for humans because it activates numerous enzymes and is a constituent of several metalloenzymes.³ Animal and limited human studies have shown that manganese through the enzymes and metalloproteins is involved in immune function, reproduction, bone health, sugar and lipid metabolism, and reactive oxygen species defense. Surprisingly, despite these activities, manganese deficiency under nonexperimental conditions has not been

identified in humans. Possible reasons for this are that most enzymes activated by manganese also are activated by other metals, especially magnesium, and because manganese is a relatively common constituent of foods.

B. Nutrient Function

Enzymes that are activated by manganese include oxidoreductases, lyases, ligases, hydrolases, kinases, decarboxylases, and transferases. However, only a limited number of enzymes are specifically activated by manganese; these are glycosyltransferases, glutamine synthetase, farnesyl pyrophosphate synthetase, and phosphoenolpyruvate carboxylase. Arginase, pyruvate carboxylase, and manganese superoxide dismutase (MnSOD or SOD2) are manganese metalloenzymes found in animals and humans.

III. MANGANESE METABOLISM

A. Digestion

Only 1%–5% of ingested manganese is absorbed.^{3,4} Arriving at this range was hampered by the fact that endogenous manganese is mostly excreted through biliary, pancreatic, and intestinal secretions.⁵ If manganese status is adequate, endogenous excretion of absorbed manganese into the gut is so rapid that it is difficult to determine the portion of fecal manganese not absorbed from the diet and portion endogenously excreted.

The absorption of manganese occurs equally well throughout the small intestine. Absorption of manganese apparently is mainly through an active transport system with the divalent metal transporter 1 (DMT1) considered mainly responsible for the transport.⁴ DMT1 transports other divalent cations, including iron. Thus, if iron intake is high, the absorption of manganese is reduced. Diffusion has also been implicated in manganese absorption, especially when manganese intakes are high.

B. Transport

A recent review⁴ comprehensively described findings that have provided knowledge about the transport of manganese in the body. This review indicates that erythrocytes distribute absorbed manganese in the blood because of various manganese transporters, including DMT1 and transferrin receptor (TfR) on the cell surface. Divalent manganese is the major form in the blood and about 84% is bound to albumin.

The liver is the primary organ for manganese storage, distribution, and elimination. Membranes of liver cells express various manganese influx transporters, including DMR1, transferrin/TfR, ZIP4, ZIP 8, ZIP 14, and citrate transporters.⁴ Efflux transporters expressed include the protein product of the protein-coding gene Solute Carrier Family 30 Member 10 (SLC30A10), ferroportin, and the

Secretory Pathway Calcium ATPase (SPCA1) protein.⁴ The passage of blood through the liver results in a small amount of manganese retained in plasma for essential functions, while more than 90%⁶ is conjugated to bile and excreted in the feces. A fraction of absorbed manganese is oxidized to trivalent manganese by ceruloplasmin and is transported in plasma bound to transferrin, albumin, and β -globulin transmanganin.⁵

Metabolically active organs (liver, kidney, and pancreas) with high numbers of mitochondria, in which manganese is predominantly found, have relatively large manganese concentrations (about 1.0 $\mu\text{g/g}$).⁴ Bone manganese concentrations also are about 1.0 $\mu\text{g/g}$. About 25% of estimated 10–20 mg of manganese in the body is in bone.^{2,5} Manganese is present in extremely low concentrations in whole blood (7.7–12.1 $\mu\text{g/L}$) and serum (0.38–1.1 $\mu\text{g/L}$).⁶

C. Elimination

As indicated above, excretion of manganese occurs mainly via the bile and thus in the feces. Very little manganese is excreted in the urine, and urinary excretion does not correlate with dietary intake.

IV. MANGANESE ROLE IN NORMAL CELL AND ORGAN FUNCTION

Some association and epidemiological studies indicate that a low manganese status may result in undesirable effects in normal cell and organ function. One review found that low dietary manganese has been associated with osteoporosis, diabetes, epilepsy, atherosclerosis, impaired wound healing, and cataracts.⁷ Since that review, additional studies have found that a low manganese status is associated with diabetes, the metabolic syndrome, and some cancers.³ Also, low maternal blood manganese concentrations have been associated with increased risk of low birth weight and fetal intrauterine growth retardation.^{4,8,9} These associations suggesting that a low manganese status can contribute to the risk for some pathological conditions indicate that attention should be given to the possibility that the occurrence of an inadequate manganese status is more prevalent than generally thought.

V. MANGANESE DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

Manganese deficiency has been induced in many animal species. Deficiency caused depressed growth, testicular degeneration, and seizures in rats; slipped tendons or perosis in chicks; osteodystrophy and severe glucose intolerance in guinea pigs, and ataxia in mice and

mink.¹⁰ In humans, only three older studies have shown some biochemical and physiological changes induced by a low manganese intake. Manganese supplementation corrected diffuse bone demineralization and poor growth exhibited by a child on long-term parenteral nutrition.¹¹ In a metabolic unit study, women fed a diet supplying manganese at about 1.0 mg/day increased mood and pain during the premenstrual phase of the estrous cycle.¹² Men fed a purified diet supplying manganese only at 0.11 mg/day for 39 days developed a finely scaling minimally erythematous rash, decreased serum cholesterol and increased serum calcium, phosphorus, and alkaline phosphatase activity.¹⁰ In the 25 years since those reports, no experimental nutritional manganese deficiency in humans has been reported.

B. Criteria of Adequacy and Assessment of Status

Because of the lack of a defined manganese deficiency in humans, it has not been possible to determine a measure indicating a low manganese status. Manganese toxicity described subsequently can be assessed by increased tissue manganese concentrations.

Finding criteria of adequacy for manganese also has not been successful. During the setting of Dietary Reference Intakes (DRIs) for manganese, it was determined

that balance data could not be used to set an Estimated Average Requirement (EAR) because of the wide range of manganese intakes that resulted in manganese balance.¹³ Collectively, the balance data indicated that a Recommended Dietary Allowance would be 2.5–3.0 mg/day, which was considered too high because of the lack of overt symptoms of manganese deficiency in the population.¹³ Therefore, dietary intake assessments were used to arrive at an Adequate Intake (AI).

C. Reference Intakes

The United States and Canada have set an AI and Tolerable Upper Intake Level (UL), which are shown in Table 29.1.¹³ Although the AIs are 1.8 and 2.3 mg/day for adult women and men, respectively, as indicated above, adults fed diets containing 0.8–1.0 mg/day under carefully controlled conditions did not show any serious differences from those fed amounts greater than the AI.¹⁴ Also, normal healthy adults fed amounts of manganese higher than the UL (20 mg/day) were not adversely affected.¹⁴

D. Basis for Individual Variability in Requirements

Because the requirement for manganese is vague, knowledge of factors affecting the requirement is not

TABLE 29.1 Dietary Reference Intakes (DRIs); Recommended Dietary Allowances (RDAs), Adequate Intakes (AIs), and Tolerable Upper Levels (ULs) for boron, manganese, molybdenum, and nickel (mg/day).

Life stage group	Boron UL	Manganese AI	Manganese UL	Molybdenum RDA	Molybdenum UL	Nickel UL
<i>Infants</i>						
0–6 Months	ND	0.003	ND	0.002 (AI)	ND	ND
7–12 Months	ND	0.6	ND	0.003 (AI)	ND	ND
<i>Children</i>						
1–3 Years	3	1.2	2	0.017	0.3	0.2
4–8 Years	6	1.5	3	0.022	0.6	0.3
<i>Males</i>						
9–13 Years	11	1.9	6	0.034	1.1	0.6
14–18 Years	17	2.2	9	0.043	1.7	1.0
19–50 Years	20	2.3	11	0.045	2.0	1.0
>50 Years	20	2.3	11	0.045	2.0	1.0
<i>Females</i>						
9–13 Years	11	1.6	6	0.034	1.7	0.6
14–18 Years	17	1.6	9	0.034	1.7	1.0
19–50 Years	20	1.6	11	0.045	2.0	1.0
>50	20	1.6	11	0.045	2.0	1.0

ND, Not determined.

Data from Institute of Medicine, Food and Nutrition Board.¹³

well established. Perhaps, other dietary factors affecting the absorption of manganese would induce variability in requirement. For example, manganese absorption decreases with increased iron intake.^{14,15}

Because manganese is cellularly transported by several metal transporters of other divalent cations, intakes of cations other than iron also could affect the manganese requirement. These elements include zinc, copper, nickel, and calcium.⁴ Also, because magnesium and manganese can interchange in numerous enzyme reactions, magnesium intake could affect the manganese requirement.

E. Dietary Sources

Plant-based foods are the richest sources of manganese.³ Whole grains, rice, and nuts contain the highest amounts of manganese. Other good sources of manganese include chocolate, shell seafood, legumes, leafy vegetables, seeds, and tea.³

The US Food and Drug Administration (FDA) has developed Daily Values (DVs) for use on food and dietary supplement labels to assist the US population in comparing products. The DVs are based on the DRIs for adults and children aged 4 years and older and for pregnant and lactating women based on a 2000 kcal diet. Originally, no DVs were established for manganese. New DVs for manganese have been established as adults and children >4 years, 2.3 mg; infants through 12 months, 0.6 mg; children 1 through 3 years, 1.2 mg; and pregnant and lactating women, 2.6 mg. These DVs were established through implementation of new labeling laws that require updated DVs on food products and dietary supplements beginning in January 2020.¹⁶

VI. ISSUES RELATED TO MANGANESE IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Manganese is receiving significant attention as a toxicological concern for individuals whose normal homeostatic mechanisms are undeveloped, bypassed, or ill-functioning.^{4,17} High retention of manganese in tissues, especially in the brain, has been reported for individuals receiving parenteral nutrition high in manganese and for individuals undergoing hemodialysis, especially when normal excretion via the hepatobiliary system is impaired. Biliary manganese excretion is not well developed in neonates and infants. Thus, high intakes of manganese may increase tissue and brain levels in infants. Iron deficiency also may exacerbate manganese toxicity because it enhances manganese absorption,¹⁴ and animal experiments indicate that iron deficiency increases manganese retention by the brain¹⁸. Toxic accumulation of manganese in the brain results in clinical features similar

to idiopathic Parkinson disease, including disequilibrium, tremor, muscle spasms, tinnitus, and hearing loss.¹⁹

B. Manganese as a Dietary Supplement

Manganese supplements exist in the form as bisglycinate chelate, glycine chelate, aspartate, gluconate, picolinate, sulfate, citrate, and chloride.²⁰ Multivitamin/mineral supplements that contain manganese typically provide 1.0–4.5 mg. Supplements of manganese only, or with a few other nutrients, usually contain 5–20 mg.

VII. MOLYBDENUM INTRODUCTION

A. Background

Molybdenum is an established essential nutrient for animals and humans because it is a component of a pterin-containing cofactor that is required for electron transfer in four known oxidation–reduction enzymes. However, no recent reports have appeared showing that nutritional molybdenum deficiencies have occurred in humans. Thus, at present, molybdenum does not receive attention as an element of public health concern. Recent research attention on molybdenum has essentially focused on rare genetic defects in enzymes that produce the molybdenum cofactor. These defects cause severe neurodegeneration and death during childhood.²¹

B. Nutrient Function

Four mammalian molybdoenzymes have been identified. These are aldehyde oxidase, sulfite oxidase, xanthine oxidase, and mitochondrial amidoxime-reducing component (mARC). Molybdoenzymes catalyze the hydroxylation of various substrates by using oxygen from water. Aldehyde oxidase oxidizes and detoxifies various pyrimidines, purines, pteridines, and related compounds. Sulfite oxidase catalyzes the transformation of sulfite, formed mainly from cysteine metabolism to sulfate. Xanthine dehydrogenase and xanthine oxidase catalyze the last two oxidative reactions converting hypoxanthine to xanthine and xanthine to uric acid, which is excreted in the urine. The enzyme mARC with cytochrome b₅ type B and NAD(H) cytochrome b₅ reductase reduces various N-hydroxylated substrates.

VIII. MOLYBDENUM METABOLISM

A. Digestion

Molybdenum, as molybdate, is readily absorbed over a broad range of intakes. Food-bound molybdenum is about 16% less bioavailable than soluble complexes such as ammonium molybdate.^{22,23} In two studies, men

absorbed 90%–94% of daily intakes of molybdenum ranging from 22 to 1490 μg .^{22,23} The oxyanion molybdate is the only known form of molybdenum that is absorbed and metabolized by the body. The sulfate anion is a competitive inhibitor of molybdenum absorption.²⁴

B. Transport

Molybdate is transported in blood loosely attached to erythrocytes.²⁵ In humans, molybdenum is transported across cell membranes by molybdate transporter type-2,²⁶ which is a member of the family of sulfate transporters.

The concentrations of molybdenum in tissues, blood, and milk vary with molybdenum intake. For example, reported plasma concentrations for men with an intake of 22 $\mu\text{g}/\text{day}$ was 0.51 $\mu\text{g}/\text{L}$; with an intake of 121 $\mu\text{g}/\text{day}$, was 1.17 $\mu\text{g}/\text{L}$; and with an intake of 1490 $\mu\text{g}/\text{day}$, was 6.22 $\mu\text{g}/\text{L}$.²⁴ The highest tissue concentrations are found in the liver, kidney, and bone (normally >1 mg/kg dry weight).²⁶ The concentrations in other tissues usually are between 0.14 and 0.20 mg/kg dry weight.

C. Elimination

After absorption, most molybdenum is removed rapidly as molybdate via the kidney.^{22,23,25} The amount and percentage of molybdenum excreted in the urine increase as dietary molybdenum increases. Thus, urinary excretion rather than regulated absorption is the major homeostatic mechanism for molybdenum. Documented cases of molybdenum deficiency and toxicity in humans may be rare because urinary excretion is able to adapt to a wide range of intakes.

IX. MOLYBDENUM ROLE IN NORMAL CELL AND ORGAN FUNCTION

As indicated below, molybdenum has a critical role in brain function. However, this brain dysfunction has only been identified in rare cases of genetic defects in molybdenum cofactor synthesis.

X. MOLYBDENUM DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy

There has been no expansion in the characterization of molybdenum deficiency in animals since those described in previous reviews.^{2,24} In rats and chicks, molybdenum deficiency in the presence of excessive tungsten depressed molybdenum enzymes, disturbed uric acid metabolism, and increased susceptibility to sulfite toxicity. In goats, molybdenum deficiency depressed

food consumption and growth and impaired reproduction characterized by infertility and elevated mortality in both mothers and offspring.²⁷

An isolated human case of nutritional molybdenum deficiency has been reported. A patient with Crohn's disease, who was receiving prolonged parenteral nutrition, exhibited high methionine and low uric acid levels in blood; high oxypurines and low uric acid in urine; very low urinary sulfate excretion; and mental disturbances progressing to coma.²⁸ The signs and symptoms were exacerbated by methionine administration. Intravenous supplementation with ammonium molybdate improved the clinical conditions, reversed the sulfur-handling defects, and normalized uric acid production. Other than this case, there have been no reports of a molybdenum deficiency induced through its low intake.

However, rare mutations in genes encoding enzymes involved in molybdenum cofactor synthesis can result in a severe deficiency.²⁹ These genetic mutations result in the simultaneous loss of activity of all human molybdenum cofactor enzymes. The diagnosis of the molybdenum cofactor deficiency is usually made shortly after birth because of failure to thrive and neonatal seizures. The disease is associated with a pronounced and progressive loss of white matter in the brain. Biochemical changes include elevated plasma S-sulfonated trans-thyretin and deficient molybdoenzyme activity in fibroblasts. Mild cases have been reported, possibly the result of a low residual activity of a mutant protein. However, most cases have been severe, and patients die in early childhood; some survive only a few days.

B. Criteria of Adequacy and Assessment of Status

Because of the lack of a defined nutritional molybdenum deficiency in humans, it has not been possible to determine criteria for adequacy and thus a status assessment measure indicating a low molybdenum status. However, it should be noted that four men adapted to consuming diets that provided 22 $\mu\text{g}/\text{day}$ exhibited decreased uric acid and increased xanthine excretion in urine in response to a load dose of adenosine monophosphate. This suggests that xanthine oxidase was decreased by the low molybdenum diet.³⁰ A test based on this finding might be a functional method to assess molybdenum status.

C. Reference Intakes

Although a dietary molybdenum deficiency has not been observed in a healthy individual, DRIs have been set for this nutrient as shown in Table 29.1.¹³ Based on 1995 balance studies²⁹ that indicated adults could achieve molybdenum balance on an intake of 22 $\mu\text{g}/\text{day}$, the requirement was estimated to be

25 µg/day (22 µg plus 3 µg to allow for losses not measured in the study). Because some foods might inhibit the bioavailability of molybdenum, an average bioavailability of 75% was used to set an EAR of 34 µg/day for adults. The RDA was set as the EAR + 30% and is 45 µg/day for adults. DRIs for children were extrapolated from the adult value, using metabolic weight ($\text{kg}^{0.75}$) as the basis for extrapolation.

D. Basis for Individual Variability in Requirements

The need for molybdenum may be influenced by the intake of other substances in the diet. Animal studies indicate that the absorption of molybdenum can be reduced by diets or foods high in sulfur because the sulfate anion is a competitive inhibitor of molybdenum absorption.²⁴ Methionine was shown to exacerbate the apparent molybdenum deficiency in a patient that responded to molybdenum supplementation.²⁸ Excessive dietary tungsten helps induce molybdenum deficiency in rats and chicks.²⁴ The bioavailability of molybdenum from soy is lower than from most other foods.³¹

E. Tolerable Upper Intake Levels

Molybdenum is of a relatively low toxicity in humans. As result, reports that show toxicity data for humans are lacking. Most reported toxicity data relevant to humans have come from nonruminant animal studies. These studies have used milligram doses, instead of the nutritional microgram intakes of molybdenum. Based largely on animal studies, UL of 2000 µg (2.0 mg)/day was established in the United States and Canada.¹³

F. Dietary Sources

Pulses, especially various forms of beans, are rich sources of molybdenum. Whole grains, cereals, and organ meats also are good sources of molybdenum. Moderate sources include dark leafy and Brassica vegetables and milk and milk products. Daily intakes have been found to usually exceed the RDAs for molybdenum. In the United States, mean intakes of 76 µg/day for women and 109 µg/day for men have been reported.³² Higher mean intakes in other countries have been reported, including 225 µg/day in Japan and 275 µg/day in France.³²

Originally, the FDA did not establish DVs for molybdenum. New DVs for molybdenum have been established through implementation of new labeling laws that require updated DVs on food products and dietary

supplements beginning in January 2020¹⁶ as adults and children >4 years, 45 µg; infants through 12 months, 3 µg; children 1 through 3 years, 16 µg; and pregnant and lactating women, 50 µg.

XI. ISSUES RELATED TO MOLYBDENUM IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Because of the low requirement for molybdenum that is usually met easily by dietary means, and because of its relatively low order of toxicity, molybdenum has not been a nutrient of public health concern. However, molybdenum in the form of tetrathiomolybdate in pharmacological amounts (~100 mg/day) has been found to have anticancer effects and used to treat Wilson's disease (a copper-accumulating toxicity).³³

B. Known Interactions

The basis for the beneficial actions of thiomolybdate is that it inhibits metal transfer functions between copper-trafficking proteins.³⁴ As a result, the reduction in copper inhibits its role in angiogenesis and inhibits copper dependent cytokines involved in inflammation.

C. Molybdenum as a Dietary Supplement

Molybdenum occurs as sodium molybdate and a variety of amino acid chelate forms in both multivitamin/mineral supplements and supplements of molybdenum only or with a few other nutrients.²⁰ These supplements usually provide between 75 and 500 µg of molybdenum per dose.

XII. BORON INTRODUCTION

A. Background

Boron has been shown to be essential for the completion of the life cycle for organisms in all phylogenetic kingdoms.^{35,36} Boron has not been firmly established as nutritionally essential for humans because it lacks a clearly defined biochemical function. However, an increasing number of cell culture, animal, and human studies indicate that boron in nutritional amounts beneficially alters several biological functions that directly or indirectly influence health and disease. The beneficial effects of boron likely occur because boric acid forms ester

complexes with several biologically important sugars, including ribose and fructose.

B. Nutrient Function and Structure

The biochemistry of boron and animal, bacteria, cell culture, and plant findings gives some insights into the possible basis for the diverse beneficial bioactivity of boron. These findings have resulted in several hypothesized mechanisms for the diverse actions of boron focused mainly on influencing cell signaling, or the formation and/or activity of an entity involved in many biochemical processes.

S-adenosylmethionine (SAM) and diadenosine phosphates have high affinity for boron.³⁷ Diadenosine phosphates function as signal nucleotides associated with neuronal response. SAM is used mainly in methylation reactions that influence the activity of DNA, RNA, proteins, phospholipids, hormones, and neurotransmitters. The methylation reactions result in the formation of S-adenosylhomocysteine that can be hydrolyzed to homocysteine. Two reports have suggested that some of the bioactivity of boron might be through affecting the formation or utilization of SAM. The bacterial quorum-sensing signal molecule, autoinducer-2, is a furanosyl borate ester synthesized from SAM.³⁸ Quorum sensing is the cell-to-cell communication between bacteria using extracellular signaling molecules (autoinducers). The other report is that a low-boron diet increased plasma homocysteine and decreased liver SAM in rats.³⁹ High circulating homocysteine and depleted SAM have been implicated in many of the disorders that boron beneficially affects.

Another molecule bound strongly by boron is oxidized nicotinamide adenine dinucleotide (NAD⁺), which binds to the plasma membrane receptor, CD38. CD38 is an adenosine diphosphate (ADP)-ribosyl cyclase that converts NAD⁺ to cyclic ADP ribose. Cyclic ADP ribose decreases endoplasmic reticulum Ca²⁺ concentrations resulting in activation of the eukaryotic initiation factor 2 α and increases activating transcription factors 4 (ATF4) and 6(ATF6). Boron in physiological concentrations is a reversible competitive inhibitor of cyclic ADP ribose and activates eIF2 α and increases downstream ATF4 and ATF6.^{40,41} ATF6 increases the genes for binding immunoglobulin protein (BiP), calreticulin, endoplasmic reticulum degradation enhancer mannosidase (EDEM), and GRP94. ATF4 and BiP are involved with osteogenesis. Calreticulin is required for tumor suppressor p53 function. BiP and EDEM prevent the aggregation of misfolded opsins, which leads to retinal degradations. A low boron status has been associated with impaired bone formation and maintenance, cancer risk, and retinal degeneration.⁴²

Boron also may be bioactive through forming diester complexes with phosphoinositide, glycoproteins, and glycolipids that contain *cis*-hydroxyl groups in membranes. Such boroesters might affect membrane receptors or signal transduction, which might be the basis for boron affecting hormone action. Boron deprivation decreased insulin sensitivity in chicks and rats³⁷; exacerbated gross bone abnormalities in chicks and rats fed a low vitamin D diet³⁶; and increased the need for exogenous thyroxine for tail resorption and made oocytes not responsive to progesterone in frogs.⁴³

XIII. BORON METABOLISM

A. Digestion

Boron apparently is mainly absorbed in two different forms. Boron in the form of boric acid is readily absorbed with about 85% of that ingested taken up by enterocytes.² Much ingested inorganic and some organic boron may be converted into boric acid and absorbed in that form because boric acid is a dominant species at the pH of the gastrointestinal tract.

Recent evidence indicates that boron occurs in food as fructoborate, a naturally occurring di-fructose-borate ester.^{44,45} Fructoborate is comprised of two fructose molecules complexed to a single boron atom. The fructoborate anion often is complexed with the calcium cation. Calcium fructoborate has been found to be the major form of boron in plant-based foods. Animal experiments indicate that calcium fructoborate is absorbed intact by the small intestine and is present in plasma.⁴⁶ Calcium fructoborate is synthesized by plants but apparently not by humans.⁴⁵

B. Transport

The transport mechanism of boric acid and calcium fructoborate across cell membranes has not been clearly established. Boric acid has the proper solubility to cross the cell membrane without the aid of facilitators.⁴⁷ Such passive diffusion has been found in plants. However, there is evidence that some aquaporins and aquaglyceroporins facilitate the uptake of boron by frog oocytes and yeast. In the past, evidence for a sodium-coupled borate cotransporter was reported.⁴⁸ However, it has been recently shown that the borate was not a substrate for human SLC4A11 transporter.⁴⁹

It has been hypothesized that plant-based organic boron compounds such as fructoborates are transported into cells via sugar-active transporters.⁴⁵

Boron is distributed throughout tissues at concentrations mostly between 1.3 and 185 μmol (14 μg and 2 mg/kg fresh weight).⁵⁰ The highest concentrations are found

in the bone. Normal fasting plasma boron concentrations are in the range of 3.14–8.79 mmol/L (34–95 ng/mL).³⁷ After absorption boron is excreted quickly via the urine, which makes urinary excretion rather than regulated absorption the major homeostatic mechanism for boron.

C. Metabolism

Once in the cell, the boron in fructoborate is converted into boric acid, the major excretory form. It also has been suggested that boron in calcium fructoborate is converted into borate outside the cell and transported into the cell through a borate transport mechanism.⁵¹

The entry of boric acid into the cell has been proposed to increase its acidity, whereas the entry of borate and its conversion to boric acid would increase alkalinity.⁵¹ The different pH changes may have an impact on the movement of signaling ions across membranes.

XIV. ROLE OF BORON IN NORMAL CELL AND ORGAN FUNCTION

A. Bone Formation, Growth, and Maintenance

A review found substantial evidence provided by animal and cell culture studies, which indicates that boron beneficially affects trabecular and alveolar bone growth and maintenance.⁴² Animal studies have also shown that the addition of boron to bioactive glasses, which are used for bone engineering and in situ bone tissue regeneration, enhances bone formation.⁴² Recent reports provide further evidence that nutritional intakes of boron promote bone health. Boron deprivation in mice decreased the density of osteocyte lacunae in femoral cortical bone tissue and decreased tibial bone rigidity.⁵² Real-time PCR analysis found that mRNA expression of four mineralization genes and five genes involved in cell proliferation and differentiation was upregulated by boron in cultured osteoblastic (NOS-1) cells.⁵³ Low concentrations of boron induced early matrix mineralization via calcium deposition and elevated alkaline phosphatase activity in differentiated rat bone marrow mesenchymal stem cells.⁵⁴ In human bone marrow stromal cells, boron increased calcium deposition and messenger RNA expression of alkaline phosphatase, osteocalcin, collagen type 1, and bone morphogenetic protein 7.⁵⁵ Boric acid administered at the site of bone fracture was found to stimulate fracture healing in rats.⁵⁶

One human study found that 6 months of providing 226 mg/day of calcium fructoborate incorporated into margarine improved bone density in 66 of 100 patients with osteoporosis.⁴⁶ Because the supplement provided only 20 mg/day calcium, the improvement was attributed

to fructoborate that provided an additional 5.65 mg/day of boron.

B. Cardiovascular Health—Inflammatory and Oxidative Stress Involvement

Inflammatory and oxidative stress are considered major risk factors for atherosclerosis and thus cardiovascular disease. Since 2008, over 25 reports have appeared indicating that boron can modulate the response to inflammatory and oxidative stress. These reports have involved studies with animals and cultured cells with induced oxidative stress and supplementation and epidemiological studies of individuals with oxidative or inflammatory stress.

Among the cell culture studies are those showing that boric acid inhibited lipopolysaccharide (LPS)-induced tumor necrosis factor- α (TNF- α) formation through a thiol-dependent mechanism in human monocytic leukemia THP-1 cells⁵⁷; calcium fructoborate decreased interleukin-1 β and interleukin-6 (IL-6) release by LPS-stimulated murine macrophage RAW 264.7 cells⁵⁸; and boric acid protected chromosome structure in human primary alveolar epithelial cells treated with nicotine.⁵⁹

Among the animal studies are those showing that boric acid increased the antioxidant capacity of spleens in rats⁶⁰ and increased serum total antioxidant activity and hepatic expression of superoxide dismutase mRNA in rats injected with sheep red blood cells.⁶¹

Among the human studies, boron supplementation as calcium fructoborate improved inflammation biomarkers of C-reactive protein (CRP), fibrinogen, and erythrocyte sedimentation rate in individuals with primary osteoporosis.⁴⁶ In several studies, boron supplementation was found to reduce serum CRP in individuals with serum concentrations greater than 3.0 mg/L, which is considered an indicator of chronic inflammatory stress. In a randomized, double-blind, parallel clinical trial, a boron supplement of 6 mg/day as calcium fructoborate significantly decreased elevated CRP in 29 patients with stable angina pectoris; 29 patients not receiving calcium fructoborate showed no change at 30 and 60 days.⁶² In another double-blind, placebo-controlled clinical study of healthy individuals with mean serum CRP concentrations >3.0 mg/L, supplements of 3 or 6 mg/day of boron as calcium fructoborate were compared to a placebo. Both boron groups exhibited significantly reduced serum concentrations of CRP, IL-6, and monocyte chemoattractant protein-1.⁶³

Boric acid activation of eIF2 α and Nrf2 also was found to increase mRNA levels of three antioxidant response element genes (ARE).⁶⁴ It was suggested that the activation of the pathway to ARE production was a basis for boron preventing oxidative DNA damage and enhancing antioxidant status.⁶⁴

C. Central Nervous System Function

In the 1990s, it was found that boron supplementation after depletion resulted in electroencephalograms indicating less drowsiness and increased mental alertness, improved motor speed and dexterity skills, and improved attention and short-term memory.⁴² It was also found that rats deprived of boron exhibited brain electrical activity similar to nonspecific malnutrition and heavy metal toxicity.⁴² In 2006, it was reported that boron-deprived rats were less active than boron-supplemented rats in a spontaneous activity evaluation.⁶⁵ Recently, it was reported that boron supplementation prevented damage to membranes of the cerebral cortex in rat pups from alcohol-treated dams.⁶⁶

D. Multifarious Cell Functions

The findings indicating that boron can affect cell signaling, and calcium ion cellular release and utilization, likely explain how boron affects various organs in the body.

Since 1981, occasional reports have appeared suggesting that boron can alleviate the symptoms of arthritis. These include animal studies finding that nutritional intakes of boron can inhibit the induction of arthritic conditions.⁴² Human studies suggesting boron may ameliorate arthritic symptoms include a report that 6 mg/d of boron as calcium fructoborate supplement reduced or eliminated the use of painkillers, decreased joint rigidity, and increased mobility in osteoarthritic patients.⁶⁷ In another study, serum boron was negatively correlated with the duration and severity of knee arthritis.⁶⁸ In a cross-sectional study, the mean serum boron concentration was significantly lower in 107 patients with rheumatoid arthritis than in 214 controls matched in age and sex.⁶⁹

Boron also has been inversely associated with prostate, cervical, lung, and breast cancer.⁴² Boron has been found to inhibit the growth or proliferations of some types of cultured prostrate cells.^{70,71}

Numerous animal studies indicate that boron affects cell function through influencing the action of various hormones, including vitamin D, estrogen, thyroid hormone, insulin, and progesterone.⁴²

after being fed a low-boron diet, a fasting serum boron concentration near or below 3.13 mmol/L (34 ng/mL) and a dietary food record indicating a boron intake less than 1.0 mg/day could be an indication that a person would benefit from an increased intake of boron.

B. Reference Intakes

Neither RDAs nor AIs have been established for boron.¹³ However, humans in depletion–repletion experiments positively responded to boron supplementation after consuming a diet supplying only 0.2–0.4 mg/day for 63 days.³⁶ Thus, a dietary boron intake greater than 0.4 mg/day may be useful for bone and brain health and in modulating inflammatory and oxidative stress. Extrapolation of data from animal experiments suggests that 1.0 mg/day of boron would meet the requirement for the beneficial effects of boron.³⁶ Both animal and human data were used by the World Health Organization⁷² to suggest that an acceptable safe range of population mean intake of boron for adults could be 1–13 mg/day.

C. Tolerable Upper Intake Levels

Upper intake levels have been set for boron (Table 29.1).¹³ The ULs were for children aged 1–3 years, 3 mg/day and aged 4–8 years, 6 mg/day. The ULs for adolescents aged 9–18 years were 17 mg/day and for adults, 20 mg/day. The World Health Organization⁷² first indicated that 13 mg/day would be a safe upper level intake but later increased this to 0.4 mg/kg body weight or about 28 mg/day for a 70-kg person.⁵⁰ The European Food Safety Authority⁷³ established an acceptable daily intake for boron, which is 0.16 mg/kg body weight.

Boron is a relatively nontoxic element. A dose of 500 mg/day of boric acid (87 mg boron) was used to show disturbances in appetite, digestion, and health in human volunteers.⁷⁴ The lack of finding adverse effects on health and fertility with drinking water with high boron concentrations and the past use of boric acid and borates as food preservatives are evidence of the low order of toxicity for boron.⁷⁴

D. Dietary Sources

Reports showing the boron content of commonly consumed foods^{75,76} indicate that boron is abundant in many foods and drinks obtained from plants, including fruits, leafy vegetables, nuts, wine, and cider. Dried fruits are especially rich in boron. However, the amount of boron in foods and beverages varies based on the environment in which they are produced. The FDA has not established DVs for boron for food or dietary supplement labeling purposes.

XV. BORON DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy, Criteria of Adequacy, and Assessment of Status

No studies have defined an indicator of boron status. Based on humans responding to boron supplementation

XVI. ISSUES RELATED TO BORON IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Boron deprivation has not been shown to cause some condition of concern. However, as described above, nutritional intakes of boron have numerous health benefits.

B. Known Interactions

As described above, boron may affect calcium function.

C. Dietary Supplement

Boron in supplements exists in the form of amino acid chelates, ascorbate, calcium fructoborate, citrate, gluconate, and picolinate.¹⁹ These supplements provide 0.25–3.0 mg per dose.

XVII. SILICON INTRODUCTION

A. Background

Silicon is nutritionally essential for some lower forms of life, including diatoms, radiolarians, some sponges, and some plants (e.g., rice).⁷⁷ Silicon has not been established as nutritionally essential for humans because it lacks a clearly defined biochemical function. However, cell culture, animal, and human studies indicate that silicon in nutritional amounts may be beneficial, especially for bone and connective tissue function. Silicon in nutritional amounts may also help in modulation of inflammatory and oxidative stress and by interacting with other minerals (e.g., aluminum) to prevent their detrimental effects.

B. Nutrient Function and Structure

Beneficial effects of silicon likely occur because it easily forms stable complexes with polyols that have at least four hydroxyl groups.⁷⁸ Such polyols include hexosamines and ascorbate used to form glycosaminoglycans, mucopolysaccharides, and collagen, which are involved in connective tissue formation and stabilization and bone formation.

The formation of silicon anions with hydroxylated molecules in the bone growth region apparently attracts calcium to form apatite. After calcification, silicon is lost, which explains the finding that silicon is highest in undermineralized bone and lowest in fully mineralized bone.^{77,79}

In plants, silicon apparently binds hydroxyl groups in proteins involved in signal transduction. A similar action in higher animals might be the basis for associating silicon with the immune or inflammatory response that

affect osteoblastogenesis and osteoclastogenesis. Silicon was found to decrease the induction of nuclear factor kappa beta by TNF- α in the signal transduction pathway that inhibits osteoblastic bone formation in vitro.⁸⁰

Some reported beneficial effects of silicon might have occurred through altering the absorption and utilization of other mineral elements involved in bone metabolism, the immune or inflammatory response, oxidative stress, or cognitive function. For example, the formation of aluminosilicates may prevent aluminum from competing for iron-binding sites that result in decreased functions involving iron.⁸¹ The iron functions include that of prolyl hydroxylase, which has a role in collagen formation.

XVIII. SILICON METABOLISM

A. Digestion

The absorption of silicon is strongly influenced by the solubility of the silicon compounds consumed. Insoluble silicates are not absorbed. Silicon as found in foods and soluble silicon compounds is relatively well absorbed.⁸² High dietary calcium, magnesium, and fiber through making insoluble complexes can decrease the amount of consumed silicon that is absorbed.⁸³

The absorption mechanism for silicon has not been clearly defined. Simple uncharged silicon species such as orthosilicic acid apparently are readily and rapidly absorbed in the proximal small intestine through a paracellular or small-pore transcellular pathway.⁸³ Charged polymeric silica species essentially do not cross the gastrointestinal tract.

B. Transport

Once absorbed, silicon is considered to exist mainly as neutral orthosilicic acid in plasma. It has been suggested that the transport of silicon into cells occurs through aquaporins, especially the aquaglyceroporin (AQGP) AQP7, AQP9, and AQP10.⁸⁴ These AQGPs are present in tissues where silicon has bioactivity, and they transcriptionally respond to changes in dietary silicon. Recently, an active efflux silicon transporter, Slc34a2, was identified in mammals.⁸⁵ This transporter was upregulated in rat kidney following chronic silicon deprivation.

Silicon is readily filtered by the renal glomerulus and eliminated with little tubular reabsorption.^{83,84} Urinary silicon exists as orthosilicic acid or magnesium orthosilicate. Peritoneal injection of silicon can elevate urinary excretion above the upper limit achieved by dietary intake.¹ Thus, urinary excretion is a major factor in silicon homeostasis. Silicon in polymeric and colloidal forms that are not readily broken down in the gastrointestinal tract is excreted in the feces.⁸⁴

C. Metabolism

Because silicon occurs both in blood and urine mainly as orthosilicic acid, it apparently is not metabolized to any great extent in the body except when it apparently forms stable polyol complexes that apparently are involved in its beneficial actions. Connective tissues, including aorta, immature bone, skin, tendon, trachea, and fingernails, contain much of the silicon retained in the body.

XIX. ROLE OF SILICON IN NORMAL CELL AND ORGAN FUNCTION

Epidemiological studies have indicated that nutritional intakes of silicon have salubrious effects on bone and connective tissue. Large differences of up to 10% in hip bone density were found in men and postmenopausal women between the highest (>40 mg/day) and lowest (<14 mg/day) quintiles of silicon intake.⁸⁶ Another study found silicon intake associated with bone mineral density at the femur neck in women using hormone replacement therapy.⁸⁷ Energy-adjusted silicon intakes have been found to be negatively associated with urinary markers of bone resorption and positively associated with a marker of bone formation in a study where the lowest quartile of silicon intake was 16 mg/day and the highest quartile was 31.5 mg/day.⁸⁷ A double-blind, placebo-controlled 12 month study of osteopenic females found supplementation with 6 and 12 mg/day of silicon as choline-stabilized orthosilicic acid increased the bone formation marker serum procollagen type 1 N-terminal propeptide.⁸⁸ Further evidence that silicon is beneficial to bone growth and maintenance are findings showing that scaffolds containing silicon enhances biomineralization, osteogenesis, and hard tissue formation.^{89,90} Support for silicon having salubrious effect on connective tissue is the finding that a silicon supplement of 10 mg/day as choline-stabilized orthosilicic acid for 20 weeks improved photodamaged skin and decreased hair and nail brittleness.⁹¹

XX. SILICON DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy, Criteria of Adequacy, and Assessment of Status

No studies have defined an indicator of silicon status. Typically, adult fasting serum concentrations are about 100 µg/L and can go 3 to 4 times higher after the intake of supplements or foods high in silicon.⁹² Epidemiological studies indicate that individuals with silicon intakes less than 25 mg/day may respond to a higher intake of silicon (see below). Thus, a fasting serum silicon

concentration well below 100 µg/L and a dietary food record indicating a silicon intake less than 25 mg/day could be an indication that a person would benefit from an increased intake of silicon.

B. Reference Intakes

Animal and human data have been judged too limited to establish DRIs for silicon.¹³ In the Framingham Offspring Cohort study, the highest bone mineral density occurred in premenopausal women and men, respectively, with quintiles intakes ranging from 30.2 to 63.2 mg/day and 34.4 to 118.0 mg/day compared to the lowest quintile intakes ranging from 7.1 to 16.7 mg/day and 7.6 to 18.8 mg/day.⁸⁶ These findings in addition to extrapolation from animal data and questionable balance data¹ suggest that a beneficial intake of silicon may be near 25 mg/day.

C. Basis for Individual Variability in Requirements

It has been found that age, sex, and estrogen status does not affect the response to beneficial intakes of silicon. The only factors that has been found that could affect the response to silicon are those that would decrease its absorption. This includes a high dietary intake of calcium, magnesium, and/or fiber.

D. Tolerable Upper Intake Levels

There are no reports of silicon toxicity through excessive dietary intake for people with normal renal function. Long-term use of silicate-containing drugs and chronic hemodialysis may cause silicon accumulation in the kidney that induces nephropathy.⁸³

E. Dietary Sources

A provisional food database indicated that the highest concentrations of silicon are found in cereal and cereal products, especially in less-refined cereals and oat-based products.⁹³ Substantial amounts of silicon also were found in beans, spinach, dried fruits, bananas, and red lentils. The silicon in barley and hops is solubilized during the brewing process, which makes beer a rich source of silicon. Drinking water, especially some bottled mineral waters, can be a significant source of silicon.⁸³ Silicon is added to processed foods in various forms for anticaking, thickening, and stabilizing purposes. These forms of silicon are not readily absorbed by the gastrointestinal tract and should not be considered providing beneficial intakes of silicon. The FDA has not established DVs for silicon for food and dietary supplements labeling purposes.

XXI. ISSUES RELATED TO SILICON IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Silicon deprivation has not been shown to cause some condition of concern. However, as described above, nutritional intakes of silicon have beneficial health effects, especially in bone and connective tissue.

B. Silicon as a Dietary Supplement

Silicon in supplements exists in numerous forms, including colloidal silicon, silicon amino acid chelates, plant (horsetail) silicon, arginine–silicate–inositol complex, and monomethylsilanetriol.¹⁹ These supplements generally provide 1–6 mg per dose.

XXII. OTHER TRACE ELEMENTS INTRODUCTION

A. Background

Numerous elements have been suggested to be of nutritional importance because of some favorable physiological or clinical finding with nutritional or supranutritional intakes in animal models or associated with some pathological condition in humans. Since the 10th edition of *Present Knowledge in Nutrition*, the nutritional and pharmacological bioactivities of nickel and vanadium have received some attention. These elements are briefly described in this review. Essentially no new findings about the beneficial effects of arsenic and fluorine in nutritional amounts have appeared recently. Newer research involving arsenic and fluorine has focused on the basis and extent of their toxicity. Several other elements with very little evidence for beneficial bioactivity in higher animals and humans in the past, including aluminum, cadmium, lead, and tin, also have received mostly toxicological research attention that is beyond the scope of this update. Essentially no new beneficial bioactivity in nutritional amounts has been reported recently for arsenic, bromine, fluorine, germanium, lithium, and rubidium. Instead of repeating the information indicating beneficial bioactivity of these elements, see previous editions of this work.^{1,2}

XXIII. NICKEL INTRODUCTION

A. Background and Nutrient Function

Nickel is essential for plants and some bacteria. In these lower forms of life, nickel is an essential component of eight different enzymes; seven of these enzymes are involved in the use or production of gases ammonia, hydrogen, carbon monoxide, carbon dioxide, methane, and oxygen.⁹⁴ The other nickel enzyme, glyoxalase I, converts methylglyoxal

to lactate. Finding enzymes in lower forms of life suggests that a similar function might be found in higher animals. However, to date, no such enzyme has been found. However, like in lower forms of life, nickel bioactivity in higher animals might involve an effect on the functions of gaseous molecules. Nickel has a stabilizing effect on hypoxia-inducible factor 1 α protein and activates hypoxia-inducible gene expression,⁹⁵ which affects glucose metabolism and osteogenesis. Nickel also potently induces the activity of heme oxygenase that produces carbon monoxide that activates guanylate cyclase to produce cyclic guanylate monophosphate (cGMP). The cGMP signal transduction system has a critical role in vision, taste, smell, blood pressure control, kidney function, and sperm motility, all of which are affected by nickel intake.⁹⁶ Nickel also might affect methyl metabolism. Nickel has been found to ameliorate an increase in serum homocysteine in vitamin B12–deficient pigs⁹⁷ and decrease homocysteine, cysteine, and S-adenosylhomocysteine production by human peripheral monocytes.⁹⁸

XXIV. NICKEL METABOLISM

Since 2012, essentially no new information about the digestion, and transport, and metabolism of nickel has come forth. The known information is described in previous reviews.^{1,2}

XXV. BENEFICIAL EFFECTS OF NICKEL IN NORMAL CELL AND ORGAN FUNCTION

Findings from animal experiments indicate that nickel supplementation has beneficial effects on a variety of organ and cell functions impaired by nickel deprivation. These impairments include decreased sperm production and motility; decreased bone strength with altered mineral composition; increased plasma lipids; decreased serum glucose; decreased iron status or utilization; exacerbation of vitamin B12 deficiency resulting in increased circulating homocysteine; exacerbation of renal damage and high blood pressure induced by a high-salt diet; and impaired special senses of vision, olfaction, and taste.^{2,96}

XXVI. NICKEL DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy, Assessment of Status, and Reference Intakes Including Tolerable Upper Intake Levels

No studies have defined an indicator of deficiency or status for nickel in humans. As a result, neither an RDA

nor an AI has been established for nickel.¹³ Based on animal findings, a beneficial intake for humans might be near 50 µg/day.¹ The UL set for nickel in the United States and Canada is 1.0 mg/day for adults (Table 29.1).¹³

B. Dietary Sources

Rich sources of nickel include chocolate, nuts, legumes, and grains. Typical daily dietary nickel intakes by adults are 70–260 µg/day.¹ The FDA has not established DVs for nickel for food and dietary supplement labeling purposes.

XXVII. VANADIUM INTRODUCTION

A. Background and Nutrient Function

Vanadium is essential for some lower forms of life where it is a component of the enzymes bromoperoxidase, iodoperoxidase, and chloroperoxidase.⁹⁹ Vanadium reacts with H₂O₂ to form a dioxygen species that reacts with halide to form an oxidized species, which is an intermediate in forming a carbon–halogen bond.⁹⁹

Vanadium has not been found to be essential for higher animals or humans but does have beneficial effects in supranutritional or pharmacological and perhaps nutritional amounts. The antagonism between vanadate and phosphorus is considered the basis for vanadium inhibiting or stimulating enzyme actions that produce beneficial responses.¹⁰⁰

The inhibition of phosphatases may be the basis for many of the reported responses to nutritional and supranutritional intakes vanadium.¹⁰⁰ However, vanadium also affects glucose-6-phosphate dehydrogenase, nicotine adenine dinucleotide, ADP, nucleoside triphosphate diphosphohyrolases, phosphodiesterases, phosphoglucosmutases, and ATPase in metabolic diseases.¹⁰⁰

B. Vanadium Metabolism

Digestion

Because urine vanadium is usually less than 0.8 µg/L and the estimated daily intake of vanadium is 12–30 µg/day, apparently less than 5% of ingested vanadium is normally absorbed.¹⁰¹ Vanadate is absorbed three to five times more effectively than vanadyl, which is the main form in the blood.

Transport

Vanadate transported into blood is quickly converted to vanadyl. Vanadyl is mainly transported by transferrin and albumin.¹⁰⁰ Vanadyl also forms complexes with ferritin in plasma and body fluids. Because vanadyl has a high affinity for iron-binding sites, transferrin

may transfer vanadium into cells through the TfR and ferritin might be a storage site for vanadium.¹⁰⁰

Metabolism

Vanadium appearing in plasma with nutritional intakes is rapidly removed and generally retained in fresh tissues in concentrations less than 10 ng/g.¹⁰⁰ Bone apparently is a major sink for excessive retained vanadium. Urine is the major excretory route for absorbed vanadium. Feces is the major excretory route for ingested vanadium.

XXVIII. BENEFICIAL EFFECTS OF VANADIUM IN NORMAL CELL AND ORGAN FUNCTION

Only limited older reports of vanadium deprivation of animals indicate that vanadium in nutritional amounts has beneficial effects. Reported deprivation signs include swollen joints, skeletal deformations, and decreased life span in goats and altered thyroid hormone metabolism, impaired reproduction, and altered bone morphology in rats.¹⁰¹ The ability of vanadium in supranutritional or pharmacological amounts to mimic insulin is the most studied beneficial effect of vanadium in humans.¹⁰⁰

XXIX. VANADIUM DIETARY REQUIREMENTS

A. Indicators of Deficiency or Inadequacy, Assessment of Status, and Reference Intakes Including Tolerable Upper Intake Levels

No studies have defined an indicator of deficiency or status for vanadium. As a result, neither an RDA nor an AI has been established for vanadium.¹³ Based on animal findings, a beneficial nutritional intake of vanadium for humans would be small; a daily intake of 10 µg probably would meet the requirement.¹⁰¹ The UL set for vanadium in the United States and Canada is 1.8 mg/day for adults.¹³

B. Dietary Sources

Good sources of vanadium include grains, mushrooms, parsley, and shellfish. Typical daily dietary vanadium intakes by adults are 12–30 µg/day.¹⁰¹ The FDA has not established DVs for vanadium for food and dietary supplement labeling purposes.

RESEARCH GAPS

Manganese: The public health concern of deficiency and toxicity for manganese requires further evaluation. Controlled long-term balance studies are needed so that evidenced-based RDAs and EARs can be established for manganese. Establishing DRIs and status indicators for manganese is needed to confirm that a low manganese status is contributing to the risk for some chronic diseases, especially for the metabolic syndrome, diabetes, and weak bones.

Molybdenum: With regard to molybdenum, clinical studies are needed to substantiate the anticancer effects of thiomolybdate. In addition, evaluation of supranutritional amounts of supplemental molybdenum might find other beneficial effects of this element. Supplemental amounts of sodium molybdate were found to prevent hyperinsulinemia and hypertension induced by fructose in rats.¹⁰² The molybdenum hydroxylases might be useful in metabolizing drugs and foreign compounds, and thus molybdenum status can affect human health through an effect on detoxifying some xenobiotic compounds.

Boron: There is a need to clearly define the mechanism/s through which boron has beneficial effects on the bone, brain, and cardiovascular health and to establish the intake of boron adequate for optimal benefits on health and well-being.

Silicon: There is a need to clearly define the mechanism/s through which silicon has beneficial effects on bone and connective tissue health and to establish the intake of silicon adequate for optimal benefits in health and well-being.

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S E C T I O N D

Other Dietary Components

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WATER

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SUMMARY

Water is critical to human health, serving multiple roles as a biochemical solvent and reactant for cellular metabolism, medium for substance exchange between tissues, and support for the circulatory system. Survival mechanisms for acquiring and conserving water to maintain cellular homeostasis and circulatory integrity have evolved in many mammals. Humans regulate body water balance remarkably well as a result of these survival mechanisms and also because food and fluid are commonly abundant. An adequate fluid intake is sufficient to replace body water losses and provide for dietary and metabolic solute excretion. Adequacy therefore depends on wide variations in physical activity and diet. Acute or chronic body water deficits do sometimes occur, most often with eccrine sweating from exercise and heat stress or gastrointestinal losses from illness. Negative effects of acute dehydration on body systems that support physical and mental performance and health are well recognized; effects of chronic dehydration on health are less clear.

Keywords: Dehydration; Fluid balance; Fluid replacement; Fluid requirements; Hyponatremia.

I. INTRODUCTION

Water plays many unique and vital roles within the body. Water is the principal chemical constituent of the human body (total body water, TBW) and serves as both a biochemical solvent and reactant supporting cellular homeostasis.^{1,2} Water is also essential for sustaining vascular volume and serves as the transport medium between tissues, supplying nutrients and removing body heat and waste. Water has unique and remarkable properties³ that serve competing organism demands of osmoregulation and thermoregulation.⁴ In addition, cell hydration state is increasingly recognized for its role in signaling and regulating metabolism and gene expression.^{1,5,6} Indeed,

our ancestral transition from aquatic to terrestrial life required mechanisms for acquiring and conserving water to maintain a cellular environment that favored macromolecular evolution.^{7–11}

Humans have evolved unique survival strategies for terrestrial living. Cells are defended from transient deviations in osmolality via both tonicity (osmotic) and volume (nonosmotic) receptors. The information they convey is integrated within the central nervous system where neuroendocrine regulation of dipsogenic behavior and renal function together increase the gain and decrease the loss of body water.^{11–13} Humans are also endotherms. Approximately 20%–25% of the total metabolic heat load at rest is balanced by the vaporization of solute-

free water from the skin (~ 0.40 mL/kcal) and respiratory tract (~ 0.12 mL/kcal).¹⁴ Solute-free respiratory water losses are typically balanced by metabolic water production (from oxidation of carbohydrate, fat, and protein) across a range of activities, thus not requiring further consideration for the discussion of body water balance, per se.^{15–17} Obligatory metabolic waste elimination (urine, feces) varies considerably but may be cautiously estimated as ~ 0.60 mL/kcal.¹⁸ Resting metabolism therefore requires ~ 1 mL/kcal or about 2.0 L of water from fluid and food sources per day for an adult.

Why then do fluid needs vary among people? The water requirement to excrete dietary solute and metabolic waste may deviate considerably from ~ 0.60 mL/kcal due to variance in dietary and physical activity practices.¹⁹ While humans are near the bottom of the mammalian spectrum in urine concentrating capacity,²⁰ our ability is impressive enough to make possible a wide range of water intakes compatible with a wide range of urine concentrations. Human life on land also requires protection against organism overheating above and beyond metabolism, such as during physical activity and especially during environmental heat exposure.² Thermoregulatory sweating affords significant evaporative heat loss (580 kcal/L),²¹ but at the cost of additional body water and salt deficits.²² Approximately 7–8 L of gastrointestinal fluids are secreted and most reabsorbed each day; however, illness can drastically increase secretion and decrease reabsorption.²³

This chapter reviews the human physiology, assessment, and intake recommendations for the nutrient water. The extent to which water imbalances affect health and physical and mental performance is also considered. The critical role of nutrient ions (electrolytes) will be discussed only in the context of body water balance. In this chapter, euhydration refers to a normal and narrowly fluctuating state of body water content, while

dehydration refers to a body water deficit that exceeds the euhydration boundary.

II. WATER METABOLISM

A. Water Acquisition and Absorption

Most fluids are consumed in the form of beverages ($\sim 50\%$) and pure water ($\sim 30\%$) with the remainder ($\sim 20\%$) coming from foods.^{2,24} While fruits and vegetables, grains, and protein are rich food sources of water, as they are composed of 75%–95% and 35%–75% water by weight, respectively,²⁵ the desire to consume fluids (thirst) is by far the most important component of water acquisition.

The evolution of thirst mechanisms was critical to survival on land.¹¹ Thirst has been referred to as “the strongest and most demanding sensation of life.”¹³ The primary stimulus for drinking is osmotically driven neural signaling^{7,13} with important secondary effects of volume deficit neural signaling (Fig. 30.1).¹² However, in modern humans, spontaneous fluid intake is governed primarily by feeding behavior as opposed to genuine thirst.²⁶ Importantly, human thirst mechanisms are subject to numerous influences unrelated to body water balance.²⁷ For example, typical stressors of body water loss (sweat, vomiting, diarrhea)^{23,28} can easily outpace voluntary oral intakes and maximal gastric emptying rates.²⁹ Peripheral osmoreceptors and oropharyngeal cues trigger thirst satiety (Fig. 30.1) well before volume is fully restored^{13,30,31} even when dehydration is substantial.^{30,32} This transitory response acts to buffer the presystemic impact of ingested fluids.⁷ This may have conferred an evolutionary advantage in humans¹³ but can lead to involuntary dehydration.^{27,32}

Fortunately, the water ingested is absorbed very efficiently along a number of connected pathways. Water is

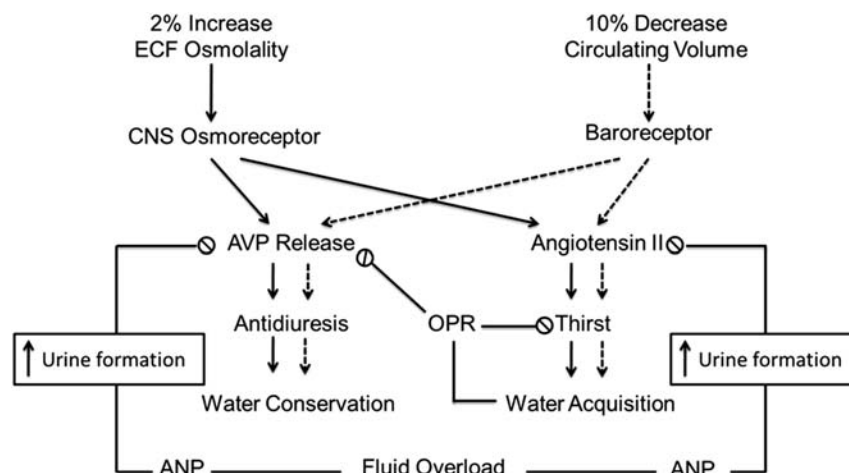


FIGURE 30.1 Schematic diagram of fluid regulation in situations of fluid deficit and surfeit. Solid lines indicate the osmotically stimulated pathways (primary), and dashed lines indicate volume stimulated pathways (secondary). ⊖ indicates negative feedback pathways; ANP, atrial natriuretic peptide; AVP, arginine vasopressin; CNS, central nervous system; ECF, extracellular fluid; OPR, oropharyngeal reflex. Adapted from Andreoli et al.¹²

absorbed by osmotic diffusion through intestinal epithelial aquaporin channels directly in parallel with sodium or indirectly in association with the potent osmotic gradient created by sodium.^{33–35} However, water acquisition and absorption are together still inadequate for survival without the aid of antidiuretic adaptations.

B. Water Retention and Compartmental Exchange

Osmotic constancy is ordinarily maintained by tonic secretion of arginine vasopressin (AVP), also known as vasopressin or antidiuretic hormone. AVP is a hormone synthesized in the hypothalamus and released into the circulation from synaptic vesicles of the posterior pituitary. AVP directly influences renal water excretion and conservation in response to ordinary body fluid shifts and ad libitum food and fluid intakes.^{36–38} Osmotic homeostasis corresponds with basal electrical activity of osmoreceptor neurons believed to encode an osmotic set point⁷ and ordinarily keeps plasma osmolality (Posm) from deviating more than 2%.^{7,12,39} The Posm threshold for AVP secretion is lower than the Posm threshold for unequivocal thirst.^{40,41} Fig. 30.1 depicts the ~2% increase in Posm (~6 mmol/kg) (as extracellular fluid osmolality) required for both compensatory renal water conservation and water acquisition (thirst),¹² which very closely approximates the Posm response to dehydration when TBW losses are $\geq 2\%$ of body mass (i.e., a loss of body water volume equal to 2% or more of the total body mass, such as 1.4 L for a 70 kg person).^{42–44}

Although the sensitivity of osmoreceptors in regulating AVP release is enhanced by relatively small losses of body water, volume-mediated water retention (and thirst) requires a larger threshold loss of blood volume than is typically observed at the 2% dehydration threshold^{12,45} (Fig. 30.1). Nonosmotic stimuli are responsible for compensatory responses to volume losses and include acute, sympathetically mediated peripheral and renal vasoconstriction and a slower, neurohumoral defense of

blood volume.^{46,47} Fig. 30.1 depicts the ~10% loss of blood volume required to fully engage the baroreceptor, angiotensin II, and nonosmotic stimulus of AVP.

A loss of body water that increases the osmolality of body fluids and decreases their volume will result in shared water exchange between intracellular (ICF) and extracellular (ECF) body fluid compartments.⁴⁸ TBW ranges from 50% to 70% of body mass (generally >40 L) with a range attributable to differences in body composition; the hydration constant for fat-free mass is 73.7%, whereas fat-mass is <10%.⁴⁹ Water exchange between ICF and ECF is driven by osmotic gradients, whereby water rapidly diffuses across cell membranes through aquaporin protein channels³⁵ to equalize osmotic pressures among cations and anions of the ICF (potassium, magnesium, proteins) and ECF (sodium, chloride, bicarbonate) spaces.² The typical distribution of TBW is ~65% ICF and 35% ECF. The ECF is further subdivided into closed (cerebrospinal, interstitial, plasma) and open (gastrointestinal, urinary, sweat glands) spaces,⁵⁰ the latter representing significant potential avenues of body water loss. Water exchange between the interstitial and plasma spaces occurs by transcapillary hydrostatic and oncotic forces; in health, filtration generally occurs at the arterial end and absorption at the venous end of the capillary.² Fig. 30.2 depicts these fluid compartments and common mechanisms for compartmental fluid exchange.⁵¹

III. CELL FUNCTION

The cell represents a nanoscopic environment where the dimensions of its water system are important for functionality. Water interacts with ions, cell membranes, the surfaces of proteins, and other molecules. For example, ultrafast infrared spectroscopy experiments reveal that protein folding is made possible by reorganization of water hydrophobic and hydrophilic regions on a picosecond scale.³ Water dynamics depends on what kind of molecule is involved, but it is universally slow

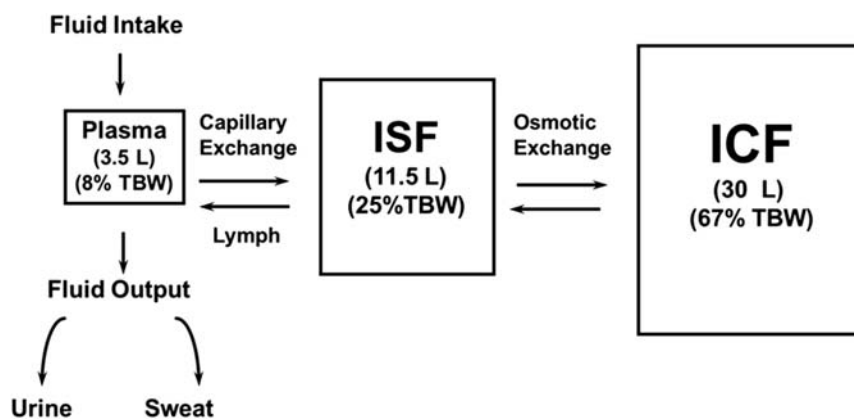


FIGURE 30.2 Schematic of the approximate volume of water and mechanisms for exchange between plasma, interstitial fluid (ISF), and intracellular fluid (ICF) compartments. Example assumes total body water (TBW) of ~45 L for a 70 kg individual. Adapted from Sawka.⁵¹

when the nanoscopic water system shrinks. The size of the water nanopool matters as more or less water molecules are present to interact with macromolecules. Therefore, the most important aspect of water dynamics within the cell depends on cell volume.³

Cell cation concentrations reflect those most prevalent in seawater, which likely favored macromolecular evolution.^{8,11} Cells swell or shrink in accordance with the prevailing osmotic gradient, and cell ionic tonicity varies inversely with its volume.⁵² The loss of cell water induces shrinkage causing macromolecular crowding and pleiotropic effects, which impair cell function by DNA damage, mitochondrial dysfunction, oxidative stress, or even cell death.^{1,5} Cell shrinkage also produces adaptations favorable to tolerance by modulating gene expression. For example, hyperosmotic stress activates genes that encode for enzymes and transporters responsible for organic osmolyte accumulation.⁶ The transcription factor NFAT5 is one of the few known mammalian osmoprotective transcription factors that increase protective organic osmolytes.^{1,5} The delayed time course of improved cellular osmotic tolerance⁵³ is consistent with organic osmolyte accumulation in the brains of rats made chronically, but not acutely, hypertonic.⁵⁴ However, the translational relevance of *ex vivo* cell experiments to mammalian *in vivo* cells remains a future challenge. The application of genomics, transcriptomics, proteomics, and metabolomics to understanding the impact of *in vivo* cell hydration changes on its function is an exciting emerging science.⁵⁵

IV. DIETARY REQUIREMENTS

A. Indicators of Adequacy

In the absence of dehydration, Posm is regulated narrowly among both terrestrial and aquatic mammals in whom water acquisition and retention are required for survival in these dehydrating environments.^{7,11} Ordinary water and solute flux are controlled by sensitive neuroendocrine adaptations and living behaviors. Mathematically, the moment-to-moment changes in Posm (~ 1 mmol/kg) and AVP (<0.5 pg/mL) in humans are smaller than what modern quantitative techniques can measure.^{39,40,46} Therefore, ECF osmotic constancy ($\pm 2\%$ deviation in Posm) is often used as a benchmark for fluid intake adequacy.² Twenty-four hour urine volume (U_{vol}), osmolality (U_{osm}), or their combination may also be used when Posm is not available, but 24 h urine measures may be impractical and can vary widely while still reflecting fluid intake adequacy.^{56–58} Fluid intake guidance in Europe and the United States is based largely on population intake survey data and its correspondence with normal blood or urine measures.^{2,59}

B. Criteria of Adequacy

In practice, fluid intake adequacy may be understood as a volume of fluid (from water, beverages, and food) sufficient to replace body water losses and provide for solute excretion.^{18,56,60} Failure to replace water losses will produce a TBW deficit, which may be detected when a reliable threshold is reached.⁶¹ It can be calculated that a body water deficit of 610–915 mL would raise Posm by ~ 5 mmol/kg,⁶² thus implicating TBW fluctuations below this range ($<1\%$ body mass) with a stable Posm ($\pm 2\%$)¹² as indicative of adequate TBW replacement. The renal potential to dilute or concentrate urine makes possible a wide range of acceptable water intakes for solute excretion purposes. In general, the kidney is designed to excrete waste with a minimal loss of water; thus, the ratio of $U_{osm}/Posm$ (U/P) is normally >1.0 . However, in times of water deficit or surplus, the kidney can concentrate or dilute urine over a 20-fold range.^{20,56} The ordinary constancy of Posm ($\pm 2\%$) is achieved by wide fluctuations in U_{osm} ; thus, a wide range of U_{osm} may be consistent with fluid intake adequacy.^{2,56}

Several good experimental examples of a wide fluid intake adequacy range are available. Perrier and colleagues⁵⁷ used biochemical measures of blood and urine to assess fluid intake adequacy in groups of habitually low (≤ 1.2 L/d) and high (2–4 L/d) volume drinkers. Average Posm measures were nearly identical between groups. AVP levels were lower in the high-volume drinkers, but remained in the normal range for both groups.⁴⁰ Differences in 24-h U_{osm} (371 mmol/kg vs. 767 mmol/kg) between high- and low-volume drinkers reflected the differences in drink volumes relative to the dietary renal solute load (RSL). Raman and colleagues⁶³ assessed water intake from food and beverages using deuterium measures of water turnover and reported a range of 1.4–7.7 L/d in a large sample of people ($n > 400$). In light of TBW turnover, the adequacy of both small and large volumes was assumed based on the balance between calculated influx and efflux measures. Clearly, a variety of fluid intakes may be considered compatible with fluid intake adequacy from both the standpoints of deficit replacement and solute excretion.

C. Reference Intakes and Sources of Variability

Recall that 1 mL of water per kcal of resting energy expenditure results in an estimated water (fluid) requirement of approximately 2.0 L/d from all beverage and food sources. European and United States population fluid intake recommendations range from 1.6 to 3.7 L/d^{2,59} for ordinary diet, temperate climate, and free-living activities in men and women. However, wide variations in diet, climatic temperature, and physical activity will markedly alter fluid needs.²

TABLE 30.1 Impact of dietary renal solute load (RSL) on required urinary volume (U_{vol}) for a given urinary osmolality (U_{osm}).

Desired 24-h U_{osm} (mmol/kg)	RDA/AI/CDRR ^a RSL required 24-h U_{vol} (L)	Median diet ^b RSL ^c required 24-h U_{vol} (L)
300	2.4	3.3
500	1.4	2.0
800	0.9	1.2

Calculations requiring body mass assume 80 kg; $RSL \approx \text{urinary urea} + (2 \times (\text{sodium} + \text{potassium}))$, where “2” accounts for negative ions.

^aDRI = Dietary Reference Intakes for protein (Recommended Dietary Allowance [RDA] = 0.8 g/kg/d⁶⁴), potassium (average Adequate Intake [AI], men and women, 3000 mg/d⁶⁴), and sodium (Chronic Disease Risk Reduction Intake [CDRR] 2300 mg/d⁶⁴).

^bMedian US intakes¹³¹: protein (1.25 g/kg), potassium (2414 mg/d), sodium (3265 mg/d).

^cRSL calculated as 720 mmol/d DRI values and 979 mmol/d USDA median intakes, where $\text{urea (mmol)} = \text{protein, g}/0.175$, $\text{potassium} = \text{mg}/39$, $\text{sodium} = \text{mg}/23$; U_{vol} (L) calculated as $= RSL/\text{desired } U_{osm}$.

D. Effect of Diet on Renal Solute Load

The dietary RSL created by food choices can vary greatly. Table 30.1 illustrates the impact of dietary RSL on the required 24-h U_{vol} at three desired U_{osm} levels. Consumption of the 2003 and 2019 Dietary Reference Intakes—recommended intakes for protein, sodium, and potassium^{64,65} provides a dietary RSL of ~ 720 mmol/d; median American intakes more typically produce an RSL of almost 1000 mmol/d (~ 979 mmol/d).^{64,65} Twenty-four hour U_{osm} represents the balance between daily water and solute excretion; thus, it can be calculated as the quotient of $RSL/24\text{-h } U_{vol}$. Similarly, the water required to produce any particular 24-h U_{osm} can be estimated by equating the water requirement for solute excretion with 24-h U_{vol} (1:1), given any particular RSL.^{56,66} As an example, the typical (median) RSL of Americans increases water needs by 30%–40% over the estimated RSL at recommended intakes for any given U_{osm} (Table 30.1). Importantly, the variability in fluid intake needed to match renal water loss requirements represented in Table 30.1 calculations is only part of the equation as they do not address variability in nonrenal water losses.

E. Effect of Nonrenal Water Losses on Water Requirements

Body size and metabolism have long been implicated in water needs. The interspecies need for water (mL) among free-living mammals is approximately proportional to an exponential log function: $0.489(g)^{0.8006}$, where g is body mass in grams.⁶⁷ Species differences can be quite large; for a 60–80 kg human, the equation gives a fairly narrow 3.3–4.1 L/d range. Insensible water losses are a linear function of metabolism and amount to differences of ~ 0.50 mL/kcal or 0.5 L/1000 kcal.¹⁴ Deliberate physical activity adds eccrine (sweat) water losses of ~ 0.5 L/d²⁸ if adhering to current physical activity

guidelines,⁶⁸ which is consistent with a $\sim 25\%$ increase in water turnover when going from sedentary to modest physical activity.⁶⁰ However, water losses from exercise can vary greatly and can easily double daily water needs in training and competing athletes.²⁸ Clearly, no single requirement for water can be made, which covers the needs of all people in all circumstances.

F. Assessment of Status

The physiological basis for many hydration status assessment techniques have been comprehensively reviewed.⁶¹ The practical limitations of many methods have also been extensively considered.⁶⁹ The Venn diagram in Fig. 30.3, known as WUT (weight, urine, thirst), is a simple tool⁷⁰ that can be used by any healthy, active, low-risk population to self-monitor day-to-day hydration status using nude (or seminude) body weight, thirst

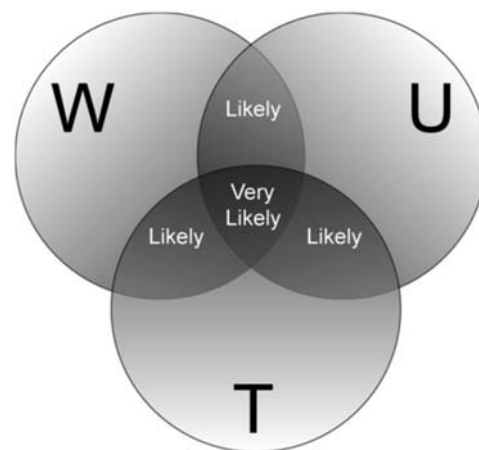


FIGURE 30.3 Likelihood that dehydration is present. W stands for “weight.” U stands for “urine.” T stands for “thirst.” When two or more simple markers of dehydration are present, it is likely that the individual is dehydrated. If all three markers are present, dehydration is very likely.⁷⁰

perception, and urine color. Emphasis must be given to the fact that all three measures must be first-morning measures to afford any interpretive power. While WUT has not yet been validated in prospectively dehydrated people, its principles are sound.^{19,71}

Decreased body weight

The use of body weight to track body water losses has a rich clinical and experimental research history. In fact, Valtin and Schafer⁷² refer to body weight change as “the most accurate, simplest, and cheapest method for assessing fluid balance.” In fact, when knowledge of a “true” dehydrated status is considered crucial for testing diagnostic procedures, an acute loss of body mass may be considered synonymous with an acute loss of TBW (1 kg = 1 L) so long as water loss (sweat, urine, ultrafiltrate)^{73,74} or water regain from an unknown premorbid state (i.v. or fluid intake)^{22,61} is quantified. If the premorbid state is unknown, a postmorbid body weight that is reduced by >2% of documented “well weights” should likewise be considered suspicious for dehydration even in the absence of a valid premorbid measurement.²² As mentioned previously, day-to-day fluctuations of ≤1% body mass represent normal variation; a day-to-day decrease in body mass >1% provides 80% probabilistic certainty that dehydration has occurred; >2% provides 95% probabilistic certainty.⁴³ A reliable bathroom scale is the minimum requirement for using WUT.

Color of urine

Acute losses of body weight (>1%) are closely related to threshold increases in renal water retention and increased urine concentration,¹⁰ which can be qualitatively related to a standardized urine color chart⁷⁵ (www.hydrationcheck.com) for use with WUT.⁷⁰

Dehydration results in dark-colored urine on the chart and correlates favorably with the level of dehydration in numerous experiments over the last 20 years. On a scale that ranges from 1 to 8, values greater than 5 arbitrary units are highly probable for dehydration.^{43,75} In practice, the use of the first morning void following an overnight fast minimizes confounding influences and maximizes measurement reliability.⁷⁶ In research, the first morning urine would be voided into a clean, clear vial or cup and the color assessed against a white background for comparison against a urine color chart.⁷⁵ It is often not appreciated that both physical activity and heat stress can reduce glomerular filtration rate⁷⁷ and influence U_{vol} and color, thus underscoring the importance of using first morning urine samples.

In everyday practice, urine color may be assessed by its general degree of darkness; pale color urine (similar to lemonade) indicates euhydration, and a darker urine color (similar to apple juice) indicates dehydration. Caution should be used if making an assessment from what has been diluted in toilet water or while in midflow. When in less than ideal conditions, urinal dilution is less problematic (men) and in the field snow can provide a suitable background. Example photos of urine color with corresponding numerical color⁷⁵ and other corresponding urine concentration measures are presented in Fig. 30.4.⁷¹

Thirst sensation

Acute losses of body weight (>1%) related to threshold increases in Posm are also related to thirst.^{12,44,78} Recall that thirst sensation is driven primarily by Posm in humans and primates,^{39,79} and it is one of the more primal human instincts critical for survival.^{13,31} The first morning measurement of thirst, in particular, is a strong predictor of dehydration as measured by loss of body



FIGURE 30.4 Hydration state and urine characteristics. Qualitative and quantitative urine color (Ucolor), urine specific gravity (USG, a.u.), and urine osmolality (Uosm, mmol/kg) from first morning urine voids in euhydrated and dehydrated volunteers.⁷¹

mass.⁸⁰ Furthermore, Engell and colleagues⁷⁸ showed that the intensity of the thirst sensation increased in association with the magnitude of dehydration. It is important to note that the reliability of first morning thirst sensation for WUT is distinct from the sensation of thirst during exercise, which is less reliable.⁸¹ Exercise alliesthesia for water is plausibly linked to hedonic minimization of the thirst sensation,⁸² conspicuous satiation,³¹ or their combination. Likert or visual analog thirst scales are used in research to quantify thirst ratings. More practical indicators of thirst include thickened saliva and difficulty or inability to easily expectorate.

Use of WUT

No single WUT marker provides enough evidence of dehydration, but the combination of any two markers makes dehydration *likely*. The combination of all three makes dehydration *very likely*.⁷⁰ Changes below these thresholds are associated with normal daily variations in body weight, Posm (thirst), and urine color and should therefore be associated with euhydration. It must be remembered that Fig. 30.3 is designed for healthy, active, low-risk populations who may or may not engage in rigorous daily exercise.^{19,21,71} It is uncertain how well Fig. 30.3 applies in chronically dehydrated, elderly, or ill populations.

V. ISSUES RELATED TO WATER IN HUMAN HEALTH

A. Dehydration, Health, and Performance

Acute dehydration

Dehydration occurs during physical exercise when sweat losses exceed fluid intake. When dehydration exceeds $\geq 2\%$ body mass (i.e., loss of TBW $\geq 2\%$ of body weight), cardiovascular strain is increased and aerobic (endurance) exercise performance becomes consistently impaired, particularly when accompanied by environmental heat stress.^{2,61} Anaerobic exercise, strength, and some aspects of cognitive performance also appear negatively affected, but the magnitude of effect is often smaller, more variable, and less consistent.^{83,84} The physiological mechanisms to explain physical performance impairments due to dehydration have been well studied but are not yet precisely resolved.^{61,85,86} The physiological mechanisms impairing cognitive-motor performance are less well studied^{86,87} but could contribute to important decision-making accidents.^{88,89} The extent to which team sports are affected invariably depends upon which aspect(s) of performance the sport relies (e.g., endurance, strength, decision-making).⁹⁰

Dehydration is also associated with significant medical costs, morbidity, and mortality across the life span.^{91,92}

Fluid and electrolyte imbalances represent a common problematic complication of acute medical treatment scenarios, particularly cardiovascular and renal perfusion problems, stemming from numerous common illnesses and disorders (e.g., gastroenteritis, hyperemesis, diuretic treatment, dialysis). Dehydration most commonly occurs from illnesses with loss of water and electrolytes through the gastrointestinal tract. Many gastrointestinal illnesses can result in ~ 2 L/d²³ of isotonic water and electrolyte losses, which is 10 times larger than the daily ~ 200 mL loss of intestinal water via feces. Cholera infection can produce losses 100 times normal and can be fatal if not treated quickly and aggressively.³⁸ Such isotonic loss of body fluids do not create an osmotic gradient for water movement from ICF to ECF space, so vascular volume contracts and can be twice that observed from sweat-induced dehydration.⁸⁵ Small children and the elderly are particularly susceptible to dehydration from diarrhea due to the rapid onset of losses and smaller TBW reserves.^{61,91,92} Restoration of circulatory and renal perfusion pressures are among the highest priorities in volume replacement.¹⁸ Guidance from several health organizations consider TBW losses between 10% and 15% of body mass as severe and possibly lethal, depending on the rate of loss.^{71,93} Based upon typical water turnover rates in minimally stressful circumstances, life without water may be limited to ~ 7 days and in highly stressful circumstances with more rapid fluid loss, perhaps <1 day.^{71,94}

Clinicians and caregivers often recommend sports beverages to replace fluid and electrolyte losses in the treatment of diarrhea.⁹⁵ However, this recommendation is not ideal to treat diarrhea. Table 30.2 provides water and electrolyte losses associated with dehydration caused by gastrointestinal or eccrine (sweat) secretions. Table 30.2 also provides the amount of electrolytes delivered by typical sports beverage and oral rehydration solutions (ORSs) when matching volume losses (standardized to 2 L over 24 h).^{23,28,96–98} Note that ORSs, which have been developed for treating diarrhea, are markedly more effective in replacing electrolyte losses than the modest correction provided by sports beverages. While sports beverages are a popular choice among caregivers for treating diarrhea,⁹⁵ they perform poorly in the treatment of diarrheal electrolyte losses and may increase the risk of developing hyponatremia.^{98,99} It should be noted, however, that sports drinks are primarily formulated to support exercise metabolism and enhance water absorption during exercise, rather than replace electrolyte losses.⁹⁷

Chronic dehydration

Chronic low-volume drinking is proposed by some to be associated with numerous medical maladies including obesity, metabolic dysregulation, kidney stones, urinary tract infections, bladder and colon cancer, constipation,

TABLE 30.2 Water, sodium (Na+), and potassium (K+) ion losses associated with two types of dehydration and their content in common rehydration beverages.

	Volume (L/d)	Na+ (mmol/d)	K+ (mmol/d)	Sum (mmol/d)
<i>Method of dehydration</i>				
Infectious diarrhea ^a	2	150	35	185
Eccrine sweat ^b	2	100	10	110
<i>Fluid replacement beverage</i>				
Oral rehydration solution ^{c,e}	2	150	40	190
Sports drink ^{d,e}	2	40	8	48

^aGennari and Weise.²³^bShirreffs and Maughan⁹⁶ and Baker et al.²⁸^cDuggan et al.⁹⁸^dMaughan.⁹⁷^eRehydration volumes set equal to anticipated daily dehydration losses (Gennari and Weise²³ and Baker et al.²⁸) to illustrate the absolute solute deficit replacement potential of ORS and sports drink beverages.

hypertension, venous thromboembolism, coronary artery disease, mitral valve prolapse, stroke, gallstones, glaucoma, and dental diseases.^{2,100–103} The strength of evidence for some of these beliefs is not robust.² Despite insufficient evidence to establish water intake recommendations to reduce any of these health risks^{2,104} or all-cause mortality,¹⁰⁵ new emphasis on research to identify low fluid intakes resulting in suboptimal health in the general population is growing.

Optimal urinary osmolality

An alternative definition of fluid intake adequacy has been proposed by Perrier and colleagues,¹⁰⁵ which includes drinking beyond TBW losses as a means of providing risk reduction for the development of various long-term pathologies.¹⁰⁶ More specifically, adequate fluid intakes associated with 24-h Uosm <500 mmol/kg¹⁰⁶ are proposed. Others have suggested ranges between 400 and 830 mmol/kg.^{107,108} The purported benefits stem from maintenance of a lower plasma AVP concentration (~1–2 pg/mL), which is speculated to decrease overall health risks.^{109,110} Indeed, there is some evidence that low-volume fluid intakes and elevated circulating AVP will exacerbate declining renal health, but far less evidence implicates the same in other disease states (e.g., cardiovascular, diabetes).¹¹¹ In addition, non-osmotic AVP stimuli related to stress, and especially hemodynamic effects of everyday living, can be substantial.¹¹² This point raises serious questions about the potential significance of small AVP elevations on developing long-term pathologies—especially when those differences remain within the normal physiological range of 1–5 pg/mL.^{40,46,112} Study of the typical biological variation of AVP is warranted.

It is conceivable that an optimal Uosm might be derived from tangential effects of an elevated Uosm on some aspects of renal health. Recommendations for slowing the progression of polycystic kidney disease and chronic kidney disease are to keep the Uosm/Posm (U/P) ratio ≤1.0 (i.e., Uosm ≤ 300 mmol/kg).^{111,113} People with more minor urine supersaturation problems, such as kidney stones, are also urged to drink more water, but no specific guidance has been offered with respect to Uosm or U/P.¹¹⁴

However, there are several reasons to refrain from this approach in healthy people. First, it is difficult to achieve U/P = 1.0 in daily life as it generally requires >3 L/d pure water intake combined with a diet low in protein and salt¹¹³ (Table 30.1). The reason for this is that urea and electrolytes are the primary solutes that contribute to body fluid osmolality. Urea also contributes proportionally more to Uosm than to Posm, thus on an ordinary diet even larger volumes of water would be needed to achieve U/P = 1.0, which intuitively goes against the primary function of the kidney to retain water and excrete waste (Uosm > Posm).^{19,20} Table 30.1 illustrates that a Uosm value of 500 mmol/kg (U/P ~ 1.7) can be achieved with U_{vol}s between 1.46 and 2.75 L/d, depending on the RSL. This intake volume (1.46 L/d) becomes the minimum water requirement for urine formation only, before adding that needed for insensible water losses and any losses related to exercise, climate, or illness stressors.

Fluid overload

A Tolerable Upper Intake Level for water intake cannot be classically determined the way it has been for some nutrients.^{2,60} However, fluid overload

problems can occur. The consumption of very large quantities of water daily is generally accompanied by copious urine production in healthy people, but if solute intake is chronically inadequate, there can still be a risk for hyponatremia (blood sodium dilution) over time.¹¹⁵ Similarly, when fluids are overconsumed or overprescribed intravenously, hyponatremia can occur. As a medical problem, hyponatremia more typically occurs when significant sodium or potassium losses take place and renal excretion of water is impaired.¹¹⁶ Hyponatremia is specifically defined as a condition of having blood sodium concentrations below <135 mmol/L. This creates a hypoosmotic ECF space and a redistribution of water into the ICF fluid space, thus resulting in cellular swelling. The most serious danger associated with hyponatremia is cerebral cell swelling and death by brain herniation.^{116,117}

Hyponatremia of exercise

A frequently discussed occurrence of hyponatremia in otherwise healthy people is during prolonged exercise (sport) or work (military). When exercise or work is especially prolonged (>4 h), the overconsumption of hypoosmotic fluids or the combined effects of ordinary hypoosmotic fluid ingestion with copious sweat electrolyte losses can result in exercise-associated hyponatremia (EAH).^{117,118} Although genetic predisposition, inappropriate AVP response, and theories regarding exchangeable sodium stores have been put forth,^{119,120} the overconsumption of fluids during prolonged exercise or work,¹¹⁷ the reduced glomerular filtration rate normally seen with exercise,⁷⁷ and the nonosmotic exercise stimulus for AVP¹²¹ combine to provide a sound explanation for the overall phenomenon.

To help prevent EAH, two types of drinking recommendations for exercise have been proposed. The first^{122,123} is to know the sweating rate and drink accordingly so that weight is never gained during exercise. One limitation to this approach is an inability to know or anticipate sweating rates in diverse conditions. The second¹²⁴ is to drink according to thirst. Although people who suffer from hyponatremia sometimes report overdrinking because of extreme thirst,¹²⁵ the disconnect between good thirst sensitivity at rest and poor thirst sensitivity during exercise⁸¹ probably makes a drink-to-thirst approach an acceptable low-risk strategy for preventing EAH. It remains unclear, however, if voluntary drinking behaviors during exercise are truly an osmotic thirst phenomenon, a dry mouth phenomenon, or a matter of behavior based on preconceived plans or even fluid availability. The fact that EAH occurs primarily in competitive races, rather than training, suggests the strong—but rarely considered—possibility that fluid availability plays a significant role.

It is a fundamental requirement of road-race event planners at local, national, and international levels to provide runners with access to fluid (water and/or sports drinks),¹²⁶ but it is ultimately the responsibility of individual athletes to choose a race drinking strategy. In fact, both individual drinking strategies^{127–129} and water station numbers (fluid availability) among road races¹²⁶ can vary greatly. Standard and rational improvements for estimating the number of water stations and the provision of race water volumes¹²⁶ are potential complements to exercise drinking recommendations^{123,124,130} and overall athlete education against developing EAH.

RESEARCH GAPS

- Humans are among many mammals to have evolved survival mechanisms against the dehydrating environments of sea and land,¹¹ yet modern humans in industrialized countries are rarely challenged by dehydration. Acute losses of body water from illness or exercise do occur and can be problematic, but they are well studied. The larger challenge to modern humans are chronic diseases that have developed at lightning speed on an evolutionary time scale. The potential interactions between chronic disease and hydration¹¹⁰ are interesting but far from settled; at present there is no evidence in developed nations between fluid intake and all-cause mortality.¹⁰⁵ Genomic, transcriptomic, proteomic, and metabolomics research may also more clearly reveal the presence or absence of a role for hydration in chronic disease risk.
- In addition to cross-sectional, cohort, and randomized controlled trial studies, experimental work focusing on dehydration that includes cell function studies is critical. Indeed, cell signaling is at the heart of organism-based homeostatic body water regulation—whether osmotic or volume-mediated. Burg and coworkers⁵ and others^{1,6} have already identified numerous molecular pathways that comprise the cellular response to dehydration. The challenge is to validate a convenient mammalian cell model(s) for study and to identify and select the best cell biomarkers for measuring relevant hydration stress.

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FIBER

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SUMMARY

Apart from starch and sugars, plant foods contain a wide variety of structural cell wall polysaccharides that resist hydrolysis by the endogenous digestive enzymes of the human alimentary tract. This complex group of carbohydrates is collectively referred to as dietary fiber. Though not, strictly speaking a nutrient, an Adequate Intake of dietary fiber is essential for maintenance of normal function in the alimentary tract and to reduce the risk of a variety of noncommunicable diseases. However, the definition of “adequate” in this context is still a matter of research and debate. This chapter reviews the definition and analysis of fiber, the behavior of its constituent polysaccharides in different regions of the alimentary tract, and the various ways in which they may interact with human physiology and support the maintenance of health.

Keywords: Colorectal cancer; Fiber; Gut health; Laxation; Metabolic health; Microbiome; Nonstarch polysaccharides; Resistant starch.

I. INTRODUCTION

A. Background

Dietary fiber is a collective term for a range of food-borne polysaccharides, varying greatly in composition from one plant food to another but united by their common property of resistance to hydrolysis by the digestive enzymes of the human gut. Even before the development of precise analytical methods for food-borne carbohydrates, chemists were aware that the proximate analysis of plant materials often left an insoluble residue, which resisted treatment with acids and alkali. This substance, defined as *crude fiber*, was of largely unknown composition, but its nutritional significance was indicated by the work of Atwater on food energy values, showing that foods with high levels of crude fiber increased fecal energy losses in both experimental animals and humans. The German chemist Rubner published a more detailed chemical analysis of crude fiber from cereal products in 1917,¹ and later, McCance and Lawrence established an important distinction between the *available carbohydrates* of plant foods, consisting of starch and sugars, and the *unavailable carbohydrates*, comprising cell wall polysaccharides that were not hydrolyzed by digestive enzymes.²

From a purely medical perspective, TL Cleave and others developed a view that many chronic noninfectious diseases of Western society were caused by consumption of highly refined carbohydrates from which nondigestible cell wall material had been removed.³ By 1971, the surgeon and medical scientist Denis Burkitt had used his own epidemiological observations to develop the hypothesis that consumption of diets rich in unrefined carbohydrate foods protected rural African populations against diverticular disease, appendicitis, and colorectal cancer (CRC).⁴ Similarly, Hugh Trowell, another physician, proposed that consumption of unrefined carbohydrates was protective against coronary heart disease (CHD),⁵ and he adopted the term dietary fiber for “*that portion of food which is derived from cellular walls of plants which are digested very poorly by human beings.*” This combination of ideas, which became known as the *dietary fiber hypothesis*, has since driven decades of research, blending food chemistry, epidemiology, microbiology, human physiology, and medicine.

B. Fiber Function and Structure

The principle constituents of dietary fiber are nonstarch polysaccharides (NSPs) containing varying

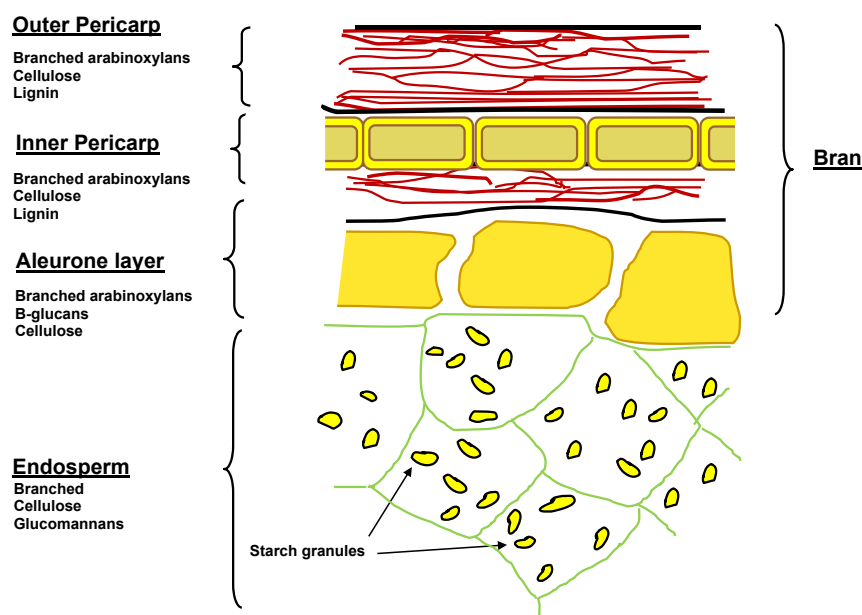


FIGURE 31.1 The general distribution of nonstarch polysaccharides and starch within the outer tissues of a wheat grain.

amounts of beta-linked saccharide units, principally glucose, galactose, arabinose, xylose, and uronic acids.⁶ Cellulose, which consists of linear polymers of 1–4 beta-linked glucose molecules, often with molecular weights of a million Daltons or more, is a major component of dietary fiber in most foods. Individual cellulose polymers aggregate by hydrogen bonding to form fibrils. A typical plant cell wall from the undifferentiated parenchyma cells of fruits and vegetables is composed of cellulose fibrils within a matrix of branching xylose polymers and water-soluble polymers of galactose and arabinose. Beta-glucans, found in fungal cell walls and in the outer layers (bran) of cereal seeds, are similar in structure to cellulose, but with the backbone composed primarily of 1–3 beta-linked glucose units.⁶ The main sources of beta-glucan in conventional human diets are oats and barley.

In vascular and supporting tissues, which require greater rigidity, the mechanical strength of the cell walls is increased by additional cellulose and matrix polysaccharides, toughened by deposition of lignin, a branching polymer of phenyl propane units. Though not itself a polysaccharide, lignin is included in most definitions of dietary fiber. A cross section of a wheat grain, showing the anatomical locations of the principal fiber constituents, is given in Fig. 31.1.

In addition to the intrinsic cell wall polysaccharides of intact plants, a variety of isolated and purified polysaccharides also resist hydrolysis in the small intestine and may be classified as dietary fiber if they meet certain other criteria, as discussed below. Guar gum, locust bean gum, and carob bean gum, for example, are galactomannan polymers, which function as energy-storage polysaccharides in certain leguminous seeds. Their solubility and high viscosity make them suitable as commercial food-thickening agents, but they also display

physiological properties that qualify them for inclusion in modern definitions of dietary fiber. Another group of carbohydrates not originally included in the definition are the beta-linked oligosaccharides with chain lengths of three or more saccharide units. These include natural food constituents such as inulin, as well as purified or synthetic galactose and fructose oligosaccharides. Finally, starch is the principle storage polysaccharide of plants, and the main source of energy from plant foods, composed of varying proportions of the straight chain glucose polymer (amylose) and a branched form (amylopectin). Despite the fact that starch contains only readily hydrolyzed alpha linkages, digestion of starch in the human small intestine often remains incomplete, and significant quantities of so-called *resistant starch* pass into the colon.⁷ By 1976, the growing understanding of both digestive physiology and analytical chemistry prompted Trowell and others to propose a broader definition of dietary fiber, which went beyond cell wall polysaccharides: *The sum of the lignin and the plant polysaccharides that are not digested by the endogenous secretions of the mammalian digestive tract.*⁸

Definitions of fiber are linked to the technical problem of its quantification in foods. Early assays were based on the so-called *neutral detergent fiber* technique,⁹ developed for the analysis of the indigestible fraction in animal feeds,¹⁰ but the harsh analytical conditions of this approach destroyed soluble polysaccharides that remain unhydrolyzed in the human gut. Recognizing this, Southgate introduced a more physiologically relevant method, employing enzymatic hydrolysis of polysaccharides followed by colorimetric analysis of the liberated sugars,¹¹ and this was used to determine the fiber values included in the fourth edition of the official UK food tables published in 1978.¹² A further refinement introduced by

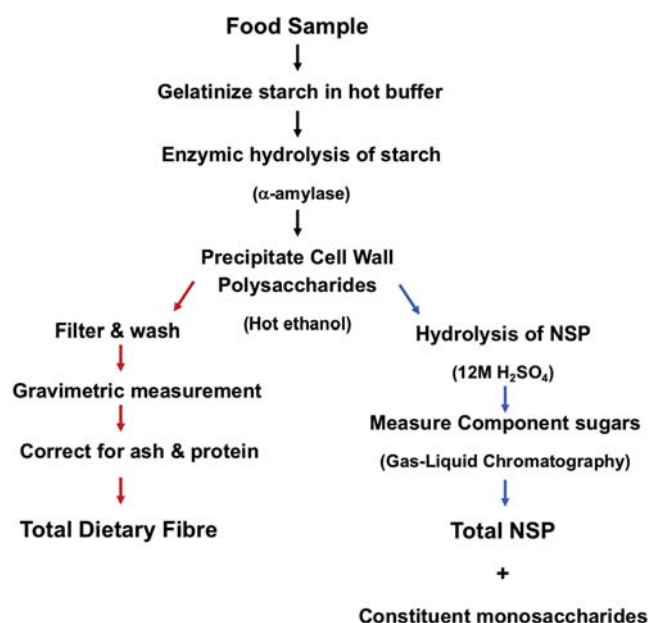


FIGURE 31.2 Simplified flow diagram to illustrate principles of empirical (red arrows) and rational (blue arrows) analyses of dietary fiber.

Englyst and colleagues¹³ enabled the physical separation of resistant starch from cell wall polysaccharides and employed gas liquid chromatography to define them in terms of their constituent monosaccharides.

The full chemical analysis of polysaccharides to the level of their monosaccharide constituents has been described as the *rational* approach to fiber analysis, as distinct from *empirical* methods, which employ sequential extraction and enzyme hydrolysis, followed by gravimetric analysis of the carbohydrate residue.¹⁴ The methods developed by Asp et al.¹⁵ and Prosky et al.^{16,17} in the 1980s employed the latter strategy and formed the basis for the Association of Official Analytical Chemists (AOAC) method¹⁸ for Total Dietary Fiber

(TDF). The essential characteristics of the two approaches are summarized in Fig. 31.2. In practice, empirical methods are faster and less expensive than those requiring chemical analysis and so are better suited to commercial applications. When substantial amounts of retrograded amylose are present, empirical methods give higher values than rational methods, but for many foods, the fiber values obtained by the two approaches are similar.¹⁴ Some values for dietary fiber obtained by the two analytical approaches applied to selected foods are given in Table 31.1. Standardized versions of the AOAC method for TDF are approved for regulatory purposes in the United States, United Kingdom, and elsewhere.¹⁹ Nevertheless, *rational* methods, though more expensive and technically demanding, may be appropriate for research or for regulatory purposes requiring precise identification of added polysaccharides.¹⁴

Both rational and empirical approaches to fiber analysis can be used to obtain estimates of *soluble fiber* as a separate component of the total figure.²⁰ The validity of this distinction is problematic, however, because it is usually not clear that the fraction defined as soluble fiber in vitro is actually solubilized in the gut lumen during digestion of foods, or indeed what the biological implications of solubility in the gut may be. A report on carbohydrates published by FAO/WHO in 1998 recommended that the distinction between insoluble and soluble fiber should be phased out,²¹ but the terminology is still widely used, both in food tables and in research papers.

In 2009, the Commission on Nutrition and Foods for Special Dietary Uses of the Codex Alimentarius Commission, which is the food regulation arm of the World Health Organization, adopted a new definition of fiber for the guidance of national governments when formulating their own legislation.¹⁹ The definition states that *dietary fiber means carbohydrate polymers with 10 or more*

TABLE 31.1 Values for total dietary fiber from some representative foods, measured using two empirical methods and a method providing chemical analysis of resistant starch and NSP.

Food source	AOAC 991.43	AOC 2009.01	Rational method (Englyst)		
	Total fiber	Total fiber	Total RS	NSP	Total fiber
Bread	5.9	7.2	2.1	2.9	6.1
Biscuit	3.2	3.0	0.2	2.1	4.0
Whole wheat cereal	11.1	12.1	1.0	7.5	10.8
Puffed rice cereal	2.3	3.2	0.5	1.1	2.2
Canned potato	10.3	13.6	5.4	6.1	11.1
Canned chick peas	22.7	22.5	6.9	11.4	19.8

NSP, nonstarch polysaccharides; RS, resistant starch.
Selected data obtained from Englyst et al.¹³

monomeric units, which are not hydrolyzed by endogenous enzymes in the small intestine of human beings and belong to the following categories:

- Edible carbohydrate polymers naturally occurring in food as consumed.
- Carbohydrate polymers, which have been obtained from raw material in food by physical, enzymatic, or chemical means and which have been shown to have physiological effect of benefit to health by generally accepted scientific evidence to competent authorities.
- Synthetic carbohydrate polymers, which have been shown to have physiological effect of benefit to health by generally accepted scientific evidence to competent authorities.

Both resistant starch and purified polysaccharides are included in the definition, subject to their having beneficial physiological effects, but oligosaccharides with degrees of polymerization between 3 and 9 are not, unless national authorities decide otherwise. Most of the techniques recommended by the Codex Alimentarius Commission for analysis of fiber are AOAC methods for TDF, but the NSP method is also recognized for research purposes.¹⁹ In Europe, a scientific panel of the European Food Safety Authority (EFSA) published an opinion on dietary fiber in 2010 recommending that dietary fiber be defined simply as *nondigestible carbohydrates plus lignin*,²² and in 2015, a similar definition, recommended by the Scientific Advisory Committee on Nutrition (SACN), replaced NSP as the definition of fiber used in the United Kingdom.²³

The most recent American definition of dietary fiber was introduced in 2018 by the US Food and Drug

Administration (FDA) and states that dietary fiber is *nondigestible soluble and insoluble carbohydrates (with three or more monomeric units), and lignin that are intrinsic and intact in plants, isolated or synthetic nondigestible carbohydrates (with three or more monomeric units) determined by FDA to have physiological effects that are beneficial to human health*.²⁴ Some examples of beneficial effects recognized by the FDA as criteria for the inclusion of a polysaccharide in the definition of fiber include lowering of blood glucose and/or cholesterol, lowering blood pressure, increasing stool frequency, and reducing energy intake. Thus, although contemporary definitions of dietary fiber may appear to differ little from that proposed by Trowell and colleagues in the 1970s, they now reflect much broader knowledge of both the chemistry of resistant carbohydrates in food and their physiological properties during digestion.⁶ A summary of the major components of dietary fiber and their sources in some foods is given in Table 31.2.

II. FIBER AND METABOLISM

A. Behavior in the Alimentary Tract

Fiber exerts important metabolic effects both within the alimentary tract and beyond, but since its components are neither digested nor absorbed by endogenous enzymes and transport pathways, its biological effects are very different from those of conventional nutrients. The many different effects of dietary fiber depend on the physical properties of its various polysaccharide

TABLE 31.2 Major components, associated substances, and sources of dietary fiber in human diets.

Food source	Nonstarch polysaccharides	Associated substances
Cereal grains	Cellulose, branched arabinoxylans, xyloglucans, beta-glucans, glucomannans	Lignin, phenolic esters
Leguminous seeds	Cellulose, xyloglucans, mannans, galactomannans, pectic substances	Oligosaccharides
Vegetables and fruits	Cellulose, xyloglucans, pectic polysaccharides (e.g., homogalacturonan, rhamnogalacturonans)	Lignin, suberin, glycoproteins
Manufactured foods	Galactomannan gums (guar, locust bean), beta-glucans, pectins, alginates, carrageenans, modified cellulose gums (carboxymethyl cellulose, methyl cellulose)	Resistant starch (e.g., retrograded amylose)

For a more detailed account, see Lovegrove et al.⁶

components, on their interactions with other food components and digestive secretions, and on their susceptibility to bacterial fermentation in the large intestine.

Oropharyngeal and gastric effects

The presence of NSPs in the alimentary tract may modify even the earliest stages of digestion and metabolism. For example, the intact cell walls of lightly processed carbohydrate foods tend to inhibit the physical disruption of food during mastication and thus slow the access of salivary amylase to starch.²⁵ After swallowing, the rate at which the stomach empties into the small intestine is partly regulated by the viscosity of gastric contents and by particle size, so that delayed disruption and dispersion of robust food fragments can slow digestion significantly. In humans, the presence of coarse wheat bran has been shown to slow gastric emptying more than finely dispersed bran,²⁶ and the rate at which glucose is absorbed from large fragments of starchy food is slower when they are swallowed whole than when they are chewed normally.²⁷ Thus, the metabolic effects of fiber present in food as intact cell walls are not necessarily predictable from a simple analytical value for the fiber content.²⁵

Small intestinal effects

The effect of dietary fiber in the small intestine is largely to slow the rate of nutrient absorption and hence to modify the metabolic response to a meal. After the physical disruption of food particles, and the partial hydrolysis of macromolecules in the stomach, the semiliquid products of the gastric phase of digestion are released intermittently into the duodenum. Starch, proteins, and triglycerides are further hydrolyzed to oligomers by pancreatic enzymes in the upper jejunal lumen. The final stages of digestion occur at the mucosal surface in the presence of the brush border hydrolases of the epithelial cells through which the newly released monosaccharides, fatty acids, peptides, and amino acids, together with water and electrolytes, are absorbed into the circulation. The peristaltic waves of the small intestinal musculature and the rhythmic contractions of the villi ensure that the partially digested chyme is well stirred. Dietary fiber exerts complex effects on transit through stomach and small intestine, which are not fully understood. For example, viscous solutions of guar gum and beta-glucan can increase the resistance to peristaltic flow, thereby slowing transit time to the large bowel, while coarse insoluble wheat bran, but not finely ground bran, has been reported to accelerate small intestinal transit.²⁶

Glycemic index is defined as the incremental area under the blood glucose response curve after consumption of a standardized food sample, expressed as a percentage of the response to an equivalent amount of

carbohydrate consumed as glucose.²⁸ Foods rich in fiber tend to have glycemic indices of less than 100%. This phenomenon is particularly well documented in beans and other leguminous seeds,²⁹ where it probably reflects the persistence of intact cell walls impeding the access of pancreatic amylase to starch in the small intestine. Much experimental research on the metabolic effects of fiber has been conducted by studying the glycemic response to glucose test meals to which various polysaccharides have been added as a supplement. In general, wheat bran and other insoluble types of fiber have little effect on glycemia when added to test meals of glucose or starch, whereas soluble, high-viscosity polysaccharides such as guar gum and oat beta-glucan suppress the rate of glucose absorption.³⁰ This may be partly due to delayed gastric emptying, but the primary mechanism of action appears to be due to the effect of increased viscosity on the effective thickness of the unstirred water layer (UWL) adjacent to the mucosal epithelial cells, from which nutrients are rapidly absorbed.³¹ To sustain a high rate of influx, the concentration of glucose in this boundary layer must be maintained, either by convective mixing with the bulk fluid phase or by the much slower process of diffusion. If the stirring effect of gut motility is inhibited by a high viscosity, the thickness of the UWL increases and the substrate concentration within the boundary layer declines. This places a limit on the rate of absorption.³²

Consumption of viscous components of dietary fiber, including guar and beta-glucan, also reduces plasma cholesterol in humans. Viscous polysaccharides do inhibit the uptake of cholesterol from mixed micelles *in vitro*,³³ but the most probable mechanism for the hypocholesterolaemic effect is inhibition of bile acid reabsorption by viscous fiber in the distal small intestine. Increased fecal bile losses probably favor a compensatory increase in hepatic bile salt synthesis that draws upon the plasma cholesterol pool.³⁴ In experimental human trials, various types of dietary fiber have been shown also to modify the lipid composition of chylomicrons entering the circulation from the small intestine after fatty test meals,³⁵ but the metabolic significance of this effect is not clear.

Colorectal effects

The human stomach and small intestine are not entirely sterile, but they are adapted to minimize bacterial colonization. In contrast, the colon and rectum have evolved to favor bacterial survival and proliferation and they contain one of the richest and most diverse bacterial ecosystems known to science. The proximal colon contains around 200 g of bacterial cells, dietary residues, and mucus in an anaerobic, semiliquid environment containing between 10^{11} and 10^{12} bacterial cells per ml. These facts have been known for many

decades, but until recently, our understanding has been limited by the difficulty of using classical microbiological methods to isolate and culture the many bacterial lineages present. Recently, however, the field has been transformed by the advent of high-throughput genomic sequencing of fecal bacterial DNA and 16S ribosomal RNA, which makes it possible to explore the full genetic diversity of the colorectal microbiota, together with the effects upon it of differences in geography, lifestyle, and diet.³⁶ Some researchers now describe the gut microbiome and its host as a coevolved “superorganism.”³⁷ Whether or not this proves to be a useful concept, the extraordinarily rich repertoire of bacterial enzymes undoubtedly carries major implications for human health.

The carbohydrate active enzymes (CAZymes), which comprise glycoside hydrolases, carbohydrate esterases, and polysaccharide lyases, have evolved to degrade the almost limitless variety of carbohydrates synthesized by plants and microorganisms.³⁸ Mammals have only a limited capacity to degrade these molecules, but the evolution of symbiotic relationships with bacteria facilitates the recovery of dietary energy from otherwise unavailable carbohydrates. Humans, for example, express only 17 endogenous CAZymes, whereas 1 single commensal bacterium from the colonic microbiota, *Bacteroides thetaiotaomicron*, expresses 260 distinct glycoside hydrolases.³⁹ Collectively, the hundreds of colonic bacterial species can degrade most of the oligo- and polysaccharides that comprise the dietary fiber complex, yielding carbon dioxide, hydrogen, and the short-chain fatty acids (SCFA) acetate, propionate, and butyrate. Total SCFA concentrations are about 10 times higher in the proximal colon (131 mMol/kg) than in the distal ileum (13 mMol/kg), but concentrations decline significantly in the descending colon (80 mMol/kg) due to uptake across the mucosa.⁴⁰ The bulk of propionate and acetate appear in the circulation, but a large fraction of butyrate is utilized by the epithelial cells as their principle source of energy.⁴¹ Propionate is metabolized primarily in the liver, and acetate is used as an energy source by various peripheral tissues.

The commonly used energy conversion factor for carbohydrate, based on the empirical work of Atwater,⁴² is 16.7 kJ/g (4 kcal/g), which assumes that all the carbohydrate is absorbed and metabolized as glucose. The energy content of SCFA released from carbohydrate by fermentation is lower, and the extents to which different NSP and components of resistant starch are fermented during passage through the colon differ significantly. If the energy content of a food or diet is calculated by including dietary fiber with the total carbohydrate content, the energy

value will be overestimated, whereas if fiber is assumed to contribute zero energy, it will be underestimated.⁴³ In practice, a figure of 8.4 kJ/g (2.0 kcal/g) is considered a realistic approximation for the energy value of fiber derived from mixed diets,⁴⁴ and this figure has been endorsed by the FAO.⁴⁵ However, the precise contribution of fiber fermentation to the energy intake of individuals differs due to variations in the composition of the colonic microflora and in the transit time of substrates through the colon.⁴⁶ A further complication is that high levels of fiber in the diet have been shown to decrease the metabolizable energy available from fat and protein.⁴⁷ On average, Western Europeans consume about 20 g of dietary fiber each day, but the range is very broad,⁴⁸ and estimates of resistant starch intake range up to 30–40 g/day.⁴⁹ Using an isotope tracer method in human volunteers, Wang et al. observed that as much as 25% of the starch in some unrefined cereal foods escaped hydrolysis in the small intestine.⁵⁰ This suggests that in rural populations consuming mainly unrefined plant-based diets, the average quantity of fermentable carbohydrate reaching the colon may commonly exceed 50 g per day,⁵¹ which would equate to about 100 kcal, or around 5% of total energy needs.

The dominant bacterial phyla of the human gut microbiome are the Firmicutes and the Bacteroidetes, which together account for about 90% of the microbiota, but the Actinobacteria, Proteobacteria, Fusobacteria, and Verrucomicrobia are also represented to varying degrees in different populations. The advent of affordable, high-throughput DNA sequencing applied to fecal samples has led recently to a rapid expansion in our understanding of gut bacterial communities down to the genera and species level. Most of the bacteria of the human colon exploit carbohydrates derived both from dietary residues and endogenous secretions as their main energy source, though some do not degrade polysaccharides directly, but instead utilize the initial degradation products of other species in the community. Many soluble polysaccharides, including pectin, beta-glucan, and inulin, are readily fermented by gut bacteria, whereas complex lignified plant tissues, such as wheat bran and the thick cell walls of legumes and other seeds, are more resistant to bacterial enzymes and tend to remain at least partially intact in the feces. The presence of readily fermentable substrates favors enrichment with bacteria that can most readily use them as fuel. It is this principle that underlies the use of isolated oligosaccharides and polysaccharides (collectively termed *prebiotics*) to provide a selective stimulus for the growth of Bifidobacteria and other bacterial lineages thought to be beneficial to human health.⁵²

In contrast to the volume of research on prebiotics, there has been relatively little work done on the effects of fiber from complex foods on the composition of the gut microbiota in humans. Some significant changes in fecal bacterial lineages in response to dietary intervention with whole grain foods have been reported.^{53,54} David et al.⁵⁵ fed two experimental diets, based primarily on meat or on plant foods, respectively, to human volunteers for 5 days. The dietary fiber intake of the subjects was not determined, but the switch away from plant-based foods to a meat-based diet was associated with a reduction in Firmicutes species that metabolize plant polysaccharides and an increase in bile-tolerant bacteria, together with changes in metabolic activity and gene expression. Diets containing a range of different sources of dietary fiber, composed of carbohydrate polymers with a variety of different monosaccharides and linkages, may increase the metabolic diversity of the colorectal bacterial community,⁵⁶ but further long-term human intervention trials will be necessary to confirm this hypothesis and to fully explore any potential health benefits.

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Functional Effects in the Upper Bowel

The rate of glucose absorption during a meal is the major determinant of insulin secretion by the pancreas. Inhibition of glucose uptake by fiber can therefore reduce insulinaemia and also modify other aspects of the endocrine response to food. The alimentary tract contains an abundance of mucosal endocrine cells that release a range of regulatory peptides including ghrelin, cholecystokinin (CCK), glucagon-like peptide 1 (GLP-1), gastric inhibitory peptide (GIP), and peptide YY (PYY) into the circulation in response to nutrients in the gut lumen. These peptide hormones help to regulate gut motility, metabolism, and probably also appetite and food intake. The release of insulin, GLP-1, and GIP is slower after ingestion of rye bread than after an equivalent test meal of white bread,⁵⁷ but the strongest effects of fiber on postprandial physiology occur in response to viscous types of fiber. For example, the presence of beta-glucan in a liquid test meal suppresses the absorption of glucose and slows the release of insulin, CCK, GLP-1, and PYY, but the effect is abolished if the viscosity of the beta-glucan is destroyed by enzymic hydrolysis, leaving the total fiber content of the beverage unchanged.⁵⁸

The distal small intestine is also richly endowed with endocrine cells (L cells) that secrete regulatory peptides, including enteroglucagon, GLP-1, GLP-2, and PYY, in

response to intraluminal signals. In contrast to its initial inhibitory effects on gut peptide release, oat beta-glucan incorporated into solid test meals has been shown raise PYY levels in a dose-related manner, 4 h after ingestion.⁵⁹ This effect is probably mediated by displacement of glucose and lipids into the more distal intestine, where they can interact with mucosal endocrine cells. Both GLP-1 and PYY function as satiety signals, which act on the central nervous system to suppress food intake. PYY in particular is implicated in the control of food intake in both experimental animals and in humans.⁶⁰

B. Fermentation and Functional Effects in the Distal Bowel

The colonic mucosa is also richly endowed with endocrine cells, but, unlike those of the small intestine, they respond primarily to the presence of fermentation products. Colorectal endocrine cells express the G-protein-coupled free fatty acid receptors GPR 41 and GPR 43 (now renamed FFAR2 and FFAR3), for which the luminal SCFAs are ligands,⁶¹ and they mediate the release of GLP-1 and PYY into the circulation. Thus, in addition to the short-term modulation of peptide release in the upper alimentary tract, dietary fiber may initiate long-term increases in regulatory peptide levels via gut fermentation. Dietary intervention with oligofructose raises GLP-1 and PYY in the peripheral circulation in human volunteers,⁶² and these effects are associated with reduced energy intake and weight loss in some trials.⁶³ When compared to a control meal of white bread, a high-fiber barley kernel-based bread leads to prolonged increases in plasma PYY and GLP-1 levels.⁶⁴

The SCFAs also exert local effects on the physiology of the colorectal epithelium. A third mucosal GPR, GPR109A/HCA2, responds to intraluminal butyrate by releasing immunoregulatory cytokines IL-18 and IL-10 and inducing regulatory T cells, potentially suppressing inflammation and regulating the balance of tolerance to commensals and immunity to pathogenic bacteria in the colonic mucosa.⁶⁵

C. Gene Regulation

The SCFA butyrate regulates mitosis, promotes differentiation, and induces programmed cell death (apoptosis) in intestinal tumor cell lines *in vitro*⁶⁶ and probably upregulates tight junction and reduces epithelial permeability *in vivo*. These effects reflect the role of butyrate as a potent histone deacetylase inhibitor.⁶⁷ Histone deacetylases are a group of enzymes involved in the epigenetic regulation of gene transcription via regulation of the protein structure of chromosomes.

TABLE 31.3 Mean and range of increased fecal output measured in response to different types of dietary fiber fed to human volunteers.

Fiber type or food source	Increase in fecal weight (g/g fiber eaten)
Wheat bran	7.0 (3.0–14.4)
Fruit and vegetables	6.0 (1.4–19.6)
Oats	3.4 (1–5.5)
Legumes	1.5 (0.3–3.1)
Pectin	1.3 (0–3.6)

Data obtained from Institute of Medicine⁷⁴ and Elia and Cummings.⁷⁹

Collectively, these mechanisms could enable butyrate derived from the fermentation of fiber to act as an intraluminal anticarcinogen, suppressing inflammation and regulating epithelial homeostasis,⁶⁸ but the extent to which such effects of butyrate and other SCFA contribute to the protective effect of fiber against CRC observed in epidemiological studies remains to be established.⁶⁹ The impact of butyrate on gene expression may reach beyond the gut to the cardiovascular endothelium. Recent work in animal models suggests that increased butyrate reduces atherogenesis through a mechanism based on suppression of inflammation.⁷⁰

D. Fecal Bulking

All components of dietary fiber increase fecal bulk to some extent, but the size of this effect varies greatly for different polysaccharides. For example, the fecal bulking capacity of wheat bran is about five times that of pectin (Table 31.3). The lignified cell wall polysaccharides of wheat bran are resistant to disruption by the gut microflora, and so the increment in stool mass results largely from water retention by intact bran particles within the fecal matrix. The weaker bulking effects of readily fermentable polysaccharides like pectin are due mainly to bacterial cell mass.⁷¹ One consequence of these differences is that the laxative effect of a fiber-rich food cannot be predicted from a single empirical analysis of its TDF content.

IV. DIETARY REQUIREMENTS

A. Reference Intakes

In the absence of any precise means of assessing status or defining objective physiological criteria for adequate fiber intake, public bodies tend to base their recommendations for fiber consumption on systematic

reviews of epidemiological studies. National and international bodies generally recommend that adults should consume 25–30 g of fiber per day, preferably obtained from a variety of different food sources.⁷² Such guidance stands in stark contrast to current consumption levels in high-income countries such as the United Kingdom, where consumption is currently around 15 g/day. According to the most recent data from the NHANES 2013–2016 *What We Eat in America* survey,⁷³ median fiber intakes for adult men were 18 g/day, while for adult women, the median intake was 14.9 g/day. Only 4% of men consumed greater than their Adequate Intake (AI), which ranges from 38 g/day for men 18–50 years of age and 30 g/day for men over 50 years. Among women, 12% consumed at or above the recommended AI of 25 g/day for those aged 18–50 years of age and 21 g/day for those of more than 50 years of age.

Advice to consume higher levels of fiber is generally justified by reference to disease prevention, though the rationale often differs between countries.⁷² For example, the current guideline AI for dietary fiber consumption by Americans and Canadians is 14 g/1000 kcal/day for all age groups. For men aged 18–50 years, consuming 2500 kcal of energy/day, this equates to 38 g/day. The recommendation is based largely on epidemiological evidence for prevention of cardiovascular disease (CVD).⁷⁴ In the United Kingdom, the SACN report of 2015 justified the recommended intake of 30 g fiber/day for adults primarily by citing new evidence for a dose-related protective effect against CRC.²³

B. Tolerable Upper Intake Levels

Dietary fiber is not known to have serious adverse effects when consumed as an intrinsic component of food, but a sudden increase in fiber consumption can cause unpleasant symptoms in some individuals. This may be due to excess production of flatus gas causing distension and pain,⁷⁵ though this may often be caused by rapid fermentation of oligosaccharides such as stachyose and verbascose present in legumes, rather than to fermentation of cell wall polysaccharides.⁷⁶ Paradoxically, in view of its proven laxative properties, higher fiber intakes can apparently be associated with worsening symptoms in patients with clinically proven constipation.⁷⁷ Esophageal obstruction by dietary fiber is occasionally reported in the clinical literature, but this potentially very serious problem appears most often to result from inadequate hydration of isolated gum preparations used as laxatives or slimming aids.⁷⁸ Thus, Tolerable Upper Intake Levels for fiber have not been established by authoritative bodies.⁷⁴

V. ISSUES RELATED TO FIBER IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

Weight gain and obesity

Foods containing high levels of dietary fiber have a lower energy density and are often bulkier than processed carbohydrate foods. These physical characteristics alone may favor satiety at lower levels of energy intake, but the emerging evidence that dietary fiber also modulates the release of gut peptides provides another plausible mechanism for an effect of fiber on appetite. However, in view of the complexity of self-selected human diets, and the difficulty of correcting for all the variables involved, this general hypothesis is difficult to test.⁷⁹ Population studies have shown that the average daily intake of dietary fiber and whole grain cereals is inversely related to risk of obesity in some populations,^{80,81} but this does not necessarily indicate that fiber consumption has any direct influence on body weight gain. Koh-Banerjee et al.⁸² studied dietary intake and self-reported weight gain over an 8-year period in 27,082 American men aged 40–75 years. Multivariate analyses indicated a statistically significant inverse association between whole grain intake and weight gain, such that for every 40-g/day increment in whole grain intake, weight gain was 0.49 kg lower. A further effect of added cereal bran was identified, such that for every 20 g/day increase in intake, weight gain was reduced by 0.36 kg. The authors concluded that the effects of whole grain were due to both dietary fiber and other constituents of cereal grains. Liu et al.⁸³ studied whole grain and dietary fiber intake in a large cohort of American women enrolled in the Nurse's Health Study and came to similar conclusions. In contrast, a study of a large Danish cohort failed to identify any relationship between weight gain and fiber intake after adjustment for covariates.⁸⁴ Howarth et al.⁸⁵ reviewed intervention studies designed to test the hypothesis that increased intakes of dietary fiber lead directly to weight loss and concluded that in most intervention studies in which energy intakes were maintained constant, an increased intake of either soluble or insoluble fiber was associated with higher postprandial satiety, which would favor weight loss. In studies where energy intake was uncontrolled, an average consumption of an additional 4 g/day fiber was associated with a reduction in energy intake and a reduction in body weight of 1.9 kg over 3.8 months. In a major systematic review of associations between carbohydrate quality and human health, Reynolds et al.⁸⁶ conducted a meta-analysis of 27 intervention trials in which the effect of

a higher intake of dietary fiber on weight gain had been investigated and estimated that weight gain was about 37% lower in subjects consuming higher levels of fiber.

Plasma cholesterol

Soluble and viscous types of dietary fiber, particularly oat beta-glucan, guar gum, and pectin, exert important effects on lipid metabolism in humans. Brown et al.⁸⁷ undertook a meta-analysis of randomized controlled intervention trials and concluded that the soluble polysaccharides most commonly used in intervention studies all modified plasma cholesterol levels to a similar extent, amounting in the case of oat beta-glucans to a reduction of about 0.13 mmol/L total cholesterol for every 3 g soluble fiber consumed per day. This is potentially significant for those with total plasma cholesterol levels close to the generally recognized desirable level of 5.2 mmol/L, and health claims relating to cholesterol metabolism are recognized for oat bran and beta-glucan in both Europe and the United States.

Type 2 diabetes and the metabolic syndrome

Interest in the role of dietary fiber in the etiology of type 2 diabetes began with the early work of Trowell and others,⁸⁸ but a definitive link has proven hard to establish. In their review of carbohydrates and human health published in 2015, the UK SACN²³ committee conducted a meta-analysis of 11 cohort studies and observed a statistically significant protective effect of higher fiber consumption, estimated as a 6% reduction in risk for every additional 7 g of fiber consumed per day. More recently, Reynolds et al.⁸⁶ conducted a systematic review of observational studies and detected a reduction in risk of type 2 diabetes among higher consumers of fiber compared to lower consumers. Statistical modeling indicated a dose-related effect such that a relative risk of 0.85 was obtained for every additional 8 g of fiber consumed.

Long-term supplementation with guar gum has been reported to improve control of blood glucose in patients with type 2 diabetes.⁸⁹ A recent meta-analysis of intervention studies shows statistically significant evidence for improved glycemic control in type 2 diabetic patients taking viscous fiber supplements (median dose 13.1 g/day).⁹⁰ It is often assumed that such improvements are mediated by physical effects on postprandial glycemia, but the guar gum and beta-glucan commonly used in such studies are also readily fermentable substrates. Recent work suggests that reductions in blood glucose levels induced by fermentable fiber in diabetic patients may be mediated by selective enrichment of the gut microbiome with specific SCFA-producing bacteria.⁹¹ Clearly, much more research will be needed

before the impact of fiber on metabolic disease is fully understood.

Blood pressure

In observational studies, higher fiber intakes are associated with lower systolic and diastolic blood pressure.⁹² In a meta-analysis of intervention trials, Streppel et al.⁹³ concluded that an average dose of 11.5 g dietary fiber per day reduced systolic blood pressure by 1.13 mm Hg (95% confidence interval −2.49 to 0.23) and diastolic blood pressure by −1.26 mm Hg (95% confidence interval −2.04 to −0.48). Evans et al.⁹⁴ carried out a meta-analysis of 18 trials. The estimated reduction in blood pressure in response to all fiber types combined (0.9 mm Hg) was not statistically significant, but analysis for effects of specific fiber types indicated that diets rich in beta-glucans reduce systolic blood pressure by 2.9 mm Hg (95% confidence interval 0.9–4.9 mm Hg) and diastolic blood pressure by 1.5 mm Hg (95% confidence interval 0.2–2.7 mm Hg) for a median difference in beta-glucans of 4 g. Similar results were obtained in a more recent meta-analysis of studies on soluble viscous fiber and blood pressure by Khan et al.⁹⁵ It has been estimated that in the US population a reduction of 2.0 mm Hg in the average systolic blood pressure would lead to reductions in annual mortality from CHD and stroke of 4% and 6%, respectively.⁹⁶

Cardiovascular disease

Since dietary fiber exerts beneficial effects on multiple risk factors for CVD and may act directly to suppress atherosclerosis via butyrate production,⁷⁰ higher levels of fiber consumption might be expected to reduce CVD at the population level.⁹⁷ Pereira and colleagues⁹⁸ pooled the data from 10 prospective studies in a systematic analysis that included 91,058 men and 245,186 women, among whom a total of 5249 incident cases of CHD and 2011 deaths occurred during the years of follow-up. In each of the studies analyzed, diets were measured using food frequency questionnaires, and intakes of total fiber and fiber from cereals, fruits, and vegetables were estimated. After correction for confounding factors, each 10 g increment in TDF intake was found to be associated with a 14% reduction in risk of coronary events and a 27% reduction in risk of death from CHD. More recently, Threapleton et al. carried out a meta-analysis of 22 prospective studies of fiber intake and risk of CHD and concluded that TDF intake was inversely associated, with a risk ratio of 0.91 per 7 g of fiber per day (95% confidence ratio 0.87–0.94).

Patients with type 2 diabetes are at increased risk of CHD. In a prospective study of 7822 diabetic American women within the Nurse's Health Study, He et al.⁹⁹ explored the relationship between whole grain consumption and death from CHD. When participants

with the highest versus the lowest fifths of intakes of whole grain, cereal fiber, bran, and germ were compared, the highest consumers experienced up to 31% lower mortality from all causes. After adjustment for confounders, a statistically significant protective effect of bran intake against both all-cause and CHD mortality was confirmed.

Although observational studies on fiber consumption and CHD indicate a protective association, they cannot establish causality. Hartley et al.¹⁰⁰ conducted a systematic review of intervention studies using dietary fiber for primary prevention of CVD but found only short-term studies on known risk factors, rather than studies on the prevention of the disease itself. Their pooled analyses confirmed evidence for reductions in total cholesterol, LDL cholesterol, and diastolic blood pressure with increased fiber intake, but they concluded that long-term, randomized controlled trials are needed to explore the potential value of dietary fiber interventions for prevention of CVD events.

Constipation and irritable bowel syndrome

Chronic constipation is a diagnosis applied to patients who regularly experience fewer than three bowel movements per week, with hard stools, a sense of incomplete evacuation, or excessive straining.¹⁰¹ The global prevalence within the community is thought to be about 14%.¹⁰² Epidemiological evidence suggests that a higher intake of dietary fiber is associated with a reduced incidence of constipation in the general population,¹⁰³ and dietary fiber has long been recommended as a frontline treatment for constipation. Patients are usually advised to increase their consumption of fiber from foods, but they may also be prescribed pharmaceutical preparations, usually based on ispaghula husk, which is derived from the seeds of *Plantago psyllium*.^{104,105} Systematic reviews confirm that intervention with fiber is moderately effective as a treatment for chronic idiopathic constipation¹⁰⁶ though, paradoxically, a worsening of symptoms has also been reported.⁷⁷

Chronic constipation is also a common feature of irritable bowel syndrome (IBS), a functional bowel disorder with a variety of symptoms, including abdominal pain and changes in bowel habit with no established cause. The condition is common in Europe and North America, where the prevalence is about 11% of the adult population.¹⁰⁷ Many general practitioners and family physicians give dietary advice for the management of the condition, and increased consumption of dietary fiber from high-fiber foods or from supplements is frequently recommended, though the clinical basis for this is not well established. Various types of dietary fiber have been used in the management of IBS, but the most common have been wheat bran and ispaghula. In a systematic review, Bijkerk et al.¹⁰⁸ analyzed 17 randomized

controlled trials and concluded that there was evidence for a statistically significant beneficial effect of dietary fiber on many symptoms of IBS, particularly in patients with constipation, but no evidence for relief from pain. The beneficial effects were confined to soluble fiber. Wheat bran had no significant beneficial effects, and in some studies, it was found to worsen symptoms slightly, an effect which has been noted before.¹⁰⁹ In a more recent analysis of placebo-controlled intervention trials, Ford et al.¹¹⁰ concluded that ispaghula was significantly more effective than placebo, whereas wheat bran was not. More recent reviews support the beneficial effects of fiber, particularly where constipation is a major feature of IBS, but also call for further randomized trials.¹¹¹

Diverticular disease

Diverticular disease is a common condition in which small protrusions called diverticulae bulge through the muscular wall of the colon at sites of penetration by blood vessels. It is much more common in western populations than in the rural African populations first studied by Burkitt and colleagues and is strongly associated with increasing age. It was one of the first disorders to be attributed to a chronic deficiency of fiber in the diet,¹¹² and it has long been thought to be caused by the increased intracolonic pressure associated with constipation. There is epidemiological evidence to support the hypothesis that a high intake of dietary fiber protects against diverticular disease in the male population of the United States,¹¹³ but there appears to be little evidence that increased consumption of dietary fiber is of any value in the management of established disease.¹¹⁴

Colorectal cancer

The mechanism proposed by Burkitt for protective effects of dietary fiber against CRC was based on the laxative properties of undigested polysaccharides.⁴ Having observed the low incidence of CRC among rural Africans consuming traditional diets, coupled with their habitually frequent passage of stool, Burkitt hypothesized that soft bulky stools are a physiological norm for human beings and that Western dietary patterns led to chronic constipation, causing prolonged exposure of the colon to fecal carcinogens. Cummings and colleagues later used observational data to show a significant inverse relationship between fecal weight and risk of CRC across population groups,¹¹⁵ but their findings do not prove a causal mechanism.

The epidemiological evidence for a protective effect of fiber against CRC has not always been consistent. For example, in 1999, a prospective study of 88,757 middle-aged North American women detected no protective effects of fiber.¹¹⁶ Similarly Pietinen et al.¹¹⁷ and

Terry et al.¹¹⁸ failed to detect any protective effect of fiber intake in prospective trials with Finnish men and Swedish women, respectively. In a systematic review of cohort studies, Park et al. pooled the results of 13 investigations to give a total of 725,628 participants and 8081 cases.¹¹⁹ Among those in the highest quintile of fiber consumption, the risk of CRC was 16% lower than among those in the lowest quintile, but this difference was reduced to only 6% and became statistically nonsignificant, when the data were adjusted for a range of other risk factors. Nevertheless, those consuming less than 10 g of fiber per day were at significantly greater risk of CRC than the remaining population.

Much clearer evidence for a protective effect of fiber consumption against CRC emerged from the European Prospective Investigation of Cancer and Nutrition (EPIC) project, a long-term prospective study based on a cohort of 519,978 individuals drawn from several geographically and culturally distinct European countries. In their original analysis of the EPIC data, Bingham et al.¹²⁰ estimated that the relative risk of CRC for the highest quintile of fiber intake (mean 35 g/day) versus the lowest (mean 15 g/day) was 0.58 (95% confidence interval 0.41–0.85). A subsequent analysis of the data, based on an increased number of cases, reinforced this conclusion and indicated a reduction in risk of approximately 9% for each quintile of higher fiber consumption ($P < .001$).¹²¹

Despite some inconsistencies in the epidemiological findings, the protective effect of higher fiber consumption against CRC is now generally accepted by both international and national advisory bodies. The WCRF report of 2007,¹²² based on a formal metaanalysis of eight prospective studies including EPIC, identified a dose-dependent protective effect, giving a reduction in risk of about 10% for each 10 g daily increment in fiber consumption. More recently, a meta-analysis performed by the UK SACN committee also identified a protective dose–response relationship between consumption of dietary fiber and CRC and used it as one justification for their recommendation that the UK population should consume higher levels of fiber.²³

CRC develops from adenomatous polyps, which are relatively common, potentially precancerous lesions of the colorectal mucosa. Removal of polyps at endoscopy is a routine preventative measure. If fiber is protective against CRC, increased fiber intake might also reduce the risk of polyp recurrence after routine polypectomy; this hypothesis has been tested in intervention studies. A recent review identified five trials, with a total of 4798 participants, in which the data were suitable for meta-analysis.¹²³ Interventions included wheat bran, ispaghula husk, and whole-diet interventions using high-fiber foods alone or in combination. No statistically

significant protective effects were observed. However, the authors questioned the overall quality of the research to date and recommended longer trials using higher levels of fiber.¹²³

Despite decades of research, the mechanisms by which fiber protects against CRC in human populations remain uncertain. The development of sporadic CRC begins with a sequence of somatic mutations in the crypt cells of the colorectal mucosa, leading to long-term abnormalities of gene expression, and the progressive disruption of normal cellular proliferation and differentiation. These changes occur over decades and are thought to have multiple causes, including both local genetic damage from fecal mutagens and adverse metabolic drivers related to overweight, insulin sensitivity, and low physical activity.¹²⁴ Higher fiber consumption may therefore be associated with a lower risk of CRC, through a variety of mechanisms including higher levels of intraluminal butyrate, coupled perhaps with a reduced prevalence of obesity and improved insulin sensitivity.¹²⁵

B. Known Interactions

Polysaccharides and the phenolic components of cell walls may, in theory, interact with and bind mineral micronutrients during digestion, thereby reducing their availability for absorption.¹²⁶ Interactions of this type certainly occur in vitro, but there is little objective evidence that dietary fiber per se has much of an adverse effect on mineral metabolism in humans. However, phytate (myo-inositol hexaphosphate), which is an organic compound present legume seeds, oats, and other cereals, does exert potent binding effects on minerals and has

been shown significantly to reduce the availability of magnesium, zinc, and calcium for absorption in humans.¹²⁷

C. As a Dietary Supplement

Official public health advice generally recommends that dietary fiber should be obtained from a variety of food sources where it exists as an intrinsic constituent.⁷² However, as discussed earlier, a large number of isolated or purified polysaccharides have now been shown in controlled intervention trials to exert beneficial physiological effects, and according to the CODEX definition of fiber, such materials may, at the discretion of national advisory or regulatory bodies, be recognized as legitimate dietary supplements. In the United States, the FDA currently recognizes several isolated or synthetic nondigestible carbohydrates containing three or more sugar units as having beneficial effects.²⁴ Initially, seven isolated or synthetic carbohydrates were included in the definition of dietary fiber (beta-glucan soluble fiber, psyllium husk, cellulose, guar gum, pectin, locust bean gum, hydroxypropylmethylcellulose). In 2018, a further eight polysaccharides were proposed for inclusion (mixed plant cell wall fibers, arabinoxylan, alginate, inulin and inulin-type fructans, high amylose starch, galactooligosaccharide, polydextrose, and resistant maltodextrin), subject to the completion of further regulatory rulemaking.²⁴ In Europe, EFSA recognizes health claims related to oat and barley beta-glucans as beneficial in relation to the maintenance of normal plasma cholesterol levels and control of blood glucose after meals.

RESEARCH GAPS

- Current reference intakes for dietary fiber are derived empirically from epidemiological observations, rather than from a deep understanding of the physiological and metabolic effects of dietary fiber. Can detailed knowledge of the human metabolic response to fiber lead to a more rational approach to dietary recommendations?
- How does the composition of the gut microbiome influence the human metabolic response to different components of dietary fiber, and how does fiber sustain and modify the complexity of the microbiome?
- We require a deeper understanding of the critical metabolic pathways through which the products of carbohydrate fermentation act on human physiology in health and disease.
- In human populations increasingly dependent on processed foods, how can optimal fiber intakes be achieved? Can fiber supplements provide the health benefits first identified in epidemiological studies on populations consuming traditional diets?

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CAROTENOIDS

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SUMMARY

Carotenoids affect a rich variety of physiological functions in nature and are beneficial for human health, serving as antioxidants and blue light filters in the human tissues. These compounds also are the dietary precursors of retinoids (vitamin A and its metabolites). In recent years, much progress has been made in identifying genes that affect carotenoid metabolism and functions. Mutations in their corresponding genes impair carotenoid metabolism and induce various pathologies in animal models. Polymorphisms in these genes alter carotenoid and retinoid homeostasis in humans as well. This chapter will provide an overview of the advanced state of knowledge about carotenoids. It will discuss intrinsic and extrinsic factors that influence the bioavailability and bioconversion of these lipids. The physiological functions of select carotenoids in cells and organs will be exemplified. Finally, the nutritional requirements for these pigments will be discussed and future directions of carotenoid research will be suggested.

Keywords: Apocarotenoids; Carotenoids; Function; Metabolism; Retinoids.

I. INTRODUCTION

A. Background

Vitamin A was discovered as a minor but vital lipid in ordinary food stuff about 100 years ago. Soon after, it was recognized that certain yellow plant pigments, the carotenoids, have the same activity.¹ A connection between diet and vision also had been acknowledged. Yudkin showed that vitamin A existed in the retina,² and Wald elegantly described the vitamin's role in the visual cycle.³ Determination of the chemical structures of β -carotenes and vitamin A by Karrer and colleagues verified the kinship between the parent compound and its metabolite.⁴ In the past decades, seminal research in the biochemical and molecular basis of the pleiotropic actions of vitamin A led to the discovery of visual G protein-coupled receptors and nuclear hormone receptors and culminated in the determination of their atomic structures.^{5,6}

Apart the role as provitamins, carotenoids have been recognized to exert additional functions in human biology as antioxidants and light filters. Large intervention trials have brought attention to the potential health benefits of

many carotenoids. However, supraphysiological doses of β -carotene have proven to be harmful in smokers and asbestos workers.⁷ These findings suggest the existence of a "therapeutic window" for carotenoids that is influenced by individual genetics and lifestyle. Despite compelling evidences for critical roles in human physiology, carotenoids are still not regarded as essential nutrients, and official recommendations for their reference intake are lacking.

B. Nutrient Function and Structure

More than 1000 different carotenoids exist in nature.⁸ The best-known and eponymous representative of these isoprenoid compounds is β -carotene (Fig. 32.1). The symmetric β -carotene possesses a central polyene chain and displays two β -ionone rings as end groups, a structural element also present in retinoids. Asymmetric carotenoids such as α -carotene display β -ionone and ϵ -ionone rings as end groups (Fig. 32.1). The enormous chemical diversity of carotenoids results from chemical modulations of the basic structure, including the shifting of conjugated double bonds and the addition of functional groups to the terminal ionone rings

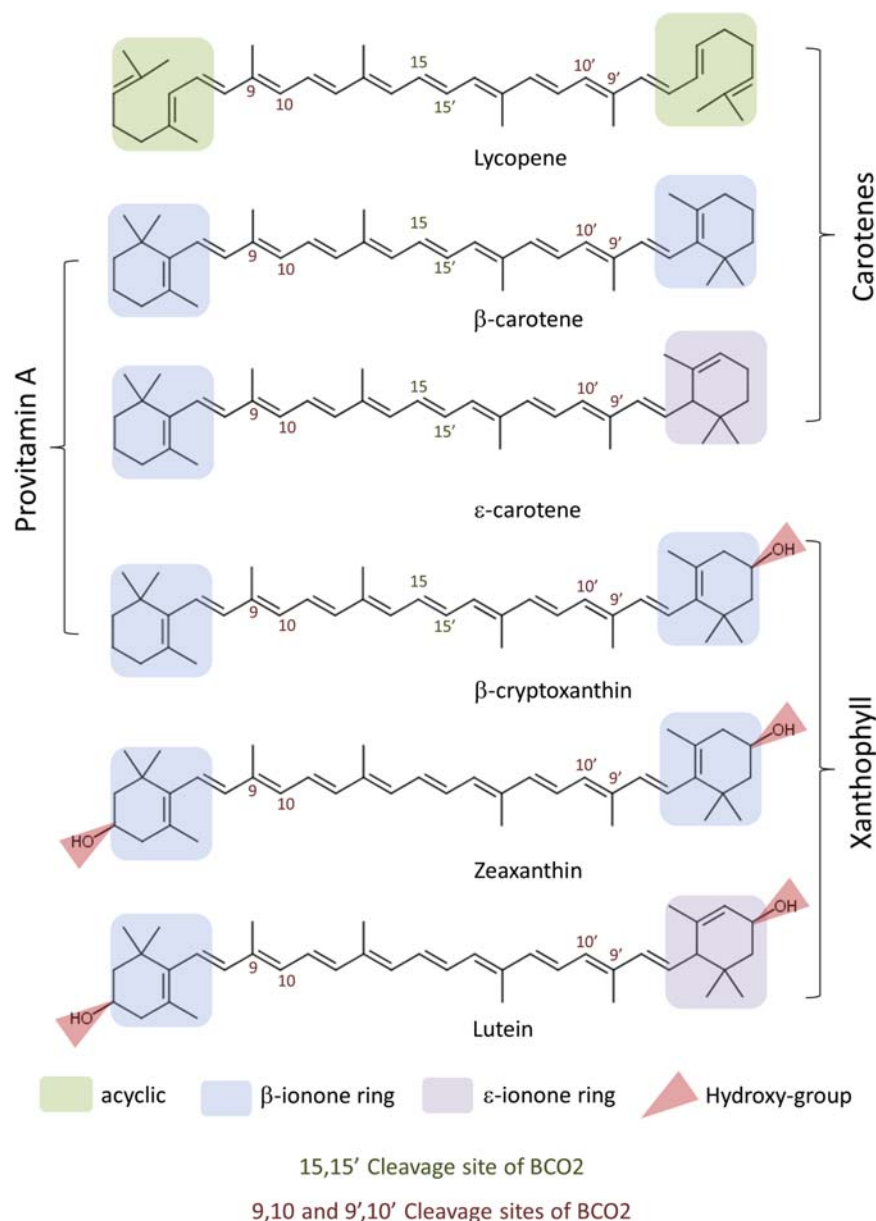


FIGURE 32.1 Chemical structures and names of the six major carotenoids in human blood. Chemical characteristics of individual carotenoids (ionone ring structures and hydroxyl groups) are highlighted.

(Fig. 32.1). Additionally, each carotenoid can exist in different geometric forms. The all-(E)-form (all-*trans*) of a carotenoid is easily isomerized to (Z)-geometric diastereomers (*cis*). The short-chain apocarotenoids also are members of the carotenoid substance class. They derive from oxidative cleavage of the parent carotenoid molecule. Trivial names for carotenoids and apocarotenoids such as β -carotene, lutein, zeaxanthin, and retinal have been in use long before their exact chemical structures were determined.

Among carotenoids, 50 are typically ingested with the diet and of these 20 are commonly found in human tissues.⁹ The six main carotenoids circulating in the human blood are β -carotene, α -carotene, β -cryptoxanthin, lutein,

zeaxanthin, and lycopene (Table 32.1). These compounds are classified in two ways: first, by their provitamin A activity and, secondly, by their oxygenation status. Pure hydrocarbons such as β -carotene and the acyclic lycopene are called carotenes, whereas their oxygenated derivatives, zeaxanthin and lutein, are called xanthophylls.

Carotenoids exist in most human tissues in varying amounts and composition (Table 32.2). Physiological functions attributed to carotenoids are related to their capability of acting as antioxidants, e.g., as free radical scavengers, in lipophilic environments such as membranes and lipoproteins.⁹ Additionally, carotenoids act as filters of short-wavelength light.¹⁰ The yellow pigmentation of the fovea in the primate retina is the

TABLE 32.1 Color, xanthophyll/carotene classification, retinol conversion factors, US mean serum and intake levels, and good food sources of selected carotenoids.

Carotenoid	Color	Structure classification	Retinol equivalent	Retinol activity equivalent	US Mean serum levels ^a (μg/dL)		US Mean dietary intake ^{b,c} (mg/d)		Carotenoid-rich foods
					Male	Female	Male	Female	
Lycopene	Red	Carotene	—	—	26.4	23.9	11.3	6.7	Tomatoes, watermelon, guava, pink grapefruit, apricots
β-Carotene	Yellow orange	Carotene	6	12	14.6	18.4	2.2	1.9	Carrots, sweet potatoes, pumpkin, spinach
α-Carotene	Light yellow	Carotene	12	24	3.8	4.4	0.4	0.4	Carrots, pumpkin
β-Cryptoxanthin	Orange	Xanthophyll	12	24	8.6	8.1	0.1	0.1	Sweet red peppers, tangerines, papaya, persimmons
Lutein	Yellow	Xanthophyll	—	—	20.1 ^c	18.2 ^c	2.1	1.9	Kale, spinach, corn, collard greens, broccoli, eggs
Zeaxanthin	Yellow	Xanthophyll	—	—					Kale, spinach, corn, collard greens, broccoli, eggs
Canthaxanthin	Red orange	Xanthophyll	—	—	—	—	—	—	Seafood
Astaxanthin	Red	Xanthophyll	—	—	—	—	—	—	Seafood

^aUS mean serum levels and food intakes were calculated using the values for the 19–30 and 30–51 age groups (DRI 2000, DRI 2001).

^bMean intake are from Holden, JM, Eldridge, AL, Beecher, GR, et al. Carotenoid content of food: an update of the database. J Food Comp Anal. 1999;12:169–196.

^cSerum and intake levels are the sum of lutein and zeaxanthin.

TABLE 32.2 Carotenoid concentrations in selected human tissues.

Tissue	Carotenoid concentrations (ng/g of tissue)					
	Lycopene	β -Carotene	α -Carotene	β -Cryptoxanthin	Lutein	Zeaxanthin
Colon	534	256	128	35	452	32
Liver	352	470	67	363	1701	591
Lung	300	226	47	121	212	90
Prostate	374	163	50	—	128	35
Retinal pigment epithelium ^a	8.64	10.8	2.97	—	18.27	4.85
Skin	69	26	8	—	26	6

^aConcentrations in approximately 0.2 g of pooled retinal pigment epithelium/choroid.

Data from Furr, HC, Clark, RM. Transport, uptake, and target tissue storage of carotenoids. In: Krinsky NI, Mayne ST, Sies, H, eds. Carotenoids in Health and Disease. New York: Marcel Dekker; 2004:229–278 and Khachik, F, Carvalho, L, Bernstein, PS, et al. Chemistry, distribution, and metabolism of tomato carotenoids and their impact on human health. Exp Biol Med (Maywood). 2002; 227:845–851.

origin of the anatomical term macula lutea, or “yellow spot.” The macula pigments have been chemically identified as lutein, zeaxanthin, and meso-zeaxanthin.¹⁰ Their light filtering properties are thought to protect against blue light damage and to reduce the adverse impact of glare disability, light scatter, and chromatic aberration, thereby optimizing contrast sensitivity of the retina.¹¹

Importantly, certain carotenoids are the natural precursors of retinoids (β -15'-apo-carotenoids), which encompass all natural and synthetic derivatives of vitamin A (all-*trans*-retinol). Vitamin A is the precursor for at least two critical metabolites, 11-*cis*-retinal, the chromophore of visual pigments in photoreceptors of the eyes,¹² and all-*trans*-retinoic acid, a hormone-like compound.¹³ All-*trans*-retinoic acid binds to ligand-activated transcription factors that control processes as diverse as cell differentiation, embryonic development, immunity, and metabolism.^{14–16}

II. NUTRIENT METABOLISM

A. Synthesis

Carotenoid synthesis takes place in plants, certain fungi, and bacteria. The initial steps follow a common scheme that is similar to sterol and isoprenoid synthesis in most species.¹⁷ The first step, exclusively devoted to carotenoid synthesis, is the condensation of two molecules of geranylgeranyl pyrophosphate (C20) to phytoene (C40) catalyzed by phytoene synthase. The conversion of the colorless phytoene to the red-colored lycopene occurs through four desaturation steps. In non-photosynthetic bacteria as well as in fungi, these steps are executed by a single enzyme. In plants, this pathway employs several proteins that catalyze a sequence of desaturation and isomerization steps. The multistep pathway of higher plants is reflected in the occurrence of mutants that accumulate intermediates, e.g., tetra-*cis*-lycopene, such as

tangerine tomatoes. Downstream in the biosynthetic pathway, all-*trans*-lycopene is the substrate for the introduction of ionone ring structures at both ends of the linear carbon chain of the carotenoid.¹⁷ These cyclization reactions yield β -carotene (β,β) and α -carotene (β,ϵ). The enormous diversity of carotenoids is achieved by modulation in length of the polyene chromophore and shifts of the conjugated double bonds that result in changes of the absorption spectra. The addition of functional groups is another notable strategy. Insertion of hydroxyl groups into the ionone rings creates lutein, the bright yellow pigment of daffodils and marigolds. Addition of ketogroups results in asthaxanthin, the vermilion color of flamingoes, salmon, and crustaceans.

B. Digestion

Human carotenoid metabolism can be portrayed in the following framework. Carotenoids must be extracted from the food matrix and dissolved in mixed micelles to become bioaccessible for absorption by enterocytes of the intestine (Fig. 32.2). Upon absorption, carotenoids are distributed in the circulation in lipoproteins and delivered to target tissues. In the case of provitamin A activity, carotenoids must be converted to vitamin A. Finally, elimination mechanisms for carotenoids must exist to maintain a normal physiological state. Several intrinsic and extrinsic factors influence the bioavailability and bioconversion of carotenoids. Clive West and colleagues¹⁸ coined the acronym “SLAMEN-GHI” (for species, linkage, amount, matrix, effectors, nutrient status, genetic factors, host-related factors, and interactions), which summarizes the factors that modulate these processes (Fig. 32.3).

The hydrophobicity of carotenoids is the key factor that affects bioaccessibility in naturally produced mixed micelles as well as their postprandial blood response.¹⁹ Polar site groups of ionone rings play a minor role though

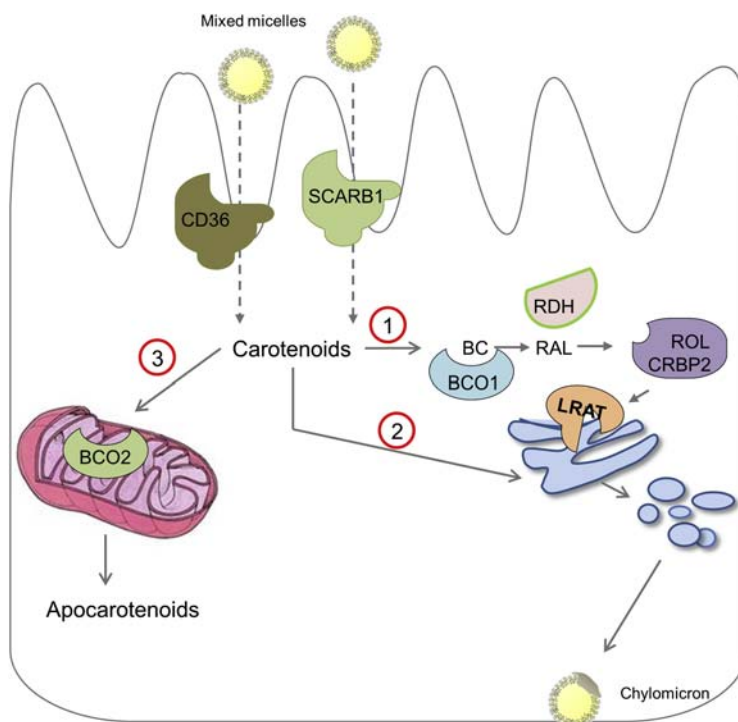


FIGURE 32.2 Schematic of carotenoid absorption in an enterocyte. Carotenoids absorption is facilitated by scavenger receptor class B type 1 (SCARB1) and cluster of differentiation 36 (CD36). Absorbed carotenoids can undergo three metabolic fates. 1, Carotenoids such as β -carotene (BC) can be converted to retinoids. The conversion of BC is catalyzed by β -carotene-15,15'-dioxygenase (BCO1). The primary product retinaldehyde (RAL) is reduced to retinol (ROL) by retinal dehydrogenases (RDHs) in a process that involves cellular retinol-binding protein (CRBP2). ROL is converted to retinyl esters (REs) by lecithin:retinol acyltransferase (LRAT) and packaged in chylomicrons for body distribution. 2, Carotenoids also can be packaged into chylomicrons for body distribution. 3, Carotenoids can be converted by mitochondrial BCO2 (β -carotene-9',10'-dioxygenase) into apocarotenoids different than retinoids.

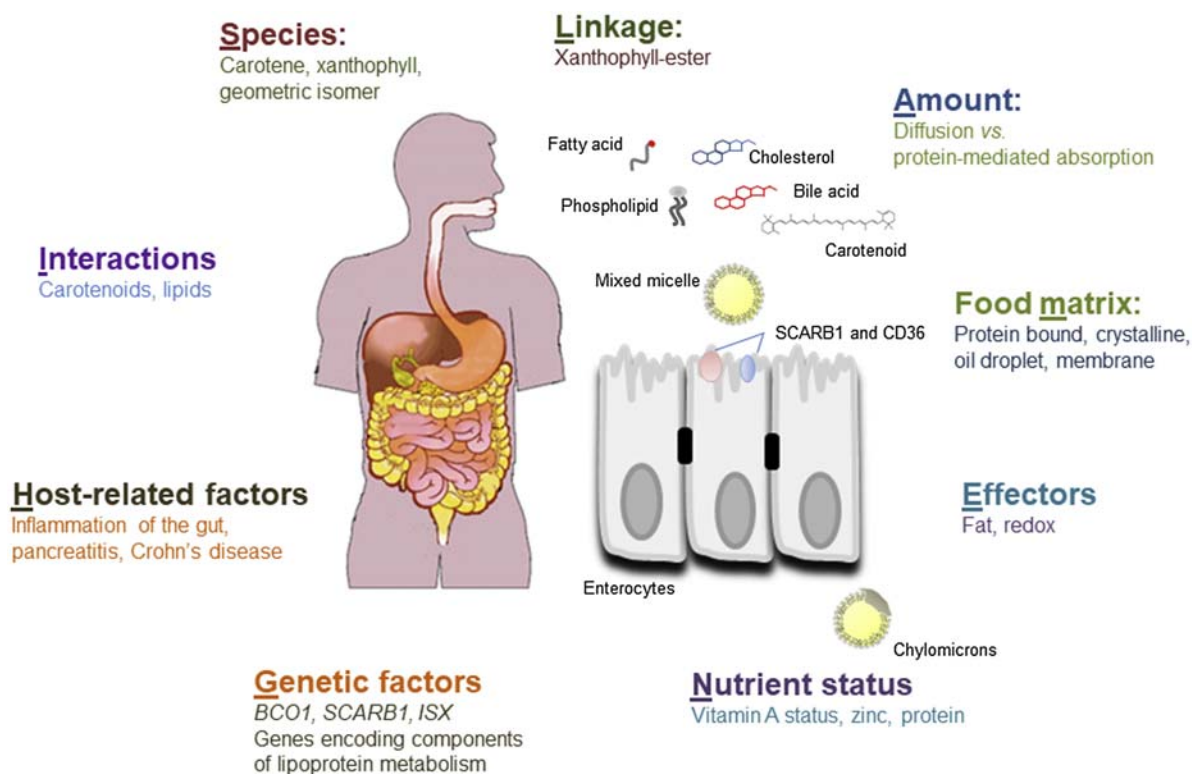


FIGURE 32.3 Extrinsic and intrinsic factors affect carotenoid bioavailability and bioconversion (SLANENGHI).

it is often stated in the literature that xanthophylls are more readily bioavailable than carotenes.²⁰ Additionally, the geometric form of carotenoids affects their bioavailability and bioconversion. The prevalent all-(E)-form of carotenoids is easily isomerized to (Z)-geometric isomers following exposure to heat and/or light. The relative absorption of (Z)-geometric isomers apparently differs between carotenoids. For example, absorption is reduced for β -carotene (Z)-geometric isomers, whereas uptake is increased for lycopene (Z)-geometric isomers compared to their respective all-(E)-form.^{21,22}

The hydroxyl groups of xanthophylls are esterified with fatty acids in many fruits and vegetables.²³ Before these carotenoids are absorbed, the fatty acid moiety is cleaved to form free xanthophylls in the gastrointestinal tract. The pancreatic carboxyl ester lipase is the major enzyme involved in this hydrolysis.²⁴ Xanthophyll ester hydrolysis appears to be an efficient process because the absorption of lutein and β -cryptoxanthin esters is similar to that of the free carotenoid forms,²⁵ while the uptake of zeaxanthin esters may be even greater than that of the parent carotenoid.²⁶

The context, or matrix, in which carotenoids is provided is an important factor affecting their bioavailability.^{18,27} Differences are likely explained by the occurrence of carotenoids in different cellular compartments (chromoplast vs. chloroplast) and aggregate forms in plant tissues. Crystalline or protein-bound carotenoids are less well solubilized in mixed micelles than carotenoids that exist in oil droplets and membranes. Mild heating and food processing increase the bioavailability of carotenoids by disrupting plant cell walls, binding proteins, and organelles, thus liberating carotenoids for uptake. Carotenoids provided in oil (e.g., red palm oil), or commercially available water-soluble beadlets, have superior bioavailability compared with fruits or vegetables.^{18,27}

There are many effectors of carotenoid absorption. Fat intake, like a modest amount of salad dressing, increases carotenoid uptake by facilitating the formation of mixed micelles.²⁸ Fiber, olestra, plant sterol, and stanol esters are dietary compounds that decrease the absorption of carotenoids. Antioxidants, such as vitamins C and E, may increase the stability of the carotenoids in the gastrointestinal tract and thereby facilitate their absorption.²⁹

Upon being made accessible in mixed micelles, carotenoids are absorbed by enterocytes. Initially, it was proposed that this absorption occurs by passive diffusion. However, genetic analysis in *Drosophila* provided evidence that carotenoid absorption is mediated by scavenger receptors similar to the mammalian scavenger receptor class B type I (SCRAB1) and cluster differentiation 36 (CD36).^{30,31} Work in human CaCo-2 cell lines confirmed the critical role of these scavenger receptors

in the cellular uptake of carotenoids.^{32,33} Accordingly, mouse lines deficient for SCRAB1 displayed highly reduced intestinal absorption of carotenes, xanthophylls, and fat-soluble vitamins.^{34–36} Subsequently, the scavenger receptor CD36 also has been shown to be involved in the absorption of carotenoids in mammalian enterocytes.³⁷

It is now acknowledged that vitamin A regulates β -carotene absorption and bioconversion efficiency via a negative feedback loop.³⁸ This regulation explains the observation that β -carotene conversion rates are influenced by the vitamin A status.³⁹ Studies in mouse models provided the molecular details and revealed that the ISX protein is the key to this regulation. This transcription factor represses the expression of SCRAB1 gene and the β -carotene-15,15'-dioxygenase (BCO1) genes^{40,41} in response to retinoic acid signaling (Fig. 32.4). This negative feedback loop adapts bioconversion of provitamin A carotenoids to the bodily requirements for vitamin A.⁴⁰ A study in Zambian children with hypervitaminosis A reports the consequences of this regulation.⁴² During mango season, a period of high provitamin A intake, these children experienced hypercarotenodermia. This might indicate that provitamin A carotenoid bioconversion to vitamin A by BCO1 is inhibited by the hypervitaminosis A. The hypercarotenodermia is elicited by the uptake of carotenoids by other receptors such as CD36, which are not regulated by ISX.

In addition, iron and zinc status as well as protein malnutrition have been reported to affect the bioconversion rate of β -carotene to vitamin A though the mechanisms are less well defined.²⁷

The vitamin A-dependent regulation of SCRAB1 expression in the intestine also explains interactions between xanthophyll and β -carotene absorption. For instance, it has been observed that dietary vitamin A reduces xanthophyll absorption in chickens.⁴³ Moreover, genetic variations contribute to interindividual variability of carotenoid absorption.⁴⁴ Genome-wide association studies and candidate gene studies identified single base pair polymorphisms (SNPs) that affect carotenoid blood and tissue levels. Most of the genetic data stem from the analyses of β -carotene, lycopene, and lutein fasting blood levels, but it is assumed that these genetic variations also modulate the bioavailability of the other carotenoids found in human blood and tissue.⁴⁴ Significant SNPs were found in the genes encoding BCO1, SCRAB1, and the transcription factor ISX.^{38,45–47} For instance, SNPs in BCO1 are associated with macula pigment density,⁴⁸ likely because of the interactions between β -carotene and xanthophyll absorption in the intestine.³⁶ Additionally, several SNPs in proteins that affect lipid and lipoprotein metabolism have been associated with carotenoid blood levels.⁴⁴ The latter association indicates significant interactions between carotenoid and lipid metabolism. Such interactions have been recently demonstrated in mouse models.⁴⁹

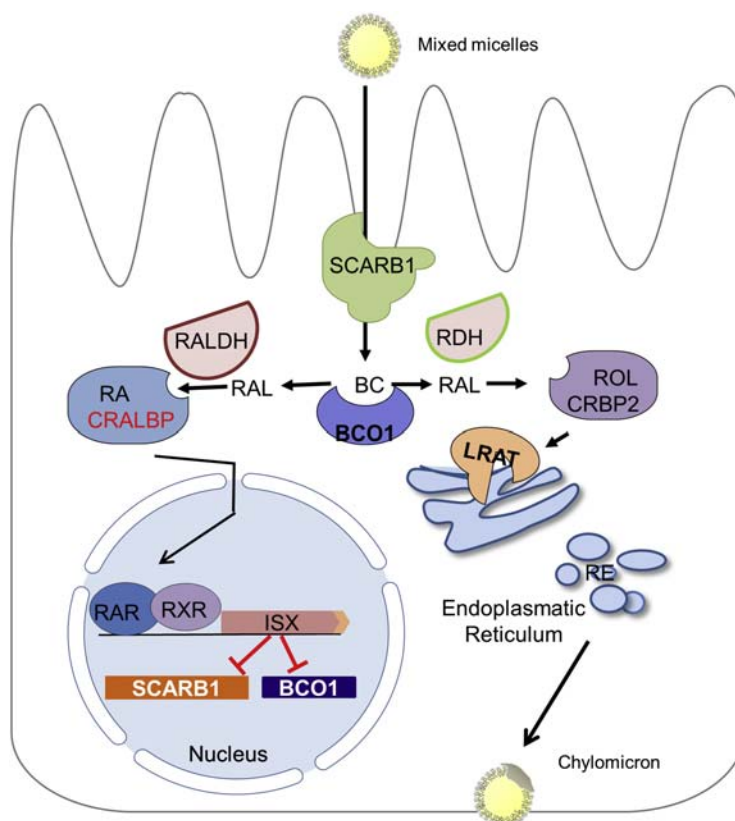


FIGURE 32.4 A diet-responsive regulatory network controls intestinal vitamin A production and carotenoid absorption. β -Carotene (BC) absorption is facilitated by scavenger receptor class B type 1 (SCARB1). The conversion of BC by β -carotene-15,15'-dioxygenase (BCO1) is an important branching point in retinoid metabolism. The primary product retinaldehyde (RAL) is reduced to retinol (ROL) by retinal dehydrogenases (RDHs) in a process that involves cellular retinol binding protein (CRBP2). ROL is converted to retinyl esters (REs) by lecithin:retinol acyltransferase (LRAT) and packed in chylomicrons for body distribution. Some BC is also converted to retinoic acid (RA) by retinal dehydrogenase (RALDH). RA binds to retinoic acid receptors (RARs) that in conjunction with retinoid X receptors (RXRs) regulate the expression of intestine-specific homeobox (ISX) transcription factor. ISX is a transcriptional repressor of the genes encoding SCARB1 and BCO1.

C. Transport

Following absorption, carotenoids with provitamin A activity are converted to retinoids. The extent of this conversion depends on individual genetic and vitamin A status of the host.^{38,50} The primary conversion product retinaldehyde is further converted to retinol and retinyl esters. Retinyl esters (REs) are incorporated into chylomicrons together with noncleaved carotenoids and other dietary lipids and then released into the lymph before entering the general circulation.^{51,52} Carotenoids and REs may be partially taken up by peripheral tissues before the remainder in chylomicron remnants is delivered to the liver.^{51,52} In the liver, vitamin A is stored in the form of REs in stellate cells.⁵² Vitamin A bound to the retinol-binding protein (holo-RBP4) is secreted from the liver into the circulation.⁵³ In the blood stream, holo-RBP4 forms a ternary complex with transthyretin.⁵⁴ The levels of this complex are homeostatic and not affected by fluctuations in dietary vitamin A supply. The uptake of retinol from holo-RBP4 into target cells depends on a membrane receptor that is encoded by the stimulated

by retinoic acid 6 gene.^{55,56} The uptake is driven by intracellular esterification of vitamin A.⁵⁷ In contrast, carotenoids are not transported bound to specific binding proteins but are repackaged into very low density lipoproteins and secreted into the blood. Carotenoids are not equally distributed among lipoproteins in the fasting circulation; the hydrophobic carotenoids are found predominantly in low-density lipoproteins (LDL), while the more hydrophilic xanthophylls tend to be found in higher amounts in high-density lipoproteins (HDL).

From circulating lipoproteins, carotenoids are taken up by LDL receptor and class B scavenger receptor, SCARB1 and CD36, mediated mechanisms.⁵⁸ Carotenoid content and composition are variable between tissues. Table 32.2 summarizes the amounts and composition of the six major carotenoids in select tissues. The high concentration of carotenoids in the fovea region of the retina of the human eyes supports the idea of active uptake and retention mechanism for these lipids. An isoform of glutathione-S-transferase, GSTP1, and a steroidogenic acute regulatory domain protein, StARD3,

were suggested as zeaxanthin- and lutein-binding proteins in the human retina.^{59,60} However, given the very high levels of xanthophylls in the macula, it is unlikely that these pigments predominantly exist in protein-bound form unless GSTP1 and StARD3 would be present in comparable amounts.

Body composition is a major factor that affects the circulating levels of carotenoids. A number of studies have reported an inverse association between body mass index and plasma carotenoid concentrations.^{27,61} Further dissection of this association suggests that carotenoid plasma concentrations are inversely related not only to fat mass but also to lean body mass. This indicates that nonfat tissues, such as muscle, may also serve as a reservoir for carotenoids.²⁷ Accordingly, people with anorexia nervosa have very high levels of plasma carotenoids, most likely due to high levels of mobilization in this catabolic condition.⁶²

In general, circulating carotenoid levels are correlated with serum triglyceride and cholesterol levels. There are reported increases in serum levels as people age, which could be related to the upward shift in the serum lipid profile of older adults. In females, the concentrations of different carotenoids peak at different points of the menstrual cycle.⁶³ The reason or significance is not conclusive, and there appears to be an inverse interaction between androgens and lycopene in the male population.⁶⁴ Smoking is associated with lower carotenoid levels, presumably owing to an increase in oxidative stress, and alcohol intake has been shown to be inversely related to β -carotene levels.⁶⁵ However, both smoking and alcohol intake are also associated with reduced dietary intakes of carotenoids.²⁷

D. Metabolism

Carotenoids can be enzymatically converted to apocarotenoids by oxidative cleavage at specific positions of their polyene chains. This conversion is mediated by carotenoid cleavage dioxygenases (CCDs).⁶⁶ These enzymes play critical roles in the production of biologically active apocarotenoids and the overall homeostasis of carotenoids, as evidenced by the phenotypes of naturally occurring mutants and polymorphisms in the corresponding genes.⁶⁷ CCDs are nonheme iron oxygenases with a characteristic tertiary structure and catalytic centers.⁶⁸ The mammalian genome encodes three different types of CCDs. BCO1 and the β -carotene-9',10'-dioxygenase (encoded by the *BCO2* gene) catalyze the oxidative cleavage across specific double bonds in the polyene backbone of carotenoids⁶⁹ (Fig. 32.5). Biochemical analyses revealed that these enzymes display ionone ring site selectivity.⁷⁰ BCO1 cleaves across the C15,C15' double bond adjacent to a β -ionone ring site. BCO2 cleaves across the C9',C10' positions adjacent of any ionone ring site but prefers carotenoids and apocarotenoids with

hydroxylated ionone ring sites.⁷⁰ The interaction with only one ring site of a bicyclic carotenoid also explains why BCO1 and BCO2 can cleave long chain ($> C20$) apocarotenoids.⁷⁰ Biochemical studies indicated that the substrate ring site selectivity of BCO1 and BCO2 is caused by the architecture of their substrate binding pockets in an overall conserved structure.⁷⁰ This architecture prevents bulky hydroxylated- β -ionone rings to enter the substrate tunnel of BCO1.⁷⁰ This characteristic ensures that retinoids are selectively produced from a relatively large pool of precursor molecules (Fig. 32.5). The promiscuity of BCO2 allows that many carotenoids and apocarotenoids can be converted by a single enzyme. Both recombinant mammalian BCO1 and BCO2 also can convert acyclic lycopene to apo-lycopenals.^{71,72}

A third family member, the retinal pigment epithelium-specific 65 kDa protein (RPE65), does not retain the double bond cleavage activity of BCO1 and BCO2 and catalyzes a combined ester cleavage and double bond isomerization reaction at position C10,C11 of REs into 11-*cis*-retinol.⁷³ Mutations in RPE65 gene causes chromophore deficiency in photoreceptors and a severe form of inherited blindness.^{74,75} Recently, RPE65 has been implicated in lutein into mesozeaxanthin conversion.⁷⁶ Further research is required to understand the molecular basis of the proposed catalytic promiscuity of this enzyme.

All three human CCDs display distinct subcellular localizations. BCO1 and RPE65 exist in the cytoplasm and endoplasmic reticulum, whereas BCO2 is found at the inner mitochondrial membrane.^{77,78} On tissue and organ levels, BCO1 and BCO2 are expressed in many cell types,^{79,80} whereas RPE65 expression is restricted to the RPE that nourishes and maintains the adjacent photoreceptor in the retina.⁸¹ This wide expression pattern of BCO1 and BCO2 indicates that most cell types and tissues can metabolize carotenoids and synthesize vitamin A. In enterocytes of the intestine, BCO1 expression is controlled by vitamin status via the transcription factor ISX.⁴⁰ In other tissues, BCO1 expression is controlled by peroxisome proliferator activator receptors.^{82,83} The control of *BCO2* gene expression is less well understood. Its cellular localization indicates high expression in mitochondria-rich cell types such as cardiomyocytes. Interestingly, BCO2 expression is induced by oxidative stress in human cell lines.⁸⁴ The later regulation may explain that carotenoid levels decrease in chronic disease states that are associated with inflammatory and stress responses.

The generation of mouse models deficient for BCO1 and BCO2 set the stage for an in-depth analysis of the metabolism of dietary carotenoids.^{85,86} These studies identified BCO1 as major β -carotene-metabolizing enzyme and key enzyme for vitamin A production.⁸⁷ In its absence, only trace amounts of β -apo-10'-carotenals were produced from supplemented β -carotene by BCO2.^{87,88} Similarly, humans with genetic BCO1

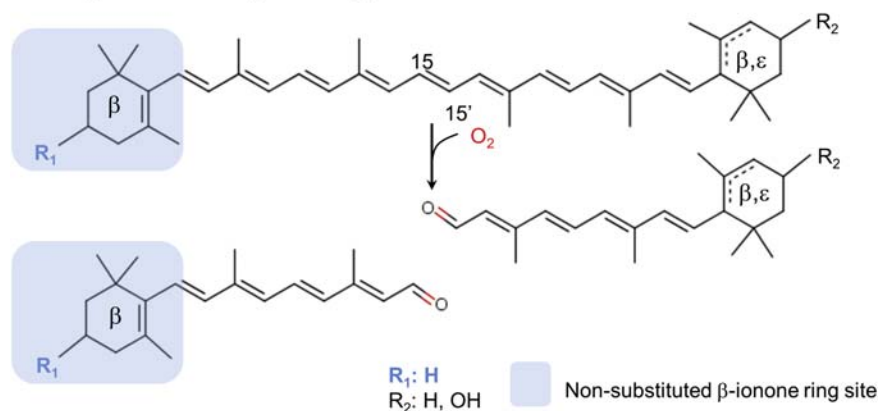
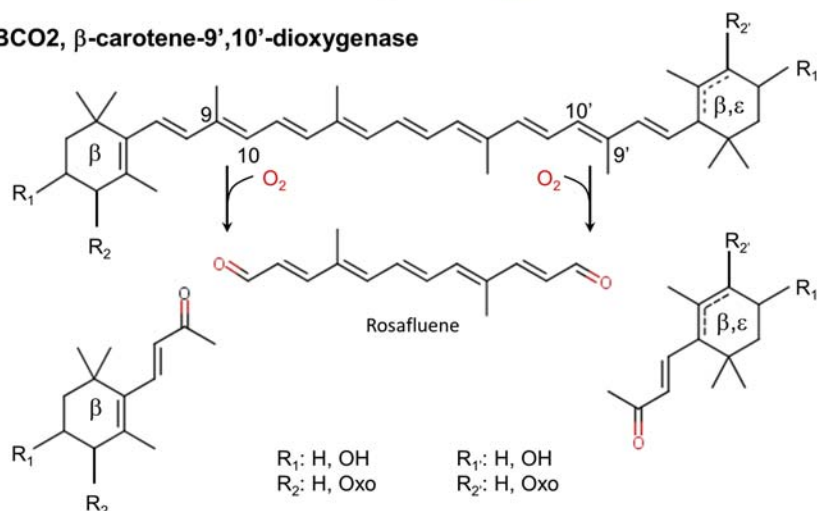
BCO1, β -carotene-15,15'-dioxygenase**BCO2, β -carotene-9',10'-dioxygenase**

FIGURE 32.5 Enzyme reactions catalyzed by BCO1 and BCO2. The two enzymes display region and ring site selectivity for the cleavage of carotenoids. BCO1 selectively interacts with an unsubstituted β -ionone ring. BCO2 displays substrate promiscuity and preferentially interacts with ionone rings with oxygen substitutions.

deficiency show elevated β -carotene and reduced vitamin A levels.⁸⁹ Consistent with BCO2's enzymatic properties and subcellular localization, xanthophylls accumulated in the mitochondria of BCO2-deficient mice.⁸⁵ This accumulation of xanthophylls increased oxidative stress in tissues and reduced mitochondrial respiration rates.^{49,85} Moreover, naturally occurring mutations in the *BCO2* gene have been associated with the yellow skin phenotype in chicken⁹⁰ and the yellow fat phenotype in cattle and sheep.^{91–93} BCO2 mice also accumulate lycopene, whereas their levels were indistinguishable from control mice in *BCO1* knockout mice.⁹⁴ These findings demonstrate that turnover of carotenoids by CCDs plays an important role in the control of their tissue levels. Notably, the primary apocarotenoid cleavage products of xanthophylls and lycopene were not detected in mice. This finding indicates that they were rapidly metabolized and excreted. The pathways for the elimination of carotenoids and apocarotenoids need to be fully elucidated in future studies.

Recent genetic studies identified additional genes encoding carotenoid-metabolizing enzymes in animals. Bird genomes encode CYP enzyme, which converts yellow to red carotenoids.⁹⁵ In mice, the occurrence of 3-oxo-carotenoids has been reported upon supplementation with xanthophyll.⁹⁶ Similar carotenoid derivatives have been described in the human circulation.⁹⁷ It remains to be investigated whether these metabolites are formed by specific enzymes and/or nonspecific redox reactions.

III. ROLE IN NORMAL CELLS AND ORGAN FUNCTIONS

A. Gene Regulation

Provitamin A carotenoids can be converted to all-*trans*-retinoic acid by endogenous metabolic pathways.¹³ The acidic form of vitamin A is the ligand of the retinoic

acid receptors α , β , and γ .⁹⁸ These ligand-activated transcription factors regulate in conjunction with the retinoid X receptors the expression of more than 500 target genes⁹⁹ in processes as diverse as cell differentiation, embryonic development, immunity, and metabolism.^{14–16} Conversion of carotenoids by BCO1 and BCO2 may give rise to apocarotenoids different than retinoids.^{70–72,100} Additionally, many apocarotenoids exist in the diet. Therefore, it is speculated that carotenoids are the precursors of additional signaling molecules. Studies in cell culture and animal models revealed that some apocarotenoids can elicit transcriptional responses.¹⁰¹ For example, pharmacological doses of apo-10'-lycopenoic acid can protect mouse liver from damage.¹⁰² β -13-Apo-carotenone acts an RAR antagonist and antagonizes vitamin A action.¹⁰³ However, tissue levels of these compounds are low, and their biochemical production in mammals has not been conclusively demonstrated.^{104,105} A mouse study investigated the effect putative β -carotene-derived long-chain apocarotenoids. For this purpose, BCO1-deficient and wild-type mice were supplemented with β -carotene. β -10'-Apo-carotenal was identified as β -carotene-derived metabolite in BCO1-deficient mice.⁸⁸ No clear common pattern in the transcriptomic changes was found between different tissues.¹⁰⁶ Gene chip analysis revealed that lungs were mainly affected by genotype, liver by genotype and gender, while the white adipose tissue was mainly affected by gender.¹⁰⁶ A more recent studies in mice suggests that BCO2 produces a bona fide signal molecule that controls carotenoid metabolism in the placenta.¹⁰⁷

Nongenomic actions of carotenoids

Carotenoids have been implicated to be important antioxidants.⁹ As an antioxidant, the compound quenches singlet molecular oxygen and scavenges reactive oxygen species, especially peroxy radicals.¹⁰⁸ Lipid peroxidation is a continuous chain reaction in membranes, affecting polyunsaturated fatty acid. Radical scavenging processes involve the donation of a hydrogen atom or an electron, which leads to the formation of a carotenoid radical or a carotenoid radical cation. Upon electron acceptance, the carotenoid radical anion is formed. All these radical species are thought to be stabilized via delocalization of electrons in the polyene chromophore of the carotenoid. These carotenoid radicals undergo rapid further reaction that can lead to the decomposition of the compounds or the formation of adducts.

Due to their physical properties, carotenoids can act as filters for short wavelength light in the skin and eyes. This action of carotenoids may protect cells against light-induced damage.¹⁰⁹ Other nongenomic action of carotenoids might be found in cell communication where certain carotenoids have been implicated in gap junction formation [Fig. 32.6](#).¹¹⁰

B. Actions in Select Tissues

Lycopene and prostate cancer

The prostate is the most frequent site of cancer in males in the United States, estimated to account for nearly one-third of all diagnosed cases in 2010 (American Cancer Society, 2010). In 1995, the Health Professional Follow-up Study, a prospective male cohort in the United States, found that lycopene intake, as well as the consumption of raw tomatoes, tomato sauce, and pizza, was significantly associated with a decreased risk of prostate cancer.¹¹¹ In a follow-up, dietary intake of lycopene was associated with reduced risk of lethal prostate cancer and with a lesser degree of angiogenesis in the tumor in the same cohort.¹¹² A recent metaanalysis of 42 studies examined the association between dietary and circulating lycopene and came to the conclusion that lycopene is inversely correlated with prostate cancer risk.¹¹³ However, lycopene was not associated with a reduced risk for advanced prostate cancer. Interestingly, the study found a linear dose response for dietary and blood lycopene levels and prostate cancer risk.

The systemic review by World Cancer Research Fund/American Institute for Cancer Research categorized lycopene as “probable for decreased prostate cancer risk,” which was revised to “limited—no conclusion” in 2014. What is the association between lycopene and prostate cancer? For reasons that are still not understood, lycopene preferentially accumulates in the prostate. In this tissue, the androgens testosterone and dihydrotestosterone play an important role in stimulating proliferation and the development of prostate cancer. There appears to be an interaction between androgens and lycopene levels, such that castration of male rats leads to a twofold increase in hepatic lycopene, an effect that is normalized by testosterone replacement.¹¹⁴ The mechanistic underlining is yet unclear, and several putative scenarios have been examined in cell culture and animal models. In principle, one has to distinguish between effects of lycopene and its putative apolycopene metabolites. For instance, mouse studies showed dietary lycopene affects a profile of testosterone-regulated genes associated with early prostate carcinogenesis as potential mechanistic targets.¹¹⁵ Genetic dissection in mouse models provided evidence that the effect of lycopene at least in part depends on BCO2 activity and apo-lycopenoids.^{116,117} Genetic polymorphisms in the BCO1 gene are associated with lycopene¹¹⁸ levels likely because of its role in controlling carotenoid uptake in the intestine. However, evidence for CCD-independent actions of lycopene also has been reported in mouse models.¹¹⁹ The effects of the parent carotenoid may involve its properties as an antioxidant, interactions with signaling molecules such as

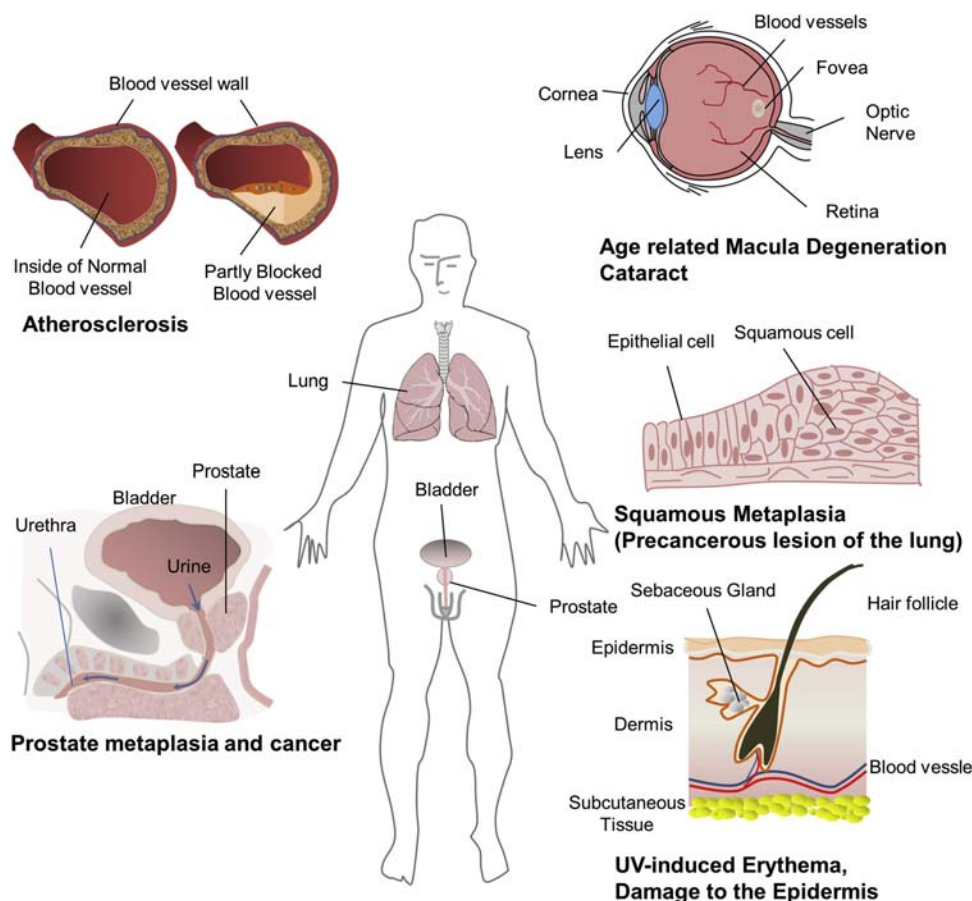


FIGURE 32.6 Health and disease states related to carotenoids.

NF κ B,¹²⁰ and interactions with the metabolism and transport of other lipids, particularly cholesterol.¹²¹ Currently, many molecular details of lycopene action remain undefined, and more research is required to analyze the distribution, metabolism, and function of lycopene in relation to prostate health.

Lutein, zeaxanthin, and eye health

In the United States and Europe, age-related macular degeneration (AMD) is the leading cause of blindness in people aged 50 or older. AMD causes damage to the macula, a small spot near the center of the retina and the part of the eye needed for high resolution color vision. The human macula uniquely concentrates three carotenoids: lutein, zeaxanthin, and meso-zeaxanthin.¹⁰ Mesozeaxanthin in enantiomer of zeaxanthin and animal studies indicate that it is synthesized from dietary lutein precursors in the eyes.¹²² There are three stages of AMD defined in part by the size and number of drusen under the retina. Drusen are generated from lipids and proteins that accumulate at the Bruch's membrane at the blood–retinal barrier. Early AMD is diagnosed by the presence of medium-sized drusen and typically not associated with vision loss. People with intermediate AMD typically have large drusen, pigment changes in

the retina, or both. Intermediate AMD may cause some vision loss, but most people will not experience any symptoms. In addition to drusen, people with late AMD have vision loss from damage to the macula. There are two types of late AMD. In geographic atrophy (dry AMD), there is a gradual breakdown of photoreceptor cells. In neovascular AMD (wet AMD), abnormal blood vessels grow underneath the retina. These vessels can leak fluid and blood, which may lead to swelling and damage of the macula.

There are several risk factors for AMD, including smoking, age, race, and genetics. A connection between carotenoids and AMD is that autopsied retinas from individuals with AMD have decreased macular pigment (MP) levels in comparison with nondiseased retinas. Nonhuman primates raised on a carotenoid-free diet lack MP and display characteristic changes in the RPE. MP is believed to be protective in two ways: by absorbing blue light and by acting as an antioxidant.¹⁰ MP serves an ocular protective role through its ability to filter photo-toxic blue light radiation and also via its antioxidant activity. These properties have led to the hypothesis that MP may protect against the development of AMD.¹⁰ Epidemiological studies support this role and revealed that individuals with lower concentrations of serum

carotenoids and MP optical density measurements are at a higher risk of developing AMD. Conversely, nutritional supplementation and diets rich in lutein and zeaxanthin readily impact MP concentrations and reduce the risk of progression to advanced AMD.¹²³

Skin

Excess sunlight can lead to ultraviolet (UV)-induced erythema, a reddening of the skin more commonly referred to as sunburn. There is evidence to suggest that carotenoids can provide modest skin protection. Studies suggest that at least 10 weeks of supplementation is needed before benefits can be expected; for example, a recent metaanalysis found that β -carotene supplementation for at least 10 weeks is associated with a significant decrease in sunburn.¹²⁴ It is also important to note that beneficial interactions occur between carotenoids and probably other dietary components such as flavonoids, meaning diets rich in these compounds will likely provide better protection against UV-induced skin damage than the individual compounds alone.¹²⁵

Metabolic and cardiovascular disease

Diseases caused by an excessive accumulation of lipids, such as atherosclerosis and obesity, are a growing health problem worldwide; thus, finding safe and effective therapeutic approaches against these disorders is of major clinical relevance. Several epidemiological studies have reported that plasma carotenoid levels are inversely correlated with the incidence of these chronic diseases.²⁷ This association has been reported for total plasma carotenoids and individual carotenoids such as lycopene, β -carotene, and lutein.¹²⁶ There are a number of potential mechanisms through which carotenoid consumption may affect the etiology of these diseases. The most commonly cited is through the antioxidant action of carotenoids, which may decrease lipid peroxidation. Additionally, local production of apocarotenoids including retinoic acid from carotenoid metabolites may influence the activity of RXRs, which are heterodimers of transcription factors such as LXRs, master regulators of macrophage lipid metabolism and atherosclerosis, and PPARs, regulators of adipocyte physiology would then decrease lipid and LDL oxidation.⁹⁸ Further research is needed to unravel the interaction of carotenoid and lipid metabolism and the carotenoids' putative beneficial effects.

IV. DIETARY REQUIREMENTS

A. Indicators for Deficiency or Inadequacy

Low plasma levels of carotenoids have been associated with an increased risk for chronic diseases, but no classical deficiencies have been defined. Considering its role as a

provitamin, low intakes of provitamin A carotenoids significantly contribute to the worldwide vitamin A deficiency problem.¹²⁷ Moreover, recent studies in primates showed that a deficiency of lutein and zeaxanthin impair vision and make the retina more susceptible to light damage.^{128,129} Thus, it can be assumed that a dietary deficiency of lutein and zeaxanthin would affect the human eye in the same way.

B. Criteria for Adequacy

At present, no recommendations for carotenoid intake or desired blood or tissue concentrations have been issued by the majority of health authorities. Thus, no recommended dietary allowance or adequate intake exists for these compounds. One would need to distinguish between provitamin A and carotenoid functions if such criteria would be defined. An increased total mortality was observed with decreasing circulating β -carotene (<1000 nM) in elderly Europeans.¹³⁰ A follow-up study of the National Health and Nutrition Examination Survey III respondents confirmed that carotenoid concentrations in serum relate to mortality, though relationships differed for specific carotenoids. This study reported that all-cause mortality was reduced until approximately 1000 nM plasma concentration with little impact on mortality at higher concentrations.¹³¹

C. Assessment of Status

Accurate assessment of the amount of the carotenoids and apocarotenoids in blood and tissues is necessary to investigate their presumed functions. HPLC is the "gold standard" technique for the determination of these compounds and can provide unambiguous identification and quantitation. However, this technique requires significant amounts of precious biological tissue. There are a variety of other methods in use to measure carotenoids in skin and eyes.¹⁰ Methods for macula pigments can be divided into psychophysical and physical methods. Of the psychophysical techniques, heterochromatic flicker photometry is the most widely used. Physical techniques include resonance Raman spectroscopy, fundus autofluorescence, and fundus reflectance. Resonance Raman spectroscopy and reflection spectroscopy are optical methods applicable to the noninvasive detection of carotenoids in human skin. Novel devices have been developed to reliably assess skin carotenoid levels in larger populations. It is yet not clear whether skin carotenoid levels allow qualitative and quantitative predication of the carotenoid levels in other tissues. However, it should be noted that skin carotenoid levels are useful biomarkers of fruit and vegetable intake and indirectly reflect a healthy lifestyle.

D. Dietary Reference Intakes

At present, Dietary Reference Intakes for carotenoids, including the provitamin A carotenoids, have not been established.¹³² When investigating published recommendations for carotenoid intake, provitamin A and non-provitamin A functions of carotenoids must be discussed separately. In 1989, the Food and Nutrition Board of the National Research Council established conversion values of 2 μg of purified all-*trans*- β -carotene from supplements to be one retinol equivalent (RE = 1 μg all-*trans*-retinol), and because of decreased bioavailability, 3 μg of dietary all-*trans*- β -carotene from food was considered to be equivalent to 1 μg of purified all-*trans*- β -carotene in oil. Thus, 6 μg of dietary all-*trans*- β -carotene was equivalent to 1 RE. Other dietary provitamin A carotenoids (mainly α -carotene and β -cryptoxanthin) were believed to have half the vitamin A activity of dietary all-*trans*- β -carotene and set at 12 μg = 1 RE (Table 12.1). In 2000, the Food and Nutrition Board reviewed the evidence for role(s) of carotenoids as antioxidants and determined that specific functions beyond that related to provitamin A activity had not yet been sufficiently identified, so no recommended intake for carotenoids was set, but that a diet high in carotenoid-rich fruits and vegetables was supported.

In 2001, the Food and Nutrition Board revised the 1989 conversion values based on human trials revealing that the bioavailability of all-*trans*- β -carotene from foods was half of what was originally believed.¹³³ To decrease confusion in making the changes, a new term was coined: "retinol activity equivalent" (RAE = 1 μg all-*trans*-retinol). The RAE ratios of conversion for purified all-*trans*- β -carotene in oil, dietary all-*trans*- β -carotene, and other dietary provitamin A carotenoids were set at 2:1, 12:1, and 24:1, respectively (Table 12.1).

Notably, these equivalency ratios do not consider the many intrinsic and extrinsic factors affecting the bioavailability and bioconversion of carotenoids. As evidenced by recent studies, the conversion rate may be significantly affected by vitamin A status and genetic makeup.⁴⁴ A consensus conference in 2009 on β -carotene in Hohenheim, Germany, concluded that the mean dietary intake of β -carotene in Europe is in the range of 1.5–1.8 mg/d and that provitamin A intake is <3 mg/d in most European countries. Since provitamin A carotenoids are the major source of vitamin A, it was recommended to increase the β -carotene consumption to 7 mg/d.¹³⁴

There are, in general, no dietary recommendations for individual non-provitamin A carotenoid intake, as it was

thought that their absence from the diet causes no specific deficiency symptoms. Existing suggestions for intake are mainly based on epidemiological and intervention studies and the apparently beneficial health effects of these compounds. Due to potential adverse effects, developing carotenoid intake recommendations will require a risk assessment for the individual carotenoids. Among non-provitamin A carotenoids, lutein and zeaxanthin have recently received considerable interest. The outcome of the Age-Related Eye Disease Study 2 (AREDS2) has provided compelling evidence for beneficial effects of low-dose zeaxanthin and lutein (2 and 10 mg) supplementation for eye health.^{123,135} While the importance of these carotenoids has been generally acknowledged,¹⁰ intakes of 10 mg/d have been advocated.^{10,136}

Regarding lycopene, regular intake from natural dietary sources in different populations is estimated to be highly variable, depending especially on the intake of tomatoes and tomato-derived products. An average intake between 0.5 and 5 mg/d has been reported.¹³⁷ However, a high consumption of fruits and vegetables, especially in the form of tomato products and watermelon, may result in intakes of 20 mg lycopene/d or higher.¹³⁸

E. Basis for Individual Variability in Requirements

Carotenoids exhibit a large interindividual variability in their bioavailability. Several extrinsic and intrinsic factors (SLAMENGHI) influence their bioavailability and bioconversion.⁴⁴ These include vitamin A status of the host and genetic makeup. The recently discovered regulation of intestinal vitamin A production indicates that populations with low dietary vitamin A supply are generally better converters than populations with high vitamin A intake.⁶⁷ This finding also suggests significant interaction between the absorption of provitamin A and other carotenoids. Disease states of the gut can be expected to affect carotenoid absorption and blood levels. Atrophic gastritis, a common condition of aging, results in insufficient gastric acid secretion and decreased carotenoid uptake, by disturbing the formation and absorption of mixed micelles.²⁰ Parasitic infections, which are prevalent in vitamin A-deficient populations, can also have a dramatic negative effect on carotenoid absorption.¹³⁹ Additionally, Crohn's disease and celiac disease can affect carotenoid levels in blood and tissues.¹⁴⁰

F. Tolerable Upper Intake Levels

The Institute of Medicine's review in 2000 did not set a Tolerable Upper Intake Level for β -carotene or the carotenoids for Canadians and Americans.¹³² More recently, the United Kingdom has proposed Tolerable Upper Intake Levels, i.e., 7 mg for β -carotene,¹⁴¹ while the German Nutrition Society recommended 2 mg/d total dietary carotenoid intake.¹⁴² Average intakes in European countries have been reported to be around 2.7 mg/d and 2.9 mg/d for men and women, respectively.¹³⁴ A subsequent expert panel recommended an intake of 7 mg/d of β -carotene for vitamin A adequacy.¹³⁴ The outcome of the AREDS2, Physician's Health Study, and the Heart Protection Study indicate that significantly higher intake amounts are tolerable in non-smokers.^{123,143,144}

Adverse health effects of high-dose β -carotene supplementation were reported in smokers in two clinical studies. The Alpha-Tocopherol, Beta-Carotene (ATBC) Prevention trial included 29,000 participants who received 20 mg of β -carotene daily for 5–8 years,¹⁴⁵ and the β -Carotene and Retinol Efficacy Trial (CARET)¹⁴⁶ included more than 18,000 participants, receiving 30 mg of β -carotene daily for 4 years. The β -carotene average plasma concentrations following supplementation were approximately 3800 and 5600 nM for the CARET and ATBC trials, respectively (normal range 90–900 nM). Both studies revealed an increased lung cancer rate in the β -carotene groups (16% ATBC trial and 28% CARET study), and overall increased mortality. Notably, this effect was not reproducible in the Physician's Health Study and the Heart Protection Study¹⁴³ where participants received 50 mg/every other day for 13 years and 20 mg/d β -carotene for 5 years, respectively. Mechanistically, the effect of β -carotene on smokers might be associated with oxidative stress, inflammation, and alterations in retinoid signaling in the lung.¹⁴⁷

G. Dietary Sources

Worldwide cultivated plants such as carrots, tomatoes, pepper, corn, squash, spinach, and broccoli contain significant amounts of carotenoids. Additionally, many yellow to red fruits peaches, oranges, grapefruit, apricots, mangos, and papayas possess high levels of these compounds (<https://ndb.nal.usda.gov>). Carotenoids also exist in plant-derived products such as fruit and vegetable juices and concentrates. In these products, the concentration of carotenoids can be significantly

higher than in the plant tissue. β -carotene and some apocarotenoids are also used as food colorants in butter, spreads, and lemonade. Animal products such as eggs and dairy products can also contain significant amounts of carotenoids. Moreover, seafoods such as salmon and shrimp are rich in the carotenoid astaxanthin. Many carotenoids in the food chain derive from chemical synthesis. Microorganisms like algae also are used to the commercial production of these compounds. More recently, gene technology has been applied to modify the carotenoid content of staples and/or to produce specific carotenoids in plants and microorganism.¹⁴⁸ A prominent example is "Golden Rice," which contains β -carotene in the kernel.¹⁴⁹

V. ISSUES RELATED TO CAROTENOIDS IN HUMAN HEALTH

A. Relevant Health Outcomes and Conditions of Concern

High supraphysiological intake of β -carotene and lycopene can lead to carotenoderma and lycopenoderma, respectively. Carotenoderma is a condition that results in a yellowing of the skin due to accumulation of high levels of β -carotene. This disorder has been reported in subjects consuming large amounts of foods such as carrots or supplementing with 30 mg β -carotene per day or more for long periods of time. Carotenoderma does not result in discoloring of the ocular sclera as occurs in jaundice, thus allowing carotenoderma to be distinguished from this malady. Similarly, lycopenoderma results from high intakes of tomatoes or lycopene and causes the skin to become orange. Another pathological condition associated with high-dose lutein supplements is crystalline maculopathy. Again, this condition is reversible when the lutein supplement is discontinued.¹⁵⁰

B. As a Dietary Supplement

The red carotenoid canthaxanthin is marketed at high doses as a natural tanning pill because it accumulates in the fat pads below the skin, giving skin a bronze color. However, the high-dose long-term supplementation needed to sustain skin's tanned color may lead to the development of canthaxanthin-induced retinopathy, a condition in which canthaxanthin crystals form in the retina. Fortunately, this condition is reversible and does not culminate in vision loss.¹⁵¹

RESEARCH GAPS

An increasing number of nutritionists and clinicians advocate for recommendations for carotenoid intake. The establishment of such criteria is based on the relationship between carotenoid intake, blood, and tissue concentrations. Such intake recommendations would need to consider the many host and environmental factors that influence this process. Studies in experimental animals provided a molecular framework on how diet and genetics affect the bioavailability and bioconversion of carotenoids. An important aspect in the coming years will be to refine our knowledge about these factors and to test their validity for human situation.

Animal studies also demonstrated a key role of CCDs in carotenoid homeostasis.¹⁰⁹ However, it is still controversial whether the function of these enzymes is fully conserved in humans.^{84,152} Furthermore, an increasing body of evidence implicates that apocarotenoids exert physiological functions different than retinoids. The molecular identification of receptor(s) for these compounds would eventually establish carotenoids as essential nutrients.

Many studies support the critical role of carotenoids in eye health. However, the pigments' putative roles in other physiological processes need to be further explored. Future studies need to take advantage of innovative methods such lipidomics, transcriptomics, and proteomics to unravel the manifold roles of carotenoids in health and disease.

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CARNITINE

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SUMMARY

Carnitine is not required in the diet of healthy young adult humans but is a conditionally essential nutrient at different stages of the life cycle and during certain pathological conditions. Ingestion of 2–5 mg/kg body weight per day comes mainly from animal products. In humans, carnitine transports carboxylic compounds to where they are needed (oxidation of long-chain fatty acids) or transports them away from where they are causing problems (several inborn errors of metabolism). Carnitine also stores readily available activated carboxylic acids and modulates the free coenzyme A to acylated coenzyme A ratio. Carnitine is an important compound to both commensal members of the gastrointestinal tract microbiome and to pathological microbes. Supplemental carnitine (often provided in doses 10 to 50 times higher than dietary intake) can have significant impacts on gut microbiota and the compounds they synthesize from carnitine, which are taken up and metabolized by human tissue.

Keywords: Acetyl-carnitine; Acylcarnitine; Carnitine; Fatty acid oxidation; Microbiome; Metabolome; Newborn screening; Trimethylamine oxide (TMAO).

I. INTRODUCTION

A. Background

Carnitine was discovered in meat in 1905. In the 1950s, it was found to be essential in the diet of the mealworm *Tenebrio molitor* and was given the name vitamin B₁₇. In the 1960s, the role of carnitine in beta-oxidation of long-chain fatty acids in the matrix of the mitochondria was described and became “the function of carnitine.” In the 1970s, the details of the carnitine shuttle in mitochondria were better delineated, and the biosynthesis of carnitine from the essential amino acids lysine and methionine via trimethyllysine in *Neurospora crassa* and protein-bound trimethyllysine in mammals was described. In the 1970s, patients with low concentrations of carnitine in plasma and muscle were identified and given oral carnitine at doses much higher than ever consumed by man (50–100 mg per kg body weight per day) with some reports of symptom improvement securing a role of carnitine as a pharmaceutical.¹ As it became clear during the 1980s and 1990s that carnitine was being used by the body to esterify and excrete carboxylic acids accumulating from a variety of inborn errors of metabolism and from metabolism of drugs,²

the emerging technique of tandem mass spectrometry of biological samples began to be used to detect elevated acylcarnitines using the procedure of collecting blood spots on filter paper started in the 1960s for the phenylketonuria Guthrie test. Today, the analyses of acylcarnitines for newborn screening is a critical component of the Centers for Disease Control and Prevention Newborn Screening and Molecular Biology Branch that ensures the accuracy of newborn screening tests in every state and more than 80 countries (<https://www.cdc.gov/nceh/dls/nsmbb.html>). The Human Microbiome Project in the 2000s brought to the attention of the nutrition community that the gastrointestinal microbiome must be included in the study of every nutrient including carnitine. Bacterial metabolism of carnitine became important not only in pathogens of infectious diseases but also in microbes that produce compounds taken up by human tissue and metabolized to compounds that increase the risk of the individual for noncommunicable chronic diseases becoming epidemic in our society.³

B. Carnitine Structure

Carnitine has a molecular weight of 161.5 Da (Fig. 33.1). It consists of carbon, hydrogen, and nitrogen. Like amino

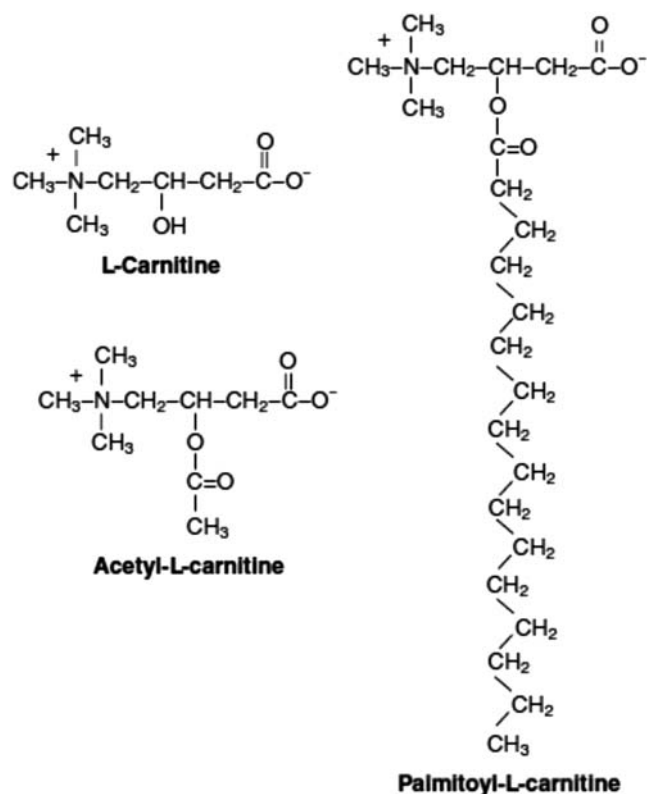


FIGURE 33.1 Structures of carnitine and representative acylcarnitine esters. From the chapter titled *L-Carnitine* by Charles J. Rebouche in *Present Knowledge in Nutrition*, 10th Edition, edited by Erdman JW, Macdonald IA, and Zeisel SH and published by Wiley-Blackwell. © 2012 International Life Sciences Institute.

acids at physiological pH, carnitine is a zwitterion, but it is not an amino acid because the nitrogen is part of a quaternary ammonium group rather than an amino group. Carnitine is a carboxylic acid itself and can become activated to carnitiny-coenzyme A in the microbiome. Carnitine has a hydroxyl group that has received most of our attention in human nutrition. Coenzyme A activated carboxylic acids form esters with the hydroxyl group of carnitine producing a variety of acylcarnitines. Acetyl-carnitine (Fig. 33.1) has the highest concentration of acylcarnitines in most human samples. Acylcarnitines are usually characterized by the chain length of the acyl moiety. Acetyl-carnitine (with an acyl group of two carbons) is a small water-soluble compound, but palmitoyl-carnitine (with an acyl group of 16 carbons) (Fig. 33.1) and stearoyl-carnitine (with an acyl group of 18 carbons) have a molecular weight of more than twice that of acetyl-carnitine and are much less water-soluble. Oleoyl-carnitine (with one double bond), linoleoyl-carnitine (with two double bonds), and alpha linolenyl-carnitine (with three double bonds) all also have acyl groups of 18 carbons. However, since the double bonds are cis, the four acylcarnitines with 18 carbons listed above have very different space-filling structures. Thus, descriptions of acylcarnitines need to include information about the acyl group in addition to carbon chain length.

II. NUTRIENT METABOLISM

A. Synthesis

In humans, the carnitine biosynthetic pathway involves four enzymes: 6-N-trimethyllysine dioxygenase (requiring ferrous iron, ascorbate, and alpha-ketoglutarate), 3-hydroxy-6-N-trimethyllysine aldolase (requiring pyridoxal phosphate), 4-N-trimethylaminobutyraldehyde dehydrogenase (requiring NAD), and γ -butyrobetaine (γ BB) dioxygenase (requiring ferrous iron, ascorbate, and alpha-ketoglutarate) (Fig. 33.2). Carnitine

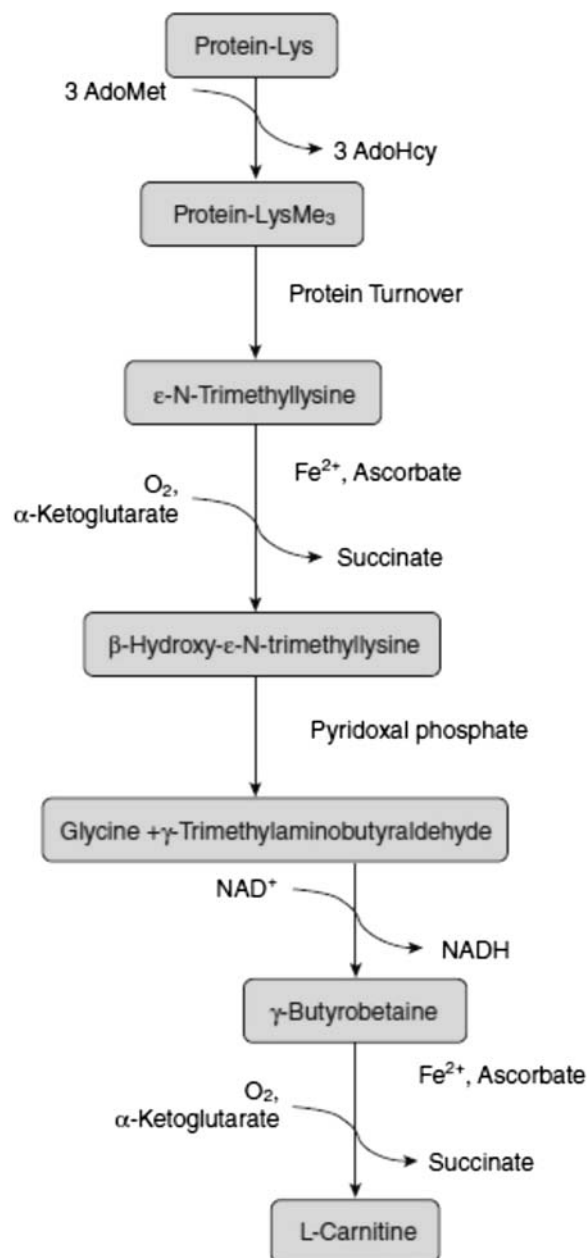


FIGURE 33.2 Pathway for carnitine biosynthesis in mammals. From the chapter titled *L-Carnitine* by Charles J. Rebouche in *Present Knowledge in Nutrition*, 10th Edition, edited by Erdman JW, Macdonald IA, and Zeisel SH and published by Wiley-Blackwell. © 2012 International Life Sciences Institute.

biosynthesis occurs in the liver, kidney, brain, and testes.¹ Although it is fairly certain that carnitine is derived from the essential amino acids lysine and methionine, the assumption that protein bound trimethyllysine is the only starting point for carnitine biosynthesis in mammals has been revised by recent data using updated analytical techniques. It has always been a mystery as to why such an energy-expensive route of protein degradation would be required for carnitine synthesis. We now know that dietary vegetables and fruits have adequate amounts of free trimethyllysine to be the substrate for carnitine biosynthesis.^{4,5}

B. Digestion and Absorption

Carnitine in food is not digested in the traditional sense. Early experiments showed that uptake of both carnitine and acetyl-carnitine was about twice as rapid in upper jejunal cells than ileal cells from animals.^{6,7} Even these early experiments found that approximately 30–60 min was required for carnitine to be taken up by the cells, acetylated (deacetylated and reacetylated in the case of acetyl-carnitine) before release to the circulation over the next several hours. Intestinal retention of administered carnitine was increased with addition of carnitine to the diet, fasting, and during the suckling period of life.⁸ As we learn more about carnitine and the gastrointestinal microbiome, these time-requiring processes should become better delineated. Cautious interpretation is needed for absorption data collected from humans before the recognition of the potential role of intestinal microbiota in the process because the calculations did not account for the numerous metabolic actions of the gut microbiota on carnitine before it reached the intestinal tissue.

C. Transport

Carnitine is transported into cells via OCTN2 (organic cation/carnitine transporter novel type 2).⁹ OCTN1 and OCTN3 also transport carnitine into cells.^{10,11} There are species differences and organ differences in the distribution and characteristics of these three carnitine transporters. Acetyl-carnitine and other acylcarnitines are also transported across cell membranes with acetyl-carnitine being more easily transported than other forms of carnitine. A study in human brain endothelial cells (hCMEC/D3), which are considered to be a model of the blood–brain barrier, indicates that OCTN2 is involved in carnitine transport at the human blood–brain barrier.⁹

D. Metabolism

Mammals usually excrete carnitine intact as free carnitine or acylcarnitine rather than metabolize it. However, the microbial metabolism of carnitine (Fig. 33.3) is now recognized as extremely important to human health. In 2019, a perspective published in JAMA¹² described how trimethylamine oxide (TMAO), in addition to the saturated fat and cholesterol content of red meat, is positively associated with heart disease and mortality. Gut bacteria in the cecum and colon convert choline (and to a lesser extent carnitine) abundant in red meat, poultry, fish, dairy, and egg yolks to trimethylamine (TMA) that is absorbed by the intestine and is transported to the liver and converted to TMAO.

Fasting endogenous plasma carnitine and γ BB levels were similar in vegans/vegetarians versus omnivores, but TMAO generation was 20-fold higher in omnivores than vegans/vegetarians. Dietary carnitine is converted into TMA via two sequential gut microbiota-dependent

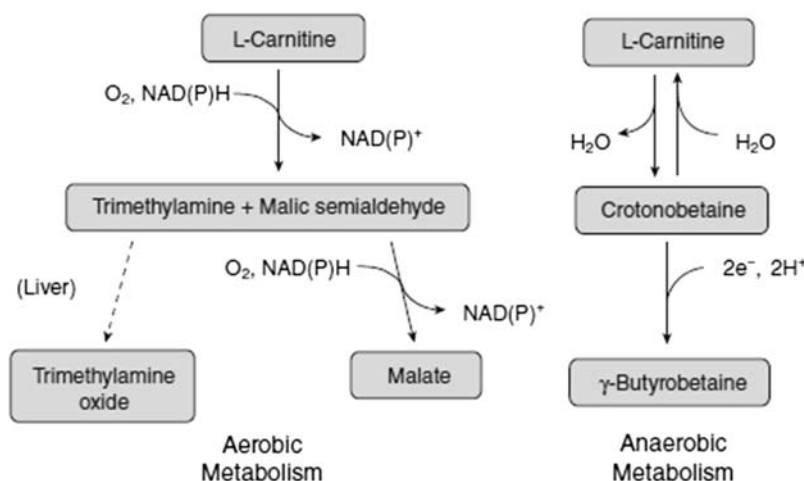


FIGURE 33.3 Catabolism of L-carnitine by microorganisms in the rat and human gastrointestinal tract. From the chapter titled *L-Carnitine* by Charles J. Rebouche in *Present Knowledge in Nutrition*, 10th Edition, edited by Erdman JW, Macdonald IA, and Zeisel SH and published by Wiley-Blackwell. © 2012 International Life Sciences Institute.

transformations.¹³ The initial generation of the atherogenic intermediate γ BB is followed by transformation into TMA via low-abundance microbiota in omnivores and to a markedly lower extent in vegans/vegetarians. Omnivores and vegans/vegetarians both rapidly convert carnitine to γ BB, but the second gut microbial transformation, γ BB $\rightarrow$ TMA, is dietary carnitine inducible. No single human fecal commensal is capable of the carnitine $\rightarrow$ TMA transformation. There are multiple community members that convert carnitine to γ BB, and only one *Clostridiales* bacterium, *Emergencia timonensis*, has been identified to convert γ BB to TMA.¹³

Elevated TMAO has been associated with several noncommunicative diseases that are showing an increased prevalence. Higher TMAO levels were associated with increased risk of first stroke in hypertensive patients¹⁴ and ischemic stroke.¹⁵ Plasma concentration of TMAO is increased in compensated and decompensated heart failure patients.¹⁶ Increased levels of TMAO are associated with increased risk of developing renal failure, metabolic syndrome, diabetes mellitus, heart failure, hypertension, atherosclerosis, dyslipidemia,^{17,18} preeclampsia,¹⁹ and disease activity for both skin and peripheral joints in psoriatic arthritis patients.²⁰ All samples of human cerebrospinal fluid collected from 58 subjects by lumbar puncture contained TMAO at concentrations ranging between 0.11 and 6.43 $\mu\text{mol/L}$.²¹ TMAO in cerebrospinal fluid is higher in individuals with mild cognitive impairment and Alzheimer's clinical syndrome dementia (AD) compared to cognitively unimpaired individuals, and

the elevated CSF TMAO is associated with biomarkers of AD pathology and neuronal degeneration.²²

The role of bacteria in our gastrointestinal tract in increasing our disease risk has begun to focus our attention on bacteria and the nutrient carnitine. Bacteria metabolize carnitine in different ways. *Pseudomonas* species assimilate carnitine as sole source of carbon and nitrogen. *Acinetobacter* species degrade only the carbon backbone forming TMA. Enterobacteriaceae species convert carnitine, via crotonobetaine, to γ BB in the presence of C and N sources and under anaerobic conditions.²³ Metabolism of carnitine by *Proteus* species proceeds at the CoA level,²⁴ and γ BB CoA synthetase has been isolated from soil bacteria.^{25,26} In *Proteus* species, crotonobetaine and carnitine induce carnitine metabolism, while γ BB and glucose repress carnitine metabolism. Other betaines were tested but were neither inducers nor repressors.²⁷ In bacteria, carnitine is an important osmoprotectant and can also enhance thermotolerance, cryotolerance, and barotolerance.²⁸ Bacteria have both aerobic and anaerobic metabolic pathways for carnitine and can even use carnitine as an electron acceptor.²⁸

III. ROLE IN NORMAL CELL AND ORGAN FUNCTION

A. Transport of Carboxylic Acids

One of the major functions of carnitine is shuttling long-chain fatty acids across the mitochondrial membrane from

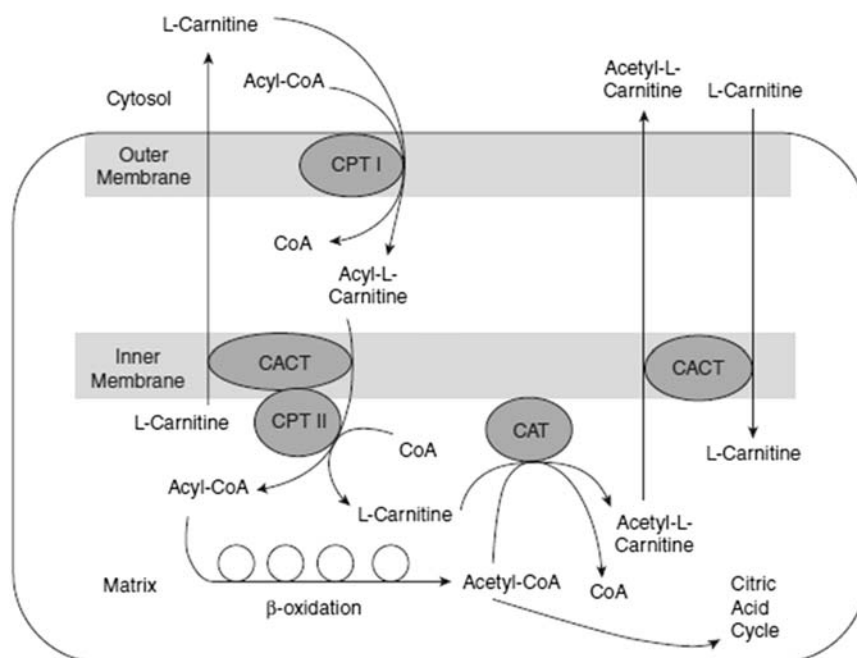


FIGURE 33.4 Facilitation of entry of long-chain fatty acids into mitochondria by carnitine. CACT, carnitine-acylcarnitine translocase; CAT, carnitine acetyltransferase; CoA, coenzyme A; CPT I, carnitine palmitoyltransferase I; CPT II, carnitine palmitoyltransferase II. From the chapter titled *L-Carnitine* by Charles J. Rebouche in *Present Knowledge in Nutrition*, 10th Edition, edited by Erdman JW, Macdonald IA, and Zeisel SH and published by Wiley-Blackwell. © 2012 International Life Sciences Institute.

the cytosol into the mitochondrial matrix for β -oxidation (Fig. 33.4). This mitochondrial carnitine–acylcarnitine cycle consists of three enzymes: carnitine palmitoyltransferase I (CPT I), carnitine-acylcarnitine translocase (CACT), and carnitine palmitoyltransferase II (CPT II).²⁹

After acyl groups enter the cell and are activated to acyl-coA by acyl-coA synthetases, carnitine CPT I transfers acyl groups from coenzyme A to carnitine, forming acylcarnitine esters at the outer mitochondrial membrane. CACT exchanges acylcarnitine esters that enter the mitochondria with mitochondrial-free carnitine. CPT II converts acylcarnitine esters back to acyl-coA esters at the inner mitochondrial membrane making them available for β -oxidation.³⁰ An ATP is required to activate the acyl group to the CoA level in the cytoplasm, but since the acyl group remains active as acylcarnitine, acyl-CoA can be reformed in the mitochondria without the requirement for ATP.

Carnitine acetyltransferase is involved in the transport of acetyl groups across membranes and in acetylation reactions that regulate gene transcription and enzyme activity. Carnitine octanoyltransferase is a peroxysomal enzyme required for the complete oxidation of very long-chain fatty acids and phytanic acid, a branched-chain fatty acid.³¹ Carnitine octanoyltransferase is found in other locations in the cell. Medium-chain triglycerides (MCT) are often assumed to be metabolized independently of carnitine. A study of metabolism in humans suggests that intravenous MCT are not metabolized independently of carnitine.³²

B. Storage of Activated Carboxylic Acids and Modulation of Free Coenzyme A to Acylated Coenzyme A Ratio

The maintenance of energy homeostasis requires the hypothalamic integration of nutrient feedback cues. Carnitine metabolism regulates hypothalamic fatty acid sensing through the actions of CPT I. The buffering of mitochondrial matrix levels of acetyl-CoA by carnitine has a role in glucose sensing since acetyl-CoA is an allosteric inhibitor of pyruvate dehydrogenase and hence glucose metabolism.³³ During the energy demanding process of cold-induced thermogenesis, the liver undergoes a metabolic switch to provide fuel for brown fat thermogenesis by producing acylcarnitines. White adipose tissue provides long-chain fatty acids for hepatic tissues to generate plasma acylcarnitines as a fuel source for peripheral tissues in endotherms to protect against a reduction in ambient temperature.³⁴ Data from rat cells in the hyperglycemic state exposed to carnitine supplementation at concentrations that do not affect growth rate or viability suggest protection of cardiomyocytes from oxidative stress–related damage, prevention of reactive oxygen species formation, and

activation of antioxidant signaling pathways.³⁵ Several long-chain acylcarnitines have anticoagulant activity related to an ability to bind and inhibit factor Xa, and patients with deep vein thrombosis and pulmonary embolism have low plasma levels of acylcarnitines.³⁶

Carnitine palmitoyltransferase IC (CPTIC) is expressed in neurons and cancer cells. Human mesenchymal stem cells widely used in regenerative medicine where they must survive under low nutrient conditions as a result of avascularity also express CPTIC. CPTIC protects them against glucose starvation, glycolysis inhibition, and oxygen/glucose deprivation via autophagy enhancement providing a greater availability of lipids to be used as fuel substrate for ATP generation.³⁷ CPTIC contributes to the control of whole-body glucose homeostasis and is predominantly expressed in the central nervous system. Steroid hormones affect CPTIC mRNA expression in human prostate.³⁸

C. Metabolomics and the Carnitinome

Metabolomics has facilitated the study of individual acylcarnitines in basic science nutrition and opened the door to precision nutrition using carnitine.³⁹ Using the carnitinome in newborn screening has completely changed the clinical care of neonates. Newborn screening has led to identification of asymptomatic adult women with carnitine transport deficiency after finding decreased free carnitine levels in their newborns.⁴⁰ Maternal carnitine deficiency with homozygous mutations that manifest during the perinatal period may result in secondary epilepsy with cerebral injury in the neonates with heterozygous mutations.⁴⁰

D. Gene Expression

Carnitine supplementation of obese Zucker rats affects 259 miRNAs in skeletal muscle that target genes involved in protein localization and transport, regulation of transcription and RNA metabolic processes, and several signal transduction pathways.⁴¹

CPT I enzyme is emerging as a potential molecular target in cancer therapeutic intervention because it activates fatty acid oxidation that would provide fuel for cancer growth, but it also functionally intertwines with other factors to regulate gene expression and apoptosis of the cancer cell.⁴² Dietary fat elevates hepatic CPT I expression via a highly coordinated transcriptional mechanism involving increased histone 3 lysine 4 dimethylation and reduced DNA methylation, upstream and within the promoter of *Cpt1a* plus increased peroxisome proliferator activated receptor alpha and CCAAT/enhancer binding protein beta binding directly downstream of the *Cpt1a* transcription start site.⁴³

Data indicate that propionate regulates intestinal lipid metabolism through the induction of PPAR α expression time-dependently and concentration-dependently in a mouse cell line.⁴⁴

E. Carnitine Beyond Fatty Acids

Because colonic microbiota can use carnitine as a source of carbon, nitrogen, or as an electron acceptor, it has an important role in maintaining the high-fiber fermentation ability in the colon. Carnitine is also utilized by the intestinal microbiota as a protective solute against different stressors. Carnitine modulates the metabolism of SCFAs by regulating normal levels of SCFAs in the cytosol and mitochondria for the host. There is an antagonistic relation between a high intestinal bacterial population and carnitine absorption.⁴⁵

A role of carnitine as a ubiquitous nutrient is consistent with conservation of carnitine metabolism genes in different bacterial taxonomic classes including soil, plant-associated, and marine bacteria. This environmental abundance of carnitine suggests this molecule might make a significant contribution to ecosystem nitrogen and carbon cycling.⁴⁶ Guts and roots are inhabited by many different bacteria, archaea, and viruses. Mammalian guts may contain thousands, while herbivores exhibit the largest bacterial diversity including probably plant-associated bacteria that, by being inside plant tissues, may survive stomach digestion. Transiting diet-borne bacteria may contribute to gut metabolic capacities. Guts and roots have large surface areas, with microvilli and folds or root hairs in some parts. Both roots and guts are structured, nonhomogenous habitats with pH, nutrient, water, and oxygen differential levels or gradients. The multiple microhabitats that exist in roots and guts contribute to high species richness. The comparison of plant and gut microbial ecologies may help to guide research toward the understanding of such complex symbioses. Plants use their “guts” (roots) outwards, and this simplifies their study in comparison to study of animal guts. Gut and root microbiotas significantly impact health, development, and fitness of their respective hosts.⁴⁷ Carnitine appears to be a nutrient and metabolite important in both guts and roots.

IV. DIETARY REQUIREMENTS

A. Reference Dietary Intakes and Indicators of Inadequacy

There are no recommended daily intakes of carnitine. Renal reabsorption maintains carnitine levels within the normal range despite variable dietary intake. Carnitine deficiency from disorders of carnitine

biosynthesis appears to be very rare because dietary carnitine and efficient renal carnitine reabsorption usually maintain normal carnitine levels with the absence of endogenously synthesized carnitine in the healthy adult.⁴⁸

B. Dietary Sources

Vegans consume much less carnitine than omnivores who consume up to 2–5 mg of carnitine per kilogram per day. Animal products are the main dietary source of carnitine.⁴⁹ The more type I fibers in meat, the greater the carnitine concentration. Thus, a chicken leg has approximately five times the concentration of carnitine than a chicken breast. Water-soluble carnitine and acetyl-carnitine are the major forms found in milk and stay with the whey fraction when milk is fractionated. Plant sources of carnitine include seeds and fermented foods such as tempeh.

C. Assessment of Status

Fed and fasted carnitine and acylcarnitine profile reference values for healthy children less than 2 years, 2–7 years, and greater than 7 years have been recently reported. Both age and fasting status alter acylcarnitine profiles.⁵⁰ Another cohort of 500 healthy children and adolescents from birth to 19 years showed that over 70% of acylcarnitines exhibited significant physiological changes during the neonatal period. Acylcarnitine profile reference values required age and sex grouping during puberty and adolescence.⁵¹

Current reference values for blood acylcarnitine concentrations are for normal weight neonates. Recent data suggest that the metabolic and nutritional status of very low birth weight and low birth weight differ from the normal birth weight neonates, and acylcarnitine cutoffs should be defined for them to reduce the risk of false-positive cases for genetic issues, when the issue is inappropriate reference values.⁵²

Primary carnitine deficiency is caused by a defect in the OCTN2 carnitine transporter and is characterized by hypoketotic hypoglycemia or cardiomyopathy in children and sudden death in children and adults.⁴⁴ OCTN2 carnitine transporter defects are autosomal recessive and characterized by decreased intracellular carnitine accumulation, increased urinary excretion of carnitine, and low serum carnitine levels. Patients presenting early in life have hypoketotic hypoglycemia and hepatic encephalopathy. Patients presenting later in life have skeletal and cardiac myopathy or sudden death from cardiac arrhythmia, usually triggered by fasting or catabolic state. This disease responds to oral carnitine. When administered in pharmacological doses, carnitine enters cells using the amino acid transporter B(0,+).⁵³ Carnitine transport deficiency ascertained by newborn screening

remains lengthy and challenging with molecular genetic testing needed for confirmation.⁵⁴

Many enzymes are involved in long-chain fatty acid oxidation. Pathogenic mutations in genes encoding these enzymes result in compromised energy homeostasis and accumulation of long-chain acylcarnitines. Symptoms arise or exacerbate during catabolic situations and include hypoketotic hypoglycemia, liver dysfunction, rhabdomyolysis, cardiomyopathy, and early demise.⁵⁵ One of the most common forms of mitochondrial fatty acid oxidation disorders is CPT II deficiency with a variety of phenotypes. Detection of all cases is difficult, and newborn screening now uses (C16 + C18:1)/C2 and C14/C3 as indices.⁵⁶

CACT deficiency is rare and characterized by hypoketotic hypoglycemia, hyperammonemia, cardiomyopathy, liver dysfunction, and muscle weakness, culminating in early death.⁵⁷

V. ISSUES RELATED TO CARNITINE IN HUMAN HEALTH

A. Health Conditions of Concern

Plasma acylcarnitine profiles of healthy newborns and newborns with hypoxic-ischemic encephalopathy showed increased medium-chain and decreased long-chain acylcarnitine concentrations. In the hypothermia group of neonates, butyryl-carnitine (C4) levels decreased in patients with favorable outcomes, but remained high in those who progressed unfavorably. C4 blood levels correlated with neuron-specific enolase levels and abnormal MRI findings.⁵⁸ Necrotizing enterocolitis is associated with abnormal fatty acid metabolism. Metabolic profiling through newborn screening indicated that 14 acylcarnitine levels and acylcarnitine ratios were associated with increased risk of developing necrotizing enterocolitis. Six acylcarnitine levels, along with birth weight and use of total parenteral nutrition, identified 89.8% of newborns with NEC in the model development cohort and 90.8% of the newborns with NEC in the validation cohort.⁵⁹

A study of patients on the continuum from healthy to subjective memory complaint and mild cognitive impairment to patients with AD found a decrease in

serum acetyl-carnitine and some other acylcarnitine levels.⁶⁰

Patients with sepsis and acute organ dysfunction with high plasma acetyl-carnitine (≥ 6000 ng/mL) had significantly increased 28-day mortality compared with those with plasma acetyl-carnitine less than 6,000 ng/mL. Plasma acetyl-carnitine significantly correlated with various plasma cytokine concentrations, with blood culture positivity, and with 28-day mortality risk.⁶¹ Data from a metaanalysis suggested that L-carnitine treatment of more than 3 months reduced inflammatory mediators.⁶² The osmotically active carnitine can replace some of the glucose in peritoneal dialysis fluid with positive effects on erythropoiesis and longevity and deformability of erythrocytes.⁶³

B. Dietary Supplement and Pharmaceutical

Patients with organic acidemias who require acute management of hyperammonemia must receive carnitine to promote urinary excretion of urinary organic acid esters at the same time they are receiving adequate calories and correction of metabolic acidosis with the substitution of bicarbonate for chloride in intravenous fluids.⁶⁴ Use of pivalate-conjugated antibiotics (especially in young children) can result in a decrease in free carnitine concomitant with an increase in pivaloyl-carnitine with acute encephalopathy and/or hypoglycemia.⁶⁵

Since carnitine synthesis is limited in fetal and placental tissue, carnitine uptake from maternal blood by the organic cation/carnitine transporters OCTN1 and OCTN2 is very important to the fetus. Studies in choriocarcinoma BeWo cells and human placenta derived models confirmed OCTN2 transport of carnitine is more active than OCTN1. Two frequently used HIV drugs, ritonavir and saquinavir, inhibited carnitine transport.⁶⁶

Dietary supplementation of acetyl-carnitine (ALC) has reported neuroprotective, neurotrophic, antidepressive, and analgesic effects in painful neuropathies. ALC also has reported antioxidant and antiapoptotic activity. It has been used in treatment of aging and neurodegenerative pathologies with reported slowing progression of mental deterioration.⁶⁷

RESEARCH GAPS

A greater understanding of the role of carnitine in the microbial world may change the nutritionist's view of the nutrient carnitine. More research with specific acylcarnitines in conjunction with our microbiome will continue to expand our understanding of important roles of carnitine that we first understood to be only a vitamin for a mealworm.

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DIETARY FLAVONOIDS

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SUMMARY

Dietary intake of flavonoids, found in a wide variety of plant-based foods and beverages, varies widely between individuals but is typically several hundred milligrams per day. For many flavonoids, such as flavanols, flavanones, and flavonols, a substantial percentage of the ingested dose is absorbed and rapidly metabolized and excreted within 24 h. Some flavonoid classes such as the oligomeric procyanidins and anthocyanins are poorly absorbed in an intact form, but their catabolites are very efficiently absorbed after microbial biotransformation in the colon. Flavonoids reduce the risk of cardiovascular disease, type 2 diabetes, and inflammatory diseases, as supported by human intervention and epidemiological studies. Although several mechanisms have been proposed, the main pathways are through modulation of sugar metabolism, reduction of blood pressure and of LDL cholesterol, improvement of platelet and endothelial function, inhibition of oxidative enzymes, and induction of antioxidant defenses. Although flavonoids are chemical antioxidants, this is not relevant to how they work in vivo. There is no current recommended intake of flavonoids, but any daily requirements are likely to vary between individuals and by health status.

Keywords: Bioavailability; Cocoa; Cardiovascular disease; Diabetes; Endothelial function; Flavonoid; Polyphenol; Tea.

I. INTRODUCTION

A. Background

Flavonoids are present in all diets containing foods or beverages derived from plants. They are synthesized to protect against stress and ultraviolet light exposure, are responsible for the color of many fruits and flowers, and also contribute to the taste sensation of red wine, tea, coffee, chocolate, and fruits. Chemically, flavonoids possess antioxidant activity in vitro and hence have been included under the general umbrella of “antioxidants” together with vitamins C and E and carotenoids. However, this chemical property is not relevant to the proposed health benefits of flavonoids, since in vivo, flavonoids are thought to act by affecting various digestive, cellular, and molecular processes. These activities combine to help reduce the risk of chronic diseases such as cardiovascular and other metabolic diseases. Over the last 25 years,

there has been an enormous and growing interest in this class of compounds, which have also been referred to as polyphenols, phytonutrients, nonnutrients, protective/food factors, dietary bioactives, and phytochemicals. It should be noted that some of these terms are purely chemical and may define origin, such as phytochemical, polyphenol, and flavonoid, whereas some terms such as phytonutrient and dietary bioactive imply a functional activity. An overview of flavonoid nomenclature with some example compounds is shown in Fig. 34.1. Although there are many thousands of flavonoids in the plant kingdom, only a handful are present in significant amounts in the diet of most humans, and this review will focus only on the flavonoids that are most abundant and important in the Western diet, and not on minor or rare compounds or those that are present in more exotic foods. Catechins (including galloylated catechins and procyanidins), selected anthocyanins, quercetin, and citrus flavanones (hesperetin and naringenin) are covered. Isoflavones

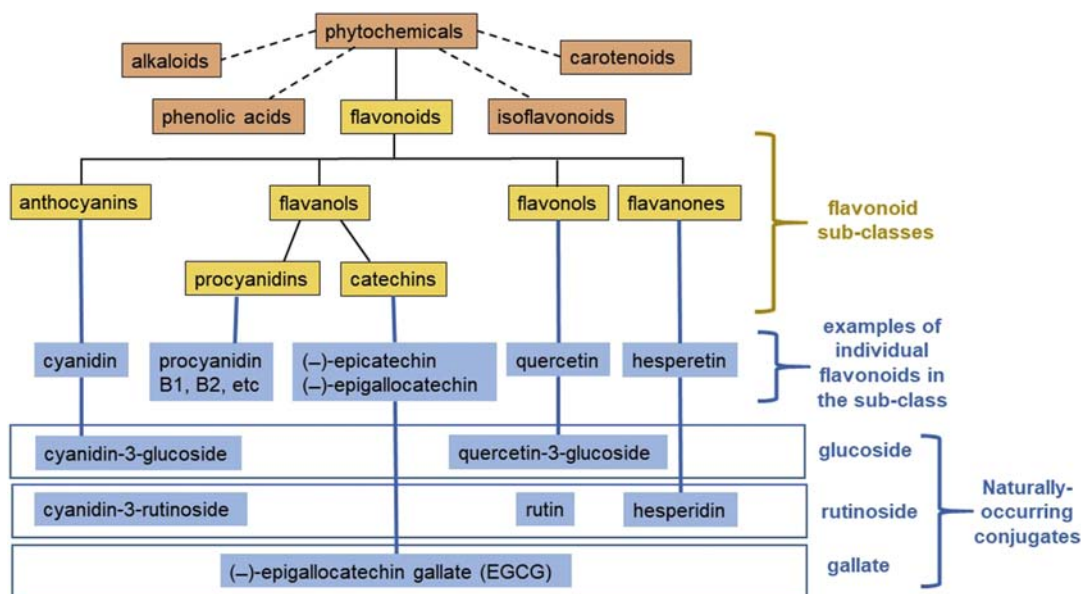


FIGURE 34.1 Overview of flavonoid nomenclature showing some subclasses and compounds with particular relevance to the human diet.

(present in soya) and phenolic compounds that are not flavonoids such as hydroxycinnamates (phenolic acids), widely present in food and drinks, especially coffee, are not covered here.

B. Classification, Structures, and Distribution in Foods

Fig. 34.2 shows the flavonoid content of foods and some representative chemical structures of the most abundant flavonoids. On a weight basis, they can be considered as minor constituents as the maximum amount in foods is usually less than 1% by weight, although the biological activity can be quite high. Fig. 34.2 does not include the higher molecular weight tannins, such as the thearubigins and theaflavins in black tea,^{1,2} and the larger procyanidins, which have been measured in apples,³ cocoa,⁴ and grape seeds.⁵ The foods richest in flavonoids include cocoa, berries, apples, onions, stone fruits, tea, citrus, and grapes, but exact intake is difficult to estimate, since these foods contain different amounts of flavonoids depending on variety, growing conditions, processing, etc., and average intake will be very variable within the population. Estimates include a median intake of total flavonoids of 239 mg/day, with a very wide range of 0.6 to 3524 mg/day, among 34,708 U.S. women.⁶ Mean pro(antho)cyanidin intake in the US population was 58 mg/day, with the main contributions coming from consumption of apples (32%), chocolate (18%), and grapes (18%).⁷ The mean intake of anthocyanins was estimated in the United States to be 12.5 mg/day.⁸ This value is surprisingly low and is subject to enormous interindividual variation, since one portion of fresh blueberries contains 300–400 mg anthocyanins per 100 g fresh weight, but the stated “average” intake is only ~1 g blueberries per day. By this reasoning, an

individual consuming 100 g blueberries in a day will consume ~30 times the average US consumption of anthocyanins.⁸ European intake of total flavonoids was estimated to be, on average, 428 ± 49 mg/d, of which 136 ± 14 mg/d were monomeric flavonoids with gal-lated flavan-3-ols (53 ± 12 mg/d) as the main contributor, but the range of intake was high.⁹ Flavonoid intakes across the world were also reported as varying widely from country to country.¹⁰

II. ABSORPTION, METABOLISM, AND EXCRETION—QUANTITATIVE AND MECHANISTIC ASPECTS

A. Flavonoids in the Gastrointestinal Tract

The principles controlling absorption and metabolism in the small intestine are now broadly understood for common flavonoids. After consumption of food, flavonoids are generally stable in the stomach¹¹ and reach the small intestine unchanged. Fig. 34.3^{12–19} shows the metabolic pathways for quercetin, epicatechin, and hesperetin in the small intestine and liver, for which the mechanisms of uptake, conjugation, and export are now well established. When the flavonoid is either unconjugated or is glucosylated in the food, absorption will normally be in the small intestine. When other sugars are present, this can affect dramatically the site of absorption and the amount absorbed. Anthocyanins, procyanidins, or other flavonoids with a sugar attached, such as rhamnose, which is not hydrolyzable by mammalian enzymes, are very poorly absorbed in the small intestine and pass through unmodified and enter the colon where, after microbial metabolism into lower molecular weight compounds,

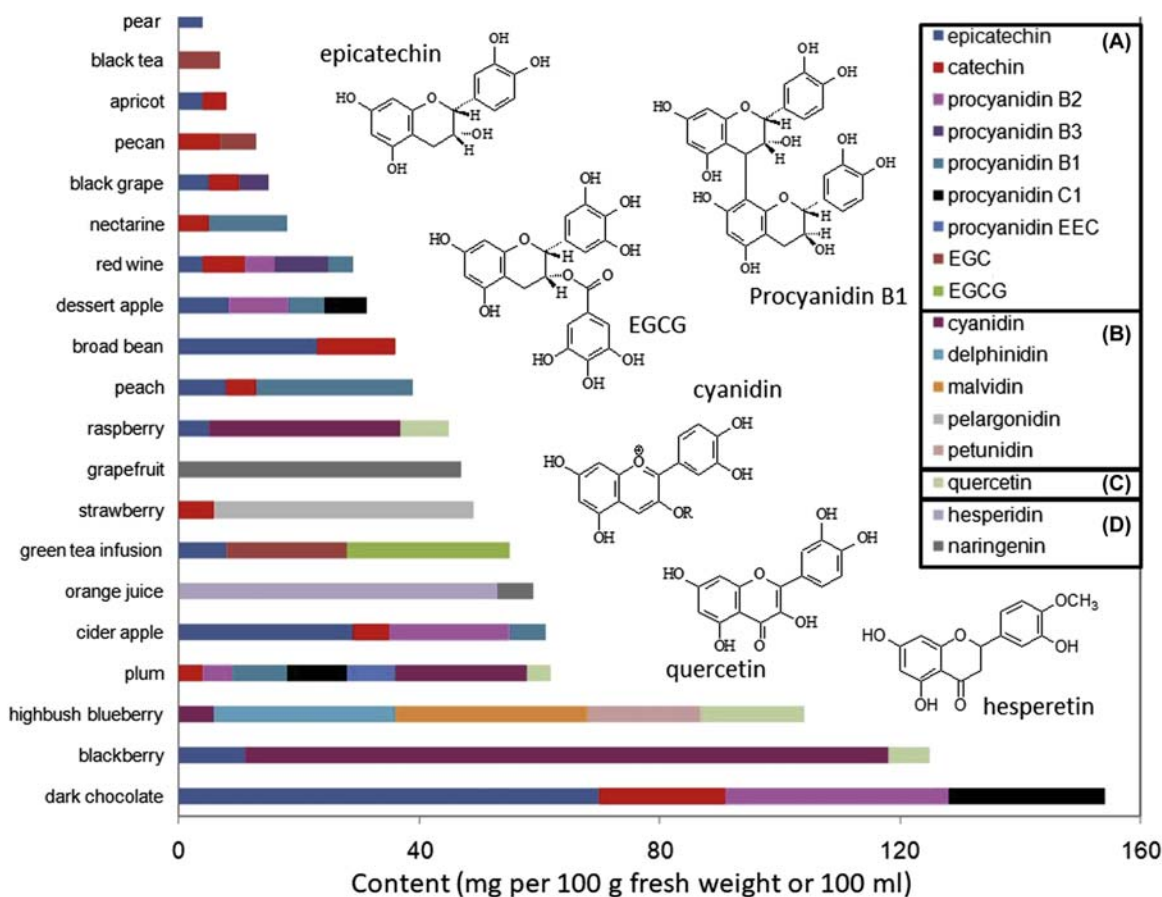


FIGURE 34.2 Representative chemical structures and content of commonly occurring flavonoid-rich foods in mg per 100 g fresh weight: A: flavanols; B: anthocyanins; C: flavonols; D: flavanones. All data are from the phenol explorer database (www.phenol-explorer.eu) and are in aglycone equivalents. Adapted from PKN 10th edition.

they are absorbed through the colonocytes into the blood.²⁰ Absorption into the blood is often measured pharmacokinetically after consumption of a single “dose,” usually in the form of a flavonoid-rich food. The $C_{\max}$ (concentration at the time point corresponding to maximum concentration of a compound) for various flavonoids has been reported in many papers.^{21–23} Epicatechin, epigallocatechin, quercetin, and hesperetin are absorbed, the latter two after deglycosylation, in the small intestine as shown by a short $T_{\max}$ (time point corresponding to $C_{\max}$). Hesperidin, naringin, and rutin (hesperetin, naringenin, and quercetin, respectively, linked to rhamnose as well as glucose) are all absorbed less efficiently with a lower $C_{\max}$ and exhibit a higher $T_{\max}$ of ~ 6 h, indicating absorption in the colon after removal of the sugars by the gut microbiota. Anthocyanins are very poorly absorbed in the intact form and appear in plasma at an extremely low level as the glucosylated form²⁴ in addition to some glucuronide forms.²⁵ Intact procyanidins are only just detectable in plasma after consumption of large quantities of the compound.²⁶ The food matrix has a substantial effect on the absorption of flavonoids, where factors such as fat content affect bioavailability.²⁷ The amount absorbed in the small

intestine has been estimated for some flavonoids using the ileostomy model. These are patients who have had their colon removed owing to a disease, and the contents of the small intestine are ejected into a pouch. The contents of this can be collected, analyzed, and the amount compared to the original dose. For quercetin glucosides in onions, the amount absorbed was estimated to be 52%, but rutin absorption was much lower.²⁸ For flavonols administered in green tea, the amount absorbed in the small intestine was $\sim 30\%$.²⁹ However, identification of metabolites shows that an additional $\sim 37\%$ of the dose was taken up into the enterocyte and effluxed back into the gut lumen as conjugates. Less than 10% of procyanidins were apparently absorbed in the small intestine in the ileostomy model,³⁰ and less than 15% for anthocyanins.³¹ However, some of the data indicated a higher level of absorption,³¹ but for anthocyanins such as cyanidin-3-O-glucoside, this could be due to inherent instability of the compound at physiological pH.³²

B. Flavonoids in the Blood After Consumption

After absorption, flavonoids circulate in the blood mostly as a mixture of glucuronide, sulfate, and/or

Small intestinal metabolism

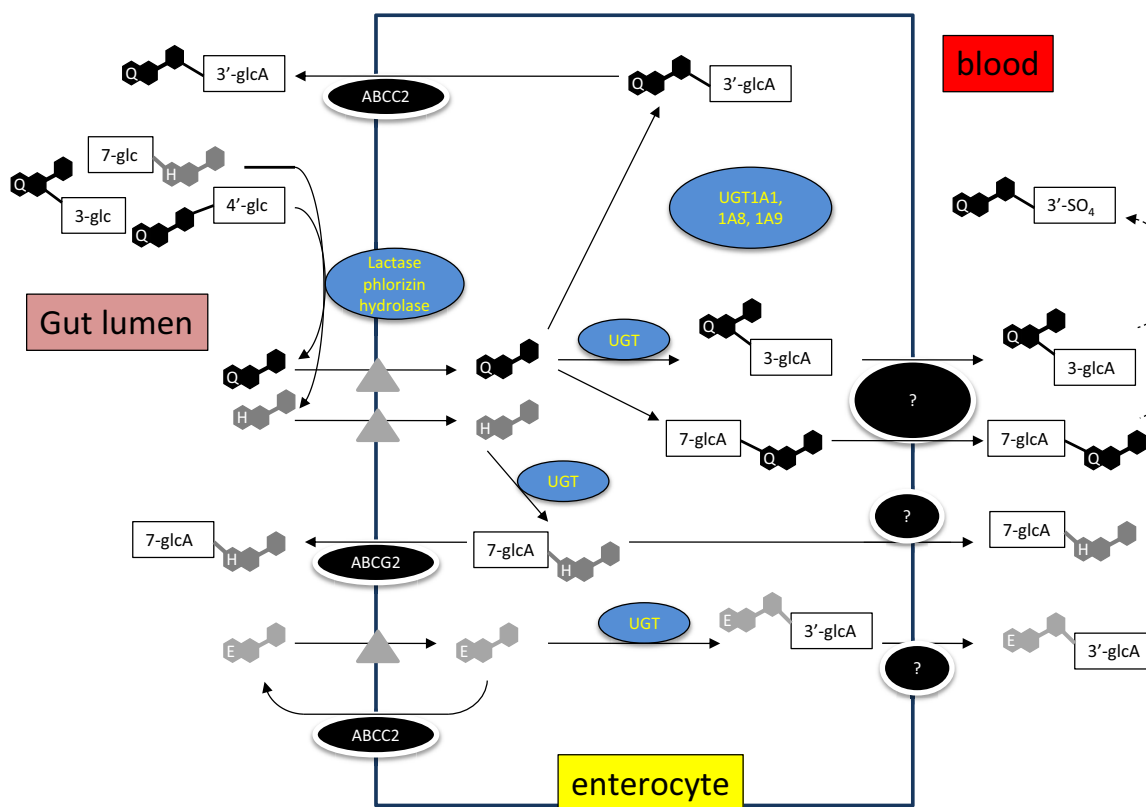


FIGURE 34.3 Pathways of absorption of the flavonoids, quercetin, hesperetin, and epicatechin. Quercetin (Q) is glucosylated (glc) in some foods and is deglycosylated by lactase-phlorizin hydrolase.^{12,13} The resulting quercetin aglycone passively diffuses across the apical membrane of enterocytes. Hesperetin-7-O-glucoside also follows the same pathway (hesperetin aglycone is shown by the letter H). Once inside the cell, quercetin and hesperetin can be conjugated with glucuronic acid by the action of defined isoforms of UDP-glucuronosyl transferases (UGT): for hesperetin, 1A1 preferentially glucuronidates position 7, whereas isoforms 1A7, 1A8, and 1A9 favor position 3'; for quercetin, 1A9 is more active at the 3 and 7 positions.^{14,15} The resulting conjugates can be absorbed by as yet unknown transporters on the basolateral membrane or effluxed back into the intestinal lumen via ABC transporters.^{16,17} Epicatechin is not glycosylated and enters the cell by passive diffusion, whereupon it can be effluxed by the ABCC2 transporter¹⁸ or alternatively can be glucuronidated and transported into the blood. It is likely that some sulfation and methylation also takes place in the enterocytes. The absorbed compounds can be further processed in the liver by further conjugation.¹⁹ Adapted from PKN 10th edition.

methyated forms.^{22,23} The pathways of absorption of some flavonoids are well understood, and for example, epicatechin is metabolized to 3'-methyl-epicatechin-5-O-sulfate, epicatechin-3'-O-sulfate, and epicatechin-3'-O-β-D-glucuronide as predominant species in the plasma, with epicatechin-7-O-β-D-glucuronide additionally present in urine.^{33,34} The chemical structures for quercetin³⁵ and hesperetin³⁶ circulating in blood in vivo are also known.

C. Excretion of Flavonoids

Flavonoids can be excreted in the urine or bile, and the amount in urine can be used as a biomarker of flavonoid intake.³⁷ The percentage excretion in urine is an indicator of the minimum amount that has been absorbed. Some flavonoids such as quercetin are well absorbed but only a few percent of the dose appears in urine, since the majority is excreted via the bile or catabolized by gut microbiota.

D. Microbial Metabolism of Flavonoids

Microbial metabolites contribute to the biological effects of dietary flavonoids.^{20,38} Flavonoids that are not absorbed in the small intestine reach the colon and are subject to extensive catabolism. Reactions occurring are dehydroxylation, reduction, ring cleavage, and hydrolysis of esters and sugars. As an example, procyanidins are hardly absorbed intact, but radiolabel studies in rodents have shown that ~80% of the microbial metabolites are absorbed and appear in the urine.³⁹ Some of the main products in the colon from flavonoid catabolism are shown in Fig. 34.4,^{22,40} using an anthocyanin as an example. Typical products of flavonoid catabolism include protocatechuic acid, phenylacetic acids, and phenylpropionic acids, where the latter are mainly in the dihydro form, especially dihydrocaffeic acid, dihydroferulic acid, phenylpropionic acid, and hydroxyphenylpropionic acid.

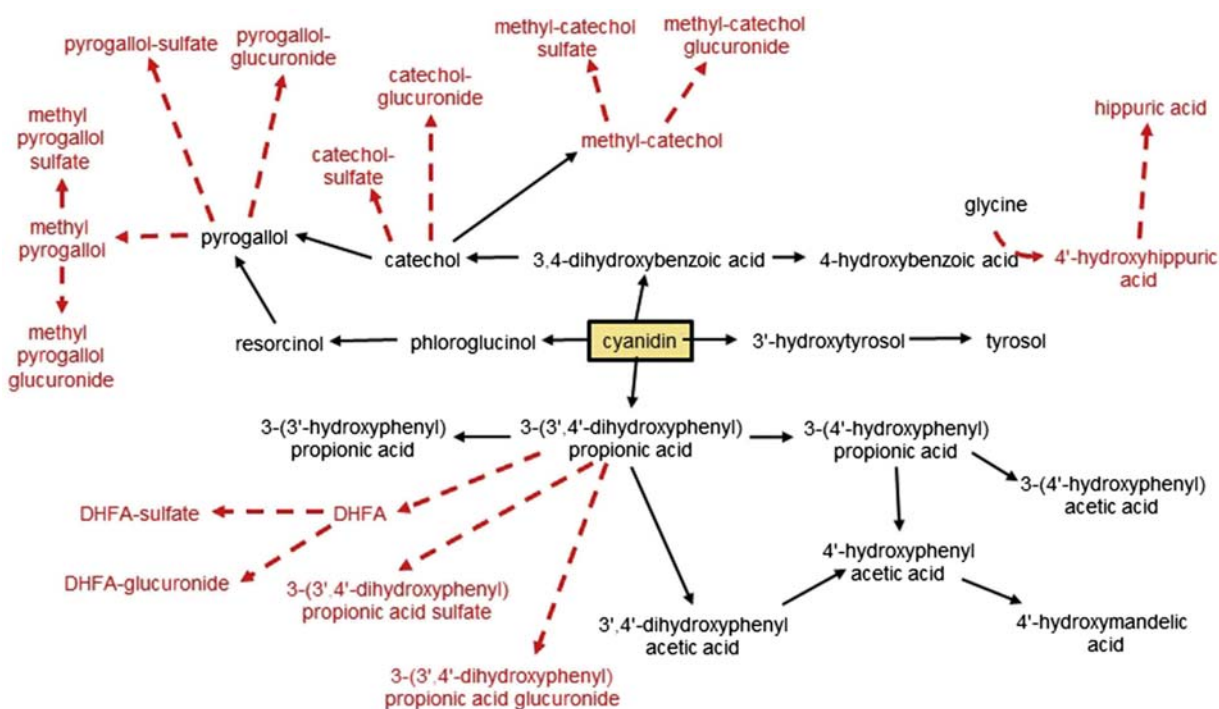


FIGURE 34.4 Metabolism of the flavonoid cyanidin by the gut microbiota. Cyanidin reaching the colon is converted to a range of lower molecular weight compounds by the gut microbiota (*solid arrows*). These enter the body across the colonocytes and circulate in the blood and are ultimately conjugated by tissues such as the liver (*dashed arrows*) and then excreted by the kidneys into the urine. Data are from Refs. 22,40.

Some of these compounds can reach high concentrations (mM) in fecal water.⁴¹ The microbial metabolism of cyanidin, shown in Fig. 34.4, illustrates the complexity of the reactions. Microbial metabolism is followed by absorption of the products across the colonocytes and subsequent conjugation in the body to give conjugated microbial metabolites that are ultimately excreted in the urine.

III. ROLE IN CELL AND ORGAN FUNCTION AND IN HUMAN HEALTH

A. Epidemiology and Meta Analyses

Historically, polyphenols were mostly of interest to plant scientists as they play many roles in plants and form part of the class of secondary metabolites or phytochemicals. For a detailed history of the flavonoid field, the reader is referred to a comprehensive review.⁴² There have now been hundreds of human intervention studies on flavonoids and flavonoid-rich foods, and these have been the subject of many reviews and meta analyses.^{43–47} The protective effect of flavonoid-rich foods on health has also been supported by many epidemiological studies comprising large total numbers of subjects. For example, tea flavonoids protect against stroke in epidemiological metaanalysis,⁴⁸ various flavonoids protect against risk of cardiovascular disease-induced death in postmenopausal women,⁴⁹ cocoa flavonoids increased flow-mediated dilatation and reduced

systolic and diastolic blood pressure,⁵⁰ and tea consumption reduced the risk of cardiovascular diseases.⁵¹ These types of analyses support a general and consistent protection against heart disease and biomarkers of cardiovascular risk and associated conditions.

B. Intervention Studies

Health benefits of polyphenols are likely derived from long-term, chronic exposure to relatively low doses, but demonstrating this in intervention studies is challenging. A 10% change in a biomarker for a year may be highly significant for health, but the necessary accuracy and precision is difficult to achieve for biomarker measurements. The approach to date has been to use higher doses in the short term, typically for 1–12 weeks.⁵² In addition, inter-individual variation also contributes substantially to biomarker variability and needs to be considered in dietary interventions.⁵³

For a flavonoid to exert a biological effect, it is necessary for the compound to reach the target site or to exert a secondary effect that subsequently affects the target site. From bioavailability studies, it is probable that flavonoids are not stored in the body, and so effects of polyphenols are more conceptually comparable to that of drugs, where an effect is produced and then the compound is excreted, rather than vitamins or minerals that are stored and can therefore be defined by a status measurement. There are many thousands of studies on the action of

flavonoids in vitro, utilizing a wide range of concentrations, but any effects measured in vitro only show potential mechanisms of action that must then be demonstrated in vivo. Since flavonoids are at low micromolar concentrations in vivo, it is usually considered that an effect should be seen in vitro at $<10\ \mu\text{M}$ in order for it to have potential dietary relevance.⁵⁴ Flavonoids can act as antioxidants by chelating iron, scavenging free radical species, donating an electron to an oxidant, scavenging peroxy radicals, and inhibiting lipid peroxidation, but these chemical characteristics are unlikely to be relevant in vivo.^{55–57} However, there are more specific and biologically relevant markers of oxidative stress⁵⁸ and mechanisms of action⁵⁹ that supersede the simple nonbiological antioxidant methods.

C. Effects in the Gut Lumen Related to Digestion of Carbohydrates

There is a high concentration of flavonoids in the lumen of the gastrointestinal tract after consumption of flavonoid-rich foods and a high concentration of lower molecular weight phenolics, derived at least partly from flavonoid catabolism, in the colon.⁴¹ A portion (100 g) of blackberries might typically contain 120 mg of flavonoids. If consumed with a small drink of $\sim 200\ \text{mL}$, then this equates to very approximately $\sim 1\ \text{mM}$ concentration of total flavonoids. This is of course diluted by digestive fluids, but the exposure of the gut tissues can therefore be several hundred micromolar, much higher than in blood.

After consumption of starch or sugar-rich foods, the resulting postprandial glucose spike can induce endothelial dysfunction⁶⁰ and oxidative stress.⁶¹ Postprandial starch digestion and sugar absorption are influenced by flavonoids, with subsequent attenuation of the glycemic response including lowered blood glucose and insulin levels. In vitro studies have shown that certain flavonoids inhibit α -amylase,⁶² interact with sugar transporters,⁶³ and inhibit sucrose-isomaltase.⁶⁴ This is supported by studies in vivo on volunteers, where effects on plasma glucose and/or insulin are also seen, involving various flavonoid-rich foods such as cranberry,⁶⁵ black tea,⁶⁶ dark chocolate,⁶⁷ and pomegranate.⁶⁸

D. Effects of Flavonoids on Risk of Type 2 Diabetes

There is evidence from intervention studies that certain flavonoids can affect markers of type 2 diabetes risk. For example, grape seed extract⁶⁹ and pycnogenol (a flavonoid-rich pine bark extract)⁷⁰ improved metabolic markers in type 2 diabetic patients. Kuna Indians with very high flavonol intake have lower rates of diabetes and heart disease,⁷¹ although not

all epidemiological studies show an effect of flavonoids at reducing diabetes risk.⁷² Regular consumption of tea is associated with reduced risk of developing type 2 diabetes in metaanalyses.^{46,73}

E. Effects of Flavonoids on Cardiovascular Disease Risk

There is now a substantial body of evidence, both in vivo and mechanistic data, to show that flavonoids affect risk factors for cardiovascular disease. Most intervention studies on flavonoids and cardiovascular disease have involved surrogate markers, such as blood pressure, endothelial function, LDL/HDL cholesterol, and chronic inflammation. A Cochrane review summarized the effects of tea, after 3–6 month intervention, as lowering blood pressure, lowering LDL cholesterol, not affecting HDL cholesterol, and with no side effects, with a grading of the evidence as low/moderate quality.⁷⁴

Effects of flavonoids on endothelial function

One of the important indicators of cardiovascular risk is endothelial function,⁷⁵ and dysfunction is increased by factors such as a high-fat diet.⁷⁶ Protective effects on endothelial function by flavonoids have been widely reported, and the mechanisms involved are summarized in Fig. 34.5.^{77–82} Epicatechin, and especially its metabolite, methylepicatechin, inhibits NADPH oxidase,⁷⁹ inhibits arginase,⁸³ and modulates nitric oxide concentration in endothelial cells, affecting cellular pathways leading to vasodilatation and improvement of endothelial function. The flavanols including epicatechin in dark chocolate improved endothelial function in hypertensive patients,⁸⁴ young soccer players,⁸⁵ and in smokers,⁸⁶ and green tea flavanols showed a similar effect in healthy women.⁸⁷ Improvement of endothelial function is seen in vivo in volunteers by red wine,⁸⁸ tea⁸⁹ and cocoa,⁹⁰ and by pure epicatechin.⁹¹ Other complementary mechanisms also operate within the cell. Xanthine oxidase produces superoxide, and this enzyme is inhibited by quercetin metabolites found in vivo.⁷⁸ When given chronically over 4 weeks, quercetin (as the more bioavailable form, quercetin-3-O-glucoside) improved endothelial function and reduced inflammation⁹² but did not affect flow-mediated dilatation or insulin resistance.⁹³ Quercetin decreased plasma uric acid in mildly hyperuricemic males⁹⁴ but did not affect blood pressure in normal weight volunteers^{95,96}.

Effect of flavonoids on blood pressure

High blood pressure is an important risk factor for cardiovascular disease⁹⁷ and is decreased by flavonoid and flavonoid-rich foods in various subjects, including glucose-intolerant, hypertensive patients.^{50,98} The mechanism

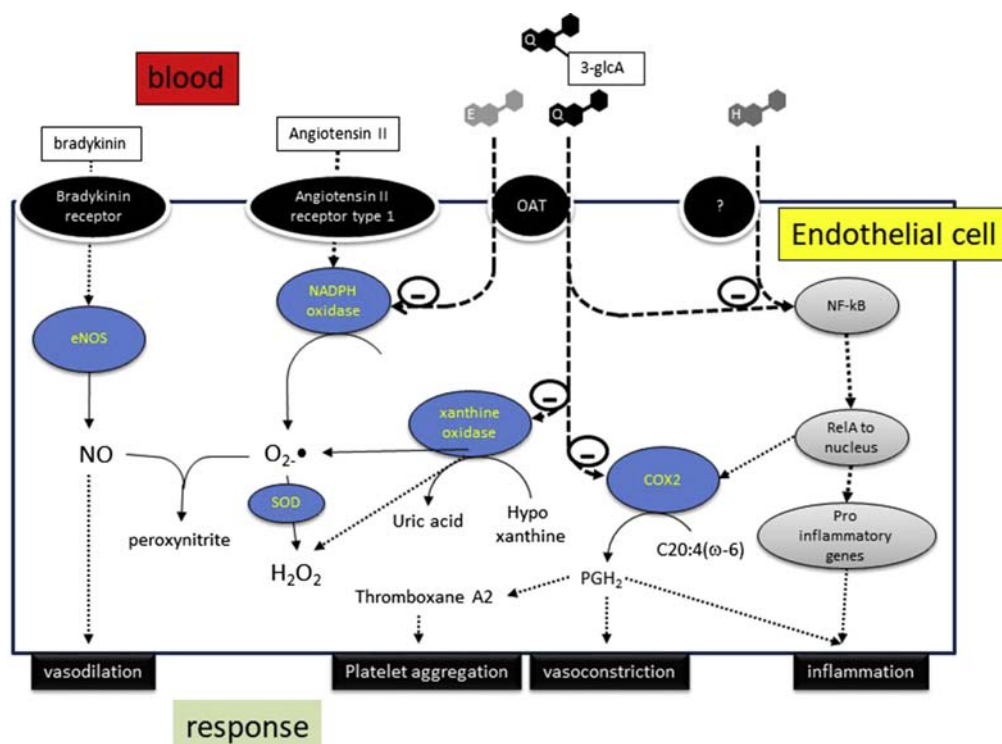


FIGURE 34.5 Effect of flavonoids on endothelial cell signaling. Several compounds in blood can affect endothelial cell function. Flavonoids and their conjugates/metabolites enter cells via OAT transporters,⁷⁷ where they can inhibit intracellular oxidative enzymes such as xanthine oxidase,⁷⁸ NADPH oxidase,⁷⁹ NF-κB,⁸⁰ and cyclooxygenase 2 (COX-2).⁸¹ Modification of these targets affects processes related to cardiovascular disease risk, such as vasodilation and endothelial function. Additional details can be found in a study by Kerimi A and Williamson G.⁸² Adapted from PKN 10th edition.

of action is shown in Fig. 34.5 and may also include inhibition of angiotensin-converting enzyme.⁹⁹ In many studies, regular consumption of cocoa, or a flavanol extract of cocoa, reduced blood pressure, blood cholesterol, F2-isoprostanes, and susceptibility of LDL to oxidation.^{50,100–106}

Effect of flavonoids on platelet function

Platelet function is another risk factor for cardiovascular disease that is affected by flavonoids.¹⁰⁷ There was a suppressive effect of cocoa flavanols on platelet reactivity in human subjects.¹⁰⁸ Red grape juice showed more effect than citrus juices on platelets in volunteers, suggesting that flavanols are more effective than flavanones¹⁰⁹; flavanols (quercetin) are also active on platelets after oral consumption in vivo.¹¹⁰

Effect of flavonoids on inflammation

Cyclooxygenase 2 (COX-2) is a key player in inflammation, and inhibition of COX-2 improves endothelial function,¹¹¹ leading to reduced inflammation, improved platelet activity and less vasoconstriction. Some flavonoids are potent COX-2 inhibitors at levels of both enzyme activity and transcription.⁸¹ Some epidemiological studies

support an effect of flavonoids against chronic inflammation, and there was an inverse association of flavonoid intake with C-reactive protein (a marker of inflammation) in a US population.¹¹² The effect of flavonoids on inflammation in humans has been reviewed.^{113,114} Some studies however show only a weak effect¹¹⁵ or no effect.¹¹⁶

Effect of flavonoids on lipid and cholesterol metabolism

Cardiovascular disease is also influenced by lipid metabolism, and LDL oxidation and cholesterol levels are well-established markers of risk.¹¹⁷ Effects on lipid metabolism include beneficial effects on cholesterol and LDL oxidation in humans and have been reviewed.²¹

F. Effect of Flavonoids on Detoxification and Carcinogenesis

Cells in vivo are continuously subject to stresses from endogenous metabolism as well as exogenous compounds, and many defense mechanisms are in place for protection; detoxification reactions are important defenses in this respect. Phase 2 enzymes catalyze the conjugation of xenobiotics, toxins, and carcinogens,

improving their solubility and excretion, and induction of these enzymes has been proposed to reduce the risk of carcinogenesis.¹¹⁸ In vitro, flavonols, especially quercetin, induce cellular antioxidant defenses via nrf2 action and gene regulation,^{119,120} and green tea–galloylated catechins also induce antioxidant defenses.¹²¹ There are very few reported studies on flavonoids and carcinogenesis in human intervention studies, and these studies are often confounded by interindividual variation.¹²² There are some epidemiological studies that show a protection, such as for black tea reducing the risk of advanced stage prostate cancer¹²³ and green tea lowering breast cancer risk,¹²⁴ but mostly the evidence for reduction in the risk of cancer by flavonoids is ambiguous^{125–128} or shows no effect.^{129,130}

IV. DIETARY RECOMMENDATIONS

There are no official recommendations for intake of flavonoids in the United States and Europe, although the idea of introducing dietary requirement advice has been mooted several times, but notably China, Russia, and South Korea are beginning to take steps to provide such information. One of the issues is that there are several different flavonoids of importance to health in the diet and so making recommendation for each compound individually would be very difficult. One solution could be to base recommended intakes based on a number of portions of fruit and vegetables per day. It is likely that there will be developments in this space in the future.

RESEARCH GAPS

There is still a need for more human intervention studies, especially using equivalent experimental foods containing a high content of flavonoids compared to a suitable placebo and for human intervention studies using intermediate biomarkers of chronic exposure and effects. In vitro experiments need to be correctly designed and appropriately interpreted, especially using combinations of flavonoids, conjugates, and microbial metabolites that could be found in vivo. In addition to reducing the risk of cardiovascular disease and type 2 diabetes, work is needed to establish if there is an effect of flavonoids on cognitive function and carcinogenesis, taking a holistic approach to healthy aging. Realistic claims and appropriate interpretation are of the utmost importance based on evidence from both in vitro data and from intervention studies.

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DIETARY SUPPLEMENTS

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SUMMARY

About half of adults and one-third of children and adolescents in the United States consume dietary supplements, which contain vitamins, minerals, herbs, and other dietary substances. However, US law limits regulatory oversight by the US Food and Drug Administration. As a result, supplement products may reach the market with limited scientific backing and potential quality concerns.

These facts, combined with the more than 50,000 dietary supplement products on the market, underscore the importance of rigorously conducted scientific research on supplements to understand their efficacy, safety, and quality. Continued development of validated biomarkers for measuring health outcomes associated with supplement use and analytical methods for characterizing supplement ingredients is also important, as is additional research on nonnutritive bioactive ingredients. Reliable sources of information on dietary supplements can help researchers, clinicians, and consumers understand the current state of knowledge and make informed decisions about their use.

Keywords: Bioactives; Botanicals; Dietary supplements; Minerals; Nutrients; Research challenges; Supplement efficacy; Supplement quality; Supplement safety; Vitamins.

I. INTRODUCTION

Dietary supplements are manufactured products containing one or any combination of ingredients from natural sources (such as plants, animals, and microorganisms) that can be extracted from these sources or chemically synthesized. These products are available and sold around the world, and they have many names, including “food supplements,” “nutraceuticals,” “therapeutic goods,” and “herbal medicines.” Different countries have different regulatory structures governing the kinds of dietary supplements available, to whom and where they may be sold, and the types of health-related claims manufacturers may make on their labels and in marketing materials.¹ Melatonin, for example, is a dietary supplement in the United States, a “natural health product” in Canada, but a prescription medicine

in Australia. Similarly, dehydroepiandrosterone is a dietary supplement in the United States but a controlled substance in other countries.

In the United States, the Dietary Supplement Health and Education Act (DSHEA) of 1994 defines dietary supplements as products “intended to supplement the diet” that contain one or more vitamins; minerals; herbs or other botanicals; amino acids; other dietary substances; or “a concentrate, metabolite, constituent, extract, or combination of” these ingredients.² The DSHEA also specifies that dietary supplements are intended for ingestion, but not as conventional foods or the only items in a meal. Anyone may purchase them without a prescription. In the United States, dietary supplements include macro- and micronutrients, botanicals (herbs), phytochemicals (such as lycopene and isoflavones from plants), zoochemicals (such as

creatine and bee pollen from animals), and other ingredients (such as probiotics and glucosamine). Dietary supplements typically contain nutritive and nonnutritive bioactive ingredients.

Unlike Australia, Canada, and many European Union countries, the United States does not require manufacturers to disclose the dietary supplements they market to government authorities.¹ Therefore, the number of unique dietary supplement products in the US market is not known; estimates range from more than 50,000 to more than 80,000.³

This chapter addresses dietary supplements as defined, marketed, and regulated in the United States. The chapter first describes dietary supplement use and reasons for their use in the United States, followed by a discussion of the three criteria—efficacy, safety, and quality—used to evaluate supplements. We illustrate the use of these criteria to evaluate the value of three supplement products available in the US market: the omega-3 fatty acids (omega-3s) eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), ginkgo, and probiotics. We then focus on the challenges of studying the value of dietary supplements and list resources that provide additional information on these products. The chapter concludes with some final considerations for investigators who study supplements and health care providers who might recommend these products to their patients.

II. DIETARY SUPPLEMENT USE

About half of adults and one-third of children and adolescents in the United States take one or more dietary supplements on a regular or daily basis.^{4,5} In 1971, when the National Health and Nutrition Examination

Survey (NHANES) began collecting nationally representative data on dietary supplement use, 32.9% of adults reported taking a dietary supplement.⁶ In the NHANES surveys between 1999 and 2012, 52% of US adults reported taking a supplement, although the products they used varied over time.⁷ For example, between 1999–2000 and 2011–2012, the rates of multivitamin–multimineral supplement use among adults decreased from 37% to 31%, but use of vitamin D supplements increased from 5.1% to 19% and of fish oil supplements from 1.3% to 12%. Vitamin and mineral products account for about half of total supplement use.¹

Based on NHANES 2011–2014 data, the proportion of adults using any supplement increases with age; about one-third of those aged 19–30 years take dietary supplements, as do three-quarters of adults aged 71 years and older (see Fig. 35.1).^{4,5} Women (58.6%) are more likely to take supplements than men (45.4%). However, among children aged 18 years and younger, boys are as likely to take supplements (32.2%) as girls (32.7%).⁵ Users are more likely to be non-Hispanic white (58.2%) than non-Hispanic Asian (53.5%), non-Hispanic black (40.3%), or Mexican or Hispanic American (35.3%).⁴ Two-thirds (65.6%) of adult users take one to two supplements daily; the remaining one-third takes more than two products every day. Supplement use increases with educational attainment: 58.1% of those with more than a high school education use supplements compared with 37.8% of those with less formal education. Users are more likely to take dietary supplements if they describe their health as excellent or very good (57.8%) than fair or poor (49.0%). Users are also more likely than nonusers to engage in healthful behaviors, such as not smoking and being physically active, and they are more likely to be financially and food secure.^{4,8}

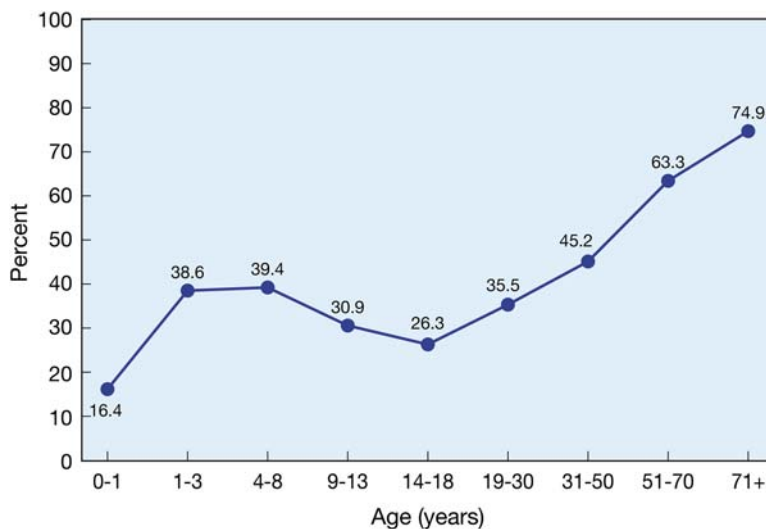


FIGURE 35.1 Prevalence of dietary supplement use by age, NHANES 2011–2014.^{4,5}

The extent of dietary supplement use can also be measured indirectly based on the amount of money spent to purchase these products. According to the *Nutrition Business Journal*, sales of dietary supplements in the United States in 2018 totaled \$46 billion or more than \$140 per year for every American.⁹ However, because this estimate includes products not labeled as dietary supplements, such as homeopathic medicines and meal replacement products, we calculate that a more accurate estimate of actual supplement sales may be about 80%–85% of this total. The *Nutrition Business Journal* also estimates that 2017 global dietary supplement sales totaled \$128 billion, suggesting that US sales account for approximately one-third of expenditures on supplements worldwide.¹⁰ In 1994, approximately 600 manufacturers produced about 4000 supplement products with total sales of at least \$4 billion in the United States.¹¹ Today, more than 50,000 products, produced by more than 6000 facilities, are available in the United States.¹ Some of the best selling dietary supplements are shown in Table 35.1.⁹

According to the 2007–2010 NHANES surveys, the main reasons adults report taking supplements are to improve overall health (45.0%), to maintain health and stay healthy (32.8%), to promote bone health (25.2%), to

supplement the diet (22.0%), and to prevent health problems (20.4%).⁸ Adults also take supplements to treat or manage health problems, conditions, and diseases.^{12–14} Manufacturers by law may not claim that supplements prevent, manage, or treat a disease, but consumers do use them, and health professionals may recommend them, for these purposes.

Dietary supplement use is largely self-directed. Only 23% of the supplements used by adults in the United States are recommended by a health care provider.⁸ However, three of four patients report that they have disclosed their use of supplements to their primary care physician voluntarily or when asked.¹⁵

III. DIETARY SUPPLEMENT EFFICACY, SAFETY, AND QUALITY

Investigations of dietary supplements (or usually of just one or two of their ingredients) assess their efficacy in providing the health benefits of interest or that manufacturers claim they have as well as their safety. However, supplement researchers should also verify the quality of the ingredient(s) they use in their studies to ensure that they actually consist of the ingredients and amounts listed on the product label.

A. Efficacy

The type of research needed to determine whether dietary supplements “work” and are safe depends on the questions asked. One of the many reasons people take supplements is to supplement the diet—as “nutritional insurance”—to help ensure that they obtain recommended amounts of vitamins and minerals when they are not sure that what they eat provides adequate amounts. In fact, the most commonly taken supplement in the United States is a multivitamin–multimineral product containing many nutrients.⁹ No research is needed to know that taking nutrient-containing supplements increases nutrient intakes, but studies can assess the contribution of dietary supplements to nutritional adequacy in different populations.¹⁶

Because the most common reasons for taking dietary supplements in the United States are to improve or maintain health, much of the research on these products focuses on whether they can achieve these goals or reduce the risk of chronic disease.⁸ However, the published research has faced several challenges and has some limitations, and much of this research has produced mixed findings. As a result, the amount of reliable information on the efficacy of various dietary supplements is quite limited.

Strong evidence does show that some dietary supplements are beneficial for certain indications. For example,

TABLE 35.1 Best-selling dietary supplements in the United States in 2018.⁹

Dietary supplement	2018 sales (millions of dollars)
Nutrients	
Multivitamins	6,228
B vitamins	2,316
Vitamin C	1,289
Calcium	1,106
Magnesium	1,010
Botanicals	
Turmeric	328
Cranberry	176
Green tea	154
Maca	150
Saw palmetto	148
Specialty	
Probiotics	2,267
Fish/animal oils	1,207
Coenzyme Q10	673
Glucosamine/ chondroitin	665
Melatonin	421

folic acid reduces the risk of neural tube defects when taken by pregnant women in the periconceptual period.¹⁷ For this reason, all women capable of becoming pregnant should consume 400 mcg of folic acid per day from supplements (and/or fortified foods) in addition to the folate they consume naturally from foods.¹⁸ Research also shows that for people who have intermediate age-related macular degeneration (AMD) or advanced AMD in one eye, taking supplements containing high doses of zinc, vitamin C, vitamin E, and other antioxidants, can delay the progression to advanced AMD in both eyes.^{19,20}

In other cases, the evidence of efficacy is more nuanced. For example, several studies have shown that vitamin D supplementation could reduce the incidence of cancer and cardiovascular disease (CVD) in men and women with a known risk of these diseases.²¹ However, many of these studies had small samples and short durations as well as other design flaws, and they often produced mixed findings.^{21,22} In addition, these studies did not assess the effects of vitamin D supplements on the primary prevention of cancer or CVD. Therefore, a large clinical trial on vitamin D was conducted. The Vitamin D and Omega-3 Trial (VITAL) included 25,871 men aged 50 and older and women aged 55 and older with no history of CVD or cancer.²³ Participants took either 2000 IU vitamin D/day (with or without an omega-3 supplement) or a placebo for a median of 5.3 years. The results for the primary outcome measures—incidence of total invasive cancer and major cardiovascular events—did not differ between the two groups.

These findings do not necessarily mean that taking supplemental vitamin D has no health benefits because it might be valuable for some subgroups, although the identity of these subgroups is not clear. Given that taking supplemental vitamin D—or at least up to 100 mcg (4000 IU) per day—is safe, it might seem reasonable to many people to consider supplementation.²² However, most people are unlikely to benefit from this strategy.

Like the VITAL trial, many other trials of dietary supplements have produced no (or very little) evidence of benefit.^{24–27} For many botanical supplements in particular, evidence of efficacy is simply lacking. In some cases, evidence from a long history of traditional use and the results of laboratory studies, animal research, and observational studies provide promising suggestions of efficacy for intended purposes. However, without evidence from well-designed clinical trials in humans, it is not possible to be certain that a supplement is of benefit.

B. Safety

The safety of a dietary supplement depends on such factors as its ingredients and their doses; the age, sex, life stage, and medication use of the supplement user;

and the product's quality. The safety of nutrient-containing dietary supplements largely depends on the amounts of their ingredients. Vitamins and minerals are considered safe at recommended intakes, which vary by age, sex, and life stage.

The National Academies of Sciences, Engineering, and Medicine has established Tolerable Upper Intake Levels (ULs) for most vitamins and minerals, and intakes higher than these amounts can cause health problems. For example, daily intakes of up to 45 mg of iron are safe for most adults, but larger amounts can cause nausea, vomiting, and decreased zinc absorption.²⁸ Extremely high doses can cause organ failure, coma, and even death. However, supplement manufacturers are not required to limit nutrient levels in their products to established ULs.¹¹ Some supplements contain, for example, doses of vitamin B6 above the UL of 100 mg, even though such intakes could cause sensory neuropathy and other adverse effects.¹⁸ Other nutrients, such as vitamin B12, have not been shown to cause harm even at high intakes and therefore do not have ULs.

Clinical trials have identified safety concerns for certain dietary supplement ingredients, including nutrients. A classic example was the Alpha-Tocopherol, Beta-Carotene (ATBC) Cancer Prevention Study in 29,133 male smokers aged 50–69 years.²⁹ Previous studies had suggested that higher intakes of alpha-tocopherol and beta carotene might decrease the risk of lung cancer. However, in the ATBC study, men who took 20 mg of beta carotene for a median of 6.1 years had an 18% higher incidence of lung cancer compared with those who did not take beta carotene. Similarly, in the Selenium and Vitamin E Cancer Prevention Trial, daily supplementation with 400 IU of vitamin E (in the form of all *rac*-alpha-tocopheryl acetate) for a median of 7 years increased the risk of prostate cancer in healthy men aged 50 and older.³⁰ These unexpected findings underscore the importance of conducting large, well-designed clinical trials of dietary supplements in diverse populations.

Safety concerns have also been raised about some nonnutrient dietary supplement ingredients, such as botanicals. Many of these concerns originated in case reports published in the literature or adverse event reports submitted to the US Food and Drug Administration (FDA). These reports cannot provide proof that supplements caused the adverse effects or identify the responsible agent(s), but they do signal a possible safety risk. Numerous herbs—green tea extract, kava, and black cohosh, for example—have been associated with liver toxicity.³¹ Identifying the cause of this toxicity is an important area of ongoing research, but it is complicated by the fact that herbs and other botanical preparations pose unique challenges related to plant identification, characterization, and standardization.^{32–34} Other issues,

such as inaccurate product labeling and contamination, can also contribute to reported adverse effects.³⁵ Investigators conducting clinical trials of herbs and other dietary supplements should characterize the ingredient(s) being studied. Investigators should also monitor safety carefully by recording adverse events throughout the trial and measuring appropriate safety biomarkers; for example, they might conduct liver function tests at both baseline and after the intervention.

Another category of safety concerns consists of interactions between dietary supplements and medications. The interaction between vitamin K and the blood thinner warfarin (Coumadin) is well known, but many others can occur.³⁶ For example, St. John's wort reduces the efficacy of some antivirals, antidepressants, immunosuppressants, contraceptives, and other medications by inducing the activity of xenobiotic metabolizing enzymes and transporters.³⁷ Other interactions, such as between proton pump inhibitors and dietary vitamin B12 absorption, might adversely affect nutritional status.³⁸ These interactions demonstrate the importance of open communication between patients and their health care providers about dietary supplement use.

C. Quality

Dietary supplements contain ingredients from around the world. Some products consist of chemically synthesized single nutrients, such as vitamin C, for which assurance of identity and potency should be simple to document. But most supplements on the market contain multiple ingredients, such as nutrients, botanicals, and other bioactive ingredients. Ensuring the identity, purity, and quantity of each ingredient and combining them properly into finished products that maintain their stability for months or years requires great attention to manufacturing and quality control procedures. Unfortunately, manufacturers do not always ensure supplement product quality, and some products do not contain the ingredients and amounts their labels list and/or contain ingredients not listed. Improperly manufactured products might not be safe or effective for their claimed purposes.

Before starting a study, researchers who study dietary supplements must independently evaluate the ingredient(s) in the product they are using to confirm their identity, potency, purity, and absence of contaminants and to ensure the study's scientific rigor and reproducibility. The National Institutes of Health (NIH), for example, often requires investigative teams submitting grant applications to demonstrate that they have the analytical expertise required to ensure the precise identity and integrity of the natural product (substance of natural origin or its synthetic alternative) they propose to test.³⁹ Considerations for selecting and characterizing

supplement products for research studies have been published.^{39,40}

IV. REGULATION, OVERSIGHT, AND ASSESSMENT OF DIETARY SUPPLEMENTS IN THE UNITED STATES

Under US law, dietary supplements are a category of "foods," and FDA has regulatory responsibility for them.⁴¹ Unlike for prescription and over-the-counter drugs, which must be approved by FDA before they can be marketed, dietary supplements do not require premarket FDA approval. The Federal Trade Commission polices health and safety claims made in advertising for dietary supplements.⁴²

A. Efficacy Considerations

FDA is not authorized to approve dietary supplements for effectiveness before they are marketed. Although supplement manufacturers are not required to give FDA evidence that can be used to substantiate the efficacy or value of their products, FDA does require them to have evidence in their files that their label claims are truthful and not misleading.⁴¹ Furthermore, no dietary supplement may be labeled or otherwise promoted as a treatment or cure for (or, with few exceptions, for prevention of) any disease; such claims make the product an unapproved, and therefore illegal, drug. Only companies marketing FDA-approved prescription and over-the-counter medications may make disease-related claims about these products.

Sellers of dietary supplements are permitted to make three types of claims about the value and efficacy of their products: nutrient content claims, health claims, and structure/function claims (see [Table 35.2](#)).^{43,44}

B. Safety Considerations

Manufacturers choose the ingredients and amounts to include in their dietary supplements and are responsible for ensuring that their products are safe.⁴¹ They must notify FDA before selling a product only if it contains a "new dietary ingredient" that was not included in supplements before October 15, 1994.⁴⁵ In this situation, the manufacturer must submit documentation to FDA on why the ingredient is "reasonably expected to be safe." Once an ingredient or product is on the market, FDA is responsible for showing that the ingredient or product is unsafe before the agency can restrict use or remove it from the market. FDA banned ephedra in supplements in 2004, for example, because this ingredient presented "an unreasonable risk of illness or injury."⁴⁶

TABLE 35.2 Types of claims allowed on dietary supplement labels.^{43,44}

Claim	Description and examples
Nutrient content claim	Characterizes the level of a nutrient in the product. For example, a label on a vitamin E supplement providing 200 mg might describe the ingredient as “high-potency vitamin E.” FDA has identified the nutrient content claims that companies may use and the nutrient levels in the product needed to use these claims.
Health claim	Describes an ingredient’s effect on reducing the risk of, or preventing, a disease or health-related condition. A health claim requires “significant scientific agreement” that the claim is true as well as FDA evaluation and authorization before it can be used. An example of a health claim is “adequate calcium and vitamin D throughout life, as part of a well-balanced diet, may reduce the risk of osteoporosis.” FDA has authorized fewer than two dozen health claims. However, FDA allows qualified health claims on dietary supplements; these claims are supported by less scientific evidence than health claims. Qualified health claims must be approved by FDA and accompanied by a disclaimer that explains the level of the scientific evidence supporting the relationship. An example of a qualified health claim is “selenium may reduce the risk of prostate cancer. Scientific evidence concerning this claim is inconclusive. Based on its review, FDA does not agree that selenium may reduce the risk of prostate cancer.”
Structure/function claim	Describes how an ingredient might affect a normal bodily structure or function. Examples include “supports the immune system” and “maintains cell integrity.” Structure/function claims may not explicitly or implicitly link the claimed effect to a disease or health state leading to a disease. Furthermore, these claims do not need to be authorized by FDA. However, structure/function claims must be accompanied by a statement saying that FDA has not evaluated the claim and that the product “is not intended to diagnose, treat, cure, or prevent any disease.” Structure/function claims are by far the most common type of health-related claims used by dietary supplement sellers.

FDA requires manufacturers and distributors of dietary supplements to share any serious adverse event reports they receive with FDA.⁴⁷ Serious adverse events are those that lead to death, a life-threatening experience, inpatient hospitalization, a persistent or significant disability or incapacity, or a congenital anomaly or birth defect. Companies must also maintain records of nonserious event reports they receive for at least 6 years and must make these records available to FDA on request. In addition, health professionals and consumers may report actual and suspected adverse effects from the use of any supplement to FDA’s MedWatch system.⁴⁸

FDA posts dietary supplement safety alerts and advisories on its website.⁴⁹ In April 2018, for example, FDA warned consumers that serious adverse effects, including death, could result from use of pure, highly concentrated caffeine supplements. FDA also makes public the warning letters it sends to supplement companies identifying such matters as poor manufacturing practices or unapproved claims for a product’s effects.⁵⁰

C. Quality Considerations

In 2007, FDA began mandating the use of Current Good Manufacturing Practices (cGMPs) in the manufacture, packaging, labeling, and storage of dietary supplements.^{51,52} The cGMPs impose minimum standards for quality control procedures throughout the manufacturing process, for example, including verification of individual ingredient identity before manufacture as well as of the

composition of finished products. These regulations are intended to ensure that products contain only the expected ingredients in the intended amounts, contain no contaminants (such as pesticides, glass, or heavy metals), and are properly packaged and labeled.

In its inspections of dietary supplement manufacturing facilities, FDA often finds noncompliance with cGMPs. In 2016, for example, common infractions FDA identified included “not establishing or following written procedures for quality control,” “not preparing or following a written master manufacturing record (i.e., a recipe) for each batch,” “not establishing product specifications for the identity, purity, strength, and/or composition of the finished dietary supplement,” and “not conducting at least one appropriate test or examination to verify the identity of a dietary ingredient.”^{50,53} Smaller manufacturing firms are more likely to fail to use cGMPs than larger, more established firms that have the resources and will to implement these standards.⁵⁴

ConsumerLab.com reports, based on its laboratory testing of thousands of dietary supplements it purchased from the retail market over 20 years, that 22% of these products are of poor quality.⁵⁵ Among the problems it has found are larger or smaller amounts of nutrients than reported on the product label, none of some nutrients listed on the label, poor-quality ingredients, spoiled oils in the product, heavy metal contamination, and (for pills) inadequate disintegration.

FDA requires supplements to provide 100% of the labeled amount of each ingredient until the product’s expiration date.⁵² In a US government investigation of

multivitamin–mineral supplements, the analytically verified content of several nutrients was substantially larger than the labeled amounts—for example, some products contained more than 20% more selenium and iodine and 10%–20% more calcium and folic acid than labeled amounts.⁵⁶ Ironically, the overage might reflect a quality control effort by manufacturers to compensate for losses during the product's shelf life.

In some cases, manufacturers adulterate dietary supplements accidentally or deliberately. The most common forms of adulteration are use of a less expensive ingredient to replace some or all the ingredients listed on the label or use of a pharmaceutical ingredient not listed on the label.⁵² For example, a saw palmetto extract might be adulterated with a vegetable oil that mimics authentic saw palmetto's fatty acid profile, or a supplement marketed for sexual enhancement might be adulterated with a Viagra-like drug.^{57,58} Supplements most likely to be adulterated with medicinal ingredients are those promoted for bodybuilding, weight loss, or sexual enhancement.⁵⁹ FDA maintains a website of "tainted products" marketed as dietary supplements.⁵⁸

Several third-party certification organizations assess the quality of marketed dietary supplements independently of FDA and make the results publicly available. In the United States, the three main certification organizations are the US Pharmacopeial Convention (USP), NSF International, and ConsumerLab.com.⁶⁰ Both the USP Dietary Supplement Verification Program and the NSF Product Certification Program test dietary supplements as finished products and conduct on-site audits of manufacturing facilities to ensure compliance with cGMPs.^{61,62} Each company lists verified products on its website, and the product label may carry the organization's certification logo. ConsumerLab.com purchases products from retail outlets and has their contents analyzed.⁵³ The organization's website lists all products that pass (and some products that fail) quality testing, and the product's label may carry the company's certification logo. Other certification organizations in the United States are the Banned Substances Control Group and the Natural Products Association's GMP Certification Program.^{63,64}

D. Dietary Supplement Labels

FDA has established regulations for dietary supplement labels. These labels must list certain information, including the following^{41,65}:

- A descriptive name of the product
- A statement that the product is a "supplement"
- The manufacturer's, packer's, or distributor's name and place of business
- A list of all ingredients, including any additives or processing aids

- A "Supplement Facts" panel that lists details on serving size, number of servings per container, and amounts of each ingredient provided by the supplement

The label may also make certain claims about a product, such as structure/function claims, but it does not need to provide information about the product's efficacy, safety, or quality.

For some ingredients, such as vitamins, minerals, and other essential nutrients, FDA has established a Daily Value (DV) to help consumers compare the amount of the ingredient in the product with approximate daily requirements for it.⁶⁶ For these ingredients, manufacturers must declare the amount per serving using units of measure specified by FDA and the percent DV. For ingredients, such as herbs, that do not have a DV, only the amount per serving must be listed, followed by a symbol, such as an asterisk, that refers to a footnote stating "Daily Value not established." Some dietary supplements contain proprietary blends that are often comprised of herbs or similar ingredients. Manufacturers of products containing these blends are not required to list the amount of each ingredient. Instead, they are required to list the total weight of all ingredients in the blend as well as the names of each ingredient in descending order by weight.

In 2016, FDA updated its labeling regulations for dietary supplements (and foods) to reflect new scientific information and to help consumers make better-informed choices about the supplements they consume.⁶⁷ The new regulations included changes in both the format and content of the information displayed on the Supplement Facts panel. FDA also established new DVs for many nutrients; eliminated the use of international units (IUs) as measures of nutrient content; and required use of new units of measure for vitamin A, vitamin D, vitamin E, and folate (see [Fig. 35.2](#)). For example, FDA changed the DV for vitamin E from 30 IU to 15 mg.

V. SAMPLE EVALUATIONS OF DIETARY SUPPLEMENT EFFICACY, SAFETY, AND QUALITY

This section describes the efficacy, safety, and quality of three dietary supplement ingredients: the omega-3s EPA and DHA, ginkgo, and probiotics. They are among the better known and studied ingredients, in addition to vitamins and minerals, in dietary supplements.

A. Omega-3 Fatty Acids EPA and DHA

EPA and DHA are long-chain omega-3s fatty acids. Cold-water fatty fish, such as salmon and sardines, and shellfish contain large amounts of these omega-3s.

Supplement Facts		
Serving Size 1 Gelcap Servings Per Container 100		
	Amount Per Serving	% Daily Value
Vitamin A (as retinyl acetate and 50% as beta-carotene)	900 mcg	100%
Vitamin C (as ascorbic acid)	90 mg	100%
Vitamin D (as cholecalciferol)	20 mcg (800 IU)	100%
Vitamin E (as dl-alpha tocopheryl acetate)	15 mg	100%
Thiamin (as thiamin mononitrate)	1.2 mg	100%
Riboflavin	1.3 mg	100%
Niacin (as niacinamide)	16 mg	100%
Vitamin B ₆ (as pyridoxine hydrochloride)	1.7 mg	100%
Folate	400 mcg DFE (240 mcg folic acid)	100%
Vitamin B ₁₂ (as cyanocobalamin)	2.4 mcg	100%
Biotin	3 mcg	10%
Pantothenic Acid (as calcium pantothenate)	5 mg	100%
Other ingredients: Gelatin, lactose, magnesium stearate, microcrystalline cellulose, FD&C Yellow No. 6, propylene glycol, preservatives (propylparaben and sodium benzoate).		

FIGURE 35.2 Example of updated supplement facts panel for a multivitamin product.⁶⁷

Dietary supplements containing EPA and DHA are being studied for many conditions, including CVD, cognitive decline, dementia, Alzheimer's disease, cancer, and rheumatoid arthritis.⁶⁸ This brief review focuses on EPA and DHA supplementation to prevent CVD.

Efficacy

Dietary supplements containing EPA and DHA improve blood lipid levels, particularly those of triglycerides.⁶⁹ Regular consumption of fish, and correspondingly higher dietary intakes and plasma levels of EPA and DHA, is linked to a lower risk of coronary heart disease and heart failure. Supplements of EPA and DHA do not appear to have these desired cardioprotective effects, although their effects might vary based on dietary omega-3 intakes and the use of cardioprotective medications, such as statins.⁷⁰

For example, in a randomized clinical trial of over 12,500 participants (mean age 64 years) in Italy with multiple CVD risk factors or atherosclerotic vascular disease, supplementation with 1 g/day omega-3s (including at least 85% EPA and DHA) for a median of 5 years failed to reduce the risk of death from cardiovascular causes or hospitalization for any cardiovascular cause compared with placebo.⁷¹ Similarly, a metaanalysis of 10 randomized clinical trials that included 77,917 patients (mean age 64 years) with a history of coronary heart disease or stroke or a high risk of CVD found that omega-3 supplementation

(376–2550 mg EPA and DHA/day) for at least 1 year did not reduce the risk of fatal coronary heart disease, nonfatal myocardial infarction, stroke, or other major vascular events.⁷² In the VITAL trial, 25,871 men aged 50 and older and women aged 55 and older with no previous heart attacks, strokes, or cancer were randomized to receive a fish oil supplement (1 g/day containing 460 mg EPA and 380 mg DHA) with or without 2000 IU/day vitamin D or a placebo for a median of 5.3 years.⁷³ The EPA and DHA supplement did not significantly reduce the rate of major cardiovascular events (a composite of myocardial infarction, stroke, and cardiovascular mortality).

Safety

Omega-3 dietary supplements are considered safe. The most commonly reported side effects, which are usually mild, include unpleasant taste, bad breath, heartburn, nausea, gastrointestinal discomfort, diarrhea, headache, and odoriferous sweat.^{74,75} Some experts have expressed concern that taking 1500 mg/day or more EPA and DHA might increase bleeding time by reducing platelet aggregation or that these doses might reduce immune function due to suppression of inflammatory responses.⁷⁶ However, other experts disagree.⁷⁷ Taking 3–6 g/day of supplemental fish oil does not alter anticoagulant status in patients taking warfarin.⁷⁸

Quality

The EPA and DHA in dietary supplements usually come from fish, krill, or algal oils. Depending on how they are extracted and processed, they might contain various forms of these fatty acids, including natural or reesterified triglycerides, free fatty acids, ethyl esters, or phospholipids. The VITAL trial, for example, used a supplement with the ethyl ester form, the form in most fish oil supplements, of EPA and DHA.⁷³ Although seafood can contain methyl mercury, a toxic heavy metal, omega-3 supplements contain no mercury because it is removed during processing and purification. ConsumerLab.com approved all 16 of the omega-3 supplements for quality that it reviewed in 2018.⁷⁹

B. Ginkgo

Ginkgo biloba (or simply ginkgo because there is only one species) is a tree whose leaves are processed into dietary supplements. Ginkgo is being studied to determine whether it improves cognition and mental performance, slows cognitive decline and Alzheimer's disease development, treats tinnitus and vertigo, and increases walking distance in people with peripheral arterial disease.^{80–82} Most research on ginkgo uses a leaf extract processed and standardized to contain 24% flavonol

glycosides and 6% triterpenes as bioactive ingredients and marker compounds. This product is manufactured by a German pharmaceutical company and known as EGb 761. This brief review focuses on use of ginkgo to prevent or treat cognitive decline, dementia, and Alzheimer's disease.

Efficacy

Prevention; In the largest clinical trial of ginkgo's ability to prevent cognitive decline, researchers gave 3069 adults at least 75 years old with normal cognition or mild cognitive impairment 120 mg of EGb 761 twice a day or a placebo for a median of 6.1 years.^{83,84} Ginkgo did not reduce the incidence of dementia or Alzheimer's disease or slow the rate of cognitive decline. One review of 36 clinical trials that used various doses of ginkgo (primarily EGb 761) to prevent dementia and cognitive impairment found no convincing evidence supporting such a preventive effect, although most of the trials were small and lasted less than 3 months.⁸⁰

Treatment; A review and metaanalysis of nine trials lasting 12–52 weeks found that compared with placebo, EGb 761 significantly improved cognition in 2372 patients with Alzheimer's disease or dementia, although the effects might not have been clinically significant because the supplements had no effect on activities of daily living; both cognition and activities of daily living were assessed using various rating scales.⁸⁵ In a randomized trial in France, 2854 adults aged 70 years and older who reported memory problems took 120 mg of EGb 761 or a placebo twice a day. After 5 years, the results showed no impact on progression to Alzheimer's disease.⁸⁶ An examination of 12 published systematic reviews of clinical trials of ginkgo found that more than 200 mg/day EGb 761 given to hundreds of participants with a diagnosis of Alzheimer's disease, vascular dementia, or both for at least 22 weeks seemed to improve cognitive performance and the ability to carry out activities of daily living.⁸⁷ However, the magnitude of the benefits was small to moderate, and the quality of the evidence was low to moderate.

Overall, clinical trial evidence to date provides weak support, at most, for using ginkgo to help prevent or treat dementia and Alzheimer's disease.

Safety

Ginkgo leaf extracts appear to be safe. In the clinical trials described above, reported side effects were not more frequent in those taking ginkgo versus a placebo. Occasional side effects reported by users are typically mild and include gastrointestinal upset, headache, dizziness, constipation, and heart palpitations.⁸⁸ Large amounts of ginkgo can cause restlessness, diarrhea,

nausea, vomiting, weakness, and reduced muscle tone. Although pharmacologic studies suggest that ginkgo can inhibit platelet aggregation and blood clotting, clinical trials have found that the supplements are not associated with more bleeding than placebo. Nevertheless, individuals taking antiplatelet or anticoagulant medications should check with their health care provider before taking ginkgo.⁸⁸

Quality

Ginkgo dietary supplements often do not contain the amount of ginkgo claimed on their labels. Ginkgo leaf extracts are expensive to prepare, and they are sometimes adulterated with less expensive mixes of ingredients containing flavonols, such as rutin (from buckwheat), quercetin, and kaempferol.⁸² [ConsumerLab.com](https://www.consumerlab.com) approved only 4 of the 10 ginkgo supplements it reviewed in 2018, and only 2 of 7 in 2008.⁸¹ When the US Department of Agriculture chemically evaluated 18 ginkgo supplements it purchased from retail stores in 2012, 7 were adulterated with rutin, quercetin, or another flavonol glycoside.⁸⁹ Although this adulteration is not a safety problem, users of adulterated ginkgo supplements do not obtain the labeled or desired amounts of ginkgo.

C. Probiotics

Probiotics are live microorganisms, primarily bacteria or yeast, that are intended to provide health benefits. They are categorized by their genus, species, subspecies (if applicable), and an alphanumeric strain designation (example: *Bifidobacterium animalis lactis* DN-173 010).⁹⁰ Probiotics mainly affect the gastrointestinal tract. For example, some of these organisms inhibit the growth of pathogenic microorganisms by fostering colonization resistance or reducing intestinal transit time. Probiotic dietary supplements are being studied for use in conditions such as diarrhea, irritable bowel syndrome, hypercholesterolemia, and atopic dermatitis. This brief review focuses on the use of probiotic dietary supplements to treat acute diarrhea.

Efficacy

The causes of acute infectious diarrhea include bacteria and viruses. Numerous clinical trials, mostly in infants and children, have investigated whether probiotics can help treat this condition, and the results are encouraging.⁹¹ In a systematic review of 10 trials in patients aged 1–48 months with diarrhea, probiotic use significantly reduced the risk that their diarrhea would last more than 3 days compared with placebo.⁹² A more recent review of 63 controlled trials in 8014 patients with acute diarrhea (in 56 trials, all participants were infants and young children only) included various

probiotic strains and dosages. The results showed that probiotic use significantly reduced the duration of diarrhea by 24.76 h and the risk of diarrhea lasting 4 or more days by 59%, and supplementation led to about one fewer stool on the second day.⁹³

Two probiotic strains—*Saccharomyces boulardii* and *Lactobacillus rhamnosus* GG (LGG)—appear to have the most promise for managing acute infectious diarrhea.⁹¹ In a systematic review of 22 controlled trials in 2440 patients aged 1 month to 15 years, treatment with *S. boulardii* (mostly 10^9 – 10^{10} colony-forming units [CFU]) for 5–10 days reduced the duration of diarrhea by 19.7 h, stool frequency on days 2 and 3, and the risk of diarrhea on days 3 and 4.⁹⁴ Similarly, LGG significantly reduced the duration of diarrhea by about 1 day in children with acute gastroenteritis, and it was more effective at a daily dose of at least 10^{10} CFU than at smaller doses, according to a metaanalysis of 11 randomized controlled trials with a total of 2444 children.⁹⁵ However, two subsequently published clinical trials found that 5-day courses of LGG (taken in one trial at 1×10^{10} CFU twice per day) and *L. rhamnosus* R0011 together with *Lactobacillus helveticus* R0052 (for a total of 4×10^9 CFU twice per day in the other) were no better than placebo at treating or improving the outcomes of acute gastroenteritis in 1770 infants and small children presenting to pediatric emergency departments.^{96,97}

Overall, the available evidence indicates that LGG or *S. boulardii*, in addition to rehydration therapy, might be useful for managing acute gastroenteritis in children.⁹⁸ Evidence supporting the use of *L. reuteri* DSM 17938 and *L. acidophilus* LB (heat-inactivated) to treat acute gastroenteritis is weak, and evidence on other strains is insufficient to support a recommendation.

Safety

Because most probiotics come from microorganisms that naturally colonize the gastrointestinal tract, they have little ability to cause harm.⁹⁰ Side effects of probiotics, which are usually minor, typically consist of gastrointestinal symptoms, such as gas. However, the clinical trials on probiotics have generally not been well designed to answer questions about their safety.^{99,100} In a few studies (mainly those involving individuals who were severely ill or immunocompromised), the use of probiotics was linked to bacteremia, fungemia, or pathological infections that resulted in severe illness.¹⁰¹ The safety of using probiotics for many months or years and the effects of probiotic use in older adults have not been adequately studied.¹⁰⁰

Quality

Probiotics are available in dietary supplements containing a wide variety of cultures and doses, often mixed

cultures of live microorganisms rather than single strains.¹⁰² The effects of most of these combinations have not been examined in research studies. The amounts of probiotic organisms in a supplement are best measured in CFUs, which indicate the number of viable cells, rather than by weight alone. Many probiotic supplements deliver 1–10 billion CFU per dose. When ConsumerLab.com evaluated 17 probiotic supplements in 2018, it found that all but one contained the number of CFUs listed on their labels.¹⁰³ To maintain quality, users need to store the products as recommended (e.g., at room temperature or in the refrigerator) on the label.

VI. CHALLENGES AND OPPORTUNITIES FOR DIETARY SUPPLEMENT RESEARCH

The challenges of studying the efficacy and safety of dietary supplements for promoting or maintaining health include the large number and array of products available and the regulations that govern their availability and marketing. In addition, any discussion of supplements can be emotive and polarizing. Views of their effectiveness and health value range widely, partly due to the variety of sources of evidence that consumers, practitioners, and marketers use when judging their health effects. These evidence sources can range from historical use or traditional medical practice to results of randomized, controlled clinical trials.

The amount of research on dietary supplements has grown substantially since the passage of the DSHEA in 1994 in response to the rising number of products in the market and their extensive use by consumers.¹ The dietary supplement industry invests little in research because of the lack of a legal requirement for the efficacy and safety of these products to be scientifically studied before they enter the market.

Most funding for research on dietary supplements comes from government sources, and NIH contributes the largest proportion of this funding by far. However, NIH funding for supplement research is limited and amounts to less than 1% of the amount of supplement sales. In 2017, for example, the NIH investment in supplements research totaled almost \$300 million, whereas supplement sales reached \$43 billion.^{9,104} The NIH Office of Dietary Supplements, which coordinates US federal research on dietary supplements, has an annual budget of about \$25 million. Because funding is limited for supplement research, choices must be made about where to invest those resources. Priorities for Office of Dietary Supplements funding are studies on the prevalence of use, mechanisms of action, positive or negative health effects, and safety concerns.

A. Clinical Trials

One challenge for clinical trials of the health effects of many nutrient-containing dietary supplements is that the group taking a supplement cannot be compared to a group with no exposure to the nutrient(s) in that supplement because everyone obtains at least a minimal amount of all of nutrients from their diets. This limitation probably reduces the size of observed health effects. Other challenges include the difficulty of choosing the intervention dose, how long the intervention should last, the types of participants to include, how to ensure that the results are generalizable to other populations, and which endpoints to choose. Of course, these issues are not unique to dietary supplements and are often true for other interventions. For most nutritional studies, not just those that employ supplements, the duration of intervention may need to be quite long before any effects on health outcomes are observed.

B. Biomarkers

Many attempts have been made to use changes in biomarkers in blood and elsewhere as evidence of the effectiveness of nutritional interventions for chronic disease prevention.¹⁰⁵ One example is that of 25-hydroxy vitamin D, which is a good biomarker of vitamin D exposure and status, but trials to support its value as a biomarker of health outcomes have generally failed.¹⁰⁶ Improving vitamin D status in those with evidence of vitamin D deficiency is valuable, but little evidence shows that doing so can reduce the risk of chronic diseases such as cancer, CVD, and diabetes. Another example consists of studies of the effect of supplement ingredients on antioxidant status. The investigators leading these studies should (but often do not) specify how they measured antioxidant status (i.e., the biomarker or biomarkers used and with acknowledgment of the myriad ways such status has been assessed in previous studies) and measure each participant's antioxidant status both before (to establish a baseline level) and after the intervention.

C. Nonnutrient Bioactive Components

Much more research is also needed on the many nonnutritive bioactive components in food that are also contained in some dietary supplements; these components can affect health but are not needed to meet basic nutritional needs.¹⁰⁷ Examples include the carotenoids lutein and lycopene, flavonoids, and long-chain omega-3s. Recommended intakes of essential nutrients are established in the United States through the Dietary

Reference Intakes, and other countries have similar frameworks, but no such framework exists for nonnutritive bioactive components.^{107–109}

Evaluating bioactive components using a standard process would enable investigators to quantify the level of evidence supporting the proposed benefit of a particular bioactive substance and identify the quantity needed for the substance to have health effects.¹¹⁰ Several criteria have been proposed for a bioactive to qualify for evaluation, including an understanding of the digestion, absorption, use, and excretion of the compound; published clinical trials on efficacy; adequate dose–response data; safety data at anticipated levels of intake; and existing methods of analyzing the bioactive.

Establishing intake recommendations for nonnutritive bioactive components poses unique challenges, however. For example, diets lacking these ingredients do not result in deficiency diseases (unlike with, for example, vitamin C and scurvy), even though their long-term absence can increase the risk of certain chronic diseases.¹¹⁰ In many cases, validated biomarkers are lacking to assess intakes of bioactive components and even of chronic disease risk.¹⁰⁹ In addition, safe levels of intake (analogous to the ULs for nutrients) cannot be established when safety data are lacking or insufficient. Therefore, the study of nonnutritive bioactive components warrants coordinated efforts among the nutrition science and public health communities and the support of funding agencies.^t

D. Measurement Issues

One critical opportunity for dietary supplement research is related to the fact that the growth in the number of ingredients in dietary supplements outpaces the availability of methods to analyze these ingredients.¹ No validated methods are available to characterize and analyze many ingredients and thus ensure their identity and quality. Efforts are being made to accelerate the development of official analytical methods and reference materials (homogenous substances used to ensure accurate, precise, and reliable measurement processes) for these ingredients, although progress has been too slow to meet the need. One such effort is led by the Dietary Supplement Analytical Methods and Reference Materials Program of the NIH Office of Dietary Supplements.¹¹¹ Supplement manufacturers, for example, can search the program's website to locate published analytical methods and available reference materials for single ingredients, plants, and matrices (e.g., ingredients in alcoholic extracts or blood plasma).

E. Systematic Reviews

Finally, more comprehensive evaluations are needed of the existing scientific evidence on dietary supplement ingredients to support public health decisions.^{112,113} In particular, the systematic review is a structured approach that evaluates, compares, and synthesizes relevant data, allowing consideration of the totality and quality of the evidence.¹¹³ Systematic reviews of published research can contribute to decisions about nutrient reference intakes, dietary guidance, clinical practice guidelines, and research agendas. Such reviews have been conducted, for example, on the use of supplements containing ephedra for weight loss and of omega-3s to treat several diseases, as well as human requirements for vitamin D and calcium.¹¹⁴ These reviews need to overcome certain challenges unique to the nutrition field, including the paucity of intervention studies with appropriate designs that assess dose–response relationships to account for baseline dietary exposure. These reviews also need to address each ingredient’s multiple and often interrelated biological activities.

VII. SOURCES OF INFORMATION ON DIETARY SUPPLEMENTS

Many reliable, evidence-based, online databases and other sources of information on dietary supplements

are designed to meet the information needs of consumers, clinicians, researchers, the supplement industry, and regulatory agencies (Table 35.3). For example, fact sheets and monographs from the NIH Office of Dietary Supplements that summarize the evidence on the safety and effectiveness of dietary supplement ingredients can help consumers and clinicians make informed decisions about the use of these products. Research and bibliographic databases, such as [ClinicalTrials.gov](https://clinicaltrials.gov) and CAM on PubMed, enable researchers to locate details on current and planned scientific studies and published literature on specific supplements. Other databases can help scientists find validated analytical methods and standard reference materials for certain dietary supplement components. Additional resources, such as ingredient, label, and product databases, can help industry and regulatory agency personnel find label and ingredient information about supplement products on the market.

VIII. ADVICE FOR INVESTIGATORS AND HEALTH CARE PROVIDERS

Based on this overview of dietary supplements, we offer some practical considerations for investigators who study these products and for health care providers who might recommend supplements to their patients.

TABLE 35.3 Fact sheets, databases, and other resources on dietary supplements

Name (Source)	URL	Description
Fact Sheets, Monographs, and Other Information Sources		
About Herbs (Memorial Sloan Kettering Cancer Center)	https://www.mskcc.org/cancer-care/diagnosis-treatment/symptom-management/integrative-medicine/herbs	Information on potential benefits, adverse effects, and interactions of herbs and other dietary supplements, primarily for patients and health care providers
ConsumerLab.com	https://www.consumerlab.com/	Results of independent quality tests of dietary supplements and other health and nutritional products and product reviews (subscription required)
Dietary Supplement Fact Sheets (Office of Dietary Supplements, NIH)	https://ods.od.nih.gov/factsheets/list-all	Fact sheets (some in Spanish) for health professionals and consumers on vitamins, minerals, herbs, and other dietary supplement ingredients
Drugs and Health Products (Health Canada)	http://webprod.hc-sc.gc.ca/nhp/nd/bdipsn/monosReq.do?lang=eng	Monographs on dietary supplements and other health products
Evidence-based Practice Centers and Evidence-Based Reports (Agency for Healthcare Research and Quality)	https://www.ahrq.gov/research/findings/evidence-based-reports/index.html	Evidence reports on clinical and health services topics, including dietary supplement ingredients
Health and Fitness (Federal Trade Commission)	https://www.consumer.ftc.gov/health	Consumer information on advertising claims about dietary supplements and other health products, with a focus on fraudulent, deceptive, and unfair business practices

TABLE 35.3 Fact sheets, databases, and other resources on dietary supplements—cont'd

Name (Source)	URL	Description
HerbMedPro™ (American Botanical Council)	http://cms.herbalgram.org/herbmedpro/index.html	Scientific data on herbs for professionals, researchers, and the public (subscription required for some searches)
Herbs and Supplements (MedlinePlus)	https://medlineplus.gov/druginfo/herb_All.html	Monographs on effectiveness, usual dosage, and drug interactions of herbs and other dietary supplements
Herbs at a Glance (National Center for Complementary and Integrative Health, NIH)	https://nccih.nih.gov/health/herbsataglance.htm	Brief fact sheets on herbs and botanicals, including common names, effectiveness, safety, side effects, and cautions
Micronutrient Information Center (Linus Pauling Institute)	https://lpi.oregonstate.edu/mic	Fact sheets on vitamins, minerals, phytochemicals, and other dietary components
Napralert® (College of Pharmacy, University of Illinois at Chicago)	https://www.napralert.org	Database of traditional medicine, pharmacological, and biochemical information on natural products (subscription required for some information)
Natural Medicines™ (Therapeutic Research Center)	https://naturalmedicines.therapeuticresearch.com	Monographs on the safety, effectiveness, and interactions of dietary supplements and other natural ingredients (subscription required)
Operation Supplement Safety (Department of Defense)	https://www.opss.org/	Information on dietary supplements for military personnel and other members of the Department of Defense community
PDQ® Cancer Information Summaries: Integrative, Alternative, and Complementary Therapies (National Cancer Institute, NIH)	https://www.cancer.gov/publications/pdq/information-summaries/cam?redirect=true	Patient and health professional versions of cancer information summaries on dietary supplements and other integrative, alternative, and complementary therapies
Safety Alerts & Advisories (FDA)	https://www.fda.gov/food/recallsoutbreaksemergencies/safetyalertsadvisories	Alerts, consumer advisories, and other safety information on dietary supplements, foods, beverages, and infant formulas
Tainted Products Marketed as Dietary Supplements (FDA)	https://www.accessdata.fda.gov/scripts/sda/sdnavigation.cfm?sd=tainted_supplements_cder	List of potentially hazardous products marketed as dietary supplements that contain ingredients not listed on their labels
Bibliographic Databases		
AGRICOLA (National Agricultural Library, US Department of Agriculture [USDA])	https://agricola.nal.usda.gov/	Agricultural publications and other resources pertaining to food and human nutrition, animal and veterinary sciences, entomology, plant sciences, and more
AGRIS (Food and Agriculture Organization of the United Nations)	http://agris.fao.org/agris-search/index.do	Multilingual publications on food and agriculture
Current Research Information System (National Institute of Food and Agriculture, USDA)	https://cris.nifa.usda.gov	Reports on USDA's agricultural, food science, human nutrition, and forestry research activities
PubMed (US National Library of Medicine, NIH)	https://www.ncbi.nlm.nih.gov/pubmed	Biomedical literature from MEDLINE, life science journals, and online books
CAM on PubMed®	https://nccih.nih.gov/research/camonpubmed	Publications on complementary and alternative medicine
Ingredient, Label, and Product Databases		
Dietary Supplement Ingredient Database (Nutrient Data Laboratory, USDA and Office of Dietary Supplements, NIH)	https://dietarysupplementdatabase.usda.nih.gov	Estimated levels, based on chemical analysis of nationally representative products, of ingredients in dietary supplement products sold in the United States

Continued

TABLE 35.3 Fact sheets, databases, and other resources on dietary supplements—cont'd

Name (Source)	URL	Description
Dietary Supplement Label Database (US National Library of Medicine and Office of Dietary Supplements, NIH)	http://dslid.nlm.nih.gov/dslid	Detailed information from the labels of thousands of dietary supplement products marketed in the United States; includes label images
Supplement Owl®, Online Wellness Library (Council for Responsible Nutrition)	https://www.supplementowl.org	Registry of information on dietary supplements voluntarily provided by dietary supplement companies
Research, Analytical Methods, and Standard Reference Materials Databases		
Analytical Methods and Reference Materials Database (Office of Dietary Supplements, NIH)	https://ods.od.nih.gov/Research/AMRMPProgramWebsite.aspx	Descriptions of validated analytical methods and reference materials for dietary supplement ingredients
ClinicalTrials.gov (US National Library of Medicine, NIH)	https://clinicaltrials.gov	Descriptions of privately and publicly funded clinical studies, some involving dietary supplements, conducted around the world
Computer Access to Research on Dietary Supplements (Office of Dietary Supplements, NIH)	https://cards.od.nih.gov	Descriptions of federally funded research on dietary supplements
Standard Reference Materials (National Institute of Standards and Technology)	https://www.nist.gov/srm	Standard reference materials for dietary supplement ingredients

Investigators who evaluate the effects of dietary supplements on health need to recognize the special challenges of studying these products and make sure that their research methods address these challenges. Before starting a study, investigators should independently confirm the identity, potency, and purity of the supplement ingredient(s) in the product they are studying and ensure that the product is free of adulterating contaminants. A full characterization of the intervention agent(s) is necessary to ensure the study's scientific rigor and reproducibility. Manuscripts should be prepared and submitted for publication for every study conducted on a supplement, even if the study results are negative or inconclusive.

Health care providers who recommend that their patients take dietary supplements should choose products, ingredients, doses, and dosing schedules used in published studies. The instructions for use on a product label might not be helpful or could even be misleading or incorrect, for the use that the

provider is suggesting. Providers should consult monographs from NIH and other reliable sources that synthesize the scientific evidence on the efficacy and safety of individual supplement ingredients (see [Table 35.3](#)). To increase the likelihood that a recommended supplement is of high quality, providers might recommend products that have been evaluated and approved by a reliable third-party certifier or that are sold by companies that are members of one or more dietary supplement trade associations (e.g., American Herbal Products Association, Consumer Healthcare Products Association, Council for Responsible Nutrition, Natural Products Association, and United Natural Products Alliance). These associations have codes of conduct and expectations regarding ethical business practices for their members. Finally, the Dietary Supplement Label Database (see [Table 35.3](#)) can help providers identify products that have the desired ingredient(s), amounts, and other qualities to recommend to their patients.

RESEARCH GAPS

Many hundreds, if not thousands, of ingredients are present in dietary supplements, both singly and in various combinations. The research opportunities to investigate the value of these single ingredients and ingredient combinations are enormous and include evaluating their mechanisms of action, studying their effects on physiological function and health outcomes, and developing analytical methods for precisely characterizing and measuring them. Some specific research gaps to be filled are the following:

- Improve dietary assessments to capture the total intake of nutrients and nonnutritive bioactive components from foods and dietary supplements.
- Improve and validate biomarkers of nutrient intake and status.
- Enhance global sharing of big data on nutrient intake and health outcomes.
- Improve the translation of preclinical models for health outcomes associated with intake of dietary supplement ingredients.
- Develop and disseminate tools to improve dietary supplement quality.
- Ensure that all studies of dietary supplement ingredients include measures of safety.

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S E C T I O N E

Cross Discipline Topics

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SYSTEMS BIOLOGY AND NUTRITION

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SUMMARY

Systems biology has established itself as a paradigm that has been adopted in order to integrate a wealth of omics data that have been generated in the last few decades. Nutrition research has gradually embraced the systems biology approach, and this holistic way of investigating biological systems is beginning to pay dividends for our understanding of diet and its relationship with health. This chapter presents an overview of the key strands of systems biology; these include genomics, transcriptomics, proteomics, metabolomics, bioinformatics, and computational modeling. I discuss how these techniques are being used to gain new insights into nutrition research. Moreover, I reveal how computational modeling is being used to unravel how aging effects cholesterol and folate metabolism. The chapter concludes by providing an insight into how systems biology will be applied to nutrition research in the future.

Keywords: Computational modeling; Nutrition; Omics technology; Systems biology.

I. INTRODUCTION AND BACKGROUND

Over the last 20 years, the bioscience community has witnessed a tectonic shift in how research is being conducted. A reductionist approach to analyzing biological systems has gradually been superseded by a more holistic methodology, which seeks to gain a deeper understanding of biological systems in time and space.^{1–3} Nutrition research has been swept along within this paradigm shift and is beginning to reap the rewards of this novel way of conducting scientific research.^{4–6} The reason nutritionists are beginning to benefit from systems biology is that previously nutrition research has tended to focus on individual components of nutrient-based systems in isolation.⁷ This has led to a gradual understanding of the biochemistry of an array of nutrient-mediated cellular and physiological systems. However, what has been lacking is a meaningful understanding of how these systems interact with each other and intersect with processes such as aging and disease. The systems biology paradigm is helping to overcome this challenge by using an array of approaches including genomics, transcriptomics, proteomics, metabolomics, bioinformatics, and computational modeling (Fig. 36.1).

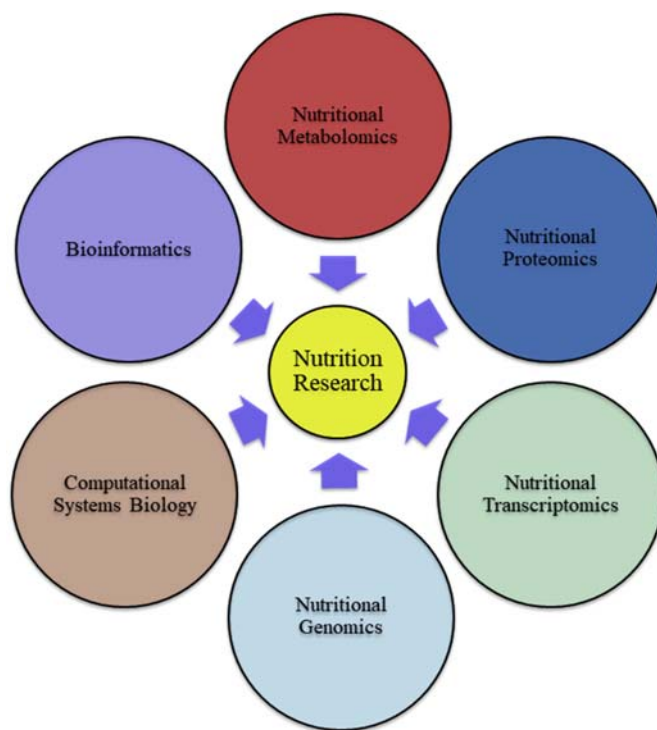


FIGURE 36.1 The key components of systems biology.

This chapter provides an overview of each of these techniques and how they are being applied to nutrition research in an integrated way. In particular, I will focus on how computational systems biology is being used to unravel the intricate biochemical mechanisms that underpin cholesterol metabolism, folate metabolism, and DNA methylation. Moreover, I discuss how computational modeling offers a unique way of understanding the intersection between these systems and the aging process. I conclude the chapter by presenting my thoughts on the future of this field, and the potential benefits that are on the horizon for nutrition research, as it continues to embrace the systems biology paradigm.

II. CURRENT STATUS OF THE FIELD

A. Nutritional Genomics

When examining nutrition research through a systems biology lens, the first key constituent in this discipline is nutritional genomics.^{8,9} As a research tool, the goal of nutritional genomics is to identify the effects of nutrients on the expression of genes. Moreover, nutritional genomics considers how different genotypes influence the metabolic behavior of the nutrients that we consume. This provides an invaluable way of capturing interindividual human variability in response to diet. The reason this field has taken off to the extent to which it has is because there is now a breath-taking array of data available, which provides the details of genomic variations.¹⁰ A key area of nutrition research which has benefitted from nutritional genomics is the area of obesity research.^{11,12} A classic example of how this technique has been applied to obesity research centers on the Nurses' Health Study¹³ and Health Professionals Follow-up Study.¹⁴ In both studies, an interaction between fried food intake and a genetic risk score based on 32 BMI-associated variants was found using genomic analysis.¹⁵ Another area which is beginning to benefit from nutritional genomics centers on investigations of the complex gene pool, which defines the human microbiome.^{16,17} Specifically, the gut microbiota has been associated with atherosclerotic cardiovascular disease (ACVD).¹⁸ In an intriguing recent metagenome-wide association study on stools from 218 individuals with ACVD, and 187 healthy controls, it was found that the gut microbiome of individuals with ACVD differed from the healthy group.¹⁹ The ACVD group had an increased abundance of *Enterobacteriaceae* and *Streptococcus* spp, and functionally, there was a difference in a number of molecules related to metabolism, which are key to cardiovascular health. However, it was uncertain if the shift in this gut microbe population was causal or a consequent of ACVD.

B. Nutritional Transcriptomics

Transcriptomics is the next step up from genomics as it centers on the entire complement of RNA transcripts in

a particular cell or tissue.²⁰ The leading techniques used for this are microarrays and RNA sequencing.²¹ Investigations into obesity emphasize the contemporary utility of transcriptomics to nutrition-focused investigations. A recent example of this is an investigation that involved feeding low-density lipoprotein receptor null (*Ldlr* $-/-$) female mice either a reference or Western diet.²² The aim of the study was to identify associations between major nonalcoholic steatohepatitis (NASH) markers and transcriptomic and lipidomic markers. Transcriptomic and lipidomics data, together with statistical analyses, were able to identify holistic changes in the quality and quantity of hepatic lipids. It was suggested these changes act as possible drivers of hepatic tissue remodeling and are potential indicators of NASH severity in mice. Another recent study that used transcriptomics to examine the relationship between diet and obesity in mice adopted RNA sequencing to generate global transcriptome profiles from brown adipose tissue fat of lean mice fed with a low-fat diet and obese mice fed with a high-fat diet (HFD).²³ Computational gene ontology analysis identified increased expression of genes related to immune responses, which enhanced processes associated with chemokine activity; decreased expression of genes involved in ion transport and muscle structure development, which reduced in channel and transporter activity and structural binding. Additional computational functional pathway analysis led to the overall suggestion that pathways associated with innate immunity were augmented by the HFD, while pathways associated with muscle contraction and calcium signaling were suppressed as a result of the HFD.

C. Nutritional Proteomics

Proteomics involves globally profiling and analyzing the interplay between the entire array of proteins, which is expressed in a cell, organ, or tissue.²⁴ Although several techniques have historically been used in proteomics research, due to its specificity of identification and sensitivity, mass spectrometry has become the primary tool used for proteomic analysis.²⁵ Mass spectrometry can identify proteins that are integral to nutrient-mediated pathways. Over the years, proteomics has been able to identify several important dietary components.²⁶ More recently, this field has been used to analyze the global proteome in a particular tissue as a result of a specific dietary regime. For example, proteomics has been used to explore caloric restriction (CR), a dietary regime that is routinely investigated in biogerontology, because of its ability to delay age-related disease and extend life span, in a wide range of organisms.²⁷ A recent study that applied proteomics to the study of CR in rats observed changes in hepatic proteins, upon age-related and diet-restricted changes in their liver.²⁸ A total of 44, 279 peptides and 3134 proteins were identified, and 260 differentially expressed proteins were identified.

Unsurprisingly, it was found that these proteins are chiefly implicated in glucose and fatty acid metabolism—related processes. It was also found that proteins regulating inflammation and gluconeogenesis were increased in old, but not young, rat liver. Intriguingly, the authors suggest that CR in old rats might not be overly advantageous due to increased liver inflammation. Another recent study with a strong aging theme used quantitative proteomic profiling, and nutrient tracing of pancreatic islets, from juvenile, and 1-year-old mice, to reveal compelling differences in the abundance of different enzymes controlling glucose metabolism. A striking and potentially useful correlation was identified, namely that alterations in protein quantity were associated with higher activities of glucose metabolic enzymes involved in coupling factor generation as well as increased activity of the coupling factor-dependent amplifying pathway of insulin secretion. In addition, an intriguing finding from nutrient tracing and metabolomics (see [Section II.D](#) for a definition and full discussion of metabolomics) was the increased accumulation of glucose-derived metabolites, and coupling factors, in islets from 1-year-old mice, which suggested that age-related alterations in glucose metabolism could help to fine-tune glucose-stimulated insulin secretion with age. Putting these observations together, and with proteomics at the center of this study, the authors concluded that they had successfully identified the age-associated metabolic rewiring of β cell function.

D. Nutritional Metabolomics

This is perhaps the newest of the omics technologies I have introduced. The aim of metabolomics is to provide a global profile of metabolism.²⁹ The techniques commonly employed are nuclear magnetic resonance spectroscopy, gas or liquid chromatography—mass spectrometry, Fourier transform infrared spectrometry, and capillary electrophoresis.³⁰ Although other fields such as toxicology research have been using this technique for a considerable period of time^{31,32}, the application of metabolomics to nutrition research has advanced significantly in the last 15 years, and some very exciting recent examples exist, which underline how metabolomics is helping to provide new insights into the influence of diet on overall metabolic state. This has implications for improving our understanding of disease states, and for tailoring personalized treatments to conditions that are growing in prevalence, such as obesity.³³ This is underlined by a recent study that used this technology to investigate whether a low—glycemic index (LGI) diet improves a set of plasma metabolites related to different metabolic diseases, when compared to a high—glycemic index (HGI) diet, and a low-fat (LF) diet.³⁴ The study utilized a parallel, randomized trial with three intervention

diets: an LGI diet, an HGI diet, and an LF diet. A total of 122 adult overweight and obese subjects were studied for 6 months. Blood samples were collected at baseline and at the end of the intervention. The plasma metabolomic profile of 102 subjects was analyzed using metabolomic analysis. It was found that serine levels were significantly higher following the LGI diet, compared to both the HGI and LF diets, whereas leucine and valine were lower in the LGI diet compared to the LF diet. A set of two sphingomyelins, two lysophosphatidylcholines, and six phosphatidylcholines were significantly modulated after the LGI diet, compared to the HGI and LF diets. Significant correlations between changes in plasma amino acids and lipid species with alterations to body weight, glucose, and insulin were also reported, leading the authors to conclude that it was the LGI diet that played a protective role against the onset of T2D, and hypertension risk. Metabolomic analysis has also been used to gain a deep insight into aging. For example, in the cross-sectional KarMeN (Karlsruhe Metabolomics and Nutrition) study, 301 healthy men and women aged 18–80 years were studied.³⁵ It was found using metabolite profiles from urine and plasma that it was possible to identify metabolic profiles for men and women, which suggested their chronological age. Key metabolites associated with this included choline in plasma and sedoheptulose in urine. For women, classification according to their menopausal status was also possible from metabolome data.

E. Bioinformatics

When a computer is used to help analyze, interpret, process, or report experimental data, associated with biological research, this constitutes bioinformatics. Specifically, within the systems biology paradigm, bioinformatics approaches are a necessity to store, manipulate, analyze, and interpret the wealth of omics data generated as a result of genomics, transcriptomics, proteomics, and metabolomics studies.³⁶ It is not possible to discuss every bioinformatics technique and how it is applied to nutrition research. Rather, I will highlight how bioinformatics approaches have been used in a number of recent studies that have illuminated prominent challenges in the field of nutrition. For instance, an area of nutrition that has come to the fore in recent years is research into vitamin D status among a variety of population groups.^{37–39} Bioinformatics workflows are one way in which computational techniques have routinely been applied to the wealth of omics data that currently exists. A bioinformatics workflow is essentially a map that has been created specifically to synthesize and conduct a sequence of computational tasks, which breaks down the vast amounts of data, which is a characteristic of systems biology.⁴⁰ In a recent study,

a bioinformatics workflow was developed to help unravel transcriptomic data from vitamin D studies.⁴¹ The vitamin D workflow was created to “join together” a number of open access bioinformatics tools. In essence, a pipeline allowed transcriptomic data to be analyzed in a straightforward manner. Bioinformatic algorithms have also been used to shed light on the putative genetic associations between Alzheimer’s disease (AD) and obesity. In one study, bioinformatics analysis aimed to detect the genetic link between AD and obesity on single-nucleotide polymorphisms (SNPs), gene, and pathway levels based on genome-wide association studies data (essentially data collected from studies focusing on a genome-wide set of genetic variants in different individuals).⁴² Using this approach, 21 SNPs were identified as being shared by AD and obesity. These were in turn pinpointed to seven genes. According to the authors, it is hoped this information can be used in experimental studies to further elucidate the mechanistic intersection between obesity and AD.

III. ISSUES RELATED SPECIFICALLY TO NUTRITION: THE GROWTH OF COMPUTATIONAL MODELING

Computational systems biology is an approach that uses computer modeling to describe and dynamically simulate the complex nature of biological systems.^{43–45} The steps involved in building a computational model

are outlined in Fig. 36.2. What these steps encapsulate is that due to its inherent ability to represent systems in an integrated manner, it can be argued that computational systems biology is the cornerstone of the systems biology paradigm. So what is a computational model and why are they so useful? A computational model is essentially a way of representing a complex biological system in a precise way using mathematics.^{46,47} Fortunately, however due to the sophisticated nature of modern-day computer software programmes, it is not a prerequisite to becoming an accomplished modeler, to hold an undergraduate degree in mathematics. In many respects, the mathematics is akin to a car engine and the computer software analogous to the car. In order to drive a car, it is not always necessary to know the exact workings of the internal combustion engine, and it is the same with modeling. However, what is needed is that the modeler has a degree of familiarity with the biological system being modeled. Importantly, if the need arises, this knowledge can be supplemented by experimentalists, who naturally will have a deeper understanding of the biological process in question.

A further skill which a modeler needs to develop is an innate ability to abstract. In other words, a good modeler should be capable of clearly defining the boundary points of a model. Moreover, a modeler should not be overwhelmed with unnecessary biological information, which they attempt to incorporate into the model. As a rule of thumb, in order to become adept at abstraction, a modeler needs to have a clearly defined hypothesis,

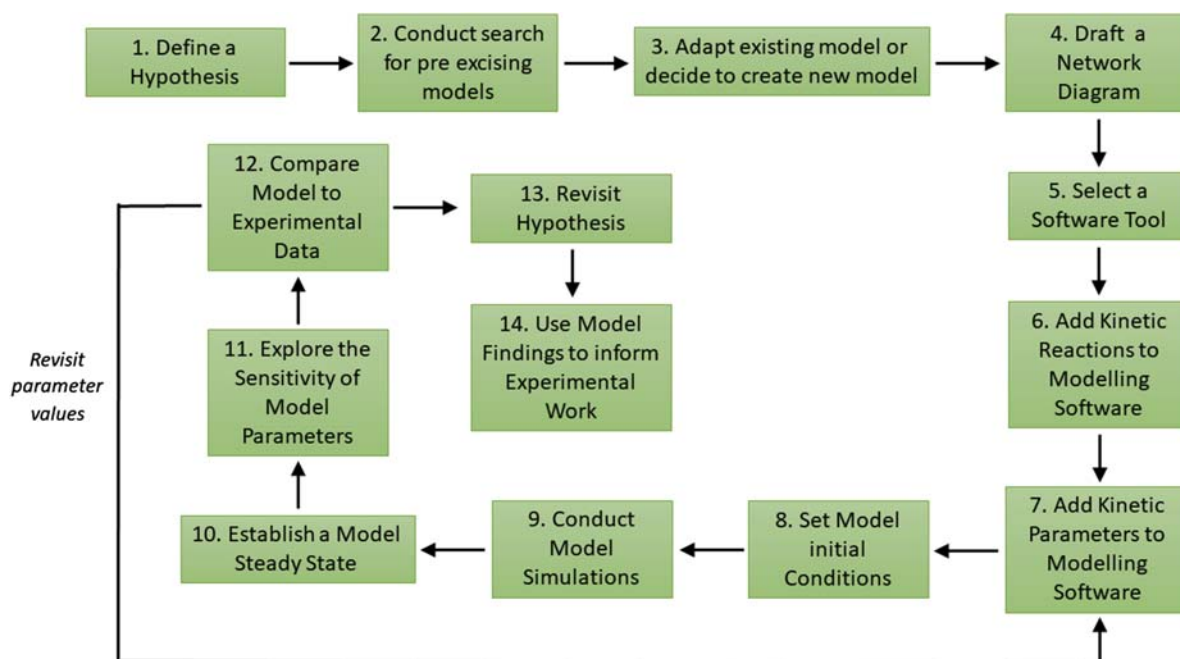
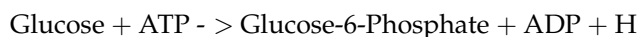


FIGURE 36.2 The steps involved in building a computational systems model. The modeling process is a cyclical process. Experimental findings are used to inform the structure of a model. Model findings are in turn used to fuel further experimental work.

which he/she wishes to use the model to test. In the absence of a clearly defined hypothesis, which helps to inform where a model starts and ends, the model can easily metamorphose into a clunky overly descriptive interpretation of a biological system. It is at this point that models can lose their utility. After many years of modeling, I have encountered models of this ilk all too often. Therefore, my advice to those new to this area is to remember that the starting point of any worthwhile model is the hypothesis/idea/question the model is trying to answer. Once a hypothesis has been established, it is then necessary to create a list of the variables that will be included in the model. In computational systems biology parlance, a variable may also be referred to as a species. Once a list of species has been created, this list acts as a template for the assembly of a network diagram. At this point, the model begins to come alive, as a network diagram captures visually the interactive reactions between model variables. In many respects, the diagram is similar to the plans a building architect would draft, or it is comparable to an electrical diagram an engineer would consult when wiring a building. In computational systems biology, it is good practise to draft a network diagram using systems biology graphical notation (SBGN).⁴⁸ SBGN is a standardized way of creating network diagrams that are routinely used by the systems biology community. This format was born from repeated observations that biological network diagrams were routinely inconsistent in nature. For example, the meaning of symbols such as arrows was often confusing or ambiguous. Thankfully, the advent of SBGN is beginning to alleviate this issue. Once a network diagram has been sketched, this serves as a road map for the model building process.

In order to assemble a model, computer software is a necessity. There are many commercial and noncommercial software tools available for this.⁴⁹ Two of the most commonly used noncommercial software for systems biology are CellDesigner⁵⁰ and COPASI.^{51,52} The latter enables the user to build models diagrammatically, model parameters and initial conditions are then added to the model, and simulations can be conducted. COPASI has a different user interface to CellDesigner, rather than using a canvas to create a diagram of the model, instead the user types in each of the model reactions. For example, if I decided to create a model of glycolysis, to create reaction one I would type into a cell in COPASI a word equation akin to the following:



In a similar fashion, the remaining nine reactions of glycolysis would be added to COPASI.

Model parameters and initial conditions then need to be entered into COPASI. To do this, it is necessary to source model parameters. This invariably means locating

enzyme kinetic information, for each of the reactions of the model. For instance, in a model of glycolysis, it would be necessary to find the $V_{\max}$ and K_m values of the enzymes, which underpin the 10 reactions involved in this system. A worthwhile place to find kinetic parameters is the online database BRENDA⁵³ or one of the other online repositories that have been established for this purpose.⁵⁴ If the database does not have the required kinetic parameters, it is necessary to perform a literature search by conventional means, using PubMed or Web of Science. It is important to be cognisant that it is not always possible to identify every parameter and gaps in models are commonplace. The important consideration is to clearly state using a table, where the kinetics has been sourced, and any assumptions that have been made about unknown parameters. Once the kinetic parameters have been added, the model can be simulated. The length of the simulation will depend on the nature of the model. For example, if the aim of the model is to explore the effects of aging, then one way to do this is to run a simulation for an extended period of time.⁵⁵

Once the model has been simulated, its output can be compared to the “normal” behavior of the biological system. If the output is unsatisfactory, it is necessary to refine the model. Typically, this involves adjusting the initial conditions/parameter values or in some cases redesigning model topology. This can be undertaken in an iterative manner until the model output is in agreement with the biology. A more formal approach to validate the model is to calibrate it using time course data. COPASI has inbuilt analysis tool box that helps with this fine tuning process. If the output and behavior of the in silico system is comparable to its in vivo counterpart, the model can be utilized for hypothesis exploration. The findings from the model can then be used to guide further laboratory experimentation, which in theory fuels additional refinements to the model. Thus, the computational modeling process is a cyclic endeavor, with each step being naturally driven by its predecessor.

A. Model Exchange and Model Archiving

Advances in systems biology have made model refinement a more straightforward and efficient process than it has been previously. This is underscored by the systems biology markup language (SBML), which has been developed to overcome the problems of sharing and updating models easily.^{56,57} For example, in the past, it was not always straightforward to update or share a model because models were invariably created using an array of software tools and languages. SBML overcomes these stumbling blocks because it provides a means of exchanging models. For instance, tools such as COPASI and CellDesigner, which support this exchange framework, are capable of generating SBML, which can then be imported

into other computer tools, which also support SBML. The SBML website provides a full list of those software tools that support this schema, and for those interested in computational modeling, it is worthwhile spending time perusing these tools from a pedagogical perspective. An adjunct to SBML is the BioModels database; essentially this database acts as a repository for models, which have been encoded using SBML.⁵⁸ The website is subdivided into those models that have been curated and those which have not. Curated means the model has been converted into SBML and validated, it comes from a peer-reviewed published article, and it is verified by the BioModels database curators to correspond to its reference description in structure and behavior. Noncurated models may have valid SBML but are awaiting to be fully curated by the BioModels database curators. It is straightforward to download the SBML code for a particular model and import it to modeling software such as those discussed previously. Therefore, it is unsurprising that over the last few years, there has been an exponential growth in the number of model archived in BioModels. Importantly, as well as containing the SBML code, and a description of the model, the BioModels database has a link to the peer-reviewed paper that contains the full details for each model. As such, BioModels offers an excellent starting point for identifying preexisting models, which have the potential to be adapted, refined, and augmented, as the biological knowledge of the system they represent evolves.

B. Computational Modeling and Its Application to Nutrition Research

It is paramount to recognize that kinetic models have been used in nutrition research for a very long time, and their use is not strictly new. For example, computer models have been used to capture the intricacies of intracellular B vitamin biochemistry over the last four decades (see [Section III.D](#)). However, it can be argued that the emergence of systems biology has introduced two important factors to computational modeling; firstly, there is now a keener emphasis on coupling modeling with experimental work, and secondly, there is a greater breadth of models that have nutrition or nutritional biochemistry as their focal point.⁴ To illustrate this point, it is worthwhile highlighting a number of recent computational models that have had nutritional biochemistry at their core. In the work by Parmar and Mendes (2019), the authors assembled a kinetic model of iron metabolism in mice.⁵⁹ Emphasizing the juxtapositioning of modeling and experimentation, the model was calibrated using ferrokinetic data and validated using data from mouse models of iron dysregulation. The predictive value of the model to simulate disease interventions was demonstrated by using it to explore the outcome of various schedules of transferrin treatment in β -thalassemia.

Other recent examples of models that focus on nutrients include intracellular models of folate, copper, zinc,⁶⁰ calcium, cholesterol,⁶¹ gut short-chain fatty acids,⁶² and hepatic fat and sugar metabolism.⁶³ In the next [sections](#), I will pay particular attention to how computational systems modeling is helping to shape our understanding of both cholesterol and folate metabolism.

C. Computational Modeling Cholesterol Metabolism

Any nutritional biochemist knows the value cholesterol has in the body. Briefly, cholesterol is an essential reinforcer of cell membranes, as well as being the precursor of bile acids, vitamin D, and steroid hormones.⁶⁴ However, despite its fundamental role in the body, the dysregulation of cholesterol metabolism presents a threat to health. The reason for this is mainly because of the correlation that exists between elevated levels of low-density lipoprotein cholesterol (LDL-C) and increased risk of ACVD.^{65–67} There is no escaping the wealth of evidence that links the LDL particle and its cargo of cholesterol with the mechanisms that illicit atherosclerosis.⁶⁵ Conversely, high-density lipoprotein cholesterol (HDL-C) is viewed as antiatherosclerotic.^{68,69} The reason is that HDL-C is the central player in reverse cholesterol transport (RCT).⁷⁰ RCT is the process whereby excess cholesterol is removed from peripheral tissue by HDLs and transported to liver so that it can ultimately be excreted from the body during enterohepatic circulation.⁷⁰ Two enzymes are key to this process, lecithin-cholesterol acyltransferase,⁷¹ which catalyzes the formation of cholesterol esters, from free cholesterol onto the HDL particle, and cholesteryl ester transfer protein (CETP), which exchanges cholesteryl esters from HDL and gives it to LDL or very-low-density lipoproteins for triglyceride.⁷²

It is natural an intense focus has been directed toward identifying the dietary factors that induce changes in LDL-C and HDL-C. This has revealed a number of insights including the fact that a diet that is high in saturated or trans fats can cause LDL-C to rise in a wide range of population groups.⁷³ Conversely, diets that contain high amounts of mono/polyunsaturated fat can reduce LDL-C levels.⁷⁴ Intriguingly, the effects of dietary cholesterol are not so clear-cut. Firstly, most individuals consume relatively small amounts of cholesterol daily. However, the response to the consumption of cholesterol can differ substantially from individual to individual. Consequently, a patient can be classified as either a hypo- or hyperresponder to dietary cholesterol.⁷⁵ These differences are perhaps not surprising when one considers the heterogeneity that exists in intestinal cholesterol absorption, which has been shown to have a range of 29.0%–80.1% in humans.⁷⁶ Adding further complexity to cholesterol metabolism is the myriad of

genetic polymorphisms that are associated with this system.⁷⁷ These facts make studying cholesterol metabolism using conventional means deeply challenging. However, there is a further striking factor that makes studying this system and its intersection with CVD even more challenging. That factor is the aging process. Aging has a profound impact on cholesterol metabolism.⁷⁸ For instance, in rodent studies, it has been observed that there is an increase in cholesterol absorption with age. In addition, evidence from rodent models suggests that there is an increase in the activity of the key rate-limiting enzyme in cholesterol biosynthesis, HMG-CoA reductase, in response to reactive oxygen species (ROS).^{79–81} ROS have long been touted as a central driver of the aging process.⁸² Finally, aging has also been shown to interfere with the activity of hepatic LDL receptors (LDLRs), thus impacting the rate of clearance of LDL-C from the circulation.

In 2012, our group set about addressing the challenges outlined in the previous paragraph by assembling a computational model of whole-body cholesterol metabolism. The initial aim was to bring together a wealth of diverse data on cholesterol metabolism and to structure it within a cohesive system-orientated framework. Our model was then utilized to answer key questions related to how aging intersects with the dysregulation of cholesterol metabolism.^{83,84} An SBGN diagram of the model is outlined in Fig. 36.3, while Fig. 36.4 provides the output from a simulation with this model. Fig. 36.4 shows key model variables reaching a steady state. It is vital that a model is in steady state before *in silico* experimentation takes place. Once in a steady state, the model was used for hypothesis exploration. Namely, the idea was tested that an increase in cholesterol absorption efficiency and a decline in the hepatic removal rate of LDL-C are responsible for the rise in LDL-C, which has

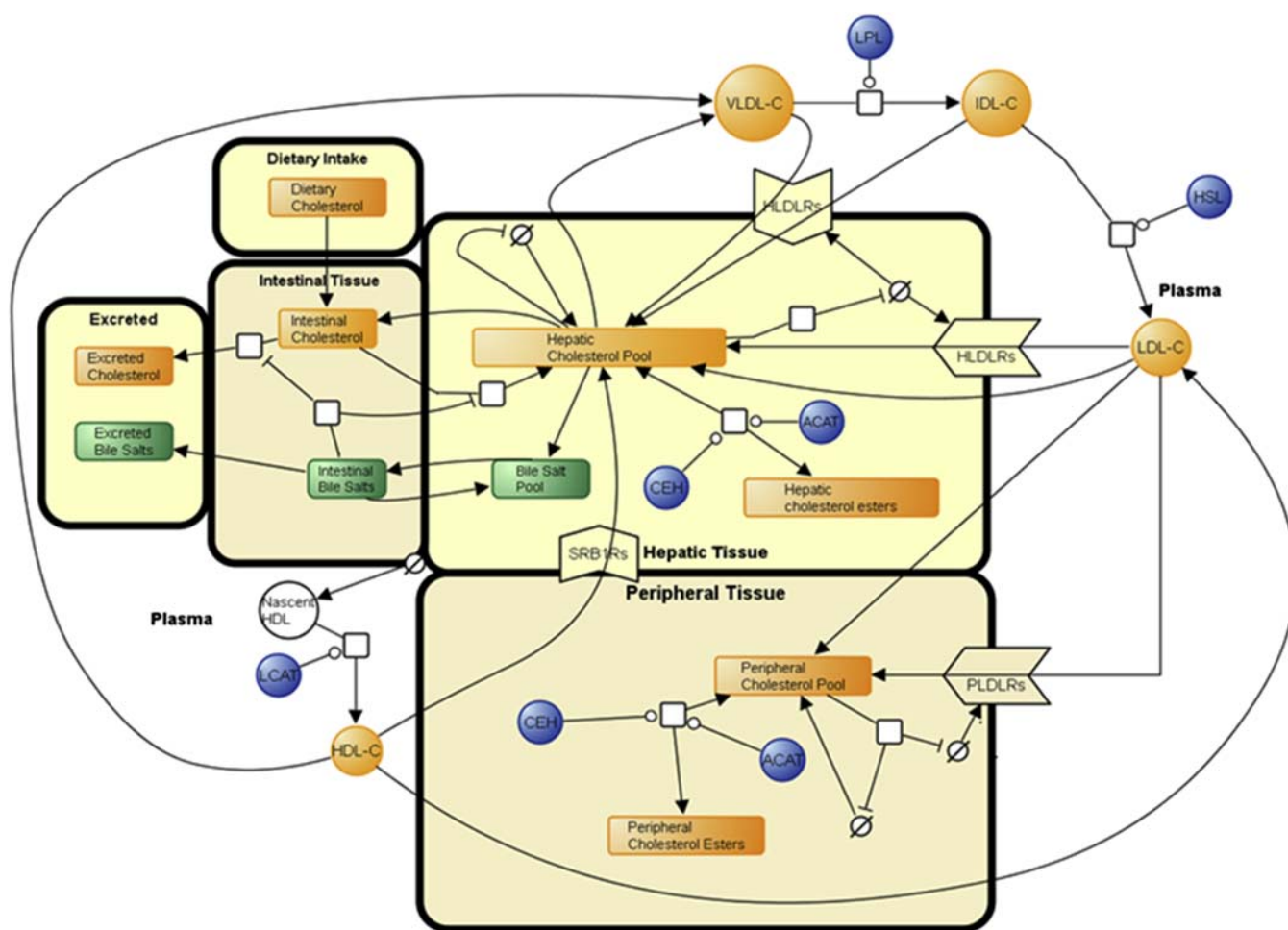


FIGURE 36.3 SBGN diagram of the Mc Auley et al. (2012) cholesterol metabolism model.⁸¹ The model is laid out in six compartments: (1) intake, (2) intestinal tissue, (3) excretion, (4) plasma, (5) hepatic, and (6) peripheral tissue. The arrows represent the flow of cholesterol around the system into its different forms. Enzymes are represented by blue spheres, and their catalytic influence on the reaction is indicated by a round arrow head coming into contact with a reaction arrow. Synthesis is represented by the mathematical symbol for an empty set, while inhibition is represented by bar-headed arrows. In summary, cholesterol from the diet and bile is formed into micelles in the small intestine. Absorbed cholesterol is then transported to the liver where it is exported into the plasma via VLDL. VLDL is in turn catabolized to LDL-C. LDL-C is in turn removed from the circulation by hepatic and peripheral tissue. Excess cholesterol from peripheral tissue is transferred to the liver via HDL. *Figure and legend taken from Ref. 83.*

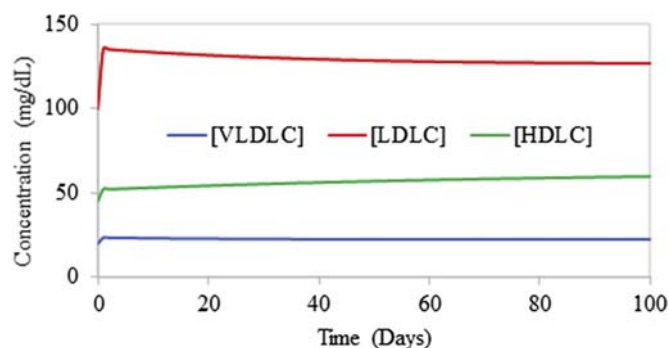


FIGURE 36.4 Model simulation of cholesterol metabolism. The figure shows key metabolic parameters, including LDL-C, VLDL, and HDL-C reaching a steady state.

been observed in several studies. Our *in silico* representation of cholesterol metabolism was able to predict that an increase in the rate of intestinal cholesterol absorption from 50% to 80% by age 65 years results in an rise in LDL-C by as much as 34 mg/dL in a hypothetical male subject. The model also demonstrated that decreasing the rate of hepatic clearance of LDL-C gradually to 50% by age 65 years resulted in an increase in LDL-C, by as much as 116 mg/dL. In 2016, we updated our computational model of whole-body cholesterol metabolism to include an additional 96 mechanisms, which we identified as being key to maintaining whole-body cholesterol balance.⁸⁵ These included processes associated with intestinal cholesterol absorption, hepatic cholesterol synthesis, RCT, and bile acid production/enterohepatic circulation. Cholesterol feeding was used to explore the effectiveness of the additional regulatory mechanisms. The new model responded to cholesterol ingestion in a hyperresponsive manner. Using the model, the effects of aging in tandem with three separate CETP genotypes were investigated. Aging in the presence of an atheroprotective CETP genotype, conferring low CETP activity, resulted in a 0.6% increase in LDL-C. This was in contrast to aging in conjunction with a genotype reflective of high CETP activity, which resulted in a 1.6% increase in LDL-C. Thus, the utility of the model was underscored by its pivotal prediction that certain CETP genotypes have a vital role to play in health span.

D. Computational Modeling Folate Metabolism

The B vitamin folate or its synthetic form folic acid is an essential mediator of one carbon metabolism.^{86,87} Folate is vital to an array of biochemical processes including methylation reactions, the homeostasis of several amino acids, and nucleotide biosynthesis. Research over many years has confirmed beyond reasonable doubt that folate is essential during embryonic development.⁸⁸ Moreover, folate intake is paramount during adult life because an insufficient intake of the vitamin leads to anemia.⁸⁹ In addition, a diet inadequate

level of this important B vitamin has been implicated in a wide range of conditions including CVD, cancer, and AD.^{90–92} A reductionist experimental approach has proved essential in identifying and determining how each of the components of folate metabolism works. However, in order to fully appreciate how each of these components interacts, it has been necessary to study the system using computational systems biology. As this system has been reasonably well characterized by nutritional biochemists, this has made building models of it more straightforward than in other areas of nutrition, where the kinetic information simply does not exist.⁹³

Prior to systems biology reaching its full potential, the first model of folate metabolism was developed in the 1970s.⁹⁴ The model represented a minimalistic interpretation of folate metabolism and was used to study the pharmacological treatment of cancer. A later incarnation of the model included nucleotide turnover.⁹⁵ The utility of computational modeling to this area was further underscored in 1989 in a model that simulated folate kinetics in human breast carcinoma cells.⁹⁶ As systems biology began to progress, this coincided with the development of a more integrated and detailed model of folate metabolism by Nijhout and colleagues in 2004.⁹⁷ The main focus of this model was on the integral role of B12 to the folate cycle. The utility of the model was emphasized by its ability to simulate “the methyl trap,” a biochemical response that is the result of an inadequate supply of methionine.⁹⁸ Thus, in 2006, there were sound reasons for integrating this model with a model of methionine biochemistry. Model simulations predicted that vitamin B12 deficiency would lead to a decrease in DNA methylation and nucleotide synthesis.⁹⁹ This model continued to develop in parallel with the burgeoning growth of systems biology, when it metamorphized into a whole-body *in silico* conceptualization of folate metabolism.¹⁰⁰ This model contained modules that represented the plasma and hepatic turnover of folate. Using this more holistic depiction of folate metabolism, it was found that tissue and plasma folates are highly correlated, but hepatic and plasma folates significantly less.

In 2016, Salcedo-Sora and Mc Auley created a model of folate metabolism in bacteria, with the goal of using it to identify hubs within microbial folate metabolism that could be used as antimicrobial targets.¹⁰¹ This model was able to identify particular targets, within the folate metabolism of bacteria, which chimed with our current understanding of how antifolates operate *in vivo*. Model predictions suggested the folinic acid substrate cycle is a key process, used by bacteria during dynamic cell proliferation. In 2017, a combined deterministic-stochastic modeling framework was used to model folate metabolism.¹⁰² This type of approach is known as a hybrid model, and the reason for using it is because certain

elements within folate metabolism were clearly identified as stochastic while others were deemed to have negligible stochasticity. The model showed that *de novo* deoxythymidine monophosphate synthesis is extremely sensitive to a common mutation in methylenetetrahydrofolate reductase, which was amplified as a result of folate deficiency. In 2019, a hybrid computational approach was again used to explore folate metabolism.¹⁰³ This model showed that 10-formyl-tetrahydrofolate inhibition of methenyltetrahydrofolate synthetase is pivotal for the regulation of purine synthesis. *In silico* experimentation also indicated that 5-methylTHF (and not 5fTHF) is the main physiological binder/inhibitor of SHMT. Interestingly, this finding resonated with the earlier modeling work of Salcedo-Sora and Mc Auley, which singled out the importance of the futile cycle in folate metabolism. This model suggested that the 5fTHF futile cycle dampens the stochastic noise in folate metabolism, which is a consequence of folate deficiency, and the common mutation in the MTHFR gene. The fact that there was commonality between these models serves to illustrate the overall utility of computational modeling in deepening our understanding of folate metabolism and in providing insights that have served to fuel further experimentation in this field.

Computational modeling DNA methylation dynamics

A key role of dietary folate is to provide the methyl groups that are a necessity for maintaining DNA methylation status.¹⁰⁴ This process involves methyl groups covalently attaching to CpG dinucleotides to generate tissue-specific methylation patterns. A CpG dinucleotide is made up of a deoxycytidine followed by a deoxyguanine, with the “p” designating the phosphate group. Covalent bonding of methyl groups occurs at the carbon-5 position of a deoxycytidines to create a methylated CpG dyad. The maintenance of DNA methylation patterns plays a fundamental role in gene expression.¹⁰⁵ Changes with age can alter the overall profile of the methylome (the complete complement of methylated

nucleic acids).¹⁰⁶ Specifically, an enduring mystery surrounds the methylome, namely, that with age globally the DNA molecule become hypomethylated, while paradoxically specific regions within the genome become hypermethylated.¹⁰⁶ Hypermethylation has been observed to occur within a variety of genes that are frequently associated with an array of different cancers.^{107–109} Moreover, this modification has recently been associated with conditions including CVD, AD,¹¹⁰ and diabetes.¹¹¹

In recent years, computational modeling has proved invaluable in expanding our understanding of the mechanisms that regulate DNA methylation at the molecular level. An intriguing model in this field was able to predict the relative abundances of unmethylated, hemimethylated, fully methylated, and hydroxymethylated CpG dyads in the DNA of cells, with fully functional methylation and demethylation enzymes.¹¹² And importantly, from a health perspective, the model was used as a tool to predict hematological malignancies. Other models have focused on the inherent variability that is associated with the methylation reactions. To this end, stochastic computational models of DNA methylation have been assembled to investigate collaboration between CpGs within gene promoters.^{113,114} Moreover, Olariu et al. (2016) complimented this work to include transcriptional regulation and DNA methylation, in a model that focused specifically on the establishment of pluripotency.¹¹⁵ In 2018, Thalheim and colleagues created a computational model that linked DNA damage and age-related promoter DNA hypermethylation in the intestine.¹¹⁶ The results of this model suggest methylation-dependent binding of maintenance enzymes to CpGs, establishing bistable DNA methylation states. This model predicted that bistability is a prerequisite to promoter hypermethylation, following DNA repair. Most recently, Zagkos et al. (2019) created models of DNA methylation dynamics.^{117,118} Intriguingly, this work was also able to consolidate experimental findings that have been observed in this area.

RESEARCH GAPS AND CONCLUSIONS

Systems biology has rapidly emerged over the last few decades. This chapter has revealed how each of the strands of systems biology is being applied to nutrition research. I revealed that systems biology is being used to drive investigations in key areas of nutrition, such as obesity and aging research. It is clear that systems biology will continue to advance in coming years and will continue to align with more conventional reductionist approaches, as a means of fusing together disparate information, to create a more holistic representation of biological systems. Consequently, a more integrated understanding of biology can help path the way to a more personalized approach to health care, and a deeper understanding of nutrition will be at the forefront of this. Using findings from the omics disciplines in tandem with computational modeling, it will be possible to provide an overview of an individuals' health trajectory. Using this information, it will be possible to tailor precise individualized dietary treatments to fine-tune an individuals' metabolism, so their health can be maintained and optimized to prevent disease.

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THE MICROBIOME AND HEALTH

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SUMMARY

It is now accepted that the gut microbiome is an important component in shaping human and animal homeostasis and disease susceptibility. Through its diverse roles in digestion and nutrient absorption, production of essential vitamins, release of novel small molecules and metabolites, stimulation of immune development and function, and broad modulation of regulatory systems in the body ranging from endocrine to neuronal, the microbiome has proven an important frontier in our understanding of animal biology and nutrition. This chapter reviews the rise and current state of microbiome science and focuses on how nutrient interactions with our microbial organ system shape nutritional health and outcomes under numerous pathophysiological states.

Keywords: Host–microbe interactions; Macronutrients; Metabolomics; Microbiome science; Microbiota; Micronutrients; Next-generation sequencing; Nutrition.

I. INTRODUCTION

A. Background

The microbiome as a virtual organ system

Long before our modern concept of the complex human microbiome and vast networks of microorganisms colonizing the body, early scientists attempted to probe microbial life. In the late 1600s, van Leeuwenhoek observed “small animals” under his self-made microscope, accurately describing bacteria, fungi, and protozoa. Centuries later, Louis Pasteur performed experiments implicating bacteria as specific sources of human illness. Pasteur’s work and his “germ theory hypothesis” set the stage for Robert Koch to subsequently prove bacilli directly caused anthrax in mice. The scientific framework used by Koch to prove or disprove specific organisms could be isolated and directly cause infection resulted in criteria now known as Koch’s postulates.¹ These leaps in microbiological understanding prompted public health and clinical professionals to take steps aimed at preventing

spread of diseases, yet tools to fully halt epidemic infections were only realized in the 1940s upon discovery of antibiotics.

By the late 20th century, microbiology remained limited to exploring cultivable microbes followed by identification using specific criteria and slow-throughput DNA sequencing methods. Microbiome science gained rapid momentum in the early 2000s with the advent of next-generation sequencing (NGS), leading to microbial community-wide interrogation of unique environments, including the gastrointestinal tract (GIT), followed by computational strategy development for large data sets. We now appreciate that the number of human body cells is matched by nearly equal numbers of microbial cells,² yet microbes harbor genetic diversity estimated to be 150-fold greater than our own.³ For instance, a single bacterial species, *Bacteroides thetaiotaomicron*, contains over 260 different hydrolase enzymes, far exceeding hydrolases encoded in the human genome.⁴ Given the gut microbiome’s genetic capacity, it is often considered a virtual organ, shaping many

aspects of digestion, gut physiology, and peripheral tissue signaling. Via our advanced technological perspective, we now have a better appreciation for the vast diversity of microorganisms (the microbiota) and their genomic potential (the microbiome) that cohabitate the human body and have begun to map out their health and nutritional influence (Table 37.1).

B. Definitions and Explanatory Relationships

TABLE 37.1 Definitions related to microbiome science

Term	Description
Microbiota	Assemblage of individual microorganisms present in a defined environment, i.e. gastrointestinal tract
Microbiome	The totality of microbes, their genetic elements (genomes), and environmental interactions in a defined environment
Multi-omic technology	Biological analysis approach using multiple layers of -omics platforms, i.e., metagenomics, metatranscriptomics, metabolomics in concert to explore roles, relationships, and actions of cells within a system in a concerted way
Amplicon sequencing	High throughput ultra deep sequencing of PCR products to analyze genetic variations from genomic DNA that can be applied to complex microbial samples to deduce taxonomic identities
16S ribosomal RNA (rRNA)	Highly conserved gene across species of bacteria and archaea ~1,500 bp in length containing hypervariable (V) regions. PCR primers can be targeted to V regions and used for amplicon sequencing
Internal Transcribed Spacer (ITS) rRNA	Spacer DNA region situated between the 16S and 23S rRNA genes. PCR primers can be targeted to V regions and used for amplicon sequencing. Commonly used to assign taxonomy to fungal microbial eukaryotes
Metagenomic sequencing	Comprehensive Next-generation sequencing (NGS) approach that allows for parallel sequencing of DNA from thousands of microorganisms from many samples. Based on short reads (~300 bp), functional genomic elements can be identified or assembly of metagenomic assembled genomes (MAGs) can be performed along with taxonomic assignment
Metatranscriptomic sequencing	Comprehensive NGS approach that allows acquisition of activity of whole genome expression profiles using microbially-derived mRNA from a complex community
Single-cell genomics	Sequencing using NGS technologies on DNA from sorted or selected single cells obtained from a complex microbial community to assign function

TABLE 37.1 Definitions related to microbiome science—cont'd

Term	Description
Culturomics	Detection and identification of a maximum number of microorganisms from select samples using combinations of classic microbiology coupled with downstream Mass spec and NGS approaches
Kyoto Encyclopedia of Genes and Genomes (KEGG)	Database collection that includes genomes, biological pathways, and chemicals for understanding function and molecular interactions of a biological system
Germ-free	Animals completely devoid of microorganisms living on or in them, allowing control of exposure to viruses, bacteria, fungi, or parasites
Gnotobiotic	Animals in which only known, defined strains of microorganisms are present, often administered to germ-free animals
Conventionally-raised	Animals raised with an undefined complex microbial community from birth
Conventionalized	Transplantation of a complex, undefined microbial community to germ-free animals, i.e. fecal microbiota transplant
Specific pathogen-free	Animals raised with an undefined complex microbial community from birth devoid of known species-specific pathogens
Organoid systems	<i>In vitro</i> 3-D cell aggregates derived from primary tissues or stem cells capable of self-renewal, self-organization, and exhibit organ functionality

II. KEY ISSUES

Despite advances in studying gut microbes, the microbiome field is still relatively young, holding a great deal of promise to unlock novel therapies in the nutritional realm. To advance the field, a thorough understanding of early research and its current status is essential. How to employ reductionist systems coupled with integration of multiomic platforms used to study complex gut microbial networks will aid in thorough dissection of host–microbe–nutrition interactions. The ultimate goal remains identifying a means to harness or modulate microbial communities to promote health and alleviate disease. Here, we provide an overview of microbiome science in a nutritional context, provide insights into technologies that have advanced our understanding of this virtual organ, highlight specific host–microbe nutritional interactions, and discuss challenges facing the field to further knowledge of how gut microbes shape nutrition in the modern age.

III. CURRENT STATUS OF THE FIELD

A. The Rise of Microbiome Research

Following initial studies published by Jeffrey I. Gordon's laboratory describing the causative role of gut microbes in driving metabolic disturbances associated with overnutrition using NGS and germ-free (GF) animal model approaches,^{3,5,6} the National Institutes of Health initiated the International Human Microbiome Project in 2007,⁷ whose initial goal was to use sequence-based approaches to characterize a "healthy cohort" human microbiome and move beyond classic cultivation methods. Since 2007, the sheer number of publications examining the microbiome in health and disease has exponentially risen annually, reaching >10,000 in 2018 (Fig. 37.1A).

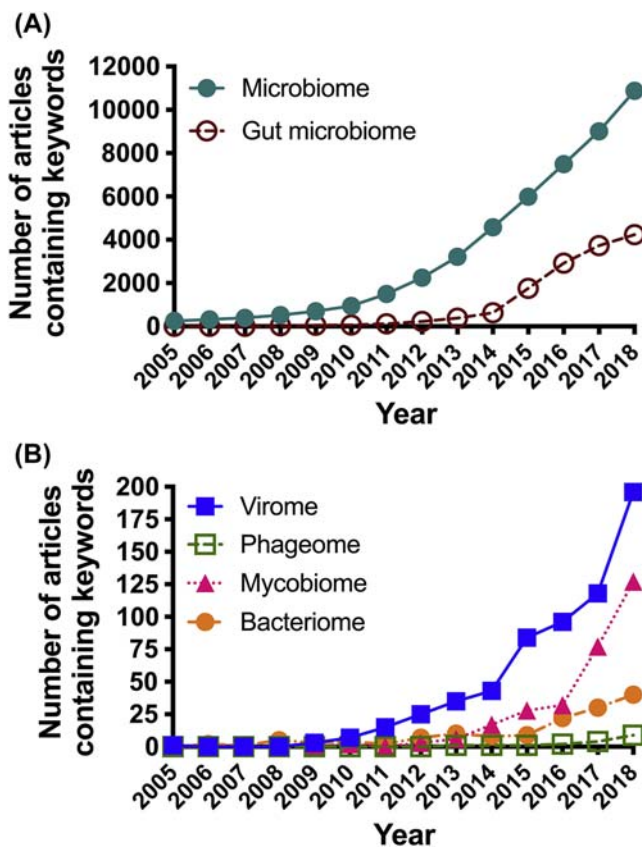


FIGURE 37.1 Expansion of microbiome-related publications. The number of NCBI Pubmed publications including search keywords "microbiome" and "gut microbiome" through 2018 has exponentially increased since 2005. Early publications revealed use of high-throughput 16S rRNA amplicon sequencing to characterize gut microbial communities, overcoming culture-based biases (A.). The number of publications examining additional and less-abundant community members, including viruses, bacteriophages, and fungi or "mycobiome", has steadily increased using amplicon sequencing, i.e., ITS or shotgun metagenomics. "Phageome" is included as a refined keyword describing a portion of the virome, while "bacteriome" is included as a refined keyword describing community membership detected via 16S rRNA sequencing (B.). Original NCBI Pubmed keyword search performed by the authors on July 10, 2019.

Albeit to a lesser degree, publication numbers examining additional gut microbial community members, such as viruses, bacteriophages, and fungi, have also recently risen (Fig. 37.1B).

Subsequent initiatives have built upon the HMP framework, including the American Gut Project,⁸ the Human Oral Microbiome Database,⁹ and Metagenomics of the Human Intestinal Tract,¹⁰ serving to curate and collect big data sets for creation of tools that move beyond characterization. Because of this, unique microbiomes have been described across different body sites, including skin, oral cavity, respiratory, urogenital, GIT, mammary duct, and cancer tumors.^{11–13} Since the GIT exhibits greatest microbial numbers, richness, and diversity and most direct interactions with nutrition, this chapter largely focuses upon this community. From mouth to anus, the GIT harbors a microbiota density gradient with bacterial estimates ranging from 10^2 bacteria/gram of contents in the upper GIT to 10^{12} bacteria/gram of contents in the distal large intestine¹⁴ (Fig. 37.2).

The two major bacterial phyla, Firmicutes and Bacteroidetes, along with less abundant phyla, including Proteobacteria, Verrucomicrobia, Actinobacteria, Fusobacteria, and Cyanobacteria, contain up to 1000 unique species based on sequencing approaches. Diversity among additional microbial community members other than bacteria has also been captured.^{10,14,15}

While considerable microbial genetic diversity is found within the GIT, it is relatively selective in microbial membership as compared to other environments. Out of ~50 environmental bacterial phyla, the GIT is commonly dominated by only 4 or 5, demonstrating GIT selectivity toward certain colonizers. To reside within the gut, microorganisms must tolerate hypoxia, high concentrations of antimicrobial compounds, and exposure to membrane-disrupting bile acids; thrive at animal body temperature; tolerate pH changes; and adapt to continuously shifting nutrient availability during the host's feeding and fasting cycles. In exchange, microbes aid in nutrient digestion through break down of indigestible macronutrients, production of essential vitamins and amino acids, stimulation of host immune homeostasis, and even drive release of metabolic regulatory molecules, such as incretins, through production of short-chain fatty acids (SCFAs) and secondary bile acids, outlined in Fig. 37.3.

The importance of these interactions throughout the life cycle and especially in a nutritional context is rapidly yielding new discoveries.

Despite technological advances enhancing our ability to sequence microbial communities, only ~50% of microbes are cultivable; identifying their causal versus associative role in disease development remains difficult. Revision of Koch's postulates has been proposed to accommodate these novel technologies, incorporating the notion that individual microbes may not cause

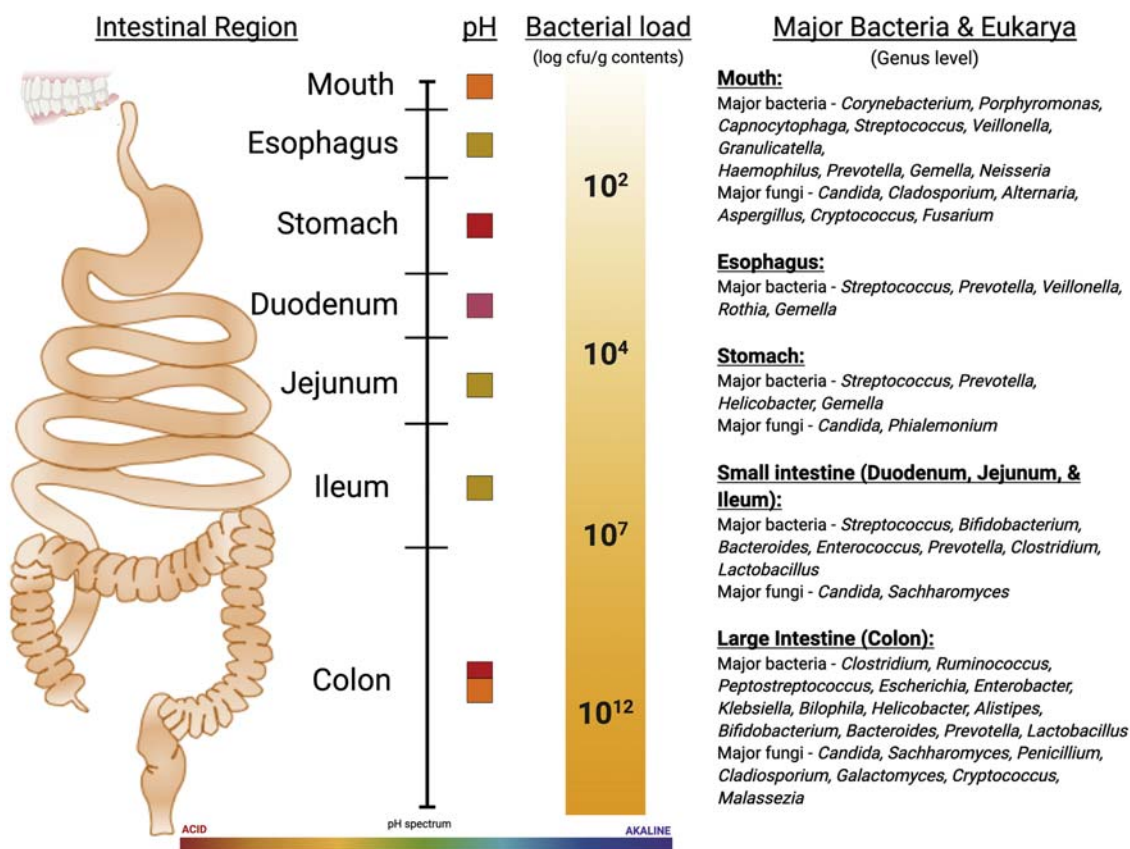


FIGURE 37.2 Microbiota abundance and membership along the GIT. Bacterial load, pH, and major bacterial and fungal taxa vary dramatically across intestinal regions from the proximal to distal GIT. *Source: Figure made using Biorender.com*

disease, but rather the collection of microbial community members is at play.^{16,17} Emerging data demonstrate that collective genes harbored by microbial communities may be more important than inclusion of specific individual members, whereby the sum of the genes and community metabolic pathways present dictate health versus disease more reliably than taxonomic identification.¹⁸ Many of these interactions are stabilized by host nutrient intake patterns and through production of secondary metabolites, quorum-sensing molecules, and nutrient acquisition molecules, such as siderophores.

As with other classic ecology examples, microbial interactions range from symbiosis to cohabitation to outright warfare. Perhaps the most well-known example of intermicrobial hostility remains Alexander Fleming's observation that the fungi *Penicillium* could kill bacteria—leading to discovery of Penicillin.¹⁹ There are now many examples of microbial toxins aimed at other microbes and examples of direct predation,^{20,21} which may be influenced by nutrient availability shaped through host nutrition. Outside Westernized societies, emerging reports suggest even more bacterial taxa and their functions remain uncharacterized in indigenous and remote populations experiencing less industrialization, seasonal changes in diet, and little to no exposure to Western

medicines or antibiotics.^{22–27} Whether and how these unique microbial community members aid in maintaining healthy microbiomes remains unknown.

Gut microbiota heterogeneity in humans and animals—nutritional implications

One common theme emerging from collective microbiome studies over the past 15 years is humans, regardless of health status, exhibit high amounts of microbial variability between individuals. Identifying a core microbiome is difficult, given only ~10% of bacteria are shared among individuals.^{10,15} Twin pair studies, with less genetic variability, revealed gut microbiota between monozygotic, genetically identical twins are more similar than between dizygotic twin pairs and related family members or unrelated individuals, suggesting some degree of gut microbiota heritability.^{22,28–30} The latter study identified that the presence of the family *Christensenellaceae* was significantly increased in lean twins and transfer of *Christensenella minuta* into ex-GF mice harboring a complex microbial community deficient in this strain promoted leanness.³⁰ Subtle differences in gut microbiomes between twin pairs associate not only with metabolic diseases³¹ but also with autoimmune diseases such as inflammatory bowel diseases²⁹ (IBDs), in part, explaining

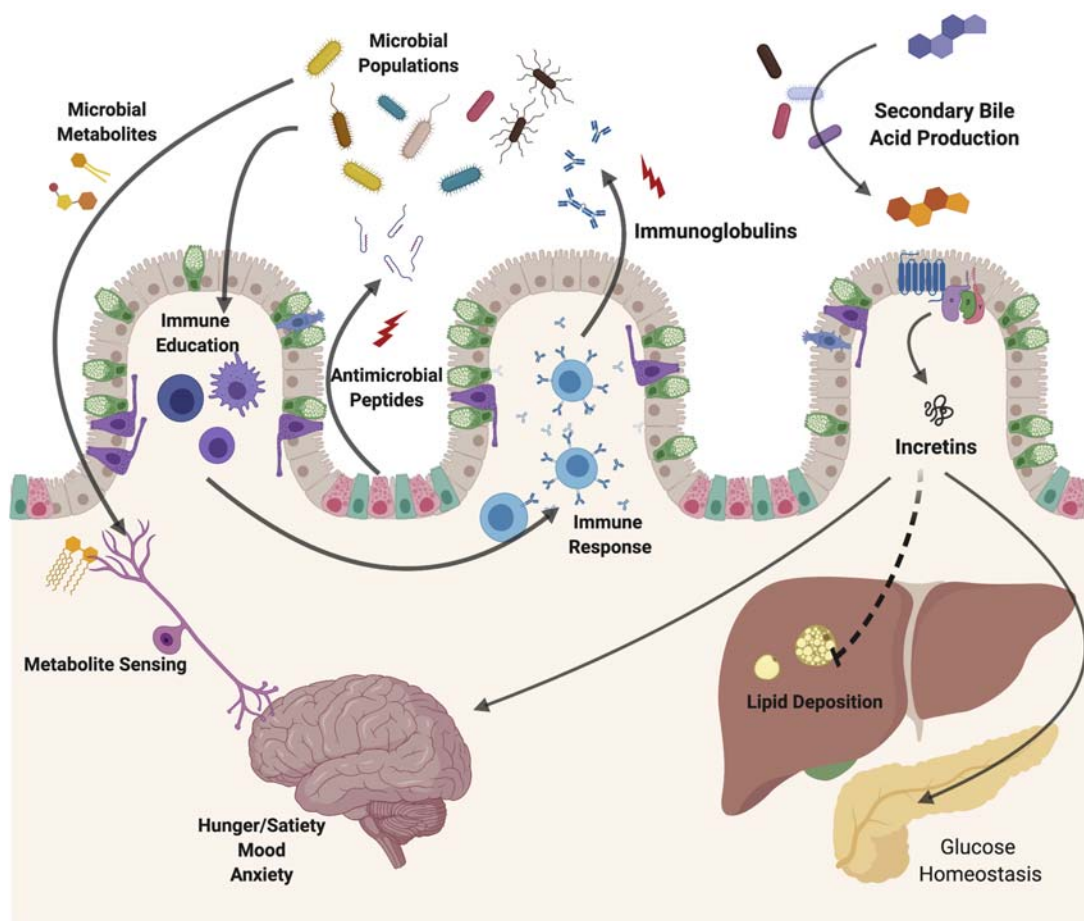


FIGURE 37.3 Schematic of host–microbiome interactions. The mammalian host directly influences gut microbes through release of mucins, antimicrobial peptides, immunoglobulins, and primary bile acids, in addition to oxygen tension, pH, and motility. These factors select microbiota capable of GIT colonization. Reciprocally, colonizing microbes stimulate host immune responses, produce small molecules and SCFAs utilizing host factors as well as factors derived from host dietary intake, and generate secondary bile acids. Together, these interactions influence host gut neuroendocrine sensory signaling resulting in metabolic modulation both within the intestine and in peripheral organs. *Source: Figure made using Biorender.com*

discordance in certain instances. Gut microbiota differences observed between discordant monozygotic twin pairs may not be solely due to bacterial community members but could involve members from other kingdoms, such as the virome.³² Further information regarding these interactions will be forthcoming in the recently published HMP2 report, which included multiomic strategies to investigate all microbial community members in diseases, including IBD and type-2 diabetes, as well as pregnancy and preterm birth.³³

In addition to findings in humans, genetically identical animal models, particularly mice, have revealed gut microbiome heterogeneity between animal vendors, and even within the same animal vivarium, which impacts physiological outcomes.^{34,35} For instance, comparisons of gut microbiota between animals from the Jackson Laboratory (Jax) versus Taconic Farms revealed presence of segmented filamentous bacteria (SFB) only in Taconic animals, which was an efficient driver of T helper (Th)

17 CD4⁺ T cells.³⁶ In both Jax and gnotobiotic mice colonized with altered schaedler flora (a defined, eight-member bacterial community) that lacked SFB, Th17 programming was significantly impaired.³⁶ Furthermore, within an animal vivarium where genetically identical animals are held under barrier-specific pathogen-free conditions, room-to-room phyla-level variations in gut microbiota profiles have been observed in lumen and mucosal samples.³⁷ Cage-to-cage variation has been identified between animals of identical genotype housed within the same room, which appeared to drive changes in the host intestinal mucus layer, immune outcomes, and metabolic responses to dietary shifts.³⁸ Various methodologies have been developed to mitigate cage variation, allowing for useful, reproducible assessments of how different diets shape gut microbes.³⁹ Together, this research highlights gut microbes can exhibit a high degree of variability contributing to dietary responses and disease outcomes that must be accounted for to

identify clear patterns following nutritional interventions. However, via careful study design, utility of correct tools, and techniques, along with conservative data interpretation, an understanding of how nutrition and diet shape gut microbes can be gained.

IV. TOOLS AND METHODS IN MICROBIOME RESEARCH

To determine community membership along with microbial community functional characteristics, a number of technologies can be used either independently or in parallel as shown in Fig. 37.4. These technologies allow for culture-independent (NGS) or culture-dependent approaches to identify and isolate novel microbial strains that may contribute either individually or as keystone members of the broader community to nutritional perturbations. Here, we discuss tools and techniques used

in the microbiome science field to elucidate the impact of and interplay between nutrition and dietary intake on gut microbes.

A. Fingerprinting Microbial Communities

Microbial gene amplicon sequencing

Studies examining health and disease associations with gut microbes in humans and animal models have extensively relied on prokaryotic 16S rRNA marker gene amplicon sequencing platforms, including Sanger-based sequencing, Roche 454, PacBio, IonTorrent, and Illumina MiSeq/HiSeq. Through unique methods, these technologies generate millions of short-read barcoded sequences. Coupled with analysis platforms, including MOTHUR⁴⁰ and Quantitative Insights Into Microbial Ecology,⁴¹ these technologies provided insight into gut bacterial diversity, allowing for relatively accurate phylogenetic and taxonomic fingerprints. Reliance on 16S

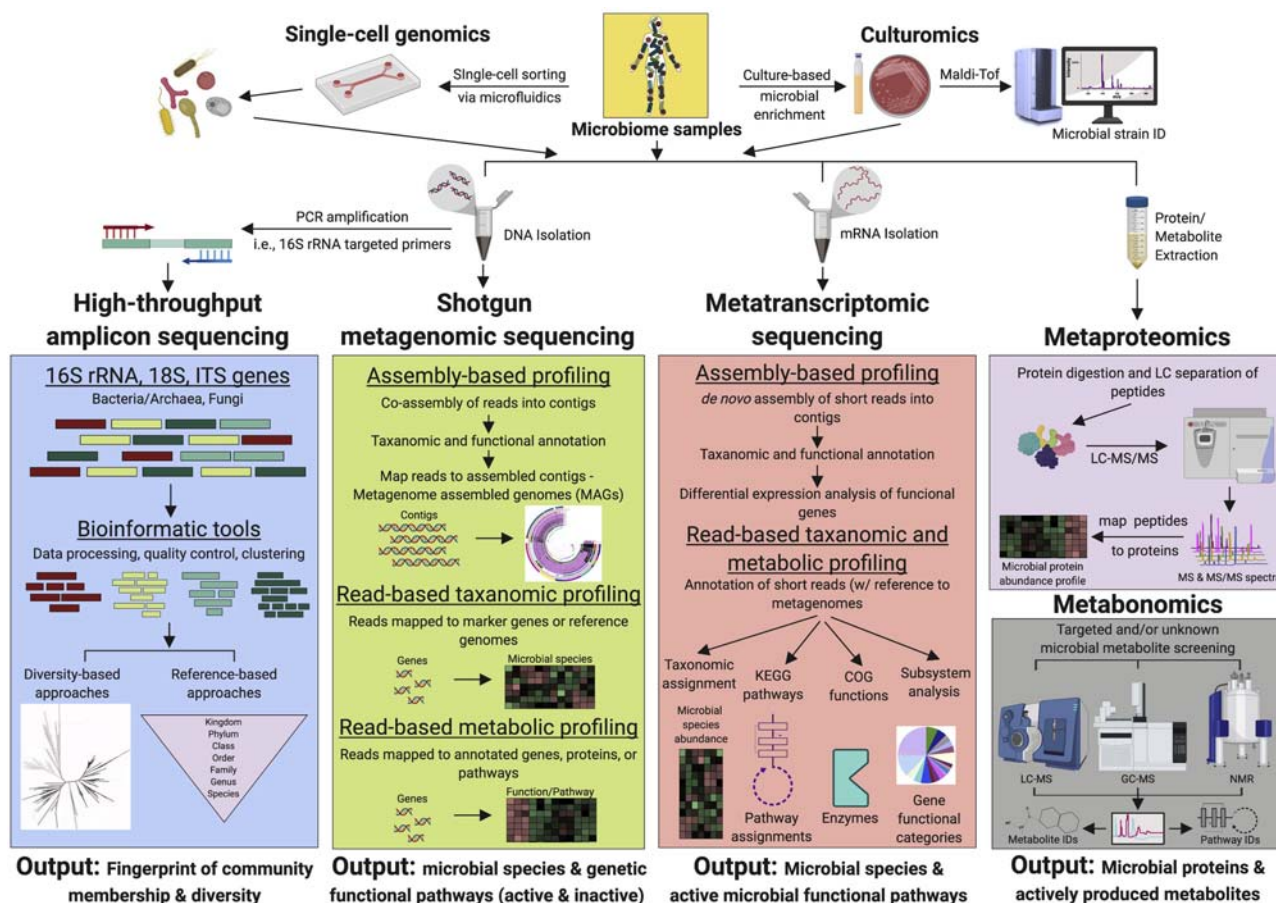


FIGURE 37.4 Overview of NGS platforms in parallel with single-cell genomics, culturomics, and metabolomics. Several sequence-based approaches are used to identify microbial community membership and function. Amplicon-based sequencing of bacteria (16S rRNA) and eukaryotes (ITS) provides fingerprints of community membership and can be coupled to upstream techniques, including single-cell genomics and culturomics-based approaches. Functional characterization of nearly all microbial community members can be determined using shotgun metagenomics (actively and inactively expressed genomic elements) or metatranscriptomics (actively expressed genomic elements) and verified using proteomics and metabolomics. Used individually or in parallel, these technologies can gain insights into multiple facets of the gut microbiome and its contribution to host nutritional and health status. Source: Figure made using Biorender.com

rRNA amplicon sequencing limits fingerprinting only to bacterial community members, which are predominant, but not the only players in health and disease.

Additional amplicon-based techniques that capture less abundant microbial community members, such as yeast, have recently been developed and become more prevalent in both human and animal studies. In contrast to studies showing relatively stable gut bacterial community membership over time in healthy individuals,⁴² fungal membership, or the mycobiota, exhibits considerably more changes, with only 20% of species found consistently between timepoints in a longitudinal analysis.⁴³ This suggests that while fungi are certainly present in the gut, their importance to the host and degree to which they exist as “core members” of the gut microbiome remains to be elucidated.

To address this, marker genes, including the 18S rRNA subunit and internal transcribed spacer 1 and 2, can be used to phylogenetically classify eukaryotic microorganisms, specifically yeast and fungi.⁴⁴ However, yeast and fungi taxonomic databases available to assign microbial identity lag considerably behind those currently available for bacteria. Furthermore, since sequencing of these eukaryotic organisms was only recently possible and many of those detected have not been cultured, it remains unclear which are true GIT residents versus nonresidents, requiring repeat introduction from diet and environment for sustained detection. It is well known that *Candida* has no known reservoir outside the mammalian gut and is a true resident; however, species of *Saccharomyces* found in many foods are potential residents.⁴⁵ *Malassezia* are yeasts and true residents, since these organisms have lost lipid synthesis capabilities, requiring the host for their metabolic needs. Greater emphasis on improving accuracy and breadth of taxonomic coverage for nonbacterial microorganisms, coupled with utilization of additional techniques, and increasing sampling depth will provide more comprehensive snapshots of undoubtedly complex interkingdom dynamics occurring in the GIT.

Transfer RNA sequencing

Beyond 16S rRNA amplicon sequencing for gut microbial fingerprinting, recent work revealed microbially derived transfer RNAs (tRNAs), which aid in decoding messenger RNA into proteins during translation, could be used to distinguish microbial communities, termed tRNA-seq with fairly high accuracy. Designed to examine pure strains in culture, novel techniques were developed for complex communities to allow for analysis of demethylated tRNA via enzymatic treatment as compared to untreated samples, permitting examination of not only tRNA transcripts but also tRNA posttranslational modifications.⁴⁶ Coupled with an open-source software platform and microbial tRNA reference libraries, this work revealed nearly identical phylogenetic

distribution in low-fat versus high-fat diet-fed mice relative to 16S rRNA amplicon sequencing, albeit with some unique distinctions.⁴⁶ Beyond phylogeny, differences in bacterial taxonomic distribution of anticodons, bacterial taxon-dependent tRNA modification level, and inferences in regard to protein expression could be identified between low-fat and high-fat samples. Together, application of tRNA-seq could serve to bridge microbial community fingerprinting and functional assessment, allowing for further insight into physiological differences between certain taxa promoted by diets.

B. In-Depth Identification and Classification of Microbial Community Members and Functions

Sequence-based approaches

Shotgun metagenomics; Despite insight into community membership gained using amplicon sequencing, very little in regard to community functional outputs can be identified. Some attempts to overcome this limitation have been developed, such as Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt), to predict KEGG ortholog functions and enzyme classification via reference databases based on 16S rRNA marker gene amplicon sequences.⁴⁷ However, for rare community members, databases are limited if they do not completely capture gene functions for microbial community members of interest,⁴⁸ particularly those of low abundance.

High-throughput shotgun metagenomic sequencing can overcome many gene amplicon sequencing limitations to examine function, since it is an untargeted, agnostic approach. Two shotgun metagenomic sequencing data analysis approaches allow for in-depth characterization of novel genetic microbial material as well as identification and classification of microbial genes. Additional nonbacterial community members, including viruses, bacteriophage, and fungi, can be explored via metagenomics. Following DNA shearing and template amplification, obtained short reads (~300 bp with Illumina sequencing) can be mapped to reference genome databases or de novo assembly tools can be applied followed by functional annotation using specialized analyses platforms to reveal metagenome assembled genomes. For mapping purposes, online platforms, including MG-RAST⁴⁹ or JCVI Metagenomics Reports (METAPREP),⁵⁰ and off-line platforms such as HUMAnN⁵¹ can be utilized. Assembly programs include khmer⁵² and novel interfaces for visualization can be used, such as A'nvio.⁵³ Several strategies have been designed to link metagenomic and 16S readouts, such as ribosomal flanking region-sequencing, which captures 16S variable regions and protein-coding genes that flank the 16S.⁵⁴ This strategy can overcome several 16S

amplicon sequencing biases, allowing for more congruent analyses of both 16S rRNA and metagenomes.

Single-cell genomics; More recently, headway has been made into using single-cell genomics to gain further insight into microbial metabolic capacity and capture rare or low-abundance community members. This involves using modalities to isolate individual microorganisms from a complex community via sorting, i.e., microfluidics, flow cytometry, or via selection i.e., pull-down of secretory IgA or IgG. Isolate identification is achieved via DNA extraction, 16S rRNA gene PCR, followed by whole genome amplification, library preparation, sequencing, and downstream analyses using programs described above. An in-depth review of techniques associated with single-cell genomics for the gut microbiome can be found in Xu and Zhao.⁵⁵

Metatranscriptomics; Metagenomic shotgun sequencing and single-cell genomics used separately or together can aid in identifying gene content and gut microbiome functions, yet microbial gene expression activity cannot be discerned from genomic DNA approaches. Metatranscriptomic shotgun sequencing (RNAseq) in combination with metagenomics offers opportunities to identify genomic potential and active microbial genomes. After total RNA isolation from microbial communities, RNA is enriched (mRNA, lincRNA, and microRNA) and fragmented and converted to cDNA via reverse transcriptase with oligo(dT) primers or random hexamers, and constructed libraries are sequenced.⁵⁶ Metatranscriptomics can be hampered by technicalities limiting its effectiveness, including compromised gut microbial RNA integrity due to sample collection and storage as well as insufficient yields of high-quality microbial RNA, limiting purification efficiency. Remnant RNA preservation solutions can interfere with downstream library preparation, biasing sequencing results. However, after overcoming these limitations, downstream metatranscriptomic data analysis utilizes similar strategies to shotgun metagenomics, including Anvi'o and HUMAnN,^{51,53,56} which allows for taxonomic assignment coupled with actively expressed gene function identification.

Long-read sequencing; While analysis platforms provide in-depth resolution of gut microbial community membership and metabolism, lack of functional annotation is problematic. To overcome annotation gaps and improve assembly, longer sequencing read lengths (>10,000 bp) are necessary. Newer technologies, such as the Pacific BioSciences' Single Molecule Real Time sequencer,⁵⁷ Illumina's Molecule,⁵⁸ and Oxford Nanopore's MinION,⁵⁹ have been used for bacteria and other microorganisms. Further technology overviews can be found in Korean and Phillippy⁶⁰ and Pollard et al.⁶¹

Together, multiomic approaches coupled with novel sequencing technologies can build upon existing gene annotations to improve identification and infer gut microbiota functionality.

Non—sequence-based approaches to identify microbial functions

To move beyond and validate microbial genomic elements resolved via sequencing, gut microbe—derived metabolites can be measured directly from GIT lumen samples as well as in serum and peripheral tissues via proteomics or metabonomics, providing valuable insights into microbial community function and activity. Advances in small-molecule discovery have enhanced comprehension of how gut microbes impact the host metabolome, which oftentimes correlates and contributes to health or disease. Many untargeted and targeted methodologies have been employed to examine transkingdom metabonomics from inorganic and organic compound perspectives, including gas chromatography—mass spectrometry, liquid chromatography—mass spectrometry, and nuclear magnetic resonance spectrometry. Coupling these techniques with additional methods provides insights into organic acids, lipids, amino acids, and both simple and complex carbohydrates.

To advance our understanding of host versus microbe metabolome contributions, conventionally raised or GF mice that have received fecal microbiota transplant (FMT) with a full microbial community have been compared to GF counterparts. Not only have differences in circulating and excreted metabolites been observed between GF and conventionally raised animals, additional peripheral tissues such as the liver, kidney, heart, and brain have also revealed a large microbial contribution.^{62–66} Classes of microbial metabolites observed are broad, including ribosome-produced and posttranslationally modified peptides, glycolipids, lipids, amino acids, secondary bile acids, oligosaccharides, nonribosomally derived peptides, terpenoids, and polyketides.⁶⁷

Culturomics-based approaches combine techniques, including classic microbiology culturing, matrix-assisted laser desorption ionization-time of flight (MALDI-TOF) mass spectrometry, and 16S rRNA gene sequencing to isolate and characterize novel gut bacteria strains from complex samples, i.e., stool. This is especially useful to identify and functionally classify novel obligate anaerobes found in the human gut⁶⁸ and for clinical diagnostic purposes.⁶⁹ By combining in-depth genomic interrogation with culturomics techniques, which reinforce and refine one another, cultivable human stool—derived bacterial strain collections have been developed, including the Culturable Genome Reference, which contains 1520 high-quality draft genomes from over 6000 indigenous isolates.⁷⁰ Via culturomics, a toxin-producing strain of *Clostridium butyricum*

was identified in neonates with necrotizing enterocolitis, whereas healthy controls did not harbor this strain.⁷¹ In a nutritional context, culturomics have been applied to kwashiorkor (a form of severe protein malnutrition) patients relative to healthy controls in Niger and Senegal, revealing 45 absent bacteria in affected patients.⁷² Identification of spectral peak outputs from mass spectrometry culturomics techniques, i.e., MALDI-TOF, can be compared to existing databases, including MetaCyc,⁷³ Human Metabolome Database,⁷⁴ SetupX, and BinBase.⁷⁵ While these tools provide insight into genomic elements and outputs of gut microbial communities or specific isolates, their direct influence on host nutrition requires further interrogation using *in vivo* and *in vitro* approaches, outlined in subsequent sections.

In vivo models for investigating microbial causality in nutrition and metabolism; Humans and many complex animals are inseparably associated from their microbiota—often beginning at birth. Therefore, capabilities to examine the role of microorganisms on host health and disease throughout the life cycle are limited to association studies. Today, the most common and standardized method to study complex effects of gut microbes on mammalian hosts in a causal way remains utilization of GF animals. The most common GF animals used are mice, due to their size, lower cost, and ability to modify

gene content. Other GF species include rats, guinea pigs, swine, poultry, zebrafish, fruit flies, and nematodes, shown in Fig. 37.5A. Beginning with a GF animal devoid of all microbial organisms, including bacteria, archaea, fungi, protists, bacteriophage, and other detectable viruses, microbial communities can be selectively reintroduced either as a complex, undefined community, in a gnotobiotic fashion (where all microorganisms in the community are known), or as a complex, undefined community at various points in the life cycle and under varying genetic disease predispositions to examine host–microbial triggers (Fig. 37.5B). Early work assessing gut microbiota impacts on both host macro- and micro-nutrient requirements, particularly related to growth, was performed in GF and gnotobiotic poultry models due to ease of rederiving these nonmammalian species under microbe-free conditions.⁷⁶

The first attempts at GF science were performed with guinea pigs when Nuttall and Thierfelder maintained sterility in GF animals for 2 weeks, demonstrating they could survive without microbial colonizers.⁷⁷ By the 1950s, following antibiotic emergence, scientific and public interest grew in understanding microbial necessity and function. James Reyniers then pioneered the first sustainable GF mouse colony by performing caesarian section on pregnant dams, transferring revived pups into isolators, and breeding them under sterile conditions for successive

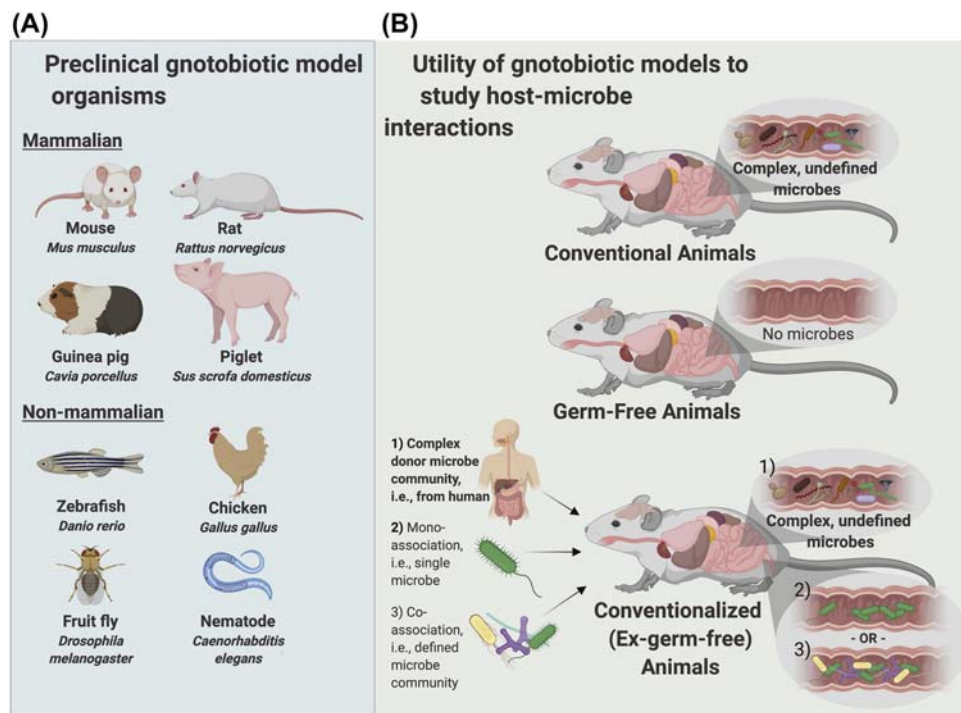


FIGURE 37.5 Germ-free (GF) and gnotobiotic animal models. Preclinical organisms studied under GF and gnotobiotic conditions (A). Several microbial states are shown (top to bottom, panel B), where most free-living animals have complex, undefined microbial communities (conventional animals). In contrast, GF animals are completely devoid of all microbes. Using donor microbiota, isolated microbial strains, or mixtures of known cultures, GF mice can be intentionally colonized, i.e., conventionalized to examine causal impacts microbes elicit upon the host (B). *Source: Figure made using Biorender.com*

generations. This method of generating GF mice with caesarean section termed 'rederivation' remains a common practice.

The use of GF and gnotobiotic pigs has reemerged as an ideal model given increased similarities of their organ systems to humans, including the GIT, brain, and immune system.⁷⁸ Piglets can be obtained via hysterectomy, maintained under GF conditions, receive conventionalization with defined communities, or be cross-fostered by a conventionally raised sow. Recent headway has been made examining host–microbe relationships in neonatal nutritional status using gnotobiotic pigs coupled with GF mouse technology.⁷⁹ Work has begun to better characterize gene expression profiles between GF and conventionalized littermate piglets revealing ~70% of the transcriptome in GI-associated and peripheral tissues is regulated by presence of gut microbes, where many differentially expressed genes are related to immunity.⁸⁰ Further work is needed to identify pathways involved in host nutrition in GF versus conventionalized piglets to improve understanding of how gut microbes impact this dynamic in this model.

Other GF animal models include zebrafish, where genetically modified embryos are placed into sterile tanks followed by colonization with known microorganisms.⁸¹ Zebrafish are attractive models because of their short life cycle, ease in generating genetic mutants, ability to image the GIT through their transparent bodies, and low maintenance costs under GF and conventionally raised conditions. Many discoveries have been made in regard to interplay between host and microbe in lipid metabolism using zebrafish.⁸² However, since fish grow at ambient temperature, much cooler than mammals, there are limits to extrapolating to mammalian biology and bacteria that thrive at these temperatures. Additional GF models include the fly species *Drosophila melanogaster*⁸³ and the nematode *Caenorhabditis elegans*,⁸⁴ offering similar benefits and limitations to zebrafish. Replication of studies in both small and large animal GF models will strengthen findings geared toward understanding roles of microbes in host nutrition and improve clinical translational capacity.

In vitro models to study the microbiome

Microbial culture-based technologies to study host–microbe interactions; Due to the large number of ways the microbiome influences the host, simplified *in vitro* methods have been developed to study effects of complex or defined microbiomes on host signaling. To examine microbial dynamics *in vitro* in response to environmental perturbations, such as pH or dietary nutrient exposure, parallel bioreactor systems made up of continuous culture vessels, also known as a chemostat, can be used.⁸⁵ While many iterations of chemostats exist, minibioreactor array systems allow for inoculation of complex or defined microbial communities in multiple replicates under

conditions that can mimic different intestinal regions, including oxygen, pH, temperature, gas mixtures, and flow rate. Using this system, day-to-day variation observed in the murine gut was recapitulated in the *in vitro* chemostat,⁸⁶ while recent work utilized this system to examine gut microbiota interactions with dietary components in the context of *Clostridium difficile*.⁸⁷ Additionally, commercially available chemostats, including the Twin Simulator of the Human Intestinal Microbial Ecosystem (TWINSHIME®), designed to mimic mucosally associated and lumenally derived ascending, transverse, and descending colon conditions can be used to examine membership and fermentation patterns in complex communities derived from human stool.⁸⁸

Cell culture model systems to study host–microbe interactions; Outside of immortalized cell lines, studying primary intestinal cells *in vitro* poses many challenges. However, development of *ex vivo* organoids from pluripotent intestinal stem cells (piSCs) obtained via biopsy or from iPS embryonic stem cells has led to various intestinal-specific organoid development.^{89–91} Small bowel piSCs isolated from any region can be used to develop enteroids, whereas stomach and colon piSCs can be developed into gastroids or colonoids. Strikingly, structures arising from different intestinal locations maintain original key features, allowing the study of various GI epithelial cell–signaling aspects in regional-specific ways.

Most organoids are grown 3D within collagen or Corning® Matrigel®. However, this results in enclosed structures, limiting access to luminal surfaces unless microinjection techniques are used. Subsequent protocols allow 2D organoid layers that polarize into luminal and basolateral surfaces, allowing introduction of macronutrients, microorganisms, their metabolites, or other stimuli to examine epithelial cell responses. Building on 2D approaches, labs have developed “gut-on-a-chip,” where epithelial cells are cultured within a dynamic platform allowing control of luminal flow, oxygen tension, temperature, and peristaltic movement, mimicking intestinal environments.⁹² These approaches, coupled with piSCs isolated from various animal mutants or patients with different intestinal disease states, have allowed unprecedented insights into roles of the microbiome, nutrition, and epithelial signaling interactions.

In addition to gut tissues, gut microbes target peripheral organs, including the liver. Using organoid approaches, liver sinusoidal cells have been differentiated into liver organoids or hepanoids.⁹³ Some advantages of hepanoids over traditional 2D primary hepatocytes are that tissues can be rapidly isolated from mutant animals or human biopsies and maintained in culture for long periods of time without dedifferentiating.

Hepanoids may also more faithfully represent complex cellular behaviors observed in vivo, including diurnal circadian rhythms in response to microbially derived metabolite exposure.⁹⁴

V. ISSUES SPECIFICALLY RELATED TO NUTRITION

The gut microbiome is dramatically shaped by dietary intake.⁹⁵ In Western societies, the rise of numerous chronic diseases is associated with dietary changes, including animal protein enrichment, increased saturated fats, processed sugars, salt, alcohol, and corn-derived fructose. Each of these dietary components is linked with gut dysbiosis, which describes deviation from optimal microbial membership and function toward a microbiome associated with disease.⁹⁶ In contrast to Western diets, diverse fruit and vegetable intake, lipids primarily from olive oil, nonrefined grains, and legumes with limited intake of red meats, characterize Mediterranean diets.⁹⁷ A growing body of work strongly supports the causal impact of a Westernized diet-induced gut microbiome upon host metabolic disruption rather than simple association. The following subsections explore how certain dietary components and nutrients influence the microbiome and health beginning early in life.

A. Early Life Microbiome Colonization and Maturation

Beginning at birth, bacteria and fungi colonization occurs with developing complexity in community structure, influenced by delivery route, antibiotic exposure, gestational age, and exposure to breast milk or formula. First colonizers are facultative anaerobes, which modify the gut environment by depleting oxygen levels required for subsequent colonization of obligate anaerobes, including *Bifidobacterium*, *Bacteroides*, *Clostridia*, and *Parabacteroides*.^{98–100} Early life microbial community importance is still under exploration, but early communities influence growth rates, immunity, and development. Early life microbiomes are shaped by delivery mode, where vaginally delivered infants display high levels of Actinobacteria, while caesarean delivered babies have greater relative abundances of Firmicutes, while premature birth leads to greater levels of Proteobacteria.^{101–103} Transfer of greater relative abundance of Lachnospiraceae of the phyla Firmicutes is also associated with increased obesity intergenerational transmission.¹⁰³ These differences are eliminated by 6 months of age, but early microbial colonization may have lasting impacts on immune education, influencing subsequent allergy or chronic disease risk.¹⁰⁴

Beyond delivery mode, feeding modality further shapes infant gut microbial communities. Early life nutrition vastly influences gut microbes with several studies revealing differences in gut microbiota community membership between breast-fed (BF) versus formula-fed (FF) infants.^{105–107} Human breast milk contains bioactive molecules, immune components, growth factors, milk oligosaccharides, and both live and dead microorganisms absent from infant formulas. Despite observations that BF and FF support unique microbial communities, heterogeneity is observed among either feeding modality, which may depend on sampling time, geographical location, cultural practices, and antibiotic exposure. Fecal samples from BF infants exhibit decreased diversity initially with more stability over time as compared to FF counterparts, where BF infants have higher relative abundance of the phyla Actinobacteria with decreased Firmicutes and Proteobacteria.¹⁰⁸ BF also supports an increased relative abundance of *Bifidobacterium* as compared to FF infants, and interestingly, a combination of FF and BF promotes increases in *Bifidobacterium* to a greater degree than FF alone; however, unique bacterial communities are observed in mixed fed infants relative to exclusively BF or FF infants.¹⁰⁸

Interestingly, metagenomic and KEGG pathway analysis revealed FF infants have a more rapidly maturing microbiota relative to BF counterparts before 24 months of age, after which this relationship inverts.¹⁰⁹ Indeed, others noted FF alters functional capacity of microbes, supporting an increased capacity for bile acid synthesis, while BF supports pathways associated with vitamin production.⁹⁹ Feeding modality and corresponding shifts in microbial membership and function impact the developing GIT transcriptome, particularly immunity and mucosal defense associated genes.¹¹⁰ Together, these data support the notion that early life feeding and nutrient exposure elicit a lasting impact on gut microbes, which ultimately impacts health later in life.

B. Macronutrient Influence on Composition and Function of the Gut Microbiome

Changes in dietary intake cause rapid shifts in microbial community structure and function. These shifts are predominantly mediated by alterations in macronutrient intake, which act as primary substrate sources for host and microbes alike. Shifts in macronutrients fundamentally alter microbial genes required to access these substrates for energy derivation, leading to different community membership. Dietary carbohydrates, fats, and proteins have been examined experimentally to understand how gut microbial communities respond to macronutrient changes. In general, strains belonging to the phylum Firmicutes are more readily capable of metabolizing dietary carbohydrates, thriving in conditions of

high carbohydrate intake, while strains belonging to the phylum Bacteroidetes utilize proteins and amino acids when dietary carbohydrates become limiting.

Gut microbiota interactions with dietary carbohydrates

Dietary carbohydrates include resistant starches, polysaccharides, oligosaccharides, and di- and monosaccharides. Carbohydrates are particularly well studied in the context of gut microbes in part because of heavy interest in understanding microbial utilization of traditionally nondigestible carbohydrates, termed microbiota-accessible carbohydrates (MACs).¹¹¹ MACs also include inulin, xylan, and cellulose, in addition to other prebiotics. Using a preclinical humanized gut microbiota mouse model, studies revealed MAC-deficient Western diets fed over multiple generations result in progressive microbial diversity reduction and MAC utilizer extinction that cannot be restored simply via MAC supplementation, requiring selective lost taxa reintroduction via FMT.¹¹² This work highlights MACs importance to maintain certain microbial membership as well as promote presence and production of downstream metabolites that broadly impact host health and nutritional status.

Microbially derived SCFA production; MACs are degraded via colonic microbial fermentation by strict anaerobes, including *Clostridia*, *Eubacteria*, and *Roseburia*, leading to metabolite synthesis including the mildly acidic 2 to 5 carbon SCFAs butyrate, propionate, and acetate.^{113,114} Total quantities of MACs reaching the colon on a daily basis are estimated at ~40 g in humans.¹¹⁵ Butyrate is produced by members of the phylum Firmicutes, including *Faecalibacterium*,¹¹⁶ whereas propionate is produced by select *Bacteroides* species and *Clostridium*. Very few species produce both butyrate and propionate. Conversely, acetate is the most abundant SCFA, produced by many bacteria. Acetate acts as an important substrate for growth of anaerobic bacteria, including *Faecalibacterium prausnitzii*, which cannot be cultured without acetate addition. Other microbially derived fermentation products include fumarate and lactate. Production of these molecules is important for microbial ecology. For instance, *Bifidobacterium longum* fermentation of fructooligosaccharides (FOS) results in lactate production, which then serves as substrate for *Eubacterium hallii*, a strain that cannot grow on FOS alone without *B. longum* present.¹¹⁷

In exchange for host carbohydrate intake, an important energy source for microbial communities, microbial production of SCFAs provides both caloric benefit and acts as signaling molecules for the host. SCFAs bind to various intestinal and adipose tissue G-protein–coupled receptors. Within the intestine, SCFAs stimulate incretin hormone release, including glucagon-like peptide-1 (GLP-1), gastric inhibitory polypeptide, and peptide

tyrosine tyrosine, which aid in regulating glucose homeostasis.^{118–120} SCFA importance was investigated in rodents fed highfat diet with and without MACs, demonstrating MACs significantly reduced food intake, which correlated with lower ghrelin levels and elevated GLP-1.¹²¹ These effects could, in part, be mediated via hypothalamic endocrine signaling and nuclei of the solitary tract as well as paracrine actions via splanchnic bed vagal nerve fibers.^{122,123}

However, not all SCFA effects are metabolically beneficial. While acetate stimulates liver and hypothalamic 5' adenosine monophosphate–activated protein kinase leading to increased thermogenesis and reduced food intake, acetate and propionate increase fat accumulation in white adipose tissues (WATs), induce antilipolytic activity, and elevate adipose differentiation.^{124,125} In contrast, butyrate stimulates lipolysis and reduces WAT adipocyte size. Additionally, succinate production by species like *Prevotella copri* is associated with increased chronic inflammation under susceptible states.¹²⁶ Therefore, specific dietary carbohydrates likely result in varying ratios of SCFAs, which differentially alter signaling in different tissues, inflammatory responses, and global homeostasis.

Certain microbial communities are capable of high-energy harvesting or low-energy harvesting, based on their SCFA production from nondigestible carbohydrates, which may have important energy balance implications.^{127,128} Human and murine studies employing metagenomics suggest increased fermentative capacity of dietary carbohydrates and SCFA production is prevalent in obese states; however, some studies failed to find this correlation. In other work, children in rural Burkina Faso, Africa, demonstrated normal growth rates when compared to age-matched European children in Italy, despite Burkina Faso children ingesting several hundred fewer calories per day than European counterparts.¹²⁹ Gut microbes of Burkina Faso children exhibited enrichment in the genera *Prevotella* and *Xylani-bacter*, which have elevated capacity to degrade xylan and cellulose. Together, these findings suggest that in addition to dietary carbohydrate type and amount, specific gut microbial communities harbored by an individual may ultimately influence total production and ratio of SCFAs, which could influence host energy balance regulation in overnutrition and undernutrition alike.

Microbially derived hydrogen production; In addition to SCFAs, lactate, and fumarate, production of hydrogen (H₂) gas is an important microbial fermentation product. Flatus is comprised of ~40% H₂ gas produced by *Bacteroides* and *Clostridium*. However, not all H₂ is released from the GIT, since it is utilized in oxidation reactions of organic substances, including sulfide, methane, and acetate generation. Sulfate obtained from endogenous mucins or diet can be reduced with available H₂ by

Desulfovibrio bacteria.¹³⁰ Archaeal genera, including *Methanobacterium*, utilize available carbon dioxide (CO₂) and H₂ to form methane.¹³¹ Finally, acetogenesis, a complex chemical reaction, utilizes four H₂ molecules and CO₂ to form acetate, utilized as an energy source by both host and bacteria. In addition to bacteria, greater carbohydrate consumption leads to expansion of *Candida*, while *Debaryomyces* and *Penicillium* appear passively related to recent milk-based product, i.e., cheese consumption. Therefore, diet may also select the gut mycobiome via introduction of organisms and provide substrate for true residents.

Interaction of gut microbes with dietary protein

Dietary proteins and their source are profoundly important for both gut microbiota selection and microbial end product diversity, including amines, indoles, thiols, H₂, H₂S, and SCFAs,¹³² the majority of which are generated by colonic microbes. Human subjects switched from plant- to animal based-protein diets lead to dramatic fecal microbiome shifts in as little as 24 h.⁹⁵ Rodent studies further corroborate this, where soy, pork, beef, chicken, fish, and casein proteins fed to rats for 14 days differentially shifted microbiota communities.¹³³ Strikingly, red versus white meat consumption produced significant microbial community differences in only 2 days, despite both being derived from animal sources.¹³³ Soy protein intake (SPI) also uniquely elevated fecal SCFA levels as well as increased relative abundance of propionate producers, *Bacteroides* and *Prevotella*, compared with animals fed red meat, white meat, or casein proteins.¹³⁴ In other work, SPI led to increased microbial diversity in rodents throughout the GIT compared to animals fed casein protein, the latter of which displayed elevated levels of *Bacteroidetes*, including Bacteroidaceae and Porphyromonadaceae families relative to SPI.¹³⁵ Conversely, SPI is associated with elevated Bifidobacteriaceae, a known stimulator of intestinal epithelial integrity and gut health. While local GIT changes are evident, SPI also significantly reduces plasma lipid level, reduces high-fat diet-induced weight gain, and increases cecal bile acid pools with altered composition compared with casein diet in rodents.^{135,136}

While various protein sources directly impact host and microbe, specific amino acid availability can lead to SCFA generation. Glutamate, cysteine, serine, and lysine fermentation generate butyrate, while alanine, threonine, and aspartate generate propionate, whereas methionine fermentation results in either butyrate or propionate. This suggests that despite similar total protein intake, specific dietary protein sources differentially influence metabolic outcomes and gut microbes may be contributory.

Leucine, isoleucine, and valine are essential branched-chain amino acids (BCAAs), which provide both structural and regulatory functions. Dietary supplements for bodybuilding often include BCAAs to stimulate protein

synthesis, yet BCAAs are positively correlated with insulin resistance and glucose intolerance.¹³⁷ In rodent models, BCAA supplementation leads to elevated muscle accumulation, inducing insulin receptor substrate-1 phosphorylation, impairing insulin signaling. BCAAs also stimulate the mammalian target of rapamycin.

While the host cannot synthesize BCAAs, many gut microbiome community members encode genes for BCAA synthesis enzymes.¹³⁸ Independent of dietary supplementation, serum BCAA levels are also correlated with microbiome changes, including elevation of *Prevotella* and *Bacteroides*.¹³⁹ Elevated levels of these genera have been observed in humans with insulin resistance, and a metaanalysis of over 20,000 patients demonstrated BCAA levels can predict diabetic risk.¹⁴⁰ Confirming this, mice colonized with *P. copri* increased circulating BCAAs and insulin resistance.¹³⁹ Together, these findings suggest certain gut bacteria may stimulate endogenous BCAA synthesis, contributing to overall protein synthesis and insulin resistance.

Interaction of gut microbes with dietary lipids

Dietary lipids represent a diverse class of molecules varying by chain length, carbon double bond number and configuration, and hydroxyl group placement. Dietary lipid type can induce diverse effects on both host and gut microbes. Often, dietary fat studies broadly compare saturated versus unsaturated or omega-3 versus omega-6 from a variety of sources. Saturated fatty acids (SFAs) typically lead to more harmful metabolic outcomes, including weight gain, elevated triglycerides, and greater insulin resistance compared with polyunsaturated fatty acid (PUFA) ingestion. Most lipid absorption takes place in the small intestine, but some fats reach the colon. Ingestion of SFAs results in greater delivery to the colon compared with unsaturated fatty acids, resulting in reduced microbial community diversity and composition.⁹⁵ In general, increased dietary fat entering the colon leads to an increased ratio of Bacteroidetes to Firmicutes,¹²⁸ although importance of this shift is still debated. Dietary fat source, as with proteins, also differentially influences gut microbes.¹⁴¹ For instance, milk fat promotes expansion of sulfite-reducing *Bilophila wadsworthia*,⁹⁵ which transforms conjugated bile acid taurine residues into H₂S. In genetically susceptible mice fed high milk fat diet, expansion of *B. wadsworthia* led to proinflammatory responses.¹⁴² Conversely, PUFA intake did not expand *B. wadsworthia*.^{95,142}

Recent findings in mice demonstrate levels of small intestinal lipid absorption are regulated by jejunum-specific microbiota.¹⁴³ Here, high-fat feeding resulted in increased jejunal *Clostridium bifermentans*, and exposure of intestinal enteroids to *C. bifermentans* conditioned media upregulated lipid reesterification genes and triglyceride uptake. Furthermore, high-fat diet-induced jejunal microbiota transplantation into GF mice led to significantly

upregulated triglyceride and cholesterol uptake compared with low-fat diet–induced jejunal microbiota, despite the exposure of both recipient groups to low-fat diet.¹⁴³ Taken together, these studies highlight the importance of direct and indirect influences dietary lipids have on host–microbe interactions, revealing differences in dietary fat source can uniquely impact these dynamics.

C. Micronutrient Influence on Gut Microbiome Composition and Function

Gut microbes and vitamin production

Vitamins are essential micronutrients that, by definition, cannot be synthesized by the host yet are required to maintain proper metabolism. Microbes and plants synthesize many vitamins required for enzymatic functions. While most vitamins must be diet-derived, gut microbes synthesize some endogenously. Gut microbes can synthesize vitamin K, in addition to thiamine, pantothenic acid, pyridoxine, riboflavin, nicotinic acid, biotin, and folates.¹⁴⁴ Lactic acid bacteria, including *Lactobacillus*, which are abundant in the gut and fermented foods, produce many of these vitamins. Accordingly, GIT or microbiome perturbations can decrease vitamin absorption.

Vitamin B12 (cobalamin) is an essential vitamin, despite its requirement for only two enzymatic functions in the body, including methyl-malonyl-CoA mutase and methionine synthase. Only a few bacteria, such as *Pseudomonas denitrificans*, *Bacillus megaterium*, and *Propionibacterium freudenreichii*, and possibly some species of *Lactobacillus* can synthesize B12, due to the roughly 30 genes required for its synthesis.¹⁴⁵ Following bariatric surgery, an intervention that alters both host anatomy and gut microbes,¹⁴⁶ B12 supplements are required since production of intrinsic factor by the gastric mucosa is reduced.¹⁴⁷

Thiamine (B1) is required for synthesis of fatty acids, nucleic acids, steroids, and aromatic amino acid precursors, as well as neurotransmitter production. B1 can be synthesized by *Lactobacillus* and *Bifidobacterium* species (especially *Bifidobacterium adolescentis*).¹⁴⁸ Folate (B11) is essential for nucleic acid synthesis, where numerous diseases as well as neural tube defects in gestational development are linked to deficiency. While the precise endogenous microbiota members that are the most important folate producers remains unknown, metagenomic analysis suggests many *Lactobacillus* species harbor most genes required for its synthesis.¹⁴⁹ However, many are missing the putative gene 4-amino-4-deoxychorismate lyase which is required for complete biosynthesis. One species of *Bifidobacterium*, *B. adolescentis*, contains all required genes and is utilized in food fermentation.

Finally, vitamin K is obtained as either plant-produced phyloquinones (K1) and to a lesser extent menaquinones (K2) generated endogenously by the microbiome,

including many *Lactobacillus* species. Prolonged antibiotic use can reduce B vitamins and vitamin K synthesis, which was initially observed soon after widespread utility of antibiotics in the 1950s.¹⁵⁰ Antibiotic-induced vitamin deficiency is especially important in infants where risk of bleeding due to low vitamin K levels, although rare, is often an indication for vitamin K prophylaxis.¹⁵¹

Gut microbial choline and carnitine modifications

An emerging important area of gut microbiota research is their interaction with dietary micronutrients choline and carnitine in cardiovascular health and metabolism. As an essential nutrient, choline functions as a methyl donor in many metabolic processes and phospholipid membrane synthesis, i.e., phosphatidylcholine and spingomyelin. Carnitine is essential for mitochondrial long-chain fatty acid transport and is fundamental for energy. Choline and L-carnitine are found in red meats, eggs, and dairy products, which are abundant in Westernized diets. Choline, L-carnitine, and lecithin microbial transformation leads to gamma-butyro-betaine and trimethylamine (TMA) production, eliciting unfortunate consequences on the host.^{152,153} Hepatic enzymes convert TMA to trimethylamine N-oxide (TMAO), which is well-associated with cardiovascular disease, increased thrombosis risk, and metabolic syndrome in humans.^{154–156} The microbiome's role was confirmed in humans by feeding hard-boiled eggs plus deuterium-labeled phosphatidylcholine with and without broad-spectrum antibiotics, demonstrating significant reductions in circulating TMAO with antibiotic exposure.¹⁵⁷ Additionally, TMAO dietary supplementation in GF mice promotes atherosclerosis, including elevated macrophage cholesterol accumulation and foam cell formation.

Interaction of gut microbes with polyphenols

Polyphenols are a diverse class of compounds produced by plants as self-defense molecules against pathogens, predators, and UV damage, which are broadly found in plant-derived food sources and beverages.^{158,159} Unlike carbohydrates, polyphenols typically contain phenyl structures with various hydroxyl and substitution groups. Flavonoids are the largest class of polyphenols, including anthocyanidins, proanthocyanidins, isoflavones, flavanols, flavonols, flavones, and flavanones, among others.¹⁶⁰ Additional polyphenols include lignin, stilbenes, and phenolic acids.

Similar to carbohydrates, gut microbes can interact with or be influenced by dietary polyphenols, conferring host health benefits, which most notably include improved metabolism, reduced inflammation, antioxidant, and even neurobeneficial properties. Many polyphenols, such as proanthocyanidins or tannins, are

completely unabsorbed from the intestine; therefore, their mechanisms of action are limited to the GI lumen or mucosal surface, interfering with host digestive processes and microbial metabolism.¹⁶¹ Large molecules, such as tannins, may delay digestive processes, reducing macronutrient absorption.¹⁶² Similarly, ingestion of other carbohydrates, such as cellulose, may provide metabolic benefits through analogous mechanisms.

Studies also suggest certain polyphenols contain antioxidant capacity and confer antiinflammatory activities.¹⁶³ However, while antioxidant mechanisms are plausible in intestinal epithelium, mechanisms in peripheral tissues may require absorption of fermentative polyphenol-derived metabolites or conversion to aglycones to mediate intestinal absorption and transport to the periphery. Smaller polyphenols may be absorbed intracellularly and confer effects at peripheral sites. Colonic fermentation of dietary polyphenols has been described for several *Bacteroides*, *Eubacterium*, and *Lachnospiraceae* species.^{164,165} Fermentative polyphenol metabolites may confer metabolic effects, including increased thermogenesis, fat oxidation, reduced glucose metabolism efficiency, and delayed lipid absorption.^{166,167}

The best-described examples of polyphenol-induced changes in gut microbes are those observed with *Akkermansia muciniphila*, a bacterial species dependent upon host epithelial goblet cell secretion of mucus. Typically found at low abundance, these bacteria spike in certain circumstances, including following bariatric surgery, dietary switch, and generally exhibit increased relative abundance in lean individual microbial communities.^{168,169} Conversely, *A. muciniphila* is found in lower abundances in diabetes, obesity, and cardiovascular disease, whereas experimental administration of *A. muciniphila* improves obesity-associated molecular metabolism.¹⁷⁰ Nonabsorbed proanthocyanidin ingestion improves mucin production and release by host GIT epithelia.¹⁷¹ Consistently, *A. muciniphila* abundance increases following proanthocyanidin intake, as well as with a variety of other supplemented polyphenols,¹⁷² improving metabolic outcomes in preclinical settings.^{173,174} Further exploring polyphenol-induced gut microbiota changes may reveal more cogent examples of how many forms of nonabsorbed polyphenol molecules improve host molecular metabolism in peripheral tissues, eliciting positive influences on global host metabolic function.

RESEARCH GAPS

The multidisciplinary and collaborative efforts across broad ranges of talented investigators, including clinicians, microbiologists, dieticians, and data scientists, have allowed for nutritionally based microbiome science advances. Gut microbiota characterization using amplicon sequencing remains a mainstay of the field. However, major goals of ongoing research efforts are to gain insight into gut microbiome functionality as a whole, in addition to individual community member isolation and characterization. Tackling several research gaps through integration of existing and novel approaches will permit elucidation of how gut microbes interact with dietary components to identify, interrogate, and target microbially derived genomic and functional outputs that promote host nutritional health.

- Improve ability to assess nutrient requirements and interactions of host and microbe separately and in parallel.
- Incorporate interventional, longitudinal one-off studies to overcome interindividual microbiota variation.
- Develop novel techniques to assess microbiome function in real time, including advanced artificial intelligence, synthetic biology tools, and biosensors.
 - Examine gut microbial community specificity throughout the GIT and regional functions.
 - Understand coassociations between bacterial species, as competition, synergism, or coevolution.
 - Identify roles of other microbial kingdoms, including fungi, archaea, and bacteriophage, in mediating nutritional health.
 - Develop cultivation techniques coupled to functional profiling to explore activity range and small molecules produced by gut microbes.

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NUTRIENT REGULATION OF THE IMMUNE RESPONSE

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SUMMARY

There is a bidirectional interaction between nutrition, infection, and immunity: undernutrition decreases immune defenses, making an individual more susceptible to infection, but the immune response to an infection can itself impair nutritional status and alter body composition. Practically all forms of immunity are affected by protein-energy malnutrition, but nonspecific defenses and cell-mediated immunity are more severely affected than humoral (antibody) responses. Micronutrients are required for an efficient immune response, and deficiencies in one or more micronutrients diminish immune function. Essential fatty acids play a role in the regulation of immune responses, since they provide precursors for the synthesis of lipid mediators. Deficiencies in essential amino acids impair immune function, but some nonessential amino acids (e.g., arginine and glutamine) may become conditionally essential in stressful situations. Probiotic bacteria enhance immune function in laboratory animals and may do so in humans. Prebiotics may also have these effects. Breast milk has a composition that promotes the development of the neonatal immune response and protects against infectious diseases.

Keywords: Antibody; Antigen presentation; Breast milk; Cytokine; Immunity; Infection; Leukocyte; Lymphocyte; Microbiota.

I. INTRODUCTION

A. Background

Associations between famine and epidemics of infectious disease have been noted throughout history: as early as 370 BCE, Hippocrates recognized that poorly nourished people are more susceptible to infectious disease. In general, undernutrition impairs the immune system, suppressing immune functions that are fundamental to host protection against pathogenic organisms.^{1–3} Undernutrition leading to impairment of immune function can be due to insufficient intake of energy and macronutrients and/or due to deficiencies in specific micronutrients. These may occur in combination. The impact of undernutrition is greatest in developing countries, but it is also

important in developed countries, especially among the elderly, individuals with eating disorders, alcoholics, patients with certain diseases, premature babies, and those born small for gestational age. Although it has proven difficult to identify the precise effects of individual nutrients on different aspects of immune function, it is now clear that many nutrients have an important role in maintaining the immune response. Thus, the functioning of the immune system is influenced by nutrients consumed as normal components of the diet, and appropriate nutrition is required for the host to maintain adequate immune defenses toward bacteria, viruses, fungi, and parasites. This chapter begins with an overview of the key components of the immune system, concentrating on the cells that participate in immune responses and the mechanisms

by which they communicate. The role of nutrients in the immune system is examined using specific examples, and the cyclic relationship between infection and nutritional status is discussed. The main body of the chapter is devoted to an evaluation of the influence of individual micronutrients and macronutrients on immune function. This chapter expands upon an earlier review of the topic.⁴

B. Definitions and Explanatory Relationships

The topics in this section are described in full in Calder.⁴

The immune system

Overview; The immune response acts to protect the host from infectious agents that exist in the environment (pathogenic bacteria, viruses, fungi, parasites) and from other noxious insults and to ensure tolerance to harmless environmental constituents like food and commensal bacteria and to the host itself. The immune response is a complex system involving various cells distributed in many locations throughout the body and moving between these locations in the lymph and the bloodstream. In some locations, the cells are organized into discrete lymphoid organs. These can be classified as primary lymphoid organs (bone marrow and thymus)

where immune cells arise and mature and secondary lymphoid organs (lymph nodes, spleen, gut-associated lymphoid tissue [GALT]) where mature immune cells interact and respond to antigens. The immune system has two general functional divisions: the innate (or natural) immune system and the acquired (also termed specific or adaptive) immune system (see Fig. 38.1).

Innate immunity; Innate immunity consists of physical barriers, soluble factors, and phagocytic cells, which include granulocytes (neutrophils, basophils, eosinophils), monocytes, and macrophages (Table 38.1). Innate immunity has no memory and is therefore not influenced by prior exposure to an organism. Phagocytic cells, the main effectors of innate immunity, express receptors that recognize certain structures of bacterial or viral origin. These receptors are termed pattern recognition receptors, and the structures they recognize are termed pathogen (or microbe)-associated molecular patterns. Binding of bacteria to surface receptors triggers phagocytosis (engulfing) and subsequent destruction of the bacterium by toxic chemicals, such as superoxide radicals and hydrogen peroxide. Natural killer cells also possess surface receptors and destroy their target cells by the release of cytotoxic proteins. In this way, innate immunity provides a rapid first line of defense against invading pathogens.

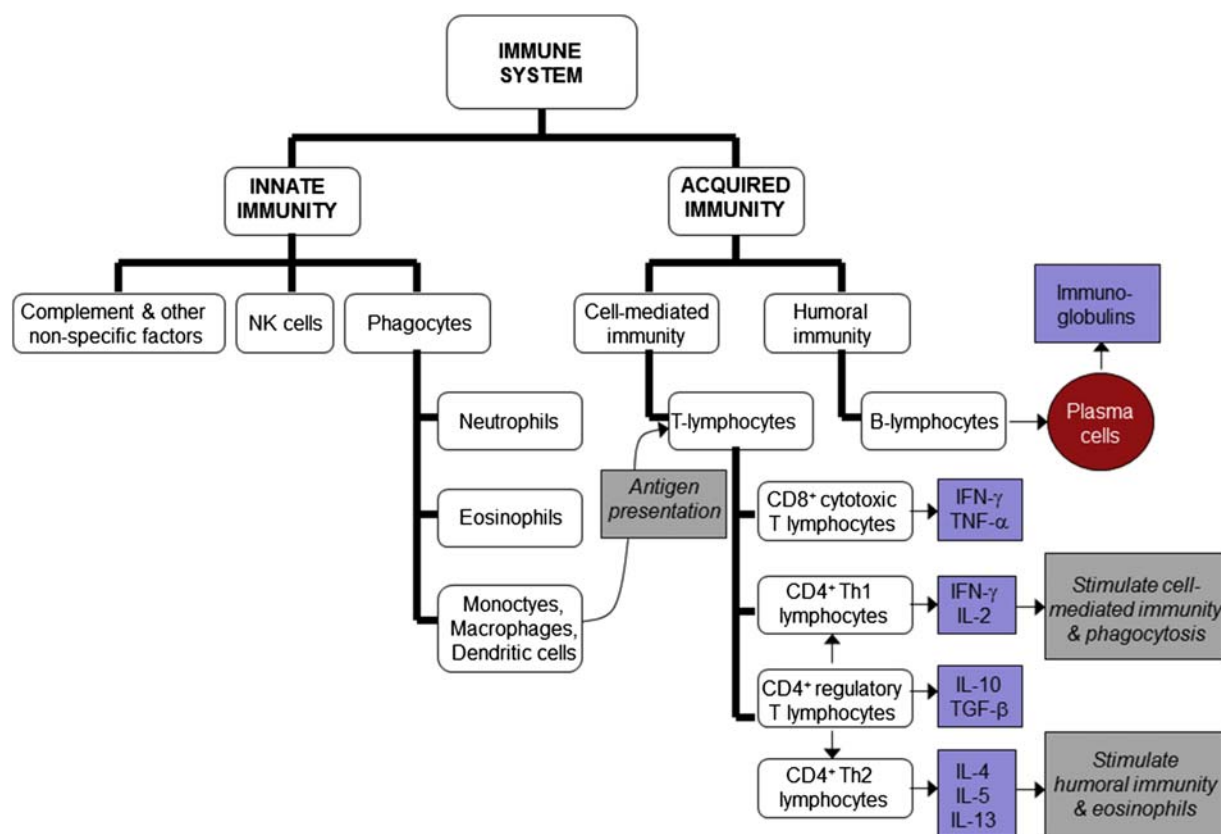


FIGURE 38.1 Schematic representation of the immune system. *IFN*, interferon; *IL*, interleukin; *NK*, natural killer; *TGF*, transforming growth factor; *Th1*, type 1 helper; *Th2*, type 2 helper; *TNF*, tumor necrosis factor.

TABLE 38.1 Components of innate and acquired immunity.⁴

	Innate	Acquired
Physicochemical barriers	Skin	Cutaneous and mucosal immune systems
	Mucus membranes	Antibodies in mucosal secretions
	Lysozyme	
	Stomach acid	
	Commensal bacteria	
Circulating molecules	Complement	Antibodies
Cells	Granulocytes	Lymphocytes (T and B)
	Monocytes/macrophages	
	Natural killer cells	
Soluble mediators	Macrophage-derived cytokines	Lymphocyte-derived cytokines

However, an immune response often requires the coordinated actions of both innate immunity and the more powerful and flexible acquired immunity.

Acquired immunity; Acquired immunity involves the specific recognition of molecules (termed antigens) on an invading pathogen, which distinguish it as being foreign to the host. Lymphocytes, which are classified into T and B lymphocytes (also called T cells and B cells), are responsible for this form of immunity (Fig. 38.1). All lymphocytes (indeed all cells of the immune system) originate in the bone marrow. B lymphocytes undergo further development and maturation in the bone marrow before being released into the circulation, while T lymphocytes mature in the thymus. From the bloodstream, lymphocytes can enter secondary lymphoid organs. Immune responses occur largely in these lymphoid organs, which are highly organized to promote the interaction of cells and invading pathogens.

Since each lymphocyte carries surface receptors for a single antigen, the acquired immune system is highly specific. However, acquired immunity is extremely diverse; the lymphocyte repertoire in humans has been estimated to be able to recognize approximately 10^{11} antigens. The high degree of specificity, combined with the huge lymphocyte repertoire, means that only a relatively small number of lymphocytes will be able to recognize any given antigen. The acquired immune system has developed the ability for clonal expansion to deal with this. Clonal expansion involves the proliferation of a lymphocyte once an interaction with its specific antigen has occurred, so that a single lymphocyte gives rise to a clone of lymphocytes, all of which have the ability to recognize

the antigen causing the initial response. The acquired immune response becomes effective over several days after the initial activation, but it also persists for some time after the removal of the initiating antigen. This persistence gives rise to immunological memory, which is also a characteristic feature of acquired immunity. It is the basis for the stronger, more effective immune response upon reexposure to an antigen (i.e., reinfection with the same pathogen) and is the rationale for vaccination. Eventually, the immune system will reestablish homeostasis using self-regulatory mechanisms that involve communication between cells.

B and T lymphocytes; B lymphocytes are characterized by their ability to produce antibodies (these are soluble antigen-specific immunoglobulins). This form of protection is termed humoral immunity. B lymphocytes also carry immunoglobulins, which are capable of binding an antigen, on their cell surface. Binding of immunoglobulin with antigen causes proliferation of the B lymphocyte and subsequent transformation into plasma cells, which secrete large amounts of antibody with the same specificity as the parent cell.

There are five major classes of immunoglobulin (IgA, IgD, IgG, IgM and IgE), each of which elicits different components of the humoral immune response. Antibodies work in several ways to combat invading pathogens. They can “neutralize” toxins or microorganisms by binding to them and preventing their attachment to host cells, and they can activate complement proteins in plasma, which in turn promote the destruction of bacteria by phagocytes. Since they have binding sites for both an antigen and for receptors on phagocytic cells,

antibodies can also promote the interaction of the two components by forming physical “bridges,” a process known as opsonization. The type of phagocytic cell bound by the antibody will be determined by the antibody class. Macrophages and neutrophils are specific for IgM and IgG, while eosinophils are specific for IgE. In this way, antibodies are a form of communication between the acquired and the innate immune response; although they are elicited through highly specific mechanisms, they are ultimately translated to a form that can be interpreted by the innate immune system, enabling it to destroy the pathogen.

Humoral immunity deals with extracellular pathogens (e.g., many bacteria). However, some pathogens, particularly viruses, but also certain bacteria, infect individuals by entering cells. These pathogens will escape humoral immunity and are instead dealt with by cell-mediated immunity, which is conferred by T lymphocytes. T lymphocytes express antigen-specific T-cell receptors (TCRs) on their surface. TCRs have an enormous antigen repertoire. However, unlike B lymphocytes, they are only able to recognize antigens that are presented to them on a cell surface (the cell presenting the antigen to the T lymphocyte is termed an antigen-presenting cell); this is the feature that distinguishes humoral from cell-mediated immunity. Activation of the TCR results in entry of T lymphocytes into the cell cycle and, ultimately, proliferation. Activated T lymphocytes also begin to synthesize and secrete

the cytokine interleukin-2 (IL-2), which further promotes proliferation and differentiation. Thus, the expansion of T lymphocytes builds up an army of antigen-specific T lymphocytes in much the same way as that of B lymphocytes. Effector T lymphocytes have the ability to migrate to sites of infection, injury, or tissue damage. There are several types of T lymphocytes including cytotoxic T cells, helper T cells, and regulatory T cells. Cytotoxic T lymphocytes carry the surface protein marker CD8 and kill infected cells and tumor cells by secretion of cytotoxic enzymes, which cause lysis of the target cell. Helper T lymphocytes carry the surface protein marker CD4 and eliminate pathogens by stimulating the phagocytic activity of macrophages and the proliferation of, and antibody secretion by, B lymphocytes. Helper T lymphocytes have traditionally been subdivided into two broad categories according to the pattern of cytokines they produce, although new categories have recently been identified (Fig. 38.2). Helper T cells that have not previously encountered antigen produce mainly IL-2 upon the initial encounter with an antigen. These cells may differentiate into a population, sometimes referred to as Th0 cells, which differentiate further into either Th1 or Th2 cells. This differentiation is regulated by cytokines: IL-12 and interferon (IFN)- γ promote the development of Th1 cells, while IL-4 promotes the development of Th2 cells. Th1 and Th2 cells have relatively restricted profiles of cytokine production:

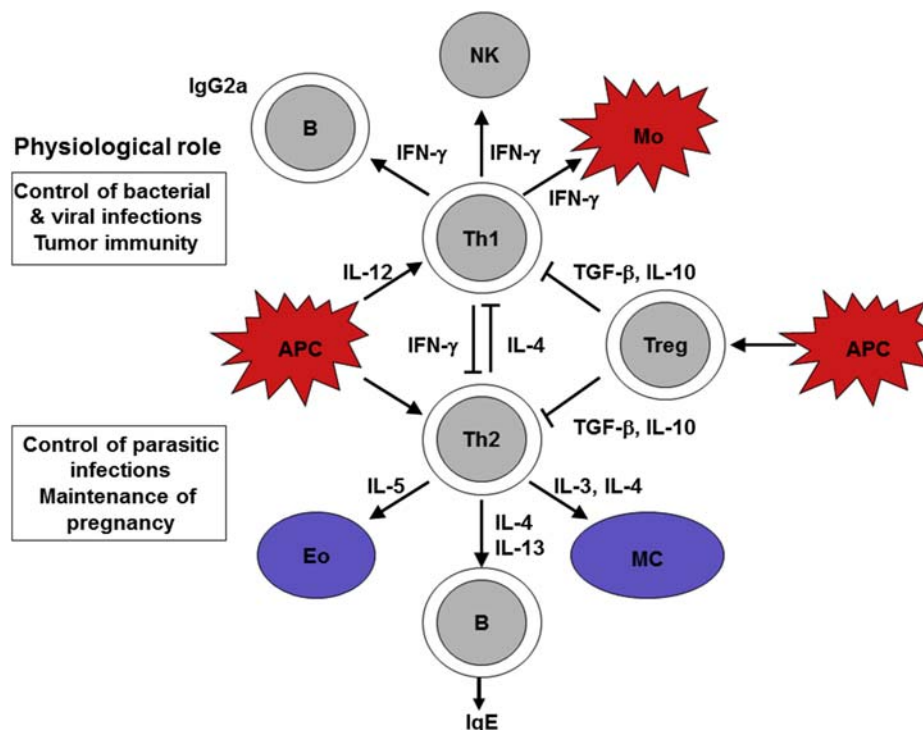


FIGURE 38.2 Schematic representation of the roles of helper T cells in regulating immune responses. APC, antigen-presenting cell; B, B cell; Eo, eosinophil; IFN, interferon; Ig, immunoglobulin; IL, interleukin; MC, mast cell; NK, natural killer cell; TGF, transforming growth factor; Th1, type 1 helper T cell; Th2, type 2 helper T cell; Treg, regulatory T cell. Adapted from Ref. 8

Th1 cells produce IL-2 and IFN- γ , which activate macrophages, natural killer cells, and cytotoxic T lymphocytes, which are the principal effectors of cell-mediated immunity. Interactions with bacteria, viruses, and fungi tend to induce Th1 activity. Th2 cells produce IL-4, which stimulates IgE production, and IL-5, an eosinophil-activating factor. Th2 cells are responsible for defense against helminthic parasites, which is due to IgE-mediated activation of mast cells and basophils. The patterns of cytokine secretion by Th1 and Th2 lymphocytes were first demonstrated in mice, and, while human helper T lymphocytes do show differences in cytokine profile, the divisions are not as clear with the majority secreting a mixture of Th1 and Th2 cytokines in differing proportions. Thus, the terms “Th1 dominant” and “Th2 dominant” are commonly used to describe the cytokine profiles of these cells. More

recently characterized categories of helper T cells include Th17 cells, which play an important role in autoimmunity (where the immune system attacks host tissues inappropriately). Regulatory T cells ($CD4^+CD25^+FoxP3^+$) produce IL-10 and transforming growth factor- β and suppress the activities of B cells and other T cells preventing inappropriate activation.

The gut-associated immune system; The immune system of the gut (sometimes termed the GALT) is extensive and includes the physical barrier of the intestinal wall, as well as components of innate and adaptive immune responses.⁵ The physical barrier includes acid in the stomach, peristalsis, mucus secretion, and tightly connected epithelial cells, which collectively prevent the entry of pathogens. The cells of the immune system are organized into specialized lymphoid tissues, termed Peyer's

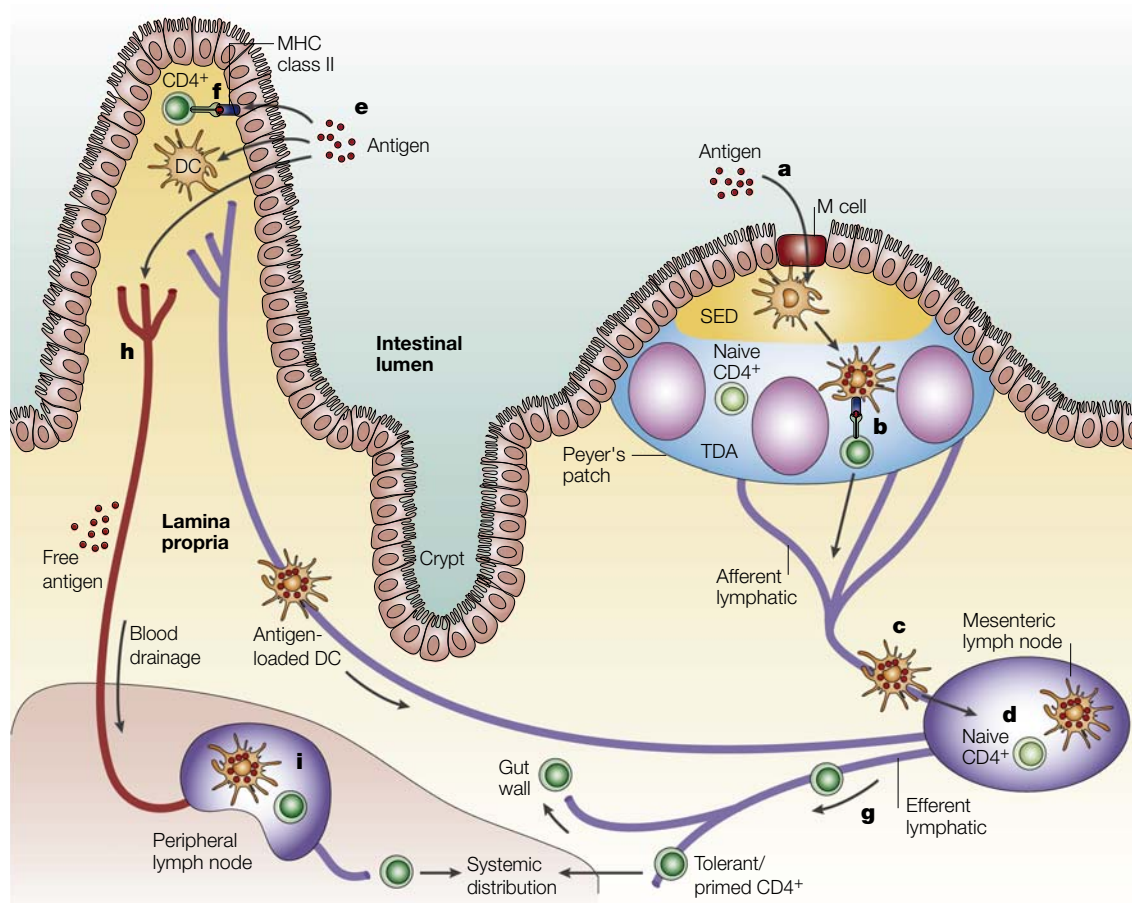


FIGURE 38.3 Structure and organization of the gut-associated lymphoid tissue. Antigen might enter through the microfold (M) cells (a), and after transfer to local dendritic cells (DCs), it might then be presented directly to T cells in the Peyer's patch (b). Alternatively, antigen or antigen-loaded DC from the Peyer's patch might gain access to draining lymph (c), with subsequent T-cell recognition in the mesenteric lymph nodes (d). A similar process of antigen or antigen-presenting cell dissemination to mesenteric lymph nodes might occur if antigen enters through the epithelium covering the lamina propria (e). In this case, there is also the possibility that enterocytes might act as local antigen-presenting cells (f). In all cases, the antigen-responsive $CD4^+$ T cells leave the mesenteric lymph nodes in the efferent lymph (g), and after entering the bloodstream through the thoracic duct, they exit into the mucosa through vessels in the lamina propria. T cells which have recognized antigen first in the mesenteric lymph node might also disseminate from the bloodstream throughout the peripheral immune system. Antigen might also gain direct access to the bloodstream from the gut (h) and interact with T cells in peripheral lymphoid tissues (i). SED, subepithelial dome; TDA, thymus-dependent area. Reprinted by permission from Ref. 5, copyright 2003.

patches, which are located directly beneath the epithelium in the lamina propria (Fig. 38.3)⁵. This region also contains M cells, which sample small particles from the gut lumen. Other lymphocytes are also present in the lamina propria, as well as associated with the epithelium itself. The GALT provides strong protective immunity against invasive pathogens, while tolerating food proteins and commensal bacteria. In order to achieve this, the GALT contains a variety of sensing and effector immune functions. Dendritic cells and M cells sample the gut content, while plasma B cells within the lamina propria produce IgA, providing protection against pathogenic organisms. The Peyer's patches allow for communication between immune cells resident within the GALT, propagation of signals to the wider systemic immune system, and the recruitment or efflux of immune cells.

The immune system in health and disease

Clearly, a well-functioning immune system is essential to health and serves to protect the host from the effects of ever-present pathogenic organisms. Cells of the immune system also have a role in identifying and eliminating cancer cells. The ability to discriminate between "self" and "nonself" is an essential requirement of the immune system and is normally achieved by the destruction of self-recognizing T and B lymphocytes before their maturation. However, since lymphocytes are unlikely to be exposed to all possible self-antigens during maturation, a second mechanism termed clonal anergy exists, which ensures that an encounter with a self-antigen induces tolerance. In some individuals, there is a breakdown of the mechanisms that normally preserve tolerance; a number of factors contribute to this, including a range of immunological abnormalities and a genetic predisposition in some individuals. As a result, an inappropriate immune response to host tissues or to normally benign environmental antigens is generated, and this can lead to autoimmune and inflammatory diseases, which are typified by ongoing chronic inflammation and a dysregulated T-cell response.

Factors influencing immune function

Many factors influence immune function and resistance to infection, leading to great variability in immune responses within the population.^{6,7} These factors include genetics, sex, early life events, age, and hormonal status. Immunological "history" also plays a role in the form of previous exposure to pathogens, vaccination history, and chronic disease burden (accumulating conditions over time). Other factors influencing immune function include stress (environmental, physiological, and psychological), exercise (acute and chronic), obesity (see below), smoking, alcohol consumption, and gut microbiota. In a newborn baby, immunologic competence is gained as the immune system encounters new antigens and so

matures and expands. Some of these early encounters with antigens play an important role in assuring tolerance, and a breakdown in this system can lead to increased likelihood of childhood atopic diseases (e.g., food allergy) and perhaps also to certain inflammatory conditions later in life.⁸ At the other end of the life course, older people experience a progressive dysregulation of the immune system, leading to decreased cell-mediated immunity and a greater susceptibility to infection.^{9,10} This age-related immune decline is termed immunosenescence. Innate immunity appears to be less affected by aging; indeed, there is a progressive increase in chronic inflammation during aging, sometimes called inflammaging.¹¹

II. KEY ISSUES

A. Infection Can Impair Nutrient Status

Undernutrition decreases immune defenses against invading pathogens and makes an individual more susceptible to infections. However, the immune response to an infection can itself impair nutritional status and alter body composition.^{2,3} Thus, there is a bidirectional interaction between nutrition, infection, and immunity (Fig. 38.4).

Infection impairs nutritional status and body composition in the following ways:^{2,3}

- Infection is characterized by anorexia. The reduction in food intake can range from as little as 5% to almost complete loss of appetite. This can lead to nutrient deficiencies even if the host is not deficient before the infection and may make apparent existing borderline deficiencies.
- Infection is characterized by nutrient malabsorption and loss. Infections that cause diarrhea or vomiting will result in nutrient loss. Apart from malabsorption, nutrients may also be lost through the feces as a result of damage to the intestinal wall caused by some infectious agents.

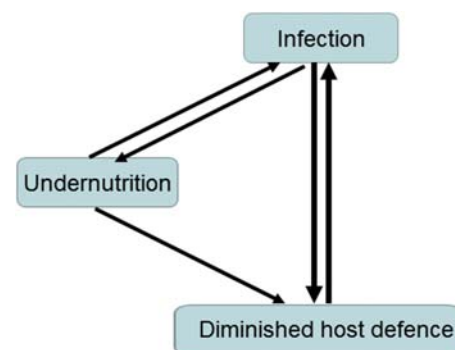


FIGURE 38.4 The interrelationship between undernutrition, immunity, and infection. Adapted from Ref. 3

- Infection is characterized by increased resting energy expenditure. Infection increases the basal metabolic rate during fever: each 1°C increase in body temperature is associated with a 13% increase in metabolic rate, which significantly increases energy requirements. This places a significant demand on nutrient supply, particularly when coupled with anorexia, diarrhea, and other nutrient losses (e.g., in urine and sweat).
- Infection is characterized by altered metabolism and redistribution of nutrients. The acute-phase response is the name given to the metabolic response to infections, and it includes the onset of fever and anorexia, the production of specific acute-phase reactants, and the activation and proliferation of immune cells. This catabolic response occurs with all infections, even when they are subclinical, and serves to bring about a redistribution of nutrients away from skeletal muscle and adipose tissue and toward the host immune system. This redistribution is mediated by the production of proinflammatory cytokines by leukocytes and associated endocrine changes. Amino acids, mobilized from skeletal muscle, are used by the liver for the synthesis of acute-phase proteins (e.g., C-reactive protein) and by leukocytes for the synthesis of immunoglobulin and cytokines. The average loss of protein over a range of infections has been estimated to be 0.6–1.2 g/kg body weight per day.¹²

It is clear that inflammatory cytokines mediate many of the effects that lead to compromised nutritional status following an infection, including anorexia, increased energy expenditure, and redistribution of nutrients, while malabsorption and maldigestion are brought about by the pathogen itself. The result is that an increased nutrient requirement coincides with reduced

nutrient intake, reduced nutrient absorption, and nutrient losses (Fig. 38.5).

B. Nutrient Intake and Status Affect Immune Function

Although the immune system is functioning at all times, specific immunity becomes activated when the host is challenged by pathogens. This activation is associated with a marked increase in the demand of the immune system for substrates and nutrients to provide a ready source of energy, which can be supplied from exogenous sources (i.e., from the diet) and/or from endogenous pools. The cells of the immune system are metabolically active and are able to utilize glucose, amino acids, and fatty acids as fuels.¹³ Energy generation involves electron carriers and a range of coenzymes, which are usually derivatives of vitamins. The final component of the pathway for energy generation (the mitochondrial electron transfer chain) includes electron carriers that have iron or copper at their active site. Activation of the immune response gives rise to the production of proteins (immunoglobulins, cytokines, cytokine receptors, adhesion molecules, acute-phase proteins) and lipid-derived mediators (prostaglandins, leukotrienes [LTs]). To respond optimally, there must be the appropriate enzymic machinery in place (for RNA synthesis and protein synthesis and their regulation) and ample substrate available (nucleotides for RNA synthesis, the correct mix of amino acids for protein synthesis, polyunsaturated fatty acids (PUFAs) for eicosanoid synthesis). An important component of the immune response is oxidative burst, during which superoxide anion radicals are produced from oxygen in a reaction linked to the oxidation of NADPH. The reactive oxygen species produced can be damaging to host tissues and

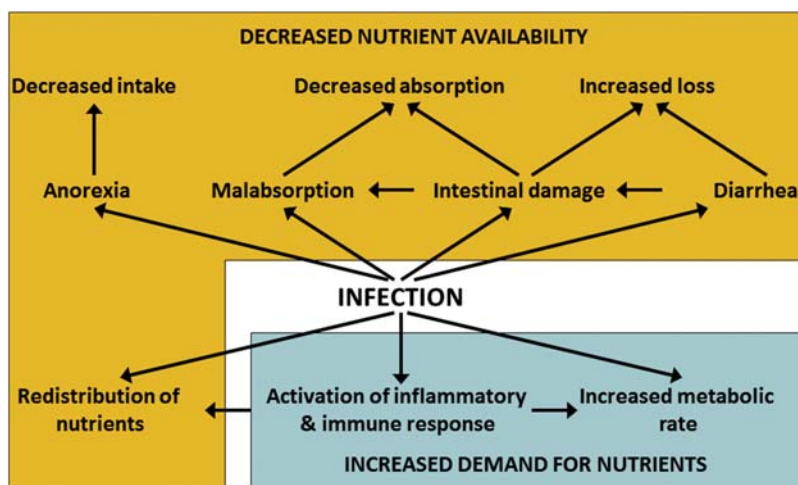


FIGURE 38.5 Effects of infection on the host which can decrease nutrient status. Adapted from Ref. 4

thus antioxidant protective mechanisms are necessary. Among these are the classic antioxidant vitamins (vitamins E and C), glutathione, the antioxidant enzymes superoxide dismutase and catalase, and the glutathione-recycling enzyme glutathione peroxidase. The antioxidant enzymes all have metal ions at their active site (manganese, copper, zinc, iron, selenium). Cellular proliferation is a key component of the immune response, providing amplification and memory: before division, there must be replication of DNA and then of all cellular components (proteins, membranes, intracellular organelles, etc.). In addition to energy, this clearly needs a supply of nucleotides (for DNA and RNA synthesis), amino acids (for protein synthesis), fatty acids, bases and phosphate (for phospholipid synthesis), and other lipids (e.g., cholesterol) and cellular components. Although nucleotides are synthesized mainly from amino acids, some of the cellular building blocks cannot be synthesized in mammalian cells and must come from the diet (e.g., essential fatty acids, essential amino acids, minerals). Amino acids (e.g., arginine) are precursors for the synthesis of polyamines, which have roles in the regulation of DNA replication and cell division. Various micronutrients (e.g., iron, folic, zinc, magnesium) are also involved in nucleotide and nucleic acid synthesis. Thus, the roles for nutrients in immune function are many and varied, and it is easy to appreciate that an adequate and balanced supply of these is essential if an appropriate immune response is to be mounted.^{1–4}

C. Assessing the Effect of Nutrition on Immune Function is Difficult

There is a wide range of methodologies that can be used to assess the impact of nutrients on immune function.^{7,14,15} Assessments can be made of cell functions *ex vivo* (i.e., of the cells isolated from animals or humans subjected to dietary manipulation and studied in short- or long-term culture) or of indicators of immune function *in vivo* (e.g., by measuring the concentrations of proteins relevant to immune function in the bloodstream or the response to an immunological challenge). Table 38.2 lists some examples of approaches used for the assessment of immune function. However, the biological relevance of many of these markers of immune function remains unclear, and there is no single marker that can be used to draw conclusions about modulation of the immune system as a whole, apart from infection as a clinical outcome. An expert group identified measures of delayed-type hypersensitivity (DTH; the cell-mediated immune response to controlled application of an antigen usually through the skin), the response to vaccination, and the production of secretory IgA as the most suitable assessments of immune function in humans.^{14,15}

TABLE 38.2 Approaches used to assess the effect of nutrition on immune function.

<i>In vivo measures</i>
Size of lymphoid organs
Cellularity of lymphoid organs
Numbers of cells circulating in bloodstream
Cell surface expression of molecules involved in immune response (e.g., antigen presentation)
Circulating concentrations of Ig specific for antigens after an antigen challenge (e.g., vaccination)
Concentration of secretory IgA in saliva, tears, and intestinal washings
Delayed-type hypersensitivity response to intradermal application of antigen
Response to challenge with live pathogens (mainly animal studies—outcome usually survival)
Incidence and severity of infectious diseases—widely used in human studies
<i>Ex vivo measures</i>
Phagocytosis by neutrophils and macrophages
Oxidative burst by neutrophils and macrophages
Natural killer cell activity against specific target cells (usually tumor cells)
Cytotoxic T lymphocyte activity against specific target cells
Lymphocyte proliferation (following stimulation with an antigen or mitogen)
Production of cytokines by lymphocytes and macrophages (usually following stimulation)
Production of immunoglobulins by lymphocytes
Cell surface expression of molecules involved in cellular activation

III. ISSUES SPECIFICALLY RELATED TO NUTRITION

A. Obesity and Immune Function

Comparisons between lean and obese individuals indicate that obesity is associated with impairments of the bacterial killing capacity of granulocytes, T-lymphocyte proliferation, natural killer cell activity, and DTH, with increased susceptibility to infection, and with poorer outcome from vaccinations and infection.¹⁶ Factors derived from adipose tissue, such as the cytokine-like hormone leptin, may play a role in immune regulation and may explain some of the effects of obesity on immunity.¹⁷ Adipose tissue is infiltrated with immune cells, notably, but not exclusively, macrophages, and releases a range of inflammatory mediators.¹⁸ As such, obesity is a state of chronic low-grade inflammation.¹⁹

B. Protein-Energy Malnutrition and Immune Function

Protein-energy malnutrition, although often considered a problem solely of developing countries, has been described in even the most affluent of countries. It is important to recognize that protein-energy malnutrition often coexists with micronutrient deficiencies, and poor outcome from intervention can result from a lack of awareness of multiple deficiencies. Practically all forms of immunity may be affected by protein-energy malnutrition but nonspecific defenses and cell-mediated immunity are more severely affected than humoral (antibody) responses.^{1,20–22} Protein-energy malnutrition causes atrophy of the lymphoid organs (thymus, spleen, lymph nodes, tonsils) in laboratory animals and humans.^{1,20–22} There is a decline in the number of circulating lymphocytes, which is proportional to the extent of malnutrition, and the proliferative responses of T lymphocytes to mitogens and antigens is decreased by malnutrition, as is the synthesis of IL-2 and IFN- γ and the activity of natural killer cells.^{1,20–22} Production of cytokines by monocytes (TNF- α , IL-6, and IL-1 β) is also decreased by malnutrition, although their phagocytic capacity appears to be unaffected.^{1,20–22} The *in vivo* skin DTH response to challenge with specific antigens is reduced by malnutrition.^{1,20–22} However, numbers of circulating B lymphocytes and immunoglobulin levels do not seem to be affected or may even be increased by malnutrition^{1,20–22}; underlying infections may influence these latter observations.

C. Individual Micronutrients and Immune Function

Overview

Much of what is known about the impact of single nutrients on immune function comes from studies of deficiency states in animals and humans and from controlled animal studies in which the nutrients are included in the diet at known levels. There is now overwhelming evidence from these studies that particular nutrients are required for an effective immune response and that deficiencies in one or more of these nutrients diminish immune function and provide a window of opportunity for infectious agents. It is logical that multiple nutrient deficiencies might have a more significant impact on immune function, and therefore resistance to infection, than a single nutrient deficiency. What is also apparent is that excess amounts of some nutrients also impair immune function and decrease resistance to pathogens. Thus, for some nutrients, there may be a relatively narrow range of intake that is associated with optimal immune function.

Vitamin A

The vitamin A (or retinoid) family includes retinol, retinal, retinoic acid, and esters of retinoic acid. There is a transitory decrease in serum retinol during the acute-phase response which follows infection, which is likely to be due to decreased synthesis of retinol binding protein (RBP) by the liver, and therefore decreased release of retinol-RBP by the liver, and also to increased vascular permeability at sites of inflammation, allowing leakage into the extravascular space.²³ For these reasons, serum retinol cannot be used as an indicator of vitamin A status in individuals with an active acute-phase response.²³

Vitamin A is essential for maintaining epidermal and mucosal integrity²⁴; vitamin A-deficient mice have histopathological changes in the gut mucosa consistent with a breakdown in gut barrier integrity and impaired mucus secretion (due to loss of mucus producing goblet cells), both of which would facilitate entry of pathogens through this route.²⁴ Vitamin A regulates keratinocyte differentiation, and vitamin A deficiency induces changes in skin keratinization, which may explain the observed increased incidence of skin infection.²⁴ Many aspects of innate immunity, in addition to barrier function, are affected by vitamin A.^{24–29} It modulates gene expression to control the maturation of neutrophils; in vitamin A deficiency, there are increased neutrophil numbers, but decreased phagocytic function.^{24–29} Macrophage-mediated inflammation is increased by vitamin A deficiency, but the ability to ingest and kill bacteria is impaired.^{24–29} Vitamin A deficiency may therefore lead to more severe infections, coupled with excessive inflammation.^{24–29} Natural killer cell activity is diminished by vitamin A deficiency.^{24–29} There is evidence that vitamin A deficiency alters the Th1/Th2 balance, decreasing the Th2 response, but often without affecting the Th1 response.^{24–29} Vitamin A also appears to be important in differentiation of regulatory T cells while suppressing Th17 differentiation, effects which have implications for control of adverse immune reactions.^{24–29} Retinoic acid seems to promote movement of T cells to the GALT,²⁶ and, interestingly, some gut-associated immune cells are able to synthesize retinoic acid.²⁶

The impact of vitamin A deficiency on infectious disease has been studied widely in the developing world.^{2,3,24,30–32} Vitamin A deficiency is associated with increased morbidity and mortality in children and appears to predispose to respiratory infections, diarrhea, and severe measles.^{2,3,24,30–32} Although vitamin A deficiency increases the risk of infectious disease, the interaction is bidirectional such that infections can lead to vitamin A deficiency: diarrhea, respiratory infections, measles, chickenpox, and human immunodeficiency virus infection are all associated with the development of vitamin A deficiency.^{2,3,24,30–32} Replenishment of vitamin

A in deficient children by provision of supplements decreases mortality in areas of the world where deficiency is a problem.^{2,3,24,30–32}

B vitamins

B vitamins participate as coenzymes in the synthesis of nucleic acids and proteins, pathways which are crucial for many aspects of immune function. There is some suggestion that folate supplementation of elderly individuals improves immune function, and in particular, natural killer cell activity, although this is not conclusive.³³ Elderly subjects supplemented for 4 months with a combination of folic acid (400 µg/day), vitamin E (120 IU/day), and vitamin B12 (3.8 µg/day) were reported to have increased natural killer cell activity and fewer infections in one study.³⁴ Elderly individuals tend to be at risk of vitamin B12 deficiency, and studies have shown that subjects >65 years with low serum vitamin B12 concentrations have impaired antibody responses to vaccination.³⁵ Patients with vitamin B12 deficiency also have decreased numbers of lymphocytes and suppressed natural killer cell activity, which may be reversed with supplementation.³⁶ Vitamin B6 deficiency in laboratory animals causes thymus and spleen atrophy and decreases lymphocyte proliferation and the DTH response. In a study in healthy elderly humans, a vitamin B6-deficient diet (3 µg/kg body weight per day or about 0.17 and 0.1 mg/day for men and women, respectively) for 21 days resulted in a decreased percentage and total number of circulating lymphocytes, decreased T- and B-cell proliferation in response to mitogens, and decreased IL-2 production.³⁷ Repletion at 15 or 22.5 µg/kg body weight per day for 21 days did not return the immune functions to starting values; however, repletion at 33.75 µg/kg body weight per day (about 1.9 and 1.1 mg/day for men and women, respectively) returned immune parameters to starting values. This comprehensive study indicates that vitamin B6 deficiency impairs human immune function and that the impairment is reversible by repletion.

Vitamin C

Vitamin C is a water-soluble antioxidant found in high concentrations in circulating leukocytes and appears to be utilized during infections.³⁸ High circulating levels of vitamin C are associated with enhanced antibody responses, neutrophil function, and antiviral activity in animal studies.³⁸ Several studies suggest a modest benefit of vitamin C supplementation, at doses ranging 1000–8000 mg/d in reducing the duration, but not the incidence, of respiratory infections.^{39,40} The potential benefits and risks of vitamin C supplementation at doses above 8000 mg/d, and the role of vitamin C in nonrespiratory infections, have not been well investigated.⁴⁰

Vitamin D

The active form of vitamin D (1,25-dihydroxy vitamin D₃) is referred to here as vitamin D. Vitamin D receptors have been identified in most immune cells,^{41,42} and in addition, some cells of the immune system, particularly macrophages, can synthesize the active form of vitamin D from its precursor,^{41,42} suggesting that vitamin D is likely to have immunoregulatory properties. Indeed, vitamin D can induce macrophages to synthesize antimicrobial peptides such as cathelicidin,⁴³ directly affecting host defense. Some reports suggest immune defects in vitamin D-deficient patients and experimental animals^{41,42,44,45} and that these translate into increased susceptibility to infections.⁴⁶ Individuals with low vitamin D status have a higher risk of viral respiratory tract infections.⁴⁷ Supplementation of Japanese schoolchildren with vitamin D (1200 U/day) for 4 months during winter decreased by about 40% the risk of influenza.⁴⁸ These studies suggest that vitamin D acts to improve immune function and host defense. However, there is, paradoxically, a large body of literature supporting an immunosuppressive role of vitamin D and related analogs.⁴⁴ The current view is that under physiological conditions, vitamin D probably facilitates immune responses and that it may also play an active role in the prevention of autoimmunity and that there may be a therapeutic role for vitamin D in some immune-mediated diseases.^{41,43–46} Vitamin D acts by binding to its receptor and regulating gene expression in target cells. Its effects include promotion of phagocytosis, superoxide synthesis, and bacterial killing, but it is also reported to inhibit T cell proliferation, production of Th1 cytokines, and B cell antibody production, highlighting the paradoxical nature of its effects.^{41,42,44,45} The role of vitamin D in autoimmunity is particularly interesting; there is increasing evidence, mainly from animal studies, that vitamin D deficiency is linked with autoimmune diseases such as multiple sclerosis, rheumatoid arthritis, and inflammatory bowel disease.⁴⁹ The inhibition of Th1-type immune activity, which underlies many autoimmune conditions, by vitamin D is thought to be key to this link.⁴⁹ Taken together, the evidence suggests that vitamin D is a selective regulator of immune function and the effects of vitamin D deficiency, vitamin D receptor deficiency, or vitamin D supplementation depend on the immunological situation (e.g., health, infectious disease, autoimmune disease).^{41,42,44,46,49}

Vitamin E

Vitamin E is the major lipid-soluble antioxidant in the body and is required for protection of membrane lipids from peroxidation. Free radicals and lipid peroxidation are immunosuppressive; thus, it is considered that vitamin E should act to optimize and even enhance the immune response. Indeed, a positive association exists

between plasma vitamin E levels and DTH responses, and a negative association has been demonstrated between plasma vitamin E levels and the incidence of infections in healthy adults over 60 years of age.⁵⁰ There appears to be particular benefit of vitamin E supplementation for the elderly,^{51,52} with studies demonstrating enhanced Th1 cell-mediated immunity at high doses. A comprehensive study demonstrated increased DTH responses in elderly subjects supplemented with 60, 200, and 800 mg vitamin E/day, with maximal effect at a dose of 200 mg/day.⁵³ This dose also increased the antibody responses to hepatitis B, tetanus toxoid, and pneumococcus vaccinations. This “optimal” dose of 200 mg vitamin E/day is well in excess of the recommended dietary intake; thus, it appears that adding vitamin E to the diet at levels beyond those normally recommended may enhance some immune functions above normal, and it has even been argued that the recommended intake for vitamin E is not adequate for optimal immune function.⁵⁴ However, RCTs do not consistently support a role for vitamin E supplementation in reducing the incidence, duration, or severity of respiratory infections in elderly populations,⁵⁵ although one large study did show benefit specifically for upper respiratory tract infections.⁵⁶

Zinc

Zinc is important for DNA synthesis and in cellular growth and differentiation and antioxidant defense. It is also a cofactor for many enzymes. Zinc deficiency impairs many aspects of innate immunity, including phagocytosis by macrophages and neutrophils, natural killer cell activity, respiratory burst, and complement activity, all of which could be important contributors to increased susceptibility to infection.^{57–62} Zinc deficiency has a marked impact on bone marrow, decreasing the number of nucleated cells and the number and proportion of cells that are lymphoid precursors.⁶³ In patients with zinc deficiency related to sickle-cell disease, natural killer cell activity is decreased, but it can be returned to normal by zinc supplementation.⁶⁴ In acrodermatitis enteropathica, which is characterized by reduced intestinal zinc absorption, thymic atrophy, impaired lymphocyte development, and reduced lymphocyte responsiveness and DTH are observed.⁶⁵ Moderate or mild zinc deficiency or experimental zinc deficiency (induced by consumption of <3.5 mg zinc/day) in humans results in decreased natural killer cell activity, lymphocyte proliferation, IL-2 production, and DTH response; all can be corrected by zinc repletion.⁶⁶

Low plasma zinc levels can be used to predict the subsequent development of lower respiratory tract infections and diarrhea in malnourished populations.⁶⁷ Indeed, diarrhea is considered a symptom of zinc deficiency, and several studies show that zinc supple-

mentation decreases the incidence, duration, and severity of childhood diarrhea.⁶⁷ Effects of zinc on respiratory tract infections are less clear. Most (but not all) studies fail to show a benefit of zinc supplementation in respiratory disease in malnourished populations and available trials on the effect of zinc on the common cold in nonmalnourished populations report conflicting results.⁶⁷ Very high zinc intakes can result in copper depletion,⁶⁸ and copper deficiency impairs immune function and increases susceptibility to infection.⁶⁸

Iron

Iron deficiency has multiple effects on immune function in laboratory animals and humans.^{69–72} However, the relationship between iron deficiency and susceptibility to infection remains controversial.^{69,73,74} Evidence suggests that infections caused by organisms that spend part of their life cycle intracellularly, such as plasmodia, mycobacteria, and invasive salmonellae, may actually be enhanced by iron therapy.^{69,72–74} In the tropics, in children of all ages, at doses of >2 mg/kg/d, iron has been associated with increased risk of malaria and other infections, including pneumonia.⁶⁹ For these reasons, iron intervention in malaria-endemic areas is not advised, particularly high doses in the young, those with compromised immunity (e.g., HIV infection), and during the peak malaria transmission season.^{69,73,74} Iron treatment for anemia in a malarious area must be preceded by effective antimalarial therapy and should be oral, rather than parenteral.⁶⁹ The detrimental effects of iron administration may occur because microorganisms require iron and providing it may favor the growth and replication of the pathogen.^{69,72–74} Indeed, it has been argued that the decline in circulating iron concentrations that accompanies infection is an attempt by the host to “starve” the infectious agent of iron. There are several mechanisms for withholding iron from a pathogen in this way.^{69,72–74} Lactoferrin has a higher binding affinity for iron than do bacterial siderophores, making bound iron unavailable to the pathogen.^{69,72–74} Furthermore, once lactoferrin reaches 40% saturation with iron, it is sequestered by macrophages.^{69,72–74} It is notable that breast milk contains lactoferrin,⁷⁵ which may protect against the use of free iron by pathogens transferred to an infant. It is important to note that oral iron supplementation has not been shown to increase risk of infection in nonmalarious countries.⁶⁹

Selenium

Selenium is essential for an effectively functioning immune system. Deficiency in laboratory animals affects both innate and adaptive immunity, particularly neutrophil function.^{76–78} It also increases susceptibility to bacterial, viral, fungal, and parasitic challenges,

and lower selenium concentrations in humans have also been linked with increased virulence, diminished natural killer cell activity, increased mycobacterial disease, and HIV progression.^{79–81} Selenium supplementation has been shown to improve various aspects of immune function in humans,^{82–85} including the elderly.⁸⁶ Selenium supplementation (50 or 100 µg/day) in adults with low selenium status improved some aspects of their immune response to a poliovirus vaccination.⁸⁷

D. Dietary Fat and Immune Function

Eicosanoids: a link between fatty acids and the immune system

Fatty acids can affect immune function through a variety of actions, including alterations in membrane structure, cell signaling mechanisms, and gene expression.⁸⁸ However, the best described mechanism of action of PUFAs is through influencing the production of a class of lipid mediators termed eicosanoids that play important roles in regulating immunity and inflammation.⁸⁹ The membranes of most immune cells contain large amounts of arachidonic acid, which is the principal precursor for eicosanoid synthesis. Arachidonic acid in cell membranes can be mobilized by various phospholipase enzymes, most notably phospholipase A₂, and the free arachidonic acid can subsequently act as a substrate for cyclooxygenase enzymes, forming prostaglandins and related compounds, or for one of the lipoxygenase enzymes, forming LTs and related compounds. Di-homo-γ-linolenic acid and eicosapentaenoic acid (EPA) are also precursors for eicosanoid synthesis and the mediators produced from each different substrate have different structures and different biological potencies. Eicosanoids are formed in a cell-specific manner and may have opposing effects to one another, so the overall physiological or pathophysiological effect will be governed by the nature of the cells producing the eicosanoids, the concentrations of the different eicosanoids, the timing of their production, and the sensitivities of target cells to their effects.

Essential fatty acid deficiency and immune function

Animal studies have shown that deficiencies in both linoleic and α-linolenic acids result in decreased thymus and spleen weight and reduced lymphocyte proliferation, neutrophil chemotaxis, macrophage-mediated cytotoxicity, and DTH response.⁹⁰ Thus, the immunological effects of essential fatty acid deficiency appear to be similar to the effects of single micronutrient deficiencies, although there are no human studies to confirm this (essential fatty acid deficiency is very rare in humans). Essential fatty acid deficiency probably has its effects

because cells of the immune system require PUFAs for membrane synthesis and as precursors for the synthesis of eicosanoids.

Amount of dietary fat and immune function

High-fat diets have been reported to result in diminished immune cell functions (both natural and cell-mediated immunity) compared with low-fat diets in both humans and experimental animals, but the precise effect depends on the exact level of fat used in the high-fat diet and its source.⁹⁰

Effect of specific fatty acids or fatty acid families on immune function

Saturated fatty acids appear to have little impact on humoral or cell-mediated immune function, but cell culture, animal model, and human epidemiological studies all indicate that some saturated fatty acids promote inflammation.⁹¹ N-6 PUFAs are also believed to promote inflammation through the actions of eicosanoids, but human studies do not fully support this contention.⁹² Cell culture and animal model studies have shown that marine n-3 PUFAs (EPA and docosahexaenoic acid [DHA]) suppress both inflammation and cell-mediated immune responses,^{88,93} acting through a variety of mechanisms including altered

- membrane structure
- membrane and intracellular signaling processes
- gene expression profiles
- production of eicosanoids.

EPA and DHA also give rise to resolvins and related lipid mediators (protectins, maresins) that have potent antiinflammatory and inflammation-resolving activities.⁹⁴

Epidemiological studies in humans and studies in which human subjects are given marine n-3 PUFAs demonstrate that these fatty acids are antiinflammatory, but less consistent effects of cell-mediated immunity are demonstrated.⁸⁸ Consistent with their antiinflammatory effects, marine n-3 PUFAs have some treatment efficacy in certain chronic inflammatory conditions.^{88,93} There is little data on fatty acids and infection in humans.

E. Dietary Amino Acids and Immune Function

Sulfur amino acids

Sulfur amino acids are essential in humans. Deficiency in methionine and cysteine results in atrophy of the thymus, spleen, and lymph nodes in mice and prevents recovery from protein-energy malnutrition.⁹⁵ When combined with a deficiency of isoleucine and valine, also essential amino acids, sulfur amino acid deficiency results in severe depletion of gut lymphoid tissue in chickens,⁹⁶ very similar to the effect of protein deprivation. Glutathione is a tripeptide that consists of

glycine, cysteine, and glutamate. It is recognized to have antioxidant properties. Glutathione concentrations in the liver, lung, small intestine, and immune cells fall in response to inflammatory stimuli (probably as a result of oxidative stress), and this fall can be prevented in some organs by the provision of cysteine in the diet.⁹⁷ Although the limiting precursor for glutathione biosynthesis is usually cysteine, the ability of sulfur amino acids to replete glutathione stores is related to the protein level of the diet.⁹⁷ Glutathione can enhance the activity of cytotoxic T cells, while depletion of intracellular glutathione diminishes lymphocyte proliferation and the generation of cytotoxic T lymphocytes.⁹⁸

Arginine

Arginine is a nonessential amino acid in humans and is involved in protein, urea and nucleotide synthesis, and adenosine triphosphate generation. It also serves as the precursor of nitric oxide, a potent immunoregulatory mediator that is cytotoxic to tumor cells and to some microorganisms. In laboratory animals, arginine decreases the thymus involution associated with trauma, promotes thymus repopulation and cellularity, increases lymphocyte proliferation, natural killer cell activity, and macrophage cytotoxicity, improves DTH, increases resistance to bacterial infections, increases survival to sepsis and burns, and promotes wound healing.^{99,100} There are indications that arginine may have similar effects in humans, although these have not been tested thoroughly.

Glutamine

Glutamine is the most abundant amino acid in the blood and in the free amino acid pool in the body; skeletal muscle is considered to be the most important glutamine producer in the body. Once released from skeletal muscle, glutamine acts as an interorgan nitrogen transporter. One important user of glutamine is the immune system.¹⁰¹ Plasma glutamine levels are lowered (by up to 50%) by sepsis, injury, and burns and following surgery.¹⁰¹ Furthermore, the skeletal muscle glutamine concentration is lowered by more than 50% in at least some of these situations.¹⁰¹ These observations indicate that a significant depletion of the skeletal muscle glutamine pool is characteristic of trauma. The lowered plasma glutamine concentrations that occur are likely to be the result of demand for glutamine (by the liver, kidney, gut, and immune system) exceeding the supply, and it has been suggested that the lowered plasma glutamine contributes, at least in part, to the impaired immune function that accompanies such situations.¹⁰¹ It has been argued that restoring plasma glutamine concentrations in these

situations should restore immune function. As with arginine, there are animal studies to support this.^{101,102} Clinical studies, mainly using intravenous infusions of solutions containing glutamine, have also reported beneficial effects for patients undergoing bone marrow transplantation and colorectal surgery, patients in intensive care and low-birth weight babies, all of whom are at risk from infection and sepsis.¹⁰³ In some of these studies, improved outcome was associated with improved immune function. In addition to a direct immunological effect, glutamine, even provided intravenously, improves gut barrier function in patients at risk of infection.¹⁰¹ This would have the benefit of decreasing the translocation of bacteria from the gut and eliminating a key source of infection.

F. Probiotics, Prebiotics, and Immune Function

Commensal bacteria are believed to contribute to the immunological protection of the host by creating a barrier against colonization by pathogenic bacteria. This barrier can be disrupted by disease and by the use of antibiotics, so allowing easier access to the host gut by pathogens. It is now believed that this barrier can be maintained by providing supplements containing live "desirable" bacteria; such supplements are termed probiotics.^{104,105} Probiotic organisms are found in fermented foods including traditionally cultured dairy products and some fermented milks. The organisms used commercially as probiotics are typically lactobacilli or bifidobacteria.¹⁰⁵ These organisms only colonize the gut temporarily, making regular consumption necessary. In addition to creating a barrier effect, some of the metabolic products of probiotic bacteria (e.g., lactic acid and a class of antibiotic proteins termed bacteriocins produced by some bacteria) may inhibit the growth of pathogenic organisms.¹⁰⁵ Probiotic bacteria may also compete with pathogenic bacteria for nutrients and may enhance the gut immune response to pathogenic bacteria.¹⁰⁵ Probiotics have various routes for internalization by the gut epithelium and contact with underlying immune tissues; it is through these interactions that probiotics are thought to be able to influence immune function.^{104,105} However, the nature of this regulation is not very well understood. A number of studies have examined the influence of various probiotic organisms, either alone or in combination, on immune function, infection, and inflammatory conditions in humans.¹⁰⁶ Probiotics appear to enhance innate immunity (particularly phagocytosis and natural killer cell activity) but have lesser effects on adaptive immunity.¹⁰⁶ Probiotics have been shown to reduce the incidence and duration of diarrheal disease in different settings.^{107–110} Probiotics may also reduce respiratory

disease in children and adults.^{111,112} One of the difficulties in interpreting results from probiotic studies is that there may be significant species and strain differences in the effects of probiotics, as well as differences in doses used, duration of treatment, and subject characteristics.

Prebiotics are typically, though not exclusively, carbohydrates that are not digestible by mammalian enzymes but which are selectively fermented by gut microbiota, leading to increased numbers of beneficial bacteria within the gut.¹¹³ Although there is growing evidence for potential immunomodulatory effects of prebiotics,¹¹⁴ it is not clear whether they are direct effects or manifested through alteration of the gut microbiota. Many types of bacteria favored by prebiotics produce short-chain fatty acids (e.g., acetate, propionate, and butyrate) as the end products of metabolism.¹¹³ These short-chain fatty acids are of a benefit to the host gut epithelium and may also have favorable systemic effects once absorbed.¹¹³

G. Breastfeeding and Immune Function

The composition of breast milk

Breast milk contains a wide range of immunologically active components, including cells (macrophages, T and B lymphocytes, neutrophils), immunoglobulins (IgG, IgM, IgD, sIgA), lysozyme (which has direct antibacterial action), lactoferrin (which binds iron, so preventing its uptake by bacteria), cytokines (IL-1, IL-6, IL-10, IFN- γ , TNF- α , transforming growth factor- β), growth factors (epidermal growth factor, insulin-like growth factor), hormones (thyroxine), fat-soluble vitamins (vitamins A, D, E), amino acids (taurine, glutamine), fatty acids, amino sugars, nucleotides, gangliosides, and prebiotic oligosaccharides.¹¹⁵ Breast milk also contains factors that prevent the adhesion of some microorganisms to the gastrointestinal tract and so prevents bacterial colonization.¹¹⁵ Human breast milk contains factors that promote the growth of useful bacteria (e.g., bifidobacteria) in the gut.¹¹⁵ The content of many factors varies among milks of different species and is different between human breast milk and infant formulas.

Breastfeeding and infection

Breastfeeding plays a key role in the prevention of infectious disease, particularly diarrhea and gastrointestinal and lower respiratory infections, in both developing

and developed countries.^{116,117} In addition to preventing infectious disease, breastfeeding enhances the antibody responses to vaccination.¹¹⁸ Breastfeeding may provide better protection against diarrhea (up to 6 months of age) than against deaths due to respiratory infections.^{116,117} There are also geographical influences on the protection afforded by breastfeeding; in some continents, protection can be observed throughout the first year of life, whereas in others, it is much more short-lived.^{116,117}

The development of the immune system in early life is influenced by both feeding practices and environmental exposures. Breastfeeding provides passive immunity to the infant, for example, via transfer of antibodies and cytokines.¹¹⁵ Breast milk components can also stimulate maturation of the GALT, with breast milk known to be rich in bifidogenic oligosaccharides and to contain its own unique microbiota.¹¹⁵ Human milk oligosaccharides (HMOs) are synthesized from lactose in the mammary gland, and the specific HMO profile will vary between individuals and across contexts and changes over the time course of lactation.¹¹⁹ These HMOs have been found to confer health benefits to infants by inhibiting the adhesion of microorganisms to the intestinal mucosa, enhancing the production of short-chain fatty acids by bacteria within the microbiome, inhibiting inflammation, and promoting immune maturation.^{119,120} Other immune active components of breast milk are also likely to be involved in immune system maturation, with studies identifying that the growth factors epidermal growth factor, fibroblast growth factor 21, and transforming growth factor 2 can change lymphocyte phenotypes in newborn rats when provided as supplements by oral gavage.¹²¹

H. Importance of the Gut Microbiota

An early point of contact between nutrients and the immune system occurs within the intestinal tract. Relatively little is known about the relationship between nutrient status and the function of the gut-associated immune system. This is of particular relevance when considering adverse reactions to foods: the role of immunoregulatory nutrients in responses to food components and in sensitization to food-borne allergens is largely unknown. An understanding of the interaction between nutrients, the types of bacteria that inhabit the gut, and gut-associated and systemic immune responses is only now beginning to emerge.^{122,123}

RESEARCH GAPS

- The effects of nutrients on newly discovered immune components (e.g., Th17 cells) are not known
- The effects of nutrient combinations, whole foods, and diets on human immune function are poorly explored
- The influence of polymorphisms in genes related to the immune response on the impact of nutrition on immune outcomes is poorly explored
- The impact of nutritional exposures in early life on later immune outcomes, susceptibility to infection, and risk of immune-mediated disease is not well described
- Whether nutrition can be used to prevent or slow immunosenescence is not known
- A deeper description of the impact of obesity on immune competence in humans is needed
- A clearer elucidation of the influence of probiotics, prebiotics, and their combination (synbiotics) on immune function through well-designed studies using robust immune outcomes is needed
- The role of the gut microbiota in shaping host immune maturation and immune responses needs better definition
- The role of nutrition in early immune maturation and the mechanisms involved are underexplored

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